OpenText™ Service Manager

Administer

Version 9.82

PDF Generated on October 12, 2026

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Administer

This section provides information on how to administer Service Manager.

Configuring installation and setup options

System configuration parameters

Smart Analytics administration

Mobile Applications administration

Service Manager maintenance tasks

Application setup

Database administration

System security

Server performance tuning

Scheduled maintenance

Status and notifications

Delta Migration Tool

Calendar administration

Time Period management

1. Configuring installation and setup options

Service Manager system installation and setup includes a number of implementation options you can configure after installation. The following table lists the options you can enable or configure from a new development environment installation.

Feature	Description	Default state in new installations
Computer Telephony Integration (CTI) with the Web client	An integration option allowing Web clients to access telephony software	Disabled
Encryption of configuration file settings	A security option that protects values listed in the configuration file	Disabled
Encryption of operator passwords	A security option that protects the passwords listed in operator records	Enabled
Event Services	A group of applications that allow Service Manager to send data to and to receive data from external systems	Disabled
Lightweight Directory Access Protocol (LDAP)	An integration option that allows Service Manager to access data stored in external directory services	Disabled
Horizontal scaling implementation	An implementation option that uses servlet implementation features to manage Service Manager instances running on multiple hosts	Disabled
Servlet implementation Single servlet implementation	An implementation option where a single servlet container process manages all client connections	Enabled
Requirements for a vertical scaling and required SSL implementation	A vertical scaling implementation option running in combination with the "Required SSL" security option where two or more servlet container processes manage all client connections on a single host	Disabled
External Web Services	An integration option that allows Service Manager to send data to and to receive data from external Web Services	Enabled
Web Services	An integration option that allows Web Services clients to access Service Manager applications	Disabled

Related topics

Servlet implementation
Server implementation options
Upgrade Service Manager

1.1. AutoPass licensing

All Service Manager installations include a bundled version of AutoPass that no longer requires a separate installation of the AutoPass software. AutoPass is a license retrieval method offered by the company. The bundled version of AutoPass is specifically for verifying the Service Manager license and as such requires you to place the `LicFile.txt` file in your server's `RUN` folder. For information about AutoPass settings and options, see *Generate and install your Service Manager licenses* in Install the Service Manager Server.

1.2. JRE support

Service Manager supports Open Java Development Kit (OpenJDK) for all SM components that require a JRE or JDK to work.

Important

Out-of-box, an OpenJDK JRE is bundled with the Windows Server, Linux Server, and Windows Client so that you don't have to manually install a JRE for them. However, if the bundled JRE has a vulnerability over time, we recommend you download the latest version of the supported JRE and replace the bundled one. For the other components, you need to manually install either an OpenJDK JRE or Oracle JRE.

You have the option to use either Oracle JRE or OpenJDK JRE with Service Manager. For the supported versions of Oracle JRE and OpenJDK JRE, see the Support matrix.

JRE bitness

For the Server (both Windows and Linux) and Windows Client: use 64-bit JRE.

For the rest of the components: use a JRE with the same bitness as the operating system.

How to use Oracle JRE or OpenJDK JRE with Service Manager

Component	Instructions
Windows Server	An OpenJDK jre folder is already bundled with this component. You can follow these steps to replace the bundled JRE with the latest version. 1. Rename the <SM Server installation path>\RUN\jre folder to create a backup copy. 2. Download the latest version of 64-bit JRE 17. 3. Copy the jre folder to the SM Server's RUN folder.
Windows Client	An OpenJDK jre folder is already bundled with this component. You can follow these steps to replace the bundled JRE with the latest version. 1. Rename the <Service Manager installation path>\Client\jre folder to create a backup copy. 2. Download an appropriate version of 64-bit JRE 17. 3. Copy the jre folder to the <Service Manager installation path>\Client folder.
Linux Server	An OpenJDK jre folder is already bundled with this component. You can follow these steps to replace the bundled JRE with the latest version. 1. Rename the <SM Server installation path>\RUN\jre folder to create a backup copy 2. Download an appropriate version of 64-bit JRE 17. 3. Copy the jre folder to the SM Server's RUN folder. 4. From the Server's RUN directory, run the following commands as the owner of the Service Manager installation directory. This will create a symbolic link for the new JRE. ○ **removeLinks.sh jre** ○ **setupLinks.sh jre**
Web tier Mobility Client	**Note** If the web tier or Mobility Client is deployed on WebSphere, you must use IBM JDK instead. 1. Download and install an appropriate version of JRE 17 according to the operating system bitness of the Tomcat host. 2. Update the JAVA_HOME variable on the Tomcat host to make it point to the new JRE.

Related topics

Deployment architecture

1.3. Hardware load balancers

Horizontal or vertical scaling implementations can now use a supported third-party hardware load balancer in place of a Service Manager load balancer process. Hardware load balancers can offer the following advantages over a Service Manager load balancer process:

- Offer a variety of load balancing algorithms
- Offer SSL acceleration or offloading capabilities
- Offer error detection and correction functionality
- Increase system resources available on a Service Manager host by removing a load balancer process
- Reduce the number communications ports required between Service Manager clients and servers
- Offers option to run Service Manager from a private network with the hardware load balancer acting as a network gateway

Note

Note F5 hardware load balancer is the certified load balancer for Service Manager. For other types of hardware load balancers, you may apply the transparent technology.

Note

Note

Some customers may choose to omit the web server component in their implementations and the exact reasons for choosing such a configuration vary. While this is a viable configuration, it's advisable to implement the SM web tier product in a standard web server with Java application server configuration. In other words, you should use a combination of products, such as an Apache HTTP Server and Apache Tomcat to host the SM web tier. This is particularly important when there is a requirement to implement `SSO` or `TSO` functionality in your environment.

Related topics

Servlet implementation options
Horizontal scaling implementation
Configuring a horizontal scaling environment

1.4. Configure an F5 hardware load balancer

The following topics describe how to configure an F5 HWLB to work with Service Manager.

Configure an HWLB to accept client requests in HTTP mode

Configure an HWLB for SSL offloading

Configure an HWLB for SSL between an F5 HWLB and a Service Manager server

Configure an HWLB Health Monitor for the Service Manager server

1.4.1. Configure an HWLB to accept client requests in HTTP mode

This topic applies to Windows clients and to Service Manager web tier clients.

Prerequisites

Before you begin this process, verify that the following conditions are true:

- The F5 management URL can be accessed through the management IP.
- A VLAN is created on the interfaces.
- Static IPs are available for the interfaces.

Step 1: Configure Secure Network Address Translation (SNAT)

If the server and client are located in the same local network, SNAT is required. To determine whether you must configure SNAT, please consult your Network Engineering team.

To configure SNAT, follow these steps:

1. Click **Local Traffic** > **SNATs** > **SNAT Pool List**.
2. Type TEST_SNAT in the **Name** field, type the IP address of the Virtual Server IP in the **IP Address** field, and then click **Add**.
3. Click **Finished**.

Step 2: Configure an iRule to enable session persistence

Configure an iRule to enable session persistence for HTTP requests, based on the JSESSIONID cookie that the Service Manager server provides. To do this, follow these steps:

1. Click **Local Traffic** > **Virtual Servers** > **iRules** > **iRule List**.
2. Click **Create**, and then type TEST_JSESSION_IRULE in the **Name** field.
3. If OneConnect transformations are enabled in the default HTTP profile, create an HTTP profile to disable the OneConnect transformations. To do this, Click **Local Traffic** > **Profiles** > **Services** > **HTTP**.
4. Copy the following text into the **Definition** area:

```
when HTTP_REQUEST { if {[HTTP::cookie "JSESSIONID"] ne ""}{ persist uie [string tolower [HTTP::cookie "JSESSIONID"]] 300 } else { set jsess [findstr [string tolower [HTTP::path]] "jsessionid=" 11 ";"] if { $jsess != "" } { persist uie $jsess 300 } } }when HTTP_RESPONSE { if {[HTTP::cookie "JSESSIONID"] ne ""} { persist add uie [string tolower [HTTP::cookie "JSESSIONID"]] 300 } }
```

5. Click **Finished**.

Step 3: Disable OneConnect transformations

If OneConnect transformations are enabled in the default HTTP profile, you must create a new HTTP profile. To do this, follow these steps:

1. Click **Local Traffic** > **Virtual Servers** > **Profiles** > **Services** > **HTTP**.
2. Click Create, and then type the TEST_HTTP in the Name field.
3. Click the **Custom** checkbox.
4. Disable the OneConnect Transformations, and then click **Finish**.

> **Note**
>
> Note
> Only complete step 4 (above) on HWLB configurations that use HTTP mode to connect to a web service.

Step 4: Create a new TCP-based health monitor

A TCP-based health monitor detects whether the Service Manager process is already listening to the port. You can also use the interface provided by Service Manager to check the RTE servlet status. For more information, see Configure an HWLB Health Monitor for the Service Manager server.

To create a new TCP-based health monitor, follow these steps:

1. Click **Local Traffic** > **Monitors**.
2. Click **Create**, and then type TEST_TCP in the **Name** field.
3. Select **tcp** in the **Type** field, and then click **Finished**.

Step 5: Create a node

To create a node, follow these steps:

1. Click **Local Traffic** > **Nodes** > **Node List**.
2. Click **Create**, and then type the SM server host IP address.

Note

Notes

- There may be many Service Manager nodes (for example, in a horizontally-scaled Service Manager system).
- You may add as many nodes as are required by your deployment.

Step 6: Configure a pool member

To configure a pool member, follow these steps:

1. Click **Local Traffic** > **Nodes** > **Pool List**.
2. Click **Create**, and then type TEST_POOL in the **Name** field.
3. Select **TEST_TCP** as the health monitor.
4. Type your server node and port, click **Add**, and then click **Finished**.
5. On the **Members** tab, select **TEST_POOL** in the **Pool** list. Verify the members, and then click **Enable**.

Step 7: Create a new virtual server

To create a new virtual server, follow these steps:

1. Click **Local Traffic** > **Virtual Servers** > **Virtual Server List**.
2. Click **Create**, and then type TEST_VS in the **Name** field.
3. Specify a virtual server IP address and port number.
4. Select **oneconnect** as the OneConnect profile.
5. Select **TEST_HTTP** as the **HTTP Profile** option.
6. Select **TEST_SNAT** as the **SNAT Pools** option.
7. Enable the **TEST_JSESSION_IRULE** iRule that you created previously.
8. Select **TEST_POOL** as the **Default Pool** option.
9. Click **Finished**.

1.4.2. Configure an HWLB for SSL offloading

You must implement the F5 HWLB's SSL offloading feature to ensure that the load balancer can correctly inspect HTTPS traffic. This is required in order to maintain session persistence.

Prerequisites

Before you begin this process, verify that the following conditions are true:

- The HWLB is configured to accept client requests in HTTP Mode. For more information about how to do this, see Configure an HWLB to accept client requests in HTTP mode.
- The required Service Manager certificates are generated. For more information about how to do this, see the following article in the knowledge base: https://softwaresupport.softwaregrp.com/km/KM00488034

Configure the virtual server to use the SSL profile (client)

To configure the virtual server to use the SSL profile for the client, follow these steps:

1. Click **Local Traffic** > **Virtual Servers** > **Virtual Server List**.
2. Select the virtual server that you use to load balance client requests to Service Manager servlets.
3. In the **SSL Profile (Client)** drop-down list, select **TEST_SSL_CLIENT**, and then click **Update**.

Note

Note

You can use the JSession Persistence iRule to maintain session persistence.

1.4.3. Configure an HWLB for SSL between an F5 HWLB and a Service Manager server

Prerequisites

Before you begin this process, verify that the following conditions are true:

- The HWLB is configured for SSL offloading. For more information about how to do this, see Configure an HWLB for SSL offloading.
- The required Service Manager certificates are generated. For more information about how to do this, see the following article in the knowledge base: https://softwaresupport.softwaregrp.com/km/KM00488034

Step 1: Create a pool that contains an HTTPS port

To create a pool that contains an HTTPS port, follow these steps:

1. Click **Pools** > **Pool list**.
2. Click **Create**, and then type TEST_SSL_SERVER in the **Name** field.
3. Select a health monitor. For more information, see Configure an HWLB Health Monitor for the Service Manager server.
4. In the **address** field, type the IP address of the Service Manager server, and then set the server port to the HTTPS port of the Service Manager server.
5. Click **Add**.

> **Note**
>
> Note
>
> You can add as many pool list items as your deployment requires.

Step 2: Configure the session persistence type

To configure the session persistence type, follow these steps:

1. Click **Local Traffic** > **Profiles** > **Persistence**.
2. Click **Create**.
3. In the **Name** field, type TEST_COOKIE_INSERT.
4. In the **Persistence Type** drop-down list, select **Cookie**.
5. Click to select the **Custom** option on the right-hand side.
6. In the **Cookie Method** drop-down list, select **HTTP Cookie Insert**.
7. Click **Finish**.

Step 3: Configure the virtual server to use the SSL profile (Server)

To enable the virtual server to use the server SSL profile, follow these steps:

1. Click **Local Traffic** > **Virtual Servers** > **Virtual Server List**.
2. Select the virtual server that you use to load balance client requests to Service Manager servlets.
3. Select **oneconnect** as the OneConnect profile.
4. In the **SSL Profile (Client)** drop-down list, select **TEST_SSL_CLIENT**.
5. In the **SSL Profile (Server)** drop-down list, select **TEST_SSL_SERVER**.
6. Set the SNAT Pool to **TEST_SNAT**, and then click **Update**.
7. On the **Resource** tab, select the default Pool that contains the HTTPS port of the Service Manager server.
8. Set the Default Persistence Profile to **TEST_COOKIE_INSERT**.

1.4.4. Configure an HWLB Health Monitor for the Service Manager server

The health monitor helps the HWLB to identify whether a Service Manager servlet can accept a new connection. This prevents the servlet returning errors if the HWLB directs a request to the servlet when the servlet is at full capacity or when Service Manager is running in Quiesce mode.

This section describes how to configure and use a health monitor with an F5 HWLB.

Prerequisites

Before you begin this process, verify that the HWLB is configured to accept client requests in HTTPS mode (if you use an SSL port) or in HTTP mode (if you use a non-SSL port). For more information about how to do this, see Configure an HWLB to accept client requests in HTTP mode.

Step 1: Create the health monitor on the F5 HWLB

To create the health monitor on the F5 HWLB, follow these steps:

1. Click **Local Traffic** > **Monitors** > **Create**.
2. Type a name for the health monitor, such as TEST_MONITOR.
3. Select **HTTP** or **HTTPS** as the type, depending on the connection type between the F5 HWLB and the Service Manager server.
4. In the **Send String** field, type:
   GET /mbeanclient-X.XX/Action?getattribute=AcceptNewConnection\r\n
5. In the **Receive String** field, type true.
6. If you use an SSL connection between the F5 HWLB and the Service Manager server, select **TEST_SSL_SERVER** in the **Client Certificate** and **Client Key** fields.
7. Click **Finished**.

Step 2: Update the pool list

1. Click **Pools** > **Pool List**.
2. Select the pool that your virtual server uses.
3. In the **Health Monitors** section, move **TEST_MONITOR** from the **Available** side to the **Active** side.
4. Click **Update**.

Step 3: Modify the sm.ini file and restart Service Manager

1. Add the following line to the `sm.ini` file:

   ```
   JVMOption999:../../lib/mbeanclient-X.XX.war
   ```

2. Restart the Service Manager servers.

Related topics

Access Service Manager through an F5 HWLB in HTTP mode

Access Service Manager on a Windows client through an F5 HWLB in HTTP mode

Access Service Manager on a web client through an F5 HWLB in HTTP mode

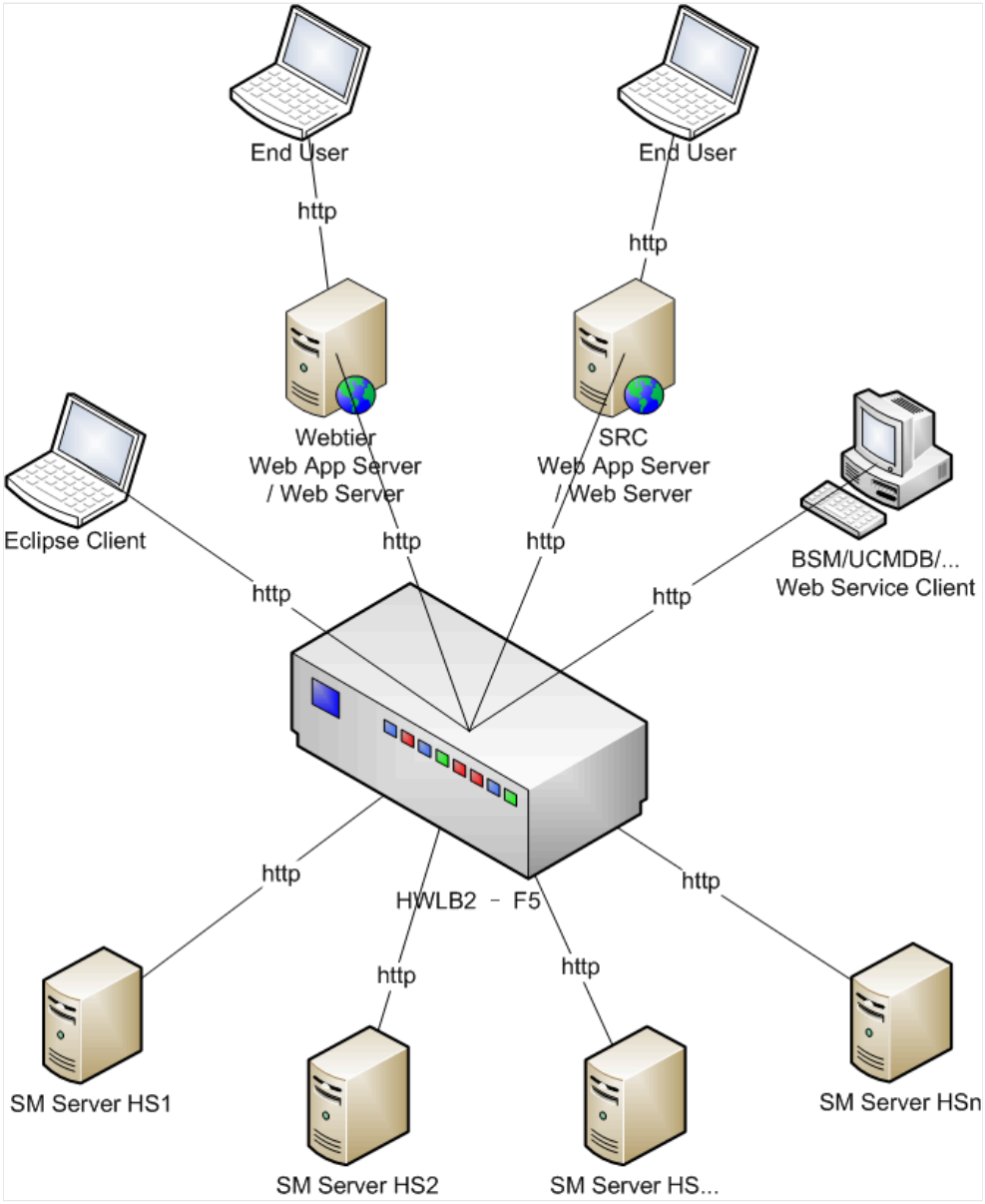
1.5. Common F5 hardware load balancer deployments

The following topics describe some common F5 HWLB deployments.

- Access Service Manager through an F5 HWLB in HTTP mode
- Access Service Manager on a Windows client through an F5 HWLB in SSL Offloading mode
- Access Service Manager on a Windows client through an F5 HWLB in Full SSL mode
- Access Service Manager in Mixed mode

1.5.1. Access Service Manager through an F5 HWLB in HTTP mode

The following figure illustrates a deployment in which users access Service Manager through an F5 HWLB in HTTP mode:



1.5.1.1. Access Service Manager on a Windows client through an F5 HWLB in HTTP mode

This deployment is suited to test Service Request Catalog and Service Manager in a development or non-production environment. As you can see from the figure in Access Service Manager through an F5 HWLB in HTTP mode, this configuration doesn't use HTTPS/SSL encrypted communication. This deployment should be used for sand box, development, or demonstration (proof of concept) purposes.

This section describes how to access Service Manager on a Windows client through an F5 HWLB in HTTP mode.

Step 1: Configure a Windows client to support an external hardware load balancer

To configure a Windows client to support an external hardware load balancer, follow these steps:

1. Install a new Windows client.
2. Start the Windows client, and then create a new network connection called `hwlb_connect`.
3. Type the IP or host name of the virtual server as the connection's server address.
4. Type the port of the HWLB virtual server as the connection's server port.
5. On the **Advanced** tab, click to select the **Connect to External Load Balancer** option.

6. Click **Apply**, and then click **Connect**.

Step 2: Enable an HWLB on a Service Manager server

To enable an HWLB on a Service Manager server, follow these steps:

1. Add the `external_lb` attribute to the `sm.ini` file on the Service Manager server.

   > **Note**
   > Note The `external_lb` attribute is a system attribute, and every servlet node on the same host should use the same attribute setting. In most cases, this attribute needs to be configured in the `sm.ini` file when you start a servlet from the operating system command prompt.
   > However, to prevent a servlet node from running in external load balancer mode, you can include `-external_lb:0` in a command when you start a servlet from the operating system command prompt. As the servlet node doesn't work in external load balancer mode, it rejects any connections from a hardware load balancer.
   > Only `-external_lb:0` can be added to a command. `-external_lb:1` and `external_lb` aren't recognized in commands.

2. Make sure that no Service Manager software load balancer (for example, `sm -loadBalancer`) is configured or running on the Service Manager server.

Related topics

Access Service Manager through an F5 HWLB in HTTP mode
Access Service Manager on a web client through an F5 HWLB in HTTP mode
F5 HWLB Health Monitor

1.5.1.2. Access Service Manager on a web client through an F5 HWLB in HTTP mode

This deployment is suited to testing Service Request Catalog and Service Manager in a development or non-production environment. As you can see from the figure in Access Service Manager through an F5 HWLB in HTTP mode, this configuration doesn't use HTTPS/SSL encrypted communication. This deployment should be used for sand box, development, or demonstration (proof of concept) purposes.

This section describes how to access Service Manager on a web client through an F5 HWLB in HTTP mode.

Step 1: Configure a web client to support an external load balancer

To configure a web client to support an external load balancer, follow these steps:

1. Set the `externalLB` parameter in the `web.xml` file to `true`, as follows:

```
<init-param>
      <param-name>externalLB</param-name>
      <param-value>true</param-value>
 </init-param>
```

2. Set the host and port to match the HWLB virtual server IP and port. To do this, configure the `web.xml` file as follows:

```
<init-param>
      <param-name>serverHost</param-name>
      <param-value><fully-qualified domain name of the HWLB virtual server></param-value>
 </init-param>
 <init-param>
      <param-name>serverPort</param-name>
      <param-value><Port of the HWLB virtual server></param-value>
 </init-param>
```

Step 2: Enable an HWLB on a Service Manager server

To enable an HWLB on a Service Manager server, follow these steps:

1. Add the `external_lb` attribute to the `sm.ini` file on the Service Manager server.

   > **Note**
   > Note The `external_lb` attribute is a system attribute, and every servlet node on the same host should use the same attribute setting. In most cases, this attribute needs to be configured in the `sm.ini` file when you start a servlet from the operating system command prompt.
   > However, to prevent a servlet node from running in external load balancer mode, you can include `-external_lb:0` in a command when you start a servlet from the operating system command prompt. As the servlet node doesn't work in external load balancer mode, it rejects any connections from a hardware load balancer.
   > Only `-external_lb:0` can be added to a command. `-external_lb:1` and `external_lb` aren't recognized in commands.

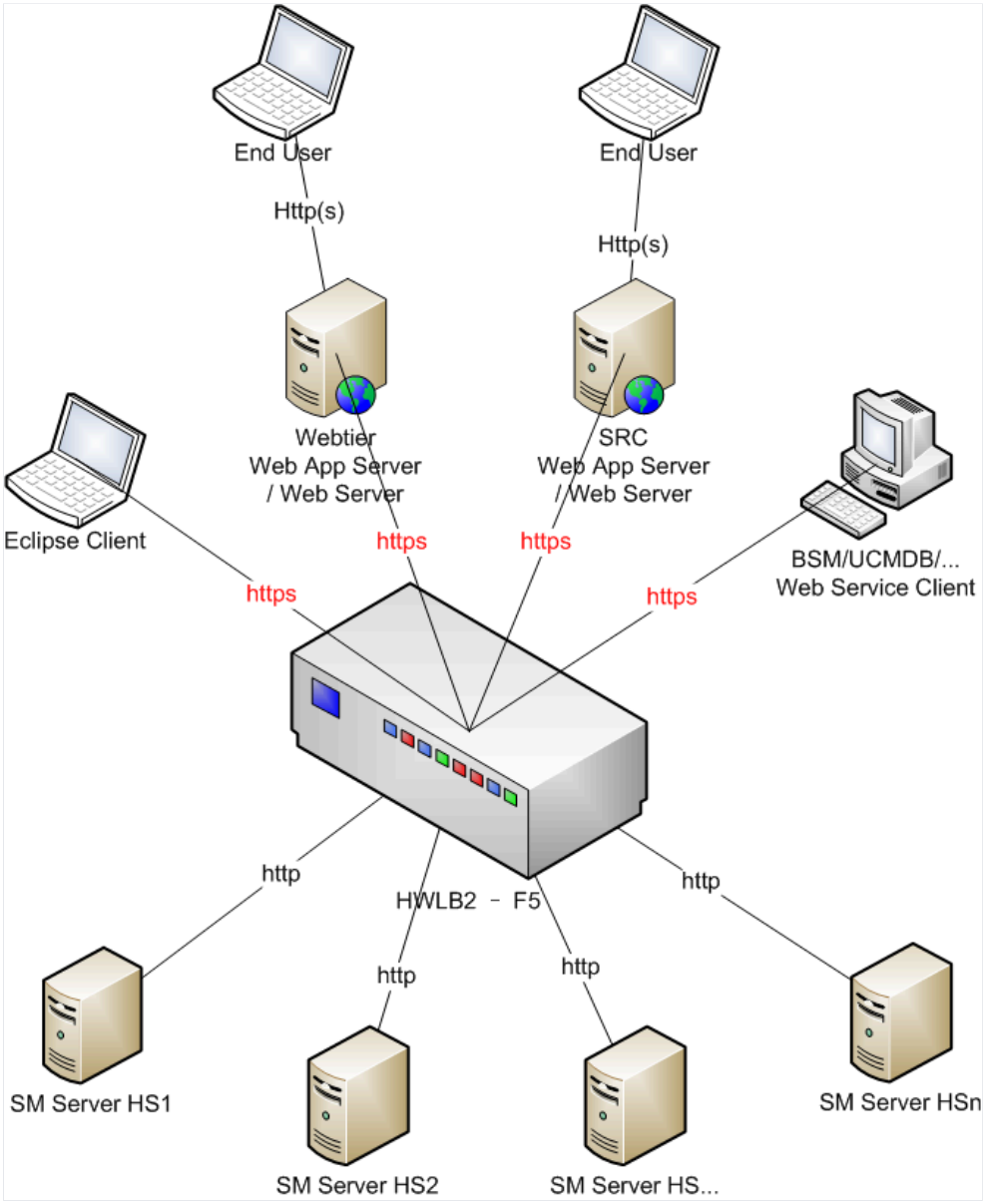
2. Make sure that no Service Manager software load balancer (for example, `sm -loadBalancer`) is configured or running on the Service Manager server.

Related topics

Access Service Manager through an F5 HWLB in HTTP mode
Access Service Manager on a Windows client through an F5 HWLB in HTTP mode
F5 HWLB Health Monitor

1.5.2. Access Service Manager through an F5 HWLB in SSL offloading mode

The following figure illustrates a deployment in which users access Service Manager through an F5 HWLB in SSL offloading mode:



1.5.2.1. Access Service Manager on a Windows client through an F5 HWLB in SSL Offloading mode

This section describes how to access Service Manager on a Windows client through an F5 HWLB in SSL Offloading mode.

Prerequisites

Before you begin this process, you must create the client certificates and private keys, as described in Configure an HWLB for SSL offloading.

Note

Note

You don't need to configure SSL on the server.

Configure the Windows client to use SSL

To access Service Manager on a Windows client through an F5 HWLB in SSL Offloading mode, you must configure the Windows client to use SSL. To do this, follow these steps:

1. Click **Windows** > **Preference** >**Service Manager** > **Security**.
2. Click **Browse** to fill the **Client Keystore** file.
3. Type clientkeystore as the password.
4. Select the `cacerts` CA Certificates file that you created earlier, and then click **OK**.
5. Restart Windows.

Related topics

Access Service Manager through an F5 HWLB in SSL Offloading mode
Access Service Manager on a web client through an F5 HWLB in SSL Offloading mode
Access Service manager on a web service client through an F5 HWLB in SSL Offloading mode

1.5.2.2. Access Service Manager on a web client through an F5 HWLB in SSL Offloading mode

This section describes how to access Service Manager on a web client through an F5 HWLB in SSL Offloading mode.

Prerequisites

Before you begin this process, you must create the client certificates and private keys, as described in Configure an HWLB for SSL offloading.

> **Note**
>
> Note
>
> You don't need to configure SSL on the Service Manager server.

Modify the web.xml file

To access Service Manager on a web client through an F5 HWLB in SSL Offloading mode, configure the following parameters in the `web.xml` file, and then restart Tomcat:

- ssl

```
<init-param>
<param-name>ssl</param-name>
<param-value>true</param-value>
</init-param>
```

- externalLB

```
<init-param>
<param-name>externalLB</param-name>
<param-value>true</param-value>
</init-param>
```

- keystore

```
<!-- If this is a relative path, it will be relative to the web application's deploy directory,
but still needs a leading slash -->
<init-param>
<param-name>keystore</param-name>
<param-value>/WEB-INF/<FQDN of This Host>.keystore </param-value>
</init-param>
<!-- Specify the password for the client's private keystore -->
<init-param>
<param-name>keystorePassword</param-name>
<param-value>clientkeystore</param-value>
</init-param>
```

Related topics

Access Service Manager through an F5 HWLB in SSL Offloading mode
Access Service Manager on a Windows client through an F5 HWLB in SSL Offloading mode
Access Service manager on a web service client through an F5 HWLB in SSL Offloading mode

1.5.2.3. Access Service Manager on a web service client through an F5 HWLB in SSL Offloading mode

This section describes how to access Service Manager on a web service client through an F5 HWLB in SSL Offloading mode.

Prerequisites

Before you begin this process, you must create the client certificates and private keys, as described in Configure an HWLB for SSL offloading.

Note

Note

You don't need to configure SSL on the server.

Create a new project in SoapUI

To access Service Manager on a web service client through an F5 HWLB in SSL Offloading mode, follow these steps:

1. Start the SoapUI tool and click **New project**.
2. Type a name for the project in the **Project Name** field.
3. In the **Initial WSDL/WADL** field, type http://*<SM Server IP>*:*<SM Server Port>*/SM/7/ChangeManagement.wsdl.
4. Click to select the **Create Requests** option, and then click **OK**.
5. Right-click **RetrieveChangeList** and select **New request**.
6. Type a name for the request.
7. Click **File** > **Preference** > **SSL Settings**.
8. Navigate to the `<client FQDN>.keystore` file.
9. Type clientkeystore as the password, and then click **OK**.

10. Modify the request as follows:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/"
xmlns:ns="http://schemas.hp.com/SM/7">
<soapenv:Header/>
<soapenv:Body>
<ns:RetrieveChangeListRequest attachmentInfo="?" attachmentData="?" ignoreEmptyElements="true"
count="2" start="0">
<!--1 or more repetitions:-->
<ns:keys query="planned.start=NULL">
</ns:keys>
</ns:RetrieveChangeListRequest>
</soapenv:Body>
</soapenv:Envelope>
```

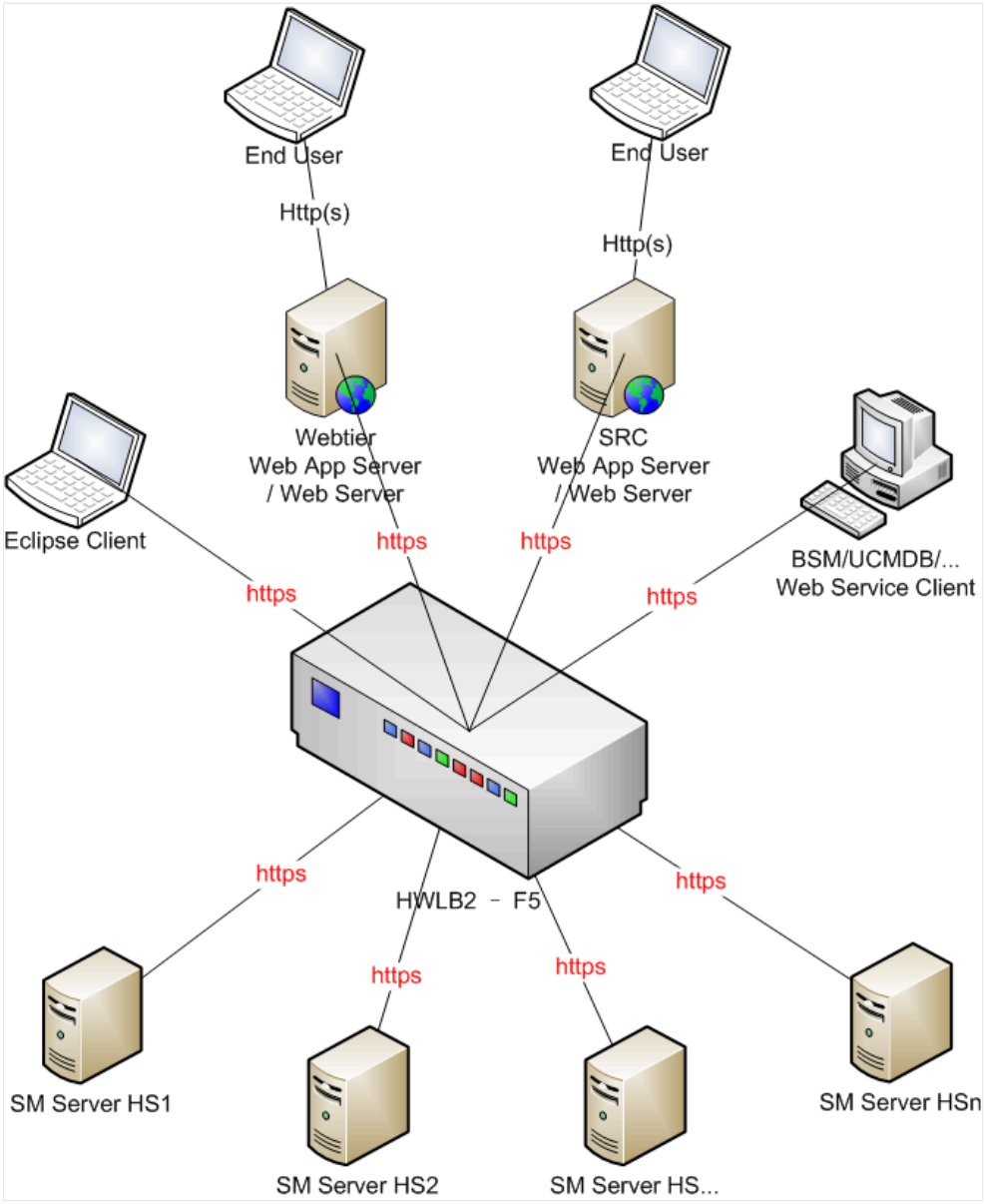
11. Click **Authenticate**.
12. Type falcon as the user name.
13. Replace **http** in the URL with **https**. For example, set the following URL: https://*<FQDN of the HWLB virtual server>*:*<Port of HWLB virtual server>*/SM/7/ws.

Related topics

Access Service Manager through an F5 HWLB in SSL Offloading mode
Access Service Manager on a Windows client through an F5 HWLB in SSL Offloading mode
Access Service Manager on a web client through an F5 HWLB in SSL Offloading mode

1.5.3. Access Service Manager through an F5 HWLB in Full SSL mode

The following figure illustrates a deployment in which users access Service Manager through an F5 HWLB in Full SSL mode:



1.5.3.1. Access Service Manager on a Windows client through an F5 HWLB in Full SSL mode

This section describes how to access Service Manager on a Windows client through an F5 HWLB in Full SSL mode.

Prerequisites

Before you begin this process, verify that the following conditions are true:

- To maintain session persistence, SSL must be configured between the F5 HWLB and the Windows client. For more information about how to configure SSL, see Configure an HWLB for SSL offloading.
- SSL is configured between the F5 HWLB and the SM server. For more information about how to do this, see Configure an HWLB for SSL between an F5 load balancer and an Service Manager server .

Modify the sm.ini file

To access Service Manager on a Windows client through an F5 HWLB in Full SSL mode, configure the following parameters in the `sm.ini` file:

```
ssl:1
ssl_reqClientAuth:2
keystoreFile:server.keystore
keystorePass:serverkeystore
ssl_trustedClientsJKS:trustedclients.keystore
ssl_trustedClientsPwd:trustedclients
truststoreFile:cacerts
truststorePass:changeit
external_lb
sslConnector:1
```

Related topics

Access Service Manager through an F5 HWLB in Full SSL mode
Access Service Manager on a web client through an F5 HWLB in Full SSL mode
Access Service Manager on a web service client through an F5 HWLB in Full SSL mode

1.5.3.2. Access Service Manager on a web client through an F5 HWLB in Full SSL mode

This section describes how to access Service Manager on a web client through an F5 HWLB in Full SSL mode.

Prerequisites

Before you begin this process, verify that the following conditions are true:

- To maintain session persistence, SSL must be configured between the F5 HWLB and the web client. For more information about how to configure SSL, see Configure an HWLB for SSL offloading.
- SSL is configured between the F5 HWLB and the SM server. For more information about how to do this, see Configure an HWLB for SSL between an F5 load balancer and an Service Manager server .

Step 1: Modify the sm.ini file

Configure the following parameters in the `sm.ini` file:

```
ssl:1
ssl_reqClientAuth:2
keystoreFile:server.keystore
keystorePass:serverkeystore
ssl_trustedClientsJKS:trustedclients.keystore
ssl_trustedClientsPwd:trustedclients
truststoreFile:cacerts
truststorePass:changeit
external_lb
sslConnector:1
```

Step 2: Modify the web.xml file

On the Service Manager web tier, locate the `web.xml` file, and then configure the following parameters:

```
<init-param>
<param-name>ssl</param-name>
<param-value>true</param-value>
</init-param>
<init-param>
<param-name>cacerts</param-name>
<param-value>/WEB-INF/cacerts</param-value>
</init-param>
<init-param>
<param-name>keystore</param-name>
<param-value>/WEB-INF/<webtier FQDN>.keystore</param-value>
</init-param>
<init-param>
<param-name>keystorePassword</param-name>
<param-value>clientkeystore</param-value>
</init-param>
<init-param>
<param-name>externalLB</param-name>
<param-value>true</param-value>
</init-param>
```

Related topics

Access Service Manager through an F5 HWLB in Full SSL mode
Access Service Manager on a Windows client through an F5 HWLB in Full SSL mode
Access Service Manager on a web service client through an F5 HWLB in Full SSL mode

1.5.3.3. Access Service Manager on a web service client through an F5 HWLB in Full SSL mode

This section describes how to access Service Manager on a web service client through an F5 HWLB in Full SSL mode.

Prerequisites

Before you begin this process, verify that the following conditions are true:

- To maintain session persistence, SSL must be configured between the F5 HWLB and SoapUI. For more information about how to configure SSL, see Configure an HWLB for SSL offloading.
- SSL is configured between the F5 HWLB and the SM server. For more information about how to do this, see Configure an HWLB for SSL between an F5 load balancer and an Service Manager server .

Step 1: Modify the sm.ini file

Configure the `sm.ini` file as follows:

```
ssl:1
ssl_reqClientAuth:2
keystoreFile:server.keystore
keystorePass:serverkeystore
ssl_trustedClientsJKS:trustedclients.keystore
ssl_trustedClientsPwd:trustedclients
truststoreFile:cacerts
truststorePass:changeit
external_lb
sslConnector:1
```

Step 2: Create a new project in SoapUI

To create a new project in SoapUI, follow these steps:

1. Start the SoapUI tool and click **New project**.

2. Type a name for the project in the **Project Name** field.
3. In the **Initial WSDL/WADL** field, type http://<*HWLB virtual server IP*>:13080/SM/7/ChangeManagement.wsdl.
4. Click to select the **Create Requests** option, and then click **OK**.
5. Right-click **RetrieveChangeList** and select **New request**.
6. Type a name for the request.
7. Click **File** > **Preference** > **SSL Settings**.
8. Navigate to the `<client FQDN>.keystore` file.
9. Type clientkeystore as the password, and then click **OK**.
10. Modify the request as follows:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/"
xmlns:ns="http://schemas.hp.com/SM/7">
<soapenv:Header/>
<soapenv:Body>
<ns:RetrieveChangeListRequest attachmentInfo="?" attachmentData="?" ignoreEmptyElements="true"
count="2" start="0">
<!--1 or more repetitions:-->
<ns:keys query="planned.start=NULL">
</ns:keys>
</ns:RetrieveChangeListRequest>
</soapenv:Body>
</soapenv:Envelope>
```

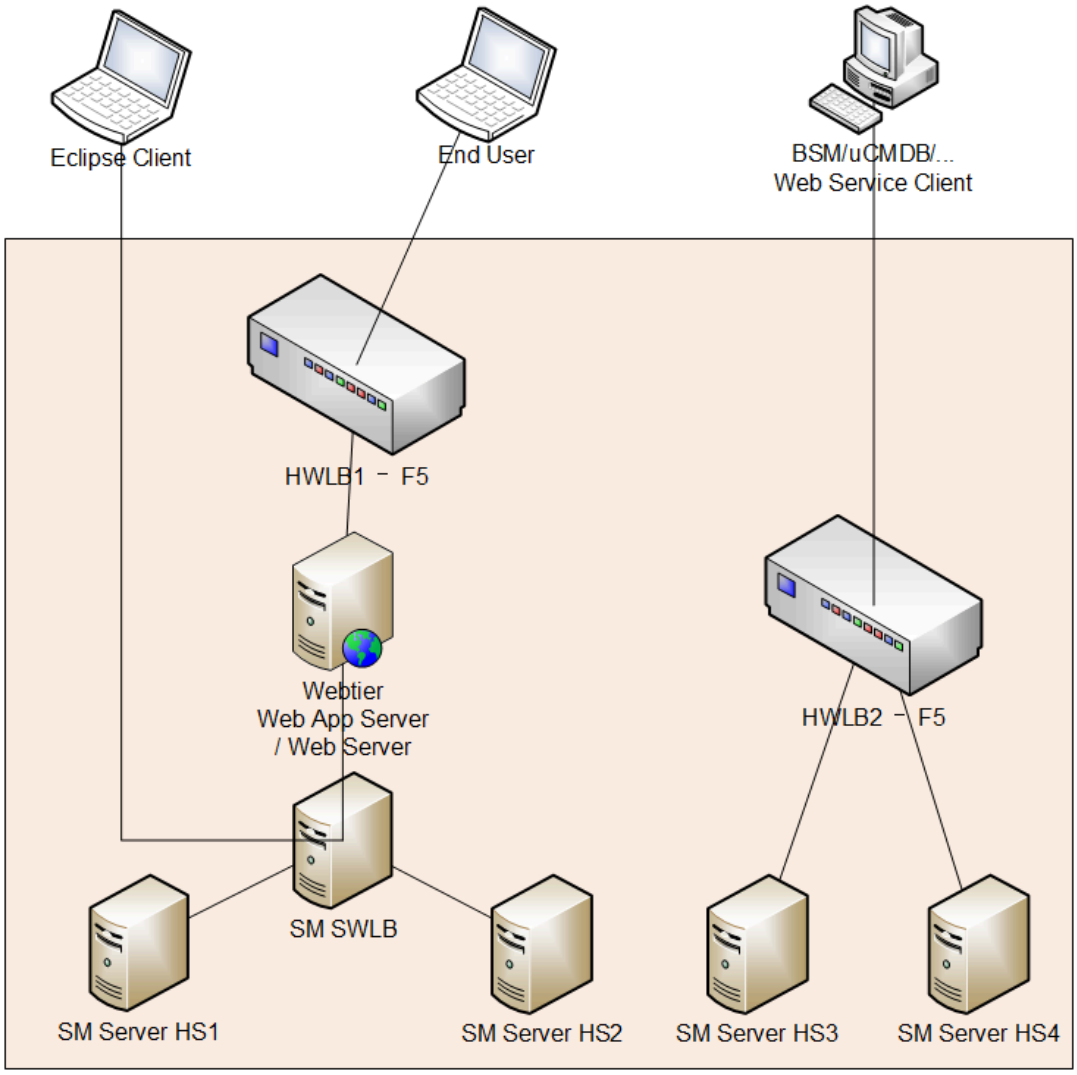
11. Click **Authenticate**.
12. Type falcon as the user name.
13. Replace **http** in the URL with **https**. For example, set the following URL: https://<*FQDN of the HWLB virtual server*>:<*Port of HWLB virtual server*>/SM/7/ws.

Related topics

Access Service Manager through an F5 HWLB in Full SSL mode
Access Service Manager on a Windows client through an F5 HWLB in Full SSL mode
Access Service Manager on a web client through an F5 HWLB in Full SSL mode

1.5.4. Access Service Manager in Mixed mode

The following figure illustrates a deployment in which users access Service Manager through an F5 HWLB in Mixed mode:



1.5.4.1. Sign in to a Service Manager server by using TSO when a HWLB is located between the browser and the webtier

This section describes how to sign in to Service Manager by using Trusted sign-on (TSO) when a HWLB is located between the browser and the web tier.

Prerequisites

Before you begin this process, you must access Service Manager on a web client through an F5 HWLB in Full SSL mode, as described in Access Service Manager on a web client through an F5 HWLB in Full SSL mode.

Configure the HWLB virtual server

To sign in to a Service Manager server by using TSO, you must configure the HWLB virtual server. To do this, follow these steps:

1. Click **Local Traffic** > **Virtual Servers** > **Virtual Server List**.
2. Select the virtual server that you want to configure.
3. In the **HTTP Profile** field, select **TEST_HTTP**.
4. In the **OneConnect Profile** field, select **oneconnect**.
5. In the **NTLM Conn Pool** field, select **ntlm**.
6. Click **Update**.
7. Add the HWLB domain URL to your browser's trusted sites list.

You can now log on to the Service Manager web tier through the HWLB. For example, type the following URL into the address bar of your browser:

http://<FQDN of HWLB virtual server>/<sm980>/index.do

Related topics

Access Service Manager in Mixed mode
Sign in to a Service Manager server by using LW-SSO when a HWLB is located between the browser and the webtier

1.5.4.2. Sign in to a Service Manager server by using LW-SSO when a HWLB is located between the browser and the webtier

This section describes how to sign in to Service Manager by using Lightweight Single Sign-On (LW-SSO) when an HWLB is located between the browser and the web tier.

Prerequisites

Before you begin this process, verify that the following conditions are true:

- The HWLB is configured to accept client requests, as described in Configure an HWLB to accept client requests in HTTP mode.
- A pool contains the members that point to the web application server (for example, Tomcat).
- The server and the web tier share the same initString setting.
- LW-SSO works without the HWLB.

> **Note**
>
> Note
> If the domain name of the HWLB is not the same as the domain name of the web tier, you must set the value of the `domain` parameter in the `lwssofmconf.xml` file on the web tier to the domain name of the HWLB.

Enable LW-SSO

1. Enable LW-SSO on the Service Manager server. For more information about how to do this, see Configure LW-SSO in the Service Manager server.
2. Enable LW-SSO on the web tier. For more information about how to do this, see Configure LW-SSO in the Service Manager Web tier.

Related topics

Access Service Manager in Mixed mode
Sign in to Service Manager by using TSO when an HWLB is located between the browser and the web tier

1.6. Server implementation options

Service Manager supports several different implementation options to manage client connections to the server. The following is a list of the most common server implementation options and the features they offer.

Feature	Single servlet implementation	Vertical scaling implementation	Horizontal scaling implementation
Specify all communication ports used?	Yes	Yes	Yes
Specify connection limit?	Yes	Yes	Yes
Load balances client connections?	No	Yes	Yes
Implementation uses multiple hosts?	No	No	Yes
Allows dynamic addition and removal of hosts to virtual group?	No	No	Yes
Allows dynamic addition and removal of servlet container processes to virtual group?	No	Yes	Yes
Options available to convert current implementation into new one	• Vertical scaling • Horizontal scaling	• Horizontal scaling	Not applicable

Related topics

Configuring installation and setup options
Horizontal scaling implementation
Single servlet implementation
Vertical scaling implementation
Vertical scaling and required SSL implementation

1.6.1. Horizontal scaling implementation

In a horizontal scaling implementation you maximize the number of client connections supported across multiple hosts.

The key features of this implementation are:

- The Service Manager system consists of multiple Service Manager servlet container processes running on separate physical hosts
- The implementation allows administrators to specify a system connection limit
- The Service Manager system can manage a number of concurrent client connections up to the number of servlet container processes times the threads per process value (For example, 6 servlet container processes supporting 50 threads per process can support up to 300 client connections)
- The implementation allows administrators to specify the communication ports the system uses
- A dedicated load balancer process manages and routes client connections to available servlet container processes
- Administrators can dynamically add and remove Service Manager instances from a virtual group

Note

Note

Service Manager doesn't support automatic switching between two hosts.

A horizontal scaling implementation is typically used in large 24 by 7 environments where system scalability and resilience is a concern. A horizontal scaling implementation can support as many client connections as the sum of the individual Service Manager instances can support (for example, if each instance can support 50 client connections and there are 6 instances then the total system can support 300 client connections). A horizontal scaling implementation has improved resilience features such as the ability to stop and start the load balancer process without causing a total system outage and the ability to add a new servlet container process to a virtual group while the system is running.

Each host in a horizontal scaling implementation can also support a vertical scaling implementation with the addition of the appropriate servlet implementation parameters. This allows each Service Manager host to support multiple servlet container processes and to make the best use of the available system resources.

Note: There can be only **one** kmupdate process running at any time regardless of the number of hosts. Starting more than one kmupdate process causes unpredictable behavior on the search engine server.

Horizontal scaling implementation diagram

The horizontal scaling implementation diagram is a reference diagram for implementers who are responsible for installing and configuring Service Manager. The guide shows the configuration settings and network connections required for multiple servers running multiple servlet containers implementation.

Related topics

Example: Setting up a horizontal scaling implementation
Server implementation options
Startup options for servlet container processes
Configuring a horizontal scaling environment

1.6.1.1. Example: Setting up a horizontal scaling implementation

The following example describes how to set up a horizontal scaling implementation that accomplishes the following:

- Maximizes the number of client connections supported across multiple hosts
- Allows an administrator to specify the communications ports the Service Manager implementation uses

You can use this example to configure the implementation depicted in the horizontal scaling implementation diagram.

1. Install Service Manager on the first host in the virtual group. For example, myserver1.
   The host must be on the same subnet as the other hosts in the virtual group.
2. Sign in to the Software Licenses and Downloads and obtain a license file for this host.

   Note
   Note After you obtain a license file for this host, it becomes the primary host of the horizontal scaling implementation. You must start this host first when starting a horizontal scaling implementation for the first time.

3. Copy the license file you obtain onto a network share accessible to the other servers that will make up the horizontal scaling implementation.
4. Sign in to the operating system of the primary host and change directories to the Service Manager Server's RUN folder.
5. Paste the license file (LicFile.txt) into the RUN folder.
6. In the RUN folder, open the Service Manager configuration file (sm.cfg) in a text editor.
7. Edit the file so that only the following lines appear:

   ```
   sm -loadBalancer -httpPort:13080
   sm -httpPort:13081 -httpsPort:13082
   sm -httpPort:13083 -httpsPort:13084
   sm system.start
   ```

   The loadBalancer parameter creates a special servlet container process to route client connection requests to other available servlet container processes. A horizontal scaling implementation only needs one load balancer.

The httpPort:13080 parameter specifies that the load balancer process listens to client connection requests on HTTP port 13080. This communications port must be unique across all hosts that you want to join the Service Manager virtual group.
The httpPort:13081 parameter specifies that the servlet container process listens to client connection requests on HTTP port 13081. This communications port must be unique on the host on which you start the servlet container.
The httpsPort:13082 parameter specifies that the servlet container process listens to client connection requests on HTTPS port 13082. This communications port must be unique on the host on which you start the servlet container.
The httpPort:13083 parameter specifies that the servlet container process listens to client connection requests on HTTP port 13083. This communications port must be unique on the host on which you start the servlet container.
The httpsPort:13084 parameter specifies that the servlet container process listens to client connection requests on HTTPS port 13084. This communications port must be unique on the host on which you start the servlet container.
The system.start parameter specifies that the host system should start all the background processes in the system.start script file. This host will be the only host in the virtual group that runs the background processes. The configuration file should already list this line.

8. Save the Service Manager configuration file.
9. Open the Service Manager initialization file (sm.ini) in a text editor.
10. Add the RDBMS connection settings. For example:

```
[oracle10]
sqldb:ora102
sqllogin:sm7user/password
sqllibrary:SQORACLE.OCI10.DLL
sqldictionary:oracle10
```

11. Add the following lines:

```
grouplicenseip:10.0.0.135
groupname:mygroup1
groupmcastaddress:224.0.1.255
groupsubnetaddress:255.255.255.0
groupport:13100
threadsperprocess:50
sessiontimeout:3
system:13080
```

Note

Note This example assumes the host system has only one network adapter card and therefore doesn't need the **groupbindaddress** parameter. If your host has multiple network adapter cards, you must add the **groupbindaddress** parameter to specify which network adapter the virtual group will use for communications.

The grouplicenseip:10.0.0.135 parameter defines the TCP/IP address of the Service Manager host with a valid AutoPass license for the virtual group. The value of this parameter must match the IP address specified in the AutoPass license.
The groupname:mygroup1 parameter creates a virtual group that servlet container processes across multiple systems can join for horizontal scaling purposes. The value is arbitrary and can be any text value without spaces.
The groupmcastaddress:244.0.1.255 parameter defines the TCP/IP address that servlet container processes can use to communicate with the load balancer process. The TCP/IP address must be consistent with the UDP multicasting protocol.
The groupsubnetaddress:255.255.255.0 parameter defines the subnet mask that servlet container processes can use to communicate with the load balancer process. The subnet mask must be consistent with the IPv4 protocol.

> **Note**
> Note You only have to define the subnet address if you are using subnet masking on your IP addresses.

The groupport:13100 parameter defines the communications port that servlet container processes can use to communicate with the load balancer process. The communications port can be any available communications port.
The threadsperprocess:50 parameter defines the number of threads each process supports. A value of 50 threads assumes that the Service Manager host has the minimum recommended system memory available for servlet container processes.
The sessiontimeout:3 parameter defines the number of minutes that the client connection can remain unresponsive before the server closes the connection. A value of 3 minutes assumes that most network latency issues are quickly and easily resolved.
The system:13080 parameter defines a numerical identifier for the system, in this case the HTTP port of the load balancer process. The value of this parameter must match for each system in the virtual group.

12. Save the Service Manager initialization file.
13. If the host runs on a Linux operating system and uses an SSL implementation, edit the /etc/hosts file to list the fully qualified domain name of each host in the virtual group. For example:

```
127.0.0.1 localhost
127.0.0.2 myserver2.mydomain.com myserver2 loghost
127.0.0.3 myserver3.mydomain.com myserver3 loghost
```

Caution: Do not edit the /etc/hosts file unless your system is running an SSL implementation.

14. Start the Service Manager server.
15. Install Service Manager on the next host in the virtual group. For example: myserver2.
The host must be on the same subnet as the other hosts in the virtual group.
16. Sign in to the operating system of the Service Manager server host and change directories to the Service Manager Server's RUN folder.
17. Open the Service Manager configuration file (sm.cfg or smstart) in a text editor.
18. Edit the file so that only the following lines appear:

```
sm -httpPort:13081 -httpsPort:13082
sm -httpPort:13083 -httpsPort:13084
sm -sync
```

> **Note**
> Note You don't need the system.start line on this host as only one host in the virtual group needs to run the background processes.

The httpPort:13081 parameter specifies that the servlet container process listens to client connection requests on HTTP port 13081. This communications port must be unique on the host on which you start the servlet container.
The httpsPort:13082 parameter specifies that the servlet container process listens to client connection requests on HTTPS port 13082. This communications port must be unique on the host on which you start the servlet container.
The httpPort:13083 parameter specifies that the servlet container process listens to client connection requests on HTTP port 13083. This communications port must be unique on the host on which you start the servlet container.
The httpsPort:13084 parameter specifies that the servlet container process listens to client connection requests on HTTPS port 13084. This communications port must be unique on the host on which you start the servlet container.
The sync parameter specifies that the host system should start the sync background process to identify and remove unused processes.

19. Save the Service Manager configuration file.
20. Open the Service Manager initialization file (sm.ini) in a text editor.

21. Add the RDBMS connection settings. For example:

```
[oracle10]
sqldb:ora102
sqllogin:sm7user/password
sqllibrary:SQORACLE.OCI10.DLL
sqldictionary:oracle10
```

22. Add the following lines:

```
grouplicenseip:10.0.0.135
groupname:mygroup1
groupmcastaddress:224.0.1.255
groupsubnetaddress:255.255.255.0
groupport:13100
threadsperprocess:50
sessiontimeout:3
system:13080
```

> **Note**
> Note This example assumes the host system has only one network adapter card and therefore doesn't need the **groupbindaddress** parameter. If your host has multiple network adapter cards, you must add the **groupbindaddress** parameter to specify which network adapter the virtual group will use for communications.

The grouplicenseip:10.0.0.135 parameter defines the TCP/IP address of the Service Manager host with a valid AutoPass license for the virtual group. The value of this parameter must match the IP address specified in the AutoPass license.
The groupname:mygroup1 parameter creates a virtual group that servlet container processes across multiple systems can join for horizontal scaling purposes. The value is arbitrary and can be any text value without spaces.
The groupmcastaddress:244.0.1.255 parameter defines the TCP/IP address that servlet container processes can use to communicate with the load balancer process. The TCP/IP address must be consistent with the UDP multicasting protocol.
The groupsubnetaddress:255.255.255.0 parameter defines the subnet mask that servlet container processes can use to communicate with the load balancer process. The subnet mask must be consistent with the IPv4 protocol.

> **Note**
> Note You only have to define the subnet address if you are using subnet masking on your IP addresses.

The groupport:13100 parameter defines the communications port that servlet container processes can use to communicate with the load balancer process. The TCP/IP address can be any available communications port.
The threadsperprocess:50 parameter defines the number of threads each process supports. A value of 50 threads assumes that the Service Manager host has the minimum recommended system memory available for servlet container processes.
The sessiontimeout:3 parameter defines the number of minutes that the client connection can remain unresponsive before the server closes the connection. A value of 3 minutes assumes that most network latency issues are quickly and easily resolved.
The system:13080 parameter defines a numerical identifier for the system, in this case the HTTP port of the load balancer process. The value of this parameter must match for each system in the virtual group.

23. Save the Service Manager initialization file.

24. If the host runs on a Linux operating system and uses an SSL implementation, edit the /etc/hosts file to list the fully qualified domain name of each host in the virtual group. For example:

```
127.0.0.1 localhost
127.0.0.2 myserver2.mydomain.com myserver2 loghost
127.0.0.3 myserver3.mydomain.com myserver3 loghost
```

> **Caution**
> Caution Do not edit the /etc/hosts file unless your system is running an SSL implementation.

25. Sign in to the operating system of the Service Manager server host and copy the primary host's license file (LicFile.txt) from the network share.
26. Change directories to this Service Manager Server's RUN folder.
27. Paste the primary host's license file into this server's RUN folder.
28. Start the Service Manager server.

> **Note**
> Note The primary host must be running in order for this host to validate the license file and start up.

29. Install Service Manager on the next host in the virtual group. For example: myserver3.
    The host must be on the same subnet as the other hosts in the virtual group.
30. Sign in to the operating system of the Service Manager server host and change directories to the Service Manager RUN folder. For example:

```
C:\Program Files\OpenText\Service Manager 9.xx\Server\RUN
```

31. Open the Service Manager configuration file (sm.cfg or smstart) in a text editor.
32. Edit the file so that only the following lines appear:

```
sm -httpPort:13081 -httpsPort:13082
sm -httpPort:13083 -httpsPort:13084
sm -sync
```

> **Note**
> Note You don't need the system.start line on this host as only one host in the virtual group needs to run the background processes.

    The httpPort:13081 parameter specifies that the servlet container process listens to client connection requests on HTTP port 13081. This communications port must be unique on the host on which you start the servlet container. The httpsPort:13082 parameter specifies that the servlet container process listens to client connection requests on HTTPS port 13082. This communications port must be unique on the host on which you start the servlet container. The httpPort:13083 parameter specifies that the servlet container process listens to client connection requests on HTTP port 13083. This communications port must be unique on the host on which you start the servlet container. The httpsPort:13084 parameter specifies that the servlet container process listens to client connection requests on HTTPS port 13084. This communications port must be unique on the host on which you start the servlet container. The sync parameter specifies that the host system should start the sync background process to identify and remove unused processes.
33. Save the Service Manager configuration file.
34. Open the Service Manager initialization file (sm.ini) in a text editor.

35. Add the RDBMS connection settings. For example:

```
[oracle10]
sqldb:ora102
sqllogin:sm7user/password
sqllibrary:SQORACLE.OCI10.DLL
sqldictionary:oracle10
```

36. Add the following lines:

```
grouplicenseip:10.0.0.135
groupname:mygroup1
groupmcastaddress:224.0.1.255
groupsubnetaddress:255.255.255.0
groupport:13100
threadsperprocess:50
sessiontimeout:3
system:13080
```

> **Note**
> Note This example assumes the host system has only one network adapter card and therefore doesn't need the groupbindaddress parameter. If your host has multiple network adapter cards, you must add the groupbindaddress parameter to specify which network adapter the virtual group will use for communications.

The grouplicenseip:10.0.0.135 parameter defines the TCP/IP address of the Service Manager host with a valid AutoPass license for the virtual group. The value of this parameter must match the IP address specified in the AutoPass license.
The groupname:mygroup1 parameter creates a virtual group that servlet container processes across multiple systems can join for horizontal scaling purposes. The value is arbitrary and can be any text value without spaces.
The groupmcastaddress:244.0.1.255 parameter defines the TCP/IP address that servlet container processes can use to communicate with the load balancer process. The TCP/IP address must be consistent with the UDP multicasting protocol.
The groupsubnetaddress:255.255.255.0 parameter defines the subnet mask that servlet container processes can use to communicate with the load balancer process. The subnet mask must be consistent with the IPv4 protocol.

> **Note**
> Note You only have to define the subnet address if you are using subnet masking on your IP addresses.

The groupport:13100 parameter defines the communications port that servlet container processes can use to communicate with the load balancer process. The TCP/IP address can be any available communications port.
The threadsperprocess:50 parameter defines the number of threads each process supports. A value of 50 threads assumes that the Service Manager host has the minimum recommended system memory available for servlet container processes.
The sessiontimeout:3 parameter defines the number of minutes that the client connection can remain unresponsive before the server closes the connection. A value of 3 minutes assumes that most network latency issues are quickly and easily resolved.
The system:13080 parameter defines a numerical identifier for the system, in this case the HTTP port of the load balancer process. The value of this parameter must match for each system in the virtual group.

37. Save the Service Manager initialization file.

38. If the host runs on a Linux operating system and uses an SSL implementation, edit the /etc/hosts file to list the fully qualified domain name of each host in the virtual group. For example:

```
127.0.0.1 localhost
127.0.0.2 myserver2.mydomain.com myserver2 loghost
127.0.0.3 myserver3.mydomain.com myserver3 loghost
```

Caution: Do not edit the /etc/hosts file unless your system is running an SSL implementation.

39. Sign in to the operating system of the Service Manager server host and copy the primary host's license file (LicFile.txt) from the network share.
40. Change directories to this Service Manager Server's RUN folder.
41. Paste the primary host's license file into this server's RUN folder.
42. Start the Service Manager server.
    Note: The primary host must be running in order for this host to validate the license file and start up.
43. Sign in to the operating system of the Service Manager Web tier host and change directories to the Service Manager WEB-INF folder. For example:

```
C:\<Tomcat>\webapps\sm\WEB-INF
```

44. Open the Web configuration file (web.xml) in a text editor.
45. Set the following parameter values:

Parameter	Default value	Description
secureLogin	true	Controls the encryption of network communication between the web application server and the web browser. Set it to false if you don't use Secure Sockets Layer (SSL) connections to the web server. **Note**: To use secure login, you must enable SSL on your web application server. For details, refer to your web application server documentation.
sslPort	8443	This parameter is needed only when secureLogin is set to true. Set it to the SSL port of the web application server.
serverHost	localhost	Specifies the name of the Service Manager host server.
serverPort	13080	Specifies the communications port number to which the Service Manager server listens.

```
...
<context-param>
<param-name>secureLogin</param-name>
<param-value>true</param-value>
</context-param>
<context-param>
<param-name>sslPort</param-name>
<param-value>8443</param-value>
</context-param>
...
<param-name>serverHost</param-name>
<param-value>myserver1</param-value>
</init-param>
<init-param>
<param-name>serverPort</param-name>
<param-value>13080</param-value>
</init-param>
```

46. Save the Web configuration file.
47. Start the Service Manager Web tier.
48. Open Service Manager Windows clients and set the Service Manager host name and communication port values:

Field	Value
Server host name	myserver1
Server port number	13080

49. Connect to the Service Manager host.

Related topics

Horizontal scaling implementation
Server implementation options
Servlet implementation
Servlet implementation options
Configuring a horizontal scaling environment
Webware software licensing center

1.6.1.2. Configuring a horizontal scaling environment

This configuration is intended for customers who:

- Want to maximize the number of client connections supported across multiple hosts
- Have multiple hosts available to manage all concurrent client connections

Terminology

- **Group**: A group of Service Manager server processes running on one or more hosts and connecting to one database to serve Service Manager server processes within a group.
- **Node**: A Service Manager server process within a group.
- **Primary host**: The host in the Service Manager horizontal scaling group whose IP is bound to the Service Manager AutoPass license.
- **Secondary host**: The host in the Service Manager horizontal scaling group whose IP is not bound to the Service Manager AutoPass license. The Service Manager processes on this host depend on running Service Manager processes on the primary host to validate the Service Manager AutoPass license.
- **Service Manager servlet process**: A Service Manager server process that embeds the Tomcat application server and serves Service Manager Windows clients and Web clients, as well as Web Services requests. It is also generally

referred to as a Service Manager process.

- **Service Manager background process**: Background processes that wake up periodically to execute a particular RAD application, or Service Manager runtime environment routine. For example: `sm -sync`.
- **Service Manager transient process**: A Service Manager server process that executes a RAD application or Service Manager runtime environment routine only once, not periodically. For example: `sm -reportlbstatus`.

Number of Service Manager hosts required

This configuration can be set up on one or more hosts.

Commands required in sm.cfg on primary host

You must set the following configuration commands.

```
sm -loadBalancer -httpPort:<value>
sm -httpPort:<value> -httpsPort:<value>
sm system.start
```

Commands required in sm.cfg on secondary host

```
sm -httpPort:<value> -httpsPort:<value>
sm -sync
```

- `sm -loadBalancer -httpPort:<value>` - Is a special Service Manager process that redirects client connections requests to other available Service Manager processes.
- `sm -httpPort:<value> -httpsPort:<value>` - Specifies the HTTP or HTTPS communications port for one Service Manager process. Each Service Manager process by default can host 50 users. You can start as many Service Manager processes as you want, as long as your server has enough physical memory and resources to start the processes. You can also change the number of users hosted on one Service Manager process by setting the `threadsperprocess` parameter.
- `sm system.start` - Starts one Service Manager process that contains all background processes.
- `sm -sync` - Identifies defunct Service Manager processes on the host and frees the shared resources that were allocated for it. The `sm -sync` process needs to start only one per host. Since the `sm system.start` process includes the `sm -sync` background process, the `sm -sync` command should not be included in the `sm.cfg` file that has the `sm system.start` command.

Parameters required in sm.ini

You must set the following initialization parameters.

```
system:<value>
grouplicenseip:<value>
groupname:<value>
groupmcastaddress:<value>
groupport:<value>
```

- `system` - Defines a unique numerical ID for the system. The value of this parameter must be identical in the `sm.ini` files for each host in the horizontal scaling group.
- `grouplicenseip` - The value should be the primary host IP.
- `groupname` - Creates a group name as the horizontal scaling group identifier that the Service Manager processes use to identify the group. The value of this parameter is only alphanumeric characters.
- `groupmcastaddress` - Defines the TCP/IP multicast address that all Service Manager processes within the horizontal scaling group use to communicate with each other.
- `groupport` - Defines the communications port that all Service Manager processes use to communicate with all other Service Manager processes within the horizontal scaling group.
- `[RDBMS Settings]` - Define the RDBMS connection and authorization parameters.

Optional parameters in sm.ini

The following initialization parameters are optional.

```
threadsperprocess:<value>
preferredFQHN:<value>
groupbindaddress:<value>
```

- `threadsperprocess` - Defines the maximum number of concurrent user sessions per Service Manager process. Use a value that maximizes the system resources of your Service Manager host. The default value for the parameter `threadsperprocess` is 250. For more sizing information using this parameter, refer to Service Manager server (servlet container).
- `preferredFQHN` - Specifies the fully qualified host name you want Service Manager clients to use when communicating with the server. Service Manager loadBalancer redirects client requests to the target host with the target host's preferredFQHN. You only need to set this parameter if your Service Manager host is identified by multiple names in the network.
- `groupbindaddress` - Defines the TCP/IP address of the network adapter you want Service Manager processes to use to communicate with other processes in a horizontal scaling group. If your Service Manager hosts contain multiple network adapters, you must specify the IP address of the network adapter you want the horizontal scaling group to use with the `groupbindaddress` parameter.

Operating system requirements

All hosts in the horizontally-scaled environment must run on the same operating system version and run at the same patch level. Minimum patch level for each operating system is specified in the support matrix. See Support Matrices on the Software Support Online site.

Caution: Running Service Manager in a horizontally-scaled environment and using different operating systems can corrupt your data. You can also run into problems when upgrading to the latest patches. Most of the time, Service Manager patches are released on all operating systems. However, in some cases, patches may be delivered to a particular operating system. When mixing operating systems, applying patches will become more complex as you need to test on multiple platforms.

Network requirements

All hosts in the horizontal scaling group must run from the same subnet on the network in order for the Service Manager processes to communicate with one another.

Memory requirements

We recommend around 1.5 GB RAM per Service Manager process. This is based on the expected user load. For example, if you have 10 processes each running with a 50-user load, the minimum required RAM would be 1.5 GB per Service Manager process, 10x1.5 GB RAM. For information on sizing, see Sizing.

Licensing requirements

Obtain an AutoPass license for one host in your horizontal scaling group. On Windows platforms, AutoPass installs as part of the server installation. On Linux platforms, you must install it manually before you can run Service Manager. For more information, see Install. This host becomes the primary host of the horizontal scaling group and AutoPass includes this host's IP address in the `LicFile.txt` file. You must start the primary host first when starting the horizontal scaling group. Copy the `LicFile.txt` file from the primary host's AutoPass directory to the AutoPass directory of each secondary host in the horizontal scaling group. Each secondary host must have a copy of the primary host's `LicFile.txt` file in order to start.

The summary steps are as follows:

1. List all hosts in your horizontal scaling environment to determine the primary server and all secondary servers.
2. Get a license.
3. Copy the primary server's IP address to get the license and save it as the primary server's `LicFile.txt` file.
4. Copy the primary server's `LicFile.txt` file on all secondary servers. The default directory for this file is as follows:

   ```
   <Service Manager server installation path>/RUN/LicFile.txt
   ```

5. Add the following parameter in the `sm.ini` file: `grouplicenseip:Primary_Servers_IpAddress`

To start the horizontal scaling group

1. Start the Service Manager server on the primary server.
   Note: The primary server must be started prior to starting any of the secondary servers.
2. Start the Service Manager server on the secondary servers, in no particular order.

To establish the horizontal scaling group, the first node of the group should be started on the primary host. When the group is established on the primary host, Service Manager server processes on secondary hosts can start to join the group. Once the whole group is established, the primary host can be brought down for maintenance, and then rejoin the rest of the running group when it is restarted. As long as there is a member Service Manager server process running in the group, Service Manager server processes on another secondary host with the same `grouplicenseip:<primary host IP>` can join the group. The primary host is not required to be running. Another secondary host can be started, even if the primary host is down for any reason. However, when the group is down (for example, there are no nodes running in the group), then the group has to be reestablished from the primary host first.

Other requirements

Install the applicable database client software on each host of the horizontal scaling group so that they can access the Service Manager RDBMS. See your RDBMS vendor documentation for instructions.

Related topics

Servlet implementation
Parameter: httpPort
Parameter: httpsPort
Parameter: loadBalancer
Parameter: grouplicenseip
Parameter: groupbindaddress
Parameter: groupmcastaddress
Parameter: groupname
Parameter: groupport
Install

1.6.1.3. Configuring SSL and LW-SSO in a horizontal scaling environment

In a horizontally scaled environment, you use the Service Manager software load balancer in front of a virtual group to redirect client requests among the server nodes in the group. The load balancer server is called the primary server, while the other servers in the group are called secondary servers.

Important Requirements

- On a horizontally scaled system, it is very important that all certificates are created from the same machine. Ensure to not copy any files to their target directories until all certificates for all server machines (the primary server and secondary server machines) in the horizontally scaled environment are created.
- For each server host (primary or secondary), you need a unique server certificate.
- For each Windows client machine you need a unique client certificate.
- For each Service Manager Web application server host you need a unique client certificate.

Best Practice recommendation

When generating each Windows or web client certificate, prefix the keystore, certificate request and certificate names with the FQDN name of the client machine to make them unique and easy to distinguish.

If a Windows client and the Web application server are on the same physical machine, it is possible to use the same cacerts and clientcerts files for both, rather than creating two sets of nearly identical keystores. In such a case, copy the files created for the Windows or Web client – whichever was created first– into either the <Service Manager Client>/plugins/com.hp.ov.sm.client.common_x.xx directory or the Service Manager/WEB-INF folder of the Web application server.

To configure SSL in a horizontal scaling implementation:

1. Create a private key and public certificate for your private certificate authority.

2. Update the Java cacerts file by importing your private certificate authority's certificate into it. The updated cacerts file will contain the certificate and private key of the certificate authority that signs each server/client certificate.
3. Create a separate signed server keystore for each server machine.
   Note: In a later step, you will add each server keystore and the updated Java cacerts file to each server's sm.ini file as the following parameters: **keystoreFile**, and **truststoreFile**.
4. Create a separate signed client certificate for each Service Manager client (web application server or Windows client) machine and import each client certificate into a trusted clients keystore.
   Note: In a later step, you will add the trusted clients keystore as the following parameter in each server's sm.ini file: **ssl_trustedClientsJKS**.
5. Copy the following keystore files to each server's RUN folder:
   - The server's server keystore file (unique)
   - The trusted clients keystore file (same for all servers)
   - The updated Java cacerts file (same for all servers and clients)
6. Copy the following keystore files to each web tier's WEB-INF folder:
   - The updated Java cacerts file
   - <clientcerts>.keystore – This keystore contains the signed certificate of your Service Manager web tier client.
7. Copy the following keystore files to each Windows client's <Windows client installation path>\plugins\com.hp.ov.sm.client.common_x.xx folder.
   - The updated Java cacerts file
   - <clientcerts>.keystore – This keystore contains the signed certificate of each Windows client.
8. Add the following SSL configuration lines to the sm.ini file for each of the primary server and secondary servers, and restart each server.
   Note: For all servers, the truststoreFile and ssl_trustedClientsJKS settings should be the same; however, each server should use its own server keystore (keystoreFile:<servercert.keystore>).

   ```
   ssl:1
   ssl_reqClientAuth:2
   sslConnector:1
   ssl_trustedClientsJKS:<trustedclients.keystore>
   ssl_trustedClientsPwd:<ClientKeyPassword>
   truststoreFile:cacerts
   truststorePass:changeit
   keystoreFile:<servercert.keystore>
   keystorePass:changeit
   ```

9. Configure each Service Manager web client to validate each server's signed certificate and present the signed client certificate.
   1. Stop the web application server running the web tier, open the web configuration file (web.xml) in a text editor.
   2. Configure the following parameters as shown below:

      ```
      <init-param>
          <param-name>cacerts</param-name>
          <param-value>/WEB-INF/cacerts</param-value>
      </init-param>
      <init-param>
          <param-name>keystore</param-name>
          <param-value>/WEB-INF/<clientcerts>.keystore</param-value>
      </init-param>
      <init-param>
          <param-name>keystorePassword</param-name>
          <param-value><<clientcerts>.keystore password></param-value>
      </init-param>
      ```

3. Set the ssl parameter to true.
4. Set the serverHost and serverPort parameters to the fully-qualified domain name and port number of the primary server. For example: myprimaryserver.mydomain.com and 13080.
5. Restart each web application server.

10. Configure each Windows client to validate each server's signed certificate and present the signed client certificate.
    1. Click **Window** > **Preferences** > **Service Manager** > **Security**.
    2. Set CA Certificates File to the cacerts keystore you copied to the <Windows client installation path>\plugins\com.hp.ov.sm.client.common_x.xx folder.
    3. Set Keystore File to the keystore containing your Windows client's signed certificate, for example <clientcerts>.keystore. You created this keystore when you created the Windows client certificate request.
    4. Set Keystore password to the password required to access the Windows client keystore. For example, ClientKeyPassword. You created this keystore password when you created the Windows client certificate request.
    5. Update your Windows client connections by selecting **Use SSL Encryption** on their Advanced tab.

To set up LW-SSO in a horizontal scaling implementation:

1. Enable LW-SSO on each server node. All server nodes should share the same initString setting. For details, see Configure LW-SSO in the Service Manager server.
   Note: Normally, all server nodes are in the same domain, so you can configure LW-SSO in one server node and then copy the server's LW-SSO configuration file (lwssofmconf.xml) to the rest of the nodes.
2. If needed, enable LW-SSO in each web tier. For details, see Configure LW-SSO in the Service Manager Web tier.
   Note: The server nodes and web tiers should share the same initString setting.

Related topics

Configure two-way SSL
Using LW-SSO with integrations
Configure LW-SSO in the Service Manager Web tier

1.6.2. Single servlet implementation

In a single servlet implementation you manage all client connections with one multithreaded process.

The key features of this implementation are:

- The Service Manager system consists of one Service Manager instance running on one physical host
- One dedicated servlet container process manages all client connections
- The implementation allows administrators to specify the communication ports the system uses
- The implementation allows administrators to specify a system connection limit
- The Service Manager system can manage a number of concurrent client connections up to the number of threadsperprocess

A single servlet implementation is a typically used for development environments or small production environments because it is easy to setup and manage.

You can convert a single servlet implementation into any of the other servlet implementations with the addition of a load balancer process and one or more additional servlet container instances.

Single servlet implementation diagram

The *Single servlet implementation diagram* is a reference diagram for implementers who are responsible for installing and configuring Service Manager. The guide shows the configuration settings and network connections required for a single server running a single servlet container implementation.

Single Servlet Implementation Overview

This implementation is intended for customers who:

- Want to manage all client connections with one multithreaded process
- Only expect up to 50 concurrent client connections
- Want to specify the communications ports the HP Service Manager implementation uses
- Want a simple configuration to test servlet features

Sample implementation

Note: Sample values are for illustration only

Capacity Planning

Desired total capacity	50
Desired capacity per host	50

Processes and Threads

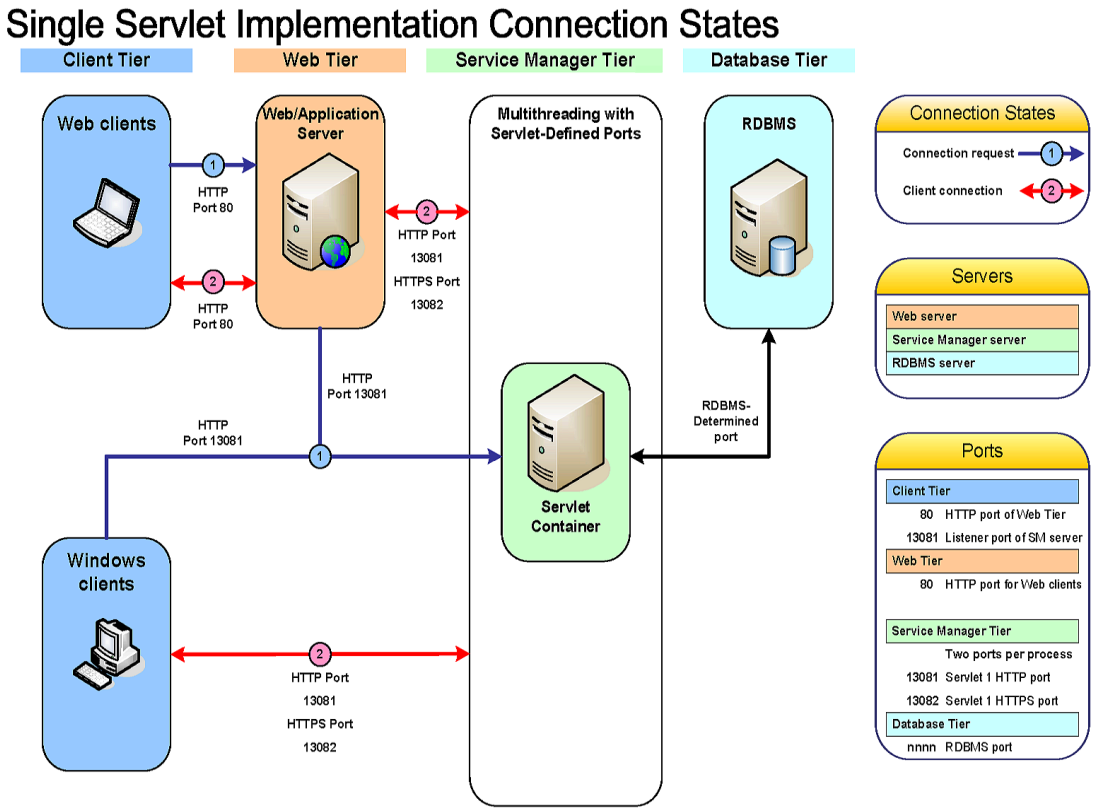
Desired Number of Threads per Process	50 Default
Number of servlet containers required per host to meet "Desired capacity per host"	1

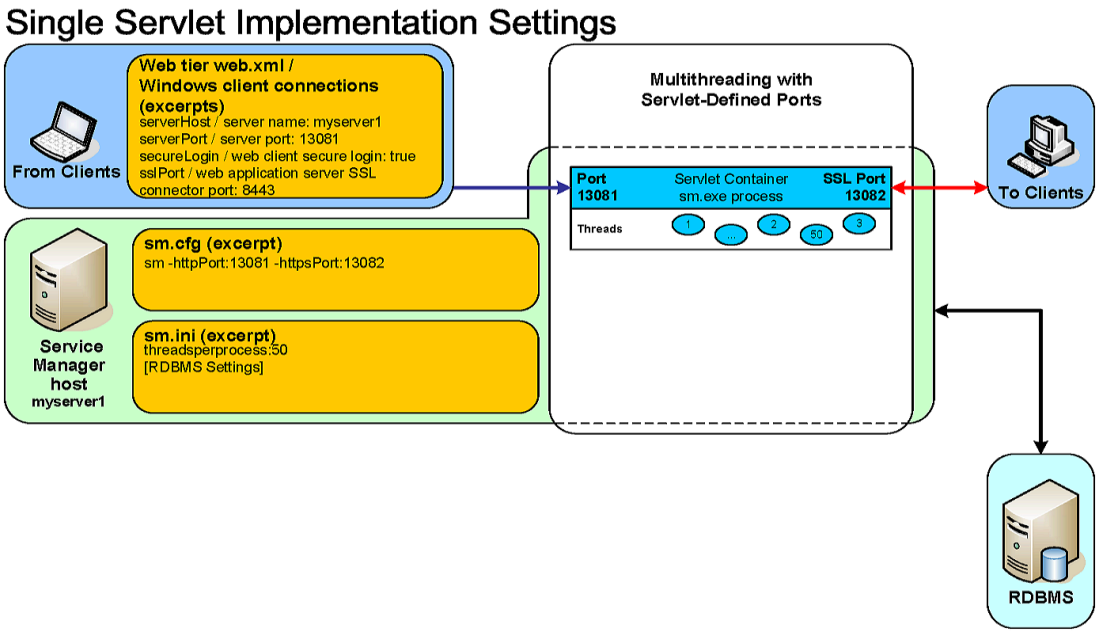
Hardware Requirements

Number of hosts required to meet "Desired total capacity"	1

Communications Ports

Number of load balancer ports required	0
Number of client connection ports required per host	2


Single Servlet Implementation Connection States
Client Tier
Web Tier
Service Manager Tier
Database Tier
Web clients
Web/Application Server
Multithreading with Servlet-Defined Ports
RDBMS
Connection States
Connection request
Client connection
HTTP Port 80
HTTP Port 80
HTTP Port 13081
HTTPS Port 13082
Servers
Web server
Service Manager server
RDBMS server
HTTP Port 13081
HTTP Port 13081
RDBMS-Determined port
Servlet Container
Ports
Client Tier
80 HTTP port of Web Tier
13081 Listener port of SM server
Web Tier
80 HTTP port for Web clients
Service Manager Tier
Two ports per process
13081 Servlet 1 HTTP port
13082 Servlet 1 HTTPS port
Database Tier
nnnn RDBMS port
Windows clients
HTTP Port 13081
HTTPS Port 13082



Related topics

Example: Setting up a single servlet implementation
Server implementation options
Requirements for a single servlet implementation

1.6.2.1. Example: Setting up a single servlet implementation

The following example describes how to set up a single servlet implementation that accomplishes the following:

- Manages all client connections with one multithreaded process
- Supports up to 50 concurrent client connections
- Allows an administrator to specify the communications ports the Service Manager implementation uses
- Provides a simple configuration to test servlet features

You can use this example to configure the implementation depicted in the *Single servlet implementation diagram*.

1. Install Service Manager on one host. For example, myserver1.
2. Sign in to the operating system of the Service Manager server host and change directories to the Service Manager RUN folder. For example:

   ```
   C:\Program Files\OpenText\Service Manager\server\RUN
   ```

3. Open the Service Manager configuration file (sm.cfg or smstart) in a text editor.
4. Add the following line:

   ```
   sm -httpPort:13081 -httpsPort:13082
   ```

   The httpPort:13081 parameter specifies that the servlet container process listens to client connection requests on HTTP port 13081. This communications port must be unique on the host on which you start the servlet container.

The httpsPort:13082 parameter specifies that the servlet container process listens to client connection requests on HTTPS port 13082. This communications port must be unique on the host on which you start the servlet container.

5. Save the Service Manager configuration file.
6. Open the Service Manager initialization file (sm.ini) in a text editor.
7. Add the RDBMS connection settings. For example:

```
[oracle10]
sqldb:ora102
sqllogin:sm7user/password
sqllibrary:SQORACLE.OCI10.DLL
sqldictionary:oracle10
```

8. Add the following lines:

```
threadsperprocess:50
sessiontimeout:3
```

The threadsperprocess:50 parameter defines the number of threads each process supports. A value of 50 threads assumes that the Service Manager host has the minimum recommended system memory available for servlet container processes.
The sessiontimeout:3 parameter defines the number of minutes that the client connection can remain unresponsive before the server closes the connection. A value of 3 minutes assumes that most network latency issues are quickly and easily resolved.

9. Save the Service Manager initialization file.
10. Start the Service Manager server.
11. Sign in to the operating system of the Service Manager Web tier host and change directories to the Service Manager WEB-INF folder. For example:

```
C:\apache-tomcat-<version>\webapps\sm\WEB-INF
```

12. Open the Web configuration file (web.xml) in a text editor.

13. Set the following parameter values:

Parameter	Default value	Description
secureLogin	true	Controls the encryption of network communication between the web application server and the web browser. Set it to false if you don't use Secure Sockets Layer (SSL) connections to the web server. **Note**: To use secure login, you must enable SSL on your web application server. For details, refer to your web application server documentation.
sslPort	8443	This parameter is needed only when `secureLogin` is set to true. Set it to the SSL port of the web application server.
serverHost	localhost	Specifies the name of the Service Manager host server.
serverPort	13080	Specifies the communications port number to which the Service Manager server listens.

```
...
<context-param>
<param-name>secureLogin</param-name>
<param-value>true</param-value>
</context-param>
<context-param>
<param-name>sslPort</param-name>
<param-value>8443</param-value>
</context-param>
...
<param-name>serverHost</param-name>
<param-value>myserver1</param-value>
</init-param>
<init-param>
<param-name>serverPort</param-name>
<param-value>13080</param-value>
</init-param>
```

14. Save the Web configuration file.
15. Start the Service Manager Web tier.
16. Open Service Manager Windows clients and set the Service Manager hostname and communication port values:

Field	Value
Server host name	myserver1
Server port number	13081

17. Connect to the Service Manager host.

Related topics

Server implementation options
Servlet implementation
Single servlet implementation
Requirements for a single servlet implementation

1.6.2.2. Requirements for a single servlet implementation

This configuration is intended for customers who:

- Want to manage all client connections with one multithreaded process
- Only expect up to 50 concurrent client connections
- Want to specify the communications ports the Service Manager implementation uses
- Want a simple configuration to test servlet features

Number of Service Manager hosts required

This implementation requires the following number of hosts.
1

Parameters required in sm.cfg

You must set the following configuration parameters.

```
sm -httpPort:<value> - httpsPort:<value>
```

- httpPort – identify the communications port that a servlet container process uses to communicate with clients using HTTP. The servlet container communications port must be unique on the host on which you start the servlet container.
- httpsPort – identify the communications port that a servlet container process uses to communicate with clients using HTTPS. The servlet container communications port must be unique on the host on which you start the servlet container.

Parameters required in sm.ini

You must set the following initialization parameters.

- sessiontimeout – define the number of minutes a client connection can remain unresponsive before the server closes the connection.
- threadsperprocess – identify the total number of threads the servlet container process supports. Use a value of threads that maximizes the system resources of your Service Manager host.
  The default value for the parameter threadsperprocess is 250. For more sizing information using this parameter, refer to Service Manager server (servlet container).

Parameters required in web.xml

You must set the following Web parameters.

- serverHost – identify the host name of the Service Manager host
- serverPort – identify the communications port on which the Service Manager host listens for client connections requests

Windows client preferences required

You must set the following preferences from the **Connection** menu.

- Server host name – identify the host name of the Service Manager host
- Server port number – identify the communications port on which the Service Manager host listens for client connections requests

Related topics

Servlet implementation
Servlet implementation options
Parameter: httpPort
Parameter: httpsPort

1.6.3. Vertical scaling implementation

In a vertical scaling implementation you maximize the number of client connections supported on a single host.

The key features of this implementation are:

- The Service Manager system consists of several servlet container processes running on one physical host
- The implementation allows administrators to specify a system connection limit
- The Service Manager system can manage a number of concurrent client connections up to the number of servlet container processes times the threadsperprocess value (For example, 6 servlet container processes supporting 50 threadsperprocess can support up to 300 client connections)
- The implementation allows administrators to specify the communication ports the system uses
- A dedicated load balancer process manages and routes client connections to available servlet container processes
- Administrators can dynamically add and remove Service Manager instances from a virtual group

A vertical scaling implementation is typically used in small to medium environments where hardware system resources are limited. A vertical scaling implementation can support as many client connections as the Service Manager host has available system resources.

You can convert a vertical scaling implementation into a horizontal scaling implementations with the addition of virtual grouping parameters and one or more additional servlet container instances installed on separate physical hosts.

Note: There can be only **one** kmupdate process running at any time regardless of the number of hosts. Starting more than one kmupdate process causes unpredictable behavior on the search engine server.

Vertical scaling implementation diagram

The Vertical scaling implementation diagram is a reference diagram for implementers who are responsible for installing and configuring Service Manager. The guide shows the configuration settings and network connections required for a single server running multiple servlet containers implementation.

Related topics

Example: Setting up a vertical scaling implementation
Server implementation options
Startup options for servlet container processes
Configuring a vertical scaling environment

1.6.3.1. Example: Setting up a vertical scaling implementation

The following example describes how to set up a vertical scaling implementation that accomplishes the following:

- Maximizes the number of client connections supported on a single host
- Allows an administrator to specify the communications ports the Service Manager implementation uses

You can use this example to configure the implementation depicted in the *Vertical scaling implementation diagram*.

1. Install Service Manager on one host. For example: myserver1.
2. Sign in to the operating system of the Service Manager server host and change directories to the Service Manager RUN folder. For example:

   ```
   C:\Program Files\OpenText\Service Manager\server\RUN
   ```

3. Open the Service Manager configuration file (sm.cfg or smstart) in a text editor.

4. Add the following lines:

```
sm -loadBalancer -httpPort:13080
sm -httpPort:13081 -httpsPort:13082
sm -httpPort:13083 -httpsPort:13084
```

   The loadBalancer parameter creates a special servlet container process to route client connection requests to other available servlet container processes. A vertical scaling implementation only needs one load balancer.
   The httpPort:13080 parameter specifies that the load balancer process listens to client connection requests on HTTP port 13080. This communications port must be unique across all hosts that you want to join the Service Manager virtual group.
   The httpPort:13081 parameter specifies that the servlet container process listens to client connection requests on HTTP port 13081. This communications port must be unique on the host on which you start the servlet container.
   The httpsPort:13082 parameter specifies that the servlet container process listens to client connection requests on HTTPS port 13082. This communications port must be unique on the host on which you start the servlet container.
   The httpPort:13083 parameter specifies that the servlet container process listens to client connection requests on HTTP port 13083. This communications port must be unique on the host on which you start the servlet container.
   The httpsPort:13084 parameter specifies that the servlet container process listens to client connection requests on HTTPS port 13084. This communications port must be unique on the host on which you start the servlet container.
5. Save the Service Manager configuration file.
6. Open the Service Manager initialization file (sm.ini) in a text editor.
7. Verify that the auth parameter has a valid authorization code.
8. Add the RDBMS connection settings. For example:

```
[oracle10]
sqldb:ora102
sqllogin:sm7user/password
sqllibrary:SQORACLE.OCI10.DLL
sqldictionary:oracle10
```

9. Add the following lines:

```
threadsperprocess:50
sessiontimeout:3
```

   The threadsperprocess:50 parameter defines the number of threads each process supports. A value of 50 threads assumes that the Service Manager host has the minimum recommended system memory available for servlet container processes.
   The sessiontimeout:3 parameter defines the number of minutes that the client connection can remain unresponsive before the server closes the connection. A value of 3 minutes assumes that most network latency issues are quickly and easily resolved.
10. Save the Service Manager initialization file.
11. Start the Service Manager server.
12. Sign in to the operating system of the Service Manager Web tier host and change directories to the Service Manager WEB-INF folder. For example:

```
C:\apache-tomcat-<version>\webapps\sm\WEB-INF
```

13. Open the Web configuration file (web.xml) in a text editor.

14. Set the following parameter values:

Parameter	Default value	Description
secureLogin	true	Controls the encryption of network communication between the web application server and the web browser. Set it to false if you don't use Secure Sockets Layer (SSL) connections to the web server. **Note**: To use secure login, you must enable SSL on your web application server. For details, refer to your web application server documentation.
sslPort	8443	This parameter is needed only when `secureLogin` is set to true. Set it to the SSL port of the web application server.
serverHost	localhost	Specifies the name of the Service Manager host server.
serverPort	13080	Specifies the communications port number to which the Service Manager server listens.

```
...
<context-param>
<param-name>secureLogin</param-name>
<param-value>true</param-value>
</context-param>
<context-param>
<param-name>sslPort</param-name>
<param-value>8443</param-value>
</context-param>
...
<param-name>serverHost</param-name>
<param-value>myserver1</param-value>
</init-param>
<init-param>
<param-name>serverPort</param-name>
<param-value>13080</param-value>
</init-param>
```

15. Save the Web configuration file.
16. Start the Service Manager Web tier.
17. Open Service Manager Windows clients and set the Service Manager hostname and communication port values:

Field	Value
Server host name	myserver1
Server port number	13080

18. Connect to the Service Manager host.

Related topics

Server implementation options
Servlet implementation
Servlet implementation options
Vertical scaling implementation
Configuring a vertical scaling environment

1.6.3.2. Configuring a vertical scaling environment

This configuration is intended for customers who:

- Want to maximize the number of client connections supported on a single host
- Have a host with enough system resources to manage all concurrent client connections

Terminology

- **Group**: A group of Service Manager server processes running on one or more hosts and connecting to one database to serve Service Manager server processes within a group.
- **Node**: A Service Manager server process within a group.
- **Service Manager servlet process**: A Service Manager server process that embeds the Tomcat application server and serves Service Manager Windows clients and Web clients, as well as Web Services requests. It is also generally referred to as a Service Manager process.
- **Service Manager background process**: Background processes that wake up periodically to execute a particular RAD application, or Service Manager runtime environment routine. For example: `sm -sync`.
- **Service Manager transient process**: A Service Manager server process that executes a RAD application or Service Manager runtime environment routine only once, not periodically. For example: `sm -reportlbstatus`.

Commands required in sm.cfg

You must set the following configuration commands.

```
sm -loadBalancer -httpPort:<value>
sm -httpPort:<value> -httpsPort:<value>
sm system.start
```

- `sm -loadBalancer -httpPort:<value>` - Is a special Service Manager servlet process that redirects client connections requests to other available Service Manager processes.
- `sm -httpPort:<value> -httpsPort:<value>` - Specifies the HTTP or HTTPS communications port for one Service Manager servlet process. Each Service Manager process by default can host 50 users. You can start as many Service Manager servlet processes as you want, as long as your server has enough physical memory and resources to start the processes. You can also change the number of users hosted on one Service Manager servlet process by setting the `threadsperprocess` parameter.

Optional parameters in sm.ini

The following initialization parameters are optional.

```
threadsperprocess:<value>
preferredFQHN:<value>
groupbindaddress:<value>
```

- `threadsperprocess` - Defines the maximum number of concurrent user sessions per Service Manager servlet process. Use a value that maximizes the system resources of your Service Manager host. The default value for the parameter threadsperprocess is 250. For more sizing information using this parameter, refer to Service Manager server (servlet container).
- `preferredFQHN` - Specifies the fully qualified host name you want Service Manager clients to use when communicating with the server. Service Manager loadBalancer redirects client requests to the target host with the

target host's preferredFQHN. You only need to set this parameter if your Service Manager host is identified by multiple names in the network.

- `groupbindaddress` - Defines the TCP/IP address of the network adapter you want Service Manager processes to use to communicate with other processes in the group.

Memory requirements

For an out-of-box environment (limited to 50 concurrent users), a servlet container requires approximately 8GB of RAM, which is distributed in the following ways:

- Java virtual machine requirement, the Java heap size is 2 GB.
- User session memory usage:
  - For Windows platform, the memory usage is 40 MB - 55 MB per user session.
  - For Linux platform, the memory usage is 55MB - 70MB per user session.
- Shared_memory setting value is equal to or greater than 512 MB (Counted only once for all servlet containers on a single computer)

For information on sizing, refer to the sizing guide.

Licensing requirements

Obtain an AutoPass license for your vertical scaling environment. On Windows platforms, AutoPass installs as part of the server installation. On Linux platforms, you must install it manually before you can run Service Manager. For more information, see Install. AutoPass includes this host's IP address in the `LicFile.txt` file.

Do the following:

1. Get a license.
2. Copy the primary server's IP address to get the license and save it as the primary server's `LicFile.txt` file. The default directory for this file is as follows:
   - **On Windows**: C:\Program Files\Common Files\Hewlett-Packard\HPOvLIC\data\LicFile.txt
   - **On Linux**: /var/opt/OV/HPOvLIC/LicFile.txt

Related topics

Servlet implementation
Servlet implementation options
Parameter: httpPort
Parameter: httpsPort
Parameter: loadBalancer
Install

1.6.4. Vertical scaling and required SSL implementation

In a vertical scaling and required SSL implementation you maximize the number of client connections supported on a single host and provided SSL-encrypted security between client and server communications.

The key features of this implementation are:

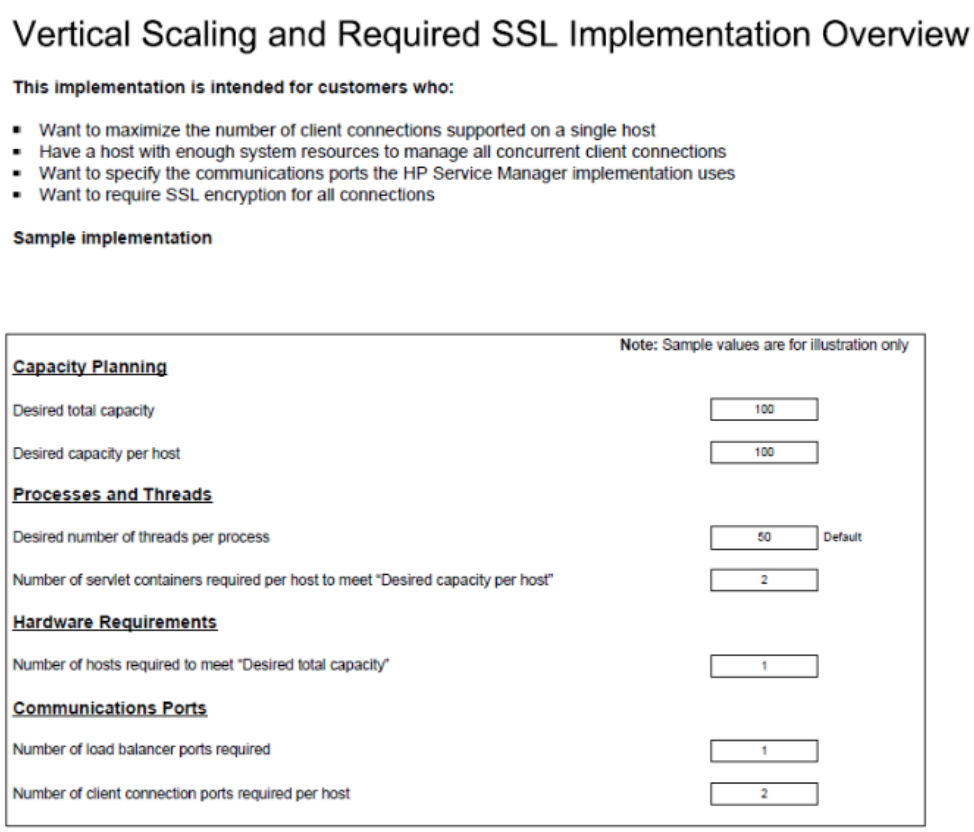
- The Service Manager system consists of several servlet container processes running on one physical host
- The implementation allows administrators to specify a system connection limit
- The Service Manager system can manage a number of concurrent client connections up to the number of servlet container processes times the threadsperprocess value (For example, 6 servlet container processes supporting 50 threadsperprocess can support up to 300 client connections)
- The implementation allows administrators to specify the communication ports the system uses
- The implementation requires SSL-encrypted communications between Service Manager clients and servers
- A dedicated load balancer process manages and routes client connections to available servlet container processes
- Administrators can dynamically add and remove Service Manager instances from a virtual group

A vertical scaling implementation is typically used in small to medium environments where hardware system resources are limited. A vertical scaling implementation can support as many client connections as the Service Manager host has available system resources.

You can convert a vertical scaling implementation into a horizontal scaling implementations with the addition of virtual grouping parameters and one or more additional servlet container instances installed on separate physical hosts.

Vertical scaling and required SSL implementation diagram

The *Vertical scaling and required SSL implementation diagram* is a reference diagram for implementers who are responsible for installing and configuring Service Manager. This diagram shows the configuration settings and network connections required for a single server running multiple servlet containers in a required SSL implementation.



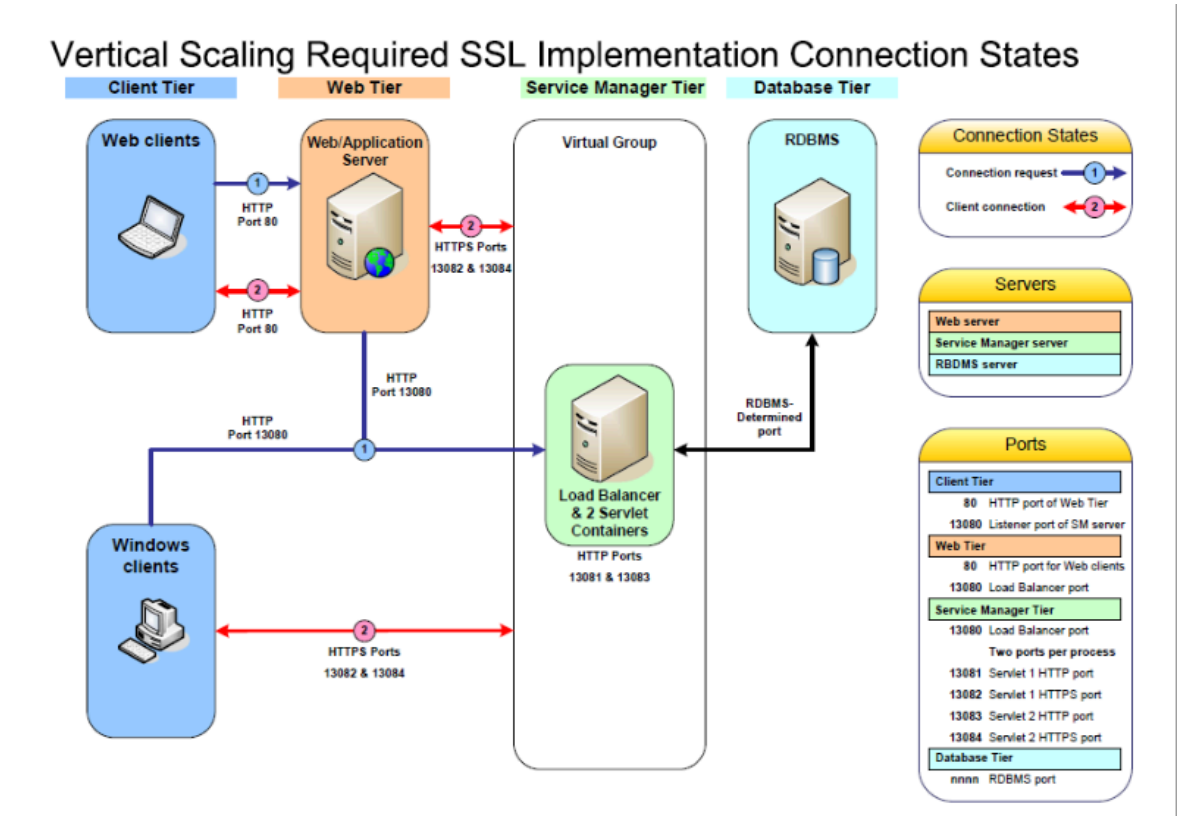
Vertical Scaling and Required SSL Implementation Overview

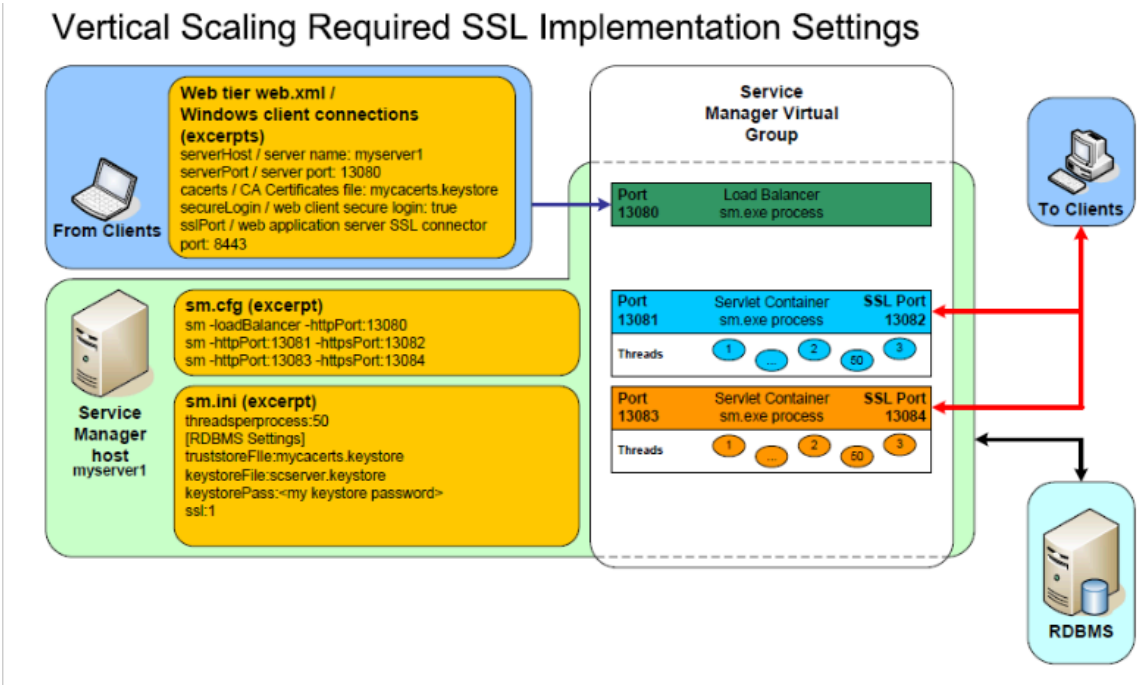
This implementation is intended for customers who:

- Want to maximize the number of client connections supported on a single host
- Have a host with enough system resources to manage all concurrent client connections
- Want to specify the communications ports the HP Service Manager implementation uses
- Want to require SSL encryption for all connections

Sample implementation

	Note: Sample values are for illustration only
Capacity Planning	
Desired total capacity	100
Desired capacity per host	100
Processes and Threads	
Desired number of threads per process	50 Default
Number of servlet containers required per host to meet "Desired capacity per host"	2
Hardware Requirements	
Number of hosts required to meet "Desired total capacity"	1
Communications Ports	
Number of load balancer ports required	1
Number of client connection ports required per host	2





Related topics

Servlet implementation
Servlet implementation options
Startup options for servlet container processes

1.6.4.1. Example: Setting up a vertical scaling and required SSL implementation

The following example describes how to set up a vertical scaling and required SSL implementation that accomplishes the following:

- Maximizes the number of client connections supported on a single host

- Allows an administrator to specify the communications ports the Service Manager implementation uses
- Require SSL encryption for all connections
- Protects against complex SSL-related attacks
- Authenticates that the Service Manager server is a valid host

You can use this example to configure the implementation depicted in the *Vertical scaling and required SSL implementation diagram*.

1. Install Service Manager on one host. For example: myserver1.
2. Sign in to the operating system of the Service Manager server host and change directories to the Service Manager RUN folder. For example:

   ```
   C:\Program Files\OpenText\Service Manager x.xx\Server\RUN
   ```

3. Open the Service Manager configuration file (sm.cfg or smstart) in a text editor.
4. Add the following lines:

   ```
   sm -loadBalancer -httpPort:13080
   sm -httpPort:13081 -httpsPort:13082
   sm -httpPort:13083 -httpsPort:13084
   ```

   - The loadBalancer parameter creates a special servlet container process to route client connection requests to other available servlet container processes. A vertical scaling implementation only needs one load balancer.
   - The httpPort:13080 parameter specifies that the load balancer process listens to client connection requests on HTTP port 13080. This communications port must be unique across all hosts that you want to join the Service Manager virtual group.
   - The httpPort:13081 parameter specifies that the servlet container process listens to client connection requests on HTTP port 13081. This communications port must be unique on the host on which you start the servlet container.
   - The httpsPort:13082 parameter specifies that the servlet container process listens to client connection requests on HTTPS port 13082. This communications port must be unique on the host on which you start the servlet container.
   - The httpPort:13083 parameter specifies that the servlet container process listens to client connection requests on HTTP port 13083. This communications port must be unique on the host on which you start the servlet container.
   - The httpsPort:13084 parameter specifies that the servlet container process listens to client connection requests on HTTPS port 13084. This communications port must be unique on the host on which you start the servlet container.
5. Save the Service Manager configuration file.
6. Open the Service Manager initialization file (sm.ini) in a text editor.
7. Add the RDBMS connection settings. For example:

   ```
   [oracle10]
   sqldb:ora102
   sqllogin:sm7user/password
   sqllibrary:SQORACLE.OCI10.DLL
   sqldictionary:oracle10
   ```

8. Add the following lines:

```
threadsperprocess:50
sessiontimeout:3
truststoreFile:cacert.keystore
truststorePass:<cacert password>
keystoreFile:scserver.keystore
keystorePass:<server certificate password>
ssl:1
```

   - The threadsperprocess:50 parameter defines the number of threads each process supports. A value of 50 threads assumes that the Service Manager host has the minimum recommended system memory available for servlet container processes.
   - The sessiontimeout:3 parameter defines the number of minutes that the client connection can remain unresponsive before the server closes the connection. A value of 3 minutes assumes that most network latency issues are quickly and easily resolved.
   - The truststoreFile:cacert.keystore parameter defines the file name and path to the keystore containing a list of trusted CA certificates. This value assumes you are using the default trust store file provided in the Service Manager server's RUN folder. This parameter must not be encrypted with the new initialization parameter encryption feature or the Java components will not be able to find the trust store file.
   - The truststorePass:<cacert password> parameter specifies the password to the trust store file. This parameter must not be encrypted with the new initialization parameter encryption feature or the Java components will not be able to read the password value.
   - The keystoreFile:scserver.keystore parameter defines the file name and path to the keystore containing the server's certificate file and private key. This value assumes you are using the default trust store file provided in the Service Manager server's RUN folder. This parameter must not be encrypted with the new initialization parameter encryption feature or the Java components will not be able to find the keystore file.
   - The keystorePass:<server certificate password> parameter specifies the password to the keystore file. This parameter must not be encrypted with the new initialization parameter encryption feature or the Java components will not be able to read the password value.
   - The ssl:1 parameter requires the Service Manager server to use a signed server certificate for SSL-encryption of all client-server communications. Each client connection validates the server's certificate against the signing certificate authority. You must also use the keystoreFile and keystorePass parameters to define the location of the server certificate and private key.
9. Save the Service Manager initialization file.
10. Start the Service Manager server.
11. Sign in to the operating system of the Service Manager Web tier host and change directories to the Service Manager WEB-INF folder. For example:

```
<Tomcat>\webapps\webtier_x.xx\WEB-INF
```

12. Open the Web configuration file (web.xml) in a text editor.

13. Set the following parameter values:

Parameter	Default value	Description
secureLogin	true	Controls the encryption of network communication between the web application server and the web browser. Set it to false if you don't use Secure Sockets Layer (SSL) connections to the web server. **Note**: To use secure login, you must enable SSL on your web application server. For details, refer to your web application server documentation.
sslPort	8443	This parameter is needed only when secureLogin is set to true. Set it to the SSL port of the web application server.
serverHost	localhost	Specifies the name of the Service Manager host server.
serverPort	13080	Specifies the communications port number to which the Service Manager server listens.

```
...
<context-param>
<param-name>secureLogin</param-name>
<param-value>true</param-value>
</context-param>
<context-param>
<param-name>sslPort</param-name>
<param-value>8443</param-value>
</context-param>
...
<param-name>serverHost</param-name>
<param-value>myserver1</param-value>
</init-param>
<init-param>
<param-name>serverPort</param-name>
<param-value>13080</param-value>
</init-param>
```

14. Save the Web configuration file.
15. Start the Service Manager Web tier.
16. Open Service Manager Windows clients and set the Service Manager hostname and communication port values:

Field	Value
Server host name	myserver1
Server port number	13080
CA certificates file	C:\Program Files\OpenText\Service Manager x.xx\Client\plugins\com.hp.ov.sm.client.common_x.xx\cacerts

17. Connect to the Service Manager host.

Related topics

Server implementation options
Servlet implementation
Servlet implementation options
Vertical scaling and required SSL implementation
Requirements for a vertical scaling and required SSL implementation

1.6.4.2. Requirements for a vertical scaling and required SSL implementation

This configuration is intended for customers who:

- Want to maximize the number of client connections supported on a single host
- Have a host with enough system resources to manage all concurrent client connections
- Want to specify the communications ports the Service Manager implementation uses
- Want to require SSL encryption for all connections
- Want to protect against complex SSL-related attacks
- Want to authenticate that the Service Manager server is a valid host

Number of Service Manager hosts required

This implementation requires the following number of hosts.
1

Certificates required

You must create or obtain the following certificates for SSL encryption.

- Certificate authority certificate
- Keystore containing the certificate authority's certificate
- Service Manager host certificate

Private keys required

You must create or obtain the following private keys for SSL encryption.

- Certificate authority's private key *
- Service Manager host private key

* This key is only necessary if you are managing your own private certificate authority.

Parameters required in sm.cfg

You must set the following configuration parameters.

```
sm -loadbalancer -httpPort:<value>
sm -httpPort:<value> - httpsPort:<value>
sm -httpPort:<value> - httpsPort:<value>
```

- loadbalancer – creates a special servlet container process to route client connection requests to other available servlet container processes
- httpPort – identify the communications port that a servlet container process uses to communicate with clients using HTTP
- httpsPort – identify the communications port that a servlet container process uses to communicate with clients using HTTPS

Parameters required in sm.ini

You must set the following initialization parameters.

- cacertpem – identify the certificate authority's certificate
- certpem – identify the Service Manager host's certificate
- pkpem – identify the Service Manager host's private key
- pkpempass – identify the password for the Service Manager host's private key
- ssl:1
- sessiontimeout – define the number of minutes a client connection can remain unresponsive before the server closes the connection.
- threadsperprocess – identify the total number of threads the servlet container process supports
  The default value for the parameter threadsperprocess is 250. For more sizing information using this parameter, refer to Service Manager server (servlet container).

Parameters required in web.xml

You must set the following Web parameters.

- cacerts – identify the keystore containing the certificate authority's certificate
- serverHost – identify the host name of the Service Manager host
- serverPort – identify the communications port on which the Service Manager host listens for client connections requests

Windows client preferences required

You must set the following preferences from the **Connection** menu.

- Server host name – identify the host name of the Service Manager host
- Server port number – identify the communications port on which the Service Manager host listens for client connections requests

You must set the following preference from the **Window** > **Preferences** > **Service Manager** > **Security** menu.

- CA certificates file – identify the keystore containing the host's certificate authority certificate

Other requirements

You must do the following additional steps to ensure that Service Manager can use your private certificates.

- Add the certificate authority's certificate to one or more key stores that your Web and Windows clients can access
- Ensure that the Service Manager server's host name matches the common name (CN) listed in the host's signed certificate

Related topics

Servlet implementation
Servlet implementation options
Parameter: httpPort
Parameter: httpsPort
Parameter: loadBalancer

1.6.5. Configure Service Manager Jgroups to use TCP instead of UDP

UDP and TCP

Service Manager JGroups can be configured to use either User Datagram Protocol (UDP) or Transmission Control Protocol (TCP). By default, Service manager uses UDP.

However, UDP has some limitations. First, it doesn't ensure that communications have reached the receiver, as the protocol doesn't incorporate the concepts of acknowledgment, time out, and retransmission. More importantly, UDP uses IP multicasting to send messages to entire clusters. Cloud service providers usually don't allow the configuration of IP multicasting, and therefore JGroups on UDP can't be run in a cloud environment.

TCP is designed to address these limitations. TCP is more reliable because it manages message acknowledgment and retransmissions in the event of lost parts. Instead of using IP multicasting, TCP sends multiple unicast messages to cluster members, so that JGroups on TCP can be run in environments that don't support IP multicasting.

Note

Note The Service Manager startup process and the **sm -reportlbstatus** report generation process are slower when you run Jgroups on TCP. This occurs because a new cluster member registers itself to the GossipRouter and queries the GossipRouter for initial cluster membership information when is starts up. Additionally, the new cluster node needs to establish TCP connections with existing members.

Configuring Service Manager Jgroups to use TCP

When you configure JGroups to use TCP, each message to all cluster members is sent as multiple unicast messages (one message to each member). Because IP multicasting can't be used to discover the initial members, this process requires a GossipRouter. Existing cluster members register by using their cluster name, and new members query the GossipRouter for initial cluster membership information.

Service Manager nodes register themselves to the GossipRouter(s) and query the GossipRouter(s) for initial cluster membership information when they start. This process may slow down the speed of the Service Manager startup process. Therefore, although using multiple GossipRouters is supported, we recommend that you use only one GossipRouter. The advantage of using multiple GossipRouters is that new members will always be able to retrieve initial membership, even if some of the GossipRouters don't work (as long as one is still running). To start multiple GossipRouters, you can add multiple commands to one sm.cfg file, or you can add one command to the sm.cfg file on multiple Service Manager hosts.

Note

Note

For more information about how to start one or more GossipRouter processes to support the use of TCP, see Configure Jgroups to use TCP in a horizontally-scaled environment and Configure Jgroups to use TCP in a vertically-scaled environment.

Related topics

Configure Jgroups to use TCP in a horizontally-scaled environment
Configure Jgroups to use TCP in a vertically-scaled environment

Parameter: jgroupstcp
Parameter: GossipRouterhosts
Parameter: asyncrefreshlbinterval
Parameter: grouptcpjointimeout
Parameter: grouptcpbindport

1.6.5.1. Configure Jgroups to use TCP in a horizontally-scaled environment

This section explains how to switch an existing horizontally-scaled environment from UDP to TCP, and how to configure a new horizontally-scaled environment to use TCP.

How to switch an existing horizontally-scaled environment from UDP to TCP

1. Add the following command to the beginning of the sm.cfg file on the Primary Host:
   `sm -GossipRouter -Gossiprouterport:<value>`

Note

Note

Make sure that you select an unused port as the value of the *Gossiprouterport* parameter. Additionally, you must add this command to the beginning of the sm.cfg file so that the GossipRouter starts before any other Service Manager processes.

2. Remove the following initialization parameters from the sm.ini file for each host:
   - `groupmcastaddress:<value>`
   - `groupport:<value>`
3. Add the following initialization parameters to the sm.ini file for each host:
   - `jgroupstcp:<value>`
   - `GossipRouterhosts:<value>`
4. (Optional) Add the following initialization parameters to the sm.ini file for each host:
   - `grouptcpbindport:<value>`

     Note

     Note

     The number of available ports within the specified range (50 by default) starting from the configured TCP bind port (7800 by default) must be greater than or equal to the number of servlets (including those running Service Manager background process). Otherwise, the horizontally-scaled Service Manager cluster can't work normally.

   - `asyncrefreshIbinterval:<value>`
   - `grouptcpjointimeout:<value>`

How to configure a new horizontally-scaled environment to use TCP

1. Add the following commands to the sm.cfg file on the Primary Host:
   - `sm -GossipRouter -Gossiprouterport:<value>`

     Note

     Note

     Make sure that you select an unused port as the value of the *Gossiprouterport* parameter. Additionally, you must add this command to the beginning of the sm.cfg file so that the GossipRouter starts before any other Service Manager processes.

   - `sm -loadBalancer -httpPort:<value>`
   - `sm -httpPort:<value> -httpsPort:<value>`
   - `sm system.start`
2. Add the following commands to the sm.cfg file on the Secondary Host:
   - `sm -httpPort:<value> -httpsPort:<value>`
   - `sm -sync`
3. Add the following initialization parameters to the sm.ini file for each host:
   - `system:<value>`
   - `grouplicenseip:<value>`
   - `groupname:<value>`
   - `jgroupstcp:<value>`
   - `GossipRouterhosts:<value>`
4. (Optional) Add the following initialization parameters to the sm.ini file for each host:
   - `groupbindaddress:<value>`
   - `grouptcpbindport:<value>`

Note

Note

The number of available ports within the specified range (50 by default) starting from the configured TCP bind port (7800 by default) must be greater than or equal to the number of servlets (including those running Service Manager background process). Otherwise, the horizontally-scaled Service Manager cluster can't work normally.

- asyncrefreshlbinterval:<value>
- grouptcpjointimeout:<value>

Example of a horizontally-scaled TCP implementation

Machine	sm.cfg (excerpt)	sm.ini (excerpt)
myserver1 10.0.0.135 (Primary Host)	sm -GossipRouter -Gossiprouterport:12001 sm -loadbalancer -httpPort:13080 sm -httpPort:13081 -httpsPort:13082 sm system.start	system:13080 grouplicenseip:10.0.0.135 groupname:mygroup1 jgroupstcp:1 GossipRouterhosts:myserver1[12001]
myserver2 10.0.0.136 (Secondary Host)	sm -httpPort:13081 -httpsPort:13082 sm -httpPort:13083 -httpsPort:13084 sm -sync	system:13080 grouplicenseip:10.0.0.135 groupname:mygroup1 jgroupstcp:1 GossipRouterhosts:myserver1[12001]
myserver3 10.0.0.137 (Secondary Host)	sm -httpPort:13081 -httpsPort:13082 sm -httpPort:13083 -httpsPort:13084 sm -sync	system:13080 grouplicenseip:10.0.0.135 groupname:mygroup1 jgroupstcp:1 GossipRouterhosts:myserver1[12001]

Related topics

Configure Service Manager Jgroups to use TCP instead of UDP
Configure Jgroups to use TCP in a vertically-scaled environment
Parameter: jgroupstcp
Parameter: GossipRouterhosts
Parameter: asyncrefreshlbinterval

Parameter: grouptcpjointimeout
Parameter: grouptcpbindport

1.6.5.2. Configure Jgroups to use TCP in a vertically-scaled environment

This section explains how to switch an existing vertically-scaled environment from UDP to TCP, and how to configure a new horizontally-scaled environment to use TCP.

How to switch an existing vertically-scaled environment from UDP to TCP

1. Add the following command to the beginning of the sm.cfg file:
   `sm -GossipRouter -Gossiprouterport:<value>`

   > **Note**
   > Note
   > Make sure that you select an unused port as the value of the *Gossiprouterport* parameter. Additionally, you must add this command to the beginning of the sm.cfg file so that the GossipRouter starts before any other Service Manager processes.

2. Add the following initialization parameters to the sm.ini file:
   - `jgroupstcp:<value>`
   - `GossipRouterhosts:<value>`
3. **Optional:** Add the following initialization parameters to the sm.ini file:
   - `grouptcpbindport:<value>`
   - `asyncrefreshlbinterval:<value>`
   - `grouptcpjointimeout:<value>`

How to configure a new vertically-scaled environment to use TCP

1. Add the following commands to the sm.cfg file:
   - `sm -GossipRouter -Gossiprouterport:<value>`

     > **Note**
     > Note
     > Make sure that you select an unused port as the value of the *Gossiprouterport* parameter. Additionally, you must add this command to the beginning of the sm.cfg file so that the GossipRouter starts before any other Service Manager processes.

   - `sm -loadBalancer -httpPort:<value>`
   - `sm -httpPort:<value> -httpsPort:<value>`
   - `sm system.start`
2. Add the following initialization parameters to the sm.ini file:
   - `jgroupstcp:<value>`
   - `GossipRouterhosts:<value>`
3. **Optional:** Add the following initialization parameters to the sm.ini file:
   - `groupbindaddress:<value>`
   - `grouptcpbindport:<value>`
   - `asyncrefreshlbinterval:<value>`
   - `grouptcpjointimeout:<value>`

Example of a vertically-scaled TCP implementation

Machine	sm.cfg (excerpt)	sm.ini (excerpt)
myserver1 10.0.0.135 (Primary Host)	sm -GossipRouter -Gossiprouterport:12001 sm -loadbalancer -httpPort:13080 sm -httpPort:13081 -httpsPort:13082 sm -httpPort:13083 -httpsPort:13084 sm system.start	jgroupstcp:1 GossipRouterhosts: myserver1[12001]

Related topics

Configure Service Manager Jgroups to use TCP instead of UDP
Configure Jgroups to use TCP in a horizontally-scaled environment
Parameter: jgroupstcp
Parameter: GossipRouterhosts
Parameter: asyncrefreshlbinterval
Parameter: grouptcpjointimeout
Parameter: grouptcpbindport

1.7. Lightweight Directory Access Protocol (LDAP)

You can integrate Service Manager to an LDAP directory service to share contact information across your network. Once you have enabled an LDAP integration to Service Manager, you can then configure Service Manager to automatically create operator records for LDAP users by either of the following methods:

- Defining a user template for LDAP logins. For details, see Create an operator template.
- Defining a system default record for all logins. For details, see Create a system default operator record.

Using either method, you can map fields in the operator record to contact information in the LDAP directory service. This mapping allows Service Manager to create an operator record with all the available contact details defined in the LDAP directory service. If you create an LDAP user template, you can make changes to all users built from this template by editing the template operator record. If you create a system default record, then you must manually make changes to each individual operator record that Service Manager creates. If you create both an operator template and a system default operator record, Service Manager uses the operator template to create new operator records.

Note

Notes

- Service Manager denies access to LDAP users unless the system administrator defines either an operator template or a system default operator record.
- The ESS access can only be picked up from the *SYSDEFAULTS template, not the template operator specified in the system information record.

The Service Manager server uses the LDAP Bind DN user that's defined in the “ldapbinddn” parameter to access LDAP. The privileges of this LDAP user determine whether the Service Manager server can add or update LDAP accounts. When a system administrator adds or updates operators in Service Manager, whether the operator changes in Service Manager can be synchronized to LDAP or not is determined by the privileges of the LDAP Bind DN user.

Note

Note Deleting an operator record doesn't cause Service Manager to delete LDAP users. Only an LDAP administrator can delete LDAP entries.

Typically, Service Manager system administrators will want to map only the operators file to an LDAP directory, however they can also map any other system table, for example, the contacts or device table, to an LDAP directory. You can map a Service Manager table to only one LDAP server at a time, although you may specify a different LDAP server for each table.

When mapping between Service Manager and LDAP directories, you can decide which data source you want to be primary. In cases where there are duplicate entries between data sources, Service Manager displays only the data listed in the primary data source.

Related topics

Enable an integration to LDAP

1.7.1. Enable an integration to LDAP

Applies to User roles: System Administrator

You must have the **SysAdmin** capability word to use this procedure.

To enable integration to LDAP:

1. Do one of the following to enable Service Manager to create operator records dynamically when LDAP users sign in.
   - Define the System Information Record operator template for LDAP users.
   - Define a system default record for the operator file. For details, see Create a system default operator record.

   These settings allow LDAP users to sign in to Service Manager without having existing operator records.
2. Define the default LDAP server and authentication base directory to which you want Service Manager to connect.
3. Define the file and field mappings you want for the LDAP directory service.
4. Set LDAP query parameters in the Service Manager initialization file.

Define the default LDAP server

You must have the **SysAdmin** capability word to use this procedure.

You must define a default LDAP server to integrate Service Manager files to an LDAP directory service.

To define the default LDAP server:

1. Click **System Administration** > **Ongoing Maintenance** > **System** > **LDAP Mapping**. The Service Manager LDAP Mapping - System Level Specification form opens.
2. Type or select the LDAP mapping information.
3. Click **Save**. Service Manager displays the message:
   Record updated in the scldapconfig file.

Define file and field-level mappings to an LDAP server

You must have the **SysAdmin** capability word to use this procedure. You must define a default LDAP server to integrate Service Manager files to an LDAP directory service.

To define file and field-level mappings to an LDAP server:

1. Click **System Administration** > **Ongoing Maintenance** > **System** > **LDAP Mapping**. The Service Manager LDAP Mapping - System Level Specification form opens.
2. Click **Set File/Field-level Mappings**. The **Service Manager LDAP Mapping - File/Field Specification** page opens.
3. In the Name field, type the name of the Service Manager file for which you want to create LDAP mappings.
4. Click **Search**. Service Manager displays a list of fields for the file.
5. Type or select the LDAP mapping settings. If necessary, press Ctrl+H to view help for each field.
6. Click **Save**. Service Manager displays the message:
   Data Policy record updated.

Set the LDAP authentication base name

You can define an operator to use a different LDAP base name than the operator name. By default, Service Manager uses the operator name to bind to the LDAP. You can define a different LDAP base name to allow users to connect to Service Manager with one name and to LDAP with a different name.

To set the LDAP authentication base name:

1. Click **System Administration** > **Ongoing Maintenance** > **Operators**.
2. Click **Search**.

3. Select the operator whose LDAP base name you want to set from the record list.
4. Click the **Security** tab.
5. In the **LDAP Base Name** field, type the name you want to use to authenticate the LDAP connection.
6. Click **Save**. Service Manager displays the message:
   Operator record updated.

Enable LDAP over SSL

You must have the **SysAdmin** capability word to use the procedures below.

By default, when you enable LDAP over SSL, you need to set the root certificate of the CA that issued the LDAP server's certificate on the Service Manager server, and then specify the location of the certificate file in the **LDAP SSL DB Path** field.

If you don't want to set the CA's root certificate on the Service Manager server, follow these steps:

1. Set the `ldapsslallownocert` parameter to `1`.
2. Sign in to Service Manager, and then click **System Administration** > **Ongoing Maintenance** > **System** > **LDAP Mapping**.
3. Set the **LDAP Server** and **LDAP Base Directory** fields, select the **LDAP SSL** checkbox. Leave the **LDAP SSL DB Path** field blank.
4. Click **Set File/Field level mapping**, enter operator in the **Name** field, and then map the `name` field of `operator` to `sAMAccountName` (for Active Directory server).
5. Restart the Service Manager server.

If you wish to authenticate Service Manager users that belong to different domains or subdomains, you can deploy multiple LDAP servers that belong to the corresponding domains, and then set up a horizontal scaled (HS) cluster. By the following configuration, users belong to different domains can share the same database while at the same time be authenticated by different domain's LDAP server over SSL.

1. Set the `ldapsslallownocert` parameter to `1`.
2. Sign in to Service Manager, and then click **System Administration** > **Ongoing Maintenance** > **System** > **LDAP Mapping**. Leave everything on this page empty.
3. Click **Set File/Field Level Mapping**, enter operator in the **Name** field, and then map the `name` field of `operator` to `sAMAccountName`.
4. Add the `ldapserver` parameter in the `sm.ini` file as the following example:
   `ldapserver1:16.183.93.217%636%cn=users,dc=swsm,dc=ind,dc=lab`
5. You can add the `ldapserver` parameter multiple times if you have more than one LDAP server.
6. Restart the Service Manager server.

Note

Note In both cases above, you still need to set the `ldapbinddn` and `ldapbindpass` parameters in `sm.ini`.

Related topics

Lightweight Directory Access Protocol (LDAP)
Create a system default operator record
Define the operator template applied to LDAP users

1.7.2. Use LDAP Proxy to integrate multiple LDAP servers

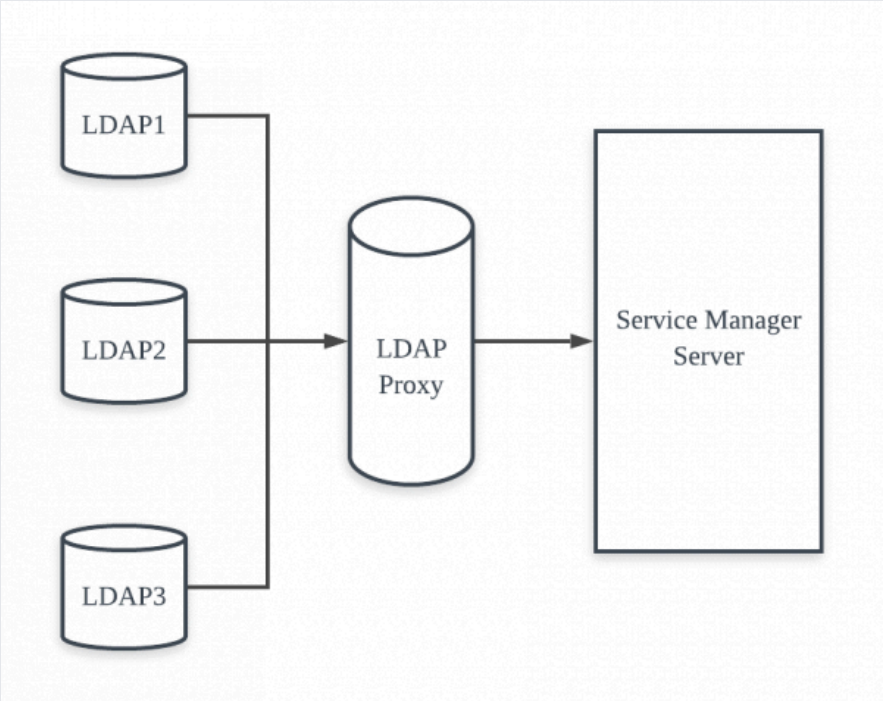
Customers usually use LDAP servers to manage users and their authentication. Some customers are using more than one LDAP server for various reasons. To allow all the users in the LDAP servers to access and use Service Manager (SM), you need to integrate all of the LDAP servers with SM. However, SM supports only one LDAP integration.

An LDAP Proxy allows an LDAP client to connect to multiple LDAP servers without the need to know their details. It combines information from multiple LDAP servers, so it can be viewed as one LDAP Server.

As an example, this document describes how to set up an LDAP Proxy with OpenLDAP Server.

Multi-LDAP support architecture

The following diagram illustrates SM's multi-LDAP support architecture.



Prerequisites

Before you proceed, you need the following:

- A Linux server with OS version: Redhat Linux Enterprise 7.2 and above
- OpenLDAP Version: 2.4

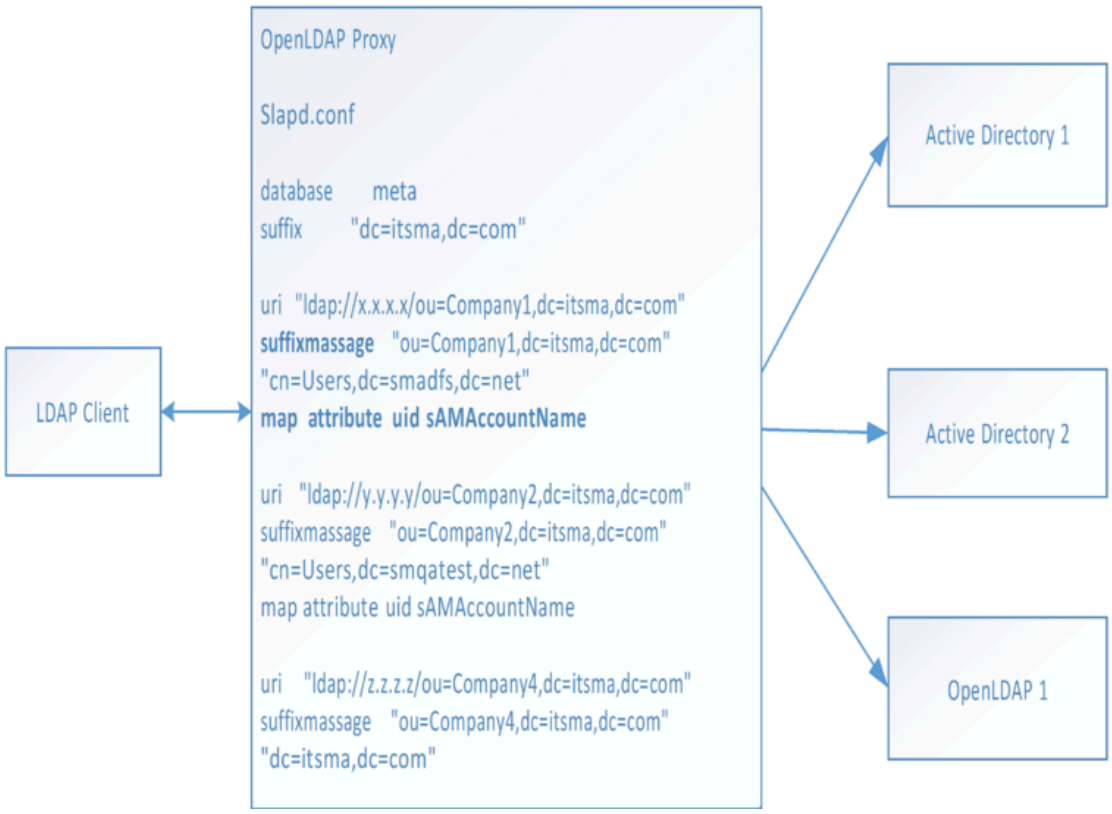
Set up OpenLDAP as a proxy

To set up OpenLDAP as a proxy, you need to prepare a Linux server and install OpenLDAP on it. You can also install OpenLDAP on an existing Redhat Linux server.

1. On Redhat Linux, install OpenLDAP.
   To set up an OpenLDAP proxy, we need the back_meta module, which is available since OpenLDAP 2.4. So always install the latest version of OpenLDAP. Use the following command to install OpenLDAP and related components:
   `yum install openldap openldap-clients openldap-servers`
2. Once the installation is successfully completed, check that the `back_meta` module is installed successfully. Its default installation folder is under `/usr/lib64/openldap`.

```
[root@SHCLITVM0184 openldap]# ls
accesslog-2.4.so.2          back_ldap-2.4.so.2.10.7    back_perl.la
accesslog-2.4.so.2.10.7     back_ldap.la               back_relay-2.4.s
accesslog.la                back_meta-2.4.so.2         back_relay-2.4.s
allop-2.4.so.2              back_meta-2.4.so.2.10.7    back_relay.la
allop-2.4.so.2.10.7         back_meta.la               back_shell-2.4.s
allop.la                    back_null-2.4.so.2         back_shell-2.4.s
auditlog-2.4.so.2           back_null-2.4.so.2.10.7    back_shell.la
auditlog-2.4.so.2.10.7      back_null.la               back_sock-2.4.so
auditlog.la                 back_passwd-2.4.so.2       back_sock-2.4.so
back_dnssrv-2.4.so.2        back_passwd-2.4.so.2.10.7  back_sock.la
back_dnssrv-2.4.so.2.10.7   back_passwd.la             check_password.s
back_dnssrv.la              back_perl-2.4.so.2         check_password.s
back_ldap-2.4.so.2          back_perl-2.4.so.2.10.7    collect-2.4.so.2
```

3. Configure the **slapd.conf** file to add information of the LDAP servers, so that the LDAP proxy server knows which LDAP servers to redirect.
   The following diagram shows a high-level architecture of the meta backend:



   The following is an example of the **slapd.conf** file:

```
#
# See slapd.conf(5) for details on configuration options.
# This file should NOT be world readable.
#
Include /etc/openldap/schema/core.schema
Include /etc/openldap/schema/corba.schema
include /etc/openldap/schema/cosine.schema
include /etc/openldap/schema/duaconf.schema
include /etc/openldap/schema/dyngroup.schema
include /etc/openldap/schema/inetorgperson.schema
include /etc/openldap/schema/java.schema
include /etc/openldap/schema/misc.schema
include /etc/openldap/schema/nis.schema
include /etc/openldap/schema/openldap.schema
include /etc/openldap/schema/collective.schema
include /etc/openldap/schema/pmi.schema
include /etc/openldap/schema/ppolicy.schema

allow bind_v2

# Define global ACLs to disable default read access.

# Do not enable referrals until AFTER you have a working directory
# service AND an understanding of referrals.
#referral  ldap://root.openldap.org

pidfile         /var/run/openldap/slapd.pid
argsfile        /var/run/openldap/slapd.args

# Load dynamic backend modules:
modulepath      /usr/lib64/openldap
moduleload      rwm.la
moduleload      back_meta.la
moduleload      back_ldap.la

loglevel 4095

########################################################################
# MDB database definitions
########################################################################

database meta
suffix "dc=itsma,dc=com"
rootdn "cn=admin,dc=itsma,dc=com"
rootpw secret

uri     "ldap://x.x.x.x/ou=Company1,dc=itsma,dc=com"

readonly no
lastmod off

suffixmassage "ou=Company1,dc=itsma,dc=com" "cn=Users,dc=smadfs,dc=net"
map attribute uid sAMAccountName
idassert-bind bindmethod=simple
     binddn="cn=administrator,cn=Users,dc=smadfs,dc=net"
    credentials="xxxxxx"
idassert-authzFrom "*"

uri     "ldap://y.y.y.y/ou=Company2,dc=itsma,dc=com"
```

```
readonly        no
lastmod  off
suffixmassage "ou=Company2,dc=itsma,dc=com" "cn=Users,dc=smqatest,dc=net"
map attribute uid sAMAccountName
idassert-bind bindmethod=simple
      binddn="CN=Administrator,CN=Users,DC=smqatest,DC=net"
      credentials="yyyyyyy"
idassert-authzFrom "*"

uri  "ldap://z.z.z.z/ou=Company4,dc=itsma,dc=com"
readonly no
lastmod off
suffixmassage "ou=Company4,dc=itsma,dc=com" "dc=itsma,dc=com"
idassert-bind bindmethod=simple
      binddn="cn=admin,dc=itsma,dc=com"
      credentials="zzzzzzzz"
idassert-authzFrom "*"
```

4. Add additional attributes in the **schema** and **ldif** files.
   You can add additional attributes to the **inetorgperson.schema** and **inetorgperson.ldif** files so that the attributes can be recognized by the OpenLDAP Proxy, especially when the LDAP Server type is not OpenLDAP. For example, when you use ActiveDirectory (AD) as the LDAP Server and OpenLDAP as the proxy, you must add AD attribute **sAMAccountName** to **inetorgperson.schema** and **inetorgperson.ldif**.
   Append the following lines in the **inetorgperson.schema** file:

   ```
   ################
   #AD schema
   ################
   attributetype ( 1.2.840.113556.1.4.221
         NAME 'sAMAccountName'
         SYNTAX '1.3.6.1.4.1.1466.115.121.1.15' SINGLE-VALUE )
   ################
   ```

   Insert the following line before the **olcObjectClasses** section in the **inetorgperson.ldif** file :

   ```
   olcAttributeTypes: ( 1.2.840.113556.1.4.221 NAME 'sAMAccountName' SYNTAX
   1.3.6.1.4.1.1466.115.121.1.15 SINGLE-VALUE )
   ```

   Add attribute **sAMAccountName** to **olcObjectClasses** as shown below:

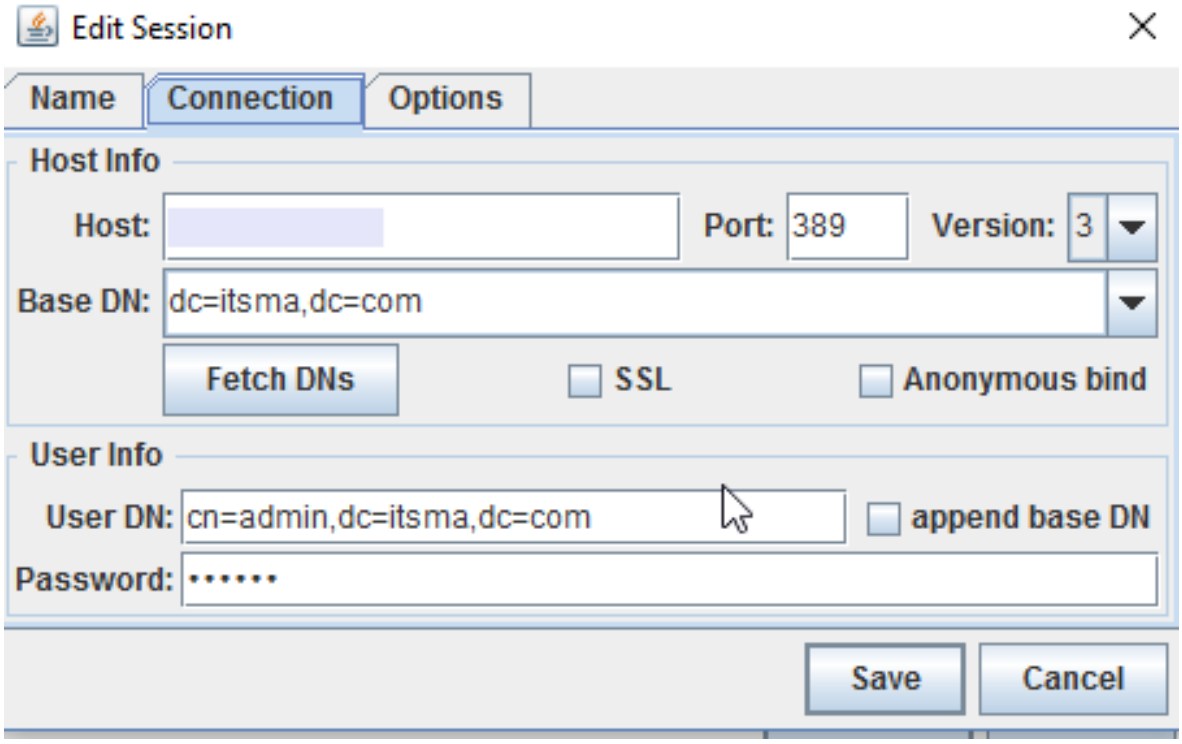
   ```
   olcObjectClasses: ( 2.16.840.1.113730.3.2.2 NAME 'inetOrgPerson' DESC 'RFC2
   798: Internet Organizational Person' SUP organizationalPerson STRUCTURAL MAY
   ( audio $ businessCategory $ carLicense $ departmentNumber $ displayName $ em
   ployeeNumber $ employeeType $ givenName $ homePhone $ homePostalAddress $ ini
   tials $ jpegPhoto $ labeledURI $ mail $ manager $ mobile $ o $ pager $ photo
   $ roomNumber $ secretary $ uid $ userCertificate $ x500uniqueIdentifier $ pre
   ferredLanguage $ userSMIMECertificate $ sAMAccountName $ userPKCS12i ) )
   ```

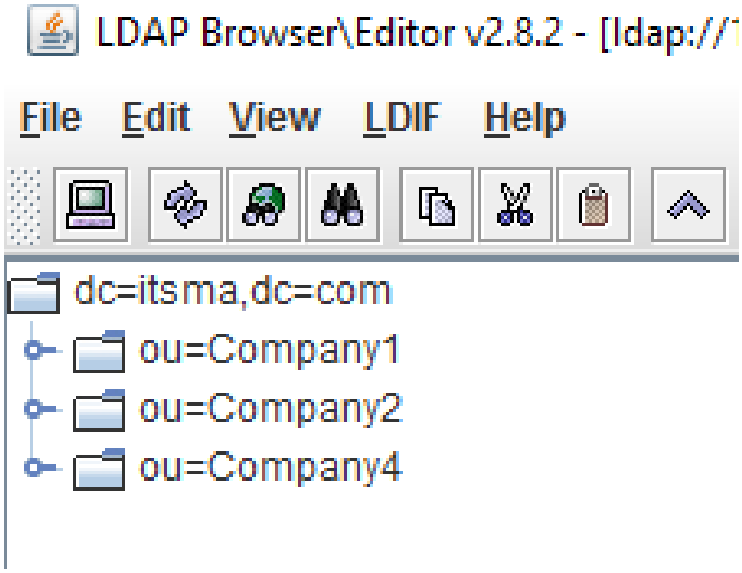
If you want to add other attributes in **inetorgperson.schema** and **inetorgperson.ldif** such as Phone Number, User Avatar, follow the same steps. Additionally, add attribute mapping such as "map attribute TelephoneNumber mobile" in the **slapd.conf** file (the MDB database definitions uri section):

```
uri "ldap://x.x.x.x/ou=Company1,dc=itsma,dc=com"
readonly no
lastmod off
suffixmassage "ou=Company1,dc=itsma,dc=com" "cn=Users,dc=smadfs,dc=net"
map attribute uid sAMAccountName
map attribute TelephoneNumber mobile
idassert-bind bindmethod=simple
          binddn="cn=administrator,cn=Users,dc=smadfs,dc=net"
          credentials="xxxxxx"
idassert-authzFrom "*"
```

5. Make slapd effective:
    1. Remove the content in slapd.d.
    2. Run the following command:
       slaptest -f /etc/openldap/slapd.conf -F /etc/openldap/slapd.d
6. Start OpenLDAP:
   slapd -d 1

7. Use an LDAP browser or another LDAP tool to check the setup:





Now, you can follow the steps in Enable an integration to LDAP document to configure the LDAP Proxy in SM. Please note that the LDAP configuration should use OpenLDAP attributes as you are using OpenLDAP as the Proxy.

1.8. IPv6 overview

As of version 9.32, Service Manager supports IPv4/IPv6 as a dual stack network. This addresses the needs of being able to run Service Manager infrastructure in an organization where both IPv6 and IPv4 are enabled in parallel.

Note

Note In a dual stack network, the required IPv4 configuration is the same as before.

Recommended Topologies[edit]

We recommend that your implementation follow these topology guidelines:

Topology 1[edit]

- Service Manager run-time environment, underlying RDBMS, and all Application/Web Servers in a pure IPv4 environment use IPv4.
- The clients in a pure IPv4 environment using IPv4.

Topology 2[edit]

- Service Manager run-time environment, underlying RDBMS, and all Application/Web Servers in a dual IPv4/IPv6 environment use IPv4.
- The clients in pure a IPv4 environment or in a dual IPv4/IPv6 environment both use IPv4.

Topology 3[edit]

- Service Manager run-time environment, underlying RDBMS, and all Application/Web Servers in a dual IPv4/IPv6 environment use IPv6.
- The clients in a pure IPv4 environment use IPv4.
- The clients in a dual (IPv4/IPv6) environment use IPv6.

Contents

- 1 Recommended Topologies
  - 1.1 Topology 1
  - 1.2 Topology 2
  - 1.3 Topology 3

1.8.1. IPv6 supported address formats

IPv6 in Service Manager supports the following IP address formats.

Text representation[edit]

Service Manager supports text representation of IPv6 formats as defined in RFC 4291 (IP Version 6 Addressing Architecture).

IPv6 addresses are 128-bit identifiers for interfaces and sets of interfaces. In general, the standard IPv6 address is `x:x:x:x:x:x:x:x` where each `x` represents 1 to 4 hexadecimal digits, as shown in the following example:

```
2001:DB8:0:0:8:800:200C:417A
```

It's not necessary to specify the leading zeros of an individual field. However, you must have at least one hexadecimal character in each field. You can use `::` to represent one or more groups of 16 bits of zeros to avoid specifying the entire IPv6 address. However, you can only use the `::` one time. You may also use `::` to compress the leading or trailing zeros of the IPv6 address. For example, the following two IPv6 addresses are equivalent:

```
FF01:0:0:0:0:0:0:101
```

```
FF01::101
```

Note

Note

You can use either of these representations when you configure IPv6 for use in Service Manager.

Address representation in mixed IPv4/IPv6 environments[edit]

According to the RFC 4291 (IP Version 6 Addressing Architecture) standard, the following format is used to handle mixed IPv4/IPv6 environments:

```
x:x:x:x:x:x:d.d.d.d
```

In this case, the *x*s are placeholders for the hexadecimal values of the six high-order 16-bit pieces of the address, and the *d*s are the decimal values of the four low-order 8-bit pieces of the address, which is the standard IPv4 representation. Therefore, you can specify an old IPv4 address in a mixed IPv4/IPv6 environment in either of the following, equivalent ways:

```
0:0:0:0:0:0:13.1.68.3
```

```
::13.1.68.3  //Uses :: to minimize the leading zeros of the IPv6 address.
```

Note

Note

In Service Manager, this mixed use IPv4/IPv6 is unsupported.

IPv6 addresses with a port number[edit]

To specify an IPv6 address together with a port number, enclose the IPv6 address in brackets as shown in the following example:

```
[2001:db8::1]:80
```

Contents

- 1 Text representation
- 2 Address representation in mixed IPv4/IPv6 environments
- 3 IPv6 addresses with a port number

1.8.2. IPv6 configuration

The following topics provide basic information on how to configure Service Manager to use IPv6:

Configure the Service Manager server for IPv6

Configure the Windows client for IPv6

Configure the Web clients for IPv6

Configure IPv6 to work with Service Manager integrations

Configure IPv6 for Mobility

1.8.2.1. Configure the Service Manager server for IPv6

To configure the Service Manager server to use IPv6, specify an IPv6 address in the `groupbindaddress` parameter in the `sm.ini` file where the Service Manager server is installed. The following example shows a valid IPv6 address:

```
groupbindaddress:2620:0:a17:e006:1060:cb7b:590a:4388
```

Note

Note

If the `groupbindaddress` parameter is specified with an IPv6 address, Service Manager will listen for both IPv4/IPv6 addresses. If no value is specified, or if an IPv4 address is specified, Service Manager will listen only for IPv4 addresses.

Configure IPv6 for horizontally-scaled environments

Make sure that you configured Service Manager as described at the beginning of this topic. To configure IPv6 for horizontally scaled environments, you must also specify the following parameters on each horizontally scaled server:

Parameter	Value
`groupbindaddress`	Specify an IPv6 address in the `groupbindaddress` parameter in the `sm.ini` file where the Service Manager server is installed.
`grouplicenseip`	Set as the valid IPv6 or IPv4 address of the machine where the licenses are stored.
`groupmcastaddress`	By default, this parameter is set to `FF02::1` after the `groupbindaddress` parameter is specified. If needed, you can modify this parameter according to the following standard: http://www.iana.org/assignments/ipv6-multicast-addresses/ipv6-multicast-addresses.xml#ipv6-multicast-addresses-1

Note

Note

All other configurations steps are the same as in an IPv4 environment.

Configure IPv6 for vertically-scaled environments

Make sure that you configured Service Manager as described at the beginning of this topic. To configure IPv6 for vertically-scaled environments, you must also specify the following variables:

Parameter	Value
groupbindaddress	Specify an IPv6 address in the `groupbindaddress` parameter in the `sm.ini` file where the Service Manager server is installed.

Note

Note

All other configuration steps are the same as in an IPv4 environment.

1.8.2.2. Configure the Windows client for IPv6

To configure the Service Manager Windows client to use IPv6, follow these steps:

1. Navigate to **File** > **Connect..** > **Connections**.
2. In the Connections dialog, enter a valid IPv6 in the "Server host name" field.

1.8.2.3. Configure the Web clients for IPv6

Application Server[edit]

The steps that you use to configure IPv6 depends on the Application server that you are using. Choose the appropriate instructions for your environment.

Tomcat 8.x[edit]

The default installation of Tomcat doesn't support listening on the IPv6 protocol. Therefore, you must download latest APR library (tcnative-1.dll) and add an `address` parameter for the connector in the `server.xml` file. To do this, follow these steps:

1. Download latest APR library as appropriate for your architecture.

Architecture	Download location:
x86	http://archive.apache.org/dist/tomcat/tomcat-connectors/native/1.1.20/binaries/win32/tcnative-1.dll
x64	http://archive.apache.org/dist/tomcat/tomcat-connectors/native/1.1.20/binaries/win64/x64/tcnative-1.dll
IA64	http://archive.apache.org/dist/tomcat/tomcat-connectors/native/1.1.20/binaries/win64/ia64/tcnative-1.dll

2. Replace the default APR library with the one you downloaded.

```
$Tomcat_Installation$\bin\tcnative-1.dll
```

3. Add an `address` parameter for the connector in the `server.xml` file as follows:

```
<Connector address="[::]" port="8081" protocol="HTTP/1.1" maxThreads="400"
connectionTimeout="20000"
redirectPort="8443" />
```

4. Add an `address` parameter for the AJP connector in the `server.xml` file as follows:

```
<Connector address="[::]" port="8008" protocol="AJP/1.3" redirectPort="8443" />
```

> **Note**
> Note Modify the `server.xml` file for each for Tomcat instance. To revert to IPv4, remove `address` parameter in the `server.xml` file.

5. Set `serverHost` parameter of the `WEB-INF\web.xml` to a valid IPv6 address:

```
<param-name>serverHost</param-name>
<param-value>2620:0:a17:e006:d024:1d09:c2f9:f03f</param-value>
```

> **Note**
> Note You don't need to use enclose the IPv6 address in the brackets ([]).

6. Access the Service Manager Web tier by using a correct URL as shown in the following example:

```
http://[2620:0:a17:e006:4a8:91c1:c9cb:93dd]:8080/webtier-9.xx/index.do
```

WebSphere 8.5[edit]

To configure WebSphere for IPv6, follow these steps:

1. Set the `serverHost` parameter of the `WEB-INF\web.xml` file to a valid IPv6 address:

```
<param-name>serverHost</param-name>
<param-value>2620:0:a17:e006:d024:1d09:c2f9:f03f</param-value>
```

> **Note**
> Note You don't need to use enclose the IPv6 address in the brackets ([]).

2. Access the Service Manager Web tier by using a correct URL as shown in the following example:

```
http://[2620:0:a17:e006:4a8:91c1:c9cb:93dd]:8080/webtier-9.xx/index.do
```

Web Server[edit]

The steps that you use to configure IPv6 depends on the Web server that you are using. Choose the appropriate instructions for your environment.

Apache 2.2 or 2.4[edit]

To configure Apache to use IPv6, you must download,modify, and recompile Apache. To do this, refer to these example steps:

1. Download and install Perl from the following web site:
   http://www.activestate.com/activeperl/downloads

2. Download and install awk95.exe to some directory, for example, `C:\Windows`. Rename the file to `awk.exe`.
3. Compile zlib. To do this, follow these steps:
    1. Download the source code for zlib from the following web site:

        ```
        http://www.zlib.net/
        ```

    2. Unzip the file to `C:\ipv6\zlib`.
    3. Run the following command to compile zlib:

        ```
        C:\ipv6\zlib\nmake -f win32\Makefile.msc
        ```

    4. If you receive an `nmake fatal error code 0xc0000135`, run the following command:

        ```
        C:\ipv6\zlib\vcvars32
        ```

4. Compile OpenSSL. To do this, follow these steps:
    1. Download the source code for OpenSSL from the following web site:

        ```
        http://www.openssl.org/
        ```

    2. Unzip the file to `C:\ipv6\openssl`.
    3. Run the following command to compile openssl:

        ```
        C:\ipv6\openssl\perl Configure VC-WIN32
        C:\ipv6\openssl\ms\do_ms
        C:\ipv6\openssl\nmake -f ms\ntdll.mak
        ```

5. Compile and modify Apache . To do this, follow these steps:
    1. Download the source code for Apache from the following web site:

        ```
        http://httpd.apache.org/download.cgi
        ```

    2. Unzip the file to `C:\ipv6\httpd`.
    3. Edit the `httpd\srclib\apr\include\apr.hw` file, and change the line `#define APR_HAVE_IPV6 0` to `#define APR_HAVE_IPV6 1`.
       Copy the `zlib` and `openssl` folders from the previous steps to `httpd\srclib`.
    4. Run the following command to compile Apache:

        ```
        C:\ipv6\httpd\nmake /F Makefile.win INSTDIR="C:\Apache" installr
        ```

Note

Note You may encounter issues if you try to compile Apache. Specifically, you may receive the following errors:

```
C:\Program Files\Microsoft SDKs\Windows\v6.0A\include\ws2def.h(91) : warning
C4005: 'AF_IPX' : macro redefinition
C:\Program Files\Microsoft SDKs\Windows\v6.0A\include\winsock.h(460) : see
previous definition of 'AF_IPX'
C:\Program Files\Microsoft SDKs\Windows\v6.0A\include\ws2def.h(127) : warning
C4005: 'AF_MAX' : macro redefinition
C:\Program Files\Microsoft SDKs\Windows\v6.0A\include\winsock.h(479) : see
previous definition of 'AF_MAX'
C:\Program Files\Microsoft SDKs\Windows\v6.0A\include\ws2def.h(163) : warning
C4005: 'SO_DONTLINGER' : macro redefinition
C:\Program Files\Microsoft SDKs\Windows\v6.0A\include\winsock.h(402) : see
previous definition of 'SO_DONTLINGER'
C:\Program Files\Microsoft SDKs\Windows\v6.0A\include\ws2def.h(206) : error C2011:
'sockaddr' : 'struct' type redefinition
C:\Program Files\Microsoft SDKs\Windows\v6.0A\include\winsock.h(485) : see
declaration of 'sockaddr'
C:\Program Files\Microsoft SDKs\Windows\v6.0A\include\ws2def.h(384) : error C2059:
syntax error : 'constant'
```

These issue are caused by an improper include order between `windows.h` and the `winsock2.h` files in the `apr` library. To resolve this, make sure that `winsock2.h` is included before `windows.h`.

IIS 8.0 or 8.5[edit]

IPv6 is unsupported for an IIS connector with Tomcat (mod_jk).

Contents

- 1 Application Server
  - 1.1 Tomcat 8.x
  - 1.2 WebSphere 8.5
- 2 Web Server
  - 2.1 Apache 2.2 or 2.4
  - 2.2 IIS 8.0 or 8.5

1.8.2.4. Configure IPv6 to work with Service Manager integrations

Knowledge Management search server

Make sure that you configured Service Manager as described in the Configure the Service Manager server for IPv6 section. To configure IPv6 in a Knowledge Management search server, follow these steps:

1. In the Service Manager client, navigate to **System Administration** > **Ongoing Maintenance** > **Smart Analytics** > **Configuration**.
2. Configure IPv6 address for Smart Analytics, Default Connector Framework Server (CFS), and Image server, including the brackets, as shown:

```
http://<[IPv6 address]>:<port number>/
```

Note

Note All other configurations steps are the same as in an IPv4 environment.

LDAP server

Make sure that you configured Service Manager as described in the Configure the Service Manager server for IPv6 section. To configure IPv6 for an LDAP server, follow these steps:

1. In the Service Manager client, navigate to **System Administration** > **Ongoing Maintenance** > **System** > **LDAP Mapping**.
2. Set `LDAP server` as the IPv6 address of the LDAP server, including the brackets, as shown:

```
[2620:0:a17:e006:4a8:91c1:c9cb:93dd]
```

Note

Note All other configurations steps are the same as in an IPv4 environment.

SMTP server

Make sure that you configured Service Manager as described in the Configure the Service Manager server for IPv6 section. To configure IPv6 for an SMTP server, follow these steps:

1. Specify the IPv6 address of the SMTP server in the `smtphost` parameter in the `sm.ini` file. The following example shows a valid IPv6 address:

```
smtphost:2620:0:a17:e006:1060:cb7b:590a:4388
```

2. Set the `smtphost` connection parameters for the `emailout` startup parameter as a valid IPv6 address:

```
sm -emailout -smtphost:[2620:0:a17:e006:4a8:91c1:c9cb:93dd] -smtpport:25
```

Note

Note All other configurations steps are the same as in an IPv4 environment.

1.8.2.5. Configure IPv6 for Mobility

To enable IPv6 support for the Service Manager Mobile Application, follow these steps:

1. Enable IPv6 support in Tomcat. To do this, see the Tomcat section of Configure the Web clients for IPv6.
2. Enable IPv6 in Service Manager. To do this, see Configure the Service Manager server for IPv6.

3. In folder where the mobile web tier is deployed, set theendpoint in the `WEB-INF\web.properties` file to a Full Qualified Domain Name or to a valid IPv6 address

```
endpoint=[2620:0:a17:e006:4a8:91c1:c9cb:93dd]
```

4. You can access the Service Manager Mobile Application by entering the valid IP address in your browser, as shown in the following example:

```
http://[2620:0:a17:e006:4a8:91c1:c9cb:93dd]:8080/sm/std
```

1.9. Print options

Service Manager offers list view or detail view print options.

Use Search to display the list of records or a detail record. Follow the instructions below to print in the windows client or the web client:

In the windows client:

1. Click **File** > **Print** or click the **Print** icon.
2. Select one of the following options:
   - **List**: Print the list of records.
     You can specify the **Start** and **Count** fields to print a number of consecutive records in the list.
   - **Detail**: Print a single record.
3. Select the printer and then click **Print**.

In the web client:

1. Do one of the following actions:
   - Print the list of records: Click the **Print** icon on the Record List toolbar or press Ctrl+Alt+P.
   - Print a single record: Click the **Print** icon on the Record Detail toolbar or press Alt+P.
2. Perform one of the following actions in the newly-opened browser window:
   - Right-click the page and then click **Print**.
   - Press Ctrl+P.
3. Select the printer and then click **Print**.

Related topics

Setting Windows client preferences from the server

1.10. Regional settings

Applies to User Roles: System Administrator

Regional settings define local values for month names and abbreviations, date format, system language, and default system currency.

Define the months in the year

To define the months in the year:

1. Click **System Administration** > **Base System Configuration** > **Miscellaneous** > **System Information Record**.
2. Click the **Month Tables** tab.

The table lists the Full Name and Abbreviation for each month.

1. Make any necessary changes.
2. Click **Save**.

Set the default date format

To set the default date format:

1. Click **System Administration** > **Base System Configuration** > **Miscellaneous** > **System Information Record**.
2. Click the **Date Info** tab.
3. In the **Format** field, type or select one of the following date formats to be the system default.
    - **mm/dd/yy** — standard U.S. date format
    - **dd/mm/yy** — standard European date format
    - **yy/mm/dd** — optional date format
    - **mm/dd/yyyy** — standard U.S. date format
    - **dd/mm/yyyy** — standard European date format
    - **yyyy/mm/dd** — optional date format
4. Click **Save**.

Set the default system currency

To set the default system currency:

1. Click **System Administration** > **Base System Configuration** > **Miscellaneous** > **System Information Record**.
2. Click the **General** tab.
3. In the **Basis Currency** field, type or select one of the options from the currency file.
4. Click **Save**.

Note: The currency setting in each individual operator record overrides the default system currency setting.

Set the default system language

Note: The language setting in each individual operator record overrides the default language setting.

To set the default system language:

1. Click **System Administration** > **Base System Configuration** > **Miscellaneous** > **System Information Record**.
2. Click the **Misc** tab.
3. In the **Language Code** field, type or select one of the following language options:
    - English
    - French
    - German
    - Italian
    - Japanese
4. Click **Save**.

Related topics

Calendars
Clocks
System information record
Time zones

1.11. Servlet implementation

Service Manager uses a servlet implementation to manage client connections to the server. The servlet implementation uses a pre-configured embedded Java servlet that doesn't require any additional installation or configuration of Java components. Administrators can manage the servlet implementation using the Service Manager configuration files and management procedures rather than Java interfaces.

A servlet implementation provides the following management features:

- The ability to specify the exact communications ports client connections use
- The ability to specify the maximum number of client connections Service Manager will accept

- Industry-standard resilience and scalability options
- Multithread processing capabilities

In a servlet implementation, administrators specify the communications ports the system uses in advance. The total number of communication ports required for a servlet implementation is based on the number of servlet container processes you want the Service Manager host to support. Each servlet container process manages a set number of client sessions as determined by the number of threads it contains.

Each servlet container process supports a set number of client connections over two communications ports, one for HTTP communications and one for HTTPS communications. Administrators set the number of client connections per process using the threading parameters. Two communications ports is sufficient for up to approximately 150 client connections (as determined by the Service Manager host's memory requirements).

A servlet implementation is ideal for administrators who want to do capacity planning and system resource management. The communications port and threading parameters allow administrators to control the resources that client connections consume. By specifying the communication ports and number of threads available for client connections, administrators can set a limit on the total number of client connections any one hardware resource supports. Setting connection limits prevent client connections from consuming more system resources than desired and allows administrators to set a server load threshold in advance. If a client attempts to connect to a Service Manager system that has reached its connection limit, the server refuses the connection and displays an error message that the server is unavailable.

Certain servlet implementations benefit from the native resilience and scalability features of grouping Java servlets. In a horizontal scaling implementation, administrators can create virtual groups where multiple Service Manager servers act as a single system. Each member in a virtual group communicates its connection availability and system resource usage to the other members of the group. Should a group member fail, the other group members are unaffected by the outage and the system can route any new client connection requests to another member in the virtual group. If the client connection routing process fails, the existing client connections are unaffected and an administrator can restart the client connection routing process.

Related topics

Server implementation options
Servlet implementation options
Servlet implementation processes
Startup options for servlet container processes

1.11.1. Servlet implementation options

The Service Manager server supports the following types of servlet implementation options. You may combine these implementation options with other Service Manager options to create your own system implementation.

- Single servlet implementation – a single host running a single servlet container implementation
- Vertical scaling implementation – a single host running multiple servlet containers implementation
- Horizontal scaling implementation – multiple hosts running multiple servlet containers implementation

A single servlet implementation doesn't require a load balancer process. Clients send connection requests directly to the HTTP communications port of the servlet container process. The servlet container's HTTP port becomes the single well-known communications port for the Service Manager system. All clients send connection requests to this one communications port. The servlet container process routes each connection request to an available thread in the process. When the number of client connections reaches the number of available threads the system has reached capacity, and the servlet container refuses any additional client connection requests. Each servlet container process can manage approximately one hundred and fifty client connections (as determined by the Service Manager host's memory requirements). Implementers can use the threadsperprocess parameter to set the total number of client connections the servlet container process manages.

A vertical scaling implementation requires a load balancer process to route client connection requests to an available servlet container process. Clients send connection requests to the HTTP communications port of the load balancer process. The load balancer's HTTP port becomes the single well-known communications port for the Service Manager system. All clients send connection requests to this one communications port. The load balancer process transparently routes client connection requests to any available servlet container processes on the local host. The end-user never sees the change in

the communications port number from the load balancer process to the servlet container process. When the number of client connections reaches the total number of available threads on all servlet container processes the system has reached capacity, and the load balancer refuses any additional client connection requests. Implementers must use the threadsperprocess parameter to set the total number of client connections each servlet container process manages.

A horizontal scaling implementation requires a load balancer process to route client connection requests to servlet containers on multiple hosts. Clients send connection requests to the HTTP communications port of the load balancer process. The load balancer's HTTP port becomes the single well-known communications port for the Service Manager system. All clients send connection requests to this one communications port. The load balancer process routes client connection requests to any available servlet container process in the group. When the number of client connections reaches the total number of available threads on all servlet containers on all available hosts in the virtual group, the system has reached capacity, and the load balancer refuses any additional client connection requests. Implementers must use the threadsperprocess parameter to set the total number of client connections each servlet container process manages. Implementers must also set the group parameters and ensure there is a valid network connection among the group members.

Related topics

Horizontal scaling implementation
Servlet implementation
Servlet implementation processes
Single servlet implementation
Vertical scaling and required SSL implementation
Vertical scaling implementation

1.11.2. Servlet implementation processes

All Service Manager servlet implementations use multithreaded processes. To control the client connections to these threads, you can configure the Service Manager server to start one of two types of processes:

- One or more servlet container processes that accept direct HTTP and HTTPS client connections
- One load balancer process to be the master thread controller for all servlet container processes in a virtual group

A servlet container process is a Service Manager sm.exe process that includes a multithreaded Java servlet container. Each servlet container process manages a pool of child threads dedicated to handling Service Manager client transactions. The Servlet container process listens for incoming client connection requests on a predetermined HTTP port and routes the requests to an available thread in the process. The thread then binds to the HTTP or HTTPS port of the parent servlet container process for the duration of the client connection. The total number of threads available determine the total number of client connections the servlet container process can manage. Each active thread in the servlet container process holds system locks and communicates this information to other servlet container processes in the same virtual group.

A load balancer process is an Service Manager sm.exe process dedicated to routing incoming client requests to servlet container processes in vertical and horizontal scaling implementations. In either scaling implementation the virtual group only requires one load balancer process. The Load balancer process doesn't have any child threads of its own, nor does it maintain any system locks or hold system resources. The load balancer listens for incoming client connection requests on a predetermined HTTP port and routes them to an available servlet container. The available servlet container may be on the same physical system or on another system in the same virtual group depending on the implementation.

Property	Servlet container process	Load balancer process
Routes client connections to servlet container processes?	No	Yes
Routes client connections to threads?	Yes	No
Creates child threads?	Yes	No
Accepts client connections?	Yes	No
Uses an HTTP port?	Yes	Yes
Uses an HTTPS port?	Yes (optional)	No

Related topics

Parameter: httpPort
Parameter: httpsPort
Parameter: loadBalancer
Parameter: groupmcastaddress
Parameter: groupname
Parameter: groupport
Parameter: sslConnector
Managing multiple servlet container processes
Servlet implementation
Servlet implementation options
Quiesce all servlet container processes in a servlet implementation
Quiesce all servlet container processes on a host
Shut down all servlet container processes in a servlet implementation
Shutdown all servlet container processes on a host

1.11.3. Startup options for servlet container processes

The only way you can start the servlet container processes in a servlet implementation is to manually start all needed servlet container processes in advance. We recommend that you start all needed servlet container processes in advance because it ensures that the host dedicates sufficient system resources to client connections, and because clients don't experience any connection delays while a servlet container process starts up. Each servlet container process requests host system resources when it first started. By starting all processes in advance you ensure that the host has sufficient resources to run your servlet container processes, and you can determine how much spare capacity, if any, the host has. In addition, clients don't have to wait for a servlet container process to start before they can connect. You can manually start servlet container processes requires using the following parameters from the command line or Service Manager configuration file (sm.cfg).

Implementation	Command line parameters or configuration file entries
Single servlet implementation	`sm -httpPort:13081 httpsPort:13082`
Horizontal or vertical scaling implementations	`sm -httpPort:13080 -loadBalancer` `sm -httpPort:13081 -httpsPort:13082`

Related topics

Servlet implementation
Servlet implementation options
Parameter: httpPort
Parameter: httpsPort
Parameter: threadsperprocess

1.11.4. Managing multiple servlet container processes

In vertical and horizontal scaling implementations, administrators have the option of managing multiple servlet container processes. Administrators can stop or quiesce individual servlet container processes from the server's operating system command line. This allows you to take servlet container processes offline for maintenance or to manage system resources on the Service Manager host.

To quiesce multiple servlet container processes you must use the command line. The quiesce form doesn't recognize servlet container implementations.

Additionally, you can set a process to monitory the heap memory and take action to prevent memory overload on any one servlet.

Related topics

Servlet implementation
Servlet implementation processes
Monitoring memory in Service Manager processes
Monitoring native heap and Java heap memory
Logging memory monitoring
Quiesce all servlet container processes in a servlet implementation
Quiesce all servlet container processes on a host
Shut down all servlet container processes in a servlet implementation
Shutdown all servlet container processes on a host
Parameter: log4jDebug

1.11.4.1. Quiesce all servlet container processes in a servlet implementation

Applies to User roles: System Administrator

You must have administrative access to the server operating system to use this procedure.

To quiesce all servlet container processes in a servlet implementation:

1. Open the server's operating system command line prompt.
   For example, to open the Windows command prompt:
   Click **Start** > **Programs** > **Accessories** > **Command Prompt**.
2. Change directories to the RUN folder of your Service Manager installation. For example:

   ```
   cd C:\Program Files\OpenText\Service Manager\server\RUN
   ```

3. Type the following command:

   ```
   sm -quiesce:2 -group
   ```

4. Press Enter.
   The server quiesces all the servlet container processes in the virtual group.

Related topics

Managing multiple servlet container processes
Servlet implementation
Servlet implementation processes
Startup parameter: quiesce
Parameter: group

1.11.4.2. Quiesce all servlet container processes on a host

Applies to User roles: System Administrator

You must have administrative access to the server operating system to use this procedure.

To quiesce all servlet container processes on a host:

1. Open the server's operating system command line prompt.

For example, to open the Windows command prompt:
Click **Start** > **Programs** > **Accessories** > **Command Prompt**.

2. Change directories to the `RUN` folder of your Service Manager installation. For example:

```
cd C:\Program Files\OpenText\Service Manager\server\RUN
```

3. Type the following command:

```
sm -quiesce:2 -host:<hostname>
```

4. Press Enter.
The server quiesces all the servlet container processes on the host.

Related topics

Managing multiple servlet container processes
Servlet implementation
Servlet implementation processes
Startup parameter: quiesce
Parameter: host

1.11.4.3. Shut down all servlet container processes in a servlet implementation

Applies to User roles: System Administrator

You must have administrative access to the server operating system to use this procedure.

To shut down all servlet container processes in a servlet implementation:

1. Open the server's operating system command line prompt.
For example, to open the Windows command prompt:
Click **Start** > **Programs** > **Accessories** > **Command Prompt**.
2. Change directories to the `RUN` folder of your Service Manager installation. For example:

```
cd C:\Program Files\OpenText\Service Manager\server\RUN
```

3. Type the following command:

```
sm -shutdown -group
```

4. Press Enter.
The server shuts down all the servlet container processes in the virtual group.

Related topics

Managing multiple servlet container processes
Servlet implementation
Servlet implementation processes
Parameter: group
Startup parameter: shutdown

1.11.4.4. Shut down all servlet container processes on a host

Applies to User roles: System Administrator

You must have administrative access to the server operating system to use this procedure.

To shut down all servlet container processes on a host:

1. Open the server's operating system command line prompt.
For example, to open the Windows command prompt:
Click **Start** > **Programs** > **Accessories** > **Command Prompt**.

2. Change directories to the `RUN` folder of your Service Manager installation. For example:

```
cd C:\Program Files\OpenText\Service Manager\server\RUN
```

3. Type the following command:

```
sm -shutdown -host:<host name>
```

4. Press Enter.
   The server shuts down all the servlet container processes on the host.

Related topics

Managing multiple servlet container processes
Servlet implementation
Servlet implementation processes
Parameter: host
Parameter: group
Startup parameter: shutdown

1.11.4.5. Monitoring memory in Service Manager processes

The Service Manager process embeds a Java Virtual Machine (JVM) so that the heap memory provided by the operating system for any process is partitioned into a Java Heap and a Native Heap. The Service Manager process self–monitors these heaps and takes action to prevent a crash.

During startup of a Service Manager process, immediately after the JVM is created, a memory monitor thread is started, which will poll for a low memory condition every 15 seconds. On every iteration, the memory monitoring thread fetches the current usage of java heap (all platforms) and native heap (Windows only) and set the process global flags `lowMemoryOnJava` and `lowMemoryOnNative` when the usage exceeds 90% of allocated memory in that category. These flags are unset again when the usage of memory drops below 70%.

To prevent server crashes due to low available heap memory and to avoid disrupting users who are logged on to a Service Manager process, the servletcontainer takes the following actions when the low memory flags are set:

- Writes warning messages to `sm.log`.
- Notifies LoadBalancer not to send that servlet any new client connections.
- Denies new client sessions (Windows client, web client or web services client) for that servlet. They are sent to another one.
- Does not allow existing clients to open new tabs. However, they can trade the existing tabs for new ones. For example, if you close two tabs, you will be allowed to open two new tabs. If you can't operate without opening any new tabs you need to logoff and login again.

Normal processing is resumed when the heap usage drops below 70%. The servletcontainer takes the following actions when the low memory flags are unset.

- Writes Info messages to the sm.log file indicating normal memory usage.
- Notifies LoadBalancer to resume sending of new client connections.

Related topics

Monitoring native heap and Java heap memory
Logging memory monitoring
Servlet parameters
Debugging parameters
Parameter: log4jDebug

1.11.4.5.1. Monitoring native heap and Java heap memory

Java heap

When a Java Virtual Machine (JVM) is started in Service Manager process, the Service Manager process defines the minimum and the maximum amount of memory that's allocated for JVM. The minimum and maximnm vary for different kinds of Service Manager processes.

- For processes with "sync", "reportlbstatus", "reportlocks" or "shutdown" parameters, the maximum value is defined as 96M;
- For processes with "load balancer" parameter, the maximum value is defined as 1500M;
- For other processeS, the maximum amount of memory is depend on the value of `threadsperprocess` parameter: When the value of `threadsperprocess` parameter is greater than 250, the maximum value is defined as 1/4th of the computer's physical memory or 4096M (whichever is smaller); When the value of `threadsperprocess` parameter is same with or less than 250, the maximum value is defined as 1/4th of the computer's physical memory or 2048M (whichever is smaller).

You can override these default values by using the JVMOption parameters. For example, you could set `JVMOption0:-Xms128M, JVMOption1:-Xmx256M`, to provide minimum and maximum values of 128M and 256M for java heap memory.

Native heap

Windows: The Windows operating system has APIs that fetch the current memory usage of a process and the maximum that can be allocated for that process. We monitor the native heap using the API's monitoring thread, and turn the low memory condition processes on or off accordingly.

Linux Systems: Not implemented.

Related topics

Monitoring memory in Service Manager processes
Logging memory monitoring
Servlet parameters
Debugging parameters
Parameter: log4jDebug

1.11.4.5.2. Logging memory monitoring

You can turn on detailed messages of the memory monitoring thread by adding the following parameter to the `sm.ini` file:

`log4jdebug:com.hp.ov.sm.common.oom.LowMemoryHandler`

Messages like these will be sent to the `sm.log` file when the parameter is turned on. Warning messages are written to the log whether detailed messages are turned on or not.

```
2120(  2232) 02/04/2009 13:25:02 JRTE I Starting Memory Monitoring thread to check for memory every 15
seconds.
2120(  5980) 02/04/2009 13:25:02 JRTE D JavaMemory Max(123928576) Used(1011872) %Used(0.0)
2120(  5980) 02/04/2009 13:25:02 JRTE D NativeMemory Max(2147352576) Used(358121472) %Used(16.0)
2120(  5980) 02/04/2009 13:28:32 JRTE W Process Low on Java Memory. Max(123928576) Used(119603800)
PercentUsed(96.0)
2120 02/04/2009 13:30:14 JRTE W Send error response: Server is running low on memory try again.
2120 02/04/2009 13:30:16 JRTE W Send error response: Server is running low on memory try again.
2120(  5980) 02/04/2009 13:30:47  RTE I Process Java Heap Memory is back to normal range.
2120(  5980) 02/04/2009 13:30:47 JRTE I Process Java Memory. Max(123928576)  Used(4256640)
PercentUsed(3.0)
```

Related topics

Monitoring memory in Service Manager processes
Monitoring native heap and Java heap memory
Servlet parameters
Debugging parameters
Parameter: log4jDebug
Parameter: memorypollinterval

1.11.4.5.3. Example: response time monitoring (RTM) output

The following sections show example output from each Response Time Monitoring setting.

RTM:2

```
248(  6112) 10/21/2013 11:09:59  RTE D Response-Total: 0.016 -- RAD:0.016  JS:0.000 Log:0.000
Database:0.000  LoadBalancer: 0.000 (CPU 0.781) Transaction - format:wizard-wiz.prompt.patchrel.unl
application:display,show.rio option:0
248(  6112) 10/21/2013 11:09:59  RTE D -Memory : D(261304) S(4011264) O(1212028) MAX(5387068) - MALLOC's
Transaction(17425) Total(397602)
```

The debugging output includes:

- **Process ID** — lists the process identification number for the transaction
- **Thread ID** — lists the thread identification number for the transaction
- **Date** — lists the date the transaction occurred
- **Time** — lists the time the transaction occurred
- **Response -Total** — lists the number of seconds taken to complete the entire transaction
- **RAD** — lists the number of seconds taken for the RAD to complete the transaction
- **JS** — lists the number of seconds taken for the JavaScript to complete the transaction
- **Log** — lists the number of seconds taken for the log to complete the transaction
- **Database** — lists the number of seconds taken for the database to complete the transaction
- **LoadBalancer** — lists the number of seconds taken for the load balancer to complete the transaction
- **(CPU n)** — lists the cumulative number of seconds the process used on the CPU
- **Transaction - format** — lists the form displayed by the transaction
- **Application** — lists the application when the transaction is completed
- **Option ID** — lists the GUI option or button ID number used in this transaction
- **D** — lists the memory delta from the previous transaction. This value could be positive or negative.
- **S** — lists the storage (in bytes) currently allocated to this session by Service Manager. This doesn't include shared memory or memory that's not directly allocated by Service Manager.
- **O** — lists the overhead associated with the internal memory manager.
- **MAX** — lists the peak value of the storage (in bytes) allocated to this session by Service Manager.
- **MALLOC's Transaction** — lists the delta number of times of Service Manager called MALLOC to allocate memory from the previous transaction
- **Total** — lists the total number of times of Service Manager called MALLOC to allocate memory

RTM:3

```
248(   364) 10/21/2013 11:06:04  RTE D RADTRACE   2781 [ 1] display                          show.rio
rio          CPU(    0    874 )
248(   364) 10/21/2013 11:06:04  RTE D Response-Total: 3.047 -- RAD:3.031  JS:0.000 Log:0.000
Database:0.016  LoadBalancer: 0.000 (CPU 0.874) Transaction - format:wizard-wiz.prompt.patchrel.unl
application:display,show.rio option:0
248(   364) 10/21/2013 11:06:04  RTE D -Memory : D(272304) S(3343528) O(945684) MAX(4440604) - MALLOC's
Transaction(18643) Total(409020)
```

The debugging output includes:

- **Process ID** — lists the process identification number for the transaction
- **Thread ID** — lists the thread identification number for the transaction
- **Date** — lists the date the transaction occurred
- **Time** — lists the time the transaction occurred
- **RADTRACE #** — lists the cumulative elapsed time for the transaction in milliseconds
- **[n]** — lists the thread number of the process
- **RAD application** — lists the RAD application used in the transaction

- **RAD panel** — lists the name of the RAD panel
- **CPU (n m)** — lists the number of milliseconds that the CPU processed the RAD panel (n) and the total number of milliseconds since the RAD trace began (m)
- **Response -Total** — lists the number of seconds taken to complete the entire transaction
- **RAD** — lists the number of seconds taken for the RAD to complete the transaction
- **JS** — lists the number of seconds taken for the JavaScript to complete the transaction
- **Log** — lists the number of seconds taken for the log to complete the transaction
- **Database** — lists the number of seconds taken for the database to complete the transaction
- **LoadBalancer** — lists the number of seconds taken for the load balancer to complete the transaction
- **(CPU n)** — lists the cumulative number of seconds the process used on the CPU
- **Transaction - format** — lists the form displayed by the transaction
- **Application** — lists the application when the transaction is completed
- **Option ID** — lists the GUI option or button ID number used in this transaction
- **D** — lists the memory delta from the previous transaction. This value could be positive or negative.
- **S** — lists the number of bytes of storage allocated by Service Manager. This doesn't include shared memory or memory that's not directly allocated by Service Manager.
- **O** — lists the overhead associated with the internal memory manager.
- **MAX** — lists the peak value of the number of bytes of storage allocated by Service Manager.
- **MALLOC's Transaction** — lists the delta number of times of Service Manager called MALLOC to allocate memory from the previous transaction
- **Total** — lists the total number of times of Service Manager called MALLOC to allocate memory

RTM:4

```
3688(  5512) 10/21/2013 17:21:25  RTE D RADTRACE    219 [ 1] display                              show.rio
rio          CPU(    0   3108 )
3688(  5512) 10/21/2013 17:21:25  RTE D Response-Total: 0.328 -- RAD:0.187  JS:0.000 Log:0.125
Database:0.141  LoadBalancer: 0.000 (CPU 3.155) Transaction - format:wizard-wiz.prompt.patchrel.unl
application:display,show.rio option:0
3688(  5512) 10/21/2013 17:21:25  RTE D -Memory : D(272624) S(3341592) O(943524) MAX(4440604) - MALLOC's
Transaction(54547) Total(879376)
3688(  5512) 10/21/2013 17:21:25  RTE D ... 88   calls made to ';' (1)
3688(  5512) 10/21/2013 17:21:25  RTE D ... 252  calls made to '=' (2)
3688(  5512) 10/21/2013 17:21:25  RTE D ... 7    calls made to ' or ' (3)
3688(  5512) 10/21/2013 17:21:25  RTE D ... 38   calls made to ' and ' (4)
3688(  5512) 10/21/2013 17:21:25  RTE D ... 16   calls made to 'not ' (5)
3688(  5512) 10/21/2013 17:21:25  RTE D ... 1    calls made to '<=' (7)
3688(  5512) 10/21/2013 17:21:25  RTE D ... 47   calls made to '=' (8)
3688(  5512) 10/21/2013 17:21:25  RTE D ... 17   calls made to '~=' (9)
3688(  5512) 10/21/2013 17:21:25  RTE D ... 10   calls made to '>' (11)
3688(  5512) 10/21/2013 17:21:25  RTE D ... 72   calls made to '+' (12)
3688(  5512) 10/21/2013 17:21:25  RTE D ... 61   calls made to 'if ' (19)
3688(  5512) 10/21/2013 17:21:25  RTE D ... 2    calls made to 'if ' (20)
3688(  5512) 10/21/2013 17:21:25  RTE D ... 8    calls made to 'for ' (22)
3688(  5512) 10/21/2013 17:21:25  RTE D ... 834  calls made to ' in ' (23)
3688(  5512) 10/21/2013 17:21:25  RTE D ... 32   calls made to 'decision' (28)
3688(  5512) 10/21/2013 17:21:25  RTE D ... 29   calls made to 'process' (42)
3688  5512) 10/21/2013 17:21:25  RTE D ... 1    calls made to 'rio' (48)
3688(  5512) 10/21/2013 17:21:25  RTE D ... 5    calls made to 'select' (50)
3688(  5512) 10/21/2013 17:21:25  RTE D ... 23   calls made to 'str' (51)
3688(  5512) 10/21/2013 17:21:25  RTE D ... 6    calls made to 'userdefn' (53)
3688(  5512) 10/21/2013 17:21:25  RTE D ... 9    calls made to 'lng' (54)
3688(  5512) 10/21/2013 17:21:25  RTE D ... 14   calls made to 'filename' (55)
3688(  5512) 10/21/2013 17:21:25  RTE D ... 1    calls made to 'operator' (56)
3688(  5512) 10/21/2013 17:21:25  RTE D ... 11   calls made to 'rinit' (58)
3688(  5512) 10/21/2013 17:21:25  RTE D ... 17   calls made to 'currec' (59)
3688(  5512) 10/21/2013 17:21:25  RTE D ... 11   calls made to '#' (60)
3688(  5512) 10/21/2013 17:21:25  RTE D ... 1    calls made to 'val' (72)
3688(  5512) 10/21/2013 17:21:25  RTE D ... 1    calls made to 'index' (78)
3688(  5512) 10/21/2013 17:21:25  RTE D ... 29   calls made to 'null' (88)
3688(  5512) 10/21/2013 17:21:25  RTE D ... 7    calls made to 'denull' (113)
3688(  5512) 10/21/2013 17:21:25  RTE D ... 1    calls made to 'contents' (117)
3688(  5512) 10/21/2013 17:21:25  RTE D ... 1    calls made to 'loop' (119)
3688(  5512) 10/21/2013 17:21:25  RTE D ... 17   calls made to 'parse' (122)
3688(  5512) 10/21/2013 17:21:25  RTE D ... 39   calls made to 'evaluate' (123)
3688(  5512) 10/21/2013 17:21:25  RTE D ... 12   calls made to 'type' (124)
3688(  5512) 10/21/2013 17:21:25  RTE D ... 67   calls made to 'nullsub' (125)
3688(  5512) 10/21/2013 17:21:25  RTE D ... 2    calls made to 'cleanup' (127)
3688(  5512) 10/21/2013 17:21:25  RTE D ... 8    calls made to '+=' (128)
3688(  5512) 10/21/2013 17:21:25  RTE D ... 8    calls made to 'same' (133)
3688(  5512) 10/21/2013 17:21:25  RTE D ... 6    calls made to 'param2' (134)
3688(  5512) 10/21/2013 17:21:25  RTE D ... 4    calls made to 'gui' (155)
3688(  5512) 10/21/2013 17:21:25  RTE D ... 1    calls made to 'sysinfo.get' (169)
3688(  5512) 10/21/2013 17:21:25  RTE D ... 9    calls made to 'strrep' (188)
3688(  5512) 10/21/2013 17:21:25  RTE D ... 14   calls made to 'scmsg' (238)
```

The debugging output includes:

- **Process ID** — lists the process identification number for the transaction
- **Thread ID** — lists the thread identification number for the transaction
- **Date** — lists the date the transaction occurred
- **Time** — lists the time the transaction occurred

- **RADTRACE #** — lists the cumulative elapsed time for the transaction in milliseconds
- **[n]** — lists the thread number of the process
- **RAD application** — lists the RAD application used in the transaction
- **RAD panel** — lists the name of the RAD panel
- **CPU (n m)** — lists the number of milliseconds that the CPU processed the RAD panel (n) and the total number of milliseconds since the RAD trace began (m)
- **Response -Total** — lists the number of seconds taken to complete the entire transaction
- **RAD** — lists the number of seconds taken for the RAD to complete the transaction
- **JS** — lists the number of seconds taken for the JavaScript to complete the transaction
- **Log** — lists the number of seconds taken for the log to complete the transaction
- **Database** — lists the number of seconds taken for the database to complete the transaction
- **LoadBalancer** — lists the number of seconds taken for the load balancer to complete the transaction
- **(CPU n)** — lists the cumulative number of seconds the process used on the CPU
- **Transaction - format** — lists the form displayed by the transaction
- **Application** — lists the application when the transaction is completed
- **Option ID** — lists the GUI option or button ID number used in this transaction
- **D** — lists the memory delta from the previous transaction. This value could be positive or negative.
- **S** — lists the number of bytes of storage allocated by Service Manager. This doesn't include shared memory or memory that's not directly allocated by Service Manager.
- **O** — lists the overhead associated with the internal memory manager.
- **MAX** — lists the peak value of the S
- **MALLOC's Transaction** — lists the delta number of times of Service Manager called MALLOC to allocate memory from the previous transaction
- **Total** — lists the total number of times of Service Manager called MALLOC to allocate memory
- **# of calls made to 'x'** — lists the number of times Service Manager called the function, panel, or operator x from the last format response to this one
- **(#)** — lists the function identification number for the function, panel, or operator from the last format response to this one

RTM:5

```
3688(  3952) 10/21/2013 17:22:45  RTE D RADTRACE    203 [ 1] display                          show.rio
rio          CPU(    0   2827 )
3688(  3952) 10/21/2013 17:22:45  RTE D Response-Total: 0.281 -- RAD:0.281  JS:0.000 Log:0.265
Database:0.000  LoadBalancer: 0.000 (CPU 2.843) Transaction - format:wizard-wiz.prompt.patchrel.unl
application:display,show.rio option:0
3688(  3952) 10/21/2013 17:22:45  RTE D -Memory : D(261512) S(3330480) O(950540) MAX(4440604) - MALLOC's
Transaction(54063) Total(878892)
3688(  3952) 10/21/2013 17:22:45  RTE D ... 595  calls made to ' in ' (23)
3688(  3952) 10/21/2013 17:22:45  RTE D ... 1    calls made to 'rio' (48)
3688(  3952) 10/21/2013 17:22:45  RTE D ... 1    calls made to 'evaluate' (123)
3688(  3952) 10/21/2013 17:22:45  RTE D ... 2    calls made to 'nullsub' (125)
3688(  3952) 10/21/2013 17:22:45  RTE D ... 2    calls made to 'gui' (155)
```

The debugging output includes:

- **Process ID** — lists the process identification number for the transaction
- **Thread ID** — lists the thread identification number for the transaction
- **Date** — lists the date the transaction occurred
- **Time** — lists the time the transaction occurred
- **RADTRACE #** — lists the cumulative elapsed time for the transaction in milliseconds
- **[n]** — lists the thread number of the process

- **RAD application** — lists the RAD application used in the transaction
- **RAD panel** — lists the name of the RAD panel
- **CPU (n m)** — lists the number of milliseconds that the CPU processed the RAD panel (n) and the total number of milliseconds since the RAD trace began (m)
- **Response -Total** — lists the number of seconds taken to complete the entire transaction
- **RAD** — lists the number of seconds taken for the RAD to complete the transaction
- **JS** — lists the number of seconds taken for the JavaScript to complete the transaction
- **Log** — lists the number of seconds taken for the log to complete the transaction
- **Database** — lists the number of seconds taken for the database to complete the transaction
- **LoadBalancer** — lists the number of seconds taken for the load balancer to complete the transaction
- **(CPU n)** — lists the cumulative number of seconds the process used on the CPU
- **Transaction - format** — lists the form displayed by the transaction
- **Application** — lists the application when the transaction is completed
- **Option ID** — lists the GUI option or button ID number used in this transaction
- **D** — lists the memory delta from the previous transaction. This value could be positive or negative.
- **S** — lists the number of bytes of storage allocated by Service Manager. This doesn't include shared memory or memory that's not directly allocated by Service Manager.
- **O** — lists the overhead associated with the internal memory manager.
- **MAX** — lists the peak value of the S
- **MALLOC's Transaction** — lists the delta number of times of Service Manager called MALLOC to allocate memory from the previous transaction
- **Total** — lists the total number of times of Service Manager called MALLOC to allocate memory
- **# of calls made to 'x'** — lists the number of times Service Manager called the function, panel, or operator x from the last panel response to this one
- **(#)** — lists the function identification number for the function, panel, or operator from the last panel response to this one

Related topics

Debugging parameters

1.12. Startup and shutdown

There are several methods you can use to start and stop the Service Manager server and associated background processes depending upon the operating system on which your system runs. See the topics associated with your Service Manager server's operating system for more information.

Related topics

Starting Service Manager on Linux
Starting Service Manager on Windows
Stopping Service Manager on Linux
Stopping Service Manager on Windows

1.12.1. Starting Service Manager on Linux

You must start the Service Manager server before users can connect with client sessions. You can also manually start the Service Manager server from the following interface:

- Linux command line

Related topics

Start the server from the Linux command line
Stop the server from the Linux command line

1.12.1.1. Start the server from the Linux command line

Applies to User roles: System Administrator

You must start the Service Manager server before users can connect with client sessions. You must have administrative access to the server operating system to use this procedure.

To start the server from the Linux command line:

1. Change directories to your Service Manager Run directory. For example:

   ```
   cd /MicroFocus/Service Manager x.xx/Run
   ```

2. Type the following command:

   ```
   smstart
   ```

3. Press ENTER.

Related topics

Stop the server from the Linux command line

1.12.2. Starting Service Manager on Windows

You must start the Service Manager server before users can connect with client sessions. By default, the Service Manager server runs as an automatically starting service on Windows. The Service Manager server will start every time that you restart your server. You can also manually start the Service Manager server from the following interfaces:

- Windows services
- Windows command prompt

Related topics

Start the server from the Windows command prompt
Start the server from Windows services
Stop the server from the Windows command prompt
Stop the server from Windows services

1.12.2.1. Configure the Service Manager service to run as a Windows user

Applies to User Roles: System Administrator

You must have administrative access to the server operating system to use this procedure.

To configure the Service Manager service to run as a Windows user:

1. Open the Services applet by doing either of the following:
   - Click **Start** > **Programs** > **Administrative Tools** > **Services**
   - Click **Start** > **Settings** > **Control Panel** > **Administrative Tools** > **Services**
2. In the services list, click **Service Manager**.
3. Click the Stop service button or click **Action** > **Stop** to stop the service.
4. Edit the service properties by doing either of the following:
   - Click the Properties button.
   - Click **Action** > **Properties**
5. Click the **Log On** tab.
6. In the **Log on as: This Account** field, click **Browse**. Windows displays a list of local domains and user accounts.
7. Select the domain and user account you want to use to run Service Manager.
8. Click **OK**.
9. Type the user account password and password confirmation.
10. Click **OK**.
11. Start the service by doing either of the following:
    - Click the Start service button
    - Click **Action** > **Start**

Related topics

Start the server from Windows services

1.12.2.2. Start a Service Manager client listener

Applies to user roles: System Administrator

You can start one or more client listeners to manage your system's Web and Windows client connection requests. The number of client listeners you need to start depends upon the type of servlet implementation you are running.

> **Note**
> Note
> The client listener can't accept connection requests from legacy integration products such as ServiceCenter Automation (SCAuto). For these SCAuto connections, you must start an SCAuto listener (scautolistener).

You must have administrative access to the server operating system to use this procedure.

To start a Service Manager client listener:

1. Stop the Service Manager server.
2. Open the Windows command prompt.
3. Change directories to the `RUN` folder of your Service Manager installation. For example:

   ```
   cd C:\Program Files\OpenText\Service Manager\server\RUN
   ```

4. Type one of the following commands:

Servlet implementation option	Command line parameters or configuration file entries
Single servlet implementation	`sm -httpPort:13081 httpsPort:13082`
Horizontal or vertical scaling implementations	`sm -httpPort:13080 -loadbalancer` `sm -httpPort:13081 -httpsPort:13082`

   Each client listener listens to client connection requests on a separate communications port. You can specify your own communications port for the example port in the table.

Related topics

Servlet implementation options
Startup and shutdown
Start the server from the Linux command line
Start the server from the Windows command prompt
Startup parameter: scautolistener

1.12.2.3. Start the server from the Windows command prompt

Applies to User roles: System Administrator

You must have administrative access to the server operating system to use this procedure.

To start the server from the Windows command prompt:

1. Open the Windows command prompt.
   Click **Start** > **Programs** > **Accessories** > **Command Prompt**.
2. Change directories to the `RUN` folder of your Service Manager installation. For example:

   ```
   cd C:\Program Files\OpenText\Service Manager x.xx\Server\RUN
   ```

3. Type the following command:

```
sm -httpPort:13080 -httpsPort:13081
```

   Tip: You can omit the **httpPort** and **httpsPort** parameters if you have specified them in the Service Manager initialization file (`sm.ini`).

4. Press Enter.
   Caution: You must leave the command prompt window open while the Service Manager server runs. Closing the command prompt window stops Service Manager immediately without cleaning up any processes or releasing any record locks.

Related topics

Start the server from Windows services
Stop the server from the Windows command prompt
Stop the server from Windows services

1.12.2.4. Start the server from Windows services

Applies to User Roles: System Administrator

You can to use this procedure to run the Service Manager Server service as an administrator user. If you need to run the Service Manager Server service as a non-administrator user, follow the steps in Run the Service Manager Server using Windows Service.

To start the server from Windows services:

1. Open the Services applet.
   - Click **Start** > **Programs** > **Administrative Tools** > **Services**
     –or–
   - Click **Start** > **Settings** > **Control Panel** > **Administrative Tools** > **Services**
2. In the services list, click **Service Manager**.
3. Start the service.
   - Click the Start Service link
     –or–
   - Click **Action** > **Start**

   Windows displays a message that the Service Manager service is starting. After several seconds, the service starts and displays Started in the Status field. If the Service Manager service doesn't start, contact customer support with any error messages.

Related topics

Start the server from the Windows command prompt
Stop the server from the Windows command prompt
Stop the server from Windows services

1.12.3. Stopping Service Manager on Linux

Stopping the Service Manager server immediately disconnects all client sessions and ends all currently running Service Manager processes. By default, the Service Manager server runs continuously and has no scheduled shutdown times.

There are two methods for stopping Service Manager:

- Server shutdown with a warning message — allows the administrator to broadcast a warning message before stopping the server
- Server stop from outside a client — allows the administrator to stop the server without first connecting to the server through a client

Server shutdown with a warning message

When Service Manager is running on a Linux server, you can issue a shutdown warning from the following interfaces:

- From a Service Manager client when you are logged in as a user with system administrator capabilities.

- From the Linux command line when you use the **smstop** script

Either shutdown option allows you to broadcast a warning message to currently connected users letting them know that you are planning to shut the server down. However, only the shutdown option from the client also allows you to specify in advance when you want to shut a server down.

There are three types of shutdown available to administrators from the client:

- Immediate — stops the server as soon as the administrator confirms the shutdown request. The server issues no warning messages before stopping.
- Delayed — stops the server after a delay specified by the administrator. The server issues a warning message every minute during the warning interval. The warning interval determines how many minutes before stopping the server that the server starts issuing warning messages.
- Scheduled — stops the server at a specified date and time. The server issues a warning message every minute during the warning interval. The warning interval determines how many minutes before stopping the server that the server starts issuing warning messages.

The shutdown option from the Linux command line doesn't allow you to delay or schedule a server shutdown.

Server stop from outside a client

You must have administrative access to the server operating system to stop the Service Manager server outside of a client. If you want to issue a warning message before stopping the server, you must use the **smstop** script. Unless you know that there are no active client connections, it is generally preferable to shutdown a server with a warning message rather than stop it.

You can stop a Service Manager server running on Linux from the following interface without using a client connection:

- Linux command line

Related topics

Start the server from the Linux command line
Stop the server from the UNIX command line

1.12.3.1. Stop the server from the Linux command line

Applies to User roles: System Administrator

You must have administrative access to the server operating system to use this procedure.

To stop the server from the Linux command line:

Note

Note

This procedure stops the server without displaying any warnings to currently connected users. If you want to broadcast a warning message before stopping the server, use one of the shutdown options available from the Service Manager clients.

1. Change directories to your Service Manager Run directory. For example:

   ```
   cd /MicroFocus/Service Manager x.xx/Run
   ```

2. Type one of the following commands:

Command	Description
sm -shutdown	Immediately stops the Service Manager server and all related processes.
sm -shutdown -group	Used in a horizontally-scaled environment. Immediately stops the Service Manager server and all related processes, including Service Manager processes on remote hosts in the same group.
smstop	The script attempts to do a normal shutdown of the Service Manager system and, if successful, stops any remaining Service Manager processes and cleans up IPC resources. **Note**: The `scdb.system` record in the `info` table acts as a Service Manager database lock, preventing multiple Service Manager groups from connecting to the same database. This record will only be removed during normal shutdown and when there are no more Service Manager processes connecting to that database. In a vertically-scaled environment, normal shutdown during `sm -shutdown`, `smstop`, or `smstop -f` removes this record. In a horizontally-scaled environment, only a normal shutdown (`sm -shutdown -group`) shuts down the whole group.
smstop -d	The script runs the shutdown script in debug mode. The system produces verbose output messages in debug mode.
smstop -f	The script forces a shutdown of the Service Manager system if the normal shutdown process fails. By default, the script will wait for up to ten minutes for the normal shutdown process to complete. To change the wait time, you can set the SMSTOP_WAIT environment variable with the number of seconds you want the script to wait for the completion of the normal shutdown process (the default value is 600 seconds). For example, to change the wait time to 15 minutes, you can use the following command: `SMSTOP_WAIT=900 smstop -f`
smstop -F	The script immediately forces a shutdown of the Service Manager system, bypassing the normal shutdown process. **Caution:** Forcing a system shutdown can result in file corruption if there is file regeneration in progress at the time of the shutdown. We recommend using the Service Manager status and system.monitor functions to determine if there are any active processes running prior to forcing a system shutdown.
smstop -h	Displays help information for the smstop script.

smstop -q	Runs the shutdown script in quiet mode. The system doesn't produce output messages in quiet mode.

3. Press ENTER.

Related topics

Start the server from the Linux command line

1.12.4. Stopping Service Manager on Windows

Stopping the Service Manager server immediately disconnects all client sessions and ends all currently running Service Manager processes. By default, the Service Manager server runs continuously and has no scheduled shutdown times.

There are two methods for stopping Service Manager:

- Server shutdown with a warning message — allows the administrator to broadcast a warning message before stopping the server
- Server stop from outside a client — allows the administrator to stop the server without first connecting to the server through a client

Server shutdown with a warning message

When Service Manager is running on a Windows server, you can only issue a shutdown warning from a Service Manager client, and then only if you are logged in as a user with system administrator capabilities. The shutdown option allows you to broadcast a warning message to currently connected users letting them know that you are planning to shut the server down. The shutdown option also allows you to specify in advance when you want to shut a server down.

There are three types of shutdown available to administrators:

- Immediate — stops the server as soon as the administrator confirms the shutdown request. The server issues no warning messages before stopping.
- Delayed — stops the server after a delay specified by the administrator. The server issues a warning message every minute during the warning interval. The warning interval determines how many minutes before stopping the server that the server starts issuing warning messages.
- Scheduled — stops the server at a specified date and time. The server issues a warning message every minute during the warning interval. The warning interval determines how many minutes before stopping the server that the server starts issuing warning messages.

Server stop from outside a client

You must have administrative access to the server operating system to stop the Service Manager server outside of a client. When Service Manager is running on a Windows server, you can't issue a warning message if you stop the server outside of a client. Unless you know that there are no active client connections, it is generally preferable to shutdown a server with a warning message rather than stop it.

You can stop a Service Manager server running on Windows from the following interfaces without using a client connection:

- Windows services
- Windows command prompt

Related topics

Start the server from the Windows command prompt
Start the server from Windows services
Stop the server from the Windows command prompt
Stop the server from Windows services

1.12.4.1. Stop the server from the Windows command prompt

Applies to User roles: System Administrator

You must have administrative access to the server operating system to use this procedure.

To stop the server from the Windows command prompt:

Warning: This procedure stops the server without displaying any warnings to currently connected users. If you want to broadcast a warning message before stopping the server, use one of the shutdown options available from the Service Manager clients.

1. Open another Windows command prompt.
   Click **Start** > **Programs** > **Accessories** > **Command Prompt**.
   Caution: Don't close the command prompt window running Service Manager. Closing the command prompt window stops Service Manager immediately without cleaning up any processes or releasing any record locks.
2. Change directories to the `RUN` folder of your Service Manager installation. For example:

   ```
   cd C:\Program Files\OpenText\Service Manager x.xx\Server\RUN
   ```

3. Type the following command:

   ```
   sm -shutdown:0
   ```

4. Press **ENTER**.

Related topics

Start the server from the Windows command prompt
Start the server from Windows services
Stop the server from Windows services

1.12.4.2. Stop the server from Windows services

You must have administrative access to the server operating system to use this procedure.

Note: This procedure stops the server without displaying any warnings to currently connected users. If you want to broadcast a warning message before stopping the server, use one of the shutdown options available from the Service Manager clients.

1. Open the services applet.
   - Click **Start** > **Programs** > **Administrative Tools** > **Services**
     –or–
   - Click **Start** > **Settings** > **Control Panel** > **Administrative Tools** > **Services**
2. In the services list, click **Service Manager**.
3. Stop the service.
   - Click the Stop Service button
     –or–
   - Click **Action** > **Stop**

   Windows displays a message that the Service Manager service is stopping. After several seconds, the service stops and displays a blank entry in the Status field. If you see any error messages, contact customer support with the message.

Related topics

Start the server from the Windows command prompt
Start the server from Windows services
Stop the server from the Windows command prompt

1.12.5. Server shutdown

A system administrator can shutdown a Service Manager server from any client. Shutting down a server differs from manually stopping a server in the following ways:

- You can schedule a shutdown in advance
- The server can broadcast warning messages to all connected users before the server stops

- The server stops all active processes before shutting down

In a production environment, it is generally preferable to shutdown a server rather than to stop it to ensure data integrity and to allow users to log off.

Note

Note

When the Service Manager server is running on Windows system and started with Windows service, only SM processes can be stopped through Windows client. The process called **smservice.exe** can only be shutted down through Windows service. Such situation doesn't occur on Linux system.

Related topics

Startup and shutdown
Do an immediate shutdown of the server
Do a delayed shutdown of the server
Do a scheduled shutdown of the server

1.12.5.1. Do an immediate shutdown of the server

Applies to User Roles: System Administrator

You must have the **SysAdmin** capability word to use this procedure.

Note: This procedure stops the server without displaying any warnings to currently connected users. If you want to broadcast a warning message before stopping the server, use the broadcast form (system.status.broadcast) before you shut down the server.

To do an immediate shutdown of the server:

1. Click **System Administration** > **Ongoing Maintenance** > **System** > **Shutdown**.
   The System Shutdown Confirmation form opens.
2. Click **Shutdown Now**.
3. Click **Confirm**.
   The Service Manager server stops disconnecting all currently connected clients and refusing any further client connections until the server has been restarted.
   Note: Service Manager also disconnects the client you used to shut down the server.

Related topics

Server shutdown
Do a delayed shutdown of the server
Do a scheduled shutdown of the server

1.12.5.2. Do a delayed shutdown of the server

Applies to User Roles: System Administrator

You must have the **SysAdmin** capability word to use this procedure.

To do a delayed shutdown of the server:

1. Click **System Administration** > **Ongoing Maintenance** > **System** > **Shutdown**.
   The System Shutdown Confirmation form opens.
2. Click **Advise and Shutdown**.
3. In the text field provided, type the number of seconds you want the system to wait before it shuts down.
4. Click **Advise and Shutdown**.
   Service Manager broadcasts the following message to all connected users:
   System will SHUTDOWN in <n> seconds; please finish and log off.
   The value <*n*> is the number of seconds you entered for the delay. After the specified number of seconds passes, Service Manager stops disconnecting all currently connected clients and refusing any further client connections until the server has been restarted.

Note: Service Manager also disconnects the client you used to shutdown the server.

Related topics

Server shutdown
Do a scheduled shutdown of the server
Do an immediate shutdown of the server

1.12.5.3. Do a scheduled shutdown of the server

Applies to User Roles: System Administrator

You must have the **SysAdmin** capability word to use this procedure.

To do a scheduled shutdown of the server:

1. Click **System Administration** > **Ongoing Maintenance** > **System** > **Schedule Shutdown**.
   The Automatic Shutdown Information form opens.
2. Click **Search**.
   Service Manager displays a list of existing shutdown records.
   Note: Service Manager can only have one active shutdown record at a time. If a shutdown record already exists, update the existing record rather than creating a new one.
3. Type or select the following information.
   - **Shutdown Date** — type the date and time you want the server to shutdown
   - **Auto-Shutdown** — select this option if you want the shutdown to be automatic
   - **Reschedule Interval** — type or select the number of days you want the server to wait for the next scheduled shutdown. If you leave this entry blank, then Service Manager uses the default value of one day.
   - **Shutdown Warning** — type or select the number of minutes you want the server to issue shutdown warnings, if any
4. Do one of the following:
   - If there is no existing shutdown record, click **Add**.
     Service Manager displays the following message:
     shutdown record added.
   - If there is an existing shutdown record, click **Save**.
     Service Manager displays the following message:
     shutdown record updated.
5. From the pull-down options menu, click **Update schd**.
   Service Manager displays the following message:
   shutdown record updated.
   At the specified date and time, Service Manager stops disconnecting all currently connected clients and refusing any further client connections until the server has been restarted.

Related topics

Server shutdown
Do a delayed shutdown of the server
Do an immediate shutdown of the server

1.13. System information record

The system information record contains common information that all Service Manager applications can use.

From the company record administrators can do the following.

Note

Note The information listed below is presented in the same order within the tabs displayed in this form.

- Add company contact information
- Select Service Manager portal
- Add URL information for Service Manager clients such as web tier and self-service
- Set general login restrictions
- Set user lockout conditions
- Define the operator template to be applied to LDAP users
- Set user account expiration conditions
- Define default menu information
- Set password reset restrictions
- Enable password history
- Set password format restrictions
- Set password lifetime conditions
- Set the time zone the system uses
- Set the date format the system uses
- Enable legacy print routines
- Define the months of the year and the abbreviations the system uses
- Enable or select miscellaneous settings
    - Set the default language the system uses
    - Set the default currency the system uses
    - Set the maximum size for a file attachment
    - Set the maximum number of inbox views that a user can create
    - Set the maximum memory all file attachments can use
    - Enable case sensitivity
    - View the system start time
    - Enable message queues
    - Enable auditing in the system log
    - Enable multi-company mode
    - Enable learning for Service Manager Knowledge Bases
    - Enable editing of the learning record
    - Enable universal search
    - Disable the pop-up window to reload records
    - Enable Calendar and Time Period Management
    - Enable Enhanced Query Hash
- List message processors the system uses

1.13.1. Add company contact information

Applies to User Roles: System Administrator

To add company contact information:

1. Click **System Administration** > **Base System Configuration** > **Miscellaneous** > **System Information Record**.
2. Type the company contact information.
3. Click **Save**.

Related topics

System information record

1.13.2. Select Service Manager portal

Applies to User roles: System Administrator

From **System Information Record**, you can specify which portal is the active end user portal.

To select the active Service Manager end user portal, follow these steps:

1. Click **System Administration** > **Base System Configuration** > **Miscellaneous** > **System Information Record**.
2. Click the **Active Integrations** tab.
3. In the **SM Portal** field, select one of the following portals from the drop-down list:
    - **ESS**
    - **Service Portal**
4. Click **Save**.

Related topics

System information record

1.13.3. Add URLs for Service Manager clients

Applies to User roles: System Administrator

You can add URLs for Service Manager clients to the system information record to make them available as system variables.

To add URLs for Service Manager clients, follow these steps:

1. Click **System Administration** > **Base System Configuration** > **Miscellaneous** > **System Information Record**.
2. Click the **Active Integrations** tab.
3. Type values for the **Web Server URL**, **ESS URL**, and **Mobility URL** fields as needed.

    Note
    Note The **ESS URL** only takes effect if the corresponding option selected as the active SM portal.

4. Click **Save**.

Related topics

System information record

1.13.4. Enable the enhanced query hash algoritm

Applies to User roles: System Administrator

Starting with version 9.41, Service Manager provides an enhanced query hash algorithm, which ensures a higher level of system security when the Service Manager web client is handling URLs. By default, this functionality is disabled for backward compatibility.

To enable the enhanced query hash algorithm, follow these steps:

1. Click **System Administration** > **Base System Configuration** > **Miscellaneous** > **System Information Record**.
2. Click the **General** tab.
3. Configure the enhanced query hash functionality as described in the following table.

Option	Default value	Description
Enable Enhanced Query Hash	false	When this option is set to true (selected), the Service Manager server generates query hashes based on the enhanced query hash algorithm. For security considerations, we recommend you to enable this option. By default, this option is disabled so that your system can continue to use the legacy algorithm and work as before.
Allow Legacy Query Hash	true	This option is available only when the **Enable Enhanced Query Hash** option is enabled. By default, this option is enabled so that the Service Manager web client can accept URLs with query hashes that are generated based on either the legacy or the enhanced algorithm. If you deselect this option, the client displays an error message when processing URLs with a legacy query hash, indicating the hash code is invalid. **Note** Note Use this option only during your transition from the legacy algorithm to the enhanced algorithm. Once no URLs with legacy query hashes exist in your system, we recommend you to disable this option.

4. Click **Save**.

Related topics

System information record
System Security
Generating Web tier URL queries

1.13.5. Set the default maximum number of login attempts per session

Applies to User Roles: System Administrator

To set the default maximum number of login attempts per session:

1. Click **System Administration** > **Base System Configuration** > **Miscellaneous** > **System Information Record**.
2. Click the **Logon Info** tab.

3. Move the slider in the **Attempts Per Login Session** field to an appropriate value. The system uses the number of login attempts as the company default.
   For more information about how to set the minimum and maximum value of the **Attempts Per Login Session** field, see Decimal control.
4. Click **Save**.

The Windows client creates a new session for each login attempt. There will never be more than one login attempt per session. To restrict login attempts from the Windows client, enable the **Use User Lockout** option and set a value for the **Attempts Until Lockout** field. Use the **Attempts Per Login Session** setting to control Web client login attempts.

Related topics

System information record
Set the default maximum number of sessions per user
Set the default password reset
Set the default user expiration interval
Set the default user inactivation interval

1.13.6. Set the default maximum number of sessions per user

Applies to User Roles: System Administrator

To set the default maximum number of sessions per user:

1. Click **System Administration** > **Base System Configuration** > **Miscellaneous** > **System Information Record**.
2. Click the **Logon Info** tab.
3. Move the slider in the **Max Logins Per User** field to an appropriate value. The system uses the number of user sessions as the company default.
   For more information about how to set the minimum and maximum value of the **Max Logins Per User** field, see Decimal control.
4. Click **Save**.

Related topics

System information record
Set the default maximum number of login attempts per session
Set the default user expiration interval
Set the default user inactivation interval

1.13.7. Set the default password reset

Applies to User Roles: System Administrator

To set the default password reset:

1. Click **System Administration** > **Base System Configuration** > **Miscellaneous** > **System Information Record**.
2. Click the **Passwords** tab.
3. Click the **Password Standards** tab.
4. Select one of the following options:
   - **Prevent Pwd Reset** — select this option to disable the password reset option.
   - **Reset to User Name** — select this option to reset passwords to the operator's login name.
   - **Prompt for Value** — select this option to have Service Manager prompt the system administrator for a new password.
     Tip: For increased security, the system administrator can also select the password expired option in the operator record. Doing so requires reset users to change their passwords at the next login.
   - **Reset to Value** — select this option to type a global password that applies to all reset users.
5. Click **Save**.

Related topics

Lockout feature
System information record

1.13.8. Set the default user expiration interval

Applies to User Roles: System Administrator

To set the default user expiration interval:

1. Click **System Administration** > **Base System Configuration** > **Miscellaneous** > **System Information Record**.
2. Click the **Logon Info** tab.
3. In the **Expiration Interval** field, type the time period that a user can remain inactive until the user account expires. Type the time period in the following format: *Dayshours*:*minutes*:*seconds*. For example, 4 03:02:01 expires the user account after 4 days, 3 hours, 2 minutes, and 1 second of inactivity.
   Important: The system administrator can't reset an expired account. The system administrator must create a new operator record for the expired account.
4. Click **Save**.

Related topics

System information record
Set the default maximum number of login attempts per session
Set the default maximum number of sessions per user
Set the default password reset
Set the default user inactivation interval

1.13.9. Set the default user inactivation interval

Applies to User Roles: System Administrator

To set the default user inactivation interval:

1. Click **System Administration** > **Base System Configuration** > **Miscellaneous** > **System Information Record**.
2. Click the **Logon Info** tab.
3. In the **Inactivation Interval** field, type the time period that a user can remain inactive until Service Manager locks out the account.
   Type the time period in the following format: *Dayshours*:*minutes*:*seconds*. For example, 4 03:02:01 locks out the user after 4 days, 3 hours, 2 minutes, and 1 second of inactivity.
4. Click **Save**.

Related topics

System information record
Set the default maximum number of login attempts per session
Set the default maximum number of sessions per user
Set the default password reset
Set the default user expiration interval

1.13.10. Setting file attachment size limits

Setting attachment size limits

You can set the following two file attachment size limits:

- Maximum single attachment size
  This value limits the maximum size of a single attachment.
- Total attachment size
  This value limits the total size of all attachments in an attachment container, such as an incident record.

You can set these attachment size limits in the following ways:

- Specify the attachment size limit for the entire system in the system information record
- Specify the attachment size limit for a particular operator in their operator record
- Specify the attachment size limit for the attachment container on a form by using Forms Designer
- Specify the attachment size limit for individual attachments by using the *MaxAttachUploadSize* and *MaxTotalAttachUploadSize* parameters.

Note

Note These parameters are located in the web.xml file. The default value of the *MaxAttachUploadSize* parameter is 10 MB, and the default value of the *MaxTotalAttachUploadSize* parameter is 50 MB.

If you don't specify a value in one of these places, there is no limit on the size of attachments, and users can add attachments as large as the database record will hold. If you specify the attachment size limits in more than one place, then the order of precedence is:

1. The value from the attachment container
2. The value from the operator record
3. The value from the company record
4. The values of the *MaxAttachUploadSize* and *MaxTotalAttachUploadSize* parameters

As an example, for the maximum size of a single attachment, if the company record limit is set to 1,000,000 bytes, the operator has a limit set to 2,000,000 bytes, and the container has a limit of 3,000,000 bytes, the person will be able to store a 3,000,000 byte attachment in the container (the attachment container property overrides the other two limits). A different operator without a limit, storing attachments in a different container with no limit set, would only be able to save a 1,000,000 byte attachment.

Note

Note File size validation is not performed at the widget level on the web service. Therefore, it is possible to attach a file that exceeds the configured size limits by using the web service.

Setting attachment count limits

You can limit the number of individual files that can be attached to each record in Service Manager. To do this, configure the *maxattachmentcount* parameter. By default, the value of this parameter is 100 files.

Related topics

System information record
Parameter: maxattachmentcount
Set the maximum file attachment size for the entire company
Set the maximum file attachment size for an operator
Set the total file attachment size for the entire company
Set the total file attachment size for an operator

1.13.11. Set the maximum file attachment size for the entire company

Applies to User Roles: System Administrator

To set the maximum file size of a single attachment for the entire company:

1. Click **System Administration** > **Base System Configuration** > **Miscellaneous** > **System Information Record**.
2. Click the **General** tab.
3. In the **Max Attachment Size** field, type the number of bytes for the maximum file attachment size.
4. Click **Save**.

You can define a file attachment maximum size limit for all users from the system wide company record. The system wide company record file attachment size limit is superseded by size limits defined in individual operator records and for attachment containers defined in Forms Designer.

Related topics

Setting file attachment limits
Set the maximum file attachment size for an operator
Set the total file attachment size for the entire company
Set the total file attachment size for an operator

1.13.12. Set the total file attachment size for the entire company

Applies to User Roles: System Administrator

To set the total attachment size in any attachment container for the entire company:

1. Click **System Administration** > **Base System Configuration** > **Miscellaneous** > **System Information Record**.
2. Click the **General** tab.
3. In the **Total Attachment Size** field, type the number of bytes for the total file attachment size.
4. Click **Save**.

You can define a total attachment size limit for all users from the system wide company record. The total attachment size limit defined in system wide company record file is superseded by size limits defined in individual operator records and for attachment containers defined in Forms Designer.

Related topics

Set the total file attachment size for an operator
Set the maximum file attachment size for the entire company
Set the maximum file attachment size for an operator
Add company contact information
Add URLs for Service Manager clients

1.13.13. Set the maximum number of inbox views

Applies to User Roles: System Administrator

To set the maximum number of inbox views that a user can create:

1. Click **System Administration** > **Base System Configuration** > **Miscellaneous** > **System Information Record**.
2. Select the **General** tab.
3. In the **Max Views Shown in the List** field, type the maximum number of inbox views that users are allowed to create.
   Notes:
   - The default number of allowable views is 100.
   - There is no limit to the number of views that can be set.
4. Click **Save**.

Related topics

System information record

1.13.14. Set the virtual group size for inbox views

Applies to User roles: System Administrator

In inbox views, records are divided into virtual groups if there are too many records in a group.

To set the number of records that are displayed as one virtual group in inbox views, follow these steps:

1. Click **System Administration** > **Base System Configuration** > **Miscellaneous** > **System Information Record**.
2. Select the **General** tab.
3. In the **Virtual Group Size** field, type the number of records that you want the system to display as one group in inbox views.
   - If the value is 0 or empty, the virtual group size is determined by the client, which is 100 by default.
   - You can set the value from 25 to 500. In addition, make sure that this group size is not greater than the value of the sqlfetchs parameter if you have specified one.
   - Make sure that you do performance testing before you set a big number in your production environment.
4. Click **Save**.

Related topics

System information record

1.13.15. Set the menu prompt

Applies to User Roles: System Administrator

The menu prompt is the text Service Manager displays in the first workspace tab. To set the menu prompt:

1. Click **System Administration** > **Base System Configuration** > **Miscellaneous** > **System Information Record**.
2. Click the **Menu Information** tab.
3. In the **Menu Prompt** field, type the text or variable to display in the first workspace tab.
4. Click **Save**.

Related topics

System information record

1.13.16. Activate the command/search line toggle button

Applies to User Roles: System Administrator

To activate the command/search line toggle button:

1. Follow the steps in Enable an operator to see the command line.
2. Click **System Administration** > **Base System Configuration** > **Miscellaneous** > **System Information Record**.
3. Click the **General** tab.
4. Mark the **Enable Universal Search** checkbox.
5. Click **Save**.
6. Restart your web applications server.
7. Click **System Status** and verify the lister background process is running. If not, click **Start Scheduler** and double-click **lister.startup** in the list to launch this process.
8. Log out of Service Manager and then log in again. The command/search line toggle button is enabled for the operator.

Related topics

Display application
Displayoptions
Selecting display options

1.13.17. Enable editing of Knowledge learning records

Applies to User Roles: System Administrator

To enable editing of Knowledge learning records:

1. Click **System Administration** > **Base System Configuration** > **Miscellaneous** > **System Information Record**.
2. Click the **General** tab.
3. Select the **Allow Editing of Learning Record** option.
4. Click **Save**.

Service Manager displays the message: Information record updated.

Related topics

Adaptive learning

1.13.18. Enable learning for Knowledge records

Applies to User Roles: System Administrator

Enabling learning for Knowledge records grants users access to the **Use Resolution** option when adding a solution to a knowledge record.

To enable learning for Knowledge records:

1. Click **System Administration** > **Base System Configuration** > **Miscellaneous** > **System Information Record**. The info.company.g form opens.
2. Click the **General** tab.

3. Select the **Use learning for Knowledge Engineering** option.
4. Click **Save**.

Service Manager displays the message: Information record updated.

Related topics

Adaptive learning
Enable editing of Knowledge learning records

1.14. Time zones

A time zone is Greenwich Mean Time (GMT) plus or minus the required offset for geographic location. Service Manager uses time zone information to display local time correctly, and calculate the correct escalation time for alerts. Service Manager uses time zone information for:

- Work schedules
- Alerts
- Request Management lead time calculations
- Service Level Agreements (SLAs) and service contracts
- Clocks
- Reminders

System administrators can define a system-wide default time zone and date format in the company record. Individual operator records can contain individual time zone values that override the system-wide default time zone.

Service Manager contains out-of-box time zone records for most regions of the world. System administrators can also create their own time zone records. Each time zone record contains:

- Local GMT offset
- Local time switch over definitions
- Local date format

Associated tables

Service Manager uses information in these tables to track time zone information.

- tzfile
- contact
- location
- device
- assignment
- cm3groups
- ocmgroups

Set the default system time zone

To set the default system time zone:

1. Click **System Administration** > **Base System Configuration** > **Miscellaneous** > **System Information Record**.
2. Click the **Date Info** tab.
3. In the **Time Zone** field, type or select the default system time zone.
4. Click **Save**.
5. Click **More** or the More Actions icon, and then select **Set Parms**.
6. Stop and restart the Service Manager server.

Add a time zone record

To add a time zone record:

1. Click **Tailoring** > **Database Manager**.
2. In the **Form** field, type `tzfile`.

3. Click **Search**.
4. Specify information for the following fields:
    - **Time Zone Name** identifies the time zone.
    - **Description** specifies the locations where the time zone applies.
    - **Default Date Format** is one of these date formats:
        - **mm/dd/yy** — standard U.S. date format
        - **dd/mm/yy** — standard European date format
        - **yy/mm/dd** — optional date format
        - **mm/dd/yyyy** — standard U.S. date format
        - **dd/mm/yyyy** — standard European date format
        - **yyyy/mm/dd** — optional date format

      The value you enter in this field overrides the default date format in the company record.
    - **Date Separator** is the character that separates the year, month, and date. Only "/","-", or "." is allowed.
    - **Time Change Definitions** defines every possible time offset that can occur within the time zone:
        - **Name (abbr)** is the abbreviation for a particular time offset.
        - **Offset from GMT** is the number of hours, minutes, and seconds the time zone is ahead or behind Greenwich Mean Time (GMT) in this format:
          plus or minushours:*minutes*:*seconds*.
          For example, +04:03:02 is ahead of Greenwich Mean Time by 4 hours, 3 minutes, and 2 seconds.
    - **Time Change Dates/Indexes** defines when time changes occur:
        - **Switchover Date** is the date and time when the offset begins.
          Type the date and time in this format: *date hour*:*minute*:*second*. In addition, the *date* must be in the *mm/dd/yy* format.
          For example, 04/15/04 02:00:00 switches over on April 15, 2004 at 2 AM.
        - **Index of Definition** is the index number of the time offset used on the switch over date.
          The index number matches the position the offset has in the Time Change Definitions. The first offset is index 0, the second offset is index 1, and so on.
5. Click **Add**.

Your user profile must include the system administration capability word to add this record.

Delete a time zone record

To add a time zone record:

1. Click **Tailoring** > **Database Manager**
2. In the **Form** field, type **tzfile**.
3. Click **Search**
4. Type or select optional search criteria.
5. Click **Search**.
6. Select the time zone to delete.
7. Click **Delete**.
8. Click **Yes** to confirm the deletion.

Update a time zone record

To update a time zone record:

1. Click **Tailoring** > **Database Manager**.
2. In the **Form** field, type `tzfile`.
3. Click **Search**.
4. Update one or more of these fields:
    - **Time Zone Name** identifies the time zone.
    - **Description** specifies the locations where the time zone applies.

- **Default Date Format** is one of these date formats:
  - **mm/dd/yy** — standard U.S. date format
  - **dd/mm/yy** — standard European date format
  - **yy/mm/dd** — optional date format
  - **mm/dd/yyyy** — standard U.S. date format
  - **dd/mm/yyyy** — standard European date format
  - **yyyy/mm/dd** — optional date format

  The value you enter in this field overrides the default date format in the company record.
- **Time Change Definitions** defines every possible time offset that can occur within the time zone:
  - **Name (abbr)** is the abbreviation for a particular time offset.
  - **Offset from GMT** is the number of hours, minutes, and seconds the time zone is ahead or behind Greenwich Mean Time (GMT) in this format:
    plus or minushours:*minutes*:*seconds*.
    For example, +04:03:02 is ahead of Greenwich Mean Time by 4 hours, 3 minutes, and 2 seconds.
- **Time Change Dates/Indexes** defines when time changes occur:
  - **Switchover Date** is the date and time when the offset begins.
    Type the date and time in this format: *date hour*:*minute*:*second*. In addition, the *date* must be in the *mm/dd/yy* format.
    For example, 04/15/04 02:00:00 switches over on April 15, 2004 at 2 AM.
  - **Index of Definition** is the index number of the time offset to use when it reaches the switch-over date. The index number matches the position the offset has in the Time Change Definitions. The first offset is index 0, the second offset is index 1, and so on.

5. Click **Save**.

Related topics

Regional settings
System information record

1.15. UTF-8 (Unicode) support

UTF-8 is part of the Unicode standard, which enables you to encode text in practically any script and language. Service Manager supports UTF-8 as the encoding method for new or existing data. It can support multiple languages that adhere to the Unicode standard on the same server.

Service Manager can also convert data to UTF-8 on-demand, thus eliminating system outages.

Prior to Unicode, languages were grouped into sets often referred to as Latin 1 (Western European language such as English, French, German), Latin 2 (Eastern European languages such as Czech, Polish, Slovak), Latin 5 (Turkish) and so on. Earlier ServiceCenter versions supported multiple languages only within a language group. Therefore, a single server instance could support French and German, English and Turkish, English and Japanese, but not Turkish and Czech, or Polish and Japanese. The ability to display data in disparate languages from a single server is attractive to any Service Manager user with an international customer base.

Consider the following point to ensure successful Unicode support:

- Ensure that your RDBMS is correctly configured for UTF-8 support. For more information, see your local Database Administrator.

2. System configuration parameters

The topics in this section describe the parameters that control your Service Manager system, and how to set them. There are two basic types of parameters: system parameters and web tier parameters.

- System parameters change the behavior of Service Manager. There are hundreds of system parameters, each of which is designed to assist system administrators and users.

- Web tier parameters control the behavior and appearance of the Service Manager web tier, a web-based client interface.

Parameters that you set when you start the Service Manager server retain their values for as long as the server is running. To reset a startup parameter, restart the server. You can set startup parameters from either the server's operating system command prompt or from the Service Manager initialization file (sm.ini).

Related topics

System parameters
Web tier parameters
Enter a parameter in the sm.ini file
Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Enter a parameter in sm.cfg or the server's OS command prompt
View parameters by function

2.1. Service Manager Server and Web Tier configuration files

The following table describes the configuration files of the Service Manager (SM) Server and Web Tier.

Component	Location	File	Description
Server	<SM installation path>/Server/RUN/	sm.ini	Service Manager Server configuration file. For more information, see Enter a parameter in the sm.ini file.
Server	<SM installation path>/Server/RUN/	sm.cfg	Service Manager Server configuration file. Parameters behave in the same way no matter whether they are specified in the server's operating system command prompt or in the sm.cfg file. For more information, see Enter a parameter in sm.cfg or the server's OS command prompt.
Server	<SM installation path>/Server/RUN/	lwssofmconf.xml	Service Manager Server LW-SSO configuration file. For more information, see Configure LW-SSO in the Service Manager server.
Web Tier	<Web Tier>/WEB-INF/	web.xml	Web Tier configuration file. For more information, see Web tier parameters.

Web Tier	• Default: <Web Tier>/WEB-INF/ • Custom: <customize-folder>/config/	webtier.properties	Web Tier configuration file. For more information, see Web tier parameters. **Important** Important The web.xml file defines lots of parameters (context-param of the web application and init-param of servlets), all of which except the **customize-folder** parameter can be overridden by the same parameters defined in the webtier.properties file. We strongly recommend that you configure your Web Tier using webtier.properties instead of web.xml to simplify future Web Tier upgrades.
Web Tier	<Web Tier>/WEB-INF/classes/	application-context.xml	Web Tier Single Sign-On configuration file (TSO, LW-SSO, and CAC)
Web Tier	<Web Tier>/WEB-INF/classes/	lwssofmconf.xml	Web Tier LW-SSO configuration file

Related topics

System configuration parameters
System parameters
Web tier parameters
Enter a parameter in the sm.ini file
Enter a parameter in the web.xml file
Enter a parameter in sm.cfg or the server's OS command prompt

2.2. System parameters

System parameters define either server or client properties.

- Other parameters don't have a value unless a system administrator sets one. For example, the *system* parameter determines the port that clients use to access the Service Manager server. The initial value for this parameter is set during installation and stored in the sm.ini file. If no value is set for the *system* parameter in the sm.ini file, then Service Manager clients don't know what port to connect to on the server.
- Some parameters affect the server and all the clients that connect to it, while others only affect a particular client or connection.
- Furthermore, a particular parameter can affect all connections or a particular connection, based on where it is set.

Parameters provide a method to handle each of the following requirements:

- In a production environment, the system administrator may want to prevent all client connections from printing or unloading data.
- In a production environment, a user may want to change the font within Service Manager.
- In a development or testing environment, the system administrator may want to grant all developers certain debugging capabilities.
- In a development or testing environment, an application developer may want additional debugging capabilities.

System parameters change the behavior of the Service Manager components. Parameters are typically designed to affect one of these components. The following table lists the components that can be influenced by system parameters.

Component	Method to set parameter value or default file location	Type of parameter
Service Manager server	• In the **C:\Program Files\OpenText\Service Manager <*x.xx*>\Server\RUN\sm.ini** file on the Service Manager server • By using the operating system command line	client and server
Windows client	• In the **C:\Program Files\OpenText\Service Manager <*x.xx*>\Server\RUN\sm.ini** file on the Service Managerserver • By using the operating system command line • In the Windows client, from the **Window** > **Preferences** menu	client
Web application server	In the **..webtier_<*x.xx*>/WEB-INF/web.xml** file on the web application server	web
Web client	• In the **..webtier_<*x.xx*>/WEB-INF/web.xml** file on the web application server • In the URL	web
Help Center	• In the Windows client, from the **Window** > **Preferences** menu • In the **..webtier_<*x.xx*>/WEB-INF/web.xml** file on the web application server	web

Because system parameters can be set from a variety of locations and by a variety of people, the values that are set may conflict. Therefore, it is important to know which parameter values take precedence when a conflict exists.

- Parameters that you set from the operating system command line or configuration file take precedence over parameters that are set in the sm.ini file on the server. You can use parameters from the command line or configuration file to override the standard sm.ini settings.
- Parameters that you set in the sm.ini file on the server take precedence over the preferences that are set in the Windows client.
- Parameters that you set in the sm.ini file on the server take precedence over the parameters that you set in the web.xml file on the web application server.
- Parameters you set in the URL of the web client take precedence over parameters that you set in the web.xml file on the web application server.

Getting help with parameters

Service Manager includes two sources of information about parameters. This Help Center is one of those sources. Additionally, you can open a command line on the Service Manager server, navigate to the **C:\Program Files\OpenText\Service Manager <*x.xx*>\Server\RUN** directory, and then run the **sm -helpall** command that is, an **sm**

command with the **helpall** parameter). Help provides detailed descriptions of each parameter and its use, while **helpall** provides a list of parameters and a brief description of each.

Related topics

System configuration parameters
Web tier parameters
Enter a parameter in the sm.ini file
Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Enter a parameter in sm.cfg or the server's OS command prompt
View parameters by function

2.3. Web tier parameters

Web parameters change the behavior of the web clients that connect to the web tier. You can set these parameters from the web tier configuration file web.xml or webtier.properties, and override some of these parameters (connection-specific parameters) from the web client login URL.

You can set web tier parameters from the following files:

- The web.xml and webtier.properties files that are provided under the <web tier>/WEB-INF folder when you deploy the Service Manager web tier.
- The custom webtier.properties file under the <customize-folder>/config folder which you defined using the `customize-folder` parameter.

To configure the `customize-folder` parameter, see Web parameter: customize-folder.

Override precedence

Except the customize-folder context parameter, you can define all other context parameters of the Web application and initialization parameters of Servlets from both the web.xml file and the webtier.properties files. The parameters configured in these files take precedence in the following order:

- Custom webtier.properties located in the <customize-folder>/config folder
- Default webtier.properties located in the <web tier>/WEB-INF folder
- Web.xml located in the <web tier>/WEB-INF folder

Parameters with higher precedence override parameters with lower precedence.

As parameters defined in custom webtier.properties (<customize-folder>/config/webtier.properties) take the highest precedence, the customize-folder parameter itself can't be overridden.

Note

Note It is recommended that you configure these parameters in the custom webtier.properties file for the ease of future webtier upgrade.

Naming convention for parameters in webtier.properties

The context parameters of the Web application and initialization parameters of the HP Service Manager Web servlet keep their names unchanged. Names for initialization parameters of other servlets should conform to the format of <servlet name>.<init-param-name>. For example:

- The name of the `jsDebug` context parameter in web.xml should be `jsDebug` in webtier.properties.
- The name of the `serverHost` initialization parameter of the `HP Service Manager Web` servlet should be `serverHost` in webtier.properties.
- The name of the `disableCompression` initialization parameter of the `FileDownload` servlet should be `FileDownload.disableCompression` parameter in webtier.properties.

Setting web tier parameters in webtier.properties

When you enter parameters into the webtier.properties file, make sure you configure the parameters in the format of <parameter name>=<parameter value>. For example, the following code sets the `rememberLang` parameter to true:

```
rememberLang=true
```

Setting web tier parameters in web.xml

When you enter parameters into the web.xml file, ensure that the values you enter conform to valid XML standards.

Note

Note All init-param tags must be children of the `<servlet>` tag and must use the following syntax:

```
<servlet>
  <init-param>
    <param-name>serverHost</param-name>
    <param-value>myhost.mydomain.com</param-value>
  </init-param>
<servlet>
```

To ensure that your web application server uses the new settings, restart your web application server whenever you make changes to the web.xml or the webtier.properties file.

Setting web tier parameters from the web browser URL

You can provide connection-specific web tier parameters in the web browser URL. These parameters override the values that are set in web.xml or webtier.properties. For example, the following URL changes the connection to host "mycompany" on port 13082 for this client session:

```
http://localhost:8080/sm/index.do?serverHost=mycompany.mydomain.com&serverPort=13082
```

Related topics

System configuration parameters
System parameters

2.4. Enter a parameter in the web client URL

Applies to User roles: System Administrator

You can enter web client parameters in the web client's URL. These parameters are only valid for the current web client session, and revert to the default values (as defined in the web.xml file) when the session closes.

To enter a parameter in the web client URL, type the parameter at the end of the URL after a question mark (?). To do this, use the following syntax:

```
http://servername:portnumber/sm/index.do?<parameter_name>=<parameter_value>
```

You can also use the ampersand (&) character to enter multiple parameters in a single URL. To do this, use the following syntax:

```
http://servername:portnumber/sm/index.do?<parameter_name>=<parameter_value>&<parameter_name>=<parameter_value>
```

Related topics

System configuration parameters
System parameters
Web tier parameters
Enter a parameter in the sm.ini file
Enter a parameter in the web.xml file
Enter a parameter in sm.cfg or the server's OS command prompt

View parameters by function
Client parameters for web clients

2.5. Enter a parameter in the sm.ini file

Applies to User roles: System Administrator

You can store parameters in the sm.ini file. To do this, follow these steps:

1. Open the sm.ini file. By default, the file is located on the Service Manager server in the **C:\Program Files\OpenText\Service Manager\Server\RUN** folder.
2. Enter the parameter and value in the file. Enter one parameter per line. Use a colon (:) to separate the parameter from the value. When you enter paths, enter them relative to the location of the Service Manager executable file. Use the following syntax:

   ```
   <parameter name>:<parameter value>
   ```

   The following excerpt is an example of parameters and their values in the sm.ini file.

   ```
   shared_memory:32000000
   ntservice: Service Manager
   log:..\logs\sm.log
   system:13080
   ```

Related topics

System configuration parameters
System parameters
Web tier parameters
Enter a parameter in the web.xml file
Enter a parameter in sm.cfg or the server's OS command prompt
View parameters by function

2.6. Enter a parameter in the web.xml file

Applies to User roles: System Administrator

You can store web parameters in the web.xml file. When you do this, make sure that you use valid XML code. Specifically, parameter tags must be children of servlet tags.

To enter a parameter in the web.xml file, follow these steps:

1. Open the web.xml file. By default, the file is located on the web application server in the ..sc/WEB-INF/web.xml directory.
2. Type the parameter information. For example, the following XML code sets value of the *serverHost* parameter to "localhost":

   ```
   <servlet>
     <init-param>
       <param-name>serverHost</param-name>
       <param-value>localhost</param-value>
     </init-param>
   <servlet>
   ```

Related topics

System configuration parameters
System parameters
Web tier parameters

Enter a parameter in the sm.ini file
Enter a parameter in the web client URL
Enter a parameter in sm.cfg or the server's OS command prompt
View parameters by function

2.7. Enter a parameter in the webtier.properties files

Applies to User roles: System Administrator

You can store web parameters in the webtier.properties files. When you do this, make sure you configure the parameters in the format of <parameter name>=<parameter value>.

To enter a parameter in webtier.properties, follow these steps:

1. Open the webtier.properties file where you want to enter the parameter:
   - Custom webtier.properties located in the <customize-folder>/config folder
   - Default webtier.properties located in the <webtier .war>/WEB-INF folder
2. Type the parameter information. For example, the following code sets the value of the `rememberLang` parameter to true:

   ```
   rememberLang=true
   ```

Related topics

System configuration parameters
System parameters
Web tier parameters
Web parameter: customize-folder
Enter a parameter in the sm.ini file
Enter a parameter in the web.xml file
Enter a parameter in the web client URL
Enter a parameter in sm.cfg or the server's OS command prompt
View parameters by function

2.8. Enter a parameter in sm.cfg or the server's OS command prompt

Applies to User roles: System Administrator

You can enter client and server parameters in the server's sm.cfg file or operating system command line. When you do this, prefix each parameter with a space and a dash (-).

To enter a parameter in sm.cfg or the server's operating system command prompt, follow these steps:

1. Open the operating system command prompt, and then navigate to the directory in which the Service Manager executable file is located. By default, the file is located in the **`C:\Program Files\OpenText\Service Manager <x.xx>\Server\RUN`** folder.
   Alternatively, open the <Service Manager installation directory>\Server\RUN\sm.cfg file in a text editor.
2. Type the command and the parameters, as demonstrated in the following examples:
   - One parameter
     sm -<parameter_name>:<parameter_value>
   - Multiple parameters
     sm -<*parameter_name*>:<*parameter_value*> -<*parameter_name*>:<*parameter_value*>

Setting startup parameters from the server's operating system command prompt

Startup parameters that you set by using the server's operating system command prompt must always follow the **sm** command. Precede startup parameters with a dash (-) when you call them from the command line.

When you call a startup parameter from the server's operating system command prompt, use the following syntax:

```
C:\<Service Manager installation directory>\RUN>sm -<start up parameter>:<setting>
```

For example, you type the following command:

```
<Service Manager installation directory>\Server\RUN>sm -httpPort:12345
```

Related topics

System configuration parameters
System parameters
Web tier parameters
Enter a parameter in the sm.ini file
Enter a parameter in the web client URL
Enter a parameter in the web.xml file
View parameters by function

2.9. View parameters by function

Applies to User roles: System Administrator

To view parameters arranged by function, follow these steps:

1. Open an operating system command prompt, and navigate to the Service Manager `RUN` folder. For example, type the following command and then press Enter.
   cd C:\Program Files\OpenText\Service Manager <*x.xx*>\Server\RUN
2. Type the following command, and then press Enter:

   ```
   sm -help<type>
   ```

   For example, to view SSL parameters, type the following command and then press Enter:

   ```
   sm -helpssl
   ```

Related topics

- System configuration parameters
- System parameters
- Web tier parameters
- Enter a parameter in the sm.ini file
- Enter a parameter in the web client URL
- Enter a parameter in the web.xml file
- Enter a parameter in sm.cfg or the server's OS command prompt

2.10. Alert parameters

The following table lists the startup parameters that you can set from the Service Manager server's operating system command prompt or from the Service Manager initialization file (sm.ini).

These parameters determine how the server handles alert messages.

Startup parameter	Description
Startup parameter: alertcpulimit	Defines the number of standard deviations from the mean CPU usage that cause the Service Manager server to issue an alert message
Startup parameter: alertfilters	Defines the names of alerts that you want the Service Manager server to filter out of alert messages
Startup parameter: alerthitratio	Defines the percentage of records inspected to records selected that cause the Service Manager server issues an alert message
Startup parameter: alertholdlimit	Defines the number of milliseconds that a process can hold a lock on a system resource before the Service Manager server issues an alert message
Startup parameter: alertquerylimit	Defines the number of milliseconds that a query can run before the Service Manager server issues an alert message
Startup parameter: alertvirtuallimit	Defines the number of kilobytes of virtual memory used that cause the Service Manager server to issue an alert message
Startup parameter: alertwaitlimit	Defines the number of milliseconds that a process can wait for a system resource before the Service Manager server issues an alert message

Related topics

System parameters
Enter a parameter in the sm.ini file

2.10.1. Startup parameter: alertcpulimit

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

alertcpulimit

Description

This parameter defines the number of standard deviations from the mean CPU usage that cause the Service Manager server to issue an alert message.

For example, suppose that a customer reports that they have set the alertperoidicschedule record to expire every 5 minutes. This means that every 5 minutes we look at the amount of CPU consumed by each process in the last 5 minutes. We take all the values and we get an average. Given all the values that make the average there is a statistical calculation called the standard deviation that returns a number that represents how far from the average each deviation is. If all of the users used between 5 and 7 cpu seconds then the standard deviation would be really small. If all the users used between 5 and 100 cpu seconds (assume the values were spread out across this range) then the standard deviation would be larger. So in the last case lets assume the average was 50 and the standard deviation was 3, then we would report on any user who used more than 65 seconds (50 + 5*3).

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

5

Possible values

Number of standard deviations

Example usage

Command line: **sm -httpPort:13080 -alertcpulimit:2**

Initialization file: `alertcpulimit:2`

Related topics

System parameters
Enter a parameter in the sm.ini file
Alert parameters

2.10.2. Startup parameter: alertfilters

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

alertfilters

Description

This parameter defines the names of alerts that you want the Service Manager server to filter out of alert messages. The Service Manager server no longer posts alert messages from alert categories you filter.

You can use an asterisk (*) at the end of the alert name to specify all alerts within that category. For example, to filter out all performance alerts, type: *alertfilters:Performance**.

You can use a semicolon to list additional alert names. For example, to specify all mapping alerts and a long running query alert against the contacts file, type: *alertfilters:Mapping*;Performance-5-contacts*.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

None

Possible values

Names of alerts

Example usage

Command line: **sm -httpPort:13080 -alertfilters:Mapping*;Performance-5-contacts**

Initialization file: `alertfilters:Mapping*;Performance-5-contacts`

Related topics

System parameters
Enter a parameter in the sm.ini file
Alert parameters

2.10.3. Startup parameter: alerthitratio

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

alerthitratio

Description

This parameter defines the percentage of records inspected to records selected that cause the Service Manager server issues an alert message. The lower hit ratio percentage you define the more inefficient queries you allow.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

None

Possible values

Percentage of query records inspected to records selected

Example usage

Command line: **sm -httpPort:13080 -alerthitratio:10**

Initialization file: `alerthitratio:10`

Related topics

System parameters
Enter a parameter in the sm.ini file
Alert parameters

2.10.4. Startup parameter: alertholdlimit

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

alertholdlimit

Description

This parameter defines the number of milliseconds that a process can hold a lock on a system resource before the Service Manager server issues an alert message. You can use this parameter in conjunction with the *alertwaitlimit* parameter to determine how the Service Manager server manages locks on system resources.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

None

Possible values

Number of milliseconds

Example usage

Command line: **sm -httpPort:13080 -alertholdlimit:1500**

Initialization file: `alertholdlimit:1500`

Related topics

System parameters
Enter a parameter in the sm.ini file
Alert parameters

2.10.5. Startup parameter: alertquerylimit

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

alertquerylimit

Description

This parameter defines the number of milliseconds that a query can run before the Service Manager server issues an alert message.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

None

Possible values

Number of milliseconds

Example usage

Command line: **sm -httpPort:13080 -alertquerylimit:3000**

Initialization file: `alertquerylimit:3000`

Related topics

System parameters
Enter a parameter in the sm.ini file
Alert parameters

2.10.6. Startup parameter: alertvirtuallimit

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

alertvirtuallimit

Description

This parameter defines the number of kilobytes of virtual memory used that cause the Service Manager server to issue an alert message.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

None

Possible values

Number of kilobytes of virtual memory

Example usage

Command line: **sm -httpPort:13080 -alertvirtuallimit:1024**

Initialization file: `alertvirtuallimit:1024`

Related topics

System parameters
Enter a parameter in the sm.ini file
Alert parameters

2.10.7. Startup parameter: alertwaitlimit

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

alertwaitlimit

Description

This parameter defines the number of milliseconds that a process can wait for a system resource before the Service Manager server issues an alert message. You can use this parameter in conjunction with the *alertholdlimit* parameter to determine how the Service Manager server manages locks on system resources.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

None

Possible values

Number of milliseconds

Example usage

Command line: **sm -httpPort:13080 -alertwaitlimit:1500**

Initialization file: `alertwaitlimit:1500`

Related topics

System parameters
Enter a parameter in the sm.ini file
Alert parameters

2.11. Application development parameters

The following table lists the startup parameters that you can set from the Service Manager server's operating system command prompt or from the Service Manager initialization file (sm.ini).

The following parameter sets a feature in the application development environment.

Startup parameter	Description
Startup parameter: agstackl	Defines the length of the stack that the Service Manager server allocates to run RAD applications

Related topics

System parameters
Enter a parameter in the sm.ini file

2.11.1. Startup parameter: agstackl

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

agstackl

Description

This parameter defines the length of the stack the Service Manager server allocates to run RAD applications. Changing the value of this parameter has a small impact on the memory allocation of the session and the process.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

1000

Possible values

Stack length from 1-2000

Example usage

Command line: **sm -httpPort:13080 -agstackl:1000**

Initialization file: `agstackl:1000`

Related topics

System parameters
Enter a parameter in the sm.ini file
Application development parameters

2.12. Background and scheduled process parameters

The following table lists the startup parameters that you can set from the Service Manager server's operating system command prompt or from the Service Manager initialization file (sm.ini).

These parameters set features for background and scheduled processes.

Startup parameter	Brief description
Special parameters: bg	Causes the Service Manager server to mark all background process as SYSTEM processes
Parameter: deadnodelocktimeout	Specifies the amount of time that must elapse before a process forcibly removes a lock from the Lock or LockShared table
Special parameters: sync	Starts the synchronization process, which identifies and removes locks owned by inactive processes and shared memory
Special parameters: synctime	Defines the number of seconds the sync process waits before identifying and removing locks owned by defunct processes and shared memory

Related topics

System parameters
Enter a parameter in the sm.ini file

2.12.1. Special parameters: bg

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

bg

Description

This parameter causes the Service Manager Server to run RAD processes in background mode (that is, no user interactions or messages are available).

This parameter affects three aspects in the RTE: the DBI (Database Interface) layer, the APPs code, and message sending.

- In the DBI layer, once this parameter is set, the following db operations will not trigger any error messages to the client any more:
  - delete:
    If DBI is trying to delete a record that was already deleted by others, the message "The record being deleted was already deleted" is not sent.
    If DBI is trying to delete a record that was already modified by others, the message "The record being deleted has been modified since read" is not sent.
  - dbUpdate:
    If DBI is trying to update a record that was already deleted by others, the message "The record being updated has been deleted" is not sent.
    If DBI is trying to update a record that was modified by others, the message "The record being updated has been modified since read" is not sent.
- When "bg" is set, the following APPs functions will fail with an error as described below:
  - fdisp: "FDISP panels can't be executed in the background."
  - rio: "RIO panels can't be executed in the background."
  - wopen: "WOPEN panels can't be executed in the background."
  - wclose: "WCLOSE panels can't be executed in the background."
  - wselect: "WSELECT panels can't be executed in the background."
- Once "bg" is set, all messages generated by RTE code and APPs code (via APPs function "scmsg") aren't sent to the client; instead, they are sent to the log files defined by the parameter "log" (by default, it is sm.log).

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

None

Possible values

None

Example usage

Command line: **sm -httpPort:13080 -bg**

Initialization file: `bg`

Related topics

System parameters
Enter a parameter in the sm.ini file
Background and scheduled process parameters

2.12.2. Special parameters: sync

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt. You can set a startup parameter from the server OS command prompt on the server machine.

Parameter

sync

Description

This parameter starts the sync process, which identifies and removes locks owned by inactive processes and shared memory. The sync process performs the following tasks:

- Checks for and removes processes that are no longer active.
- Checks for and release semaphores that the server has held for more than 60 seconds.
- Checks for and removes cached items in shared memory that are no longer in use.

The sync process checks Service Manager shared memory for processes that no longer exist.

When a Service Manager process starts, it stores data structures in shared memory indicating the process name and PID. You can see this information using the system status utility. When a process terminates, the information is removed.

The sync process checks the user chain for processes that are still active but no longer exist in the operating system. The process takes the PID and queries the operating system to see if this process still exists. If it doesn't, it terminated unexpectedly.

The sync process cleans all outstanding resource locks and semaphore locks and removes the processes from the user thread. By doing so, semaphores and locks become available again and other processes waiting for the resources begin to respond.

Note

Note You must run the sm -sync command on every host in the clustering environment in order to clean user sessions. Thread level detection only occurs on the local host and each user runs as a thread. With no sync processing running, the local host is unable to terminate the expired user session.

Valid if set from

The server's operating system command prompt

Requires restart of the Service Manager server?

No

Default value

None

Possible values

None

Example usage

Command line: **sm -sync**

Related topics

System parameters
Enter a parameter in the sm.ini file
Background and scheduled process parameters

2.12.3. Special parameters: synctime

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

synctime

Description

This parameter defines the number of seconds the sync process waits before identifying and removing locks owned by defunct processes and shared memory.

Valid if set from

Server's operating system command prompt

Requires restart of the Service Manager server?

No

Default value

1200 seconds

Possible values

Number of seconds

Example usage

Command line: **sm -synctime:1300**

Related topics

System parameters
Enter a parameter in the sm.ini file
Background and scheduled process parameters

2.12.4. Parameter: deadnodelocktimeout

Parameter

deadnodelocktimeout

Description

This parameter specifies the amount of time that must elapse before a process forcibly removes a lock from the Lock or LockShared table. By default, this parameter is set to 10, which indicates that 10 minutes must elapse before a record is forcibly removed. 10 minutes is also the minimum value for this parameter.

Valid if set from

Initialization file (sm.ini)

Requires restart of server?

No

Default value

10 (minutes)

Possible values

>=10 (minutes); 10 minutes is also the minimum value.

Example usage

Initialization file: `deadnodelocktimeout:10`

Related topics

System parameters
Lock management

2.13. Client parameters for web clients

These parameters override web client preferences. Any client parameter that you set on the server overrides the local client preference and prevents users from changing it. You can set some of these parameters from the web client login URL, some from the web tier configuration file (web.xml), and some from the webtier.properties file. In the following table, a check mark (√) indicates that you can set the parameter from the URL or web.xml file, and an em dash (-) indicates the opposite.

Web parameter	URL	web.xml	webtier.properties	Description
Web parameter: acceptBrowserLang	-	√	√	Specifies if the browser's language preference is used for trusted sign-on login when the *lang* parameter is not set in the web client login URL
Web parameter: allowed	-	√	-	Specifies the allowed list of file extensions that may be attached to a Service Manager record, such as an Incident record.
Web parameter: antiCSRFEnabled	-	√	√	Enables tokenization for the web client
Web parameter: applicationIcon	-	√	√	Enables you to modify the company's logo to display a custom company icon in the URL address bar
Web parameter: autoCloseMaskDelay	-	√	√	Defines the delay in seconds before the web tier mask is automatically closed
Web parameter: autoCompleteDelayTime	-	√	√	Defines the delay time in milliseconds to trigger auto complete after the user stops typing characters
Web parameter: autoCompleteListSize	-	√	√	Indicates the record number size fetched from the server side at the first time
Web parameter: autoCompleteMinChars	-	√	√	Indicates the number of characters that are required to trigger the auto complete feature
Web parameter: combo.autoCompleteMode	-	-	√	Defines the Auto Complete mode in Combo Box fields.

Web parameter: autoCompleteSkipCachingChar	-	√	√	The previous response will not be cached if this character is included in the user input
Web parameter: autoSubmit	√	√	√	Enables users to submit forms when they press the Enter key
Web parameter: cacerts	-	√	√	Defines the path to the `cacerts` file required for Web clients to make SSL connections to the Web tier
Web parameter: CACLogin	-	√	√	Enabling this parameter causes all Web clients to present a Common Access Card (CAC) certificate as authentication information and use SSL connections to the Web tier.
Web parameter: comfillAutoComplete	-	√	√	Enables the auto complete for the comfill widgets globally
Web parameter: compactLayout	-	√	√	Enables the compact layout of the web client
Web parameter: compress_soap	-	√	√	Requires the Web tier to compress SOAP messages to Web clients
Web parameter: cookie.httpOnly	-	-	√	Defines whether to set the HttpOnly flag for HTTP cookies (such as JSESSIONID, mode, and lang).
Web parameter: cookie.secure	-	-	√	Defines whether to set the Secure flag for HTTP cookies (such as JSESSIONID, mode, and lang).

Web parameter: customize-folder	-	√	√	Defines the location to save the branding configurations
Web parameter: detailViewMinWidth	√	-	-	Increases the min-width value of a web page if the system displays some texts incompletely after you enlarge the browser text size or font size.
Web parameter: disableJumpAddress	-	√	√	Enables use of the Quick Jump component on the toolbar
Web parameter: disableKeyHelp	-	√	√	Allows to disable the contextual help.
Web parameter: disableLoginAutoComplete	-	√	√	Controls password auto-complete for the web client login page.
Web parameter: enableExitConfirmMessage	-	√	√	Displays a confirmation message when users click the Close (x) button of the SM web client page or manually refresh the entire web client page.
Web parameter: enableGetOriginalUrl	-	√	√	Enables the web tier to obtain the original URL from the browser before a user logs in (in TSO or LWSSO mode).
Web parameter: enableListFrameStateRetain	-	√	√	Enables the list detail page to retain the expand-collapse state of the List Pane when the list detail page refreshes.

Web parameter: enableMoreContextMenu	-	√	√	Enables a contextual "More" menu that's displayed when you right-click lists on the details pane of a record in the web client.
Web parameter: enableRecordlistOddEvenRowStyle	-	√	√	This parameter determines if the odd rows and the even rows in the record list have the same background color or not.
Web parameter: enableSidebarMenu	-	√	√	This parameter specifies if the System Navigator of the web client can use the sidebar mode and icons.
Web parameter: enableTableMagnify	-	√	√	Specifies whether the **Magnify** button appears for editable tables in the web client.
Web parameter: enableTableEllipsis	-	√	√	For certain read-only controls in record details area, you can enable this parameter to display '...' at the end of the text fields when the input value is too long to be shown completely.
Web parameter: encryptPasswordInBrowser	-	-	√	Determines whether to encrypt the user password in the web browser before the system sends it to the web tier.
Web parameter: expandCommandLine	-	-	√	Specifies whether the command/search line is expanded by default or not.
Web parameter: expandSmartSearch	-	-	√	Specifies whether the Smart Search field is expanded by default or not.

Web parameter: externalForgotPasswordUrl	-	√	√	Defines an external forgot password URL for the Forgot Password link on the Service Manager web client login page.
Web parameter: helpServerContext	-	√	√	Defines the context path when deploying the Service Manager help on a web server (for example, Apache)
Web parameter: helpServerHost	√	√	√	Defines the name of the Help server host listening for client requests
Web parameter: helpServerPort	√	√	√	Defines the communications port number on which the Help server listens to client requests
Web parameter: honorUrlHost	-	√	√	Determines whether Web client users can specify the name of the Service Manager server from the URL
Web parameter: honorUrlPort	-	√	√	Determines whether Web client users can specify the communications port of the Service Manager server from the URL
Web parameter: isCustomAuthenticationUsed	-	√	√	Determines how web client users are authenticated
Web parameter: JCEProviderClassName	-	√	√	The class name of a third-party JCE provider that you want to use when the web tier is running in non-FIPS mode.
Web parameter: JCEProviderName	-	√	√	The name of a third-party JCE provider that you want to use when the web tier is running in non-FIPS mode.

Web parameter: jsDebug	-	√	√	Enables the web client to display to end users an error message that includes full stack trace.
Web parameter: lockWaitTimeout	-	√	√	Specifies the time after which a Web tier application server thread will cease trying to acquire a lock
Web parameter: log4jConfiguration	-	√	√	Defines the path to the Web tier's logging properties file
Web parameter: labelbgunderline	-	√	√	Defines the form-level label background display if a color other than white was selected but the Background Style property was left blank
Web parameter: maskMessageDelay	-	√	√	Specifies the number of seconds before the Web tier mask message is displayed after the mask is shown.
Web parameter: maxattachuploadsize	-	√	√	Specifies the maximum size limit for individual attachments in a record.
Web parameter: maxtotalattachuploadsize	-	√	√	Specifies the maximum total size of all the attachments in a record.
Web parameter: maxGroupCacheSize	-	√	√	Defines the maximum number of records that can be cached in the user's web browser when the user navigates through the groups in a record list

Web parameter: maxMessageNumber	-	-	√	Defines the maximum number of messages displayed in Messages And Alerts.
Web parameter: message display properties	-	√	√	Indicate how different types of messages are displayed
Web parameter: openTabsSoftLimit	-	√	√	Defines the maximum number of tabs a user can open in a browser before a warning message displays to advise them to close some tabs.
Web parameter: preventDuplicatedAttachmentName	-	√	√	This parameter controls whether to prevent an attachment that has the same file name with existing ones from being uploaded to Service Manager.
Web parameter: querySecurity	-	√	√	Embeds a security key in all queries generated by the Web client
Web parameter: recordListArrayDisplayStyle	-	√	√	Defines how array column values are displayed in record lists
Web parameter: recordListArrayMaxChars	-	√	√	Defines the maximum allowed field length of any array fields that appear in record lists
Web parameter: refreshMessages	√	√	√	Enables the Web client to query the Service Manager server for messages at a regular interval

Web parameter: refreshMessagesInterval	√	√	√	Defines the interval the Web client waits before checking for messages on the Service Manager server
Web parameter: rememberLang	-	-	√	Enables the Web client to remember the browser's language preference
Web parameter: searchMinChars	-	-	√	Defines the minimum number of characters you must type in the search field to trigger a System Navigator search.
Web parameter: secureLogin	-	√	√	Controls the encryption of network communication between the Web application server and the Web browser
Web parameter: serverHost	√	√	√	Defines the name of the Service Manager host listening for client requests
Web parameter: serverPort	√	√	√	Defines the communications port number on which the Service Manager server listens to client requests
Web parameter: session-timeout	-	√	√	Defines the number of minutes that the server waits for a client heartbeat signal before the server assumes that the client session has timed out and closes the connection
Web parameter: showAddFavButton	-	√	√	Enables use of the Add Favorite button to add favorites in Favorites and Dashboards

Web parameter: showFavMgrButton	-	√	√	Enables use of the Favorite Manager button to manage favorites in Favorites and Dashboards
Web parameter: showHelp	-	√	√	Displays the **Online Help** option when you click the **Help** button, which allows you to access the help server
Web parameter: showReloadNavButton	-	√	√	Enables use of the Reload button on the navigator toolbar. The Reload button refreshes the list of favorites in Favorites and Dashboards
Web parameter: sideBarButtonNumberToCollapse	-	√	√	Defines the number of side bar buttons to be collapsed.
Web parameter: socketReadTimeout	-	√	√	Specifies the time that a thread on the Web tier application server will wait for a socket IO operation.
Web parameter: sortNavItems	-	√	√	Defines the sort order of the System Navigator menu items.
Web parameter: sortNavFavorites	-	√	-	Defines the sort order of the favorites.
Web parameter: ssl	-	√	√	Requires all Web clients to use SSL connections to the Web tier.
Web parameter: sslPort	-	√	√	Defines the SSL port of the Web application server; required only when secureLogin is set to true.
Web parameter: sso.failure.httpStatus	-		√	Specifies the HTTP status code to be returned when TSO, LW-SSO, IdM or CAC fails.

Web parameter: startDayOfWeek	-	√	√	Defines the first day of the week in the date picker.
Web parameter: tooltipAutoHide	√	√	√	Controls whether to automatically hide a tooltip after the pointer leaves the target element. Only the tooltip that is triggered by pointer hover and is not controlled by the browser (that is, not the tooltip by the title attribute) is controlled by this parameter.
Web parameter: tracesoap	√	√	√	Logs SOAP messages in the Web server's sm.log file (< Web server home directory>/sm.log)
Web parameter: unmaskDeferMillis	-	√	√	Defers the unmasking of a browser window after an unmask request is issued.
Web parameter: useJavaDynamicVersioning	-	√	√	Enables use of the system registry to locate the latest version of Java installed on a user's system.
Web parameter: useservertabs	√	√	√	Displays tabs in the order the server has them saved on the form
Web parameter: viewactivenotes	√	√	√	Displays messages in a message box as well as in the status bar
Web parameter: viewcontexthelp	√	√	√	Displays the same context-sensitive help debug information (field) in field help as the Windows client does.

Web parameter: viewpromptforsave	√	√	√	Displays a warning message when users cancel records with unsaved changes
Web parameter: viewrecordlist	√	√	√	Displays the record list/detail view for search results
Web parameter: virtualGroupSize	-	√	√	Defines the number of records that are displayed as one virtual group in inbox views.
Web parameter: webtierFullUrl	-	-	√	Describes the full URL used to access the web tier.

Related topics

System parameters
Web tier parameters
Enter a parameter in the web client URL
Enter a parameter in the web.xml file

2.13.1. Web parameter: acceptBrowserLang

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

acceptBrowserLang

Description

This parameter specifies if the browser's language preference is used for single sign-on (including TSO, SAML SSO, and LW-SSO) and CAC Sign-On when the *lang* parameter is not set in the web client login URL. By default, the browser's language preference has a higher priority than the language defined in the operator's profile. If you want to use the operator's profile language setting instead of the browser's language preference in this case, set this parameter to false.

> **Note**
>
> Note If the user includes the "lang" parameter in the Web client URL, the URL language setting take precedence over the acceptBrowserLang setting.

Valid if set from

Web tier configuration file (web.xml and webtier.properties)

Requires restart of the web applications server?

Yes

Default value

true

Possible values

true (enabled)

false (disabled)

Example usage

```
<init-param>
<param-name>acceptBrowserLang</param-name>
<param-value>true</param-value>
</init-param>
```

Related topics

Client parameters for web clients

2.13.2. Web parameter: allowed

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

allowed

Description

During attachment file processing on the Service Manager Web tier client, it is possible to configure a white list (allowed list) of file extensions that may be uploaded by users. This parameter specifies the allowed list of file extensions that may be attached to a Service Manager record, for example, an Incident record.

Note

Note This parameter only takes effect when the server-side attachment allowlist is disabled. However, it is a recommended best practice to enable the server-side attachment allowlist.

This parameter exists in the following servlets in the web tier configuration file (web.xml) file:

- File upload servlet
- Image upload servlet
- Attachment upload servlet
- Attachment upload servlet for AJAX request
- Attachment upload servlet for HTML editor AJAX request
- Unique attachment upload servlet

Valid if set from

Web tier configuration file (web.xml)

Requires restart of the web applications server?

Yes

Default value

`bmp,jpg,jpeg,png,gif,txt,log,unl,csv`

Note

Note The default value of the Image upload servlet is a subset of this list.

Possible values

A list of comma-separated file extensions

- To deny all file extensions, set the value to empty.
- To allow all file extensions, set the value to * or remove this parameter.

Example usage

The following example allows only the listed file types. As a result, other file types (for example, JSP files) are disallowed.

```
<init-param>
        <param-name>allowed</param-name>
        <param-
value>bmp,jpg,jpeg,png,gif,doc,xls,rtf,txt,docx,xlsx,ppt,pptx,pdf,msg,zip,tar,gz,tgz</param-value>
</init-param>
```

Related topics

System parameters
Web tier parameters
Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Client parameters for web clients

2.13.3. Web parameter: antiCSRFEnabled

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

antiCSRFEnabled

Description

This parameter enables tokenization for the web client. By default, this parameter is enabled.

Valid if set from

Web tier configuration file (web.xml and webtier.properties)

Requires restart of the web applications server?

Yes

Default value

true

Possible values

true or false

Example usage

```
<init-param>
    <param-name>antiCSRFEnabled</param-name>
    <param-value>true</param-value>
</init-param>
```

Related topics

Tokenization
Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Client parameters for web clients

2.13.4. Web parameter: applicationIcon

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

applicationIcon

Description

By default, the Service Manager application displays the company logo icon. This parameter enables you to modify the company logo to display a custom company icon in the URL address bar.

Using the ICO file format for icons, you can publish an icon that contains at least one 16x16 bitmap with a bit depth of 8, 16, 24, or 32. You can provide a relative URL (without the context root) and save it in your Favorites.

When not specified in this parameter, the default icon displayed in the Service Manager application will be the company logo icon.

Valid if set from

Web tier configuration file (web.xml and webtier.properties)

Requires restart of the web applications server?

Refer to your web server documentation

Default value

/cwc/images/HPLogoSolidBlue.ico

Possible values

Relative path to the company icon file

Example usage

```
<init-param>
  <param-name>applicationIcon</param-name>
  <param-value>/cwc/images/MyCompanyIcon.ico</param-value>
</init-param>
```

In this example, *<MyCompanyIcon.ico>* represents your company logo image.

Related topics

System parameters
Web tier parameters
Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Client parameters for web clients

2.13.5. Web parameter: autoCloseMaskDelay

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

autoCloseMaskDelay

Description

This parameter specifies the delay in seconds before which the mask is automatically closed. This parameter value must be at least 60 seconds. If no value is specified, the default value (180 seconds) is used.

Valid if set from

Web tier configuration file (web.xml and webtier.properties)

Requires restart of the web applications server?

Yes

Default value

180 seconds

Possible values

60-300 seconds

Example usage

```
<context-param>
<param-name>autoCloseMaskDelay</param-name>
<param-value>180</param-value>
</context-param>
```

Related topics

System parameters
Web tier parameters
Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Client parameters for web clients
SSL parameters

2.13.6. Web parameter: autoCompleteDelayTime

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

autoCompleteDelayTime

Description

This parameter defines the delay time in milliseconds to trigger the auto complete feature after you stop typing characters in a Comfill field.

Valid if set from

Web tier configuration file (web.xml and webtier.properties)

Requires restart of the web application server?

Yes

Default value

200

Possible values

0 to unlimited

Note

Note If you specify an invalid value, the default value is used.

Example usage

```
<context-param>
<param-name>autoCompleteDelayTime</param-name>
<param-value>200</param-value>
</context-param>
```

Related topics

System parameters
Web tier parameters
Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Client parameters for web clients
Web parameter: comfillAutoComplete

Web parameter: autoCompleteListSize
Web parameter: autoCompleteMinChars

2.13.7. Web parameter: autoCompleteListSize

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

autoCompleteListSize

Description

This parameter defines the maximum number of records that are fetched from the server side at one time and returned to an auto complete selection list. If the total number of matching records is larger than this specified list size, a **More** icon displays at the bottom of the list. Clicking this icon will load more records to the list.

For example, if this parameter is set to 10 (default) and the total number of matching records is 100, the auto complete selection list displays 10 matching records initially with a **More** icon the bottom. If you click the **More** icon, another 10 records are loaded to the list. Clicking **More** repeatedly will load the rest of the records.

Valid if set from

Web tier configuration file (web.xml and webtier.properties)

Requires restart of the web application server?

Yes

Default value

10

Possible values

1 to 100

Note

Note If you specify an invalid value, the default value is used.

Example usage

```
<context-param>
<param-name>autoCompleteListSize</param-name>
<param-value>10</param-value>
</context-param>
```

Related topics

System parameters
Web tier parameters
Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Client parameters for web clients
Web parameter: comfillAutoComplete
Web parameter: autoCompleteDelayTime
Web parameter: autoCompleteMinChars

2.13.8. Web parameter: autoCompleteMinChars

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

autoCompleteMinChars

Description

This parameter defines the minimum number of characters you must type in a Comfill widget to trigger the auto complete feature.

Valid if set from

Web tier configuration file (web.xml and webtier.properties)

Requires restart of the web application server?

Yes

Default value

3

Possible values

1 or greater

Note

Note If you specify a value of zero or a negative number, the default value is used.

Example usage

```
<context-param>
  <param-name>autoCompleteMinChars</param-name>
  <param-value>3</param-value>
</context-param>
```

Related topics

Client parameters for web clients
Web parameter: comfillAutoComplete
Web parameter: autoCompleteDelayTime
Web parameter: autoCompleteListSize

2.13.9. Web parameter: combo.autoCompleteMode

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

combo.autoCompleteMode

Description

This parameter defines the Auto Complete mode for drop-down fields. In addition to the combo widgets, this parameter is also applied to other combo-like widgets, such as the Tags widgets and the smart search filter option fields.

- startsWith (default) - The drop-down list is filtered for items that begin with the search string.
- contains - The drop-down list is filtered for items that contain the search string.

Note

Note
For the select-only fields:
In the `startsWith` mode, the field value is auto-selected with the first corresponding item.
In the `contains` mode, the first item in the list will only be auto-selected if the item also begins with the search string.

Valid if set from

Web tier configuration file (webtier.properties)

Requires restart of the web application server?

Yes

Default value

startsWith

Possible values

- startsWith
- contains

Example usage

```
combo.autoCompleteMode=contains
```

2.13.10. Web parameter: autoCompleteSkipCachingChar

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

autoCompleteSkipCachingChar

Description

A Comfill field's link record may define query expressions that split the Comfill field's values to multiple levels, using a special character (such as a "/") as the separator. In this case, you must specify the separator in this parameter, so that the Auto Complete feature returns the same query results as the **Fill** operation when the user input contains this separator character.

The following table shows an example of such Comfill fields.

Comfill field	Field value	Level 0 value (Company)	Level 1 value (Location)	Separator
Location	advantage/Africa	advantage	Africa	/
Location	advantage/Asia	advantage	Asia	/

In this example, the Link record of the Location field uses the following expressions, where "/" is used as a separator:

```
$find.skip=true
```

```
if (index("/", location.full.name in $File)=0) then ($ln.query="location.full.name#\""+location.full.name in $File+"\"";$ln.query=nullsub($ln.query, "true")+" and level=0")
```

```
if (index("/", location.full.name in $File)>0) then ($L.tempx=0;for $L.ix = 1 to lng(location.full.name in $File) do (if (substr(location.full.name in $File, $L.ix, 1)="/") then $L.tempx+=1);$ln.query="location.full.name#location.full.name in $File and level="+$L.tempx)
```

Note

Note If the Link records of your Comfill fields use a different character as the separator, set this parameter to that character (for example, "|"). This is a global setting, which means all Comfill fields that have Auto Complete enabled have to use the same separator.

If a Comfill field allows values that contain the specified character (for example, "E-mail / Webmail (Africa)"), but its Link record doesn't use the specified character in its query expressions, Auto Complete will retrieve the correct query results no matter whether the user input contains the specified character or not.

Valid if set from

Web tier configuration file (web.xml)

Requires restart of the web applications server?

Yes

Default value

/ (forward slash)

Possible values

Any ASCII character

Example usage

```
<context-param>
  <param-name>autoCompleteSkipCachingChar</param-name>
  <param-value>/</param-value>
</context-param>
```

Related topics

System parameters
Web tier parameters
Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Client parameters for web clients

2.13.11. Web parameter: autoSubmit

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

autoSubmit

Description

When this parameter is enabled, users can submit forms on the web client by pressing the Enter key. Enabling this parameter from the web tier configuration file forces all web clients to submit data with the Enter key. Enabling the this parameter from the web browser URL only submit data with the Enter key on that particular web client. Typically, you only need to disable this parameter on Japanese systems.

Valid if set from

Web browser URL

Web tier configuration file (web.xml and webtier.properties)

Requires restart of the web applications server?

Yes

Default value

true

Possible values

true (Enabled)

false (Disabled)

Example usage

```
<init-param>
  <param-name>autoSubmit</param-name>
  <param-value>false</param-value>
</init-param>
```

Related topics

System parameters
Web tier parameters
Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Client parameters for web clients

2.13.12. Web parameter: cacerts

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

cacerts

Description

This parameter defines the path to the cacerts file that's required for web clients to make SSL connections to the web tier. You must also install a Certificate Authority (CA) certificate file on the web tier server and enable the *ssl* web parameter to enable SSL connections.

Valid if set from

Web tier configuration file (web.xml and webtier.properties)

Requires restart of the web applications server?

Yes

Default value

WEB-INF/cacerts

Possible values

Relative path and file name to the keystore containing the certificate authority's certificate

Example usage

```
<init-param>
  <param-name>cacerts</param-name>
  <param-value>WEB-INF/mycacerts</param-value>
</init-param>
```

Related topics

System parameters
Web tier parameters
Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Client parameters for web clients
SSL parameters

2.13.13. Web parameter: CACLogin

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

CACLogin

Description

Enabling this parameter causes the web client to present a Common Access Card (CAC) certificate as authentication information and to use SSL connections to the web tier.

Valid if set from

Web tier configuration file (web.xml)

Requires restart of the web applications server?

Yes

Default value

false

Possible values

true (Enable CAC authentication)

false (Disable CAC authentication)

Example usage

```
<init-param>
    <param-name>CACLogin</param-name>
    <param-value>true</param-value>
</init-param>
```

Related topics

Common Access Card (CAC) sign-on
Example: enabling CAC sign-on
Requirements for CAC sign-on

2.13.14. Web parameter: cms.truststore

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

cms.truststore

Description

The trust store of the UCMDB Gateway.

Valid if set from

Web tier configuration file (webtier.properties)

Requires restart of the web applications server?

Yes

Default value

Empty

Possible values

Relative or absolute path and file name of the UCMDB Gateway's trust store

Example usage

```
cms.truststore=/WEB-INF/cmsrootca.p12
```

Related topics

Web parameter: cms.truststore.password

2.13.15. Web parameter: cms.truststore.password

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

cms.truststore.password

Description

The password of the UCMDB Gateway's trust store.

The value of this parameter is automatically encrypted after the Service Manager server starts. If you need to update the value later, remove the asterisk symbol (*) before the parameter name.

Valid if set from

Web tier configuration file (webtier.properties)

Requires restart of the web applications server?

Yes

Default value

Empty

Possible values

Password of the UCMDB Gateway's trust store

Example usage

```
cms.truststore.password=changeit
```

Related topics

Web parameter: cms.truststore

2.13.16. Web parameter: comfillAutoComplete

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

comfillAutoComplete

Description

This parameter enables auto complete for the Comfill control. If you want to disable auto complete globally in Service Manager, set the parameter to false.

To enable auto complete for a specific Comfill field, you must further enable the **Auto Complete** property of this Comfill control in Forms Designer. If this property is disabled, auto complete is disabled for this Comfill field regardless of the global *comfillAutoComplete* setting.

Valid if set from

Web tier configuration file (web.xml and webtier.properties)

Requires restart of the web application server?

Yes

Default value

true

Possible values

true (Enabled)

false (Disabled)

Example usage

```
<context-param>
  <param-name>comfillAutoComplete</param-name>
  <param-value>true</param-value>
</context-param>
```

Related topics

Client parameters for web clients
Web parameter: autoCompleteDelayTime
Web parameter: autoCompleteListSize
Web parameter: autoCompleteMinChars

2.13.17. Web parameter: compactLayout

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

compactLayout

Description

This parameter enables the compact layout of the web client.

Service Manager 9.34 introduced the new user interface (UI) style, in which the size of some UI elements is bigger than those of prior versions of Service Manager (for example, the height of the text widget, and the vertical space between two adjacent widgets). The out-of-box forms have been adjusted to match the new size. However, there is a possibility that the new size may cause some display issues in your tailored forms. For example, you may experience truncated text or overlapping UI elements.

You can use the compact layout of the web client to fix these display issues. When you enable the compact layout, the web client uses the same sizes as that of Service Manager 9.33, so the display issues caused by new sizes will disappear.

Note

Note We recommend that you use the compact layout of the web client in the following scenarios:

- Your forms have display issues caused by new sizes.
- You can't easily change the impacted forms to solve the display issues.

Valid if set from

Web tier configuration file (web.xml and webtier.properties)

Requires restart of the web applications server?

Yes

Default value

false

Possible values

true (Enabled)

false (Disabled)

Example usage

```
<context-param>
  <param-name>compactLayout</param-name>
  <param-value>false</param-value>
</context-param>
```

Related topics

System parameters
Web tier parameters
Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Client parameters for web clients

2.13.18. Web parameter: compress_soap

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

compress_soap

Description

Enabling this parameter causes the web tier to compress SOAP messages to web clients.

Valid if set from

Web tier configuration file (web.xml and webtier.properties)

Requires restart of the web applications server?

Yes

Default value

true

Possible values

true (Enabled)

false (Disabled)

Example usage

```
<init-param>
  <param-name>compress_soap</param-name>
  <param-value>true</param-value>
</init-param>
```

Related topics

System parameters
Web tier parameters
Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Client parameters for web clients

2.13.19. Web parameter: cookie.httpOnly

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

cookie.httpOnly

Description

This parameter determines whether to set the HttpOnly flag for HTTP cookies (such as JSESSIONID, mode, and lang). We strongly recommend you to set the HttpOnly flag for cookies.

Notes:

- For the LWSSO_COOKIE_KEY cookie, you need to set the HttpOnly flag in the lwssofmconf.xml file.

- For the JSESSIONID cookie in WebSphere, you need to set the HttpOnly flag in the WebSphere admin console.

Valid if set from

Web tier configuration file (webtier.properties)

Requires restart of the web applications server?

Yes

Default value

true

Possible values

true (Enabled)

false (Disabled)

Example usage

```
cookie.httpOnly=true
```

Related topics

System parameters
Web tier parameters
Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Client parameters for web clients

2.13.20. Web parameter: cookie.secure

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

cookie.secure

Description

This parameter determines whether to set the Secure flag for HTTP cookies (such as JSESSIONID, mode, and lang). We recommend you to set the Secure flag for cookies if the users access the web tier by using the HTTPS protocol.

Notes:

- For the LWSSO_COOKIE_KEY cookie, you need to set the Secure flag in the lwssofmconf.xml file.
- For the JSESSIONID cookie in WebSphere, you need to set the Secure flag in the WebSphere admin console.

Valid if set from

Web tier configuration file (webtier.properties)

Requires restart of the web applications server?

Yes

Default value

false

Possible values

true (Enabled)

false (Disabled)

Example usage

```
cookie.secure=false
```

Related topics

System parameters
Web tier parameters
Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Client parameters for web clients

2.13.21. Web parameter: customize-folder

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

customize-folder

Description

This parameter specifies the absolute path to a folder in which your web client branding files and web client custom webtier.properties file are stored. You must have write access to this directory to rebrand the web client or configure the web client custom webtier.properties file.

You need to place your branding image files in this folder. When you perform branding in the branding interface, Service Manager saves your branding settings in a **branding.xml** file and in a **branding-images** subfolder under this folder.

You can configure the customize-folder parameter by using either of the following:

- The default web.xml located in the <web tier>/WEB-INF folder
- The JVM option -Dhpe.sm.customizefolder in the application server

Note

Note The customize-folder parameter configured in the web.xml file has higher precedence.

Caution

Caution This parameter is required to enable the branding interface. If it's not specified, when you click **Tailoring** > **Branding**, the branding interface is not launched and an error message is displayed instead.

Valid if set from

Web tier configuration file (web.xml)

JVM option -Dhpe.sm.customizefolder

Requires restart of the web applications server?

Yes

Default value

Empty

Possible values

An absolute directory

Example usage

Configure from web.xml:

```
<context-param>
  <param-name>customize-folder</param-name>
  <param-value>C:/customize</param-value>
</context-param>
```

Configure from JVM option:

Add `-Dhpe.sm.customizefolder=C:/customize` to the java options of the Tomcat configuration.

Related topics

System parameters
Web tier parameters
Encryption of client keystore passwords
Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Client parameters for web clients

2.13.22. Web parameter: detailViewMinWidth

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

detailViewMinWidth

Description

This parameter increases the min-width value of a web page if the system displays some texts incompletely after you enlarge the browser text size or font size.

Note

Note

- Due to this min-width configuration, you may get the horizontal scroll bar in the detail view even if it is unnecessary.
- This parameter has no effect on the record list view and the pop-up view.

Valid if set from

The URL of the Service Manager web client accessible view

Requires restart of the Service Manager server?

No

Default value

None

Possible values

Any integer between 40 and 90

Example usage

When accessing the accessible view of the Service Manager web client, use the following URL to set the min-width to 72em.

http://<servername>:<portnumber>/sm/accessible.do?detailViewMinWidth=72

The new value takes effect immediately after the page is reloaded.

Related topics

System parameters
Web tier parameters
Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Enter a parameter in the webtier.properties files
Client parameters for web clients

2.13.23. Web parameter: disableJumpAddress

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

disableJumpAddress

Description

This parameter controls whether the Quick Jump component is displayed. The Quick Jump component enables you to navigate to a group section quickly.

Valid if set from

Web tier configuration file (web.xml)

Requires restart of the web applications server?

Refer to your web server documentation

Default value

false

Possible values

true

false

Example usage

```
<context-param>
  <param-name>disableJumpAddress</param-name>
  <param-value>false</param-value>
</context-param>
```

Related topics

System parameters
Web tier parameters
Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Client parameters for web clients
SSL parameters

2.13.24. Web parameter: disableKeyHelp

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

disableKeyHelp

Description

By default, Service Manager displays contextual help for each field on a form. This parameter allows you to disable the contextual help.

Valid if set from

Web tier configuration file (web.xml and webtier.properties)

Requires restart of the web applications server?

Yes

Default value

false

Possible values

true (Disabled)

false (Enabled)

Example usage

```
<context-param>
  <param-name>disableKeyHelp</param-name>
  <param-value>true</param-value>
</context-param>
```

Related topics

System parameters
Web tier parameters
Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Client parameters for web clients

2.13.25. Web parameter: disableLoginAutoComplete

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

disableLoginAutoComplete

Description

This parameter allows administrators to enable password auto-complete for the SM web client login page.

By default, this parameter is set to true, which prevents the browser from saving the user's Service Manager login password. Therefore, even if password auto-complete is enabled in the browser, password auto-complete is disabled for the web client login page.

When this parameter is set to false, as long as password auto-complete is enabled in the browser, the browser will save the user's login password and automatically complete it at login.

Caution

Caution Be aware that setting this parameter to false may increase the risk of allowing unauthorized access to Service Manager through the exploitation of stored passwords.

Note

Note Due to a known issue in web browsers, if a user has already enabled password auto-complete and saved the password in the web browser before an administrator changes this parameter to true, the user needs to manually remove the saved password; otherwise the password will still be auto-completed even if this parameter is set to true.

For example, to remove a saved password in Firefox, do the following:

1. Click **Tools** > **Options** > **Security**, and click the **Saved Password** button.
2. Select the password, and then click **Remove**. Note that clicking the **Remove All** button will remove all saved passwords for all accessed web sites.

Valid if set from

Web tier configuration file (web.xml and webtier.properties)

Requires restart of the web applications server?

Yes

Default value

true

Possible values

true: Disable password auto-complete

false: Enable password auto-complete

Example usage

```
<context-param>
  <param-name>disableLoginAutoComplete</param-name>
  <param-value>true</param-value>
</context-param>
```

Related topics

System parameters
Web tier parameters
Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Client parameters for web clients

2.13.26. Web parameter: enableExitConfirmMessage

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

enableExitConfirmMessage

Description

If this parameter is enabled (set to "true"), when the user clicks the Close ("x") button of the Service Manager web client page or manually refreshes the entire web client page, a confirmation message is displayed that indicates the user might have unsaved data. The message asks the user to confirm whether to stay on the current page or leave the page.

Valid if set from

Web tier configuration file (web.xml and webtier.properties)

Requires restart of the web applications server?

Yes

Default value

false

Possible values

true (Enabled)

false (Disabled)

Example usage

```
<context-param>
  <param-name>enableExitConfirmMessage</param-name>
  <param-value>true</param-value>
</context-param>
```

Related topics

System parameters
Web tier parameters

Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Client parameters for web clients

2.13.27. Web parameter: enableGetOriginalUrl

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

enableGetOriginalUrl

Description

When this parameter is enabled, the webtier obtains the original URL from the browser before a user logs in (in TSO or LWSSO mode).

This enables backend servers to identify and use the original URL when a firewall, proxy, or redirect has altered it. For example, when a client connects to the web tier through a hardware load balancer.

Valid if set from

Web tier configuration file (web.xml and webtier.properties)

Requires restart of the web applications server?

Yes

Default value

False

Possible values

True, False

Example usage

```
<init-param>
  <param-name>enableGetOriginalUrl</param-name>
  <param-value>True</param-value>
</init-param>
```

Related topics

Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Client parameters for web clients

2.13.28. Web parameter: enableListFrameStateRetain

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

enableListFrameStateRetain

Description

This parameter enables the list detail page to retain the expand-collapse state of the List Pane when the list detail page refreshes. If you set this parameter to false, the List Pane is always in expanded state once the list detail page refreshes.

Note

Note With `enableListFrameStateRetain` enabled, if you open a page ("destination page") from an existing page ("source page") and the two pages have the same file name (for example, `ocmq`), the expand/collapse state of the list frame in the source page is not retained when you return to it.

Valid if set from

Web tier configuration file (web.xml and webtier.properties)

Requires restart of the web applications server?

Yes

Default value

false

Possible values

true (Retain the expand-collapse state of the List Pane)

false (Do not retain the expand-collapse state of the List Pane)

Example usage

```
<context-param>
  <param-name>enableListFrameStateRetain</param-name>
  <param-value>false</param-value>
</context-param>
```

Related topics

System parameters
Web tier parameters
Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Client parameters for web clients
SSL parameters

2.13.29. Web parameter: enableMoreContextMenu

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

enableMoreContextMenu

Description

When this parameter is enabled, a contextual "More" menu is displayed when you right-click lists on the details pane of a record in the web client.

Note

Note If you right-click in an area outside of a list or the details pane, the browser's contextual menu is displayed instead.

Valid if set from

Web tier configuration file (web.xml and webtier.properties)

Requires restart of the web applications server?

Yes

Default value

false

Possible values

true (Enabled)

false (Disabled)

Example usage

```
<context-param>
  <param-name>enableMoreContextMenu</param-name>
  <param-value>true</param-value>
</context-param>
```

Related topics

System parameters
Web tier parameters
Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Client parameters for web clients
SSL parameters

2.13.30. Web parameter: enableRecordlistOddEvenRowStyle

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

enableRecordlistOddEvenRowStyle

Description

This parameter determines if the odd and even rows in the record list have a different background color.

Valid if set from

Web tier configuration file (web.xml and webtier.properties)

Requires restart of the web applications server?

Yes

Default value

false (odd and even rows in the record list have the same background color)

Possible values

true or false

Example usage

```
<context-param>
  <param-name>enableRecordlistOddEvenRowStyle</param-name>
  <param-value>false</param-value>
</context-param>
```

Related topics

Enter a parameter in the web.xml file
Web parameter: message display properties
Client parameters for web clients

2.13.31. Web parameter: enableSidebarMenu

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

enableSidebarMenu

Description

This parameter specifies if the System Navigator of the web client can use the sidebar mode and icons.

The icons in the System Navigator require the 9.40 version of the Service Manager application and Run-Time Environment (RTE). When you use an earlier version of the Service Manager application or RTE, we recommend that you set the *Enablesidebarmenu* parameter to `false` to disable the icons and the sidebar mode of the System Navigator. Otherwise, the Service Manager web client uses the default icon for all the menu items in the System Navigator.

Valid if set from

Web tier configuration file (web.xml and webtier.properties)

Requires restart of the web applications server?

Yes

Default value

true

Possible values

true (Enabled)

false (Disabled)

Example usage

```
<context-param>
  <param-name>enableSidebarMenu</param-name>
  <param-value>true</param-value>
</context-param>
```

Related topics

System parameters
Web tier parameters
Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Client parameters for web clients

2.13.32. Web parameter: enableTableEllipsis

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

enableTableEllipsis

Description

For the following read-only controls in record details area, you can enable this parameter to display '**...**' at the end of the text fields when the value is too long to be shown completely:

- tables
- comfill with array
- combobox with array
- text fileds with array

Valid if set from

Web tier configuration file (web.xml and webtier.properties)

Requires restart of the web applications server?

Yes

Default value

false

Possible values

true, false

Example usage

```
<context-param>
  <param-name>enableTableEllipsis</param-name>
  <param-value>true</param-value>
</context-param>
```

Related topics

System parameters
Web tier parameters
Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Client parameters for web clients

2.13.33. Web parameter: enableTableMagnify

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

enableTableMagnify

Description

This parameter specifies whether the **Magnify** button is visible for editable tables in the web client. If this value is set to true, the Magnify button appears on the top-right corner of editable tables in the web client. If this value is set to false, the Magnify button doesn't appear in the web client.

Valid if set from

Web tier configuration file (web.xml and webtier.properties)

Requires restart of the web applications server?

Yes

Default value

true

Possible values

true, false

Example usage

```
<init-param>
  <param-name>enableTableMagnify</param-name>
  <param-value>true</param-value>
</init-param>
```

Related topics

System parameters
Web tier parameters
Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Client parameters for web clients

2.13.34. Web parameter: encryptPasswordInBrowser

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

`encryptPasswordInBrowser`

Description

This parameter determines whether to encrypt the user password in the web browser before the system sends it to the web tier.

Note

Note

This parameter has a limitation in the Chrome browser. The encryption leverages the `WebCrypto` API, whereas Chrome restricts the access to the `WebCrypto` API to secure origins (the https:// pages). This parameter takes effect in Chrome only if you access the Web client using the secure scheme (such as `https://smwebtier.yourdomain.net/webtier/index.do`). If you access the Web client using a non - secure scheme (such as `http://smwebtier.yourdomain.net/webtier/index.do`) in Chrome, the system still sends the password to the web tier as clear text.

Valid if set from

Web tier configuration file (`webtier.properties`)

Requires restart of the web applications server?

Yes

Default value

false

Possible values

true (enabled)

false (disabled)

Example usage

```
encryptPasswordInBrowser=true
```

Related topics

System parameters
Web tier parameters
Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Client parameters for web clients

2.13.35. Web parameter: externalForgotPasswordUrl

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

externalForgotPasswordUrl

Description

This parameter defines an external forgot password URL for the Forgot Password link on the Service Manager web client login page.

When the Service Manager server uses an authentication method other than database authentication and it has the forgot password function, you can configure this parameter to redirect the users to an external forgot password page.

Valid if set from

Web tier configuration file (webtier.properties)

Requires restart of the web applications server?

Yes

Default value

empty

Possible values

A valid URL

Example usage

```
externalForgotPassword=https://ldap.yourdomain.net/forgotpassword
```

Related topics

System parameters
Web tier parameters
Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Client parameters for web clients

2.13.36. Web parameter: expandComnandLine

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

expandCommandLine

Description

This parameter specifies whether to expand the command/search line by default on the top right area in the web client. If you set this parameter to true, the command/search line is expanded by default when you log on to the web client, and you don't need to click on the command/search line icon to expand it.

Valid if set from

Web tier configuration file (webtier.properties)

Requires restart of the web applications server?

Yes

Default value

false

Possible values

true (the command/search line is expanded by default)

false (the command/search line is not expanded by default)

Example usage

```
expandCommandLine=true
```

Related topics

System parameters
Web tier parameters

Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Client parameters for web clients

2.13.37. Web parameter: expandSmartSearch

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

expandSmartSearch

Description

This parameter specifies whether to expand the Smart Search field by default on the top right area in the web client. If you set this parameter to true, the Smart Search field is expanded by default when you log on to the web client, and you don't need to click the Smart Search field icon to expand it.

Valid if set from

Web tier configuration file (webtier.properties)

Requires restart of the web applications server?

Yes

Default value

false

Possible values

true (the Smart Search field is expanded by default)

false (the Smart Search field is not expanded by default)

Example usage

```
expandSmartSearch=true
```

Related topics

System parameters
Web tier parameters
Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Client parameters for web clients

2.13.38. Web parameter: helpServerContext

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

helpServerContext

Description

This parameter defines the context path when the Service Manager help center is published on a web server. Service Manager will launch the help center using this URL: http://<helpServerHost>:<helpServerPort>/<helpServerContext>, where <helpServerContext> refers to the context path.

Important

Important The Service Manager help center is published at the following web site: **https://docs.microfocus.com**. No help server installation package is available for you to deploy the help center on your own server. To enable your end users to access the help center from the web client UI, you must set this parameter to the context path part of the help center home page URL. First, find this URL from the **https://docs.microfocus.com** website. For example, if the help center home page URL is **/itom/Service_Manager:9.xx/Home**, set this parameter to itom/Service_Manager:9.xx/Home.

Note

Note

By default, this parameter is not included in the web.xml file, and Service Manager will directly open this URL: http://<helpServerHost>:<helpServerPort>/help.

Valid if set from

Web tier configuration file (web.xml and webtier.properties)

Requires restart of the web applications server?

Yes

Default value

help

Possible values

Context path part of the help center home page URL

Example usage

In the web.xml file:

```
<init-param>
  <param-name>helpServerContext</param-name>
  <param-value>itom/Service_Manager:9.xx/Home</param-value>
</init-param>
```

In the webtier.properties file:

```
helpServerContext=itom/Service_Manager:9.xx/Home
```

Related topics

Web parameter: showHelp
Web parameter: helpServerHost
Web parameter: helpServerPort

2.13.39. Web parameter: helpServerHost

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

helpServerHost

Description

This parameter defines the name of the Help server host that listens for client requests. You must define this parameter if your end users need to access the Service Manager help center from the web client.

Important

Important The Service Manager help center is published at the following web site: **https://docs.microfocus.com**. No help server installation package is available for you to deploy the help center on your own server. To enable your end users to access the help center from the web client UI, you must set this parameter to **docs.microfocus.com**.

Valid if set from

Web browser URL

Web tier configuration file (web.xml and webtier.properties)

Requires restart of the web applications server?

Yes

Default value

None

Possible values

Host name of Help Server

Example usage

```
<init-param>
  <param-name>helpServerHost</param-name>
  <param-value>docs.microfocus.com</param-value>
</init-param>
```

Related topics

Web parameter: showHelp
Web parameter: helpServerPort
Web parameter: helpServerContext

2.13.40. Web parameter: helpServerPort

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

helpServerPort

Description

This parameter defines the communications port number on which the Help server listens to client requests. You must define this parameter if your end users need to access the Service Manager help center from the web client.

Important

Important The Service Manager help center is published at the following web site: **https://docs.microfocus.com**. No help server installation package is available for you to deploy the help center on your own server. Since the help center website uses the default port, you must leave this parameter empty or set it to 80.

Valid if set from

Web browser URL

Web tier configuration file (web.xml and webtier.properties)

Requires restart of the web applications server?

Yes

Default value

None

Possible values

Communications port number

Example usage

```
<init-param>
  <param-name>helpServerPort</param-name>
  <param-value></param-value>
</init-param>
```

Related topics

Web parameter: showHelp
Web parameter: helpServerHost
Web parameter: helpServerContext

2.13.41. Web parameter: honorUrlHost

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

honorUrlHost

Description

This parameter determines whether web client users can specify the name of the Service Manager server from the URL. Setting this parameter to false prevents web client users from changing the value of the *serverHost* parameter from the URL.

Valid if set from

Web tier configuration file (web.xml and webtier.properties)

Requires restart of the web applications server?

Yes

Default value

false

Possible values

true

false

Example usage

```
<init-param>
  <param-name>honorUrlHost</param-name>
  <param-value>true</param-value>
</init-param>
```

Related topics

System parameters
Web tier parameters
Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Client parameters for web clients

2.13.42. Web parameter: honorUrlPort

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

honorUrlPort

Description

This parameter determines whether web client users can specify the communications port of the Service Manager server from the URL. Setting this parameter to false prevents web client users from changing the value of the *serverPort* parameter from the URL.

Valid if set from

Web tier configuration file (web.xml and webtier.properties)

Requires restart of the web applications server?

Yes

Default value

false

Possible values

true

false

Example usage

```
<init-param>
  <param-name>honorUrlPort</param-name>
  <param-value>true</param-value>
</init-param>
```

Related topics

System parameters
Web tier parameters
Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Client parameters for web clients

2.13.43. Web parameter: isCustomAuthenticationUsed

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

isCustomAuthenticationUsed

Description

By default, Service Manager authenticates web client users by comparing the user name and password that the user enters on the Service Manager login page to a matching operator record in the system.

To enable single sign-on (Trusted Sign-On, LW-SSO, or SAML SSO) and CAC Sign-On), you must disable the default authentication method by setting this parameter to **false**.

For example, for trusted sign-on, this causes Service Manager to send the current user name in the HTTP header. Trusted sign-on uses the user name to determine if a web client is already authenticated or not.

Caution

Caution You should only disable this parameter if you are using any of the above-described sign-on configurations. Disabling this parameter without such a sign-on configuration will prevent your web client users from logging in to Service Manager.

Valid if set from

Web tier configuration file (web.xml and webtier.properties)

Requires restart of the web applications server?

Yes

Default value

true

Possible values

true (Enabled)

false (Disabled)

Example usage

```
<context-param>
  <param-name>isCustomAuthenticationUsed</param-name>
  <param-value>false</param-value>
</context-param>
```

Related topics

System parameters
Web tier parameters
Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Client parameters for web clients

2.13.44. Web parameter: JCEProviderClassName

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

JCEProviderClassName

Description

This parameter specifies the class name of a third-party Java Cryptography Extension (JCE) provider when the web tier is running in non-FIPS mode (that is, when the **fipsMode** parameter is set to **false** in the webtier.properties file).

Note

Note In FIPS mode, this parameter is ignored.

Valid if set from

Web tier configuration file (web.xml and webtier.properties)

Requires restart of the web applications server?

Yes

Default value

None

Possible values

JCE provider class names

Example usage

```
<init-param>
  <param-name>JCEProviderClassName</param-name>
  <param-value>com.rsa.jsafe.provider.JsafeJCE</param-value>
</init-param>
```

Related topics

System parameters
Web tier parameters
Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Client parameters for web clients

2.13.45. Web parameter: JCEProviderName

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

JCEProviderName

Description

This parameter specifies the name of a third-party Java Cryptography Extension (JCE) provider that you plug in when the web tier is running in non-FIPS mode.

Note

Note In FIPS mode, this parameter is ignored.

Valid if set from

Web tier configuration file (web.xml and webtier.properties)

Requires restart of the web applications server?

Yes

Default value

None

Possible values

JCE provider names

Example usage

```
<init-param>
  <param-name>JCEProviderName</param-name>
  <param-value>JsafeJCE</param-value>
</init-param>
```

Related topics

System parameters
Web tier parameters
Enter a parameter in the web client URL

Enter a parameter in the web.xml file
Client parameters for web clients

2.13.46. Web parameter: jsDebug

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

jsDebug

Description

This parameter enables the web client to load unminified JavaScript files. However, when this parameter is enabled, the web client also displays to end users an error message that includes full stack trace. For this reason, don't enable this parameter in production environments to avoid disclosure of sensitive information.

Valid if set from

Web tier configuration file (web.xml and webtier.properties)

Requires restart of the web applications server?

Yes

Default value

false

Possible values

true, false

Example usage

```
<init-param>
  <param-name>jsDebug</param-name>
  <param-value>false</param-value>
</init-param>
```

Related topics

System parameters
Web tier parameters
Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Client parameters for web clients

2.13.47. Web parameter: labelbgunderline

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

labelbgunderline

Description

This parameter defines the label background display if the user has selected a color other than white as the Background Color property for a label in a form, but has left the Background Style property blank. If set to true, the empty Background Style property defaults to underline. If set to false, the empty Background Style property defaults to a solid background.

Valid if set from

Web tier configuration file (web.xml and webtier.properties)

Requires restart of the web applications server?

Yes

Default value

true

Possible values

true (default to underline)

false (default to solid background)

Example usage

```
<init-param>
  <param-name>labelbgunderline</param-name>
  <param-value>true</param-value>
</init-param>
```

Related topics

System parameters
Web tier parameters
Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Client parameters for web clients

2.13.48. Web parameter: lockWaitTimeout

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

lockWaitTimeout

Description

The lockWaitTimeout parameter prevents a web tier application server thread from waiting infinitely to acquire a lock. After the specified time is reached, the thread ceases trying to acquire a lock, and the user receives the following warning message:

The session is busy, please try again later.

Valid if set from

Web tier configuration file (web.xml and webtier.properties)

Requires restart of the web applications server?

Yes

Default value

15 seconds

Possible values

1-300 seconds (A value of less than or equal to 0 will indicate the maximum value 300 is used)

Example usage

```
<init-param>
  <param-name>lockWaitTimeout</param-name>
  <param-value>15</param-value>
</init-param>
```

Related topics

System parameters
Web tier parameters

Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Client parameters for web clients
SSL parameters

2.13.49. Web parameter: log4jConfiguration

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

log4jConfiguration

Description

This parameter defines the path to the web tier's logging properties file, which is a standard log4j2 logging properties file. You can use the logging properties file to define the default logging level, the logging level for specific web tier components, the path to the log file, and to specify what output formats and actions logging should take (such as, send an e-mail over SMTP, send the log file over FTP, or print the log file to the event viewer). See the log4j web site for more information on configuring this file.

Valid if set from

Web tier configuration file (web.xml)

Note: This parameter is not included in web.xml by default.

Requires restart of the web applications server?

Yes

Default value

/WEB-INF/log4j2.properties

Possible values

Absolute path to logging properties file

Relative path to logging properties file

URL to logging properties file

Example usage

```
<context-param>
<param-name>log4jConfiguration</param-name>
<param-value>/WEB-INF/log4j2.properties</param-value>
</context-param>
```

Related topics

System parameters
Web tier parameters
Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Log4j Web site
Client parameters for web clients

2.13.50. Web parameter: maskMessageDelay

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

maskMessageDelay

Description

The *maskMessageDelay* parameter specifies the number of seconds before the mask message is displayed after the mask is shown. The minimum value is 1 second and the maximum is 30 seconds. A default value of 5 seconds will be used if an invalid value is specified.

Valid if set from

Web tier configuration file (web.xml and webtier.properties)

Requires restart of the web applications server?

Yes

Default value

5 (seconds)

Possible values

Any integer value from 1 second to 30 seconds. Any invalid value will use the default value of 5 seconds.

Example usage

```
<context-param>
  <param-name>maskMessageDelay</param-name>
  <param-value>5</param-value> \
</context-param>
```

Related topics

System parameters
Web tier parameters
Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Client parameters for web clients
Web parameter: recordListArrayDisplayStyle

2.13.51. Web parameter: maxGroupCacheSize

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

maxGroupCacheSize

Description

This parameter defines the maximum number of records that can be cached in the user's web browser when the user navigates through the groups in a record list. The default value is 300.

Valid if set from

Web tier configuration file (web.xml)

Requires restart of the web applications server?

Yes

Default value

300

Possible values

No less than 300

Example usage

```
<context-param>
  <param-name>maxGroupCacheSize</param-name>
  <param-value>1000</param-value>
</context-param>
```

Related topics

System parameters
Web tier parameters
Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Client parameters for web clients

2.13.52. Web parameter: maxattachuploadsize

Web parameters change the behavior of web clients that connect to the web tier. You can set some of these parameters from the web client login URL, some from the web tier configuration file (web.xml), and some from both.

Parameter

maxattachuploadsize

Description

This parameter specifies the maximum size limit for individual attachments in a record.

Valid if set from

Web tier configuration file (web.xml and webtier.properties)

Requires restart of the web applications server?

Yes

Default value

10 MB

Possible values

Unlimited

Example usage

```
<context-param>
  <param-name>MaxAttachUploadSize</param-name>
  <param-value>10240</param-value>
</context-param>
```

Related topics

System parameters
Web tier parameters
Setting file attachment limits
Enter a parameter in the web.xml file
Enter a parameter in the webtier.properties files
Web parameter: maxtotalattachuploadsize

2.13.53. Web parameter: maxtotalattachuploadsize

Web parameters change the behavior of web clients that connect to the web tier. You can set some of these parameters from the web client login URL, some from the Web tier configuration file (web.xml), and some from both.

Parameter

maxtotalattachuploadsize

Description

This parameter specifies the maximum total size of all the attachments in a record.

Valid if set from

Web tier configuration file (web.xml and webtier.properties)

Requires restart of the web applications server?

Yes

Default value

50 MB

Possible values

Unlimited

Example usage

```
context-param>
  <param-name>MaxTotalAttachUploadSize</param-name>
  <param-value>51200</param-value>
</context-param>
```

Related topics

System parameters
Web tier parameters
Setting file attachment limits
Enter a parameter in the web.xml file
Enter a parameter in the webtier.properties files
Web parameter: maxattachuploadsize

2.13.54. Web parameter: maxMessageNumber

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

maxMessageNumber

Description

This parameter defines the maximum number of messages displayed in Messages And Alerts.

Valid if set from

webtier.properties

Requires restart of the Service Manager server?

Yes

Default value

50

Possible values

Any positive integer between 1 and 500

Example usage

```
maxMessageNumber=50
```

Related topics

System parameters
Web tier parameters

Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Client parameters for web clients

2.13.55. Web parameter: maxRequestPerSecond

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

maxRequestPerSecond

Description

This parameter defines the maximum number of requests that's allowed per second for one user session. The default value for this parameter is 50. To allow unlimited requests, set the value less than or equal to zero.

Valid if set from

Web tier configuration file (web.xml and webtier.properties)

Requires restart of the web applications server?

Yes

Default value

50

Possible values

Unlimited

Note

Note

The parameter value easily reaches the maximum value if you use too many DVD select statements in a form. In this case, you must adjust the parameter value accordingly.

The system may have performance issue that leads to significant increase of form loading time if there are too many DVD select statements added to a form.

Example usage

```
<context-param>
  <param-name>maxRequestPerSecond</param-name>
  <param-value>-1</param-value>
</context-param>
```

Related topics

System parameters
Web tier parameters
Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Client parameters for web clients
Web parameter: recordListArrayDisplayStyle

2.13.56. Web parameter: message display properties

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

messageBarError

messageBarInfo

messageBarWarning

Description

These parameters determine how different types of message are displayed. The parameter value consists of three parts: the first part defines if this type of message is displayed in the message bar, the second part defines if the message automatically disappears, and the third part defines the duration in seconds before the message disappears.

The latter part of the parameter value is dependent on the previous part. For example, if the first part of value is set as false, which means that this type of message is not displayed on the top of the detail window, then the following two parts of parameter values are meaningless and thus ignored. If the second part is set as false, which means that this type of message doesn't automatically disappear, then the last value defining the duration before the message disappears is ignored.

Valid if set from

Web tier configuration file (web.xml and webtier.properties)

Requires restart of the web applications server?

Yes

Default value

messageBarError	true:false
messageBarInfo	true:true:4
messageBarWarning	true:true:6

Possible values

true (displayed)/false (not displayed): true (disappear automatically)/false (not disappear): 0 to unlimited

Note

Note If you specify an invalid number for the third part of the value, a default value of 6 seconds is used.

Example usage

- ```
  <context-param>
  <param-name>messageBarError</param-name>
  <param-value>true:false</param-value>
  </context-param>
  ```

  The default value means that the error type of message is displayed on the top of the detail window, and doesn't automatically disappear.

  Note

  Note If the user changes the second value to `true` to make the error message bar automatically disappear, then the user has to set the last value to decide the duration before it disappears.

- ```
  <context-param>
  <param-name>messageBarInfo</param-name>
  <param-value>true:true:4</param-value>
  </context-param>
  ```

  The default value means that the information type of message is displayed on the top of the detail window, and it automatically disappears after four seconds.

- ```
  <context-param>
  <param-name>messageBarWarning</param-name>
  <param-value>true:true:6</param-value>
  </context-param>
  ```

  The default value means that the warning type of message is displayed on the top of the detail window, and it automatically disappears after six seconds.

Related topics

System parameters
Web tier parameters
Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Enter a parameter in the web.xml file
Client parameters for web clients

2.13.57. Web parameter: openTabsSoftLimit

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

openTabsSoftLimit

Description

This parameter defines the number of tabs users can open in their browsers before a warning message displays to let them know of the possible risks and advise them to close some tabs. The message goes away when the number of open tabs is at or below the limit. This parameter is not a hard limit; it doesn't prevent the user from opening more tabs than it is set for. Also, it's not related to the RAD thread number delimiter defined on the Service Manager server (appthreadspersession).

Valid if set from

Web tier configuration file (web.xml and webtier.properties)

Requires restart of the web applications server?

Yes

Default value

15

Possible values

Number of tabs

Example usage

```
<init-param>
  <param-name>openTabsSoftLimit</param-name>
  <param-value>10</param-value>
</init-param>
```

Related topics

System parameters
Web tier parameters
Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Client parameters for web clients

2.13.58. Web parameter: preventDuplicatedAttachmentName
```

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

preventDuplicatedAttachmentName

Description

This parameter controls whether to prevent an attachment that has the same file name with existing ones from being uploaded to Service Manager. If this parameter is enabled, a pop-up message box prompts you to change the name of the attachment when you upload an attachment with a file name that already exists.

Valid if set from

Web tier configuration file (web.xml and webtier.properties)

Requires restart of the web applications server?

Yes

Default value

false

Possible values

true (Do not allow the upload of attachments with duplicate names)

false (Allow the upload of attachments with duplicate names)

Example usage

```
<context-param>
  <param-name>preventDuplicatedAttachmentName</param-name>
  <param-value>true</param-value>
</context-param>
```

Related topics

System parameters
Web tier parameters
Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Enter a parameter in the webtier.properties files
Client parameters for web clients

2.13.59. Web parameter: querySecurity

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

querySecurity

Description

Enabling this parameter causes the Service Manager web tier to verify the security key of a URL query generated by the Service Manager server, and, if valid, authorize the query. Disabling this parameter allows any user with log on permissions, the skills to create a query, and access to the Service Manager URL to extract data from any Service Manager table.

Note

Note

This parameter is tied to the security parameter "querysecurity" (spelled with all lowercase letters). To disable the security hash feature, you must disable querySecurity in the web.xml file and querysecurity in the sm.ini file.

Valid if set from

Web tier configuration file (web.xml and webtier.properties)

Requires restart of the web applications server?

Yes

Default value

true

Possible values

true (Enabled)

false (Disabled)

Example usage

```
<init-param>
  <param-name>querySecurity</param-name>
  <param-value>false</param-value>
</init-param>
```

Related topics

System parameters
Web tier parameters
Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Client parameters for web clients

2.13.60. Web parameter: recordListArrayDisplayStyle

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

recordListArrayDisplayStyle

Description

This parameter defines how array fields are displayed in record lists (for example, To-Do queue lists, lists in the list pane of list/detail pages, and lists generated by clicking the Fill button), and it takes no effect on table objects (for example, tables in the detail pane of a list/detail page). The parameter can be set to one of the following values:

- 1 (default): The original content length of each array column value is truncated to the value defined in the *recordListArrayMaxChars* parameter.
- 2 (recommended): Array type columns aren't displayed in record lists. This option is recommended because displaying array fields in record lists can have a negative impact on performance.
- 3: Keep the original content length of each array column value in record lists. This is the old behavior when this parameter was not introduced.

Note

Note The setting of this parameter has no effect on the Export to Text File functionality. When you export a record list to a text file, array fields are always exported with their original content.

Valid if set from

Web tier configuration file (web.xml and webtier.properties)

Requires restart of the web applications server?

Yes

Default value

1

Possible values

1, 2, and 3

Example usage

```
<context-param>
  <param-name>recordListArrayDisplayStyle</param-name>
  <param-value>3</param-value>
</context-param>
```

Related topics

System parameters
Web tier parameters
Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Client parameters for web clients
Web parameter: recordListArrayMaxChars

2.13.61. Web parameter: recordListArrayMaxChars

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

recordListArrayMaxChars

Description

This parameter defines the maximum allowed field length of any array fields that appear in record lists. It takes effect only when the *recordListArrayDisplayStyle* parameter is set to 1

Valid if set from

Web tier configuration file (web.xml and webtier.properties)

Requires restart of the web applications server?

Yes

Default value

100 (bytes)

Possible values

No less than 1

Example usage

```
<context-param>
  <param-name>recordListArrayMaxChars</param-name>
  <param-value>50</param-value>
</context-param>
```

Related topics

System parameters
Web tier parameters
Enter a parameter in the web client URL
Enter a parameter in the web.xml file

Client parameters for web clients
Web parameter: recordListArrayDisplayStyle

2.13.62. Web parameter: refreshMessages

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

refreshMessages

Description

Enabling this parameter causes the web client to query the Service Manager server for messages at a regular interval. You can define the interval that the web client waits between checking for messages by using the *refreshMessageInterval* web parameter.

Valid if set from

Web browser URL

Web tier configuration file (web.xml and webtier.properties)

Requires restart of the web applications server?

Yes

Default value

true

Possible values

true (Enabled)

false (Disabled)

Example usage

```
<init-param>
  <param-name>refreshMessages</param-name>
  <param-value>true</param-value>
</init-param>
```

Related topics

System parameters
Web tier parameters
Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Client parameters for web clients

2.13.63. Web parameter: refreshMessagesInterval

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

refreshMessagesInterval

Description

This parameter defines the interval that the web client waits before checking for messages on the Service Manager server. You must also enable the *refreshMessages* web parameter to have web clients check the Service Manager server for messages.

Valid if set from

Web browser URL

Web tier configuration file (web.xml and webtier.properties)

Requires restart of the web applications server?

Yes

Default value

15000

Possible values

Number of milliseconds

Example usage

```
<init-param>
  <param-name>refreshMessagesInterval</param-name>
  <param-value>30000</param-value>
</init-param>
```

Related topics

System parameters
Web tier parameters
Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Client parameters for web clients

2.13.64. Web parameter: rememberLang

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

rememberLang

Description

This parameter specifies if the browser's language preference is remembered. By default, Service Manager webtier remembers the browser's language preference in the persistent cookie. If you don't want the language preference to be remembered, set this parameter to false.

Valid if set from

webtier.properties

Requires restart of the web applications server?

Yes

Default value

true

Possible values

true (enabled)

false (disabled)

Example usage

```
rememberLang=true
```

Related topics

System parameters
Web tier parameters

Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Client parameters for web clients

2.13.65. Web parameter: searchMinChars

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

searchMinChars

Description

This parameter defines the minimum number of characters you must type in the search field to trigger a System Navigator search.

Valid if set from

Web tier configuration file (webtier.properties)

Requires restart of the web application server?

Yes

Default value

2

Possible values

Any meaningful integer greater than or equals to 1

Note: If you specify a value of zero or a negative number, the default value 2 is used.

Example usage

```
searchMinChars=2
```

Related topics

Client parameters for web clients

2.13.66. Web parameter: secureLogin

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

secureLogin

Description

This parameter controls the encryption of network communication between the Web application server and the web browser. Enabling this parameter causes web browsers to use SSL connections to the web application server.

Note

Note

To use secure login, you must enable SSL connections on the web application server. We recommend that you not disable secure login.

Valid if set from

Web tier configuration file (web.xml and webtier.properties)

Requires restart of the web applications server?

Yes

Default value

true

Possible values

true (Enabled)

false (Disabled)

Example usage

```
<context-param>
  <param-name>secureLogin</param-name>
  <param-value>true</param-value>
</context-param>
```

Related topics

System parameters
Web tier parameters
Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Client parameters for web clients
Web parameter: sslPort

2.13.67. Web parameter: serverHost

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

serverHost

Description

This parameter defines the name of the Service Manager host that listens for client requests. You must define this parameter if you run the web tier on a separate server than the Service Manager server.

Valid if set from

Web browser URL

Web tier configuration file (web.xml and webtier.properties)

Requires restart of the web applications server?

Yes

Default value

localhost

Possible values

Host name of Service Manager server

Example usage

```
<init-param>
  <param-name>serverHost</param-name>
  <param-value>myserver.mydomain.com</param-value>
</init-param>
```

Related topics

System parameters
Web tier parameters

Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Client parameters for web clients

2.13.68. Web parameter: serverPort

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

serverPort

Description

This parameter defines the communications port number on which the Service Manager server listens to client requests.

Valid if set from

Web browser URL

Web tier configuration file (web.xml and webtier.properties)

Requires restart of the web applications server?

Yes

Default value

12670

Possible values

Communications port number

Example usage

```
<init-param>
  <param-name>serverPort</param-name>
  <param-value>13080</param-value>
</init-param>
```

Related topics

System parameters
Web tier parameters
Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Client parameters for web clients

2.13.69. Web parameter: session-timeout

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

session-timeout

Description

This parameter defines the default session timeout interval for all sessions created in the Service Manager web tier.

Note

Note It is best practice to lower the session-timeout value to 2 minutes if you experience Service Manager web client sessions lingering for a longer period of time than expected.

Valid if set from

Web tier configuration file (web.xml)

Requires restart of the web applications server?

Yes

Default value

15

Possible values

Minutes

Example usage

```
<session-config>
 <session-timeout>2</session-timeout>
</session-config>
```

Related topics

System parameters
Web tier parameters
Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Client parameters for web clients

2.13.70. Web parameter: showAddFavButton

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

showAddFavButton

Description

Enables the Add Favorite button in the Record List toolbar and the Detail List toolbar to add favorites.

Valid if set from

Web tier configuration file (web.xml and webtier.properties)

Requires restart of the web applications server?

Yes

Default value

true

Possible values

true (Enabled)

false (Disabled)

Example usage

```
<context-param>
  <param-name>showAddFavButton</param-name>
  <param-value>true</param-value>
</context-param>
```

Related topics

System parameters
Web tier parameters
Enter a parameter in the web client URL

Enter a parameter in the web.xml file
Client parameters for web clients

2.13.71. Web parameter: showFavMgrButton

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

showFavMgrButton

Description

Enables use of the Favorite Manager button to manage favorites in Favorites and Dashboards.

Valid if set from

Web tier configuration file (web.xml and webtier.properties)

Requires restart of the web applications server?

Yes

Default value

true

Possible values

true (Enabled)

false (Disabled)

Example usage

```
<context-param>
  <param-name>showFavMgrButton</param-name>
  <param-value>true</param-value>
</context-param>
```

Related topics

System parameters
Web tier parameters
Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Client parameters for web clients

2.13.72. Web parameter: showHelp

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

showHelp

Description

If you enable this parameter, the web client displays the **Online Help** option when you click the **Help** button. You can click the **Online Help** option to open the Help Server that you define in the *helpServerHost* and *helpServerPort* parameters. Web client users can't access your Help Server unless you enable this parameter and also define the help server host, port number, and context path.

Valid if set from

Web tier configuration file (web.xml and webtier.properties)

Requires restart of the web applications server?

Yes

Default value

false

Possible values

true (Enabled)

false (Disabled)

Example usage

```
<context-param>
  <param-name>showHelp</param-name>
  <param-value>true</param-value>
</context-param>
```

Related topics

Web parameter: helpServerHost
Web parameter: helpServerPort
Web parameter: helpServerContext

2.13.73. Web parameter: showNavIcon

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

showNavIcon

Description

If this parameter is set to true, each System Navigator menu item will display an icon that indicates the type of menu item. If the parameter is set to false or not present, the navigator menu items don't display icons.

Valid if set from

Web tier configuration file (web.xml and webtier.properties)

Requires restart of the web applications server?

Yes

Default value

false

Possible values

true (Enabled)

false (Disabled)

Example usage

```
<context-param>
  <param-name>showNavIcon</param-name>
  <param-value>false</param-value>
</context-param>
```

Related topics

System parameters
Web tier parameters
Enter a parameter in the web client URL

Enter a parameter in the web.xml file
Client parameters for web clients

2.13.74. Web parameter: showReloadNavButton

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

useJavaDynamicVersioning

Description

Enables use of the system registry to locate the latest version of Java installed on a user's system.

Valid if set from

Web tier configuration file (web.xml and webtier.properties)

Requires restart of the web applications server?

Yes

Default value

false

Possible values

true (Enabled)

false (Disabled)

Example usage

```
<context-param>
  <param-name>useJavaDynamicVersioning</param-name>
  <param-value>true</param-value>
</context-param>
```

Related topics

System parameters
Web tier parameters
Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Client parameters for web clients

2.13.75. Web parameter: sideBarButtonNumberToCollapse

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

sideBarButtonNumberToCollapse

Description

This parameter defines the number of side bar buttons to be collapsed. If the number of side bar buttons is bigger than or equal to this value, these buttons will be collapsed.

Valid if set from

Web tier configuration file (web.xml and webtier.properties)

Requires restart of the web application server?

Yes

Default value

4

Possible values

1 to 100

Note

Note If you specify an invalid value, the default value is used.

Example usage

```
<context-param>
  <param-name>sideBarButtonNumberToCollapse</param-name>
  <param-value>4</param-value>
</context-param>
```

Related topics

System parameters
Web tier parameters
Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Client parameters for web clients
Web parameter: comfillAutoComplete
Web parameter: autoCompleteDelayTime
Web parameter: autoCompleteMinChars

2.13.76. Web parameter: socketReadTimeout

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

socketReadTimeout

Description

The *socketReadTimeout* parameter prevents a thread on the web tier application server from waiting infinitely for a socket IO operation. After the specified time is reached, the socket read operation will terminate.

Valid if set from

Web tier configuration file (web.xml and webtier.properties)

Requires restart of the web applications server?

Yes

Default value

180 seconds

Possible values

Any value is valid. A value of less than or equal to 0 will indicate no time out.

Example usage

```
<init-param>
  <param-name>socketReadTimeout</param-name>
  <param-value>180</param-value>
</init-param>
```

Related topics

System parameters
Web tier parameters
Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Client parameters for web clients
SSL parameters

2.13.77. Web parameter: sortNavItems

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

sortNavItems

Description

This parameter sets the sort order System Navigator menu items. By default, this parameter is set to true, which means the System Navigator menu items are sorted in ascending, alphabetical order. When this parameter is set to false, the menu items display in the order they arrive from the server. If this parameter is deleted, System Navigator menu items are sorted in ascending, alphabetical order.

Valid if set from

Web tier configuration file (web.xml and webtier.properties)

Requires restart of the web applications server?

Yes

Default value

true

Possible values

true (enabled)

false (disabled)

Example usage

```
<context-param>
  <param-name>sortNavItems</param-name>
  <param-value>true</param-value>
</context-param>
```

Related topics

System parameters
Web tier parameters
Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Client parameters for web clients

2.13.78. Web parameter: sortNavFavorites

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

sortNavFavorites

Description

This parameter sets the sort order of favorites. By default, this parameter is set to `false`. Hence, the system displays the private favorites section on top of the public favorites section. Within each section, the folders are on top of the views. The folders and views are sorted in ascending alphabetical order, respectively.

When this parameter is set to `true`, all private and public favorites folders and views are mixed together in ascending alphabetical order. If this parameter is deleted, the system displays the favorites the same as you set this parameter to `false`.

Valid if set from

Web tier configuration file (web.xml)

Requires restart of the web applications server?

Yes

Default value

false

Possible values

true (enabled)

false (disabled)

Example usage

```
<context-param>
  <param-name>sortNavFavorites</param-name>
  <param-value>true</param-value>
</context-param>
```

Related topics

System parameters
Web tier parameters
Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Client parameters for web clients

2.13.79. Web parameter: ssl

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

ssl

Description

Enabling this parameter causes all web clients to use SSL connections to the web tier. You must also install a Certificate Authority (CA) certificate file on the Web tier server and define the *cacerts* web parameter to enable SSL connections.

Valid if set from

Web tier configuration file (web.xml and webtier.properties)

Requires restart of the web applications server?

Yes

Default value

false

Possible values

true (Enabled)

false (Disabled)

Example usage

```
<init-param>
  <param-name>ssl</param-name>
  <param-value>true</param-value>
</init-param>
```

Related topics

System parameters
Web tier parameters
Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Client parameters for web clients
SSL parameters

2.13.80. Web parameter: sslPort

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

sslPort

Description

This parameter defines the SSL port of the web application server. It is required only when the *secureLogin* parameter is set to true.

Valid if set from

Web tier configuration file (web.xml and webtier.properties)

Requires restart of the web applications server?

Yes

Default value

8443

Possible values

SSL port number of the web application server

Example usage

```
<context-param>
  <param-name>sslPort</param-name>
  <param-value>8443</param-value>
</context-param>
```

Related topics

System parameters
Web tier parameters
Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Client parameters for web clients
Web parameter: secureLogin

2.13.81. Web parameter: sso.failure.httpStatus

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

sso.failure.httpStatus

Description

This parameter specifies the HTTP status code to be returned when TSO, LW-SSO, IdM or CAC fails.

Valid if set from

Web tier configuration file (webtier.properties)

Requires restart of the web applications server?

Yes

Default value

403

Possible values

401
403

Example usage

```
sso.failure.httpStatus=403
```

Related topics

System parameters
Web tier parameters
Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Client parameters for web clients

2.13.82. Web parameter: startDayOfWeek

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

startDayOfWeek

Description

This parameter defines the first day of the week in the date picker. However, if you configure the **First Day of Week** field in the **Login Profiles** for an operator, then this parameter is not used for that user. For more information, see Set the first day of week for an operator.

Valid if set from

Web tier configuration file (web.xml and webtier.properties)

Requires restart of the web applications server?

Yes

Default value

0

Possible values

0 (Sunday)
1 (Monday)
2 (Tuesday)
3 (Wednesday)
4 (Thursday)

5 (Friday)
6 (Saturday)

Example usage

```
<context-param>
  <param-name>startDayOfWeek</param-name>
  <param-value>0</param-value>
</context-param>
```

Related topics

System parameters
Web tier parameters
Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Client parameters for web clients

2.13.83. Web parameter: tooltipAutoHide

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

tooltipAutoHide

Description

This parameter controls whether to automatically hide a tooltip after the pointer leaves the target element. Only the tooltip that is triggered by pointer hover and is not controlled by the browser (that is, not the tooltip created through use of the HTML title attribute) is controlled by this parameter.

By default, after you move the pointer out of a target element, the tooltip is not dismissed if the pointer is moving toward the tooltip. The tooltip is dismissed if the pointer is not moving toward the tooltip. After you move the pointer out of the tooltip, the tooltip is dismissed.

If you set this parameter to `false`, the web client displays a close button (X) on the top right corner of the tooltip. You can either manually click the close button or press the ESC key to dismiss the tooltip if the keyboard focus is in the same iframe as the target element of the tooltip.

Valid if set from

Web browser URL

Web tier configuration file (manually add this parameter into `web.xml` or `webtier.properties`)

Note

Note When you define this parameter using all three methods, the one set from the web browser URL takes the highest precedence. However, if you set this parameter from the web browser URL, it takes effect immediately after reloading the page and requires no re-login.

Requires restart of the web applications server?

No if set from the browser URL

Yes if set from the web tier configuration file

Default value

`true`

Possible values

`true` or `false`

Example usage

```
http://<your web tier host>:<port>/<web tier>index.do?tooltipAutoHide=true
```

Related topics

System parameters
Web tier parameters
Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Enter a parameter in the webtier.properties files
Client parameters for web clients

2.13.84. Web parameter: tracesoap

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

tracesoap

Description

This parameter defines whether to log SOAP traffic in the web tier. If you add the `tracesoap=true` parameter to the web browser URL when you log in, all request/response SOAP messages during the current session are logged until you log out or the session expires.

The SOAP information is logged in the web tier. By default, it is logged in the sm.log file located in the current working directory of the application server process. For example, `C:\Program Files\Apache Software Foundation\Tomcat 9.0\sm.log` if the Tomcat is run as a service. The SOAP message logging is defined in `WEB-INF` / `log4j2.properties` as follows:

```
logger.TraceSoap.level=debug
```

Note

Note By default, the logging level is DEBUG. It is recommended that you NOT change this setting. The root logger level setting in `log4j.properties` has no effect on the SOAP logging level. For example, no matter whether the root logger level is `info`, `warn`, or `error`, Service Manager always logs SOAP messages in the web tier.

SOAP messages are logged with a `DEBUG` prefix, as in the following examples:

```
DEBUG http-8080-4 TraceSoap - request:
DEBUG http-8080-4 TraceSoap - response:
```

Valid if set from

Web browser URL

Web tier configuration file (web.xml or webtier.properties)

Note

Note When this parameter is defined using all of the three methods, the one set from web browser URL takes the highest precedence. However, if set from web browser, this parameter is valid only when set at login, and once set at login it remains valid until you log out or the session times out.

Requires restart of the web applications server?

No if set from the login URL

Yes if set from the web tier configuration file

Default value

false

Possible values

true or false

Example usage

```
http://localhost:8080/webtier/index.do?tracesoap=true
```

Related topics

System parameters
Web tier parameters
Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Enter a parameter in the webtier.properties files
Client parameters for web clients

2.13.85. Web parameter: unmaskDeferMillis

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

unmaskDeferMillis

Description

Defer the unmasking of a browser window after an unmask request is issued. This parameter relates to how much time the client browser needs to finish rendering after a page is loaded. The client user may feel low application performance if set it to a value greater than 2000. Additionally, this value should not be set to less than 550 milliseconds unless system performance is well characterized.

Valid if set from

Web tier configuration file (web.xml and webtier.properties)

Requires restart of the web applications server?

Yes

Default value

750 milliseconds

Possible values

350-3000 milliseconds

Example usage

```
<context-param>
  <param-name>unmaskDeferMillis</param-name>
  <param-value>750</param-value>
</context-param>
```

Related topics

System parameters
Web tier parameters
Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Client parameters for web clients
SSL parameters

2.13.86. Web parameter: useJavaDynamicVersioning

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

useJavaDynamicVersioning

Description

Enables use of the system registry to locate the latest version of Java installed on a user's system.

Valid if set from

Web tier configuration file (web.xml and webtier.properties)

Requires restart of the web applications server?

Yes

Default value

false

Possible values

true (Enabled)

false (Disabled)

Example usage

```
<context-param>
  <param-name>useJavaDynamicVersioning</param-name>
  <param-value>true</param-value>
</context-param>
```

Related topics

System parameters
Web tier parameters
Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Client parameters for web clients

2.13.87. Web parameter: useServertabs

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

useservertabs

Description

This parameter causes web clients to display tabs in the order that the server has them saved on the form. By default, this parameter is disabled, and web clients display tabs from right to left and top to bottom.

Valid if set from

Web browser URL

Web tier configuration file (web.xml and webtier.properties)

Requires restart of the web applications server?

Yes

Default value

false

Possible values

false

true

Example usage

```
<init-param>
  <param-name>useservertabs</param-name>
  <param-value>false</param-value>
</init-param>
```

Related topics

System parameters
Web tier parameters
Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Client parameters for web clients

2.13.88. Web parameter: viewactivenotes

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

viewactivenotes

Description

Enabling this parameter causes web clients to display messages in a message box as well as in the status bar. Enabling this parameter from the web tier configuration file forces all web clients to display messages. Enabling this parameter from the web browser URL only displays messages on that particular web client.

Valid if set from

Web browser URL

Web tier configuration file (web.xml and webtier.properties)

Requires restart of the web applications server?

Yes

Default value

false

Possible values

true (Enabled)

false (Disabled)

Example usage

```
<init-param>
  <param-name>viewactivenotes</param-name>
  <param-value>true</param-value>
</init-param>
```

Related topics

System parameters
Web tier parameters
Enter a parameter in the web client URL

Enter a parameter in the web.xml file
Client parameters for web clients

2.13.89. Web parameter: viewcontexthelp

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

viewcontexthelp

Description

This parameter enables the web client to display the same context-sensitive help debug information (field) in field help as the Windows client. To enable this feature, append *viewcontexthelp=true* to the web client URL for current user or set it in the web tier configuration files (web.xml or webtier.properties) for all users.

Note

Note This feature is only supported for the Power User mode (index.do).

Valid if set from

Web browser URL

Web tier configuration file (web.xml and webtier.properties)

Note

Note When this parameter is defined using all of the three methods, the one set from web browser URL takes the highest precedence. However, if set from web browser, this parameter is valid only when set at login, and once set at login it remains valid until you log out or the session times out.

Requires restart of the web applications server?

No if set from the login URL (only requires a re-login)

Yes if set from the web tier configuration file

Default value

false

Possible values

true or false

Example usage

http://<*your web tier host*>:<*port*>/<*web tier*>index.do?viewcontexthelp=true

Related topics

System parameters
Web tier parameters
Client parameters for web clients

2.13.90. Web parameter: viewpromptforsave

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

viewpromptforsave

Description

Enabling this parameter causes web clients to display a warning message when users cancel records with unsaved changes. Enabling this parameter from the web tier configuration file forces all web clients to display the save warning message. Enabling the this parameter from the web browser URL only displays the save warning message on that particular web client.

Valid if set from

Web browser URL

Web tier configuration file (web.xml and webtier.properties)

Requires restart of the web applications server?

Yes

Default value

true

Possible values

true (Enabled)

false (Disabled)

Example usage

```
<init-param>
  <param-name>viewpromptforsave</param-name>
  <param-value>false</param-value>
</init-param>
```

Related topics

System parameters
Web tier parameters
Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Client parameters for web clients

2.13.91. Web parameter: viewrecordlist

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

viewrecordlist

Description

Enabling this parameter causes web clients to display the record list/detail view for search results. Enabling this parameter from the web tier configuration file forces all web clients to display the record list/detail view. Enabling the this parameter from the web browser URL only displays the record list/detail view on that particular web client.

Valid if set from

Web browser URL

Web tier configuration file (web.xml and webtier.properties)

Requires restart of the web applications server?

Yes

Default value

true

Possible values

true (Enabled)

false (Disabled)

Example usage

```
<init-param>
  <param-name>viewrecordlist</param-name>
  <param-value>false</param-value>
</init-param>
```

Related topics

System parameters
Web tier parameters
Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Client parameters for web clients

2.13.92. Web parameter: virtualGroupSize

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

virtualGroupSize

Description

In inbox views, records are divided into virtual groups if there are too many records in a group. This parameter defines the number of records that are displayed as one virtual group in inbox views.

Note

Note

- You can set the value from 25 to 500. In addition, make sure that this group size is not greater than the value of the sqlfetchs parameter if you have specified one.
- The virtual group size set in the system information record takes precedence over this parameter.
- Make sure that you do performance testing before you set a big number in your production environment.

Valid if set from

Web tier configuration file (webtier.properties)

Requires restart of the web application server?

Yes

Default value

100

Possible values

25 to 500

Example usage

```
virtualGroupSize=200
```

Related topics

System parameters
Web tier parameters

Enter a parameter in the web client URL
Enter a parameter in the web.xml file

2.13.93. Web parameter: webtierFullUrl

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

`webtierFullUrl`

Description

This parameter describes the full URL used to access the web tier. For example, `https://smwebtier.yourdomain.net/webtier-9.x`.

Suppose you have deployed the web tier behind a reverse proxy or a load balancer and used the form - based authentication. In this case, you must set this property if the reverse proxy or the load balancer doesn't pass the appropriate `X-Forwarded-*` HTTP headers to the web tier.

Note: The `webtierFullUrl` parameter takes precedence over the `X-Forwarded-*` headers. If you choose to use the `X-Forwarded-*` headers, don't configure this parameter.

Valid if set from

Web tier configuration file (`webtier.properties`)

Requires restart of the web applications server?

Yes

Default value

N/A

Possible values

Full URL of the web tier

Example usage

```
webtierFullUrl=https://smwebtier.yourdomain.net/webtier-9.x
```

Related topics

System parameters
Web tier parameters
Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Client parameters for web clients

2.13.94. Web parameter: xff.trusted.proxies

Web parameters change the behavior of the web clients that connect to the web tier. For more information, see Web tier parameters.

Parameter

xff.trusted.proxies

Description

You can add your trusted proxy IPs (separated by comma) in the xff.trusted.proxies parameter.

If the value in xff.trusted.proxies is not empty and the X-Forwarded-For header in the request is not empty, the X-Forwarded-For header will be searched to find the original end user IP, removing the matched trusted proxies. Then, the system can store the original end user IP instead of the proxy IP in the syslog table when you use proxies or have a load balancer environment.

For example, with the following configuration, the value "192.168.2.2, 192.168.0.10" in the X-Forwarded-For header returns the end user IP as "192.168.2.2".

xff.trusted.proxies=192.168.3.3,192.168.0.10

Valid if set from

Web tier configuration file (webtier.properties)

Requires restart of the web applications server?

Yes

Default value

Empty

Possible values

IP addresses (separated by comma)

Example usage

```
xff.trusted.proxies=192.168.3.3,192.168.0.10
```

Related topics

System parameters
Web tier parameters
Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Client parameters for web clients

2.14. Client parameters for Windows clients

These parameters override Windows client preferences. Any client parameter that you set on the server overrides the local client preference and prevents users from changing it. You can set these parameters from the Service Manager server's operating system command prompt or from the Service Manager initialization file (sm.ini).

Startup parameter	Description
Startup parameter: chartrefresh	Specifies the length of time that Service Manager waits between refreshing charts
Startup parameter: clientformcache	Enables Windows clients to cache forms
Startup parameter: clientsideunload	Enables client-side unloading of Service Manager files
Parameter: servicemanager.disableclientprinting	Removes the Print and Print Preview menu items from the menu in Windows client
Parameter: heartbeatinterval	Controls the client heartbeat frequency
Startup parameter: sessiontimeout	Defines the number of minutes that the server waits for a client heartbeat signal before the server assumes that the client session has timed out and closes the connection
Startup parameter: useservertabs	Causes Service Manager clients to display tabs in the order the server has them saved on the form
Startup parameter: viewactivenotes	Causes Service Manager clients to display server messages in the Messages view
Startup parameter: viewpromptforsave	Causes Service Manager clients to prompt the user to save changes when they close forms
Startup parameter: viewrecordlist	Causes Service Manager clients to display the record list and record details simultaneously

Related topics

System parameters
Enter a parameter in the sm.ini file

2.14.1. Startup parameter: chartrefresh

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

chartrefresh

Description

This parameter specifies the interval Service Manager waits between refreshing charts. More frequent intervals increase network traffic, less frequent intervals delay the updated chart information. Windows client users can also set this parameter on their local clients from **Windows > Preferences**.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

None

Possible values

Number of seconds

Example usage

Command line: **sm -httpPort:13080 -chartrefresh:60**

Initialization file: `chartrefresh:60`

Related topics

System parameters
Enter a parameter in the sm.ini file
Client parameters for Windows clients

2.14.2. Startup parameter: clientformcache

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

clientformcache

Description

Enabling this parameter causes the Windows client to cache forms. Form caching improves server responsiveness by reducing the need to load forms.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

1

Possible values

0 (Disable)

1 (Enable)

Example usage

Command line: **sm -httpPort:13080 -clientformcache:1**

Initialization file: `clientformcache:1`

Related topics

System parameters
Enter a parameter in the sm.ini file
Client parameters for Windows clients

2.14.3. Startup parameter: clientsideunload

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter====

clientsideunload

Description

Enabling this parameter allows client-side unloading of Service Manager files. Windows client users can also set this parameter on their local clients from **Windows > Preferences**.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

0

Possible values

0 (Disable)

1 (Enable)

Example usage

Command line: **sm -httpPort:13080 -clientsideunload:1**

Initialization file: `clientsideunload:1`

Related topics

System parameters
Enter a parameter in the sm.ini file
Client parameters for Windows clients

2.14.4. Parameter: disableclientprinting

This parameter overrides Windows client preferences. Any client parameter you set on the server overrides the local client preference and prevents users from changing it.

Parameter

servicemanager.disableclientprinting

Description

Enabling this parameter removes the Print and Print Preview menu items.

Valid if set from

Client configuration file ({SM_HOME}\Client\configuration\config.ini)

Requires restart of server?

Yes

Default value

false

Possible values

- true (Disable client side printing)
- false (Enable client side printing)

Example usage

Client configuration file: `servicemanager.disableclientprinting:true`

Related topics

Setting Windows client preferences from the server
Client parameters for Windows clients

2.14.5. Startup Parameter: heartbeatinterval

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

heartbeatinterval

Description

This parameter controls the client heartbeat frequency. If the server doesn't receive a heartbeat from the client within the time-out limit as defined by the *sessiontimeout* parameter, the server terminates the client. All unsaved data is lost and the client must establish a new connection.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

15 seconds

Possible values

Number of seconds

Example usage

Command line: **sm -httpPort:13080 -heartbeatinterval:60**

Initialization file: `heartbeatinterval:60`

Related topics

System parameters
Enter a parameter in the sm.ini file
Client parameters for Windows clients

2.14.6. Startup Parameter: sessiontimeout

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

sessiontimeout

Description

This parameter defines the number of minutes that the server waits for a client heartbeat signal before the server assumes that the client session has timed out and closes the connection.

The client sends a heartbeat signal to the server every 15 seconds or as defined by the *heartbeatinterval* parameter.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

30

Possible values

Number of minutes

Example usage

Command line: **sm -httpPort:13080 -sessiontimeout:30**

Initialization file: `sessiontimeout:30`

Related topics

System parameters
Enter a parameter in the sm.ini file
Client parameters for Windows clients

2.14.7. Startup parameter: useservertabs

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

useservertabs

Description

Enabling this parameter causes Service Manager clients to display tabs in the order the server has them saved on the form. By default, this parameter is disabled and Service Manager clients display tabs from right to left and top to bottom. Windows client users can also set this parameter on their local clients from **Windows > Preferences**.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

false

Possible values

true (Enabled)

false (Disabled)

Example usage

Command line: **sm -httpPort:13080 -useservertabs:1**

Initialization file: `useservertabs:1`

Related topics

System parameters
Enter a parameter in the sm.ini file
Client parameters for Windows clients

2.14.8. Startup parameter: viewactivenotes

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

viewactivenotes

Description

Enabling this parameter causes Service Manager clients to display server messages in the Messages view. Windows client users can also set this parameter on their local clients from **Windows > Preferences**.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

0

Possible values

0 (Disable)

1 (Enable)

Example usage

Command line: **sm -httpPort:13080 viewactivenotes:1**

Initialization file: `viewactivenotes:1`

Related topics

System parameters
Enter a parameter in the sm.ini file
Client parameters for Windows clients

2.14.9. Startup parameter: viewpromptforsave

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

viewpromptforsave

Description

Enabling this parameter causes Service Manager clients to prompt the user to save changes when they close forms. Windows client users can also set this parameter on their local clients from **Windows > Preferences**.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

1

Possible values

0 (Disable)

1 (Enable)

Example usage

Command line: **sm -httpPort:13080 -viewpromptforsave:1**

Initialization file: `viewpromptforsave:1`

Related topics

System parameters
Enter a parameter in the sm.ini file
Client parameters for Windows clients

2.14.10. Startup parameter: viewrecordlist

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

viewrecordlist

Description

Enabling this parameter causes Service Manager clients to display the record list and record details simultaneously. Windows client users can also set this parameter on their local clients from **Windows > Preferences**.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

1

Possible values

0 (Disable)

1 (Enable)

Example usage

Command line: **sm -httpPort:13080 -viewrecordlist:1**

Initialization file: `viewrecordlist:1`

Related topics

System parameters
Enter a parameter in the sm.ini file
Client parameters for Windows clients

2.15. Connection parameters

The following table lists the startup parameters that you can set from the Service Manager server's operating system command prompt or from the Service Manager initialization file (sm.ini).

These parameters determine how the server handles client connections.

Startup parameter	Brief description
Special parameters: allowwsdlretrieval	Allows WSDL retrieval for users that don't have SOAP API licenses
Special parameters: internalURI	Enables Service Manager to load Web Service (WSDL) when the server can't access Internet.
Startup parameter: compress_soap	Compresses SOAP messages between the Service Manager server and clients
Parameter: soap_timezone	Specifies the timezone (UTC timezone or SM timezone) in the soap response for the datetime type.
Parameter: disableWSEss	Controls whether or not a non-ESS user consumes a floating or named license when logging in to Service Manager through Service Request Catalog
Startup parameter: emailout	Causes the Service Manager server to start the JavaMail application and create mail events
Parameter: enablecoredump	Enables Service Manager to log information generated by a system crash. The operating system of the Service Manager server determines what core file gets generated, if any, during a system crash
Parameter: group	Shuts down or quiesces all Service Manager server members of a virtual group
Parameter: groupmcastaddress	Defines the TCP/IP address that servlet container processes can use to communicate with the load balancer process in a horizontal scaling implementation
Parameter: groupname	Creates a virtual group that servlet container processes across multiple systems can join for horizontal scaling purposes
Parameter: groupport	Defines the communications port that servlet container processes can use to communicate with the load balancer process in a horizontal scaling implementation
Parameter: host	Specifies the host name of the Service Manager server you want to shutdown or quiesce
Parameter: httpPort	Defines the communications port that a servlet container process uses to communicate with clients using HTTP
Parameter: httpsPort	Defines the communications port that a servlet container process uses to communicate with clients using HTTPS (SSL-encrypted HTTP)
Parameter: licensefile	Defines the path to the AutoPass license file that contains the list of modules and features that are enabled on the Service Manager server
Parameter: loadBalancer	Creates a load balancer process that listens to incoming client requests on the communications port that's specified by the httpPort parameter

Startup parameter: ntservice	Defines the name of the Windows service that you want to run Service Manager.
Parameter: pid	Terminates a Service Manager process with the given process ID.
Parameter: preferredFQHN	Allows you to specify the fully qualified host name you want Service Manager clients to use when communicating with the server.
Startup parameter: quiesce	Causes Service Manager to restrict logins to system administrators (when set to 1) or no users (when set to 2)
Parameter: reportlbstatus	Displays information about available ports and threads in horizontal and vertical scaling servlet implementations.
Parameter: requestviaproxy	Specifies if the Service Manager Windows client is allowed to connect to the server through a proxy.
Parameter: scauto	Specifies the SCAuto service name.
Startup parameter: scautolistener	Causes the Service Manager server to listen for connection requests from the legacy SCAuto integration on the communications port specified
Startup parameter: shutdown	Causes the Service Manager server to shut down immediately.
Parameter: sslConnector	Defines whether servlet container processes have an HTTPS (SSL-encrypted HTTP) communications port available
Startup parameter: system	Defines a numerical identifier for the Service Manager system that becomes part of the unique identifier is a horizontal scaling implementation
Parameter: threadTerminateWait	Defines the length of time (in seconds) that the Service Manager server continues to handle a client request after the client session is closed
Parameter: useHostinWSDL	When this parameter is enabled, the Service Manager server replaces the host name in the schema location in the webservice response with the host name in the web service request URL.
Parameter: useIPinWSDL	When the web services client application can't reach the Service Manager server with the host name, this parameter specifies that Service Manager can automatically obtain the IP address from the local host and use that IP address instead of the host domain name in the WSDL endpoint location URL
updatelicense	Automatically loads a new AutoPass license file into the existing one.

Startup parameter: webservices_sessiontimeout	Defines the number of seconds that the server waits for a WebService API client request before the server assumes that the client session has timed out and closes the connection
Parameter: ws_endpoint	This parameter specifies the web services end point.
Parameter: checkExtaccess	Enables field checking so that only the fields defined in the extaccess record can be used in SOAP requests.
Startup parameter: connectionTimeout	This parameter defines the number of milliseconds that the server waits for a Service Manager client request before the server assumes that the client session has timed out and closes the connection.
Parameter: smartemailTimeout	This parameter defines the number of milliseconds that the Smart Email process waits for a connection to the mail server to be established.

Related topics

System parameters
Enter a parameter in the sm.ini file

2.15.1. Special parameters: allowwsdlretrieval

Parameter

allowwsdlretrieval

Description

This parameter allows WSDL retrieval for users that don't have SOAP API licenses.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

None

Possible values

None

Example usage

Command line: **smenter -allowwsdlretrieval**

Initialization file: `allowwsdlretrieval`

Related topics

System parameters
Enter a parameter in the sm.ini file
System performance parameters

2.15.2. Special parameters: internalURI

Parameter

internalURI

Description

This parameter enables you to load Web Service (WSDL) when your server can't access Internet.

Valid if set from

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

None

Possible values

None

Example usage

Initialization file: `internalURI`

Related topics

System parameters
Enter a parameter in the sm.ini file
System performance parameters

2.15.3. Startup parameter: compress_soap

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

compress_soap

Description

Enabling this parameter compresses SOAP messages between the Service Manager server and clients.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

1

Possible values

0 (Disable)

1 (Enable)

Example usage

Command line: **sm -httpPort:13080 -compress_soap:1**

Initialization file: `compress_soap:1`

Related topics

System parameters
Enter a parameter in the sm.ini file
Client parameters for Windows clients

2.15.4. Parameter: soap_timezone

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

soap_timezone

Description

This parameter specifies the timezone (UTC timezone or SM timezone) in the soap response for the datetime type.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No, just log out and log back in the user

Default value

0

Possible values

0 (UTC +00:00 timezone)

1 (SM timezone, which is set in Date Info of System Information Record or Login Profile of the operator)

Example usage

Command line: **sm -httpPort:13080 -soap_timezone:1**

Initialization file: `soap_timezone:1`

Related topics

System parameters
Enter a parameter in the sm.ini file
Client parameters for Windows clients

2.15.5. Parameter: disableWSEss

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

disableWSEss

Description

This parameter controls whether or not a non-ESS user consumes a Floating or Named login license when logging in to Service Manager through Service Request Catalog:

- By default, a non-ESS user consumes only a self-service license (Unlimited) instead of a Floating or Named login license;
- If this parameter is set in sm.ini or the server's operating system command prompt (no value needed), a non-ESS user then consumes a Floating or Named login license.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

None

Possible values

None

Example usage

Command line: **sm -httpPort:13080 -httpsPort:13081 -disableWSEss**

Initialization file: `disableWSEss`

Related topics

System parameters
Enter a parameter in the sm.ini file
Connection parameters

2.15.6. Startup parameter: emailout

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

emailout

Description

Defining this parameter causes the Service Manager server to start the JavaMail application and create mail events. The JavaMail application allows you to send Service Manager mail to any mail system that supports Simple Mail Transfer Protocol (SMTP).

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

None

Possible values

JavaMail connection parameters

- ***keepmail*** - preserves mail events in event queues.
  Example of command line: `sm -emailout -keepmail`
- ***sleep*** - waits specified number of seconds before checking for mail events.
  Example of command line: `sm -emailout -sleep:30`
- ***debug*** - displays additional information about mail events in the Service Manager log file.
  Example of command line: `sm -emailout -debug`
- ***clean*** - removes extraneous headers in mail, for example, Service Manager Operator: falcon.
  Example of command line: `sm -emailout -clean`
- ***mailFrom*** - specifies the descriptive name or other identifier of the sender of an e-mail. This parameter should be set in the format of email address.
  Example of command line: `sm -emailout -mailFrom:admin@mymailserver.com`
- ***smtpEnableSSL*** - defines whether SSL should be used for SMTP operations.
  Example of command line: `sm -emailout -smtpEnableSSL:1`
- ***smtpTLS*** - defines whether SMTP requires Transport Layer Security (TLS) authentication to send emails.
  Example of command line: `sm -emailout -smtpTLS:0`
- ***smtphost*** - specifies the name of the SMTP server host for client requests. The value for the parameter can be the IP address, machine name, or DNS name of the SMTP server used for sending notifications.
  Example of command line: `sm -emailout -smtphost:192.168.255.255`

- ***smtppassword*** - identifies the password the Service Manager server uses to bind to the SMTP server.
  Example of command line: `sm -emailout -smtppassword:administrator`

  > **Note**
  > According to Google's announcement, as of end of September 2024, user can no longer use a password to access Gmail. For customers who are using Gmail as SMTP Server, currently the workaround is to use "app password" instead of user password for this field.

- ***smtpport*** - defines the communications port the SMTP server uses.
  Example of command line: `sm -emailout -smtpport:25`
- ***smtpSSLPort*** - defines the port number for SSL connection.
  Example of command line: `sm -emailout -smtpSSLPort:991`
- ***smtpusername*** - defines the account name of the SMTP server.
  Example of command line: `sm -emailout -smtpusername:admin@example.com`
- ***mailThreadCount*** - specifies the number of threads to send emails. The default value is 10.
  Example of command line: `sm -emailout -mailThreadCount:10`

  > **Note**
  > The main email thread now reads email records from the database and places them in a queue, and a number of other threads (N=mailThreadCount) send the emails from the queue. However, there is only one "eventout" entry in System Status, and the logged-in users are counted as one. This is because the mail-sender threads are internal and therefore should not be displayed or counted.

- ***SMIMEKeyAlias***- specifies the alias of the S/MIME private key.
  Example of command line: `sm -emailout -SMIMEKeyAlias:smemailkeyalias`
- ***SMIMEKeystore***- specifies the S/MIME keystore in PKCS12 format.
  Example of command line: `sm -emailout -SMIMEKeystore: smemailkey.p12`
- ***SMIMEKeystorePass***- specifies the pass phrase for the S/MIME keystore.
  Example of command line: `sm -emailout -SMIMEKeystorePass: smemailkeystorepass`
- ***mailBCC***- specifies a list of blind carbon copy (BCC) recipients to all emails. The value of this parameter is a comma-delimited list of user email accounts.
  Example of command line: `sm -emailout -mailBCC:admin@mymailserver.com,user@ mymailserver.com`
- ***mailFrom***- specifies the descriptive name or other identifier of the sender of an e-mail. This parameter should be set in the format of email address. Service Manager processes this parameter as follows:
  - If the mailFrom parameter is configured in the `sm.ini` or `sm.cfg` file, Service Manager always uses this value as the Mail From address for all outbound emails.
  - If the mailFrom parameter is not configured in the `sm.ini` or `sm.cfg` file, Service Manager always uses the value of the Mail From value of the eventout record as the Mail From address, without checking if the value is a valid email address.
  - If there is no mailFrom value configured in `sm.ini` or `sm.cfg` and no Mail From value in the eventout record, Service Manager logs an exception in sm.log.

  Example of command line: `sm -emailout -mailFrom:admin@mymailserver.com`

Related topics

System parameters
Parameter: emailoutConfigFromSmisOnly
Enter a parameter in the sm.ini file
Connection parameters

2.15.7. Parameter: enablecoredump

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

enablecoredump

Description

This parameter enables Service Manager to log information generated by a system crash. The operating system of the Service Manager server determines what core file gets generated, if any, during a system crash. Core file names are specific to the operating system.

For example:

Linux: `Core.<process_id>`

Windows: `sm.<process_id>_<thread_id>.mini.dmp`

When you see system messages in the sm.log file that indicate the Core Generation is disabled and ignoring Generate Core dump requests, start the processes to log information by enabling the parameter with a setting of "1" (*enablecoredump:1*).

> **Note**
>
> Note By default, the file is generated in Service Manager's RUN directory. You can choose an alternate location by providing the corepath parameter.

Valid if set from

Server's operating system command prompt

Initialization (sm.ini) file

Requires restart of the Service Manager server?

No

Default value

1

Possible values

0 (Disable)

1 (Enable)

Example usage

Command line: **sm -enablecoredump:1 -corepath:/sm/corefiles/**

Initialization file: `enablecoredump:1`

Related topics

System parameters
Enter a parameter in the sm.ini file
Debugging parameters

2.15.8. Parameter: groupmcastaddress

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

groupmcastaddress

Description

This parameter defines the TCP/IP address that servlet container processes can use to communicate with the load balancer process in a horizontal scaling implementation. The servlet container processes talk to one another using User Datagram

Protocol (UDP) multicasting and must use a TCP/IP address consistent with that protocol. You must enable UDP multicasting traffic on your network to use Service Manager virtual grouping.

It is best practice to place this parameter in the Service Manager initialization file so that all servlet container processes started on the same host share the TCP/IP address specified by this parameter.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

None

Possible values

Any TCP/IP address valid for UDP multicasting (addresses 224.0.1.0 to 239.255.255.255, inclusive.)

Example usage

Initialization file:

```
groupname:mygroup1
groupmcastaddress:224.0.1.255
groupport:13100
```

Related topics

Servlet implementation
System parameters
Enter a parameter in the sm.ini file
Connection parameters
Servlet parameters

2.15.9. Parameter: groupname

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

groupname

Description

This parameter creates a virtual group that servlet container processes across multiple systems can join for horizontal scaling purposes. Systems that list the same virtual group name are part of the same group. This parameter requires the use of the *groupmcastaddress* and *groupport* parameters to define the resources group members can use to talk to the load balancer process.

It is best practice to place this parameter in the Service Manager initialization file so that all servlet container processes started on the same host share the virtual group name specified by this parameter.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

None

Possible values

Any alphanumeric name without spaces

Example usage

Initialization file:

```
groupname:mygroup1
groupmcastaddress:224.0.1.255
groupport:13100
```

Related topics

Servlet implementation
System parameters
Enter a parameter in the sm.ini file
Connection parameters
Servlet parameters

2.15.10. Parameter: groupport

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

groupport

Description

This parameter defines the communications port that servlet container processes can use to communicate with the load balancer process in a horizontal scaling implementation. The servlet container processes talk to one another using User Datagram Protocol (UDP) multicasting and must use a common communications port available on all member systems of the virtual group. You must enable UDP multicasting traffic on your network to use Service Manager virtual grouping.

It is best practice to place this parameter in the Service Manager initialization file so that all servlet container processes started on the same host share the communications port specified by this parameter.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

12346

Possible values

Any communications port valid for UDP multicasting

Example usage

Initialization file:

```
groupname:mygroup1
groupmcastaddress:224.0.1.255
groupport:13100
```

Related topics

Servlet implementation
System parameters

Enter a parameter in the sm.ini file
Connection parameters
Servlet parameters

2.15.11. Parameter: host

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

host

Description

This parameter is intended for horizontally scaled implementations in which there are multiple servers. This parameter allows you to specify the member of the horizontal scaled group you want to shut down or quiesce.

Valid if set from

Server's operating system command prompt

Requires restart of the Service Manager server?

No

Default value

None

Possible values

A valid Service Manager host name

Example usage

The System Administrator can run the following commands to shut down or quiesce Service Manager server members in the horizontal scaled group.

Note

Note When the host option is missing in the commands, the default is taken as the localhost or currenthost.

Example: To shut down the default host (localhost) in the horizontal scaled group:

Command line: **sm -shutdown**

Example: To shut down a specific host (not the default host) in the horizontal scaled group:

Command line: **sm -shutdown -host:myserver1**

Example: To shut down all hosts in the horizontal scaled group:

Command line: **sm -shutdown -group**

Example: To restrict Service Manager processes to administrators only on the local host.

Command line: **sm -quiesce:1**

Example: To restrict Service Manager processes for all users in the entire cluster.

Command line: **sm -quiesce:2 -group**

Example: To bring the system back to running mode (unquiesce). All users can now connect to the processes.

Command line: **sm -quiesce:0 -group**

Related topics

Servlet implementation
System parameters
Enter a parameter in the sm.ini file
Connection parameters

Servlet parameters
Startup parameter: quiesce

2.15.12. Parameter: httpPort

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

httpPort

Description

This parameter defines the communications port that a servlet container process uses to communicate with clients using HTTP. A servlet container process can only have one HTTP port open at a time.

It is best practice to use this parameter from the command line or configuration file for each individual servlet container process you start. This practice makes it easy to identify the HTTP communications port each servlet container uses.

Valid if set from

Server's operating system command prompt

Start up file (sm.cfg or smstart)

Requires restart of the Service Manager server?

No

Default value

None

Possible values

Any valid communications port number

Example usage

Command line: **sm -httpPort:13081**

Related topics

Servlet implementation
System parameters
Enter a parameter in the sm.ini file
Connection parameters
Servlet parameters

2.15.13. Parameter: httpsPort

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

httpsPort

Description

This parameter defines the communications port that a servlet container process uses to communicate with clients using HTTPS (SSL-encrypted HTTP). A servlet container process can only have one HTTPS port open at a time. Servlet container processes can only use an HTTPS communications port if the *sslConnector* parameter is enabled. This parameter requires the use of the *sslConnector* parameter.

It is best practice to use this parameter from the command line or configuration file for each individual servlet container process you start. This practice makes it easy to identify the HTTPS communications port each servlet container uses.

Valid if set from

Server's operating system command prompt

Start up file (sm.cfg or smstart)

Requires restart of the Service Manager server?

No

Default value

None

Possible values

Any valid communications port number

Example usage

Command line: **sm -httpsPort:13081**

Related topics

Servlet implementation
System parameters
Enter a parameter in the sm.ini file
Connection parameters
Servlet parameters

2.15.14. Parameter: licensefile

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

licensefile

Description

This parameter defines the path to the AutoPass license file containing the list of enabled modules and features for your Service Manager server.

Tip

Tip If you need to renew your license, you can use the updatelicense command to load a new license file into your existing one.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

Service Manager installation directory/RUN/LicFile.txt

Possible values

Path and file name of the AutoPass license file.

In the command line usage, if there is a space in the path name, you need to enclose path names in quotation marks.

Example usage

Command line: **sm -httpPort:13080 -httpsPort:13081 -licensefile:"c:/CommonFiles/LicFile.txt"**

Initialization file: `licensefile:c:/CommonFiles/LicFile.txt`

Related topics

System parameters
Enter a parameter in the sm.ini file
Parameter: updatelicense

2.15.15. Parameter: loadBalancer

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

loadBalancer

Description

This parameter creates a load balancer process that listens to incoming client requests on the communications port specified by the httpPort parameter. The load balancer process forwards client connection requests to an available thread on a servlet container process. This parameter requires at least one servlet container process to which to forward client requests. The servlet container process can be on the local Service Manager system or on another system that's part of the same virtual group.

Valid if set from

Server's operating system command prompt

Start up file (`sm.cfg` or `smstart`)

Requires restart of the Service Manager server?

No

Default value

None

Possible values

None

Example usage

Command line: **sm -loadBalancer -httpPort:13080**

Related topics

Servlet implementation
System parameters
Enter a parameter in the sm.ini file
Connection parameters
Servlet parameters

2.15.16. Startup parameter: ntservice

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

ntservice

Description

This parameter defines the name of the Windows service you want to run Service Manager. You can set this parameter from the Configure server utility.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Configure server utility

Requires restart of the Service Manager server?

Yes

Default value

None

Possible values

Name of the Windows service

Example usage

Command line: **sm -httpPort:13080 -ntservice: Service Manager server**

Initialization file: `ntservice: Service Manager server`

Related topics

System parameters
Enter a parameter in the sm.ini file
Connection parameters

2.15.17. Parameter: pid

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

pid

Description

This parameter terminates a Service Manager process with the given process ID. This parameter must be used with the **-shutdown** command.

Valid if set from

Server's operating system command prompt

Requires restart of the Service Manager server?

No

Default value

None

Possible values

The ID of the process to terminate

Example usage

Command line: **sm –shutdown –pid:12345**

Related topics

System parameters
Enter a parameter in the sm.ini file
Connection parameters

2.15.18. Parameter: preferredFQHN

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

preferredFQHN

Description

This parameter allows you to specify the fully qualified host name you want Service Manager clients to use when communicating with the server. You only need to set this parameter if your Service Manager host is identified by multiple names on the network.

During startup, the Service Manager process determines the host name and registers it with the load balancer for later use. The load balancer process manages and routes internal client connections to the host name, which is specified as `http://hostname:httpPort`. When the load balancer receives a client request, it redirects the client to the previously determined host name of the process.

This parameter applies to both horizontally-scaled and vertically-scaled environments. In a horizontally-scaled environment, this parameter specifies the preferred host name of the local system for client connections. For example, your system manages two concurrent client connections, prodhost.corp.com and prodhost.emea.corp.net. You need the load balancer process to manage and route internal client connections through prodhost.emea.corp.net. Set the parameter in the initialization file as follows:

`preferredFQHN:prodhost.emea.corp.net`

Valid if set from

Server's operating system command prompt

Start up file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

None

Possible values

None

Example usage

Command line: **preferredFQHN:prodhost.emea.corp.net**

Related topics

Servlet implementation
System parameters
Enter a parameter in the sm.ini file
Parameter: loadBalancer
Connection parameters
Servlet parameters

2.15.19. Startup parameter: quiesce

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

quiesce

Description

This parameter sets login restrictions to prevent users from logging on to Service Manager processes. This gives System Administrators a way to stop users from logging on to Service Manager processes, and wait for existing users to gracefully log off before starting system maintenance or testing tailoring activities.

There are three levels of login restrictions. Quiesce level 1 restricts all users, except System Administrators, from logging on to Service Manager processes. Quiesce level 2 restricts all users from logging on to Service Manager processes. Quiesce level 0 (zero) sets Service Manager processes to allow user logins. In the vertical scaled or horizontal scaled environment, Service Manager loadBalancer doesn't forward any client connection requests to Service Manager processes that are in quiesce level 1 or 2. If System Administrators want to connect to a Service Manager process in quiesce level 1, they must connect directly to the Service Manager process without connecting through Service Manager loadBalancer.

For detailed information on when quiesce mode can be set and how logins are restricted in a vertical scaled environment and a horizontal scaled environment, see the related topics.

> **Note**
>
> Note If the "-group" or "-host" option is not specified, the quiesce command applies to all Service Manager processes on the local host.

Valid if set from

Server's operating system command prompt

Requires restart of the Service Manager server?

No

Default value

0

Possible values

0 (Allows all users, no restrictions)

1 (Restricts all user logins, except System administrators)

2 (Restricts all user logins)

Example usage

Example 1: The System Administrator runs the following command to set quiesce mode 1 on all Service Manager processes on the local host. All users are restricted from these processes, except the System Administrator.

Command line: **sm -quiesce:1**

Example 2: The System Administrator runs the following command to set quiesce mode 1 on the host whose IP address is 15.28.113.121 in the horizontal scaled group.

Command line: **sm -quiesce:1 -host:15.28.113.121**

Example 3: The System Administrator runs the following command to set quiesce mode 2 for all Service Manager processes in the horizontal scaled group. This restricts all users from accessing Service Manager processes in the entire cluster.

Command line: **sm -quiesce:2 -group**

Example 4: The System Administrator has completed system maintenance and runs the following command to bring the system back to non-quiesce mode. All Service Manager processes in the horizontal scaled group that were quiesced can now accept user logins.

Command line: **sm -quiesce:0 -group**

Example 5: The System Administrator runs the following command to set Service Manager processes on host 15.28.113.121 to non-quiesce mode. Service Manager processes on host 15.28.113.121 now accept user logins.

Command line: **sm -quiesce:0 -host:15.28.113.121**

Related topics

System quiesce: Login restrictions
Enter a parameter in the sm.ini file
Connection parameters

2.15.20. Parameter: reportlbstatus

Parameter

reportlbstatus

Description

This parameter displays information about available ports and threads in horizontal and vertical scaling servlet implementations. You can capture the output of this report in a text file using standard output (*stdout*) parameters

appropriate to your operating system.

The following example shows connection statistics for a vertical scaling servlet implementation on one host.

Note

Note

- This example report has been reformatted for print display.
- The output in this example has been truncated (this is not the full output from the command).

```
Load Balancer Status:Sat Jun 23 09:58:52 PDT 2012
HP Service Manager LoadBalancer Running on Host:myhost.myserver.com Port:34567
List of Hosts:
HostName: myhost.myserver.com
-------------------------------------------------------------------Server Instances
ProcessID     ClusterAddress     HttpPort  HttpsPort  Sessions  DbgMode  QMode  LB
7872     15.178.177.72:33285    34568       0       (3/50)      N       N     N
7881     15.178.177.72:33283    34569       0       (1/50)      N       N     N
7837     15.178.177.72:33278    34567       0       (0/50)      N       N     Y
-------------------------------------------------
State         LowMem     JAVA_USED/MAX/PERCENT
WR[6d22h39m]  N       (4404408/238616576/1.8458097)
QR[9m]        N       (2983152/238616576/1.2501864)
RUN           N       (2601080/238616576/1.0900668)
-----------------------------------------------------------Non Server Instances---
ProcessID     ClusterAddress         State      LowMem     JAVA_USED/MAX/PERCENT
8510     15.178.177.72:33329       RUN          N         (0/89522176/0.0)
7931     15.178.177.72:33299       RUN          N   (1502256/59703296/2.5162027)
7880     15.178.177.72:33280       RST          N   (1539248/59703296/2.5781624)
7930     15.178.177.72:33291       SHT          N   (1514024/59703296/2.5359137)
--------------------------------------
Command Line parameters
-reportlbstatus
-sync -log:../logs/sm.sync.log
system.start -log:../logs/sm.start.log
```

The State column lists one of the following values:

State	Description
RUN	The process is currently running.
WR[<time>]	The process is in restart waiting interval. The value in brackets is the remaining waiting interval expressed in d days, h hours, and m minutes.
QR[<time>]	The process is in quiesce mode prior to restarting. The value in brackets is the time remaining until the process restarts expressed in d days, h hours, and m minutes.
RST	The process is currently restarting.
SHT	The process is currently shutting down.

The report output will appear in the log directory with a file name of sm.log_datetime.log, where datetime is in the format mmddyyhhmmss and represent the time at which the command was submitted.

Valid if set from

Server's operating system command prompt

Requires restart of the Service Manager server?

Yes

Default value

None

Possible values

The number of seconds in which the report should execute again

Example usage

Command line: **sm -reportlbstatus status.txt**

Related topics

Monitoring native heap and Java heap memory
Servlet implementation
System parameters
Enter a parameter in the sm.ini file
Connection parameters
Servlet parameters

2.15.21. Parameter: requestviaproxy

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

requestviaproxy

Description

This parameter specifies if the Service Manager Windows client is allowed to connect to the server through a proxy. Enable this parameter only if, for security reasons, your production environment requires each Windows client to connect to the server through a proxy. This functionality works as described in the following:

- When **requestviaproxy** is enabled, the Windows client can connect to the server either with or without a proxy.
- When **requestviaproxy** is disabled, the Windows client can connect to the server successfully only if the client is not behind a proxy. An error message pops up when the Windows client is attempting to connect to the server through a proxy.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

0 (Disabled)

Possible values

0 (Disabled)

1 (Enabled)

Example usage

Command line: **sm -httpPort:13080 -requestviaproxy:1**

Initialization file: `requestviaproxy:1`

Related topics

Enter a parameter in the sm.ini file
Connection parameters
Servlet parameters
Startup parameter: quiesce

2.15.22. Parameter: scauto

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

scauto

Description

This parameter specifies the SCAuto service name.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

12690

Possible values

A user-provided service name when starting "scautod".

If this CR is not specified, the system uses 12690.

Example usage

Command line: **sm –scauto:myservice**

Initialization file: `scauto:myservice`

Related topics

System parameters
Enter a parameter in the sm.ini file
Connection parameters

2.15.23. Parameter: scautolistener

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

scautolistener

Description

Defining this parameter causes the Service Manager server to listen for connections requests from the legacy SCAuto integration on the communications port specified.

Important: The SCAuto listener can't accept connection requests from Windows or web clients. You must start a client listener to manage client connections.

Valid if set from

Server's operating system command prompt

Requires restart of the Service Manager server?

Yes

Default value

12690

Possible values

Communications port number

Example usage

Command line: **sm -scautolistener:12690**

Related topics

System parameters
Enter a parameter in the sm.ini file
Connection parameters

2.15.24. Startup parameter: shutdown

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

shutdown

Description

Use this parameter at the operating system command prompt to perform the following actions:

- Immediately shut down the Service Manager server.
- Shut down a specific Service Manager process by specifying the hostname of the Service Manager server and system process (PID) that you want to shut down.

Note

Note If the host is not specified, the default is localhost.

Warning: Using the *shutdown* parameter immediately shuts down the Service Manager server without any warning to connected users and could result in data loss or corruption. We recommend you first quiesce a system before shutting it down.

Valid if set from

Server's operating system command prompt

Requires restart of the Service Manager server?

No

Default value

None

Possible values

None

Example usage

Shut down the local Service Manager server.

Command line: **sm -shutdown**

Shut down a specific Service Manager process.

Command line: **sm -shutdown -host:myhost.myserver.com -PID:1234**

Related topics

System parameters
Enter a parameter in the sm.ini file
Connection parameters

2.15.25. Parameter: sslConnector

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

sslConnector

Description

This parameter defines whether servlet container processes have an HTTPS (SSL-encrypted HTTP) communications port available. A servlet container process can only have one HTTPS port open at a time. Servlet container processes can only use an HTTPS communications port if the sslConnector parameter is enabled. This parameter requires the use of the httpsPort parameter.

It is best practice to place this parameter in the Service Manager initialization file so that you enable or disable the HTTPS port for all servlet containers on the same system.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

1

Possible values

0 (Disable)

1 (Enable)

Example usage

Initialization file: `-sslConnector:0`

Related topics

Servlet implementation
System parameters
Enter a parameter in the sm.ini file
Connection parameters
Servlet parameters

2.15.26. Startup parameter: system

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

system

Description

This parameter defines a numerical identifier for the Service Manager system that becomes part of the unique identifer in a horizontal scaling implementation. This parameter is typically set to the communications port that the Service Manager server uses. You can set this parameter from the Configure server utility.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Configure server utility

Requires restart of the Service Manager server?

Yes

Default value

12670

Possible values

Port number

Example usage

Command line: **sm -httpPort:13080 -system:13080**

Initialization file: `system:13080`

Related topics

System parameters
Enter a parameter in the sm.ini file
Connection parameters

2.15.27. Parameter: threadTerminateWait

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

threadTerminateWait

Description

This parameter defines the length of time (in seconds) the Service Manager server continues to handle a client request after the client session is closed (for example, the client session is manually disconnected or is timed out). When the specified time is reached, the server terminates the server thread that handles the client request.

If you specify a value of 0 (zero), the server immediately terminates the server thread when the client session is closed (which is the same behavior as in versions 9.32 through 9.33).

This parameter gives the server an extra period of time to continue handling requests to help prevent "Session already invalidated" errors from occurring. It works on requests from all clients: Windows, web, and web services.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No (Only requires a client re-login)

Default value

3 (seconds)

Possible values

Number of seconds no less than zero

Example usage

Command line: **threadTerminateWait:24**

Initialization file: *threadTerminateWait:24*

Rlated topics

System parameters
Enter a parameter in the sm.ini file
Connection parameters

2.15.28. Parameter: updatelicense

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

updatelicense

Description

This command automatically loads a new AutoPass license file into the existing one, which is specified in the AutoPass licensing.

> **Note**
>
> Note The new license file must be in .txt format.

Valid if set from

Server's operating system command prompt

Requires restart of the Service Manager server?

Yes

Default value

None

Possible values

Path and file name of an AutoPass license file.

In the command line usage, if there is a space in the path name, you need to enclose path names in quotation marks.

Example usage

Command line: **sm -updatelicense:"c:/CommonFiles/NewLicFile.txt"**

Related topics

System parameters
Parameter: licensefile

2.15.29. Parameter: useHostinWSDL

Parameter

useHostinWSDL

Description

When this parameter is enabled, the Service Manager server replaces the host name in the schema location in the webservice response with the host name in the web service request URL.

This parameter is required for web service requests when multiple HWLBs are located before the Service Manager RTE.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

None

Possible values

None

Example usage

Initialization file: `useHostinWSDL`

Command line: **sm -useHostinWSDL**

Related topics

System parameters
Web tier parameters
Enter a parameter in the web client URL
Enter a parameter in the web.xml file
Client parameters for web clients
SSL parameters

2.15.30. Parameter: useIPinWSDL

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

useIPinWSDL

Description

Web Service Description Language (WSDL) is used to connect to external applications, such as Asset Manager. By default, Service Manager builds a WSDL endpoint location URL with a fully qualified domain name (host name). If the web services client application can't reach the Service Manager server with the host name, you can specify the *useIPinWSDL* parameter in the sm.ini file. Service Manager then automatically gets the IP address from the local host, and uses that IP address instead of the host domain name in the WSDL endpoint location URL.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

0

Possible values

0 (Disable)

1 (Enable)

Example usage

Initialization file: `useIPinWSDL:1`

Related topics

System parameters
Enter a parameter in the sm.ini file
Connection parameters

2.15.31. Parameter: ws_endpoint

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

ws_endpoint

Description

This parameter specifies the web services end point. This parameter overwrites *useIPInWSDL* and *preferredFQHN*.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

None

Possible values

https://<lbhost>:<lbport>

or

http://<lbhost>:<lbport>

For example, **https://<FQDN of the HWLB virtual server>:<Port of HWLB virtual server>**/SM/7/ws/ChangeManagement.wsdl. In this example, the *ws_endpoint* parameter refers to the boldface part.

Example usage

Command line: **sm -ws_endpoint:https://mysite.com:13080**

Initialization file: `ws_endpoint:https://mysite.com:13080`

Related topics

System parameters
Enter a parameter in sm.cfg or the server's OS command prompt
Parameter: preferredFQHN
Connection parameters

2.15.32. Startup parameter: webservices_sessiontimeout

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

webservices_sessiontimeout

Description

This parameter defines the number of seconds that the server waits for a WebService API client request before the server assumes that the client session has timed out and closes the connection. If a Service Manager connection is kept alive, a value that's 30 seconds shorter than the value of the connectionTimeout parameter is automatically applied for the timeout limit of the web service session. Otherwise, the value set in the webservices_sessiontimeout parameter is applied.

Note

Note If a Service Manager connection is kept alive, the minimum timeout limit (15 seconds) is then applied for the web service session when the timeout limit for the Service Manager connection is less than 45 seconds.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

1800

Possible values

For example, 3600

Example usage

Command line: **sm -httpPort:13080 -webservices_sessiontimeout:1800**

Initialization file: `webservices_sessiontimeout:1800`

Related topics

System parameters
Enter a parameter in the sm.ini file
Client parameters for Windows clients
Startup parameter: connectionTimeout

2.15.33. Startup parameter: checkExtaccess

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

checkExtaccess

Description

This parameter enables field checking so that only the fields defined in the extaccess record can be used in SOAP requests.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

0

Possible values

0 (Disable)

1 (Enable)

Example usage

Command line: **sm -checkExtaccess:1**

Initialization file: `checkExtaccess:1`

Related topics

System parameters
Enter a parameter in the sm.ini file

2.15.34. Startup parameter: connectionTimeout

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

connectionTimeout

Description

This parameter defines the number of milliseconds that the server waits for a Service Manager client request before the server assumes that the client connection has timed out and closes the connection.

Note

- The timeout limit for the connectionTimeout parameter should be longer than that for the webservices_sessiontimeout parameter. Otherwise, the client session could not be reused. It is recommended that you set the value of the connectionTimeout parameter no less than 60000 milliseconds.
- A value that's 30 seconds shorter than the value of the connectionTimeout parameter is automatically applied for the timeout limit of a web service if the Service Manager connection is kept alive.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

60000

Possible values

At least 45000

Example usage

Command line: **sm -httpPort:13080 -connectionTimeout:80000**

Initialization file: `connectionTimeout:80000`

Related topics

System parameters
Enter a parameter in the sm.ini file
Client parameters for Windows clients

Startup parameter: webservices_sessiontimeout

2.15.35. Parameter: smartemailTimeout

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

smartemailTimeout

Description

This parameter defines the number of milliseconds that the Smart Email process waits for a connection to the mail server to be established. When the defined timeout value is reached, the Smart Email process stops trying to connect to the mail server until the next time when the Service Manager Smart Email scheduler starts again.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

45000

Possible values

Number of milliseconds

Example usage

Command line: **sm -smartemailTimeout:30000**

Initialization file: `smartemailTimeout:30000`

Related topics

System parameters
Enter a parameter in the sm.ini file
Client parameters for Windows clients
Startup parameter: webservices_sessiontimeout

2.16. Database parameters

The following table lists the startup parameters that you can set from the Service Manager server's operating system command prompt or from the Service Manager initialization file (sm.ini).

These parameters determine how the Service Manager server manages the RDBMS database systems.

Startup parameter	Brief description
Startup parameter: allowunsupporteddbversion	Specifies whether the unsupported versions of the Oracle database are allowed when Service Manager connects to the database server
Parameter: dbcachequery	Creates a cache for a comma separated list of table names.
generateIndexDDL	Generates the DDL of indexes and constraints for your database in a DDL file: <Server installation directory>/RUN/ddls/index.ddl.
SQL parameters: oracle	Creates a section header in the initialization file for connection parameters to an Oracle database
SQL parameters: oracle_replicate	Creates a section header in the initialization file for connection parameters to a replicated Oracle database
SQL parameters: pgschema	Specifies the schema when you use a PostgreSQL database
SQL parameters: postgresql	Creates a section header in the initialization file for connection parameters to a PostgreSQL database
SQL parameters: sqlserver	Creates a section header in the initialization file for connection parameters to a Microsoft SQL Server database
SQL parameters: sqlserver_replicate	Creates a section header in the initialization file for connection parameters to a replicated Microsoft SQL Server database
SQL parameters: sqladhoccachesize	Limits the memory consumption of the ad hoc SQL cache
Parameter: sqlalwayson	Enables SQL Server AwaysOn when Service Manager is running on a SQL Server database cluster.
SQL parameters: sqlautosort	Causes the Service Manager server to automatically sort query results based on any keys within the query
SQL parameters: sqlbatchcount	Defines the number of records the Service Manager server retrieves from the RDBMS during a single select call
Parameter: sqlcitext	Enables case-insensitive when you use a PostgreSQL database
SQL parameters: sqldb	Defines the host name of the RDBMS to which you want the Service Manager server to connect
Parameter: sqlddldirectory	Defines the path where the Service Manager server saves DDL files
Parameter: sqlddllogging	Enables creation of DDL files when database dictionary records change
SQL parameters: sqldebug	Enables the Service Manager server to write information about RDBMS connections to the sm.log file
SQL parameters: sqldictionary	Defines the name of the RDBMS where the Service Manager database dictionary resides

SQL parameters: sqldictkey	Defines the primary key of the RDBMS column where the Service Manager database dictionary resides
SQL parameters: sqldicttable	Defines the name of the RDBMS table where the Service Manager database dictionary resides
SQL parameters: sqldirect	Determines how the Service Manager server formats direct SQL commands to an RDBMS using the SQLexecute RAD function
SQL parameters: sqlfetchrows	Determines the number of rows Service Manager retrieves during one connection to the RDBMS
SQL parameters: sqlfetchs	Determines the number of primary keys that Service Manager retrieves during one connection to the RDBMS
SQL parameters: sqljoinsok	Specifies a comma-delimited list of tables for which outer joins are allowed
SQL parameters: sqllibrary	Defines the file name of library containing your RDBMS drivers
SQL parameters: sqllimit	Defines the time limit for requests to the RDBMS
SQL parameters: sqllockretry	Defines the maximum number of retries the Service Manager server makes attempting to lock a system resource
SQL parameters: sqllockwait	Defines the time in seconds that the Service Manager server waits between attempts to lock a system resource
SQL parameters: sqllogin	Defines the user name and password that Service Manager uses to authenticate connections to the RDBMS
SQL parameters: sqlloginretry	Defines the maximum number of times that the Service Manager server retries to authenticate a connection to the RDBMS
SQL parameters: sqlloginwait	Defines the time in seconds that the Service Manager server waits to connect to the RDBMS
SQL parameters: sqlnullclause	Allows using nulls first clause in SQL statements.
SQL parameters: sqlouterjoins	Provides options on how two tables are joined together when being queried
SQL parameters: sqlreuseablesql	Enables the RDBMS to cache query results for use later on Oracle databases
SQL parameter: sqlstats	Logs statistics from an Oracle database in the sm.log file
SQL parameters: sqltextdateformat	Defines the format that the Service Manager server uses to write out dates to strings
SQL parameter: sqltimeoutlimit	When the query execution time for expert search or advanced filter is longer than the time set in the `sqltimeoutlimit` parameter, the query will be terminated in Oracle server.

SQL parameters: sqltz	Defines the time zone that the Service Manager server uses to control storing of date and time values in the RDBMS
Parameter: sqlupper	Converts new tables in a case-sensitive Oracle database to use UPPER indexes and enables a case-insensitive search behavior in the case-sensitive Oracle database
Parameter: system_addconstraint	Adds a Not Null constraint to the first unique key or converts the unique key to a primary key for the specified database tables, depending on usage
Parameter: system_createupperindex	Converts specified tables in an Oracle database to use UPPER indexes
Startup parameter: unlockdatabase	Allows cleaning up the database lock record, and then starting the Service Manager system on a backup server in a vertical scaled implementation.
Parameter: wildcardcharacters	Defines the wildcard characters that you want the Service Manager server to use during searches
Startup parameter: alignradsensitivitywithdb	Specifies whether RAD case sensitivity for string comparing is aligned with that of the Service Manager database.

Note

Note The database parameters in the following list are section-dependent. For example, if you specify `sqldebug:1` in the [sqlserver] section but not in the [sqlserver_replicate] section, then sqldebug is enabled only in database connections to [sqlserver].

- `sqldb`
- `sqllogin`
- `sqllibrary`
- `sqldebug`
- `sqlfetchs`
- `sqlbatchcount`
- `sqlfetchrows`
- `sqlfilejoins`
- `sqllockretry`
- `sqllockwait`
- `sqllimit`
- `sqlloginretry`
- `sqlloginwait`
- `sqlnullclause`
- `sqlreuseablesql`
- `sqlouterjoins`
- `sqlstats`
- `sqlalwayson`
- `sqlcitext`

Related topics

System parameters
Enter a parameter in the sm.ini file

2.16.1. alignradsensitivitywithdb

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

alignradsensitivitywithdb

Description

This parameter specifies whether RAD case sensitivity for string comparing is aligned with that of the Service Manager database. By default, this parameter is disabled (value set to 0).

Valid if set from

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

0

Possible values

0 or1

Example usage

Initialization file: `alignradsensitivitywithdb:0`

Related topics

System parameters
Enter a parameter in the sm.ini file
Database parameters

2.16.2. allowunsupporteddbversion

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt. You can always set a startup parameter from the server's operating system command prompt.

Parameter

allowunsupporteddbversion

Description

This parameter specifies whether unsupported versions of Oracle Database are allowed or not when Service Manager connects to the Oracle Database server. Service Manager connects to the Oracle Database and prints the "Connection to dbtype" message, Service Manager checks the Oracle server and client versions. If the Oracle Database version is unsupported and the *allowunsupporteddbversion* parameter is configured to "0," Service Manager disconnects the Oracle Database server.

Note

Note

This parameter applies to the Oracle Database server only. You can't use this parameter to allow unsupported versions of the Oracle database client.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

0

Possible values

0: Do not allow unsupported versions of the Oracle Database server.

1: Allow unsupported versions of the Oracle Database server.

Example usage

Command line: **sm -allowunsupporteddbversion:0**

Initialization file: `allowunsupporteddbversion:0`

Related topics

System parameters
Enter a parameter in the sm.ini file
Database parameters

2.16.3. dbcachequery

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

dbcachequery

Description

This parameter creates a cache for a comma separated list of table names.

> **Note**
>
> Note It is recommended to use this parameter for dbdicts with only static contents. Additionally, when the query results include more than 1,000 records, caching is disabled automatically.

Valid if set from

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

None

Possible values

Any valid file names

Example usage

Initialization file: `dbcachequery:extaccess`

Related topics

System parameters
Enter a parameter in the sm.ini file

2.16.4. generateIndexDDL

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

generateIndexDDL

Description

This command enables the Service Manager server to generate the DDL of indexes and constraints for your database and save the DDL to the following file: <Server installation directory>/RUN/ddls/index.ddl.

You can use this command to identify which indexes and constraints are needed for your database.

Caution

Caution This command doesn't work when sqlupper:1 is specified.

Valid if set from

Server's operating system command prompt

Requires restart of the Service Manager server?

No

Default value

None

Possible values

None

Example usage

```
sm -generateIndexDDL
```

Related topics

Oracle case-sensitivity
Enter a parameter in the sm.ini file

2.16.5. hidedberrorfromui

Parameter

hidedberrorfromui

Description

This parameter determines whether to hide the database-related error messages in sm.log. You can set this parameter to the following values:

- 1: Hide the database-related error messages in sm.log.
- 0: Display the database-related error messages in sm.log.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

1

Possible values

0 (Disabled)

1 (Enabled)

Example usage

Command line: **sm -httpPort:13080 -hidedberrorfromui:1**

Initialization file: `hidedberrorfromui:1`

2.16.6. [oracle]

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

[oracle]

Description

This parameter creates a section header in the initialization file for information about an Oracle™ database. You only need to provide this parameter if you have set *sqldictionary* and are using an Oracle database.

Valid if set from

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

None

Possible values

None

Example usage

Initialization file: `[oracle]`

Related topics

System parameters
Enter a parameter in the sm.ini file
Database parameters

2.16.7. [oracle_replicate

Error in getting page content from source.

2.16.8. pgschema

Error in getting page content from source.

2.16.9. [postgresql]

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

[postgresql]

Description

This parameter creates a section header in the initialization file for information about a PostgreSQL database. You only need to provide this parameter if you have set **sqldictionary** and are using a PostgreSQL database.

Valid if set from

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

None

Possible values

None

Example usage

Initialization file: `[postgresql]`

Related topics

System parameters
Enter a parameter in the sm.ini file
Database parameters

2.16.10. [sqlserver]

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

[sqlserver]

Description

This parameter creates a section header in the initialization file for information about an Microsoft™ SQL Server database. You only need to provide this parameter if you have set **sqldictionary** and are using an SQL Server database.

Valid if set from

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

None

Possible values

None

Example usage

Initialization file: `[sqlserver]`

Related topics

System parameters
Enter a parameter in the sm.ini file
Database parameters

2.16.11. [sqlserver_replicate

Error in getting page content from source.

2.16.12. sqladhoccachesize

Error in getting page content from source.

2.16.13. sqlalwayson

Parameter

sqlalwayson

Description

This parameter is used to enable SQL Server AwaysOn when Service Manager is running on a SQL Server database cluster. You can set this parameter to the following values:

- 1: SQL Server AlwaysOn is enabled. The Service Manager server turns on SQL_COPT_SS_MULTISUBNET_FAILOVER and sets SQL_COPT_SS_APPLICATION_INTENT as ReadWrite.
- 0: SQL Server AlwaysOn is disabled.

For more information, see Support of SQL Server AlwaysOn.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

0

Possible values

0 (Disabled)

1 (Enabled)

Example usage

Command line: **sm -httpPort:13080 -sqlalwayson:1**

Initialization file: `sqlalwayson:1`

2.16.14. sqlautosort

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

sqlautosort

Description

Enabling this parameter causes the Service Manager server to automatically sort query results based on any keys within the query.

The following table describes different system behaviors based on the value of this parameter.

Value	Description
1	Enable
0 or empty	Disable **Note** Note By default, this value is not specified and a value of "0" is used.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

0 (Disable)

Possible values

0 (Disable)

1 (Enable)

Example usage

Command line: **sm -httpPort:13080 -sqldb:dbserver -sqlautosort:1**

Initialization file: `sqlautosort:1`

Related topics

System parameters
Enter a parameter in the sm.ini file
Database parameters

2.16.15. sqlbatchcount

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

sqlbatchcount

Description

This parameter defines the number of records the Service Manager server retrieves from the RDBMS during a single select call. You can use this parameter to reduce the number of interactions between Service Manager and your RDBMS. Rather than selecting and returning each record from a record list individually, Service Manager can select and return a batch of records in a single *SELECT* statement.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

64

Possible values

Number of records

Example usage

Command line: **sm -httpPort:13080 -sqldb:dbserver -sqlbatchcount:72**

Initialization file: `sqlbatchcount:72`

Related topics

System parameters
Enter a parameter in the sm.ini file
Database parameters

2.16.16. sqlcitext

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

sqlcitext

Description

By default, Service Manager doesn't support case-insensitive searching when you use the PostgreSQL database. If you want to use case-insensitive searching, set this parameter to sqlcitext:1. Because this parameter takes effect when initializing the database, if you need to change the setting, you must load the SM data again after change the setting.

Valid if set from

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

0

Possible values

0 (Disabled)

1 (Enabled)

Example usage

Initialization file: `sqlcitext:1`

2.16.17. sqldb

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

sqldb

Description

This parameter defines the host name of the RDBMS to which you want the Service Manager server to connect.

- For Oracle this is the Network Service Name in tnsnames.ora. You can also enter a connection string to specify a particular connection driver.
- For SQL Server this is the ODBC DSN name.
- For PostgreSQL, this includes the PostgreSQL host, port, and database name.

The format is as follows:
`host=<db_server_host> port=<db_port> dbname=<db_name>`
Or
`//<db_server_host>:<db_port>/<db_name>`

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

None

Possible values

Host name of RDBMS

Example usage

Command line: **sm -httpPort:13080 -sqldb:dbserver**

Initialization file: `sqldb:dbserver`

Tip

Tip Example for PostgreSQL: `sqldb:host=192.0.2.10 port=5432 dbname=dbserver`

Related topics

System parameters
Enter a parameter in the sm.ini file
Database parameters

2.16.18. sqlddldirectory

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter[edit]

sqlddldirectory

Description[edit]

This parameter defines the path where you want the Service Manager server to save database definition language (DDL) files with every change to a database dictionary record. You can enable DDL logging to keep a record of all your database dictionary changes or as a means to deliver proposed changes to an RDBMS administrator who will actually implement the changes. This parameter requires the use of the *sqlddllogging* parameter to enable the creation of DDL files.

Valid if set from[edit]

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?[edit]

Yes

Default value[edit]

Service Manager server installation folder\RUN

Possible values[edit]

Path to DDL files relative to the server's RUN directory

Example usage[edit]

Command line: **sm -sqlddllogging:1 -sqllddldirectory:ddl**

Initialization file:

```
sqlddllogging:1
sqllddldirectory:ddl
```

Contents

- 1 Parameter
- 2 Description
- 3 Valid if set from
- 4 Requires restart of the Service Manager server?
- 5 Default value
- 6 Possible values
- 7 Example usage

2.16.19. sqlddllogging

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

sqlddllogging

Description

This parameter defines whether the Service Manager server produces database definition language (DDL) files with every change to a database dictionary record. You can enable DDL logging to keep a record of all your database dictionary changes or as a means to deliver proposed changes to an RDBMS administrator who will actually implement the changes. This parameter requires the use of the *sqlddldirectory* parameter to define the path where DDL files will be written.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

0

Possible values

0 (Disable)

1 (Enable)

Example usage

Command line: **sm -sqlddllogging:1 -sqlddldirectory:ddl**

Initialization file:

```
sqlddllogging:1
sqlddldirectory:ddl
```

Related topics

Database dictionary
Database dictionary records
Enter a parameter in the sm.ini file
Database parameters

2.16.20. sqldebug

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Note

Note:
If you use an Oracle database, this parameter requires the following additional privileges for $sm_db_user:

```
grant select on sys.col$ to $sm_db_user;
grant select on sys.obj$ to $sm_db_user;
grant select on sys.icol$ to $sm_db_user;
grant select on sys.ind$ to $sm_db_user;
grant select on sys.all_users to $sm_db_user;
```

Parameter

sqldebug

Description

This parameter causes the Service Manager server to write information about RDBMS connections to the sm.log file. If enabled, the server writes the time to login to the RDBMS (*sqllogintime*) and the time it takes to perform a query request (*sqlquerytime*).

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

0

Possible values

0 (Disable)

1 (Enable)

Example usage

Command line: **sm -httpPort:13080 -sqldb:dbserver -sqldebug:1**

Initialization file: `sqldebug:1`

Related topics

System parameters
Enter a parameter in the sm.ini file
Database parameters

2.16.21. sqldictionary

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

sqldictionary

Description

This parameter defines the name of the RDBMS where the Service Manager database dictionary file resides.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

None

Possible values

Name of RDBMS

Example usage

Command line: **sm -httpPort:13080 -sqldb:dbserver -sqldictionary:dbserver**

Initialization file: `sqldictionary:dbserver`

Related topics

System parameters
Enter a parameter in the sm.ini file
Database parameters

2.16.22. sqldictkey

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

sqldictkey

Description

This parameter defines the primary key of the RDBMS column where the Service Manager database dictionary resides. You only need to specify this parameter if you are using the *sqldictionary* parameter and you are also using a primary key other than the **name** field.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

name

Possible values

Primary key of the RDBMS column containing the `name` field of the database dictionary

Example usage

Command line: **sm -httpPort:13080 -sqldb:dbserver -sqldictionary:dbserver -sqldictkey:id**

Initialization file: `sqldictkey:id`

Related topics

System parameters
Enter a parameter in the sm.ini file
Database parameters

2.16.23. sqldicttable

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

sqldicttable

Description

This parameter defines the name of the RDBMS table where the Service Manager database dictionary resides. You only need to specify this parameter if you are using the *sqldictionary* parameter and you are also using a table other than dbdictm1.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

dbdictm1

Possible values

Name of the RDBMS table that contains the database dictionary

Example usage

Command line: **sm -httpPort:13080 -sqldb:dbserver -sqldictionary:dbserver -sqldicttable:dbdict**

Initialization file: `sqldicttable:dbdict`

Related topics

System parameters
Enter a parameter in the sm.ini file
Database parameters

2.16.24. sqldirect

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

sqldirect

Description

This parameter determines where the Service Manager server sends a direct SQL command to an RDBMS by using the SQLexecute RAD command panel. The database type you specify here must match the RDBMS database type that's set up as a section in your sm.ini file.

The Service Manager server uses shared libraries to interface with any supported RDBMS client. For example, the SQORACLE.OCI.10.dll shared library on Windows enables communication with Oracle 11 and 12 clients. When Service Manager opens a file that's mapped to an RDBMS, the Service Manager SQL layer automatically loads the appropriate shared library into memory.

The SQL Query tool also allows you to query tables that aren't Service Manager files. To do this, add the *sqldirect:<dbtype>* parameter to the sm.ini file. Set the dbtype to the name of an existing SQL section in your sm.ini file. For example for an Oracle system, add the line `sqldirect:oracle` to the sm.ini file. This loads the appropriate .DLL file for the backend RDBMS and enables direct SQL queries at any time.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

If this parameter is not set, but a RAD panel tries to execute a SQL Statement through SQLexecute, the statement will be sent to the RDBMS that was used last by the RAD (for a query, or an update, and so on.) Therefore if the whole file system is mapped to one and only one RDBMS, that's the RDBMS sqldirect will use.

Possible values

```
sqldirect:oracle11
[oracle]
sqldb:SomeOraDB
sqllogin:sm/sm
```

Example usage

Command line: **sm -httpPort:13080 -sqldb:<dbserver> -sqldirect:<oracle>**

Initialization file:

```
sqldirect:oracle11
[oracle]
sqldb:SomeOraDB
sqllogin:sm/sm
[oracle11]
sqldb:SomeOtherOracleDB
sqllogin:sm/sm
```

Related topics

System parameters
Enter a parameter in the sm.ini file
Software Support
Database parameters

2.16.25. sqlfetchrows

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

sqlfetchrows

Description

This parameter determines the number of rows the Service Manager server retrieves during one connection to the RDBMS. You can only use this parameter with an Oracle OCI RDBMS. The server converts any value greater than 500 to the maximum limit of 500 rows.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

0

Possible values

Number of rows to fetch up to 500

Example usage

Command line: **sm -httpPort:13080 -sqldb:dbserver -sqlfetchrows:250**

Initialization file: `sqlfetchrows:250`

Related topics

System parameters
Enter a parameter in the sm.ini file
Database parameters

2.16.26. sqlfetchs

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

sqlfetchs

Description

This parameter determines the number of primary keys Service Manager retrieves during one connection to the RDBMS.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

500

Possible values

Number of primary keys to fetch

Example usage

Command line: **sm -httpPort:13080 -sqldb:dbserver -sqlfetchs:750**

Initialization file: `sqlfetchs:750`

Related topics

System parameters
Enter a parameter in the sm.ini file
Database parameters

2.16.27. sqljoinsok

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

sqljoinsok

Description

The values for this parameter comprise a comma-delimited list of tables for which outer joins are allowed.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

None

Possible values

A comma-delimited list of table names

Example usage

Command line: **sm -httpPort:13080 -sqldb:dbserver -sqljoinsok:problem, incident, contact, operator**

Initialization file: `sqljoinsok:problem, incidents, contacts, operator`

Related topics

System parameters
Enter a parameter in the sm.ini file
Database parameters

2.16.28. sqllibrary

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

sqllibrary

Description

This parameter defines the file name of the library containing your RDBMS drivers. Service Manager provides a collection of database drivers for supported vendors and versions.

Valid if set from

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

One of the following RDBMS drivers:

Oracle

- Windows – SQORACLE.OCI10.DLL
- Linux – libsqoracle19.so

Microsoft SQL Server

- Windows – SQMSSQL.ODBC.DLL

Possible values

File name of library containing RDBMS drivers:

> **Note**
>
> Note Driver names are case-sensitive. Be sure to use the correct case.

Windows drivers

- SQDB2.CLI.DLL
- SQMSSQL.ODBC.DLL
- SQORACLE.OCI10.DLL

Linux drivers

- libsqoracle19.so

Example usage

If you are running on the Linux system and have the Oracle 19 client, the initialization file entry is as follows:

```
-sqllibrary:libsqoracle19.so
```

Related topics

System parameters
Enter a parameter in the sm.ini file
Database parameters

2.16.29. sqllimit

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

sqllimit

Description

If a particular request exceeds the time limit defined by this parameter, then the Service Manager server writes a message to the Service Manager log file documenting the name of the user making the request, how long it took, the actual SQL query, and the information of the database on which the query was running slow.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

5

Possible values

Number of seconds

Example usage

Command line: **sm -httpPort:13080 -sqldb:dbserver -sqllimit:10**

Initialization file: `sqllimit:10`

Related topics

System parameters
Enter a parameter in the sm.ini file
Database parameters

2.16.30. sqllockretry

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

sqllockretry

Description

This parameter defines the maximum number of retries the Service Manager server makes attempting to lock a system resource. If a particular request exceeds the retry limit defined by this parameter, then the Service Manager server writes a message to the Service Manager log file documenting the name of the user making the request, the number of lock attempts made by the request, and the actual SQL query.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

5

Possible values

Number of retries

Example usage

Command line: **sm -httpPort:13080 -sqldb:dbserver -sqllockretry:7**

Initialization file: `sqllockretry:7`

Related topics

System parameters
Enter a parameter in the sm.ini file
Database parameters

2.16.31. sqllockwait

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

sqllockwait

Description

This parameter defines the time in seconds the Service Manager server waits between attempts to lock a system resource.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

0

Possible values

Number of seconds

Example usage

Command line: **sm -httpPort:13080 -sqldb:dbserver -sqllockwait:2**

Initialization file: `sqllockwait:2`

Related topics

System parameters
Enter a parameter in the sm.ini file
Database parameters

2.16.32. sqllogin

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

sqllogin

Description

This parameter defines the user name and password that Service Manager uses to authenticate connections to the RDBMS. You must use a slash character to separate the user name and password. If you omit this parameter, then the server attempts to authenticate the connection using the user name and password of the user who started the Service Manager server, however this feature requires the Service Manager server and the RDBMS server to use the same operating system. If the Service Manager server and the RDBMS server use different operating systems, then you must specify a *sqllogin* value.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

None

Possible values

User name/password

Example usage

Command line: **sm -httpPort:13080 -sqldb:dbserver -sqllogin:dbadmin/password**

Initialization file: `sqllogin:dbadmin/password`

Related topics

System parameters
Enter a parameter in the sm.ini file
Database parameters

2.16.33. sqlloginretry

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

sqlloginretry

Description

This parameter defines the maximum number of retries the Service Manager server makes attempting to authenticate a connection to the RDBMS. If a particular request exceeds the retry limit defined by this parameter, then the Service Manager server writes a message to the Service Manager log file documenting the name of the user making the request and the number of retry attempts made by the request.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

3

Possible values

Number of retries

Example usage

Command line: **sm -httpPort:13080 -sqldb:dbserver -sqlloginretry:3**

Initialization file: `sqlloginretry:3`

Related topics

System parameters
Enter a parameter in the sm.ini file
Database parameters

2.16.34. sqlloginwait

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

sqlloginwait

Description

This parameter defines the time in seconds that the Service Manager server waits to connect to the RDBMS.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

15

Possible values

Number of seconds

Example usage

Command line: **sm -httpPort:13080 -sqldb:dbserver -sqlloginwait:30**

Initialization file: `sqlloginwait:30`

Related topics

System parameters
Enter a parameter in the sm.ini file
Database parameters

2.16.35. sqlnullclause

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

sqlnullclause

Description

This parameter allows to use nulls first clause in SQL statements.

Valid if set from

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

0

Possible values

0 or 1

Example usage

Initialization file: `sqlnullclause:1`

Related topics

System parameters
Enter a parameter in the sm.ini file
Database parameters

2.16.36. sqlouterjoins

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

sqlouterjoins

Description

When querying two tables at the same time, Service Manager by default generates a SELECT statement with a JOIN clause to join the tables together. This parameter provides options on how two tables are joined together when being queried.

Caution: In some cases, generating a LEFT OUTER JOIN when querying two tables at the same time causes performance degradation.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

2

Possible values

0 (Generates a LEFT OUTER JOIN clause only when querying a multi-row array table and one other table in the same query. Otherwise, generates just a JOIN clause.)

1 (Always generates a LEFT OUTER JOIN clause when querying two tables in the same query.)

2 (Never generates a LEFT OUTER JOIN clause; always generates just a JOIN clause.)

Example usage

Using option 2 (default) to generate a JOIN clause when querying tables M1 and A3 (SELECT * FROM M1 JOIN A3 ON...).

Command line: **sm -httpPort:13080 -sqldb:dbserver -sqlouterjoins:2**

Initialization file: `sqlouterjoins:2`

Related topics

System parameters
Enter a parameter in the sm.ini file
Database parameters

2.16.37. sqlreuseablesql

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

sqlreuseablesql

Description

This parameter is only valid for Oracle databases. When this parameter is enabled, the RDBMS sends SQL statements with bind variables instead of in-text-values (“literals”). It enables the RDBMS to cache the parsed SQL statement for use later to

reduce the number of parses; it also enables the RDBMS to prepare an execution plan for the statement only once and then reuse the plan.

Using bind variables not only significantly reduces the Oracle shared memory used, but also prevents SQL injection attacks.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

1

Possible values

0 (Disable)

1 (Enable)

Example usage

Command line: **sm -httpPort:13080 -sqldb:dbserver -sqlreuseablesql:1**

Initialization file: `sqlreuseablesql:1`

Related topics

System parameters
Enter a parameter in the sm.ini file
Database parameters

2.16.38. sqlstats

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

sqlstats

Description

This parameter allows statistics from an Oracle database to log in the sm.log file.

Valid if set from

Initialization file (sm.ini). This parameter is only used when using an Oracle database.

Requires restart of the Service Manager server?

Yes

Default value

0 (Disabled)

Possible values

0 (disabled)
1 (enabled)

Example usage

Initialization file: `sqlstats:1`

Command line: **sm -httpPort:13080 –sqlstats:1**

Related topics

System parameters
Enter a parameter in the sm.ini file

Database parameters

2.16.39. sqltextdateformat

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

sqltextdateformat

Description

You may find the `unable to convert value of field xxx from type STRING2 to type ARRAY.` error message in the sm.log file when both of the following conditions are met:

- The format of the input date/time data as a character string doesn't match the current user's date/time format setting
- The system needs to convert the string to an array field of Service Manager

To solve this problem, use the *sqltextdateformat* parameter to specify the date/time format of this input string.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

None

Possible values

Date formats

- YYYY/MM/DD
- YY/MM/DD
- MM/DD/YYYY
- MM/DD/YY
- DD/MM/YYYY
- DD/MM/YY

Example usage

Command line: **sm -httpPort:13080 -sqldb:dbserver -sqltextdateformat:MM/DD/YYYY**

Initialization file: `sqltextdateformat:MM/DD/YYYY`

Note

Note You should only use this parameter prior to conversion. Using it or changing it doesn't work for a converted system.

Related topics

System parameters
Enter a parameter in the sm.ini file
Database parameters

2.16.40. sqltimeoutlimit

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

`sqltimeoutlimit`

Description

When you executes a query, for example, search for an interaction or an incident, the query can take a long time. If the time to process the query exceeds the session expiration time, the system will automatically log you out. So there needs to be a way to stop long queries. For example, stop the query if the query takes more than X seconds.

The `sqltimeoutlimit` parameter is introduced for this issue, which is implemented on expert search and advanced filter. When the query execution time for expert search or advanced filter is longer than the time set in the `sqltimeoutlimit` parameter, the query will be terminated in Oracle server and display the following error message:
`Query terminated: Your query execution time exceeded the time set in the query timeout configuration.`

Prerequisites:

- The feature is only applicable to Oracle Database.
- Oracle Instance Client version must be 19.3 or above.

Valid if set from

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

0

Possible values

- 0, which means this function is disabled.
- At least 30 (seconds). If you set the parameter to a value less than 30, the system will override this value and set it to 30 instead.

Example usage

Initialization file: `sqltimeoutlimit:60`
In this example, if your query execution time is longer than 60 seconds, the query will be terminated automatically.

2.16.41. sqltz

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

sqltz

Description

This is an optional parameter that the Service Manager server uses to control storing of date and time values in the RDBMS. This parameter defines the time zone to use as a base for all date and time values. The time zone is specified as the name of the time zone record in the Service Manager tzfile. The default time zone is Greenwich/Universal (GMT).

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

Greenwich/Universal (GMT)

Possible values

Time zone name as defined in Service Manager time zone record (tzfile)

Example usage

Command line: **sm -httpPort:13080 -sqldb:dbserver -sqltz:Canada/Atlantic**

Initialization file: `sqltz:Canada/Atlantic`

Related topics

System parameters
Enter a parameter in the sm.ini file
Database parameters

2.16.42. sqlupper

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

sqlupper

Description

This parameter enables the Service Manager server to convert new tables in an Oracle database to use UPPER indexes and to implement case-insensitive search behavior in Service Manager. This feature can implement case-insensitive search behavior on a case-sensitive Oracle database, and therefore eliminates the need to use a case-insensitive Oracle database which might slow down your system performance.

In sqlupper mode, the records are divided into groups according to their names when you use grouping. If the names of two records differ only in the case of the letter, then the records will be divided into different groups. For example, the "application" records and "Application" records in the database are divided to the "Application" group and the "application" group. To get a better view, you can unify the record names as "application" or "Application" in the database.

Caution

Caution We recommend using Oracle's case-insensitive solution instead of this parameter. For more information, see Oracle case-sensitivity and Prepare an Oracle database.

To use this feature in an existing Service Manager system, follow these steps:

1. Make sure your Oracle database is case-sensitive. If not, convert your Oracle database from case-insensitive to case-sensitive.

   Note

   Note To make Service Manager treat your Oracle database as case-sensitive, make sure you have not NLS_SORT=BINARY_CI and NLS_COMP=LINGUISTIC configured in your Oracle. For example, you can configure NLS_SORT=BINARY and NLS_COMP=BINARY to make it case-sensitive. For more information, see Oracle case-sensitivity.

2. Make sure all Service Manager server nodes are stopped.
3. Add sqlupper:1 to the sm.ini file of each server node.
4. In a server's command prompt, run the **system_createupperindex:all** command to convert all existing tables to use UPPER indexes.

   Note

   Note Every time after you run the **system_createupperindex** command, a message appears in the console that indicates if any tables aren't successfully converted.

5. If any tables fail to be converted, run the **system_createupperindex:**<*list of tables, separated by a comma*> command to convert these tables again.

> **Note**
> Note You can run this command multiple times as needed until all tables are successfully converted.

6. Restart the server nodes.

To use this feature in a new Service Manager system, follow these steps:

1. Prepare a case-sensitive Oracle database.
2. Install the Service Manager server.
3. Add sqlupper:1 to the sm.ini file.
4. Run the server configuration utility to connect to the database.
5. Load the Service Manager out-of-box tables and demonstration data.
   All tables are loaded with UPPER indexes. All new tables that you may create later will also use UPPER indexes.

Valid if set from

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

0

Possible values

0 (Disabled)

1 (Enabled)

Example usage

Initialization file: `sqlupper:1`

Related topics

Oracle case-sensitivity
Enter a parameter in the sm.ini file
Parameter: system_createupperindex

2.16.43. system_addconstraint

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

system_addconstraint

Description

The *system_addconstraint* parameter adds a Not Null constraint to the first unique key or converts the unique key to a primary key for the specified database tables, depending on usage. The *system_addconstraint* parameter takes the following two arguments:

- The target tables: Specify "all" to update all tables or specify a list of tables separated by commas. If no tables are specified, running this command will don'thing but bring up the parameter help.
- Behavior:
  - `0` adds a "Not Null" constraint on the first unique key of the specified tables (if no unique key or primary key exists, Service Manager adds the constraint to the RECORD_KEY field).
  - `1` converts the first unique key of the specified tables to primary keys (if there is no unique or primary key, Service Manager converts the RECORD_KEYfield to primary key) .
  - `2` converts the first unique key of the specified tables to primary keys by using the full-table copy mechanism. Running this command may take a very long time.

Note

Note This parameter can only be set from an operating system command prompt.

Valid if set from

Server's operating system command prompt

Requires restart of the Service Manager server?

You should run this command only when the Service Manager server is shutdown.

Default value

0 (By default, this parameter doesn'thing.)

Possible values

`all:0` - Adds a "Not Null" constraint to the first unique key of all tables in the system.

`all:1` - Converts the first unique key of all tables in the system to a primary key.

`all:2` - Force a full table copy to convert the first unique key of all tables to a primary key in case you can't convert the first unique key to a primary key in an MS SQL Server database.

[A comma separated list of tables]:0 - Adds a "Not Null" constraint only to the first unique key of the specified tables.

[A comma separated list of tables]:1 - Converts the first unique key of the specified tables to primary keys.

[A comma separated list of tables]:2 - Converts the first unique key of the specified tables to primary keys by using the full-table copy mechanism. Running this command may take a very long time..

Example usage

Command line: **sm -system_addconstraint:all:1**
Command line: **sm -system_addconstraint:probsummary,incidents:0**
Command line: **sm -system_addconstraint:operator:2**

Related topics

System parameters

Working with primary keys
Enter a parameter in the sm.ini file

2.16.44. system_createupperindex

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

system_createupperindex

Description

This parameter enables the Service Manager server to convert specified tables in an Oracle database to use UPPER indexes.

Note

Note When running this command, error messages that resemble the following may occur, which can be safely ignored:

`3952( 4916) 09/18/2014 11:48:15 RTE E could not find key addl.parents in ocml`

In this example, the key field (addl.parents) is an array, and mapped to a separate table on SQL. The error message occurs because the index can't be generated on the main table (OCMLM1). The index will be generated later on the new table (OCMLA1).

Valid if set from

Server's operating system command prompt

Requires restart of the Service Manager server?

Yes

Default value

None

Possible values

`all` (Convert all tables in the Oracle database)

list of table names separated by a comma (Convert specified tables in the Oracle database)

Example usage

```
sm -system_createupperindex:all
```

```
sm -system_createupperindex:cm3r,cm3t
```

Related topics

Oracle case-sensitivity
Enter a parameter in the sm.ini file

2.16.45. wildcardcharacters

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

wildcardcharacters

Description

This parameter defines the wildcard characters you want the Service Manager server to use during searches.

Valid if set from

Server's operating system command prompt

Initialization (sm.ini) file

Requires restart of the Service Manager server?

No

Default value

*?\

where * defines the place holder for any 0 to N characters

where ? defines the place holder for any one character

where \ defines the escape character to search literally for any of the wildcard characters

Possible values

These characters are only used in queries using the LIKE operator. For example: "*al?n*" matches "alen"

These characters aren't used when using the = or # operator. For example:

- name="*al?n*" would only match "*al?n*"
- name#"*al?n*" would match "*al?n*" and "*al?n*e" [starts with]

Note

Note When Service Manager generates a query from what is typed on the search screen, the data is scanned for these characters and if found, a LIKE operator is generated automatically instead of the default "starts with" (#) operator.

Some wildcard character search examples are listed below:

- Wildcard character "*" to search for any number of characters. For example: "*ev*" would match "kevin" but not "kendel"
- Wildcard character "?" to search for a single character. For example, "l?n" would match "alen" and "alan" but not "alain"
- Wildcard character "\" to search for text containing special or wildcard characters. For example: "Help\?" would match "Help?" but not "Help!"

Example usage

Command line: **sm -listener -wildcardcharacters:%?/**

Initialization file: `wildcardcharacters:%?/`

Related topics

System parameters
Enter a parameter in the sm.ini file

2.17. Debugging parameters

The following table lists the startup parameters that you can set from the Service Manager server's operating system command prompt or from the Service Manager initialization file (sm.ini).

These parameters determine how the Service Manager server manages the debugging environment.

Startup parameter	Description
Parameter: corepath	Defines the path to the core file generated by a system crash
Parameter: dbmonitorfiles	Causes the Service Manager server to monitor the table you specify and write database information in the log file
Parameter: dbstats	Causes the Service Manager server to gather database usage statistics
Parameter: dbtriggertrace	Causes the Service Manager server to monitor the tables you specify and to write the indicated level of trigger information to the log file
Parameter: debugattachments	Causes the Service Manager server to write detailed attachment information to the log file
Parameter: debugadhocsql	Enables adhocsql debugging messages.
Parameter: debugca	Causes the Service Manager server to write cache information at the indicated logging level to the log file
Parameter: debugdbquery	Causes the Service Manager server to write database queries, timings, and results information to the log file
Parameter: debugdbtypes	Causes the Service Manager server to perform data type checking during SQL mapping
Parameter: debugdiagnostics	Enables or disables the Diagnostic Service
Parameter: debugfileio	Causes the Service Manager server to write detailed debugging information about file input and output to the log file
Parameter: debughttp	Causes the Service Manager server to write HTTP SOAP requests and responses to the log file
Parameter: debugjni	Provides detailed debugging in the Java Native Interface implementation.
Parameter: debugjavascript	Causes the Service Manager server to write JavaScrip debugging messages to the log file
Parameter: debuglk	Causes the Service Manager server to write detailed debugging messages about locks to the log file
Parameter: debugnode	Enables a servlet container process to join a Service Manager virtual group without the load balancer process assigning any client requests to the servlet container.
Parameter: debugprocesses	Causes the Service Manager server to write process creation and termination messages to the log file
Parameter: debugrest	Enables the Service Manager server to write detailed log trace to the server log for RESTful web services diagnostics
Parameter: debugrs	Causes the Service Manager server to write detailed messages about resource locks to the log file

Parameter: debugscauto	Causes the Service Manager server to write detailed messages about SCAuto connections to the log file
Parameter: debugshutdown	Causes the Service Manager server to write shutdown messages to the log file
Parameter: debugstartup	Causes the Service Manager server to write start up messages to the log file
Parameter: dryrun	Enables the Service Manager administrators to run a load test script or test case to assess the usage of shared memory and process memory for a typical user
Parameter: log4jDebug	Defines the location of the log file that's used for java debugging
Parameter: log4jError	Specifies a list of comma-delimited packages to set the log level to Error.
Parameter: logdebuglevel	Defines the log level of the JRTE codes.
Parameter: rtm	Causes the Service Manager server to write Response Time Monitor (RTM) performance statistics to the log file
Parameter: enablecoredump	Enables Service Manager to log any information generated by a system crash
Parameter: debugvmmap	Causes the Service Manager server to print a map of the virtual memory should the server fail to attach to shared memory
Parameter: tracememerror	Enables the Service Manager to print memory trace information and generate coredump when the memory is handled incorrectly

Related topics

System parameters
Enter a parameter in the sm.ini file

2.17.1. Parameter: corepath

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

corepath

Description

This parameter defines the path to the core file that's generated by a system crash. The operating system of the Service Manager server determines which core file is generated, if any, during a system crash. Core file names are specific to the operating system.

For example:

Linux: Core.<*process_id*>

Windows: sm.<*process_id*>_<*thread_id*>.mini.dmp

Note

Note

- In Linux, this parameter is available only when the GCORE command is available.
- You must set the value of this parameter to a location that already exists. Service Manager will not create the directory structure in which to write the core file.

Valid if set from

Server's operating system command prompt

Initialization (sm.ini) file

Requires restart of the Service Manager server?

No

Default value

The Service Manager Server RUN directory

Possible values

Path to the operating system core file

Example usage

Command line: **sm -enablecoredump:1 -corepath:/sm/corefiles/**

Initialization file (two lines):

```
enablecoredump:1
corepath:/sm/corefiles/
```

Related topics

System parameters
Enter a parameter in the sm.ini file
Debugging parameters
Parameter: enablecoredump

2.17.2. Parameter: dbmonitorfiles

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

dbmonitorfiles

Description

This parameter causes the Service Manager server to monitor the table you specify and to record the following information in the log file when an update occurs:

- The application that'ssued the update request
- The list of records updated
- The user who initiated the update

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

None

Possible values

Comma separated list of table names to monitor

Example usage

Command line: **sm -httpPort:13080 -dbmonitorfiles:problem,probsummary**

Initialization file: `dbmonitorfiles:problem,probsummary`

Related topics

System parameters
Enter a parameter in the sm.ini file
Debugging parameters

2.17.3. Parameter: dbstats

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

dbstats

Description

This parameter causes the Service Manager server to gather database usage statistics.

Valid if set from

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

0

Possible values

0, 1, 2

Example usage

Set `dbstats:2` in the sm.ini file to display the statistics better.

Related topics

System parameters
Enter a parameter in the sm.ini file
Debugging parameters

2.17.4. Parameter: dbtriggertrace

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

dbtriggertrace

Description

This parameter causes the Service Manager server to monitor the tables you specify and write the indicated level of trigger information to the log file.

Valid if set from

Server's operating system command prompt

Initialization (sm.ini) file

Requires restart of the Service Manager server?

Yes

Default value

None

Possible values

Comma separated list including tracing level (1, 2, 3, or 4) followed by table names to monitor.

For example: *dbtriggertrace:<level1>,<file1>,<file2>,<file3>*

Example usage

Command line: **sm -httpPort:13080 -dbtriggertrace:3,probsummary,cm3r,scmessage**

Initialization file: `dbtriggertrace:3,probsummary,cm3r,scmessage`

Related topics

System parameters
Enter a parameter in the sm.ini file
Debugging parameters

2.17.5. Parameter: debugadhocsql

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

debugadhocsql

Description

This parameter enables adhocsql debugging messages.

Valid if set from

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

None

Possible values

No value

Example usage

Initialization file: `debugadhocsql`

Related topics

System parameters
Enter a parameter in the sm.ini file
Debugging parameters

2.17.6. Parameter: debugattachments

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

debugattachments

Description

This parameter causes the Service Manager server to record detailed attachment information in the log file.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

None

Possible values

None

Example usage

Command line: **sm -httpPort:13080 -debugattachments**

Initialization file: `debugattachments`

Related topics

System parameters
Enter a parameter in the sm.ini file
Debugging parameters

2.17.7. Parameter: debugca

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

debugca

Description

This parameter causes the Service Manager server to write cache information at the indicated logging level to the log file.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

None

Possible values

Logging level

- 1 (Minimal)
- 2 (Verbose)

Example usage

Command line: **sm -httpPort:13080 -debugca:1**

Initialization file: `debugca:1`

Related topics

System parameters
Enter a parameter in the sm.ini file
Debugging parameters

2.17.8. Parameter: debugdbquery

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

debugdbquery

Description

This parameter causes the Service Manager server to record database queries, timings, and results information that exceed the time threshold you specify in this parameter value in the log file.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

None

Possible values

Number of seconds or 999 for a full query debug.

- To show all database access, use the following syntax: `debugdbquery:999`.
- To show all queries that exceed *n* number of seconds, use the following syntax: `debugdbquery:n/`.

Example usage

Command line: **sm -httpPort:13080 -debugdbquery**

Initialization file: `debugdbquery`

Output from the debugdbquery parameter writes to the sm.log file. The messages written to the sm.log file contain several fields, each separated by a caret (^) character. The sm.log file is located in the following directory:

`..\..\OpenText\Service Manager\Server\logs`

Sample debugdbquery 999 output to the sm.log file:

```
223 02/05/2007 17:39:51 DBFIND^F^scmessage(Oracle)^1^0.000000^F^0^0.000000^"syslanguage="en"
and class="us" and message.id="1""^ ^0.000000^0.000000 ( [ 0] apm.get.inbox.by.name start )
223 02/05/2007 17:39:51 DBQUERY^F^probsummary(Oracle)^18^1.000000^F^0^0.020000^"hot.tic#true"^
{"category"}^0.000000^0.000000 ( [ 0] sc.manage select )
```

The following table describes the output fields for the database debug query (debugdbquery) parameter.

Field	Description
who	DBFIND or DBQUERY
where	F = foreground or B = background
file	The file name followed by database type: Oracle, SQL server or LDAP, or JOIN. If you add a letter *I* suffix, then the file name is case-insensitive.
key	The number of the selected key.
weight	The calculated weight for the key Service Manager selects for that specific query. For additional information, see the Key selection algorithms Help topic.
keytype	P = Partially keyed, I = IR expert search
record count	The number of records that satisfy the query. The system adds the DBQUERY entry to the log after processing the select panel and then returns the first 128 records that satisfy that query.
seconds till result came back	The amount of time required to satisfy the query.
query	The actual query from the user.
sortfields	The sort order in which the records are requested.
extracttime	The time required to read data records and extract the key values needed for sorting. This is only necessary if a key satisfying the requested sort order doesn't exist.
sorttime	The time required to sort all data records matching the query. This is only necessary if a key satisfying the requested sort order doesn't exist.

Related topics

System parameters
Enter a parameter in the sm.ini file
Debugging parameters

2.17.9. Parameter: debugdbtypes

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

debugdbtypes

Description

This parameter causes the Service Manager server to perform data type checking during SQL mapping. The server writes results to the log file.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

None

Possible values

None

Example usage

Command line: **sm -httpPort:13080 -debugdbtypes**

Initialization file: `debugdbtypes`

Related topics

System parameters
Enter a parameter in the sm.ini file
Debugging parameters

2.17.10. Parameter: debugdiagnostics

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

debugdiagnostics

Description

This parameter enables administrators to disable the Diagnostic Service. The Diagnostic Service when enabled could cause performance issues to large Service Manager systems. This parameter allows system administrators to enable or disable the Diagnostic Service as needed.

> **Note**
>
> Note By default, the Diagnostic Service is disabled. Before running the reportdiagnostics command, administrators must specify debugdiagnostics:1 to enable the Diagnostic Service.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

0

Possible values

0 (Disabled)

1 (Enabled)

Example usage

Command line: **sm -httpPort:13080 -debugdiagnostics:1**

Initialization file: `debugdiagnostics:1`

Related topics

System parameters
Automatic selection of shared memory address
Enter a parameter in the sm.ini file

Debugging parameters
Special parameters: reportdiagnostics

2.17.11. Parameter: debugfileio

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

debugfileio

Description

This parameter causes the Service Manager server to write detailed debugging information about file input and output to the log file.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

None

Possible values

None

Example usage

Command line: **sm -httpPort:13080 -debugfileio**

Initialization file: `debugfileio`

Related topics

System parameters
Enter a parameter in the sm.ini file
Debugging parameters

2.17.12. Parameter: debughttp

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

debughttp

Description

This parameter causes the Service Manager server to write HTTP SOAP requests and responses to the following log files.

- logs\sm.log
- logs\HTTP.log

Important

Important

Enabling this parameter significantly reduces available system resources because the log files the server produces contain all HTTP traffic, including HTTP headers and attachments. For this reason, we recommend that you not enable this parameter on production systems, but rather in test environments only.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

0 (Disable)

Possible values

1 (Enable)

Example usage

Command line: **sm -httpPort:13080 -debughttp:1**

Initialization file: `debughttp:1`

Related topics

System parameters
Enter a parameter in the sm.ini file

2.17.13. Parameter: debugjni

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

debugjni

Description

This parameter provides detailed debugging in the Java Native Interface implementation.

Valid if set from

Start up file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

None

Possible values

None

Example usage

Initialization file: `debugjni`

Related topics

System parameters
Enter a parameter in the sm.ini file
Debugging parameters

2.17.14. Parameter: debugjavascript

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

debugjavascript

Description

This parameter causes the Service Manager server to write JavaScript debugging messages to the log file. Its value can be 1, 2, or 3.

- 1: Prints log information when an object is compiled, created, or disposed.
- 2: In addition to the output of *debugjavascript:1*, prints log information when the garbage collector of the server's internal JavaScript engine is started or stopped.
- 3: Forces the JavaScript engine to run garbage collection before running any JavaScript scripts.

Caution

Caution
This setting is intended only for developers, and must not be used in production environments.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

None

Possible values

1, 2, and 3

Example usage

Command line: **sm -httpPort:13080 -debugjavascript:1**

Initialization file: `debugjavascript:1`

Related topics

System parameters
Enter a parameter in the sm.ini file
Debugging parameters

2.17.15. Parameter: debuglk

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

debuglk

Description

This parameter causes the Service Manager server to write detailed debugging messages about locks to the log file. Service Manager only writes messages about locks that exceed the lock time threshold you specify.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

None

Possible values

Number of milliseconds

Example usage

Command line: **sm -httpPort:13080 -debuglk:1000**

Initialization file: `debuglk:1000`

Related topics

System parameters
Enter a parameter in the sm.ini file
Debugging parameters

2.17.16. Parameter: debugnode

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

debugnode

Description

When a process is started, it registers itself through loadBalancer so that loadBalancer can forward clients to the process. When System Administrators need to debug issues, they can start a process in debug mode to register with loadBalancer that this process is being debugged. LoadBalancer doesn't forward any clients to the process while debugnode is active.

The following table describes the behavior based on the value of this parameter.

Value	Behavior
1 or empty	The process is in debug mode.
0	The process is not in debug mode. **Note** Note By default, this parameter is not specified and value "0" is used.

Caution

Caution Do not add this parameter to the sm.ini file. If you do this, all the sm processes that share the same sm,ini file are set as a debug node.

Valid if set from

Server's operating system command prompt

Requires restart of the Service Manager server?

No

Default value

0 (this is the value used when this parameter is not specified)

Possible values

0, 1, or empty

Example usage

To set a process as a debug node: **sm -httpPort:13081 -debugnode:1** (or **sm -httpPort:13081 -debugnode**)

Related topics

System parameters
Enter a parameter in the sm.ini file
Debugging parameters

2.17.17. Parameter: debugprocesses

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

debugprocesses

Description

This parameter causes the Service Manager server to write process creation and termination messages to the log file.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

None

Possible values

None

Example usage

Command line: **sm -httpPort:13080 -debugprocesses**

Initialization file: `debugprocesses`

Related topics

System parameters
Enter a parameter in the sm.ini file
Debugging parameters

2.17.18. Parameter: debugrest

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

debugrest

Description

This parameter enables the Service Manager server to write detailed log trace for RESTful web services diagnostics. by default, this feature is disabled.

To use this debugging parameter, set it to 1, then restart the Service Manager server and re-run the RESTful web service application.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

0

Possible values

0: Disable

1: Enable

Example usage

Command line: **sm -httpPort:13080 -debugrest:1**

Initialization file: `debugrest:1`

Related topics

System parameters
Enter a parameter in the sm.ini file
Debugging parameters

2.17.19. Parameter: debugrs

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

debugrs

Description

This parameter causes the Service Manager server to write detailed messages about resource locks to the log file. Service Manager only writes messages about resource locks that exceed the lock time threshold you specify. Enabling this parameter can cause your log file to increase in size very quickly. Only do so at the direction of Customer Support.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

None

Possible values

Number of milliseconds

Example usage

Command line: **sm -httpPort:13080 -debugrs:1000**

Initialization file: `debugrs:1000`

Related topics

System parameters
Enter a parameter in the sm.ini file
Debugging parameters

2.17.20. Parameter: debugscauto

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

debugscauto

Description

This parameter causes the Service Manager server to write detailed messages about SCAuto connections to the log file.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

None

Possible values

None

Example usage

Command line: **sm -httpPort:12690 -debugscauto**

Initialization file: `debugscauto`

Related topics

System parameters
Enter a parameter in the sm.ini file
Debugging parameters

2.17.21. Parameter: debugshutdown

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

debugshutdown

Description

This parameter causes the Service Manager server to write shutdown messages to the log file.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

None

Possible values

None

Example usage

Command line: **sm -httpPort:13080 -debugshutdown**

Initialization file: `debugshutdown`

Related topics

System parameters
Enter a parameter in the sm.ini file
Debugging parameters

2.17.22. Parameter: debugstartup

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

debugstartup

Description

This parameter causes the Service Manager server to write start up messages to the log file.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

None

Possible values

None

Example usage

Command line: **sm -httpPort:13080 -debugstartup**

Initialization file: `debugstartup`

Related topics

System parameters
Enter a parameter in the sm.ini file
Debugging parameters

2.17.23. Parameter: debugvmmap

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

debugvmmap

Description

This parameter causes Service Manager servers running on Windows operating systems to list the following information in the sm.log file:

- a list of memory address ranges large enough to contain shared memory
- a map of virtual memory

You can use the list of memory address ranges to specify a value for the *shared_memory_address* parameter. We recommend you only specify a *shared_memory_address* value if Service Manager fails to automatically find one.

You can use the virtual memory map to troubleshoot shared memory problems and to identify a potential address for the *shared_memory_address* parameter. The virtual memory map lists the areas of free and used memory as well as the name of the resource using the address range.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

0

Possible values

0 (Disable)

1 (Enable)

Example usage

Command line: **sm -httpPort:13080 -debugvmmap:1**

Initialization file: `debugvmmap:1`

Example excerpt of available shared_memory_address values

```
2724(  3572) 03/05/2010 11:06:38  RTE I List of all possible shared_memory_address parameter settings for
a shared memory size of 67108864 (64 MB)
2724(  3572) 03/05/2010 11:06:38  RTE I # Free range 0x7FFF0000-0xFFFAD000 size=2147209216 (2047 MB)
2724(  3572) 03/05/2010 11:06:38  RTE I Preferred        : shared_memory_address:0x80000000
2724(  3572) 03/05/2010 11:06:38  RTE I Lowest available : shared_memory_address:0x80000000
2724(  3572) 03/05/2010 11:06:38  RTE I Highest available: shared_memory_address:0xFBFA0000
2724(  3572) 03/05/2010 11:06:38  RTE D # Free range 0x122B5000-0x30000000 size= 500477952 ( 477 MB)
2724(  3572) 03/05/2010 11:06:38  RTE D                   shared_memory_address:0x20000000
2724(  3572) 03/05/2010 11:06:38  RTE D Lowest available : shared_memory_address:0x122D0000
2724(  3572) 03/05/2010 11:06:38  RTE D Highest available: shared_memory_address:0x2C000000
...
2724(  3572) 03/05/2010 11:06:38  RTE D # Free range 0x77DF0000-0x7C360000 size=  72810496 (  69 MB)
2724(  3572) 03/05/2010 11:06:38  RTE D                   shared_memory_address:0x77E00000
2724(  3572) 03/05/2010 11:06:38  RTE D Lowest available : shared_memory_address:0x77E00000
2724(  3572) 03/05/2010 11:06:38  RTE D Highest available: shared_memory_address:0x78360000
2724(  3572) 03/05/2010 11:06:38  RTE D # Free range 0x039B0000-0x10000000 size= 207945728 ( 198 MB)
2724(  3572) 03/05/2010 11:06:38  RTE D                   shared_memory_address:0x039C0000
2724(  3572) 03/05/2010 11:06:38  RTE D Lowest available : shared_memory_address:0x039C0000
2724(  3572) 03/05/2010 11:06:38  RTE D Highest available: shared_memory_address:0x0C000000
2724(  3572) 03/05/2010 11:06:38  RTE W !!! Use the preferred shared_memory_address parameter above
2724(  3572) 03/05/2010 11:06:38  RTE W !!! or remove the shared_memory_address parameter completely.
2724(  3572) 03/05/2010 11:06:38  RTE E HP Service Manager is unable to start. Failed to initialize or
attach to shared memory environment
```

Example excerpt of the virtual memory map

```
2388(  5572) 02/16/2010 15:23:34  RTE D Heap#00 Extension#0 - 0x00580598-0x00680000 (0x000FFA68)
2388(  5572) 02/16/2010 15:23:34  RTE D Heap#01 Extension#0 - 0x00940590-0x00950000 (0x0000FA70)
2388(  5572) 02/16/2010 15:23:34  RTE D Heap#01 Extension#1 - 0x023C0048-0x024C0000 (0x000FFFB8)
2388(  5572) 02/16/2010 15:23:34  RTE D Heap#02 Extension#0 - 0x00B40590-0x00B80000 (0x0003FA70)
2388(  5572) 02/16/2010 15:23:34  RTE D Heap#03 Extension#0 - 0x003D0590-0x003E0000 (0x0000FA70)
2388(  5572) 02/16/2010 15:23:34  RTE D Heap#03 Extension#1 - 0x007B0048-0x008B0000 (0x000FFFB8)
2388(  5572) 02/16/2010 15:23:34  RTE D Heap#04 Extension#0 - 0x02270598-0x02280000 (0x0000FA68)
2388(  5572) 02/16/2010 15:23:34  RTE D Heap#04 Extension#1 - 0x02110050-0x02210000 (0x000FFFB0)
2388(  5572) 02/16/2010 15:23:34  RTE D Heap#05 Extension#0 - 0x023B0590-0x023C0000 (0x0000FA70)
2388(  5572) 02/16/2010 15:23:34  RTE D Heap#05 Extension#1 - 0x02280048-0x02380000 (0x000FFFB8)
2388(  5572) 02/16/2010 15:23:34  RTE D Heap#06 Extension#0 - 0x02600598-0x02640000 (0x0003FA68)
2388(  5572) 02/16/2010 15:23:34  RTE D Heap#07 Extension#0 - 0x02800590-0x02810000 (0x0000FA70)
2388(  5572) 02/16/2010 15:23:34  RTE D Heap#07 Extension#1 - 0x024C0048-0x025C0000 (0x000FFFB8)
2388(  5572) 02/16/2010 15:23:34  RTE D Heap#08 Extension#0 - 0x02770578-0x027B0000 (0x0003FA88)
2388(  5572) 02/16/2010 15:23:34  RTE D Heap#09 Extension#0 - 0x00500580-0x00540000 (0x0003FA80)
2388(  5572) 02/16/2010 15:23:34  RTE D Heap#10 Extension#0 - 0x008F0580-0x00930000 (0x0003FA80)
2388(  5572) 02/16/2010 15:23:34  RTE D Heap#11 Extension#0 - 0x026E0580-0x02720000 (0x0003FA80)
2388(  5572) 02/16/2010 15:23:34  RTE D Heap#12 Extension#0 - 0x029E0578-0x02A20000 (0x0003FA88)
2388(  5572) 02/16/2010 15:23:34  RTE D Heap#13 Extension#0 - 0x02BD0578-0x02C10000 (0x0003FA88)
2388(  5572) 02/16/2010 15:23:34  RTE D Heap#14 Extension#0 - 0x02DD0580-0x02E10000 (0x0003FA80)
2388(  5572) 02/16/2010 15:23:34  RTE D Heap#15 Extension#0 - 0x02880578-0x028C0000 (0x0003FA88)
2388(  5572) 02/16/2010 15:23:34  RTE D Heap#16 Extension#0 - 0x02960578-0x029A0000 (0x0003FA88)
2388(  5572) 02/16/2010 15:23:34  RTE D Heap#17 Extension#0 - 0x02680578-0x026C0000 (0x0003FA88)
2388(  5572) 02/16/2010 15:23:34  RTE D Heap#18 Extension#0 - 0x03000578-0x03040000 (0x0003FA88)
2388(  5572) 02/16/2010 15:23:34  RTE D Heap#19 Extension#0 - 0x035D0590-0x035E0000 (0x0000FA70)
2388(  5572) 02/16/2010 15:23:34  RTE D Heap#20 Extension#0 - 0x02950590-0x02960000 (0x0000FA70)
2388(  5572) 02/16/2010 15:23:34  RTE D Heap#21 Extension#0 - 0x03540598-0x03580000 (0x0003FA68)
2388(  5572) 02/16/2010 15:23:34  RTE D Heap#22 Extension#0 - 0x02840598-0x02880000 (0x0003FA68)
2388(  5572) 02/16/2010 15:23:34  RTE D **Process Memory Map** - User address space extends from
0x00010000 to 0xFFFEFFFF
2388(  5572) 02/16/2010 15:23:34  RTE D   Start      End        (Size      ) Name
2388(  5572) 02/16/2010 15:23:34  RTE D   0x00010000-0x00020000 (0x00010000) rw-   s [UNKNOWN]
2388(  5572) 02/16/2010 15:23:34  RTE D   0x00020000-0x00030000 (0x00010000) rw-   s [UNKNOWN]
2388(  5572) 02/16/2010 15:23:34  RTE D   0x00030000-0x00031000 (0x00001000) r--   s [UNKNOWN]
2388(  5572) 02/16/2010 15:23:34  RTE D   0x00031000-0x00040000 (0x0000F000)         [FREE]
2388(  5572) 02/16/2010 15:23:34  RTE D   0x00040000-0x00041000 (0x00001000) r--   i
\Device\HarddiskVolume2\Windows\System32\apisetschema.dll
2388(  5572) 02/16/2010 15:23:34  RTE D   0x00041000-0x00050000 (0x0000F000)         [FREE]
2388(  5572) 02/16/2010 15:23:34  RTE D   0x00050000-0x00089000 (0x00039000)         [RESERVED]
2388(  5572) 02/16/2010 15:23:34  RTE D   0x00089000-0x0008C000 (0x00003000) rw- g   [UNKNOWN]
2388(  5572) 02/16/2010 15:23:34  RTE D   0x0008C000-0x00090000 (0x00004000) rw-     [UNKNOWN]
2388(  5572) 02/16/2010 15:23:34  RTE D   0x00090000-0x0016D000 (0x000DD000)         *** HP Service
Manager stack: thread#5572 ***
```

Related topics

System parameters
Automatic selection of shared memory address
Enter a parameter in the sm.ini file
Debugging parameters

2.17.24. Parameter: dryrun

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

dryrun

Description

This parameter enables Service Manager administrators to run a load test script or test case to assess the usage of shared memory and process memory for a typical user. You can use this parameter for capacity planning during the development or test phases of implementation. This parameter creates a log file in the server's RUN folder.

Valid if set from

Server's operating system command prompt

Initialization (sm.ini) file

Requires restart of the Service Manager server?

No

Default value

None

Possible values

None

Example usage

Command line: **sm -dryrun -httpPort:13080**

Sample dry run report

```
1572(  5900) 06/20/2008 15:18:11  RTE D ####### Dry Run Report #######
1572(  5900) 06/20/2008 15:18:11  RTE I ------ Shared Memory ------
1572(  5900) 06/20/2008 15:18:11  RTE I
1572(  5900) 06/20/2008 15:18:11  RTE I Shared Memory Release     7.10
1572(  5900) 06/20/2008 15:18:11  RTE I
1572(  5900) 06/20/2008 15:18:11  RTE I Current Size              64000000
1572(  5900) 06/20/2008 15:18:11  RTE I
1572(  5900) 06/20/2008 15:18:11  RTE I Segment Allocation         2886304
1572(  5900) 06/20/2008 15:18:11  RTE I Large Block Allocation     4255744
1572(  5900) 06/20/2008 15:18:11  RTE I
1572(  5900) 06/20/2008 15:18:11  RTE I Unused Space              56857952  (88%)
1572(  5900) 06/20/2008 15:18:11  RTE I Free Space                57071600  (89%)
1572(  5900) 06/20/2008 15:18:11  RTE I
1572(  5900) 06/20/2008 15:18:11  RTE I
1572(  5900) 06/20/2008 15:18:11  RTE I Shared Memory Type  Allocations    Frees     Allocated
1572(  5900) 06/20/2008 15:18:11  RTE I ------------------  ----------- ----------- -----------
1572(  5900) 06/20/2008 15:18:11  RTE I Not named                    11           0        2832
1572(  5900) 06/20/2008 15:18:11  RTE I User blocks                   3           0        6144
1572(  5900) 06/20/2008 15:18:11  RTE I Messages                      0           0           0
1572(  5900) 06/20/2008 15:18:11  RTE I Resource locks                1           0          16
1572(  5900) 06/20/2008 15:18:11  RTE D  Sessions PeakMemoryUsage: 5416720, MemoryUsage: 203581,
MemoryOverhead:11232, TotalAllocations:405289
1572(  5900) 06/20/2008 15:18:11  RTE D  Memory Leak Report is available in file memdebug.1572.5900.log
1572(  5900) 06/20/2008 15:18:11  RTE D ####### End DryRun Report #######
```

Related topics

System parameters
Startup parameter: shutdown
Enter a parameter in the sm.ini file

2.17.25. Parameter: log4jdebug

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt. You can always set a startup parameter from the server's operating system command prompt.

Parameter

log4jDebug:com.hp.ov.sm.common.oom.LowMemoryHandler

Description

This parameter enables certain java packages to be started in debug mode.

Valid if set from

Server's operating system command prompt

Start up file (sm.ini or smstart)

Requires restart of Service Manager server

Yes

Default value

None -- By default none of the java packages will be run in debug mode.

Possible values

com.hp.ov.sm.common.oom.LowMemoryHandler

Example usage

Command line: **sm -httpPort:13080 -log4jDebug:com.hp.ov.sm.common.oom.LowMemoryHandler**

Related topics

Monitoring memory in Service Manager processes
Monitoring native heap and Java heap memory
Logging memory monitoring
Servlet parameters
Debugging parameters

2.17.26. Parameter: log4jError

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

log4jError

Description

This parameter specifies a list of comma-delimited packages to set the log level to Error.

Valid if set from

Server's operating system command prompt

Start up file (sm.ini or smstart)

Requires restart of Service Manager server

No

Default value

Package List.

Possible values

Any comma-delimited Service Manager JRTE class names.

log4jError:com.hp.ov.sm.server.utility.Scheduler,com.hp.ov.sm.server.utility.GenericThreadManager

log4jError:com.hp.ov.sm.server.utility.SoapServer,com.hp.ov.sm.server.appserver.servlet

log4jError:com.hp.ov.sm.common.utility.DAOThread

log4jError:com.hp.ov.sm.common.resource.ResourceBase

Example usage

Command line: **SM -log4jError:com.hp.ov.sm.common.utility.DAOThread**

Initialization file: `log4jError:com.hp.ov.sm.common.utility.DAOThread`

Related topics

System parameters
Servlet parameters
Debugging parameters

2.17.27. Parameter: logdebuglevel

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt. You can always set a startup parameter from the server's operating system command prompt.

Parameter

logdebuglevel

Description

This parameter defines the log level of the JRTE codes. If the value of this parameter equals to or larger than 1, all JVM options are logged in sm.log when the Service Manager server is starting.

Valid if set from

Server's operating system command prompt

Start up file (sm.ini or smstart)

Requires restart of Service Manager server

Yes

Default value

1

Possible values

0 = DEBUG

1 = INFO (default)

2 = WARN

3 = ERROR

4 = FATAL

Example usage

Command line: **SM -logdebuglevel:1**

Initialization file: `logdebuglevel:1`

Related topics

System parameters
Servlet parameters
Debugging parameters

2.17.28. Parameter: rtm

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

rtm

Description

This parameter causes the Service Manager server to write Response Time Monitor (RTM) performance statistics to the log file.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

None

Possible values

Logging level

- 2 (Displays the response and transaction times)
- 3 (Displays the same information as *rtm:2*, and also provides a panel by panel trace of the RAD application)
- 4 (Displays the same information as *rtm:2* and *rtm:3*, and also provides details about the usage of each RAD operator and command panel from the last format response to this one)
- 5 (Displays the same information as *rtm:2* and *rtm:3*, and also provides details about the usage of each RAD operator and command panel from the last panel response to this one)

Example usage

Command line: **sm -httpPort:13080 -rtm:2**

Initialization file: `rtm:2`

Related topics

System parameters
Example: response time monitoring (RTM) output
Enter a parameter in the sm.ini file
Debugging parameters

2.17.29. Parameter: tracememerror

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's OS command prompt.

Parameter

tracememerror

Description

This parameter specifies whether to print memory trace information in the sm.log file and generate coredump when the memory is handled incorrectly.

Valid if set from

Server's operating system command prompt

Initialization (sm.ini) file

Requires restart of the Service Manager server?

No

Default value

0

Possible values

0 (Disable)

1 (Print call stack)

2 (Print call stack and generate coredump)

Example usage

Command line: **sm -tracememerror:1**

Initialization file: `tracememerror:1`

Related topics

System parameters
Enter a parameter in the sm.ini file
Debugging parameters

2.18. Help parameters

These parameters display Service Manager help information.

Help parameter	Description
Parameter: help	Lists Service Manager general help topics.
Parameter: helpalert	Lists help for Service Manager alert parameters.
Parameter: helpall	Lists help for all topics.
Parameter: helpanubis	Lists help for Service Manager Anubis monitor thread options.
Parameter: helpdebug	Lists help for Service Manager debug options.
Parameter: helpemailout	Lists help for Service Manager emailout options.
Parameter: helpgroup	Lists help for Service Manager cluster options.
Parameter: helpir	Lists help for Service Manager IR Expert.
Parameter: helplocale	Lists help for Service Manager international locale options.
Parameter: helpplugin	Lists help for Service Manager plug-in options.
Parameter: helprestart	Lists help for Service Manager process restart options.
Parameter: helpservlet	Lists help for Service Managerservlet mode options.
Parameter: helpshutdown	Lists help for Service Manager shutdown options.
Parameter: helpspecial	Lists help for Service Manager special options.
Parameter: helpsql	Lists help for Service Manager SQL interface startup options.
Parameter: helpssl	Lists help for Service Manager SSL options.
Parameter: helpstart	Lists help for Service Manager start up options.
Parameter: helpthreading	Lists help for Service Manager threading options.

Related topics

System parameters
Enter a parameter in the sm.ini file
Enter a parameter in sm.cfg or the server's OS command prompt

2.18.1. Parameter: help

Help parameters display Service Manager help information. You can always set a help parameter from the server's operating system command prompt.

Parameter

help

Description

This parameter lists Service Manager general help topics.

Valid if set from

Server's operating system command prompt

Requires restart of the Service Manager server?

No

Example usage

Command line: **sm -help**

Related topics

System parameters
Help parameters

2.18.2. Parameter: helpalert

Help parameters display Service Manager help information. You can always set a help parameter from the server's operating system command prompt.

Parameter

helpalert

Description

This parameter lists help for Service Manager alert parameters.

Valid if set from

Server's operating system command prompt

Requires restart of the Service Manager server?

No

Example usage

Command line: **sm -helpalert**

Related topics

System parameters
Help parameters

2.18.3. Parameter: helpall

Help parameters display Service Manager help information. You can always set a help parameter from the server's operating system command prompt.

Parameter

helpall

Description

This parameter lists help for all topics.

Valid if set from

Server's operating system command prompt

Requires restart of the Service Manager server?

No

Example usage

Command line: **sm -helpall**

Related topics

System parameters
Help parameters

2.18.4. Parameter: helpanubis

Help parameters display Service Manager help information. You can always set a help parameter from the server's operating system command prompt.

Parameter

helpanubis

Description

This parameter lists help for Service Manager Anubis monitor thread options.

Valid if set from

Server's operating system command prompt

Requires restart of the Service Manager server?

No

Example usage

Command line: **sm -helpanubis**

Related topics

System parameters
Help parameters

2.18.5. Parameter: helpdebug

Help parameters display Service Manager help information. You can always set a help parameter from the server's operating system command prompt.

Parameter

helpdebug

Description

This parameter lists help for Service Manager debug options.

Valid if set from

Server's operating system command prompt

Requires restart of the Service Manager server?

No

Example usage

Command line: **sm -helpdebug**

Related topics

System parameters
Help parameters

2.18.6. Parameter: helpemailout

Help parameters display Service Manager help information. You can always set a help parameter from the server's operating system command prompt.

Parameter

helpemailout

Description

This parameter lists help for Service Manager emailout options.

Valid if set from

Server's operating system command prompt

Requires restart of the Service Manager server?

No

Example usage

Command line: **sm -helpemailout**

Related topics

System parameters
Help parameters

2.18.7. Parameter: helpgroup

Help parameters display Service Manager help information. You can always set a help parameter from the server's operating system command prompt.

Parameter

helpgroup

Description

This parameter lists help for Service Manager cluster options.

Valid if set from

Server's operating system command prompt

Requires restart of the Service Manager server?

No

Example usage

Command line: **sm -helpgroup**

Related topics

System parameters
Help parameters

2.18.8. Parameter: helpir

Help parameters display Service Manager help information. You can always set a help parameter from the server's operating system command prompt.

Parameter

helpir

Description

This parameter lists help for Service Manager IR Expert.

Valid if set from

Server's operating system command prompt

Requires restart of the Service Manager server?

No

Example usage

Command line: **sm -helpir**

Related topics

System parameters
Help parameters

2.18.9. Parameter: helplocale

Help parameters display Service Manager help information. You can always set a help parameter from the server's operating system command prompt.

Parameter

helplocale

Description

This parameter lists help for Service Manager international locale options.

Valid if set from

Server's operating system command prompt

Requires restart of the Service Manager server?

No

Example usage

Command line: **sm -helplocale**

Related topics

System parameters
Help parameters

2.18.10. Parameter: helpplugin

Help parameters display Service Manager help information. You can always set a help parameter from the server's operating system command prompt.

Parameter

helpplugin

Description

This parameter lists help for Service Manager plug-in options.

Valid if set from

Server's operating system command prompt

Requires restart of the Service Manager server?

No

Example usage

Command line: **sm -helpplugin**

Related topics

System parameters
Help parameters

2.18.11. Parameter: helprestart

Help parameters display Service Manager help information. You can always set a help parameter from the server's operating system command prompt.

Parameter

helprestart

Description

This parameter lists help for Service Manager process restart options.

Valid if set from

Server's operating system command prompt

Requires restart of the Service Manager server?

No

Example usage

Command line: **sm -helprestart**

Related topics

System parameters
Help parameters

2.18.12. Parameter: helpservlet

Help parameters display Service Manager help information. You can always set a help parameter from the server's operating system command prompt.

Parameter

helpservlet

Description

This parameter lists help for Service Manager servlet mode options.

Valid if set from

Server's operating system command prompt

Requires restart of the Service Manager server?

No

Example usage

Command line: **sm -helpservlet**

Related topics

System parameters
Help parameters

2.18.13. Parameter: helpshutdown

Help parameters display Service Manager help information. You can always set a help parameter from the server's operating system command prompt.

Parameter

helpshutdown

Description

This parameter lists help for Service Manager shutdown options.

Valid if set from

Server's operating system command prompt

Requires restart of the Service Manager server?

No

Example usage

Command line: **sm -helpshutdown**

Related topics

System parameters
Help parameters

2.18.14. Parameter: helpspecial

Help parameters display Service Manager help information. You can always set a help parameter from the server's operating system command prompt.

Parameter

helpspecial

Description

This parameter lists help for Service Manager special options.

Valid if set from

Server's operating system command prompt

Requires restart of the Service Manager server?

No

Example usage

Command line: **sm -helpspecial**

Related topics

System parameters
Help parameters

2.18.15. Parameter: helpsql

Help parameters display Service Manager help information. You can always set a help parameter from the server's operating system command prompt.

Parameter

helpsql

Description

This parameter lists help for Service Manager SQL interface startup options.

Valid if set from

Server's operating system command prompt

Requires restart of the Service Manager server?

No

Example usage

Command line: **sm -helpsql**

Related topics

System parameters
Help parameters

2.18.16. Parameter: helpssl

Help parameters display Service Manager help information. You can always set a help parameter from the server's operating system command prompt.

Parameter

helpssl

Description

This parameter lists help for Service Manager SSL options.

Valid if set from

Server's operating system command prompt

Requires restart of the Service Manager server?

No

Example usage

Command line: **sm -helpssl**

Related topics

System parameters
Help parameters

2.18.17. Parameter: helpstart

Help parameters display Service Manager help information. You can always set a help parameter from the server's operating system command prompt.

Parameter

helpstart

Description

This parameter lists help for Service Manager start up options.

Valid if set from

Server's operating system command prompt

Requires restart of the Service Manager server?

No

Example usage

Command line: **sm -helpstart**

Related topics

System parameters
Help parameters

2.18.18. Parameter: helpthreading

Help parameters display Service Manager help information. You can always set a help parameter from the server's operating system command prompt.

Parameter

helpthreading

Description

This parameter lists help for Service Manager threading options.

Valid if set from

Server's operating system command prompt

Requires restart of the Service Manager server?

No

Example usage

Command line: **sm -helpthreading**

Related topics

System parameters
Help parameters

2.19. Information Retrieval (IR) expert parameters

The following table lists the startup parameters you can set from the Service Manager server's OS command prompt or from the Service Manager initialization file (sm.ini).

These parameters determine how Service Manager indexes and retrieves information using IR Expert.

Startup parameter	Brief description
Information Retrieval (IR) expert parameters: ir_disable	Allows you to disable the IR keys on your existing Service Manager system, so that the upgrade process runs faster
Information Retrieval (IR) expert parameters: ir_language	Defines the language of the text that you want Information Retrieval to index
Information Retrieval (IR) expert parameters: ir_pending_secs	Determines the maximum pending period (in seconds) before an IR index gets synchronized after the base table is updated.
Information Retrieval (IR) expert parameters: ir_stoplist	Specifies your own stoplist if you don't want to use the default database stoplist.

Related topics

System parameters
Enter a parameter in the sm.ini file

2.19.1. ir_disable

Parameter

ir_disable

Description

This parameter allows you to disable the IR keys on your existing Service Manager system, so that the upgrade process runs faster.

Note

Note After the application upgrade succeeds, you can enable the IR keys again by removing the `ir_disable:1` entry from the sm.ini file.

The following table describes different system behaviors based on the value of this parameter.

Value	Description
1 or empty	The IR keys are disabled.
0	The IR keys aren't disabled. **Note** Note By default, this parameter is not specified and value "0" is used.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

0

Possible values

1 or empty (Disable the IR keys)

0 (Enable the IR keys)

Example usage

Command line: **sm ir_disable:1**

Initialization file: `ir_disable:1`

Related topics

System parameters
Enter a parameter in the sm.ini file
Information Retrieval (IR) expert parameters

2.19.2. ir_language

Parameter

ir_language

Description

This parameter defines the language of the text that you want Information Retrieval to index.

Note

Note For the detailed information regarding the supported languages, you may refer to the official site for your database.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Note

Note Peform an IR Regen operation after setting or changing this parameter.

Default value

English

Possible values

English

German

Example usage

Command line: **sm -httpPort:13080 -ir_language:German**

Initialization file: `ir_language:German`

Related topics

System parameters
Enter a parameter in the sm.ini file
Information Retrieval (IR) expert parameters

2.19.3. ir_pending_secs

Parameter

ir_pending_secs

Description

This parameter is only for the Oracle database. Use this parameter to control the maximum pending period (in seconds) before an IR index gets synchronized after the base table is updated. The default and minimum value is 3600 seconds. The bigger the value, the later IR search can search out the new text terms. However, a bigger value also means a smaller impact on the memory and other resources of the DB server.

Note

Note After changing this parameter, make sure that you run `sm -upgradeir` for the change to take effect as it only works after IR re-generates the existing indexes, or for newly created IR indexes.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

3600

Possible values

Number of seconds greater than or equal to 3600

Example usage

Command line: **ir_pending_secs:7200**

Initialization file: `ir_pending_secs:7200`

Related topics

System parameters
Enter a parameter in the sm.ini file
Information Retrieval (IR) expert parameters

2.19.4. ir_stoplist

Parameter

ir_stoplist

Description

This parameter specifies your own stoplist if you don't want to use the default database stoplist.

Important

Important:

Start from Service Manager p1 and 9.70 version, this parameter is only available for PostgresSQL databases. As for Oracle and MS SQL Server, you need to follow the instrustions in the Manage IR Stop Words page to migrate your existing stoplist (if you have specified in previous versions) and maintain your own stoplist using the IR Stop List Management tool.

To create or update the PostgresSQL database stoplist, see the following documentation:

https://www.postgresql.org/docs/11/textsearch-dictionaries.html#TEXTSEARCH-STOPWORDS

After you create or update the stoplist, make sure that you run the `sm -upgradeir` command to regenerate IR indexes.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

None

Possible values

The file name of your database stoplist.

Example usage

Command line: **sm - ir_stoplist:mysmstoplist**

Initialization file: `sm - ir_stoplist:mysmstoplist`

Related topics

System parameters
Enter a parameter in the sm.ini file
Information Retrieval (IR) expert parameters

2.20. LDAP parameters

These parameters determine how the server manages connections to an LDAP directory service. You can set these parameters from the Service Manager server's operating system command prompt or from the Service Manager initialization file (sm.ini).

Startup parameter	Brief decription
Startup parameter: ldapauthenticateonly	Requires the Service Manager server to use LDAP for login authentication only
Startup parameter: ldapbinddn	Defines the name that the Service Manager server uses to bind to the LDAP server
Startup parameter: ldapbindpass	Defines the password the Service Manager server uses to bind to the LDAP server
Startup parameter: ldapdisable	Disables the LDAP login authentication procedure
Startup parameter: ldapdisconnect	Specifies when the Service Manager server disconnects from the LDAP server
Startup parameter: ldapmaxrecords	Specifies the maximum number of records to return from an LDAP query
Startup parameter: ldapnosizelimitmsg	Disables the "Size limit exceeded" message from the LDAP server, so that this informational message doesn't display in each client and confuse users
Startup parameter: ldapnostrictlogin	Allows users with valid operator records to log into to Service Manager without also having a valid LDAP record
Startup parameter: ldapquerymapping	Defines LDAP attribute mapping when the system builds the LDAP query during login
Startup parameter: ldapsearchscope	Specifies whether to search the LDAP base directory only or the LDAP base directory and subtrees
Startup parameter: ldapserver	Defines any back up LDAP servers that the Service Manager server can connect to if the primary LDAP server is unavailable
Startup parameter: ldapsslallownocert	When this parameter is enabled, the RTE doesn't check a CA's root certification. In this situation, you don't need to set a root certificate for the CA that'ssued the LDAP server's certificate on the Service Manager server in order to enable LDAP over SSL in Service Manager
Startup parameter: ldapstats	Causes Service Manager to record LDAP query statistics in the sm.log file.
Startup parameter: ldaptimelimit	Defines the maximum number of seconds that Service Manager waits for a response to an LDAP query.

Related topics

System parameters
Enter a parameter in the sm.ini file

2.20.1. Startup parameter: ldapauthenticateonly

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

ldapauthenticateonly

Description

This parameter requires the Service Manager server to use LDAP for login authentication only.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

None

Possible values

None

Example usage

Command line: **sm -httpPort:13080 -ldapserver1:ldapparent -ldapauthenticateonly**

Initialization file: `ldapauthenticateonly`

Related topics

System parameters
Enter a parameter in the sm.ini file
LDAP parameters

2.20.2. Startup parameter: ldapbinddn

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

ldapbinddn

Description

This parameter defines the name the Service Manager server uses to bind to the LDAP server.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

None

Possible values

LDAP bind name

Example usage

Command line: **sm -httpPort:13080 -ldapserver1:ldapparent -ldapbinddn:uid=falcon,ou=People,dc=<*company*>,dc=com**

Initialization file: `ldapbinddn:uid=falcon,ou=People,dc=<company>,dc=com`

Related topics

System parameters
Enter a parameter in the sm.ini file
LDAP parameters

2.20.3. Startup parameter: ldapbindpass

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

ldapbindpass

Description

This parameter defines the password the Service Manager server uses to bind to the LDAP server.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

None

Possible values

LDAP password

Example usage

Command line: **sm -httpPort:13080 -ldapbindpass:ldappassword**

Initialization file: `ldapbindpass:ldappassword`

Related topics

System parameters
Enter a parameter in the sm.ini file
LDAP parameters

2.20.4. Startup parameter: ldapdisable

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

ldapdisable

Description

This parameter disables the LDAP login authentication procedure. Although you don't have to restart the server for this parameter to work, the server will write connection error messages to the sm.log file until you have restarted the server.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

1

Possible values

0 (Disable)

1 (Enable)

Example usage

Command line: **sm -httpPort:13080 -ldapserver1:ldapparent -ldapdisable:1**

Initialization file: `ldapdisable:1`

Related topics

System parameters
Enter a parameter in the sm.ini file
LDAP parameters

2.20.5. Startup parameter: ldapdisconnect

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

ldapdisconnect

Description

This parameter specifies when the Service Manager server disconnects from the LDAP server.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

0

Possible values

0 (After client log out)

1 (After every transaction)

Example usage

Command line: **sm -httpPort:13080 -ldapserver1:ldapparent -ldapdisconnect:1**

Initialization file: `ldapdisconnect:1`

Related topics

System parameters
Enter a parameter in the sm.ini file
LDAP parameters

2.20.6. Startup parameter: ldapmaxrecords

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

ldapmaxrecords

Description

This parameter specifies the maximum number of records to return from an LDAP query. If the LDAP server has a more restrictive query limit, then that limit determines the actual number of records returned by a query.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

None

Possible values

Number of records

Example usage

Command line: **sm -httpPort:13080 -ldapserver1:ldapparent -ldapmaxrecords:10**

Initialization file: `ldapmaxrecords:10`

Related topics

System parameters
Enter a parameter in the sm.ini file
LDAP parameters

2.20.7. Startup parameter: ldapnosizelimitmsg

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

ldapnosizelimitmsg

Description

When limiting LDAP query with *ldapmaxrecords:100* (for example), if LDAP query results are bigger than 100 records, users receive the following message:

`Message from LDAP server: Size limit exceeded.`

This informational message may mislead users to think that something is not working correctly.

This parameter, when set to 1, allows you to disable this informational message so that it will not display to and therefore not disturb end users.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

0

Possible values

0 (Enable)

1 (Disable)

Example usage

Command line: sm -httpPort:13080 -ldapserver1:ldapparent -ldapnosizelimitmsg:1

Initialization file: `ldapnosizelimitmsg:1`

Related topics

System parameters
Enter a parameter in the sm.ini file
LDAP parameters

2.20.8. Startup parameter: ldapnostrictlogin

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

ldapnostrictlogin

Description

Enabling this parameter allows users with valid operator records to log into Service Manager without also having a valid LDAP record. By default, a user is required to have both a valid LDAP record and a valid Service Manager operator record.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

0

Possible values

0 (Disable)

1 (Enable)

Example usage

Command line: `sm -httpPort:13080 -ldapserver1:ldapparent -ldapnostrictlogin:1`

Initialization file: `ldapnostrictlogin:1`

Related topics

System parameters
Enter a parameter in the sm.ini file
LDAP parameters

2.20.9. Startup parameter: ldapquerymapping

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

`ldapquerymapping:<original_mapped_LDAP_attribute>=<target_mapped_LDAP_attribute>`

Description

If you set this parameter, when the system builds the LDAP query during login, the target value replaces the original value.

Valid if set from

Server's operating system command prompt

Initialization file (`sm.ini`)

Requires restart of the Service Manager server?

Yes

Default value

None

Possible values

A string mapping the original attribute to the target attribute: `<original_mapped_LDAP_attribute>=<target_mapped_LDAP_attribute>`

Example usage

The following is an example of replacing the user name with the email address. By setting the parameter as follows, the system uses the user's email address and password for login, instead of user name and password.

Command line: **sm -httpPort:13080 -ldapquerymapping:sAMAccountName=userPrincipalName**

Initialization file: `ldapquerymapping:sAMAccountName=userPrincipalName`

Related topics

System parameters
Enter a parameter in the sm.ini file
LDAP parameters

2.20.10. Startup parameter: ldapsearchscope

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

ldapsearchscope

Description

This parameter specifies whether to search the LDAP base directory only or the LDAP base directory and subtrees.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

0

Possible values

0 (Search LDAP base directory and subtrees

1 (Search LDAP base directory only)

Example usage

Command line: `sm -httpPort:13080 -ldapserver1:ldapparent -ldapsearchscope:1`

Initialization file: `ldapsearchscope:1`

Related topics

System parameters
Enter a parameter in the sm.ini file
LDAP parameters

2.20.11. Startup parameter: ldapserver

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt. You can always set a startup parameter from the server's operating system command prompt.

Parameter

ldapserver[Numerical ID]

Description

This parameter defines any backup LDAP servers that the Service Manager server can connect to if the primary LDAP server is unavailable.

You can add this parameter multiple times by adding a number to the end of the parameter. The Service Manager server attempts to connect to each LDAP server in turn. For example:

```
ldapserver1:...
ldapserver2:...
ldapserver<n>:...
```

The parameter uses the following syntax and attributes to define the LDAP server settings.

```
<hostname>%<port>%<base directory>%<certificate file path>
```

Attribute	Required?	Description
<hostname>	Yes	The data source name (DSN) or IP address of the LDAP server.
<port>	No	The communications port on which the LDAP server listens to connection requests.
<base directory>	No	The base directory on the server where you want LDAP searches to start.
<certificate file path>	No	The path to the LDAP server SSL certificate.

The following entries illustrate how to properly use the parameter attributes:

```
ldapserver1:ldapparent%389%cn=users,dc=ldapparent,dc=microfocus,dc=com%c:\certificates\sslcert.pem
ldapserver2:16.183.93.217%636%cn=users,dc=swsm,dc=ind,dc=lab
```

Note

Note If you configure LDAP servers over an SSL connection, you may wish to enable the *ldapsslallownocert* parameter. When this parameter is enabled, the Service Manager server doesn't check a CA's root certification.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

None

Possible values

hostname, port, base directory, certification file path

Append the numerical ID of LDAP server to the end of the parameter (values from 1-99 are possible)

Example usage

Command line: `sm -httpPort:13080 -ldapserver1:ldapparent`

Initialization file: `ldapserver1:ldapparent`

Related topics

System parameters
Enter a parameter in the sm.ini file
LDAP parameters
Startup parameter: ldapsslallownocert

2.20.12. Startup parameter: ldapsslallownocert

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt. You can always set a startup parameter from the server's operating system command prompt.

Parameter

ldapsslallownocert:<*numerical ID*>

Description

When this parameter is enabled, the Service Manager server doesn't check a CA's root certification. In this situation, you don't need to set a root certificate for the CA that'ssued the LDAP server's certificate on the SM server in order to enable LDAP over SSL in SM.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

0

Possible values

0 (Service Manager checks the CA's root certificates)

1 (Service Manager doesn't check the CA's root certificates)

Example usage

Command line: `sm -httpPort:13080 -ldapsslallownocert:1`

Initialization file: `ldapsslallownocert:1`

Related topics

Enable an integration to LDAP
LDAP parameters

2.20.13. Startup parameter: ldapstats

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

ldapstats

Description

Enabling this parameter causes Service Manager to record LDAP query statistics in the `sm.log` file.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

0

Possible values

0 (Disable)

1 (Enable)

Example usage

Command line: `sm -httpPort:13080 -ldapserver1:ldapparent -ldapstats:1`

Initialization file: `ldapstats:1`

Related topics

System parameters
LDAP parameters

2.20.14. Startup parameter: ldaptimelimit

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

ldaptimelimit

Description

This parameter defines the maximum number of seconds that Service Manager will wait for a response to an LDAP query. If this parameter is not provided or the value is set at 0 (zero), then there is no time limit.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

0

Possible values

Number of seconds

Example usage

Command line: `sm -httpPort:13080 -ldapserver1:ldapparent -ldaptimelimit:60`

Initialization file: `ldaptimelimit:60`

Related topics

LDAP parameters

2.21. Localization parameters

The following table lists the startup parameters you can set from the Service Manager server's OS command prompt or from the Service Manager initialization file (sm.ini).

These parameters determine how the Service Manager server displays language and regional settings.

Localization Parameter	Brief description
Localization parameters: Language identifiers table	List of valid language and regional identifier codes
Parameter: localizedecimalpoint	Enables the decimal symbol in numeric fields to be localized.

Related topics

System parameters
Enter a parameter in the sm.ini file

2.21.1. Language identifiers table

Language	Identifier	Unix code page	Windows code page
Albanian	sq		
American		819	1252
Australian		819	1252
Basque	eu	819	1252
Bulgarian	bg	915	1251
Canadian		819	1252
Catalan	ca	819	1252
Chinese	zh	Chinese doesn't use code pages	Chinese doesn't use code pages
Croatian	hr	912	1250
Cyrillic	sr	915	1251
Czech	cs	912	1250
Danish	da	819	1252
Dutch	nl	819	1252
English	en	819	1252
Estonian	et	819	1257
Finnish	fi	819	1252
French	fr	819	1252
German	de	819	1252
Greek	el	813	1253
Hungarian	hu	912	1250
Icelandic	is	819	1252
Italian	it	819	1252
Japanese	ja	EUC	SJIS
Korean	ko	949	949
Latvian	lv	912	1257
Lithuanian	lt	912	1257
Norwegian	no	819	1252
Polish	pl	912	1250
Portuguese	pt	819	1252
Romanian	ro		

Russian	ru	915	1251
Serbian	sh		
Slovak	sk	912	1250
Slovenian	sl	912	1250
Spanish	es	819	1252
Swedish	sv	819	1252
Swiss		819	1252
Thai	th	8741	874
Turkish	tr	920	1254
Ukrainian	uk	1251	1251
utf8	utf8		

Note: Service Manager supports Japanese only if you install a multi-byte version.

Related topics

Localization parameters

2.21.2. Parameter: localizedecimalpoint

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

localizedecimalpoint

Description

This parameter enables users to use a pre-configured decimal symbol (for example, the comma) when completing numeric fields. By default, this parameter is disabled (set to "0"), and users can only enter a period as a decimal symbol. However, in Europe a comma is normally used. When this parameter is enabled, users can enter numeric values using the Decimal Symbol (either a period or comma) defined for one of the following currencies, listed from the highest to lowest priority:

- The currency defined in the **Display Currency** field in the user's operator record
- The currency defined in the **Basis Currency** field on the **General** tab of the System Information Record
- The **US Dollar** currency

Tip

Tip To view the Decimal Symbol setting for a specific currency, click **System Administration** > **Base System Configuration** > **Currencies**, and then click **Search**.

Note

- Whether this parameter is enabled or disabled, when a user views a cost value, Service Manager automatically converts the value according to the above currency settings. For example, when an operator whose currency is British Pounds is ordering an item from the Service Catalog, Service Manager displays the item's cost in British Pounds (for example, £ 50,00). However, there are exceptions (for example, the Total Cost value of an Order in Request Management always display in US Dollars.)
- When creating an unload file in .txt format, Service Manager always uses a period as the decimal symbol even if the original decimal symbol is a comma, and it can successfully process the unload when loading it back. If you need to edit the unload file before loading it back, always use a period as the decimal symbol; otherwise errors will occur when the unload is loaded back.
- When using the Export to Excel feature, each decimal symbol (for example, a comma) will be exported as is. For this reason, be sure not to select a comma as the delimiter when exporting data.

Caution

Caution When the Service Manager server is running on a Linux host, set the locale to en_US.utf8. If you use other locale settings such as de_DE.utf8 or fr_FR.utf8, it's not possible to assign a number with decimal places to a variable in RAD expressions or JavaScript. If you try to assign something like "$number=1.456" in RAD Debugger, the result is 1.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

0 (Disabled)

Possible values

0: Disabled

1: Enabled

Related topics

System parameters
Enter a parameter in the sm.ini file
Localization parameters

2.22. Regional settings parameters

These parameters determine the regional settings that the server uses. You can set these parameters from the Service Manager server's operating system command prompt or from the Service Manager initialization file (sm.ini).

Startup parameter	Brief description
Startup parameter: servertz	Causes server processes to use the listed time zone

Related topics

System parameters
Enter a parameter in the sm.ini file

2.22.1. Startup parameter: servertz

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

servertz

Description

Defining this parameter causes server processes to use the listed time zone. The Service Manager server only uses this time zone value if it can't find time zone information in the user's operator record or in the System Wide Company Record. Typically, only external processes such as *SCAUTO*, *SCBUTIL*, and *JavaMail* use this time zone entry.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

None

Possible values

Time zone name in the tzfile table

Example usage

Command line: **sm -httpPort:13080 -servertz:US/Pacific**

Initialization file: `servertz:US/Pacific`

Related topics

System parameters
Enter a parameter in the sm.ini file
Regional settings parameters

2.23. Security parameters

These parameters determine the access requirements for the server. You can set these parameters from the Service Manager server's operating system command prompt or from the Service Manager initialization file (sm.ini).

Security parameter	Brief description
Startup parameter: changeencrkey	Changes the database encryption key by updating all the encrypted fields
Startup parameter: denyattachwithoutext	Enables the Service Manager server to check contents of attachments without file extensions submitted through web services.
Startup parameter: encryptionkey	Defines the encryption key that's used to encrypt fields in the database
Parameter: fipsmode	Determines if the Service Manager server runs in FIPS 140-2 compliant mode
Startup parameter: ntlogin	Defines the account and password of the Windows Service you want to run Service Manager server.
Startup parameter: ntsecuritygroup	Restricts access to the Service Manager application on the server
Parameter: querysecurity	Determines if the Service Manager server requires a security hash with web tier URL queries
Startup parameter: restaccessviabrowser	Determines if RESTful access to the Service Manager server through a web browser is allowed.
Startup parameter: soap12binding	Enables or disables SOAP 1.2 binding on the server
Startup parameters: JavaScript access	Specifies access restrictions from JavaScript code for file access and command execution
Startup parameter: upgradeencralg	Upgrades the database encryption algorithm from DES to AES by updating all encrypted fields with a new 256-bit key

Related topics

System parameters
Enter a parameter in the sm.ini file

2.23.1. Startup parameter: changeencrkey

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

changeencrkey

Description

This parameter changes the database encryption key by updating all encrypted fields. The value of this parameter must be either 8 characters long in non-FIPS mode or 32 characters (256 bits) long in FIPS mode, and consist of alphanumeric characters.

Valid if set from

Server's operating system command prompt

Requires restart of the Service Manager server?

No

Default value

xxxxxxxx in non-FIPS mode

xxxxxxxxxxxxxxxxxxxxxxxx in FIPS mode

Possible values

8 character alphanumeric string in non-FIPS mode

32 character alphanumeric string in FIPS mode

Example usage

Command line: **sm -httpPort:13080 -changeencrkey:1234567890abcdefghijklmnopqrstuv**

Related topics

FIPS mode
Security parameters

2.23.2. Startup parameter: denyattachwithoutext

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

denyattachwithoutext

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Description

By default, this parameter is set to 0. Attachments without file extensions submitted through web services to Service Manager are checked by Tika against the Service Manager allowlist saved in a DB table in DBMS. If matching content types are found, the attachments can be submitted successfully. If not, the attachments are denied.

When this parameter is set to 1, attachments without file extensions submitted through web services to Service Manager are denied immediatedly by the Service Manager server without checking the content.

Requires restart of the Service Manager server?

No

> **Note**
>
> Note For the configuration to take effect, you need to log off and then log back on to the clients where attachments are submitted through web services.

Default value

0

Possible values

0, 1

Example usage

Command line: **sm -httpPort:13080 -denyattachwithoutext:0**

Initialization file: `denyattachwithoutext:0`

Related topics

Attachments

Support for allowing attachments with certain file extensions
Enter a parameter in the sm.ini file

Security parameters

2.23.3. Startup parameter: encryptionkey

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

encryptionkey

Description

This parameter defines the encryption key for encrypting fields in the database. The value of this parameter must be either 8 characters (64 bits) long in non-FIPS mode or 32 characters (256 bits) long in FIPS mode, and consist of alphanumeric characters.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

xxxxxxxx in non-FIPS mode

xxxxxxxxxxxxxxxxxxxxxxxx in FIPS mode

Possible values

32 character alphanumeric string

Example usage

Command line: **sm -httpPort:13080 -encryptionkey:1234567890abcdefghijklmnopqrstuv**

Initialization file: `encryptionkey:1234567890abcdefghijklmnopqrstuv`

Related topics

FIPS mode
Security parameters

2.23.4. Startup parameter: jsaccesscheckforadmin

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

`jsaccesscheckforadmin`

Description

The parameter determines whether the system administrator (sysadmin) users can execute OS commands through JavaScript. When setting this parameter to 1, the system administrator users will also follow the access restrictions specified by `jsaccesscmdregex`. See Startup parameters: JavaScript access.

Valid if set from

Initialization file (`sm.ini`)

Requires restart of the Service Manager server?

Yes

Default value

0

Possible values

0 (System administrator users are allowed unrestricted access to the file system and command line)

1 (System administrator users will follow the access restrictions specified by `jsaccesscmdregex`)

Example usage

Initialization file: `jsaccesscheckforadmin:1`

2.23.5. Startup parameters: JavaScript access

Startup parameters change the behavior of the server.

Parameter

jsaccessfilereadregex

jsaccessfilewriteregex

jsaccesscmdregex

Description

These parameters specify access restrictions from JavaScript code for file access and command execution. This is required so that users with Process Designer tailoring rights aren't able to use the available Service Manager JavaScript APIs to directly access the underlying host file system and command line. These restrictions are described below:

- System administrator (sysadmin) users are allowed unrestricted access to the file system and command line. Note that if the jsaccesscheckforadmin parameter is set to 1, the the system administrator users will also follow the access restrictions specified by the `jsaccesscmdregex` parameter below.
- Non-sysadmin users are allowed unrestricted access as well by default (none of these configuration parameters is present); if any of these parameters is present, access is restricted as follows:
  - `jsaccessfilereadregex:<regex>` : Files read using the readFile() and uncompressFile() (compressed file only) JavaScript functions must have an absolute path name that matches the regular expression <regex>.
  - `jsaccessfilewriteregex:<regex>` : Files written using the writeFile(), deleteFile(), makeDir(), uncompressFile() (output dir only), and writeAttachmentToFile() JavaScript functions must have an absolute path name that matches the regular expression <regex>.
  - `jsaccesscmdregex:<regex>` : Operating system commands executed by using the sysExec() JavaScript function (including parameters) must match the regular expression <regex>.

Regular expressions must use proper syntax for Java-style regex (similar to Perl-style) (for example, '.' means any character and backslashes need to be doubled). File path names in `jsaccessfilereadregex` and `jsaccessfilewriteregex` must be absolute and use proper delimiters ('/' for UNIX, and '\' for Windows). They are case-sensitive for UNIX and case-insensitive for Windows.

File path names used as parameters to JavaScript functions can still be any legal OS-specific file path names. Absolute and relative path names are allowed as well as the usage of '.' and '..'. On Windows, delimiters can be either '/' or '\', while UNIX only allows '/'. However, path names from JavaScript commands are normalized first (relative paths are made absolute, the '.' and '..' are eliminated, and on Windows '/'s are transformed into '\'s) before they are pattern matched against their corresponding regular expression.

Command line strings in the sysExec() JavaScript function are taken as is and compared to the `jsaccesscmdregex` regular expression. The following table lists the strings that aren't allowed in the command line run by sysExec(). If administrators need to use any of these strings in the command line, they can put the real command line in a .sh or .bat file, and then run the .sh/.bat file in sysExec() instead. When any JavaScript attempts to run unauthorized commands or use these forbidden strings in the command line run by sysExec(), audit warning messages will be logged in the server log (sm.log). The messages include the commands to be run, and the user's login name.

Forbidden string in command line	Comment
||	Logical OR in shell.
|	The Pipe symbol.
;	Shell statement ending; the shell interpreter will run the command that follows.
``	A pair of backquotes. The shell interpreter will run the command between `` first, and then insert its STDOUT output to the command line of the outside command.
&&	Logical AND in shell
&	Run in the background; the shell interpreter will run the command that follows.
>>	File appending; it will write to file.
>	File redirection; it will write to file.

Valid if set from

Server's operating system command promt

Initialization file (sm.ini)

Requires restart of server?

Yes

Default value

None (Unrestricted file access and command execution)

Possible values

Regular expressions

Example usage

- `jsaccessfilereadregex:C:\\Users\\joe\\Documents\\.+|C:\\SM-Install\\server-dist\\RUN\\(\w)+\.js`
  Read access is limited to any files under the C:\Users\joe\Documents directory and only for .js files under the C:\SM-Install\server-dist\RUN directory.
- `jsaccessfilewriteregex:^$`
  Only matches the empty string. This is used to deny all write access to the file system.
- `jsaccesscmdregex:dir .*|copy "C:\\Users\\spinu\\Documents\\SM\\Special Dir\\from\\(\w)+" "C:\\Users\\spinu\\Documents\\SM\\Special Dir\\to\\(\w)+"`
  Command execution is limited to the `dir` command with any parameters or the `copy` command, but only from a file in the `C:\Users\spinu\Documents\SM\Special Dir\from` directory to a file in the `C:\Users\spinu\Documents\SM\Special Dir\` directory.

Related topics

System parameters
Enter a parameter in the sm.ini file
Security parameters

2.23.6. Startup parameter: mbeanclientlogin

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

`mbeanclientlogin`

Description

This parameter defines the account and password for the MBean client.

You must use a slash character to separate the user name and password.

Valid if set from

Initialization file (sm.ini)

Requires restart?

This parameter requires a restart of the Service Manager process.

Default value

None

Possible values

The MBean user account and password

Example usage

`mbeanclientlogin:User/Password`

2.23.7. Startup parameter: ntlogin

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

`ntlogin`

Description

This parameter defines the account and password of the Windows Service you want to run Service Manager server.

You must use a slash character to separate the user name and password. If you omit this parameter, Service Manager will use the Local System as its account by default.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart?

This parameter requires a restart of the Service Manager process.

Default value

None

Possible values

The Windows user account and password

Example usage

`smservice -install -ntlogin:User/Password`

Related topics

System parameters
Enter a parameter in the sm.ini file
Security parameters

2.23.8. Startup parameter: ntsecuritygroup

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

ntsecuritygroup

Description

Defining this parameter restricts access to the Service Manager application on the server. Only users who are members of the Windows user group defined by this parameter can start or stop the Service Manager application.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

None

Possible values

Name of the Windows user group

Example usage

Command line: **sm -httpPort:13080 -ntsecuritygroup:SM Users**

Initialization file: `ntsecuritygroup:SM Users`

Related topics

System parameters
Enter a parameter in the sm.ini file
Security parameters

2.23.9. Parameter: querysecurity

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

querysecurity

Description

The parameter determines whether the Service Manager server generates a query hash when calling the makeSCWebURL server function. If this parameter is set to 1, a query hash value is appended to URLs generated by makeSCWebURL.

Note This parameter is tied to the *querySecurity* (spelled with an uppercase "S") web parameter. To disable the security hash feature, you must disable both the web.xml and sm.ini versions of this parameter.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

1

Possible values

0 (Disable)

1 (Enable)

Example usage

Command line: **sm -httpPort:13080 -querysecurity:0**

Initialization file: `querysecurity:0`

Related topics

Generating Web tier URL queries
SSL parameters

2.23.10. Startup parameter: restaccessviabrowser

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

restaccessviabrowser

Description

This parameter determines if RESTful access to the Service Manager server through a web browser is allowed. You can specify 0 or 1 for this parameter:

- 0: Access to Service Manager through REST from a web browser is not allowed
- 1: Access to Service Manager through REST from a web browser is allowed

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

0

Possible values

0, 1

Example usage

Command line: **sm -httpPort:13080 -restaccessviabrowser:1**

Initialization file: `restaccessviabrowser:1`

Related topics

Security parameters

2.23.11. Startup parameter: soap12binding

Startup parameters change the behavior of the server.

Parameter

soap12binding

Description

By default, Service Manager generates WSDLs that conform to soap 1.1 binding. However, some Web Services clients accept both soap 1.1 and soap 1.2. In such cases, you can enable *soap12binding* by providing the parameter *soap12binding:1*.

Valid if set from

Server's operating system command prompt
Initialization file (sm.ini)

Requires restart of server?

Existing connections will honor the current value. New connections will honor the new value.

Default value

0 (Disabled)

Possible values

0 (Disabled)

1 (Enabled)

Example usage

Command line: **sm -httpPort:13080 –debugnode:1 –soap12binding:0**

Initialization file: `soap12binding:0`

Related topics

Security parameters

2.23.12. Startup parameter: upgradeencralg

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

upgradeencralg

Description

This parameter upgrades the database encryption algorithm from DES to AES by updating all encrypted fields with a new 256-bit key. The value of this parameter must be either empty or a 32 characters long alphanumeric string. The specified value will be used as the encryption key of AES, and a default 256-bit key will be used if you specify an empty value.

Valid if set from

Server's operating system command prompt

Requires restart of the Service Manager server?

No

Default value

xxxxxxxxxxxxxxxxxxxxxxxx

Possible values

32 character alphanumeric string

Example usage

Command line: **sm -upgradeencralg:1234567890abcdefghijklmnopqrstuv**

Related topics

Security parameters

2.24. Servlet parameters

The following table lists the startup parameters you can set from the Service Manager server's operating system command prompt or from the Service Manager initialization file (sm.ini).

These parameters determine how a Service Manager server manages client connections.

Servlet Parameter	Description
Parameter: appthreadspersession	Gives a site some control over the amount of server memory that a user can consume by limiting the number of application tabs that can be opened through the client
Parameter: asyncrefreshlbinterval	Determines the interval (measured in seconds) at which the status of the loadbalancer is refreshed.
Parameter: attachedByName	Specifies whether the operator name or the contact name is displayed in the **Attached By** field from the attachment list in the web client.
checkdatetimevalue	Determines whether Service Manager validates the date fields, such as the **Planned Start** and **Planned End** date fields.
Parameter: disableXrs	Enables and disables the **-Xrs** JVM option
Parameter: emailoutConfigFromSmisOnly	Set the **emailoutConfigFromSmisOnly:1** server parameter to use SMIS to configure emailout settings. After this parameter is enabled, the parameters defined in the emailout process through sm.cfg and sm.ini won't work. Instead, those parameter values such as smtphost will only be get from SMIS configuration.
Parameter: fetchnotnullsystemp	Specifies whether or not to fully enable the template merge functionality for the operator table
Parameter: GossipRouterhosts	provides a list of available GossipRouters. Service Manager nodes read this parameter to identify all GossipRouters (this behavior requires the addition of command lines in the sm.cfg file).
Parameter: group	Shuts down or quiesces all Service Manager server members in a virtual group
Parameter: groupbindaddress	Defines the TCP/IP address of the network adapter you want servlet container processes to use to communicate with other processes in a virtual group
Parameter: grouplicenseip	Defines the TCP/IP address of the Service Manager host with a valid AutoPass license for the virtual group
Parameter: groupmcastaddress	Defines the TCP/IP address that servlet container processes can use to communicate with the load balancer process in a horizontal scaling implementation
Parameter: groupname	Creates a virtual group that servlet container processes across multiple systems can join for horizontal scaling purposes
Parameter: groupport	Defines the communications port that servlet container processes can use to communicate with the load balancer process in a horizontal scaling implementation
Parameter: groupsubnetaddress	Defines the subnet mask that the multicasting IP address resides on in a horizontal scaling implementation

Parameter: grouptcpbindport	Specifies the bind port for each group member to communicate with each other. By default, Service Manager tries to assign port 7800 for each member. If this is impossible, it tries the next higher port (7801) and so on, until it finds an unused port.
Parameter: groupudpjointimeout	Determines the number of seconds within which a Service Manager node successfully joins a Jgroups cluster before the operation times out when using the UDP protocol.
Parameter: grouptcpjointimeout	Determines the number of seconds within which a Service Manager node must successfully join a Jgroup before the operation times out.
Parameter: grouptimeout	Defines the timeout value (in milliseconds) for Jgroups remote procedure calls
Parameter: host	Specifies the host name of the Service Manager server that you want to shut down or quiesce
Parameter: httpPort	Defines the communications port that a servlet container process uses to communicate with clients using HTTP
Parameter: httpsPort	Defines the communications port that a servlet container process uses to communicate with clients using HTTPS (SSL-encrypted HTTP)
Parameter: idmsigningkey	Specifies a key that's used to sign Identity Manager (IdM) tokens.
Parameter: idmssoenabled	Determines whether Identity Manager (IdM) Single Sign-On is enabled for the Service Manager Server.
Parameter: JVMOption	Specifies a JVM option.
Parameter: jgroupstcp	Determines whether Service Manager Jgroups run on Transmission Control Protocol (TCP) or on User Datagram Protocol (UDP) .
Parameter: keystoreFile	Defines the path and file name to the Java keystore that contains the certificate of the Service Manager server
Parameter: keystorePass	Defines the password to the Java keystore that contains the certificate of the Service Manager server
Parameter: loadBalancer	Creates a load balancer process that listens to incoming client requests on the communications port specified by the *httpPort* parameter
Parameter: logperprocess	Defines whether to create a log file for each vertical scaled Service Manager process.
Parameter: maxattachmentcount	Specifies the maximum number of files that can be attached to an individual record.
Parameter: queryhashcode	Enables all Service Manager processes to consistently generate a unique hash code, which can be accepted and properly decoded by the web tier

Parameter: reportgroup	Displays the status of ports, processes, and threads in horizontal and vertical scaling servlet implementations
Parameter: reportlbstatus	Displays information about available ports and threads in horizontal and vertical scaling servlet implementations
Parameter: sslConnector	Defines whether servlet container processes have an HTTPS (SSL-encrypted HTTP) communications port available
Parameter: threadsperprocess	Defines the number of threads that you want the Service Manager server to start when it starts a new process
Parameter: ssl_trustedClientsJKS	Defines the path and file name to the Java keystore that contains the list of trusted client certificates from which the Service Manager server allows connection requests
Parameter: ssl_trustedClientsPwd	Defines the password to the Java keystore that contains the list of trusted client certificates from which the Service Manager server allows connection requests
Parameter: truststoreFile	Defines the path and file name of the Java keystore that contains the list of trusted certificate authorities
Parameter: truststorePass	Defines the password to the Java keystore that contains the list of trusted CA certificates

Related topics

System parameters
Enter a parameter in the sm.ini file

2.24.1. Parameter: appthreadspersession

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

appthreadspersession

Description

This parameter gives a site some control over the amount of server memory that a user can consume by limiting the number of application tabs that can be opened through the client. If a user performs an action that will result in opening a new application tab that exceeds the limit (for example, the maximum number of tabs is set at three and the user performs an action that would open a fourth tab), a message is displayed letting the user know that the maximum number of application tabs has been exceeded and the action is denied.

This parameter can also be used during load tests to prevent a common problem where a script that's recorded never closes the application tab, and then results in out-of-memory errors.

Note

Note A user can't open the last allowed tab by executing a command via the command line. For example, the *appthreadspersession* parameter is set to 5, a user has four open tabs (including the first tab). The user tries to open the fifth tab by executing a command (for example, the **db** command) via the command line. In this scenario, the fifth tab can't be opened. However, the user can open the fifth tab from the System Navigator.

This is a known limitation caused by the mechanism that calculates the number of open tabs.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

0 (Unlimited number of application tabs.)

> **Note**
>
> Note If you are running in scripting mode, the default value is 5.

Possible values

No maximum is enforced.

Example usage

Command line: **sm -appthreadspersession:3**

Initialization file: `appthreadspersession:3`

Related topics

System parameters
Enter a parameter in the sm.ini file

2.24.2. Parameter: asyncrefreshlbinterval

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

asyncrefreshlbinterval

Description

This parameter determines the interval (measured in seconds) at which the status of the loadbalancer is refreshed.

Valid if set from

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

10 seconds

Possible values

1-Any

Example usage

`asyncrefreshlbinterval:25`

Related topics

Servlet implementation
System parameters
Enter a parameter in the sm.ini file
Servlet parameters

2.24.3. Parameter: attachedByName

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

attachedByName

Description

This parameter specifies whether the operator name or the contact name is displayed in the **Attached By** field from the attachment list in a record. By default, this parameter is set to 0, and the operator name is used for the **Attached By** field. If you want to use the contact name instead of the operator name for the **Attached By** field, set this parameter to 1.

Note

Note This parameter takes effect in the web client only.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

0

Possible values

0 (Display the operator name in the **Attached By** field)

1 (Display the contact name in the **Attached By** field)

Example usage

Command line: **sm -attachedByName:1**

Initialization file: `attachedByName:1`

Related topics

System parameters
Enter a parameter in the sm.ini file
Debugging parameters

2.24.4. Parameter: checkdatetimevalue

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

checkdatetimevalue

Description

This parameter determines whether Service Manager validates the date fields, such as the **Planned Start** and **Planned End** date fields.

- If the value of checkdatetimevalue is 1, the system checks the date/time value. The system doesn't allow time-only values and displays an error message (for example, `the value is invalid`). Meanwhile, the system accepts date-only values and adds default time 00:00:00 to the date.

  Note

  Note The date format is strictly followed. For example, if the date format is YYYY/MM/DD, the system doesn't accept either 18/04/24 or 2018/04. In this case, the correct time value is 2018/04/24.

- If the value of checkdatetimevalue is 0, the system doesn't check the date/time value. Both date-only and time-only values are accepted.

Valid if set from

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No. Re-login Service Manager after you update the parameter value.

Default value

0

Possible values

0 (The system doesn't check the date/time value)

1 (The system checks the date/time value)

Example usage

Initialization file: `checkdatetimevalue:1`

Related topics

System parameters
Enter a parameter in the sm.ini file

2.24.5. Parameter: disableXrs

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

disableXrs

Description

The Service Manager process creates a JVM and provides default option of -Xrs on startup to reduce the signals to be handled by the JVM. However, you may want to generate thread dumps for debugging purposes, and then you would want the JVM to receive these signals. Use the *-disableXrs* parameter in sm.ini, so that the Service Manager process will not provide this option to JVM.

Notes:

- This parameter is honored only in Windows environment for debugging purposes.
- If you use Remote Desktop and start a Service Manager process using *disableXrs:1*, and then close the session or log off, the Service Manager process will be shutdown.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

0

Possible values

0 = enable JVM option -Xrs

1 = disable JVM option –Xrs

Example usage

Command line: **sm -httpPort:13080 -disableXrs:1**

Initialization file: `disableXrs:1`

Related topics

System parameters
Enter a parameter in the sm.ini file
Debugging parameters

2.24.6. Parameter: emailoutConfigFromSmisOnly

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

emailoutConfigFromSmisOnly

Description

The Emailout instance is available in SMIS integration from Service Manager 9.81.

The existing email out settings in sm.ini configured through Startup parameter: emailout are still available.

If you choose to use SMIS to configure emailout settings, you need set the **emailoutConfigFromSmisOnly:1** server parameter. After this parameter is enabled, the parameters defined in the emailout process through sm.cfg and sm.ini won't work. Instead, those parameter values such as smtphost will only be get from SMIS configuration.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes.

Once SMIS configurations are changed, you need to restart the whole SM Server Cluster to apply the updates. Restarting SMIS instance only won't take effect. Restarting one host will apply the new changes on this host only, while other hosts won't get the updated values.

Default value

0

Possible values

0 (Use Startup parameter: emailout to configure emailout settings)

1 (Use SMIS to configure emailout settings)

Example usage

Command line: **sm -emailout -emailoutConfigFromSmisOnly:1**

Initialization file: `emailoutConfigFromSmisOnly:1`

Related topics

System parameters
Enter a parameter in sm.cfg or the server's OS command prompt

2.24.7. Parameter: fetchnotnullsystemp

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

fetchnotnullsystemp

Description

Prior to version 9.31, records that have a non-NULL systemplate field were passed to the query condition filter before they merged the template, and were thus incorrectly filtered out. As of version 9.31, these records will merge the template record before they are passed to the query condition filter. You need to set it to *fetchnotnullsystemp:1* to fully enable the template merge functionality. However, it may cause performance issues if the table has more than 10K records whose systemplate field is not NULL.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

0

Possible values

0 (Do not fully enable the template merge functionality)

1 (Fully enable the template merge functionality)

Example usage

Command line: **sm -fetchnotnullsystemp:1**

Initialization file: `fetchnotnullsystemp:1`

Related topics

System parameters
Enter a parameter in sm.cfg or the server's OS command prompt

2.24.8. Parameter: GossipRouterhosts

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

GossipRouterhosts

Description

This parameter provides a list of available GossipRouters. Service Manager nodes read this parameter to identify all GossipRouters (this behavior requires the addition of command lines in the sm.cfg file).

Valid if set from

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

None

Possible values

Any comma-delimited list of GossipRouters.

Example usage

`GossipRouterhosts:HostA[12001],HostB[12001]`

In this example, there are two GossipRouters (one on host A and one on host B), both at port 12001.

Related topics

Servlet implementation
System parameters
Enter a parameter in the sm.ini file
Servlet parameters

2.24.9. Parameter: group

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

group

Description

This parameter is intended for horizontal scaled implementations where there are multiple servers. This parameter allows the System Administrator to shut down or quiesce all Service Manager server members of the horizontal scaled group.

Valid if set from

Server's operating system command prompt

Requires restart of the Service Manager server?

Yes

Default value

None

Possible values

None

Example usage

The System Administrator can run the following commands to shut down or quiesce Service Manager server members in the horizontal scaled group.

Example: To shut down the default host (localhost) in the horizontal scaled group:

Command line: **sm -shutdown**

Example: To shut down a specific host (not the default host) in the horizontal scaled group:

Command line: **sm -shutdown -host:otherhost**

Example: To shut down all hosts in the horizontal scaled group:

Command line: **sm -shutdown -group**

Example: To restrict Service Manager processes to administrators only on the local host.

Command line: **sm -quiesce:1**

Example: To restrict Service Manager processes for all users in the entire cluster.

Command line: **sm -quiesce:2 -group**

Example: To bring the system back to running mode (unquiesce). All users can now connect to the processes.

Command line: **sm -quiesce:0 -group**

Related topics

Servlet implementation
System parameters
Enter a parameter in the sm.ini file
Connection parameters
Servlet parameters
Startup parameter: quiesce

2.24.10. Parameter: groupbindaddress

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

groupbindaddress

Description

This parameter defines the TCP/IP address of the network adapter you want servlet container processes to use to communicate with other processes in a virtual group. You only need to define this parameter if the Service Manager host contains more than one network adapter.

It is best practice to place this parameter in the Service Manager initialization file so that all servlet container processes started on the same host share the TCP/IP address specified by this parameter.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

None

Possible values

Any TCP/IP address

Example usage

Initialization file:

```
groupbindaddress:10.0.0.124
```

Related topics

Servlet implementation
System parameters
Enter a parameter in the sm.ini file
Servlet parameters

2.24.11. Parameter: grouplicenseip

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

grouplicenseip

Description

Service Manager uses the AutoPass software to validate the license. This parameter defines the TCP/IP address of the Service Manager host with a valid AutoPass license for the virtual group. This parameter contains the TCP/IP address of the Service Manager primary host, or a list of comma-seperated IP addresses of the primary and backup hosts for license verification. When you request a license, AutoPass prompts you for the IP address of the system where you are going to install the software. Use this IP address as the value for the *grouplicenseip* parameter. During startup, Service Manager needs the *grouplicenseip* parameter for the following conditions:

- You are running Service Manager on a system with multiple Network Interface Cards (NICs) and multiple IP addresses.
- You are running Service Manager in a horizontally-scaled system and the primary host has multiple NICs and multiple IP addresses.

- You are running Service Manager in a horizontally-scaled system and you are configuring the `sm.ini` file on a secondary host.

It is best practice to place this parameter in the Service Manager initialization (sm.ini) file, so that all servlet container processes started on the same host share the TCP/IP address specified by this parameter.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

None

Possible values

Any TCP/IP address

Example usage

Example 1: Initialization file:

```
grouplicenseip:10.0.0.135
```

Example 2: Initialization file:

```
grouplicenseip:10.0.0.135,10.0.0.137,10.0.0.139
```

Related topics

Horizontal scaling implementation
System parameters
Enter a parameter in the sm.ini file
Servlet parameters

2.24.12. Parameter: groupsubnetaddress

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

groupsubnetaddress

Description

This parameter defines the subnet address of a Service Manager server node in a horizontal scaling implementation. It is used to identify a cluster when the system uses multiple network adapter cards.

Service Manager attempts to automatically compute the subnet address from the IP address you provide in the *groupbindaddress* parameter and the subnet mask of any currently running servlet container processes in the same group. However, if the system can't automatically compute the subnet address, the server will fail to start and fail to join the group. You can then use this parameter to provide the proper subnet address.

Caution

Caution

Only provide the subnet address if one of your group members fails to start because the system could not automatically determine the proper subnet address.

Note

Note

A subnet address is a computed value derived from a bitwise AND operation on the binary values of the IP address and the subnet mask. You can use an online calculator to compute this value if your network administrator can't provide it.

It is best practice to place this parameter in the Service Manager initialization file rather than the command line.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

None

Possible values

A subnet address on the host system

Example usage

Initialization file:

```
groupname:mygroup1
groupbindaddress:192.168.1.2
groupsubnetaddress:192.168.1.0
groupport:13100
```

Related topics

Servlet implementation
System parameters
Enter a parameter in the sm.ini file
Connection parameters
Servlet parameters

2.24.13. Parameter: grouptcpbindport

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

grouptcpbindport

Description

This parameter specifies the bind port for each group member to communicate with each other. By default, Service Manager tries to assign port 7800 for each member. If this is impossible, it tries the next higher port (7801) and so on, until it finds an unused port.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

7800

Possible values

None

Example usage

Initialization file:

```
grouptcpbindport:7800
```

Related topics

Servlet implementation
System parameters
Enter a parameter in the sm.ini file
Servlet parameters

2.24.14. Parameter: groupudpjointimeout

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

groupudpjointimeout

Description

This parameter determines the number of seconds within which a Service Manager node successfully joins a Jgroups cluster before the operation times out when using the UDP protocol.

Valid if set from

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

2 (seconds)

Possible values

Number of seconds greater than 2

Example usage

`groupudpjointimeout:3`

Related topics

Servlet implementation
System parameters
Enter a parameter in the sm.ini file
Servlet parameters

2.24.15. Parameter: grouptcpjointimeout

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

grouptcpjointimeout

Description

This parameter determines the number of seconds within which a Service Manager node must successfully join a Jgroup before the operation times out.

Valid if set from

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

300 seconds

Possible values

1-Any

Example usage

`grouptcpjointimeout:350`

Related topics

Servlet implementation
System parameters
Enter a parameter in the sm.ini file
Servlet parameters

2.24.16. Parameter: grouptimeout

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

grouptimeout

Description

This parameter allows you to configure the timeout value (in milliseconds) for Jgroups remote procedure calls. You can tune this value to keep the number of available nodes stable.

It is recommended to set this parameter to a minimal value as long as it can keep the number of the available nodes stable. Meanwhile, the network latency between the web server and the Service Manager server should be small, otherwise the load balancer may forward a request to a node that's not able to handle the request due to network latency.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

5000 (milliseconds)

Possible values

Number of milliseconds from 300 to 10000

Example usage

Command line: **sm -grouptimeout:1500**

Initialization file: `grouptimeout:1500`

Related topics

System parameters
Enter a parameter in the sm.ini file
Servlet parameters

2.24.17. Parameter: idmsigningkey

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

idmsigningkey

Description

This parameter specifies a key that is used to sign Identity Manager (IdM) tokens when SAML SSO is enabled for the Service Manager Server.

Note

Note

- This key must match the idm.encryptedSigningKey parameter value that you configure in the IdM service. For more information, see Install Identity Manager (IdM).
- When the Service Manager Server is started, this key is automatically encrypted. When you want to reset this key, remember to remove the asterisk (*) before the parameter name.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

None

Possible values

A text string

Example usage

Command line: **sm -httpPort:13080 -idmsigningkey:1Qaz2wsx3edc**

Initialization file: `idmsigningkey:1Qaz2wsx3edc`

Related topics

SAML Single Sign-On

2.24.18. Parameter: idmssoenabled

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

idmssoenabled

Description

This parameter determines whether Identity Manager (IdM) Single Sign-On is enabled for the Service Manager Server. By default, IdM SSO is disabled. For more information, see SAML-based Single Sign-On.

Note

Note

If this parameter is set to 1, you need to additionally specify the idmsigningkey parameter.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

0

Possible values

0 (Disable)

1 (Enable)

Example usage

Command line: **sm -httpPort:13080 -idmssoenabled:1**

Initialization file: `idmssoenabled:1`

Related topics

SAML-based Single Sign-On

2.24.19. Parameter: jgroupstcp

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

jgroupstcp

Description

This parameter determines whether Service Manager Jgroups run on Transmission Control Protocol (TCP) or on User Datagram Protocol (UDP) .

When the value of the parameter is set to 1, JGroups run on TCP. When the value is set to 0, JGroups run on UDP.

Valid if set from

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

0

Possible values

0 (UDP)

1 (TCP)

Example usage

`jgroupstcp:1`

Related topics

Servlet implementation
System parameters
Enter a parameter in the sm.ini file
Configure Jgroups to use TCP in a horizontally-scaled environment
Configure Jgroups to use TCP in a vertically-scaled environment
Servlet parameters

2.24.20. Parameter: JVMOption

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

You can specify at most 48 JVM options.

Parameter

JVMOption

Description

This parameter specifies a JVM option.

Valid if set from

Server's operating system command prompt

Start up file (sm.cfg or smstart)

Requires restart of the Service Manager server?

Yes

Default value

JVM Option

Possible values

All JVM options

Example usage

Command line: **SM -JVMOption0:-Xmx256M**

Command line: **SM -JVMOption1:-Djava.net.preferIPv4Stack=true**

Initialization file: `JVMOption0:-Xmx256M`

Initialization file: `JVMOption1:-Djava.net.preferIPv4Stack=true`

Related topics

Servlet implementation
System parameters
Enter a parameter in the sm.ini file
Connection parameters
Servlet parameters

2.24.21. Parameter: keystoreFile

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

keystoreFile

Description

This parameter defines the path and file name to the Java keystore that contains the certificate of the Service Manager server. Service Manager uses this keystore to encrypt and decrypt messages sent to clients in servlet implementations. All

servlet implementation options require access to certificates stored in Java keystore formats.

Valid if set from

- Server's operating system command prompt
- Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

None

Possible values

Path and file name of the Java keystore containing the Service Manager server's signed certificate

Example usage

Command line: **sm -httpPort:13080 -httpsPort:13081 -keystoreFile:server.keystore -keystorePass:changeit**

Initialization file: `keystoreFile:server.keystore`

Related topics

System parameters
Enter a parameter in the sm.ini file
SSL parameters
Servlet parameters

2.24.22. Parameter: keystorePass

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

keystorePass

Description

This parameter defines the password to the Java keystore containing the certificate of the Service Manager server. Service Manager uses this keystore to encrypt and decrypt messages sent to clients in servlet implementations. All servlet implementation options require access to certificates stored in Java keystore formats.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

None

Possible values

Password to the Java keystore containing the Service Manager server's signed certificate

Example usage

Command line: **sm -httpPort:13080 -httpsPort:13081 -keystoreFile:server.keystore -keystorePass:changeit**

Initialization file: `keystorePass:changeit`

Related topics

System parameters
Enter a parameter in the sm.ini file

SSL parameters
Servlet parameters

2.24.23. Parameter: logperprocess

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

logperprocess

Description

This parameter defines whether to create a log file for each vertical scaled Service Manager process.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

0

Possible values

0 = all process will write to 1 log file

1 = Each process will write to its own log file

Example usage

Command line: **sm –logperprocess:1**

Initialization file: `logperprocess:1`

Related topics

System parameters
Enter a parameter in the sm.ini file
Servlet parameters

2.24.24. Parameter: maxattachmentcount

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

maxattachmentcount

Description

This parameter specifies the maximum number of files that can be attached to an individual record. If this limit is exceeded, the attachment upload process fails and the following error is displayed in the user interface:

The total number of attachments in this record exceeds the maximum allowed number for a single record.

The same error message is returned in the response if you attempt to add attachments through a SOAP or RESTful call and the number of attachments exceeds this limit.

> **Note**
>
> Note If an existing record has already more attachments than allowed by *maxattachmentcount*, the existing attachments are retained, but uploading of new attachments will fail.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

100

Possible values

0 to unlimited

Example usage

Command line: **sm -httpPort:13080 -maxattachmentcount:100**

Initialization file: `maxattachmentcount:100`

Related topics

System parameters
Web tier parameters
Setting file attachment limits
Web parameter: maxattachuploadsize
Web parameter: maxtotalattachuploadsize

2.24.25. Parameter: queryhashcode

Startup parameters change the behavior of the server.

Parameter

queryhashcode:hostname:port

Description

This parameter enables all Service Manager processes to consistently generate a unique hash code, which can be accepted and properly decoded by the web tier. If you provide the Fully Qualified Domain Name (FQDN) of a server host in the web.xml file, you need to provide the same value in this parameter.

The value of this parameter should be the same on all your horizontally-scaled systems.

Note

Note The value of this parameter is used only when the `querysecurity` parameter is enabled..

Valid if set from

Command line

Initialization file (sm.ini)

Requires restart of server?

All new connections will take the new value.

Sessions which are already running will continue to use the old value. In this case, we recommend restarting the server.

Default value

None

Possible values

A valid Service Manager hostname and communication port

Example usage

If no FQDN is specified in the web.xml, the following example is valid:

No FQDN provided in web.xml:

```
<serverhost>hostname</serverhost>
```

```
<serverport>13080</serverport>
```

Enter the following in the Initialization (sm.ini) file:

```
queryhashcode:hostname:13080
```

When the FQDN of a server host is provided in the web.xml file, you need to provide the same value in this parameter, as follows:

FQDN provided in web.xml:

```
<serverhost>hostname.com</serverhost>
```

```
<server port>13080</serverport>
```

Enter the following in the Initialization (sm.ini) file:

```
queryhashcode:hostname.com:13080
```

Note

Note You only need to provide one entry in the sm.ini file.

Related topics

System parameters
Enter a parameter in the sm.ini file
Servlet parameters
Parameter: querysecurity

2.24.26. Parameter: reportgroup

Parameter

reportgroup

Description

This parameter displays the status of ports, processes, and threads in horizontal and vertical scaling servlet implementations.

Important: The reportgroup process requires every other process to send it a report. This means that every process must consume some amount of system memory to generate its report, which decreases the amount of memory available to run the system. While the reporting information may be valuable, you need to balance your data tracking requirements with the amount of memory being used. To reduce the performance impact of running this process, you may want to run it on a development or testing system, or choose a longer interval between running reports.

Note

Note If you have opted to run this process continuously (for example, every minute non-stop), you have the option to shut down the process. To shut down a specific Service Manager process, specify the hostname of the Service Manager server and system process (PID) you want to shut down. On a command line, enter **sm -shutdown -host:myhost.myserver.com -PID:1234**.

Some examples of the information that's reported include:

- Service Manager locks that are currently being held.
- Service Manager threads waiting to acquire locks. This is useful for lock analysis.

- Thread dumps of all Service Manager processes on all hosts. This is useful for detecting deadlocks, if any. Thread dumps repeat for each process and host.

The information is sent to a separate log file.

Valid if set from

Server's operating system command prompt

Requires restart of the Service Manager server?

No

Default values

Run the report every 20 seconds.

Repeat the process 15 times.

Possible values

Number of seconds between reports and the number of times to run a report.

Example usage

Run the report every 20 seconds and repeat the process 15 times.

sm -reportgroup

sm -reportgroup:20,15

Recommended: Run the report every minute and repeat the process 20 times. This provides less load on the cluster.

sm -reportgroup:60,20

Log File Examples

The following sample is an example of the information that's displayed in a log file for a vertical scaling servlet implementation on one host.

Note

Note

- This example report has been reformatted for printed display.
- The output in this example has been truncated (this is not the full output from the command).

```
Load Balancer Status:Sat Jun 23 09:58:52 PDT 2012
HP Service Manager LoadBalancer Running on Host:myhost.myserver.com Port:34567
List of Hosts:
HostName: myhost.myserver.com
-------------------------------------------------------------------Server Instances
ProcessID     ClusterAddress     HttpPort  HttpsPort  Sessions  DbgMode  QMode  LB
7872     15.178.177.72:33285    34568       0       (3/50)      N       N     N
7881     15.178.177.72:33283    34569       0       (1/50)      N       N     N
7837     15.178.177.72:33278    34567       0       (0/50)      N       N     Y
-------------------------------------------------
State        LowMem     JAVA_USED/MAX/PERCENT
WR[6d22h39m] N       (4404408/238616576/1.8458097)
QR[9m]       N       (2983152/238616576/1.2501864)
RUN          N       (2601080/238616576/1.0900668)
-----------------------------------------------------------Non Server Instances---
ProcessID     ClusterAddress         State    LowMem     JAVA_USED/MAX/PERCENT
8510     15.178.177.72:33329       RUN        N         (0/89522176/0.0)
7931     15.178.177.72:33299       RUN        N   (1502256/59703296/2.5162027)
7880     15.178.177.72:33280       RST        N   (1539248/59703296/2.5781624)
7930     15.178.177.72:33291       SHT        N   (1514024/59703296/2.5359137)
------------------------------------
Command Line parameters
-reportgroup:1,1
-sync -log:../logs/sm.sync.log
system.start -log:../logs/sm.start.log
```

```
Locks from all nodes in the cluster.
Waiting locks
Thread Dumps from all nodes In the cluster
Thread Dump from process: Host: myhost.myserver.com PID: 7872
ChannelAddress: 15.178.177.72:33285
SM_LowMemoryHandler
java.lang.Thread.sleep(Native Method)
comp.hp.ov.sm.common.oom.LowMemoryHandler.run(LowMemoryHandler.java:181)
java.lang.Thread.run(Thread.java:619
```

Related topics

Servlet implementation
System parameters
Startup parameter: shutdown
Connection parameters
Servlet parameters

2.24.27. Parameter: threadsperprocess

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

threadsperprocess

Description

This parameter defines the number of threads that you want the Service Manager server to start when it starts a new process. Each thread can support one client connection or one background process.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

250

Possible values

Number of threads to start

Example usage

Command line: **sm -httpPort:13080 -threadsperprocess: 250**

Initialization file: `threadsperprocess:250`

Related topics

System parameters
Enter a parameter in the sm.ini file

2.24.28. Parameter: ssl_trustedClientsJKS

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

ssl_trustedClientsJKS

Description

This parameter defines the path and file name to the Java keystore containing the list of trusted client certificates from which the Service Manager server allows connection requests. When you enable the *ssl_reqClientAuth* parameter with any of the servlet implementations, only the clients with signed certificates in this keystore can connect to the Service Manager server. This keystore contains a copy of each client's signed certificate. All servlet implementation options require access to certificates stored in Java keystore formats.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

None

Possible values

Path and file name of the Java keystore containing the list of trusted client certificates

Example usage

Command line: **sm -httpPort:13080 -httpsPort:13081 -ssl_trustedClientsJKS:trusted.jks -ssl_trustedClientsPwd: <password>**

Initialization file: `ssl_trustedClientsJKS:trusted.jks`

Related topics

System parameters
Enter a parameter in the sm.ini file
SSL parameters
Servlet parameters

2.24.29. Parameter: ssl_trustedClientsPwd

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

ssl_trustedClientsPwd

Description

This parameter defines the password to the Java keystore containing the list of trusted client certificates from which the Service Manager server allows connection requests. When you enable the *ssl_reqClientAuth* parameter with any of the servlet implementations, only the clients with signed certificates in this keystore can connect to the Service Manager server. This keystore contains a copy of each client's signed certificate. All servlet implementation options require access to certificates stored in Java keystore formats.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

The keystorePass value

Possible values

Password to the Java keystore containing the signed client certificates of trusted clients

Example usage

Command line: `sm -httpPort:13080 -httpsPort:13081 -ssl_trustedClientsJKS:trusted.jks -ssl_trustedClientsPwd:<password>`

Initialization file: `keystorePass:<password>`

Related topics

System parameters
Enter a parameter in the sm.ini file
SSL parameters
Servlet parameters

2.24.30. Parameter: truststoreFile

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

truststoreFile

Description

This parameter defines the path and file name of the Java keystore containing the list of trusted certificate authorities. Service Manager requires this keystore to validate the server and client certificates in servlet implementations only. Service Manager implementations using classic listeners can't use this parameter and must use the *cacertpem* parameter instead.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

A null value causes the Service Manager server to look for the trusted keystore file *JAVA HOME*\lib\security\cacerts

Possible values

Path and file name of the Java keystore containing the list of trusted CA certificates relative to the Service Manager RUN folder

Example usage

Command line: **sm -httpPort:13080 -httpsPort:13081 -truststoreFile:mycacerts -truststorePass:changeme**

Initialization file: `truststoreFile:mycacerts`

Related topics

System parameters
Enter a parameter in the sm.ini file
SSL parameters
Servlet parameters

2.24.31. Parameter: truststorePass

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

truststorePass

Description

This parameter defines the password to the Java keystore containing the list of trusted CA certificates. Service Manager uses this keystore to validate server and client certificates in servlet implementations. All servlet implementation options require access to trusted certificates stored in Java keystore formats.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

changeit

Possible values

Password to the Java keystore containing the Service Manager server's signed certificate

Example usage

Command line: **sm -httpPort:13080 -httpsPort:13081 -truststoreFile:mycacerts -truststorePass:changeme**

Initialization file: `truststorePass:<password>`

Related topics

System parameters
Enter a parameter in the sm.ini file
SSL parameters
Servlet parameters

2.25. SSL parameters

The following table lists the startup parameters you can set from the Service Manager server's operating system command prompt or from the Service Manager initialization file (sm.ini).

These parameters determine how the Service Manager server manages SSL connections.

SSL Parameter	Brief description
Parameter: acceptsharedcert	Defines how the Service Manager server handles signed SSL certificates from incoming client requests in a Trusted Sign-On configuration
Web parameter: cacerts	Defines the path to the cacerts file required for web clients to make SSL connections to the web tier
Parameter: cacsignon	Defines whether trusted clients can log on to the Service Manager server with a Common Access Card (CAC), without entering log-on information
Parameter: ciphers	Defines an array of cipher suites used for SSL connections for the embedded Tomcat of the Service Manager Server.
Parameter: keystoreFile	Defines the path and file name to the Java keystore that contains the certificate of the Service Manager server
Parameter: keystorePass	Defines the password to the Java keystore that contains the certificate of the Service Manager server
Parameter: onewayssl4ws	Disables or enables one-way/anonymous SSL for web services clients in a Trusted Sign-On (TSO) or Common Access Card (CAC) configuration
Parameter: pgsslEnableCiphers	Defines an array of cipher suites used for PostgreSQL SSL connections.
Parameter: pgsslEnableProtocols	Defines the supported SSL protocols for PostgreSQL SSL connections.
Web parameter: secureLogin	Controls the encryption of network communication between the web application server and the web browser.
Web parameter: ssl	Causes all web clients to use SSL connections to the web tier
Parameter: ssl	Defines whether the Service Manager server requires SSL connections from all incoming client requests that are sent from Service Manager's Windows client or web client.
Parameter: sslConnector	Defines whether servlet container processes have an HTTPS (SSL-encrypted HTTP) communications port available.
Web parameter: sslPort	Defines the SSL port of the web application server (this is required only when the *secureLogin* parameter is set to true)
Parameter: sslEnabledProtocols	Defines the supported SSL protocols for HTTPS connections to the Service Manager Server.
Parameter: ssl_reqClientAuth	Defines whether the Service Manager server requires signed certificates from all incoming client requests

Parameter: ssl_trustedClientsJKS	Defines the path and file name to the Java keystore that contains the list of trusted client certificates from which the Service Manager server allows connection requests
Parameter: ssl_trustedClientsPwd	Defines the password to the Java keystore that contains the list of trusted client certificates from which the Service Manager server allows connection requests
Parameter: trustedsignon	Defines whether trusted clients can log in to the Service Manager server without having to provide login information

Related topics

System parameters
Enter a parameter in the sm.ini file

2.25.1. Parameter: acceptsharedcert

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

acceptsharedcert

Description

This parameter defines how the Service Manager server handles signed SSL certificates from incoming client requests in a Trusted Sign-On configuration.

Caution

Caution This functionality works only for the Service Manager Windows and web clients; it doesn't work for web service client connections, such as Mobility, and other third-party web service integrations.

When this parameter is set to 0 (default), the Service Manager server validates the signed SSL client certificates by using standard best practices. The validation procedure is described in Secure Sockets Layer (SSL) encryption.

Note

Note We recommend that you run the Service Manager server with the default value set for this parameter (*acceptsharedcert:0*), as it is the most secure mode of operation. Before you modify the default behavior, consider the following alternative workarounds:

- Do not use the Service Manager Windows client. Instead, use only the Service Manager web tier, as it doesn't incur the additional maintenance overhead or complexity that's associated with managing numerous signed client SSL certificates.
- If you must use the Service Manager Windows client in your environment, consider limiting the distribution of this client to a small number of users. This minimizes the additional overhead costs associated with managing numerous Service Manager Windows clients and their unique signed client SSL certificates.
- Use as many Service Manager Windows clients as are needed, but disable Trusted Sign-On functionality for these users. This eliminates the requirement to generate unique signed client SSL certificates.

When the parameter is enabled (*acceptsharedcert:1*), the Service Manager server allows Trusted Sign-On connections by using a so-called "shared certificate." TheService Manager server validates the shared certificate using only the following checks:

- Whether the certificate is issued by a trusted certificate authority

- Whether the Common Name attribute of the certificate is in the Service Manager Server's trusted clients keystore

This parameter is provided primarily for use in customer environments where the following conditions are true:

- There is a requirement to allow access to Service Manager through Trusted Sign-On for a large number of Service Manager Windows clients.
- Creating and maintaining the required signed SSL client certificates adds too much maintenance overhead and complexity to IT operations.

By using *acceptsharedcert:1*, only one client SSL certificate (the "shared certificate") needs to be created and maintained. This significantly minimizes the maintenance overhead costs and complexity that are associated with managing signed SSL client certificates. However, bear in mind the following considerations:

- You must still copy and distribute the shared certificate to individual Service Manager Windows clients before you can successfully use Trusted Sign-On access.
- By using *acceptsharedcert:1* you will have minimized your maintenance overhead and complexity of your IT operations at the cost of reduced security in Service Manager. This is due to the two simple "shared certificate" validation checks that the Service Manager server performs when it runs with *acceptsharedcert:1*. Running the Service Manager server with the recommended default value for the *acceptsharedcert* parameter provides the most secure method for enabling Trusted Sign-On features because the Service Manager server performs additional validation checks against the client SSL certificate. It is also possible, though unlikely, that if a malicious user obtains the "shared certificate" that user may be able to gain unauthorized access to Service Manager if they can then also defeat the NTLM-based implementation of Trusted Sign-On on the Service Manager Windows client.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

0

Possible values

0 (Disabled)

1 (Enabled)

Example usage

Command line: **sm -httpPort:13080 -acceptsharedcert:1**

Initialization file: `acceptsharedcert:1`

Related topics

Trusted sign-on
Enter a parameter in the sm.ini file
SSL parameters
Requirements for trusted sign-on
Parameter: ssl_reqClientAuth

2.25.2. Parameter: cacsignon

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

cacsignon

Description

This parameter defines whether trusted clients can sign in to the Service Manager server by using a Common Access Card (CAC) and without entering sign-in information. Enable this parameter to allow trusted clients to bypass the Service Manager sign-in screen and directly sign in with a valid certificate stored in a CAC.

Once you have enabled *cacsignon*, the following parameters are automatically (and implicitly) set to 1:

Parameter	Value	Notes
ssl	1	Any other value explicitly specified in `sm.ini` are ignored.
sslConnector	1	Any other value explicitly specified in `sm.ini` are ignored.
ssl_reqClientAuth	1	Any other value explicitly specified in `sm.ini` are ignored except *ssl_reqClientAuth:2*.

Valid if set from

Server's operating system command prompt

Initialization file (`sm.ini`)

Requires restart of the Service Manager server?

Yes

Default value

0

Possible values

0 (Disable CAC sign-on)

1 (Enable CAC sign-on)

Example usage

Command line: **sm -httpPort:13080 -cacsignon:1**

Initialization file: `cacsignon:1`

Related topics

Common Access Card (CAC) sign-on
Example: enabling CAC sign-on
Requirements for CAC sign-on

2.25.3. Parameter: ciphers

Parameter

ciphers

Description

This parameter defines an array of cipher suites used for SSL connections for the embedded Tomcat of the Service Manager Server.

- If you specify this parameter, Service Manager only enables the listed ciphers
- If you don't specify this parameter, Service Manager uses a default list of ciphers (including TLSv1.2 and TLSv1.3).

Valid if set from

Server's operating system command prompt

Initialization file (`sm.ini`)

Requires restart of the Service Manager server?

Yes

Default value

TLS_AES_256_GCM_SHA384
TLS_AES_128_GCM_SHA256
TLS_ECDHE_ECDSA_WITH_AES_256_GCM_SHA384
TLS_ECDHE_ECDSA_WITH_AES_128_GCM_SHA256
TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384
TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256

Possible values

An array of cipher suites listed below:

TLSv1.3 ciphers:

TLS_AES_256_GCM_SHA384
TLS_AES_128_GCM_SHA256
TLS_AES_128_CCM_8_SHA256
TLS_AES_128_CCM_SHA256
TLS_CHACHA20_POLY1305_SHA256

TLSv1.2 ciphers:

TLS_ECDHE_ECDSA_WITH_AES_128_GCM_SHA256
TLS_ECDHE_ECDSA_WITH_AES_256_GCM_SHA384
TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256
TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384
TLS_DHE_RSA_WITH_AES_128_GCM_SHA256
TLS_DHE_RSA_WITH_AES_256_GCM_SHA384
TLS_ECDHE_ECDSA_WITH_CHACHA20_POLY1305_SHA256
TLS_ECDHE_RSA_WITH_CHACHA20_POLY1305_SHA256

Example usage

Command line:

```
sm -httpport:13080 -
ciphers:TLS_AES_256_GCM_SHA384,TLS_AES_128_GCM_SHA256,TLS_DHE_RSA_WITH_AES_256_GCM_SHA384,TLS_ECDHE_RSA_W
ITH_AES_256_GCM_SHA384,TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256,TLS_ECDHE_ECDSA_WITH_AES_256_GCM_SHA384,TLS_
ECDHE_ECDSA_WITH_AES_128_GCM_SHA256
```

Initialization file:

```
ciphers:TLS_AES_256_GCM_SHA384,TLS_AES_128_GCM_SHA256,TLS_DHE_RSA_WITH_AES_256_GCM_SHA384,TLS_ECDHE_RSA_W
ITH_AES_256_GCM_SHA384,TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256,TLS_ECDHE_ECDSA_WITH_AES_256_GCM_SHA384,TLS_
ECDHE_ECDSA_WITH_AES_128_GCM_SHA256
```

Related topics

FIPS mode
Startup parameter: encryptionkey
Startup parameter: changeencrkey

2.25.4. Parameter: fipsmode

Parameter

fipsmode

Description

This parameter determines whether the Service Manager Server runs in FIPS 140-2 compliant mode ("FIPS mode"). This parameter can have one of the following values:

Caution

Caution In a horizontal scaling environment, you must set this parameter to the same value in all server nodes.

- 0 (default): FIPS mode is disabled. Service Manager uses the 64-bit DES encryption algorithm, and SSL connections to the Service Manager server don't use TLS protocols that are implemented by using a FIPS validated provider. By default, FIPS mode is disabled in Service Manager.
- 2: FIPS mode is enabled. Service Manager uses the 256-bit AES data encryption algorithm, and SSL connections to the Service Manager Server use TLS protocols that are implemented by using a FIPS validated provider. Additionally, Service Manager must use FIPS compliant Lightweight Single Sign-On (LW-SSO) and random number generation algorithms.

Note

Note For backward compatibility, if you used "fipsmode:1" to enable FIPS compliant AES-256 data encryption only, you can continue to use this configuration without any problems.

When FIPS mode is enabled on the Service Manager Server side, all clients (Windows Client, Web Tier Client, Mobility Client, and Web Service integrations) and LDAP Server can communicate with the Service Manager Server through only TLS (either FIPS validated TLS or normal TLS); however, the recommended best security practice is to run all of them in FIPS mode as well.

For more information, see FIPS mode.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

0

Possible values

0, 2

Example usage

Command line: **sm -httpport:13080 -fipsmode:2**

Initialization file: `fipsmode:2`

Related topics

FIPS mode

Configuring FIPS mode in Service Manager

2.25.5. Parameter: keystoreType

Parameter

keystoreType

Description

This parameter defines the keystore type for the server keystore and trusted clients kyestore of the Service Manager Server. This parameter must be set to one of the following values:

- JKS: when Service Manager is running in non-FIPS mode (fipsmode:0)

- BCFKS: when Service Manager is running in FIPS mode (fipsmode:2)

If not specified, the default value is "JKS"; however, when "fipsmode:2" is specified in the Server configuration file (sm.ini), "BCFKS" is used by default.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

JKS

Possible values

JKS, or BCFKS

Example usage

Command line: **sm -httpport:13080 -keystoreType:BCFKS**

Initialization file: `keystoreType:BCFKS`

Related topics

FIPS mode

Configuring FIPS mode in Service Manager

2.25.6. Parameter: onewayssl4ws

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's OS command prompt.

Parameter

onewayssl4ws

Description

This parameter provides the option to use one-way/anonymous SSL for web services clients in a Trusted Sign-On (TSO) or Common Access Card (CAC) configuration. When enabled, this parameter allows web services clients to access SM RTE interfaces without a certificate.

This parameter is used only for web service integrations, and therefore doesn't affect the TSO behavior of the Service Manager Windows and web clients. If you can vouchsafe your internal security policies that govern your Web Services clients that are integrated with the Service Manager server, this parameter enables you to deploy SSO/TSO without the need to specify the stringent *ssl_reqClientAuth:2*. You can start specific servlets by using the *onewayssl4ws*, *ssl_reqClientAuth:0*, and *debugnode* parameters for a web services integration.

By default, this parameter is disabled.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of Service Manager server?

Yes

Default value

0

Possible values

0: Disable one-way/anonymous SSL for web services clients

1: Enable one-way/anonymous SSL for web services clients

Related topics

System parameters
Enter a parameter in the sm.ini file
SSL parameters
Connection parameters
Servlet parameters

2.25.7. Parameter: pgsslEnableProtocols

Parameter

pgsslEnableProtocols

Description

This parameter defines a comma separated list of SSL protocols to support SSL connections for the PostgreSQL database. For security considerations, it is advisable that you use TLSv1.2 or TLSv1.3.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

TLSv1.3,TLSv1.2

Possible values

Supported combinations of the following values (comma-separated): TLSv1.2, TLSv1.3

Example usage

Command line:

```
sm -pgsslEnableProtocols:TLSv1.2,TLSv1.3
```

Initialization file:

```
pgsslEnableProtocols:TLSv1.2,TLSv1.3
```

2.25.8. Parameter: pgsslEnableCiphers

Parameter

pgsslEnableCiphers

Description

This parameter defines an array of cipher suites used for PostgreSQL SSL connections.

- If you specify this parameter, Service Manager only enables the protocols listed and supported by the SSL implementation.
- If you don't specify this parameter, Service Manager uses a default list of ciphers (including TLSv1.2 and TLSv1.3).

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

```
TLS_AES_128_GCM_SHA256,TLS_AES_256_GCM_SHA384,TLS_AES_128_CCM_SHA256,TLS_AES_128_CCM_8_SHA256,DHE-RSA-AES256-GCM-SHA384,ECDHE-RSA-AES256-GCM-SHA384,ECDHE-RSA-AES128-GCM-SHA256,DHE-RSA-AES256-SHA256,DHE-RSA-AES128-SHA256,ECDHE-RSA-AES256-SHA384,ECDHE-RSA-AES128-SHA256
```

Possible values

Any combination of the following cipher suites (separated by commas):

```
TLS_AES_128_GCM_SHA256,TLS_AES_256_GCM_SHA384,TLS_AES_128_CCM_SHA256,TLS_AES_128_CCM_8_SHA256,DHE-RSA-AES256-GCM-SHA384,ECDHE-RSA-AES256-GCM-SHA384,ECDHE-RSA-AES128-GCM-SHA256,DHE-RSA-AES256-SHA256,DHE-RSA-AES128-SHA256,ECDHE-RSA-AES256-SHA384,ECDHE-RSA-AES128-SHA256,ECDHE-ECDSA-AES128-SHA256,ECDHE-ECDSA-AES128-GCM-SHA256,ECDHE-ECDSA-AES256-SHA384,ECDHE-ECDSA-AES256-GCM-SHA384,AES256-GCM-SHA384,AES256-SHA256,AES128-SHA256
```

Example usage

Command line:

```
sm -pgsslEnableCiphers:ECDHE-RSA-AES128-GCM-SHA256,ECDHE-RSA-AES256-GCM-SHA384
```

Initialization file:

```
pgsslEnableCiphers:ECDHE-RSA-AES128-GCM-SHA256,ECDHE-RSA-AES256-GCM-SHA384
```

2.25.9. Parameter: ssl

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

ssl

Description

This parameter defines whether the Service Manager server requires SSL connections from all incoming client requests that are sent from Service Manager's Windows client or Web client.

Note

Note This SSL connection enforcement also applies to Web Services clients.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

0

Possible values

0 (Disable)

1 (Enable)

Example usage

Command line: **sm -httpPort:13080 -ssl:1**

Initialization file: `ssl:1`

Related topics

System parameters
Enter a parameter in the sm.ini file
SSL parameters
Connection parameters
Servlet parameters

2.25.10. Parameter: sslEnabledProtocols

Parameter

sslEnabledProtocols

Description

This parameter defines a comma separated list of SSL protocols to support for HTTPS connections of the Service Manager Server.

- If you specify this parameter, Service Manager only enables the protocols listed and supported by the SSL implementation.
- If you don't specify this parameter, Service Manager uses a default list of ciphers (including TLSv1.2 and TLSv1.3).

Note

Note You are recommended to use the default value, which is `TLSv1.2,TLSv1.3`. If you want to use another value, keep in mind that TLSv1.2 and TLSv1.3 is always mandatory.

Important

Important By selecting to change the allowed SSL protocols, you are disabling or bypassing security features, thereby exposing the system to increased security risks. By using this option, you understand and agree to assume all associated risks and hold the company harmless for the same.

Important

Important In case SSL protocols are changed from their default configuration (which is `TLSv1.2,TLSv1.3`), you may need to add relevant protection measures to protect against risks associated with the SSL protocols which were added, which is not provided by the company. By not implementing relevant protection measures you may be exposing the system to increased security risks. You understand and agree to assume all associated risks and hold the company harmless for the same. It remains at all times the Customer's sole responsibility to assess its own regulatory and business requirements. The company doesn't represent or warrant that its products comply with any specific legal or regulatory standards applicable to Customers in conducting Customer's business.

Valid if set from

Server's operating system command prompt

Initialization (sm.ini) file

Requires restart of the Service Manager server?

Yes

Default value

`TLSv1.2,TLSv1.3`

Possible values

Supported combinations of the following values (comma-separated): `TLSv1.2,TLSv1.3` (see the note in the description)

Example usage

```
sslEnabledProtocols:TLSv1.2,TLSv1.3
```

2.25.11. Parameter: ssl_reqClientAuth

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

ssl_reqClientAuth

Description

This parameter defines whether the Service Manager server requires signed certificates from all incoming client requests. Enable this parameter to limit access to the Service Manager server to only those clients that present signed certificates.

When enabled, clients can no longer connect to the Service Manager server using the server's certificate for anonymous SSL. Each client must have its own signed certificate. If you enable this parameter with the value *ssl_reqClientAuth:2*, in addition to presenting client certificates, the server validates each client certificate against a list of trusted clients as defined by the `ssl_trustedClientsJKS` parameter. Using *ssl_trustedClientsJKS* with the value *ssl_reqClientAuth:2* is required when using the Trusted Sign-On features of Service Manager (*trustedsignon:1*). The server only allows connections from clients with certificates in the trusted clients list.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

0

Possible values

0 (Disable)

1 (Enable - require client certificates)

2 (Enable - require client certificates and require clients to be on the list of trusted clients)

Example usage

Command line: **sm -httpPort:13080 -ssl_reqClientAuth:1**

Initialization file: `ssl_reqClientAuth:1`

Related topics

System parameters
Enter a parameter in the sm.ini file
SSL parameters
Servlet parameters

2.25.12. Parameter: trustedsignon

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

trustedsignon

Description

This parameter defines whether trusted clients can sign in to the Service Manager server without having to provide sign-in information. Enable this parameter to allow trusted clients to bypass the Service Manager sign-in screen. Users must already have signed in to a trusted authentication source for *trustedsignon* to succeed.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

0

Possible values

0 (Disable)

1 (Enable)

Example usage

Command line: **sm -httpPort:13080 -trustedsignon:1**

Initialization file: `trustedsignon:1`

Related topics

Trusted sign-on
System parameters
Enter a parameter in the sm.ini file
SSL parameters
Servlet parameters

2.25.13. Parameter: truststoreType

Parameter

truststoreType

Description

This parameter defines the keystore type for the Service Manager Server's truststore. This parameter must be set to one of the following values:

- JKS: when Service Manager is running in non-FIPS mode (fipsmode:0)
- BCFKS: when Service Manager is running in FIPS mode (fipsmode:2)

If not specified, the default value is "JKS"; however, when "fipsmode:2" is specified in the Server configuration file (sm.ini), "BCFKS" is used by default.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

JKS

Possible values

JKS, or BCFKS

Example usage

Command line: **sm -httpport:13080 -truststoreType:BCFKS**

Initialization file: `truststoreType:BCFKS`

Related topics

FIPS mode

Configuring FIPS mode in Service Manager

2.26. System information parameters

These parameters display information about the server.

Startup parameter	Description
Startup parameter: appversion	Displays the version information for the Service Manager server
Parameter: certExpireWarningDays	Specifies the number of days before the user's certificate expires when Service Managerstarts to issue a warning message at user login
Startup parameter: clustername	Specifies a Microsoft cluster for failover and replaces the subnetaddress and host name with the clustername value
convertweek53	Determines whether a Service Manager report includes the last week of a year as the first week of the next year when this week is the 53rd week that straddles two years
Parameter: corepath	Defines the path to the core file that's generated by a system crash
Parameter: enablecoredump	Enables Service Manager to log any information generated by a system crash
esstoolbarbutton	Enables toolbar buttons on ESS pages.
Special parameters: encrypt	Encrypts a parameter value that can only be used in the command line.
Special parameters: mergeRoles	When you add multiple security roles to an operator, if you change the original sequence of the security roles, the restrictions set in configuration management area for specific CI types doesn't work. You can add the **mergeRoles** in the sm.ini file to avoid this issue.
Special parameters: filterQueryOperators	Enables Service Manager to check the character of the fields. If a field contains query operators (<, >, =, ~, #) as first character, an error message is displayed. If you ignore the error and insist to save the record anyway, the system will remove the query operator.
Special parameters: instantOn	Installs an AutoPass Temporary License.
Special parameters: language	Creates locale for the system running Service Manager.
Parameter: licensefile	Defines the path to the AutoPass license file that contains the list of enabled modules and features for your Service Manager server
Startup parameter: log	Defines the path to the Service Manager log file
Parameter: log4jDebug	Enables certain java packages to be started in debug mode
Startup parameter: logappend	Causes Service Manager to add log messages to the existing log file
Parameter: logqueuesize	Controls the size of the log queue, specified by the number of messages

Startup parameter: maxlogsize	Defines the maximum size to which a log file can grow before Service Manager rotates to a new log file
Special parameters: maxreportpages	Defines the maximum number of pages that users can print from server-side printers
Special parameters: msgid	Forces the Service Manager server to display the message ID with all messages
Special parameters: msglog	Defines the maximum number of messages the Service Manager server displays in the log file
Parameter: numberoflogfiles	Specifies the number of the log files.
Special parameters: reportcache	Displays basic Service Manager CACHE statistics to the command line, which you can capture in a text file by using standard output (stdout) parameters appropriate to your operating system
Special parameters: reportdiagnostics	Displays the counter diagnostics report, which contains information about web services transaction count, session login/logout count, database client transmission count, heartbeat interval - request response count, event-in / event-out count, and email-in / email-out count **Note** Note By default, the Diagnostic Service is disabled. Before running the reportdiagnostics command, administrators must specify *debugdiagnostics:1* to enable the Diagnostic Service.
Special parameters: reportipc	Displays a diagnostic report of semaphore use, which you can capture in a text file by using standard output (stdout) parameters appropriate to your operating system
Parameter: reportlbstatus	Displays information about available ports and threads in horizontal and vertical scaling servlet implementations
Parameter: reportLBInterval	Periodically and automatically logs information (every *n* seconds) about available ports and threads in horizontal and vertical scaling servlet implementations
Special parameters: reportlanguages	Lists all the code pages that Service Manager supports **Note** Note Only those code pages that start with "mswin" are used in the context of SQL Server

Special parameters: reportlic	Displays the Service Manager License Report, which you can capture in a text file by using standard output (stdout) parameters appropriate to your operating system
Special parameters: reportlocks	Displays resource locks, which you can capture in a text file by using standard output (stdout) parameters appropriate to your operating system.
Special parameters: reportsem	Displays the semaphore report, which you can capture in a text file by using standard output (stdout) parameters appropriate to your operating system.
Special parameters: reportshm	Displays the shared memory report, which you can capture in a text file using standard output (stdout) parameters appropriate to your operating system.
Special parameters: reportstatus	Displays a shared memory, semaphore and process report, which you can capture in a text file using standard output (stdout) parameters appropriate to your operating system.
Special parameters: reporttolog	Allows the `reportstatus`, `reportsem`, and `reportshm` reports to be written to the console or to the sm.log file.
Special parameters: str_export	Exports a *.str file for localization.
Special parameters: str_generate	Generates the scmessage file localization strings.
Special parameters: str_import	Imports a localized *.str file.
Special parameters: str_language	Imports a localized *.str file for a certain language.
Parameter: usedmemcompmode	Determines the method that's used to calculate memory usage
Special parameters: verifylic	Verifies whether a valid license is found or not.
Startup parameter: version	Causes the Service Manager server to display version information

Related topics

Enter a parameter in the sm.ini file
Enter a parameter in sm.cfg or the server's OS command prompt

2.26.1. Parameter: certExpireWarningDays

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

certExpireWarningDays

Description

This parameter specifies the number of days before the user's certificate expires when Service Manager starts to issue a warning message at user login.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

30

Possible values

Number of days

> **Note**
>
> Note If the value you set is longer than your certificate expiration day, the warning message appears immediately at user login. If the value you set is shorter than your certificate expiration day, the warning message appears exactly on the day as you set at user login.

Example usage

Command line: `sm -certExpireWarningDays:30`

Related topics

System parameters
Log switching
Enter a parameter in the sm.ini file
System information parameters

2.26.2. Parameter: esstoolbarbutton

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

esstoolbarbutton

Description

By default, the action buttons on ESS pages are displayed at the bottom of the pages. You can use this parameter to enable toolbar buttons on ESS pages.

To enable toolbar buttons, set the parameter to *esstoolbarbutton:1*, restart your web tier application server, and then re-login ESS. After you enable this parameter, the action buttons are displayed on the toolbar at the top of ESS pages instead of at the bottom.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No. Restart your web tier application server.

Default value

0

Possible values

0 (Do not enable ESS toolbar buttons)

1 (Enable ESS toolbar buttons)

Example usage

Command line: **esstoolbarbutton:1**

Initialization file: `esstoolbarbutton:1`

Related topics

System parameters
Enter a parameter in the sm.ini file

2.26.3. Parameter: numberoflogfiles

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt. You can always set a startup parameter from the server's operating system command prompt.

Parameter

numberoflogfiles

Description

This parameter specifies the number of the log files.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of Service Manager server?

Yes

Default value

0

Possible values

0 to 100

0 means “don't rotate” and the system generates one log file only.

If this value is set to larger than 100, the system resets this value to 7.

Example usage

Command line: **sm -httpPort:13080 -numberoflogfiles:3**

Initialization file: `numberoflogfiles:3`

Related topics

System parameters
Servlet parameters
Debugging parameters

2.26.4. Parameter: reportLBInterval

Parameter

reportLBInterval

Description

This parameter defines the interval (in seconds) at which Service Manager automatically reports the information about available ports and threads in horizontal and vertical scaling servlet implementations. When this parameter is specified, Service Manager automatically logs the information every *n* seconds to the following log file located in the same directory of the server log (sm.log): lbstatus_<*pid*>.log, where <*pid*> is the ID of the load balancer process.

This parameter reports the same information when you manually run the reportlbstatus parameter. For details about such information, see Parameter: reportlbstatus.

Note

Note By default, this parameter is disabled; if a value less than 60 is specified, it is automatically reset to 60.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Startup file (sm.cfg)

Requires restart of the Service Manager server?

Yes

Default value

None

Possible values

Numbers no less than 60

Example usage

Command line: **sm -loadbalancer -reportLBInterval:180 -httpPort:13080**

Initialization file: `reportLBInterval:180`

Startup file: `sm -loadbalancer -reportLBInterval:180 -httpPort:13080`

Related topics

Servlet implementation
System parameters
Enter a parameter in the sm.ini file
Connection parameters
Servlet parameters

2.26.5. Parameter: usedmemcompmode

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

usedmemcompmode

Description

This parameter determines the calculation method for how memory usage is calculated.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of Service Manager server?

Yes

Default value

1 (The default is 1, which is the normal Service Manager method)

Possible values

0: Specifies that used memory will be the sum of committed memory plus the reserved memory

1: Specifies that used memory will be equal to the committed memory without a log entry

2: Specifies that used memory will be equal to the committed memory with a log entry

Example usage

Initialization file: `usedmemcompmode:1`

(This setting will sum only the committed memory when computing the used memory, and will not generate a line of calculation log in sm.log.)

Command line: **sm -httpPort:13080 – usedmemcompmode:0**

Related topics

System parameters
Enter a parameter in the sm.ini file
List: System Information Parameters

2.26.6. Startup parameter: appversion

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

appversion

Description

Entering this parameter on the command line displays the version information for the Service Manager server.

Valid if set from

Server's operating system command prompt

Requires restart of the Service Manager server?

No

Default value

None

Possible values

None

Example usage

Command line: **sm -appversion**

Related topics

System parameters
Enter a parameter in the sm.ini file
System information parameters

2.26.7. Startup parameter: clustername

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

clustername

Description

Enabling this parameter specifies a Microsoft cluster for failover and replaces the subnetaddress and host name with the value of the *clustername* parameter.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

CLUSTER

Possible values

<NAME>

Example usage

Command line: **sm -clustername:<*NAME*>**

Initialization file: `clustername:<NAME>`

Related topics

System parameters
Log switching
Enter a parameter in the sm.ini file
System information parameters

2.26.8. Startup parameter: convertweek53

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

convertweek53

Description

This parameter determines whether a Service Manager report includes the last week of a year as the first week of the next year when this week is the 53rd week that straddles two years.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

0

Possible values

- 0 (Do not convert the 53rd week of a year to be the 1st of the next year)
- 1 (Convert the 53rd week of a year to be the 1st of the next year)

Example usage

Command line: **sm -convertweek53:1**

Initialization file: `convertweek53:1`

Related topics

System parameters
Enter a parameter in the sm.ini file

2.26.9. Startup parameter: log

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

log

Description

This parameter defines the path to the Service Manager log file.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

../logs/sm.log

Possible values

Path to log file

Example usage

Command line: `sm -httpPort:13080 -log:log.log`

Initialization file: `log:log.log`

Related topics

System parameters
Log switching
Enter a parameter in the sm.ini file
System information parameters

2.26.10. Startup parameter: logappend

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

logappend

Description

Enabling this parameter causes Service Manager to add log messages to the existing log file. Disabling this parameter causes Service Manager to overwrite existing log messages.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

1

Possible values

0 (Disable)

1 (Enable)

Example usage

Command line: `sm -httpPort:13080 -logappend:1`

Initialization file: `logappend:1`

Related topics

System parameters
System information parameters

2.26.11. Startup parameter: maxlogsize

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

maxlogsize

Description

This parameter defines the maximum size to which a log file can grow before Service Manager rotates to a new log file. The system creates a new log when the current file reaches the indicated size. This parameter requires the use of the *numberoflogfiles* parameter.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

5242880

Note

Note We recommend that you don't set the value of the *maxlogsize* parameter lower than the default value.

Possible values

Size in bytes

Example usage

Command line: **sm -httpPort:13080 -numberoflogfiles:3 -maxlogsize:5242880**

Initialization file: `maxlogsize:5242880`

Related topics

System parameters
Enter a parameter in the sm.ini file
System information parameters

2.26.12. Startup parameter: logqueuesize

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

logqueuesize

Description

This parameter controls the number of log messages the server caches when writing to the log file. Each RTE process has a single thread dedicated for writing to the log file.Each thread writes log messages to an in-memory queue which is then read by the log file writer thread. This parameter controls how big that queue is, measured in the number of messages. Modify the logqueuesize parameter if users experience slower response times during log rotation of a large file.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart?

This parameter requires a restart of the Service Manager process.

Default value

1000

Possible values

Number of messages

Example usage

Command line: `sm -httpPort:13080 -numberoflogfiles:3 -maxlogsize:5242880 -logqueuesize:2000`

Initialization file: `logqueuesize:2000`

Related topics

System parameters
Enter a parameter in the sm.ini file
System information parameters

2.26.13. Startup parameter: version

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

version

Description

Using this parameter at the operating system command prompt causes the Service Manager server to display the Service Manager server (RTE) version, host server, operating system, and network information. If you use a value of 2 (*sm -version:2*), this command also reports the Service Manager database type and version, application version, and a listing of all Java jars used by the RTE and their corresponding versions, when available.

Valid if set from

Server's operating system command prompt

Requires restart of the Service Manager server?

No

Default value

0

Possible values

0, 1 - Returns server (RTE) version, host server, operating system, and network information.

or

2 - Returns database type and version, applications, and jar version information in addition to server (RTE) version, host server, operating system, and network information.

Example usage

Command line: **sm -version:2**

Related topics

System parameters
Enter a parameter in the sm.ini file
System information parameters

2.26.14. Special parameters: encrypt

Parameter

encrypt

Description

This parameter encrypts a parameter value that can only be used in the command line.

Valid if set from

Server's operating system command prompt

Requires restart of the Service Manager server?

No

Example usage

Command line: **sm -encrypt:<*parameter name*>:<*parameter value*>**

If the string containing the *<parameter name>:<parameter value>* includes spaces, you must enclose the entire parameter name and value in quotation marks. For example, **sm -encrypt:"sqllogin:RDBMS user/my password"**.

Related topics

System information parameters

2.26.15. Special parameters: mergeRoles

Parameter

mergeRoles

Description

When you add multiple security roles to an operator, if you change the original sequence of the security roles, the restrictions set in configuration management area for specific CI types doesn't work.

You can add the **mergeRoles** in the sm.ini file to avoid this issue.

Valid if set from

Initialization (sm.ini) file

Requires restart of the Service Manager server?

Yes

Default value

None

Possible values

None

Example usage

Initialization file: **mergeRoles**

2.26.16. Special parameters: filterQueryOperators

Parameter

filterQueryOperators

Description

Enabling this parameter causes Service Manager to check the character of the fields. If a field contains query operators (<, >, =, ~, #) as first character, an error message is displayed. If you ignore the error and insist to save the record anyway, the system will remove the query operator.

Note:

- When this parameter is enabled, in SOAP Request, if Request XML contains attribute with query operators (<, >, =, ~, #) as first character, these characters will be removed by the system. But in this case, SOAP Request will fail with validation failure.
- For the two-character query operators, for example, ">=", both ">" and "=" will be removed by the system if the the first two characters are ">=".
- **Exception:** When this parameter is enabled, if the first characters of the field is "**<**" and this field also include "**>**", the record can be saved successfully.
- This parameter doesn't work on JS and REST

- If an existing template include a field with query operators (<, >, =, ~, #) as first character, the character will not be removed by the system when you apply the template. But when you update a template, if the updated field contains query operators as first character, the character will be automatically removed by the system.

Valid if set from

Initialization (sm.ini) file

Requires restart of the Service Manager server?

No

Default value

0

Possible values

0 (Disable)

1 (Enable)

Example usage

Initialization file: `filterQueryOperators:0`

Related topics

System parameters
Enter a parameter in the sm.ini file
Debugging parameters

2.26.17. Special parameters: instantOn

Parameter

instantOn

Description

This parameter installs an AutoPass Temporary License.

Valid if set from

Server's operating system command prompt

Requires restart of the Service Manager server?

No

Default value

None

Possible values

None

Example usage

Command line: **sm -instantOn**

Related topics

System information parameters

2.26.18. Special parameters: language

Parameter

language

Description

This parameter creates locale for the system running Service Manager.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

English-US

Possible values

All Service Manager supported language codes.

Example usage

Command line: **SM -language:LANGUAGE_LOCALE**

Initialization file: `language:LANGUAGE_LOCALE`

Related topics

System information parameters

2.26.19. Special parameters: maxreportpages

Parameter

maxreportpages

Description

This parameter defines the maximum number of pages that users can print from server-side printers.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

1000

Possible values

Number of pages

Example usage

Command line: **sm -httpPort:13080 -maxreportpages:1500**

Initialization file: `maxreportpages:1500`

Related topics

System information parameters

2.26.20. Special parameters: msgid

Parameter

msgid

Description

This parameter forces the Service Manager server to display the message ID with all messages.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

None

Possible values

None

Example usage

Command line: **sm -httpPort:13080 -msgid**

Initialization file: `msgid`

Related topics

System information parameters

2.26.21. Special parameters: msglog

Parameter

msglog

Description

This parameter defines the maximum number of messages the Service Manager server displays in the log file. These include only messages designated for the current operator. That is, the messages that are addressed to operator() or any field whose value equals operator().

> **Note**
>
> Note If this parameter is not explicitly specified (either in sm.ini or in the server's operating system command prompt), only the messages that aren't sent to the clients are logged (for example, the messages generated by the background processes), and the default value (50000) is used. If you want the system to log the messages that are sent to the clients, you must explicitly specify this parameter.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

50000

Possible values

Number of log messages to display

Example usage

Command line: **sm -httpPort:13080 -msglog:75000**

Initialization file: `msglog:75000`

Related topics

System information parameters

2.26.22. Special parameters: reportcache

Parameter

reportcache

Description

This parameter displays basic Service Manager CACHE statistics to the command line, which you can capture in a text file using standard output (stdout) parameters appropriate to your operating system.

Valid if set from

Server's operating system command prompt

Requires restart of the Service Manager server?

No

Default value

None

Possible values

None

Example usage

Command line: **sm -reportcache**

Related topics

System parameters
Enter a parameter in the sm.ini file
System information parameters

2.26.23. Special parameters: reportdiagnostics

Parameter

reportdiagnostics

Description

This parameter displays the counter diagnostics report. This report contains the following diagnostic information, which is also available from the server log (default: sm.log):

- Web services transaction count / size (max size of a web service transaction / average size)
- Session login / logout count
- Database client transmission volume / size (network traffic)
- Heartbeat interval - request response count
- Event-in / event-out count
- Email-in / email-out count

Note

Note By default, the Diagnostic Service is disabled. Before running the reportdiagnostics command, administrators must specify debugdiagnostics:1 to enable the Diagnostic Service.

Valid if set from

Server's operating system command prompt

Requires restart of the Service Manager server?

No

Default value

None

Possible values

None

Example usage

Command line: **sm -reportdiagnostics**

Related topics

System parameters
Enter a parameter in sm.cfg or the server's OS command prompt
View diagnostic counter information
System information parameters
Parameter: debugdiagnostics

2.26.24. Special parameters: reportipc

Parameter

reportipc

Description

This parameter displays a diagnostic report of semaphore use, which you can capture in a text file using standard output (*stdout*) parameters appropriate to your operating system. The semaphores roughly match the categories of shared memory. That is, each shared memory category has a semaphore (some shared memory categories share a semaphore so it's not quite one to one).

Valid if set from

Server's operating system command prompt

Requires restart of the Service Manager server?

No

Default value

None

Possible values

None

Example usage

Command line: **sm -reportipc**

Related topics

System parameters
Enter a parameter in the sm.ini file
System information parameters

2.26.25. Special parameters: reportlanguages

Parameter

reportlanguages

Description

This parameter lists all possible language settings for Service Manager, as well as information regarding which code pages those settings will really use.

Valid if set from

Server's operating system command prompt

Requires restart of the Service Manager server?

No

Default value

None

Possible values

None

Example usage

Command line: **sm -reportlanguages**

Sample output:

Language	CP	Description
iso8859-1	819	ISO 8859-1 (Western European Latin-1)
8859-1	819	ISO 8859-1 (Western European Latin-1)
iso8859-2	912	ISO 8859-2 (Central European Latin-2)
8859-2	912	ISO 8859-2 (Central European Latin-2)
iso8859-5	915	ISO 8859-5 (Cyrillic)
8859-5	915	ISO 8859-5 (Cyrillic)
iso8859-7	813	ISO 8859-7 (Greek)
8859-7	813	ISO 8859-7 (Greek)
iso8859-9	920	ISO 8859-9 (Turkey Latin-5)
8859-9	920	ISO 8859-9 (Turkey Latin-5)
iso8859-11	8741	ISO 8859-11 (Thai)
8859-11	8741	ISO 8859-11 (Thai)
iso8859-15	922	ISO 8859-15 (Euro, Finnish, Estonian Latin-9)
8859-15	922	ISO 8859-15 (Euro, Finnish, Estonian Latin-9)
koi8-r	921	Kod Obmena Informatsiey, Russian, Bulgarian
koi8-u	1124	Kod Obmena Informatsiey, Ukranian
utf-8	884	UTF-8
utf8	884	UTF-8
mswin874	874	MS cp874 Thai
mswin932	9932	MS cp932 Japanese (almost Shift-JIS)
mswin936	936	MS cp936 Simplified Chinese
mswin949	949	MS cp949 Korean
mswin950	950	MS cp950 Traditional Chinese
mswin1250	1250	MS cp1250 Central European
mswin1251	1251	MS cp1251 Russian, Bulgarian, Serbian
mswin1252	1252	MS cp1252 Latin/Western European

mswin1253	1253	MS cp1253 modern Greek
mswin1254	1254	MS cp1254 Turkish
mswin1257	1257	MS cp1257 Estonian, Latvian, Lithuanian
sjis	932	Shift-JIS
american	1252	MS cp1252 Latin/Western European
australian	1252	MS cp1252 Latin/Western European
basque	1252	MS cp1252 Latin/Western European
bosnian	1250	MS cp1250 Central European
bulgarian	1251	MS cp1251 Russian, Bulgarian, Serbian
canadian	1252	MS cp1252 Latin/Western European
catalan	1252	MS cp1252 Latin/Western European
chinese_simplified	936	MS cp936 Simplified Chinese
chinese_traditional	950	MS cp950 Traditional Chinese
croatian	1250	MS cp1250 Central European
czech	1250	MS cp1250 Central European
danish	1252	MS cp1252 Latin/Western European
dutch	1252	MS cp1252 Latin/Western European
english	1252	MS cp1252 Latin/Western European
estonian	1257	MS cp1257 Estonian, Latvian, Lithuanian
finnish	1252	MS cp1252 Latin/Western European
french	1252	MS cp1252 Latin/Western European
german	1252	MS cp1252 Latin/Western European
greek	1253	MS cp1253 modern Greek
hungarian	1250	MS cp1250 Central European
icelandic	1252	MS cp1252 Latin/Western European
italian	1252	MS cp1252 Latin/Western European
japanese	9932	MS cp932 Japanese (almost Shift-JIS)
korean	949	MS cp949 Korean
latvian	1257	MS cp1257 Estonian, Latvian, Lithuanian

lithuanian	1257	MS cp1257 Estonian, Latvian, Lithuanian
new_zealand	1252	MS cp1252 Latin/Western European
new_zealand	1252	MS cp1252 Latin/Western European
norwegian	1252	MS cp1252 Latin/Western European
polish	1250	MS cp1250 Central European
portuguese	1252	MS cp1252 Latin/Western European
russian	1251	MS cp1251 Russian, Bulgarian, Serbian
slovak	1250	MS cp1250 Central European
slovenian	1250	MS cp1250 Central European
slovak	1250	MS cp1250 Central European
spanish	1252	MS cp1252 Latin/Western European
swedish	1252	MS cp1252 Latin/Western European
swiss	1252	MS cp1252 Latin/Western European
thai	874	MS cp874 Thai
turkish	1254	MS cp1254 Turkish
ukrainian	1251	MS cp1251 Russian, Bulgarian, Serbian

Related topics

System parameters
Enter a parameter in the sm.ini file
System information parameters

2.26.26. Special parameters: reportlic

Parameter

reportlic

Description

This parameter displays the Service Manager License Report, which you can capture in a text file using standard output (*stdout*) parameters appropriate to your operating system. The license report includes applications and platforms enabled, user seats, and expiration dates.

Valid if set from

Server's operating system command prompt

Requires restart of the Service Manager server?

No

Default value

None

Possible values

None

Example usage

Command line: **sm -reportlic**

Related topics

System parameters
Enter a parameter in the sm.ini file
System information parameters

2.26.27. Special parameters: reportlocks

Parameter

reportlocks

Description

This parameter displays resource locks, which you can capture in a text file using standard output (*stdout*) parameters appropriate to your operating system.

Valid if set from

Server's operating system command prompt

Requires restart of the Service Manager server?

No

Default value

None

Possible values

None

Example usage

Command line: **sm -reportlocks**

Related topics

System parameters
Enter a parameter in the sm.ini file
System information parameters

2.26.28. Special parameters: reportsem

Parameter

reportsem

Description

This parameter displays the semaphore report, which you can capture in a text file using standard output (*stdout*) parameters appropriate to your operating system. The semaphores roughly match the categories of shared memory. That is, each shared memory category has a semaphore (some shared memory categories share a semaphore so it's not quite one to one).

Valid if set from

Server's operating system command prompt

Requires restart of the Service Manager server?

No

Default value

None

Possible values

None

Example usage

Command line: **sm -reportsem**

Related topics

System parameters
Enter a parameter in the sm.ini file
System information parameters

2.26.29. Special parameters: reportshm

Parameter

reportshm

Description

This parameter displays the shared memory report, which you can capture in a text file using standard output (*stdout*) parameters appropriate to your operating system.

Valid if set from

Server's operating system command prompt

Requires restart of the Service Manager server?

No

Default value

None

Possible values

None

Example usage

Command line: **sm -reportshm**

Related topics

System parameters
Enter a parameter in the sm.ini file
System information parameters

2.26.30. Special parameters: reportstatus

Parameter

reportstatus

Description

This parameter displays a shared memory, semaphore and process report, which you can capture in a text file using standard output (*stdout*) parameters appropriate to your operating system. The semaphores roughly match the categories of shared memory. That is, each shared memory category has a semaphore (some shared memory categories share a semaphore so it's not quite one to one).

This parameter includes a sleep attribute; for example, *sm -reportstatus -sleep:nn* executes the reportstatus report every *nn* seconds.

Valid if set from

Server's operating system command prompt

Requires restart of the Service Manager server?

No

Default value

None

Possible values

None

Example usage

Command line: **sm -reportstatus -sleep:30**

This will execute the reportstatus report every thirty seconds.

Related topics

System parameters
Enter a parameter in the sm.ini file
System information parameters

2.26.31. Special parameters: reporttolog

Parameter

reporttolog

Description

This parameter allows the reportstatus, reportsem, and reportshm reports to be written to the console or sm.log file.

Valid if set from

The server's operating system command prompt

The initialization (sm.ini) file.

Requires restart of the Service Manager server?

No

Default value

0

Possible values

0 (Output to the console)

1 (Output to the sm.log file)

Example usage

Command line: **sm -reportstatus -reporttolog:1**

Related topics

System parameters
Enter a parameter in the sm.ini file
System information parameters

2.26.32. Special parameters: str_export

Parameter

str_export

Description

This parameter exports a *.str file for localization.

Note

Note For html templates, this parameters exports a *.str file and several *.html files into the `htmltemplates` folder.

Valid if set from

The server's operating system command prompt

Requires restart of the Service Manager server?

No

Possible values

all

format

message

help

unitofmeasure

htmltemplates

> **Note**
>
> Note The value “all” includes all the other four kinds of tables. All these values refer to Service Manager database tables that contain information to be localized for Service Manager UI (such as button and label) and messages (such as error and warning).

Example usage

Command line: **sm –str_export:all**

Related topics

System parameters
Enter a parameter in the sm.ini file
System information parameters

2.26.33. Special parameters: str_generate

Parameter

str_generate

Description

This parameter generates the scmessage file localization strings.

The Service Manager localization team usually uses this command to extract the English message strings from the scml.h RTE source code, and then export these strings to a text file named scml.str

. The internationalization team will translate these English message strings to other languages and then insert the result to scmessage database files by using the **sm_import** command.

Valid if set from

The server's operating system command prompt

Requires restart of the Service Manager server?

No

Example usage

Command line: **sm –str_generate**

Related topics

System parameters
Enter a parameter in the sm.ini file
System information parameters

2.26.34. Special parameters: str_import

Parameter

str_import

Description

This parameter imports a localized *.str file.

> **Note**
>
> Note For HTML templates, this parameters imports a *.str file and several *.html files into the `htmltemplates` folder.

Valid if set from

The server's operating system command prompt

Requires restart of the Service Manager server?

No

Possible values

all

format

message

help

unitofmeasure

htmltemplates

> **Note**
>
> Note The value “all” includes all the other four kinds of tables. All these values refer to Service Manager database tables that contain localized information for Service Manager UI (such as button and label) and messages (such as error and warning).

Example usage

Command line: **sm –str_import:all**

Related topics

System parameters
Enter a parameter in the sm.ini file
System information parameters

2.26.35. Special parameters: str_language

Parameter

str_language

Description

This parameter imports a localized *.str file for a certain language. This parameter must be used with **–import** to specify the target language for which the localized messages are imported.

Valid if set from

The server's operating system command prompt

Requires restart of the Service Manager server?

No

Example usage

Command line: **sm –str_import: all –str_language:Hans**

Related topics

System parameters
Enter a parameter in the sm.ini file
System information parameters

2.26.36. Special parameters: verifylic

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

verifylic

Description

This parameter verifies whether a valid license is found or not.

Valid if set from

Server's operating system command prompt

Requires restart of the Service Manager server?

No

Example usage

Command line: **sm -verifylic**

Related topics

System parameters
Enter a parameter in the sm.ini file

2.27. System performance parameters

These parameters determine how the server manages system memory and fail over. You can set these parameters from the Service Manager server's operating system command prompt or from the Service Manager initialization file (sm.ini).

Startup parameter	Description
Startup parameter: agstackl	Defines the length of the stack the Service Manager server allocates to run RAD applications
Startup parameter: arraysizelimit	Defines the maximum number of entries in an array.
Parameter: attachmentsegmentsize	Determines the maximum number of bytes that each segment record in the SYSATTACHMENTS table can use
Startup parameter: cache_clean_interval	Defines the interval that the Service Manager server waits before clearing the cache of infrequently used entries
Startup parameter: cache_locks	Defines the number of locks that the Service Manager server allocates to the cache table
Startup parameter: cache_slots	Defines the number of hash slots that the Service Manager server allocates to the cache table
Startup parameter: dao_sessiontimeout	Specifies the number of seconds that the Service Manager server waits before terminating RESTful threads
Startup parameter: dao_threadsperprocess	Specifies the maximum number of threads allowed to run concurrently in the process for a RESTful web service application
Startup parameter: anubisPollInterval	Specifies the time interval (in seconds) at which the anubis monitor checks to see if any registered scheduler is alive
Startup parameter: enableAnubisMonitor	Starts the anubis monitor to check the status of background scheduler threads
Startup parameter: KMSearchEngineTimeout	Defines a timeout value in seconds for all Knowledge Management search server hosts
Startup parameter: maxgroupsperview	Defines the maximum number of groups that can be used in a view
Startup parameter: maxhttpreqresponse	Defines the maximum response size (in bytes) for HTTP requests
Startup parameter: maxKeepAliveRequests	Allows you to override the Apache Tomcat server limit of processing 100 requests before ending a session
Startup parameter: maxloginspercluster	Enables or disables Max Logins at the cluster level
Startup parameter: maxmsgsharedmemory	Defines the maximum number of bytes of shared memory that the Service Manager server can use for unviewed messages
Startup parameter: maxpagesize	Defines the maximum number of records that the Windows client requests from the server when displaying a record list
Startup parameter: memmanager	Specifies the memory manager to be used
Startup parameter: precision	Defines the number of decimal places at which the Service Manager server rounds numeric fields

Startup parameter: recordlistcount	Sets the default number of records displayed in a record list.
Startup parameter: shared_memory	Defines the amount of memory the Service Manager server sets aside for system caches and lock tables
Parameter: semaphoreWaitTime	Defines the number of seconds the Service Manager server waits for a semaphore before generating a core dump
Startup parameter: shared_memory_address	Ensures that the shared memory allocation is kept in an area that will not interfere with other applications on the server or the operating system
Startup parameter: shutipc	Causes the Service Manager server to remove all IPC shared memory and semaphores
Parameter: strongtypecheck	Specifies the strong type checking times for database records
Startup parameter: tmpdirectory	Defines the path where the Service Manager server can create and find temporary files for any file operation with Client Side Load/Unload and attachment handling
Startup parameter: usealtsignalstack	Causes processes to handle StackOverFlow exceptions correctly
Startup parameter: usemembar	Used by Service Manager to create a JVM
Parameter: vj_record_limit	Sets the maximum number of records a virtual join may return on a subform
Parameter: vj_record_warning	Specifies when the system writes a message to log file about virtual join record requests

Related topics

System parameters
Enter a parameter in the sm.ini file

2.27.1. Startup parameter: arraysizelimit

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

arraysizelimit

Description

Defines the maximum number of entries in an array (for example, a drop-down box).

Valid if set from

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

About 171 millions (this is the value used when this parameter is not specified or specified with an empty value)

Possible values

Any positive integer

Example usage

Initialization file: `arraysizelimit:20`

Related topics

System parameters
Enter a parameter in the sm.ini file
System information parameters

2.27.2. Parameter: attachmentsegmentsize

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

attachmentsegmentsize

Description

This parameter determines the maximum number of bytes that each segment record in the SYSATTACHMENTS table can use. You can use this parameter to reduce the number of segments that the Service Manager server divides attachments into when it stores them in the SYSATTACHMENTS table. No single attachment can exceed 2147483647 bytes.

Note

Note If the size of an attachment is less than the *attachmentsegmentsize* value, the Service Manager server allocates memory based on the actual attachment size instead of the *attachmentsegmentsize* value.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

512,000 (bytes)

Possible values

Number of bytes up to 2147483647 (2 GB) maximum

Example usage

Command line: **sm -httpPort:13080 -attachmentsegmentsize:65536**

Initialization file: `attachmentsegmentsize:65536`

Related topics

System parameters
Enter a parameter in the sm.ini file
System performance parameters

2.27.3. Parameter: strongtypecheck

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

strongtypecheck

Description

This parameter specifies the strong type checking times for database records.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

0

Possible values

0, 1, 2, 3

The bigger value, the more checks.

Example usage

Command line: **sm -strongtypecheck:3**

Initialization file: `-strongtypecheck:3`

Related topics

System parameters
Enter a parameter in the sm.ini file
System performance parameters

2.27.4. Startup parameter: anubisPollInterval

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

anubisPollInterval

Description

This parameter specifies the time interval (in seconds) at which the anubis monitor checks to see if any registered scheduler is alive.

This parameter should be used in conjunction with the *enableAnubisMonitor* parameter.

Important: The ThreadControllerId-background thread, which can be started no matter whether *enableAnubisMonitor* is enabled or not, has been modified to additionally include the anubis monitor functionality and thus eliminated the need for the anubis process. You are recommended to use the new anubis parameters instead of the old anubis process. If an anubis agent record and anubis schedule record, which were used to restart stopped background processes automatically, already exist in your system, before enabling *enableAnubisMonitor* you need to disable the original anubis RAD applications as follows:

1. Delete the existing anubis agent record.
    1. Type **info** on the Service Manager command line, and click **Search**.
    2. Delete the anubis agent record.
2. Delete the existing anubis schedule record.
    1. Type **sch** on the Service Manager command line, and click **Search**.
    2. Delete the anubis schedule record.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

15 (seconds)

Possible values

Number of seconds

Example usage

Command line: **sm scheduler agent 30 4 -anubisPollInterval:15**

Initialization file: `anubisPollInterval:15`

> **Note**
> Note
> To use this parameter in a horizontally scaled environment, you must add this parameter to all sm.ini files.

Related topics

System parameters
Enter a parameter in the sm.ini file
Enter a parameter in sm.cfg or the server's OS command prompt
Restart stopped background processes automatically
System performance parameters
Startup parameter: enableAnubisMonitor

2.27.5. Startup parameter: cache_clean_interval

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

cache_clean_interval

Description

This parameter defines the interval the Service Manager server waits before clearing the cache of infrequently used entries. This parameter works in conjunction with the *-sync* parameter.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No. Requires a restart of the *sm -sync* process.

Default value

3600

Possible values

Number of seconds

Example usage

Command line: **sm -sync -cache_clean_interval:4800**

Initialization file: `cache_clean_interval:4800`

Related topics

System parameters
Enter a parameter in the sm.ini file
System performance parameters

2.27.6. Startup parameter: cache_locks

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

cache_locks

Description

This parameter defines the number of locks the Service Manager server allocates to the cache table. By default, the lock granularity is one lock for each hash slot.

The lock count you specify must be bigger than zero (0) and no more than the number of hash slots that's specified in Startup parameter: cache_slots. If you specify an invalid value for this parameter, the default value is used.

Note

Note This parameter solves the performance issue in versions earlier than 9.33 that the system may encounter slowness at peak load because earlier versions use the entire hash table as the lock granularity.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

9973

Possible values

Greater than 0 and no more than the number of hash slots

Example usage

Command line: **sm -httpPort:13080 -cache_locks:2000**

Initialization file: `cache_locks:2000`

Related topics

System parameters
Enter a parameter in the sm.ini file
System performance parameters

2.27.7. Startup parameter: cache_slots

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

cache_slots

Description

This parameter defines the number of hash slots the Service Manager server allocates to the cache table. You can use the *reportcache* parameter to gauge the effectiveness of your cache allocation.

Note

Note If this parameter is not set or set to be greater than 100,000 or less than 300, the default value (9973) is used; if it is set to a non-prime number, the smaller one of its neighboring prime numbers is used instead.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

9973

Possible values

300 - 100000

Example usage

Command line: **sm -httpPort:13080 -cache_slots:2000**

Initialization file: `cache_slots:2000`

Related topics

System parameters
Enter a parameter in the sm.ini file
System performance parameters

2.27.8. Startup parameter: dao_sessiontimeout

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

dao_sessiontimeout

Description

This parameter specifies the seconds Service Manager server to wait before terminating RESTful threads. Unless the client sends subsequent requests within the timeout, the Service Manager server will recycle the session for re-use and re-allocate it on demand.

If there is large divergence, we recommend that you connect several servlets with different thresholds.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

15 (seconds)

> **Note**
>
> Note Change this default value to another proper value according to the Restful API call interval. For example, if the Restful API call interval is 30 seconds, set this value to a little bigger than 30.

Possible values

Number of seconds

Example usage

Command line: **sm -httpPort:13080 -dao_sessiontimeout:20**

Initialization file: `dao_sessiontimeout:20`

Related topics

System parameters
Enter a parameter in the sm.ini file
System performance parameters

2.27.9. Startup parameter: dao_threadsperprocess

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

dao_threadsperprocess

Description

This parameter specifies the maximum number of threads allowed to run concurrently in the process for a RESTful Web Service application.

It is good to start enough server threads to handle requests. We recommend that you maintain a buffer of 30% to 40% spare capacity.

> **Note**
>
> Note The dao_threadsperprocess parameter is limited by the threadsperprocess parameter (default value: 240). If you use the dao_threadsperprocess parameter, make sure that you also update the threadsperprocess parameter to a value not less than the value of dao_threadsperprocess. For example, you need to change the value of threadsperprocess to 250 when you enable dao_threadsperprocess with its default value.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

240

Possible values

Number of threads

Example usage

Command line: **sm -httpPort:13080 -dao_threadsperprocess: 240**

Initialization file: `dao_threadsperprocess:240`

Related topics

System parameters
Enter a parameter in the sm.ini file
System performance parameters

2.27.10. Startup parameter: dashboard_export_path

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

dashboard_export_path

Description

This parameter specifies the path to export the generated report and dashboard files.

> **Note**
>
> Note To export the generated reports, you must start a background schedule process by running the **sm report.export** command.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

None

Possible values

A directory on the Service Manager Server host

Example usage

Initialization file: `dashboard_export_path:C:\Reports`

Related topics

System parameters
Enter a parameter in the sm.ini file
Database parameters

2.27.11. Startup parameter: dashboardquerycache_enable

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

dashboardquerycache_enable

Description

This parameter defines whether caching query results in the database is enabled or disabled.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

1 (enabled)

Possible values

0 (disabled)

Example usage

Command line: **sm -httpPort:13080 -dashboardquerycache_enable:1**

Initialization file: `dashboardquerycache_enable:1`

Related topics

System parameters
Enter a parameter in the sm.ini file
Database parameters

2.27.12. Startup parameter: dashboardquerycache_dbtime

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

dashboardquerycache_dbtime

Description

This parameter defines the time threshold in milliseconds for queries against the database when a report is generated. If the time exceeds this threshold, Service Manager saves the query results to the reporting cache.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

100

Possible values

Number of milliseconds

Example usage

Command line: **sm -httpPort:13080 -dashboardquerycache_dbtime:100**

Initialization file: `dashboardquerycache_dbtime:100`

Related topics

System parameters
Enter a parameter in the sm.ini file
Database parameters

2.27.13. Startup parameter: dashboardquerycache_expire

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

dashboardquerycache_expire

Description

This parameter defines the expiration time in minutes of one database query in reporting cache.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

10

Possible values

Number of minutes

Example usage

Command line: **sm -httpPort:13080 -dashboardquerycache_expire:10**

Initialization file: `dashboardquerycache_expire:10`

Related topics

System parameters
Enter a parameter in the sm.ini file
Database parameters

2.27.14. Startup parameter: dashboardonreplicatedb

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

dashboardonreplicatedb

Description

This parameter defines whether queries against a replicated database for dashboard reporting are enabled.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

None

Possible values

None

Example usage

Initialization file: `dashboardonreplicatedb`

Related topics

System parameters
Enter a parameter in the sm.ini file
Database parameters

2.27.15. Startup parameter: enableAnubisMonitor

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

enableAnubisMonitor

Description

This parameter starts the anubis monitor to check the status of background scheduler threads. By default, this parameter is disabled.

This parameter should be used in conjunction with the *anubisPollInterval* parameter, which specifies the time interval (in seconds) at which the anubis monitor checks to see if any registered scheduler is alive (default: 15 seconds).

The following table describes different system behaviors based on the value of this parameter.

Value	Description
1 or empty	Enabled
0	Disabled **Note** Note By default, this parameter is not specified and a value of "0" is used.

Important: The ThreadControllerId-background thread, which can be started no matter whether enableAnubisMonitor is enabled or not, has been modified to additionally include the anubis monitor functionality and thus eliminated the need for the anubis process. You are recommended to use the new anubis parameters instead of the old anubis process. If an anubis agent record and anubis schedule record, which were used to restart stopped background processes automatically, already exist in your system, before enabling *enableAnubisMonitor* you need to disable the original anubis RAD applications. To do this, follow these steps:

1. Delete the existing anubis agent record.
    1. Type **info** on the Service Manager command line, and then click **Search**.
    2. Delete the anubis agent record.
2. Delete the existing anubis schedule record.
    1. Type **sch** on the Service Manager command line, and then click **Search**.
    2. Delete the anubis schedule record.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

0 (Disabled)

Possible values

0: Disabled

1 or empty: Enabled

Example usage

Command line: **sm scheduler agent 30 4 -enableAnubisMonitor:1**

Initialization file: `enableAnubisMonitor:1`

Note
Note
To use this parameter in a horizontally scaled environment, you must add this parameter to all sm.ini files.

Related topics

System parameters
Enter a parameter in the sm.ini file
Enter a parameter in sm.cfg or the server's OS command prompt
Restart stopped background processes automatically

System performance parameters
Startup parameter: anubisPollInterval

2.27.16. Startup parameter: KMSearchEngineTimeout

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

KMSearchEngineTimeout

Description

This parameter defines a timeout value in seconds for all Knowledge Management search server hosts (including indexing hosts, search hosts, and load balancer hosts). The default is 20 seconds, and the minimum is 10 seconds.

This parameter is used to prevent overloaded KM search servers from causing Service Manager to hang. When a search server host hangs (for example, because of 100% CPU usage), the Service Manager server tries to connect to the search server repeatedly until the timeout of the connection. The Service Manager server then returns an error.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of Service Manager server?

Yes

Default value

20 (seconds)

Possible values

No less than 10 seconds (if you specify a value less than 10, 10 is used.)

Example usage

Command line: **sm -httpPort:13080 -KMSearchEngineTimeout:25**

Initialization file: `KMSearchEngineTimeout:25`

Related topics

System parameters
Enter a parameter in the sm.ini file
System performance parameters

2.27.17. Startup parameter: maxgroupsperview

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

maxgroupsperview

Description

This parameter defines the maximum number of groups that can be used in a view. When a user selects a view from the View list whose group count exceeds the limit, only the maximum allowed number of groups are displayed in the view, and the following message displays in the client UI:

"Maximum number of groups (xxxx) exceeded. Please modify the view definition to reduce the number of groups."

Note: This limit would be useful if a user runs a poorly created view that causes the servlets to consume too much CPU and memory and terminate the servlets.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

5000 (groups)

Possible values

500 or greater. If a value less than 500 is specified, Service Manager will ignore the value and use 500 instead. A warning message will also occur in the sm.log file: "A maxgroupsperview value less than 500 specified and ignored. 500 is used."

Example usage

Command line: `sm -httpPort:13080 -maxgroupsperview:6000`

Initialization file: `maxgroupsperview:6000`

Related topics

System parameters
Enter a parameter in the sm.ini file
System performance parameters

2.27.18. Startup parameter: maxhttpreqresponse

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

maxhttpreqresponse

Description

This parameter defines the maximum response size (in bytes) for HTTP requests that are issued by using the doHTTPRequest JavaScript global method. Huge HTTP responses may result in high CPU utilization and memory consumption, and subsequently impact the performance of Service Manager. If the response size exceeds the value of maxhttpreqresponse, the doHTTPRequest call fails and an exception is generated.

A zero value disables the feature, which means there is no size limit.

Note

Note This parameter doesn't limit the response sizes of doSOAPRequest calls.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

20971520 (20MB)

Possible values

0: No size limit

Any size limit (in bytes)

Related topics

System parameters
Enter a parameter in the sm.ini file

JavaScript global method: doHTTPRequest
JavaScript global method: doSOAPRequest

2.27.19. Startup parameter: maxKeepAliveRequests

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

maxKeepAliveRequests

Description

This parameter allows you to override the Apache Tomcat server limit of processing 100 requests before ending a session. You can set a higher limit on the number of requests to be processed before the Service Manager server closes the connection.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

100

Example usage

Command line: `sm -maxKeepAliveRequests:250`

Initialization file: `maxKeepAliveRequests:250`

Related topics

System parameters
Enter a parameter in the sm.ini file
System performance parameters

2.27.20. Startup parameter: maxloginspercluster

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

maxloginspercluster

Description

In a horizontal scaling implementation, the maximum number of logins for operators is enforced at the cluster level. This parameter allows administrators to turn off this feature. By default, this parameter is set to 1, which means maximum number of logins for operators is enforced by counting each operator's logins on all cluster hosts. When this parameter is set to 0, only logins on the local host are counted.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of Service Manager server?

Yes

Default value

1

Possible values

0 (Disable Max Logins at the cluster level)

1 (Enable Max Logins at the cluster level)

Example usage

Command line: **sm -httpPort:13080 -maxloginspercluster:0**

Initialization file: `maxloginspercluster:0`

Related topics

System parameters
Enter a parameter in the sm.ini file
System performance parameters

2.27.21. Startup parameter: maxmsgsharedmemory

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

maxmsgsharedmemory

Description

This parameter defines the maximum number of bytes of shared memory that the Service Manager server can use for unviewed messages.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

Ten percent (10%) of the shared memory or 1 GB (1073741824), whichever is less.

Possible values

Size in bytes

Example usage

Command line: **sm -httpPort:13080 -maxmsgsharedmemory:5120000**

Initialization file: `maxmsgsharedmemory:5120000`

Related topics

System parameters
Enter a parameter in the sm.ini file
System performance parameters

2.27.22. Startup parameter: maxpagesize

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

maxpagesize

Description

This parameter defines the maximum number of records that the Windows client requests from the server when the client displays a record list. This helps to avoid high memory consumption by getList requests when the **Record list request**

count option in the Windows client preferences is set to a large value.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No (only requires a re-login)

Default value

8000

Possible values

0: No size limit

8000 to 80,000

Example usage

Command line: **sm -httpPort:13080 -maxpagesize:10000**

Initialization file: `maxpagesize:10000`

Related topics

System parameters
Enter a parameter in the sm.ini file
System performance parameters

2.27.23. Startup parameter: memmanager

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

memmanager

Description

Specifies the memory manager to be used.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

utpool

Possible values

utpool, malloc

Example usage

Command line: **sm -memmanager:utpool**

Service Manager Initialization (sm.ini) file: `memmanager:utpool`

Related topics

System parameters
Enter a parameter in the sm.ini file
System performance parameters

2.27.24. Startup parameter: precision

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

precision

Description

This parameter defines the number of decimal places at which the Service Manager server rounds numeric fields. By default, the Service Manager server rounds to six decimal places.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

6

Possible values

Number of decimal places at which to round

Example usage

Command line: **sm -httpPort:13080 -precision:5**

Initialization file: `precision:5`

Related topics

System parameters
Enter a parameter in the sm.ini file
System performance parameters

2.27.25. Startup parameter: recordlistcount

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

recordlistcount

Description

Set the default number of records displayed in a record list.

Valid if set from

Start up file (sm.ini)

Requires restart of server?

No

Default value

None

Possible values

Any positive integer

Note

Note You can only set the value to 25, 50 and 100 for the Service Manager web tier. Any value other than 25, 50 and 100 will be reset to 50.

Example usage

Start up file: `recordlistcount:50`

Related topics

Setting Windows client preferences from the server
Client parameters for Windows clients

2.27.26. Parameter: semaphoreWaitTime

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

semaphoreWaitTime

Description

This parameter defines the number of seconds the Service Manager server waits for a semaphore before generating a core dump. The default is 0 (no core dump will be generated).

Note

Note This parameter takes effect on Windows servers only.

To use this functionality, be sure to set the **enablecoredump** parameter to 1. The following table describes the server behavior when these two parameters are set to 1 or 0.

Enablecoredump value	semaphoreWaitTime value	Behavior
1	n	The server generates a core dump after waiting n seconds for a semaphore.
0	n	The server doesn't generate a core dump.
1 or 0	0	The server doesn't generate a core dump.

Valid if set from

Server's operating system command prompt

Initialization (sm.ini) file

Requires restart of the Service Manager server?

Yes

Default value

0

Possible values

0 or n seconds

Example usage

Command line: **sm -enablecoredump:1 – semaphoreWaitTime:10**

Initialization file: semaphoreWaitTime:10

Related topics

System parameters
Enter a parameter in the sm.ini file
Debugging parameters
Parameter: enablecoredump

2.27.27. Startup parameter: shared_memory

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's OS command prompt.

Parameter

shared_memory

Description

This parameter defines the amount of memory the Service Manager server sets aside for system caches and lock tables. The Service Manager add-on product, IR Expert, also relies heavily upon shared memory. You can set this parameter from the Configure server utility.

Notes:

- We recommend that you set this parameter to 268,435,456 (256 MB) or greater. If this parameter has a value less than 256 MB, a warning message occurs in the server log file (sm.log). To remove this warning, change the value to 256 MB or greater.
- You can read and use the sysctl.conf file to control this parameter. To set proper OS parameters, copy the `<Service Manager server installation path>/RUN/97-sm-sysctl.conf` file into /etc/sysctl.d directory.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of Service Manager server?

Yes

Default value

536,870,912 bytes (512 MB)

Possible values

Number of bytes

Example usage

Command line: sm -httpPort:13080 -shared_memory:536870912

Initialization file: `shared_memory:536870912`

Related topics

System parameters
Enter a parameter in the sm.ini file
System performance parameters
Shared memory

2.27.28. Startup parameter: shared_memory_address

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Note

Note You should only use this parameter when instructed to do so by the company's Customer Support.

Parameter

shared_memory_address

Description

This parameter can be used on all operating systems to ensure that shared memory allocation is kept in an area that will not interfere with shared libraries, stacks, heaps or other memory related data structures. You can use this parameter to specify a new address that has enough contiguous memory for the Service Manager process.

If you encounter a shared memory allocation failure, the address that the Service Manager process is using doesn't have enough contiguous memory. On Windows, the sm.log file lists recommendations for possible *shared_memory_address* parameter values. If the *debugvmmap* parameter is active, the log also contains a map of virtual memory at the time of the failure.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

On Linux: 0x50000000

On Windows: Dynamically determined

Possible values

A memory address in a decimal, hexadecimal, or octal format.

Example usage

Command line: **sm -httpPort:13080 -shared_memory_address:0x80000000**

Initialization file: `shared_memory_address:0x80000000`

Example sm.log file that's generated when you encounter a shared memory allocation failure

```
6116(  2716) 03/05/2010 11:06:41  RTE E shmat: MapViewOfFileEx failed
6116(  2716) 03/05/2010 11:06:41  RTE E Error 487 in call MapViewOfFileEx - Attempt to access invalid
address.
6116(  2716) 03/05/2010 11:06:41  RTE E sm_init: shmat( 516, 0x73000000, 0 ) failed for size 67108864 and
key 0x2A02E500
6116(  2716) 03/05/2010 11:06:41  RTE I List of possible shared_memory_address parameter settings for a
shared memory size of 67108864 (64 MB)
6116(  2716) 03/05/2010 11:06:41  RTE I # Free range 0x7FFF0000-0xFFFAD000 size=2147209216 (2047 MB)
6116(  2716) 03/05/2010 11:06:41  RTE I Preferred        : shared_memory_address:0x80000000
6116(  2716) 03/05/2010 11:06:41  RTE I Lowest available : shared_memory_address:0x80000000
6116(  2716) 03/05/2010 11:06:41  RTE I Highest available: shared_memory_address:0xFBFA0000
6116(  2716) 03/05/2010 11:06:41  RTE I For a complete list of available shared_mempory_address settings,
run with the debugvmmap parameter.
6116(  2716) 03/05/2010 11:06:41  RTE W !!! Use the preferred shared_memory_address parameter above
6116(  2716) 03/05/2010 11:06:41  RTE W !!! or remove the shared_memory_address parameter completely.
6116(  2716) 03/05/2010 11:06:41  RTE E HP Service Manager is unable to start. Failed to initialize or
attach to shared memory environment
```

Related topics

System parameters
Enter a parameter in the sm.ini file
System performance parameters
Shared memory

2.27.29. Startup parameter: shutipc

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

shutipc

Description

Using this parameter at the operating system command prompt causes the Service Manager server to remove all IPC shared memory and semaphores.

Note

Note This command is only for Linux systems. On a Windows system this command doesn'thing and is not needed.

Valid if set from

Server's operating system command prompt

Requires restart of the Service Manager server?

No

Default value

None

Possible values

None

Example usage

Command line: **sm -shutipc**

Related topics

System parameters
Enter a parameter in the sm.ini file
System performance parameters

2.27.30. Startup parameter: tmpdirectory

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

tmpdirectory

Description

This parameter defines the path where the Service Manager server can create and find temporary files for any file operation with Client Side Load/Unload and attachment handling.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of the Service Manager server?

No

Default value

Linux: /var/tmp

Windows: The possible existence of environment variables are:

- The path specified by the TMP environment variable.
- The path specified by the TEMP environment variable.
- The path specified by the USERPROFILE environment variable.
- The Windows directory.

For more information about the tmpdirectory parameter, see GetTempPath.

Possible values

Path to temporary files

Example usage

Command line: **sm -httpPort:13080 -tmpdirectory:temp**

Initialization file: `tmpdirectory:temp`

Related topics

System parameters
Enter a parameter in the sm.ini file
System performance parameters

2.27.31. Startup parameter: unlockdatabase

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

unlockdatabase

Description

When a system is started either in a vertically-scaled implementation or in a horizontally-scaled implementation, a unique key is generated and saved in the database as a lock record. The lock record prevents another Service Manager system starting, so as not to use the same database. However, this causes the backup server to fail to start when the primary server is abruptly brought down. This parameter allows the System Administrator to clean up the database lock record by issuing the **sm -unlockdatabase** command, and then starting the backup server.

Note

Note The unlock database functionality is needed only when you are switching the Service Manager product from one host to another host.

Valid if set from

Server's operating system command prompt

Requires restart of the Service Manager server?

Yes

Default value

None

Possible values

None

Example usage

Command line: **sm -unlockdatabase**

> **Note**
>
> Note Do not add this parameter to the `sm.ini` file.

Related topics

System parameters
Enter a parameter in the sm.ini file
System performance parameters

2.27.32. Startup parameter: usealtsignalstack

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

usealtsignalstack

Description

By default, the Service Manager processes register the alternate signal handler to handle signals generated by the operating system. When *usealtsignalstack* is enabled, the processes handle stack overflow exceptions properly. The signal is delivered on an alternate stack, and the process has a chance to terminate the offending thread and continue to requests of other threads that exist on the process. Older Linux kernels may not honor the alternate signal stack. If you are running on older Linux kernels, disable this feature.

> **Note**
>
> Note This parameter applies to UNIX only.

Valid if set from

Initialization file (sm.ini)

Requires restart of the Service Manager server?

Yes

Default value

1

Possible values

0 (Disabled)

1 (Enabled)

Example usage

Command line: **sm -httpPort:13080 -usealtsignalstack:1**

Initialization file: `usealtsignalstack:1`

Related topics

System parameters
Enter a parameter in the sm.ini file
Client parameters for Windows clients

2.27.33. Parameter: vj_record_limit

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

vj_record_limit

Description

This parameter sets the maximum number of records a virtual join may return on a subform. You can use this parameter to set a threshold on virtual joins so they don't affect overall system performance. When a search request reaches the virtual join record limit, the system displays a warning message to the user indicating the record limit and the subform requesting the records. The system also writes the warning message to the log file.

Note

Note

- If a request reaches the virtual join record limit, the search will not return all matching records. The search results will exclude any search results past the virtual join record limit.
- If you set the parameter value to zero, the virtual join record limit reverts to 32000 records.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of Service Manager server?

Yes

Default value

2000 records

Possible values

0 (Disable)

The number of records

Example usage

Initialization file: `vj_record_limit:1000`

Related topics

System parameters
Enter a parameter in the sm.ini file
Parameter: vj_record_warning

2.27.34. Parameter: vj_record_warning

Startup parameters change the behavior of the Service Manager server. You can always set a startup parameter from the server's operating system command prompt.

Parameter

vj_record_warning

Description

This parameter specifies when the system writes a message to log file about virtual join record requests. When a record request meets or exceeds the virtual join record warning limit, the system writes a message to the log file. The system will write a new message to the log file for every 1000 records over the warning limit.

Valid if set from

Server's operating system command prompt

Initialization file (sm.ini)

Requires restart of Service Manager server?

Yes

Default value

1000

Possible values

Number of records

Example usage

Initialization file: `vj_record_warning:500`

Related topics

System parameters
Enter a parameter in the sm.ini file
Parameter: vj_record_limit

3. Smart Analytics administration

Built on Service Manager (SM) and using an OEM-licensed version of IDOL, SM Smart Analytics heralds the debut of the "Big Data" edition of Service Manager.

Smart Analytics offers the following capabilities: Smart Ticket, Hot Topic Analytics, and Smart Search. For more information, see the Smart Analytics overview help topic.

The following topics help you administer Smart Analytics after initial installation and configuration. For information on installing Smart Analytics, see Install Smart Analytics.

Configure Smart Analytics

Administer Smart Analytics

Best practices

Tailoring

Limitations

Smart Analytics APIs

3.1. Configure Smart Analytics

To configure Smart Analytics in Service Manager, follow these steps:

1. Set up data cleansing configuration. See Configure data cleansing.
2. Configure Smart Ticket. See Configure Smart Ticket.
3. Configure Hot Topic Analytics. See Configure Hot Topic Analytics.
4. Set up Smart Search connectors. See Configure connectors.
5. Configure Smart Search. See Configure Smart Search.
6. Add the "idol.assistant" capability word to the operator records. See Add Smart Analytics capability word for power users.

Validate the Smart Analytics configurations when you finish these steps. For more information, see Validate Smart Analytics configurations.

3.1.1. Configure data cleansing

User Role: Administrator

The purpose of data cleansing is to remove unwanted contents from the Smart Analytics source data set that's used to train and index into Smart Analytics as well as in runtime processing.

Note

Note

- For Smart Ticket and Hot Topic Analytics features, data cleansing is only applied to the "Title Field" or "Content Fields" that are defined in configurations.
- For Smart Search, data cleansing can be applied to any field individually which you can configure. For detailed information, see the data cleansing description in Manage Smart Search Knowledgebases.
- All modification for data cleansing will take effect from next round of indexing.

To add a data cleansing configuration, follow these steps:

1. From the System Navigator, click **System Administration** > **Ongoing Maintenance** > **Smart Analytics** > **Data Cleansing**.
2. Select a module. For example, Interaction.

   Note

   Note In the module drop-down list there is a Global option. This means that it's a global data cleansing record for all modules which are using Hot Topic Analytics and Smart Ticket.

3. Select one of the following actions:
   - **Remove**: Remove the matched texts and index the rest to SM Smart Analytics.
   - **Include**: Extract and index the texts between the start pattern and the end pattern exclusively.
   - **Exclude**: Exclude the texts that match the pattern (including start, end, and all the words between them) and index the rest to SM Smart Analytics.
   - **ExtractFromTemplate**: Extract the content from template that's configured as regular expressions. The capturing groups that are matched by the Regular Expressions are extracted and returned.
4. Enter the text or pattern for the action that you selected. For the **Remove** action, you only need to type the text string to be removed. For the **Include** and **Exclude** actions, the start pattern is the text string that you need to specify while the end pattern can be one of these options: a text string that you specify, end of line, or end of document.
   The processing of the **ExtractFromTemplate** action is of first priority. The Data cleansing actions are processes in the following order:
   1. ExtractFromTemplate
      If there are matched texts found, then return. Otherwise, perform the **Include** action.
   2. Include
      If there are matched texts found, then perform the **Remove** action. Otherwise, perform the **Exclude** action..
   3. Exclude
      If there are matched texts found, then perform the **Remove** action.
   4. **Remove**

   To learn how the text or pattern takes effect, see the following examples.

- Example of the **Remove** action:

Original content	`[telephone communication history with customer]: Microsoft Office keeps asking for installation of additional components / language packs.`
Specified text to be removed	`[telephone communication history with customer]:`
After cleansing	`Microsoft Office keeps asking for installation of additional components / language packs`

- Example of the **Remove** action by using Regex:

Original content	`From: Sender<123456@yourcompany.com>`
	`Sent: Thursday, December 21, 2023 6:43 PM`
	`To: Receiver<654321@yourcompany.com>`
	`Subject: Email title`
	
	`Email body`

Specified text to be removed	`\b[A-Za-z0-9._%+-]+@yourcompany\.com\b`

After cleansing	`From: Sender<>`
	`Sent: Thursday, December 21, 2023 6:43 PM`
	`To: Receiver<>`
	`Subject: Email title`
	
	`Email body`

- Examples of the **Include** action:

Original content	`Description of the issue: Sent items aren't being sent by Outlook. Actions suggested by help desk agent: asked customer to check network connection status, shows connection is OK`
Start pattern	`description of the issue:`
End pattern	`actions suggested by help desk agent:`
After cleansing	`Sent items aren't being sent by Outlook.`

Original content	`Description of the issue: Items aren't sent by Outlook.`
	`Actions suggested by help desk agent:`
	`asked customer to check network connection status,`
	`shows connection is OK`

Start pattern	`description of the issue:`
End pattern	End of line
After cleansing	`Items aren't sent by Outlook.`

Original content	`Description of the issue:`
	`Sent items aren't being sent by Outlook.`
	`Actions suggested by help desk agent:`
	`asked customer to check network connection status,`
	`shows connection is OK`

Start pattern	`description of the issue:`
End pattern	End of document

After cleansing	`Sent items aren't being sent by Outlook.`
	`Actions suggested by help desk agent:`
	`asked customer to check network connection status,`
	`shows connection is OK`

- Examples of the **Exclude** action:

Original content	`SQL Server is down and can't be restarted.`
	`[appendix: error log] Details:`
	`Xxxxxxxxxxxxxxx`
	`[end of appendix]`

Start pattern	`[appendix: error log]`
End pattern	`[end of appendix]`
After cleansing	`SQL Server is down and can't be restarted.`

Original content	`SQL Server is down and can't be restarted.`
	`[appendix: error log] Details:`
	`Xxxxxxxxxxxxxxx`
	`[end of appendix]`

Start pattern	`[appendix: error log]`
End pattern	End of line

After cleansing	`SQL Server is down and can't be restarted.`
	`Xxxxxxxxxxxxxxx`
	`[end of appendix]`

Original content	`SQL Server is down and can't be restarted.`
	`[appendix: error log] Details:`
	`Xxxxxxxxxxxxxxx`
	`[end of appendix]`

Start pattern	`[appendix: error log]`
End pattern	End of document
After cleansing	`SQL Server is down and can't be restarted.`

- Example of the **ExtractFromTemplate** action:

Example configuration	Brief description of the problem: ([^]*)FirstName: ([^]*)LastName:([^]*)Phone:([^]*)
Data before cleansing	**Brief description of the problem**: The user called because eDocs could not be accessed. The user was able to find the eDocs administrator but that person doesn't work for the company anymore First Name : Herr Maximo Christian Last Name : Graf Phone : 01 234 567
Data after cleansing	The user called because eDocs could not be accessed. The user was able to find the eDocs administrator but that person doesn't work for the company anymore Herr Maximo Christian Graf 01 234 567

Note

Note Regular expression is supported only for the **ExtractFromTemplate** action.

5. Select the **Match Case** checkbox if you only want to find the texts that match the case of the text or pattern that you entered.
6. Select the **Active** checkbox to activate this configuration.

7. Click **Add**. The new data cleansing configuration is now added.

3.1.2. Configure Smart Ticket

User Role: Administrator

Smart Ticket provides the following two out-of-box Smart Ticket (auto-classification) configurations:

- Standard category field
- Service category field

These out-of-box configurations are best practices based on the out-of-box data. You can use or modify these configurations, or you can add new configurations that best reflect your business needs.

Add a new Smart Ticket task

In the out-of-box system, two Smart Ticket configuration tasks ("Standard category field" and "Service category field") are used for Smart Ticket by default. You can choose to use these out-of-box Smart Ticket tasks, or you can create new Smart Ticket tasks.

To add a new Smart Ticket task, follow these steps:

1. From the System Navigator, click **System Administration** > **Ongoing Maintenance** > **Smart Analytics** > **Smart Ticket**.
2. Select **Blank** from the drop-down list, and then click **Add**.

   Note

   You can also select one of the out-of-box templates ("Category" or "Affected Service") from the drop-down list, and then click **Add** to create a new Smart Ticket task based on the template.

3. Type the task name for the new Smart Ticket configuration.
4. Go to the **Configurations** tab.
5. Select a module for auto-classification. For example, Interaction.
6. In the **Training Sample Query** field, define a query to refine the sample data. The default value is `category~="service catalog"`, which means the data that's not in the "service catalog" category can be selected as the training samples.
7. Select the target fields to be automatically filled by SM Smart Analytics. You can select up to three levels. For example, Category, Subcategory, and Area.
8. Select the source fields that the auto-classification is based on. For example, title and description.
9. In the **Training Optimization** tab, modify the settings for training optimization.

   Note

   We recommend that you keep the default settings. For more information on improving accuracy for Smart Ticket, see Improving accuracy for Smart Ticket.

Setting	Description
Training Samples Per Category	The maximum records to be used as the training samples for each category. Default: 200
Test Data Coverage	The percentage of records out of the total source data that are used to test the trained system. Default: 5
Source Data Coverage	The percentage of records out of the total source data that are used to train the system. Default: 90

10. Click **Add**. The new auto-classification task is now added to the **Current Configuration List**.

11. Modify the Smart Ticket form (idol.quick.new.interaction) to use the new Smart Ticket task that you just created.

> **Note**
> This step is only required for operators to create Smart Interaction from index.do. The new auto-classification task will take effect directly on the user requests from ESS after a training is performed.

Perform training and testing

To perform a training for a Smart Ticket task, follow these steps:

1. From the System Navigator, click **System Administration** > **Ongoing Maintenance** > **Smart Analytics** > **Smart Ticket**.
2. Click the task name of a Smart Ticket configuration. The **Smart Ticket Task** screen appears.
3. Click the **Training** button to start training this auto-classification.

> **Tip**
> You can click **Refresh Status** to view the latest training status.

4. When the training is done, click **Testing**. When the testing is finished, you can view an estimated result of the accuracy for this auto-classification in the **Testing Result** field. For example,

```
First hit count: 7; correct rate: 50.00%
Second hit count: 0; correct rate: 50.00%
Third hit count: 1; correct rate: 57.14%
Missed count: 6; missed rate: 42.86%
"***************** Total samples: 14 ******************
```

If you perform testing in the Multi-Company mode, the **Testing Result** field displays the following message when the testing is finished:
`Click the Multiple Company tab to check the testing status of each company, and then click Visualize to check their testing results`

> **Note**
> - Smart Analytics doesn't support duplicated category names that are different only in text cases (For example, two categories named as "AVAILABILITY" and "availability"). If such categories exist during the training, the system logs an error and aborts the training process.
> - The testing status is `Not start yet` after upgrade to Service Manager 9.70. To get the latest test result based on the new tuning mechanism provided by Service Manager 9.70, a system administrator need to click the **Testing** button to test the Smart Ticket tasks again.
> - Suppose you are working with Smart Analytics in the single company mode. After you upgrade to Service Manager 9.70 and enable the multi-company mode, the training Status is `Finished training 0 company`. To get the latest test result based on the new tuning mechanism provided by Service Manager 9.70, a system administrator need to click the **Training** button to refresh the training status.
>
> If you are unable to test or train a Smart Ticket task, see the related topic in Troubleshoot Smart Analytics.

> **Tip**
> - The quality of the sample data is critical to the accuracy of the auto-classification. To refine your sample data, you can define a query in the **Training Sample Query** field under the **Configurations** tab. For more best practices to improve accuracy, see Improving accuracy for Smart Ticket.
> - If you disable or enable the Multi-Company mode for Service Manager, you need to delete the existing Smart Ticket configuration tasks and re-create them before you perform training.

Apply a rule-based training

You can append the rule-based analysis on top of the meaning based analysis. The typical scenario is that if one particular record has the same relevancy within several categories, you can append a rule to one specific category to improve the categorization accuracy.

"Rule Field Name" is where you can specify the field based on which you define the rule.

"Apply Rule" lists all the categories, where you can choose the target category and set the value for the rule you want to append.

For example, suppose there are two affected services, "printer_San Diego" and "printer_Shanghai". You can define the rule field as "Primary Contact Location City". Then, set value "San Diego" to the "printer_San Diego", and set value "Shanghai" to the "printer_Shanghai". With this rule, if the contact person for the new coming record is from the San Diego office, the record will be automatically filled with "printer_San Diego" as the affected service.

To apply a rule-based training for an auto-classification, follow these steps:

1. From the System Navigator, click **System Administration** > **Ongoing Maintenance** > **Smart Analytics** > **Smart Ticket**.
2. Click a task name of a Smart Ticket configuration. The **Smart Ticket Task** screen appears.
3. Go to the **Rule Base** tab.
4. In the **Rule Field Name** field, specify the field name based on which you define the rule.
5. Click **Apply Rule**, and then click **Search**. A list containing all the categories appears, where you can choose the target category and set the value for the rule that you want to apply.
6. Click a category.
7. In the **Rule Field Value** field, set the value for the rule that you want to apply.
8. Click the **Apply Rule** button.

Perform tuning in the Smart Ticket definition

Another way to improve the accuracy of Smart Ticket is to perform tuning continually for the Smart Ticket definition.

To perform tuning in the Smart Ticket definition, follow these steps:

1. Service Desk agents select tuning candidates during their daily work:
    1. In an interaction record, update the fields suggested by Smart Ticket if the suggested values are incorrect, such as category or affected service.
    2. After the interaction is closed, from the interaction record, click **More** > **Add to Tuning Records** to add this record as a tuning candidate for Smart Ticket.

    Note

    The **Add to Tuning Records** option is only available when an interaction is in the "Closed" status.

2. A system administrator tunes Smart Ticket after a period of time to increase the accuracy.
    1. From the System Navigator, click **System Administration** > **Ongoing Maintenance** > **Smart Analytics** > **Smart Ticket**.
    2. Click a task name of a Smart Ticket configuration. The **Smart Ticket Task** screen appears.
    3. Click the **Visualize** button to check the Confusion Matrix and check the category detail.
    4. Click **Apply Change** to implement your changes.

    For more information about the Confusion Matrix, see Use the Confusion Matrix tool.

Configure Smart Ticket for multi-company

Service Manager Smart Analytics supports multi-tenancy. When the multi-company mode is enabled in Service Manager, you can configure specific Smart Ticket task to apply to multiple companies when applicable. The Smart Ticket configuration takes effect on these companies individually by segregating their data in Smart Analytics database.

To specify the companies in a Smart Ticket configuration, follow these steps:

1. From the System Navigator, click **System Administration** > **Ongoing Maintenance** > **Smart Analytics** > **Smart Ticket**.
2. Click a task name of a Smart Ticket configuration. The **Smart Ticket Task** screen appears.
3. Click the **Multiple Company** tab, and then do one of the following:
    - Click **Add Company** to add companies to this configuration.

      Note
      A training is needed if you add a new company.

    - Click **Remove Company** to remove companies from this configuration.

Tip
If you are unable to see the **Multiple Company** tab, see the related topic in Troubleshoot Smart Analytics.

Note

- If you have enabled multiple company, you must restart SMIS to make sure that the SMIS scheduler receives the multiple company information.
- You can only use the company values as defined in the company table.

Configure Smart Ticket to support mandanten

Service Manager Smart Analytics supports mandanten. When mandanten is enabled in Service Manager, you can configure specific Smart Ticket task to apply to multiple mandanten when applicable. The Smart Ticket configuration takes effect on these mandanten individually by segregating their data in Smart Analytics database.

Follow these steps to configure Smart Ticket if mandanten is enabled in Service Manager:

1. From the System Navigator, click **System Administration** > **Ongoing Maintenance** > **Smart Analytics** > **Smart Ticket**.
2. Click a task name of a Smart Ticket configuration. The **Smart Ticket Task** screen appears.
3. Click the **Mandant Value List** tab, and then do one of the following:
    - Click **Add Mandant Value** to add a new mandant to this configuration.
    - Click **Remove Mandant Value** to remove a selected mandant from this configuration.

Note

- If mandanten is configured in “restricting query” only and not configured in “filed restrictions”, the current Smart Ticket implementation doesn't support this use case.
- You can only use the mandanten values as defined in the mandanten security groups.
- Smart Ticket mandanten doesn't support SMA-SM Service Portal (SMA-SM SP).

Configure Smart Ticket for OCR

1. From the System Navigator, click **System Administration** > **Ongoing Maintenance** > **Smart Analytics** > **Smart Ticket**.
2. From the toolbar, click **More** or the **More Actions** button, and then select **Tailoring**. The Smart Ticket Settings page opens.
3. Update the following fields as needed.

Field Name	Description
Use FFT (Fast Fourier Transform) to remove the grid noise during OCR	The default value is true. When selected, the system uses Fast Fourier Transform (FFT) to remove the grid noise during OCR. However, using FFT consumes around 400 MB memory for each image and makes the OCR process slower. You can disable this feature if you want to speed up the system performance.
Maximum image size allowed to perform OCR without resizing (pixel):	The default value is 2000 pixels. It is recommended that you keep the default value unchanged. Specifies the maximum pixels of an image that's allowed to perform OCR without resizing. Images that larger than the specified value will be resized before Smart Analytics performs OCR.
The images size after resizing (pixel):	The default value is 700 pixels. Specifies the number of pixels to which an image is re-sized when the image size exceeds the maximum image size allowed to perform OCR without resizing. To ensure best OCR quality, the recommended value for this field is from 600 to 1200.

Note

The Smart Analytics Media server OCR supports *.png, *.jpeg, *.gif, *.bmp, and *.tiff images.

Related topics

Administrator tasks

Improving accuracy for Smart Ticket

3.1.3. Configure Hot Topic Analytics

User Role: Administrator

To configure Hot Topic Analytics, follow these steps:

1. From the System Navigator, click **System Administration** > **Ongoing Maintenance** > **Smart Analytics** > **Hot Topic Analytics**.
2. Select and open a Hot Topic Analytics configuration record from the configuration list. For example, Incident.

Note

Note In the out-of-box system, four Hot Topic Analytics configuration records are provided (for the Interaction, Problem, Incident, and Survey modules). If you want to add a new Hot Topic Analytics configuration for another module, select a file from the **Add Configuration** drop-down list, and then click **Add**. For more information, refer to Enable Hot Topic Analytics for other modules.

Important

Important As of Service Manager version 9.61, the index function is removed from Hot Topic Analytics configuration. Instead, Hot Topic Analytics analyzes the indexed data in Smart Search. You must firstly make sure that all modules defined for Hot Topic Analytics have been indexed in Smart Search. In particular, since there is no knowledge base for the **Internal survey** module in Smart Search, you must add a knowledge base for this module.

3. From the **Analytic Corpus** tab, modify the following settings as needed:
    - **Title Field**: Select a field to define the title when viewing an individual record in Hot Topic Analytics. The title field is also an important data source for hot topic hunting.
    - **Contents Fields**: Select the data source for Hot Topic Analytics. Be sure to use only text fields such as description and solution.

> **Tip**
> Tip Since HTA analyzes the indexed data in Smart Search, you must make sure that all fields (title field, content fields, filter fields) configured in HTA have been defined in the Smart Search knowledge bases. To add more fields for HTA, you must add these new fields in Smart Search knowledge base configuration if they're not there, and then add these new fields in HTA configuration. In particular, you must set the **Match** property to `true` for the filter fields in Smart Search configuration. Otherwise, it will not be treated as a filter in HTA.

4. From the **Filter Fields** tab, modify the following settings as needed:
    - **Timestamp Field**: Select a field to indicate the time stamp for filtering.
    - **Properties Fields**: Select fields that can be used for advanced filtering in Hot Topic Analytics. For example, you can define Category, Priority, or Source as a filter.
5. From the **Advanced** tab, modify the following settings as needed:
    - **Max Return Results**: Define the maximum number of records returned from Hot Topic Analytics. We have tested with Intel XEON E7-8867 10 Core CPU and 32G memory and got the following results:

Scenarios	Concurrent users	Response time	Service Manager used memory
Max Return Results: 5000 Analyze synchronously	1	25 Second	About 100M
Max Return Results: 5000 Analyze synchronously	10	40 Second	About 300M
Max Return Results: 5000 Analyze asynchronously	1	15 Second	About 100M
Max Return Results: 5000 Analyze asynchronously	10	20 Second	About 400M

    - **Analyze Asynchronously**: When this option is selected, HTA runs in the background and allows you to perform other tasks in the Service Manager system. If you don't select this option and use the synchronous analytics mode, you can test in your own environment to find a proper number that can have an acceptable response time. For detailed instructions, see Configure asynchronous analysis.
    - **Analytics Action**: Specify the query condition for the analytics action.

> **Note**
> Note In the out-of-box system, the **Analytics Action** setting is only available in the Hot Topic Analytics configuration record for incidents with the following three action queries: Set Parent, Create Problem, and Create Change/Article. If you want to add more custom action queries in the **Analytics Action** section, see Add more "Analytics Action" in the Hot Topic Analytics for Incidents.

Since HTA analyzes the indexed data in Smart Search, you must make the following modifications to make the Set Parent type query conditions work:

1. Click **Smart Analytics** > **Smart Search**.
2. Click **Incident_Library** in the Current Knowledgebase List.
3. Add the following fields to the Field Definitions tab:

Field Name	Alias	Type	Display In Results	Index Weight	Match	Sort	Data Cleansing
master.incident	master.incident	String	false	Default	true		false
parent.incident	parent.incident	String	false	Default	true		false

4. Click **Save** and **OK**.
5. Click **Full Reindex** to perform a full reindex for the knowledge base.

Caution

Caution The query condition doesn't work for the "Create Problem" and "Create Change/Article" type in this release.

6. Click **Save** to save your modification.

3.1.4. Configure connectors

User Role: Administrator

To enable search actions among different data sources, you need to configure different connectors and servers, and monitor their working status. You can get the URL information from the `respective .cfg` file of the connectors after you have configured them on your servers.

Configure SharePoint Connector

Configure Web Connector

Configure File System Connector

Configure Oracle Connector

Configure Confluence Connector

3.1.4.1. Configure SharePoint Connector

To configure SharePoint Connector, follow these steps:

1. Make sure that the OMNI Group Server and SharePoint Connector components were included when Smart Analytics was installed.

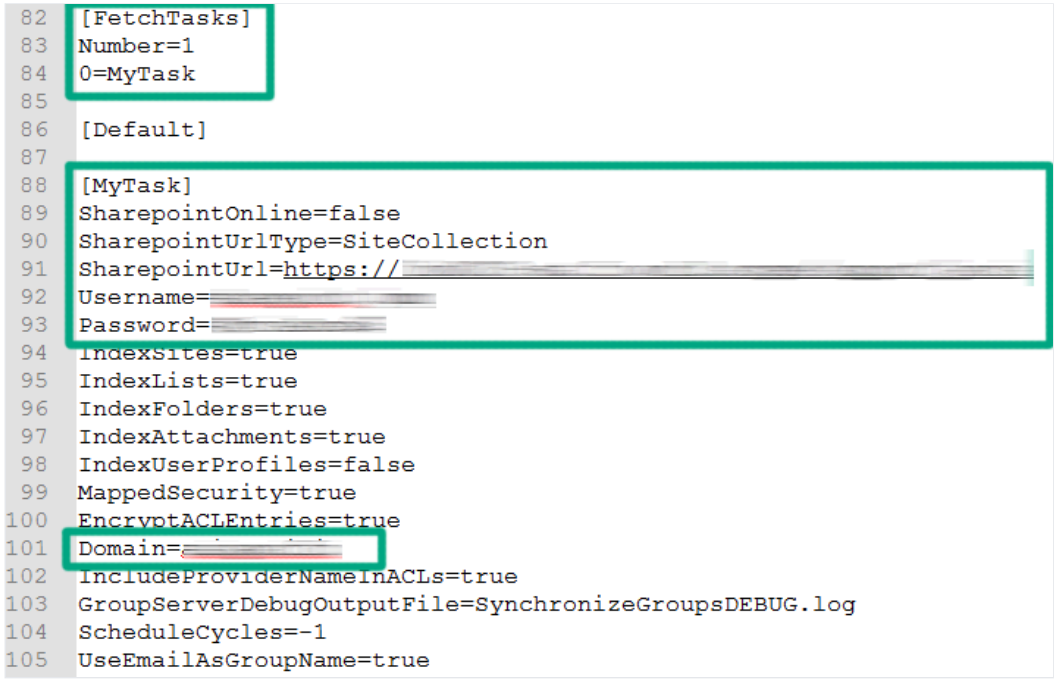
Note

Note You can install OMNI Group Server and SharePoint connector either along with other components by using the All in One install template, or install them separately by using the Customize install template through advanced installation.

2. Go to <*Smart Analytics Installation*>/SharepointRemoteConnector, locate the `SharepointRemoteConnector.cfg` file, and then configure the `[FetchTasks]` and `[MyTask]` sections as needed.
   Sample code:

```
[FetchTasks]
Number=1
0=MyTask
[Default]
[MyTask]
SharepointOnline=false
SharepointUrlType=SiteCollection
SharepointUrl=http://<SharePoint URL>/
Username=<username>
Password=<password>
IndexSites=true
IndexLists=true
IndexFolders=true
IndexAttachments=true
IndexUserProfiles=false
MappedSecurity=true
EncryptACLEntries=true
//Domain=DOMAIN
IncludeProviderNameInACLs=true
GroupServerDebugOutputFile=SynchronizeGroupsDEBUG.log
ScheduleCycles=-1
UseEmailAsGroupName=true
```

   Sample screenshot:



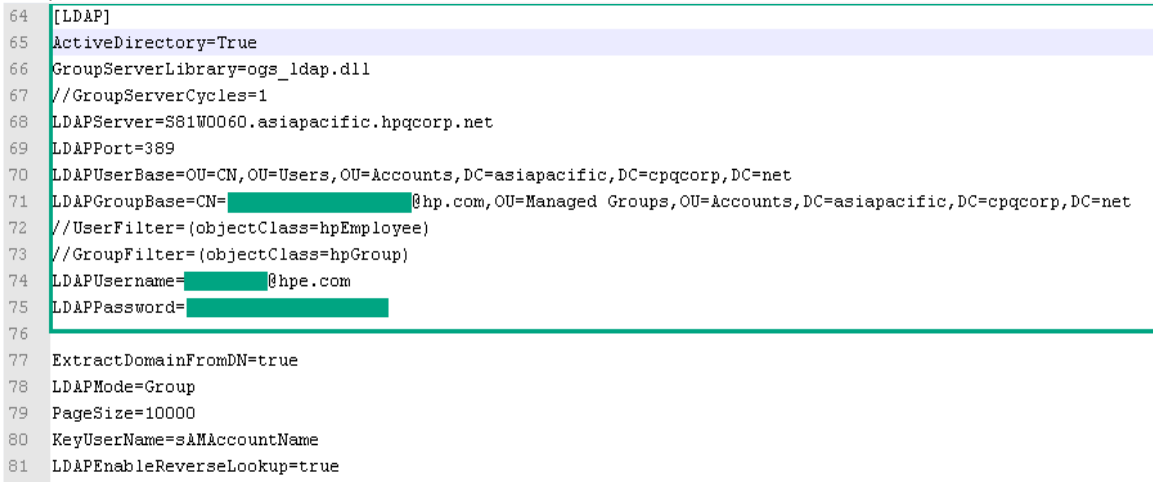
3. Restart your Smart Analytics SharePoint Connector service.
4. If you want to use OMNI Group Server, continue the following steps:
   1. Go to <*Smart Analytics installation*>/OmniGroupServer, locate the `OmniGroupServer.cfg` file, and then configure the `[LDAP]` and `[Sharepoint]` sections as needed

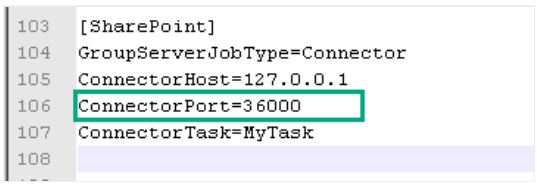
Sample code:

```
[LDAP]
ActiveDirectory=True
GroupServerLibrary=ogs_ldap.dll
//GroupServerCycles=1
LDAPServer=abc.asciapacific.example.net
LDAPPort=389
LDAPUserBase=OU=CN,OU=Users,OU=Accounts,DC=asicapacific,DC=example,DC=net
LDAPGroupBase=CN=PDL@example.com, OU=Managed
Groups,OU=Accounts,DC=asicapacific,DC=example,DC=net
//UserFilter=(objectClass=<objectClass>)
//GroupFilter=(objectClass=<objectClass>)
LDAPUsername=user@example.com
LDAPPassword=<password>
ExtractDomainFromDN=true
LDAPMode=Group
PageSize=10000
KeyUserName=sAMAccountName
LDAPEnableReverseLookup=true
```

```
[SharePoint]
GroupServerJobType=Connector
ConnectorHost=<ConnectorHost>
ConnectorPort=36000
ConnectorTask=MyTask
```

Sample screenshots:





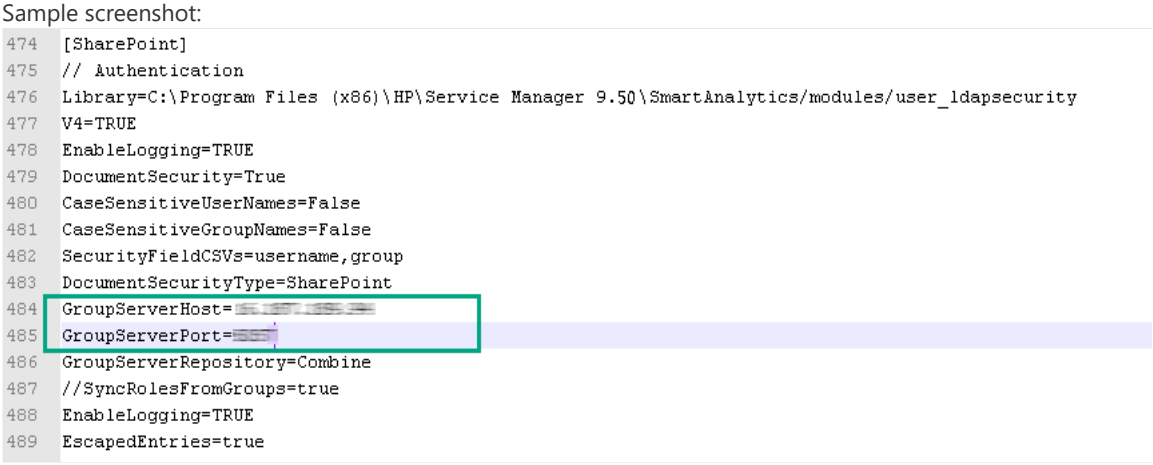
Note

Note Make sure that the value of `ConnectorTask` is consistent with `<TaskName>` in the `[FetchTasks]` section of the `SharepointRemoteConnector.cfg` file.

2. Go to <*Smart Analytics installation*>/IDOL, locate the `IDOLServer.cfg` file, uncomment `GroupServerHost` and `GroupServerPort`, and then specify OMNI Group Server Host and Port as needed.
   Sample code:

```
[SharePoint]
//Authentication
Library=C:\Program Files (x86)\OpenText\Service Manager
9.xx\SmartAnalytics/modules/user_ldapsecurity
V4=TRUE
EnableLogging=TRUE
DocumentSecurity=true
CaseSensitiveUserNames=False
CaseSensitiveGroupNames=False
SecurityFieldCSVs=username,group
DocumentSecurityType=SharePoint
GroupServerHost=<GroupServerHost>
GroupServerPort=5057
GroupServerRepository=Combine
//SyncRolesFromGroups=true
EnableLogging=TRUE
EscapedEntries=true
```

   Sample screenshot:



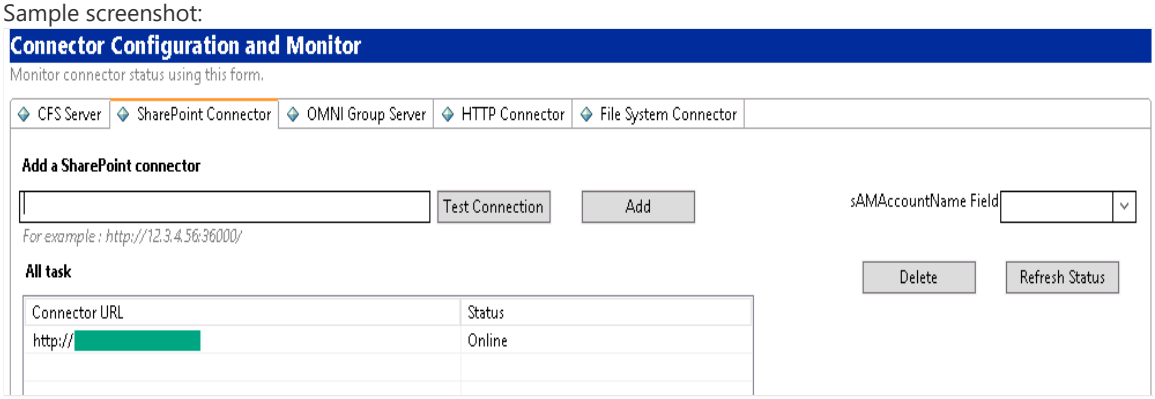
3. Restart the Smart Analytics Server and the OMNI Group Server services.

5. Sign in to Service Manager, and then click **System Administration** > **Ongoing Maintenance** > **Smart Analytics** > **Smart Search**. The Smart Search configuration page appears.
6. Click the **Connector Configuration** link to open the connector configuration page.
7. From the **CFS Server** tab, a list of all CFS server URLs is provided. You can click the **Refresh Status** button to refresh the URL list. Connectors need CFS servers to transfer data, so after you add a new connector, the corresponding CFS server information is added to this list.

> **Note**
> Note If there two or more connectors that are installed on the same machine and share one CFS server, there is no new URLs added to the list.

8. Go to the **SharePoint Connector** tab, perform the following actions:
   - **Add a SharePoint connector**: Type a configured SharePoint connector URL here. The format is http://<*Smart Analytics server address*>:<*SharePoint Connector server port*>.For example, http://192.168.255.255:36000/. You can click **Test connection** to test the URL connection status, and then click **Add** to add this URL to the current list.

Sample screenshot:



Note

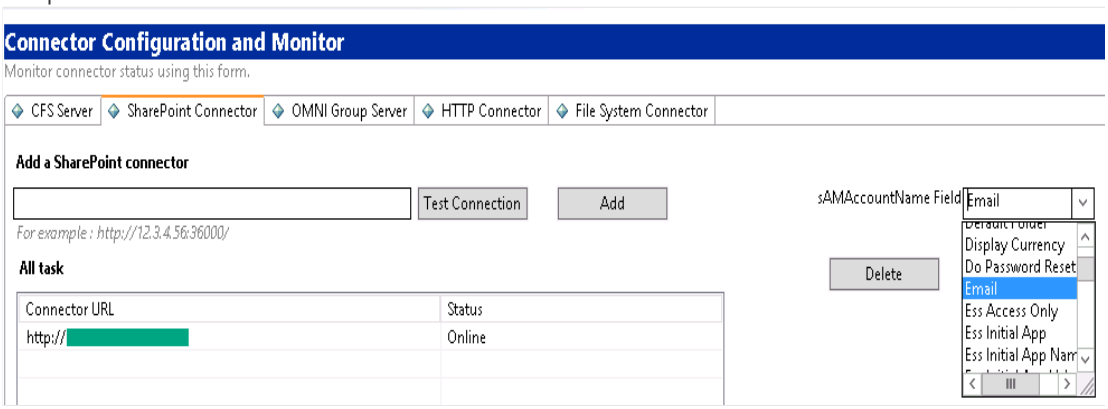
Note

- The "/" at the end of the URL is mandatory.
- Make sure the status for SharePoint is online.
- To get the URL information for the SharePoint connector you have configured, check the following configuration file:
  <Smart Analytics Installation>/SharepointRemoteConnector/SharepointRemoteConnector.cfg

- **sAMAccountName Field**: Choose the field type from the drop-down list. This field is the mapping field of SharePoint and SM users.

  If SharePoint on premise is used, configure this field to the domain account field of the SM operator table. If SharePoint Online is used only, there is no need to set this field as operator email field is used for user mapping.

  Sample screenshot:



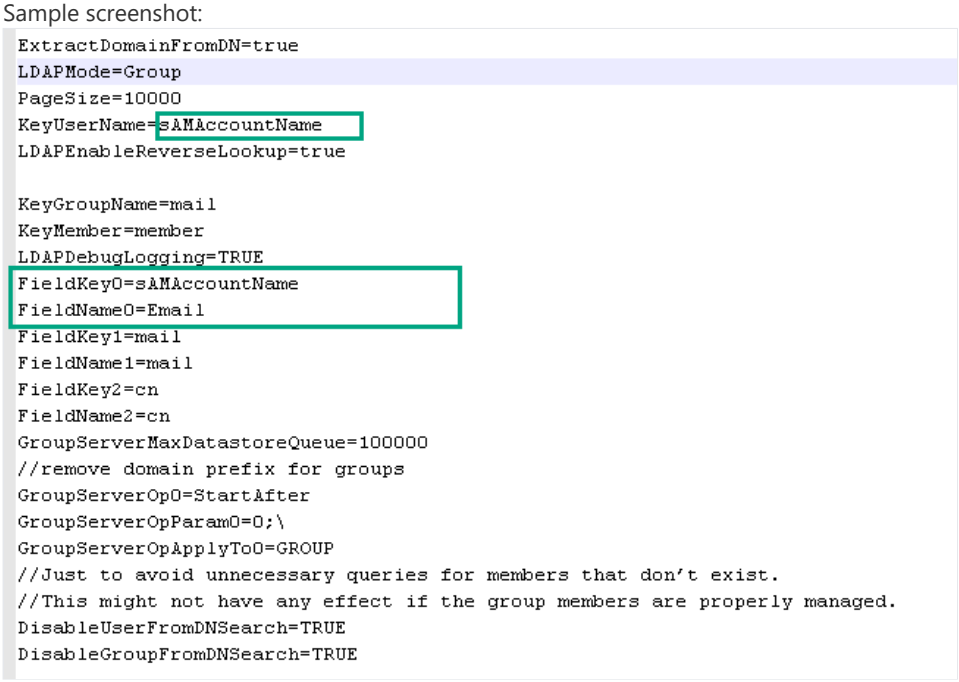
- **Delete**: Select a SharePoint connector URL and then click this button to delete it from the list.
- **Refresh Status**: Click to refresh the status of the URL list.

9. From the **OMNI Group Server** tab, a Repository list of all OMNI group servers is provided. OMNI Group Server provides the LDAP configuration information which may be required for SharePoint sign in. You can also check the target task of a repository and its working status. You can click the **Refresh Status** button to refresh the URL list.

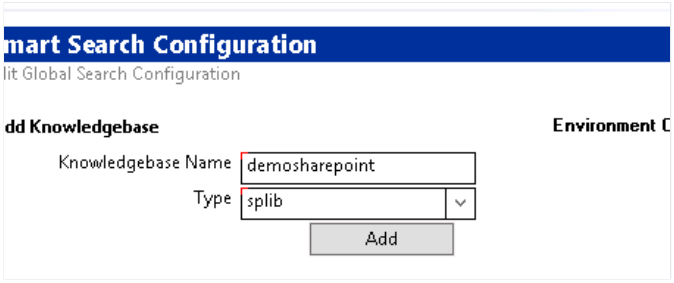
10. (For Security SharePoint), go to *<Smart Analytics installation>*/OmniGroupServer, locate the `OmniGroupServer.cfg` file, and specify `FiledName0` to the option you selected for the **sAMAccountName** Field.
    Sample code:

```
ExtractDomainFromDN=true
LDAPMode=Group
PageSize=10000
KeyUserName=sAMAccountName
LDAPEnableReserveLookup=true
KeyGroupName=mail
KeyMember=member
LDAPDebugLogging=TRUE
FieldKey0=sAMAccountName
FieldName0=Email
FieldKey1=mail
FieldName1=mail
FieldKey2=cn
FieldName2=cn
GroupServerMaxDatastoreQueue=100000
//remove domain prefix for groups
GroupServerOp0=StartAfter
GroupServerOpParam0=0;\
GroupServerOpApplyTo0=GROUP
//Just to avoid unnecessary queries for members that don't exist.
//This might not have any effect if the group members are properly managed.
DisableUserFromDNSearch=TRUE
DisableGroupFromDNSearch=TRUE
```

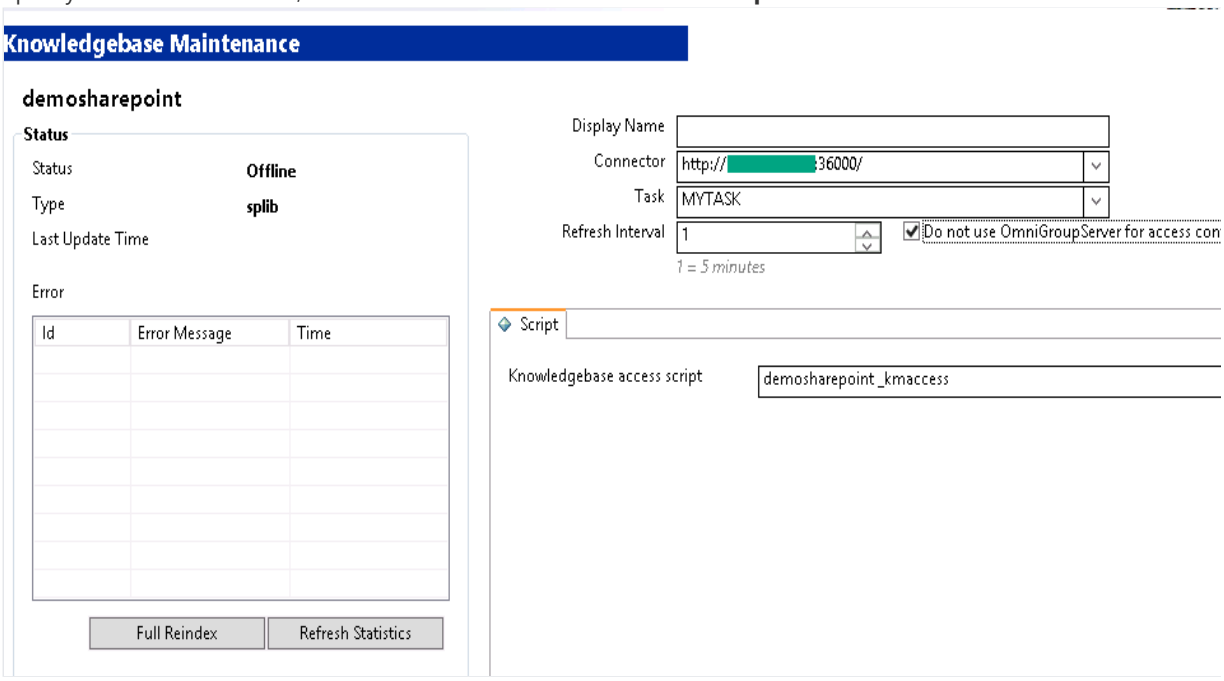
    Sample screenshot:



11. Restart the Smart Analytics OMNI Group Server service.
12. Do the following to add a splib library for the sharepoint connector:
    1. go to Smart Search Configuration page.
    2. Specify Knowledgebase Name,
    3. Select **splib** for **Type**
    4. Click **Add**. The Knowledgebase Maintenance page opens.



13. Specify **Connector** and **Task**, and then select the **Do not use OmniGroupServer for access count** checkbox.



14. Click **Save**.
15. Click **Full Reindex** and **Refresh Status**.
16. You can perform the search when the status changes to Indexing and Doc Count for this library is greater than 1.

> **Note**
> Note Sign out and then sign in to your Service Manager again if you can't find the library in your Smart Search library list.

3.1.4.2. Configure Web Connector

To configure the Web Connector, follow these steps:

1. Make sure that the Web Connector component was included when Smart Analytics was installed.

> **Note**
> Note
> You can install Web Connector either along with other components by using the All in One install template or install the connector separately by using the Customize install template through advanced installation. For more details, see section of Advanced Installation in Install Smart Analytics

2. Go to *<Smart Analytics Installation>*/WebConnector, locate the `WebConnector.cfg` file, and then configure the `[FetchTasks]` and `[MyTask]` sections as needed.
   See IDOL 12.3 Documentation for detailed information.

Sample code for one task:

```
[FetchTasks]
SSLMethod=SSLV23
Number=1
0=MyTask
WkoopPath=WKOOP/WKOOP.exe

[MyTask]
Url=http://www.example.com/
SpiderUrlCantHaveRegex=.*\.css|.*\.js|.*.js\?.*
StayOnSite=True
Depth=99
ProxyHost=<ProxyHost>
ProxyPort=8080
...
```

To configure multiple tasks, for example, two tasks, configure the .cfg file as follows:

```
[FetchTasks]
SSLMethod=SSLV23
Number=2
0=MyTask1
1=MyTask2
WkoopPath=WKOOP/WKOOP.exe

[MyTask1]
...

[MyTask2]
...
```

The following table describes the parameters of the `[MyTask]` section in the webconnector.cfg file. If you want to configure multiple tasks for one connector, you just need to copy the content in the `[MyTask]` section and rename the section.

Parameter	Description
`Url`	The URL of the page to start crawling from. You can specify multiple URLs to start crawling from by setting a comma-separated list of values or using a numbered list of parameters (Url0=, Url1=, and so on).
`SpiderUrlCantHaveRegex`	A regular expression to restrict the pages that are crawled by the connector. The full URL of a page must match the regular expression, otherwise it is not downloaded, crawled, or ingested.
`UrlMustHaveRegex`	A regular expression to restrict the content retrieved by the connector. The full URL of a page must match the regular expression, otherwise the page is not ingested. This parameter applies to all file types, for example HTML pages, images, text documents, and so on.
`UrlCantHaveRegex`	A regular expression to restrict the content retrieved by the connector. If the full URL of a page matches the regular expression, the page is not ingested. This parameter applies to all file types, for example HTML pages, images, text documents, and so on.
`PageMustHaveRegex`	A regular expression to restrict the content retrieved by the connector. The content of a page must match the regular expression, otherwise the page is not ingested. This parameter applies to all file types.
`PageCantHaveRegex`	A regular expression to restrict the content retrieved by the connector. If the content of a page matches the regular expression, the page is not ingested. This parameter applies to all file types.
`ContentTypeMustHaveRegex` `ContentTypeCantHaveRegex`	The connector can also accept or reject pages based on the `Content-Type` header that is returned with the page. This a good way to filter out certain types of content, such as scripts. The `ContentTypeCantHaveRegex` and `ContentTypeMustHaveRegex` parameters are also set as regular expressions. The following example is a good starting point that attempts to filter out stylesheets, XML, and javascript files: `ContentTypeCantHaveRegex=(application|text)/(javascript|xml|x-javascript|css)(;.*)?`
`PageDelay`	Specify a time interval that a synchronize thread must wait between requests. If you set a delay of 5 seconds (`PageDelay= 5s`), each synchronize thread must wait at least one second between requests.
`StayOnSite`	When `StayOnSite=TRUE`, the connector doesn't follow links to other sites, including other sub-domains of the same site.

SiteDuration	Maximum ammount of time to spend crawling, 0 indicating unlimited.
MaxPages	Maximum number of pages to ingest per synchronize run, 0 indicating unlimited.
MinPageSize	Minimum size that a page must be for it to be ingested.
MaxPageSize	Maximum size that a page must be for it to be ingested, 0 indicating unlimited
MaxLinksPerPage	Maximum number of links a page can have before it is not ingested, 0 indicating unlimited.
FollowRobotProtocol	If FollowRobotProtocol=true, the system will follow robots.txt and ROBOTS meta tag commands.
UserAgent	The UserAgent sent in the request headers. For example, UserAgent=Autn-WKOOP
Depth	The maximum depth to which the connector follows links when crawling. For example, to index all pages that can be reached from the Url by following no more than three links, set Depth=3. If the value of this parameter is set to -1, there is no limit for the maximum depth.
FollowFlash	If FollowFlash=true, the system will follow links found in flash content.
LinkElementMustHaveRegex LinkElementCantHaveRegex	Regexes to restrict which element types are crawled for links. For example, you can use the LinkElementMustHaveRegex and LinkAttributes parameters to follow only certain types of hyperlinks (in this example, <a href="...">) in the page source. LinkElementMustHaveRegex is set as a regular expression and LinkAttributes accepts a comma-separated list of attribute names: LinkElementMustHaveRegex=^a$ LinkAttributes=href
LinkAttributes	HTML tag attribute names to be treated as containing a link to be followed. For example, LinkAttributes=href
Clipped	This parameter is used to enable web page clipping. To clip pages automatically, add Clipped=TRUE to your task configuration.
ClipPageUsingCssSelect ClipPageUsingCssUnselect	If Clipped=true , the area of the page that is kept/removed can be specified using css2 selectors. If these parameters are left empty, the SmartPrint algorithm is used to determine which parts of the page are kept/removed.

IgnoreCanonicalLinks	Set IgnoreCanonicalLinks=true to ignore canonical links in web page headers or meta tags.
RedirectAsLink	If RedirectAsLink=true, the system will treat redirection as a page with a single link that should be followed, but not ingested so that content from a single redirection target from multiple source pages is not duplicated across multiple documents.
IncludeMetaTagsInMetadata	Set IncludeMetaTagsInMetadata=true to include a metadata field for each <meta name=".." content="..." /> element in the html head.
IncludeLinksInMetadata	Set IncludeLinksInMetadata=true to include a LINK metadata field for each link found on a page.
MetadataSelector0 MetadataFieldName0	Extract custom metadata from each page using css2 selectors.
MetadataSelectorExtractPlainText	Set MetadataSelectorExtractPlainText=true to extract plain text from the HTML elements identified by the MetadataSelector parameters above.
RemoveScripts	Set RemoveScripts=true to remove all <script> elements from indexed pages.
RemoveNoscripts	Set RemoveNoscripts=true to remove all <noscript> elements from indexed pages.
RemoveNoframes	Set RemoveNoframes=true to remove all <noframes> elements from indexed pages.
RankPages	If RankPages=true, the system will add a AUTN_RANK field to indexed pages, giving each a rank based on a number of factors.
ProxyHost	The hostname or IP address of the proxy server to use. For example, ProxyHost=PROXY.COM
ProxyPort	The port of the proxy server to use. For example, ProxyPort=80
ProxyUsername	The user name to use to authenticate with the proxy server. For example, ProxyUsername=username
ProxyPassword	The password to use to authenticate with the proxy server. For example, ProxyPassword=password
AuthUser AuthPassword	To sign in to a Web site that uses Basic, HTTP Digest, or NTLM version 2 authentication, specify a username and password by setting the configuration parameters AuthUser and AuthPassword. For example: AuthUser=user AuthPassword=pass

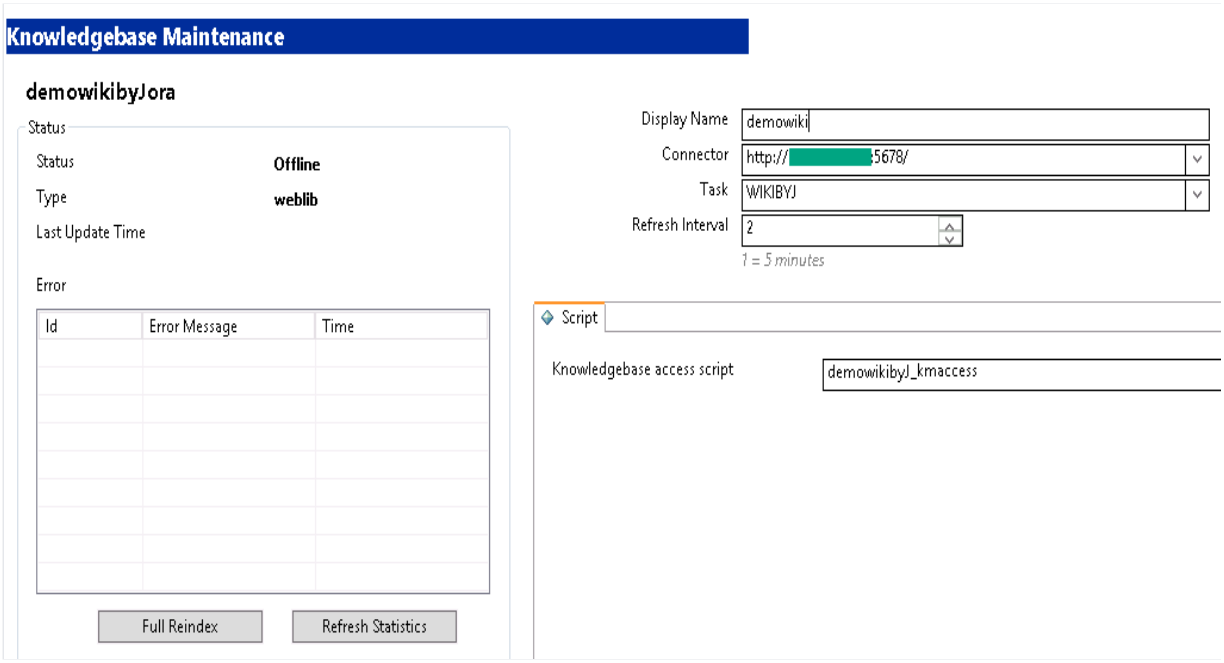
3. Restart the Smart Analytics Web Connector service.
4. Sign in to Service Manager, and then click **System Administration** > **Ongoing Maintenance** > **Smart Analytics** > **Smart Search**. The Smart Search configuration page appears.
5. Click the **Connector Configuration** link to open the connector configuration page.
6. From the **Web Connector** tab, a list of all connector URLs and their status is provided. You can perform the following actions:
    - **Add a web connector**: Type a new Web connector URL here. The format is http://<*Smart Analytics server address*>:<*Web Connector server port*>. For example, http://192.168.255.255:5678/. You can click **Test connection** to test the URL connection status, and click **Add** to add this URL to the current list.

      Note

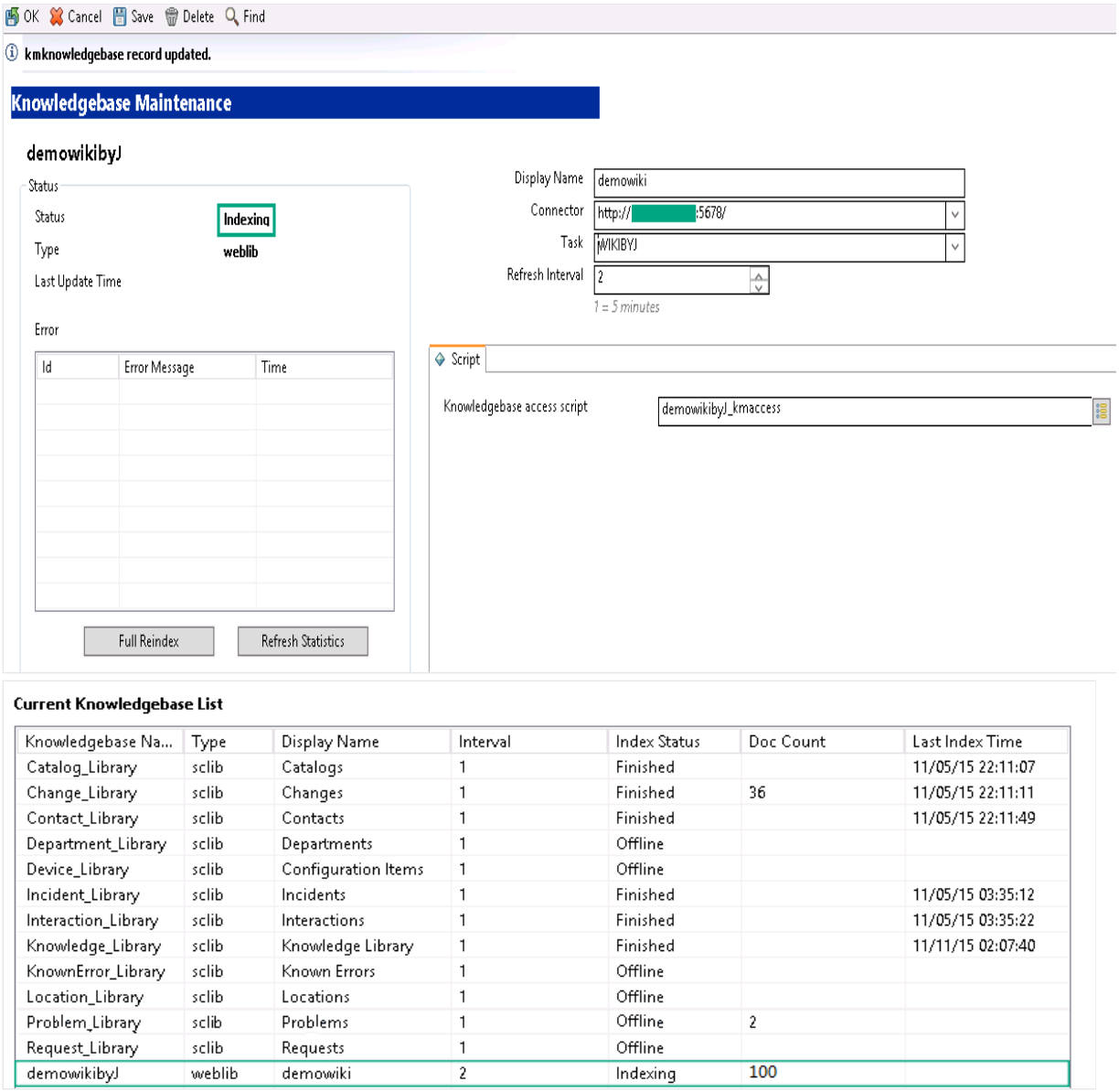
      Note

      - The "/" at the end of the URL is mandatory.
      - Make sure the status for the web connector URL is online.
      - To get the URL information for the web connector you have configured, check the following configuration file:
        <Smart Analytics Installation>/WebConnector/WebConnector.cfg

    - **Delete**: Select a web connector URL, and then click this button to delete it from the list.
    - **Refresh Status**: Click to refresh the status of the URL list.
7. Do the following to add a weblib library for the web connector:
    1. Go to Smart Search Configuration page.
    2. Specify Knowledgebase Name
    3. Select **weblib** for Type,
    4. Click **Add**. The Knowledgebase Maintenance page opens.
8. Specify **Connector** and **Task**.
9. Click **Save**.
10. Click **Full Reindex** and **Refresh Status**.



11. You can perform the search when the status changes to Indexing and Doc Count for this library is greater than 1.



Current Knowledgebase List

Knowledgebase Na...	Type	Display Name	Interval	Index Status	Doc Count	Last Index Time
Catalog_Library	sclib	Catalogs	1	Finished		11/05/15 22:11:07
Change_Library	sclib	Changes	1	Finished	36	11/05/15 22:11:11
Contact_Library	sclib	Contacts	1	Finished		11/05/15 22:11:49
Department_Library	sclib	Departments	1	Offline		
Device_Library	sclib	Configuration Items	1	Offline		
Incident_Library	sclib	Incidents	1	Finished		11/05/15 03:35:12
Interaction_Library	sclib	Interactions	1	Finished		11/05/15 03:35:22
Knowledge_Library	sclib	Knowledge Library	1	Finished		11/11/15 02:07:40
KnownError_Library	sclib	Known Errors	1	Offline		
Location_Library	sclib	Locations	1	Offline		
Problem_Library	sclib	Problems	1	Offline	2	
Request_Library	sclib	Requests	1	Offline		
demowikibyJ	weblib	demowiki	2	Indexing	100	

Note

Note

Sign out and then sign in to your Service Manager again if you can't find the weblib library in your Smart Search library list.

3.1.4.3. Configure File System Connector

To configure File System Connector, follow these steps:

1. Make sure that the Smart Analytics installation includes the File System Connector component.

Note

Note You can install File System Connector either along with other components by using the All in One install template, or install the connector solely by using the Customize install template through advanced installation.

2. Go to `<Smart Analytics Installation>/FileSystemConnector`, locate the `filesystemconnector.cfg` file, and then configure the `[FetchTasks]` and `[MyTask]` sections as needed.
   Sample code:

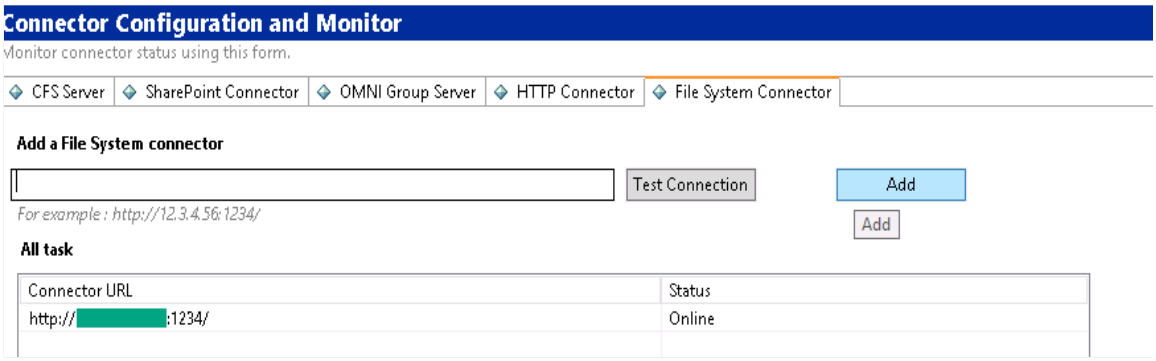
```
[FetchTasks]
Number=1
0=MYTASK
[MYTASK]
//specifies the interval (in seconds) between scheduled synchronize actions.
ScheduleRepeatSecs=300
//specifies whether the connector searches subfolders.
DirectoryRecursive=TRUE
//The DirectoryPathCSVs parameter specifies a comma-separated list of folders to search for files.
DirectoryPathCSVs=\\JORA7\ShareDirectory
//A regular expression that specifies the folders to search for files. The connector only searches
folders
//that are within the location specified by DirectoryPathCSVs
//where the full path of the folder matches the regular expression
//where the full path of all parent folders (up to the folder specified by DirectoryPathCSVs) match
the regular expression.
PathCrawlRegex=.*
//A regular expression that specifies the folders to ignore. The connector ignores any folders
where the path matches the regular expression.
//If a folder is ignored, all of its subfolders are also ignored.
//PathCrawlRegex=
//The DirectoryFileMatch parameter limits the files that are retrieved by the connector. The value
of this parameter is a wildcard expression
//The filename of a file must match the wildcard expression, otherwise the file is ignored. default
to all files
DirectoryFileMatch=*.pdf,*.doc,*.ppt,*.log
```

   Sample screenshot:

```
59  [FetchTasks]
60  Number=1
61  0=MyTask
62
63  [MyTask]
64  //specifies the interval (in seconds) between scheduled synchronize actions.
65  ScheduleRepeatSecs=300
66  //specifies whether the connector searches sub-folders.
67  DirectoryRecursive=TRUE
68  //The DirectoryPathCSVs parameter specifies a comma-separated list of folders to search for files.
69  DirectoryPathCSVs=\\J7\ShareDirectory
70  //A regular expression that specifies the folders to search for files. The connector only searches folders:
71  //that are within the location specified by DirectoryPathCSVs
72  //where the full path of the folder matches the regular expression
73  //where the full path of all parent folders (up to the folder specified by DirectoryPathCSVs) match the regular expression.
74  PathCrawlRegex=.*
75  //A regular expression that specifies the folders to ignore. The connector ignores any folders where the path matches the regular expression.
76  //If a folder is ignored, all of its subfolders are also ignored.
77  //PathNoCrawlRegex=
78  //The DirectoryFileMatch parameter limits the files that are retrieved by the connector. The value of this parameter is a wildcard expression
79  //The filename of a file must match the wildcard expression, otherwise the file is ignored. default to all files
80  DirectoryFileMatch=*.pdf,*.doc,*.ppt,*.txt,*.log
81
```

3. Save your changes, and then restart the File System Connector service.
4. Sign in to Service Manager, and then click **System Administration** > **Ongoing Maintenance** > **Smart Analytics** > **Smart Search**. The Smart Search configuration page appears.
5. Click the **Connector Configuration** link to open the connector configuration page.

6. The **File System Connector** tab provides a list of all connector URLs and their status. You can perform the following actions:
    - **Add a File system connector**: Type a new file system connector URL here. The format is `http://<Smart Analytics server address>:<File System Connector server port>`. For example, `http://192.168.255.255:1234/`. You can click **Test connection** to test the URL connection status, and click **Add** to add this URL to the current list.

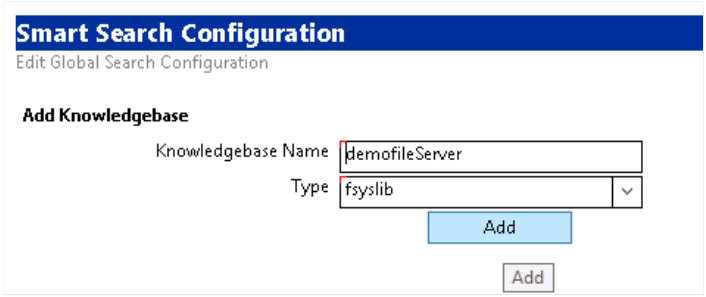


> **Note**
>
> Note
>
> - The "/" at the end of the URL is mandatory.
> - Make sure the task status is online.
> - To get the URL information for the file system connector you have configured, check the following configuration file:
>
> `<Smart Analytics Installation>/FileSystemConnector/filesystemconnector.cfg`

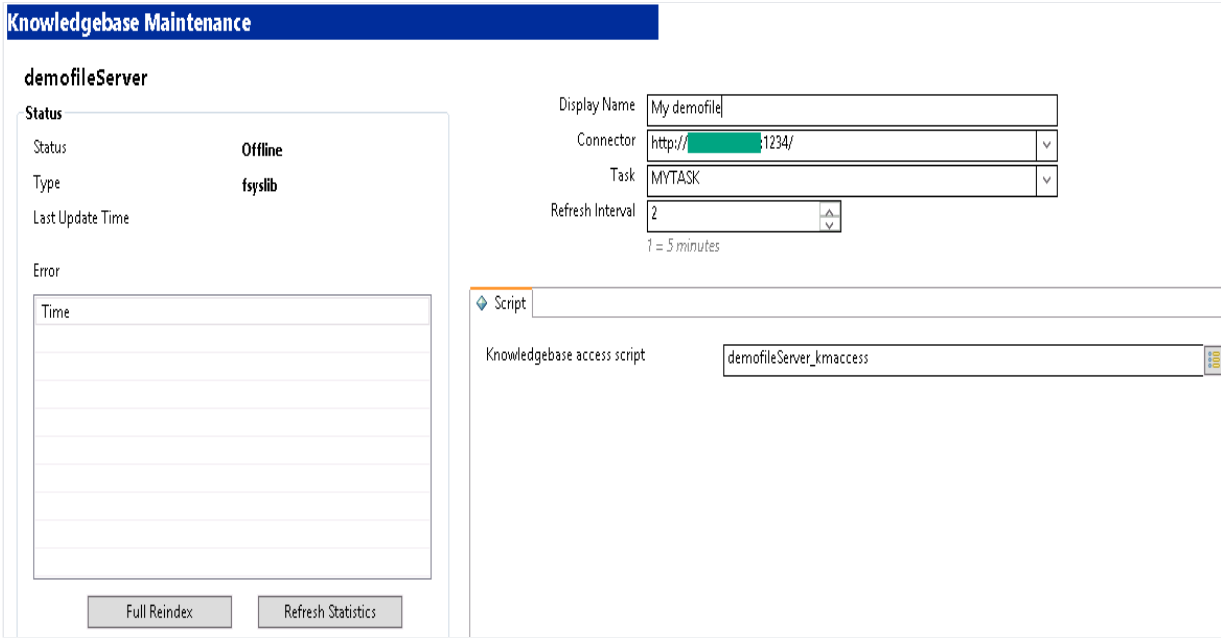
    - **Delete**: Select a file system connector URL, and then click this button to delete it from the list.
    - **Refresh Status**: Click to refresh the status of the URL list.

> **Note**
>
> Note When the fileserver connectors (including CFS server) and the fileserver share folders are on the same machine, Service Manager supports the Uniform Naming Convention (UNC) path (`DirectoryPathCSVs=\\path\to\shared\folder`) by using the web browser.

7. Do the following to add a fsyslib library for the file system connector:
    1. Go to Smart Search Configuration page.
    2. Specify **Knowledgebase Name**.
    3. Select **fsylib** for **Type**.
    4. Click **Add**. The Knowledgebase Maintenance page opens.



8. Specify **Connector** and **Task**.
9. Click **Save**.
10. Click **Full Reindex** and **Refresh Status**.



11. You can perform a search when the status changes to Indexing and Doc Count for this library is greater than 1.

> **Note**
> Note Sign out and then sign in to your Service Manager again if you can't find weblib library in your Smart Search library list.

3.1.4.4. Configure Oracle Connector

To install and configure the Oracle Connector, follow these steps:

1. Stop Service Manager and Smart Analytics.

2. Follow these steps to install the Oracle Client:

> **Note**
> Note
> - This step uses the Linux operating system as an example.
> - To retrieve information from an Oracle database, you must install Oracle Client version 11 on the same computer where you have installed the IDOL Oracle connector. The connector can use the Oracle Client version 11 libraries to retrieve information from Oracle Database version 9, 10, or 11.

   1. Sign in to the Linux server where you have installed Smart Analytics.
   2. Download the Oracle Client 11.1.0.7.0 (64-bit) zip file for Linux from the following Oracle website: https://www.oracle.com/database/technologies/instant-client/linux-x86-32-downloads.html
      You can only use this Oracle Client version for the IDOL Oracle Connector.
   3. Unzip `instantclient-basic-linux-11.1.0.7.0.zip` and `instantclient-sqlplus-linux-11.1.0.7.0.zip` into the same directory. For example, `../OracleClient/` .
   4. Navigate to `../OracleClient/instantclient_11_1/` , and then create the `network/admin directory` . Copy the `tnsnames.ora` file from Oracle database to `../OracleClient/instantclient_11_1/network/admin/` .
   5. Configure the environment variables for the Oracle client.
      For example, modify `/etc/profile` as follows if you are using CentOS, and then run the **source /etc/profile** command.

      ```
      export ORACLE_HOME=/root/OracleClient/instantclient_11_1
      export LD_LIBRARY_PATH=LD_LIBRARY_PATH:$ORACLE_HOME
      export PATH=$PATH:${K8S_HOME}/bin:$ORACLE_HOME
      ```

   6. Check LD_LIBRARY_PATH by running the **echo $LD_LIBRARY_PATH** command. The output is `/root /OracleClient/instantclient_11_1` .
   7. Verify the Oracle Client installation.
      For example, the following screenshot illustrates a successful Oracle Client installation:

3. Install the IDOL Oracle Connector.

> **Note**
> Note This step uses the Linux operating system as an example.

   1. Sign in to the Linux server where you have installed Smart Analytics.
   2. Refer to Install Smart Analytics on Linux to install the Oracle connector. Make sure that you select **Advanced install** as your installation type, and then select **Oracle Connector** in the **Choose Distributed Components** step.
   3. Navigate to the `../OracleConnector` directory, and then open the `OracleConnector.cfg` file. You can update the `ConnectionString` value as necessary. For more information, refer to the Documentation.zip file in the Oracle Connector installation directory.
   4. Update the SQL query according to your requirements.

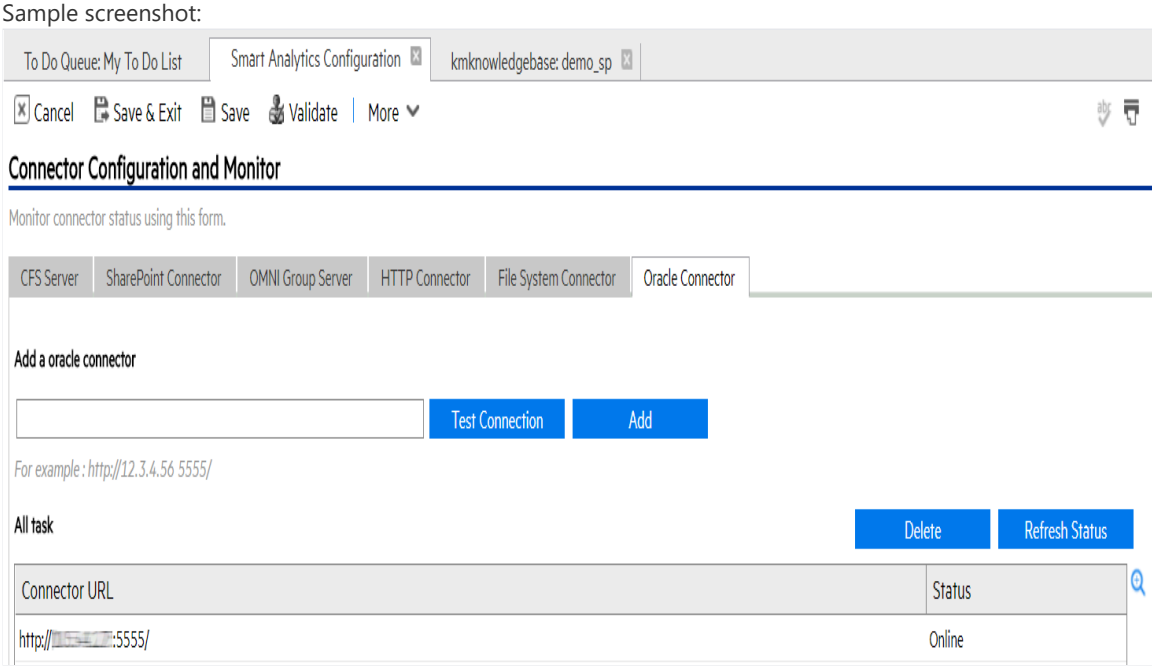
5. You also can add a task for another query. Refer to the following sample codes:

```
Number=2
0=mytask
1=mytask1
[mytask1]
IngestActions=LUA:okmparse.lua
ConnectionString = username/passwords@net_service_name
#used your own Oracle username,passwords and net_service_name.
SQL=select field_names from table_name
PrimaryKeys=PK_field
Template=template.tmpl
SubTables= sub_table
[sub_table]
Template=sub.tmpl
SQL=sub_query
```

4. Start the IDOL Oracle Connector.

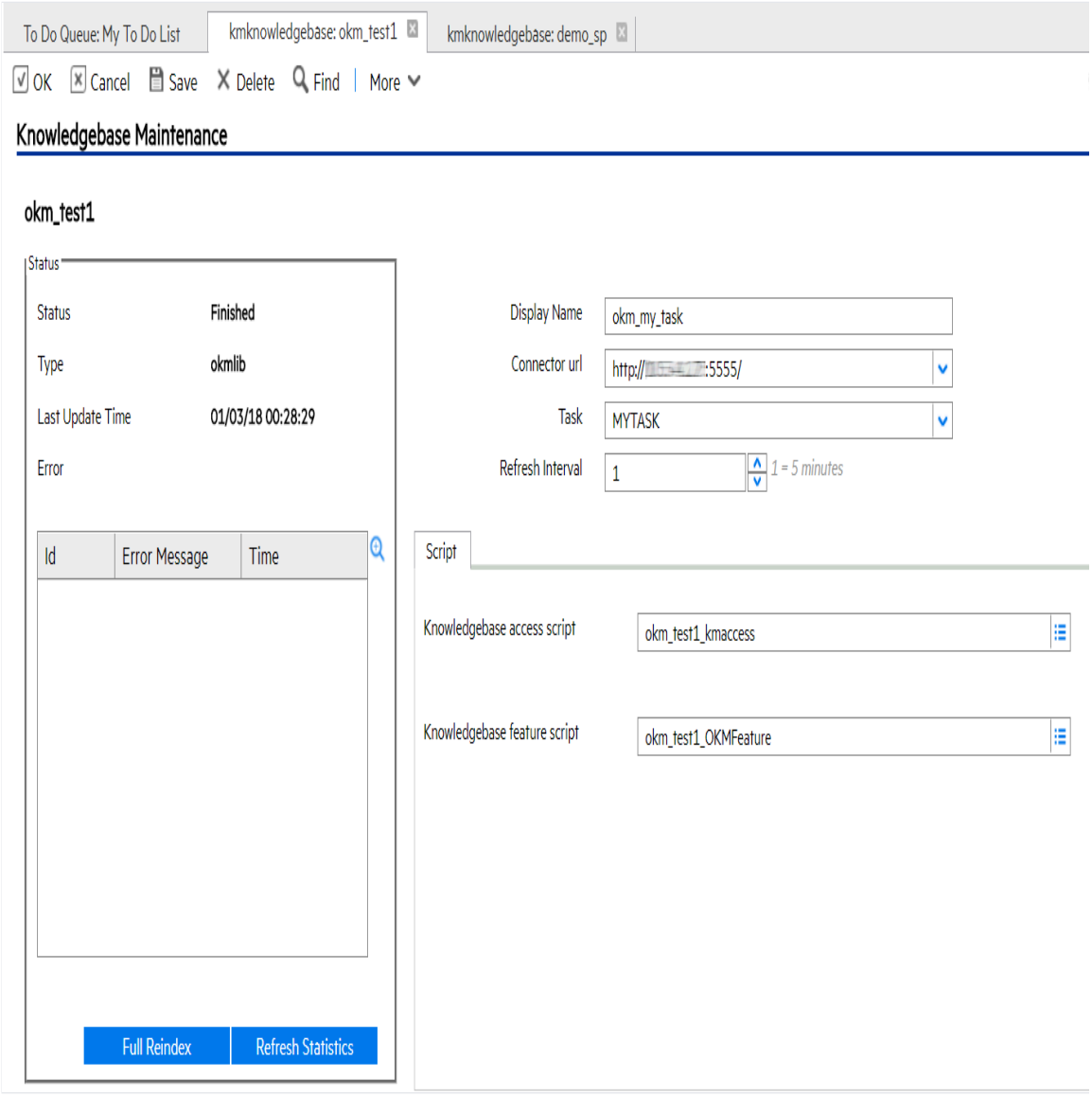
> **Note**
> Note This step uses the Linux operating system as an example.

   1. Start Service Manager and Smart Analytics.
   2. Navigate to the `../OracleConnectorCFS` directory, and then run the **StartOracleConnector.sh** command to start the Oracle Connector Connector Framework Server (CFS).
5. Configure the IDOL Oracle Connector.
   1. Sign in to Service Manager as a system administrator, and then click **System Administration** > **Ongoing Maintenance** > **Smart Analytics** > **Smart Search**. The Smart Search configuration page appears.
   2. Click the **Connector Configuration** link to open the connector configuration page.
   3. Go to the **Oracle Connector** tab, type a configured Oracle connector URL in the **Add an Oracle connector** field. For example, `http://<hostname.domain>:5555/`. You can click **Test connection** to test the URL connection status, and then click **Add** to add this URL to the current list. Note that the connector's status must be **Online**. Sample screenshot:



6. Do the following to add an **okmlib** library for the Oracle connector:
   1. Go to the Smart Search Configuration page.

2. Specify the Knowledgebase Name.
3. Select **okmlib** for Type.
4. Click **Add**. The Knowledgebase Maintenance page opens.
5. Specify **Connector URL**, **Task**, and **Refresh Interval**.



6. Click **Save**.

7. (Optional) Modify the Knowledgebase access script if you want to further control access permissions to this Oracle knowledge base.
   For example, you can add the following codes to prevent the ESS users from accessing the okm_test1 library:

```
if (system.functions.nullsub(vars.$G_ess,false))
          return false;
else
return true;
```

8. Modify the Knowledgebase feature script to further configure the current Oracle knowledge base.
   1. The `Oracle KM view base url` is the base URL for the article page in the Oracle KM system. For example: `https://<FQDN>/infocenter/index`. You can configure the Oracle KM application base URL in the `getOKMViewUrl` function.

2. You can configure the Lua script in the `getOKMParseScriptName` function. For more information about how to apply the Lua script to manipulate documents, refer to the `Documentation.zip` file in the Oracle Connector installation directory.

9. Click **Full Reindex** and **Refresh Status**.

3.1.4.5. Configure Confluence Connector

To configure Confluence Connector, follow these steps:

1. Make sure that the Confluence Connector components are included when you install Smart Analytics.

   > **Note**
   > Note You can install the Confluence connector either along with other components by using the All in One install template or install them separately by using the Customize install template through advanced installation. For more details, see the **Advanced Installation** section in Install Smart Analytics.

2. Go to the *<Smart Analytics Installation>*/ConfluenceRESTConnector directory. Locate the `ConfluenceRESTConnector.cfg` file, and then configure the `[FetchTasks]` and `[MyTask]` sections as needed. Sample code:

```
[FetchTasks]
Number=2
0=CloudConfluence
1=OnPremisesConfluence

[CloudConfluence]
MappedSecurity=false
UseSecureConnection=true
ConfluenceHost=<cloud Confluence url(For example, mfli.atlassian.net)>
ConfluencePort=443
BasicUsername=<cloud Confluence user name>
BasicPassword=<cloud Confluence api token>
//if we need to use proxy
//ProxyHost=
//ProxyPort=
KMSourceIdentityName=confluenceCloud
SSLConfig=SSLSettings
ConfluenceApiRoot=wiki

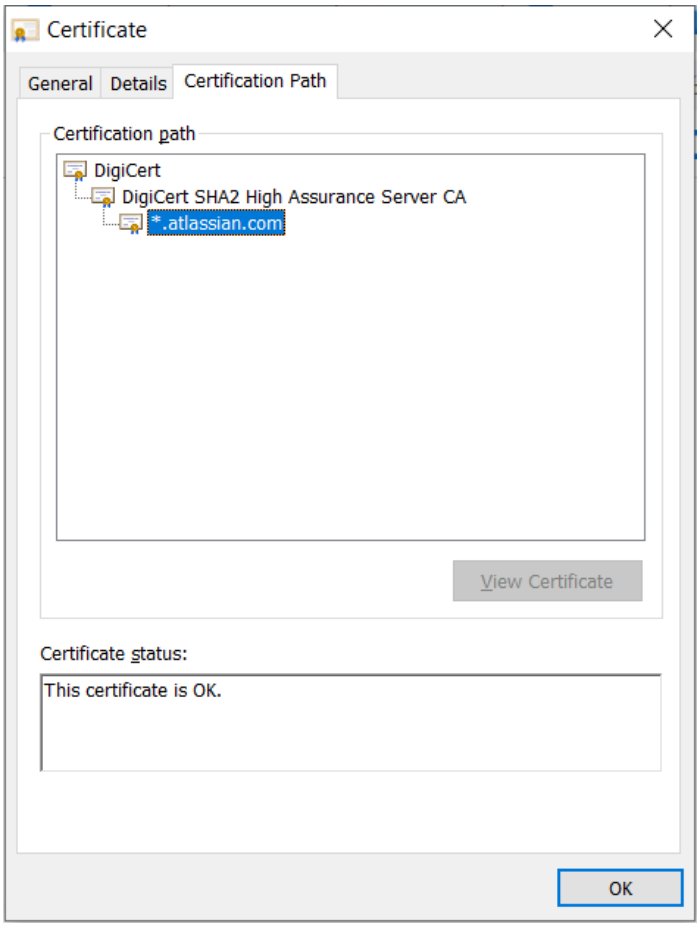
[SSLSettings]
SSLMethod=TLSV1.3
SSLCACertificate=cloud.cer

[OnPremisesConfluence]
MappedSecurity=false
UseSecureConnection=false
ConfluenceHost=<Confluence IP or host>
ConfluencePort=<Confluence port. Default is 8090>
BasicUsername=<Confluence user name>
BasicPassword=<Confluence password>
//ignore all attachment
AttachmentCantHaveRegex=.*.*
```

   In this example, you can configure one for more tasks for cloud or on-premises Confluence.
   The `basicpassword` value for cloud is the API token. You need to visit https://id.atlassian.com/manage-profile/security/api-tokens to create an API token, and then use this API token as the password. Meanwhile, you can encrypt the `BasicPassword` value. For detailed instructions, download the Confluence REST Connector document and then refer to the *Encrypt Passwords* section.

The cloud Confluence also needs SSL configurations and `SSLCACertificate` is a certificate file that locates in *<Smart Analytics Installation>*/ConfluenceRESTConnector. In the above example, the value for `SSLCACertificate` is `cloud.cer`. To generate this file, follow these steps:

1. Open the cloud Confluence URL in a web browser. For example, mfli.atlassian.net.
2. Click the **View site information** (lock) icon from the address bar and then select **Certificate**.
3. Click the **Certification Path** tab to view the certificate chain. You need to export all certificates except the last one. In this example, you need to export two certificates: DigiCert and DigiCert SHA2 High Assurance Server CA.



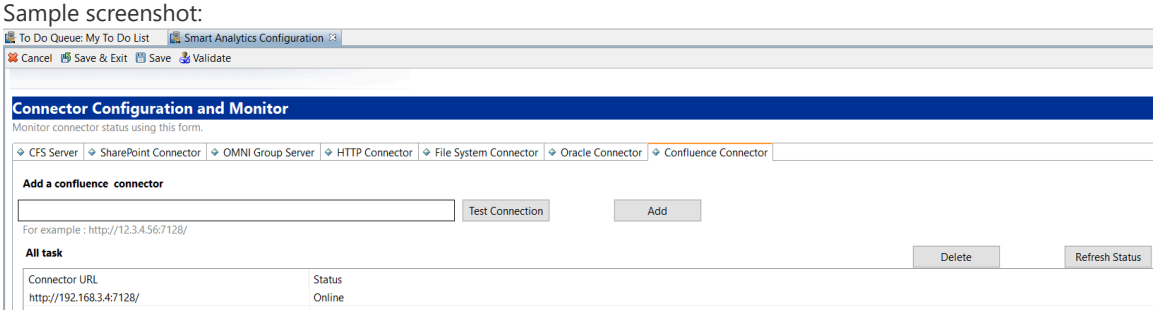
   First, export the DigiCert certificate. Select the DigiCert certificate, click **View Certificate**, and then click **Copy to File...** on the **Details** tab.
   Click **Next**, and then select the **Base-64 encoded X.509(.CER)** format to export the certificate to a file. For example, `01.cer`.
   Similarly, export the second certificate to a file. For example, `02.cer`.
4. Open the two certificate files in a text editor. Copy and append the content of `01.cer` to the content of `02.cer`, and then save the `02.cer` file.
   Now, the `02.cer` file is the certificate file you need. Rename it to `cloud.cer` if you are using `cloud.er` in the cfg file.
5. Copy `cloud.cer` to the ;*<Smart Analytics Installation>*/ConfluenceRESTConnector directory.

3. Restart your Smart Analytics Confluence Connector service.
4. Go to *<Smart Analytics installation>*/IDOL, and then open the `IDOLServer.cfg` file. Add the following line under the `[Server]` section.

```
HasConfluence=true
```

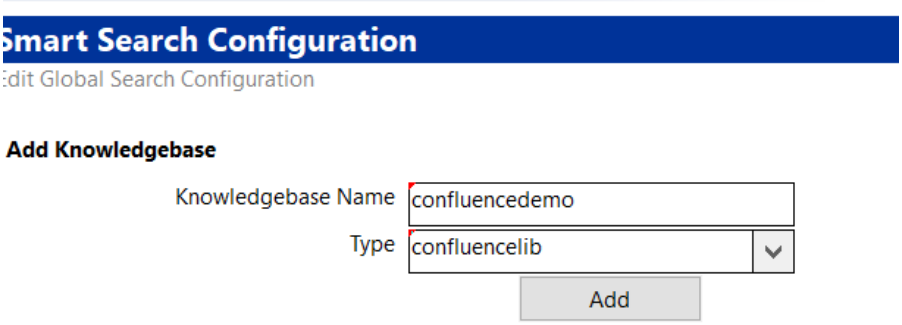
   Save your changes. Don't restart IDOL. Sign out and then sign in to Service Manager again.

5. Sign in to Service Manager, and then click **System Administration** > **Ongoing Maintenance** > **Smart Analytics** > **Smart Search**. The Smart Search configuration page opens.
6. Click the **Connector Configuration** link to open the connector configuration page.
7. Go to the **Confluence Connector** tab, perform the following actions:
    - **Add a Confluence connector**: Type a configured Confluence connector URL here. The format is http://<*Confluence Connector server address*>:<*Confluence Connector server port*>. For example, http://192.168.3.4:7128/. You can click **Test connection** to test the URL connection status, and then click **Add** to add this URL to the current list.
      Sample screenshot:

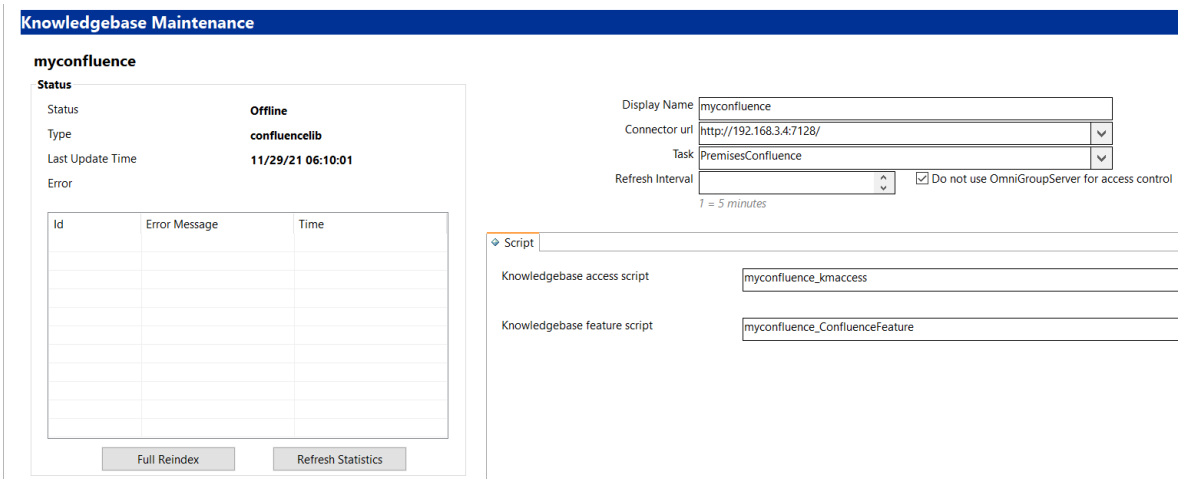


> **Note**
>
> Note
>
> - The "/" at the end of the URL is mandatory.
> - Make sure the status for Confluence is online.
> - To get the URL information for the Confluence connector you have configured, check the <*Smart Analytics Installation*>/ConfluenceRESTConnector/ConfluenceRESTConnector.cfg configuration file.

    - **Delete**: Select a Confluence connector URL and then click this button to delete it from the list.
    - **Refresh Status**: Click to refresh the status of the URL list.
8. Do the following to add a confluencelib library for the Confluence connector:
    1. Go to Smart Search Configuration page.
    2. Specify Knowledgebase Name,
    3. Select **confluencelib** for **Type**
    4. Click **Add**. The Knowledgebase Maintenance page opens.



9. Specify **Connector** and **Task**, and then select the **Do not use OmniGroupServer for access control** checkbox.



For cloud Confluence, you must select the **Do not use OmniGroupServer for access control** checkbox for access control.

For on-premises confluence, you can unselect the **Do not use OmniGroupServer for access control** checkbox. This implies that you will use OmniGroupServer for access control. The system reads user information from OmniGroupServer and only allows these users to search for documents. If you unselect select this checkbox, you must perform the following steps:

1. Change `MappedSecurity=true` in the Confluence connector configuration file.
2. Install and configure OmniGroupServer by following similar procedures for the Sharepoint connector.
3. Add GroupServer configuration to the Confluence connector configuration file. Refer to the Confluence REST Connector document for detailed instructions.
4. Install the security plugin for on-premises Confluence. You can find the plugin at `<Smart Analytics Installation>/ ConfluenceRESTConnector/MicroFocusConfluenceConnectorSecurityPlugin.jar`.

10. Click **Save**.
11. Click **Full Reindex** and **Refresh Status**.
12. You can perform the search when the status changes to Indexing and Doc Count for this library is greater than 1.

3.1.5. Configure Smart Search

User Role: System Administrator

Note: As of Service Manager version 9.6x, Smart Search can be used without a Smart Analytics license. However, to enable Smart Search without the need for a Smart Analytics license on Service Manager version 9.5x and 9.41 PD, you must apply the appropriate unload files based on the applications version of Service Manager. These unload files are available at ITOM Marketplace. See *Step 2: Apply unload files to Service Manager* in Set up SMA-SM for more information about these unload files.

To configure Smart Search, follow these steps:

1. From the System Navigator, click **System Status** , click **Start Scheduler**, and then click **KMReindex** to start the KMReindex process.

   Note
   Note Make sure that you start only one KMReindex process in the same Service Manager cluster.

2. From the System Navigator, click **System Administration** > **Ongoing Maintenance** > **Smart Analytics** > **Smart Search**.
   The **Smart Search Configuration** page opens.
3. From the **Add Knowledgebase** section, specify the following settings as needed:
   - **Knowledgebase name**: Specifies the name of the library you want to add.
   - **Knowledgebase type**: Specifies the type of the library you want to add.
   - **Add**: Adds the library you specified as a new knowledgebase.

Once you click the **Add** button, the new IDOL knowledgebase record is added and a new knowledgebase maintenance page is displayed. Configure the required knowledgebase information on the maintenance page. For detailed information about different types and configuration steps of knowledgebases, see Manage Smart Search Knowledgebases.

4. From the **Environment Configuration** section, modify the following settings as needed:
   - **Expiry days**: Smart Search removes the data that was indexed longer than the setting in this field from search.
   - **Assign the default knowledge view group to all operators**: If this option is selected, the default knowledge view group is assigned to all operators.
   - **Connector configuration**: Click to configure and monitor the status of the connectors and servers. For detailed information, see Configure connectors.
5. When you first install the IDOL search engine after having used the Solr search engine, you need to do the following: Check libraries in the **Current Knowledgebase List**. From the kmsearchsecurity script of each library, delete the code which you updated to enable Mandanten Security for libraries when using Solr, and then replace the deleted code with the following:

```
function getSecurityInfo(user, record)
{
return "";
}
```

Note

Note

- Starting from Service Manager 9.41, Mandanten security control is introduced as a built-in feature in Smart Search. Users no longer need to implement it manually by updating the KM search security script. If you keep the KM Search security script which was updated for Mandanten security, the built-in Mandanten control in Smart Search doesn't work.
- Smart Search and HTA don't support array type of mandanten in this release.

6. Click **Save** to save your modification.
7. From the **Smart Search Configuration** page, select the knowledagebase libraries that you want to index, and then click **Full Reindex**.
   For the libraries in the "sclib" type, you can configure additional settings such as scheduling the full re-index or changing the maximum worker threads before you start indexing. For more information, see Index Smart Search knowledgebases and Manage Smart Search Knowledgebases.
   When you click the **Full Reindex** button, the IDOL search engine performs a full re-index of the selected libraries. However, because indexing runs as a background process, the search engine doesn't begin indexing until the specified refresh interval is reached.

Tip

Tip To perform a full re-index on large-scale knowledgebases, see Perform indexing of large-scale knowledgebases.

8. Click the **Refresh Status** button to view the index status of the knowledgebases.

Tip

Tip To modify settings for any current knowledgebase, click the respective link from the list. For detailed information about knowledgebase settings, see Manage Smart Search Knowledgebases.

9. Log off and then log onto Service Manager for the changes to work.
10. Perform a Smart Search to verify your configuration.

If you encounter session time out or long response time when performing the first Smart Search after indexing, set a higher value for the `TermCachePersistentKB` parameter in the `[TermCache]` section for all content server configuration files(content<N>.cfg) and then restart your content servers.

> **Note**
>
> Note The default value for this parameter is 512000. You can configure the parameter value according to system memory availability and desired querying speed. For example, if you selected All-in-one install template when deploying Smart Analytics, you can keep the default value unchanged. If you deployed your content servers on multiple machines, it is recommended you modify the value to 2097152.

3.1.6. Add Smart Analytics capability word for power users

User Role: Administrator

To enable power users such as Service Desk Agent or Problem Coordinator to use the Smart Analytics features, you need to add the "idol.assistant" capability word to their operator records. The operators with this capability word can see Smart Analytics menus and use these features.

> **Note**
>
> Note ESS self-service users are able to submit Smart Request records after you enable SM Smart Analytics. No additional capability word is needed.

To add the "idol.assistant" capability word to an operator record, follow these steps:

1. From the System Navigator, click **System Administration** > **Ongoing Maintenance** > **Operators**.
2. Enter or select your search criteria, and then click **Search**.
3. Select an operator from the record list to view the operator record.
4. Click the **Startup** tab.
5. Add idol.assistant in the **Execute Capabilities** section.

3.2. Smart Analytics administrator tasks

The following sections describe tasks for a Smart Analytics administrator:

Work with Smart Search

Validate Smart Analytics configurations

Add a content server for Smart Ticket

Configure TSL/SSL for two-way authentication

Get ports occupied by Smart Analytics

Use Smart Analytics Assistant

Check Smart Analytics log files

Manage Smart Analytics background schedule in SMIS

Back up and restore content data

Back up and restore connector data

Back up and restore category data

Perform indexing of large-scale knowledgebases

Encrypt Smart Analytics index data

Modify Stop Words for IDOL search engine

Transfer Smart Analytics intelligence between systems

Set stop phrases for Hot Topic Analytics

Configure user assess to stop phrases for Hot Topic Analytics

Configure asynchronous analysis

Work with Smart Ticket tuning

3.2.1. Work with Smart Search

To work with Smart Search, refer to the following topics:

Manage Smart Search Knowledgebases

Index Smart Search knowledgebases

View index job status

Customize Smart Search actions and settings

Customize context-aware search

Add a content server for Smart Search

Remove a content server for Smart Search

Modify Smart Search content servers and weight for data

3.2.1.1. Manage Smart Search Knowledgebases

User Role: Administrator

You can add five types of Smart Search knowledgebases to Smart Analytics: an sclib knowledgebase, an fsyslib knowledgebase, a weblib knowledgebase, an splib knowledgebase, and an okmlib knowledgebase. An sclib knowledgebase is created from a table in Service Manager. For example, the out-of-the-box Incident_Library uses the probsummary table. A weblib knowledgebase is created by using web crawling to browse and index an external web site. The system creates an fsyslib knowledgebase when it crawls a file system. An splib knowledgebase is created to browse and index a SharePoint connector. An okmlib knowledgebase is created to browse and index an Oracle connector.

Add a new Smart Search knowledgebase

To add a new Smart Search knowledgebase, follow these steps:

1. From the System Navigator, click **System Administration** > **Ongoing Maintenance** > **Smart Analytics** > **Smart Search**.
   The **Smart Search Configuration** page opens.
2. From the **Add Knowledgebase** section, type a unique name for the new Smart Search knowledgebase in the Knowledgebase name field (required).
3. Select the knowledgebase type from the Type field drop-down list.
4. Click **Add**. The new IDOL knowledgebase record is added and a new knowledgebase maintenance page is displayed.
5. Configure the required knowledgebase information. The field availability varies for different knowledgebase types.

Field	Description
Display Name	Types a display name for the IDOL knowledgebase.
Refresh Interval	Displays the number of interval unit used to update the selected knowledgebase index. One interval unit is 5 minutes. Default number of interval is 1. You may increase the interval, which extends time delay between updates, by increasing this number. Setting this number to 0 (zero) disables updates to the index. To re-start indexing, reset the interval to a value greater than zero.
Connector (not for sclib)	Selects the specified connector from the configured connectors according to the knowledgebase type: ○ weblib: Web connector ○ splib: SharePoint connector ○ fsylib: File System connector ○ okmlib: Oracle connector The connector selected here should be the be consistent with the configurations in the .cfg files of related remote connectors. For more information about configuring Smart Analytics connectors, see Configure connectors.
Task (not for sclib)	Selects the task that connects the Smart Analytics server and other data sources. The task selected here should be consistent with the configurations in the .cfg files of related remote connectors. For more information about Smart Analytics connectors, see Configure connectors.
Table Name (only for sclib)	The table that will be indexed. A valid Service Manager table is required.
Skip These Extensions (only for sclib)	A semicolon-separated list of file extensions that should not be indexed or extracted. Certain file types either can't be indexed, or provide no relevance. By providing these extensions, you can increase index performance. **Note** Sample gif;jpg without any spaces.
Document ID Field (only for sclib)	Every table in Service Manager has a unique ID field and this field identifies the field name of the ID field. The indexer uses this field to uniquely identify each document in the index. This is a required field for indexing a knowledgebase.
Index Attachments (only for sclib)	If the table being indexed has attachments, select this checkbox to have the attachments indexed.

Index Chat History (only for sclib)	If you have Microsoft Teams integrated, select this checkbox to index the conversation history in Teams. This will allow you to search for the keywords in the chat history by using Smart Search.
Max threads for full reindex (only for sclib)	Specifies the number of threads to process the records in the library for Smart Search indexing. The default value is 1, which means the system creates one back-end process (worker) to send the data in the library from Service Manager to Smart Analytics. **Note** For performance considerations, we recommend that the total number of threads for all the libraries is less than 20.
Full reindex batch size (only for sclib)	Specifies the number of documents that the SM server sends to IDOL at a time. The default value is 1000. If an out-of-memory issue occurs during indexing, you can reduce this value to a smaller size, such as 500 or 250.
Table Query (only for sclib)	Specifies a Service Manager style query to limit what records in the table are indexed. For example, a query to return only documents that are neither draft nor retired in the kmdocument table: status ~= "draft" and status ~="retired". A blank query indicates that all records will be indexed.

Status

Field	Description
Full Reindex:	When clicked, the search engine performs a full re-index of a knowledgebase. If the index doesn't exist in IDOL, an index will be created. If the index does exist, Smart Analytics replaces the existing database with the new index.
Refresh Statistics	When clicked, the search engine refreshes the statistics for this library.

Field Definitions (only for sclib)

Field	Description
Field Name	Specifies the field name in the Service Manager table to be included in the index.
Alias	Specifies the name that the field is to be indexed as.
Type	Indicates whether the field is a plain text string, a rich text string, or a date type. The indexer ignores HTML markup in rich text strings and indexes plain text strings completely. Rich text: When the field type is **Rich Text**, it means the field may contain HTML code. HTML tags will be removed during index.. Reference: Select **Reference** only when the current field is a CI logical name. Thus the Display Name of a CI can be indexed.
Display in Results	Defines what fields are available in the search results. Fields marked as "true" in this column are available to be included on a search results. The information for these field is displayed at the bottom of the search result.
Index weight	Defines the sequence of the search results which are displayed on the search hitlist. You can assign different index weight to adjust the display sequence. If the sort value of a field is assigned as level 4, the search results of this field is displayed at the top of the list. No index means that this field is not used for search. **Note** We recommend you to tune this field according to your own business use cases.
Match	By setting the field in this column to "true," the system indexes the field's content as an advanced search filter for Smart Search in this knowledgebase. The search engine searches this individual field with the Advanced Search filter. For example, if you set the Category field as true, this field is used as an advanced search filter. If you leave this value as empty, it applies the false value.
Sort	Defines the sequence of the search results which are displayed on the search hitlist. If you set multiple fields as hitlist, you can decide the display sequence by assigning different sort values for these fields. If the sort value of a field is assigned as 1, the search results of this field is displayed at the top of the list.
Data cleansing	Used to remove text which matches the data cleansing rules from the table before indexing.

Script

Field	Description
Knowledgebase access script	This script specifies the script the system uses to determine if a particular user has rights to access the knowledgebase. See the default script for detailed information.
Category index script	This script processes the document category so that the indexer can translate the document's category into a string that the search engine can use later to find the document based on the user's category access.
Search security script	This script is used to build and return a string of library-specific query values that were entered by the user under the tabs in the Advanced Search screen. Tailor this script when a knowledgebase has a tab in the Advanced Search screen and you wish to modify the fields available for Advanced Search.
Default Locale	Specifies the default language used by the search engine when searching and indexing. By default, the language code is English.

Language Settings (only for Catalog Library)
The settings only work when you enter values for both the default language and available languages.

Field	Description
Default Language	Only English is available.
Available Languages	You may configure non-English languages that users may set when log in to Service Portal or SM web client. The language settings affect the search results as below: ○ If a user logs in to Service Portal or SM web client with one of the languages configured in the available language list, when the user performs a search, ■ an English document is displayed if a document is only available in English. ■ a document in the user's login language is displayed if the document in available in English and the user's login language. ○ If a user logs in to Service Portal or SM web client with a language that's not configured in the available language list, when the user performs a search, ■ no document is displayed if a document is only available in English. ■ a document in the user's login language is displayed if the document in available in English and the user's login language. For example, if Spanish and German are configured in the available language list, after a full reindex process completes, when a user logs in to Service Portal or SM web client with language set as German and performs a search, English document is displayed if a document is available in English and Spanish.

6. Click **Save** to save your modifications and perform a full reindex.

Modify a current Smart Search knowledgebase

To modify a current Smart Search knowledgebase, follow these steps:

1. Click **System Administration** > **Ongoing Maintenance** > **Smart Analytics** > **Smart Search**. The **Smart Search Configuration** page opens. The current knowledgebase list is displayed on the **Smart Search Configuration** page.
2. Click a Smart Search knowledgebase to open the knowledgebase maintenance page.
3. Configure the required knowledgebase information. The field availability varies for different knowledgebase types.
4. Click **Save** to save your modifications.

Sample configuration of the Smart Search knowledgebase for survey

This section provides a configuration example of the Smart Search knowledgebase for survey.

Document ID Field:

number

Field Definitions tab:

Field Name	Alias	Type	Display In Results	Index Weight	Match	Sort	Data Cleansing
reference.number	reference_number	String	true	Level 1	false		false
comments	comments	String	false	Level 4	false		false
status	status	String	true	Default	true		false

Script tab:

Knowledgebase access script	Survey_Library_kmaccess
Category index script	Survey_Library_kmcategoryidxscript
Search security script	Survey_Library_kmsearchsecurity
Default Locale	English

3.2.1.2. Index Smart Search knowledgebases

User Role: Administrator

To index Smart Search knowledgebases, follow these steps:

1. Make sure that you have backed up your content data before performing a full reindex on the knowledgebases. For more information, see Back up content data.
2. Make sure that the KMReindex process is running in **System Status**.
3. From the System Navigator, click **System Administration** > **Ongoing Maintenance** > **Smart Analytics** > **Smart Search**.
   The **Smart Search Configuration** page is displayed.
4. (Only for the sclib libraries) Do one of the following to schedule the full reindex.
   - If you want to start indexing immediately, select the **Run full reindex immediately** checkbox.
   - If you want to start indexing based on a pre-defined schedule, clear the **Run full reindex immediately** checkbox, and then click the **Schedule Full Reindex** button to update the time period record:
     1. Add rules. For how to define a rule, see Create a time period.
     2. Approve the time period. For details, see Submit a time period for approval and Approve a time period.
        In this case, the full reindexing process only runs between the start time and the end time of each occurrence for the time period.

     > **Note**
     > Note
     > - When you create the time period, the total time for all the occurrences should be long enough to index the selected libraries based on the data column. If all the occurrences have finished but the system doesn't complete indexing, you can update the time period record to add more occurrences with more time.
     > - If two occurrences are overlapping, the system only runs the indexing process in the first occurrence and skips the second one.
5. Click the **Save** button.
6. From the **Current Knowledgebase List** section, select the knowledgebases that you want to index, and then click the **Full Reindex** button.

Tip

Tip Alternatively, you can click the knowledgebase name in the list to open the **Knowledgebase Maintenance** page of a knowledgebase, configure the index conditions as needed, and then click the **Full Reindex** button on the **Knowledgebase Maintenance** page.
On the **Knowledgebase Maintenance** page, you can also configure some advanced settings such as the number of threads for indexing the sclib library. For more information, see Manage Smart Search Knowledgebases.

Tip

Tip If you want to index multiple libraries at one time, we recommend that you select the libraries with similar data volume.

After you click the **Full Reindex** button, for the sclib libraries, the system creates one or multiple back-end processes (workers) that send the data in batches (jobs) from Service Manager to the Smart Analytics database for indexing, starting with the latest records.

7. To make sure that the data is successfully indexed to the Smart Analytics database, check the overall index status for each knowledgebase from the **Current Knowledgebase List** section of the **Smart Search Configuration** page.
   From the **Current Knowledgebase List** section, you can check the following information.

Field	Description
Knowledgebase Name	Specifies the name of the Smart Search knowledgebase. **Note** Note When the administrator adds a new library, users can only see this library available on the list after next login.
Type	Specifies the type of the Smart Search knowledgebase.
Display Name	Specifies the display name for the Smart Search knowledgebase. You can change this value from a knowledgebase details page.
Interval	Displays the current interval used to update the selected knowledgebase index. Each interval unit is 5 minutes (default). You can change this value from a knowledgebase details page.
Index Status	Displays the current index status of the library. A library has the following four index status: ○ Offline: the library is newly added. ○ Not started: the **Full Reindex** button is clicked for the library and the indexing is in the queue. If the indexing is not started for the library, users can still use Smart Search for this library based on the previous indexed result. ○ Indexing: the indexing for the library is ongoing. **Tip** Tip If you want to stop indexing before the process is finished, click the library link to open the **Knowledgebase Maintenance** page, and then click **More** > **Force Stop**. ○ Finished: the indexing for the library is finished.

Doc Count	Displays the current number of records in the library. The number of the records is affected by the replica settings. When a content server is disabled, the number may be incorrect before you redistribute the server data. In addition, the doc count keeps changing during indexing process or even after the Index Status is displayed as "Finished". For example, if the full reindex has completed but later you successfully perform indexing for some failed jobs, the Doc Count will increase. **Tip** Tip If "Doc Count" is empty, check the status of the Smart Search server to see if it is offline.
Thread Count	Displays the number of worker threads to process the records in the library.
Last Index Time	Displays the time when the library completes its last indexing.

Tip

Tip You can also check the index job status for the knowledgebase libraries in the "sclib" type. For more information, see View index job status.

3.2.1.3. View index job status

User Role: Administrator

You can check the status for each index job of a Smart Search knowledgebase library in the "sclib" type.

To view index job status for a Smart Search knowledgebase, follow these steps:

1. Click **System Administration** > **Ongoing Maintenance** > **Smart Analytics** > **Smart Search**.
   The **Smart Search Configuration** page is displayed.
2. From the **Current Knowledgebase List** section, click the knowledgebase library that's in the the "sclib" type. Then, the **Knowledge Maintenance** page is displayed.
3. In the **Status** section, click the **Jobs** tab, and then click the **Refresh Statistics** button. The following information is displayed.

Note

Note The **Job** tab is only available for the libraries in the "sclib" type, not for the external libraries.

Status	Description
Total	Indicates the number of total jobs that Smart Analytics creates to index this knowledgebase. One job can index a maximum of one thousand records.
Planned	Indicates the number of jobs that are created and queued in Service Manager.
Sending	Indicates the number of jobs that are sending data out to the Smart Analytics database from Service Manager.
Indexing	Indicates the number of jobs that have sent out data from Service Manager, and the data is being indexed by Smart Analytics.
Completed	Indicates the number of jobs that successfully index the knowledgebase data in the Smart Analytics database.
Failed	Indicates the number of jobs that aren't sent out from Service Manager, or fail to index the knowledgebase data in the Smart Analytics database. **Note** Note The Failed category also includes the jobs in the "Plan to Transfer" and "Transferred" statuses. That is, if you manually retry a failed job, the job will still remain in the Failed category.

4. Click the number area for the jobs in one status to view the detailed information for each job such as query condition, status, start time, end time, and failure message.
   If a job fails, you can perform a manual retry:
   - When the knowledgebase library is still in the progress of indexing, manually change the job status from "Failed" to "Planned" and save your change.
   - When the knowledgebase library has completed indexing, manually change the job status from "Failed" to "Plan to Transfer" and save your change. When the retry completes, the job status changes to "Transferred", which means the recods previously in the failed job are added to the incremental indexing queue.

     Note
     Note To retry a failed job after the knowledgebase library has completed indexing, be sure that the KMReindex process is still running in **System Status**.

3.2.1.4. Customize Smart Search actions and settings

User Role: Administrator

Service Manager provides out-of-box pre-defined filters and actions for different libraries. The information of pre-defined filters and actions is saved in the smartsearchpreconditions and smartsearchaction tables. To perform Smart Search for a record, you can customize the Smart Search pre-defined filters to narrow down the library scope, and you can customize the Smart Search actions to pre-define the possible actions based on your search conditions and results.

Customize Smart Search actions

To customize Smart Search actions, follow these steps:

1. Click **Smart Analytics** > **Smart Search** > **More** > **Smart Search Action**.
   The **Smart Search Action** page opens.
2. Click **Search** to open all the available action records.
3. Double-click a record to view or change it. Service Manager displays each record in its appropriate form.
4. Update any field within the record.

   Note

   Note The integration with Operations Orchestration Flow in context-aware search is supported in Incident only.

5. If you make changes, click **Save**.

Customize Smart Search settings

To customize Smart Search settings, follow these steps:

1. Click **Smart Analytics** > **Smart Search** > **More** > **Tailoring**.
   The **Smart Search Settings** page opens.
2. Update the following fields as needed.

Field Name	Description
Automatically detect language in search terms	The default value is true. If you select this checkbox, Smart Search automatically detects the language to use. If you clear this checkbox, Smart Search performs a search based on the **DefaultLanguageType** parameter defined in the **content.cfg** file. Example usage: To improve the search accuracy in Japanese, you can uncheck this option and update the related configurations as instructed here.
Enable phrase queries	The default value is false. If you select this checkbox, Smart Search automatically adds quotations to the search criteria. Example usage: If you select this checkbox and then enter mobile phone in the search box, Smart Search will send out search request with the query text of `mobile phone` and then displays the results that exactly match the search criteria on top of the result list.
Enable instant search	The default value is true. If you select this checkbox, Smart Search is automatically triggered as you type in the search text box or the specified field of a record, for example the **Title** field of a km record. If you clear this checkbox, you need to press **Enter** to trigger a Smart Search.

Ignore certain special characters	The default value is false. If you select this checkbox, Smart Search interprets special elements as normal characters instead of a query syntax. These elements include asterisks (*), question mark (?), colon (:), double quotation marks ("), brackets, boolean, and proximity operators such as AND, NOT, OR, EOR, XOR, NEAR, DNEAR, WNEAR, BEFORE, and AFTER. Select this checkbox to disable wildcards, phrase queries, field restrictions and boolean operations. Example usage: If you select this checkbox and then enter mobile AND phone in the search box, Smart Search displays the search results which contain either `mobile` or `phone`. If you clear this checkbox and then enter mobile AND phone in the search box, Smart Search displays the search results which contain both `mobile` and `phone`. **Note** Note If you select this checkbox, the system ignores the setting of **Enable phrase queries** and doesn't add quotations to the query text.
Instant search delay time (ms)	The default value is 300. Specifies the time that Smart Search waits for when the typing stops, before it processes the search query.
Smart Search behavior for high-frequency queries	The default value is "wait for previous query to respond." Indicates the Smart Search reaction in the case of high-frequency queries.
Page size for drop-down list of library field filters	The default value is 100. Specifies the number of items that can be displayed in the drop-down list of a field when you click the **Set Filters** icon for a selected library in the general Smart Search page.
Search query timeout limit (ms):	The default value is 3000. Specifies the time duration after which a search query times out.
Maximum number of characters displayed in a Smart Search result	The default value is 300. Specifies the maximum number of characters that each general Smart Search result can contain.
Search result page size	The default value is 10. Specifies the number of search results that can be displayed.
Search string spell check character threshold	The default value is 50. The spell check is disabled when the number of characters in a search string exceeds this limit.

SuggestOnText character threshold	The default value is 400. The SuggestOnText function is triggered when the number of characters in a search string exceeds this limit.
Minimum search results relevance threshold	The default value is 30. The value must between 0 and 100. Negative numbers or any numbers that are greater than 100 aren't allowed. Specifies the minimum percentage of the relevance that the search results must have to the query.
The displayevent that opens records in the search results	Read-only. The value in the current setting is se.global_open. Indicates the displayevent that's triggered to open a record in the search results.
The displayevent that supports Smart Search actions	Read-only. The value in the current setting is se.global_context_aware. Indicates the displayevent that's triggered to support Smart Search actions. **Note** Note Only available when you select **Use displayevent (se.global_context_aware)** for **Smart Search action handler**.
Smart Search delta data recycle time (days)	The default value is 11. This field is used for content data backup and restore. You can set a time to clear data for the history index action. The history index action data will be deleted once it exists over the defined time.
Connector daily backup time	The default value is 19:00. Specifies the daily backup time of the connectors.
Connector daily backup folder	The default value is c:\BackupFolderName. Specifies the backup folder of the connectors.

3. Click **Save**.
4. Log off and then log onto Service Manager for the changes to work.

Note

Note To make **Enable phrase queries**, **Ignore certain special characters**, and **Minimum search results relevance threshold** effective for Smart Email search, you must follow these additional steps after *step 4*:

1. Click **Tailoring** > **Integration Manager**, and then restart the Smart Email integration instance.
2. Click **System Status**, and then restart the Emailout.startup, SmartEmail.Processor, and the Smart Email records.

3.2.1.5. Customize context-aware search

User Role: Administrator

Context-aware search enables users to view records and search results in the same window. Users can copy a solution from the search results to the working record or trigger automated tasks. Depending on configurations of **Smart Search action handler** in the Smart Search Tailoring page, you can enable the context-aware search icon inside a record by performing either of the following :

- Add the 9527 display option to the specific display screen.
- Customize the **Enable Condition** field in the Smart Search Predefined Filters page.

Customize Smart Search pre-defined filters[edit]

To customize Smart Search pre-defined filters, follow these steps:

1. Click **Smart Analytics** > **Smart Search** > **More** > **Predefined Filters**.
   The **Smart Search Predefined Filters** page opens.
2. Click **Search** to open all the available pre-defined filter records.
3. Double-click a record to view or change it. Service Manager displays each record in its appropriate form.
4. Update any field within the record.
5. If you make changes, click **Save**.

Note

Note Make sure that the filename record is already defined as a Knowledge Management Mapping Record. Otherwise, you must define the file name from **Knowledge Management** > **Configuration** > **Integration Mapping**.

Customize context-aware search related settings[edit]

To customize context-aware search related settings in Smart Search, follow these steps:

1. Click **Smart Analytics** > **Smart Search** > **More** > **Tailoring**.
   The **Smart Search Settings** page opens.
2. Update the following fields as needed.

Field Name	Description
Smart Search action handler	The default value is "Use displayoption (9527)." If you select the "Use displayoption (9527)" option, you need to configure the 9527 display option for each displayscreen of a module to enable the context-aware search icon inside a record. If you select the other option "Use displayevent (se.global_context_aware)", you can enable the context-aware search icon for all displayscreens of a record in a specified module by configuring the **Enable Condition** field in the Smart Search Predefined Filters page only.
Display the number of records that match filter conditions	The default value is true. Displaying count number can impact performance. Only disable counting when performance is low. If this checkbox is selected, you can see the following: ○ For context-aware search, the number of records that match filter conditions is displayed next to the selected knowledgebase libraries. ○ For Smart Search, the number of records that match filter conditions is displayed next to the selected knowledgebase libraries. Smart Search automatically filters the search results based on the field filters you select under the corresponding library. If this checkbox is cleared, you can see the following: ○ For context-aware search, the number of records that match filter conditions is not displayed next to the selected KM libraries. ○ For Smart Search, the number of records that match filter conditions is displayed next to the selected KM libraries only when you click the **Set Filters** icon on the right and is automatically hidden when you start a new search. When you select a field filter under a selected library, you need to manually filter the search results by clicking the **Search** button at the bottom of the field filters window.
Enable context-aware search in ESS	The default value is true. If you select this checkbox, context-aware search is enabled in self-service user view (ess.do).
Manually trigger context-aware search	The default value is false. If you select this checkbox, context-aware search is not automatically triggered as you type in the specified field of a record, for example the **Title** field of a km record. You need to trigger context-aware search manually by clicking the Context-aware Search icon.

Maximum number of characters displayed in a context-aware search result	The default value is 200. Specifies the maximum number of characters that each context-aware search result can contain.
Maximum number of characters to display in expanded view	The default value is 600. Specifies the maximum number of characters that can be displayed in the expanded view of a context-aware search result.
Maximum number of characters to display in the field area:	The default value is 200. Specifies the maximum number of characters that can be displayed in the fields area at the bottom of a context-aware search result.
Maximum number of records whose context-aware search results can be stored:	The default value is 50. Specifies the maximum number of records for which Smart Search can store context-aware search results. When you switch between open records for which you used context-aware search, the respective search result for the current open record is retrieved and displayed on the right panel. If you switch from an open record for which you used context-aware search to an open record for which you did not use context-aware search, the context-aware search window is hidden.

3. If you make any changes, click **Save**.
4. Log off and then log onto Service Manager for the changes to work.

Contents

- 1 Customize Smart Search pre-defined filters
- 2 Customize context-aware search related settings

3.2.1.6. Add a content server for Smart Search

User Role: System administrator

The initial setting deploys one content server at the main level, and two content servers at the Smart Search level. If the default or existing smart search content servers aren't enough for the indexed data, you may need to add additional content server(s).

To add a content server for Smart Search, follow these steps:

1. Install a content server by using Smart Analytics installer. For details about installing a content server, see Install Smart Analytics on Windows or Install Smart Analytics on Linux.

   Note

   Note
   Make sure the content server configured the Service Manager server and Smart Search proxy server as its client. These configuration are set in content's configuration file.
   Make sure Smart Search proxy also configured the new content server as its client (in the `level2proxy\IDOLServer.cfg` file). Service Manager Server uses load balance, therefore, you need to specify all slave Service Manager addresses in these configuration files and corresponding sections.
   The following are configuration items about clients:

   ```
   [Service]
   ServiceStatusClients=127.0.0.1, SmarSearchProxyAddress,
   SMServerAddress,SmartAnalytivsServerAddress.*
   ServiceControlClients=127.0.0.1, SmarSearchProxyAddress,
   SMServerAddress,SmartAnalytivsServerAddress.*
   [AdminRole]
   Clients=127.0.0.1,SmarSearchProxyAddress,SMServerAddress,SmartAnalyticsServerAddress.*
   [QueryRole]
   Clients=127.0.0.1,SmarSearchProxyAddress,SMServerAddress,SmartAnalyticsServerAddress.*
   [IndexRole]
   Clients=127.0.0.1,SmarSearchProxyAddress,SMServerAddress,SmartAnalyticsServerAddress.*
   ```

2. Make sure the Smart Search proxy server has configured the new content servers as its clients. You need to configure the IPv4 and IPv6 addresses for the new content server in the `level2proxy\IDOLServer.cfg` file for the following items.
   `ServiceStatusClients`

   `ServiceControlClients`

   `[AdminRole]...Clients`

   `[QueryRole]...Clients`

   `[IndexRole]...Clients`
   You may need to restart the Smart Search proxy server after you update the configuration file.

   Note

   Note
   If you use FQDN instead of IP addresses, make sure that your DNS is correctly configured to revolve the FQDN.

3. After you start a new content server, go to the Smart Analytics Assistant page to add it to the Smart Search Content by clicking the **Add** link in the **Smart Search Content** section. For details about using Smart Analytics Assistant, see Use Smart Analytics Assistant.
4. Repeat the previous steps if you want to add more content servers.

5. Click the **Redistribute** link to balance data distribution. This step is **mandatory** to mark the new content server active.

> **Note**
> Note
> We recommend that you back up your existing content servers before you run the Redistribute action. It may take a long time to redistribute your data among all Smart Search content servers.

3.2.1.7. Remove a content server for Smart Search

User Role: Administrator

To remove a content server for Smart Search, follow these steps:

1. Update the data weight of the content server that you want to remove to 0.
   Action example:
   `http://<Smart Search Host>:<dih port>/action=EngineManagement&EngineAction=Edit&ID=2&Weight=0`
2. Redistribute data among the remaining contents servers.
   Action example:
   `http://<Smart Search Host>:<dih port>/DREREDISTRIBUTE`
3. Remove the content server.
   Action example:
   `action=EngineManagement&EngineAction=Remove&ID=2`

3.2.1.8. Modify Smart Search content servers and weight for data

User Role: Administrator

Smart Search uses **consistent hashing** for distribution of data. In consistent hashing mode, you can add, remove, or change the weight of child servers in your DIH without having to reindex all your content.

In this mode, DIH distributes documents among a large, fixed number of virtual nodes (4096), and then assigns these nodes to one or more child servers. When you change the number or weight of the child servers, you can use the **DREREDISTRIBUTE** index action to redistribute the virtual nodes between the new set of servers without reindexing all the data.

In consistent hashing mode, all content is distributed in virtual nodes, which the DIH assigns to one or more child servers. When you run the **DREREDISTRIBUTE** index action, DIH checks whether the child server architecture has changed. If there is a change, DIH automatically exports and indexes the content in virtual nodes to redistribute your data evenly between the available child servers.

You can check current Smart Search contents and their ratio of data (including the information of data weight and vnode) by sending an action to the Smart Search proxy DIH server (20080 by default):

Action example:

`http://<Smart Search Host>:<dih port>/action=enginemanagement&engineaction=showstatus`

To modify the Smart Search content servers and the weight for data distribution, use the **EngineMangement&EngineAction=Edit** action.

- Change the weight of data distribution.
  Action example:
  `http://<Smart Search Host>:<dih port>/action=EngineManagement&EngineAction=Edit&ID=2&Weight=2`
- Update only this content server (receives document updates but doesn't receive new documents):
  Action example:
  `http://<Smart Search Host>:<dih port>/action=EngineManagement&EngineAction=Edit&ID=2&UpdateOnly=true`

- Disable this content server (mark it as offline):

  > **Note**
  > Note When a child server is disabled, the DIH continues to assign documents to it, and queue up index actions, but it doesn't attempt to send them until the child server is enabled again.

  Action example:
  `http://<Smart Search Host>:<dih port>/action=EngineManagement&EngineAction=Edit&ID=2&Disabled=true`

3.2.2. Validate Smart Analytics configurations

User Role: Administrator

Starting from Service Manager 9.51, you can validate your Smart Analytics configurations when the setup is finished, to make sure that Smart Analytics works properly.

To validate Smart Analytics configurations, follow these steps:

1. Click **System Administration** > **Ongoing Maintenance** > **Smart Analytics** > **Configuration**.
   The **Smart Analytics Configuration** page opens.
2. From the toolbar, click the **Validate** button.
3. Follow the on-screen instructions to fix issues if the validation fails.

3.2.3. Add a content server for Smart Ticket

User Role: System Administrator

The initial setting deploys one content server at the main level, and two content servers at the Smart Search level. It's advisable to store no more than 3 million records for each content server. You may need to add one or more content servers if your data size is huge for Smart Ticket.

To add a content server for Smart Ticket, follow these steps:

1. Install a content server by using the Smart Analytics installer. For details about installing a content server, see Install Smart Analytics on Windows or Install Smart Analytics on Linux.
2. After you start the new content server, add it to the Smart Analytics main server by editing the `<Smart Analytics Installation>/IDOL/IDOLServer.cfg` file as follows:
   a. Increase the value of **Number** by 1 in the **DistributionIDOLServers** section:

   ```
   [DistributionIDOLServers]
   Number=3
   ```

   b. Add a new section of **[IDOLServer*N*]**, and then replace `N` with the number of your server.

   ```
   [IDOLServerN]
   Name=ContentN
   Host=ContentHostAddress
   Port=ContentPort
   DistributeByFieldsValues=CONTENTN
   ```

   c. Map virtual databases to the new server as needed, and then replace `N` with the number of your server.

```
[VDB<x>]
DbName=my_database
Type=Combinator
MapsTo=N:my_database
```

Note

Note

Do not map virtual databases of one library to different content servers.

3. Restart the **OpenText SM Smart Analytics Main Server** service.

3.2.4. Configure TSL/SSL for two-way authentication

User Role: Administrator

TLS/SSL creates encrypted connections that allow private and sensitive information to be transmitted without the risk of eavesdropping, data tampering, or message forgery. We recommend setting up a TLS/SSL connection between Service Manager and the Smart Analytics main server, the Connector Framework Server (CFS)/connectors, or the Media Server. In the TLS/SSL connection, Service Manager acts as the client, and Smart Analytics acts as the server.

To do this, see the following steps for different scenarios.

Between Service Manager and Smart Analytics

To Configure TSL/SSL for two-way authentication between Service Manager and Smart Analytics, follow these steps as an example:

Step 1 Create certificates

Create a signed Service Manager server certificate and Smart Analytics certificate using the certificate generation toolkit. See work with the certificate generation toolkit.

After you generate the certificates, do the following:

1. Open the `smartanalytics.key` file.
2. Remove the redundant contents before the `-----BEGIN PRIVATE KEY-----` section.
3. Save this file as `exported_rsa.key`.
4. Open the `smartanalytics.crt` file. Save this file as `smartanalytics.pem`.
5. Open the `smrootca.crt` file. Save this file as `smrootca.pem`.

Step 2 Configure SM server

Configure the Service Manager server to use the server certificate and to trust the client certificate.

1. Copy the following files to the Service Manager server host and put them under the RUN directory:
   `\certs\cacerts`
   `\certs\trustedclients.p12`
   `\certs\smserver.p12`
2. Go to the RUN directory of the SM server, then add below configuration into `sm.ini` file:

```
sslConnector:1
ssl:1
ssl_reqClientAuth:2
sslEnabledProtocols:TLSv1.2,TLSv1.3
ciphers:TLS_AES_256_GCM_SHA384,TLS_AES_128_GCM_SHA256,TLS_DHE_RSA_WITH_AES_256_GCM_SHA384,TLS_ECDHE_
RSA_WITH_AES_256_GCM_SHA384,TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256,TLS_ECDHE_ECDSA_WITH_AES_256_GCM_S
HA384,TLS_ECDHE_ECDSA_WITH_AES_128_GCM_SHA256

keystoreFile:smserver.p12
keystorePass:serverkeystore
truststoreFile:cacerts
truststorePass:changeit
ssl_trustedClientsJKS:trustedclients.p12
ssl_trustedClientsPwd:trustedclients
```

The ciphers parameter in this example allows strong cipher suites only. If you remove this parameter, the system allows all cipher suites of TLSv1.2 and TLSv 1.3.
If you don't want to restrict access to the server to only those clients whose certificates are in a list of trusted clients, set the ssl_reqClientAuth parameter to 1, and you don't need to configure the ssl_trustedClientsJKS and ssl_trustedClientsPwd parameters.

3. Comment out the old value `sslConnector:0` save the `sm.ini` file.
4. Restart SM Server.

Step 3 Configure Smart Analytics components

Configure the Smart Analytics components to use the client certificate and to trust the server certificate.

1. Create a new folder named sslCertificate in the <*Smart Analytics Installation*> directory.
2. Copy the following files to the <*Smart Analytics Installation*>\sslCertificate folder on your Smart Analytics local machine.
   `certs\smartanalytics.pem`

   `certs\smrootca.pem`

   `exported_rsa.key`
3. Configure all content components to use the certificates by setting the <*Smart Analytics Installation*>\Content#\Content#.cfg file. Search the file for the sections listed below, and then configure the parameters under them. If a section already exists, update the section; otherwise add the section. The same rule applies for the parameters. When copying lines, be sure not to create duplicate sections or parameters. Note that the following example uses the TLS 1.3 protocol. You may change `SSLMethod` to `TLSV1.2` if required.
   ```
   [SSLOption1]
   SSLMethod=TLSV1.3
   SSLCertificate=<Smart Analytics Installation>\sslCertificate\smartanalytics.pem
   SSLPrivateKey=<Smart Analytics Installation>\sslCertificate\exported_rsa.key
   SSLCACertificate=<Smart Analytics Installation>\sslCertificate\smrootca.pem
   SSLCheckCertificate=TRUE
   [IndexServer]
   SSLConfig=SSLOption1
   [Server]
   SSLConfig=SSLOption1
   ```
   Note If a section (such as the `[IndexServer]` section) doesn't exist in the out-of-box configuration files, you need to add the section manually with the settings described in this help topic.
4. Configure the Smart Search proxy to use the certificates by setting the <*Smart Analytics Installation*>\level2proxy\IDOLServer.cfg file. Note that the following example uses the TLS 1.3 protocol. You may change `SSLMethod` to `TLSV1.2` if required.
   ```
   [SSLOption1]
   ```

```
SSLMethod=TLSV1.3
SSLCertificate=<Smart Analytics Installation>\sslCertificate\smartanalytics.pem
SSLPrivateKey=<Smart Analytics Installation>\sslCertificate\exported_rsa.key
SSLCACertificate=<Smart Analytics Installation>\sslCertificate\smrootca.pem
SSLCheckCertificate=TRUE
[IndexServer]
SSLConfig=SSLOption1

    [Service]

    SSLConfig=SSLOption1
[Server]
SSLConfig=SSLOption1
SSLIDOLComponents=TRUE
[IDOLServerN]
SSLConfig=SSLOption1
```

5. Configure the Smart Analytics main server to use the certificates by setting the *<Smart Analytics Installation>*\IDOL\IDOLServer.cfg file. Note that the following example uses the TLS 1.3 protocol. You may change `SSLMethod` to `TLSV1.2` if required.

```
[SSLOption1]
SSLMethod=TLSV1.3
SSLCertificate=<Smart Analytics Installation>\sslCertificate\smartanalytics.pem
SSLPrivateKey=<Smart Analytics Installation>\sslCertificate\exported_rsa.key
SSLCACertificate=<Smart Analytics Installation>\sslCertificate\smrootca.pem
SSLCheckCertificate=TRUE
[IndexServer]
SSLConfig=SSLOption1
[DataDRE]
SSLConfig=SSLOption1
[CatDRE]
SSLConfig=SSLOption1
[AgentDRE]
SSLConfig=SSLOption1
[Server]
SSLConfig=SSLOption1
SSLIDOLComponents=TRUE
[IDOLServerN]
SSLConfig=SSLOption1
[Agent]
SSLConfig=SSLOption1
```

6. Change the *<Smart Analytics Installation>*\IDOL\agentstore\agentstore.cfg file. Note that the following example uses the TLS 1.3 protocol. You may change `SSLMethod` to `TLSV1.2` if required.

```
[SSLOption1]
SSLMethod=TLSV1.3
SSLCertificate=<Smart Analytics Installation>\sslCertificate\smartanalytics.pem
SSLPrivateKey=<Smart Analytics Installation>\sslCertificate\exported_rsa.key
SSLCACertificate=<Smart Analytics Installation>\sslCertificate\smrootca.pem
SSLCheckCertificate=TRUE
[IndexServer]
SSLConfig=SSLOption1
[Server]
SSLConfig=SSLOption1
SSLIDOLComponents=true
```

7. Configure the Connector Framework Server (CFS) to use the certificates by setting the *<Smart Analytics Installation>*\CFS\CFS.cfg file. Note that the following example uses the TLS 1.3 protocol. You may change `SSLMethod` to `TLSV1.2` if required.
   `[SSLOption1]`
   `SSLMethod=TLSV1.3`
   `SSLCertificate=<Smart Analytics Installation>\sslCertificate\smartanalytics.pem`
   `SSLPrivateKey=<Smart Analytics Installation>\sslCertificate\exported_rsa.key`
   `SSLCACertificate=<Smart Analytics Installation>\sslCertificate\smrootca.pem`
   `SSLCheckCertificate=TRUE`
   `[Server]`
   `//to make CFS ACI port ssl encrypted.`
   `SSLConfig=SSLOption1`
   `[idolserver]`
   `SSLConfig=SSLOption1`
8. Restart the corresponding services for the Smart Analytics components that you modified.

Between Service Manager and CFS/connectors

To Configure TSL/SSL for two-way authentication between Service Manager and the CFS/connectors, follow these steps as an example:

Step 1 Create certificates

Make sure you have performed the above steps in **Between Service Manager and Smart Analytics > Step 1 Create certificates**.

Step 2 Configure SM server

Configure the Service Manager server to use the server certificate and to trust the client certificate.

1. Copy the following files to the Service Manager server host and put them under the RUN directory:
   `\certs\cacerts`
   `\certs\trustedclients.p12`
   `\certs\smserver.p12`
2. Go to the RUN directory of the SM server, then add below configuration into `sm.ini` file:

   ```
   sslConnector:1
   ssl:1
   ssl_reqClientAuth:2
   sslEnabledProtocols:TLSv1.2,TLSv1.3
   ciphers:TLS_AES_256_GCM_SHA384,TLS_AES_128_GCM_SHA256,TLS_DHE_RSA_WITH_AES_256_GCM_SHA384,TLS_ECDHE_
   RSA_WITH_AES_256_GCM_SHA384,TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256,TLS_ECDHE_ECDSA_WITH_AES_256_GCM_S
   HA384,TLS_ECDHE_ECDSA_WITH_AES_128_GCM_SHA256

   keystoreFile:smserver.p12
   keystorePass:serverkeystore
   truststoreFile:cacerts
   truststorePass:changeit
   ssl_trustedClientsJKS:trustedclients.p12
   ssl_trustedClientsPwd:trustedclients
   ```

   The ciphers parameter in this example allows strong cipher suites only. If you remove this parameter, the system allows all cipher suites of TLSv1.2 and TLSv1.3.
   If you don't want to restrict access to the server to only those clients whose certificates are in a list of trusted clients, set the ssl_reqClientAuth parameter to 1, and you don't need to configure the ssl_trustedClientsJKS and ssl_trustedClientsPwd parameters.
3. Comment out the old value `sslConnector:0` save the `sm.ini` file.
4. Restart SM Server.

Step 3 Configure Smart Analytics CFS/connectors

Configure the Smart Analytics Connector Framework Server (CFS) or connectors to use the client certificate and to trust the server certificate.

1. Copy the following files to the *<Smart Analytics Installation>*\sslCertificate folder on your Smart Analytics local machine:
   `certs\smartanalytics.pem`

   `certs\smrootca.pem`

   `exported_rsa.key`
2. Configure the Connector Framework Server (CFS) to use the certificates by setting the *<Smart Analytics Installation>*\CFS\CFS.cfg file. Note that the following example uses the TLS 1.3 protocol. You may change `SSLMethod` to `TLSV1.2` if required.
   ```
   [SSLOption1]
   SSLMethod=TLSV1.3
   SSLCertificate=<Smart Analytics Installation>\sslCertificate\smartanalytics.pem
   SSLPrivateKey=<Smart Analytics Installation>\sslCertificate\exported_rsa.key
   SSLCACertificate=<Smart Analytics Installation>\sslCertificate\smrootca.pem
   SSLCheckCertificate=TRUE
   ```
3. ```
   [Server]
   //to make CFS ACI port ssl encrypted.
   SSLConfig=SSLOption1
   [IdolServer]
   SSLConfig=SSLOption1
   ```
4. Configure the connectors to use the certificates by setting the *<connector>*.cfg file. Note that the following example uses the TLS 1.3 protocol. You may change `SSLMethod` to `TLSV1.2` if required.
   ```
   [SSLOption1]
   SSLMethod=TLSV1.3
   SSLCertificate=<Smart Analytics Installation>\sslCertificate\smartanalytics.pem
   SSLPrivateKey=<Smart Analytics Installation>\sslCertificate\exported_rsa.key
   SSLCACertificate=<Smart Analytics Installation>\sslCertificate\mycacert .pem
   SSLCheckCertificate=TRUE
   [Server]
   SSLConfig=SSLOption1
   [Ingestion]
   //If CFS ACI port is ssl encrypted
   IngestSSLConfig=SSLOption1
   ```
5. Restart the corresponding CFS and connector services.

Between Service Manager and Media Server

To Configure TSL/SSL for two-way authentication between Service Manager and the Media Server, follow these steps as an example.

Step 1 Create certificates

Make sure you have performed the above steps in **Between Service Manager and Smart Analytics > Step 1 Create certificates**.

Step 2 Configure SM server

Configure the Service Manager server to use the server certificate and to trust the client certificate.

1. Copy the following files to the server host and put them under the RUN directory:
   `\certs\cacerts`

   `\certs\trustedclients.p12`
   ```

`\certs\smserver.p12`

2. Go to the RUN directory of the SM server, then add below configuration into `sm.ini` file:

```
sslConnector:1
ssl:1
ssl_reqClientAuth:2
sslEnabledProtocols:TLSv1.2,TLSv1.3
ciphers:TLS_AES_256_GCM_SHA384,TLS_AES_128_GCM_SHA256,TLS_DHE_RSA_WITH_AES_256_GCM_SHA384,TLS_ECDHE_
RSA_WITH_AES_256_GCM_SHA384,TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256,TLS_ECDHE_ECDSA_WITH_AES_256_GCM_S
HA384,TLS_ECDHE_ECDSA_WITH_AES_128_GCM_SHA256

keystoreFile:smserver.p12
keystorePass:serverkeystore
truststoreFile:cacerts
truststorePass:changeit
ssl_trustedClientsJKS:trustedclients.p12
ssl_trustedClientsPwd:trustedclients
```

   The ciphers parameter in this example allows strong cipher suites only. If you remove this parameter, the system allows all cipher suites of TLSv1.2 and TLSv1.3.
   If you don't want to restrict access to the server to only those clients whose certificates are in a list of trusted clients, set the ssl_reqClientAuth parameter to 1, and you don't need to configure the ssl_trustedClientsJKS and ssl_trustedClientsPwd parameters.
3. Comment out the old value `sslConnector:0` save the `sm.ini` file.
4. Restart SM Server.

Step 3 Configure Smart Analytics Media Server

Configure the Smart Analytics Media Server to use the client certificate and to trust the server certificate.

1. Copy the following files to the <*Smart Analytics Installation*>\sslCertificate folder on your Smart Analytics local machine:
   `certs\smartanalytics.pem`

   `certs\smrootca.pem`

   `exported_rsa.key`
2. Configure the Media Server to use the certificates by setting the <*Smart Analytics Installation*>\MediaServer1\MediaServer1.cfg file. Note that the following example uses the TLS 1.3 protocol. You may change `SSLMethod` to `TLSV1.2` if required.
   `[SSLOption1]`
   `SSLMethod=TLSV1.3`
   `SSLCertificate=<Smart Analytics Installation>\sslCertificate\smartanalytics.pem`
   `SSLPrivateKey=<Smart Analytics Installation>\sslCertificate\exported_rsa.key`
   `SSLCACertificate=<Smart Analytics Installation>\sslCertificate\smrootca.pem`
   `SSLCheckCertificate=TRUE`
3. `[Server]`
   `SSLConfig=SSLOption1`
4. Restart the Media Server service.

3.2.5. Get ports occupied by Smart Analytics

To get ports occupied by Smart Analytics, follow these steps:

Systems based on Windows

Using PowerShell

1. Run the script shown below to obtain the process IDs of `<Smart Analytics>IDOL\IDOLServer.exe` and `<Smart Analytics>\level2proxy\IDOLServer.exe`.
   `gwmi win32_process -filter "Name LIKE '%IDOL%'"|select ProcessID,Name|Sort-Object ProcessID | ft -AutoSize`
   The following information appears on the console:

   ```
   ProcessID             Name
   ---------             ----
   <PID1>                IDOLServer.exe
   <PID2>                IDOLSE~1.EXE
   ```

2. Run the script shown below to find all process IDs for components called by `<Smart Analytics>IDOL\IDOLServer.exe` and `<Smart Analytics>\level2proxy\IDOLServer.exe`.
   `gwmi win32_process -filter "ParentProcessID=<PID>"|select ProcessID,ParentProcessID,Name |where {$_.ParentProcessID -eq <PID>}|Sort-Object ProcessID | ft -AutoSize`

   Note
   Note `<PID>` here is the process ID obtained in Step 1.

   The following information appears on the console:

   ```
   ProcessID         ParentProcessID         Name
   ---------         ---------------         ----
   <PID1>            <PID>                   view.exe
   <PID2>            <PID>                   agentstore.exe
   <PID3>            <PID>                   dah.exe
   <PID4>            <PID>                   community.exe
   <PID5>            <PID>                   dih.exe
   <PID6>            <PID>                   category.exe
   <PID7>            <PID>                   conhost.exe
   ```

3. Run: `netstat -anot | findstr <PID>` to get ports occupied by the components found in step 2:

   Note
   Note `<PID>` here is the process ID you got from both step 1 and step 2.

   The following information appears on the console:

   ```
   TCP    0.0.0.0:<PORT1>        0.0.0.0:0        LISTENING       <PID>      InHost
   TCP    0.0.0.0:<PORT2>        0.0.0.0:0        LISTENING       <PID>      InHost
   TCP    0.0.0.0:<PORT3>        0.0.0.0:0        LISTENING       <PID>      InHost
   TCP    [::]:<PORT1>           [::]:0           LISTENING       <PID>      InHost
   TCP    [::]:<PORT2>           [::]:0           LISTENING       <PID>      InHost
   TCP    [::]:<PORT3>           [::]:0           LISTENING       <PID>      InHost
   ```

   The ports listed here are used by Smart Analytics.

If PowerShell isn't installed, you can find all the process IDs from Task Manager.

Systems based on Linux

Using Bash

Run: `ps -ef | grep IDOL [OR: Content, Connector, etc.]`

You will get a list similar to the following:

```
<USER>     <PID1>     1  0 14:06 pts/0    00:00:00
/opt/MicroFocus/ServiceManager9.xx/SmartAnalytics/level2proxy/dah/dah.exe -idolcomponent -configfile
/opt/MicroFocus/ServiceManager9.xx/SmartAnalytics/level2proxy/IDOLServer.cfg -aciport <PORT1> -
serviceport <PORT2>
<USER>     <PID2>     1  0 14:06 pts/0    00:00:01
/opt/MicroFocus/ServiceManager9.xx/SmartAnalytics/level2proxy/dih/dih.exe -idolcomponent -configfile
/opt/MicroFocus/ServiceManager9.xx/SmartAnalytics/level2proxy/IDOLServer.cfg -aciport <PORT3>  -indexport
<PORT2>  -serviceport <PORT3>
```

All the ports listed here are used by Smart Analytics.

3.2.6. Use Smart Analytics Assistant

Applies to User Role: System administrator

Smart Analytics Assistant (SAA) is a built-in tool that enables administrators to interactively perform IDOL administrative actions in Smart Analytics. For example, you can use this tool for content server maintenance, system status checking, and troubleshooting.

In the SAA interface, there is a command line to enable the administrator to send IDOL actions to IDOL components. The most frequently used actions are included in the drop-down list. You can get more actions and information from IDOL reference guides.

Service Manager provides five blocks to fulfill different functionality. For each component in all blocks, you can click the underlined name to get detail status.

Block	Description
Main Proxy	Top level proxy status that shows the component name, host, port, and status. If any component is down, you need to perform further investigation.
Smart Ticket and Hot Topic content	By default, one content store is used for Smart Ticket and Hot Topic Analytics. It must always be online. If you need more content stores to host more databases, you can click the “Add” link. Each content store to be added to the system has to be installed and running on another machine.
Smart Search Proxy	The second level proxy that's dedicated to Smart Search. Only DAH and DIH are included in this proxy.
Smart Search Content	By default, two content stores are used for Smart Search. You can add more content stores by clicking the “Add” link. Each content store to be added to the system has to be installed and running on another machine.
Media Server	By default, one media server is installed for OCR purposes. If you want to improve the performance, you can add more media servers. Each media server to be added to the system has to be installed and running on another machine. If you want to use two or more media servers, you need to install a media proxy server too.

To use Smart Analytics Assistant, follow these steps:

1. Type saa in the Service Manager command line, and then press Enter. The Smart Analytics Assistant interface is launched.

2. Use Smart Analytics Assistant to perform the administrative tasks . For example, you can select an action from the drop-down list.

> **Note**
> Note
> Replace <*Host*>, <*port*>, <*variable_value*> in the following action examples with the corresponding values that you can find from the corresponding status blocks.

   - Backup Component
     Creates a backup that can be used to restore the component's state. You can use this action for the Content, Category, Agent and Community components, but you must send the action to the component ACI port, rather than to the IDOL Proxy port. The backup file is stored in the path that you specified.
     For example, to backup category,
     `http://<Host>:<CategoryPort>/action=BackupServer&path=c:\backup_category`

     > **Note**
     > Note
     > By default, the Smart Analytics server CategoryPort is 9020.

   - Backup Database
     Exports all the index documents of a database from the Smart Analytics content server to a series of compressed files in the defined backup directory. This action is appropriate for backing up an individual database. If you want to back up all databases on a content server, use the action Backup Component as mentioned above.
     Action example:
     `http://<Host>:<indexPort>/DREEXPORTIDX?filename=c:/BackupFolderName/FilePrefix&DatabaseMatch=<Database_name>&HostDetails=true`

     > **Note**
     > Note
     > In this example, the filename is a combination of the path and basic file name. Make sure you specify `parameter DatabaseMatch=DatabaseName` .

   - Restore Component or Category step 1
     Imports backup files of component that was backed up by using the Backup Component action. It could be the Content, Category, or Community component. For the Category component, you need an extra step to synchronize it.
     Action example:
     http://<Host>:<CategoryPort>/action=RestoreServer&filename=c:\backup_category***.zip
   - Restore Category step 2 - Synchronize
     After you restore the category component, you need to synchronize and build the category.
     Action example:
     http://<Host>:<MainProxyACIPort>/action=CategorySyncCatDRE

     > **Note**
     > Note
     > By default, the port number is 9000.

   - Restore Database
     Restores the index IDX exported before. If no DREDbName is specified, use the dbname of the indexed file.
     Action example:
     `http://<MainProxyHost>:<IndexPort>/DREADD?c:\\path\to\<xxx.idx>&DREDbName=<variable_value>&CreateDatabase=True`

> **Note**
> Note
> By default, the port number is 9001.

- View Index Status
  Checks the status of index actions in the Smart Analytics index queue.
  Action:
  `http://<Host>:<port>/action=indexerGetStatus`
- View Action History
  Displays a log of requests, including the date and time that a request was made, the IP address of the client that made the request, and the internal thread that handled the action.
  Action example:
  `http://<Host>:<port>/action=GRL&format=xml`
- View Root Category Detail
  Displays the root categories after training.
  Action example:
  `http://<Host>:<port>/action=CategoryGetHierDetails`
- View Status
  Requests details about all components. Indicates whether all components are up and running; indicates how many documents are stored in each database.
  Action example:
  `http://<Host>:<port>/action=GetStatus`

  > **Note**
  > Note
  > If any component is offline, check the corresponding log file and fix it to bring it online. During indexing, the DIH periodically checks whether all the connected content servers are available. If a content server is unavailable, the DIH queues the data that this content server normally receives, and when the content server starts operating again, the DIH indexes the queued data into it.

  Show Smart Search content server status
  Displays the status of the Smart Search content server.
  Action example:
  `http://SmartSearchProxyHost:DIHPort(20080)/action=EngineManagement&EngineAction=showstatus`

  > **Tip**
  > Tip
  > To view the status for different Smart Analytics components, instead of using the action command, you can easily click the component link on the Smart Analytics Assistant form.

- Add Smart Search Content servers dynamically
  To add a Smart Search content server dynamically, follow these steps:
  1. Run the following action:
     `http://SmartSearchProxyHost:DIHPort(20080)/action=EngineManagement&EngineAction=Add&Host=ContentServerHost&Port=ContentServerPort&weight=1`
  2. Run the following action:
     `http://SmartSearchProxyHost:DAHPort(20070)/action=EngineManagement&EngineAction=EngineAdd&EngineHost=ContentServerHost&EnginePort=ContentServerPort`
  3. Repeat previous steps until all content servers are added.
  4. After you add the new content servers, click the **Redistribute** link in **saa** to redistribute data among all Smart Search content servers.

3. Click **Run**. The result of the action is displayed.

3.2.7. Check Smart Analytics log files

The log files from the Smart Analytics servers provide details to help you identify possible problems or invalid configurations. By default, you can find these log files under corresponding folders in which different Smart Analytics components are installed, such as <SmartAnalytics Installation>\IDOL, <SmartAnalytics Installation>\MediaServer, and <SmartAnalytics Installation>\Content1. Each connector also has its own log folder. All the log files that are related to content are now in the corresponding content folder.

Some log files in the IDOL Server are listed here as examples for your better understanding of the details:

File name	Description
action.log	Logs all the actions on IDOL Server.
agentstore_application.log	Logs general application errors, warnings and information relating to the agent index.
agentstore_index.log	Logs messages relating to the indexing, deletion and updating of agents.
agentstore_query.log	Logs messages relating to the querying of agents.
application.log	Logs general application errors, warnings and information relating to indexes.
category_application.log	Logs general application errors, warnings and information relating to the category index.
category_category.log	Logs messages relating to category actions that read or manipulate the categories, including errors, warnings and progress information.
category_cluster.log	Logs messages relating to cluster actions, including errors, warnings and progress information.
category_schedule.log	Logs messages relating to the running of the Analysis Schedules that are specified in the configuration file.
category_taxonomy.log	Logs messages relating to the TaxonomyGenerate action, including errors, warnings and progress information.
community_action.log	Logs all the actions relating to communities.
community_application.log	Logs general application errors, warnings and information relating to the communities.
community_term.log	Logs terms as they are added to profiles and agents.
community_user.log	Logs when users and agents are added or deleted (or when this fails).
dah_action.log	Logs all the actions relating to DAH.
dah_application.log	Logs general application errors, warnings and information relating to DAH.
dih_application.log	Logs general application errors, warnings and information relating to DIH.
dih_index.log	Logs messages relating to the indexing, deletion and updating of DIH.
dih_query.log	Logs messages relating to the querying of DIH.
index.log	Logs the index actions that the Smart Analytics server receives.

The following items are the default logging configuration of the Smart Analytics servers:

```
LogArchiveDirectory=./logs/archive
LogDirectory=./logs
LogTime=TRUE
LogEcho=FALSE
LogLevel=normal
LogExpireAction=Day
LogMaxOldFiles=7
LogOldAction=delete
LogMaxSizeKBs=102400
```

Note

Note By default, the log files are compressed and deleted as follows:

1. Smart Analytics keeps the log files in the `./logs` folder. A log file is compressed into a .zip file under any of the following conditions:
   - The size of the log file reaches 102400 KBs.
   - The size of the log file is less than 102400KBs, and there are log entries in the subsequent day. Then, the log file is compressed at the beginning of the second day.
   - The size of the log file is less than 102400KBs, and there are no log entries in the subsequent day. Then, the log file is compressed at the time when new entries are logged.
2. A maximum of 6 zipped log files and 1 log file are stored in the `./logs` folder. The .zip file's name starts with the log file's name and ends with the current timestamp in the format of yyyy.mm.dd. When there are more than 1 .zip file compressed from a log file, the timestamp in the new .zip files is followed by a two-digit number from 00 to 05 depending on the sequence of the .zip file. The oldest .zip file is automatically deleted when the number of such.zip files reaches 7. The 7 .zip file names are reused in sequence during the process.

You can customize the settings according to your requirements. For example, if you want to delete history log files automatically whenever the number of log files is more than 100, you can configure `LogOldAction=Delete` and `LogMaxOldFiles=100` in the .cfg file.

You can find these configuration items in the [Logging] section of the configuration file under the servers folders. For example, the .cfg file of IDOL Server is located in the following path by default:

`<SmartAnalytics Installation>\IDOL\IDOLServer.cfg`

The following table lists the description for these configuration items:

Parameter	Description
LogArchiveDirectory	Path to log archive directory. Type the directory in which you want the application to archive old log files when *LogOldAction* is set to `Move`.
LogDirectory	Path to log directory.
LogTime	Displays time with each log entry. Enable this parameter to display the current time next to each log entry in the log file. Possible values: `TRUE` or `FALSE`
LogEcho	Displays logging messages on the console. Possible values: `TRUE` or `FALSE`
LogLevel	The log levels are hierarchical from least logging to most logging. You can use the *LogLevelMatch* parameter to specify which messages are reported relative to the specified *LogLevel*. For example, if `LogLevelMatch=LessThan` and `LogLevel=Warning`, "Normal" and "Full" message types are reported. The following are the possible values for this parameter: • `Always` Basic processes are logged. **Note** Note This produces only minimal logging and no errors are logged. • `Error` Errors are logged. • `Warning` Errors and warnings are logged. • `Normal` Errors, warnings, and basic processes are logged. • `Full` Every occurrence is logged. **Note** Note This produces a large log file and can affect performance.

LogExpireAction	Determines how log files are handled when they exceed the maximum size. Type one of the following to determine how log files are handled when they exceed the *MaxLogSizeKBs* size: • `Compress` The log file's name is appended with a timestamp, compressed and saved in the log directory. By default, this is a ZIP file. Use the *LogCompressionMode* parameter to specify another compression format. • `Consecutive` The log file's name is appended with a number and saved in the log directory. When the next log file reaches its *LogMaxSizeKBs* size, it is appended with the next consecutive number. • `Datestamp` The log file's name is appended with a timestamp and saved in the log directory. • `Previous` The log file's name is appended with .previous and saved in the log directory. Every time a log file reaches its *LogMaxSizeKBs* size, it is given the same postfix so that it overwrites the old log file. • `Day` Only one log file is created for each day and is appended with the current timestamp. Log files are archived after they reach the *LogMaxSizeKBs* size. **Note** Note The *LogMaxSizeKBs* parameter takes precedence over the *LogExpireAction* parameter. Therefore, if you set *LogExpireAction* to Day, and the value for *LogMaxSizeKBs* results in more than one log file, multiple log files are generated for each day.
LogOldAction	Determines how log files are handled when the maximum number of log files is exceeded. Type one of the following to determine how log files are handled when the *LogDirectory* has reached the maximum number of log files, as determined by the *LogMaxOldFiles* parameter: • `Delete` The log files are deleted. • `Move` The log files are moved to the specified *LogArchiveDirectory*.

LogMaxSizeKBs	Maximum log file size (in kilobytes). If you don't want to restrict the log file size, type `-1`. The *LogExpireAction* parameter determines how a log file is handled after it has reached its maximum size.
LogMaxOldFiles	Maximum number of log files in the log directory. The maximum number of log files the specified *LogDirectory* can store before the application runs the specified *LogOldAction*. If you don't want to restrict how many log files the *LogDirectory* can store, type `-1`.

3.2.8. Manage Smart Analytics background schedule in SMIS

In Smart Analytics, some background processes are managed by Integration Manager, which is a plug-in based platform called Service Manager Integration Suite (SMIS). In the out-of-box system, Smart Analytics have three types of SMIS instances:

- SMIDOL
  The SMIDOL instance is for the training, testing, index, and tuning processes, which are triggered in **Smart Analytics Configuration**. Normally, one Service Manager server only has one SMIDOL instance named as SMIDOL*. The asterisk sign is the sequence number, which is usually 0.
- SMIDOLOCR
  The SMIDOLOCR instance is for the image analyzing process, which is triggered when users submit requests from the ESS portal (including the Mobility client). For performance consideration, you can have multiple SMIDOLOCR instances for one Service Manager server. The number of instances equals to the thread number of the Media Servers. If the Media Servers can't be connected, only one instance is created. The instance name is SMIDOLOCR* (The asterisk is the sequence number). The Optical Character Recognition (OCR) task will be added to one of these instances after an interaction is created in ESS, depending on which instance has the least tasks in queue.
- SMIDOLAutoFill
  The SMIDOLAutoFill instance is for the auto-fill process. If an interaction is submitted to ESS without an attachment, a task will be added into this instance. In addition, if SMIDOLOCR is done, a SMIDOLAutoFill task will be also added . For performance consideration, one SM server has multiple SMIDOLAutoFill instances. The number of instances equals to the thread number of the IDOL Server. If the IDOL Server can't be connected, only one instance will be created. The instance name is SMIDOLAutoFill* (The asterisk is the sequence number).

You can check the following topics on troubleshooting Smart Analytics background schedule in SMIS:

Manually create and enable an instance in SMIS

The SMIDOL and SMIDOLAutoFill instances are created and enabled automatically when you enable Smart Analytics in the **Smart Analytics Configuration** menu. The SMIDOLOCR is created and enabled automatically when you enable the Media Server in the **Smart Analytics Configuration** menu. These instances are also created when corresponding processes are triggered, such as the training, testing, tuning, index, or image process.

If the instance is not created automatically or you want do some troubleshooting, you can manually start up these SMIS instances for Smart Analytics.

To create and enable a SMIS instance for Smart Analytics manually, follow these steps:

1. Do one of the following to access Integration Manager:
   - Click **Tailoring** > **Integration Manager**.
   - Type smis in the command line, and then press Enter.
2. Click **Add**.
3. Select the **SMIDOL** , **SMIDOLAutoFill**, or **SMIDOLOCR** template, and then click **Next**. The Integration Instance Information screen is displayed.
4. Change the configuration as needed, and then click **Next**. The Parameters screen is displayed.
5. Do not change anything and click **Next**. The Fields screen is displayed.

6. Do not change anything and click **Next**. The Mapping screen is displayed.
7. Click **Finish**. The main SMIS configuration screen is displayed.
8. Click **Enable** to enable the instance.

Manually disable and remove an instance in SMIS

The SMIDOL and SMIDOLAutoFill instances are removed when you disable Smart Analytics while the SMIDOLOCR instances are removed when you disable the Media Server in the **Smart Analytics Configuration** menu.

If you want to manually disable or delete an SMIS instance, follow these steps:

1. Do one of the following to access Integration Manager:
   - Click **Tailoring** > **Integration Manager**.
   - Type smis in the command line, and then press Enter.
2. Select an instance, and then click **Disable** or **Remove**.

Configure the SMIS instances

If the SMIDOL, SMIDOLAutoFill, and SMIDOLOCR instances are created automatically, the default configuration is defined as follows:

Name:	SMIDOL*, SMIDOLAutoFill*, or SMIDOLOCR* **Note** Note The asterisk sign (*) is the sequence number starting from 0.
Interval Times:	30s
Max Retry Times:	5
Log Level:	INFO
Log File Directory:	N/A
Run as system startup:	true

To change the settings for performance tuning or troubleshooting, follow these steps:

1. Do one of the following to access Integration Manager:
   - Click **Tailoring** > **Integration Manager**.
   - Type smis in the command line, and then press Enter.
2. Select an instance, and then click **Disable**.
3. Select the disabled instance, and then click **Edit**.
4. Change the configuration, and then click **Finish**.

Check task logs in SMIS

To view all the task logs, click **Tailoring** > **Integration Manager**, and then click **Log**.

For training, testing, tuning, and index operations, check the log for the SMIDOL* instance to identify the corresponding task.

For image operation, you have to check all the SMIDOLOCR* instances to identify the corresponding task.

Tip

Tip

- Besides the task log, the program executed by SMIS prints log to `sm.log` by default. You can define another file to print all SMIS logs by specifying the log file directory in the SMIS instance.
- By default, if SMIS fails to execute a task after five retries, the task is set as expired and will never be triggered automatically. However, you can retry the task manually.
- If a SMIS instance is always running and obviously no background process of this instance is running, this instance may be dead due to some exception that can't be caught. In this case, you can't disable or delete the instance from the SMIS configuration page. The workaround is to kill the corresponding SMIS process from the Service Manager System Status page.

3.2.9. Back up and restore content data

User Role: Administrator

This section describes how to backup and restore content data, and how to reindex delta data.

Topics in this section include:

Back up content data

Restore content data

Reindex delta data

3.2.9.1. Back up content data

User Role: Administrator

To ensure that you always have up-to-date copies of the data that the IDOL Server stores, we recommend that you back up the IDOL Server at regular intervals. This section describes two methods to back up and restore content data:

Enabling scheduled backup

To enable scheduled backup, follow these steps:

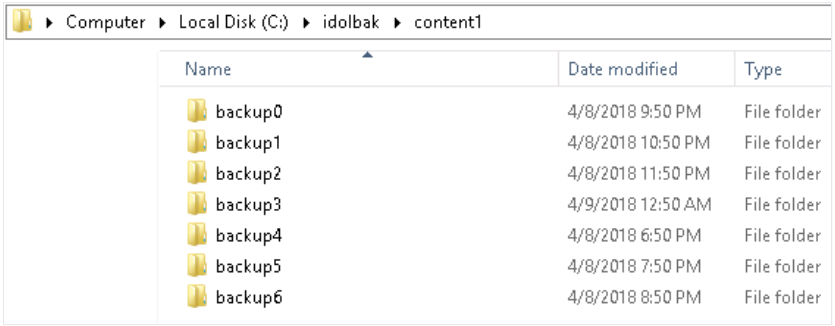
1. Open the `<Smart Analytics Installation>\Content<x>\Content<x>.cfg` files of the content servers.
2. Locate the `[schedule]` section. Insert the following lines, and then update the values if needed.

```
[Schedule]
Backup=true
BackupTime=10:50
//0 means daily backups
BackupInterval=0
BackupRetryAttempts=3
BackupRetryPause=5
NumberOfBackups=7
BackupDir0=C:\idolbak\content1\Backup0
BackupDir1=C:\idolbak\content1\Backup1
BackupDir2=C:\idolbak\content1\Backup2
BackupDir3=C:\idolbak\content1\Backup3
BackupDir4=C:\idolbak\content1\Backup4
BackupDir5=C:\idolbak\content1\Backup5
BackupDir6=C:\idolbak\content1\Backup6
```

Parameter	Description
BackupInterval	The interval that the system creates backup files. If you set the value of this parameter to 0, the system will automatically create a backup every 24 hours.
BackupRetryAttempts	The number of retry attempts when a backup task fails.
BackupRetryPause	The interval (seconds) between retry attempts when a backup task fails.
NumberOfBackups	The number of backup files that are created by the system. If you set the value of this parameter to 7, the first 7 backup schedules create new backup files respectively. The eighth backup operation overwrites the first backup, the nineth backup operation overwrites the second backup, and so on.

3. Restart the content servers.

After you enable scheduled backup, the system automatically creates a series of backup files for the content server. See the following screenshot as an example:



Using commands

To back up content data, you can export all contents index and category files with commands. To do this, run the following actions by using the Smart Analytics Assistant (SAA) utility:

To export contents index:

`http://<idolhost>:<MainIDOLServerIndexPort>/DREEXPORTIDX?filename=c:\BackupFolderName\File&HostDetails=true`

To export category files:

`http://<idolhost>:<MainIDOLServerPort>/action=CategoryExportToXML&Output=File&FileName=c:\\<folder>\\<subfolder>\\<xxx.xml>&IncludeCatID=true`

Note

Note

- By default, the port number for < *MainIDOLServerIndexPort* > is 9001.
- By default, the port number for `<MainIDOLServerPort>` is 9000.

To restore the content data backed up using this method, you can run the following action by using Smart Analytics Assistant (SAA) utility:

`http://<idolhost>:<ACIPort>/DREADD?/<path>/<xxx.idx.gz>&CreateDatabase=True`

For example:

`http://<idolhost>:<Content1indexport>/DREADD?/home/Jora/BackupFolderName/FilePrefix-SGDLITVM0772-10010-0.idx.gz&CreateDatabase=True`

`http://<idolhost>:<Level2proxyindexport>/DREADD?/home/Jora/BackupFolderName/FilePrefix-SGDLITVM0772-30010-0.idx.gz&CreateDatabase=True`

`http://<idolhost>:<Level2proxyindexport>/DREADD?/home/Jora/BackupFolderName/FilePrefix-SGDLITVM0772-30020-0.idx.gz&CreateDatabase=True`

Note

Note

- By default, the port number for < *Content1IndexPort* > is 10011.
- By default, the port number for `<Level2proxyIndexPort>` is 20011.

3.2.9.2. Restore content data

User Role: Administrator

Important

Important The content data restore process described in this document doesn't apply to the following scenario:

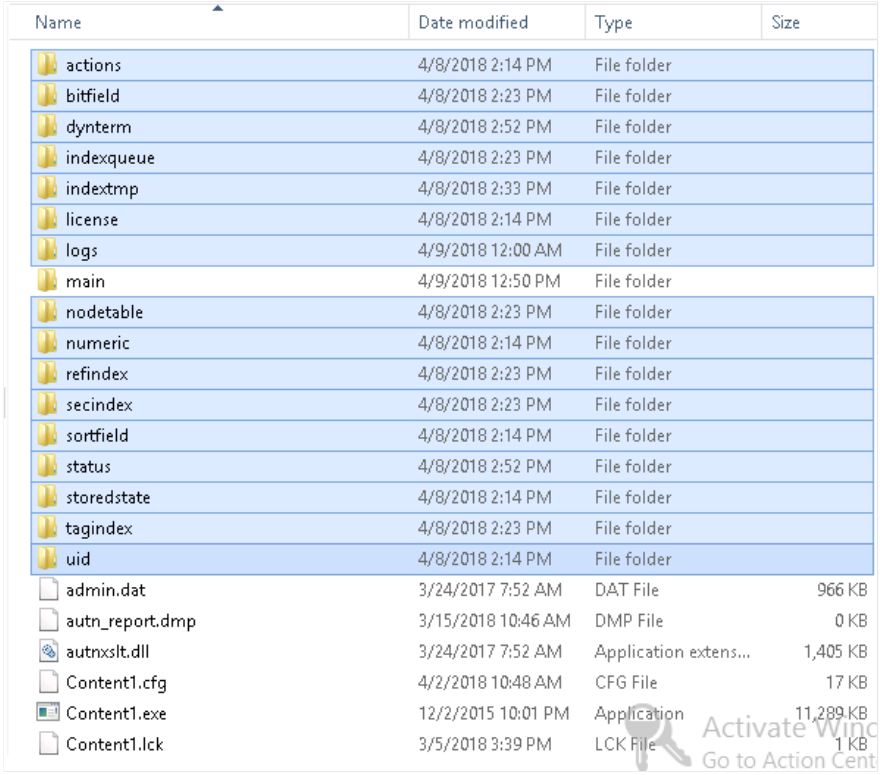
1. The original content sever is running alive and the system performs a scheduled backup
2. You click the **Full Reindex** button, the system performs a full re-index
3. The content server crashes before the next scheduled backup

In this case, skip both this topic and Reindex delta data, and then perform a full re-index again to restore data on the content server.

To restore content data during a disaster recovery process, follow these steps:

1. Depending on which content server is crashed, you may need to stop either the KMUpdate scheduler or the SMIDOL* in SMIS:
   - If the crashed content server is related to Hot Topic Analytics or Smart Ticket (by default, Content1), stop the SMIDOL* in SMIS by following the instructions as described in *Service Manager Help Center > Troubleshooting: Smart Analytics background schedule in SMIS*.
   - If the crashed content server is related to Smart Search (by default, Content2 or Content3), stop the KMUpdate scheduler by following the instructions as described in *Service Manager Help Center > Stop a process*.

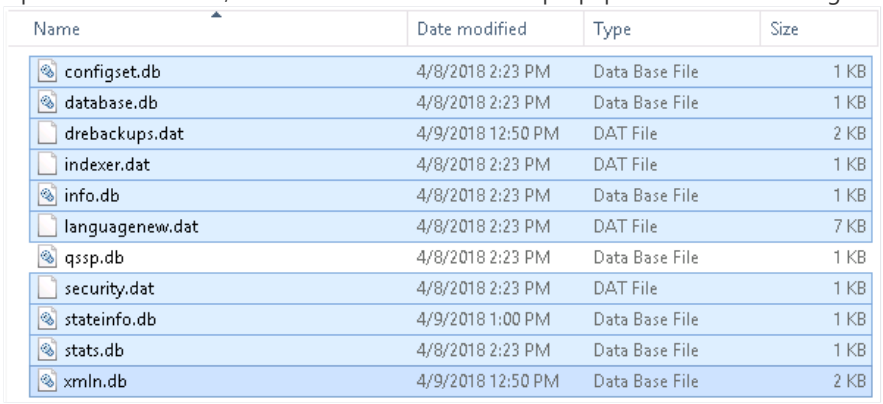
2. Clear the corrupted data on the crashed content server.
    1. Open the crashed content server (for example, <Smart Analytics Installation>/Content1), and then remove all folders except the main folder. See the following screenshot as an example:



Name	Date modified	Type	Size
actions	4/8/2018 2:14 PM	File folder	
bitfield	4/8/2018 2:23 PM	File folder	
dynterm	4/8/2018 2:52 PM	File folder	
indexqueue	4/8/2018 2:23 PM	File folder	
indextmp	4/8/2018 2:33 PM	File folder	
license	4/8/2018 2:14 PM	File folder	
logs	4/9/2018 12:00 AM	File folder	
main	4/9/2018 12:50 PM	File folder	
nodetable	4/8/2018 2:23 PM	File folder	
numeric	4/8/2018 2:14 PM	File folder	
refindex	4/8/2018 2:23 PM	File folder	
secindex	4/8/2018 2:23 PM	File folder	
sortfield	4/8/2018 2:14 PM	File folder	
status	4/8/2018 2:52 PM	File folder	
storedstate	4/8/2018 2:14 PM	File folder	
tagindex	4/8/2018 2:23 PM	File folder	
uid	4/8/2018 2:14 PM	File folder	
admin.dat	3/24/2017 7:52 AM	DAT File	966 KB
autn_report.dmp	3/15/2018 10:46 AM	DMP File	0 KB
autnxslt.dll	3/24/2017 7:52 AM	Application extens...	1,405 KB
Content1.cfg	4/2/2018 10:48 AM	CFG File	17 KB
Content1.exe	12/2/2015 10:01 PM	Application	11,289 KB
Content1.lck	3/5/2018 3:39 PM	LCK File	1 KB

Activate Win
Go to Action Cent

    2. Open the main folder, and then remove all files except qssp.db. See the following screenshot as an example:



Name	Date modified	Type	Size
configset.db	4/8/2018 2:23 PM	Data Base File	1 KB
database.db	4/8/2018 2:23 PM	Data Base File	1 KB
drebackups.dat	4/9/2018 12:50 PM	DAT File	2 KB
indexer.dat	4/8/2018 2:23 PM	DAT File	1 KB
info.db	4/8/2018 2:23 PM	Data Base File	1 KB
languagenew.dat	4/8/2018 2:23 PM	DAT File	7 KB
qssp.db	4/8/2018 2:23 PM	Data Base File	1 KB
security.dat	4/8/2018 2:23 PM	DAT File	1 KB
stateinfo.db	4/9/2018 1:00 PM	Data Base File	1 KB
stats.db	4/8/2018 2:23 PM	Data Base File	1 KB
xmln.db	4/9/2018 12:50 PM	Data Base File	2 KB

3. Start the service of the crashed content server in Windows services. For example, SM Smart Analytics Content1.
4. From the System Navigator, click **System Administration** > **Ongoing Maintenance** > **Smart Analytics** > **Configuration**.
5. Click **More** > **Restore Smart Analytics Data**.

6. Click **Add and Start Task** to open the Add a New Restore Task wizard. You need to provide the following information:

Field	Description
Task name	Specify a name for the restore task.
Endpoint	Specify the host name of the corresponding IDOL content server. For example, http://abc.softwarelab.net/. Note that this URL must end with a slash (/). **Tip** Tip You can check the host name by using Smart Analytics Assistant.
Port	Specify the port number of the crashed content server. **Tip** Tip You can check the port number by using Smart Analytics Assistant. Follow these steps: 1. Type SAA in the Service Manager command line, and then press Enter. The system displays configuration and status of all IDOL components. 2. Check the corresponding port number in **Smart Ticket & Hot Topic Content** and **Smart Search Content**.
Restore File Name	Specify the name and path of the latest restored file. For example, C:\idolbak\content1\backup5.
Task Type	Select **Restore Content** from the drop-down list.

7. Click **Next**.
8. Review the restore task details, and then click **Finish**.
9. The system displays the restore task on top of the task list.
10. Start the KMUpdate scheduler, or start the SMIDOL* instance in SMIS.

Note

Note If the restored data consists of knowledge articles that are expired after the latest backup date, you need to run the **DREEXPIRE** command (such as http://<*IDOLhost*>:<*indexPort*>/DREEXPIRE) to manually retire these articles. Alternatively, you can wait till the system performs the next expiration schedule.

Note

Note Consider the following scenario:

1. The original content severs (content2 and content3) are running alive and the system performs a scheduled backup.
2. You install another content server (for example, content4) so that the data is distributed in the three content servers.
3. Content2 or Content3 crashes before the next scheduled backup and you perform data restore.
4. The data is restored to content2 or content3 only. This is not consistent with the situation before data restore.

In this case, do the following:

1. Clear the data on the three content servers.
2. Restore content data to content2 and content3.
3. Redistribute the data on the three content servers.

Note

Note The current version of IDOL has a limitation that the system doesn't respond to the status checking request during the restore process. In this case, you have to check the IDOL log to get the restore status. Follow these steps:

1. Open the content.cfg file with a text editor, and then update `LogLevel=normal` to `LogLevel=full`.
2. Navigate to the %SmartAnalytics directory%/content<X>/logs directory, and then open the application.log file with a text editor.
3. Check the following messages:
   `06/06/2018 16:35:34 [1] 00-Always: Recovering from backup...`
   `06/06/2018 16:35:34 [1] 10-Full: Copying "Main" files from C:\IDOLBackup\main to C:\IDOLInstallation\main` <- These messages indicate the progress.
   `06/06/2018 16:35:34 [1] 10-Full: Copying "Status" files from C:\IDOLBackup\status to C:\IDOLInstallation\status`
   `06/06/2018 16:35:34 [1] 10-Full: Copying "Dynterm" files from C:\IDOLBackup\dynterm to C:\IDOLInstallation\dynterm`
   `06/06/2018 16:36:20 [1] 10-Full: Copying "Nodetable" files from C:\IDOLBackup\nodetable to C:\IDOLInstallation\nodetable`
   `06/06/2018 16:36:59 [1] 10-Full: Copying "Refindex" files from C:\IDOLBackup\refindex to C:\IDOLInstallation\refindex`
   `06/06/2018 16:36:59 [1] 10-Full: Copying "Numeric" files from C:\IDOLBackup\numeric to C:\IDOLInstallation\numeric`
   `06/06/2018 16:37:00 [1] 10-Full: Copying "Bitfield" files from C:\IDOLBackup\bitfield to C:\IDOLInstallation\bitfield`
   `06/06/2018 16:37:00 [1] 10-Full: Copying "Geospatial" files from C:\IDOLBackup\geoindex to C:\IDOLInstallation\geoindex`
   `06/06/2018 16:37:00 [1] 10-Full: Copying "Secindex" files from C:\IDOLBackup\secindex to C:\IDOLInstallation\secindex`
   `06/06/2018 16:37:00 [1] 10-Full: Copying "TagIndex" files from C:\IDOLBackup\tagindex to C:\IDOLInstallation\tagindex`
   `06/06/2018 16:37:01 [1] 10-Full: Copying "SortField" files from C:\IDOLBackup\sortfield to C:\IDOLInstallation\sortfield`
   `06/06/2018 16:37:01 [1] 10-Full: Copying "IndexQueue" files from C:\IDOLBackup\indexqueue to C:\IDOLInstallation\indexqueue`
   `06/06/2018 16:37:01 [1] 10-Full: Copying "storedstate" files from C:\IDOLBackup\storedstate/current to C:\IDOLInstallation\storedstate\current`
   `06/06/2018 16:37:01 [1] 10-Full: Copying "storedstateref" files from C:\IDOLBackup\storedstate to C:\IDOLInstallation\storedstate`
   `06/06/2018 16:37:01 [1] 30-Normal: Recovered from backup successful.` <-This message doesn't indicate a full completion because the engine starts after this. It may take a long time or a short while depending on your configuration.
4. Navigate to the %SmartAnalytics directory%/content<X>/logs directory, and then open the index.log file with a text editor.
5. Check the following messages:
   `06/06/2018 16:35:31 [1] 30-Normal: Index Queue Command: DREINITIAL?Path=D:\IDOLBackup`
   `06/06/2018 16:37:36 [1] 30-Normal: Index command finished. Took 124.91 s` <- This message indicates a full completion.

3.2.9.3. Reindex delta data

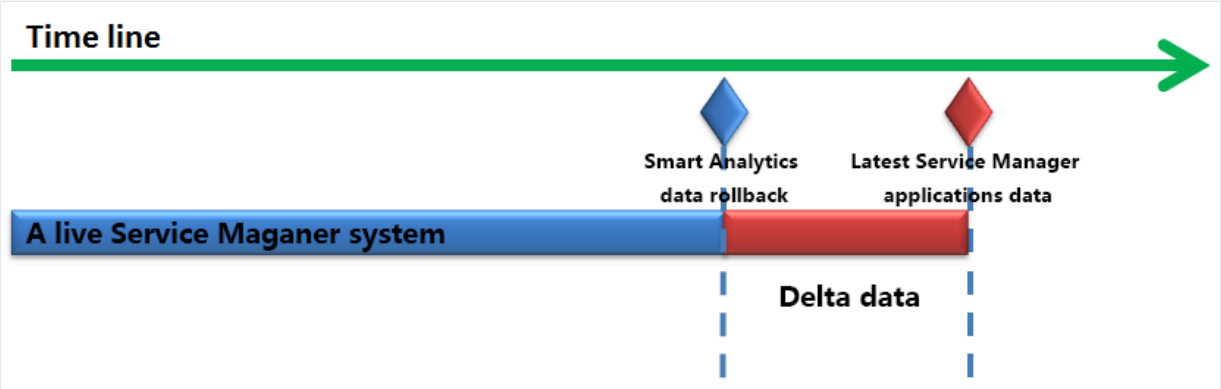
Applies to User Role: Administrator

Service Manager provides the Reindex delta data option to reindex the “delta data”. Delta data refers to the gap between the selected backup data in Smart Analytics and the latest Service Manager data during a disaster recovery process. Consider the following scenario:

- The original IDOL system is running alive and you enable scheduled backup

- The original IDOL system is down
- You roll back the Smart Analytics data from the latest backup
- The end users are making changes from the time the original IDOL system is down to the time the new IDOL system goes live

The following figure depicts this scenario:



To reindex the delta data for the Smart Analytics content server, follow these steps:

1. Form the System Navigator, click **System Administration** > **Ongoing Maintenance** > **Smart Analytics** > **Configuration**.
2. Depending on which content server crashes, click **Hot Topic Analytics** for the Content1 server, or click **Smart Search** for the Content2/Content3 server.
3. Click **More** > **Reindex Delta Data**.
4. If you click **Hot Topic Analytics** in *step 2*, you must select a start date which is earlier than that of the latest backup data you've specified in Restore content data > *step 6*. This procedure ensures that all delta data created after the content restore is reindexed into IDOL.
   For example, if the latest backup file C:\idolbak\content1\backup5 is created on 04/08/18 13:57:00, you can do either of the following:
   - Select a time earlier than 04/08/18
   - Select 04/08/18 and then update the start time to 13:50:00

   If you click **Smart Search** in *step 2*, you must select a start date which is earlier than that of the latest backup data you've specified in Restore content data > *step 6*. Meanwhile, your selected date must be in the delta data recycle time scope.
   For example, if the latest backup file C:\idolbak\content1\backup5 is created on 04/08/18 13:57:00, and you set the value of the **Smart Search delta data recycle time (days):** option to 7, you can do either of the following:
   - Select a time between 04/08/18 and 04/01/18.
   - Select 04/08/18 and then update the start time to 13:50:00

   Important
   Important
   In case the IDOL server and the Service Manager server are running in different time zones, you must calculate the time difference when selecting a start date in this step.
5. Click **Finish**. The delta data will be reindexed the next time the KMUpdate scheduler or the SMIDOL* instance runs.
6. Perform global search or Hot Topic Analytics search to check whether the delta data is reindexed. Otherwise, check the sm.log file for related debugging messages.

3.2.10. Back up and restore connector data

User Role: Administrator

Back up connector data

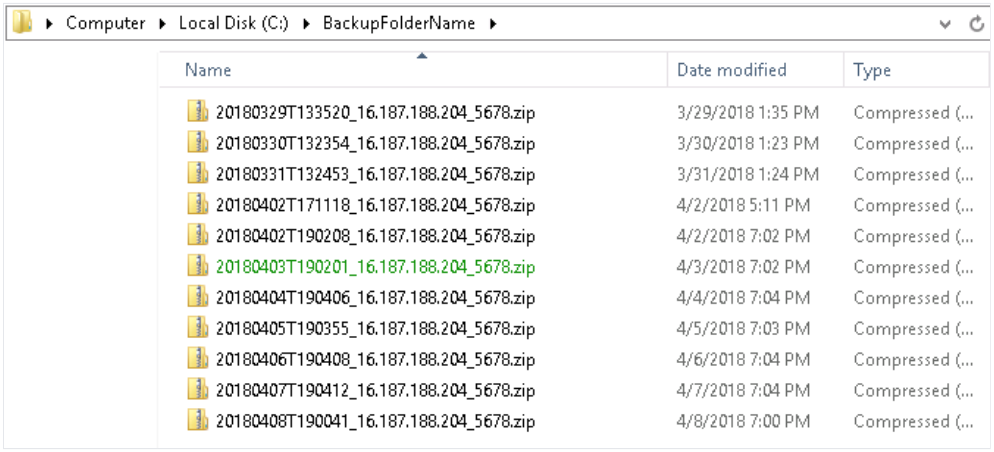
To enable scheduled backup, follow these steps:

1. Click **System Administration** > **Ongoing Maintenance** > **Smart Analytics** > **Smart Search** > **More** > **Tailoring**.
2. Type a daily backup time in the **Connector daily backup time** field. For example, `12:00`.
3. Specify a backup folder in the **Connector daily backup folder** field. For example, `C:\\BackupFolder`.

> **Note**
>
> Note You must have read and write permissions to the specified connector data backup folder. On the Linux system, it's advisable that this data backup folder is at the same level as the IDOL installation directory.

4. Click **Save** and **OK**.

After you enable scheduled backup, the system automatically creates a series of zipped backup files for the connectors. See the following screenshot as an example:

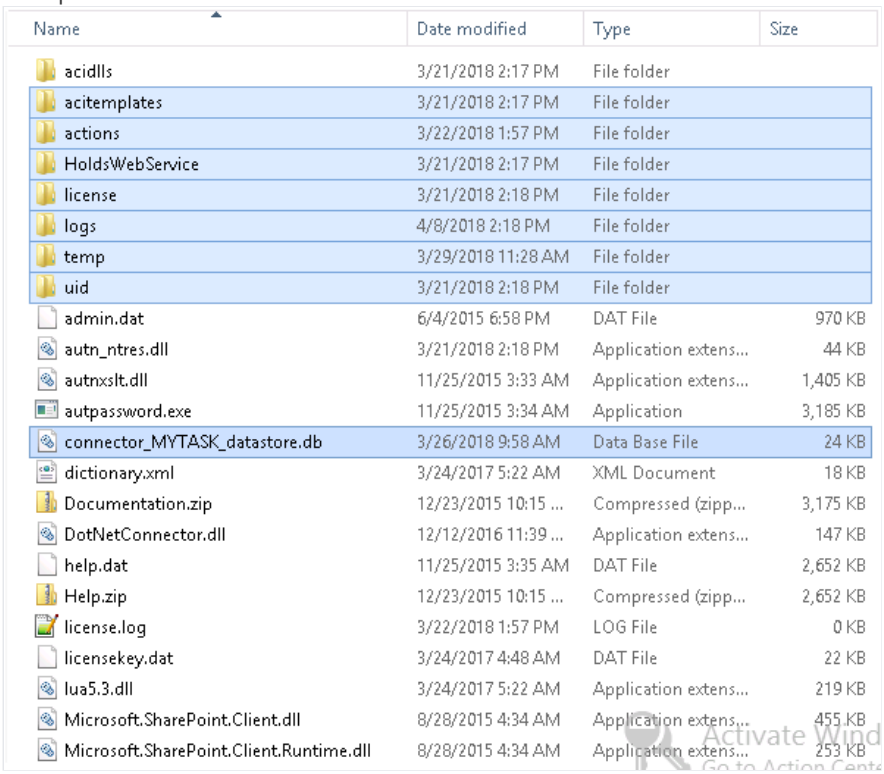


Restore connector data

To restore connector data during a disaster recovery process, follow these steps:

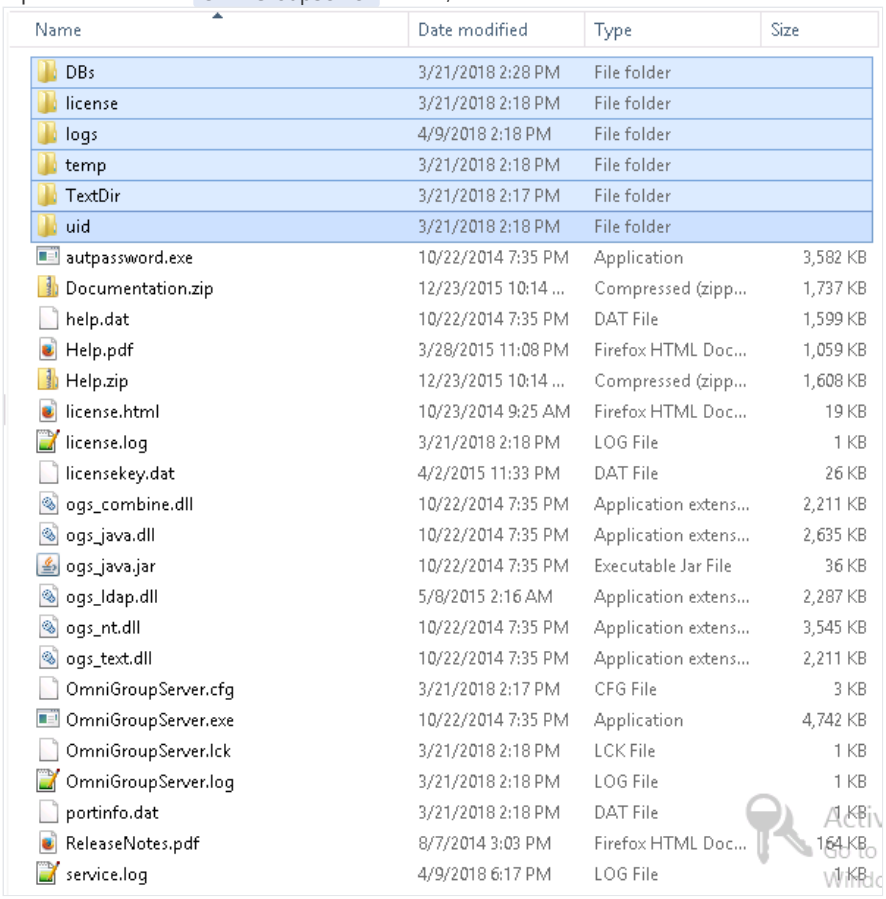
1. Stop the **KMUpdate** scheduler.

2. Clear the corrupted data on the crashed connector server.
   SharePoint connector, Web connector, File System connector, Oracle connector (for Service Manager 9.60 and later), and Confluence connector:
   Open the crashed connector folder (for example, `<Smart Analytics Installation>/SharepointRemoteConnector`), and then remove all folders except the `acidlls` folder. In addition, remove the *.db files. See the following screenshot as an example for the SharePoint connector:



OmniGroupServer:

Open the crashed `OmniGroupServer` folder, and then remove all folders. See the following screenshot as an example:



Name	Date modified	Type	Size
DBs	3/21/2018 2:28 PM	File folder	
license	3/21/2018 2:18 PM	File folder	
logs	4/9/2018 2:18 PM	File folder	
temp	3/21/2018 2:18 PM	File folder	
TextDir	3/21/2018 2:17 PM	File folder	
uid	3/21/2018 2:18 PM	File folder	
autpassword.exe	10/22/2014 7:35 PM	Application	3,582 KB
Documentation.zip	12/23/2015 10:14 ...	Compressed (zipp...	1,737 KB
help.dat	10/22/2014 7:35 PM	DAT File	1,599 KB
Help.pdf	3/28/2015 11:08 PM	Firefox HTML Doc...	1,059 KB
Help.zip	12/23/2015 10:14 ...	Compressed (zipp...	1,608 KB
license.html	10/23/2014 9:25 AM	Firefox HTML Doc...	19 KB
license.log	3/21/2018 2:18 PM	LOG File	1 KB
licensekey.dat	4/2/2015 11:33 PM	DAT File	26 KB
ogs_combine.dll	10/22/2014 7:35 PM	Application extens...	2,211 KB
ogs_java.dll	10/22/2014 7:35 PM	Application extens...	2,635 KB
ogs_java.jar	10/22/2014 7:35 PM	Executable Jar File	36 KB
ogs_ldap.dll	5/8/2015 2:16 AM	Application extens...	2,287 KB
ogs_nt.dll	10/22/2014 7:35 PM	Application extens...	3,545 KB
ogs_text.dll	10/22/2014 7:35 PM	Application extens...	2,211 KB
OmniGroupServer.cfg	3/21/2018 2:17 PM	CFG File	3 KB
OmniGroupServer.exe	10/22/2014 7:35 PM	Application	4,742 KB
OmniGroupServer.lck	3/21/2018 2:18 PM	LCK File	1 KB
OmniGroupServer.log	3/21/2018 2:18 PM	LOG File	1 KB
portinfo.dat	3/21/2018 2:18 PM	DAT File	1 KB
ReleaseNotes.pdf	8/7/2014 3:03 PM	Firefox HTML Doc...	164 KB
service.log	4/9/2018 6:17 PM	LOG File	1 KB

3. Start the crashed connector server in Windows services. For example, SM Smart Analytics SharePoint connector.
4. From the System Navigator, click **System Administration** > **Ongoing Maintenance** > **Smart Analytics** > **Configuration**.
5. Click **More** > **Restore Smart Analytics Data**.

6. Click **Add and Start Task** to open the Add a New Restore Task wizard. You need to enter the following information:

Field	Description
Task name	Specify a name for the restore task.
Endpoint	Specify the corresponding connector's URL. For example, `http://abc.softwarelab.net/`. Note that this URL must end with a slash (/). **Tip** Tip You can check this URL in **Smart Search Configuration** > **Connector Configuration**.
Port	Specify the port number of the crashed content server. **Tip** Tip You can check the port number in **Smart Search Configuration** > **Connector Configuration** as well.
Restore File Name	Specify the name and path of the latest restored file. For example, `C:\20180708T190041_16.188.188.123_5678.zip`.
Task Type	Select **Restore Connector** from the drop-down list.

7. Click **Next**.
8. Review the restore task details, and then click **Finish**.
9. The system displays the restore task on top of the task list. Select this task and then click Refresh to check the task status.

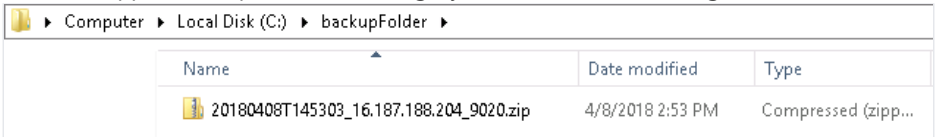
3.2.11. Back up and restore category data

User Role: Administrator

Back up category data

To enable scheduled backup, follow these steps:

1. Click **System Administration** > **Ongoing Maintenance** > **Smart Analytics** > **Smart Ticket** > **More** > **Tailoring**.
2. Specify a folder to store daily backups in the Category backup folder field. For example, c:/backupFolder.
3. Click **Save** and **OK**.
4. Select a task in the Current Configuration List, and then perform either training or tuning. The system automatically creates a zipped backup file for the category server. See the following screenshot as an example:

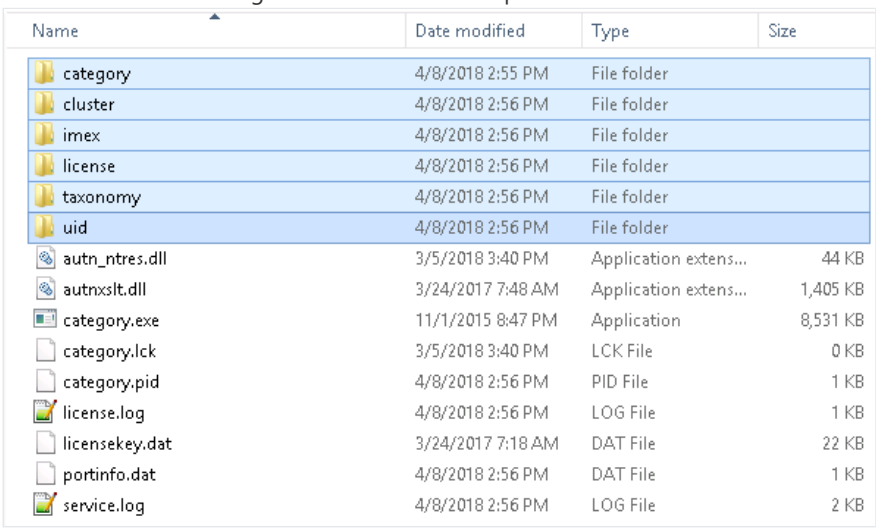


Restore category data

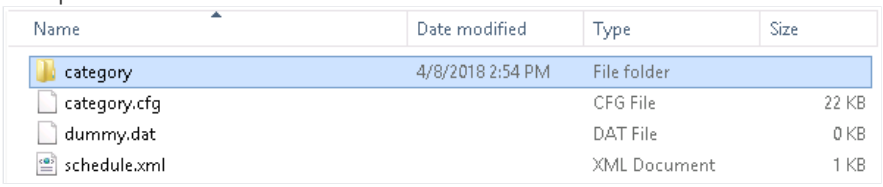
To restore category data during a disaster recovery process, follow these steps:

1. Stop the main server.

2. Clear the corrupted data on the crashed category server.
   Open the crashed category folder (for example, <Smart Analytics Installation>/IDOL/category), and then remove all folders. See the following screenshot as an example:



3. Navigate to the category backup folder, and then extract the latest restored file.
4. Copy the category folder back to <Smart Analytics Installation>/IDOL/category. See the following screenshot as an example:



5. Start the main server.

3.2.12. Perform indexing of large-scale knowledgebases

User Role: Administrator

Performing a full index of a large knowledgebase may take hours or even days to complete depending on the scale of your knowledgebases, which may have a significant impact on system resources. As of version 9.50, Service Manager provides a ScriptLibrary AddToKMBUTable, allowing you to perform incremental index of knowledgebases. If you plan to perform full index of a large knowledgebase, it is recommended that you perform full indexing on a small amount of records first, and then perform several incremental indexing tasks for the rest records by using the ScriptLibrary AddToKMBUTable.

The following section is an example of how to perform full indexing of a large scale knowledgebase.

Suppose there are 4 million records in your Interaction_Library which were created in the past 12 months, and you want to perform full indexing on all these records. To reduce potential downtime, you can perform the full index as follows:

1. Perform a full index on records created within 3 months.
   1. Open the Smart Search Configuration page.
   2. In the Current Knowledgebase List section, click the corresponding link to open Interaction_Library. Then, the Knowledgebase maintenance page is displayed.
   3. In the **Table Query** field, specify a Service Manager style query so that only records created within 3 months are indexed, as shown below:

      ```
      open.time>='05/25/16'
      ```

      Modify other fields as necessary.
   4. Click the **Full Reindex** button.

5. Wait for Service Manager to finish indexing the records that match specified conditions.

2. Perform incremental index on records created within 6 months but older than 3 months.
    1. Once the index status turns to **Finished**, re-specify the **Table Query** to set the time frame to 6 months since now, as shown below:

        ```
        open.time>='02/25/16'
        ```

    2. Open the AddToKMBUTable ScriptLibrary.
    3. Locate the following function:

        ```
        function addRecordToKBU(filename,query)
        ```

        Then, update the value to perform incremental index on records that have not been full re-indexed while you have re-specified in the Table Query field, as shown below:

        ```
        function addRecordToKBU("incidents",open.time>='02/25/16' and open.time<='05/25/16')
        ```

    4. Click **Execute**
3. Perform incremental index on records created within 9 months but older than 6 months.
4. Perform incremental index on records created within 12 months but older than 9 months.

3.2.13. Encrypt Smart Analytics index data

User Role: Administrator

You can choose whether to configure the IDOL content component to encrypt your index data, 256-bit AES encryption is used and the following data are encrypted:

- document data in the index
- temporary data in the index cache
- value mapping files used by the parametric, match, and security indexes
- the term dictionary
- the unstemmed term tree
- the geospatial index
- any data sent with index actions that are stored in your index queue before processing

Caution

Caution
If you lose your encryption keys after you enable the encryption, you can't recover your data.

Enable index encryption

1. Create a key file (.ky) with a key of length is 64, for example: D15B643D5332BB9B9871EB1828D91367FA5419FD7179C8254AA4CCB647AB8009
2. Open the Smart Analytics Content component configuration file in a text editor one by one such as content1.cfg, content2.cfg, content3.cfg...
   Add a [DataEncryption] section to the configuration file one by one:
   `[DataEncryption]`

   `AESKeyFile=<key file path>`
   For example:
   `[DataEncryption]`

   `AESKeyFile=C:\idoldata\keys\indexkey.ky`

3. Save and close the configuration file.
4. Restart the Smart Analytics Content component for your changes to take effect.

Note

Note
When the AES encryption is enabled in an index with contents:

- Smart Analytics IDOL Server encrypts the value mapping and geospatial index files at startup.
- Document data encryption applies only to new data. Any unprocessed data in your index queue remains unencrypted.
- Smart Analytics IDOL Server does not encrypt the unstemmed term tree unless you regenerate it (see Regenerate with a Server Restart).
- Smart Analytics IDOL Server encrypts the term dictionary when it next flushes the terms to disk during index, or when you run index compaction.

To ensure that all your data are encrypted, you need to enable the encryption in an empty index.

Check Your Encryption Status

You can check the encryption status of your Content index by sending the GetStatus action.

The response for Content includes a data_encryption section, which can have one of the following values:

- false. The index is not encrypted.
- true. All data in the index are encrypted.
- partial. Some data in the index are encrypted, but there are also some unencrypted data.

For example:

```
<data_encryption>
  <activated>true</activated>
  <nodetable_encrypted>partial</nodetable_encrypted>
  <indexqueue_encrypted>partial</indexqueue_encrypted>
  <diskindex_encrypted>true</diskindex_encrypted>
  <unstemmed_encrypted>false</unstemmed_encrypted>
  <geoindex_encrypted>true</geoindex_encrypted>
</data_encryption>
```

Each time you start Content, it verifies your encryption key to ensure that it matches the existing settings. The following table describes the errors you can find in the `application.log` in the content folder when the content can't be started.

Error	Cause
`Key file '<path>indexkey1.ky' not found.`	The path for the key file is wrong.
`Encryption key from <path>indexkey.ky is the wrong length.`	The length of the key is wrong, the correct length is 64.
`Encryption key from <path>indexkey.ky failed to match previous data encryption configuration.`	The key is mismatched.
`Error: Data encryption was previously enabled, but no encryption key found.`	The encryption can't be turned off when the content index isn't empty.

Similarly, when you attempt to use the DREINITIAL index action with a backup path, Content checks the encryption settings in the backup directory before it restores the index. If the target index has incompatible encryption settings, the DREINITIAL index action fails with a `Bad Parameter` error.

Note

Note
Content stores only a hash generated from the encryption key on disk, so you can't recover the original encryption keys from the saved information.

When you export data from Content by using DREEXPORTIDX or DREEXPORTXML, the exported content is unencrypted. We recommend that you use appropriate authorization roles to ensure that only authorized users can export content from your index.

If you want to turn off encryption or change your encryption settings, you must export your data and index into an empty content index as follows:

1. On the Smart Analytics server, open a browser and go to http://localhost:<index_port>/DREEXPORTIDX? (by default, content1 index port is 10011, content2 index port is 30011, and content3 index port is 30021). This index action exports all IDX documents to a series of compressed files in the content directory. The file names are `AUTN-IDX-EXPORT-<host name>-<content server port>-<dd.mm.yy>-<hh-mm-ss>-0.idx.gz`, `AUTN-IDX-EXPORT-<host name>-<content server port>-<dd.mm.yy>-<hh-mm-ss>-1.idx.gz` and so on.
2. Stop the server and remove all the folders under Content.
3. Remove the [DataEncryption] section from the configuration files to turn off the encryption or modify the [DataEncryption] section in the configuration files to change the encryption settings.
4. Start the server and open a browser, run `http://localhost:<index_port>/DREADD?Filename=<path of the file you exported in step 1>`, for example, `http://localhost:index_port/DREADD?Filename=C:\Program%20Files%20(x86)\Micro%20Focus\Service%20Manager%209.64\SmartAnalytics\Content2\AUTN-IDX-EXPORT-SHCLITVM0030-30020-25.03.2021-06.24.17-0.idx.gz`

3.2.14. Modify Stop Words for IDOL search engine

User role: System Administrator

A stop-word list is a list of terms that can be ignored when the search engine is searching or indexing. Typically, stop-word lists include short and common words or prepositions, such as "a," "the," or "with" in English. However, they may also include longer words, such as long number strings, or words that are too common to be useful as search targets, such as the term "internet." Stop words are removed from words entered in the "Search for" box unless they are enclosed in double quotes (phrase search). They're not removed during indexing to allow for phrase searching.

Smart Search applies both the Smart Analytics stop words and the Service Manager stop words. However, some stop words used in Service Manager conflict with the stop words logic used in Smart Search. For example, "before" is a stop word defined in Service Manager, which means that "before" in a search string is ignored. However, in Smart Search, it supports the search string as "A BEFORE B", which means that Smart Search returns the results in which A comes before B. If "before" is not removed from the Service Manager stop words list, the returned results aren't as expected. To avoid this problem,

when you enable Smart Analytics, some of the English stop words are removed. To obtain a list of the removed stop words, search for "stop_words_to_remove" in the "IDOL_EnableKM" script library.

If you want to keep all the Service Manager stop words when you upgrade your Smart Analytics server, back up the stop words list before upgrade and import it after upgrade.

When you search by the IDOL search engine, if the search word is a stop word defined in IDOL search engine only, a warning message is displayed.

Stop words are stored in Service Manager in lists by specific language. Not all languages support stop words (for example, Japanese and Chinese). Adjust the list of stop words by either adding or removing words from this list.

The stop-word list for your log-in language is used by default, and is loaded once when you first log in. Changing the query language parameter on the advanced search screen changes the stop-word list used. The new stop-word list is loaded each time you search in a language other than your log-in language. This may cause a delay in your search being submitted as the stop-word list is loaded. If you need to perform extensive searches in a language other than your log-in language, we recommend that you log out and then log back in with the other language to reduce this delay.

To modify Service Manager stop words:

1. Click **Smart Analytics** > **Smart Search** > **More** > **Stop Words**.
2. Click **Search**.
3. Select the record for the language code you wish to change.
4. Add a new word or modify an existing word.
5. Click **Save**.

To modify Smart Analytics stop words:

1. Open the <*Smart Analytics Installation*>/lanfiles folder.
   The stop words lists are saved as the <language name>.dat file in this folder.
2. Open the language file that you wish to change.
3. Add a new word or modify an existing word.
4. Click **Save**.

3.2.15. Transfer Smart Analytics intelligence between systems

User Role: Administrator

As an administrator, you may want to transfer the intelligence in Smart Analytics from one environment to another environment. For example, when you finish testing Smart Analytics in your test environment, you may want to migrate the configured Smart Analytics to your production environment.

Precondition

If you need to keep changes to production Service Manager in Smart Search during Smart Analytics intelligence testing and migration, do the following in your production Service Manager immediately after you start copying data to test Service Manager for Smart Analytics testing:

- If you have enabled Solr search engine, stop the backend schedule for KM index updating.
- If you have enabled IDOL search engine, make sure the libraries status is NOT offline, and then stop the backend schedule for KM index updating.
- If you have not enabled Knowledge Management, add the following section in script KMLanguages:

```
if(vars.$G_kmsearchengine ==null) {
                vars.$G_kmsearchengine = "IDOL";
        }
```

Then, add a script which contains the following code and run it:

```
function knowledgebase() {

function getKBs(kbnames) {
var sql = 'kbname isin ' + kbnames + '';
if(kbnames==null || '' == kbnames) {
sql = 'true';
}
var f = lib.FileUtil._file('kmknowledgebase').select(sql);
return f;
};
function changeStatus(kbnames, status) {
var kbs = getKBs(kbnames);
kbs.iterate(function(kb){
kb['indexstatus'] = status;
print('Status of ' + kb.kbname + ' change to ' + status );
kb.doUpdate();
}) ;
};
return {'getKbBs':getKBs, 'changeStatus':changeStatus}
}
var knows = new knowledgebase();
// NOTE: Change the libraries you want to change the status here.
// Status value could be 'indexing', 'not started', 'offline', 'finished'
var list = ['Incident_Library', 'Interaction_Library','Knowledge_Library'];
knows.changeStatus('{"'+ list.join('","') + '"}','finished');
```

Prepare the test Service Manager

1. In your test Service Manager, configure, index, tune and test Smart Analytics as needed.

2. In your test Service Manager, click **Tailoring > Unload Script Utility**. Search for the out-of-box unload script IDOL_KM_Config_Export or add an new unload script using the following query:

File name	Query
idolcategorytest	true
idolserverinfo	true
number	name="cate2idolid" or name="cate2idoltestid" or name#"idol"
cate2idol2	true
counters	true
idolDataFilter	true
idoladapter	true
idolindex	true
idolpbmhunter	true
idolsecgroup	true
idoltestresult	true
idoltestsample	true
idoltuning	true
kmknowledgebaseupdates	true
kmknowledgebase	true
SmartTicketCategorySnapshot	true
SmartTicketTestResults	true
SmartTicketTestRounds	true

3. Run the unload script to get a .unl file. For details, see Run an unload script.

Transfer the configured Smart Analytics from test environment to production environment

If you use your test Smart Analytics server as the production Smart Analytics server, follow these steps to transfer Smart Analytics from test environment to production environment:

1. Import the .unl file into your production Service Manager by using Unload Manager. For details, see Load an unload file.

Note

Note

- You need to resolve any conflicts that are detected.
- Make sure that the kmknowledgebaseupdates table contains the latest data from the database of production Service Manager instead of the test Service manager.

2. Log off and then log back onto your production Service Manager.

3. In your production Service Manager, configure Smart Analytics to connect to the address of your production Smart Analytics server. Log off and then log back onto the production Service Manager
4. In the file system of your Smart Analytics server, configure production Service Manager IP address for section `Service` in the configuration files of idolserver, level2server, content and other components.
5. Synchronize category. To do this, run the following commands by using Smart Analytics Assistant (SAA) utility in your production Service Manager: `http://<idolhost>:<MainProxyPort>/action=CategorySyncCatDRE`
6. After you finish transferring the configured Smart Analytics, delete the following section from script KMLanguages if you added it in Precondition, and then run script KMLanguages again:

```
if(vars.$G_kmsearchengine ==null) {
                vars.$G_kmsearchengine = "IDOL";
}
```

7. Start the backend schedule for KM index updating in the Service Manager production environment.

If you use a different server other than your test Smart Analytics server as the production Smart Analytics server, follow these steps to transfer Smart Analytics from test environment to production environment:

1. Export all contents index and category files. To do this, run the following actions by using the Smart Analytics Assistant (SAA) utility in your test Service Manager:
   To export contents index:
   `http://<idolhost>:<MainIDOLServerIndexPort>/DREEXPORTIDX?filename=c:\BackupFolderName\File&HostDetails=true`
   To export category files:
   `http://<idolhost>:<MainIDOLServerPort>/action=CategoryExportToXML&Output=File&FileName=c:\\<folder>\\<subfolder>\\<xxx.xml>&IncludeCatID=true`

   Note

   Note

   - `<idolhost>` is the address of the test Smart Analytics server.
   - The file path should be specified to a location in the file system of the test Smart Analytics server.
   - By default, the port number for < *MainIDOLServerIndexPort* > is 9001.
   - By default, the port number for `<MainIDOLServerPort>` is 9000.

Tip

Tip

In the file system of the test Smart Analytics server, properly configure the IDOL server configuration file (`/IDOL/IDOLserver.cfg`) to make sure that the value for `AllowedOutputDirectoryCSVs` is valid as shown below:

`AllowedOutputDirectoryCSVs=C:\<folder>\<subfolder>`

Otherwise, you might receive the following error message:

```
<autnresponse xmlns:autn="http://schemas.autonomy.com/aci/">
<action>CATEGORYEXPORTTOXML</action>
<response>ERROR</response>
<responsedata>
<error>
<errorid>IDOLPROXYCATEGORYEXPORTTOXML-2147483386</errorid>
<rawerrorid>0x80000106</rawerrorid>
<errorstring>Bad value in parameter FILENAME</errorstring>
<errordescription>Output to the specified file is forbidden by configuration.
</errordescription>
<errorcode>ERRORPERMISSION</errorcode>
<errortime>27 Apr 16 15:02:13</errortime>
</error>
</responsedata>
</autnresponse>
```

For information about how to use the SM SAA utility, see Use Smart Analytics Assistant.

2. Import the .unl file into your production Service Manager by using Unload Manager. For details, see Load an unload file.

Note

Note

- You need to resolve any conflicts that are detected.
- Make sure that the kmknowledgebaseupdates table contains the latest data from the database of production Service Manager instead of the test Service manager.

3. Log off and then log back onto the Service Manager production environment.
4. In your production Service Manager, configure Smart Analytics to connect to the address of your production Smart Analytics server, if needed. Log off and then log back onto the production Service Manager
5. Import all content index that you exported in *step 1*. To do this, run the following action by using Smart Analytics Assistant (SAA) utility in your production Service Manager:

   `http://<idolhost>:<ACIPort>/DREADD?/<path>/<xxx.idx.gz>&CreateDatabase=True`

   For example:

```
http://<idolhost>:<Content1indexport>/DREADD?/home/Jora/BackupFolderName/FilePrefix-SGDLITVM0772-
10010-0.idx.gz&CreateDatabase=True
http://<idolhost>:<Level2proxyindexport>/DREADD?/home/Jora/BackupFolderName/FilePrefix-
SGDLITVM0772-30010-0.idx.gz&CreateDatabase=True
http://<idolhost>:<Level2proxyindexport>/DREADD?/home/Jora/BackupFolderName/FilePrefix-
SGDLITVM0772-30020-0.idx.gz&CreateDatabase=True
```

If you have exported more than one file in *step 1*, namely xxx-0.idx.gz, xxx-1.idx.gz, and xxx-2.idx.gz, the example may look as follows:

```
http://<idolhost>:<Content1indexport>/DREADD?/home/Jora/BackupFolderName/FilePrefix-SGDLITVM0772-10010-0.idx.gz&CreateDatabase=True
http://<idolhost>:<Content1indexport>/DREADD?/home/Jora/BackupFolderName/FilePrefix-SGDLITVM0772-10010-1.idx.gz&CreateDatabase=True
http://<idolhost>:<Content1indexport>/DREADD?/home/Jora/BackupFolderName/FilePrefix-SGDLITVM0772-10010-2.idx.gz&CreateDatabase=True

http://<idolhost>:<Level2proxyindexport>/DREADD?/home/Jora/BackupFolderName/FilePrefix-SGDLITVM0772-30010-0.idx.gz&CreateDatabase=True
http://<idolhost>:<Level2proxyindexport>/DREADD?/home/Jora/BackupFolderName/FilePrefix-SGDLITVM0772-30010-1.idx.gz&CreateDatabase=True
http://<idolhost>:<Level2proxyindexport>/DREADD?/home/Jora/BackupFolderName/FilePrefix-SGDLITVM0772-30010-2.idx.gz&CreateDatabase=True

http://<idolhost>:<Level2proxyindexport>/DREADD?/home/Jora/BackupFolderName/FilePrefix-SGDLITVM0772-30020-0.idx.gz&CreateDatabase=True
http://<idolhost>:<Level2proxyindexport>/DREADD?/home/Jora/BackupFolderName/FilePrefix-SGDLITVM0772-30020-1.idx.gz&CreateDatabase=True
http://<idolhost>:<Level2proxyindexport>/DREADD?/home/Jora/BackupFolderName/FilePrefix-SGDLITVM0772-30020-2.idx.gz&CreateDatabase=True
```

Note

Note

- `<idolhost>` is the address of the production Smart Analytics server.
- The file path should be specified to a location in the file system of the production Smart Analytics server.
- By default, the port number for < *Content1IndexPort* > is 10011.
- By default, the port number for `<Level2proxyIndexPort>` is 20011.

6. Copy and paste the following VDB settings to the configuration file (`/IDOL/IDOLserver.cfg`) in the file system of your production Smart Analytics server:
   - The `[vdb#]` settings that are located at the end of the configuration file (`/IDOL/IDOLserver.cfg`) in the file system of your test Smart Analytics server.
   - The value of VirtualDatabases for `[DistributionSetting]` section of the configuration file (`/IDOL/IDOLserver.cfg`) in the file system of your test Smart Analytics server.
7. Import category files that you exported in step 1. To do this, run the following action by using Smart Analytics Assistant (SAA) utility in your production Service Manager:
   `http://<idolhost>:<MainIDOLServerPort/action=CategoryImportFromXML&ImportFilename=c:\\<folder>\\<subfolder>\\<xxx.xml>&BuildNow=true`
8. Check the consistence of category in your test Service Manager and production Service Manager.
   `http://<idolhost>:<MainIDOLServerPort>/action=CategoryGetHierDetails&expand=all`
   Make sure the results are the same. Otherwise, repeat this step.
9. After you finish transferring the configured Smart Analytics, delete the following section from script KMLanguages if you added it in Precondition, and then run script KMLanguages again:

```
if(vars.$G_kmsearchengine ==null) {
                vars.$G_kmsearchengine = "IDOL";
}
```

10. Start the backend schedule for KM index updating in the Service Manager production environment.

Now, your production Service Manager works for Smart Search, Smart Ticket, and Hot Topic Analytics as your test Service Manager does.

3.2.16. Set stop phrases for Hot Topic Analytics

User Role: Administrator

Adding stop phrases is a way to hide particular words from appearing in the Hot Topic Analytics topic map. The stop phrases are still retained in the Smart Analytics data set.

To add stop words for Hot Topic Analytics, follow these steps:

1. Click **System Administration** > **Ongoing Maintenance** > **Smart Analytics** > **Hot Topic Analytics**.
2. Click **Stop Phrase**. The **Configure Stop Phrase** page opens.
3. Add the words that you identified as stop phrases to this page. These words must exactly match the key words to find hot topics, and each word must start with a new line.

   Note

   Note You can't use wildcards in stop phrases.

4. Click **Save**. The system automatically saves the stop words in alphabetical order. Alternatively, you can click **Save & Exit** to close the **Configure Stop Phrase** page.
5. Run Hot Topic Analytics again. Now, those stop phrases are no longer displayed in the topic map.

3.2.17. Configure user assess to stop phrases for Hot Topic Analytics

Hot Topic Analytics provides many out-of-the-box stop words. Only the authorized Smart Analytics users can add some stop phrases to an analysis by using the **Add Stop Phrase** button on a Hot Topic Analytics form.

Follow these steps to configure users' assess to stop words:

1. Click **Tailoring** > **Script Library**.
2. Type IDOL_Help in the Name field, and then click **Search**.
3. Locate the **canAddStopWords** function.
4. Configure a proper capability word in the line. For example, `return system.functions.index("SysAdmin", vars['$lo.ucapex'])>0;` .
   The default configuration is `SysAdmin` .
5. Click **Save** and **OK**.

3.2.18. Configure asynchronous analysis

Hot Topic Analytics supports asynchronous analysis, which means that HTA will run in the background and allows you to perform other tasks in the Service Manager system. This feature helps to avoid an unresponsive system especially when you have huge volumes of data for analysis. You can decide to run HTA asynchronously or synchronously according to the actual needs.

Follow these steps to enable asynchronous analysis:

1. Click **System Administration** > **Ongoing Maintenance** > **Smart Analytics** > **Hot Topic Analytics**.
2. Select a module in the Current Configuration List, and then click the **Advanced** tab.
3. Select the **Analyze Asynchronously** checkbox.
4. Click **Save & Exit**.
5. Add a SMIS (Integration Instance Manager) scheduler.

1. Open SMIS page.
2. Click **Add**.
3. Select **HTALDA** in the Integration Template drop-down list.
4. Fill in the mandatory fields. In particular, you must configure the **Interval Time (s)** field to control the running frequency of the back ground HTA job.
5. Accept other default values, and then click **Finish** to create the scheduler.

Tip

Tip If you don't select the **Analyze Asynchronously** checkbox and use the synchronous analytics mode, you can test in your own environment to find a proper number which can have an acceptable response time.

3.2.19. Work with Smart Ticket tuning

The company provides a Smart Ticket tuning toolkit to assist in tuning your Smart Ticket configuration and improving the Smart Ticket classification accuracy. This toolkit includes a data distribution analysis tool that can provide reports on your data distribution and a visualization tool that can generate a confusion matrix based on a set of sample test data. Additionally, this toolkit can help you achieve better results by modifying certain IDOL configuration parameters from the Smart Ticket configuration UI.

To work with the Smart Ticket tuning toolkit, refer to the following topics:

Use the data distribution analysis tool

View data distribution analysis results

Configure the best Source Data Coverage percentage

Prepare your training samples

Use the Confusion Matrix tool

Access Confusion Matrix data

Understand Confusion Matrix data

How to use Confusion Matrix

New configurations

Updated configurations

Move trained intelligence from staging to production

3.2.19.1. Use the data distribution analysis tool

In a Smart Ticket configuration, on the **Training Optimization** tab, the **Source Data Coverage** parameter has a default value of 90, which means 90% of the total source data records are used to build a classification system.

The default value of 90 is the recommended value, which means with 90 percent of the training samples, the system can capture the best description of your data to build the intelligence, which will be used to categorize new tickets later. However, it may not always work well for all customer scenarios, and this is where the data distribution analysis tool can help. The data distribution analysis tool can help you visualize your tickets distribution among all of your categories, and render the results in descending order.

3.2.19.1.1. View data distribution analysis results

Follow these steps to view data distribution results:

1. Go to **System Administration** > **Ongoing Maintenance** > **Smart Analytics** > **Smart Ticket**.
2. Click **Search**.

3. Select a Smart Ticket task from the list, and then click **View Distribution**.

The following figure shows an example based on the out-of-box data.

#	Category	Subcategory	Area	Amount	Accumulation
1	incident	performance	performance degradation	21	15.22%
2	incident	hardware	hardware failure	19	28.99%
3	incident	failure	function or feature not working	19	42.75%
4	incident	failure	error message	13	52.17%
5	incident	failure	job failed	12	60.87%
6	incident	access	authorization error	7	65.94%
7	complaint	service delivery	functionality	6	70.29%
8	complaint	service delivery	performance	6	74.64%
9	incident	failure		6	78.99%
10	incident	security	security event/message	6	83.33%
11	incident	access	login failure	3	85.51%
12	incident			2	86.96%
13	complaint	service delivery	availability	2	88.41%
14	incident	data	storage limit exceeded	2	89.86%
15	incident	facilities	supplies	2	91.30%
16	incident	security	virus alert	2	92.75%
17	incident	performance	system or application hangs	2	94.20%
18	incident	security	security breach	1	94.93%
19	request for change	service portfolio	upgrade / new release	1	95.65%
20	service request	Request for Administration		1	96.38%
21	complaint	service delivery	availabilty	1	97.10%
22	incident	data	data or file corrupted	1	97.83%
23	incident	data	data or file incorrect	1	98.55%

The columns of **Category**, **Subcategory**, and **Area** are different levels in the category hierarchy of the interactions in the OOB system, the "Amount" column is the number of tickets of each category, and the column of "Accumulation" is the accumulated percentage of interactions of the total records.

We can see that 90% means the categories of the first 14 rows constitute the majority of the OOB interactions. So, with this data distribution analysis tool, you can get a better understanding of your tickets, which can in turn help you better train your Smart Analytics system from the following perspectives.

You can perform the following tuning tasks based on your data distribution results.

3.2.19.1.2. Configure the best Source Data Coverage percentage

In the visualized results, you can have a clear understanding of which of the 90% training samples are mapped to which categories and how many. And you can adjust this configuration according to your own situation. Take the previous screenshot of the OOB data for example, we can find, from row 6 to row 14, those categories don't contain many tickets. We can regard them as minor categories, so we can set the "Source data coverage" to 61%. In this way, we can cover the major categories and reduce noises as well.

3.2.19.1.3. Prepare your training samples

If some of your categories are too similar, they're not good training samples. For instance, “laptop” and “desktop” are too similar to be distinguished from each other. However, in most customer systems, they are normally treated as two different categories and most likely linked to the same supporting group. Therefore, training the system with this kind of data will lead to low accuracy test results, but has no impacts on the ticket assignment.

For such situations, we recommend you consider combining these similar categories before starting training.

3.2.19.2. Use the Confusion Matrix tool

When working with Smart Ticket, you may want to know more about how Smart Analytics works. The most frequently asked questions are: Why our tickets get classified into certain categories? Why our Smart Ticket has a lower testing accuracy? How do I tune Smart Ticket to get a better categorization accuracy?

The Confusion Matrix tool aims to help you understand what happens under the hood, and to guide you to further tune Smart Ticket to achieve a satisfactory categorization accuracy.

A Confusion Matrix is basically a table to show users one specific test result. You can find out which categories are good or bad in this test, and then tune the system accordingly.

3.2.19.2.1. Access Confusion Matrix data

The Confusion Matrix tool analyzes data against a Smart Ticket task. Follow these steps to view confusion matrix data for a Smart Ticket task:

1. Go to **System Administration** > **Ongoing Maintenance** > **Smart Analytics** > **Smart Ticket**.
2. Select a Smart Ticket task from the list, and then click **Visualize**.
   If your Service Manager is in the multi-company mode and you have added multiple companies to the Smart Ticket task, you need to select a company after you click **Visualize**.

   Note

   Note
   This Visualize functionality is available only after you run a test against the categorizer (Smart Ticket task).

3. If needed, modify the following settings for Confusion Matrix.

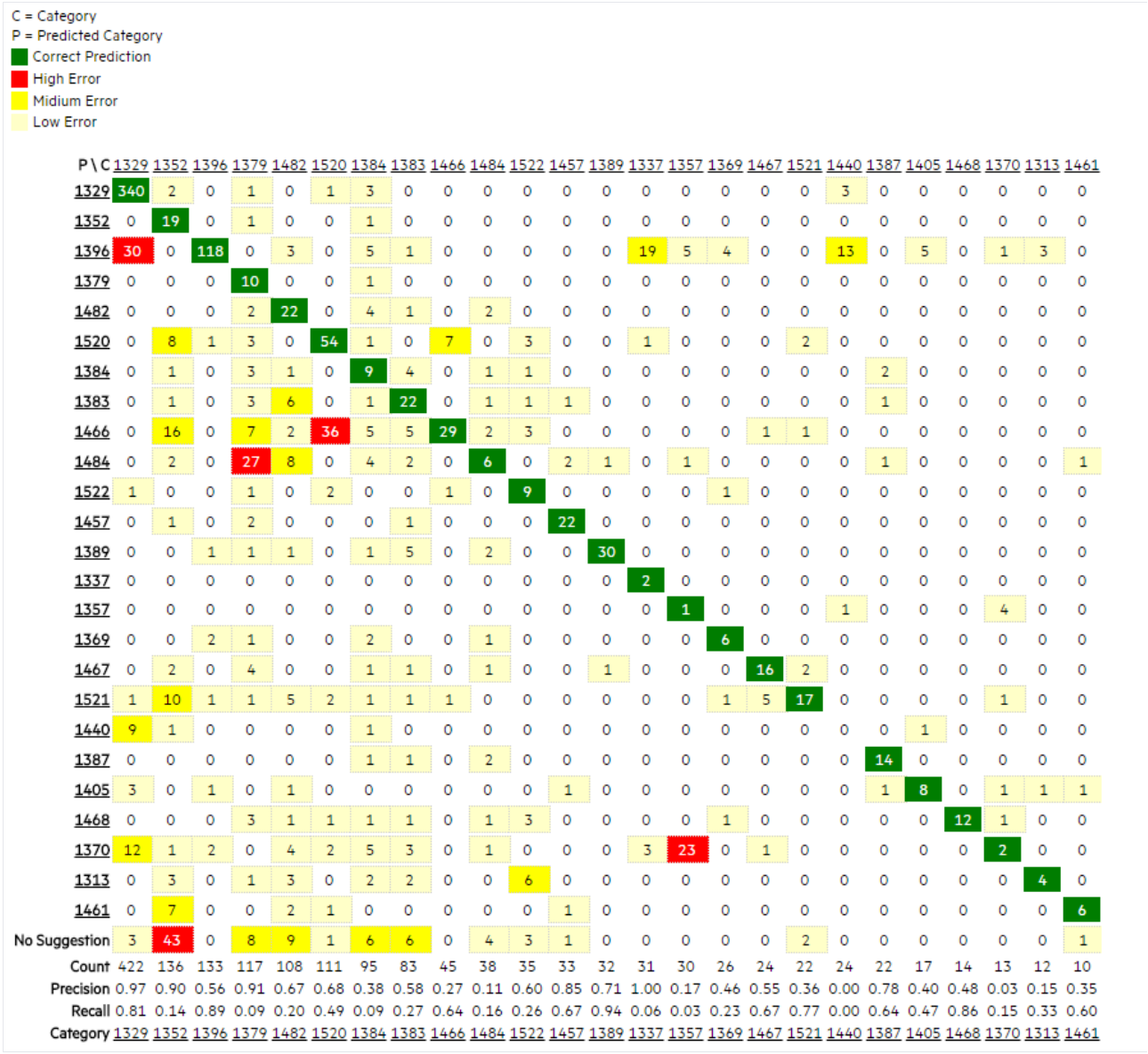
Field	Description
Top N Categories	This field indicates how many categories you want to analyze with the Confusion Matrix tool. For example, if you select a value of 10, only the top 10 categories in your data distribution analysis report will be analyzed. **Caution** Caution Selecting a large number of categories may lead to slow page rendering. Normally, you need to take care of your major categories only. **Note** Note Both the row number and the column number in a Confusion matrix are less than the minimum value of the current predictable category count and the **Top N Categories** value. If the previous testing version has unpredictable categories, the number of rows and columns in the Confusion Matrix based on the previous test result will not match the **Top N Categories** settings. For example, there are 13 categories in the first round testing result. The system administrator changes the data source coverage from 90 to 50, and then only 7 categories are predictable. Consequently, after training and testing, the system displays a 7*7 Confusion Matrix table for both the first round and the second round testing result even though the system administrator sets the **Top N Categories** value to 13.
Display Format	There are two options: • **By Error Count:** Show the confusion matrix data by number of error tickets. • **By Error Ratio (%):** Show the confusion matrix data by test error ratio (percentage).

Medium Error Threshold	This is the minimum error count or error ratio (depending on the display format) that you want to assign a medium severity. **Note** Note In the Confusion Matrix table, numeric values between this field value and the High Error Threshold value are highlighted in yellow.
High Error Threshold	This is the minimum error count or error ratio (depending on the display format) that you want to assign a high severity. **Note** Note In the Confusion Matrix table, numeric values no less than this field value are highlighted in red.

4. Click **Confusion Matrix** to view the Confusion Matrix results.

3.2.19.2.2. Understand Confusion Matrix data

The following figure shows an example Confusion Matrix.



P \ C	1329	1352	1396	1379	1482	1520	1384	1383	1466	1484	1522	1457	1389	1337	1357	1369	1467	1521	1440	1387	1405	1468	1370	1313	1461
1329	340	2	0	1	0	1	3	0	0	0	0	0	0	0	0	0	0	0	3	0	0	0	0	0	0
1352	0	19	0	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1396	30	0	118	0	3	0	5	1	0	0	0	0	0	19	5	4	0	0	13	0	5	0	1	3	0
1379	0	0	0	10	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1482	0	0	0	2	22	0	4	1	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1520	0	8	1	3	0	54	1	0	7	0	3	0	0	1	0	0	0	2	0	0	0	0	0	0	0
1384	0	1	0	3	1	0	9	4	0	1	1	0	0	0	0	0	0	0	0	2	0	0	0	0	0
1383	0	1	0	3	6	0	1	22	0	1	1	1	0	0	0	0	0	0	0	1	0	0	0	0	0
1466	0	16	0	7	2	36	5	5	29	2	3	0	0	0	0	0	1	1	0	0	0	0	0	0	0
1484	0	2	0	27	8	0	4	2	0	6	0	2	1	0	1	0	0	0	0	1	0	0	0	0	1
1522	1	0	0	1	0	2	0	0	1	0	9	0	0	0	0	1	0	0	0	0	0	0	0	0	0
1457	0	1	0	2	0	0	0	1	0	0	0	22	0	0	0	0	0	0	0	0	0	0	0	0	0
1389	0	0	1	1	1	0	1	5	0	2	0	0	30	0	0	0	0	0	0	0	0	0	0	0	0
1337	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0
1357	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	0	0	0	4	0	0
1369	0	0	2	1	0	0	2	0	0	1	0	0	0	0	0	6	0	0	0	0	0	0	0	0	0
1467	0	2	0	4	0	0	1	1	0	1	0	0	1	0	0	0	16	2	0	0	0	0	0	0	0
1521	1	10	1	1	5	2	1	1	1	0	0	0	0	0	0	1	5	17	0	0	0	0	1	0	0
1440	9	1	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0
1387	0	0	0	0	0	0	1	1	0	2	0	0	0	0	0	0	0	0	0	14	0	0	0	0	0
1405	3	0	1	0	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	8	0	1	1	1
1468	0	0	0	3	1	1	1	1	0	1	3	0	0	0	0	1	0	0	0	0	0	12	1	0	0
1370	12	1	2	0	4	2	5	3	0	1	0	0	0	3	23	0	1	0	0	0	0	0	2	0	0
1313	0	3	0	1	3	0	2	2	0	0	6	0	0	0	0	0	0	0	0	0	0	0	0	4	0
1461	0	7	0	0	2	1	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	6
No Suggestion	3	43	0	8	9	1	6	6	0	4	3	1	0	0	0	0	0	2	0	0	0	0	0	0	1
Count	422	136	133	117	108	111	95	83	45	38	35	33	32	31	30	26	24	22	24	22	17	14	13	12	10
Precision	0.97	0.90	0.56	0.91	0.67	0.68	0.38	0.58	0.27	0.11	0.60	0.85	0.71	1.00	0.17	0.46	0.55	0.36	0.00	0.78	0.40	0.48	0.03	0.15	0.35
Recall	0.81	0.14	0.89	0.09	0.20	0.49	0.09	0.27	0.64	0.16	0.26	0.67	0.94	0.06	0.03	0.23	0.67	0.77	0.00	0.64	0.47	0.86	0.15	0.33	0.60
Category	1329	1352	1396	1379	1482	1520	1384	1383	1466	1484	1522	1457	1389	1337	1357	1369	1467	1521	1440	1387	1405	1468	1370	1313	1461

A Confusion Matrix is a two-dimensional matrix: each column header is a true Category ID and each row header is a predicted Category ID. Each intersection cell lists the number (or ratio) of records that actually belong to the column category but are predicted as the row category.

Each value on the diagonal line indicates the number (or ratio) of correctly predicted tickets for this category, and is highlighted in green.

Each of other values indicates the number (or ratio) of incorrectly predicted tickets for the category, and is highlighted in light yellow, dark yellow, or red, based on the **Medium Error Threshold** and **High Error Threshold** values that you have defined.

You can find the following information from the header of Confusion Matrix form.

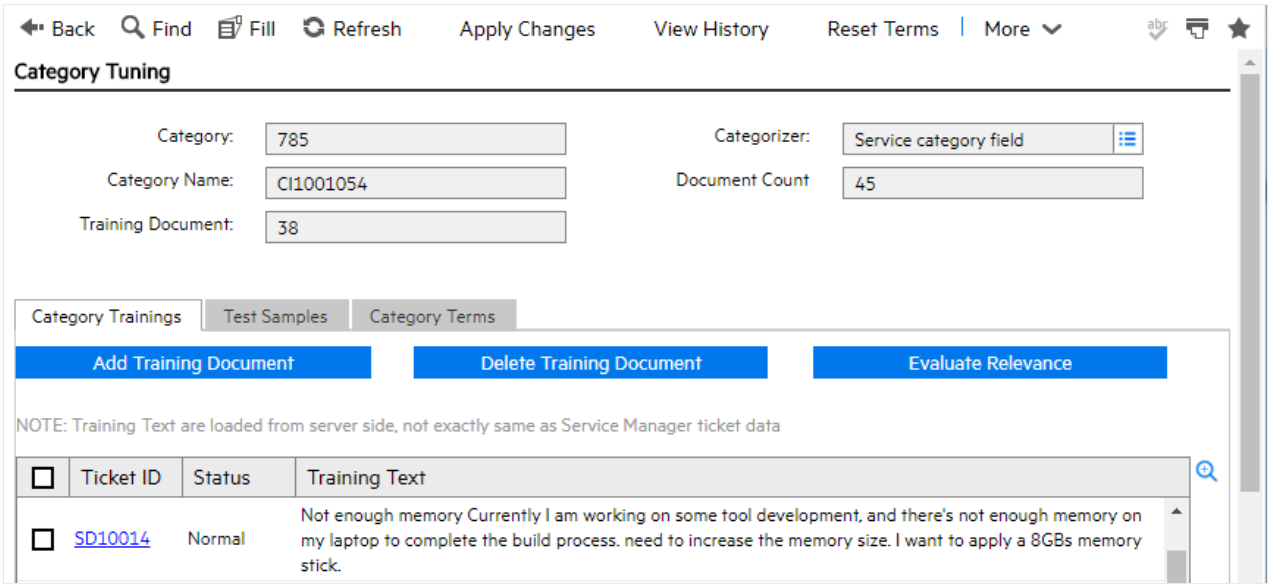
Item	Description
Comment	In this field, you can add comments for this test round.
Test Time	Timestamp of this test round.
Test Result	The accuracy results of this test round, including the following: • **First hit count**: number of tickets that succeeded in the first recommended category **Correctness Ratio**: first hit count/total number of test samples (%) • **Second hit count**: number of tickets that succeeded in the second recommended category **Correctness Ratio**: sum of first and second hit counts/total number of test samples (%) • **Third hit count**: number of tickets that succeeded in the third recommended category **Correctness Ratio**: sum of the first, second and third hit counts/total number of test samples (%) • **Missed count**: number of tickets that have no recommended categories **Missed Ratio**: Missed count/total number of test samples (%)

Also, at the bottom of the Confusion Matrix table, there are several other things, which are aligned to each category:

Item	Description
No suggestion	This is the number of tickets in this category that the system did not provide a suggested category.
Count	This is the total number of test tickets in this category, including successful, failed, and no suggestion ones.
Precision	A precision value indicates the ratio of the number of correctly predicted records in a row (the green number in a row) to the sum of all predicted records in the same row (sum of the row).
Recall	A recall value indicates the ratio of correctly predicted records (the green number in a column) to the sum of all records in this category (sum of the column).

Category Tuning form

In a Confusion Matrix, category IDs in both the column headers and the row headers are clickable. You can click a category ID in a Confusion Matrix to open a Category Tuning form.



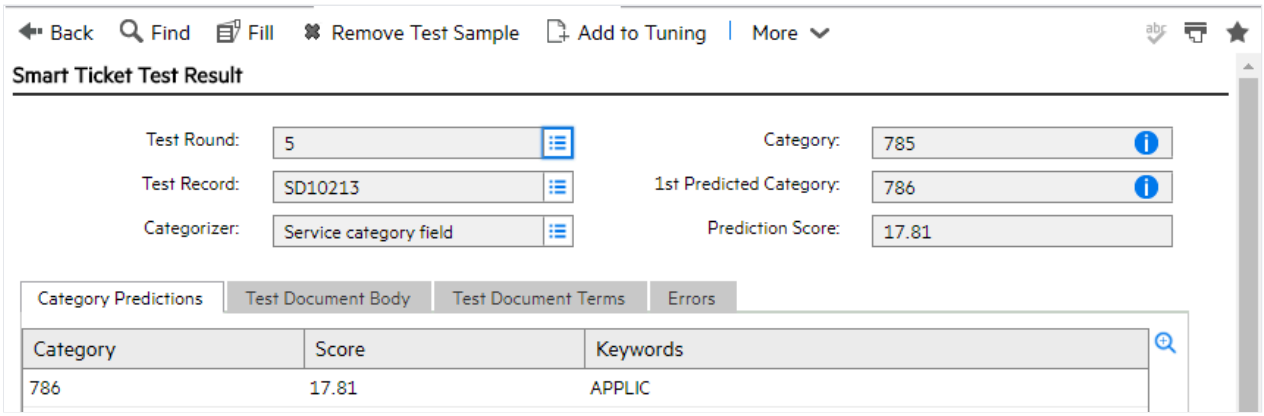
This form contains the following fields or tabs.

Field / Tab	Description
Category	The ID of the current category.
Categorizer	The Smart Ticket task that's used for this test round.
Category Name	The name of the current category.
Document Count	The total number of documents within this category.
Training Document	The number of documents used to train this category.
Category Trainings	In this tab, you can find all training documents for this category, as well as their ticket ID, status and training text. • You can use the **Add Training Document** button to add more documents or the **Delete Training Document** button to delete existing documents, and then click the **Apply Changes** button to tune this category. • The Confusion Matrix tool can help you decide which tickets should be deleted by calculating the relevance of chosen tickets against the current category. You can select training tickets and click the **Evaluate Relevance** button to get the results.
Testing Samples	This tab contains all tickets used to test this category, as well as their ticket ID, title, and body. • You can use the **Add Test Sample** button to add more documents to trigger a new round of test. • You can also use the **Delete Test Sample** button to delete any existing test samples that are appropriate for this category.

Category Terms	This tab contains all terms related to the current category, and their weights. • You can add new terms to the system with a specified weight by using the **Add Term** button. • You can modify the weight for existing terms by specify a new weight in the **Modified Term Weight** column and the clicking **Save Term Weights**. Such modifications will adjust the classification results regarding this category. • You can use the **Reset Terms** button to revert the weights of all terms to their generated term weights. This will revert your manual term weight changes. • You can choose one term and click the **View Selected Term Occurrence** button to see the exact places where this term occurs. • You can use the **View History** button to view all category tuning snapshots of the current category. **Note** Note • The **Weight** column indicates the current weight of the term, the **Generated Term Weight** column indicates the system-generated weight for the term, and the **Modified Term Weight** column allows you to specify a new weight for the term. • Term weights may change accordingly after training with new training samples; however, if you adjust a term weight manually, it becomes a fixed value and will not be automatically changed. • If a term is a stop word after stemming, the system displays the following error message when you click this term: `No valid query text supplied All terms were invalid, through being stop words, too short, or incorrectly formatted .`

Smart Ticket Test Result form

From within the confusion matrix table, if you hover over a non-zero number, a tip message is displayed to explain the expected category and actual recommended category in this test round. Clicking the number will direct you to the Smart Ticket Test Result form:



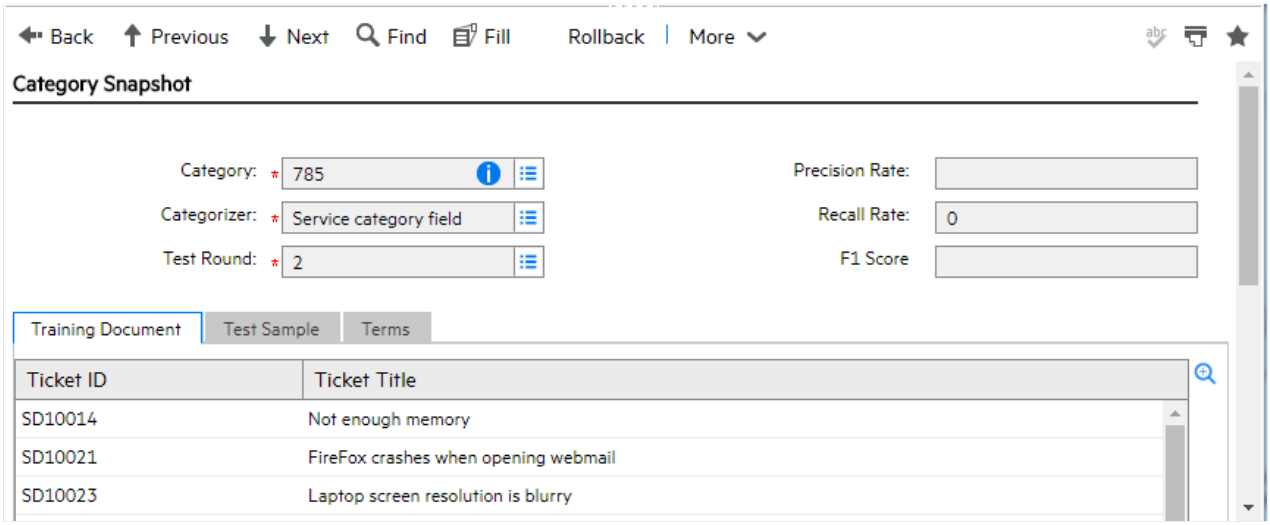
The first half of this page is a list of test documents, and the second half is the detail form of the selected test document in the list.

The test document detail form provides the following information.

Field	Description
Test Round	The ID of this test.
Test Record	The ID of the current test document.
Categorizer	The name of the categorizer (Smart Ticket task) used for the current test.
Category	The expected category for the current test document.
Predicted Category	The actual recommended category for the current test document.
Prediction Score	The confidence score of this prediction.
Category Predictions	This tab contains a list of all predicted categories from Smart Ticket for the current test document, sorted by prediction score. You can also find the key words used to match the current test document to each predicted category.
Test Document Body	This tab contains the test document body text. Out of box, the text comes from the title and description of the ticket.
Test Document Terms	This tab contains a list of terms extracted from the current test document, as well as their following information: • Weight: term weight • Document Occurrences: number of occurrences in the current document • Total Occurrences: total number of occurrences in the system
Errors	This tab contains log information if any errors occur during the test.

Category Snapshot form

Click the **View History** button on the Category Tuning form to drill down to the Category Snapshot form.



On this form, you can find all of the tuning rounds for the current category that are recorded in the system. You can choose the best round and click the **Rollback** button to change the category training state to that tuning round.

> **Note**
>
> Note The rollback action affects the current selected category only, and you can only roll back the modifications made by the system administrator. If an agent uses the **Add to Tuning Records** option to add a record to the training sample and the system administrator has applied this change, this modification can't be rolled back.

3.2.19.2.3. How to use Confusion Matrix

Training document management[edit]

Training documents are the source of the intelligence Smart Analytics learns from. That said, the better you choose the training documents, the higher categorization accuracy you will achieve. Good training document management will eventually improve your IT help desk working efficiency.

In the Category Tuning form, the Category Trainings tab displays all of the documents/tickets that the system uses to train the current category.

If the categorization accuracy is unacceptable, you can do the following:

1. Check the Confusion Matrix table to identify the test values highlighted in red, and perform the following steps for each corresponding expected category.
2. Click the corresponding expected category to drill down to the category tuning page.
3. Review all current training documents. If you find any unreasonable training documents, delete them by clicking the **Delete Training Document** button to improve the training quality. If you find there aren't adequate number of training documents for the category, add more by using the **Add Training Document** button to tune this category.
4. Click **Apply Changes** to make your training changes take effect.

Testing data management[edit]

The Confusion Matrix tool can keep all your test tickets, which enqables you to review the test accuracy results and then troubleshoot the causes of bad test results.

The following is one possible scenario:

1. Find one red value in the Confusion Matrix.
2. Click the expected category in the same column where the red value is located to drill down to the Category Tuning page for the expected category.
3. Click the **Test Samples** tab to review all of the test documents for the current category and identify those unreasonable test tickets. In reality, there might be some tickets mistakenly categorized by IT agents, and these tickets aren't appropriate for testing the current category.
4. Select these unreasonable tickets and then click **Delete Test Sample** to delete them.

Another way to identify those bad test tickets is as follows:

1. Find one red value in the Confusion Matrix.
2. Click the predicted category in the same row where the red value is located to drill down to the “Smart Ticket Test Result” page, where you can see a whole list of test tickets that failed the test.
3. Review the detail information of these test tickets.
4. If a test ticket demonstrates some characteristics that are unknown to the system, click the **Add to Tuning** button to add this ticket to the training sample pool.
5. If you find that a test ticket was put in this category by mistake, click the **Remove Test Sample** button to delete it to improve the test accuracy.

Term management[edit]

Sometimes you may know the typical terms for one specific category. In this case, you can add these terms directly to the system and bind them to this category.

It may also happen that when you review the terms for one specific category in the **Category Terms** tab, you know exactly that some terms should not have been related to this category. In this case, you can change its weight to 0 to remove it from the categorization process.

Category similarity analysis[edit]

With the Confusion Matrix tool, you can find out that some test errors are caused by the high similarity of some categories that the system can hardly distinguish.

Normally, if you notice some red blocks in the Confusion Matrix table, it means a category has a lot of test errors. You can then click those two relevant categories to drill down to the pages where you can find the terms that Smart Analytics extracted for them. Most likely, most of those terms are the same, which means the training samples that you provide to Smart Analytics for those two categories have very similar descriptions. You need to think about whether it is proper to keep them as two separate categories.

Contents

- 1 Training document management
- 2 Testing data management
- 3 Term management
- 4 Category similarity analysis

3.2.19.3. New configurations

This section describes the new configurations introduced by the Smart Ticket tuning tool.

Suggestion Threshold[edit]

Smart Ticket classifies every ticket with a prediction score, which indicates how confident Smart Ticket is when making this prediction (suggestion).

The Suggestion Threshold setting is introduced for you to control whether Smart Ticket makes a suggestion based on the prediction score or not. If you have configured the Suggestion Threshold value, the system automatically adds the *Threshold* parameter to the `CategorySuggestFromText` action when making a prediction. All the returned categories include the related prediction scores (the weight value). If the prediction score of a predicted category is less than this threshold value, Smart Ticket will ignore this predicted category.

Follow these steps to configure this setting:

1. Go to **System Administration** > **Ongoing Maintenance** > **Smart Analytics** > **Smart Ticket**.
2. Select a task from the list.
3. On the **Configurations** tab, enter an integer between 0 and 100 in the **Suggestion Threshold** field.
4. Click **Save**.

The Suggestion Threshold value in Smart Ticket tasks is null by default, which means that the system will not add the *Threshold* parameter when making a prediction. In this case, the default value is 0.

Prefilled parent category testing[edit]

Note

Note The Prefilled checkbox is used only for testing Smart Ticket.

By default, Smart Ticket assigns Categories to generic support tickets, and uses this information in the prediction process. However, the Category, Subcategory, and Area fields of tickets submitted by users are often already filled. To more closely replicate the results of end-user prediction, you can select the first Prefilled checkbox so that Smart Ticket uses information entered by end-users instead of the automatically-assigned information. However, this will make the results of Smart Ticket prediction less accurate.

Follow these steps to improve test results:

1. Go to **System Administration** > **Ongoing Maintenance** > **Smart Analytics** > **Smart Ticket**.
2. Select a task from the list.
3. On the **Configurations** tab, select the **Prefilled** checkbox to the right of the **Level 1** and **Level 2** fields.
   If Subcategory is not prefilled, don't select the checkbox for Level 2.

Contents

- 1 Suggestion Threshold
- 2 Prefilled parent category testing

3.2.19.4. Updated configurations

This section describes the configuration changes that you need to make because of the introduction of this Smart Ticket tuning toolkit.

Transliteration

In the `content.cfg` file, remove the `Transliteration=false` line:

```
[LanguageTypes]
GenericTransliteration=TRUE
```

Define the removed line only in a specific language section if you don't want that language to use "transliteration".

You can find a detailed description of this configuration in the following document:

`<Smart Analytics install directory>/IDOL/help/IDOLServerReference#Configuration/LanguageTypes/Languages/Transliteration.html`

Stemming

In the `content.cfg` file, use the following configuration for stemming:

```
GenericStemming=true
GenericFullStemming=true
```

You can find a detailed description of this configuration in the following document:

`<Smart Analytics install directory>/IDOL/help/IDOLServerReference#Configuration/LanguageTypes/Languages/Stemming.html`

Propername

The `ProperNames` configuration parameter controls whether terms are created from pairs of consecutive words in index fields.

In the `content.cfg` file, configure the following parameter:

```
ProperNames
```

You can find a detailed description of this configuration in the following document:

`<Smart Analytics install directory>/IDOL/help/IDOLServerReference#Configuration/LanguageTypes/Languages/ProperNames.html`

Hyphenchars and AugmentSeparators

In the `content.cfg` file, you can configure how Smart Analytics should handle hyphens. Configure the following parameters:

- `HyphenChars`
- `AugmentSeparators`

You can find detailed descriptions of this configuration in the following documents:

- `<Smart Analytics install directory>/IDOL/help/IDOLServerReference#Configuration/LanguageTypes/Languages/Hyphenchars.html`
- `<Smart Analytics install directory>/IDOL/help/IDOLServerReference#Configuration/LanguageTypes/Languages/AugmentSeparators.html`

Number index

You can find a detailed description of this configuration in the following document:

`<Smart Analytics install directory>/IDOL/help/IDOLServerReference#Configuration/LanguageTypes/Languages/IndexNumbers.html`

3.2.19.5. Move trained intelligence from staging to production

For information on how to perform this task, refer to Transfer Smart Analytics intelligence.

3.3. Smart Analytics best practices

This section provides the following best practices on how you can configure and use Smart Analytics so that it brings more value to your business:

Improving accuracy for Smart Ticket

Improve Smart Analytics query performance

3.3.1. Improve accuracy for Smart Ticket

Smart Analytics provides several methods to help you to increase Smart Ticket accuracy according to different data sets. This section provides some best practices for you to improve the accuracy of Smart Ticket.

- Set up data cleansing rules
  Data cleansing can help you prepare the data that you want to send to Smart Analytics for indexing, training, and analyzing. By setting up proper data cleansing rules, you can have better data quality, which is critical to the best accuracy of auto-suggestion. To set up data cleansing rules, see Configure data cleansing.
- Choose best sample data
  In the definition for Smart Ticket or Hot Topic Analytics, you can specify a sample data query, through which you can decide what kind of data that you want to use as sample data to teach Smart Analytics to build the intelligence out of your large data volume.
  For example, you may have an "Other" category in your Service Manager implementation to accommodate the interactions for which the Service Desk agents can't find a better, more accurate category. Normally, the interactions in this "Other" category aren't considered as good sample data for Smart Ticket. We recommend that you add a filtering clause such as `category~="Other"` into the **Training Sample Query** field to exclude those records.
  Another example is that you can choose the records logged by Subject Matter Experts (SME) as the training samples for Smart Tickets. In this way, you can have sample data with better quality for categorization.
- Apply rule based training for Smart Ticket
  The basic training of Smart Ticket is meaning based training, which means Smart Analytics builds its intelligence based on the text information of your data. On top of meaning based intelligence, Smart Analytics also supports you to add rule based training to the Smart Ticket. Those rules will further increase the suggestion accuracy, especially in the case that multiple suggestion results have the same relevancy with the new record. The typical scenario is that if one particular record has the same relevancy within several categories, you can append a rule to one specific category to improve the categorization accuracy.
  For how to apply a rule to the Smart Ticket task definition, see Apply a rule-based training.
- Optimize your training for Smart Ticket
  Several advanced parameters defined in the Smart Ticket task definition are used to optimize the accuracy of auto suggestion. Note that these settings are tradeoffs between training time and accuracy, which means higher accuracy is achieved at the cost of longer training time. Listed below are some best practices for these optimization configurations.
  - Training by documents or training by terms
    Choose “best term” for a faster training process if you have huge data volume; choose "training documents" for a higher accuracy with a slower training process.
  - Training sample per category
    The maximum records for each category, normally more training sample per category leads to higher accuracy but longer training time.
  - Source data coverage
    The percentage of records out of the total source data that are used to create categories. Normally higher percentage means higher accuracy, but there is a threshold point. When the training source data percentage exceeds the threshold, the margin contribution will be lowered remarkably. The out-of-box value for this configuration is 90%, which is a best number tested in the lab. You can use the “source data coverage calculator” tool to find the best number for your data set.
  - Document weight and term weight
    Enable "Adjust term weight from test result" to automatically adjust the term weight for some terms in some categories based on the testing result, which may help improve accuracy.
    Enable "Remove low weight document" to help reduce the disturbance of low relevance training samples and improve accuracy.
    By default, these two parameters are disabled in the out-of-box environment.
    These advanced features need your experiment to get best results. You may enable either one or both.
- Perform tuning periodically

Tuning the training result is a mechanism to continuously improve the accuracy of auto suggestion. For information about how to tune the training result, see Perform tuning in the Smart Ticket definition.

3.3.2. Improve Smart Analytics query performance

User Role: Administrator

Caching some match fields into memory can accelerate the query speed of Smart Analytics. To achieve this, you can modify the configuration files and corresponding sections of the content servers for which you want to accelerate the query speed.

To configure cache settings for match fields, follow these steps:

1. Open the `<SmartAnalytics>/Content<x>/Content<x>.cfg` files of the content servers for which you want to accelerate the query speed.
2. Configure the cache settings for match fields as follows:

```
[server]
NodeTableCacheFields=True
NodeTablePrepopulateCache=True
//-------------------------- Cache Settings --------------------------------//
//Maximum size of the field cache (in KB). Increasing the size of the cache can improve
performance. However, you must ensure the host machine has sufficient available memory.
[MemoryCache]
EvictWhenFull=True
MaxSize=1024000
[FieldProcessing]
23=SetCachedFields
[SetCachedFields]
Property=CachedFields
PropertyFieldCSVs=*/*_MATCH,*/IMPORTMAGICEXTENSION
//-------------------------Properties-----------------------------//
[CachedFields]
MemCachedType=True
```

3. Restart the content servers for which you modified the configuration files.

Note

Note It takes a while for the content servers to get restarted. For example, it takes about 35 minutes if you restart content servers on an environment with 1G memory cache. You can monitor the restart status by checking the `<Smart Analytics>/Content<x>/logs/application.log` file. If the caching is successful, you can see log information similar to the following:

```
22/09/2016 16:45:50 [1] 30-Normal: Indexing nodetable memory information for 2268213
documents.
22/09/2016 16:46:26 [1] 30-Normal: Indexing information from documents 10% complete.
22/09/2016 16:47:35 [1] 30-Normal: Indexing information from documents 20% complete.
22/09/2016 16:49:31 [1] 30-Normal: Indexing information from documents 30% complete.
22/09/2016 16:53:39 [1] 30-Normal: Indexing information from documents 40% complete.
22/09/2016 16:58:10 [1] 30-Normal: Indexing information from documents 50% complete.
22/09/2016 17:02:05 [1] 30-Normal: Indexing information from documents 60% complete.
22/09/2016 17:06:28 [1] 30-Normal: Indexing information from documents 70% complete.
22/09/2016 17:10:51 [1] 30-Normal: Indexing information from documents 80% complete.
22/09/2016 17:15:11 [1] 30-Normal: Indexing information from documents 90% complete.
22/09/2016 17:19:34 [1] 30-Normal: Populated nodetable memory cache.
22/09/2016 17:19:34 [1] 30-Normal: Writing out stateinfo
22/09/2016 17:19:34 [1] 00-Always: Initialized memory.
```

To improve the Smart Search response performance, you need to update the following configuration files in all content servers and the Smart Analytics Main server:

- <SmartAnalytics>/Content1/Content1.cfg
- <SmartAnalytics>/Content2/Content2.cfg
- <SmartAnalytics>/Content3/Content3.cfg
- <SmartAnalytics>/IDOL/IDOLServer.cfg
- Scaled content server

Follow these steps:

1. Update the existing parameters as highlighted in bold, or add the highlighted parameters if they don't exist:
   - `IndexCacheMaxSize=512000`
     Sample code:

     ```
     //--------------------------- Cache Settings -----------------------------//
     [IndexCache]
     IndexCacheMaxSize=512000
     ```

   - `PropertyFieldCSVs=*/*_MATCH,*/EDK_*,*/*_MATCH_INDEX, */*_MATCH_WEIGHT*,*/*_BOOLEAN,*/IMPORTMAGICEXTENSION`
     Sample code:

     ```
     [SetMatchFields]
     // Specify fields whose entire value is frequently used as a restriction in a search and are
     thus
     // stored for optimised matching.
     // NB Such fields can't be numeric and can only be parametric if
     parametricnumericmapping=false.
     Property=MatchFields
     PropertyFieldCSVs=*/*_MATCH,*/EDK_*,*/*_MATCH_INDEX,
     */*_MATCH_WEIGHT*,*/*_BOOLEAN,*/IMPORTMAGICEXTENSION
     ```

2. Locate the `[DateFields]` section under the `Properties` separation line, and then add a new parameter `Index=TRUE`:

   ```
   //---------------------------Properties------------------------------//
   [IndexFields]
   Index=TRUE
   [IndexWeightFields2]
   Index=TRUE
   Weight=2
   [IndexWeightFields4]
   Index=TRUE
   Weight=4
   [IndexWeightFields8]
   Index=TRUE
   Weight=8
   [IndexWeightFields16]
   Index=TRUE
   Weight=16
   [SectionFields]
   SectionBreakType=TRUE
   [DateFields]
   DateType=TRUE
   Index=TRUE
   ```

3. Restart the content server and the IDOL server.
4. Perform a full re-index for Smart Search.

To improve Smart Search performance, you need to update the following configuration files:

- <SmartAnalytics>/Content1/Content1.cfg
- <SmartAnalytics>/Content2/Content2.cfg
- <SmartAnalytics>/Content3/Content3.cfg
- Scaled content server

Follow these steps:

1. Update the `Schedules` section as follows:

```
//-------------------------- Schedules ----------------------//
//expire and compact data everyday
[Schedule]
Expire=TRUE
ExpireTime=23:00
ExpireInterval=24:00
Compact=True
CompactTime=00:00
CompactInterval=24:00
```

2. Restart the content servers.

3.4. Smart Analytics tailoring

This section provides the following tailoring best practices for Smart Analytics:

Extend Smart Ticket to other modules

Execute OCR in other processes

Implement OCR in a field other than the Description field

Add filters to the Hot Topic Analytics form

Enable Hot Topic Analytics for other modules

Customize actions in hot topic map

Configure Smart Analytics to support more languages

3.4.1. Extend Smart Ticket to other modules

User Role: Administrator

In the out-of-box system, Smart Ticket (auto-classification) is only enabled for the Service Desk module. If you want to use this feature in other modules, you can tailor your Service Manager system.

For example, if you want to automatically classify Change category with Process Designer enabled, follow these steps:

1. Define the Smart Ticket task for the Change Management module:
    1. From the System Navigator, click **System Administration** > **Ongoing Maintenance** > **Smart Analytics** > **Configuration** > **Smart Tickets**.
    2. Select **Blank**, and then click the **Add** button.
    3. In the **Add Smart Ticket Task** form, fill in the task name with any words meaningful to you.
    4. In the **Configuration** tab, choose **Change** as **Module Name**.
    5. In the **Training Sample Query** field, provide a query by using Query Builder. This query decides which Change records will be sent to Smart Analytics for Smart Ticket training.
    6. Enter the fields in the Change module that you want to be automatically filled by Smart Ticket. For example, category.

7. Enter the fields in the Change module that you want to use as the inputs for Smart Analytics to provide suggestions. For example, description.

2. Create new Smart Ticket rule sets for the Change workflow by referring to the following out-of-box Smart Ticket rule sets as examples.

Rule Set	Description
sd.idol.ocr	This rule triggers Optical Character Recognition (OCR) and auto-classification to fill the fields defined in the Smart Ticket definitions.
sd.idol.tuning.action	This rule sends the record to the tuning list.

Tip

Tip For information about how the design workflows and rule actions, refer to the Process Designer documents in the Service Manager help center.

3. Add the Smart Ticket rule sets that you created to the Change workflow:
   1. From **Change Management** > **Configuration** > **Change Workflows**, select the workflow to which you want to apply the Smart Ticket rule. For example, standard change.
   2. Select the **Registration and Categorization** phase.
   3. Click the **Rule Sets** tab in the property section.
   4. Select the **On Exit** event.
   5. Select the Smart Ticket rule that you created.
4. Create a new form or update the existing form to use the new Smart Ticket task for the Change Management module. You can refer to the "Create Smart interaction" form (idol.quick.new.interaction) as an example.

3.4.2. Execute OCR in other processes

User Role: Administrator

Optical Character Recognition (OCR) is an out-of-box feature for Smart Ticket, which can extract texts from images, and then put the text into the interaction records when submitting requests by using Smart Ticket.

If you want to execute OCR when proceeding with other processes such as escalating, saving, or closing records, you can refer to the following example to tailor your Service Manager.

The following example describes how to execute OCR when escalating an interaction to an incident in different Service Manager modes. You need to make changes accordingly if you want to execute OCR on other actions in different modules.

Execute OCR when escalating an interaction to an incident in the Service Manager Classic or Hybrid mode[edit]

To execute OCR when escalating an interaction to an incident in Service Manager Classic, follow these steps:

1. Add a new ScriptLibrary, such as IDOL_OCR_Esclation, as follows:

```
function processOCRNonPD(interaction, language) {
var ocrResult = lib.IDOL_Utilities.processOCR(interaction, 'description', language);
if (ocrResult) {
interaction.doSave();
system.functions.rtecall("refresh", vars['$L.errorcode'], interaction);
}
}
```

2. Update the "cc.first.log2" process:
    1. Open the "cc.first.log2" process, and then go to the **RAD** tab.
    2. Locate the "cc.save" RAD application.
    3. In the **Post RAD Expressions** section of the "cc.save" RAD application, add the following statement:
    `if ($L.exit="normal") then ($L.void=jscall("IDOL_OCR_Esclation.processOCRNonPD", $L.file, $G.my.language))`

Execute OCR when escalating an interaction to an incident in the Service Manager Codeless mode[edit]

To execute OCR when escalating an interaction to an incident in Service Manager Codeless, follow these steps:

1. Add a new ScriptLibrary, such as IDOL_OCR_Esclation, as follows:

```
function processOCRPD(interaction, language) {
var ocrResult = lib.IDOL_Utilities.processOCR(interaction, 'description', language);
if (ocrResult) {
interaction.doSave();
system.functions.rtecall("refresh", vars['$L.errorcode'], interaction);
}
}
```

2. Update the "sd.escalate" process:
    1. Open the "sd.escalate" process, and then go to the **RAD** tab.
    2. Locate the "se.view.engine" RAD application.
    3. In the **Post RAD Expressions** section of the "se.view.engine" RAD application, add the following statement:
    `if ($L.es.action={"added", "normal", "resetrec"}) then ($L.void=jscall("IDOL_OCR_Esclation.processOCRPD", $L.file, $G.my.language))`

Contents

- 1 Execute OCR when escalating an interaction to an incident in the Service Manager Classic or Hybrid mode
- 2 Execute OCR when escalating an interaction to an incident in the Service Manager Codeless mode

3.4.3. Implement OCR in a field other than the Description field

User Role: System Administrator

Optical Character Recognition (OCR) is an out-of-box feature for Smart Ticket, which can extract texts from images, and then put the text into the interaction records when you submit requests by using Smart Ticket.

In an out-of-box Service Manager system, the Smart Ticket OCR feature saves and displays the extracted image content in the **Description** field. The subsequent Smart Interaction suggestions are also based on the **Description** field. However, if you want to use a field other than the Description field for the Smart Interaction OCR feature, you can follow the instructions in this section to tailor your system.

To use a field other than the Description field for the Smart Interaction OCR feature, follow these steps:

1. Add your new field to the "incidents" table:
    1. From the System Navigator, click **Tailoring** > **Database Dictionary**.
    2. Search for and open the "incidents" table.
    3. Add a new field in the table. For example, you can use "description.ocr" as the name of the new field.

        Note

        Note
        To ensure the new field can store the strings extracted from images, we recommend that when you add the new field, select the array data type first, and then select the character data type.

    4. Save the table.
2. Add new Smart Ticket configuration tasks for your new field:
    1. From the System Navigator, click **System Administration** > **Ongoing Maintenance** > **Smart Analytics** > **Smart Ticket**.
    2. Delete the out-of-box "Standard category field" and "Service category field" configuration tasks.
    3. Add a new "Category" configuration, and then add "Description Ocr" to the **Content Fields** list on the **Configurations** tab.
    4. Add a new "Affected Service" configuration, and then add "Description Ocr" to the **Content Fields** list on the **Configurations** tab.
3. Add the new "description.ocr" field to the OCR related Java scripts. To do this, follow the steps that are appropriate for your environment:
   Service Manager Codeless
    1. From the System Navigator, click **Tailoring** > **Process Designer** > **Rule Sets**.
    2. Search for and clone the "sd.idol.ocr" rule set.

        Note

        Note The out-of-box rule sets and workflows are **Proprietary**, you need to clone the related rule sets and workflows, and then make changes as necessary so that your tailoring can take effect.

    3. From the cloned the rule set, open the OCR rule, and then add the *description.ocr* parameter as appropriate to the "lib.IDOL_SMIS.addOCRTask" JS call.

```
if (vars.$G_src && (system.vars.$lo_idol_img_enabled || system.vars.$lo_idol_enabled)) {
        if (vars['$G.multi']) {
        company = vars['$lo.operator']['company'];
        lib.IDOL_SMIS.addOCRTask(vars['$L.file'], 'OCR', company, 'description.ocr');
} else {
        lib.IDOL_SMIS.addOCRTask(vars['$L.file'], 'OCR', '', 'description.ocr');
}
} else {
        var attachments = record.getAttachments();
        var attLen = 0;
        for (var i = 0; i < attachments.length; i++) {
                if (lib.IDOL_Utilities.isImage(attachments[i])){
                        attLen++;
                        break;
                }
        }
        //call OCR interface
        if (vars.$G_ess && (system.vars.$lo_idol_img_enabled || system.vars.$lo_idol_enabled)
&& attLen > 0)
        {
        if (vars['$G.multi']) {
                company = vars['$lo.operator']['company'];
                lib.IDOL_SMIS.addOCRTask(record, 'OCR', company, 'description.ocr');
        } else {
                lib.IDOL_SMIS.addOCRTask(record, 'OCR', '', 'description.ocr');
        }
        } else if ((vars.$G_ess && (system.vars.$lo_idol_img_enabled ||
system.vars.$lo_idol_enabled) && attLen === 0)) {
                if (vars['$G.multi']) {
                company = vars['$lo.operator']['company'];
                lib.IDOL_SMIS.addOCRTask(record, 'AutoFill', company, 'description.ocr');
        } else {
                lib.IDOL_SMIS.addOCRTask(record, 'AutoFill', '', 'description.ocr');
        }
        }
}
```

4. Search for and clone the “sd.idol.ocr.action” rule set. Then, from the cloned rule set, in the Process OCR rule, add the “description.ocr” parameter to the “lib.IDOL_Utilities.processOCRFG” JS call.

```
lib.IDOL_Utilities.processOCRFG(record, 'description.ocr')
```

5. Make other changes as necessary so that the updated rule sets can take effect. For example, clone the related workflow, and then insert the updated rules sets to the cloned workflow.

Service Manager Classic or Hybrid

1. From the System Navigator, click **Tailoring** > **Document Engine** > **Processes**.
2. Search for and open the “cc.first.log” process.
3. Go to the **Final Javascript** tab.
4. Add the *description.ocr* parameter to the “lib.IDOL_SMIS.addOCRTask” JS call.

```
if (vars['$L.action'] == 'logcatalog'){
return;
}

if (vars.$lo_idol_img_enabled || vars.$lo_idol_enabled) {
//if is src, add to smis task
if (vars.$G_src) {
if (vars['$G.multi']) {
company = vars['$lo.operator']['company'];
lib.IDOL_SMIS.addOCRTask(vars['$L.file'], 'OCR', company, 'description.ocr');
} else {
lib.IDOL_SMIS.addOCRTask(vars['$L.file'], 'OCR', '', 'description.ocr');
}
return;
}
var attachments = vars['$L.file'].getAttachments();
var attLen = 0;
for (var i = 0; i < attachments.length; i++) {
if (lib.IDOL_Utilities.isImage(attachments[i])){
attLen++;
break;
}
}
//call OCR interface
if (vars.$G_ess && attLen > 0) {
if (vars['$G.multi']) {
company = vars['$lo.operator']['company'];
lib.IDOL_SMIS.addOCRTask(vars['$L.file'], 'OCR', company, 'description.ocr');
} else {
lib.IDOL_SMIS.addOCRTask(vars['$L.file'], 'OCR', '', 'description.ocr');
}
} else if (vars.$G_ess && attLen === 0) {
if (vars['$G.multi']) {
company = vars['$lo.operator']['company'];
lib.IDOL_SMIS.addOCRTask(vars['$L.file'], 'AutoFill', company, 'description.ocr');
} else {
lib.IDOL_SMIS.addOCRTask(vars['$L.file'], 'AutoFill', '', 'description.ocr');
}
}
}
```

4. Re-train the two new Smart Ticket configurations that you added in step 2.
5. When the training is finished, try the OCR related feature. The Smart Ticket OCR feature now saves and displays the extracted image content in the new **Description Ocr** field.

3.4.4. Add filters to the Hot Topic Analytics form

User Role: Administrator

In the out-of-box system, the Hot Topic Analytics form contains some pre-defined filter fields. If you want to add more filter fields, follow these steps:

1. From the System Navigator, click **System Administration** > **Ongoing Maintenance** > **Smart Analytics** > **Hot Topic Analytics**.
2. Select and open a Hot Topic Analytics configuration record from the configuration list. For example, Incident.
3. Go to the **Filter Fields** tab.
4. In the **Properties Fields** section, select fields that can be used for advanced filtering in Hot Topic Analytics. For example, you can define **Category** or **Priority** as filter.

Note: Start from 9.52p3 and 9.61 release, the following two filters are introduced so that HTA can generate a better Hot Topic Analytics map:

- **Min doc of topic**: Controls the minimal number of records in each topic. This filter helps you configure the minimal number of records to generate the topic.
- **Topic number**: Controls the number of topic blocks in the current Hot Topic Analytics map.

5. Click **Save** to save your modification.
6. Click the **Start Index** button to re-index the module for the new filter fields to take effects.
7. Go to Forms Designer, and then add the new filter fields into the "Hot Topic Analytics" forms that are used for the specific module. You can search for "idol.hta" in Forms Designer to get a list of the "Hot Topic Analytics" forms. For example, the format name of the Hot Topic Analytics for incidents is "idol.hta.im.pd.advcontent" for Service Manager Codeless or Hybrid, and "idol.hta.im.advcontent" for Service Manager Classic.

3.4.5. Enable Hot Topic Analytics for other modules

User Role: Administrator

In the out-of-box system, Hot Topic Analytics is available in the Service Desk, Incident Management, Problem Management, and Survey modules. If you want to use Hot Topic Analytics to help you identify hot topics in other modules such as Change Management, you can tailor your Service Manager system.

The following section provides two examples of how to enable Hot Topic Analytics for a module, one for Change Management and the other for Knowledge Management.

Enable Hot Topic Analytics for KM feedback

If you want to enable Hot Topic Analytics to help you identify hot topics in KM feedback and create incident records and KM articles from suggested candidates, follow these steps:

1. Add a new Hot Topic Analytics configuration record for the for kmfeedback, and index.
    1. From the System Navigator, click **System Administration**> **Ongoing Maintenance** > **Smart Analytics** > **Hot Topic Analytics**.
    2. In the **Add Configuration** drop-down list, select **kmfeedback**, and then click **Add**.
    3. Enter information for all the required fields such as the title, contents, time stamp, and properties fields, as follows.

Analytic Corpus section

Field	Value
Title Field	Kmdocumentid
Contents Fields	Feedbacktext
Index Condition	True **Note:** You can define index conditions per business requirements.

Filter Fields section

Field	Value
Timestamp Field	Create Time
Properties Field	Author Kmdocumentid Useful **Tip:** To fill in these fields, you can refer to the out-of-box Hot Topic Analytics configuration, and then use the same or similar settings.

4. Click **Add**
5. Click **Start Index** to start indexing.

2. Add a menu item to the System Navigator to display Hot Topic Analytics for kmfeedback.
    1. From the System Navigator, click **Tailoring** > **Tailoring Tools** > **Menus**.
    2. In the **Menu Name** field, type the parent menu where you want to add the new **Hot Topic Analytics for Kmfeedback** menu item, and then click **Search** to open the menu.

        Tip

        Tip For example, you can type HOME to add the new **Hot Topic Analytics for Kmfeedback** menu item to the main menu.

    3. Add a new row at the end of the menu table by entering information for the following settings:

Setting	Value
Description	Hot Topic Analytics for Kmfeedback
Application	launch.idol.problem.hunter
Parameter Name	{"name","query"}
Parameter Value	{"kmfeedback","true"}
Condition	true

    4. Click **Save**.

3. Tailor the `IDOLModuleConfiguration` script file so that it can work for the knowledge feedback in the Knowledge Management module:
    1. Click **Tailoring** > **Script Library**.
    2. In the **Name** field, type IDOLModuleConfiguration, and then click **Search**. The detailed script is displayed.

3. Add the following code in the `MODULE_CONFIGS` section:

```
//km feedback
"kmfeedback": {
actions : [{
id: 'idol-action-list',
text: scmsg(1033, 'idol'),
aria: scmsg(1033, 'idol'),
process: 'idol.hot.topic.view'
},{
id: 'idol-action-link-incident',
text: scmsg(1025, 'idol'),
aria: scmsg(1025, 'idol'),
process: function() {
return PDE.isIncidentEnabled() ? 'idol.hot.topic.create.pd.incident' :
'idol.hot.topic.create.incident';
}
},{
id: 'idol-action-link-article',
text: scmsg(1029, 'idol'),
aria: scmsg(1029, 'idol'),
process: function() {
return 'idol.hot.topic.create.document';
}
}],
displayscreen : "idol.hta.kmfeedback",
next : function() { return generateTicketID("kmfeedback"); }
}
```

Note

Note In out-of-box systems, the `IDOLModuleConfiguration` script file enables Hot Topic Analytics only for incidents, problems, interactions and surveys.

4. Create a message List Kmfeedback for the **List Kmfeedback** button that you plan to display on the Hot Topic Analytics for Kmfeedback page:
    1. Click **Tailoring** > **Database Manager**. Or, type `db` in the command line, and then press **Enter**.
    2. Type `scmessage` in the **Table** field, and click the **Search** icon.
    3. Open the `message.edit` format. The Search Message Records page opens.
    4. Type `idol` in the **Class** field, and then click **Search**.
    5. Specify the text and message number which was specified for the **List Kmfeedback** button in the script file in step 3.c. Click **Add**.
5. Create a new display screen idol.hta.kmfeedback by copying the idol.hta.pm display screen:
    1. Type `ds` in the command line, and then press **Enter**.
    2. In the **Screen ID** field, type idol.hta.pm, and then click **Search**.
    3. Rename Screen ID as idol.hta.kmfeedback , click **Add**, then you will be asked if you would also like to copy display options and display event. Select **Yes**. The system automatically creates display events and display options for the idol.hta.kmfeedback display screen.

Then, modify the display screen accordingly. For the **Initialization Expressions** tab, modify as follows:

```
$L.title=scmsg(605, "idol")
$L.start.search=nullsub($L.start.search, false)
if ($L.start.search=true) then ($hta.analytics.action=nullsub($hta.analytics.action, "action"))
if ($L.start.search=true) then ($idol.ph.is.advfiler=false;if (not  (null($search.field.fromdate)))
then ($idol.ph.is.advfiler=true);if (not  (null($search.field.todate))) then
($idol.ph.is.advfiler=true))
if ($L.start.search=true) then if (not  (null($search.field.advquery.cond))) then
($idol.ph.is.advfiler=true)
if ($L.start.search=true) then ($L.format="idol.hta.kmfeedback.content";if
($idol.ph.is.advfiler=true) then ($L.format="idol.hta.kmfeedback.advcontent"))
if ($L.start.search=true) then (if null($search.field.keyword) then
($search.field.keyword="*");$L.void=jscall("IDOL_Help.clickSearch4ProblemHunter"))
$L.start.search=false
```

6. Create a message Hot Topic Analytics for Kmfeedback for the idol.hta.kmfeedback display screen. The message number should be consistent with that defined in the **Initialization Expressions** tab of the idol.hta.kmfeedback display screen described in step 5, , as shown below:

   ```
   $L.title=scmsg(605, "idol")
   ```

7. Create two formats idol.hta.kmfeedback.content and idol.hta.kmfeedback.advcontent for the kmfeedback HTA filters and advanced filters by copying the idol.hta.pm.content and the idol.hta.pm.pd.advcontent. formats respectively:
   1. Type `fd` in the command line, and then press **Enter**.
   2. In the **Form** field, type idol.hta.pm.content, and then click **Search**.
   3. Click **More** or the **More Actions** icon, and then select **Copy/Rename**. Rename the format as idol.hta.kmfeedback.content and then click **Copy**.
   4. Repeat step **a** to **c** to create the idol.hta.kmfeedback.advcontent format.

   Modify the controls in the two kmfeedback formats accordingly:
   - Rename the caption name as Hot Topic Analytics for Kmfeedback.
   - (For idol.hta.kmfeedback.advcontent only) Delete the original filter controls and add filter controls for Kmfeedback.

     Note

     Note Filter fields defined here should be consistent with those defined in Properties Fields under the Filter Fields section of the Hot Topic Analytics configuration record, as described in step 1.

   - (For idol.hta.kmfeedback.advcontent only) In the properties window of the Edit Query control, configure both the **Query Table Name** and the **Current Record Table Name** properties as kmfeedback.

8. Modify the following configurations for the display options created in step 5:

Unique ID (Alternative text/label)	Value in RAD > Pre Rad Expressions
idol.hta.kmfeedback_advfilter (Adv Filters)	`$idol.ph.is.advfiler=true` `$L.format="idol.hta.kmfeedback.advcontent"`
idol.hta.kmfeedback_hideadv (hide advanced search)	`$idol.ph.is.advfiler=false` `$L.format="idol.hta.kmfeedback.content"`

idol.hta.kmfeedback_clearfilter (Clear filters)	Modify the expressions as follows:
	`$idol.ph.is.advfiler.old=$idol.ph.is.advfiler`
	`$idol.ph.has.advfiler.value=false`
	`if (not (null($search.field.fromdate)) or not (null($search.field.todate)) or not (null(author in $L.file)) or not (null(kmdocumentid in $L.file))) then ($idol.ph.has.advfiler.value=true)`
	`$search.field.fromdate=NULL`
	`$search.field.todate=NULL`
	`$search.field.advquery.cond=NULL`
	`$search.field.advquery.conddesc=NULL`
	`$xmlString2=NULL`
	`$L.company=NULL`
	`author in $L.file=NULL`
	`kmdocumentid in $L.file=NULL`
	`useful in $L.file=NULL`
	`$idol.ph.is.advfiler=true`
	`$L.format="idol.hta.kmfeedback.advcontent"`
	`$L.format="idol.hta.kmfeedback.advcontent"`

idol.hta.kmfeedback_findhottopics (Find Hot Topics)	`$idol.ph.is.advfiler=false`
	`if (not (null($search.field.fromdate))) then ($idol.ph.is.advfiler=true)`
	`if (not (null($search.field.todate))) then ($idol.ph.is.advfiler=true)`
	`if (not (null(useful in $L.file))) then ($idol.ph.is.advfiler=true)`
	`if (not (null(kmdocumentid in $L.file))) then ($idol.ph.is.advfiler=true)`
	`if (not (null(author in $L.file))) then ($idol.ph.is.advfiler=true)`
	`if null($search.field.keyword) then ($search.field.keyword="*")`
	`$L.void=jscall("IDOL_Help.clickSearch4ProblemHunter")`

9. Create a new screlconfig record for recording relationship between km feedback and incidents.
   You can create the record by copying an existing one in the system, as follows:
   1. Type `db` in the command line, and then press **Enter**.
   2. In the **Table** field, type screlconfig, and then click **Search**. A list of screlconfig records is displayed.
   3. Open a rootcause record whose open.application type is document.new and view.application type is se.search.engine, rename the record as kmfeedbackand then click **Add**.

   Modify the record as needed and ensure of the following:.
   - File names and table names defined in the record are kmfeedback.

For example, the table name for the View Names and View Values under the View tab should be kmfeedback:

View names	Value
text	`kmfeedback`
query	`$L.query`

- Unique field name in the queries and script snippets here is consistent with that specified in the **Unique Field Name** field.
  For example, if you specify Id in the **Unique Field Name** field, the Query and script in the expressions should be similar to the following:

Field	Value
Query	`"id isin "+str($L.unique.array)`
Post Open Expressions	`depend in $L.relation=str(id in $L.related)`

Pre View Expressionss	`if (filename($L.primary)={"kmfeedback"})`
	`then ($L.number=id in $L.primary)`
	`if (filename($L.related)={"kmfeedback"})`
	`then ($L.number=id in $L.related)`
	`$L.query=nullsub("id=\""+$L.number+"\"",`
	`"true"`

Active Condition	`True`

- The filters defined for the **Descriptions** field are consistent with those defined in **Properties Fields** under the **Filter Fields** section of the Hot Topic Analytics configuration record, as described in step 1.

10. Modify the vj.number.1 source field of the probsummary link file to display related kmfeedback in incident records:
    1. Type `link` in the command line, and then press **Enter**. The Link File search page is displayed.
    2. In the **Name** field, type probsummary, and then click **Search**. A list of source field records is displayed.
    3. Move the cursor to the vj.number.1 record, right-click the mouse and then click **Select Line**. The record opens.
    4. In the Query field, add the following code at the end:
       `or depend.filename="kmfeedback"`
11. Add a kmmapping record of kmfeedback for opening km article from knowledge feedback:
    1. Type `db` in the command line, and then press **Enter**. The Link File search page is displayed.
    2. In the **Table** field, type kmmapping, and then click **Search**. The Knowledge Management Mapping Record search page is displayed.
    3. In the **Map Name** field, type kmfeedback and then click **New**.The record is created.
    4. Configure the **Search** and the **Create Solutions** tabs accordingly, as follows:
       For the **Search** tab:

Source Field in Current Record	Target Search Field
feedbacktext	`query`
id	`contexttag`

For the **Create Solutions** tab:

Document Type	Source Field in Current Record	Target Search Field
all	feedbacktext	`tite`
reference	feedbacktext	`reference`

12. Log off and log back on to Service Manager. Now you can access and use Hot Topic Analytics for kmfeedback from the System Navigator.

Enable Hot Topic Analytics for Change Management

If you want to enable Hot Topic Analytics to help you identify hot topics in Change Management, follow these steps:

1. Add a new Hot Topic Analytics configuration record for the Change Management module:
    1. From the System Navigator, click **System Administration**> **Ongoing Maintenance** > **Smart Analytics** > **Hot Topic Analytics**.
    2. In the **Add Configuration** drop-down list, select **cm3r**, and then click **Add**.
    3. Enter information for all the required fields such as the title, contents, time stamp, and properties fields, and then click **Add**.

    > **Tip**
    > Tip To fill in these fields, you can refer to the out-of-box Hot Topic Analytics configuration, and then use the same or similar settings.

    4. Click **Start Index** to start indexing.
2. Add a menu item to the System Navigator to display Hot Topic Analytics for Changes:
    1. From the System Navigator, click **Tailoring** > **Tailoring Tools** > **Menus**.
    2. In the **Menu Name** field, type the parent menu where you want to add the new **Hot Topic Analytics for Changes** menu item, and then click **Search** to open the menu.

    > **Tip**
    > Tip For example, you can type HOME to add the new **Hot Topic Analytics for Changes** menu item to the main menu.

    3. Add a new row at the end of the menu table by entering information for the following settings:

Setting	Value
Description	Hot Topic Analytics for Changes
Application	launch.idol.problem.hunter
Parameter Name	{"name","query"}
Parameter Value	{"cm3r","true"}
Condition	true

    4. Click **Save**.
3. Tailor the `IDOLModuleConfiguration` script file so that it can work for the change records in the Change Management module:
    1. Click **Tailoring** > **Script Library**.
    2. In the **Name** field, type IDOLModuleConfiguration, and then click **Search**. The detailed script is displayed.

3. Modify the script accordingly.

> **Note**
> Note In out-of-box systems, the `IDOLModuleConfiguration` script file enables Hot Topic Analytics only for incidents, problems, interactions and surveys.

> **Tip**
> Tip You can search for `rootcause` in the script file to see where you need to make changes for the script to support other modules.

4. Create a message for the button that you plan to create on the display screen of Hot Topic Analytics for Changes.
    1. Click **Tailoring** > **Database Manager**. Or, type `db` in the command line, and then press **Enter**.
    2. Type `scmessage` in the **Table** field, and click the **Search** icon.
    3. Open the `message.edit` format. The Search Message Records page opens.
    4. Type `idol` in the **Class** field, and then click **Search**.
    5. Specify the message number and text as needed. Click **Add**.
5. Create a new display screen for Change module in the following way:
    1. Type `ds` in the command line, and then press **Enter**.
    2. Specify the fields on the definition form as needed.
    3. Click **Add**.

> **Tip**
> Tip You refer to the `idol.hta.pm` display screen for configuration.

6. Create display options for the display screen which is added in step 5.
    1. Type `do` in the command line, and then press **Enter**.
    2. Specify the fields on the definition form as needed.
    3. Click **Add**.

> **Tip**
> Tip You can refer to the **Options** tab in the `idol.hta.pm` display screen for the definitions of options.

7. Create display events for the display screen which is added in step 5.
    1. Type `de` in the command line, and then press **Enter**.
    2. Specify the fields on the definition form as needed.
    3. Click **Add**.

> **Tip**
> Tip You can refer to the **Events** tab in the `dol.hta.pm` display screen for the definitions of events.

8. Create formats for the Change module filters and advanced filters respectively. For detailed information, see Create a form using the Form Wizard.

> **Tip**
> Tip You can refer to the `idol.hta.pm.content` and `idol.hta.pm.advcontent` formats as examples for your tailoring.

9. Create processes which you used for Change Module in the `IDOLModuleConfiguration` script file as described in step 3. For detailed information, see Create and modify a Process.

Note

Note Ignore this step if you used OOB processes when modifying the `IDOLModuleConfiguration` script file.

10. Log off and log back on to Service Manager. Now you can access and use Hot Topic Analytics for Changes from the System Navigator.

3.4.6. Customize actions in hot topic map

User Role: Administrator

If you want to customize the actions available in the hot topic map, you can tailor the out-of-box Hot Topic Analytics. For example, for the out-of-box Hot Topic Analytics for Incidents, the Create Problem action (button) is available. You can change this action or add a new action according to your business needs.

To customize the actions in hot topic map, follow these steps:

1. Click **Tailoring** > **Script Library**.
2. In the **Name** field, type IDOLModuleConfiguration, and then click **Search**. The detailed script is displayed.
3. Locate the "MODULE_CONFIGS" Javascript Object. Inside MODULE_CONFIGS, you can see the configuration entries for different modules. The entry "key" must correspond to the table name of an SM module. For example: "probsummary."
   The "actions" configuration section is a Javascript array. The items defined in this array will be rendered as "the action button" in the Hot Topic Analytics chart. For example, you can see the following lines in the script:

```
   actions : [{
id: 'idol-action-list',
text: scmsg(1023, 'idol'),
aria: scmsg(1023, 'idol'),
process: 'idol.hot.topic.view'
}, {
id: 'idol-action-link-incident',
text: scmsg(1025, 'idol'),
aria: scmsg(1025, 'idol'),
process: function() {
return PDE.isIncidentEnabled() ? 'idol.hot.topic.create.pd.incident' :
'idol.hot.topic.create.incident';
}
}]
```

4. Modify the "actions" configuration section to add the definition for the new action. The parameters that you need to specify are described in the following table.

Parameter	Description
id	A unique value to identify which "action button" is clicked. The "id" value also has the CSS style attached. Currently, two styles are available: ○ "idol-action-list" has an "Open" icon. ○ "idol-action-link" has a "Link" icon.
text	The "action button" label that's displayed on the chart
aria	Texts used for accessibility
process	Represents the SM process name. The process is invoked when the "action button" is clicked. The configuration value can be either a text string or a function that returns a text string. When the process is invoked, two additional variables are available: ○ "$L.idol.selected.filename": the current module table name ○ "$L.idol.selected.query": the SM query to retrieve the selected records in the Hot Topic Analytics chart

For example, you can modify the following lines accordingly and add them to the script:

```
{
id: 'idol-action-link-others',
text : 'Link to some other module',
aria : 'Link to some other module',
process : 'idol.link.to.other.module'
}
```

5. Click **Save** to save your change in the script file.
6. Create the process that you use for the new action button. For example, you need to create the "idol.link.to.other.module" process that you specified in step 4. In the process, you can define the behaviors when the action button is clicked.

Add more "Analytics Action" in the Hot Topic Analytics for Incidents[edit]

In the out-of-box system, the **Analytics Action** setting in the Hot Topic Analytics for incidents includes the following three out-of-box actions: Set Parent, Create Problem, and Create Change/Article.

If you want to add more custom action queries in the Hot Topic Analytics for Incidents, follow these steps:

1. Tailor the form of the Hot Topic Analtyics for Incidents to add one more option to the drop-down list of "Analytics Action".
2. Add the new action to the "IDOL HTA Action Type" globallist that's used in the Hot Topic Analytics configuration record for the Incident module.
3. In the **Advanced** tab of the Hot Topic Analytics configuration record for Incidents, select the new action that you added, and then define the query for this newly added action.

Tip

Tip The query also supports jscall function.

4. From the script library, open "IDOLModuleConfiguration" and then configure which actions can be shown in the action form. For example:

```
 if (vars['$hta.analytics.action']==="createproblem") {
return [list, createProblem];
}
```

Contents

- 1 Add more "Analytics Action" in the Hot Topic Analytics for Incidents

3.4.7. Configure Smart Analytics to support more languages

User Role: Administrator

With the out-of-box configuration, Smart Analytics supports the same language list that Service Manager supports. If the texts entered in the records (such as in title and description) aren't in a language that's included in the supported language list of Service Manager, these texts can't be indexed and analyzed by Smart Analytics.

However, Smart Analytics can be configured to index and analyze texts in other languages in addition to the out-of-box supported languages. To do this, follow these steps:

1. In the IDOL configuration file (<Smart Analytics>/IDOL/IDOLServer.cfg), add the desired language type into the [LanguageTypes] section. For example, if you want to support Vietnamese, you can add the following line into the [LanguageTypes] section:

   ```
   19=Vietnamese
   ```

2. Add encoding format and stoplist into the [LanguageTypes] section. For example, you can add the flowing configurations for Vietnamese:

   ```
   [vietnamese]
   Encodings=UTF8:vietnameseUTF8
   Stoplist=vietnamese.dat
   ```

   Note

   Note The stoplist file is a text file placed in the <Smart Analytics>/langfiles folder, which contains words that you don't want to be treated as keywords in the query requests to IDOL.

3. Save the file.

Note

Note The IDOLServer.cfg configuration file doesn't affect the language support for the Optical Character Recognition (OCR) feature. OCR recognizes images based on the language setting specified in the contact record of a user in Service Manager. If no language is set in the user's contact record, English is used as the default value in image recognition by Smart Analytics.

3.5. Limitations

Smart Analytics contains the following limitations in the current release:

- Smart Ticket
  If your Service Manager is upgraded from Service Manager non-PD environment, IT agents are unable to use the Optical Character Recognition (OCR) feature when creating interactions for users in the out-of-box environment. However, when Process Designer is enabled, IT agents can use the "Image2Text" feature to copy and paste messages to the interaction records.
- Smart Search
  In this release, Service Manager indexes Service Catalog capability words into IDOL content server. When an end user searches for catalog items, IDOL returns only catalog items with capability words that match the current login user. However, when there are catalog items with expressions specified in Access Filter that IDOL is not able to evaluate, such catalog items are also returned to Service Manager for further permission checking in the applications server. If the end user has no permission to such catalog items, Service Manager doesn't return them to the end user. Therefore, sometimes the search result returns empty, or has less than 10 records, but still has next page.

3.6. Smart Analytics APIs

Smart Analytics provides the following out-of-box APIs:

Smart Analytics RESTful APIs

Auto-classification Javascript API

APIs for Hot Topic Analytics indexing

3.6.1. Smart Analytics RESTful APIs

Smart Analytics provides the following RESTful APIs:

- Smart classification
- Hot topic
- Smart search
- Search
- Smart classification (by specifying IDOL adapters)

You must add the capability word **RESTful API** to a user's operator record for a user to be able to execute a RESTful API request.

1. From the System Navigator, click **System Administration** > **Ongoing Maintenance** > **Operators**.
2. Enter or select your search criteria, and then click **Search**.
3. Select an operator from the record list to view the operator record.
4. Click the **Startup** tab.
5. Add RESTful API in the **Execute Capabilities** section.

3.6.1.1. Smart classification

This smart classification RESTful API is used to get the classifications suggested by Smart Analytics.

Request:

```
POST http://<SM Server Address>:13080/SM/9/rest/SmartAnalytics/1/action/suggest
```

An example of the request body:

```
{
SmartAnalytics:{
in:'{"filename": "incidents", "title": "test", "description": ["test"],"callback_contact": "AARON, JIM"}'
}
}
```

Response example:

Note

Note You can ignore the messages in the response as these messages are printed by the Restful framework.

```
{
"Messages": [
"Invalid syntax for query.  Failed parsing",
"in=\"1\"",
"Invalid syntax for query.  Failed parsing",
"in=\"1\""
],
"ReturnCode": 0,
"SmartAnalytics": {
"in": "{\"filename\": \"incidents\", \"title\": \"test\", \"description\":
[\"test\"],\"callback_contact\": \"AARON, JIM\"}",
"out": "{\"category\":\"incident\",\"subcategory\":\"performance\",\"product.type\":\"performance
degradation\",\"affected.item\":null}"
}
}
```

3.6.1.2. Hot topic

This hot topic RESTful API is used to get the hot topics suggested by Smart Analytics.

Request:

```
POST http://<SM Server Address>:13080/SM/9/rest/SmartAnalytics/1/action/hottopic
```

An example of the request body:

```
{
SmartAnalytics:{
in:'{filename: "probsummary",  keyword:"email"}',
}
}
```

Response example:

Note

Note

You can ignore the messages in the response as these messages are printed by the Restful framework.

```
{
"Messages": [
"Invalid syntax for query.  Failed parsing",
"in=\"1\"",
"Invalid syntax for query.  Failed parsing",
"in=\"1\""
],
"ReturnCode": 71,
"SmartAnalytics": {
"in": "{filename: \"probsummary\",  keyword:\"email\"}"
"out": ‘{"recommends":["Shared MailBox"," Unable to access"," Email receiving"],"subtopics":
[{"topic":"Shared MailBox","subtopics":[{"topic":"New mail sending","docs":["IM10122","IM10056"],"id":2},
{"topic":"box is not accessible on outlook","docs":["IM10121"],"id":4}],"docs":
["IM10121","IM10056"],"id":1},{"topic":"Unable to access","subtopics":[{"topic":"group mailbox's","docs":
["IM10131","IM10128"],"id":7,"subtopics":[{"id":8,"topic":"CHAR-Intranet / Internet (North
America)","raw":"Intranet / Internet (North America)","trivial":true,"docs":["IM10131"]},
{"id":9,"topic":"CHAR-Applications","raw":"Applications","trivial":true,"docs":["IM10128"]}]},
{"topic":"Communication mail","docs":["IM10084","IM10128"],"id":10,"subtopics":[{"id":11,"topic":"CHAR-E-
mail / Webmail (North America)","raw":"E-mail / Webmail (North America)","trivial":true,"docs":
["IM10084"]},{"id":12,"topic":"CHAR-Applications","raw":"Applications","trivial":true,"docs":
["IM10128"]}]}],"docs":["IM10131","IM10021"],"id":6},{"topic":"Email receiving","subtopics":
[{"id":14,"topic":"CHAR-E-mail / Webmail (North America)","raw":"E-mail / Webmail (North
America)","trivial":true,"docs":["IM10081","IM10125","IM10021"]},{"id":15,"topic":"CHAR-Handheld, PDA &
Telephony","raw":"Handheld, PDA & Telephony","trivial":true,"docs":["IM10012"]}],"docs":
["IM10081","IM10125","IM10021","IM10012"],"id":13},{"topic":"mail issues","subtopics":
[{"id":17,"topic":"CHAR-Applications","raw":"Applications","trivial":true,"docs":["IM10126"]},
{"id":18,"topic":"CHAR-E-mail / Webmail (North America)","raw":"E-mail / Webmail (North
America)","trivial":true,"docs":["IM10084","IM10081","IM10056"]}],"docs":
["IM10126","IM10084","IM10081","IM10056"],"id":16},{"topic":"User","subtopics":[{"id":20,"topic":"CHAR-E-
mail / Webmail (Asia)","raw":"E-mail / Webmail (Asia)","trivial":true,"docs":["IM10186"]},
{"id":21,"topic":"CHAR-E-mail / Webmail (North America)","raw":"E-mail / Webmail (North
America)","trivial":true,"docs":["IM10091","IM10039","IM10081","IM10021"]},{"id":22,"topic":"CHAR-E-mail
/ Webmail (Africa)","raw":"E-mail / Webmail (Africa)","trivial":true,"docs":["IM10059"]},
{"id":23,"topic":"CHAR-Applications","raw":"Applications","trivial":true,"docs":["IM10105"]}],"docs":
["IM10186","IM10091","IM10059","IM10039","IM10105","IM10081","IM10021"],"id":19},
{"topic":"outlook","subtopics":[{"id":25,"topic":"CHAR-
Applications","raw":"Applications","trivial":true,"docs":["IM10126","IM10011","IM10105","IM10128"]},
{"id":26,"topic":"CHAR-E-mail / Webmail (North America)","raw":"E-mail / Webmail (North
America)","trivial":true,"docs":["IM10091","IM10039","IM10121","IM10084"]},{"id":27,"topic":"CHAR-
Intranet / Internet (North America)","raw":"Intranet / Internet (North America)","trivial":true,"docs":
["IM10131"]}],"docs":
["IM10126","IM10091","IM10039","IM10011","IM10121","IM10105","IM10084","IM10131","IM10128"],"id":24}],"su
mmaries":{"IM10011":{"reference":"IM10011","title":"problems with the send mails from
outlook","summary":"problems with the send mails from outlook. problems with the send mails from
outlook\n","weight":84.67,"type":"Applications"},"IM10012":{"reference":"IM10012","title":"Email
receiving error when using mobile phone as email client.","summary":"Email receiving error when using
mobile phone as email client.. report error about 'connection fail' when attempt to connect to mail
server.\n","weight":83.83,"type":"Handheld, PDA & Telephony"},"IM10021":
{"reference":"IM10021","title":"User did not receive the reset password email
notification","summary":"User did not receive the reset password email notification. I was unable to
access Spend Management as it would not accept my password. I chose Forgot  Password and I still have not
received an e-mail with the correct password.\n","weight":83.83,"type":"E-mail / Webmail (North
America)"},"IM10039":{"reference":"IM10039","title":"outlook flags have disappeared from the e-
mails","summary":"outlook flags have disappeared from the e-mails. user uses the flags to organize the e-
mails, flags have disappeared from the mailbox.\n","weight":84.67,"type":"E-mail / Webmail (North
America)"},"IM10056":{"reference":"IM10056","title":"Shared MailBox issue","summary":"Shared MailBox
issue. We are receiving the delivery failure from Generic mailbox while sending new
mails.\n","weight":83.83,"type":"E-mail / Webmail (North America)"},"IM10059":
{"reference":"IM10059","title":"User is not recieving mails on blackberry","summary":"User is not
```

```
recieving mails on blackberry. User is not recieving mails on blackberry and still not work after
restarting.\n","weight":84.67,"type":"E-mail / Webmail (Africa)"},"IM10081":
{"reference":"IM10081","title":"mail issues","summary":"mail issues. User not receiving salary slips on
email\n","weight":84.46,"type":"E-mail / Webmail (North America)"},"IM10084":
{"reference":"IM10084","title":"Communication mail issue","summary":"Communication mail issue. I’ am not
getting any communication email to my outlook email id.\n","weight":84.46,"type":"E-mail / Webmail (North
America)"},"IM10091":{"reference":"IM10091","title":"Users aren't rcving mails on all the campaings of
the COR on Altitude and Outlook both","summary":"Users aren't rcving mails on all the campaings of the
COR on Altitude and Outlook both. Users aren't rcving mails on all the campaings of the COR on Altitude
and Outlook both. Please assist asap.\n","weight":84.67,"type":"E-mail / Webmail (North
America)"},"IM10105":{"reference":"IM10105","title":"User unable to send receive mails on
outlook","summary":"User unable to send receive mails on outlook. outlook state keeps changing from
trying to connect to connected...browser setttings are fine and owa link is also not
working\n","weight":84.46,"type":"Applications"},"IM10121":{"reference":"IM10121","title":"Shared mail
box is not accessible on outlook","summary":"Shared mail box is not accessible on outlook. I am not able
to access shared mailbox from outlook and Lync since Thursday after the migration.\nTried adding and
removing the mailbox but still it's not working.\n","weight":84.46,"type":"E-mail / Webmail (North
America)"},"IM10122":{"reference":"IM10122","title":"New mail sending error","summary":"New mail sending
error. When I tried to send new mail from GSRPA mailbox I’m receiving the below error msg. please do the
needful.\n","weight":84.67,"type":"E-mail / Webmail (North America)"},"IM10125":
{"reference":"IM10125","title":"Not able to send emails","summary":"Not able to send emails. Not able to
send emails..mails are stucking in outbox.user is able to receive emails how ever mails are stucking in
the Outbox.so raising the request for the same.\n","weight":83.83,"type":"E-mail / Webmail (North
America)"},"IM10126":{"reference":"IM10126","title":"outlook mail delivery issue","summary":"outlook mail
delivery issue. I am receiving e-mail in batches, about once an hour. Also, my smartphone keeps
synchronizing e-mail, even though I have set the Sync setting to
'Manual'.\n","weight":84.88,"type":"Applications"},"IM10128":{"reference":"IM10128","title":"Outlook
problem","summary":"Outlook problem. Communication mail for my Group mailbox\n\nError for your reference
–\n\n\"Cannot Expand the folder\"\n","weight":83.83,"type":"Applications"},"IM10131":
{"reference":"IM10131","title":"Unable to access group mailbox's archive","summary":"Unable to access
group mailbox's archive. CSCS group mailbox is recently migrated on outlook and we are unable to access
our group mailbox archive mails. Please do the needful ASAP.\n","weight":83.83,"type":"Intranet /
Internet (North America)"},"IM10186":{"reference":"IM10186","title":"Not getting mail from external
customer","summary":"Not getting mail from external customer. user report can't getting mail from
external customer\n","weight":84.67,"type":"E-mail / Webmail (Asia)"}},"docs":[],"id":0,"topic":"e-
mail"}’
}
}
```


3.6.1.3. Smart search

Service Manager uses the hot topic RESTful API to get the search results suggested by Smart Analytics.

Request:

```
POST http://<SM Server Address>:13080/SM/9/rest/SmartAnalytics/1/action/smartsearch
```

An example of the request body:

```
{
SmartAnalytics:{
in:'{keyword: "test",  start:1, pagesize:2}',
}
}
```

Response example:

Note

Note You can ignore the messages in the response as these messages are printed by the Restful framework.


```
{
"Messages": [
"Invalid syntax for query.  Failed parsing",
"in=\"1\"",
"Invalid syntax for query.  Failed parsing",
"in=\"1\""
],
"ReturnCode": -1,
"SmartAnalytics": {
"in": "{keyword: \"test\",  start:1, pagesize:2}",
"out": "<autnresponse xmlns:autn=\"http://schemas.autonomy.com/aci/\"><action>QUERY</action>
<response>SUCCESS</response><responsedata>\n<autn:numhits>2</autn:numhits>
<autn:totalhits>18</autn:totalhits><autn:totaldbdocs>346</autn:totaldbdocs>
<autn:totaldbsecs>346</autn:totaldbsecs><autn:hit>
<autn:reference>scactivelink:%2F%2Fkmdocument:id%3D%26quot%3BKM0580%26quot%3B%26nbsp%3Band%26nbsp%3Bkbnam
e%3D%26quot%3BKnowledge_Library%26quot%3B</autn:reference><autn:id>1588</autn:id>
<autn:section>0</autn:section><autn:weight>57.09</autn:weight><autn:links>TEST</autn:links>
<autn:database>Knowledge_Library</autn:database><autn:title>test error message by Jora(Title)
</autn:title><autn:summary>&lt;_em_&gt;test&lt;/_em_&gt; error message by Jora(Title). 2015/06/22
14:07:46. falcon. external. errormsg. &lt;_em_&gt;test&lt;/_em_&gt; error message by Jora(Error Message).
&lt;_em_&gt;test&lt;/_em_&gt; error message by Jora(cause). &lt;_em_&gt;test&lt;/_em_&gt; error message
by Jora(Wordaround/Fix).</autn:summary><autn:content><DOCUMENT>
<KMSEARCHCATEGORY>KM01CAT:KM03CAT:KM05CAT</KMSEARCHCATEGORY><KMSTATUS>external</KMSTATUS>
<DOCTYPE>errormsg</DOCTYPE><ID>KM0580</ID><SYSMODTIME_DATE>2015/06/22 14:47:00</SYSMODTIME_DATE>
<CATEGORY>Service Manager&gt;Knowledge Management&gt;Document Maintenance and Lifecycle;</CATEGORY>
<DRETITLE>test error message by Jora(Title)</DRETITLE></DOCUMENT></autn:content></autn:hit><autn:hit>
<autn:reference>scactivelink:%2F%2Fkmdocument:id%3D%26quot%3BKM0580%26quot%3B%26nbsp%3Band%26nbsp%3Bkbnam
e%3D%26quot%3BKnowledge_Library%26quot%3B</autn:reference><autn:id>1588</autn:id>
<autn:section>0</autn:section><autn:weight>57.09</autn:weight><autn:links>TEST</autn:links>
<autn:database>Knowledge_Library</autn:database><autn:title>test error message by Jora(Title)
</autn:title><autn:summary>&lt;_em_&gt;test&lt;/_em_&gt; error message by Jora(Title). 2015/06/22
14:07:46. falcon. external. errormsg. &lt;_em_&gt;test&lt;/_em_&gt; error message by Jora(Error Message).
&lt;_em_&gt;test&lt;/_em_&gt; error message by Jora(cause). &lt;_em_&gt;test&lt;/_em_&gt; error message
by Jora(Wordaround/Fix).</autn:summary><autn:content><DOCUMENT>
<KMSEARCHCATEGORY>KM01CAT:KM03CAT:KM05CAT</KMSEARCHCATEGORY><KMSTATUS>external</KMSTATUS>
<DOCTYPE>errormsg</DOCTYPE><ID>KM0580</ID><SYSMODTIME_DATE>2015/06/22 14:47:00</SYSMODTIME_DATE>
<CATEGORY>Service Manager&gt;Knowledge Management&gt;Document Maintenance and Lifecycle;</CATEGORY>
<DRETITLE>test error message by Jora(Title)</DRETITLE></DOCUMENT></autn:content></autn:hit><autn:engines>
<autn:used>0</autn:used><autn:unused>1</autn:unused></autn:engines><autn:warning>Engine 0: Engine 0: At
least one of the fields in a parameter (probably 'printfields') doesn't exist</autn:warning>
<autn:warning>Engine 0: Engine 0: One of the fields used in the field text doesn't exist</autn:warning>
<autn:warning>Engine 0: Engine 0: The languagetype of the text could not be determined. The default
languagetype was used.</autn:warning><autn:warning>Engine 0: Engine 1: At least one of the fields in a
parameter (probably 'printfields') doesn't exist</autn:warning><autn:warning>Engine 0: Engine 1: One of
the fields used in the field text doesn't exist</autn:warning><autn:warning>Engine 0: Engine 1: The
languagetype of the text could not be determined. The default languagetype was used.</autn:warning>
</responsedata></autnresponse>"
}
}
```


3.6.1.4. Search

This search RESTful API is used to get the search results suggested by Smart Analytics.

Request:

```
POST http://<SM Server Address>:13080/SM/9/rest/SmartAnalytics/1/action/search
```

An example of the request body:

```
{
SmartAnalytics:{
query:'mobile',
start:1,
count:100,
filters:'{"Knowledge_Library":{"status":["external","internal"],"doctype":["howto"]},"Incident_Library":
{}}',
max_characters:500,
startTag: '<_em_>',
endTag: '</_em_>'
}
}
```

Response example:

Note

Note You can ignore the messages in the response as these messages are printed by the Restful framework.

```
{
"@count": 2,
"@start": "1",
"Messages": [],
"ResourceName": "SmartAnalytics",
"ReturnCode": 0,
"content": [
{"SmartAnalytics": {
"content": "ADJUST THE TIME AFTER WHICH <em>MOBILE<\/em> ENTERS THE SLEEP MODE. howto. How to adjust the
time after which <em>mobile<\/em> enters the sleep mode in Android 4.2.2 Operating System.. To adjust the
time after which <em>mobile<\/em>",
"id": "KM0122",
"title": "ADJUST THE TIME AFTER WHICH MOBILE ENTERS THE SLEEP MODE"
}},
{"SmartAnalytics": {
"content": "Email receiving error when using <em>mobile<\/em> phone as email client.. Email receiving
error when using <em>mobile<\/em> phone as email client.. report error about &#x27;connection fail&#x27;
when attempt to connect to mail server..",
"id": "IM10012",
"title": "Email receiving error when using mobile phone as email client."
}}
]
}
```

3.6.1.5. Smart classification (by specifying IDOL adapters)

This section introduces another way to perform auto-classification by using two RESTful APIs.

Use the first API to query all the categorization configurations, and then use your desired configuration (adapter id) to call the second API, which will return the category values.

Normally, you only need to use the first API once to get the configuration while you can use the second API to do categorization jobs as many times as you need.

API-1: Get all the configurations/adapters definitions

Request:

```
GET http://<SM Server Address>:13930/SM/9/rest/idoladapters?view=expand
```

Response example:

```
{
"@count": 2,
"@start": 1,
"@totalcount": 2,
"Messages": [],
"ResourceName": "idoladapter",
"ReturnCode": 0,
"content": [
{"idoladapter": {
"adapter.id": 5,
"level1.field": "affected.item",
"source": [
"title",
"description"
]
}},
{"idoladapter": {
"adapter.id": 7,
"level1.field": "category",
"level2.field": "subcategory",
"level3.field": "product.type",
"source": [
"title",
"description"
]
}}
]
}
```

API-2: Get the categorization result based on adapter id

For example: id=137, company=abc, text="my laptop is broken", return top 2 suggestions.

Request:

```
POST http://<SM Server Address>:13080/sm/9/rest/idoladapters/{adapter.id}
```

An example of the request body:

```
{
  "idoladapter":{
"source":["Critical CPU temp. BIOS error message", Maybe I work too hard, but the temperature of my
computers CPU is critical according to the error message it displays in the message"],
"company" :"abc",
"top":2
}
}
```

Response example:


```
{
"Messages": [],
"ReturnCode": 0,
"idoladapter": {
"level1.field": "category",
"level2.field": "subcategory",
"level3.field": "product.type",
"suggest.candidates": [
{
"level1.value": "request for service",
"level2.value": "app. infrastructure",
"level3.value": "shared web hosting"
},
{
"level1.value": "request for service",
"level2.value": "infrastructure",
"level3.value": "server"
}
]
}
}
```


3.6.2. Auto-classification Javascript API

This Javascript API returns an array of arrays that contains the suggested categories.

Syntax

`lib.acicategory.getCategoryByContent(file,adapterid,numresult,company)`

Arguments

Name	Data type	Required	Description
file	Datum object	Yes	This argument contains the object that holds the input source data. For example, an "incidents" file with title and description.
adapterid	Integer	Yes	This argument contains the id you use to call createAndTrainingOneCategory, or adapter.id in idoladapter.
numresult	Integer	Yes	This argument controls the number of suggested categories to return.
company	String	No	The company that the categories belong to. Note If you use an *adapterid* with the multi-company setting configured, you must specify this parameter in your request.

Return values

An array of arrays that contains the suggested categories. For example:

`[["incident","hardware","hardware failure"],["incident","performance","performance degradation"],["incident","failure","job failed"]]`

Example

This example assumes that a category group for interactions is configured as the following:

id: 201

Source fields: title and description

Category fields: category, subcategory, product_type

Then, the Javascript API can be used as in the following example:

```
var f = new SCFile("incidents");
f.title = "my pc is broken";
f.description=["Starting from yesterday, my pc can't be started. Both battery and power adapter looks
good. The pc itself was very hot before it is broken."]
f.category="incident";
print(lib.JSON.json().stringify(lib.acicategory.getCategoryByContent(f,201,3,"es")))
```

If the input parameter (file) includes value of level 1 and level 2 fields that are configured in the idoladapter, this function will use their values as the parent schema to suggest child category to get better accuracy. In this example, the level 1 field is specified as `f.category="incident"`, which means the returned suggestion will all belong to the "incident" category.

3.6.3. APIs for Hot Topic Analytics indexing

If you want to tailor the index schedule of Hot Topic Analytics, you can use the following out-of-box Javascript APIs to control when to run the index task:

- lib.aciindex.manualFullReindex
  You can use this API to manually run a full index for the specified file that's used by Hot Topic Analytics.
- lib.aciindex.manualIncrementalIndex
  You can use this API to manually run an incremental index for the specified file that's used by Hot Topic Analytics.

These two functions will run in synchronous mode. We recommend that you run these two functions in the background schedule of Service Manager.

Syntax

```
lib.aciindex.manualFullReindex(file)
lib.aciindex.manualIncrementalIndex(file)
```

Arguments

Name	Data type	Required	Description
file	String	Yes	This argument is the Service Manager table (for example, "probsummary") that holds the input source data. Note Make sure that the fields of this "file" have been added into the **Filter Fields** section in the Hot Topic Analytics configuration (**System Administration** > **Ongoing Maintenance** > **Smart Analytics** > **Hot Topic Analytics**).

4. Mobile Applications administration

This section provides details about how to tailor Service Manager Mobile Applications to support your business processes.

This section includes:

- Tailoring Mobile Applications in Service Manager
- Protecting communications between Mobile Applications and the Service Manager server
- Mobile Applications Form Widgets
- Mobile Applications for Service Manager Process Designer

4.1. Tailoring Mobile Applications in Service Manager

When you have finished installing Mobile Applications, you can tailor the following in Service Manager:

- Set up email notifications to include URL links
- Configure the prefix of a record
- Add a view for Mobile Applications
- Add a form for Mobile Applications
- Customize the fields on Mobile Applications form
- Customize the action bar options

- Configure the maxRequestPerSecond parameter

4.1.1. Set up email notifications to include URL links

A Service Manager System Administrator can set up email notifications to include the mobility URL. Hence, when tickets are assigned, field engineers can receive email notifications and click on the URL links in the emails to go directly to the assigned ticket.

Service Manager Mobile Applications automatically synchronizes users' mobile data with information in the Service Manager database. After sending an email notification to a field engineer, Service Manager Mobile Applications searches for the record by name and then displays it. If the record isn't in the cached database, Mobile Applications queries Service Manager to fetch the record.

To set up email notifications, follow these steps:

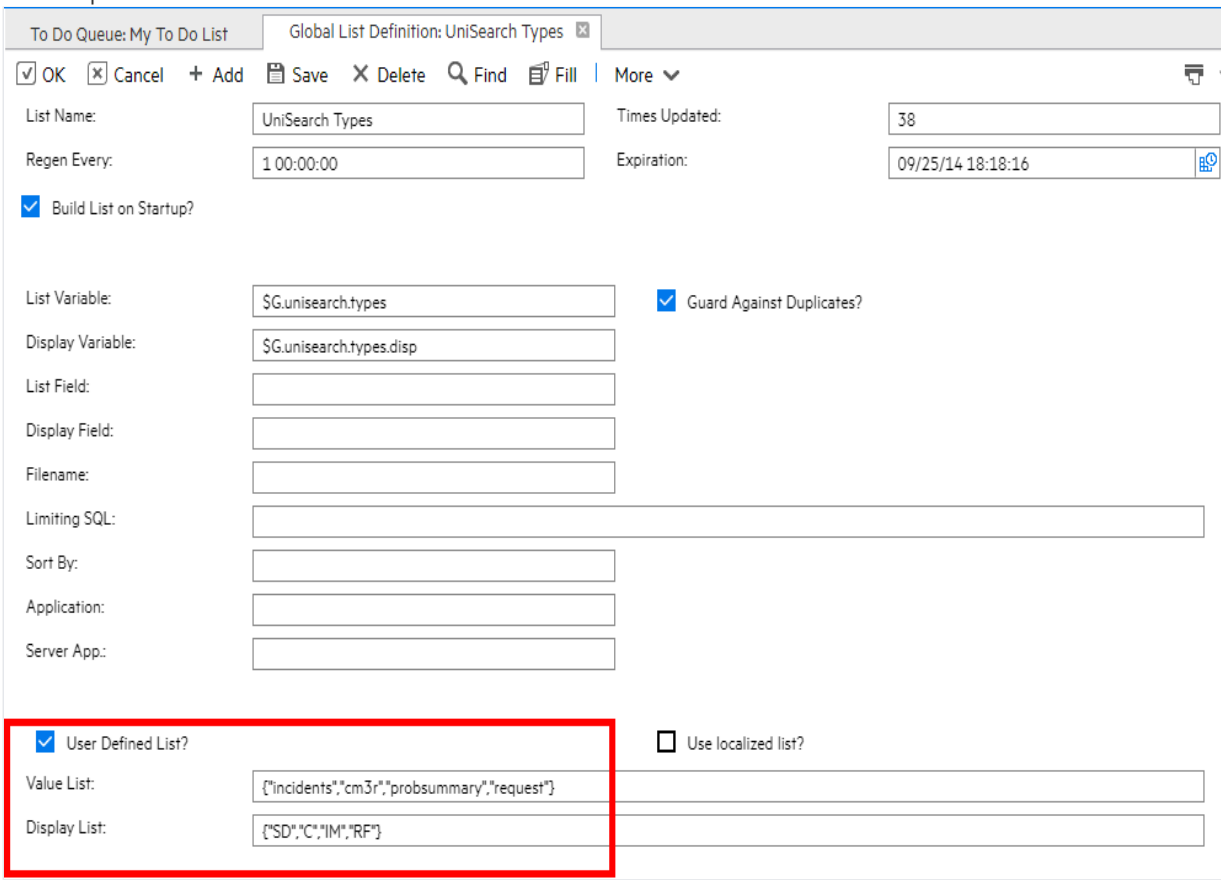
1. Restart the Service Manager Mobile Applications server.
2. Sign in to the Service Manager server as a System Administrator.
3. Click **System Administration** > **Base System Configuration** > **Miscellaneous** > **System Information Record**.
4. Select the **Active Integrations** tab.
5. In the **Mobility URL** field, type the fully qualified URL to the Service Manager server from the Mobile Applications client.
   For example: `http://<servername>:<portnumber>/<appname>/std/`
   Where:
   `servername` is the host name or IP address of the Service Manager server.
   `portnumber` is the port number. For example, 8080.
   `appname` is the name of the deployed Service Manager Mobile Applications client `WAR` or `EAR` file. For example, `webapp-x.xx.xxxx.war`.
   The server stores the value of this field in the `$L.mobility.url` global variable.
6. Save your changes.

4.1.2. Configure the prefix of a record

To configure the prefix of an incident record or a change request which already has a prefix, follow these steps:

1. Sign in to the Service Manager server as a System Administrator.
2. Type `gl` in the Service Manager command line field, and then press **Enter**.
3. Type `UniSearch Types` in the **List Name** field, and then click **Search**.

4. Verify and configure the values in the **Value List** field and the **Display List** field. Refer to the following screenshot as an example:

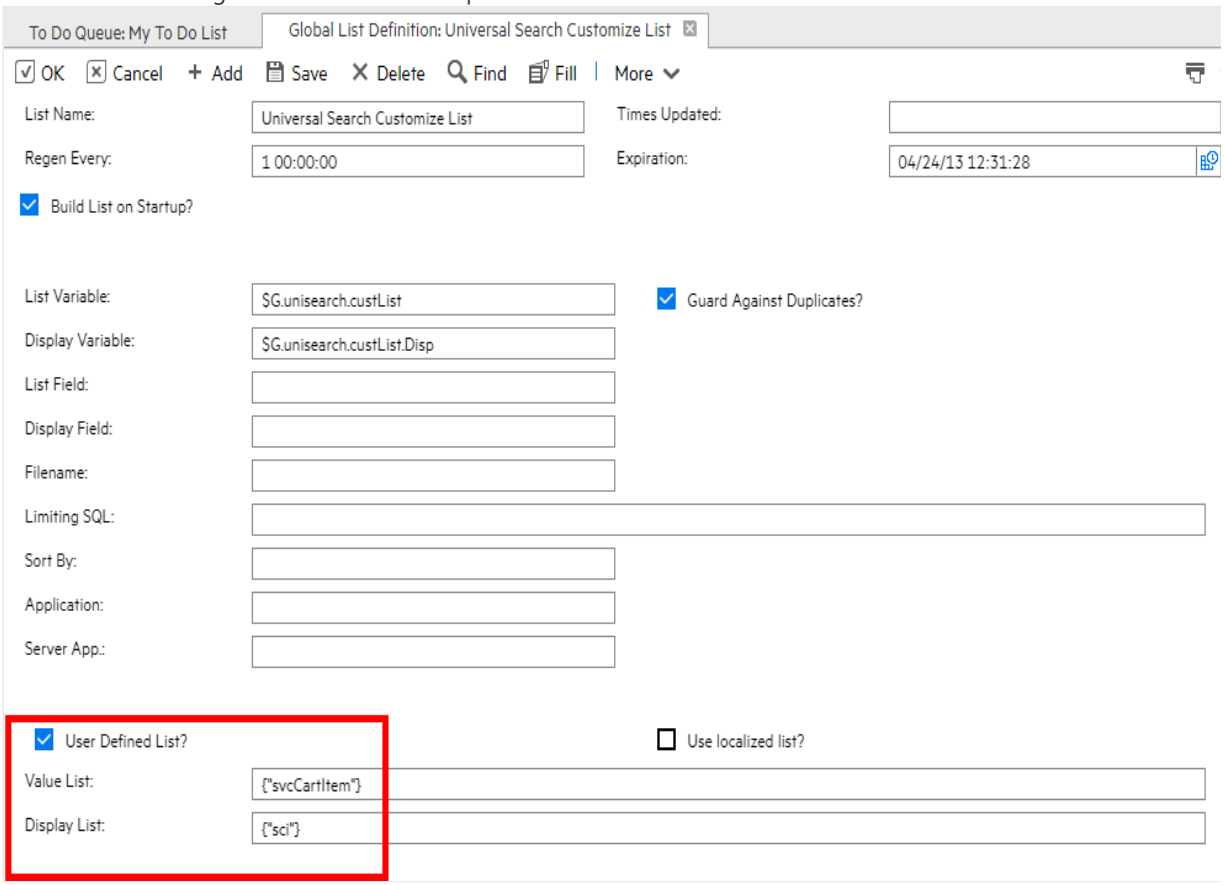


5. Click **Save**, and then click **OK**.

To configure the prefix of an incident record or a change request which doesn't have a prefix, follow these steps:

1. Sign in to the Service Manager server as a System Administrator.
2. Type `gl` in the Service Manager command line field, and then press **Enter**.
3. Type `Universal Search Customize List` in the **List Name** field, and then click **Search**.
4. Add customized record type to the **Value List** field and then add the corresponding prefix to the **Display List**.

Refer to the following screenshot as an example:



The out-of-the-box record is `svcCartItem` with the prefix `sci`. When you search for `sci1`, the List view displays the `svcCartItem` record with the ID of 1.

5. Click **Save**, and then click **OK**.

4.1.3. Add a view for Mobile Applications

Service Manager Mobile Applications provides several out-of-box views to access individual and group records. For example, the default views of Incidents are **Assigned to Me** and **Assigned to My Groups**. As the system administrator, you can add more views for the end-users. For example, if a group of users regularly searches on the same query, you can provide them with a shared query view as their default whenever they sign in to Service Manager from the Mobile Applications client.

To display an existing view in the Mobile Applications client, follow these steps:

1. Sign in to Service Manager using the Windows client as a System Administrator.
2. From the System Navigator, click **Favorites and Dashboards**.
3. Right-click the favorite that you want to edit, and then select **Properties**. The View Definition form opens.
4. Select the **Mobile Accessible** checkbox.
5. In the **Audience** tab, update the audience which is exposed to the view, and then click **Save**.

Tip

Tip You may need to configure the access to browse for certain views:

1. From the System Navigator, click **System Administration** > **Base System Configuration** > **Miscellaneous** > **View/Favorites**.
2. Select the view and click the **Audience** tab.
3. Select **Selected Groups** and update the Groups list.

To create a new view for the Mobile Applications client, follow these steps:

1. Sign in to Service Manager using the Windows client as a System Administrator.
2. Click **System Administration** > **Base System Configuration** > **Miscellaneous** > **Views/Favorites**.
3. Click **New**. The New View wizard opens.
4. Select the Area to create the view and then click **Next**.
5. Specify the name and type for the view you wish to create, and then click **Next**.
6. Select the fields you want in your view.
   Click **Fields**. To add fields, select items from the table menu and click **Add to List**.

   > **Note**
   > Note If you select a field that has a link to another table, select from the secondary menu to add fields you want in the view.

7. Select properties for Group By, Sort By, Filter, and Autoformat as desired.
8. Click **Finish**.
9. Select the **Mobile Accessible** checkbox.
10. In the **Audience** tab, select an audience for the view and then click **Save**.
11. Open the To Do Queue, select the area in the Queue field as you defined in *step 4* and check which form is used by the new view.
12. Open Form Designer and search for the form.
13. Click **More Actions** > **Copy/Rename** to copy the form with a mobile suffix.

    > **Note**
    > Note When copying the forms, don't omit the .g suffix which is used by some forms. For example, you can copy sc.manage.cmr.g to sc.manage.cmr.g.mobile.

To remove a view from the Mobile Applications client, follow these steps:

1. Sign in to Service Manager using the Windows client as a System Administrator.
2. From the System Navigator, click **Favorites and Dashboards**.
3. Right-click the favorite that you want to edit, and then select **Properties**. The View Definition form opens.
4. In the **Audience** tab, update the audience which is to be excluded from the view; or you can clear the **Mobile Accessible** checkbox to remove the view from Mobile Applications.
5. Click **Save**.

4.1.4. Add a form for Mobile Applications

After you added a new view that enables you to navigate through lists of records, you need to add forms for these records in the List view.

To add a form for the Mobile Applications, follow these steps:

1. Sign in to Service Manager using the Windows client as a System Administrator.
2. Select the area in the **Queue** field as you defined in *step 4* of To create a new view for the Mobile Applications client.
3. Select the view you created in the **View** field.
4. Open a record and check which form the record uses.
5. Click **Tailoring** > **Forms Designer** and search for the form.
6. Click **More Actions** > **Copy/Rename** to copy the form with a mobile suffix. For example, you can copy IM.close.incident to IM.close.incident.mobile.

> **Note**
>
> Note The form used by a record may vary with different status or phases of that record. You need to add a form for each status or phase respectively. For example, the form associates with an Incident can be either IM.close.incident or IM.update.incident. If you have made any omissions, the Mobile Applications client will automatically synchronize data with the original form in the Service Manager database after you logged on to the system.

4.1.5. Customize the fields on Mobile Applications form

For a better appearance of a record's Detail view on the screen of your smartphone, you may need to add, rearrange or remove the fields on a Mobile Applications form.

For detailed information about the widgets on Service Manager Mobile Applications forms, see Mobile Applications Form Widgets.

To customize the fields on a Mobile Applications form, follow these steps:

1. Follow *step 1* to *step 4* as described in Add a form for Mobile Applications.
2. Click **Tailoring** > **Forms Designer** and search for the form with the .mobile suffix.
3. Click **Design** to open design mode.
4. Remove the fields which aren't exposed to be displayed on the form. The field groups will be displayed as corresponding tabs in the Detail view on Mobile Applications client.

   > **Note**
   >
   > Note
   >
   > - Do not nest field groups.
   > - For usability and performance issues, don't include more than ten fields in a group.
   > - If there is one field group defined on a form at a minimum, the widgets which aren't included in this group will not be displayed on a Mobile Applications form.

5. Specify the **Label for** property of every lable to match with the **Name** property of the widget next to it. Take the IM.close.incident.mobile form for example. If the **Name** property of the Incident ID text box is `id`, the **Label for** property of the Incident ID lable must be specified to `id`.
6. Before adding a field, make sure the field exists in the Database Dictionary. Otherwise, you must firstly add this new field to the Database Dictionary. See Add a field to a table for detailed steps.
7. Click **OK** to save your customization.

4.1.6. Customize the action bar options

To customize the action bar options on the Mobile Applications client, follow these steps:

1. Sign in to Service Manager using the Windows client as a System Administrator.
2. Select the area in the **Queue** field as you defined in *step 4* of To create a new view for the Mobile Applications client.
3. Open a record and check which display screen the record uses. For example, cm.view.display.
4. From the System Navigator, click **Tailoring** > **Tailoring Tools** > **Display Screens** and search for the display screen.
5. Click **Show in Mobility**.
6. Select the area in the **File Name** field.
7. Select **Mobile** in the **Client Type** field, and then click **Search**. IDs of the action bar options are displayed in the **Button Ids** field.
8. Refer to the option IDs and the actions listed in the **Options** tab, add or remove the IDs.
9. Click **Save** to save your customization.

4.1.7. Configure the maxRequestPerSecond parameter

The *maxRequestPerSecond* parameter in the Service Manager Mobile Applications configuration file (web.xml) defines the maximum allowed request per second for one user session from the Service Manager Mobile Applications client. The default

value for this parameter is 10. To allow unlimited requests, set the value to -1.

4.2. Localize Service Manager Mobile Applications

The officially supported languages of Service Manager can meet most localization requirements. However, if you need to localize Mobile Applications to support additional languages, follow these steps:

1. Service Manager configurations.
    1. Localize the Service Manager server (including messages and formats) to the new language.
    2. Refer to Add a form for Mobile Applications to add new forms for Mobile Applications.
2. Service Manager Mobile Applications configurations.
    1. Copy <mobile web application directory>\WEB-INF\classes\appLoginBundle.properties and save this file as <mobile web application directory>\WEB-INF\classes\appLoginBundle_<Service Manager language ID>.properties.

        Tip

        Tip Refer to the Service Manager language table for a list of language IDs. For example, if the new language is Malay, change appLoginBundle_<Service Manage Language ID>.properties to appLoginBundle_z3.properties.

    2. Translate each text string in appLoginBundle_<Service Manager language ID>.properties to the new language and then save this file.
    3. Using native2ascii or other tools to convert appLoginBundle_<Service Manager language ID>.properties to one with ASCII and/or Unicode escapes.

        Tip

        Tip For detailed usage of native2ascii, refer to the following Oracle web page: http://docs.oracle.com/javase/7/docs/technotes/tools/windows/native2ascii.html.

    4. Copy <mobile web application directory>\WEB-INF\classes\appMainBundle.properties and save this file as <mobile web application directory>\WEB-INF\classes\appMainBundle_<Service Manager language ID>.properties.
    5. Translate each text string in appMainBundle_<Service Manager language ID>.properties to the new language and then save this file.
    6. Using native2ascii or other tools to convert appMainBundle_<Service Manager language ID>.properties to one with ASCII and/or Unicode escapes.
    7. Restart the web applications server and launch Service Manager Mobile Applications again.

4.3. Protecting communications between Mobile Applications and the Service Manager server

To protect communications between Mobile Applications and the Service Manager server, you can perform the following tasks:

- Set up Secure Sockets Layer (SSL)
- Set up trusted sign-on (TSO)
- Set up Lightweight Single Sign-On (LW-SSO)

4.3.1. Set up Secure Sockets Layer (SSL)

Service Manager supports Secure Hypertext Transfer Protocol (HTTPS), which encrypts and decrypts message requests and responses. Service Manager uses Secure Sockets Layer (SSL) for encryption only and relies on the server to authenticate each operator's user name and password. Service Manager supports SSL for the following connections:

- SSL on the Service Manager server to encrypt all communications between clients and the server

- SSL on the Mobile Applications client to verify the client's identity and limit server connections to these identified clients

For detailed information about how to enable SSL in Service Manager, refer to Configure two-way SSL.

Enable SSL on the Service Manager server

Service Manager clients send the operator's user name and password in each request as part of an HTTP Basic Authorization header. In order to protect these operators' user names and passwords, you can enable SSL on the Service Manager server.

Enabling SSL on the Service Manager server includes three major steps:

1. Purchase a certificate for the Service Manager server from a certificate authority, or create a certificate by yourself.
2. Create a server keystore. For detailed steps, refer to Configure two-way SSL.
3. Add SSL parameters to the sm.ini file.

> **Note**
> Note Ignore the SSO parameters in the table at this point.

Enable SSL on the Mobile Applications client

In order to restrict access to the server to only clients that are known and identified by the server, you can enable SSL on Service Manager clients. Enabling SSL on clients requires creating or purchasing signed certificates for each Service Manager client. Service Manager Mobile Applications can share a single signed certificate for all handset connections. If you enable SSL on the client, we recommend that you also enable SSL on the server to encrypt all communications between clients and the server.

To enable SSL on the Mobile Applications client, follow these steps:

1. Purchase a certificate for the Mobile Applications client host from the same certificate authority for the Service Manager server certificate, or create a certificate by yourself.
2. Create a client keystore. For detailed steps, refer to Configure two-way SSL.
3. Import the client certificate to a trusted clients keystore. For example, trustedclients.keystore.
4. Copy the trusted clients keystore (trustedclients.keystore) to the Service Manager server's RUN folder.
5. Open webapp-9.70.xxxx.war in an archive management program.
6. Copy the cacerts file to the WEB-INF directory.
7. Extract the web.properties file from the WEB-INF directory to your local system, and then open this file in a text editor.
8. Locate the following codes and configure the parameters as follow:

```
endpoint=https://full.qualified.domain.name:13443/SM/ui
cacerts=
keystore=
keystorePassword=
cacertsPassword=
```

9.

Required parameters for Mobile Applications client SSL configurations

Parameter	Default value	Description
endpoint	`https://full.qualified.domain.name:13443/SM/ui`	Change `full.qualified.domain.name` to the domain name of your Service Manager server. Change `13443` to your SSL port number.
cacerts	The default value is null.	The keystore that contains your server's certificate authority. For example, /WEB-INF/cacerts.
keystore	The default value is null.	The keystore that contains your mobility's signed certificate. For example, /WEB-INF/smmobile.p12.
keystorePassword	The default value is null.	The password of the keystore file. We recommend adding a strong password for the keystore. After you restart the Mobile Applications client, the keystore password is encrypted and the original `keystorePassword=` code is automatically updated to `*keystorePassword=<encrypted string>`. **Note** Note The preceding asterisk (*) implies that the keystore password is encrypted. To change the password, remove the asterisk and replace the encrypted string with the new password.

cacertsPassword	The default value is null.	The password of the cacerts file. We recommend adding a strong password for the cacerts. After you restart the Mobile Applications client, the password is encrypted and the original `cacertsPassword=` code is automatically updated to `*cacertsPassword=<encrypted string>`. **Note** Note The preceding asterisk (*) implies that the cacerts password is encrypted. To change the password, remove the asterisk and replace the encrypted string with the new password.

10. Save the file and add the updated file back to the WEB-INF directory of the webapp-9.61.xxxx.war archive.
11. Restart the Service Manager Mobile Applications client.

Set up SSL between the smartphone browser and Mobile Applications

You can set up SSL to allow smartphone browsers and web servers to communicate over a secure connection. The data being sent is encrypted by one side, and decrypted by the other side before processing. This is a two-way process, meaning that both the server and the browser encrypt all traffic before sending out the data.

After SSL is set up on the web servers, browsers which support secure flag only send cookies with the secure flag when the request is going to an HTTPS page, that is, the browser doesn't send a cookie with the secure flag set over an unencrypted HTTP request.

You need to set up SSL on the web server or web application server where the Mobile Applications client is deployed, and update the port number mapping for the Mobile Applications client as necessary.

Set up SSL on you web server

For details, refer to the documentation of your web server.

Note

Note It is recommended to enable HTTPOnly and Secure cookies on your web application server to help prevent malicious JavaScript injection and make the browser (or other http clients) only send cookies over SSL connections. For more information, see https://softwaresupport.softwaregrp.com/km/KM02233778.

Set up SSL on Tomcat

Configuring Tomcat to use SSL is only necessary when Tomcat is run as a stand-alone web server. When Tomcat is primarily run as a Servlet/JSP container behind another web server, such as Apache or Microsoft IIS, it is necessary to configure the primary web server to handle the SSL connections from users. Typically, this server negotiates all SSL-related functionality, and then pass on any requests destined for the Tomcat container only after decrypting those requests. Likewise, Tomcat returns cleartext responses that are encrypted before being returned to the user's browser. In this case, Tomcat knows that communications between the primary web server and the client are taking place over a secure connection, but it doesn't participate in the encryption and decryption processes.

For information about how to set up SSL on Tomcat, refer to *Apache Tomcat SSL Configuration HOW-TO* document.

Note

Note Since you have already got your certificates for the server as described in the previous step in the *Enable SSL on the Mobile Applications client* section, you only need to perform the steps in the **Edit the Tomcat Configuration File** section in Apache Tomcat SSL Configuration HOW-TO document.

Update the port number mapping for the Mobile Applications client

To update the port number mapping for the Mobile Applications client, follow these steps:

1. Go to Tomcat webapps directory of Mobile Applications.
2. Open WEB-INF\spring\security.xml with a text editor.
3. Locate the following codes, and then update the default security ports as necessary.

```
<security:http auto-config='true'>
<security:port-mappings>
<security:port-mapping http="8080" https="8443"/>
</security:port-mappings>
<security:intercept-url pattern="/**"  requires-channel="https"/>
</security:http>
```

Note

Note Remember to update the port number to the same as the configuration in your web server or web application server.

4. Save the changes.

4.3.2. Set up trusted sign-on (TSO)

Trusted sign-on (TSO) or Single sign-on is an optional Mobile Applications configuration that relies on a working SSL configuration, and integration with a trusted authentication source such as CA SiteMinder, IBM Webseal, and Integrated Windows Authentication. It also requires a web server to accept the pre-authenticated HTTP header information from your authentication source.

For more information, refer to Example: Configuring Single Sign-on with third party authentication on the web tier.

When you enable trusted sign-on (TSO), Mobile Applications will used the username of the user represented by the Principal to bypass the Service Manager log-on screen, and then enter the application directly.

Enable TSO on the Service Manager server

In the sm.ini file, add the following parameter:

```
trustedsignon:1
```

Enable TSO on the Mobile Applications client

To enable TSO on the Mobile Applications client, follow these steps:

1. Open webapp-x.xx.xxxx.war in an archive management program.
2. Extract the web.properties file from the WEB-INF directory to your local system, and then open this file in a text editor.
3. Locate the `isCustomAuthenticationUsed` script and set the value to `false`:

```
# Set false to enable Trusted Sign-on
isCustomAuthenticationUsed=false
```

4. Save the file and add the updated file back to the WEB-INF directory of the webapp-x.xx.xxxx.war archive.

5. Extract the security.xml file from the WEB-INF/spring directory to your local system, and then open this file in a text editor.
6. Locate the following scripts:

   ```
   <!-- <security:custom-filter ref="preAuthenticationFilter" after="SECURITY_CONTEXT_FILTER"/>
   <security:custom-filter ref="lwSsoFilter" before="BASIC_AUTH_FILTER"/>
   <security:custom-filter ref="springSecurity2lwssoIntegrationFilter" position="LAST"/> -->
   ```

   Uncomment the first line as follows to enable TSO:

   ```
   <security:custom-filter ref="preAuthenticationFilter" after="SECURITY_CONTEXT_FILTER"/>
   <!--  <security:custom-filter ref="lwSsoFilter" before="BASIC_AUTH_FILTER"/>
   <security:custom-filter ref="springSecurity2lwssoIntegrationFilter" position="LAST"/> -->
   ```

   Or,
   For HTTP header pre-authentication, locate the following scripts and uncomment it:

   ```
   <!--<security:custom-filter ref="httpHeaderPreAuthenticationFilter"
   after="SECURITY_CONTEXT_FILTER"/>-->
   ```

7. Locate the httpHeaderPreAuthenticationFilter bean definition, and then at least change the principalRequestHeader setting, corresponding to the Header your Identity Management solution uses. For Webseal, iv-user is used as the value of principalRequestHeader, Siteminder often uses sm_user or sm_universalid. Note that this setting is case sensitive.
8. Save the file and add the updated file back to the WEB-INF/spring directory of the webapp-x.xx.xxxx.war archive.
9. Configure the deployment environment to support the mobile system to receive the customized user information for authentication.

4.3.3. Set up Lightweight Single Sign-On (LW-SSO)

If Lightweight Single Sign-On (LW-SSO) is enabled on both the Service Manager server and the Mobile Applications, Service Manager authentication will be bypassed if you have logged into another web application which also has LW-SSO enabled. The detailed bypass critieria includes both the protected domain/IP/DNS name and the initial LW-SSO string.

Enable LW-SSO on the Service Manager server

For detailed information about how to enable LW-SSO on Service Manager server, refer to the *Configure LW-SSO in the Service Manager server* topic in the help center.

Enable LW-SSO on the Mobile Applications client

To enable LW-SSO on the Mobile Applicationsclient, follow these steps:

1. Open webapp-x.xx.xxxx.war in an archive management program.
2. Extract the web.properties file from the <Tomcat>/webapps/<Mobility>/WEB-INF directory to your local system, and then open this file in a text editor.
3. Update the web.properties file.
   1. Change the value of context parameter **isCustomAuthenticationUsed** to false.

      ```
      isCustomAuthenticationUsed=false
      ```

   2. Save the file.
4. Extract the security.xml file from the WEB-INF/spring directory to your local system, and then open this file in a text editor.

5. Locate the following scripts:

```
<!-- <security:custom-filter ref="preAuthenticationFilter" after="SECURITY_CONTEXT_FILTER"/>
<security:custom-filter ref="lwSsoFilter" before="BASIC_AUTH_FILTER"/>
<security:custom-filter ref="springSecurity2lwssoIntegrationFilter" position="LAST"/> -->
```

   Uncomment the second line and the third line as follows to enable LW-SSO:

```
<!-- <security:custom-filter ref="preAuthenticationFilter" after="SECURITY_CONTEXT_FILTER"/> -->
<security:custom-filter ref="lwSsoFilter" before="BASIC_AUTH_FILTER"/>
<security:custom-filter ref="springSecurity2lwssoIntegrationFilter" position="LAST"/>
```

6. Save the file and add the updated file back to the WEB-INF/spring directory of the webapp-x.xx.xxxx.war archive.
7. Extract the lwssofmconf.xml file from the WEB-INF/classes directory to your local system, and then open this file in a text editor.

8. Configure the parameters as follows:

```
<?xml version="1.0" encoding="UTF-8"?>
<lwsso-config
xmlns="http://www.hp.com/astsecurity/idmenablmentfw/lwsso/2.0">
<enableLWSSO enableLWSSOFramework="true" enableCookieCreation="true"
cookieCreationType="LWSSO" enableSAML2Support="false"/>
<webui>
<validation>
<in-ui-lwsso>
<lwssoValidation  id="ID000001">
<domain>yourdomain.net</domain>
<crypto cipherType="symmetricBlockCipher"
engineName="AES" paddingModeName="CBC" keySize="256"
encodingMode="Base64Url"
initString="Please put your string here"></crypto>
</lwssoValidation>
</in-ui-lwsso>
</validation>
<creation>
<lwssoCreationRef id="ID000002">
<lwssoValidationRef refid="ID000001"/>
<expirationPeriod>60</expirationPeriod>
</lwssoCreationRef>
</creation>
<logoutURLs>
<url>.*/std/logout</url>
<url>.*/std/goodbye</url>
</logoutURLs>
<multiDomain>
<trustedHosts>
<!-- <DNSDomain>companydomain1.com</DNSDomain>
<DNSDomain>companydomain2.com</DNSDomain>
<NetBiosName>myserver1</NetBiosName>
<NetBiosName>myserver2</NetBiosName>
<IP>192.168.12.13</IP>
<IP>192.168.12.14</IP>
<FQDN>myserver1.companydomain1.com</FQDN>
<FQDN>myserver2.companydomain2.com</FQDN>  -->
</trustedHosts>
</multiDomain>
</webui>
</lwsso-config>
```

Required parameters for Mobile Applications client LW-SSO configurations

Parameter	Default value	Description
enableLWSSOFramework	`false`	Change this value to `true` to enable the LW-SSO framework.
domain	`yourdomain.net`	Change `yourdomain.net` to the domain name of the server host where you deploy your Mobile Applications client.
initString	`Please put your string here`	Enter your initial string. This initial string must be same as the initString value in the Service Manager LW-SSO configuration (minimum length: 12 characters). For example, smintegrationlwsso.

Note

Note Beginning from the `<creation>` section to the end of the lwssofmconf.xml file, the variables are subject to change according to your actual deployment.

9. Save the file and add the updated file back to the WEB-INF/classes directory of the webapp-x.xx.xxxx.war archive.

4.4. Mobile Applications Form Widgets

This section provides detailed information about the widgets on Service Manager Mobile Applications forms.

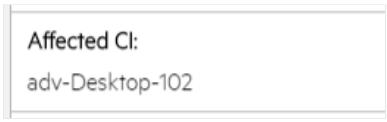
This section includes:

- Label control
- Text control
- Text area control
- Date control
- Combo Box control
- Comfill control
- Group control
- Button control
- Check box control
- Table control
- Table column control
- Subform control
- Dynamic Form control
- Notebook control
- Notebook tab control

4.4.1. Label control

Use this control to add a label. A label is a single line of text you can use to give titles to forms, give labels to objects within the form, or otherwise place text on the form.

Refer to the following screenshot as an example of the label control widget on a Mobile Applications form:



Property	Usage
Name	(Optional) Specify a unique identifier for the object on the screen. This name is used by external applications, such as RAD, to dynamically change the properties of the object.
Caption	(Required) Specify a text caption for the object.
Input	(Optional) Specify the database field or variable to associate with this control. When input is specified, the Label will be displayed as a read-only Text.
Label For	(Optional) Specify the widget which the label belongs to.

Note

Note A label which is neither referred by any field or variable nor belongs to any other widget will not be displayed on a Mobile Applications form.

4.4.2. Text control

Use this control to add a text box that displays the contents of a text field and conditionally enables users to enter or modify its contents.

Refer to the following screenshot as an example of the text control widget on a Mobile Applications form:

Property	Usage
Name	(Optional) Specify a unique identifier for the object on the screen. This name is used by external applications, such as RAD, to dynamically change the properties of the object. This name is also used by the Label For property to specify the label of this widget.
Input	Specify the database field or variable to associate with this control.
Read-Only	Select this option to disable editing capabilities and provide only viewing access to the field.
Mandatory	Setting Mandatory to **Yes** by selecting the checkbox indicates that the field is required and therefore an asterisk indicating a required field is displayed. This is a visible change only. To make the field mandatory for any form it appears on, use the System Definition Utility. To make the field mandatory for a small number of forms only, use Format Control.

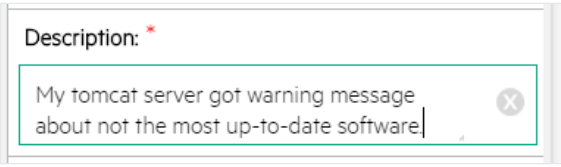
Note

Note If the text is too long for a mobile device to display in one line, use a Text area control instead.

4.4.3. Text area control

Use this control to add a text area that displays the contents of a text field and conditionally enables users to input several lines of data. This object contains scroll bars and allows text wrapping.

Refer to the following screenshot as an example of the text area control widget on a Mobile Applications form:

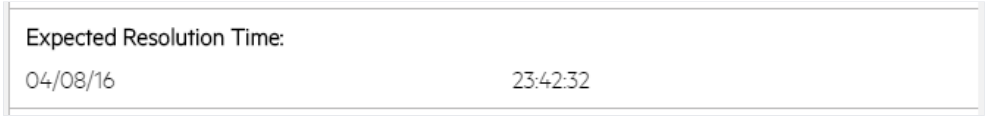


Property	Usage
Name	(Optional) Specify a unique identifier for the object on the screen. This name is used by external applications, such as RAD, to dynamically change the properties of the object. This name is also used by the Label For property to specify the label of this widget.
Input	Specify the database field or variable to associate with this control.
Read-Only	Select this option to disable editing capabilities and provide only viewing access to the field.
Mandatory	Select this option so that a red asterisk indicating a required field is displayed. This is a visible change only. • To make the field mandatory for any form it appears on, use the System Definition Utility. • To make the field mandatory for a small number of forms only, use Format Control.

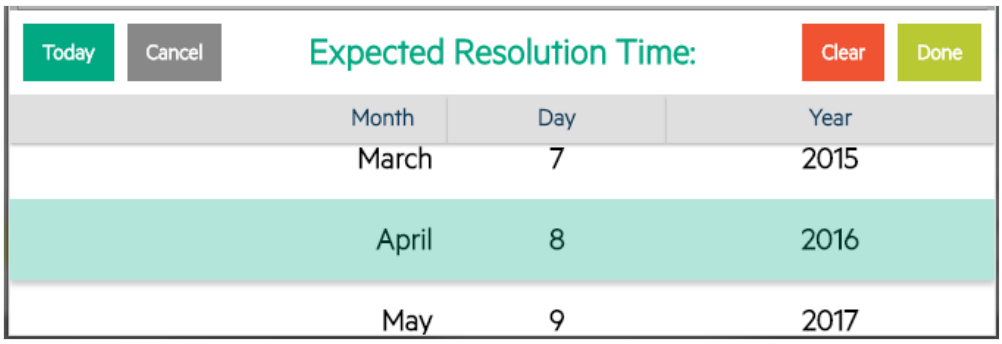
4.4.4. Date control

Use this control to add a text box that displays and enables users to update the contents of a date field.

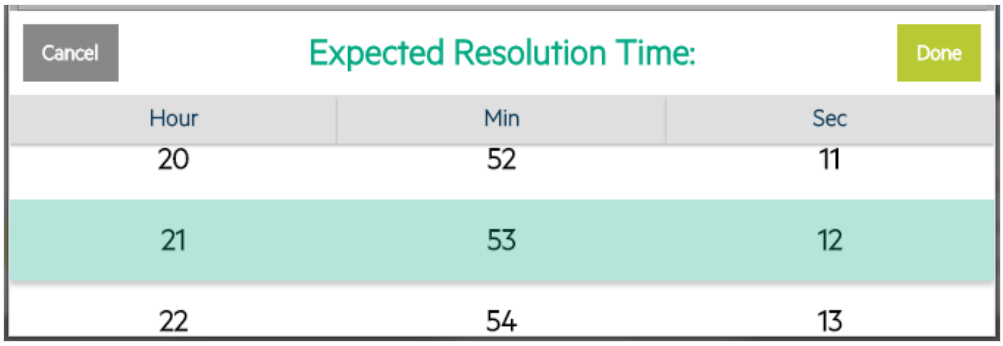
Refer to the following screenshot as an example of the date control widget on a Mobile Applications form:



Refer to the following screenshot as an example of the date selector on a Mobile Applications form:



Refer to the following screenshot as an example of the time selector on a Mobile Applications form:

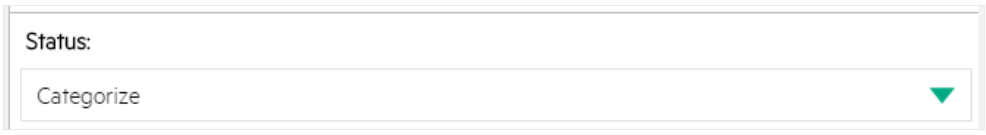


Property	Usage
Name	(Optional) Specify a unique identifier for the object on the screen. This name is used by external applications, such as RAD, to dynamically change the properties of the object. This name is also used by the Label For property to specify the label of this widget.
Input	Specify the database field or variable to associate with this control.
Read-Only	Select this option to disable editing capabilities and provide only viewing access to the field.
Mandatory	Select this option so that a red asterisk indicating a required field is displayed. This is a visible change only. • To make the field mandatory for any form it appears on, use the System Definition Utility. • To make the field mandatory for a small number of forms only, use Format Control.

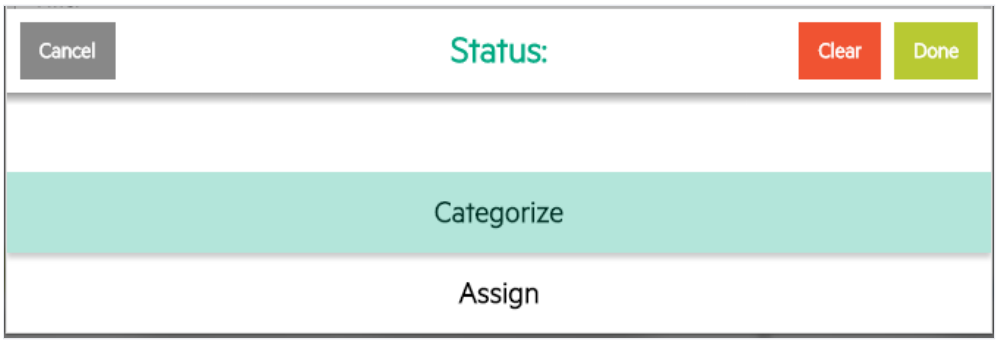
4.4.5. Combo Box control

Use this control to add a Combo Box that enables users to click a button and select from a drop-down list. The items in the list are associated with a database field or variable. Users can also type in a value if the checkbox in the Select Only property is cleared (set to false).

Refer to the following screenshot as an example of the Combo Box control widget on a Mobile Applications form:



Refer to the following screenshot as an example of the Combo Box selector on a Mobile Applications form:



Property	Usage
Name	(Optional) Specify a unique identifier for the object on the screen. This name is used by external applications, such as RAD, to dynamically change the properties of the object. This name is also used by the Label For property to specify the label of this widget.
Input	Specify the database field or variable to associate with this control.
Read-Only	Select this option to disable editing capabilities and provide only viewing access to the field. Give read–only fields a tab–stop value of –1 to prevent users from tabbing into them.
Mandatory	Select this option so that a red asterisk indicating a required field is displayed. This is a visible change only. • To make the field mandatory for any form it appears on, use the System Definition Utility. • To make the field mandatory for a small number of forms only, use Format Control.

Value List	In conjunction with the Input property, defines how the Display List values are identified in the database. Value and Display Lists are entered using the Edit List dialog box. You can enter hard coded entries for each list, or you can supply a variable as the first and only entry. The run time values of the variable are used to populate these lists. **Note** Note When defining Value List and Display List properties, you should avoid using the following reserved characters: • semicolon (;) • tab key • newline (carriage return) • form feed • backspace • equal sign (=) **Caution** Caution Do not use keys (such as backspace). The system can't process their ASCII representation entries correctly, which may lead to unpredictable results.

Display List	Specify the values that appear in the drop-down list at run time. There must be a one–to–one correspondence between the values for Value List and for Display List. If the Display List is the only populated property, the display values are written to the database. **Note** Note When defining Value List and Display List properties, you should avoid using the following reserved characters: • semicolon (;) • tab key • newline (carriage return) • form feed • backspace • equal sign (=) **Caution** Caution Do not use keys (such as backspace). The system can't process their ASCII representation entries correctly, which may lead to unpredictable results.

Note

Note Combo box is always select-only on Service Manager Mobile Applications.

4.4.6. Comfill control

Use this control to add a combination Combo Box and Fill button. Comfill has all the properties of a Combo Box, plus the capability for Fill button. You can select which comfill buttons to display. For example, to display just a Fill button, set the Third Button Visible property to true by selecting the checkbox and set the Combo Button Visible property to false, by clearing the checkbox.

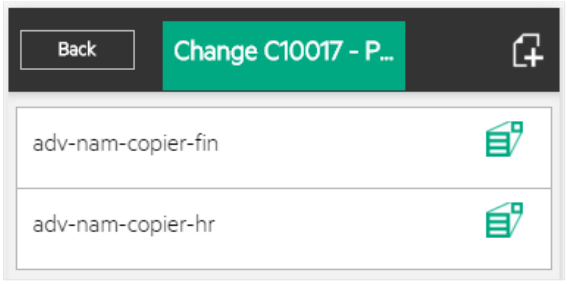
Note

Note As of Service Manager 9.41, the display name of a CI can be displayed in comfill for mobility. You can fill a CI by its display name.

Refer to the following screenshot as an example of the non-array comfill widget with popup form enabled on a Mobile Applications form:

Refer to the following screenshot as an example of the array comfill widget on a Mobile Applications form:

Refer to the following screenshot as an example of the array comfill widget after clicking:



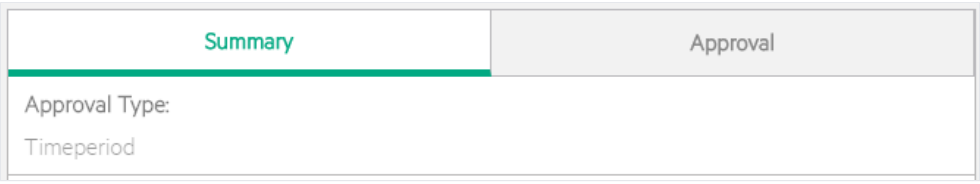
Property	Usage
Name	(Optional) Specify a unique identifier for the object on the screen. This name is used by external applications, such as RAD, to dynamically change the properties of the object. This name is also used by the Label For property to specify the label of this widget.
Input	Specify the database field or variable to associate with this control.
Read-Only	Select this option to disable editing capabilities and provide only viewing access to the field.
Mandatory	Select this option so that a red asterisk indicating a required field is displayed. This is a visible change only. • To make the field mandatory for any form it appears on, use the System Definition Utility. • To make the field mandatory for a small number of forms only, use Format Control.
Array Length	Specify the size of the scrolling region used to view array entries. A scroll bar appears beside the fields to allow users to view the array entries. • If a field is assigned an Array Length of 5, the form stacks five fields vertically to allow users to view the five array entries. • If Array Length is set to -1, the form dynamically creates as many fields as there are entries in the array. • If the Input data type is scalar, only a single text box appears. **Note** Note The screen object must be associated with an array data structure. The default is 0, which means one vertical line of information appears.
Value List	In conjunction with the Input property, defines how the Display List values are identified in the database. Value and Display Lists are entered using the Edit List dialog box. You can enter hard coded entries for each list, or you can supply a variable as the first and only entry. The run time values of the variable are used to populate these lists.

Display List	Specify the values that appear in the drop-down list at run time. There must be a one–to–one correspondence between the values for Value List and for Display List. If the Display List is the only populated property, the display values are written to the database.
Combo Button Visible	Select this option to make the Combo Button visible when the form opens. **Note** Note When this property is enabled, the Fill Button Visible property is no longer effective.
Fill Button ID	Specify a Control ID to transmit when clicked.
Fill Button Visible	Select this option to make the Fill Button visible when the form opens.
Popup Subform Format	Specify the form to display.
Popup Subform Input	Specify the database field or variable to associate with this control.
Popup Subform Enabled	Select this option to enable the Popup Subform for this control.

4.4.7. Group control

Use this control to add a container that enables you to logically group associated items. In the Windows client the Group has a rectangular border with a text label at the top.

Refer to the following screenshot as an example of the group control widget on a Mobile Applications form:



Property	Usage
Name	(Optional) Specify a unique identifier for the object on the screen. This name is used by external applications, such as RAD, to dynamically change the properties of the object.
Caption	(Required) Specify a text caption for the object.

Note

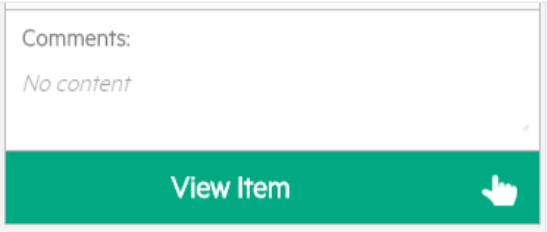
Note

- Do not nest field groups.
- For usability and performance issues, don't include more than ten fields in a group.
- If there is one field group defined on a form at a minimum, the widgets which aren't included in this group will not be displayed on a Mobile Applications form.

4.4.8. Button control

Use this control to add an input field that displays and enables users to update the contents of a numeric field and offers spinner buttons to increase or decrease a value.

Refer to the following screenshot as an example of the button control widget on a Mobile Applications form:



Property	Usage
Name	(Optional) Specify a unique identifier for the object on the screen. This name is used by external applications, such as RAD, to dynamically change the properties of the object.
Caption	Specify a text caption for the object.
Button ID	Specify a numeric identification that specifies a Control ID to transmit when clicked.

4.4.9. Check box control

Use this control to add a checkbox that displays and enables users to update the contents of a Boolean (logical) field, which can evaluate to true, false, unknown, or null.

Refer to the following screenshot as an example of the checkbox control widget on a Mobile Applications form:

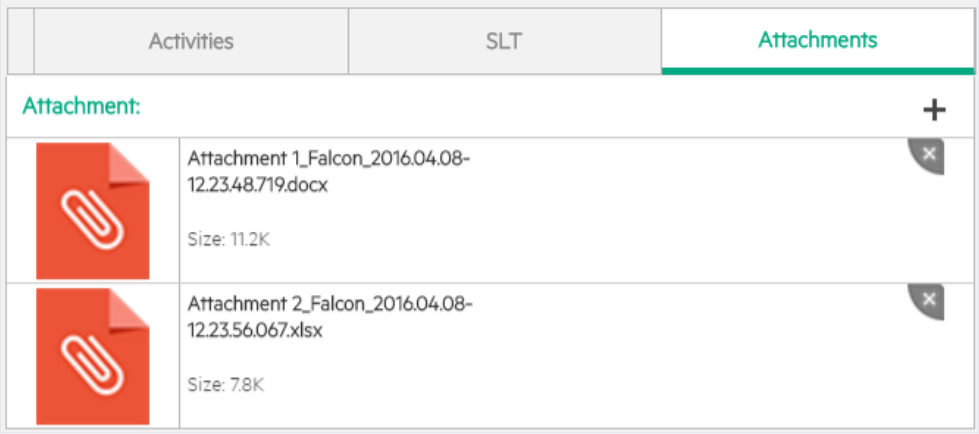
Property	Usage
Name	(Optional) Specify a unique identifier for the object on the screen. This name is used by external applications, such as RAD, to dynamically change the properties of the object.
Caption	Specify a text caption for the object.
Input	Specify the database field or variable to associate with this control.
Read-Only	Select this option to disable editing capabilities and provide only viewing access to the field. Give read–only fields a tab–stop value of –1 to prevent users from tabbing into them.

4.4.10. Attachments control

Use this control to add a box into which users can place non-Service Manager documents (For example, from Microsoft Word or Microsoft Excel).

To place an attachment container on a form, click **Attachments**, and then click the form.

Refer to the following screenshot as an example of the Attachments control widget on a Mobile Applications form:



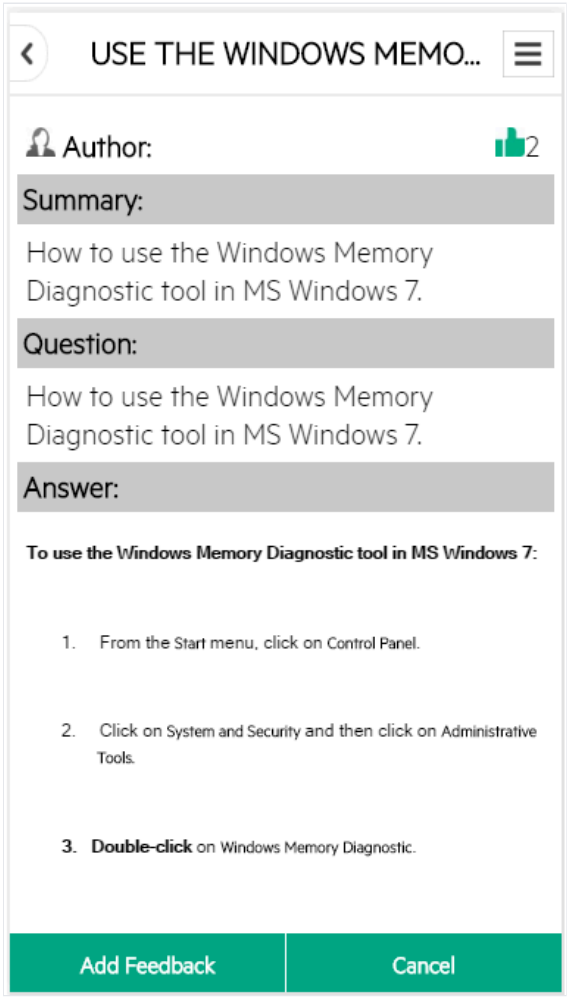
Property	Usage
Name	(Optional) Specify a unique identifier for the object on the screen. This identifier is used by external applications, such as RAD, to dynamically change the properties of the object.
Visible	Select this option to make the object visible on the form. Clear the checkbox to hide the object from view on the form.
Read-Only	Disable the editing capability of the control. Users can only view the attached documents when this option is selected. To prevent users from using the Tab key to enter the control, you can assign the value of -1 to the Read-Only field.
Maximum Attachment Size	Specify the maximum size of an attachment, in bytes. **Note** Note Size limits defined in the Maximum Attachment Size field in a user's operator record override the size limit specified in the Company record. A non-zero value overrides any values specified in the Company or Operator record.
Total Attachment Size	Specify the maximum amount of memory (in bytes) that all attachments in a form can use. The total size of all attachments must be lower than this threshold or the form no longer accepts additional attachments.
Maximum Attachments Allowed	Select whether you want to allow one or any number of attachments. **Known issue**: This property is unsupported in Mobile Applications.

4.4.11. HTML Viewer control

Use this control to add an HTML Viewer that enables users to view the HTML created by using the HTML Editor.

To place an HTML Viewer on a form, click **HTML Viewer**, and then click the form. Be sure to put the name of the database field or variable that you want to associate with this control into the Input property.

Refer to the following screenshot as an example of the HTML Viewer control widget on a Mobile Applications form:

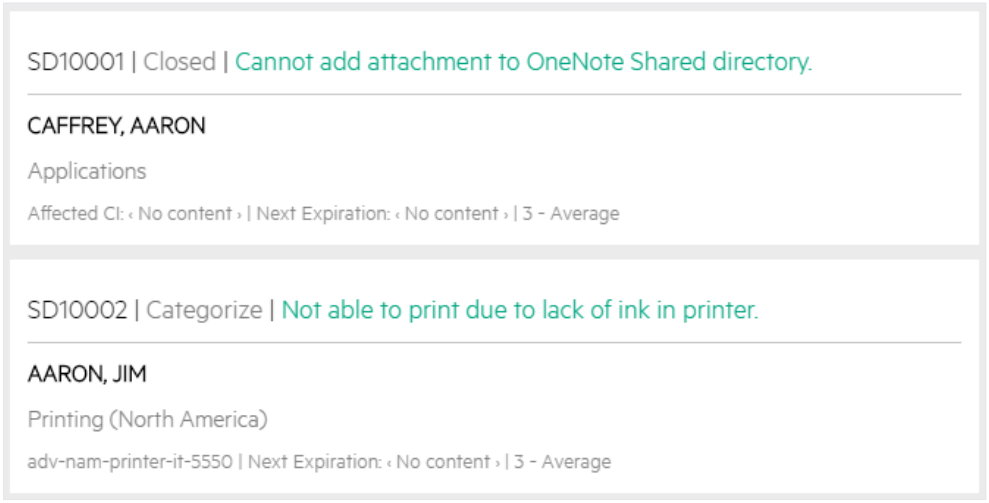


Property	Usage
Input	Specify the database field or variable to associate with this control.
Name	(Optional) Specify a unique identifier for the object on the screen.
Visible	Make the object visible or invisible on the form. Selecting the checkbox makes the object visible and clearing the checkbox makes it invisible.

4.4.12. Table control

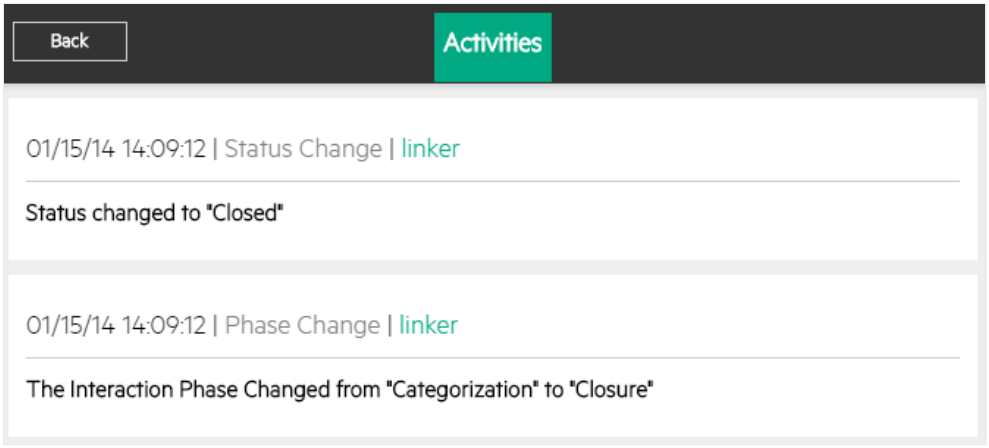
Use this control to add a table that displays one or more columns of data in a scrollable pane. The tables you create in the Form Designer appear as a rectangular region subdivided by rows and columns. The look and feel mimics tables in Microsoft Windows applications like Excel.

Refer to the following screenshot as an example of the table control widget in List view:



Refer to the following screenshot as an example of the table control widget in Detail view:

Refer to the following screenshot as an example of the table control widget after clicking:



Property	Usage
Name	(Optional) Specify a unique identifier for the object on the screen. This field is required if you enable the Multiple Selection property. This name is used by external applications, such as RAD, to dynamically change the properties of the object. This name is also used by the “Label For” property to specify the label of this widget.
Read-Only	Always read-only.
Columns	Specify a list of columns in the table.

4.4.13. Table column control

Use this control to add a column to an existing table.

Property	Usage
Name	(Optional) Specify a unique identifier for the object on the screen. This name is used by external applications, such as RAD, to dynamically change the properties of the object.
Caption	Specify a text caption for the object.
Input	Specify the database field or variable to associate with this control.
Field	In the case where the array specified in the input property is an array of structure, it specifies the field in the structure for this object.
Display List	Specify the values that appear in the drop-down list at run time. There must be a one–to–one correspondence between the values for Value List and for Display List. If the Display List is the only populated property, the display values are what were written to the database.
Read-Only	Always read-only.

4.4.14. Subform control

Use this control to display a subform on another form.

Property	Usage
Name	(Optional) Specify a unique identifier for the object on the screen. This name is used by external applications, such as RAD, to dynamically change the properties of the object.
Format	Specify the form to display. Be sure you type the exact form name.
Virtual Join	Select this option to associate virtual join run time processing with the subform object. Clear the checkbox to use a same-file join.
Display Using Table	Select this option to display the subform in table format.
Input	Specify the database field or variable to associate with this control.

4.4.15. Dynamic Form control

Use this control to add a dynamic form that becomes visible when populated by XML from a RAD application or JavaScript. The aspect and content of the dynamic form depend on the XML.

To place a dynamic form on a form, click **Dynamic Form** and then click the form.

Be sure to put the name of the database field or variable that you want to associate with this control into the Input property.

Property	Usage
Name	(Optional) Specify a unique identifier for the object on the screen. This name is used by external applications, such as RAD, to dynamically change the properties of the object.
Visible	Select this option to make the object visible on the form. Clear the checkbox to hide the object from view on the form.
Read-Only	In this release, we doesn't support change user options from mobile. All the contents inside the dynamic form will be read-only.
Input	Specify the database field or variable to associate with this control.

4.4.16. Notebook control

Use this control to subdivide the contents of a screen into logical groups or categories. Notebooks provide an aesthetic way of organizing large amounts of data into small spaces.

Click to the right of the last tab on the notebook to open the notebook properties. Click the tab and then the blank area below the tab to open the notebook tab properties.

To place a notebook on a form, click **Notebook** and then click the form.

Property	Usage
Name	(Optional) Specify a unique identifier for the object on the screen. This name is used by external applications, such as RAD, to dynamically change the properties of the object.
Visible	Select this option to make the object visible on the form. Clear the checkbox to hide the object from view on the form.

4.4.17. Notebook tab control

Use this control to add a tab to an existing notebook. Navigate to each page of the notebook by selecting its tab.

Click to the right of the last tab on the notebook to open the notebook's properties. Click the **notebook** tab and then the blank area below the tab to open the notebook tab's properties.

Property	Usage
Name	(Optional) Specify a unique identifier for the object on the screen. This name is used by external applications, such as RAD, to dynamically change the properties of the object.
Caption	(Required) Specify a text caption for the object.
Visible	Select this option to make the object visible on the form. Clear the checkbox to hide the object from view on the form.

4.5. Mobile Applications for Service Manager Process Designer

Multiple Mobile Applications forms are designed for Service Manager Codeless. Refer to the following tables for detail.

Change module

Workflow	Phase	Display Option/Action/Transition	Forms for Mobile Applications
Emergency	E-CAB Approval	Approve, Deny, Retract, and Save	chm.emergency.approval.mobile chm.subform.activity.mobile
Standard	Authorization CAB	Approve, Deny, Retract, and Save	chm.standard.approval.mobile
Normal	T-CAB Approval D-CAB Approval	Approve, Deny, Retract, and Save	chm.normal.dcab.approval.mobile chm.normal.tcab.approval.mobile
Change Proposal	Authorization CAB	Approve, Deny, Retract, and Save	chm.proposal.tcab.approval.mobile

Incident module

Workflow	Phase	Display Option/Action/Transition	Forms for Mobile Applications
Incident	Categorization Investigation Review Recovery Closure	Save and Close	im.incident.categorization.mobile im.incident.investigation.mobile im.incident.recovery.mobile im.incident.review.mobile im.incident.closure.mobile im.incident.close.mobile

Interaction module

Workflow	Phase	Display Option/Action/Transition	Forms for Mobile Applications
Service Catalog	Categorization Work In Progress Review Closure	Approve, Deny, Retract, Update, and Close	sd.interaction.svc.categorization.mobile sd.interaction.svc.workinprogress.mobile sd.interaction.svc.review.mobile sd.interaction.svc.closure.mobile
Service Desk	Categorization Work In Progress Review Closure	Update, Close	sd.interaction.categorization.mobile sd.interaction.closure.mobile sd.interaction.workinprogress.mobile sd.interaction.review.mobile sd.interaction.subform.detail.mobile
Streamlined Complaint or Compliment	Open In Progress Close Canceled	Update, Close	sd.streamlined.detail.mobile sd.streamlined.closure.mobile
Streamlined Service Catalog	Open Pending Approval Work In Progress Close Canceled	Approve, Deny, Retract, Update, and Close	sd.streamlined.svc.detail.mobil sd.streamlined.svc.closure.mobile
Streamlined Service Desk	Open Work In Progress Close Canceled	Update, Close	sd.streamlined.itil.detail.mobil sd.streamlined.itil.closure.mobile

Request module

Workflow	Phase	Display Option/Action/Transition	Forms for Mobile Applications
Request	Logging Authorization Fulfillment Review Closure	Update, Close, Approve, and Deny	rm.request.authorization.mobile rm.request.close.mobile rm.request.closure.mobile rm.request.fulfillment.mobile rm.request.logging.mobile rm.request.logging.subform.detail.mobile rm.request.review.mobile rm.request.review.subform.detail.mobile

Note

Note Being widely used in PD forms, the Dynamic View Dependencies (DVD) conditions for widgets aren't supported by Mobile Applications. On the Mobile Applications client, errors may occur when you are updating a record that uses PD forms with DVD conditions. The errors include a blank title on the Related Records tab, some invisible fields in Service Manager becoming visible on the Mobile Applications client, and so on. We recommend that you use customized forms on the Mobile Applications client and avoid the DVD conditions.

PD Change module

A typical usage of your Mobile Applications is to approve or deny a change on the smartphone regardless of the change's category. Based on the phases and workflows introduced by PD, a series of chm.*.mobile sample forms are designed for Mobile Applications to address the Emergency, Standard, Normal changes and the Change Proposals. For Normal or Standard changes, or Emergency changes in a phase other than E-CAB Approval , you need to define your own forms. Otherwise, the original PD forms will be exposed on Mobile Applications client.

You need to move the created Emergency change to the E-CAB Approval phase in Service Manager before utilizing them on Mobile Applications client. For any Emergency change in the E-CAB Approval phase, Mobile Applications client displays Change Main, Updates, and Approval groups as corresponding tabs in the Detail view. You are able to approve, deny and update a change in this phase. In addition, you are also able to search for any approved or denied change by using global search and then retract the change.

PD Incident module

For a newly created incident in the Categorization phase, Incident Details, Categorization and Assignment, Major & Escalation, Activities, Proposed Solution and Related Records groups are displayed as corresponding tabs in the Detail view on Mobile Applications client. You are able to update the incident in this phase.

Change the incident's status to Work in Progress, the incident moves to the Investigation phase. You are able to update the incident in this phase.

After updating the proposed solutions and save the record, the incident moves to the Recovery phase. You are able to update and close the incident in this phase.

After changing the incident's status to Resolved, the incident moves to the Review phase. The Proposed Solution tab is renamed to Recovery Action and you are able to update and close the incident in this phase.

After closing the incident on Mobile Applications client, the incident moves to the Closure phase and all tabs become read-only.

PD Interaction module

If an interaction is ordered from the catalog or opened by Service Desk and includes pending request level approvals, it is read-only on Mobile Applications client before all approvals are approved. You are able to approve or deny the interaction in this phase. In addition, you are also able to search for any approved or denied interaction by using global search and then retract the interaction.

After approving all pending approvals, the interaction moves to the Work In Progress phase. You are able to update and withdraw the interaction in this phase.

After updating the proposed solutions and save the record, the interaction moves to the Review phase. You are able to update and close interaction in this phase.

After closing the interaction on Mobile Applications client, the interaction moves to the Closure phase and all tabs become read-only.

Note

Note Streamlined Interaction is disabled by default in Service Manager. You need to manually enable it before accessing the interactions from the Mobile Applications client. After you adopted Streamlined Interaction, the workflow is slightly different after interaction approval. For more information, see Service Desk (Codeless).

PD Request module

For a newly created request in the Authorization phase, Request Details, Activities, Approvals, Additional Properties, and Attachments are displayed as corresponding tabs in the Detail view on Mobile Applications client. You are able to update, approve, or deny the request in this phase.

After moving the request to the Fulfillment phase, you are still able to update the request in this phase.

After reviewing and updating the information in the request, the request moves to the Review phase. You are able to update and close the request in this phase.

After closing the request on Mobile Applications client, the request moves to the Closure phase and all tabs become read-only.

5. Service Manager maintenance tasks

One of the user roles within Service Manager is the System Administrator, who is usually responsible for installing, configuring, and maintaining the Service Manager system to ensure proper availability, performance, and security.

This section describes a set of tasks the System Administrator needs to perform to ensure stability, performance, and security. The aim is to list typical Service Manager maintenance tasks touching all system-relevant operations.

Note

Note Because Service Manager is typically run as a tailored and integrated system, the list of tasks provided can neither be complete nor fit every system in all details. The intention of this section is to assist administrators to get an overall picture of their routine work and refine their current practice.

Frequency or trigger events to run maintenance tasks

In order to provide a complete concept for maintenance tasks, each task described in this section is assigned a frequency or trigger event, assuming the task is run manually. The frequency provided is only a suggestion to the System Administrator. If a task can be automated, a higher frequency may be desirable. To ensure a concrete Service Manager system, you may need to adjust these frequencies.

Automation of maintenance tasks

In order to effectively administrate a Service Manager system, the System Administrator should automate repetitive tasks as much as possible by using tools such as Application Performance Management (APM) or Operations Orchestrations (OO), or by using shell scripts.

Automated tasks cause system load. Even when automation allows a higher frequency for administrative tasks, the additional system load must be anticipated.

When automating a task using external tools or a scheduled operating system script, be sure to handle the startup or shutdown of the Service Manager system or single servlets for maintenance. Frequent monitoring requests that Service Manager receives during shutdown or startup may cause instabilities. For example, switching the schedule for OS scripts on or off as a part of the Service Manager startup or shutdown script is recommended.

Reports and dashboards for the Service Manager administrator

The Service Manager Reporting module allows scheduled queries to run on the data, present the query results as dashboards, and send emails. Use this module to automate monitoring tasks.

For more information, see Service Manager Reports introduction.

Related topics

Knowledge required for system maintenance
Areas of maintenance tasks
Document all automated administrative tasks
Service Manager Reports introduction

5.1. Knowledge required for system maintenance

To maintain a Service Manager system, the System Administrator must have the product knowledge described in this section.

Configuration files

Service Manager logs

System reports

Server side integrations

Queue dbdicts

Performance of dbdicts and tables

Sequential numbers

Related topics

Service Manager maintenance tasks

5.1.1. Configuration files

The following describes the Service Manager Server, web tier, and Windows Client configuration files.

Server configuration files

The following table describes the server configuration files.

File	Location	Description
sm.ini	<SM server installation>/RUN	This file contains the default values for all servlet parameters.
sm.cfg	<SM server installation>/RUN	This is a batch file that's executed at system startup. Enter all servlet and background process startup commands in this file. The Service Manager load balancer is started unless a hardware load balancer is used. Configuration parameters set in the sm.ini file can be overwritten by this file.
lwssofmconf.xml	<SM server installation>/RUN	This is the file for configuring LW-SSO in the Service Manager server.

JGroups configuration files

The JGroups configuration files, udp.xml or udpcluster.xml or tcpcluster.xml, are located under the `<SM server installation>/RUN` directory.

It is strongly recommended to not modify these files, unless advised by Software Support.

Note

Note:
In Service Manager 9.70 release, the JGroups configuration file is upgraded for 4.2.2. If you have modified this file in previous releases, you can merge your customized configurations to the new file. Before you merge the contents, see following changes of the JGroups configuration file in 9.70 for your attention:

- RSVP protocol is removed. If you made customized configurations in previous releases, don't merge your configurations to the new file.
- UNICAST3 protocol is used to replace UNICAST2 or UNICAST.
- MERGE3 protocol is used to replace MERGE2.
- The following attributes of protocols are obsoleted. If you made customized configurations in previous releases, don't merge your configurations to the new file.

```
<UDP
        loopback="true"
       max_bundle_timeout="30"
        enable_bundling="false"
        timer_type="new"
        timer.min_threads="2"
        timer.max_threads="4"
        timer.keep_alive_time="3000"
        timer.queue_max_size="500"
        thread_pool.queue_enabled="true"
        thread_pool.queue_max_size="100"
        thread_pool.rejection_policy="Run"
        oob_thread_pool.enabled="true"
        oob_thread_pool.min_threads="3"
        oob_thread_pool.max_threads="5"
        oob_thread_pool.keep_alive_time="5000"
        oob_thread_pool.queue_enabled="true"
        oob_thread_pool.queue_max_size="100"
        oob_thread_pool.rejection_policy="Run"/>
<TCP
        max_bundle_timeout="30"
        use_send_queues="true"
        timer_type="new3"
        timer.min_threads="4"
        timer.max_threads="10"
        timer.keep_alive_time="3000"
        timer.queue_max_size="500"
        thread_pool.queue_enabled="true"
        thread_pool.queue_max_size="10000"
        thread_pool.rejection_policy="Run"
        oob_thread_pool.enabled="true"
        oob_thread_pool.min_threads="1"
        oob_thread_pool.max_threads="8"
        oob_thread_pool.keep_alive_time="5000"
        oob_thread_pool.queue_enabled="true"
        oob_thread_pool.queue_max_size="100"
        oob_thread_pool.rejection_policy="Run"/>
<PING timeout="5000" num_initial_members="100"/>
<pbcast.NAKACK2  max_msg_batch_size="500"/>
<UNICAST2 max_bytes="20M" stable_interval="2000" max_msg_batch_size="500"/>
<UNICAST  max_msg_batch_size="500"/>
<pbcast.STABLE cap="0.1"/>
<pbcast.GMS view_bundling="true"
```

```
                    max_bundling_time="1000"
                    resume_task_timeout="15000"/>
```

Web tier configuration files

The following table describes the web tier configuration files.

File	Location	Description
web.xml	<context>/WEB-INF	Basic configuration file for the web tier. For more information, see Enter a parameter in the web.xml file.
webtier.properties	Depending on whether the **customize-folder** parameter is specified in the web.xml file: • If specified: <customize-folder>/config • If not specified: <context>/WEB-INF	This file overrides web.xml parameters, such as for custom web client branding. For more information, see the following topics: • Web tier parameters • Enter a parameter in the webtier.properties files • Web parameter: customize-folder
log4j2.properties	Location as specified by the log4jConfiguration parameter in the web.xml or webtier.properties file. Default: <context>/WEB-INF	This file configures log parameters for the web tier. For more information, see the following topics: • Web parameter: log4jConfiguration • Running traces on the Service Manager web client

Windows client configuration files

You can't edit Windows client configuration files directly; instead, you need to use the **Window** > **Preferences** menu options in the Windows client application.

The Windows client stores the Windows client configuration in multiple files in the following directory:

```
%USERPROFILE%\ServiceManager\workspace
```

Related topics

Web client

Windows client

Running traces on the Service Manager web client

Web parameter: log.properties

Web parameter: customize-folder

Configure LW-SSO in the Service Manager server

Enter a parameter in the web.xml file

Enter a parameter in the webtier.properties files

5.1.2. Service Manager logs

Service Manager frequently logs information to the following places.

Server log files

Dbdicts containing log information

Client log files

Related topics

Service Manager maintenance tasks
Knowledge required for system maintenance

5.1.2.1. Server log files

The Service Manager system is running as one or multiple servlets on one or multiple hosts. All of the servlets share the same binaries, but may be configured differently. Each servlet is configured to write information to a log file on its host, which it may share with other servlets.

For different approaches to implementing custom logging, see Implementing custom logging.

Server log file entries

Running traces on the Service Manager server

Trace parameters that can be proactively activated

Related topics

Service Manager maintenance tasks
Knowledge required for system maintenance
Client log files

5.1.2.1.1. Server log file entries

A server log file entry has the following format:

```
PID (TID) date time component log-level message
```

The following is an example:

```
2276( 8160) 11/10/2013 23:22:26 RTE I User xyz has logged out.
```

The following table explains each portion of a log entry.

Portion	Description
PID	The process id identifying the servlet
TID	The thread id identifying the session thread
date	Date when the message was logged
time	Time when the message was logged
component	The binary component writing the message. Typically, these components are: • RTE: The C-RTE of Service Manager. This component performs most internal processing. • JRTE: The Java-RTE of Service Manager. This component is responsible for communication of the servlet with a client or other servlets. • RAD: The RAD component. It is part of C-RTE and interprets code written in the system language code. • JS: The JS engine. It is compiled as part of C-RTE and interprets JavaScript code.
log level	Information about the severity of the log message: • F (Fatal) – fatal errors, which cause risk to the system • E (Error) – Errors that may still allow the system to run • W (Warning) – potential harmful situations • I (Information) – informational messages only • D (Debug) – fine-grained informational events that are most useful to debug the system; typically used only when a trace is enabled • A (Alert) – hints to the administrator about a misconfiguration or corruption. See System alerts.
message	Any string, which is typically a single line. Java run-time messages (component: JRTE) may be multi-line (that is, JRTE stack trace messages), in contrast to RTE stack trace messages that are written as a sequence of single-line messages. If the length of a single message exceeds 8000 characters, the message will be truncated.

Related topics

Service Manager maintenance tasks
Knowledge required for system maintenance
System alerts

5.1.2.1.2. Running traces on the Service Manager server

Service Manager provides parameters to produce detailed trace information for different analysis requirements. The most used trace parameters are rtm, debugdbquery, sqldebug, and debughttp.

When generating a trace in a production system, it should be limited to as few sessions as possible in order to avoid system load and limit the size of the log data to analyze to a minimum. In most cases, only tracing a single user session or background scheduler is required.

Starting a servlet with trace parameters and connecting to it for reproduction

When using the Service Manager load balancer (SMLB), use the **debugnode** parameter to start a servlet for tracing purposes. This parameter prevents the SMLB from assigning sessions to this servlet. Only when bypassing the SMLB and connecting directly to the servlet port, the session is traced.

To bypass the SMLB with Windows client, simply specify the port of the traced servlet instead of the SMLB port in the connection.

For the web client, the serverHost and serverPort parameters can also be specified in the URL, and by doing this the load balancer is bypassed.

Note

Note By default, direct connection to a servlet by specifying the serverHost and serverPort parameters in the web client URL is not allowed; to allow direct connection, set the honorUrlHost and Web parameter: honorUrlPort parameters in the web.xml file to true.

When a hardware load balancer (HWLB) is used, it doesn't assign a servlet a port that is not set up as the target in its configuration. However, typically a HWLB also uses a network gateway, and hence direct access to the servlet bypassing the HWLB may not be possible. In this case, additional configuration by the HWLB administrator is required.

Activating dynamic debugging options in Status Monitor

Another option is to activate dynamic debugging for already running sessions or schedulers using the status monitor. By this method, a message is sent to the target session to start tracing, and the tracing becomes active as soon as the target session is in a state to receive the message.

Starting with Service Manager 9.50, dynamic debugging can also be activated programmatically for the current session by using rtecall("setdebug"). This allows conditional activation of dynamic debugging, that is, for web service integrations from expressions in the extaccess dbdict.

Activating traces of the current session in RAD Debugger

The **rtm**, **sqldebug**, and **debugadhocsql** trace parameters can be activated for the current session using the RAD Debugger. This is convenient during online debugging.

Related topics

Dynamic debugging of user sessions or schedulers
Hardware load balancers
RAD Debugger
Debugging parameters
Parameter: debugnode
Web parameter: honorUrlHost
Web parameter: honorUrlPort
Parameter: loadBalancer
rtecall ("setdebug") function
Enter a parameter in the web client URL

5.1.2.1.3. Trace parameters that can be proactively activated

Activating a trace parameter in a production system causes additional load and causes log files to grow quickly, and therefore reduces the time interval covered by log rotation.

The following trace parameters can be activated in production environments even for a long time.

Parameter	Description
msglog:1	Writes all on-screen messages to the log file as well
sqllimit:2 (or higher)	Writes a message about the slow execution of any SQL statement (SELECT, INSERT, UPDATE, DELETE or others)
debugdbquery:2 (or higher)	Writes a message about slow SM queries taking longer than the specified number of seconds. The adding, updating, or deleting of records is NOT logged. **Caution** Caution Do not set it to 999 because this is a special and verbose trace.
rtm:2	Writes information about how long a user transaction took. This is basically the time from a user action until the next format is displayed. This parameter also causes information about native heap consumption to be printed to the log file so that sessions consuming a lot of memory can be identified.
log4jdebug: com.hp.ov.sm.common.oom.LowMemoryHandler	The log4jdebug parameter is used to enable debugging messages for different java modules. The LowMemoryHandler module may be activated during production to monitor for a low memory situation of the servlets.
dbmonitorfiles:<dbdict1>[,<dbdict2>..]	Logs a message whenever a record of the listed dbdicts is inserted, updated or deleted. The message contains the following information: • The application that'ssued the update request • A list of records that were updated • The user who initiated the update

The following are examples of verbose trace parameters that must NOT be activated by default in a production system:

- rtm:3|4|5
- debugdbquery:999
- sqldebug:1|2
- debughttp

Activate these trace parameters only for special case analysis. It is recommended to contact Software Support if an issue can't be analyzed otherwise. Refer to Running traces on the Service Manager server to identify the method to activate these parameters that creates the least additional system load and log file size.

Related topics

Server log files
Logging memory monitoring
Log switching
Dynamic debugging of user sessions or schedulers
Parameter: debugdbquery

Parameter: rtm
Special parameters: msglog
SQL parameters: sqllimit
Parameter: debugnode

5.1.2.2. Dbdicts containing log information

Log information for background processes, or integrations that are implemented in Service Manager applications is often written into dbdicts. There are log dbdicts in an out-of-box Service Manager system, and tailored implementations may also create such dbdicts.

Maintenance tasks include reviewing these log files for availability monitoring, capacity planning, and performance monitoring, as well as cleaning up old records so that the amount of records has no negative impact on the system performance.

Dbdicts with log information that serves different purposes are described in the following table.

Dbdict	Area	Description
msglog	Availability monitoring	The msglog file gives helpful hints when a scheduled application fails, as well as provides other information.
syslog	Availability monitoring	The syslog file contains sign in and sign out information for all users, and information about the start and termination of background processes.
SMISTaskLog	Availability monitoring	The SMIS log contains messages regarding integrations using the Service Manager Integration Suite (SMIS).
reportLog	Performance monitoring	This is an internal log for report execution. When using the Reporting module, reporting tasks create reports and dashboards, and send notifications. Because these tasks are executed in the background, long running tasks may not be observed by users, but slow down the system unnoticed.
stathistory	Capacity planning	This is for daily tracking of module license usage.
systemperform systemtotals	Performance monitoring	This is for system load tracking. This is started and stopped from **System Status** > **System Monitor** > **Capture**. **Caution** Caution If the Capture functionality is activated for a long time, slow system performance may occur.

Related topics

Stathistory table

5.1.2.3. Client log files

Each Service Manager client writes a log file. For more information, see Client log file.

As the Windows client is executed on each client computer, no constant monitoring is possible. The client log file is mainly used to troubleshoot functional misbehavior only.

The web client log file resides on the web application server. It contains information for all client sessions, and is therefore relevant for availability and performance monitoring.

Related topics

Client log file
Running traces on the Service Manager web client

5.1.2.3.1. Running traces on the Service Manager web client

Specify trace parameters in the log4j2.properties file. This allows flexible configuration according to analysis requirements. In order to activate the modifications, the web application server must be restarted.

Example: A standard web tier trace configuration referred to as "sm debug trace"

```
log4j.logger.com.peregrine.eclipse.web=DEBUG
log4j.logger.com.hp.ov.cwc=DEBUG

appender.console.type=Console
appender.console.name=STDOUT
appender.console.layout.type=PatternLayout
# Pattern to output the caller's file name and line number.
appender.console.layout.pattern=%5p [%t] (%F:%L) - %m%n

appender.R.type=RollingFile
appender.R.name=RollingFile
appender.R.fileName= sm.log
appender.R.filePattern=sm-%i.log
appender.R.layout.type=PatternLayout
appender.R.layout.pattern=%d %-5p [%t] [%X{username}-%X{sessionHash}] [%c] %m%n
appender.R.policies.type=Policies
appender.R.policies.size.type=SizeBasedTriggeringPolicy
appender.R.policies.size.size=5000KB
appender.R.strategy.type=DefaultRolloverStrategy
# Keep backup file
appender.R.strategy.max=9
#log for trace soap message, don't modify it.
loggers=TraceSoap
logger.TraceSoap.name=TraceSoap
logger.TraceSoap.level=debug
logger.TraceSoap.additivity=true
logger.TraceSoap.appenderRef.TraceSoap.ref=RollingFile
rootLogger.level=debug
rootLogger.appenderRefs=R
#uncomment next line to output to console
#rootLogger.appenderRef.console.ref=STDOUT
rootLogger.appenderRef.R.ref=RollingFile
```

There is rarely the need for a web tier trace. However, in these cases, typically it is required to also trace browser communications using third-party tools (such as Fiddler and Wireshark) and the Service Manager server.

Related topics

Web tier configuration files

Client log files

Knowledge required for system maintenance

5.1.3. System reports

System reports provide information of the system at the very moment they are run.

A system report is started from the command line and its output is written to the console. Use the reporttolog parameter to write system reports to the log file, or redirect stdout by OS means.

Because every system report starts a new session, don't run system reports too frequently.

Status report (reportstatus)

Servlet and session reports (reportlbstatus, reportLBInterval, and reportgroups)

Shared memory report (reportshm)

Shared memory cache report (reportcache)

Semaphore report (reportsem)

System lock report (reportlocks)

License report (reportlic)

Related topics

Service Manager maintenance tasks
Knowledge required for system maintenance
Special parameters: reporttolog

5.1.3.1. Status report (reportstatus)

The status report provides information about processes in Service Manager, as well as related information.

To generate this report, in the Service Manager Server's RUN directory, run this command from the OS command prompt:

```
sm -reportstatus
```

Tip

Tip For more information, see Special parameters: reportstatus.

Example output:

```
---- Shared Memory ----
Release  : 9.50.0
Mapped at: 0x80000000
Size     : 0x05B8D800 - 96000000 bytes
Resource Name: Global\SM.SHRM.13080
Owner: BUILTIN\Administrators
Group: SMHOST\None
Grant: NT AUTHORITY\Authenticated Users rw-     ReadControl WriteDacl WriteOwner
Grant: NT AUTHORITY\SYSTEM       rw-     ReadControl WriteDacl WriteOwner
---- Semaphores ----
Semaphore Name: Global\SM.LOCK.13080.0
Owner: BUILTIN\Administrators
Group: SMHOST\None
Grant: NT AUTHORITY\Authenticated Users ---     ReadControl WriteDacl WriteOwner
Grant: NT AUTHORITY\SYSTEM       ---     ReadControl WriteDacl WriteOwner
---- Processes ----
Owner           User            UID  PTID     PID    NTID  Name
Idle     Command
Administrators  Administrator   1    -1       6932   7328  ThreadControllerId-13
080 30:11:18 sm  -httpPort:13080
Administrators  Administrator   2    9096     6932   9096  falcon
00:00:33 sm  -httpPort:13080
Administrators  Administrator   4    5728     6932   5728  TRCLIENT_LOCKHANDLER
28:50:30 sm  -httpPort:13080
Administrators  Administrator   5    6476     6932   6476  sync
00:00:09 sm  -httpPort:13080
Administrators  Administrator   13   -1       8948   8316  scstatus
00:00:00 sm  -reportstatus
```

Related topics

Knowledge required for system maintenance
System reports
Special parameters: reportstatus

5.1.3.2. Servlet and session reports (reportlbstatus, reportLBInterval, and reportgroups)

Reportlbstatus and reportgroups display the status of ports, processes, and threads in horizontal and vertical scaling servlet implementations.

The following are example commands that you can run from the OS command prompt in the Service Manager Server's RUN directory:

```
sm -reportgroup:60,10
sm –reportlbstatus
sm –loadbalancer –reportLBInterval:60
```

Related topics

Service Manager maintenance tasks
Knowledge required for system maintenance
Parameter: reportlbstatus
Parameter: reportLBInterval
Parameter: reportgroup

5.1.3.3. Shared memory report (reportshm)

The shared memory report provides information about the shared memory size, the available space, and the size of shared memory pools.

Shared memory exists once on each host on which Service Manager runs, and hence running a shared memory report on each host is required to get a full picture. For more information, see Special parameters: reportshm.

In the Service Manager Server's RUN directory, run this command from the OS command prompt:

```
sm -reportshm
```

Example output:

```
------ Shared Memory ------

Shared Memory Release     9.63.2

Current Size             256000000

Segment Allocation         9346088
Large Block Allocation    11288576

Unused Space             235365336  (91%)
Free Space               235542624  (92%)


Shared Memory Type  Allocations    Frees     Allocated
------------------  ----------- ----------- -----------
Not named                   26           3      759744
User blocks                 24           3       43008
Messages                     0           0           0
Database Services          630           0       28848
Cache overhead              26           3       42656
Application cache         2158         703     1938832
DBDICT cache             11630        1146    15424064
SQL descriptor cache       530           0     1046432
Join/ERD/Type cache        402           0     1032320
String Type cache         1080         208      141472
Global cache                 0           0           0
Diagnostic Services          0           0           0
```

The “Free Space” value should be monitored: For a stable system, it should always be between 25 percent and 75 percent of the shared memory size. If this range is exceeded, the shared_memory parameter should be adjusted in the sm.ini file.

Note

Note The percentage in this report is based the total shared memory size.

Related topics

Knowledge required for system maintenance
System reports
Shared memory
Startup parameter: shared_memory
Special parameters: reportshm

5.1.3.4. Shared memory cache report (reportcache)

The shared memory cache report provides aggregated information about the shared memory cache.

The shared memory exist once on each host on which Service Manager runs, hence running a shared memory cache report on each host is required to get a full picture. For more information, see Special parameters: reportcache.

In the Service Manager RUN directory, run this command from the OS command prompt:

```
sm -reportcache
```

Example output:

```
---------------------- Cache Statistics      ----------------------
Slot use: 9%; Average Slot Depth: 1; Maximum Slot Depth: 3
```

The shared memory cache is organized as an array of slots containing a list of cached objects. A hash function determines which slot an object uses.

The Average and Maximum slot depths are important for the system performance.

If the average slot depth exceeds 20, you need to increase the number of slots by using the cache_slots parameter.

Related topics

Knowledge required for system maintenance
System reports
Startup parameter: cache_slots
Special parameters: reportcache

5.1.3.5. Semaphore report (reportsem)

A semaphore is an operating system provided mechanism to control concurrent access to resources, and prevent race conditions.

In the Service Manager Server's RUN directory, run this command from the OS command prompt:

```
sm -reportsem
```

Related topics

Service Manager maintenance tasks
Knowledge required for system maintenance
System reports
Special parameters: reportsem

5.1.3.6. System lock report (reportlocks)

Service Manager manages locks in the LOCKM1 and LOCKSHAREDM1 tables in the database management system, and therefore monitoring lock files can also be done externally on these log files. Because the efficiency of access to these tables is critical to Service Manager system performance, be sure to not run too frequent or too complex selects.

In the Service Manager Server's RUN directory, run this command from the OS command prompt:

```
sm -reportlocks
```

Example output:

```
--- Resource Locks ---
HostName                  SID          PID          TID  Type       Resource Name
------------------- -------- ------------ ------------  ---------  ------------------------------
host03sm25                  3         7024         5020  Shared     agent:TEST
There are currently 1 lock(s).
--- Threads waiting for locks ---
No thread is waiting for locks
```

You can also monitor locks from the system navigator by clicking **System Status** > **Show Locks**.

Related topics

Knowledge required for system maintenance
Lock management

System reports
Special parameters: reportlocks

5.1.3.7. License report (reportlic)

The License report provides information about application licenses, login licenses, and the expiration date of the current license.

In the Service Manager Server's `RUN` directory, run this command from the OS command prompt:

```
sm -reportlic
```

Related topics

Service Manager maintenance tasks
System reports
Special parameters: reportlic
Check license expiration
Check license usage
AutoPass licensing

5.1.4. Server side integrations

Service Manager provides different technologies to integrate with external applications. This section focuses on server-side integrations because they are relevant to queue monitoring.

Web services

A Web service is an integration mechanism, where the service publisher publishes its API as a Web Service Definition Language (WSDL) file in XML format. This file defines the format of requests and responses, which are also sent as XML documents. The service consumer must implement its requests according to the WSDL.

RESTful API

The Service Manager RESTful API works with different HTTP methods. Data for requests may be included in the URL for requests using the GET method or as a JSON document when using POST methods. Service Manager also responds with JSON documents to REST requests.

Service Manager Integration Suite (SMIS)

SMIS provides a framework to implement scheduled-based or UI-based integrations using the web services API or RESTful API.

With the Integration Manager, SMIS provides a graphical user interface to manage integrations.

A UI-based integration means synchronous execution when the user clicks a button or menu option.

For each scheduled-based integration, a background process is assigned and a record in the schedule dbdict is created, which is rescheduled after each execution. The execution reads the expired integration tasks from the smisTaskQueue dbdict and performs the required action.

Event services

An event service is an integration mechanism used by integrations such as SC Automate, Connect-it, and JavaMail.

The eventin and eventout dbdicts contain incoming and outgoing events, respectively.

Events use an event code that refers to an event registration record containing all information about how the event should be processed.

Unlike other queue dbdicts, eventin records may be processed synchronously or asynchronously depending on the state of the **Process input events synchronously?** flag in the eventregister record. Synchronous processing is implemented by the “eventin.after.add” trigger.

Note

Note Changing the state of the “Process input events synchronously?” flag in the eventregister record is not effective immediately, but requires these global lists: “Sync Event Types” (variable $G.sync.events) and “Sync Event Types” (variable $G.sch.events).

Asynchronous events use the **evtime** field and therefore may be scheduled. Incoming events can be configured to automatically create an outgoing event as the response. Successfully processed events are automatically deleted (if not configured otherwise, for example, for debugging purposes), while events that were not processed successfully aren't deleted automatically to allow error analysis.

Related topics

Service Manager integration methods and tools
Valid URLs for Service Manager
RESTful API
Integration Manager
Event Services
Troubleshooting: Event Services

5.1.5. Queue dbdicts

Queue dbdicts contain records that are created by any session, and processed by a dedicated background process, which finally deletes it. Queues implement asynchronous task execution, and are also used to schedule task execution.

Queue dbdicts require monitoring in order to prevent system performance degradation, and require background processes to process the queue records.

The out-of- box queue dbdicts are described in the following table.

Dbdict name	Description
schedule	Scheduled tasks of many different kinds, such as output of on-screen messages, or sla processing.
eventin	Queue for incoming events
eventout	Queue for outgoing events
SMISTaskQueue	Service Manager Integration Suite (SMIS) provides a management framework for schedule-based and UI-based integrations using the web services or RESTful API. With SMISTaskQueue, schedule-based integrations are realized.
reportscheduleTask	Files for scheduled tasks of the Reporting module
spool spoolheader mail	These queue dbdicts were used for server-side printing and the internal mail system, and aren't used in most systems anymore.

Because some queue dbdicts have a high frequency of record creation and deletion, efficient indexing of the tables on the RDBMS is important. Also, preparing for temporary steep increases or decreases in the number of records is important.

Note

Note

It is possible that multiple background processes work on the same queue file (for example, schedule or eventin). In this case, the sequence of processing is possibly no more identical to the sequence of creation. So additional design effort is required to prevent an update event for a record from being processed before its creation event.

Schedule dbdict

The **schedule** dbdict contains records that are tasks for most background processes (the so-called "schedulers"). All schedulers execute the same main RAD application named **scheduler.process**, while the schedule record contains the implementation.

Schedule records are used to achieve the following goals:

- Schedule a task into the future (for example, SLA alerting) using the expiration time field;
- Move time consuming processing from foreground sessions to background processes to increase response times, or
- Implement periodically interval-based rescheduled tasks

Other queue managers such as the event scheduler and the SMIS background tasks are implemented as schedulers using periodically rescheduled schedule records.

Important fields in a schedule record are described in the following table.

Field	Description
Expiration time	This field contains the time after which the schedule record is to be processed.
Class	This field contains the name of the background scheduler to process this schedule record when the task is scheduled. While a background process is processing it, however, it is empty.
scheduled class	This field contains the name of the background scheduler while it processes the schedule record. For a rescheduled schedule record, this field is also filled even if the record is only scheduled.
Status	This is a free-text field used to provide status information about the schedule record by the scheduler.process RAD application.

When the background scheduler wakes up, it looks for expired schedule records whose **Class** field contains its own name. It processes all these records, deletes each of them after they are processed successfully, and goes to sleep again. The sleep time is configured in its startup command (in the **info** dbdict using the **info.startup** format, or the sm.cfg file). It is recommended to not set the sleep time to be less than 15 seconds. The default value for most schedulers is 60 seconds.

Only when the scheduler is started for the first time, it checks for all schedule records that have a status of "application running" and a scheduled class that contains its own name. This is to ensure the task that was processed by a scheduler is completed before it is terminated prematurely. Because this is also the case for hanging schedule records, the system administrator needs to edit the hanging schedule record before restarting the background process.

Typical scenarios of failures in schedule record processing are described in the following table.

Status	Description
scheduled or empty status field	This is a normal state as long as the expiration time has not passed. If expiration time has passed, corrective actions are required.
application failed due to error - check msglog for possible messages	An error occurred while processing a schedule record and the schedule record gets into an “Application error” status. The background scheduler continues to pick up the next schedule record and continues to process its tasks.
locked	A schedule record could not be processed because resources required were locked. The schedule record gets into a “locked” status and is rescheduled for a few minutes later. The background schedule continues to pick up the next schedule record and processes this record at the rescheduled time.
application running	This is a temporary state while the schedule record is being processed. If this temporary state persists, it is likely that the scheduler processing this record got into an endless loop, which is also referred to as “hanging”.

The background processes can be monitored by the status monitor. The important column in the following table is **Idle Time**.

Idle Time	Interpretation and administrative action
> Sleep time	The scheduler is not running properly. Normally, killing and restarting the background process should resolve this issue.
00:00:00	If the idle time is exactly 00:00:00, press the **Refresh** button several times and watch if the idle time is increasing. If not, the schedule record is very busy or hanging. To identify this, check schedule records with a status of "application running” for a Scheduled Class field value that contains the process name. If there is no hanging schedule record, while the idle time stays 00:00:00, there is a load issue. Inspect all schedule records for this background process (the Class field) to understand the root cause of the load and then take adequate actions.

Eventin dbdict

The eventin dbdict contains incoming events. Eventin records may be processed synchronously or asynchronously depending on the state of the “Process input events synchronously?” flag in the eventregister record.

Synchronous processing is implemented by the “eventin.after.add” trigger.

Asynchronous events use the evtime field. Eventin records are selected by the event background process based on a query specified in the schedule record of the **event** class, which is always rescheduled.

Note

Note

Changing the state of the "Process input events synchronously?" flag in the eventregister record is not effective immediately, but requires these global lists: "Sync Event Types" (variable $G.sync.events), and "Sync Event Types" (variable $G.sch.events).

Incoming events can be configured to automatically create an outgoing event as the response. Successfully processed events are automatically deleted (if not configured otherwise, for example, for debugging purposes), while events that were not processed successfully aren't deleted automatically to allow error analysis.

Eventout dbdict

The eventout dbdict contains outgoing events, which will be picked up by a background processor and then sent to an external application such as an email service.

SMISTaskQueue dbdict

When using Service Manager Integration Suite (SMIS), the queue of pending tasks can be inspected using Integration Manager. Repeated inspections help identify if the task queue is stuck and if a corrective action is required.

ReportscheduleTask and reportscheduleHistory dbdicts

The Reporting module queues tasks in the reportscheduleTask dbdict, and moves them into reportscheduleHistory after processing. The Reporting module also takes care of the cleanup of old records.

The Reporting module has its own cleanup mechanism implemented.

Related topics

Integration Manager
Service Manager Reports introduction
IR Expert
Event Services

5.1.6. Performance of dbdicts and tables

The definitions of dbdicts in Service Manager have a huge impact to system performance and response time. The following are the aspects in a dbdict definition:

- Mapping of the dbdict to tables in the relational database management system (RDBMS)
- Indexes on the tables in the RDBMS
- The number of fields in the dbdict. This increases the size of messages exchanged between the client and server.

The mapping of the dbdict to tables defines how many tables the dbdict is mapped to, and whether the fields are mapped to SQL column types interpretable by the RDBMS (such as VARCHAR, NUMBER, and FLOAT) or not (such as IMAGE, and CLOB).

Using non-interpretable fields may improve the performance mildly for reading and writing records, and they can be used neither by the index on a table, nor in the WHERE clause of a SQL SELECT statement. Service Manager relaxes queries that use these fields by replacing this clause with "(1=1)" and re-selects among the records in its own memory. Using non-interpretable fields in queries therefore causes the RDBMS to fetch more records and send them over the network, and causes Service Manager to load more records into its own memory for reselection.

Dbdicts support array and structure-type fields. These fields can't be natively mapped to SQL column types, and must be mapped either to non-interpretable column types or as tables for the array elements using the primary key (or the first unique key) fields and the RECORD_NUMBER field (which contains the position of each element in the array).

Array fields mapped as tables of interpretable column types can be used efficiently in queries , because Service Manager is then translating the queries into a SELECT statement on the join of multiple tables.

Note

Note Arrays nested in others can't be mapped to tables, and need to be mapped to non-interpretable column types.

As the database operations are executed at the RDBMS, dbdict keys have no direct impact on database performance. Their relevance is as follows:

- The primary (first unique) key is relevant as the record identifier during query processing: Service Manager first selects a list of key values that match the query, before selecting the fields displayed in the record list view or all fields for the record displayed in the detail format.
- The primary (first unique) key is relevant for the mapping of array fields.
- All dbdict keys are translated into indexes on the RDBMS when a dbdict is initially mapped to tables or remapped. Because remapping means creating new tables according to the modified mapping and replacing the existing tables, all modifications of the tables on the RDBMS side aren't maintained.
- When a new key is created in the dbdict, an index is automatically created on the table.

Indexing all fields used in queries that are executed frequently by the out-of-box implementation or a tailored implementation is required for an efficient system. User-defined queries that are executed from search formats, view definitions or reports typically can't be all indexed. Training and highlighting indexed fields in a search format (that is, by using bold font for field labels) will enable the user to formulate queries efficiently and achieve short response times.

Note

Note The debugdbquery trace parameter can be turned on during the prototyping of Service Manager to verify that user and background queries are well supported with the corresponding keys. For more information, see Trace parameters that can be proactively activated.

Related topics

Keys and indexes in Service Manager
Trace parameters that can be proactively activated

5.1.7. Sequential numbers

Sequential numbers refer to the sequential identifiers assigned to the records of a dbdict, such as an interaction or incident number. Sequential numbers are controlled by the **numbers** or **counters** dbdict.

Call the system language function named rtecall("getnumber") to increase and update the last number in these dbdicts and get the number returned.

Both the **numbers** and **counters** dbdicts contain a batch size setting, which allows the Service Manager Server to reduce the number of SELECT and UPDATE operations on these dbdicts. The mechanism is that a servlet doesn't request one sequential number, but a batch of sequential numbers. It then assigns the numbers in this batch to any request for a number by its sessions and accesses the sequential number dbdict only when all numbers have been assigned. This mechanism allows high-frequency assignment of numbers without a huge load at the RDBMS side.

Note

Note When using the batch size feature, a number being requested is not necessarily larger than a number requested before by another servlet. Additionally, when a servlet is terminated, unassigned numbers in its current batch will not be assigned anymore.

Related topics

Service Manager maintenance tasks
Knowledge required for system maintenance
Reset sequential numbers
Define a batch size for sequential number records

5.2. Areas of maintenance tasks

System maintenance tasks described in this section are categorized into reactive and proactive areas.

Reactive areas

Reactive maintenance areas are described in the following table.

Area	Description
Availability monitoring	Monitors the availability of the Service Manager system or a service that it provides
Performance monitoring	Monitors acceptable performance of the Service Manager system or a service that it provides
Functional monitoring	Monitors the correct execution of functionalities
Queue monitoring	Monitors queue tables: Queues may get into a stuck state, or be processed slowly. Also, queue items may fail to be processed.
Web tier health monitoring	Monitors the Service Manager web tier health status.
Operation	These are tasks for starting a Service Manager instance, shutting it down, or running it in quiescence mode for maintenance.

Proactive areas

Proactive maintenance areas are described in the following table.

Area	Description
Configuration	Setup and base configuration of the Service Manager instance, including setup regarding performance and security
Fallback strategy	Fallback strategies to address situations of partial or total loss of service
Documentation	Documentation of system configuration, changes, and fallback strategies
Change control	Applying patches, hotfixes, and new tailoring
Capacity planning	The anticipation of changes in load or functionality enhancements
Cleanup	Cleanup activities that guarantee an efficient and easy-to-maintain system
Online support	• Using the Software Support website • Staying up to date with the company's security alerts and news • Sharing knowledge with and learning from user communities

Related topics

Service Manager maintenance tasks

5.2.1. Availability monitoring tasks

The following table lists maintenance tasks that a System Administrator needs to perform in the Availability Monitoring area. For information about other areas, see Areas of maintenance tasks.

Event or frequency	Task	Automatable
Daily	Monitor log information for availability related messages	Yes
Daily	Monitor background processor activities	Yes
Daily	Monitor web tier health	No
Daily	Check servlet utilization	No
Daily	Check system reports	Yes
Daily	Monitor background processor activities	Yes
Daily	Check servlet utilization	
Weekly	Check license expiration	Yes
Quarterly	Check SSL certificate expiration	No
After web tier update	Check signatures expiration of web client jar files	Yes

Related topics

Areas of maintenance tasks

5.2.1.1. Monitor log information for availability related messages

This section provides guidelines on how to monitor the server log and dbdicts containing log information for availability related messages.

Monitor the server log for availability related messages

Monitor the following types of messages:

- Alert messages
- Error messages about exhausted resources
- Proactive Memory pool messages

Alert messages

Service Manager writes configurable alert messages to the log file for availability monitoring. For more information, see System alerts.

Error messages about exhausted resources

There are two kinds of memory related error messages:

- Out of memory error messages: A session requests memory inside one of the memory pools; however, as resources are exhausted, the request can't be served. This causes the RAD thread, the session, or the entire servlet to fail immediately.

Shared memory
RAD stack
Java Heap
Process Native Heap

- Proactive memory related messages. These messages are either pure informational or related to proactive mechanisms to prevent out of memory situations.

Shared memory
RAD stack
Java Heap
Process Native Heap

For more information, see Monitoring native heap and Java heap memory.

Resource exhaustion problems include the following:

- Shared memory resources exhausted
  When a block of shared memory is requested, there is no more shared memory available, the session can't continue and terminates. The sm.log file contains messages about the failed allocation and a shared memory report.
  For more information, see Special parameters: reportshm.
- Rad stack exhausted
  The Rad Stack is required by System Language and resides in process private memory. Each RAD thread (that relates to open tabs in the Service Manager Windows or Web client) maintains a RAD stack. The **agstackl** parameter controls the number of stack frames.
  When the Rad Stack is exhausted, the RAD thread is terminated. The cause is most probably a recursive implementation. Increasing the number of stack frames typically only causes the issue to appear less frequently, while removing the recursive implementation is required for a permanent fix.
  The log file is showing a message of the failed request and an RAD application stack trace.
  For more information, see Startup parameter: agstackl.
- Process private memory exhausted
  When a RAD application requests a block of memory, which can't be served, the session terminates with RAD application stack trace.
  When a JavaScript function running out of memory, the function is not reaching the return statement, so the returned value is null. A message is printed to the log file, including JavaScript and RAD stack. The session, however, is not terminated.
  Example:

  ```
  JS W Print ScriptTrace for OutOfMemory
  JS W MemWaster@'MemWaster':2,
  RAD W
  display:process.option:1,se.view.engine:display.go,se.list.engine:call.view.engine,se.search.engine
  :call.list.engine,database.engine:call.search.engine.1,database:call.database.engine,
  ```

  For more information, see the following help topics:
  Service Manager Programming
  Parameter: log4jDebug
  Monitoring memory in Service Manager processes
  Monitoring native heap and Java heap memory
- Java Heap exhausted
  When a Java heap memory request can't be served, the session terminates with java exception stack.

The java heap is used by the servlet to communicate to clients, integrations, or other servlets. The size of the request or response document is the biggest driver for java heap memory requirements.
Resolving these issues requires analysis of the actual document and the implementation. A typical cause is that huge requests or response documents are required by integrations. A mechanism implemented to replace huge requests is web services pagination.
For more information, see Retrieve data from Service Manager using Pagination.

Proactive Memory pool messages

Proactive memory pool messages include the following:

- Shared memory warnings
  When the first process of Service Manager is started on a host, it creates the shared memory. At this time, it compares the configured size in sm.ini with the minimum acceptable value of 8000000 (and automatically adjusts) and the recommended size.
  Example messages:

  ```
  RTE W Shared memory size of 4000000 too small, adjusted to 8000000
  RTE W The shared memory size 8000000 is less than the recommended size 134217728
  ```

- RAD stack warnings
  Service Manager logs warning messages when the RAD stack allocation exceeds a threshold.
  Example:

  ```
  RTE I RAD stack is 72% used, please exit out of current application.
  ```

- Process private memory warnings
  Service Manager monitors the native heap memory allocation and writes proactive messages to the log file before all resources are exhausted. In addition, the servlet will switch to a low memory mode to avoid total exhaustion and the servlet from completely terminating.
  When a servlet moves into low memory mode and is no more accepting new sessions and application threads for native memory until the garbage collection has released memory again.
  When the allocation of native memory increases for a session, information about the allocated memory size is printed to the log file.
  Example:

  ```
  RTE I -Memory : S(392328) O(165668) MAX(1001004) - MALLOC's Total(4136)
  ```

  Here, the MAX value is relevant: Average here is for typical user sessions below 20M; if 40M is exceeded by a single session, investigations should be started. Possible causes are that users open a very large number of tabs in the client, or memory-intensive JavaScript implementations, or other problems.
  More frequent trace information is written to sm.log when trace parameter rtm:2 is turned on.

  Note

  Note Process native memory monitoring and the proactive messages aren't implemented in Service Manager running on unix operating system.

  For more information, see Avoiding variable declaration inside a loop.
- Java Heap warnings
  Service Manager monitors the native heap memory allocation and writes proactive messages to the log file before all resources are exhausted. In addition, the servlet will switch to a low memory mode to avoid total exhaustion and the servlet from completely terminating.
  When a servlet moved into low memory mode and is no more accepting new sessions and application threads for java heap memory until garbage collection released memory again.

In order to monitor java heap for low memory, set the following parameter in sm.ini:

```
log4jdebug:com.hp.ov.sm.common.oom.LowMemoryHandler
```

Example of low memory handler log messages:

```
JRTE I Starting Memory Monitoring thread to check for memory every 15 seconds.
JRTE D JavaMemory Max(123928576) Used(1011872) %Used(0.0)
JRTE D NativeMemory Max(2147352576) Used(358121472) %Used(16.0)
JRTE W Process Low on Java Memory. Max(123928576) Used(119603800) PercentUsed(96.0)
JRTE W Send error response: Server is running low on memory try again.
JRTE W Send error response: Server is running low on memory try again.
RTE I Process Java Heap Memory is back to normal range.
JRTE I Process Java Memory. Max(123928576)  Used(4256640) PercentUsed(3.0)
```

Note

Note For detailed explanation of Garbage collection and Memory Pools, refer to the Oracle documentation: https://www.oracle.com/java/technologies/javase/javase-core-technologies-apis.html

Monitor dbdicts with log information for availability related messages

Review on a daily basis the following dbdicts, which contain log information, for error messages:

- msglog
- syslog
- SMISlog

For more information, see Dbdicts containing log information.

Related topics

Availability monitoring tasks

Introduction

Monitoring memory in Service Manager processes

Monitoring native heap and Java heap memory

Parameter: log4jDebug

Dbdicts containing log information

5.2.1.2. Monitor background processor activities

When checking which system processes are active in System Status, it is a good idea to maintain a list of those background processes that should always be running. If one of these processes is not running, check the sm.log file to see if there are any messages as to what caused the interruption.

The background processes can be monitored by the status monitor.

The important column in the following table is **Idle time**.

Idle time	Interpretation and administrative action
> Sleep time	The process is not continuing after its sleep interval. Normally, killing and restarting the background process should resolve this issue.
00:00:00	If the idle time is exactly 00:00:00, press the **Refresh** button several times to see if the idle time is increasing. If not, the process is very busy or hanging.

Background processes executing the tasks specified in records of the **schedule** dbdict are called background schedulers. For more information, see Monitor the schedule dbdict and background schedulers.

Note

Note The background process called "sync" is especially important for availability: It deallocates resources of other sessions when they terminate. If this process is not running, Service Manager will still run for a while, resource requests can't be served anymore as all resources are allocated. Exactly one instance of the "sync" process should be running on each host of the Service Manager system.

Related topics

Availability monitoring tasks
Scheduled processes

5.2.1.3. Monitor web tier health

The following table lists the API URLs that a System Administrator can use to monitor the web tier health status.

API URL	Task	Automatable
/api/healthz	Use this URL to return the health status of the Service Manager web tier. The HTTP Status 200 message indicates that the services are ready. If you have deployed a load balancer (such as the F5 HWLB) for the web tier, you can use this URL to monitor the web tier health status.	No
/api/healthz/profile	Use this URL to return the web tier profile in the *.json format. A system administrator can access this URL only after he/she logs in to Service Manager.	No
/api/healthz/profile/html	Use this URL to return the web tier profile in the *.html format. The content automatically refreshes in every five seconds. A system administrator can access this URL only after he/she logs in to Service Manager.	No

Related topics

Service Manager maintenance tasks

5.2.1.4. Monitor server health

The following table lists the API URLs that a System Administrator can use to monitor the Service Manager server health status.

API URL	Task	Automatable
/mbeanclient-<*version*>/Action?getattribute=AcceptNewConnection	Use this URL to check whether Service Manager can accept new connections or not. When returning `true`, Service Manager can accept a new connection. Sample usage: /mbeanclient-<*version*>/Action?getattribute=AcceptNewConnection **Note:** This API only checks the connection status of the node that the URL is connecting to. Do not connect to the load balancer node.	No
/mbeanclient-<*version*>/Action?getattribute=State	Use this URL to return the health status of the Service Manager services. `RUN` indicates that the services are ready. Sample usage: /mbeanclient-<*version*>/Action?getattribute=State	No

Note

Note

To make the mbeanclient API URL effective, you must add the **JVMOption999:../../lib/mbeanclient-<*version*>.war** parameter to the sm.ini file.

Related topics

Service Manager maintenance tasks
Configure an HWLB Health Monitor for the Service Manager server

5.2.1.5. Check servlet utilization

Review the `sm.log` file or system status from time to time to check the number of concurrent users or connected processes and the servlet utilization. If necessary, add another servlet to the group dynamically or add another servlet line to the `sm.cfg` file for the next recycle of Service Manager. On the Windows operating system, when starting a servlet dynamically from a DOS prompt, it's imperative to not sign out of the server machine. The reason is that this would stop the servlet that you just started. Ensure you have set the system to lock the desktop and requires authentication to unlock.

Related topics

Availability monitoring tasks

5.2.1.6. Check system reports

The System Administrator should execute these system reports multiple times a day or use a scheduled OS script to execute them, to monitor the current system status:

- sm -reportshm: Verify that the “Free Space” doesn't fall below 25 percent to avoid shared memory related outage.
- sm -reportcache

Related topics

Availability monitoring tasks
System reports

Special parameters: reportshm
Special parameters: reportcache

5.2.1.7. Check license expiration

The license expiration of the Service Manager server may cause issues when restarting the system.

At any time, the license report provides information about license expiration:

```
sm -reportlic
```

When the expiration date is approaching, Service Manager proactively sends warning messages at user login to warn for this situation:

Related topics

Availability monitoring tasks
License report (reportlic)

5.2.1.8. Check SSL certificate expiration

An expired SSL certificate can cause outage. The administrator should periodically verify that the certificate is not about to expire in order to take proactive actions.

Related topics

Availability monitoring tasks

5.2.1.9. Check signatures expiration of web client jar files

The code files of the Service Manager web client (jar files) are code signed. Depending on the security level for java in the browser, jar files with expired signature certification can't run. The recommended fix for this issue is to upgrade the web client to the latest version.

For that reason, the Service Manager administrator should proactively get a list of these expiration dates.

You can use the jarsigner tool of the Java JDK to display the expiration date of a jar file.

```
<JDK directory>\bin\jarsigner.exe -certs -verify -verbose <web application server
directory>\webapps\webtier-9.80\ext\<jar file>
```

Only the first listed certificate expiration date is relevant.

For information on how to verify whether a Java certificate has expired or not, refer to the Oracle documentation.

Related topics

Availability monitoring tasks

5.2.2. Performance monitoring tasks

The following table lists maintenance tasks that a System Administrator needs to perform in the Performance Monitoring area. For information about other areas, see Areas of maintenance tasks.

Event or frequency	Task	Automatable
Daily	Review the server log for performance messages	Yes
Daily	Analyze dbdicts containing log information	Yes
Weekly	Check view and favorite creation	No
Weekly	Check the usergrid table for changes to user's record lists	Yes
Daily	Monitor system reports for performance related information	Yes
Daily	Performance monitoring tasks	Yes
Quarterly	Monitor the dbdicts purging progress	Yes

Review logs for performance messages

This task includes the following subtasks.

Review the server log for performance messages

The Service Manager server log files contain a lot of information about system availability and performance.

Review the output produced by alert messages and proactively activated trace parameters. The alert log messages provide information about potential performance issues, or other specific situations that require attention.

Analyze dbdicts containing log information

Service Manager maintains multiple dbdicts with log information for different purposes. Among them, the reportLogs dbdict is important for performance monitoring.

It is important for the System Administrator to review the performance log file and see if a task is slow. Each reporting query may take a different amount of time to execute, and therefore manual review is required.

Look for report queries that are often more complex than other queries and are expected to take more time for execution. It is therefore not possible to define a one-fits-all threshold.

It is important to identify long-running reports, and learn their business requirements and implementation. Based on this knowledge, the System Administrator can decide on how to act on excessive report execution times or the trends that are identified.

Check view and favorite creation

If users are given the ability to create views and favorites (public or private), the System Administrator will need to put into place a process where the administrator can check the records in the inbox table regularly to ensure that the queries and sort sequences used in the views and favorites are properly indexed. If a sort sequence is specified, the fields used in the sort sequence should be part of the index.

Additionally, if the administrator finds multiple users creating the same private inbox, the administrator needs to consider creating a public inbox for all users in a query group (all users that have this inbox created). By doing this, the administrator will be able to keep the number of records in the inbox table under control. For performance tuning of queries, refer to Performance of dbdicts and tables.

Knowledge required for system maintenance > Performance of dbdicts and tables.

Check the usergrid table for changes to user's record lists

Check the usergrid table to see what users are saving as their qbe formats. If you find that many users have added the same field to a particular QBE, you may want to consider adding the field to the default QBE format. Remember to delete the appropriate records from the usergrid file once you have modified the format.

Monitor system reports for performance related information

Monitor the following system report on a daily basis.

Parameter	Description
sm - reportcache	If the average slot depth exceeds 20, consider reconfiguring the shared memory cache size by the cache_slots parameter. For more information, see Startup parameter: cache_slots.

Monitor the server's performance statistics

Check for memory usage and CPU usage peaks to ensure smooth operation. Over time, with more concurrent users logging in, it may be necessary to add more memory or more CPU, or adding more servlets to serve the increasing user community.

Monitor the dbdicts purging progress

Data, queue, or log dbdicts containing a large number of records can impact system performance, and therefore archiving and purging should be configured.

Monitor the number of records in these dbdicts frequently to supervise cleanup activities.

Related topics

System alerts
Trace parameters that can be proactively activated
Dbdicts containing log information
Activate proactive trace parameters
Purge or archive data records
Purge queue records
Purge logs

5.2.3. Functional monitoring tasks

The following table lists maintenance tasks that a System Administrator needs to perform in the Functional Monitoring area. For information about other areas, see Areas of maintenance tasks.

Event or frequency	Task	Automatable
Quarterly	See *Validate timezone records* below	No
Quarterly	See *Update the calholiday table* below	No
Weekly	See *Review the reportscheduleHistory dbdict* below	No

Validate timezone records

Review and update timezone records for approaching changes to the start or end dates of daylight savings. Additionally, verify that the tzfile records for your time zone are built far enough into the future.

Update the calholiday table

Verify that the holidays table (the **calholiday** dbdict) is maintained. As holidays are used to calculate SLA alerts and breaches, make sure this table is always maintained at a minimum from now until the longest SLA agreement.

Review the reportscheduleHistory dbdict

When using the Reporting Module, reporting tasks create reports and dashboards and send notifications. All these tasks are recreated in the reportscheduleTasks dbdict and after execution moved into the reportscheduleHistory dbdict.

Review this dbdict for tasks that were not executed successfully. It is helpful to add the "Succeed" field by using the **Modify Columns** option to the record list view. After review, decide on corrective actions and cleanup.

Related topics

Areas of maintenance tasks
Time zones
Holiday records

5.2.4. Queue Monitoring tasks

The following table lists maintenance tasks that a System Administrator needs to perform in the Queue Monitoring area. For information about other areas, see Areas of maintenance tasks.

Event or frequency	Task	Automatable
Daily	Monitor the schedule dbdict and background schedulers	Yes
Daily	Monitor the eventin and eventout dbdicts	Yes
Daily	Monitor the SMIS task queue	Yes
Daily	Monitor reportscheduleTasks and reportscheduleHistory	Yes

Related topics

Service Manager maintenance tasks
Areas of maintenance tasks

5.2.4.1. Monitor the schedule dbdict and background schedulers

Background processes that execute the tasks specified in records of the **schedule** dbdict are called background schedulers.

Monitor the **schedule** dbdict and the background schedulers multiple times a day, and take appropriate actions.

Idle time	Administrative action
> Sleep time	Restart the background process.
00:00:00	The scheduler is very busy or hanging. To identify the schedule record processed, check schedule records with a status of "application running" for a Scheduled Class field that contains the process name. Rerun the query a few times and see if the schedule record returned is varying. If varying, this is a load issue. If not, the background process was hanging during the processing of this schedule record.

Monitor the schedule records as described in the following table.

Status	Administrative action
Scheduled or empty status field	If a schedule record is not picked up although the expiration time has passed, the problem may be caused by any of the following: • The assigned scheduler is not running, or • There is too much load for the assigned scheduler, or • The scheduler is hanging on another record (see the *application running* status below).
application failed due to error - check msglog for possible messages	The System Administrator is required to analyze what made the processing of this task fail, decide if corrective actions are required to data records not being updated correctly, and take proactive actions to prevent this from happening to other similar tasks. Schedule records in this status aren't automatically removed for error analysis. Over time, these records may have negative impact to system performance, and need to be removed either manually by the administrator or by a scheduled purging task. See Purge queue records
locked	This is typically a state during normal operations. However, if a schedule record stays in this state for a long time, the System Administrator is required to analyze and resolve this situation.
application running	Refresh the search for schedule records in "application running" state and verify by field ID, if always the same record is displayed. If a background scheduler hangs, kill it from the status monitor and modify the Scheduled Class field of the schedule record that's not matching any background process name: This record requires to be analyzed later. Then restart the background scheduler, so that it continues to process other scheduled tasks assigned to it.

Related topics

Schedule dbdict
Queue Monitoring tasks

5.2.4.2. Monitor the eventin and eventout dbdicts

Monitoring eventin or eventout records is mainly to ensure the number of records is small and not constantly increasing. Old records should be reviewed because they are artifacts of failure, and then removed.

If the number of eventin records is large, verify that the **event** background scheduler is running and not hanging. As a large number of eventin records means large load, the idle time is expected to be 00:00:00. So it is important to understand whether it is running well or hanging under this load (see Monitor the schedule dbdict and background schedulers).

If Service Manager requires a longer time to process an event, the external application sending the eventin record may time out and resend the request. Service Manager is then completing the initial request and the resent one, and then causes

another timeout. Depending on the timeout or resending configuration of the external application, this may increase the load for the **event** background scheduler so much that it can't handle it.

Related topics

Queue Monitoring tasks
Event Services
Eventin dbdict
Eventout dbdict

5.2.4.3. Monitor the SMIS task queue

The task queue for integrations using the Service Manager Integration Suite (SMIS) framework is written to the smisTaskQueue dbdict. Monitor this dbdict to identify potential issues during task execution.

Related topics

Queue Monitoring tasks
Service Manager Integration Suite (SMIS)
Monitor failover tasks

5.2.4.4. Monitor reportscheduleTasks and reportscheduleHistory

When using the Reporting module, reporting tasks create reports and dashboards and send notifications. All these tasks are recreated in the reportscheduleTasks dbdict and moved into the reportscheduleHistory dbdict after execution.

Review this dbdict for tasks that were not executed successfully. It also helps to add the **Succeed** field to the record list view by using the **Modify Columns** option. After review, decide on corrective and cleanup actions.

Related topics

Queue Monitoring tasks
Report schedule and distribution

5.2.5. Operation tasks

The following table lists maintenance tasks that a System Administrator needs to perform in the Operation area. For information about other areas, see Areas of maintenance tasks.

Event or frequency	Task	Automatable
On Demand	Start Service Manager	No
On Demand	Shut down Service Manager	No
On Demand	Switch servlets to quiesce mode	No

Related topics

Areas of maintenance tasks

5.2.5.1. Start Service Manager

To start a Service Manager system, do one of the following:

- Start Windows service "Service Manager <version> Server" on Windows
- Execute the **smstart** script on Linux systems

Single servlets can be started by command line (the executable **sm** in the RUN directory) for a limited amount of time.

Service Manager is using a JGroups open source component to build a reliable cluster of Service Manager servlets. In case Service Manager must be stopped completely, the JGroups cluster is shutdown as well.

When a servlet is started, it tries to join an already existing JGroups cluster, otherwise it will create a new cluster that others can join. To create a cluster, a valid license must exist for the host and the **grouplicenseip** parameter must include the IP address of the host.

Related topics

Operation tasks
Startup and shutdown
Parameter: grouplicenseip

5.2.5.2. Shut down Service Manager

To shut down a Service Manager system, do one of the following:

- Stop Windows service "Service Manager <version> Server" on Windows
- Execute the **smstop** script on Linux systems

Alternatively, you can also use the shutdown commands to shutdown the complete Service Manager system, a group of servers, all servlets running on a host, or single servlets identified by process id.

```
sm -shutdown [-group:<group>][-host:<host>][-pid:<pid>]
```

Related topics

Operation tasks
Startup and shutdown

5.2.5.3. Switch servlets to quiesce mode

In quiesce mode, all users or all users except those with the **SysAdmin** capability may not connect. You can set quiesce mode for a single, multiple or all servlets. This allows the System Administrator to apply maintenance changes without being interfered by or interfering with other sessions.

Command line:

```
sm –quiesce:<mode> [-group:<group>][-host:<host>]
```

Whether the servlet that the current session is running on is in quiesce mode or not can by tested by the sysinfo.get("Quiesce") system language function.

The following table describes different restriction levels of quiesce mode.

Restriction level	Description
Level 0	Service Manager has no login restrictions and accepts all logins normally.

Level 1	Service Manager restricts login to operators who have the SysAdmin capability word. Service Manager denies login to all other operators and displays the message:
	`System quiesced, you can't login at this time.`
	When you issue **sm -quiesce:1 -group** on a primary or secondary host, all Service Manager processes (except the Service Manager load balancer) are set to quiesce mode 1. The Service Manager load balancer updates its list of available nodes to 0. If users try to connect to the Service Manager load balancer at this time, they receive the following message, because there are no Service Manager processes currently available:
	`Max sessions exceeded.`

Level 2	Service Manager denies login to all operators and displays the message: `System quiesced, login restricted at this time.`

Related topics

Operation tasks
System quiesce: Login restrictions
Startup parameter: quiesce
RAD function: sysinfo.get("Quiesce")
Quiesce all servlet container processes in a servlet implementation
Quiesce all servlet container processes on a host

5.2.6. Configuration tasks

The following table lists maintenance tasks that a System Administrator needs to perform in the Configuration area. For information about other maintenance areas, see Areas of maintenance tasks.

Event or frequency	Task	Automatable
Before going to production	Log file configuration	No
Before going to production	Set up one Service Manager account for each incoming integration	No
Before going to production	Configure SSL encryption for Service Manager clients and integrations	No
Before going to production	Restrict JavaScript access to the operating system	No
Before going to production	Review security configuration options and the security guide	No
After modifying credentials	Encrypt all credentials	No
Before going to production	Define a batch size for sequential number records	No
Before going to production	Set up the certExpireWarningDays parameter	No
Before going to production	Review known issues	No

Related topics

Service Manager maintenance tasks
Areas of maintenance tasks

5.2.6.1. Log file configuration

Log files grow over time and consume disc space. If no disc space is available components may start failing.

Configure the log files especially for Service Manager server and web tier. The web tier log file configuration depends on the web application server. For instructions, refer to your web application server documentation.

Server log file configuration

The server log file configuration includes the following tasks.

Separate the log files

By default, all servlets running on a host write to a single log file (sm.log) in the **logs** subdirectory of the Service Manager installation, which is specified in the "log" parameter in the sm.ini file. For more information, see Startup parameter: log.

At least for servlets dedicated for integrations or background processes, you should write a separate log file. To configure this, in the sm.cfg file of the servlet, add to the startup command the "-log" parameter with the log file path relative to the RUN directory.

Here is an example:

```
sm -startup -log:..\logs\background.log
```

Tip

Tip A common practice is to write a separate log file for each servlet.

Set up log file rotation

As the log files grow large over time, it is recommended to set up log rotation. This guards against a log file size generating disc space shortage. For more information, see Log switching.

When setting up log rotation, calculate the required disc space for all servlets and make sure it is available with a buffer for the unforeseen. For example:

```
sm -httpPort:13080 -numberoflogfiles:10 -maxlogsize:52428800 -logqueuesize:2000
```

This command will start a servlet keeping up to 10 log file copies. Log file switch will happen when the log file size reaches 50MB. So the total amount of disc space required for this servlet is 500MB.

The **-logqueuesize** parameter controls the number of messages buffered in memory before being written to the log file. It helps to avoid performance impact of switching larger log files to user sessions. For more information, see Parameter: logqueuesize.

Activate proactive trace parameters

Review the list of proactive trace parameters, and decide which are helpful to monitor and maintain the administrated system. For details, see Trace parameters that can be proactively activated.

Tip

Tip It is recommended to activate only traces whose information is reviewed periodically.

Web tier log file configuration

The web tier log file configuration includes the following.

- Web server and web application server logging
  Refer to the documentation of the web server and the web application server regarding log file configuration and set it up as appropriate.
- Logging for the Service Manager web client application
  The Service Manager web client application uses the <Webtier_Home>\WEB-INF\log4j2.properties configuration file, where you can specify web application specific log parameters.

Related topics

Service Manager maintenance tasks
Configuration tasks

5.2.6.2. Setup one Service Manager account for each incoming integration

When external systems access Service Manager, provide valid credentials for the account they are running in. It is best security practice to not reuse an account and password, but create a unique account for each integration. This helps managing account privileges, tracking access, and analyzing issues in log files.

Related topics

Service Manager maintenance tasks
Configuration tasks

5.2.6.3. Configure SSL encryption for Service Manager clients and integrations

Make sure the clients and integrations are using SSL encryption for security reasons.

Related topics

Configuration tasks

Secure Sockets Layer (SSL) encryption

TLS 1.2 support and configuration

Enable SSL encryption for published Web Services

5.2.6.4. Review security configuration options and the security guide

The system administrator is responsible for operational security of the installation, and is required to be familiar with security requirements and configurations.

For more information, see Service Manager Security Guide white paper.

Especially, planning and setting up the TLS/SSL security between the components of the Service Manager Environment is required. In an out-of-box installation, there is no TLS/SSL security configured by default. For more information, see Secure Sockets Layer (SSL) encryption and TLS 1.2 support and configuration.

Related topics

Service Manager maintenance tasks
Configuration tasks
System Security
TLS 1.2 support and configuration
Service Manager Security Guide

5.2.6.5. Restrict JavaScript access to the operating system

JavaScript implementation can access operating system resources using the following global functions: readFile(),uncompressFile() , writeFile(),writeFile(), deleteFile(), makeDir(), writeAttachmentToFile(), and sysExec().

This access can be restricted using the following sm.ini parameters: jsaccessfilereadregex , jsaccessfilewriteregex, and jsaccesscmdregex. For more information, see Startup parameters: JavaScript access.

Define exactly which operating system resources are required to be accessed from implementations inside the Service Manager applications.

Tip

Tip For operating system command line access that's required, write shell scripts in the Service Manager Server's RUN directory and set up the jsaccesscmdregex parameter to only allow access to these shell scripts.

Related topics

Configuration tasks
Startup parameters: JavaScript access

5.2.6.6. Encrypt all credentials

Parameters in configuration files for credentials must not be stored in clear text for security reasons. The same applies to the location in the database to store credentials for external applications that Service Manager integrates with.

Credentials stored in configuration files

Service Manager encrypts credentials in configuration files stored in the file system automatically using a randomly generated master key. For more information, see Randomly generated master keys.

The Service Manager administrator must be familiar with this concept, and the risk of making the system inaccessible by accident. The administrator needs to know how to make Service Manager recreate the master key, and the importance to take precautions and store credentials at a safe place. See also Document credentials in a safe place.

Credentials stored in dbdicts

Credentials of external applications that Service Manager integrates with are stored in the RDBMS. Examples include Service Manager Integration Suite (SMIS) integrations, or integrations implemented in JavaScript (both based on web services or REST).

Credentials of external applications used by Service Manager are stored in the database, and transmitted in HTTP message headers. The passwords are required to be protected in both.

Passwords for external applications using SMIS are stored as encrypted in the database. In the wizard to set up an integration instance, these parameters appear on the **Secure parameters** tab.

When implementing a web service or RESTful integration directly, the implementer is responsible for storing the credentials securely. When using the RunWSDL2JS wizard to produce JavaScript code to interface with an external application, the credentials get stored as plain text in the generated ScriptLibrary record.

The transmission of credentials in HTTP messages is also not secure: The Authorization header property is not securely encrypted. Secure transmissions require TSL/SSL configuration.

Related topics

Service Manager maintenance tasks
Configuration tasks
Randomly generated master keys
Document credentials in a safe place

5.2.6.7. Define batch size for sequential number records

For a dbdict using the sequential numbering mechanism to assign unique numbers to an identifier field, define a batch size for sequential number generation when the dbdict has a high creation rate of new records.

Note

Note When using a batch size, only the uniqueness of the numbers is guaranteed. Later requested numbers aren't necessarily greater numbers.

Related topics

Sequential numbers

5.2.6.8. Set up certExpireWarningDays parameter

Expired SSL certificates may cause system unavailability. Service Manager prints a warning before certificate expiration. The timing of the warning is configured in the certExpireWarningDays parameter.

Related topics

Configuration tasks
Parameter: certExpireWarningDays

5.2.6.9. Review known issues

Review the list of known issues documented for the Service Manager version used. This list can be found in the Service Manager Help Center for Service Manager 9.50 or later.

Note

Note For releases earlier than 9.50, the known issues can be found in their specific release notes in PDF format. See Service Manager Document Matrix for their download links.

Related topics

Configuration tasks
Known issues
Service Manager Document Matrix

5.2.7. Fallback strategy tasks

The following table lists tasks that a System Administrator needs to perform in the Fallback Strategy area. For information about other maintenance areas, see Areas of maintenance tasks.

Event or Frequency	Task	Automatable
Before going to production	Unload falcon and fail-safe operator records	No
Before going to production	Set up backup strategies	No
Quarterly	Test fallback procedures	No

Create a "fail-safe" administrative operator

The Database Manager (RAD application: database) allows an operator with the **SysAdmin** capability to access all contents of the Service Manager system, even if menus and the system navigator should not be available.

Therefore, a copy of the **falcon** operator with startup application **database** is a valuable fallback for emergency scenarios.

Unload falcon and fail-safe operator records

In case operator records get corrupted or deleted, an unload of an operator record with the **SysAdmin** capability will enable you to gain back access to the system.

To load this file from the operating system command line (the RUN directory), use following command:

```
sm file.load operator.unl NULL NULL winnt -bg
```

or

```
sm file.load operator.unl NULL NULL unix
```

Set up backup strategies

As any computer system may fail with the risk of data loss, it is important to take precautions for the data managed by Service Manager system, as well as all information to run it.

This includes:

- Data stored in the RDBMS system. Work with the database administrator to set up a backup strategy that can meet the business requirement, and to document the restoration practice.
- All documentation about the system such as configuration diagrams, configuration settings, tailoring, and maintenance concept. Work with the IT department to set up a strategy about the backup and restoration of this kind of data.

Test fallback procedures

In case of an outage, the prepared fall back strategies must work in order to prevent data loss and ensure quick service reestablishment. Therefore, it is important that all fall back procedures are tested periodically.

Related topics

Service Manager maintenance tasks
Areas of maintenance tasks

5.2.8. Documentation tasks

The following table lists maintenance tasks that the System Administrator needs to perform in the Documentation area. For information about other areas, see Areas of maintenance tasks.

Event or frequency	Task	Automatable
Before going to production	Document fallback procedures	No
On demand	Document the application of patches, hotfixes, and custom tailoring	No
On demand	Document all automated administrative tasks	No
After modifying credentials	Document credentials in a safe place	No

Document fallback procedures

All fallback strategies must be documented to cover their applicable scenarios, precautions to be taken, and restoration procedures, so that they are available when needed.

Document the application of patches, hotfixes, and custom tailoring

The administrator is responsible for documenting all changes applied to the system, such as the application of patches and hotfixes, or custom tailoring releases. This includes the following information:

- Date and time of the application
- Information about the change
- Status after the application (error or warning messages, and results of initial functionality tests)

This document is required for analysis if an issue is observed.

> **Note**
>
> Note The standard question from Software Support engineers about the *last change before first observation* is directly referring to this document.

Document all automated administrative tasks

Diligence in implementation, documentation, and maintenance of these automated tasks is important because changes of the system configuration, or tailoring may directly result in a requirement to modify automated maintenance tasks as well.

Document credentials in a safe place

Service Manager automatically encrypts credentials in configuration files and in the Service Manager Integration Suite (SMIS) configuration. Therefore, credentials can no more be looked up from these files or SMIS.

These credentials include credentials for the relational database, LDAP server, SSL keystore files, or external applications that Service Manager integrates with.

To address the risk of not being able to access required resources because of failure, document all credentials in a secure place.

Related topics

Areas of maintenance tasks
System information record
Encrypt all credentials

5.2.9. Change control tasks

The following table lists maintenance tasks that a System Administrator needs to perform in the Change Control area. For information about other areas, see Areas of maintenance tasks.

Event or frequency	Task	Automatable
On demand	Apply Service Manager Server patches	No
On demand	Apply Service Manager application patches	No
On demand	Apply Service Manager web tier patches	No
On demand	Apply JRE updates on the Service Manager Server	No
On demand	Apply operating system patches or hardware updates	Yes

Related topics

Areas of maintenance tasks
JRE support
Operation tasks

5.2.9.1. Apply Service Manager Server patches

Updating the Service Manager server by a patch or hotfix is the replacement of executable files in the Service Manager server's RUN directory on each host. The files replaced include among others the **sm** executable, and the contents of the RUN/lib directory.

The java archive files (*.jar) contained in the RUN/lib directory are the company's or third-party files with a version number in the file name. For this reason, only copying the patch into an existing RUN/lib directory is not replacing all existing files when the version has been updated, and the file with the older number has to be removed manually. A good practice therefore is to rename the RUN/lib directory before patch application.

When a parameter in sm.ini refers to a Java archive file in the RUN/lib directory, the parameter needs to be updated when the version of this file changes. An example is the Service Manager mbeanclient JAR file, which is required when Service Manager is running with a hardware load balancer:

```
JVMOption999:./lib/mbeanclient-<version>.war
```

Note

Note Always follow the installation advice provided with the patch.

Related topics

Change control tasks

5.2.9.2. Apply Service Manager application patches

Application patches or hotfixes update the implementation data stored in the database. Customization and tailoring implementations may be affected by application of a patch. For this reason, conflicts between the current application and the patch need to be identified and resolved. The Application Patch Manager utility supports this process.

Note

Note Always follow the installation advice provided with the patch.

Related topics

Change control tasks

5.2.9.3. Apply JRE updates on the Service Manager server

Each Service Manager servlet starts a Java Runtime Environment (JRE). The installation directory of this JRE is:

<Service Manager installation path>/RUN/jre

When installing Service Manager on Windows, a JRE is installed as well.

When a JRE update is required, verify first that Service Manager supports the Java version by consulting the Support Matrix.

Related topics

Change control tasks
JRE support

5.2.9.4. Apply Service Manager web tier patches

Service Manager web tier patches or hotfixes are shipped as web application archive, and therefore a complete replacement of the existing Service Manager web tier files. The web tier patch must be applied on all web application server instances of the Service Manager installation.

Only save the files you made configuration changes before such as in the web.xml file in order to apply the same configuration to the new web application deployment.

Note

Note Always follow the installation advice provided with the patch.

Related topics

Change control tasks
Upgrade the Web Tier

5.2.9.5. Apply operating system patches or hardware updates

The operating system (OS) administrator is responsible for applying OS patches.

The application of an OS patch may require a restart of the server. Because of that, there are some considerations regarding the Service Manager systems.

If Service Manager is installed as a horizontally scaled environment, it may be shut down on a host for maintenance, while it is still accessible on other hosts. The only requirement is to not shut down all hosts concurrently, which keep the JGroups cluster of the Service Manager servlets running.

Related topics

Change control tasks
Start Service Manager

5.2.10. Capacity planning tasks

The following table lists maintenance tasks that a System Administrator needs to perform in the Capacity Planning area. For information about other areas, see Areas of maintenance tasks.

Event or frequency	Task	Automatable
Before going to production and ongoing	Reserve and monitor disc space for log files	Yes
Before going to production	Reserve and monitor sufficient RDBMS size limits	Yes
Quarterly	Setting up the inactivity timer	No
Monthly	Check license usage	No
Quarterly	Determine a list of named users and update their operator records	No

Related topics

Service Manager maintenance tasks

5.2.10.1. Reserve and monitor disc space for log files

A system administrator needs to monitor the disc space, and reserve enough space for the log files of the Service Manager Server and web tier.

Server log files

Plan the disc space for server log files when configuring the log files initially, or when the configuration changes (for example, by starting more servlets). Using log rotation guarantees a maximum size for log files so that the required space can be calculated.

When the system grows, more servlets may need to be started, and starting temporary traces to generate traces for problem analysis should be possible. Be sure to have more disc space available than needed by the initial configuration.

Web tier log files

The web tier log file configuration depends on the web application server. Refer to the specific documentation for your web application server.

Related topics

Service Manager maintenance tasks
Log file configuration

5.2.10.2. Reserve and monitor sufficient RDBMS size limits

When configuring Service Manager systems, consider the required database sizes for a definite time frame and monitor the data growth.

The tailored implementation and large number of tables in a Service Manager system make predictions on required database sizes difficult, and it is therefore very important to monitor the trends.

Related topics

Service Manager maintenance tasks
Capacity planning tasks

5.2.10.3. Setting up the inactivity timer

The inactivity timer is used to terminate inactive user sessions after a specified idle time. This is important to prevent inactive sessions from still consuming licenses and important for security reasons.

The inactivity timer is controlled by the servlet itself. There is no background process required to be started. This implementation avoids the need to request status update by each servlet for idle time.

For for information, see Inactivity timer.

Related topics

Inactivity timer
Capacity planning tasks

5.2.10.4. Determine a list of named users and update their operator records

If named user licenses have been purchased, determine which users should log into the system as named users and update their operator records with the **Named User** checkbox.

Related topics

Named users
Capacity planning tasks

5.2.10.5. Check license usage

For capacity planning, check your license usage in the following ways.

Monitor the license usage by the stathistory dbdict

Generate a Service Manager License report, especially during peak hours, and analyze the usage patterns and trends in order to prepare for any needed license increases.

Tracking of USER and SYSTEM connections in stathistory

User connections and system connections are tracked in the stathistory table. The following table provides an example.

Name	Exact Time	Value	Month	Day	Year
Connected System Processes	09/04/16 06:22:51	23	9	4	2016
Connected Users	09/04/16 06:22:51	1292	9	4	2016

Two schedule records come with the out-of-box installation:

- Count USER Connections
- Count SYSTEM Connections

Tip

Tip If you don't need this information, rename the class field of these schedule records to "agentDISABLED" to assign them for processing to the nonexistent background scheduler "agentDISABLED". This removes the execution of processes() function which queries information from all servlets over the network.

Tracking of module license usage in stathistory

Module licenses are tracked by updating stathistory during the login process of each session.

The following table provides an example of a record list.

Name	Exact Time	Value	Month	Day	Year
Asset Contract Management - Floating	09/02/16 14:17:51	1	9	2	2016
Change Management - Floating	09/02/16 14:17:50	213	9	2	2016
Incident Management - Floating	09/02/16 14:17:51	987	9	2	2016
Knowledge Management - Floating	09/02/16 14:17:51	1131	9	2	2016
Problem Management - Floating	09/02/16 14:17:51	312	9	2	2016
Request Management - Floating	09/02/16 14:17:51	25	9	2	2016
Service Desk - Floating	09/02/16 14:17:51	1292	9	2	2016
SLA - Floating	09/02/16 14:17:51	1292	9	2	2016
Smart Analytics - Floating	09/02/16 14:17:51	1292	9	2	2016

Related topics

System information record
License tracking
Stathistory table
AutoPass licensing

5.2.11. Cleanup tasks

The following table lists maintenance tasks that a System Administrator needs to perform in the Cleanup area. For information about other areas, see Areas of maintenance tasks.

Event or frequency	Task	Automatable
Before going to production	Clean up out-of-box data records	No
Before going to production	Reset sequential numbers	No
Before going to production	Reset the "Times updated" field in globalists	No
Before going to production	Disable unused background processes	No
Monthly	Purge or archive data records	Yes
Daily	Purge queue records	Yes
Weekly	Purge logs	Yes
Before startup	Clean up server side log files	Yes
Weekly	Purge dbdicts containing log information	Yes
Weekly	Clean up AUDIT* records if auditing backup is turned on	Yes

Related topics

Areas of maintenance tasks

5.2.11.1. Clean up out-of-the-box data records

Before the system goes live, delete all test data records and out-of-the-box demonstration data. It's advisable that you use the predefined **PurgeOutofBoxData** unload script record to remove these data records.

To purge the data, follow these steps:

1. Go to Database Manager.
2. Open the **unload** table.
3. Select the **PurgeOutofBoxData** record.
4. Make sure to purge all the listed tables.
   • To avoid purging a table, remove the line from the unload script record.
   • You can add or modify a query (for example, add name="failsave" to the operator line) as needed.
   It's advisable that you add **knownerrortasks** and **rootcausetasks** to the list of tables to purge if you have entered information during testing.

Related topics

Cleanup tasks
Purging out-of-the-box data

5.2.11.2. Reset sequential numbers

Before going to production, when all demonstration and test data is purged, reset the last number in the related **number** or **counters** dbdict.

Related topics

Cleanup tasks
Purging out-of-box data
Sequential numbers

5.2.11.3. Reset the "Times updated" field in globalists

Global lists are records to specify value lists to be retrieved from the database and stored in a global variable during run-time. By this, access to the database while preparing data to display is reduced, and the response time is reduced. Global

lists require to be refreshed from time to time to ensure that the most current data is displayed.

By resetting the "Times updated" field in a globalist record to 0 when setting up the system, this field can be used to understand the frequency of updates.

Related topics

Cleanup tasks

5.2.11.4. Disable unused background processes

Background processes that aren't being used should be removed from the **info.startup** record. The "startup" record contains a list of all background processes that are initiated by the **sm system.start** line in the sm.cfg file. For example, remove the despooler, agent and marquee lines, because they are no longer being used.

Caution

Caution Do not remove the **Sync** process, because it is very important for cleaning up sessions, memory, and so on.

Related topics

Cleanup tasks

5.2.11.5. Purge or archive data records

How to purge or archive data records depends on the requirements of your corporate data retention policy or local laws. User data includes interactions, incidents, problems, changes, and so on.

A typical purge or archive strategy is to archive and delete all incidents that were closed more than <x> months ago.

Business data (for example, an incident) is stored in many tables: The main information is stored in **probsummary** records; however, there are clock records, file attachments in the SYSATTACHMENTS dbdict, and relation to interactions, change requests, and problems in the screlation dbdict. Additionally, dbdicts as specified in the Object record may also contain information, for example, for activities and so on.

There are multiple options to implement purging or archiving: the Purge/Archive utility, custom implementation inside Service Manager, and custom implementation by an RDBMS implementation. The Purge/Archive utility allows you to set up data maps to archive records along with all of their related records.

Related topics

Cleanup tasks
Purging and archiving records
Archiving and viewing archived Service Manager data using RDBMS Methods

5.2.11.6. Purge queue records

Clean up old records from the dbdicts described in the following table.

Dbdict	Description
schedule	Old schedule records that are in application error status and no longer required for issue analysis should be removed. Executing schedule records that were not processed with a longer delay may not be adequate, and they therefore should be purged. For example "message processor records"? are used to send on-screen messages to a user session. Other records, however, require to be executed still.
eventin, eventout	Event services queue for email and some other integrations

Related topics

Cleanup tasks
Purging and archiving records
Archiving and viewing archived Service Manager data using RDBMS Methods

5.2.11.7. Purge logs

Log information in server and client log files, dbdicts containing log information, and web tier log files require to be cleaned up frequently to prevent log files from growing to an unmanageable size, and to avoid Disc Full situations or performance impacts because of huge tables on the relational database system.

The cleanup of web tier log files is managed by the web application server.

Clean up server side log files

It is good practice to remove the log files completely or move them to a separate folder or even another network location before every startup of Service Manager.

Note

Note Some countries have laws regarding which information may be captured in log files and how long log files may by stored before they need to be destroyed. The Service Manager Administrator shall assume responsibility to comply with the local rules.

Purge dbdicts containing log information

Old records in dbdicts containing log information should be cleaned up frequently. The following table describes out-of-box dbdicts containing log information.

Dbdict	Description
msglog, syslog	Internal log files
stathistory	Used for daily tracking of module license usage
systemperform, systemtotals	Used for system load tracking. This is started and stopped from **System Status** > **System Monitor** > **Capture**. **Caution** Caution Keeping this activated for a long time may cause slow performance.

The dbdicts can be purged automatically by scheduling an archive or purge. The out-of-box system provides a background purge to clear the msglog and syslog records that are over 7 days old. Modify the purge interval if needed. To do this, go to the **schedule** table from the Database Manager and search for "background purge/archive". You can modify the purge/archive parameters on the **Strings** tab.

Related topics

Cleanup tasks
Purging and archiving records
Dbdicts containing log information
Purge the SMIS task log records

5.2.11.7.1. Clean up AUDIT* records if auditing backup is turned on

If you have backups turned on for auditing changes, clean up all AUDIT* forms, links, format controls, and so on.

Note

Note To turn on auditing backup, go to **Tailoring** > **Audit** > **Turn Auditing On/Off**, and then select the **Do you want to keep backups of Changes** option.

Important

Important We strongly recommend turning off "keep backups of changes" in a production system. You can use this feature on development, but should purge the AUDIT* records there at the end of each development cycle.

Related topics

Cleanup tasks
Purging and archiving records
Dbdicts containing log information
Purge the SMIS task log records

5.2.12. Online support tasks

The following table lists maintenance tasks that a System Administrator needs to perform in the Online Support area. For information about other areas, see Areas of maintenance tasks.

Event or frequency	Task	Automatable
Before going to production	Register to Software Support	No
Weekly	Subscribe to Service Manager product news	No
Before going to production	Bookmark the Service Manager security bulletins	No
Before going to production	Bookmark the Service Manager Document Matrix	No
Before going to production	Check for news on the Support web site and community	No

Related topics

Areas of maintenance tasks

5.2.12.1. Register to Software support

The Software Support web site is a communication channel to Software Support to log support requests, research the knowledge base, download software, and join the community.

Register for a Software Passport and enter your Support Agreement IDs (SAID). Additionally, bookmark the Software Support Site Reference and User Guides:

https://softwaresupport.softwaregrp.com/site-reference-user-guides

Related topics

Online support tasks

Contact support

5.2.12.2. Subscribe to Service Manager product news

The Software Support web site allows you to subscribe to product news sent through email notifications. Use the link below to configure the notification service according to your requirements.

https://portal.microfocus.com/s/subscriberproduct

Related topics

Online support tasks
Software Support

5.2.12.3. Bookmark the Service Manager security bulletins

In order to keep informed about security alerts for the Service Manager product, be sure to bookmark the Service Manager security bulletins:

https://www.microfocus.com/en-us/support/security/

Related topics

Online support tasks
Service Manager Security Bulletins
Service Manager Document Matrix

5.2.12.4. Bookmark the Service Manager Document Matrix

Bookmark the Service Manager Document Matrix for quick access to all Service Manager related documentation:

https://portal.microfocus.com/s/article/KM000002518

Related topics

Online support tasks
Service Manager Security Bulletins

5.2.12.5. Check for news on the Support web site and community

Administering a Service Manager installation is a daily routine for many administrators. Participating in a forum to learn experiences and best practices from the community is a great opportunity to grow.

We provide an internet forum where administrators can discuss issues and share knowledge:

https://community.softwaregrp.com/t5/SMA-Service-Manager-Suite/ct-p/sws-SMgr

Related topics

Online support tasks
Service Manager Support Customer Forum
Service Manager Security Bulletins
Service Manager Document Matrix

6. Application Setup

Administrators enable users to access applications by creating security rights, security area, security role, role, and operator records. Administrators control the following application tables and related administrative tasks:

- Security rights
- Security roles
- User roles
- Capability words
- Adding users
- Purging and archiving records
- Lockouts
- Login restrictions

Related topics

Application profiles
Environment record
User roles
Capability word model
Adding users
Checklist: Adding a new user
Controlling user access and security
Creating operator records
Defining named users
Operator passwords
Operator records
Operator templates
Lockout feature
System quiesce: Login restrictions

6.1. Processes and Best Practices

In order for you to make optimal use of the functionality of Service Manager, we have created Service Manager Processes and Best Practices based on practical experience with service management implementations with many customers of various sizes.

The *Service Manager Processes and Best Practices* provides best practices, management overviews, process flows, and workflow diagrams for different Service Manager modules.

The *Service Manager Processes and Best Practices* documents the practice workflows that are standard with out-of-box Service Manager applications. It includes high-level process workflow diagrams, and step-by-step procedural guidelines.

Related topics

Processes

6.1.1. Processes

The Service Manager processes are based on ITIL theory and are referenced in the ITIL core.

The Service Manager best practices cover the following processes in the ITIL Service Transition and Service Operation documents. For information on these processes, see the *Service Manager Processes and Best Practices* linked in the related topics.

ITIL core volume	process number	process reference	process reference
Service Operations	0	User Interaction Management overview	SO0
Service Operations	2	Incident Management overview	SO2
Service Operations	3	Request Fulfillment overview	SO3
Service Operations	4	Problem Management overview	SO4
Service Transition	2	Change Management overview	ST2
Service Transition	3	Configuration Management overview	ST3

Related topics

Processes and Best Practices

6.1.2. User Interaction Management overview

The Service Manager User Interaction Management process helps you handle User Interactions either reported by using Self-Service web pages or directly to your Service Desk. Though not an ITIL process, this process enables the separation between information requests and service requests. The result is that multiple user interactions may all be linked to one incident record in the tool. This process will streamline the Service Desk activities and thereby decrease the workload for second line support teams.

The Service Manager Service Desk application supports the service desk function of the Information Technology Infrastructure Library (ITIL) with its User Interaction Management processes for the IT service and the customer base. It provides a single point of entry to the other Service Manager applications and enables you to document and track all calls received by your service desk. It incorporates the essential concepts of ITIL to ensure that the best practices of IT service management in place. The Service Desk application helps you to aid end customers, ensure data integrity, and streamline communication channels in the organization.

For complete information on the User Interaction Management (Service Desk) process overview and workflows, see *Service Manager Processes and Best Practices* in the related topics.

Related topics

Processes and Best Practices

6.1.3. Incident Management overview

The Service Manager Incident Management process enables you to categorize and track various types of incidents (such as service unavailability or performance issues and hardware or software failures). It helps you to ensure that incidents are resolved within agreed on service level targets.

The Service Manager Incident Management application supports the Incident Management process. It incorporates the essential concepts of Information Technology Infrastructure Library (ITIL) to ensure that the best practices of IT service management in place. The Incident Management application helps you to restore normal service operation as quickly as possible and minimize the adverse impact on business operations.

For complete information on the Incident Management process overview and workflows, see *Service Manager Processes and Best Practices* in the related topics.

Related topics

Processes and Best Practices

6.1.4. Request Fulfillment overview

The Service Manager Request Fulfillment process enables you to effectively manage all user requests for products and services. It provides your users a mechanism to request and receive standard services based on a pre-defined approval and qualification process. The Service Manager Request Fulfillment application supports the Request Fulfillment process. It incorporates the essential concepts of Information Technology Infrastructure Library (ITIL) to ensure that the best practices of IT service management in place. The Request Fulfillment application gives you an increased level of control for your organization's services and helps you source and deliver the necessary components for requested standard services. For complete information on the Request Fulfillment process overview and workflows, see the *Service Manager Processes and Best Practices*.

Related topics

Processes and Best Practices

6.1.5. Problem Management overview

The Service Manager Problem Management process allows you to find, fix, and prevent problems in the IT infrastructure, processes, and services. It helps you to prevent problems and their resulting incidents, eliminate recurring incidents, and minimize the impact of incidents that can't be prevented.

The Service Manager Problem Management application supports the entire Problem Management process. It incorporates the essential concepts of Information Technology Infrastructure Library (ITIL) to ensure that the best practices of IT service management in place. The Problem Management application helps you to maximize system availability, improve service levels, reduce costs, and improve customer convenience and satisfaction.

Note

Note ITIL defines a Problem as the unknown cause of one or more Incidents.

For complete information on the Problem Management process overview and workflows, see *Service Manager Processes and Best Practices* in the related topics.

Related topics

Processes and Best Practices

6.1.6. Change Management overview

The Service Manager Change Management process enables you to control changes to baseline service assets and configuration items across the entire service life cycle. It helps you to control the process to request, manage, approve, and control changes that modify your organizational infrastructure. This includes assets, such as network environments, facilities, telephony, and resources.

The Service Manager Change Management application supports the Change Management process. It incorporates the essential concepts of Information Technology Infrastructure Library (ITIL) to ensure that the best practices of IT service management in place. The Change Management application help you to ensure that changes are requested, evaluated, and implemented according to your business processes and needs.

Note

Note To control changes, we recommend that you handle all standard changes (that is, changes of predefined types) by using the Service Catalog request fulfillment process.

For complete information on the Change Management process overview and workflows, see *Service Manager Processes and Best Practices* in the related topics.

Related topics

Processes and Best Practices

6.1.7. Configuration Management overview

The Service Manager Configuration Management process ensures that selected components of a complete IT service, system, or product (the Configuration Item) are identified, baselined, and maintained and that changes to them are controlled. It enables you define and control the components of services and infrastructure, and to maintain accurate configuration information about the historical, planned, and current state of services and infrastructure.

The Service Manager Configuration Management application supports the Configuration Management process. It incorporates the essential concepts of Information Technology Infrastructure Library (ITIL) to ensure that the best practices of IT service management in place. The Configuration Management application supports your business and control objectives, and helps you to track your configuration items through their lifecycle.

For complete information on the Configuration Management process overview and workflows, see *Service Manager Processes and Best Practices* in the related topics.

Related topics

Processes and Best Practices

6.2. Adding users

Each person (user) who logs onto Service Manager must have a personal information record stored in the operator table. Information associated with a user includes personal data, such as name, address, phone numbers, login name, and password for Service Manager. Operator records also store capability words for a given user. Without an operator record, a user can't log onto Service Manager. A user can belong to a group with a single profile or have a unique profile.

Note

Note: Based on the RFC 2617 authentication scheme, the colon (":") character is not allowed to be used in usernames.

Related topics

Checklist: Adding a new user
Controlling user access and security
Creating operator records
Defining named users
Operator passwords
Operator records
Operator templates

6.2.1. Checklist: Adding a new user

To add a user, follow these steps:

1. Create a contact record for the user.

Caution

Caution You can't create an operator without a contact Id.

2. Create an operator record for the user.
3. Select security settings for the user. Do one of the following:
   • Select a user role, which includes predefined security roles, application profiles (for Knowledge Management and Configuration Management modules), and capability words.
   • Select security roles, application profiles and capability words.
4. Create a startup menu for the user.
5. Create a menu record for the user.

Related topics

Adding users
Controlling user access and security
Creating operator records
Defining named users
Operator passwords
Operator records
Operator templates

6.2.2. Creating operator records

Service Manager provides the following methods for creating an operator record:

- Create the operator record from the operator.g form, which enables you to:
  - Create an operator record starting from a blank operator record
  - Create an operator by copying properties from an existing operator
- Create the operator record from Ongoing maintenance (CAU) User Quick Add Utility, which enables you to:
  - Create an operator record starting from a blank operator record
  - Create an operator by copying properties from an existing operator
  - Create a contact record starting from a blank contact form
  - Create a contact by copying properties from an existing contact

Service Manager includes several predefined operator records that you can use as templates to create your own users. Search your operator records to see a complete list of sample operators.

Related topics

Application Setup
Adding users
Checklist: Adding a new user
Controlling user access and security
Defining named users
Operator passwords
Operator records
Operator templates

6.2.3. Operator records

Each Service Manager user must have an operator record to log on and use Service Manager applications. The operator record defines the access rights and security settings that a user has. The following fields are required for an operator record:

- Login Name
- Resource Type

You can also define the following optional information in an operator record:

Operator tab	Fields and sections	Purpose
General	• Login Name — Required field Full Name • Default Company • Contact ID • Application Profile • User Role • Application profiles • Security Roles	Use this tab to add identifying information and application profiles to an operator record. These application profiles determine which features and views the operator can access in Service Manager applications.
Security	• Password Information • User Session Information • Login Information • User Lockout Information • LDAP Information • Template Information • Password History	Use this tab to manage how an operator accesses Service Manager. These settings determine whether an operator can log on to Service Manager.

Login Profiles	Login Profile — This section contains: • The operators login profile • Licensing Information	The login profile includes: • Language: Specifies the login language for the operator. • Time Zone: Specifies the time zone for the operator. • Date Format: Specifies the preferred format for the date, such as mm/dd/yy or dd/mm/yy. • First Day of Week: Specifies the first day of the week in the calendar and date picker for the operator. • Display Currency: Specifies what currency displays for catalog items when the operator orders from the Service Catalog or approves catalog requests. Defaults to the basis currency in the System Information Definition record for the company when left blank. • Message Level: Specifies the level of messages to store in the message queue for this operator. • Max Attachment Size: Specifies the maximum size of a file that the operator can insert into an attachment object. • Total Attachment Size: Specifies the maximum size of all the attachment files that the operator can insert into an attachment object.
Groups	• Knowledge Groups to which the operator belongs • Assignment Groups to which the operator belongs	Use this tab to determine which groups to which the operator belong. Assignment groups are used for both work assignment and approval.
Startup	• Initial Application • Home Page • Queue • Dashboard • Execute Capabilities • Query Groups • Months	Use this tab to determine which forms the operator sees at startup and also to customize the capability words the operator has in addition to those granted by a user role.

Notification	• Standard Notifications • Client Printing	Use this tab to record contact information about the operator.
Rate	• Hourly Rate • Rate Currency	Use this tab to determine the operator's labor cost rate.
Self Service	• Self Service Access Only • Self Service Menu • Self Service Starting Page	Use this tab to authorize self service access to an operator. This gives the user access to the self service Web client.

Related topics

Application Setup
Checklist: Adding a new user
Controlling user access and security
Creating operator records
Defining named users
Operator templates
Adding users
Add an operator record
Add an operator record with the User Quick Add utility
Add an operator to a security group
Create a system default operator record
Define a startup menu for an operator
Define the distinguished name an operator uses to log in to an LDAP directory service
Delete an operator record
Enable an operator to see the command line
Set the maximum file attachment size for an operator
Set the time zone for an individual operator
Update an operator record
View an existing operator record
Use Mass Update with operator records

6.2.3.1. Add an operator record

Applies to User roles: System Administrator

Note: To create an operator, you need to first create a contact record to match the operator record.

To add a contact record:

1. Click **System Administration** > **Base System Configuration** > **Contacts**.
2. Specify the **Contact Name**.
3. Specify the **Full Name**.
4. Add any other necessary information.
5. Click **Add**, and then click **OK**.

You are now ready to create an operator record.

To add an operator record:

1. Click **System Administration** > **Ongoing Maintenance** > **Operators**.
2. Specify a **Login Name**.
   Note: We recommend that you only use ASCII characters in **Login Name**. In Service Manager Web Service integrations, non-ASCII operator login names aren't supported.
3. Specify a **Contact ID**, or select the contact name that you just created.
4. Click **Add**.

5. On the Security tab, type the name of the **Template** from which you want to create the new operator record. The information from the template will be populated in the new operator record.
6. Add any other necessary information for this new operator record.
7. Click **Save**.

Related topics

Application Setup
Operator passwords
Operator records
Operator templates
Add an operator record with the User Quick Add utility
Add an operator to a security group
Create a system default operator record
Define a startup menu for an operator
Define the distinguished name an operator uses to log in to an LDAP directory service
Delete an operator record
Enable an operator to see the command line
Set the maximum file attachment size for an operator
Set the time zone for an individual operator
Update an operator record

6.2.3.2. Add an operator record with the User Quick Add Utility

Applies to User roles: System Administrator

To add an operator record with the User Quick Add utility:

1. Click **System Administration** > **Ongoing Maintenance** > **User Quick Add Utility**.
2. Fill in the required fields.
3. Click **Next**.
4. Type the name of a user to clone or use the Fill icon to find a user to clone.
5. Click **Finish**.
6. Update the new Operator record as required.

Related topics

Application Setup
Operator passwords
Operator records
Operator templates
Add an operator record
Add an operator to a security group
Create a system default operator record
Define a startup menu for an operator
Define the distinguished name an operator uses to log in to an LDAP directory service
Delete an operator record
Enable an operator to see the command line
Set the maximum file attachment size for an operator
Set the time zone for an individual operator
Update an operator record
View an existing operator record

6.2.3.3. Add an operator to a security group

Applies to User roles: System Administrator

To add an operator to a security group:

1. Click **System Administration** > **Ongoing Maintenance** > **Operators**.
2. Fill in optional search criteria, and then click **Search**.

3. Select the operator record to be changed.
4. Click the **Groups** tab.
5. In the groups array, type the names of the security groups for the new operator on separate lines.
6. Click **Save**.

Related topics

Application Setup
Operator passwords
Operator records
Operator templates
Add an operator record
Add an operator record with the User Quick Add utility
Create a system default operator record
Define a startup menu for an operator
Define the distinguished name an operator uses to log in to an LDAP directory service
Delete an operator record
Enable an operator to see the command line
Set the maximum file attachment size for an operator
Set the time zone for an individual operator
Update an operator record
View an existing operator record

6.2.3.4. Create a system default operator record

Applies to User roles: System Administrator

Information in the system default operator record becomes part of all operator records at the time you create the operator record and whenever you query the operator table. The operator table displays the field values in the system default record unless there is another entry that overrides it. For example, if the system default record defines the Default Company field with the value DEFAULT, then all operator records without a defined value for this field display the value DEFAULT.

To create a system default operator record, follow these steps:

1. Create a contact record for the system default operator record.
    1. Type contacts in the command line, and press Enter.
    2. In the Contact Name and Full Name fields, enter *SYSDEFAULTS.
    3. Click **Add**.
2. Click **System Administration** > **Ongoing Maintenance** > **Operators**.
3. In the Login Name and Full Name fields, type *SYSDEFAULTS.
4. In the Contact ID field, enter *SYSDEFAULTS and click **Fill**.
5. Click **Add**.
6. Provide additional information that you want to use as default values for creating new operator records. For example, Time Zone, Language, Default Company, and Security Roles.
7. Click **Save**.

Related topics

Application Setup
Operator passwords
Operator records
Operator templates
Add an operator record
Add an operator record with the User Quick Add utility
Add an operator to a security group
Define a startup menu for an operator
Define the distinguished name an operator uses to log in to an LDAP directory service
Delete an operator record

Enable an operator to see the command line
Set the maximum file attachment size for an operator
Set the time zone for an individual operator
Update an operator record
View an existing operator record

6.2.3.5. Define a startup menu for an operator

Applies to User roles: System Administrator

The default home page for a Service Manager user is the Dashboard home page or To Do list specified for the operator or user role. You can define an old style startup menu or any other RAD application so that it is the initial application that Service Manager displays when an operator logs in. You define an initial application in the user's operator record.

Note

Note Specifying a Dashboard home page in a user's operator record overrides the To Do list and menus for that operator when they log in using the web client, even if the flag for menus has been set.

To define a startup menu for an operator, follow these steps:

1. Click **System Administration** > **Ongoing Maintenance** > **Operators**.
2. Type or select optional search criteria.
3. Click **Search**.
4. Select the operator record to change.
5. Click the **Startup** tab.
6. In the **RAD Name** field, type the name of the startup menu or initial application to run when the operator logs on.
7. Type any parameter names and parameter values required by your startup form. For example, specify the following values for old style menus.
    - Parameter Names: boolean1
    - Parameter Values: true
8. Click **Save**.

Related topics

Application Setup
Operator passwords
Operator records
Operator templates
Menus
To Do list
Add an operator record
Add an operator record with the User Quick Add utility
Add an operator to a security group
Create a system default operator record
Define the distinguished name an operator uses to log in to an LDAP directory service
Delete an operator record
Enable an operator to see the command line
Set the maximum file attachment size for an operator
Set the time zone for an individual operator
Update an operator record
View an existing operator record

6.2.3.6. Define the distinguished name an operator uses to log into an LDAP directory service

Applies to User roles: System Administrator

To define the distinguished name an operator uses to log in to an LDAP directory service, follow these steps:

1. Click **System Administration** > **Ongoing Maintenance** > **Operators**.

2. Click **Search** to view a list of operators.
3. Select the operator to update.
4. Click the **Security** tab.
5. In the **LDAP User DN** field, type the distinguished name you want the operator to use to bind to the LDAP directory service.
6. Click **Save**.
7. Click **OK**.

Related topics

Application Setup
Operator passwords
Operator records
Operator templates
Add an operator record
Add an operator record with the User Quick Add utility
Add an operator to a security group
Create a system default operator record
Define a startup menu for an operator
Delete an operator record
Enable an operator to see the command line
Set the maximum file attachment size for an operator
Set the time zone for an individual operator
Update an operator record
View an existing operator record

6.2.3.7. Delete an operator record

Applies to User roles: System Administrator

To delete an operator record, follow these steps:

1. Click **System Administration** > **Ongoing Maintenance** > **Operators**.
2. Use search or advanced search to find one or more records.
3. Select the operator record that you want to delete.
4. Click **Delete**.
5. Click **Yes** to confirm the deletion.
6. When you are prompted whether to remove a certain associated record, for example, an associated contact record, click **Yes** to delete the associated record or click **No** to keep the record.

Related topics

Application Setup
Operator passwords
Operator records
Operator templates
Add an operator record
Add an operator record with the User Quick Add utility
Add an operator to a security group
Create a system default operator record
Define a startup menu for an operator
Set the dashboard home page for a user role
Define the distinguished name an operator uses to log in to an LDAP directory service
Enable an operator to see the command line
Set the maximum file attachment size for an operator
Set the time zone for an individual operator
Update an operator record
View an existing operator record

6.2.3.8. Enable an operator to see the command line

Applies to User roles: System Administrator

To enable an operator to see the command line, follow these steps:

1. Click **System Administration** > **Ongoing Maintenance** > **Operators**.
2. Type or select optional search criteria.
3. Click **Search**.
4. Select an operator record from the record list.
5. Click the **Startup** tab.
6. Select **Activate Command Line on Startup**.
7. Click **Save**.

Related topics

Application Setup
Operator passwords
Operator records
Operator templates
Add an operator record
Add an operator record with the User Quick Add utility
Add an operator to a security group
Set the dashboard home page for a user role
Create a system default operator record
Define a startup menu for an operator
Define the distinguished name an operator uses to log in to an LDAP directory service
Delete an operator record
Set the maximum file attachment size for an operator
Set the time zone for an individual operator
Update an operator record
View an existing operator record

6.2.3.9. Set database access for an operator

Applies to User roles: System Administrator

Database access is a feature that gives you the ability to limit or grant access to database records, such as contacts, company, and regions. You can add or modify the existing out-of-box database access settings per user role or operator.

To set database access for an operator, follow these steps:

1. Click **System Administration** > **Ongoing Maintenance** > **Operators**.
2. Type or select optional search criteria, and then click **Search**.
3. Select an operator record from the record list.
4. Select the **Data Access** tab in the **General** tab.
5. To add a new Data Access record, follow these steps:
    1. Click **Add new Data Access Record**.
       The **Database Manager Data Access** form opens.
    2. Click **Fill** in the **Database Table Name** field to select a table.
    3. The operator name you initially selected is in the **Operator Name** field. If you want to add access to a different operator name than the one you selected, clear the field and click **Fill** to a select a different operator name.
    4. To set the database access, select one of the following:
        - **Allow DB access**: The operator specified is granted access to the table specified.
        - **Prohibit DB access**: The operator specified is denied access to the table specified.
    5. When access has been granted to a table, click **Fill** in the **View Format** field to select the table view.
    6. Click **Add**.
       The new database access record is added.

6. To modify an existing database access record, follow these steps:
    1. Select the **Data Access** tab in the **General** tab.
    2. Double-click the Table Name of the existing database access record you want to modify.
       The **Database Manager Data Access** form opens.
    3. Make the necessary edits.
    4. Click **Save**.
       Your changes are saved.
7. Click **OK**.

Related topics

Operator passwords
Operator records
Operator templates
Add an operator record
Add an operator record with the User Quick Add utility
Add an operator to a security group
Create a system default operator record
Define a startup menu for an operator
Define the distinguished name an operator uses to log in to an LDAP directory service
Delete an operator record
Set the maximum file attachment size for an operator
Set the time zone for an individual operator
Update an operator record
View an existing operator record

6.2.3.10. Set the first day of week for an operator

Applies to User roles: System Administrator

To set the first day of week for an operator, follow these steps:

1. Click **System Administration** > **Ongoing Maintenance** > **Operators**.
2. Type or select optional search criteria.
3. Click **Search**.
4. Select the operator record that you want to change.
5. Click the **Login Profiles** tab.
6. In the **First Day of Week** field, select a day from the drop-down list.
7. Click **Save**.

This setting applies to the calendar and the date picker.

If you don't configure this field, the first day of the week for the operator is as follows:

- In the date picker, the first day of the week is the day that's configured by the **startDayOfWeek** parameter. For more information about the **startDayOfWeek** parameter, see Web parameter: startDayOfWeek.
- In the calendar, the first day of the week is the day that's configured in calendar settings. For more information about the calendar settings, see Configure calendar settings

If you configure this field, the first day of the week in the date picker and the calendar for the operator will be the day that's specified in this field.

Related topics

Setting file attachment limits
Operator passwords
Operator records
Operator templates
Set the maximum file attachment size for the entire company
Set the total file attachment size for the entire company

Set the total file attachment size for an operator
Add an operator record
Add an operator record with the User Quick Add utility
Add an operator to a security group
Create a system default operator record
Define a startup menu for an operator
Define the distinguished name an operator uses to log in to an LDAP directory service
Delete an operator record
Enable an operator to see the command line
Set the time zone for an individual operator
Update an operator record
View an existing operator record

6.2.3.11. Set the maximum file attachment size for an operator

Applies to User roles: System Administrator

To set the maximum file size of a single attachment for an operator, follow these steps:

1. Click **System Administration** > **Ongoing Maintenance** > **Operators**.
2. Type or select optional search criteria.
3. Click **Search**.
4. Select the operator record that you want to change.
5. Click the **Login Profiles** tab.
6. In the **Max Attachment Size** field, type the number of bytes for the maximum file attachment size.
7. Click **Save**.

You can define a file attachment maximum size limit for a particular operator from the operator record. The operator file attachment size limit supersedes the maximum file size set in the system wide company record but is itself subordinate to size limits defined for an attachment container in the Form Designer.

Related topics

Setting file attachment limits
Operator passwords
Operator records
Operator templates
Set the maximum file attachment size for the entire company
Set the total file attachment size for the entire company
Set the total file attachment size for an operator
Add an operator record
Add an operator record with the User Quick Add utility
Add an operator to a security group
Create a system default operator record
Define a startup menu for an operator
Define the distinguished name an operator uses to log in to an LDAP directory service
Delete an operator record
Enable an operator to see the command line
Set the time zone for an individual operator
Update an operator record
View an existing operator record

6.2.3.12. Set the total file attachment size for an operator

Applies to User roles: System Administrator

To set the total attachment size in any attachment container for an operator, follow these steps:

1. Click **System Administration** > **Ongoing Maintenance** > **Operators**.
2. Type or select optional search criteria.

3. Click **Search**.
4. Select the operator record that you want to change.
5. Click the **Login Profiles** tab.
6. In the **Total Attachment Size** field, type the number of bytes for the total file attachment size.
7. Click **Save**.

You can define a total attachment size limit for a particular operator from the operator record. The total attachment size defined in the operator record supersedes the total attachment size set in the system wide company record but is itself subordinate to size limits defined for a specific attachment container in the Form Designer.

Related topics

Setting file attachment limits
Operator passwords
Operator records
Operator templates
Set the total file attachment size for the entire company
Set the maximum file attachment size for the entire company
Set the maximum file attachment size for an operator
Add an operator record
Add an operator record with the User Quick Add utility
Add an operator to a security group
Create a system default operator record
Define a startup menu for an operator
Define the distinguished name an operator uses to log in to an LDAP directory service
Delete an operator record
Enable an operator to see the command line
Set the time zone for an individual operator
Update an operator record
View an existing operator record

6.2.3.13. Set the time zone for an individual operator

Applies to User roles: System Administrator

To set the time zone for an individual operator, follow these steps:

1. Click **System Administration** > **Ongoing Maintenance** > **Operators**.
2. Type or select optional search criteria.
3. Click **Search**.
4. Select the operator record to change from the record list.
5. In the **Time Zone** field, select the new time zone for the operator.
6. Click **Save**.

Related topics

Operator passwords
Operator records
Operator templates
Add an operator record
Add an operator record with the User Quick Add utility
Add an operator to a security group
Create a system default operator record
Define a startup menu for an operator
Define the distinguished name an operator uses to log in to an LDAP directory service
Delete an operator record
Enable an operator to see the command line
Set the maximum file attachment size for an operator

Update an operator record
View an existing operator record

6.2.3.14. Update an operator record

Applies to User roles: System Administrator

To update an operator record, follow these steps:

1. Click **System Administration** > **Ongoing Maintenance** > **Operators**.
2. Type or select optional search criteria.
3. Click **Search**.
4. Select the operator that you want to update from the record list.
5. Type or select the new operator details. For example, change the **Application Profile** on the **General** tab, or add new capability words to the **Execute Capabilities** table on the **Startup** tab.
6. Click **Save**.
7. Click **OK**.

Related topics

Operator passwords
Operator records
Operator templates
Add an operator record
Add an operator record with the User Quick Add utility
Add an operator to a security group
Create a system default operator record
Define a startup menu for an operator
Define the distinguished name an operator uses to log in to an LDAP directory service
Delete an operator record
Enable an operator to see the command line
Set the maximum file attachment size for an operator
Set the time zone for an individual operator
View an existing operator record

6.2.3.15. View an existing operator record

Applies to User roles: System Administrator

To view an existing operator record, follow these steps:

1. Click **System Administration** > **Ongoing Maintenance** > **Operators**.
2. Type or select optional search criteria.
3. Click **Search**.
4. Select an operator from the record list to view the operator record.

Related topics

Operator passwords
Operator records
Operator templates
Add an operator record
Add an operator record with the User Quick Add utility
Add an operator to a security group
Create a system default operator record
Define a startup menu for an operator
Define the distinguished name an operator uses to log in to an LDAP directory service
Delete an operator record
Enable an operator to see the command line
Set the maximum file attachment size for an operator

Set the time zone for an individual operator
Update an operator record

6.2.3.16. Use Mass Update with operator records

Applies to User roles: System Administrator

> **Note**
>
> **Note:** If you exit the **Mass Update** template form or **Complex Update** form without clicking **Execute**, all of your changes are cleared on the respective form.

To update several records at the same time, use one of the following methods.

Mass Update template

1. Click **System Administration** > **Ongoing Maintenance** > **Operators**.
2. Type or select optional search criteria, and then click **Search**.
3. Select the records in the record list that you want to update.
4. Click **Mass Update**.
5. Type the value for the field in the text box or use the Find feature to display a list of potential values for the field.

   > **Note**
   >
   > Note The value(s) you enter in the form are propagated to all of the records you selected.

6. Click **Next**.
7. Continue updating fields and click **Execute** when you complete your updates.

Simple Update

1. Click **System Administration** > **Ongoing Maintenance** > **Operators**.
2. Type or select optional search criteria, and then click **Search**.
3. Select the records in the list that you want to update.
4. Click **Mass Update**.
5. Click **Simple Update**.
6. Type new values into the applicable fields on the form.
7. Click **Execute**.

Complex Update

1. Click **System Administration** > **Ongoing Maintenance** > **Operators**.
2. Type or select optional search criteria, and then click **Search**.
3. Select the records in the list that you want to update.
4. Click **Mass Update**.
5. Click **Complex Update**.
6. Type the RAD code in the **Complex Update** form.
7. Click **Execute**.

Related topics

Operator records
Update multiple records with Mass Update

6.2.3.17. Synchronization between contact and operator records

The contact and operator records are synchronized so that if you change the Service Manager ID field in the contacts record, it triggers the same change in the Contact ID field in the operator record. The reverse is also true. In order to make the synchronization possible, every user in the system must have both a contact record and an operator record.

Several redundant fields in the operator table were removed in order to leverage existing fields in the contacts table. See the Service Manager Upgrade section for a list of the fields removed from the operator table.

Related topics

Operator records
Create an operator record from a contact record
Create an operator record from an operator template
Mass create contact records from operator records
Mass create operator records from contact records

6.2.3.17.1. Mass create contact records from operator records

Applies to User roles: System Administrator

To mass create contact records from operator records, follow these steps:

1. Click **System Administration** > **Ongoing Maintenance** > **Operators**.
2. Type or select optional search criteria.
3. Click **Search**.
4. In the record list, open the **More Actions** menu.
5. Click **Mass Create Contact**.
6. Follow the prompts to create the contacts.

Note

Note You can run the mass create function to create contact records only once. If you run the function again, it stops when it encounters the first operator record for which a contact already exists.

Related topics

Synchronization between contact and operator records
More Actions menu
Mass create operator records from contact records
Create an operator record from a contact record
Create a contact record from an operator record

6.2.3.17.2. Mass create operator records from contact records

Applies to User roles: System Administrator

To mass create operator records from contact records, follow these steps:

1. Click **Support** > **Contacts**.
2. Type or select optional search criteria to find the contacts for which you want to create operator records.
3. Click **Search**.
4. In the record list, open the **More Actions** menu.
5. Click **Mass Create Operators**.
6. Follow the prompts to create the operators.

Note

Note You can run the mass create function to create operator records only once. If you run the function again, it stops when it encounters the first contact record for which an operator record already exists.

Related topics

Synchronization between contact and operator records
More Actions menu
Mass create contact records from operator records

Create an operator record from a contact record
Create a contact record from an operator record

6.2.3.17.3. Create an operator record from a contact record

Applies to User Roles: System Administrator

The contact and operator records are synchronized so that if you change the Service Manager ID field in the contacts record, it triggers the same change in the Contact ID field in the operator record. The reverse is also true. In order to make the synchronization possible, every user in the system must have both a contact record and an operator record.

To create an operator record from a contact record, follow these steps:

1. Click **System Administration** > **Base System Configuration** > **Contacts**.
2. Type or select optional search criteria to find the contact for which you want to create an operator record.
3. Click **Search**.
4. If the search returned a list of records, select the contact from the list.
5. In the contact record, click **More** or the More Actions icon.
6. Click **Create Operator**.
7. Follow the prompts to create the operator record.

Related topics

Synchronization between contact and operator records
More Actions menu
Mass create contact records from operator records
Mass create operator records from contact records
Create a contact record from an operator record

6.2.3.17.4. Create an operator record from a operator template

Applies to User roles: System Administrator

Caution
Caution
Any changes that you make to an operator template automatically propagate to operator records based on that template. These automatic updates can consume substantial system resources.

To create an operator record from an operator template, follow these steps:

1. Click **System Administration** > **Ongoing Maintenance** > **Operators**.
2. In the **Login Name** field, type the new operator name.

Note
Note
- Operator record login names are case-sensitive.
- We don't recommend including any colons (:) in the login name.

3. Click the **Security** tab.
4. In the **Template** field, type the login name of the operator record to be used as an operator template.
5. Click **Add**.

Note

Note

If an operator record has group1 in its security groups array when its template is NULL and group1 is in the security groups array of template_1 but not in that of template_2, when you assign template_1 to the operator record, the operator record has group1 in its security groups array; however, if you switch its template to template_2, group1 is lost in its security group array. To work around this issue, clear the template field, remove template_1L, save it, change the template to template_2, and then save.

Related topics

Application Setup
Adding users
Checklist: Adding a new user
Controlling user access and security
Creating operator records
Defining named users
Operator passwords
Operator records
Operator templates
Create an operator template
Define the operator template applied to LDAP users

6.2.3.17.5. Create an operator template

Applies to User roles: System Administrator

System Administrators can design operator templates for creating operator records for users that share common information and settings. For example, you can design one operator template for managers and another template for first-level Service Desk Agents. An operator template is an operator record containing all the common information that applies to operator records created from the template.

To create an operator template, follow these steps:

Note

Note You can specify as many details in the new operator record template, as necessary.

1. Click **System Administration** > **Ongoing Maintenance** > **Operators**.
2. Enter the search criteria to find the operator record for which you want to create an operator template.
3. Fill in the necessary information, including application profiles, and proper permissions and settings.
4. In the Security tab, select the option **Template Operator**.
5. Click **Add**.

Related topics

Application Setup
Adding users
Checklist: Adding a new user
Controlling user access and security
Creating operator records
Defining named users
Operator passwords
Operator records
Operator templates
Create an operator record from an operator template
Define the operator template applied to LDAP users

6.2.3.17.6. Define the operator template applied to LDAP users

Applies to User roles: System Administrator

To define the operator template applied to LDAP users, follow these steps:

1. Click **System Administration** > **Base System Configuration** > **Miscellaneous** > **System Information Record**.
2. In the **Operator Template** field, specify the operator record that you want to use as the company default for LDAP users.
3. Click **Save**.

Related topics

Application Setup
Adding users
Checklist: Adding a new user
Controlling user access and security
Creating operator records
Defining named users
Operator passwords
Operator records
Operator templates
Create an operator record from an operator template
Create an operator template

6.2.3.17.7. Create a contact record from an operator record

Applies to User roles: System Administrator

The contact and operator records are synchronized so that if you change the Service Manager ID field in the contacts record, it triggers the same change in the Contact ID field in the operator record. The reverse is also true. In order to make the synchronization possible, every user in the system must have both a contact record and an operator record.

To create a contact record from an operator record, follow these steps:

1. Click **System Administration** > **Ongoing Maintenance** > **Operators**.
2. Enter and select the search criteria to find the operator for which you want to create a contact record.
3. Click **Search**.
   A list of operator records is opened.
4. Select an operator from the list.
5. In the selected operator record, replace the **Contact ID** with the new contact ID.
6. Click **More** or the **More Actions** icon, and then **Create Contact** option.
7. Click **Create Contact**.
8. Follow the prompts to finish creating the contact.
9. Click **Save**.

Related topics

Synchronization between contact and operator records
More Actions menu
Mass create contact records from operator records
Mass create operator records from contact records
Create an operator record from a contact record

6.2.4. Operator templates

System Administrators can design operator templates for creating operator records for users that share common information and settings. For example, you can design one operator template for managers and another template for first-level Service Desk Agents. An operator template is an operator record containing all the common information that applies to operator records created from the template. Typically, this includes information such as:

- Application profile
- Default Company

- Date Information
- Time Limits
- User Session Information
- LDAP Information
- Security Groups
- Security Roles

To distinguish the operator template from a user's operator record, select the **Template Operator** option in the template. Users aren't attached to templates and, therefore, can't log in to Service Manager using an operator template.

When you have created the new operator template with all the desired settings, you can then create the operator records you need, based on that new template. When you create operator records, fill in the **Template** field with the name of the template from the operator template, and then all the information in the template will be applied to the new operator records.

Any changes you make to an operator template automatically apply to all operator records you created from the template. If you later decide to change a setting in the template, you modify the template, and then the setting will automatically be applied to all the operator records that use the template.

Related topics

Adding users
Checklist: Adding a new user
Controlling user access and security
Creating operator records
Defining named users
Operator passwords
Operator records
Create an operator record from an operator template
Create an operator template
Define the operator template applied to LDAP users

6.3. ProcessDesigner

Process Designer provides a graphical interface to develop workflows that you can use to control the flow of a single record throughout its lifecycle within Service Manager. The Process Designer enables an implementer to graphically create or update a workflow without being an expert in RAD programming.

- Define a workflow process in a graphical editor.
- Create business rules in a user-friendly interface to drive the workflow logic.
- Provide a platform to model complex processes that require branching and looping.
- Lower Total Cost of Ownership of Service Manager by enabling an intuitive and simplified configuration.
- Simplify future upgrades by enabling standardization and clarity of business logic.

Process Designer includes the following components:

- Workflow designer: Design and update a workflow using a graphical user interface.
- Rules editor: Create rules to enforce business logic in workflows and forms.
- Security model: Provide a common role-based security model.

A workflow is a collection of phases with transitions from one phase to another. Each phase represents the state of the workflow linked to a form for data capture.

Transitions

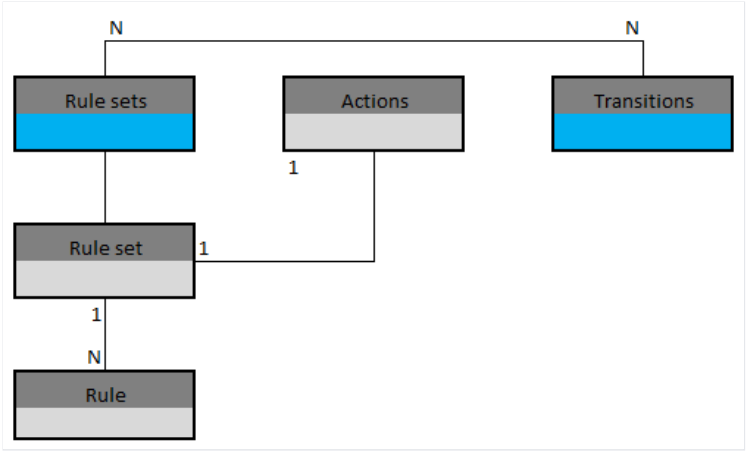
Transitions can connect phases in any manner to create a workflow with many branches and loops back to previous phases. Transitions may be manual, automatic, or default. A manual transition requires the workflow user to make a manual decision to move to the next workflow phase. An automatic transition moves the workflow to another phase based on data in the workflow record. A default transition is a special case that moves the workflow automatically, when no other automatic transition condition applies.

Rule sets

The rules editor enables administrators, such as business process owners, implementers, and developers, to add or remove the out-of-box rules and actions of a workflow to match particular business requirements. You can group individual rules into a rule set to allow their re-use within the process. You can configure a rule set to execute on events, such as saving or updating a business table record. Rules can also be set up to run on a workflow transition.

Actions

You can also configure rules as actions, which allows operators to run rules on demand rather than wait for workflow processing. When an operator initiates an action (such as pressing a button in the system tray), the system runs the associated rule sets for that phase of the workflow. The following diagram shows the relationship between rules, rule sets, actions, and transitions:



Condition Editor

With the Condition Editor, you can apply conditions to rules. When a rule condition is met, the rule executes. You can also edit conditions for a rule.

Process Designer security model

The Process Designer security model provides a consistent method of assigning permissions to users across all areas of the system. You can use security model to configure out-of-box rights for a specified role within an area. It also provides standardized methods to manage user rights.

Process Designer modules

Process Designer features are implemented for Knowledge Management, , Change Management, Request Fulfillment, Help Desk (which is comprised of Service Desk, Incident Management, and Problem Management) and Service Level Management. To access the Process Designer features from the system navigator, click **Tailoring** > **Process Designer**. A typical Process Designer flow broadly comprises the following major tasks:

- Create rule sets with rules using out-of-the-box rule types to enforce business logic on traditional Service Manager forms.
- Create a workflow with phases and transitions to build a process flow.
- From workflow phases and transitions, associate forms, rule sets, approvals, and alerts to implement your business process.

6.3.1. Create a rule set

Applies to User roles: System Administrator, Implementer

A rule set contains a list of rules that you may run against a record. Rules implement business logic to drive a workflow or a process. Rules can help perform calculations, validate fields based on data or rule sets, set required fields, and more. A rule set uses role-based security.

You can re-use rule sets in many processes when you require the same rules in many places. rule sets simplify the effort of implementing business logic. You can apply rule sets based on conditions, or configure them to run during an action. rule sets are most often associated with specific phases of a workflow.

To create a rule set, follow these steps:

1. Click **Tailoring** > **Process Designer** > **Rule Sets** > **New**.
2. Type a unique ID for the rule set. For example, kmdocument_draft.

> **Note**
> Note The name must be unique within the rule sets records.

3. Select **Available as action** if you want to offer this rule set as an action within a workflow phase. For example, Service Manager interaction with an external system.
4. Type the display name for the rule set, for example, Saving a draft.
5. Select the table(s) from the Table name list that you want the rule sets to apply to, for example, select kmdocument.

> **Note**
> Note If the list is blank, the rule set can be applied to all tables.

6. Click **Save** to add the rule set.

> **Note**
> Note If you want to edit a **Proprietary** rule set, create a copy of this **Proprietary** rule set. To add Rules to a rule set, you must first save the rule set.

Related topics

Adding a rule
Clone an existing Rule Set

6.3.2. Clone an existing rule set

Applies to User roles: System Administrator, Implementer

You can create an editable copy of an existing Rule Set by using the cloning option. Cloning a read-only Rule Set marked as **Proprietary** creates an editable copy of the Rule Set.

To clone an existing Rule Set, follow these steps:

1. In the System Navigator, click **Tailoring** > **Process Designer** > **Rule Sets**.
2. Click **Search**.
3. Select the Rule Set you want to clone.
4. Click **More** > **Clone**. A copy of the Rule Sets with both the ID and Name prefixed <CLONE> is displayed.
5. Rename both the ID and Name prefixed <CLONE> fields by typing something unique.
6. Click **Save**.

Related topics

Process Designer security model
Create a rule set

6.3.3. Adding a rule

A rule defines a singular piece of functionality, such as making a field mandatory. A rule can perform calculations, validate fields based on data or lists, set required fields, and more. Rules can also specify conditions that determine when the rule should execute or who can execute rules run in the order they appear in the rule set. The order can be changed using the **Move Up** and **Move Down** buttons. If there is an error executing a rule, rule processing will stop. It is important to order the rules such that validation rules come first in the list, and then any rules that will perform actions. For example, to validate data and send an HTML email, place rules to validate data first and then the rule to send an HTML email. If the rules were reversed, the HTML email will be sent with invalid data or an invalid workflow state if the rules are part of a transition.

All rule sets in a phase, except the "On Enter" and "On Exit" rule sets, run before a record proceeds from one phase to another. The "On Exit" rule sets run before a transition occurs. "On Enter" rule sets run when a record moves to that phase.

The "On Exit" rules are the best place for notifications when a record leaves a phase especially when multiple transitions are used or when integration notifications are triggered. Transitions are used to add validations that are required before leaving to the current phase. For example, in a change phase, the start date and end date are entered before proceeding to the Approval phase and no input is required if you are abandoning the change. In this case, if the "On Exit" rules are run before the transition rules, you will be sending out notifications but the actual transition would not have happened. Therefore, the order of execution of rule is Transition > On Exit > On Enter.

For example, rules may run in the following situations:

- When a record enters or exits a specific phase of the workflow
- When a record is created, updated, or deleted in a certain phase of the workflow because a workflow action is invoked
- When a workflow transition occurs
- When a workflow action is invoked

The following rule types are available in out-of-box Service Manager deployments.

Rule type	Description
Assignment	Automatically distribute records (such as tasks or records) to the groups and individuals who are most able to process them
Call a process	Call a Service Manager process record
Case Exchange	Trigger certain activities for the Case Exchange integration
Clear Fields	Clear the specified field and related fields
JavaScript Validation	Use JavaScript to perform actions and validations
Launch a Script	Launch a Service Manager script
Launch a URL	Call a URL to launch a web page
Mandatory Fields	Set fields as mandatory
Mandatory Variables	Set variables as mandatory
Popup Message Box	Create and configure popup message boxes that appear to end users
Run Action	Run actions on records that have a specified relationship to the record that triggers the rule
Run JavaScript	Use JavaScript to perform actions and validations
Run Scheduled Action	Run actions on the current record after a specified length of time has passed
Run a Wizard	Run a Service Manager wizard
Send HTML Email	Send an HTML Email to users or a group
Send Notifications	Send Service Manager notifications
Send Teams Message	Send a Microsoft Teams message to users or a group
Set Field	Set a field value using JavaScript
Set Field from Number	Set field based on a number record
Start or Stop Clock	Start and stop a Service Manager clock
Validate against List	Validate a field against a list (global or defined)
Validate against Table	Validate a field against a field in another table
Validate Date	Validate a date against a date range
Validate Text/Number	Validate a field against a range of text or a number in another field of the same table

Note

Note

- Out-of-box rule sets are labeled as **Proprietary** and you can edit user-defined rule sets only.
- Many rules have a default description that contains tokens. The tokens are replaced with values when you define the rule. For example, the default description of the Set Mandatory Fields rule is **<fields> are Mandatory**. When you define the rule, the **<fields>** token is replaced by the fields that you select in the rule. You can edit the default description.
- If you don't specify a condition, rules are always triggered to perform the action specified in the rule.

Related topics

Process Designer
Create a rule set

6.3.3.1. Add a Launch a URL rule

Applies to User roles: System Administrator, Implementer

This rule enables you to launch a remote web address by using its uniform resource locator (URL). This rule helps you to integrate Service Manager with other products.

To add a Launch a URL rule, follow these steps:

1. In the System Navigator, click **Tailoring** > **Process Designer** > **Rule Sets**.
2. Click **Search** to open the **Rule Set** form, and then select the rule set to which you want to add the rule.

   Note

   Note Out-of-box rule sets are labeled as **Proprietary** and you can edit user-defined rule sets only.

3. Click **Add Rule** to open the **Select Rule Type** page, and then click **Launch a URL**.
4. Type a description in the **Rule Description** field.
5. Click **Edit** to add conditions to the rule.

   Note

   Note If you don't specify a condition, the value defaults to **Always**.

6. In the **URL** field, type the URL address of the remote resource that you want to launch, and then click **OK**.

Related topics

Process Designer
Process Designer security model
Create a rule set

6.3.3.2. Add a Run a Macro rule

Applies to User roles: System Administrator, Implementer

This rule enables you to run a Service Manager macro that performs a distinct action. Service Manager macros are specific actions driven by predefined conditions. For example, if you want to configure a macro to send an email, select the Service Manager macro type **Mail 1 Person**. You can configure this rule to send an email to an intended recipient when an incident record moves from one phase to another.

To add a Run a Macro rule, follow these steps:

1. In the System Navigator, click **Tailoring** > **Process Designer** > **Rule Sets**.
2. Click **Search** to open the **Rule Set** form, and then select the rule set to which you want to add the rule.

Note

Note Out-of-box rule sets are labeled as **Proprietary** and you can edit user-defined rule sets only.

3. Click **Add Rule** to open the **Select Rule Type** page, and then click **Run a Macro**.
4. Type the description in the **Rule Description** field.
5. Click **Edit** to add a condition.

Note

Note If you don't specify a condition, it will default to **Always**.

6. Select a macro name from the drop-down list.

Note

Note The drop-down list has macros related to the current table of the rule set.

7. Click **OK** to add the new rule to the rule set.

Related topics

Process Designer
Create a rule set

6.3.3.3. Add a Call a Process rule

Applies to User roles: System Administrator, Implementer

This rule enables users to call a Service Manager process. You have to specify the process name and the conditions that call the process.

Caution

Caution

You can't call a Service Manager process that expects an input, because there is no provision to pass the input through the rule. Calling a process that requires an input causes the rule functionality to fail and may generate an unrecoverable error message.

To add a Call a Process rule type, follow these steps:

1. In the System Navigator, click **Tailoring** > **Process Designer** > **Rule Sets**.
2. Click **Search** to open the **Rule Set** form, and then select the rule set to which you want to add the rule.

Note

Note
Out-of-box rule sets are labeled as **Proprietary** and you can edit user-defined rule sets only.

3. Click **Add Rule** to open the **Select Rule Type** page, and then click **Call a Process**.
4. In the **Rule Description** field, type a description of your new rule.
5. Click **Edit** to add conditions to the rule.

Note

Note If you don't specify a condition, the value defaults to **Always**.

6. Click **Fill Field Process**, and then select the process that you want to call.
7. Click **OK** to add the new rule to the rule set.

Related topics

Process Designer
Process Designer security model
Adding a rule
Create a rule set

6.3.3.4. Add a Case Exchange rule

Applies to User roles: System Administrator, Implementer

This rule enables you to trigger certain activities for the Case Exchange integration.

To add a Case Exchange rule, follow these steps:

1. In the System Navigator, click **Tailoring** > **Process Designer** > **Rule Sets**.
2. Click **Search** to open the **Rule Set** form, and then select the rule set to which you want to add the rule.

   Note
   Note
   Out-of-box rule sets are labeled as **Proprietary** and you can edit user-defined rule sets only.

3. Click **Add Rule** to open the **Select Rule Type** page, and then click **Case Exchange**.
4. In the **Rule Description** field, type a description of your new rule.
5. Click **Edit** to add conditions to the rule.

   Note
   Note
   If you don't specify a condition, the value defaults to **Always**.

6. In the rule from the **Instance Name** drop-down list, select the Case Exchange integration instance that you want to apply.
7. In the **Event** drop-down list, select an event from, and then select the fields you want to add.
8. Click **Finish** to add the new rule to the rule set.

Related topics

Using the Condition Editor
Launch the Condition Editor from a rule
Launch the Condition Editor from a workflow

6.3.3.5. Add a Run a Wizard rule

Applies to User roles: System Administrator, Implementer

This rule enables you to run a Service Manager wizard. The form helps you to specify the wizard to run when the rule executes.

To add a Run a Wizard rule, follow these steps:

1. In the System Navigator, click **Tailoring** > **Process Designer** > **Rule Sets**.
2. Click **Search** to open the **Rule Set** form, and then select the rule set to which you want to add the rule.

   Note
   Note Out-of-box rule sets are labeled as **Proprietary** and you can edit user-defined rule sets only.

3. Click **Add Rule** to open the **Select Rule Type** page, and then click **Run a Wizard**.
4. In the **Rule Description** field, type a description of your new rule.
5. Click **Edit** to add conditions to the rule.

> **Note**
> Note If you don't specify a condition, the value defaults to **Always**.

6. Click **Fill Field Wizard to run**, and then select the wizard to run when the rule is executed.
7. Click **OK** to add the new rule to the rule set.

Related topics

Process Designer
Process Designer security model
Adding a rule
Create a rule set

6.3.3.6. Add a Clear Fields rule

Applies to User roles: System Administrator

From a rule set, you can clear the value of specified fields in a record. Optionally, the rule can also clear the value in fields related to the specified fields.

To add a Clear Fields rule, follow these steps:

1. In the System Navigator, click **Tailoring** > **Process Designer** > **Rule Sets**.
2. Click **Search** to open the **Rule Set** form, and then select the rule set to which you want to add the rule.

> **Note**
> Note
> Out-of-box rule sets are labeled as **Proprietary** and you can edit user-defined rule sets only.

3. Click **Add Rule** to open the **Select Rule Type** page, and then click **Clear Fields**.
4. In the **Rule Description** field, type a description of your new rule.
5. Click **Edit** to add conditions to the rule.

> **Note**
> Note
> If you don't specify a condition, the value defaults to **Always**.

6. In the **Field Name** column, select the fields to clear when the rule is executed.
7. In the **Clear Related Fields** column, select whether to clear the value of related fields of the specified fields.
8. Click **OK** to add the new rule to the rule set.

Related topics

Process Designer
Process Designer security model
Adding a rule
Create a rule set

6.3.3.7. Add a JavaScript Validation rule

Applies to User roles: System Administrator, Implementer

This rule allows you to validate any JavaScript code a user enters.

To add a JavaScript Validation rule, follow these steps:

1. In the System Navigator, click **Tailoring** > **Process Designer** > **Rule Sets**.
2. Click **Search** to open the **Rule Set** form, and then select the rule set to which you want to add the rule.

> **Note**
> Note Out-of-the-box rule sets are labeled as **Proprietary** and you can edit user-defined rule sets only.

3. Click **Add Rule** to open the **Select Rule Type** page, and then click **JavaScript Validation**.
4. In the **Rule Description** field, type a description of your new rule.
5. Click **Edit** to add conditions to the rule.
   • If you don't specify a condition, it will default to **Always**.
   • You may reference the current record as **record** in the JavaScript. For example, if the field in a form is a category, Service Manager may referred to it as follows: `record.category=="xyz"`
   • You may reference the original copy of the record (before the user make any changes) as **oldRecord** in the JavaScript (that's `oldRecord.category`).
   • You may set the **returnCode** variable in the JavaScript to specify whether this rule was successful or whether the system should treat this rule as a failed validation. A return code of 0 (zero) means successful (it's the default), any other code is unsuccessful. The return code must be a numeric value.
   • You may set the **message** variable in the JavaScript to specify a message that displays to the user if the rule is unsuccessful (returns a code that isn't zero).
   • You may set the **cursorPosition** variable in the JavaScript to specify a field on the current form where the cursor locates if the rule is unsuccessful.
6. Enter the JavaScript in the text box.
7. Click **OK** to add the new rule to the rule set.

Related topics

Process Designer
Process Designer security model
Create a rule set

6.3.3.8. Add a Run JavaScript rule

Applies to User roles: System Administrator, Implementer

This rule enables you to perform actions and validations by using JavaScript.

To add a run JavaScript rule:

1. Click **Tailoring** > **Process Designer** in the System Navigator.
2. Click **Rule Sets** > **Search** to open the Rule Set page.
3. Select the Rule Set to which you want to add the rule.
4. Click **Add Rule** to open the Select Rule Type page.
5. Click **Run JavaScript**.
6. Type the Rule Description.
7. Click **Edit** to add a condition.

> **Note**
> Note If you don't specify a condition, it will default to **Always**.

8. Type the JavaScript to run in the text box.
9. Click **OK** to add the rule.

Related topics

Process Designer
Create a rule set

6.3.3.9. Add a Set Mandatory Fields rule

Applies to User roles: System Administrator, Implementer

This rule enables you to make one or more fields mandatory and to specify a default value for each field, which will be set if the mandatory fields are empty.

To add a JavaScript Validation rule, follow these steps:

1. In the System Navigator, click **Tailoring** > **Process Designer** > **Rule Sets**.
2. Click **Search** to open the **Rule Set** form, and then select the rule set to which you want to add the rule.

   Note
   Note
   Out-of-box rule sets are labeled as **Proprietary** and you can edit user-defined rule sets only.

3. Click **Add Rule** to open the **Select Rule Type** page, and then click **Set Mandatory Fields**.
4. In the **Rule Description** field, type a description of your new rule.
5. Click **Edit** to add conditions to the rule.

   Note
   Note
   If you don't specify a condition, the value defaults to **Always**.

6. Select the error message type (**Pop-up** or **Screen**) that's displayed during validation.
7. Click the **Show All Error Messages Together** checkbox if you want to display all error messages together during validation.
8. Select the field name you want to set as mandatory.
9. You can choose a default value, which will be set if the mandatory fields are empty.
10. Click **OK** to add the new rule to the rule set.

Related topics

Process Designer
Process Designer security model
Adding a rule
Create a rule set

6.3.3.10. Add a Set Mandatory Variables rule

Applies to User roles: System Administrator, Implementer

This rule enables you to set global and thread variables as mandatory. If mandatory variables are empty, then a default value is set if specified in the Rule.

Global or thread variables begin with the $ symbol. However, the rule doesn't accept local variables beginning with $L. Global variables are set when the operator logs on and the server automatically cleans them up when the session ends.

Thread variables don't have a consistent naming scheme. They are only valid for the current RAD thread. If the RAD thread terminates, the server automatically cleans up all thread variables.

To add a Mandatory Variables rule, follow these steps:

1. In the System Navigator, click **Tailoring** > **Process Designer** > **Rule Sets**.
2. Click **Search** to open the **Rule Set** form, and then select the rule set to which you want to add the rule.

   Note
   Note Out-of-box rule sets are labeled as **Proprietary** and you can edit user-defined rule sets only.

3. Click **Add Rule** to open the **Select Rule Type** page, and then click **Set Mandatory Variables**.
4. In the **Rule Description** field, type a description of your new rule.
5. Click **Edit** to add conditions to the rule.

> **Note**
> Note If you don't specify a condition, the value defaults to **Always**.

6. Select the error message type (**Pop-up** or **Screen**) that's displayed during validation.
7. Click the **Show All Error Messages Together** checkbox if you want to display all error messages together during validation.
8. Select a global or thread variable.

> **Note**
> Note If a mandatory variable is empty, then the rule sets it to a default value.

9. Type the variable of the default value.
10. Click **OK** to add the new rule to the rule set.

Related topics

Process Designer
Process Designer security model
Adding a rule
Create a rule set

6.3.3.11. Add a Send Notifications rule

Applies to User roles: System Administrator, Implementer

This rule enables you to send notifications using a Service Manager notification record. The rule is typically mapped to a workflow or workflow phase. For example, you can use the rule to send a notification when a phase is entered. The rule type links to an existing notification record.

To configure a notification, follow these steps:

1. In the System Navigator, click **Tailoring** > **Process Designer** > **Rule Sets**.
2. Click **Search** to open the **Rule Set** form, and then select the rule set to which you want to add the rule.

> **Note**
> Note Out-of-box rule sets are labeled as **Proprietary** and you can edit user-defined rule sets only.

3. Click **Add Rule** to open the **Select Rule Type** page, and then click **Send Notifications**.
4. In the **Rule Description** field, type a description of your new rule.
5. Click **Edit** to add conditions to the rule.

> **Note**
> Note If you don't specify a condition, the value defaults to **Always**.

6. Click **Fill Field Notification Name**, and then select the existing notification record that you want to send when this rule is executed.
7. Click **OK** to add the new rule to the rule set.

Related topics

Process Designer
Process Designer security model
Adding a rule
Create a rule set

6.3.3.12. Add a Send Teams Message rule

Applies to User roles: System Administrator, Implementer

This rule enables you to send Teams messages using pre-configured Teams message templates. Recipients can be particular users or a group, either explicitly named or based on data on the form. The rule is typically mapped to a workflow or workflow phase. You can use this rule to send a Teams message at specific points in a workflow.

To add a Send Teams Message rule, follow these steps:

1. In the System Navigator, click **Tailoring** > **Process Designer** > **Rule Sets**.
2. Click **Search** to open the **Rule Set** form, and then select the rule set to which you want to add the rule. Alternatively, you can create a new rule set as well.

   Note

   Note

   Out-of-box rule sets are labeled as **Proprietary** and you can edit user-defined rule sets only.

3. Click **Add Rule** to open the **Select Rule Type** page, and then click **Send Teams Message**.
4. In the **Rule Description** field, type a description of your new rule.
5. Click **Edit** to add conditions to the rule.

   Note

   Note

   If you don't specify a condition, the value defaults to **Always**.

6. In the **Message Template** field, enter a Teams message template.
7. Select one of the four recipient options: **Users**, **Group**, **Set Using JavaScript**, or **Current Record**. Then, follow the appropriate steps to configure the recipient:
   - **Users**: Select a user type (**Operators** or **Contacts**), and then select the operators or contacts accordingly.
   - **Group**: Select one of the following group types: **Assignment Group** (assignment), **Change Management Group** (cm3groups), **Knowledge Management Group** (kmgroup), or **Request Management Group** (ocmgroups). Then, follow the appropriate steps to configure the group:
     - **Assignment Group**: Enter a group name in the **Group** field, and then select **Operators**, **Manager**, **Approvers** or **All** in the **Send To** drop-down list.
     - **Change Management Group**: Enter the group name in the **Group** field, and then select **Members**, **Approvers**, or **All** in the **Send To** drop-down list.
     - **Knowledge Management Group**: Enter the group name in the **Group** field, and then select **Operators**, **Manager**, or **All** in the **Send To** drop-down.
     - **Request Management Group**: Enter the group name in the **Group** field. Also, select **Members**, **Approvers**, or **All** in the **Send To** drop-down list.

     The system separately sends Teams messages to each member as specified in the **Send To** drop-down list of a selected group.
   - **Set Using JavaScript**: In the script field that appears, set the recipients in the **users** array variable by using valid operator login names or contact names. For example, you type the following:

     ```
     var users = ["jennifer"];
     ```

   - **Current Record**: Choose a field from the current record, and then specify if the field is the ID of an operator, contact, assignment, cm3groups, kmgroup, or ocmgroup in the **Field Type** drop-down list.
     If you select any field type apart from **Operator** or **Contact**, you have to select from **Send To** field. For example, if you select **Assignment Group**, you have to select **Operators**, **Manager**, **Approvers** or **All** from the **Send To** drop-down list.
8. Click **OK** to add the new rule to the rule set.

Related topics

Process Designer
Process Designer security model

Create a rule set
Set up Teams notification

6.3.3.13. Add a Launch a Script rule

Applies to User roles: System Administrator, Implementer

This rule enables you to launch a Service Manager script. Scripts enable you to interrupt the normal screen flow and gather the prerequisites before the form is displayed. For example, you can use a script to automatically close an incident in five days. The script closes the incident.

To add a Launch a Script rule, follow these steps:

1. In the System Navigator, click **Tailoring** > **Process Designer** > **Rule Sets**.
2. Click **Search** to open the **Rule Set** form, and then select the rule set to which you want to add the rule.

   Note
   Note Out-of-box rule sets are labeled as **Proprietary** and you can edit user-defined rule sets only.

3. Click **Add Rule** to open the **Select Rule Type** page, and then click **Launch a Script**.
4. In the **Rule Description** field, type a description of your new rule.
5. Click **Edit** to add conditions to the rule.

   Note
   Note If you don't specify a condition, the value defaults to **Always**.

6. Click the **Fill Field Script** icon in the **Script text** box to select a Service Manager script.

   Note
   Note If you type an invalid script name, an error message appears.

7. Click **OK** to add the new rule to the rule set.

Related topics

Process Designer
Adding a rule
Create a rule set
Build a condition

6.3.3.14. Add a Send HTML Email rule

Applies to User roles: System Administrator, Implementer

This rule enables you to send a formatted HTML email to intended recipients using pre-configured HTML templates. Recipients can be particular users or a group, either explicitly named or based on data in the form. This rule is used to send emails at specific points in a workflow as opposed to create a general Service Manager notification outside of Process Designer that, for example, would apply for all Change Requests.

To add a Send HTML Email rule, follow these steps:

1. In the System Navigator, click **Tailoring** > **Process Designer** > **Rule Sets**.
2. Click **Search** to open the **Rule Set** form, and then select the rule set to which you want to add the rule.

   Note
   Note Out-of-the-box rule sets are labeled as **Proprietary** and you can edit user-defined rule sets only.

3. Click **Add Rule** to open the **Select Rule Type** page, and then click **Send HTML Email**.
4. In the **Rule Description** field, type a description of your new rule.

5. Click **Edit** to add conditions to the rule.

> **Note**
> Note If you don't specify a condition, the value defaults to **Always**.

6. In the **HTML Template** field, enter an HTML template.
7. On the **Recipients To** tab, select one of the four recipient options: **Users**, **Group**, **Set Using JavaScript**, or **Current Record**. Then, follow the appropriate steps to configure the recipient:
   - **Users**: Select a user type (**Operators** or **Contacts**), and then select the operators or contacts accordingly.
   - **Group**: Select one of the following group types: **Assignment Group** (assignment), **Change Management Group** (cm3groups), **Knowledge Management Group** (kmgroup), or **Request Management Group** (ocmgroups). Then, follow the appropriate steps to configure the group:
     - **Assignment Group**: Enter a group name in the **Group** field, and then select **Operators**, **Manager**, **Approvers** or **All** in the **Send To** drop-down list.
     - **Change Management Group**: Enter the group name in the **Group** field, and then select **Members**, **Approvers**, or **All** in the **Send To** drop-down list.
     - **Knowledge Management Group**: Enter the group name in the **Group** field, and then select **Operators**, **Manager**, or **All** in the **Send To** drop-down.
     - **Request Management Group**: Enter the group name in the **Group** field. Also, select **Members**, **Approvers**, or **All** in the **Send To** drop-down list.
   - **Set Using JavaScript**: In the script field that appears, set the recipients in the **users** array variable by using valid operator login names or contact names. For example, you type the following:

     ```
     var users = ["jennifer"];
     ```

   - **Current Record**: Choose a field from the current record, and then specify if the field is the ID of an operator, contact, assignment, cm3groups, kmgroup, or ocmgroup in the **Field Type** drop-down list.

     If you select any field type apart from **Operator** or **Contact**, you have to select from **Send To** field. For example, if you select **Assignment Group**, you have to select **Operators**, **Manager**, **Approvers**, or **All** from the **Send To** drop-down list.
8. (Optional) On the **Recipients Cc** and **Recipients Bcc** tabs, select either of the two recipient options: **Users** or **Group**.
   - **Users**: Select a user type (**Operators** or **Contacts**), and then select the operators or contacts accordingly.
   - **Group**: Select one of the following group types: **Assignment Group** (assignment), or **Knowledge Management Group** (kmgroup). Then, follow the appropriate steps to configure the group:
     - **Assignment Group**: Enter a group name in the **Group** field, and then select **Operators**, **Manager**, **Approvers** or **All** in the **Send To** drop-down list.
     - **Knowledge Management Group**: Enter the group name in the **Group** field, and then select **Operators**, **Manager**, or **All** in the **Send To** drop-down list.

> **Note**
> Note If a Smart Email consists of tokens, the Cc and Bcc functions don't work.

9. Click **OK** to add the new rule to the rule set.

Tip

Tip

Suppose you have set a valid group email of members/approvers in the Assignment Groups definition and your selection meets either of the following combinations. In that case, the system sends emails to this group email address when sending notifications.

- Select **Current Record** as the Recipients option, and then select **Assignment Group** as the Field option.
- Select **Group** as the Recipients option, and then select **Assignment Group** as the Group Type option.

Note that this group email function doesn't support all the Smart Emails with action buttons (such as the **Approve**, **Deny**, or **Close** buttons, and so on). The system separately sends the Smart Emails with action buttons to each member in the assignment group even though you have set a valid group email address in the **Approver Group Email** field.

Related topics

Process Designer
Process Designer security model
Create a rule set

6.3.3.15. Add a Start or Stop Clock rule

Applies to User roles: System Administrator, Implementer

This rule enables you to start or stop a Service Manager clock to measure elapsed time. For example, suppose you hope to know the time taken for an Incident to move from the Validation to the Risk and Impact Analysis phase. You can configure a rule to start a clock when the record enters the Validation phase. Then, configure another rule to stop the same clock when the incident record moves to the Risk and Impact Analysis phase. The clocks table stores the time difference along with start and stop times.

To add a Start or Stop Clock rule to start a clock, follow these steps:

1. In the System Navigator, click **Tailoring** > **Process Designer** > **Rule Sets**.
2. Click **Search** to open the **Rule Set** form, and then select the rule set to which you want to add the rule.

   Note

   Note Out-of-the-box rule sets are labeled as **Proprietary** and you can edit user-defined rule sets only.

3. Click **Add Rule** to open the **Select Rule Type** page, and then click **Start or Stop Clock**.
4. In the **Rule Description** field, type a description of your new rule.
5. Click **Edit** to add conditions to the rule.

   Note

   Note If you don't specify a condition, the value defaults to **Always**.

6. Select **Start Clock**, and then click **Next**.
7. In the **Name of Clock** pane, choose one of the following methods to define the clock name:
   • Select the **Fixed Name** option to identify a specific clock. Type a clock name to add a new clock, or select an existing clock name from the drop-down list.
   • Select the **Set Using JavaScript** option to set a clock name by using JavaScript. For example, use the `clockName = "New Clock";` JavaScript to set a clock.
8. In the **Use this schedule** pane, select a predefined schedule or configure a schedule by using JavaScript.
9. In the **Using this time zone** pane, select the time zone that the clock uses. Options include the location of the system server, technician, customer, or CI. You can also enter a fixed value (a specific time zone) or configure the time zone by using JavaScript.
10. Click **Finish** to add the new rule to the rule set.

Once you have created a rule to start a clock, you can add another rule to stop the clock. To do this, follow these steps:

1. In the System Navigator, click **Tailoring** > **Process Designer** > **Rule Sets**.
2. Click **Search** to open the **Rule Set** form, and then select the rule set to which you want to add the rule.
3. Click **Add Rule** to open the **Select Rule Type** page, and then click **Start or Stop Clock**.
4. In the **Rule Description** field, type a description of your new rule.
5. Click **Edit** to add conditions to the rule.

   Note
   Note If you don't specify a condition, the value defaults to **Always**.

6. Select **Stop Clock**, and then click **Next**.
7. In the **Name of Clock** pane, choose one of the following methods to define the clock name:
   • Select the **Fixed Name** option to identify a specific clock. Type a clock name to add a new clock, or select an existing clock name from the drop-down list.
   • Select the **Set Using JavaScript** option to set a clock name by using JavaScript.

   Note
   Note The name of the stop clock rule should be the same as the corresponding start clock rule.

8. Click **Finish** to update the rule.

Related topics

Process Designer
Process Designer security model
Adding a rule
Create a rule set

6.3.3.16. Add a Set Field rule

Applies to User roles: System Administrator, Implementer

This rule enables you to set a field with a value that's determined by a JavaScript. From a rule set, you can select a common field from multiple tables and set its value based on a JavaScript expression.

Note
Note Make sure that the field is set to the contents of the **value** variable. If it's not set to the **value** variable, the JavaScript can't be processed.

To configure a field based on a JavaScript, follow these steps:

1. In the System Navigator, click **Tailoring** > **Process Designer** > **Rule Sets**.
2. Click **Search** to open the **Rule Set** form, and then select the rule set to which you want to add the rule.

   Note
   Note Out-of-box rule sets are labeled as **Proprietary** and you can edit user-defined rule sets only.

3. Click **Add Rule** to open the **Select Rule Type** page, and then click **Set a Field**.
4. In the **Rule Description** field, type a description of your new rule.
5. Click **Edit** to add conditions to the rule.

   Note
   Note If you don't specify a condition, the value defaults to **Always**.

6. Select an applicable field name from the drop-down list.
7. Type the JavaScript that sets the variable **value** to the desired value for the field in the text box.
8. Click **OK** to add the new rule to the rule set.

Related topics

Process Designer
Process Designer security model
Adding a rule
Create a rule set

6.3.3.17. Add a Set Field from Number rule

Applies to User roles: System Administrator, Implementer

This rule enables you to set a field based on an Service Manager number record. When this rule is executed, unique sequential numbers for the records in the database are generated.

When a new record is added, the unique sequential number is incremented or decremented based on the prefix or suffix provided. For example, you can create a number record for the cm3r class with **Increment By** as **+1**, **Prefix** as **C**, and**Last Number** as **10**. Whenever a new change request is added in the system, it assigns the change request with a unique change ID C10, the next change request you add will have the change ID C11.

To add a Set Field from Number rule, follow these steps:

1. In the System Navigator, click **Tailoring** > **Process Designer** > **Rule Sets**.
2. Click **Search** to open the **Rule Set** form, and then select the rule set to which you want to add the rule.

> **Note**
> Note Out-of-box rule sets are labeled as **Proprietary** and you can edit user-defined rule sets only.

3. Click **Add Rule** to open the **Select Rule Type** page, and then click **Set Field from Number**.
4. In the **Rule Description** field, type a description of your new rule.
5. Click **Edit** to add conditions to the rule.

> **Note**
> Note If you don't specify a condition, the value defaults to **Always**.

6. Select the field from the **Field Name** drop-down list.
7. Select the class from the **Class of Number Record** drop-down list.

> **Note**
> Note You can select the **Overwrite When Record Is Saved** checkbox to overwrite the number when you save a record.

8. Click **OK** to add the new rule to the rule set.

Related topics

Process Designer
Create a rule set
Build a condition

6.3.3.18. Add a Validate Date rule

Applies to User roles: System Administrator, Implementer

This rule enables you to perform validation between a date field and an absolute date or a range of relative dates. For example, you can add a rule to validate whether a document expiration date is greater than or equal to the document creation date.

To add a Validate Date rule, follow these steps:

1. In the System Navigator, click **Tailoring** > **Process Designer** > **Rule Sets**.
2. Click **Search** to open the **Rule Set** form, and then select the rule set to which you want to add the rule.

   Note
   Note Out-of-the-box rule sets are labeled as **Proprietary** and you can edit user-defined rule sets only.

3. Click **Add Rule** to open the **Select Rule Type** page, and then click **Validate Date**.
4. In the **Rule Description** field, type a description of your new rule.
5. Click **Edit** to add conditions to the rule.

   Note
   Note If you don't specify a condition, the value defaults to **Always**.

6. Select the error message type: **Pop-up** or **Screen** to display during validation.
7. Select the field to validate from the drop-down list.

   Note
   Note The drop-down list displays all fields with the Date data type from the current table.

8. Select the comparison operator from the drop-down list.
9. Select the **Not** option if you want to negate the comparison.
10. Click **Relative Date/Time** or **Absolute Date/Time** to validate the date field.

    Note
    Note
    • If you select **Relative Date/Time**, select a date field or Current Date/Time from the drop-down list.
    • You can add or subtract an offset time of days, hours, and minutes from the relative date/time by selecting **+** or **-**. For instance, an offset `+04:03:02` indicates the relative date/time is ahead by 4 days, 3 hours, and 2 minutes.
    • If you select **Absolute Date/Time**, click the **Calendar** icon to select the date and time.

11. Click **OK** to add the new rule to the rule set.

Related topics

Adding a rule
Create a rule set
Build a condition

6.3.3.19. Add a Validate against List rule

Applies to User roles: System Administrator, Implementer

This rule enables you to select a field to validate with a list of values. This list may be global or manually defined.

To add a **Validate against List** rule, follow these steps:

1. In the System Navigator, click **Tailoring** > **Process Designer** > **Rule Sets**.
2. Click **Search** to open the **Rule Set** form, and then select the rule set to which you want to add the rule.

   Note
   Note
   Out-of-the-box rule sets are labeled as **Proprietary** and you can edit user-defined rule sets only.

3. Click **Add Rule** to open the **Select Rule Type** page, and then click **Validate against List**.

4. In the **Rule Description** field, type a description of your new rule.
5. Click **Edit** to add conditions to the rule.

> **Note**
> Note
> If you don't specify a condition, the value defaults to **Always**.

6. Select **Pop-up** or **Screen** to set the type of error message that's displayed during validation.
7. In the **Field to Validate** drop-down list, select the field to validate.
   If you select **Global List**, select the global list from the drop-down list.
   If you select **Manual List**, type the manual list names in the Value table.
8. Click **OK** to add the new rule to the rule set.

> **Note**
> Note
> If the field value doesn't match a value in the specified list, the system displays a message and sets the cursor to that field when the rule isn't followed.

Related topics

Process Designer
Process Designer security model
Adding a rule
Create a rule set

6.3.3.20. Add a Validate against Table rule

Applies to User roles: System Administrator, Implementer

This rule enables you to validate a field against a field in another table and fill data into other fields. You can also filter the data you are validating against, and fill data into other fields.

> **Note**
> Note The Validate against Table rule doesn't include the Set Mandatory Fields rule, so that you can't use the Validate against Table rule to verify whether a field is empty or not. Before you create a Validate against Table rule, it is recommended that you first add a Set Mandatory Fields rule to verify whether the field is empty or not.

To add a Validate against Table rule, follow these steps:

1. In the System Navigator, click **Tailoring** > **Process Designer** > **Rule Sets**.
2. Click **Search** to open the **Rule Set** form, and then select the rule set to which you want to add the rule.

> **Note**
> Note Out-of-box rule sets are labeled as **Proprietary** and you can edit user-defined rule sets only.

3. Click **Add Rule** to open the **Select Rule Type** page, and then click **Validate against Table**.
4. In the **Rule Description** field, type a description of your new rule.
5. Click **Edit** to add a condition.

> **Note**
> Note If you don't specify a condition, it will default to **Always**.

6. Select **Pop-up** or **Screen** to set the type of error message that's displayed during validation.

7. Select the field to validate from the drop-down list.
8. Select the name of the table you want to validate against.
9. Select the name of the field you want to validate against.
10. You can filter the data that you are validating against by using a standard RAD query.
    For example, the following field entries will validate only for cm3r with the current field name ("Demo String2," in this example) of the workflow:
    Current Field Name field: **Demo String2**
    Validate Against Table field: **ModuleStatus**
    Validate Against Field field: **Status**
    Filter field: **module="cm3r"**
11. Click **Click Here to Add Fill Information** to enter Fill From and Fill To fields.

> **Note**
> Note The **Fill From** field represents the fields of the validation table against which it validates. The **Fill To** field represents the fields in the current table being validated.

12. (Optional) If you want the field to be validated whether its value changes or not, select the **Always Check Validity** checkbox.
    If this option is not selected, the system validates the field only when its value changes.
13. Click **OK** to add the new rule to the rule set.

Related topics

Process Designer
Create a rule set

6.3.3.21. Add a Validate Text/Number rule

Applies to User roles: System Administrator, Implementer

This rule enables you to validate a field against a range of text or a number in another field of same table. You can define fixed values or select a field from the current table to validate against a field. For example, the field to validate is **Demo Number**, field to validate against is **Demo String 1**, and the comparison operator is **Is greater than or equal to**.

> **Note**
> Note If the field to validate against is blank, the rule is ignored. If you need to enforce the rule, then use a Set Mandatory Fields rule on the filed you need to validate against.

To add a Validate Text/Number rule, follow these steps:

1. In the System Navigator, click **Tailoring** > **Process Designer** > **Rule Sets**.
2. Click **Search** to open the **Rule Set** form, and then select the rule set to which you want to add the rule.

> **Note**
> Note Out-of-box rule sets are labeled as **Proprietary** and you can edit user-defined rule sets only.

3. Click **Add Rule** to open the **Select Rule Type** page, and then click **Validate Text/Number**.
4. In the **Rule Description** field, type a description of your new rule.
5. Click **Edit** to add conditions to the rule.

> **Note**
> Note If you don't specify a condition, the value defaults to **Always**.

6. Select **Pop-up** or **Screen** to set the type of error message that's displayed during validation.
7. Select the field to validate from the drop-down list.

8. Select the comparison operator from the drop-down list.
9. Select the **Not** checkbox if you want to negate the comparison.
10. To validate against a field or manually defined value, follow the appropriate steps:
    To validate against a field, select **Field** under **Values**, and then select the field you want to validate against from the drop-down list.
    To validate against a manually defined value, select **Custom** under **Values**, and then type the value to validate against the selected field. For example, to validate whether the "Risk Assessment" value is between 1 and 5, select the **Is between** comparison operator, and then type 1 and 5 in the **Custom** text box.
11. Click **OK** to add the new rule to the rule set.

Related topics

Process Designer
Process Designer security model
Adding a rule
Create a rule set

6.3.3.22. Add a Popup Message Box rule

Applies to User roles: System Administrator, Implementer

This rule enables you to create and configure popup message boxes that appear to end users.

To add a Popup Message Box rule, follow these steps:

1. In the System Navigator, click **Tailoring** > **Process Designer** > **Rule Sets**.
2. Click **Search** to open the **Rule Set** form, and then select the rule set to which you want to add the rule.

Note

Note Out-of-the-box rule sets are labeled as **Proprietary** and you can edit user-defined rule sets only.

3. Click **Add Rule** to open the **Select Rule Type** page, and then click **Popup Message Box**.
4. In the **Rule Description** field, type a description of your new rule.
5. Click **Edit** to add conditions to the rule.

Note

Note If you don't specify a condition, the value defaults to **Always**.

6. In the **Using this message** section, select whether you want to set a fixed message or configure the message text by using JavaScript. Then, enter the message or JavaScript in the relevant text box.

Note

Note If you select a fixed message, you can additionally enter localized versions of the message. If you select to set the message text by using JavaScript, the JavaScript implementation handles the message localization.
To localize a fixed message, first save the rule, and then open it for editing. An **Edit localized labels** button is now displayed next to the **Fixed Message** field. Click the button to configure the language code, message ID, and text of the localized message.

7. Select the message box type. The following options are available.

Option	Description
OK only	This type of message box provides end users with the option to click "OK" only. When you select this rule type, you must select the message level and set the behavior when the user clicks "OK."
Yes, No	This type of message box provides end users with the option to click "Yes" or "No." When you select this rule type, you must set the behavior when the user clicks "Yes" or "No."
Yes, No, Cancel	This type of message box provides end users with the option to click "Yes," "No," or "Cancel." When you select this rule type, you must set the behavior when the user clicks "Yes," "No," or "Cancel."

8. Click **OK** to add the new rule to the rule set.

Related topics

Process Designer
Process Designer security model
Adding a rule
Create a rule set

6.3.3.23. Add an Assignment rule

Applies to User roles: System Administrator, Implementer

This rule enables you to automatically distribute records (such as tasks or records) to the groups and assignees who are most able to process them.

To add an Assignment rule, follow these steps:

1. In the System Navigator, click **Tailoring** > **Process Designer** > **Rule Sets**.
2. Click **Search** to open the **Rule Set** form, and then select the rule set to which you want to add the rule.

Note

Note
Out-of-box rule sets are labeled as **Proprietary** and you can edit user-defined rule sets only.

3. Click **Add Rule** to open the **Select Rule Type** page, and then click **Assignment**.
4. In the **Rule Description** field, type a description of the rule.
5. Click **Edit** to add conditions to the rule.

Note

Note
If you don't specify a condition, the value defaults to **Always**.

6. In the **Group Field Name**, specify the field that stores the group name when a record is assigned according to the assignment rules.
7. In the **Default Group** field, specify the group to which the record is assigned if the remaining rules don't return a specific group.
8. In the **Service Field Name** field, specify the field that contains the related service of the record.
9. In the **Contact Field Name** field, specify the field that contains the contact information of the record.
10. From the **Group Assignment** tab, configure the group to whom the record is assigned when the rule is triggered (that is, the value that is inserted into the field which you specified in the **Group Field Name** field).

To do this, follow these steps:

1. Select one of the following assignment types:
   - **Automatic - take first:** Service Manager automatically assigns the record to the first group that is identified.
   - **Manual - let the user choose:** Service Manager presents a list of potential assignment groups to the user who has triggered the rule.

   Note
   Note
   The system uses **Automatic - take first** by default for the backend processes. The **Manual - let the user choose** option only takes effect for the UI processes.

2. Select the **Consider Workschedule of Group** checkbox if you want to assign the record only to the groups that are available according to the calendar defined in the assignment groups.

   Note
   Note
   If the calendar is not specified in the assignment group definition, the system uses "24x7" by default.
   In addition, the system also uses the time zone defined in the assignment group definition. If the time zone is empty in the assignment group definition, the system time zone defined in system information record (**System Information Record** > **Date Info**) is used instead.

3. In the **Assignment Rule** pane, select one of the following options to configure the assignment rule:
   - **Predefined:** Select the following options to define the assignment rule.
     - **Service Based**
     - **Location Based**
     - **Department Based**
     - **Language Based**
     - **Company Based** (Only available when the multi-company mode is enabled.)

     The system returns groups by matching the information in the record and the information in the definitions of assignment groups. If multiple options are selected, only the groups that match all the selected options are returned.
     In addition, you can click **Edit Query** to add additional query to further filter the assignment groups.
   - **Fixed:** Specify a specific group in the **Group** field. The system automatically inserts this group into the field that you specified in the **Group Field Name** field.
   - **Set Using JavaScript:** Determines the group to whom the record is assigned based on a JavaScript.

   Tip
   Tip
   For more information about configuring assignment rules for groups, see Configure assignment rules.

11. From the **Individual Assignment** tab, configure the assignee to whom the record is assigned when the rule is triggered.

    Note
    Note
    Assigning tasks or records to specific assignees is optional. If you want to assign records to a group only, select the **None** option.

    To do this, follow these steps:

    1. In the **Assignee Field Name** drop-down list, select the field in which the name of the assignee to whom you want to assign the record is stored.

2. Select one of the following options to configure the assignment rule:
    - **None:** Do not assign to any individual.
    - **Fixed:** Automatically inserts a fixed value into the field that you specified in the **Assignee Field Name** field.
    - **Group Member:** Automatically assigns the record to a member of the configured group. Records can be assigned on a round robin basis or to the assignee who is currently assigned the lowest number of records. To do this, select an option from the **Member Type** drop-down list.
        - If you select the **Round robin** option, enter the field in which the time used for the calculation is stored in the **Assignment Time Field Name** field. Then, select the number of days that are taken into consideration when the assignment is calculated. The default value is 60 days.
        - If you select the **Number of Assigned Tickets** option, click **Edit Query** to open the condition editor and create the filter that determines the assignee.
    - **Coordinator:** Assigns the record to a coordinator of your choice in the specified group.
    - **JavaScript:** Determines the assignee to whom the record is assigned based on a JavaScript.

Tip

Tip

For more information about configuring assignment rules for individuals, see Configure assignment rules.

12. Click **OK** to add the new rule to the rule set.

Prioritizing assignment rules

Service Manager supports prioritizing multiple assignment rules by introducing a new return variable "$isDefaultGroupAssigned". You need to define multiple assignment rules in the same rule set. Then, after the first assignment rule, use the $isDefaultGroupAssigned as the condition in the subsequent assignment rules for the system to decide whether or not it continues to execute the latter assignment rules:

- If the returned value of $isDefaultGroupAssigned is false, it means there are groups identified by the current assignment rule.
- If the returned value of $isDefaultGroupAssigned is true, it means there is no group identified by the current assignment rule.

The system executes the latter rule only if no assignment group is identified in the previous assignment rule.

Use the following instructions when you want to use the $isDefaultGroupAssigned variable to prioritize multiple assignment rules in a combined rule set:

- The first assignment rule in the rule set is the most specific one to identify the best fit.
- The subsequent rules in this rule set may contain assignment logic with less specific queries to ensure records are assigned.
- If the system identifies groups in the former rules (not default group), the latter rules will be ignored.

6.3.3.24. Add a Run Action rule

Applies to User roles: System Administrator, Implementer

This rule automatically runs actions (defined by rule sets and/or backend transitions) on records that have a specified relationship to the record that triggers the rule. For example, you can use this rule to change the workflow phase of related records under specific conditions.

To add a Run Action rule, follow these steps:

1. In the System Navigator, click **Tailoring** > **Process Designer** > **Rule Sets**.
2. Click **Search** to open the **Rule Set** form, and then select the rule set to which you want to add the rule.

Note

Note Out-of-box rule sets are labeled as **Proprietary** and you can edit user-defined rule sets only.

3. Click **Add Rule** to open the **Select Rule Type** page, and then click **Run Action**.
4. Type a description in the **Rule Description** field.
5. Click **Edit** to add conditions to the rule.

   Note

   Note If you don't specify a condition, the value defaults to **Always**.

6. Select the type of record on which you want to run the action.
   - Select **Related Records** to change the workflow status of records related to the record that triggers the rule.
   - Select **Other Records** to change the workflow status of records that have a relationship with the current record that's built by a manually-defined query string.
   - Select **Current Record** to change the workflow status of the record that triggers the rule.

     Note

     Note This option enables you to run a backend transition directly on the current record.

7. Configure the relationship between the records on which you want to run the action and the record that triggers the rule.
   - If you selected the **Related Records** option, select a relationship type from the **Relation Type** drop-down list.
   - If you selected the **Other Records** option, select a table from the **Table Name** drop-down list.
   - If you selected the **Current Record** option, there is no need to define a relationship.
8. Click **Edit Query** if you want to define a subset of the related records on which the actions will be run.

   Note

   Note
   - If you select the **Other records** option but don't specify a query, no records are queried. This prevents the rule from running the action on every record in the defined table.
   - This step doesn't apply if you selected to run the actions on the current record only.
   - The Query Editor widget functions similarly to the Condition Editor. However, note the following differences:
     - In the left-hand side (LHS) of the query, you can only select a field in the record on which you want to run the action. Cross table fields aren't supported.
     - Only the following data types are supported on the right-hand side of the query:
       - Blank/Nul
       - Value
       - Variable
       - Current Record
       - Saved Record
9. In the **Run Rule Set** field, select the rule set that's applied to the target records when this rule is triggered.
10. In the **Action after Rule Set** field, select the action that's performed on the target records after the rule set is applied. You can select to do nothing, save the record, or apply a backend transition. The list of available backend transitions is retrieved from the workflows of the target record's table name (depending on the Relation Type that you selected).
11. Click **OK** to add the new rule to the rule set.

Note

Note This rule doesn't take locked records into account. If a record is locked when the rule is run, the action is not automatically rescheduled.

Related topics

Process Designer
Process Designer security model
Adding a rule
Create a rule set

6.3.3.25. Add a Run Scheduled Action rule

Applies to User roles: System Administrator, Implementer

This rule automatically runs an action (defined by rule sets and/or backend transitions) on records after a specified length of time has passed. For example, you can use this rule to automatically close incident records that have been in a Resolved state for a certain number of days.

To add a Run Scheduled Action rule, follow these steps:

1. In the System Navigator, click **Tailoring** > **Process Designer** > **Rule Sets**.
2. Click **Search** to open the **Rule Set** form, and then select the rule set to which you want to add the rule.

   Note

   Note Out-of-box rule sets are labeled as **Proprietary** and you can edit user-defined rule sets only.

3. Click **Add Rule** to open the **Select Rule Type** page, and then click **Run Scheduled Action**.
4. Type a description in the **Rule Description** field.
5. Click **Edit** to add conditions to the rule.

   Note

   Note If you don't specify a condition, the value defaults to **Always**.

   When these conditions are met, a schedule instance is created and monitored by a backend scheduler.
6. Select one of the following options to set the calculation method of the schedule instance:
   • **Use field in record + interval:** Enables you to configure when the rule is triggered by setting a time and an interval. To do this, you must specify a field in a record (to set the time) and a length of time (to set the interval). Enter the interval time in the **Calc Interval** field, and use the `ddd hh:mm:ss` time format:

   Note

   Note This option supports negative intervals. That is, you can specify the length of time before a time that's set in a field.

   • **Use javascript to set variable actionExecutionTime:** Enables you to use JavaScript to set the time when the backend schedule is executed.
7. Click **Edit** to set a further set of conditions that are applied once the time that's defined in the schedule instance is reached.
8. In the **Run Rule Set** field, select the rule set that's applied to the current records when the backend schedule is executed.
9. In the **Action after Rule Set** field, select the action that's performed on the current records after the rule set is applied. You can select to do nothing, save the record, or apply a backend transition. The list of available backend transitions is retrieved from the workflows of the current record's table name.
10. Click **OK** to add the new rule to the rule set.

Note

Note This rule takes locked records into account. If a record is locked when the rule is run, the action is automatically rescheduled.

Caution

Caution A scheduled action is triggered based on the interval that's configured in the rule. If a user manually updates a field that the scheduled action will update later, the user's update will be overwritten by the scheduled action. For example, the im.schedule.unlinkAllChild Rule Set defines a scheduled action that will clear the Parent Incident field of all child incidents. If you click **Unlink All** in a parent incident, this scheduled action is triggered; before the scheduled action is completed, if a user manually changes the Parent Incident value of a child incident, the user's update will be cleared later.

Related topics

Process Designer
Process Designer security model
Adding a rule
Create a rule set

6.3.3.26. Group rules

Applies to User roles: System Administrator, Implementer

You can group multiple rules into a rule group with an overall condition. The system runs all the rules in the rule group if the condition matches.

To group rules, follow these steps:

1. Click **Tailoring** > **Process Designer**.
2. Click **Rule Sets** > **Search** to open the **Rule Set** form.
3. Select the rule set to which you want to add a group, and then click **Add Group**.
4. Type a rule group name. For example, kmrules.
5. Click **Edit** to add a condition.

   Note

   Note If you don't enter a condition, it will default to **Always**

6. Click **OK** to add the group name.
7. Highlight the group by selecting it.
8. Click **Add Rule**.
9. Select a rule.
10. Type the attributes in the Rule form.
11. Repeat steps 8 to 11 if you want to add more rules.
12. Click **OK**to create the rule group.

Note

Note You can add one or more rule groups within another rule group.

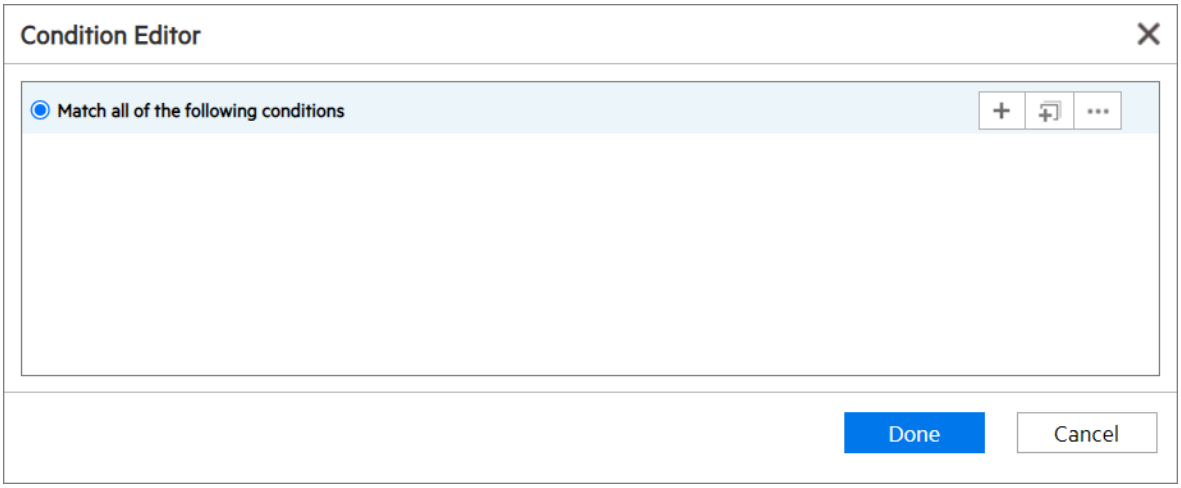
Related topics

Adding a rule
More Actions menu
Create a rule set

6.3.4. Using the Condition Editor

The Service Manager Condition Editor enables you to build a condition or set of conditions without any programming language knowledge. Conditions always evaluate to True or False. When a condition evaluates to True, the system runs the rule or applies an action that the condition controls. For example, you can use conditions to determine if a workflow transition should occur or if a rule should be executed.

A condition item is an expression built from fields, operators, and values. A condition item has a Left-Hand-Side (LHS) and a Right-Hand-Side (RHS), which are separated by an operator. There can be a combination of variables, types, and values on either side of the condition item. The Condition Editor uses a tabular format in which you populate each element of the condition item (RHS, LHS, and operator) by using a series of drop-down lists.



You can also build complex conditions with the Condition Editor. Complex conditions are composed of groups of condition items that are connected by logical operators.

After you build a condition, Service Manager displays it in a user-readable form. See the following example:

```
((description in Category = 100 OR display.name in Category = "Test") AND ((assoc.published.doc in
CurrentFile
!=true AND activity.mandatory.msg in Object <=20) OR error in SavedFile Starts With 101 OR ((the
"edit.adaptive.learning" value in the "Knowledge Document" Security Area <= 123 AND the "update" value in
the
"Tailoring" Security Area = "always") OR test!=12)))
```

You can use the Condition Editor to perform the following tasks:

- Build a condition
  For more information, see Build a condition.
- Create a group of condition items
  For more information, see Create a group of condition items
- Edit a condition
  To do this, open the Condition Editor and click on the condition item that you want to edit. You can then edit the existing values.
- Delete a condition item or group of condition items
  To do this, open the Condition Editor, and then click the Delete Condition icon (**-**) beside the condition item or group of condition items that you want to delete.
- Clear all condition items and groups
  To do this, open the Condition Editor, click the More Actions icon (**...**), and then click **Clear**.

Related topics

Process Designer
Build a condition

Launch the Condition Editor from a rule
Launch the Condition Editor from a workflow

6.3.4.1. Launch the Condition Editor from a rule

Applies to User roles: System Administrator, Implementer

The Condition Editor enables you to build a condition without any programming language knowledge.

> **Note**
>
> Note You can launch the Condition Editor in the web client only.

To launch Condition Editor from a new rule, follow these steps:

1. In the System Navigator, click **Tailoring** > **Process Designer** > **Rule Sets**.
2. Click **Search** to open the **Rule Set** form, and then select the rule set to which you want to add the rule.

   > **Note**
   >
   > Note You can edit user-defined rule sets only. Out-of-box rule sets are labeled as **Proprietary** and can't be modified.

3. Click **Add Rule** to open the **Select Rule Type** page, and then select the type of rule that you want to open.
4. Type a description in the **Rule Description** field.
5. Click **Edit** to open the Condition Editor and add conditions to the rule.

Related topics

Process Designer security model
Using the Condition Editor
Launch the Condition Editor from a workflow

6.3.4.2. Launch the Condition Editor from a workflow

Applies to User roles: System Administrator, Implementer

The Condition Editor enables you to build a condition without any programming language knowledge.

> **Note**
>
> Note You can launch the Condition Editor in the web client only.

To launch the Condition Editor from a workflow, follow these steps:

1. Click **Process Designer** > **Workflows** in the System Navigator.
2. Select an existing workflow or create a new workflow.
3. You can launch the Condition Editor from workflow phases and workflows transitions.
   - To launch the Condition Editor from a phase, select a phase, and then use one of the following methods:
     - In the **Details** tab, click the **Form Edit Condition** field.
     - In the **Actions** tab, click **Add**, and then click the **Action Condition** field.
     - In the **Approvals** tab, click the **Reset Condition** or **Recalculate Condition** field.
     - In the **Alerts** tab, click the **Reset Condition** or **Recalculate Condition** field.
   - To launch the Condition Editor from a transition, select a transition, and then click the **Condition** field in the **Details** tab.

> **Note**
>
> Note To close the Condition Editor, click **Cancel**.

Related topics

Process Designer security model
Using the Condition Editor
Launch the Condition Editor from a rule

6.3.4.3. Build a condition

Applies to User roles: System Administrator, Implementer

A condition has a Left-Hand-Side (LHS) and a Right-Hand-Side (RHS) that are separated by an operator. There can be a combination of variables, types, and values on either side of the condition.

> **Note**
>
> Note The data type in the RHS must match the data type in the LHS. For example, if you select a field that has a Boolean data type in the LHS, you must select a matching Boolean field in the RHS. Service Manager automatically validates the data types in conditions. If you don't enter matching data types, an error message is displayed in the appropriate drop-down list, and you can't save the condition until you correct the issue.
>
> If you decide to choose a value rather than a field in the RHS, Service Manager provides you with the option to type or select a value that's the same data type as the field in the LHS.

To build a condition, follow these steps:

1. Open the Condition Editor.
   For example, click **Edit** in the **Condition** field to open the Condition Editor when you create or edit a rule.
2. Click **Match all of the following conditions**, and then select the logical operator for the conditions. The following operators are available:
   - **Match all** (this option corresponds to an AND operator)
   - **Match any** (this option corresponds to an OR operator)
   - **Not match all** (this option corresponds to negating all conditions)
   - **Not match any** (this option corresponds to negating any of the specified conditions)

   Match all is selected by default.
3. In the left-most drop-down list, select a data type for the LHS. The following table describes the available data types for the LHS.

Field	Description
Category	A field in the associated category definition of the current record **Note** Note Only displayed when the category table is defined in the object record.
CurrentRecord	A field in the name of the current record
CurrentRecord Authorization	A field in the name of the current record **Note** Note Only listed in the condition editor in the workflow editor.
CurrentWorkflowPhase	A field in the associated workflow phase definition of the current workflow
Object	A field in the Object definition of the current record
Phase	A field in the associated phase definition of the current record **Note** Note Only displayed when the phase table is defined in the object record.
SavedRecord	A field in the original copy of the current record (before the user made any changes)
SavedWorkflowPhase	A field in the associated workflow phase definition of the original copy of the current workflow (before the user made any changes)
RAD Expression	Evaluates a RAD expression typed in the text area. For example: `evaluate(cls.control in $L.phase) and evaluate($L.tableAccess.delete) and open in $L.file=true and nullsub($G.ess, false)=false and approval.status in $L.file="approved" and (category in $L.file~="KM Document" or category in $L.file="KM Document"` **Note** Note You can't combine this condition with any other type of condition. Therefore, once you enter a RAD expression, click **Done** to build the condition. You can ignore all remaining steps.

Security	An operator's security rights. You can choose rights from the drop-down list of defined values, such as: Change, Change Management Configuration, Change Tasks, Knowledge Management, Knowledge Administration, Security, or Tailoring.
UserOption	A user-selectable option for a Service Catalog item **Note** Note ○ You must type the name of the user option. ○ The User Options field only supports string and boolean values.
Variable	Evaluates the value of a global, local, or thread variable condition. For example: $G.test $L.testNumber

4. Select the field for comparison.

Note

Note

- If a "one to one" or "many to one" cross table relationship is defined for a field, the cross table fields are available for selection in a second-level drop-down list. If relationships are defined with more than one field in the same table, the second-level drop-down enables you to specify the relationship, and a third-level drop-down list enables you to select the field.
- To quickly navigate to a field, move the focus to the field drop-down list, and then type the name of the desired field. To quickly navigate to a field in a second-level list, move the focus to the first-level item in the drop-down list, and then type the name of the desired field.

5. Select an operator. The following operators are available:
   - **Equals**
   - **Not Equals**
   - **Greater Than**
   - **Greater Than or Equals**
   - **Less Than**
   - **Less Than or Equals**
   - **Starts With**

Note

Note For Boolean conditions that evaluate to `!=true`, you can use the following RAD expression:

```
( nullsub(Boolean in $L.file,false) != true)
```

6. Select a value type for the RHS. The following table describes the available value types for the RHS.

Field	Description
Blank/NULL	A blank or null value
Category	A field in the associated category definition of the current record
CurrentUser	The user currently logged in to Service Manager
CurrentRecord	A field in the name of the current record
CurrentWorkflowPhase	A field in the associated workflow phase definition of the current workflow
Object	A field in the Object definition of the current record
Phase	A field in the associated phase definition of the current record
SavedRecord	A field in the original copy of the current record (before the user made any changes)
SavedWorkflowPhase	A field in the associated workflow phase definition of the original copy of the current workflow (before the user made any changes)
Security	An operator's security rights. You can choose rights from the drop-down list of defined values, such as: Change, Change Management Configuration, Change Tasks, Knowledge Management, Knowledge Administration, Security, or Tailoring. **Note** Note The Knowledge Management value is displayed as "Knowledge Document" in the back-end RDBMS and after you build the condition.
UserOption	A user-selectable option for a Service Catalog item **Note** Note ○ You must type the name of the user option. ○ The User Options field only supports string and boolean values.

Value	Allows you to type a value that's appropriate for the data type on the LHS. **Note** Note If a Global List is defined for the field that you specified on the LHS, a drop-down list of values is available for selection on the RHS. These values are defined in the Value List field of the Global List.
Variable	Evaluates the value of a global, local, or thread variable condition.

7. Select the field for comparison.

Note
Note To quickly navigate to a field, move the focus to the field drop-down list, and then type the caption of the desired field. To quickly navigate to a field in a second-level list, move the focus to the first-level item in the drop-down list, and then type the caption of the desired field.

8. Click **Done** to create the condition. Once you add a condition, Service Manager displays a user readable version.

Note
Note

- To edit an existing condition, open the condition by using the Condition Editor, and then click anywhere in the Condition Editor to enable editing.
- To delete an existing condition, click the Delete Condition icon (**-**) beside the condition that you want to delete.
- To cancel the creation of a condition, click **Cancel** to exit the Condition Editor.

Related topics

Process Designer
Launch the Condition Editor from a rule
Launch the Condition Editor from a workflow
Create a group of condition items

6.3.4.4. Create a group of condition items

Applies to User roles: System Administrator, Implementer

You can create complex conditions by grouping condition items. A separate logical operator applies to each group. You can add child and sibling groups to a complex condition.

To add a group of condition items, click the Add Condition Group icon in the Condition Editor. To add a child group of condition items, click the Add Condition Group icon in the parent group.

To remove a group of condition items, click the Delete Condition icon (**-**) in the group.

Note
Note This action deletes all condition items within the group.

You can easily edit complex conditions by dragging groups and condition items. You can drag a condition item into any group within the complex condition. You can drag a group into any other group, apart from its own child.

Related topics

Process Designer
Launch the Condition Editor from a rule
Launch the Condition Editor from a workflow
Build a condition
Copy a condition

6.3.4.5. Copy a condition

Applies to User roles: System Administrator, Implementer

You can copy and paste conditions between Condition Editors. This saves you time if you have a complex condition that you need to use multiple times within your environment.

To copy a condition, follow these steps:

1. Open the Condition Editor in the rule or workflow that contains the condition that you want to copy.
2. Click the More Actions icon (**...**), and then select **Copy**.
3. Close the Condition Editor, and then open the Condition Editor in the rule or workflow in which you want to paste the condition.
4. Click the More Actions icon (**...**), and then select **Paste** to copy the condition into the rule or workflow.

Note

Note You can copy whole conditions only. You can't copy the individual simple conditions or groups of conditions that comprise a complex condition.

Related topics

Process Designer
Launch the Condition Editor from a rule
Launch the Condition Editor from a workflow
Create a group of condition items
Build a condition

6.3.4.6. Enable deprecated fields and system fields in the Condition Editor

User Roles: System Administrator

A deprecated field is a field that's assigned the `Deprecated` usage type in the data policy. A system field is a field that's assigned the `System` data type in the data policy.

To display deprecated fields and system fields in the Condition Editor, follow these steps:

1. Click **Tailoring** > **Process Designer** > **Configuration**> **Settings** to display the Condition Editor Settings screen.
2. Select the **Display Deprecated Fields in Current Record** checkbox if you want to display deprecated fields in Current Record. This option is selected by default.
3. Select the **Display System Fields in Current Record** checkbox if you want to display system fields in Current Record. This option is selected by default.
4. Click **Save**.

The selected field types will now appear in the drop-down lists in the Condition Editor.

Related topics

Process Designer
Launch the Condition Editor from a rule
Launch the Condition Editor from a workflow

6.3.5. Task Planner

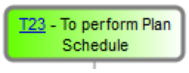
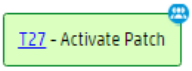
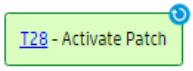
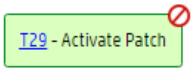
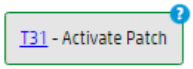
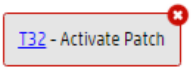
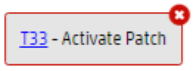
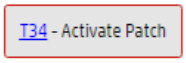
Task Planner enables you to schedule tasks in Service Manager modules such as Change Management and Request Fulfillment.

Task Planner graphical interface

Task Planner graphical interface supports panning and zooming.

- To zoom in or zoom out Task Planner graphical interface, click **Zoom in** or **Zoom out** in the toolbar.
- To reposition the workflow within the graphical interface, move the outline box at the top right of the graphical interface, or click and drag the workflow background.

The following table describes the various statuses of a task and their appearance in the graphical interface.

Task status	Graphic style	Remarks
0-Waiting		
0-Planned (opened)		This style appears when a task is created but still in Planned status because it is dependent on another task.
1-Ready		
2-Assigned		
3-In Progress		
4-Blocked		
20-Completed		
21-Completed with problem		
30-Canceled		
31-Withdrawn		
32-Failed		

Related topics

Process Designer
Process Designer security model
Create a rule set

6.3.5.1. Plan a task with Task Planner

Applies to User Roles: System Administrator, Implementer

You can use Task Planner to plan tasks for current or future phases of records that support tasks.

To plan a task by using Task Planner, follow these steps:

1. Open the record for which you want to plan a task.
   For example, click **Change Management** > **Change Queue**, and then select a Change record from the queue. Or, click **Request Fulfillment** > **Search Requests**, enter your search criteria, and then click **Search**.

2. Click the **Tasks** tab in the record, and then click **Edit** in the Planned Tasks pane to open Task Planner.

   Note

   Note If you open Task Planner before a record is created (that is, a new record that you have not yet saved) or after a record is closed, Task Planner is read-only. In this situation, the **Add Task** and **Save** buttons are invisible.

3. Click **Add Task**, and then do the following:
   1. Type the task title in the **Description** text box.
   2. Select task category from the **Task Category** drop-down list.
   3. Select a phase to start the task from the **Open In Phase** drop-down list.

      Note

      Note

      - If you plan a large number of tasks for a single phase in a workflow, you may experience slow performance. Therefore, we recommend that you don't create more than ten tasks that start in the same phase.
      - If the start phase in the task is the same as the current phase of the parent record, the task is created with a task number assigned. Otherwise, the task remains in the planned status until the parent record reaches the start phase (the "open in" phase) defined for this task.

   4. Select a phase to end the task from the **Close By Phase** drop-down list.

      Note

      Note The end phase must be no earlier than the start phase. If left blank, the end phase matches the start phase.

4. To add another task, click **Add Task**. To add a dependent task, hover the mouse on an existing task until you see a hand symbol, and then drag the mouse pointer to add the new sequential task.
5. Click **Save** to save the planned tasks.

   Note

   Note

   - After you finish adding tasks, you can click **Auto Layout** to automatically arrange the tasks and dependencies in the workflow.
   - Once the task record is created, you can't delete the task or modify its properties in Task Planner. You can click the task number in the graphical interface to modify the details of a created task record.
   - The **Mark as required/Set properties as read-only in change** field can only be set for tasks that are defined in Task Planner in the change model and request model. These fields are read-only in Task Planner, and are displayed for your reference.
   - If a task is not marked as required in a change model or request model and has not been created, you can modify the properties of this planned task, or you can click **Delete** to remove the task. However, you can't move the "open in" phase of a planned task to a phase prior to the current phase of the record.

Related topics

Process Designer
Process Designer security model

6.3.5.2. Setting the input and output data in Task Planner

The Input and Output fields in Task Planner enable you to configure the information that's pulled from the parent record into a task record, or pushed from the task record to the parent record.

Task Planner shares the following types of information between the parent and task records:

- Values in fields in the parent record
- User options
- Additional properties

Information that's pulled into the task record from the parent record ("input") is displayed in a read-only field in the **Additional Properties** tab of the task record. If you configure "output" (information that's pushed from the task record to the parent record), an editable field is displayed in the **Additional Properties** tab of the task record. Values that you enter into this field are shared with the parent record.

To configure the input and output of planned tasks, follow these steps:

1. Open the record for which you want to plan a task.

   For example, click **Change Management** > **Change Queue**, and then select a Change record from the queue. Or, click **Request Fulfillment** > **Search Requests**, enter your search criteria, and then click **Search**.
2. Click the **Tasks** tab in the record, and then click **Edit** in the Planned Tasks pane to open Task Planner.
3. Select the task for which you want to configure input or output, and then open the **Input-Output** tab.

   Note

   Note You can't configure the input and output of active tasks.

4. To configure input, click the icon at the right of the **Input** field, and then select one or more values from the drop-down list. If the value that you require doesn't appear in the list, you must create it as an additional property. For more information about how to do this, see the "" section.
   To configure output, click the icon at the right of the **Output** field, and then select one or more values from the drop-down list.

   Note

   Note The output field supports additional properties only.

5. Click **Save** in the toolbar to save your changes.

Related topics

Process Designer
Process Designer security model
Configure the additional properties in Task Planner

6.3.5.3. Configure the recommended maximum number of tasks

Service Manager processes planned tasks sequentially. Therefore, if you plan a large number of tasks for a single phase in a workflow, you may experience slow performance when a record moves to that phase and the tasks are processed.

To help prevent users from planning too many tasks and degrading the system performance, you can configure the recommended maximum number of tasks for phases. When a record that has more than the maximum recommended number of tasks is saved, a warning message is displayed to the current user.

To configure the recommended maximum number of tasks, follow these steps:

1. Click **Tailoring** > **Process Designer** > **Configuration** > **Settings**.
2. In the **Suggested Maximum Task Number in Task Planner** field, enter the desired maximum number of tasks for each phase. By default, the value is 10.

> **Note**
> Note We don't recommend that you configure a value over 10.

3. Click **Save** in the toolbar to save your changes and exit.

6.3.5.4. Configuring Task Planner

You can perform the following tasks to configure Task Planner to your requirements:

- Add fields to Task Planner
- Edit additional fields in Task Planner
- Remove additional fields from Task Planner
- Configure the status mapping between the task file and Task Planner
- Configure the additional properties in Task Planner

Related topics

Process Designer
Process Designer security model

6.3.5.4.1. Add fields to Task Planner

You can add fields to Task Planner, and determine whether they are mandatory or not. Additional fields are associated with a specific type of task file (for example, the change task file), and can be configured at both the file and task category level. This determines whether the additional field is displayed whenever Task Planner is started for that type of task file, or whether the field is displayed once the user has entered a value in the Task category field.

> **Note**
> Note You can't remove or modify the fields that are present by default (that is, the Title, Task Category, Open In Phase, Close By Phase, Task template, Task Condition, and Assignment Rule fields).

The following example adds the **Assignment Group** field in the **requestTask** table to Task Planner.

To add a field to Task Planner, follow these steps:

1. Click **Tailoring** > **Process Designer** > **Configuration** > **Predefined Additional Task Fields**.
2. Select **requestTask** from the **File Name** drop-down list, and then click the **New** icon.
3. If you want to configure an additional field at the file level, leave the **Task Category Name** field empty. If you want to configure an additional field that applies only to a specific task category, click **Fill** to select a task category in the **Task Category Name** field.

   > **Note**
   > Note If one field is defined at both the file and category levels, the setting in the category level overrides the setting in the file level.

4. Click **Add Field**.
5. In the **Field Name** drop-down list, select the field that you want to add to Task Planner. For example, select **Assignment Group**.
6. If you want this additional field to be mandatory, select the **Mandatory** option.
7. If this field has a link definition defined, select the **Is Record in Table?** checkbox. For example, a link definition exists for the **Assignment Group** field because the values of this field should be retrieved from the **assignment** table.
   The Record Information area appears. If only one relationship record of this field exists in Relationship Manager, this area is automatically populated with the relationship information. If multiple relationship records exist, you need to click the Fill icon in the **Relationship Id** field to select the correct relationship record. If no relationship records of this field exist in Relationship Manager, no relationship information is displayed.
   See the following table for more information.

Field	Description	Example value
Relationship Id	This is the Id of the relationship record, which you can find in Relationship Manager by performing a search. For example, for the **Affected CI** (logical.name) field in cm3t, enter table name cm3t and field name logical.name, and then click **Search**.	5914
Table	This is the name of the other table in the relationship record.	assignment
Field	This is the name of mapping field in the other table.	name

You can define a query by clicking **Edit Query** so that only values that meet the specified conditions will be available for selection in Task Planner. For example, you can specify this query:
(Time Zone = "US/Pacific")

Caution

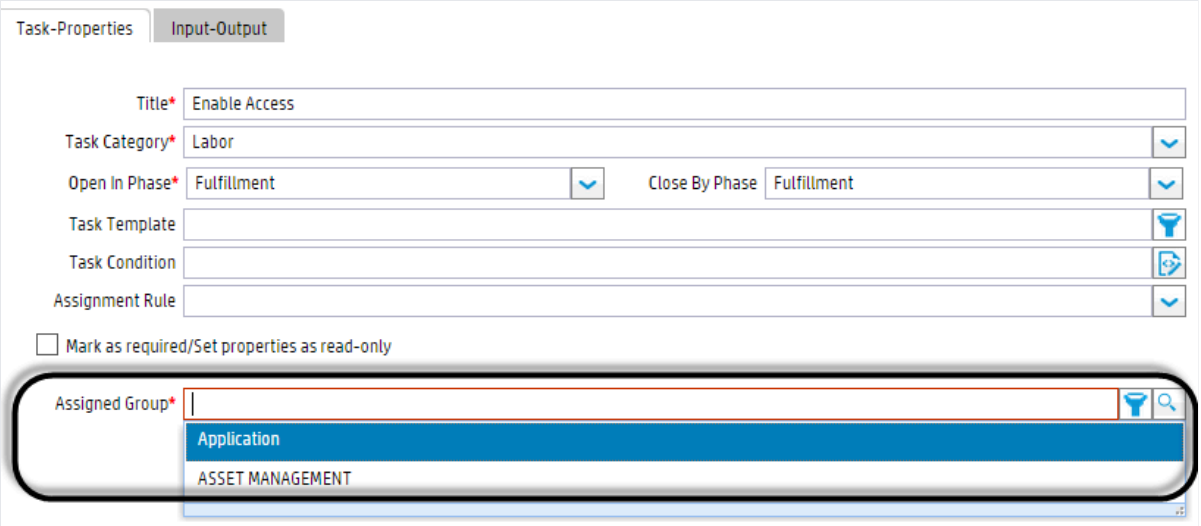
Caution If you want to add an array field (for example, Affected CI) to Task Planner, don't select the **Is Record in Table** option. Additionally, in Task Planner, you can't select values for the field by clicking Fill; instead, you can only manually enter values by using one of the following formats depending on the array type:
• array character {"a","b"}
• array number {1,2}
• array datetime {'09/02/15 13:35:42'}
• array boolean {true, false}

8. Click **OK**. The field is now added.

Tip

Tip You can click **Edit Field** to edit the field or click **Delete Field** to delete the field. You can also click **Add Field** to add more fields.

9. Click **Save** to save the record.
10. Verify that the field is added to Task Planner.
a. Go to **Request Fulfillment** > **Configuration** > **Request Models**, and open a request model record by performing a search. For example, open the out-of-box Application Access request model.
b. Go to the Task section and click **Edit** to open Task Planner.
c. Select a task. For example, select the out-of-box Enable Access task. The field you added (Assignment Group in this example) is displayed on the **Task-Properties** tab. If you click Fill in this field, a list of values based on the query conditions you set is displayed.



d. If you select a value for the field and then save the request model record, the task record is automatically populated with this field value when created later.

Related topics

Process Designer
Process Designer security model

6.3.5.4.2. Edit additional fields in Task Planner

To edit an additional field in Task Planner, follow these steps:

1. Click **Tailoring** > **Process Designer** > **Configuration** > **Predefined Additional Task Fields**.
2. Select a task file type from the **File Name** drop-down list, and then click the **Search** icon to display the additional fields that are configured for that type of task file.
3. In the table of additional fields, select the row that describes the field that you want to edit, and then click **Edit Field**.
4. Make your desired changes, and then click **OK**.
5. Click **OK** in the toolbar to save your changes.

Note

Note You can't remove or modify fields in Task Planner that are present by default. These fields are Title, Task Category, Open In Phase, Close By Phase, Task template, Task Condition, and Assignment Rule.

6.3.5.4.3. Remove additional fields from Task Planner

To remove an additional field from Task Planner, follow these steps:

1. Click **Tailoring** > **Process Designer** > **Configuration** > **Predefined Additional Task Fields**.
2. Select a task file type from the **File Name** drop-down list, and then click the **Search** icon to display the additional fields that are configured for that type of task file.
3. In the table of additional fields, select the row that describes the field that you want to remove, and then click **Delete Field**.
4. Click **OK** in the toolbar to save your changes.

Note

Note You can't remove or modify fields in Task Planner that are present by default. These field are Title, Task Category, Open In Phase, Close By Phase, Task template, Task Condition, and Assignment Rule.

6.3.5.4.4. Configure the mapping between the parent file and the task file

The different statuses through which tasks in Service Manager pass don't correspond exactly to the different statuses that Task Planner uses to describe the progress of tasks. In order for the Task Planner graphical interface to display the correct

status for each task, you must map the statuses of each task file type to the statuses in Task Planner.

Tip

Tip Mapping records for both Change Tasks and Request Tasks are included in out-of-box Service Manager systems.

For more information about the Task Planner graphical interface and a list of the task statuses, see Task Planner.

To create a new map between the statuses of a task and the statuses displayed in Task Planner, follow these steps:

1. Click **Tailoring** > **Process Designer** > **Configuration** > **Task Planner Mapping**.
2. Click **New** in the toolbar.
3. In the **Parent File Name** drop-down list, select the type of record with which the task is associated. For example, select **request**.
4. In the **Model File Name** field, select the model level for the task. For example, **requestModel**.
5. In the **Task File Name** field, select the name of the task file. For example, **requestTask**.
6. In the **Task Status Field Name**, select the field in the task record that contains the task status. For example, select **Status**. The left-hand **Task Status** column in the table is populated with all the values that are defined in the global list for that field.
7. In the right-hand **Task Planner Status** column, select the Task Planner status to which you want to map the corresponding task status. For example, map the **CMDB Update** task status to the **In Progress** Task Planner status. Repeat this until you have mapped all task statuses to a Task Planner status.
8. Click **Save** in the toolbar to save your changes.

Related topics

Process Designer
Process Designer security model

6.3.5.4.5. Configure the additional properties in Task Planner

The additional properties in Task Planner enable you to share information between task records and parent records, and also between task records. When you configure the input and output of a planned task, the options available for selection are typically user options or fields in the parent record. In order to make other options available, you must configure the additional properties.

Add an additional property

To add an additional property, follow these steps:

1. Open the record for which you want to plan a task.

   For example, click **Change Management** > **Change Queue**, and then select a Change record from the queue. Or, click **Request Fulfillment** > **Search Requests**, enter your search criteria, and then click **Search**.
2. Click the **Tasks** tab in the record, click **Edit**, and then click **Additional Properties** to open the Additional Properties information page.
3. In the toolbar, click **Add Additional Properties** to open the New Additional Property page.
4. In the **ID** field, type an identifier for the additional property.

   Note

   Note The identifier can't start with either a number or a punctuation mark.
5. In the **Display Label** field, type the name of the field as you want it to appear in the Additional Properties tab of the task record.
6. In the **Type** drop-down list, select the type of data that you want to share, and then click **Next**.
   Depending on the type of data that you choose, you may have to enter further information. For example, if you select **Record**, you must then specify the table and the field in which the record is located.

> **Note**
>
> Note If you select **Date/Time** in the **Type** drop-down list, you can only enter a static date/time value, for example: `07/31/2019 00:00:00`. The value you entered will be parsed according to the Date Format defined in your operator record.

7. (Optional) If the additional property requires validation, enter the validation script into the **Validation Script** field.
8. Click **Finish** to save your changes.

The additional property is now available for selection when you configure input and output values in Task Planner.

Edit an additional property

To edit an additional property, follow these steps:

1. Open the record for which the task that contains the additional property is configured.
   For example, click **Change Management** > **Change Queue**, and then select a Change record from the queue. Or, click **Request Fulfillment** > **Search Requests**, enter your search criteria, and then click **Search**.
2. Click the **Tasks** tab in the record, click **Edit**, and then click **Additional Properties** to open the Additional Properties information page.
3. In the toolbar, click **Edit Additional Properties** to open the New Additional Property page.
4. Make the required changes to the additional properties definition.
5. Click **Finish** to save your changes.

Delete an additional property

To delete an additional property, follow these steps:

1. Open the record for which the task that contains the additional property is configured.
   For example, click **Change Management** > **Change Queue**, and then select a Change record from the queue. Or, click **Request Fulfillment** > **Search Requests**, enter your search criteria, and then click **Search**.
2. Click the **Tasks** tab in the record, click **Edit**, and then click **Additional Properties** to open the Additional Properties information page.
3. In the toolbar, click **Edit Additional Properties** to open the New Additional Property page.
4. Click **Remove This Additional Property**, and then click **Yes** to confirm the deletion.

Related topics

Process Designer
Process Designer security model
Setting the input and output data in Task Planner

6.3.5.4.6. Configure child task behavior when the parent task is not created

If you have a number of interdependent tasks in Service Manager, a situation may arise in which a parent task is not created. You can configure the behavior of the child task in this situation.

For example, assume that you use Task Planner to add three tasks (T1, T2, and T3) to a Request model. T2 is dependent on T1, and T3 is dependent on T2. Additionally, T2 includes a condition that evaluates to "false," so that when you create a Request by using this Request model, T1 is created, T2 is not created, and T3 is created in the "Waiting" state. If you then close T1 and refresh the Request, T3 is moved to the "Active" state by default. However, you can configure Task Planner to leave T3 in the "Waiting" state in this situation.

To configure the behavior of child tasks when the parent task is not created, follow these steps:

1. In System Navigator, click **Tailoring** > **Process Designer** > **Configuration** > **Settings**.
2. Select the **Enable Strong Task Dependency in Task Planner** option if you want the child task to remain in the "Waiting" state.
   Clear the **Enable Strong Task Dependency in Task Planner** option if you want the child task to move to the "Active" state.

Note

Note The **Enable Strong Task Dependency in Task Planner** option is not selected by default.

3. Click **Save**.

Related topics

Process Designer
Create a workflow
Create a request model
Create a request task

6.3.6. Process Designer workflows

A workflow is a collection of phases with transitions from one phase to another. Each phase represents the state of the workflow linked to a form for data capture. Transitions can connect phases in any manner such as creating parallel branches or loop backs to previous branches.

A workflow enables an implementer to graphically layout the entire lifecycle of a process or record without any programming knowledge. The end user of the workflow views the same graphical workflow to see the current phase and past transitions.

Before you create a new workflow, you must add rules and rule sets to the phases and transitions of your intended workflow.

Related topics

Workflow phases
Workflow transitions
Add a phase

6.3.6.1. Create a workflow

Applies to User roles: System Administrator, Implementer

Note

Note You can create a workflow only in the web client.

After you have a basic workflow, you can then modify it by adding additional phases, transitions, rules, and actions to match the business process.

You can click **Zoom in** or **Zoom out** in the toolbar either to zoom in or zoom out on the workflow page.

To create a workflow, follow these steps:

1. Click **Tailoring** > **Process Designer** > **Workflows**.
2. Click **New**.
3. Type a name for the workflow. For example, demo_Workflow.

Note

Note You can't modify a workflow name once it is set.

4. Type a description for the workflow.
5. Select an Service Manager table from the drop-down list to associate the table with the new workflow, for example kmdocument.

Note

Note You can use the kmdocument table to manage the Knowledge Management document process. You can use the WorkflowDemo table to explore Process Designer capabilities.

6. Click **OK**. The new workflow **demo_Workflow** appears in the Workflows list.

Note

Note

- The **Proprietary** workflows are accompanied by a company logo in the workflow list. Non-proprietary workflows aren't accompanied by a logo.
- If you want to edit a **Proprietary** workflow or use it as a model, create a copy of this Proprietary workflow.

After you created a basic workflow, modify the Object record of the table for the new workflow.

Note

Note Modifying the Object record is only required for the Knowledge Management module. All other modules that use Process Designer set the workflows in each of the Category records. For other modules using Process Designer, update the category record.

To modify the object record, follow these steps:

1. Click **Tailoring** > **Document Engine** > **Objects**.
2. Under File name, provide the table name of the associated workflow. For example, "kmdocument."
3. Select **In Object** in **Workflow Location** drop-down list.
4. Select the workflow name in Workflow drop-down list.
5. Click **Save**.

Related topics

Workflow transitions
Workflow phases
Add a phase

6.3.6.2. Copy an existing workflow

Applies to User roles: System Administrator, Implementer

You can clone an existing workflow to use it in another business process or if you need to make changes to a **Proprietary** workflow. You also have an option to copy the rule sets of the existing workflow.

To copy an existing workflow, follow these steps:

1. Click **Tailoring** > **Process Designer** > **Copy Existing Workflow**.
2. In the Clone a Workflow form, select the workflow that you want to copy.
3. Type a new workflow name.
4. Click **Copy rule sets** checkbox if you want to copy rule sets.
5. Type a rule set prefix.
6. Click **Next**.
7. Click **Workflows** . The newly-copied workflow is displayed.

6.3.6.3. Export a Workflow

You can export a workflow from Process Designer into a Service Manager unload file. You can then use that unload file to import the workflow into another Process Designer-based Service Manager system.

Note

Note The export only includes the related records in the Workflow, WorkflowPhase, and RuleSet tables. Processes and Java Scripts called from rules sets can't be exported.

To export a workflow to an unload file, follow these steps:

1. Navigate **Tailoring** > **Process Designer** > **Export Workflow**.
2. In the Export/Unload a Workflow tab, select a Process Designer Table from the **Table** drop-down.
3. Select the workflow from the **Workflow Name** field.
4. Enter an appropriate name in the **External File Name** box.
5. Select or clear the **Append to file** checkbox as appropriate.

   Note

   Note: If you wish to append this workflow to an existing unload file, enter the unload file name, and make sure that the **Append to file** checkbox is checked. You can use this mechanism to build a single unload file that contains multiple workflows that you can later import into another Process Designer-based Service Manager system.

6. Click **Ok**.

After the file is created, save the file to an appropriate location. You can then import the generated unload file to in another Process Designer-based Service Manager system.

6.3.6.4. Workflow phases

Workflow phases show the state of a record in the workflow and provide processing that must occur for the record to move to the next phase. Phases use transitions to move from one phase to another phase. Some of this processing can be manual, or it can be triggered by the rules or actions specified for that specific phase.

Within a workflow, you can specify a form to display when that phase is active. Use different forms on different phases to allow different information to be displayed and captured at various stages of the workflow.

Rule sets are defined in the phase or transition to enforce business logic, such as checking if the user has entered the required data or has proper security to perform a transition.

Related topics

Workflow transitions
Add a phase

6.3.6.4.1. Add a phase

Applies to User roles: System Administrator, Implementer

You can add a phase to an existing workflow by using the graphical interface of the workflow editor.

Note

Note

After you add a phase and click **Save**, you can't modify its name. You can modify only its display name.

To add a phase to a workflow, follow these steps:

1. Click **Tailoring** > **Process Designer** > **Workflows** from the System Navigator to display the workflows list.
2. Select the applicable workflow for which you want to add a phase.
3. Click **Add phase**.
4. Drag the mouse to add a new transition and a phase.
5. Enter the details described in the tables.

Details

Phase Order	The Phase Order field gives a numerical order number to each phase of a workflow. These numbers are used in calculations for Service Level Targets (SLTs) and similar metrics, so that the number and timestamps of entries and exits from specific phases can be tracked. For example, an SLT calculation can determine the time of entry to the final phase and therefore determine whether a breach has occurred. As best practice, you should specify your starting phase as 1, and your closing phase as the highest number. We also recommend that these numbers should be roughly sequential from phase to phase. However, some workflows may loop multiple times through a sequence or take divergent paths.
Name	Type the name of the tab.
Display name	Type the display name of the tab.
Table name	The selected table name during workflow creation. You can't modify it.
Form Edit condition	If the condition evaluates to true for a user, who will be able to edit the form. If it doesn't, the form will be read-only.
Records in this phase are active	Select the checkbox if you want the records in this phase to be active.
Make this the first phase	Click **Make this the first phase** if you want it to be the first phase.
Make this the default phase	Click **Make this the default phase** if you want it to be the default phase. Click **Save**. **Note** Note If the current phase of a record is set to a phase that doesn't exist in the current workflow, it will be moved to the default phase. This may occur if a phase is removed from a workflow or if data is imported from another source that did not share the same workflow.

Additional Phase Information	Click **Additional Phase Information** to open the **Extended Phase Information** page to modify phase information. **Note** Note Only the Change Management module supports this feature. You can't edit or delete a phase name from the Extended Phase Information page or the cm3rcatphase.main form. Change Management workflows have unique workflow phases but they will share change phases if the workflow phases have the same name. For example, if the Workflow1 and Workflow2 each have a phase named 'Build and Test' they will share the same change phase record.
Description	Enter a description of the workflow phase. This field supports hyperlinks.

Forms

Display form	The display form is the primary method to capture and display data. If you don't specify a form, the system will look for a form that has the same name as the current table name. To add a display form, follow these steps: 1. Select the display form from the **Default Display form** drop-down list. 2. Click **Save**. To add a display form that's selected by a condition or set of conditions, follow these steps: 1. In the Additional/Display forms section, click **Add**. 2. In the **Name** drop-down list, select the form name. 3. In the **Description** field, type a description of the form. 4. Click the **Form Condition** field and use the condition editor to enter a condition. 5. In the **Type** drop-down list, select **Display Form**. 6. Click **Update** to add the additional form. **Note** Note You may also specify a form in a RAD expression, for example “display.form in $L.category”.
Additional Forms	You can add, edit, and delete additional forms for a workflow phase. To add an additional form to a workflow record, follow these steps: 1. Click **Add**. 2. In the **Name** drop-down list, select the table name. 3. In the **Description** field, type a description of the form. 4. Click the **Form Condition** field and enter a condition. 5. In the **Type** drop-down list, select **Additional Form**. 6. Click **Update** to add the additional form.

Rule sets

Add	To add rule sets at various stages: 1. Click the **On enter**, **After successful enter**, **On exit**, **Initialization**, **On display**, **On update**, or **After successful update** tab. Form more information, see the note after step 4. 2. Click **Add**. 3. Select the appropriate rule sets checkbox you want to add. 4. Click **Save**.

Note

Note

You may also specify a form in a RAD expression, for example “display.form in $L.category”.

- **On enter** – Runs when the record tries to move from another phase into this phase. For example, the rule set can set the time at which the record first entered the phase or send notifications that the record has entered the phase.
- **After successful enter** – Runs after the record successfully moves from another phase into this phase.
- **On exit** – Runs when the record moves out of this phase. For example, the rule set can set the time at which the record left the phase or send notifications that the record has left the phase.
- **Initialization** – Runs once just before the record is displayed to the user. For example, the rule set can set up variables for display that aren't meant to change while the user is viewing the record.
- **On display** – Runs each time the record is displayed after a user action. For example, if the user uses the “fill” function to populate a field, the display rules will run after the action is completed and before the form is displayed. A possible use may be to populate a variable that's displayed on the form, which is calculated based on other values in the form.
- **On update** – Runs immediately before the record is updated in the database (or created if this is a new record). For example, the rule set can validate field data (the record will not be updated if validations fail), set default values, or perform calculations on existing fields.
- **After successful update** – Runs immediately after the record is updated in the database (or created if this is a new record). For example, the rule set can send notifications of the record update or update related records based on changes to this record.

Actions

Add	Actions perform a task for the phase. In Knowledge Management, Actions are used to preview the document. Actions refer to rule sets that are marked as **Available as Action.** 1. Click **Add** to add an action, which runs associated rule sets for that phase of the workflow. 2. Type the identification name, which is how the action will appear in the Tray, More Options List, or on the button. 3. Select the action from the drop-down list. 4. Select the location of the action: **Tray**, **More Options List**, or **Button**. **Note** Note If you select **Button**, then manually add a button to the form with the option number of the action specified. 5. Click **Action Condition** to add a condition if required. 6. Select **Add**, **Save**, or **Delete** for action when complete. 7. Select **Requires lock** checkbox if you want to lock the record before the action can be performed. **Note** Note If another user has the record locked, you will not be able to perform the action. 8. Click **Save**. **Note** Note You can add up to 30 phase actions of the same type.

Approvals

Reset condition or Recalculate condition	1. Click **Reset Condition** or **Recalculate Condition**. 2. Click **Save**.
Add approvals	1. Click **Add** to add approvals. 2. Select desired approvals for the phase. 3. Click **Save**.

Alerts

Reset condition or Recalculate condition	1. Click **Reset Condition** or **Recalculate Condition**. 2. Click **Save**.

Note

Note

- To reposition the phase, select the phase, hover your mouse until you see a crosshairs symbol, and then move the phase to the desired location on the workflow page.
- To delete a phase, select the phase and either click the trashcan symbol next to the phase or **Delete** from the toolbar. You can't delete the default or first phases.
- To move multiple phases at the same time, press **Ctrl** and select multiple phases, and then move the phases or section of the workflow to a new location.
- In the **Rule Sets** tab, select a rule set and either double-click the rule set or click the **View** icon to open the rule set page.
- In the **Approvals** tab, select an approval name and either double-click the approval name or click the **View** icon to open the Approval Definition page.
- In the **Alerts** tab, select an alert name and either double-click the alert name or click the **View** icon to open the Alert Definition page.

6.3.6.4.2. Copy a phase

Applies to User roles: System Administrator, Implementer

You can copy a phase in an existing workflow by using the graphical interface of the workflow editor. Every attribute of the phase is copied, apart from the name and display name. This feature enables you to create workflow phases that have similar attributes quickly.

To copy a phase in a workflow, follow these steps:

1. Click **Tailoring** > **Process Designer** > **Workflows** from the System Navigator to display the workflows list.
2. Open the workflow in which you want to copy a phase.
3. Select the phase that you want to copy, click **Copy** in the toolbar, and then click **Paste**. Alternatively, you can press Ctrl+C and Ctrl+V after you select the phase to copy.
   The duplicate phase appears overlapping the copied phase. By default, the name of the new phase is "<*copied phase name*>_copy."
4. Modify the attributes of the phase as required.
5. Drag the copied phase to a convenient location in the workflow, and then add the necessary transitions.
6. Click **Save** to save your changes.

6.3.6.5. Workflow Viewer

The following sections explain the workflow viewer and how to integrate it into a knowledge document.

6.3.6.5.1. View a workflow in Workflow Viewer

Applies to User roles: System Administrator, Implementer

The Workflow Viewer enables you to view a graphical layout of the entire lifecycle of a process or record. The phases and transitions of the workflow are color-coded as follows.

Phase/transition
Current phase
Past phase
Future phase
Inactive phase
Past transition
Future transition

If a phase in a workflow contains approvals, an Approvals icon is displayed in the phase in the Workflow Viewer.

You can view information about specific parts of the workflow by hovering the mouse over that component in the Workflow Viewer. For example, hover the mouse over a transition between two phases to display the description of that transition. Or, hover the mouse over the Approvals icon to display the approval definition list (in future phases), the approvals information in the Approval table (in the current phase), or the approval history (in past phases).

To view a workflow, open a record, and then click the **Workflow** tab.

Note

Note In order to view a workflow in the Workflow Viewer, you must integrate a Workflow widget into any form that's associated with a Process Designer-enabled table. This is currently available for Knowledge Management (`kmdocument`), Change Management (`cm3r` and `cm3t`), and `WorkflowDemo` , Service Desk (`incidents`), Incident Management (`probsummary` and `imTask`), Problem Management (`rootcause` and `rootcausetask`), and Request Fulfillment (`request` and `requestTask` .

The procedure to integrate the viewer into each module is very similar.

Related topics

Add a phase

6.3.6.5.2. View the Workflow Properties

Applies to User roles: System Administrator, Implementer

You can use the Workflow Properties button to view the properties that are related to an entire workflow. You can see all the workflow-based Rule Sets, Actions, and Backend Transitions that are related to that workflow.

To view the Workflow properties, follow these steps:

1. Click **Tailoring** > **Process Designer** > **Workflows**. Or, from within a Process Designer module, click **Configuration** and then **Workflows**.
2. Double-click on any workflow.
3. After the Workflow tab opens, click the **Workflow Properties** button.
4. Click on any of the following tabs:
    - Workflow Properties
    - Workflow Based Rule Sets
    - Workflow Based Actions

- Workflow Backend Transitions

> **Caution**
>
> **Caution:** If you modify a rule set that you have accessed through the Workflow Properties button, those changes are applied to the rule set directly. Therefore, any other workflow that also uses this rule set will also be affected.

6.3.6.5.3. Integrate Workflow Viewer into a new form

Applies to User roles: System Administrator, Implementer

The Workflow Viewer icon is available in Forms Designer. The following example will help you integrate the Workflow Viewer widget into the `kmdocument.document` form. After the integration, you can see a graphical view of the workflow in the **Contribute Knowledge** > **External** > **Workflow** tab. Using similar steps, you can integrate the Workflow Viewer with other forms.

Follow these steps to integrate the Workflow Viewer widget into the `kmdocument` form.

1. Sign in as a System Administrator. In the System Navigator, click **Tailoring** > **Forms Designer** to open Forms Designer in the Windows client.
   For information on accessing Forms Designer and updating a form, see Access Forms Designer and Update a form.
2. In the Form field, type `kmdocument.external` and click **Search**. The Contribute Knowledge form opens with the External Document Upload tab selected.
3. Add the **Workflow** tab on the form.
4. Set the visible condition to `[$showWF]=true` in the Workflow tab.

   > **Note**
   >
   > Note The condition hides the tab in the Windows client, but is visible in the web client.

5. Add the Workflow Viewer component to the new Workflow tab.
6. In the Workflow Viewer component, set the following variables in the properties.
   `Workflow Name:  $L.wfgWFName`
   `Workflow Table: $L.wfgWFTable`
   `Workflow RecordID: $L.wfgWFRecId`
   `Workflow Current Phase: $L.wfgWFPhase`
7. Add four text boxes at the bottom of the Workflow Viewer.
8. Set their input values in the properties to:
   `$L.wfgWFName`
   `$L.wfgWFTable`
   `$L.wfgWFRecId`
   `$L.wfgWFPhase`

   > **Note**
   >
   > Note The four text boxes display workflow name, table name, document id, and phase of the record.

9. Set the Workflow Viewer variables in the **Main** > **Initialization** tab of display screens for `kmdocFlow.open` and `kmdocFlow.view` :
   `$L.wfgWFName=workflowName in $L.wfPhase`
   `$L.wfgWFTable=tableName in $L.wfPhase`
   `$L.wfgWFPhase=phaseName in $L.wfPhase`
   `$L.wfgWFRecId=id in $L.file`
   `if (sysinfo.get("environment")~="scguiwswt") then ($showWF="true") else ($showWF="false")`

Note

Note To go to the display screen, type `ds` in Service Manager command prompt and search for the `kmdocument.document` form.

10. Continue to follow *step 1* through *step 7* to add Workflow View in the other Knowledge Management forms: `kmdocuments` (`kmdocument.probsol` , `kmdocument.howto` , `kmdocument.reference` , and `kmdocument.errormsg`)

6.3.6.6. Workflow-based rule sets, actions, and transitions

Applies to User roles: System Administrator, Implementer

Process Designer allows you to apply rule sets, actions and transitions to a workflow as a unit, which are collectively referred to as workflow-based rule sets, actions and transitions. You can access the workflow-based rule sets, actions, or transitions by following the steps in View the workflow properties.

The following sections describe special considerations when you apply a rule sets, actions or transitions to an entire workflow.

6.3.6.6.1. Workflow-based rule sets

Applies to User roles: System Administrator, Implementer

You can add workflow-based rule sets to an entire workflow by viewing the **Workflow Properties**.

A workflow-based rule set is invoked on each phase of the workflow as long as the triggering events are triggered in the back-end.

A workflow-based rule set is comprised of six sub-rule sets.

Sub-rule set	Triggering action	Execution Order
On Add	Record Creation (Adding Record)	1. Add (before record is created in the database) 2. After Successful Add (after record is created in the database) 3. Initialize 4. On Display
After Successful Add	Record Creation (Adding Record)	1. Update (before record is updated in the database) 2. After Successful Add (after record is created in the database) 3. Initialize 4. On Display
On Enter	Phase Change	1. From phase On Exit 2. To phase On Enter 3. To phase After successful Enter
After Successful Enter	Phase Change	1. From phase On Exit 2. To phase On Enter 3. To phase After successful Enter
On Exit	Phase Change	1. From phase On Exit 2. To phase On Enter 3. To phase After successful Enter
Initialization	Search and access existing record	1. Initialize 2. On Display
On display	Current page refresh (caused by link execution or other action that still keeps at current format)	1. On Display
On Update	Record Update	1. Update (before record is updated in the database) 2. After Successful Update 3. Initialize 4. On Display
After Successful Update	Record Update	1. Update (before record is updated in the database) 2. After Successful Update 3. Initialize 4. On Display

6.3.6.6.2. Workflow-based actions

Applies to User roles: System Administrator, Implementer

In earlier versions of Process Designer, actions were applied to individual phases. In certain circumstances, the same actions had to be duplicated in each phase. For example, in cases where a certain button needed to be present in each phase, the workflow designer had to add the button manually to each phase.

In the current version of Process Designer, you can add workflow-based actions to an entire workflow by viewing the **Workflow Properties**.

A workflow-based action appears in each phase of the workflow when its condition is satisfied.

Note

Note

You can add up to 30 workflow-based actions of the same type.

6.3.6.6.3. Workflow transitions

Process Designer workflow transitions occur when a record moves from one phase to another phase. Transitions can happen manually, automatically, or by default. An automatic transition is taken when its condition is met. For example, the condition might check if a field value is a specific value. In this case, the transition is taken automatically when the record is saved. An example of a manual transition is when an operator clicks the "Request Validation" button in the toolbar of a form. A default transition is a special transition type that moves the workflow automatically only when no other automatic transition conditions are satisfied.

Note

Note Transitions govern the flow of the workflow. However, in Change Management processes, an operator with Administrator rights can use the "Change Phase" menu option to bypass the workflow and jump to any phase.

After you add a transition, you can reposition it to avoid transitions overlapping with phases or other transitions. To reposition the transition, select the transition and move the green rectangular dot to achieve a desired layout.

You can also move a transition from one phase to another. To move the transition, select the transition, hover your mouse near the blue rectangular dot to see a hand symbol, and then move it to another phase.

Tip

Tip Workflow phases are connected by transitions to move from one phase to another phase. However, if you want to move to another phase from the current phase, you can use backend transitions. For more information, see the Process Designer Tailoring Best Practices.

Create a default workflow transition

Applies to User roles: System Administrator, Implementer

A default transition moves the workflow automatically only when no other transition conditions are satisfied.

Note

Note You can add rule sets to the default transition.

To create a default workflow transition, follow these steps:

1. Click **Tailoring** > **Process Designer** > **Workflows**.
2. Open the applicable workflow from the Workflows list.
3. Click the phase where the default transition will begin.

4. Hover your mouse until you see a hand symbol and then drag the mouse to add another phase.
5. Enter the following information.

Field	Description
Transition Type	Evaluates to one of the following transition types: **Manual**, **Automatic**, or **Default**. Select the **Default** transition type.
Description	Type a description of the transition. This description is displayed when you hover the mouse over the transition in the Workflow Viewer.
Rule Sets	Evaluates to the rule sets you created for the transition. 1. Click **Add** and then select the ID of the rule set you want to add. 2. Click **OK**.

6. Click **Save**.

You have added a default transition.

Create an automatic workflow transition

Applies to User roles: System Administrator, Implementer

An automatic transition moves the workflow to another phase based on data in the workflow record.

Note

Note You can add a condition and rule sets to the automatic transition.

To create an automatic workflow transition, follow these steps:

1. Click **Tailoring** > **Process Designer** > **Workflows**.
2. Open the applicable workflow from the Workflows list .
3. Click the phase where the automatic transition will begin.
4. Hover your mouse until you see a hand symbol and then drag the mouse to add another phase.
5. Enter the following information.

Field	Description
Transition Type	Evaluates to one of the following transition types: **Manual**, **Automatic**, or **Default**. Select the **Automatic** transition type.
Description	Type a description of the transition. This description is displayed when you hover the mouse over the transition in the Workflow Viewer.
Condition	Evaluates to a condition that you can add to the transition.
Rule Sets	Evaluates to the rule sets you created for the transition. 1. Click **Add** and then select the ID of the rule set you want to add. 2. Click **OK**.

6. Click **Save**.

You have added an automatic transition.

Create a manual workflow transition

Applies to User roles: System Administrator, Implementer

When you set up a workflow, you can require an operator action to move a record from one phase to another. This type of transition, in which an operator must press a button or otherwise trigger an action, is a manual transition.

To create a manual workflow transition, follow these steps:

1. Click **Tailoring** > **Process Designer** > **Workflows**.
2. Open the applicable workflow from the Workflows list.
3. Click the phase where the manual transition will begin.
4. Hover your mouse until you see a hand symbol and then drag the mouse to add another phase.
5. Enter the following information.

Field	Description
Transaction Type	Evaluates to one of the following transition types: **Manual**, **Automatic**, or **Default**. Select the **Manual** transition type.
Command Name	Evaluates to command location: **Tray**, **More Options List**, or **Button**.
Description	Type a description of the transition. This description is displayed when you hover the mouse over the transition in the Workflow Viewer.
Condition	Evaluates to a condition that you can add to the transition.
Rule Sets	Evaluates to the rule sets you created for the transition. 1. Click **Add** and then select the ID of the rule set you want to add. 2. Click **OK**.
Save record prior executing the transition	Select this option if you want the record to be saved before the transition occurs.

6. Click **Save**.

You have added a manual transition.

Note

Note Click the **Localize Command Label** icon in the **Command Name** text box to open the **Service Manager Message** form of a transition. On this page, you can add a message record for a new transition or view the message record of an existing transition.

Related topics

Workflow phases

6.3.7. Add or edit a module status

Applies to User roles: System Administrator

Records in the ModuleStatus table determine the values that appear in the **Status** drop-down list of records. You can use the ModuleStatus table to:

- Add new statuses
- Edit existing statuses
- Alter the position in which statuses appear in the drop-down list
- Prevent statuses from being displayed in the drop-down list
- Attach additional conditions to statuses

> **Note**
>
> Note To ensure that module statuses work as expected, you must also correctly configure the rule set that determines which statuses to use in a workflow.
>
> For example, the Incident workflow is configured with an out-of-box "im.status.list" rule set that uses the statuses assigned to the Incident module.

Add a module status[edit]

To add a module status, follow these steps:

1. Click **Tailoring** > **Database Manager**.

   > **Note**
   >
   > Note Alternatively, you can open Database Manager from the command line, by typing db, and then pressing **Enter** or clicking the **Execute** icon.

2. Type ModuleStatus in the **Table** field, and then click the **Search** icon.
3. Select a module from the **Module** drop-down list, and then type a name for the status in the **Status** field.

   > **Note**
   >
   > Note Optionally, you can also enter a value for the sort order of the status, assign the status to specific phases of a workflow, and mark the status as a background status.

4. In the menu, click **Add** to create the status.

   > **Note**
   >
   > Note Once the status is created, you can click **Edit** to set a condition for the status.

Edit a module status[edit]

To edit a module status, follow these steps:

1. Click **Tailoring** > **Database Manager**.

   > **Note**
   >
   > Note Alternatively, you can open Database Manager from the command line, by typing db, and then pressing **Enter** or clicking the **Execute** icon.

2. Type ModuleStatus in the **Table** field, and then click the **Search** icon.
3. Enter search criteria for the module status that you want to edit, and then click **Search**. For example, select the Incident module from the drop-down list, and then click **Search** to display all available statuses for the Incident module.
4. Edit the status as required:
   - Use the Module drop-down list to change the module to which the status applies.

- Use the **Status** field to edit the display name of the status.
- Use the **Sort Order** field to set where in the status drop-down list this status is displayed.
- Use the **Workflow** drop-down list to set the workflow to which this status applies.
- Use the **Workflow Phase** drop-down list to set the workflow to which this status applies.
- Select the **Is Background Status** option to prevent the status from being displayed in the drop-down list.
- Click **Edit** to open the Condition Editor and set any additional conditions for the status.

5. Click **Save** to save your changes.

Contents

- 1 Add a module status
- 2 Edit a module status

6.3.8. Process Designer security model

The Process Designer security model provides a consistent method of assigning permissions to users across all facets of Service Manager data and accounts for out-of-box rights that can be configured for a specified role within an area. It also provides standardized methods to manage user rights.

> **Note**
>
> Note The Process Designer security model is implemented for Knowledge Management, Change Management, Service Desk, Incident Management, Problem Management, Request Fulfillment, and Service Level Management modules. For all other areas and modules, the traditional security features for Service Manager still apply.

The Process Designer security model includes the following components:

- **Area**: An area defines a specific functional area or module within Service Manager, such as Knowledge Management or Knowledge Management administration. Each area definition includes default rights that are copied to the role whenever a new role is created. In addition to the out-of-box areas, system administrators are able to define additional areas.
  In an out-of-box system, the following three security areas are shared by several modules: Tailoring, Common Configuration, and Security. These areas contain the default security rights and settings that apply to the Change Management, Service Desk, Incident Management and Problem Management modules:
  - The Tailoring area is used to set the permissions that control operator access to Workflows.
  - The Common Configuration area is used to set the permissions that control access to common configurations, including Alert Definitions, Approval Definitions, Assignment Groups, Service Desk/Problem Solution Matching, and Environment.
  - The Security area is used to set the permissions that control operator access to Security configurations.

  > **Note**
  >
  > Note The rights to access Settings are controlled by the separate Configuration area in each module.

- **Rights**: The system includes a set of rights such as view, new, update, and delete that control an operator's data access. When an administrator creates a role, the default rights from each area are used to set the rights for that role. Rights can be modified for a specific area and role by an administrator that has update rights for the area and role. Each combination of role and area creates a collection of rights.
  Security Rights also include the following configurations:
  - **Allowed Statuses**: This field displays the list of statuses that are available to operator when they access records. A System Administrator specifies the allowed statuses for a role within an area. When this list is populated, the role may only update the records that are in one of the listed statuses. If a record has a status that's not in the list, the role will not be able to modify it. However, when the role updates records, the statuses that are available for selection aren't limited to the list. If no statuses are listed, the role may modify records in any status.
  - **Allowed Categories**: This field displays the list of categories that are available to the operator when they access category data. A System Administrator specifies the allowed categories for a role within an area. When this list is populated, only the listed categories are available for selection when the role creates a new record or updates a record in the Category field. If no categories are listed, all categories are available for the role within the area.
  - **Admin**: This option indicates if the current Role has Admin rights for the current Area.
  - **Expert**: This option indicates if the current Role has Expert rights for the current Area.
  - **Modify Template**: This option indicates if the current Role is able to modify Template records for the current Area.
  - **ESS Only**: If this option is selected, the current Role has the specified rights for the current Area only when logged in to the ESS web client (ess.do). When this option is not selected, the specified rights take effect for

both the standard and ESS web clients (index.do and ess.do).

- **Settings**: Settings are configurable security extensions such as an initial view, a format to display a list, or an approval checkbox. Settings are added for an area. The types of settings include number, string, Boolean, date/time, global list, manual list, record, and condition.
- **Security Folders**: If Folder Entitlement is enabled in the system, a System Administrator must select the security folders that each security role can access. If a role is not granted rights to a specific folder, operators associated with that role will not be able to access records in that folder.
- **Roles**: A role has a set of rights and settings assigned to it. Each operator is assigned a role or roles which, along with area, determine the access rights for the operator. Whenever the roles on an operator record are updated, the operator must log out and then log in for the changes to take effect.

  Note

  Note The out-of-box system includes a default role for the Security area that can't be deleted.

- **Data Policy records**: The data policy records include an Area field used to specify the area associated with the table. An area needs to be associated with a Data Policy record in order to access the information from the table.

Related topics

Multiple security roles
Add a security role
Add an area

6.3.8.1. Out-of-box role rights in the Common Configuration area

The Common Configuration area is used to set the permissions that control access to common configurations, including Alert Definitions, Approval Definitions, Assignment Groups, Service Desk/Problem Solution Matching, and Environment.

Based on the mapping rules, the rights and settings in previous security profiles are mapped to the rights and settings in the Common Configuration area specified in the corresponding security roles. See the table below for the out-of-box security rights in the Common Configuration area. This table only lists the new security roles that have different settings with the default rights.

Area	Role	View	New	Update	Delete/Close	Modify Template	Expert	Admin
Common Configuration	change analyst change	True	True	Always	Always	False	False	False
	change analyst tasks	True	True	Always	Always	False	False	False
	change approver	True	True	Always	Always	False	False	False
	change coordinator change	True	True	Always	Always	False	False	False
	change coordinator tasks	True	True	Always	Always	False	False	False
	change manager	True	True	Always	Always	False	False	False
	configuration auditor	True	True	Always	Always	False	False	False
	emergency group	True	True	Always	Always	False	False	False
	incident analyst	True	True	Always	Always	False	False	False
	incident coordinator	True	True	Always	Always	False	False	False
	incident manager	True	True	Always	Always	False	False	False
	initiator	True	True	Always	Always	False	False	False

	operator	True	True	Always	Always	False	False	False
	problem analyst	True	True	Always	Always	False	False	False
	problem coordinator	True	True	Always	Always	False	False	False
	problem manager	True	True	Always	Always	False	False	False
	reviewer	True	True	Always	Always	False	False	False
	SD agent/manager	True	True	Always	Always	False	False	False
	service desk agent	True	True	Always	Always	False	False	False
	service desk manager	True	True	Always	Always	False	False	False
	system administrator	True	True	Always	Always	True	True	True
	Requestor	True	False	Never	Never	False	False	False
	Request Approver	False	False	Never	Never	False	False	False
	Request Coordinator	True	False	Never	Never	False	False	False
	Request Analyst	False	False	Never	Never	False	False	False
	Request Manager	True	True	Always	Always	False	False	False
	Request Process Owner	True	True	Always	Always	False	False	False
	Stock Manager	False	False	Never	Never	False	False	False

6.3.8.2. Multiple security roles

When an operator has multiple security roles, the operator's data access rights are combined to give the operator the greatest data access.

When the rights don't include specific allowed statuses and allowed categories, all statuses and categories are available for the role within the area. Refer to the following examples:

- If an operator has view rights for one role in an area and update rights for another role in that area, then the operator has view and update rights for the area.
- If an operator has allowed statuses and allowed categories specified in one role and not in another role, then the operator is able to access all categories and statuses for the area.
- If an operator has multiple roles that include different allowed statuses and allowed categories, the allowed statuses and allowed categories are merged so that the operator has all allowed categories and allowed statuses. No allowed statuses and allowed categories are duplicated.
- If an operator has multiple settings because the operator has been assigned multiple roles, the operator only has the actual settings for the first settings in the operator record .

Related topics

Add a security role
Add an area

6.3.8.2.1. Add a security role

Applies to User roles: System Administrator

When you create a security role, the rights from each area defined in the system are used to set the rights for that role record.

To add a role record, follow these steps:

1. Click **System Administration** > **Security** > **Roles**.
2. Click **New**.
3. Type the name of the role.
4. Type a description for the role.
5. Click **Save & Exit**.

Related topics

Process Designer security model
Roles in the operator record
Assign a role or roles to an operator

6.3.8.2.2. Add security roles and settings

Applies to User roles: System Administrator

To create a security role and assign rights and settings, follow these steps:

1. Click **System Administration** > **Security** > **Roles**.
2. Click **New**. The **Security Role** form opens.
3. Type the security role name in the **Name** field.
4. Type the security role description in the **Description** field.
5. Click **Save**.
6. Select the security area.
   The **Security Rights and Settings** form opens.
7. Under **Rights**, select the rights to be assigned to the security role. For example, set **Expert** rights for the security role. The **Expert** security right enables the operator to view alert log, opened tasks, affected services, and clocks of change request. It also enables you to set reminders, send notifications, create hot news, and associate change request to changes, incidents, interactions, requests, and know errors.
8. Under **Settings**, add required settings.
9. Under **Folders**, add folder permissions to the security role.

The out-of-box security folders available in Service Manager are **DEFAULT** and **advantage**. You can also create security folders to meet your business needs. By default, all security folders are assigned to a new security role created. Once a role is created and rights are configured, you can modify the security rights for a role within an area.

Related topics

User roles

6.3.8.3. Roles in the operator record

System Administrators assign roles to an operator in the Security Roles field of the operator record. When a System Administrator creates an operator record, the system assigns the default role to the operator if the System Administrator doesn't add a role for the operator. The System Administrator can update the operator record to change a role or add additional roles.

When an operator has multiple roles, the operator has the rights that provide the greatest data access. However, if each role has additional settings, the rights from the first role listed in the Security Rights field is used to determine an operator's access rights.

Related topics

Process Designer security model
Assign a role or roles to an operator

6.3.8.3.1. Assign a role or roles to an operator

Applies to User roles: System Administrator

You assign a role or multiple roles to an operator to provide the operator access to Service Manager data. You can also update the operator's role in the operator record.

To assign a role or roles to an operator record, follow these steps:

1. Click **System Administration** > **Ongoing Maintenance** > **Operators**.
2. Enter the applicable search criteria to find an operator record and then click **Search**.
3. In **Security Roles**, select the role or roles you want to assign to the operator.
4. Click **Save**.

Related topics

Process Designer security model
Multiple security roles
Add a security role

6.3.8.3.2. Add an area

Applies to User roles: System Administrator

When you create an area and define the rights and settings for the area, all roles will inherit the rights and settings for the area.

To add an area, follow these steps:

1. Click **System Administration** > **Security** > **Areas**.
2. Type the name for the role.
3. Click **New**.
4. Select the module for the area.
5. Type a description for the area.
6. Define the rights for the area.
7. Define the settings for the area.
8. Click **Save** to add the record.

Add a setting within an area

Applies to User roles: System Administrator

You can add a new setting, such as record or format, in the area. A role inherits the new setting from the area when the new setting for the area is not already defined in the role. For example, this is useful when you want to have role within an area to have additional access to system data or another form.

When you add a new setting, it is only inherited when you add a new role.

To add a setting within an area, follow these steps:

1. Click **System Administration** > **Security** > **Areas**.
2. Add optional search criteria and then click **Search**.
3. Click the area to update.
4. Click **More** or the More Actions icon and then select **Administration** > **Add New Setting**.
5. In the **New Setting Info** wizard, provide the following:
    - **Id**: Uniquely identifies the setting so that it can be referenced programmatically.
    - **Display Label**: Defines display label for the setting.
    - **Description**: Describes the setting.
    - **Type**: Select the type of data used by the setting to match the type of setting you are creating. For the type selections, the format of the field on the form depends on the type you select. For example, if you choose boolean, then the setting displays a checkbox and if you select record, the setting displays as a text field. When you select some of the types, the system prompts you for additional information. For example:
        - If you select **Record**, you are also prompted to specify the table and field name for the record.
        - If you select **Manual List**, you are prompted to provide a display type and a name - value pair to display in the list. The Value is the field name in the table and Display Value provides the list of items in the drop-down list for the field.
        - If you select **Global List**, you are prompted to provide a Global List and Display Type for the Global List.
6. Specify whether or not the setting is mandatory. When checked, a value is required for the setting you are adding.
7. Click **Next**.
8. In Adding Settings Validation, add an optional validation script.
9. Click **Finish**.

Validation script for a new setting

When you add an additional setting you have the ability to create simple or complex validations for the setting using JavaScript. The JavaScript automatically sets up an XML variable named `result`, which has a child node named `message`. If the value of the result node is 0 (zero), the validation has succeeded. Any non-zero result is interpreted as a failure. When a validation fails, the contents of the message node display. The value of the setting being validated is set to a variable named `value`.

Example:

This example validates that the setting is a number between 1 and 10.

```
if ( value < 1 || value > 10 )
{
result.setValue( -1 );
message.setValue(“The value must be between 1 and 10.”);
}
```

Update a setting within an area

Applies to User Roles: System Administrator

Once you create a setting for an area, you can update the definition of the setting or delete it. You can also edit any labels that have been localized.

When you add a new setting to an area, it is only inherited when you add a new role.

To edit a setting within an area, follow these steps:

1. Click **System Administration** > **Security** > **Areas**.
2. Add optional search criteria and then click **Search**.
3. Click the area to update.
4. Click **More**or the **More Actions** icon and then select **Administration** > **Edit/Delete Setting**.
5. Select the setting to edit or delete.

6. On the **Edit Setting Info** wizard, update the following:
    - **Id**: Uniquely identifies the setting so that it can be referenced programmatically.
    - **Display Label**: Defines a display label for the setting.
    - **Description**: Describes the setting.
    - **Type**: Select the type of data used by the setting to match the type of setting you are creating. For the type selections, the format of the field on the form depends on the type you select. For example, if you choose boolean, then the setting displays a checkbox and if you select record, the setting displays as a text field. When you select some of the types, the system prompts you for additional information. For example:
        - If you select **Record**, you are also prompted to specify the table and field name for the record.
        - If you select **Manual List**, you are prompted to provide a name-value pair to display in the list. You must also enter Display Type (Radio Button or Drop Down List).
        - If you select **Global List**, you are prompted to provide a Global List and Display Type for the Global List.
7. Specify whether or not the setting is mandatory.
8. Click **Edit localized labels** to update localized labels.
9. For each language available, you can localize the label text. After you localize the text, click **Save** and then **OK** to return to the wizard.
10. Click **Next**.
11. As an option, you can add or edit a validation script for the setting in Adding Settings Validation.
12. Click **Finish**.

Localize an additional setting

Applies to User Roles: System Administrator

Once you add a new setting, such as record or format, you can localize the setting for all of the languages that are active in the system.

To add a setting within an area, follow these steps:

1. Click **System Administration** > **Security** > **Areas**.
2. Add optional search criteria and then click **Search**.
3. Click the area to update.
4. Click **More** or the More Actions icon and then select **Administration** > **Edit/Delete Setting**.
5. In the **Edit Setting Info** wizard, click **Edit localized label**. A list of the messages displays for the user's current language.
6. Click the item you want to localize and then type the applicable text to localize the label you created for the setting.
7. Click **Save**.
8. Continue selecting items until you have localized all the labels for the languages active in the system.
9. To localize the data in the settings for global lists or manual lists, use the record tag localization utility.

Modify the rights for a role within an area

Applies to User roles: System Administrator

Once a role is created and rights are configured, an administrator updates the rights for a role within an area.

To modify the rights for a role within an area, follow these steps:

1. Click **System Administration** > **Security** > **Roles**.
2. Add optional search criteria and then click **Search** to find the role you want to modify.
3. Select the role, and then select the area that you want to modify.
4. Make the changes you want for default rights, folders, and settings.

Note

Note The values for the "update" and "delete/close" rights of security areas are given the following priorities (from highest to lowest):

- Always
- When assigned to workgroup
- When assigned
- Never

Higher values include the permissions of lower values. For example, if you select "When assigned to workgroup," the role will also have the permissions associated with "When assigned."
If you are assigned two security roles, and if these two roles have different values for the same right in the same area, the higher value will override the lower one. For example, if one role has a right set to "When assigned to workgroup" and another role has the same right set to "When assigned," then you have the "When assigned to workgroup" permissions for this right.

5. Click **Save & Exit**.
6. For any other area, repeat steps 3 through 5 to update the rights for the selected role.

Update rights to display allowed categories and allowed statuses

Applies to User roles: System Administrator

A System Administrator can add additional rights to a role within an area by adding the **Allowed Categories** and **Allowed Statuses** fields for an area. Once these fields are available, an administrator can then specify the particular categories and statuses for which the role has rights. If an area has multiple tables that support category and status, the drop-down list in **Allowed Categories** and **Allowed Statuses** will have a combination of all categories and statuses from all tables.

The example below uses the data policy record for the cm3r table for Change Management to demonstrate how to display the **Allowed Categories** and **Allowed Statuses** fields.

Note

Note When you add a new setting, it is only inherited when you add a new role.

To add a setting within an area, follow these steps:

1. Click **Tailoring** > **Data Policy**.
2. Use search to find the data policy record for the cm3r table.
3. Select an area from the drop-down list in the **Area** field. This list displays all the areas available for the selected data policy record.
4. Click **Save**. The **Allowed Categories** and **Allowed Statuses** fields now appear in the **Default Rights** section for the roles within the area you specified in step 3.

Modify allowed categories and allowed statuses

Applies to User Roles: System Administrator

For both roles and areas, an administrator can specify in the Allowed Categories and Allowed Statuses fields the categories and statuses for which a role has rights within an area. Before you can do this, you must first update the applicable data policy record. These rights give a role in combination with the other rights access to the specific categories and statuses selected. When no selections are made for Allowed Categories or Allowed Statuses, the role has access to all categories or statuses within the area.

You must first ensure that the data policy record for the role within an area has been updated to include the **Allowed Categories** and **Allowed Statuses** fields.

Note

Note When you add a new setting, it is only inherited when you add a new role.

To modify Allowed Categories and Allowed Statuses, follow these steps:

1. Click **System Administration** > **Security** > **Roles**.
2. Use search to display a list of roles.
3. Select the role to update.
4. Select the area you want to update.
5. Select the allowed categories and the allowed Statuses from the drop-down lists.
6. Click **Save**.

Related topics

Process Designer security model
Roles in the operator record
Add a security role
Assign a role or roles to an operator

6.3.8.3.3. Check security rights by using Java Script or variables

Applies to User roles: System Administrator

In the Process Designer security framework, you can use Java Script functions or use variables to check the security rights of an operator.

Check security rights using Java Script[edit]

In the Process Designer security framework, you can use the "security.getRights" Java Script function to check security rights for an operator. For how to use this function, see the following description and examples.

security.getRights(<*area*>, <*right*>)

This function checks and returns the specific right (or setting) of the current role in a given Area. The return value is a string.

<*area*> - The Security Area that's to be checked, such as Incident or Change.

<*right*> - The right to check, such as new or update.

Example 1

var temp = lib.security.getRights("Change", "update")

Depending on the specific right that the current user has, the function may return one of the following values: "always", "never", "assigned" or "workgroup".

If the current user has no update right for the Change module, the return value is "false".

Example 2

var temp = lib.security.getRights("Change", "allowCategory")

The return value is a group of specific categories. For example: {"Emergency Change", "CI Group"}

Example 3

var temp = lib.security.getRights("Change", "view")

It returns "true" if the current user has the view right for the Change module. Otherwise, it returns "false" if the current user doesn't have the view right for the Change module.

Note

Note You can also use the "security.getToken(<area>, <right>)" function to check rights (or settings).

The security.getToken function is an alias of security.getRights.

Check record right with variables[edit]

You can use the following variables to check whether a user can access a record. The variables are calculated based on new, view, update, delete, allowed status, and folder setting from secRights with each record.

- $L.tableAccess.new

- $L.tableAccess.view
- $L.tableAccess.update
- $L.tableAccess.delete
- $L.tableAccess.expert
- $L.tableAccess.admin

Contents

- 1 Check security rights using Java Script
- 2 Check record right with variables

6.4. Controlling user access and security

The System Administrator can control user access to Service Manager in the following five ways:

1. Setting individual access restrictions in operator records
    - Assigning capability words to operators
    - Assigning operators to application profiles
    - Assigning roles to operators
    - Assigning Security Roles to operators for Process Designer enabled modules
    - Assigning operators to assignment groups
    - Assigning operators to security groups
    - Assigning operators to user role descriptions
    - Creating a login profile
    - Modifying field-level rights
    - Enabling application time limits
    - Enabling file attachment restrictions
    - Enabling password requirements
    - Enabling printing restrictions
    - Enabling user session restrictions
2. Setting access restrictions by tailoring the application layer
    - Creating displayoptions and displayscreens
    - Creating document engine objects, states, and processes
    - Creating formatctrl
    - Creating menus
    - Defining DVD controls
3. Setting global access restrictions in the System Wide Company record
    - Enabling account expiration times
    - Enabling active integrations to external applications
    - Enabling application time limits
    - Enabling login restrictions
    - Enabling password requirements
4. Setting global access restrictions to application tables in Mandanten
    - Defining security group access to database tables
    - Filtering records visible to users by security groups
    - Limiting access to records by adding security group restricting queries
5. Setting global access restrictions in the initialization file:
    - Defining named users and restricting login to these named users only
    - Enabling Secure socket layer (SSL) connections between the server and clients
    - Enabling shared Mandanten file restrictions

In some cases, these five levels of access restrictions describe the same settings. In such cases, Service Manager uses the following precedence to determine what access restrictions apply:

1. Settings in the initialization file override all other security settings
2. Tailoring settings override security settings in individual operator records and the System Wide Company Record
3. Settings in an individual operator records and Mandanten override security settings in the System Wide Company Record
4. Settings in the System Wide Company Record apply to all operators that have no other security settings

Related topics

Application Setup
Adding users
Creating operator records
Defining named users
Operator passwords
Operator records
Operator templates

6.4.1. Proprietary records

Some proprietary records in Service Manager are identified by the words **Proprietary** and are read-only. Having these records as read-only will help future Service Manager upgrades.

Related topics

Tailoring
Create a form using the Form Wizard

6.4.2. Ongoing maintenance

The Ongoing Maintenance menu enables a System Administrator to access an operator record to view user and contact information, application profile privileges, and the Mandanten utility. If you use the ongoing maintenance menu and have administrative rights, you can also access and control several users or a group from a single application.

As an Application Administrator, you can add or edit users and manage user profiles from within Service Manager applications. You can restrict certain user rights and control the forms that users see when they access different parts of an application.

6.4.3. Environment record

Each application has an Environment record, which defines options that affect the functionality of an application for all users. Some of the typical options stored in the Environment record include:

- Relationship model
- Access rights
- Default category

Related topics

Application Setup
Application profiles

6.4.3.1. Service Manager record relationship models

Service Manager Service Desk relationship models are methods to control the relationships between records inside Service Manager. The Service Desk relationship models affect four record types: Service Desk interaction records, Incident Management incident records, Change Management change records, and Request Fulfillment records. A Service Desk Agent can:

- Open an incident, complaint, or request for information in Incident Management
- Open a request for change in Change Management
- Open a Request Fulfillment record to order a product or service
  For example, if a user complains that a printer has stopped working and the Service Desk Agent determines that an older printer is not compatible with a new laptop, the agent can open a Request Fulfillment record to order a new printer for the user. Once the order is placed, the Service Desk Agent can close the interaction record and add the Request Fulfillment record number for the new printer to the resolution note of the interaction. The Request Fulfillment record is also associated to the interaction record, so that even though the interaction is at a closed status the agent can refer back to the closed interaction to check the status of the related Request Fulfillment record.

The following can be set in the Service Desk Environment Profile record and are models for managing Service Desk record relationships:

- **Full Service Desk Model**: In this model, the state of a Service Desk interaction record changes when each related record closes. The closed state of an interaction record depends on the notification value chosen for the "Notify By" field in the interaction record. For example, if the notification value is "Telephone," the interaction record has a required action before it is closed. This action describes why the customer must be contacted. It also prevents the interaction record from closing until all required actions are complete. In this case, the interaction record goes into the Callback state before it is closed. For more information, see Full Service Desk model.
- **All Records close Independently**: In this model, all Service Desk interaction records close independently. The state of related records doesn't affect closing an interaction record, and closing the interaction record doesn't affect related records.
- **Close Interactions when Related Record closes**: In this model, when the last related record closes, the Service Desk interaction record closes.
- **Cannot close Related Record until Interactions are closed**: In this model, records related to a Service Desk interaction can't close until the interaction record is closed.
- **Cannot close Interactions until Related Records are closed**: In this model, a Service Desk interaction record can't close until all related Incident Management records, Change Management records, and Request Management records are closed.

Related topics

Full Service Desk model
Environment record

6.4.3.1.1. Full Service Desk model

In this model, the state of a Service Desk interaction record changes when each related record closes, depending on the value of the **Notify By** field in the interaction record. The following notification options are available.

Option	Description
None	The interaction record closes.
E-mail	Service Desk sends an email to the contact listed in the interaction record. The email informs the contact that the related record is closed. Service Desk closes the interaction record.
Page	Service Desk sends a page to the contact listed in the interaction record. Service Desk closes the interaction record.
Telephone	The interaction record has a required action. This action describes why the customer must be contacted. It also prevents the interaction record from closing until all required actions are complete. The interaction record goes into the Callback state.

Related topics

Configure the Service Desk environment

6.4.3.2. Status progression

Service Manager applications have a logical status progression as a Service Desk Interaction, Change, Change Task, Problem, Problem Task, Request, or Request Task moves through its individual life cycle. The following table describes the out-of-box status values that Service Manager assigns to these applications.

Application	Status progression
Change Management (Change)	Initial Closed
Change Management (Change Task)	Planned Ready Assigned In Progress Completed Completed with Problems Canceled Withdrawn Failed Blocked
Incident Management	Categorize Assign Work In Progress Pending Customer Pending Evidence Pending Vendor/Supplier Pending Other Resolved Suspended Closed
Problem Management (Problem)	Open Categorize Assign Work In Progress Deferred Pending Resolved Closed

Problem Management (Problem Task)	Planned Ready Assigned Work In Progress Pending Pending Review Closed
Service Desk	Open Categorize Assign In Progress Dispatched Resolved Suspended Withdrawal Requested Closed
Request Fulfillment	Open In Progress Pending Customer Suspended Fulfilled Closed Ordering Closed
Request Fulfillment (Request Task)	Planned Ready In Progress Pending Customer Pending Vendor/Supplier CMDB Update Pending Review Canceled Closed

Related topics

Service Manager record relationship models

6.4.4. Application profiles

Application profiles are security settings that decide which features a user can access from a particular Service Manager application. Each of the seven Service Manager applications has a set of application profiles that decide which features a user can see. An application profile defines the access settings that a particular business function or role has to the application. Typically, System Administrators assign application profiles as part of user role descriptions. However, the administrator can also assign an individual application profile that overrides the default settings of a user role.

Users must have an application profile in their operator record to access any application. Each out-of-the-box application has a profile record named Default to use when a profile doesn't exist. Each application also has a setting to enable access to the application using only the Default profile.

For example, records in the `smenv` table store Service Desk rights and privileges information, such as whether a user can close a service desk interaction record. Profiles also store information that may affect the way an application looks and behaves, such as defining a personal search form for a user.

Note

Note Application profiles aren't used for modules using Process Designer (other than Knowledge Management). For more information, see Process Designer security model.

Application profile authentication

When a user attempts to access one of the applications, Service Manager does the following:

1. Retrieves the profile name from the operator record and accesses the profile record for the specific application.
2. If Service Manager can't find a user profile, it uses the Default profile.
3. If Service Manager can't find a user profile and the setting to use the Default profile isn't enabled, it denies the user's access to the application.

Related topics

Application Setup
Environment record

6.4.5. Enabling approvals

Service Manager Codeless supports approvals in the out-of-box system for the following objects:

- Interaction
- Change and Change Task
- Request
- Service Catalog Item
- Survey
- Time Period

For the applications based on these objects, you can directly use the approval related features from the out-of-box system.

Tip

Tip For more information on the out-of-box approval features for a specific application, see the help topics for the corresponding module.

If you want to enable approval for a Process Designer based object that doesn't offer approvals in the out-of-box system, such as problem, you need to configure approval from the Process Designer workflow, and then manually add the approval.status field to the related forms. After the approval enablement, users who have the required privileges can use the approval features for the application based on the approval enabled object. The following table describes the approval features available after the approval enablement and the privileges that are required for using the corresponding approval features.

Approval feature	Required security privileges for users in a security area
Approve/deny/retract records in the record menu	Users who have the "Can Approve" right or are delegated with the approval authority can approve, deny, and retract records.
Mass approve records in the list view	Users who have the "Expert" right can mass approve records in the list view.
Approval Inbox	Users who are in the approver list and have the "Can Approve" right or are delegated with the approval authority can see the record list for an area and perform approval actions from the Approval Inbox. **Note** Note Users must also have the privilege to run Approval Inbox.
Approval delegation	User who have the "Can Delegate Approvals" right can delegate their approval rights to other qualified operators.

For instructions on enabling approval for a Process Designer based object that doesn't offer approvals in the out-of-box system, see Enable approval for a Process Designer based object.

Related topics

Approval delegation Enable approval for a Process Designer based object
Delegate approvals to another operator
Disable an active approval delegation
Update an active approval delegation

6.4.5.1. Enable approval for a Process Designer based object

To enable approval for a Process Designer based object that doesn't offer approvals in the out-of-the-box system, follow these steps:

1. Configure approval in the workflow that's associated with a Process Designer based object.
    1. Sign in to Service Manager with a System Administrator account from the web client.
    2. Click **Tailoring** > **Process Designer** > **Workflows** from the System Navigator to display the workflows list.
    3. Open the workflow associated with the object for which you want to enable the approval related features.

        Note
        Note The out-of-the-box workflows are **Proprietary**. You can clone the out-of-the-box workflows, and then make changes as necessary.

    4. Select a phase, and then go to the **Approval** tab.
    5. Click **Add**, select one or more approval definitions, and then click **OK**.
    6. (Optional) Configure the approval reset condition and the recalculate condition if necessary.
    7. Click **Save**.

Suppose you have configured approval from the workflow associated with a Process Designer based object that doesn't offer approvals in the out-of-the-box system. The system automatically creates or updates the following items to enable the approval related features for the records based on the object:

Item	Changes after enabling approval
Menu items	Menu items are created in record menu options, such as **Approve**, **Deny**, **Retract**, and **Mass Approve**.
Approval Inbox	Approval Inbox can display the pending approvals for the records corresponding to the approval enabled object.
Approval delegation wizard	SM adds the module associated with the object to the approval delegation wizard.
Security area	▪ Security settings The "Can Approve" and "Can Delegate Approval" security settings are added for the security area that's associated with the object. With the "Can Approve" right for an area, an operator can see the approval records in the Approval Inbox for all the Service Manager tables that are associated with the area. With the "Can Delegate Approvals" right for an area, an operator can delegate approvals from the approval delegation wizard for all the Service Manager tables that are associated with the area. ▪ Security folder SM creates a new folder right with the name of <Record module name> approver for this security area. The system sets the folder to "DEFAULT" with the view authority only.
Security role and security rights	SM creates a new security role in the name of <Record module name> approver. The system also sets default security rights (such as "Expert") and default settings (such as "Can Approve" and "Can Delegate Approvals") to this role.

2. Edit the necessary forms to add the new approval field ("approval.status").
    1. Sign in to Service Manager with a System Administrator account from the Windows client.
    2. Click **Tailoring** > **Forms Designer** from the System Navigator.
    3. Search for and open a form in which you want to display the approval status.
    4. Add the "approval.status" to the form as appropriate.
    5. Click **Save**.
    6. Repeat this task to modify other forms as needed.
3. Assign the new security role ("<Record module name> approver") to operators who need to perform the approval tasks for the new approval enabled module. To do this, see Assign a role or roles to an operator.
   Now the operators with the security role can use the approval features as described in Enabling approvals.

Related topics

Enabling approvals
Approval delegation
Delegate approvals to another operator

6.4.6. Approval delegation

Approval delegation is an optional feature that enables users with approval rights to temporarily delegate their approval authority to another qualified operator. Operators with the **Delegate Approvals** or **Can Delegate Approvals** option enabled in their application profiles can delegate some or all their approvals by using the Approval Delegation wizard.

Using the **Approval Delegation** wizard, an operator can grant another qualified operator the right to temporarily view and act on items in that operator's approval queue. The wizard offers the following delegation options:

- Delegate all approvals to another qualified operator
- Delegate approvals from a particular application to another qualified operator
  - Delegate approvals directly assigned to you as an operator
  - Delegate approvals assigned to you as a member of an approval group
- Delegate approvals from a specified start date to a specified end date

The **Approval Delegation** wizard enables an operator to create any number of approval delegation combinations, including delegating the same approvals to multiple operators at the same time. **Delegators** can also update an existing approval delegation to change the delegation start/end dates and change the delegate's name.

Note

Note Service Manager tracks all changes to approval delegations using the standard field auditing capability.

When delegates log on to Service Manager, they see both their own and any delegated approvals in their approval list. For security reasons, delegates always retain their original application profiles and operator records. Service Manager determines what temporary rights delegates have when they view or act on an approval.

For more information about the tasks that relate to approval delegation, refer to the following topics:

What happens when I delegate approval authority?
Temporary rights of an approval delegate
What happens when I receive delegated approval authority?
Copy an approval delegation
Delegate approvals to another operator
Disable an active approval delegation
Update an active approval delegation
Global variables available for approval delegation
Views available for approval delegation

6.4.6.1. Administering approval delegation

In the out-of-box system, Service Manager supports approval delegation for the following applications :

- Change Management
- Request Fulfillment
- Service Catalog

To enable approval delegation for one of these applications, an Administrator must edit one of the application's security role records and select the **Can Delegate Approvals** option. Next, the Administrator must grant this security role to the operators who will be given approval delegation authority.

Note

Note If you want to support approval delegation for an application that doesn't support approvals out-of-box, you must first enable approvals for the application. To enable approvals, see Enabling approvals.

Approval delegation never changes a delegate's original security role or operator record. Service Manager only changes a delegate's approval groups in memory when the following conditions occur.

- When the system notifies the delegate
- When a delegate views or acts on an approval

By default, Service Manager sends an email notification to the approval delegate when the following conditions occur:

- A new approval delegation is assigned to the delegate
- A new approval arrives in the approval owner's queue

Note

Note Service Manager also sends an email notification to the approval owner when a new approval arrives in the owner's queue. System Administrators can change the notification behaviors for approval delegation directly from the notification engine.

A delegated approval always retains its original operator assignment. Service Manager records the delegate's actions separately from the owner in the following new fields:

Field name	Label in approval record	Description
approved.by	Operator	The name of the operator who acted on the approval.
approved.for	Approved For	The name of operator who delegated the approval.

Note

Note Delegators and delegates can view these fields from the approval record and the approval log. However, in some cases, only a delegator may have the approval group necessary to view the records in the approval log. A delegate's temporary rights don't include viewing approvals in the log. In order to view approvals in the approval log, a delegate's application profile must include the required approval groups.

Related topics

Approval delegation
What happens when I receive delegated approval authority?
What happens when I delegate approval authority?
Enabling approval delegation
Delegate approvals to another operator
Disable an active approval delegation
Update an active approval delegation
Global variables available for approval delegation
Views available for approval delegation

6.4.6.1.1. Enabling approval delegation

The **Can Delegate Approvals** security role setting controls whether an operator can view the **Approval Delegation** wizard.

Note

Note It is a best practice to only enable the **Delegate Approvals** or **Can Delegate Approvals** option for operators who can also view and approve objects in the application.

Refer to the following example for how to enable approval delegation for Change Management.

1. Sign in to Service Manager as a System Administrator.

2. Click **System Administration** > **Security** > **Roles**.
3. In the **Name** field, type the name of the security role you want to grant approvals. For example, change approver.
4. Click **Search**. The **Security Roles** form opens.
5. Click **Change** in the **Area** column.
6. Under **Settings**, select the **Can Delegate Approvals** checkbox.
7. Click **Save**.

Related topics

Administering approval delegation
Approval delegation
What happens when I delegate approval authority?
What happens when I receive delegated approval authority?
Copy an approval delegation
Delegate approvals to another operator
Disable an active approval delegation
Update an active approval delegation
Global variables available for approval delegation
Views available for approval delegation

6.4.6.1.2. Global variables available for approval delegation

Service Manager provides global variables for approval delegation. Administrators can use these global variables to create their own custom queries or views.

Global variable	Description
$G.delegated.cm3r.groups	Stores the assignment groups that are associated with Change Management changes that the current operator can view and act on as an approval delegate.
$G.delegated.cm3t.groups	Stores the assignment groups that are associated with Change Management tasks that the current operator can view and act on as an approval delegate.
$G.delegated.ocmq.groups	Stores the assignment groups that are associated with Request Management that the current operator can view and act on as an approval delegate.
$G.delegated.svc.groups	Stores the assignment groups that are associated with Service Catalog that the current operator can view and act on as an approval delegate.

Related topics

Approval delegation
What happens when I delegate approval authority?
What happens when I receive delegated approval authority?
Copy an approval delegation
Delegate approvals to another operator
Disable an active approval delegation
Update an active approval delegation
Views available for approval delegation

6.4.6.2. Enabling approval delegation for custom approvals

If you want to enable approval delegations for an application that doesn't offer approvals in the out-of-box system, you must first create and enable custom approvals for the application. For more information, see related topics.

Enabling approvals and approval delegation requires administrative access to the system database dictionary, the Forms Designer utility, application profiles, and the script library. You should also know which tables and fields the application uses.

Refer to the following checklist for how to set up an application to enable approvals and approval delegation for Problem Management.

Note

Note

- To complete this checklist, follow the step-by-step procedures described in the following examples found in the related topics.
- You must use the Windows client whenever you need to add a new field/key to a database dictionary table.

1. Edit the object definition for the application to enable approvals by category.
2. Edit the database dictionary for Problem Management to add an approval status field.
3. Edit the database dictionary for the Problem Management categories to add an approval field.
4. Edit the necessary forms to add the new approval fields.
5. Edit the database dictionary for the Problem Management profile record to add to an approval groups field.
6. Create a new approval definition record to manage the Problem Management approvals.
7. Edit the Problem Management category definitions to add your new approval definition.
8. Edit the Problem Management profile records that you want to have access to your custom approval group.
9. Edit the script that controls the approval inbox to add your custom approvals to the list of approvals the system displays.
10. Enable approval delegations for the custom approvals.

For more information, see Example: Enabling Approval Delegation for custom Problem Management approvals and Example: Enabling custom approvals for Problem Management.

Related topics

Approval delegation
Example: Enabling Approval Delegation for custom Problem Management approvals
Example: Enabling custom approvals for Problem Management
What happens when I delegate approval authority?
What happens when I receive delegated approval authority?
Copy an approval delegation
Delegate approvals to another operator
Disable an active approval delegation
Update an active approval delegation
Global variables available for approval delegation
Views available for approval delegation

6.4.6.2.1. Example: Enabling custom approvals for Problem Management

The following example illustrates how to enable approvals for Problem Management. At the end of this example, you will be able to:

- Create a problem record that requires an approval
- Assign the problem record to the approval queue
- View and act upon the problem approval

Note

Note In order to enable custom approvals for Problem Management, you must perform the entire sequence of example tasks. You must be a System Administrator, or have equivalent system privileges, to perform the steps in this example. Also, you must use the Windows client whenever you need to add a new field/key to a database dictionary table.

Task 1: Edit the object definition for the application to enable approvals by category.

1. Sign in to Service Manager with a System Administrator account.
2. Click **Tailoring** > **Document Engine** > **Objects**.
3. Type rootcause in the **File Name** field, and then click **Search**.
   The **Object Definition** form opens.
4. Click the **Approvals** tab, and then type the following field values.

Field	Value
Approval condition	true
Approval location	Category
Approval field name	approvals
Approval status field	approval.status
Approval type	All must approve
Approval process	rootcause.save
Preapprove on open	false

Note

Note The fields approvals and approval.status fields don't yet exist. You will add the field name to the Problem Management database dictionary in a later step.

5. Click **Save**.

Task 2: Edit the database dictionary for the application to add an approval status field.

1. Click **Tailoring** > **Database Dictionary**.
2. Type rootcause in the **File Name** field, and then click **Search**.
3. Select the **descriptor** row in the **Fields** tab, and then click **New Field/Key**.
4. Type the following field attributes.

Field attribute	Value
Name	approval.status
Type	character

5. Click **Add Field**.
6. Click **OK** to save the database dictionary.
   Service Manager prompts you to confirm how you want to change the tables in the RDBMS.
7. If Service Manager has write permissions to the RDBMS, click **SM Alters**. Otherwise, click **User Alters**.
   Service Manager displays the database dictionary.

Note
Note Check your DDL output folder for DDL text you can use to update your RDBMS manually.

Task 3: Edit the database dictionary for the application's categories to add an approval field.

1. Click **Tailoring** > **Database Dictionary**.
2. Type rootcausecat in the **File Name** field, and then click **Search**.
3. Select the **descriptor** row in the **Fields** tab, and then click **New Field/Key**.
4. Type the following field attributes.

Field attribute	Value
Name	approvals
Type	array

5. Click **Add Field**. Service Manager prompts you to select the array's data type.
6. Select **character**, and then click **Add Field**.
7. Click **OK** to save the database dictionary.
   Service Manager prompts you to confirm how you want to change the tables in the RDBMS.
8. If Service Manager has write permissions to the RDBMS, click **SM Alters**. Otherwise, click **User Alters**.
   Service Manager displays the database dictionary.

Note
Note Check your DDL output folder for DDL text you can use to update your RDBMS manually.

Task 4: Edit the necessary forms to add the new approval fields.

1. Click **Tailoring** > **Forms Designer**.
2. Type problemcat in the **Form** field, and then click **Search**.
   The **Problem Control Category Definition** form opens.
3. Click **Design**.
4. Click **Group**.
5. Click on an empty space on the form to add the group.
6. Edit the properties of the group as follows.

Property	Value
Caption	Approval Requirements
Width	152

7. Click **Comfill**, and then click within the **Approval Requirements** group.
8. Edit the properties of the combo fill field as follows.

Property	Value
Input	approvals
Array Length	5

9. Position the **Approval Requirements** group as desired.
   For example, you may want to position the group between the **Active?** checkbox and the **Phase name** table.
10. Click **OK** to exit design mode.

11. Click **OK** to save the form and return to the **Forms Designer** search screen.
12. Type PM.pc.investigate.and.diag in the **Form**field, and then click **Search**.
    The **Problem Control - Problem Investigation and Diagnosis** form opens.
13. Click **Design**.
14. Click **Label**, and then click in the open space beneath the **Record Number** label.
15. Edit the properties of the label as follows.

Property	Value
Caption	Approval Status:
X	2
Y	5
Width	34

16. Click **Text**, and then click to the right of the **Approval Status:** label.
17. Edit the properties of the text field as follows.

Property	Value
X	36
Y	5
Width	36
Input	approval.status
Read-Only	Selected

18. Click **OK** to exit design mode.
19. Click **OK** to save the form and return to the **Forms Designer** search screen.

Task 5: Edit the database dictionary for the application's profile record to add to an approval groups field.

1. Click **Tailoring** > **Database Dictionary**.
2. Type rcenv in the **File Name** field, and then click **Search**.
3. Select the **descriptor** row in the **Fields** tab, and then click **New Field/Key**.
4. Type the following field attributes.

Field attribute	Value
Name	approval.groups
Type	array

5. Click **Add Field**. Service Manager prompts you to select the array's data type.
6. Select **character**.
7. Click **Add Field**.
8. Click **OK** to save the database dictionary.
   Service Manager displays the database dictionary.

Task 6: Create a new approval definition record to manage the application's approvals.

1. Click **Tailoring** > **Forms Designer**.
2. Type rc.profile.g in the **Form** field, and then click **Search**.

The **Problem Management Security Profile** form opens.

3. Click **Design**.
4. Click **Label**, and then click in the open space beneath the **Complex Mass Update** checkbox.
5. Edit the properties of the label as follows.

Property	Value
Caption	Approval Groups:
X	2
Y	31
Width	28

6. Click **Comfill**, and then click to the right of the **Approval Groups:** label.
7. Edit the properties of the combo fill field as follows.

Property	Value
X	31
Y	31
Input	approval.groups
Array Length	5
Value List Condition	select("name", "assignment", "name", "*")

8. Click **OK** to exit design mode.
9. Click **OK** to save the form and return to the Forms Designer search screen.

Task 7: Edit the application's category definitions to add your new approval definition.

1. Click **Tailoring** > **Tailoring Tools** > **Links**.
2. Type rc.profile in the **Name** field, and then click **Search**.
3. Add the following link entry at the end of the list of links.

Property	Value
Source Field name	approval.groups
Target File name	assignment
Target Field name	name

4. Click **Save**.
5. Click **Change Management** > **Maintenance** > **Approvals**.

Note

Note You are using a Change Management menu link because by default, Problem Management doesn't have a menu item for Approvals.

6. Type the following field values.

Field	Value
Name	Problem Approval
Approval Condition	true
Group/Oper	HELPDESK
Sequence	1
Condition	true

Caution

Caution The **Group/Oper** field is case sensitive.

7. Click **Add**.
8. Click **Problem Management** > **Administration** > **Problem Control Categories**.
   Your customized **problemcat** form opens.
9. In the **Name** field, type the name of a Problem Management category that you want to trigger your custom approval process. For example, ITIL.
   Note: You can repeat this step for each Problem Management category you want to trigger approvals.
10. Click **Search**.
    Service Manager displays the current ITIL category definition.
11. In the **Approval Requirements** array, type Problem Approval.
12. Click **Save**.

Task 8: Edit the application profile records that you want to have access to your custom approval group.

1. Click **System Administration** > **Ongoing Maintenance** > **Profiles** > **Problem Management Profiles**.
   The **Problem Management Security Profile** form opens.
2. In the **Profile Name** field, type the name of a profile you want to grant approvals. For example, SYSADMIN.
3. Click **Search**.
   Service Manager displays the current Problem Management Security Profile.
4. In the **Approval Groups** array, type the name of the approval group you defined to manage approvals. For example, HELPDESK.
5. Click **Save**.
6. Sign out of the current operator and sign in to Service Manager with an operator account that has the SYSADMIN Problem Management profile. For example, System.Admin.
7. Click **Problem Management** > **Problem Control** > **Create New Problem**.
8. Create a new problem record and note the record number. For example, PM0002.

Note

Note The **Category** field on the **Problem Control** form maps to the Incident Management category, not the Problem Management category. Since there is only one approval category for ITIL Problem Management, any problem you create will automatically be part of the ITIL Problem Management category.

Task 9: Edit the script that controls the approval inbox to add your custom approvals to the list of approvals the system displays.

1. Click **Tailoring** > **Database Manager**.
2. Type Approvals in the **Table** field, and then click **Search**.
3. Select the **Approval** form from the list.
4. In the **Id** field, type the record number for the problem you created. For example, PM0002.

5. Click **Search** to display the pending approval.
6. Click **Tailoring** > **Script Library**.
7. Type ApprovalCustomPlugin in the Name field, and then click **Search**.
8. Locate the appendCustomApprovalSql() function.
9. Remove the comment lines /* and */ at lines 49 and 55 to enable the function.
10. Click **Save**.
11. Click **Compile**.
12. Click **Service Catalog** > **Approve Requests**.
    Service Manager displays the list of pending approvals. Check the list for your custom approval. For example, PM0002.

Related topics

Approval delegation
Enabling approval delegation for custom approvals
Example: Enabling Approval Delegation for custom Problem Management approvals
What happens when I delegate approval authority?
What happens when I receive delegated approval authority?
Delegate approvals to another operator
Disable an active approval delegation
Update an active approval delegation
Global variables available for approval delegation
Views available for approval delegation

6.4.6.2.2. Example: Enabling Approval Delegation for custom Problem Management approvals

The following example shows how to delegate the custom Problem Management approvals you created in a previous example. At the end of this example, you will be able to:

- Delegate Problem Management approvals to another operator
- View Problem Management approvals in a delegate's approval queue

Note

Note In order to enable approval delegation for custom Problem Management approvals, you must perform the entire sequence of example tasks. You must be a System Administrator, or have equivalent system privileges, to perform the steps in this example. Also, you must use the Windows client whenever you need to add a new field or key to a database dictionary table.

Task 1: Add a new approval delegation field to the database dictionary.

1. Sign in to Service Manager with a System Administrator account.
2. Click **Tailoring** > **Database Dictionary**.
3. Type rcenv in the **File Name** field, and then click **Search**.
4. Select the **descriptor** row in the **Fields** tab, and then click **New Field/Key**.
5. Type the following field attributes.

Field attribute	Value
Name	can.delegate
Type	logical

6. Click **Add Field**.
7. Click **OK** to save the database dictionary.

Task 2: Edit the Problem Management Security Profile form.

1. Click **Tailoring** > **Forms Designer**.

2. Type rc.profile.g in the **Form** field, and then click **Search**.
   The **Problem Management Security Profile** form opens.
3. Click **Design**.
4. Select the **assig.groups** comfill, and change its property as follows.

Property	Value
Y	23

5. Select the **Assignment Groups** label, and change its property as follows.

Property	Value
Y	21

6. Click **Checkbox**.
7. Click on the empty space between the **Assignment Groups** label and the **Skip Query Warning** checkbox.
8. Edit the properties of the checkbox.

Property	Value
X	80
Y	19
Caption	Can Delegate?
Input	can.delegate

9. Click **OK** to exit the design mode.
10. Click **OK** to save the form and return to the Forms Designer search screen.

Task 3: Edit the custom Problem Management Profile form.

1. Click **System Administration** > **Ongoing Maintenance** > **Profiles** > **Problem Management Profiles**.
   The **Problem Management Security Profile** form opens.
2. Type the name of a profile you want to grant approval delegation rights in the **Profile Name**field. For example, SYSADMIN.
3. Click **Search**.
   Service Manager displays the current Problem Management Security Profile.
4. Click **Can Delegate?** to enable it.
5. Click **Save**.

Task 4: Edit the ApprovalCustomPlugin functions in the script library.

1. Click **Tailoring** > **Script Library**.
2. In the **Name** field, type ApprovalCustomPlugin, and then click **Search**.
3. Locate the checkProfileForModule(filename, operator) function, and then remove the /* and */comment lines at lines 6 and 17 to enable the function.
   This function makes problem records a valid choice in the approval type list of the **Approval Delegation** wizard.
4. Locate the getCustomApprovalGroups(module, operator) function, and then do the following:
   1. Delete the // comment characters at line 25 to activate the line.
   2. Remove the /* and */ comment lines at lines 27 and 40 to enable the function.

   This function returns all the approvals groups defined for the profile of the current application.
5. Locate the getAllCustomGroups(aDele) function, and then do the following:
   1. Verify that the line 100 lists **profile_rootcause** as the profile definition.

To change the profile type, type a different profile definition.

2. Verify that line 113 lists SCFile("rcenv") as the type.
   To use another application, type a different profile table name between the quotation marks.
3. Verify that line 143 lists "Problem" as the module condition.
   To use another application, type a different approval module name.
4. Verify that line 161 lists \"rootcause\" as the file name.
   To use another application, type a different file name.

6. Click **Save**.
7. Click **Compile**.
8. Click **OK** to return to the Script Library search screen.

Task 5: Edit the ApprovalDelegationGroups functions in the Script Library.

1. Click **Tailoring** > **Script Library**.
2. In the **Name** field, type `ApprovalDelegationGroups`, and then click **Search**.
3. Locate the getDelegationSql function.
4. Delete the // comment characters on line 29 to activate the line.
5. Delete the // comment characters on line 31 to activate the line.
6. Type the // comment characters at the beginning of line 32 to deactivate the line.
7. Click **Save**.
8. Click **Compile**.
9. Click **OK** to return to the Script Library search screen.

Task 6: Add a new approval delegation for the new Problem Management custom approvals.

1. Click **Approval Delegation**.
   The **Approval Delegation** wizard opens.
2. Add a new delegation with the following properties.

Property	Value
Delegated modules	Selected
Modules delegated	Problem Management
Assignment Groups and Operator delegated	HELPDESK, System.Admin
Delegate	Problem.Manager
Delegate from and to dates	From today to tomorrow

3. Sign out of the current operator and sign in to Service Manager as the delegate. For example, Problem.Manager.
4. Click **Approval Inbox**.
   You should see the Problem approval you created previously. For example, PM0002.

Related topics

Approval delegation
Enabling approval delegation for custom approvals
Example: Enabling custom approvals for Problem Management
What happens when I delegate approval authority?
What happens when I receive delegated approval authority?
Delegate approvals to another operator
Disable an active approval delegation
Update an active approval delegation
Global variables available for approval delegation
Views available for approval delegation

6.4.6.3. What happens when I receive delegated approval authority?

If operators delegate their approval authority to you, Service Manager sends an email to notify you of the new approval delegation. You are also notified when a new approval arrives in your approval queue.

Viewing approvals

When you log on to Service Manager, you will see both your own and any delegated approvals in your approval queue.

- To view approvals that another operator has delegated to you, you can use the **Active approval delegations assigned to me** view.
- To view approvals delegated to you in the past, you can use the **Past approval delegations assigned to me** view.
- To see which items in the approval queue are due to an active approval delegation, you can open the **Approve Requests** view from the System navigator. In this view, Service Manager indicates which approvals are in the queue due to an active delegation by displaying a value of **YES** in the As Delegate? column. You can use this view to view, approve, or deny approval requests.

Tracking approval actions

As a delegate, when you act on an approval, Service Manager tracks your actions by adding both your operator name and the delegator's operator name to the approval record.

- Service Manager lists your name in the Operator column of the Completed Approval Actions table.
- Service Manager lists the delegator's name in the Approve For column of the Completed Approval Actions table.

After the approval delegation expires

When an approval delegation expires, you are no longer considered a temporary member of the delegator's approval groups. This means that you can no longer view or act on items that belong exclusively to the delegator's approval groups.The restriction includes any approvals that you previously acted on during the delegation period. In some cases, this may mean that only the delegator can see a particular approval record.

Related topics

What happens when I delegate approval authority?
Delegate approvals to another operator
Disable an active approval delegation
Update an active approval delegation
Global variables available for approval delegation
Views available for approval delegation

6.4.6.3.1. Temporary rights of an approval delegate

An approval delegate temporarily gains the rights for the assignment group of the delegating operator while the approval delegation is active. After the approval delegation period ends, the delegate's temporary rights for the assignment group revert to their original status.

Approval delegation never changes a delegate's original application profile or operator record. Service Manager only changes a delegate's assignment group rights in memory when the following conditions occur.

- When the system notifies the delegate
- When a delegate views or acts on an approval

For example, a manager wants to delegate approval authority to a technician. When the manager delegates approval authority to the technician, the technician temporarily becomes a member of all of the assignment groups that the manager is an approver of.

Related topics

What happens when I delegate approval authority?
What happens when I receive delegated approval authority?
Copy an approval delegation
Delegate approvals to another operator
Disable an active approval delegation
Update an active approval delegation

Global variables available for approval delegation
Views available for approval delegation

6.4.6.4. What happens when I delegate approval authority?

When you delegate approval authority to a qualified operator, the delegate receives an email notification. Delegates are also notified when a new approval arrives in their approval queues.

As a delegator, you always retain your normal approval authority. Both you and any delegates you authorize have the ability to approve items while an approval delegation is active.

Viewing approvals

As a delegator, when you log on to Service Manager, you will see both your own and any delegated approvals in your approval queue.

- To view your active approval delegations, you can use the **Approval Delegation** wizard or the **My active approval delegations** view.
- To view your past delegations, you can use the **Copy Approval Delegation** wizard or the **My past approval delegations** view.

Tracking approval actions

When a delegate acts on an approval, Service Manager tracks the delegate's actions by adding both the delegate's operator name and your operator name to the approval record.

- Service Manager lists the delegate's name in the Operator column of the Completed Approval Actions table.
- Service Manager lists the delegator's name in the Approve For column of the Completed Approval Actions table.

After the approval delegation expires

When an approval delegation expires, a delegate is no longer considered a temporary member of your approval groups. This means that the delegate can no longer see or act on items that belong exclusively to your assignment groups. The restriction includes any approvals that the delegate previously acted on during the delegation period. In some cases, this may mean that only you as delegator can see a particular approval record.

Related topics

What happens when I receive delegated approval authority?
Delegate approvals to another operator
Disable an active approval delegation
Update an active approval delegation
Global variables available for approval delegation
Views available for approval delegation

6.4.6.4.1. Delegate approvals to another operator

Applies to User roles: System Administrator and other users with approval delegation authority

You can only delegate approvals to another operator if a System Administrator enables the **Delegate Approvals** or **Can Delegate Approvals** option for you in your security role.

To delegate approvals to another operator, follow these steps:

1. Click **Approval Delegation**. The **Approval Delegation** wizard opens and displays any active approval delegations assigned to you.
2. To create a new approval delegation, click **Add New Delegation**.
3. Select whether to delegate all your approvals or to select approvals.
4. If you are selecting approvals, make the following choices:
    1. Choose which application's approvals you want to delegate.
    2. Choose how you want delegate approvals assigned to you:
        - Assigned as part of an assignment group
        - Assigned directly to you as an operator

Note

Note You can select multiple assignment groups or operators as needed.

5. Select the delegate to whom you want to grant approval authority.

Note

Note Service Manager only displays operators who are eligible approval delegates. If you don't see a particular operator listed as a potential delegate, it means that the operator doesn't have one or more of the rights required to be eligible for approval delegation. Consult your System Administrator if you want to assign additional rights to a particular operator.

6. Select the date range during which the approval delegation will be active.

Related topics

What happens when I delegate approval authority?
What happens when I receive delegated approval authority?
Copy an approval delegation
Disable an active approval delegation
Update an active approval delegation
Global variables available for approval delegation
Views available for approval delegation

6.4.6.4.2. Update an active approval delegation

Applies to User roles: System Administrator and other users with approval delegation authority

You can change the delegate, the start date, or the end date of any currently active approval delegation by using the Approval Delegation wizard. If you want to use a delegation as a template for a new delegation, use the Copy Approval Delegation wizard.

Note

Note To change approvals delegated by a specified assignment group or operator name, you must disable the current delegation and create a new one with the new assignment groups and operator names. Service Manager requires a new delegation in order to determine which operators are qualified to be delegates.

To update an active approval delegation, follow these steps:

1. Click **Approval Delegation**. The **Approval Delegation** wizard opens and displays any active approval delegations assigned by you.
2. Select the approval you want to edit from the list of active delegations.
3. Click **Edit Current Delegation**.
4. Select the new start and end dates for the approval delegation.

Note

Note Service Manager dates always default to midnight (00: 00: 00) of the selected day. If you want to set a different start time, manually type in the new start time using the twenty-four hour: minute: second format notation. For example 23: 59: 59 represents 11: 59 PM and 59 seconds.

5. Click **Next** to save your changes and close the wizard.

Related topics

What happens when I delegate approval authority?
What happens when I receive delegated approval authority?
Copy an approval delegation
Delegate approvals to another operator

Disable an active approval delegation
Global variables available for approval delegation
Views available for approval delegation

6.4.6.4.3. Disable an active approval delegation

Applies to User roles: System Administrator and other users with approval delegation authority

You can disable any currently active approval delegation. You can't disable an inactive past delegation.

To disable an active approval delegation, follow these steps:

1. Click **Approval Delegation**.
   The **Approval Delegation** wizard opens and displays any active approval delegations assigned by you.
2. Select the approval you want to disable from the list of active delegations.
3. Click **Edit Current Delegation**.
4. Clear the **Enabled** checkbox.
5. Click **Next** to save your changes and close the wizard.

Note

Note To view your past delegations or delegations assigned to you, use one of the default approval delegation views.

Related topics

What happens when I delegate approval authority?
What happens when I receive delegated approval authority? Copy an approval delegation
Delegate approvals to another operator
Update an active approval delegation
Global variables available for approval delegation
Views available for approval delegation

6.4.6.4.4. Copy an approval delegation

Applies to User roles: System Administrator and other users with approval delegation authority

You can use an existing approval delegation as a template to create a new approval delegation. The wizard copies the values from the existing delegation and allows you to change the delegate and the delegation dates. You can't change the application module, delegated approval groups, or operator when copying an approval delegation. If you want to change these values, you must create a new approval delegation.

To copy an approval delegation, follow these steps:

1. From the **To Do** view, select **Approval Delegation** from the **Queue** list. Service Manager displays the Approval Delegation view.
2. From the **View** list, select either **My active approval delegations** or **My past approval delegations**.
3. Select the approval you want to copy from the list of approval delegations.
4. Click **Copy Approval Delegation**. Service Manager displays the Copy Approval Delegation wizard and automatically fills in the delegate name, the delegated module, approval groups, and operator.
5. Select the new delegate if needed.
6. Select the new delegation start and end dates.
7. Select **Enabled**.
8. Click **Next** to create a new approval delegation.

Related topics

Delegate approvals to another operator
Disable an active approval delegation
Update an active approval delegation Global variables available for approval delegation
Views available for approval delegation

6.4.6.5. Views available for approval delegation

Service Manager provides default views for delegators and delegates to manage approval delegations. Administrators can also view approval delegation records directly from the Database Manager by viewing the ApprovalDelegation table.

View	Description	Available Actions
My active approval delegations	A list of the currently active approvals that you delegated to other operators. This view doesn't display future delegations because they're not currently active.	• Start the Approval Delegation wizard • Export • Print
My past approval delegations	A list of the inactive approvals that you delegated to other operators in the past.	• Start the Copy Approval Delegation wizard • Export • Print
Active approvals assigned to me	A list of the currently active approvals delegated to you. This view only displays approvals where you are the active delegate. Use the other views to display your past or pending delegations.	• Export • Print
Past approvals assigned to me	A list of the inactive approvals that other operators have delegated to you in the past.	• Export • Print

Note

Note If you create or update an approval delegation record you may need to use the Service Manager **Refresh** option in order for the views listed above to display the new delegation record.

Related topics

Approval delegation
What happens when I receive delegated approval authority?
What happens when I delegate approval authority?
Copy an approval delegation
Delegate approvals to another operator
Disable an active approval delegation
Update an active approval delegation

6.4.7. User roles

The Service Management process can meet best practices when employees involved in the process are assigned user roles in your IT organization. For information on the Service Management organizational model of user roles for best practices, see *Service Manager Processes and Best Practices* in the related topics.

A user role is a template that combines a collection of application profiles, security roles, and capability words into a single record. Service Manager provides out-of-box user role descriptions with appropriate capability words and application profiles that define a variety of business functions or roles. By defining and assigning user role descriptions, a System Administrator can grant an operator the capability words and application profile required for a particular job.

The Service Management process consists of the following user roles:

Related topics

Add a user role record
Delete a user role record
Search for a user role record
Processes and Best Practices

6.4.7.1. Add a user role record

Applies to User roles: System Administrator

A user role defines a set of application profiles, capability words, and query groups that you can apply to any operator record. For example, the self-service role enables the self-service user to open, view, update, and close service requests.

To add a user role record, follow these steps:

1. Click **System Administration** > **Ongoing Maintenance** > **User Roles**.
2. Enter a User Role or click **Search** to select a role from a record list.
3. Add optional information to the form. If necessary, press **Ctrl + H** to view help for each field.
4. Click **Add**.

> **Note**
>
> Note Service Manager provides an out-of-box self-service user role record.

Related topics

Delete a user role record
Search for a user role record
Set the dashboard home page for a user role

6.4.7.2. Delete a user role record

Applies to User roles: System Administrator

To delete a user role record, follow these steps:

1. Click **System Administration** > **Ongoing Maintenance** > **User Roles**.
2. Type or select optional search criteria.
3. Click **Search**.
4. Select the user role that you want to delete.
5. Click **Delete**.
6. Click **Yes** to confirm the deletion.

Related topics

Add a user role record
Search for a user role record
Set the dashboard home page for a user role

6.4.7.3. Search for a user role record

Applies to User roles: System Administrator

To search for a user role record, follow these steps:

1. Click **System Administration** > **Ongoing Maintenance** > **User Roles**.
2. Click **Search** to select a role from a record list.

Related topics

Add a user role record
Delete a user role record

6.4.7.4. Set database access for a user role

Applies to User roles: System Administrator

Database access is a feature that gives you the ability to limit or grant access to database records, such as contacts, company, and regions. You can add or modify the existing out-of-box database access settings per user role or operator.

To set database access for a user role, follow these steps:

1. Click **System Administration** > **Ongoing Maintenance** > **User Roles**.
2. Type or select optional search criteria, and then click **Search**.
3. Select a user role from the record list.
4. Select the **Data Access** tab.
5. To add a new Data Access record, follow these steps:
    1. Click **Data Access Records**.
       The **Database Manager Data Access** form opens.
    2. Click **Fill** in the **Database Table Name** field to select a table.
    3. The user role you initially selected is in the **User Role** field. If you want to add access to a different user role than the one you selected, clear the field and click **Fill** to select a different user role.
    4. To set the database access, select one of the following:
        - **Allow DB access**: The user role specified is granted access to the table specified.
        - **Prohibit DB access**: The user role specified is denied access to the table specified.
    5. When access has been granted to a table, click **Fill** in the **View Format** field to select the table view.
    6. Click **Add**.
       The new database access record is added.
6. To modify an existing database access record, follow these steps:
    1. Select the **Data Access** tab.
    2. Double-click the Table Name of the existing database access record you want to modify.
       The **Database Manager Data Access** form opens.
    3. Make the necessary edits.
    4. Click **Save**.
       Your changes are saved.
7. Click **OK**.

Related topics

Add a user role record
Delete a user role record
Search for a user role record

6.4.7.5. Set the dashboard home page for a user role

Setting a dashboard home page for a user role enables you to auto-fill operator records with a dashboard home page, along with the other default settings for the user role. If no dashboard home page is set, the user sees a To Do list upon login.

Note

Note: Specifying a dashboard home page in a user's operator record overrides the default To Do list and menus for that operator when they log in using the web client, even if the flag for menus has been set.

To set a dashboard home page for a user role, follow these steps:

1. Click **System Administration** > **Ongoing Maintenance** > **User Roles**.
2. Type or select optional search criteria.
3. Click **Search**.
4. Select a user role from the record list.
5. Click the **Startup** tab.

6. Select a dashboard home page from the drop-down list.
7. Click **Save**.

Related topics

To Do list
Add a user role record
Delete a user role record
Search for a user role record

6.4.8. Capability word model

Capability words provide a security mechanism to control access to Service Manager applications by enabling or disabling parts of the interface. You can add capability words to a user role or individual operator to control access to Service Manager. In some cases capability words are redundant to the privileges and views that are already provided by application profiles. In cases where capability words and application profiles overlap, capability words take precedence.

Service Manager stores capability words in the capability table, which you can access by opening the capability form in Database Manager, or by clicking **System Administration** > **Ongoing Maintenance** > **Capability Words**. To limit access, choose a subordinate capability word. To grant a broad range of permissions, choose a parent capability word, such as SQLAdmin or SysAdmin. If you create your own capability words or modify the default permissions, you can use the **checkPermission()** JavaScript function that's available in the script library to determine if an operator has a particular capability word. You must use this function to enforce the capability word model.

Related topics

Application Setup
Capability words
Add a capability word
Delete a capability word
Search for capability words

6.4.8.1. Capability words

Service Manager provides a hierarchy of out-of-box capability words.

Note

Note Capability words are case-sensitive.

Capability	Primary subordinate	Secondary subordinate	Application	Description
EditContacts			Any application mainly for admins to edit contacts (Administration)	Enables editing of contact records.
EditOperators			Any application mainly for admins to edit operators (Administration)	Enables editing of the password in operator records. If you're not SysAdmin, without this capability word even if you could edit some other operator information you wouldn't be able to change the password.
HTMLTemplatesAdmin			Applications that use the HTML email feature.	Enables a user to manage or create HTML templates.
ODBC			Server Access	Grants access via ODBC32.DLL for reporting.
RESTful API			Web Services	Enables a user to execute a RESTful API request.
SOAP API			Web Services	Enables the user to sign in to Service Manager and execute a SOAP API request.
SQLAdmin			SQL	Enables SQL administrator authority.
SysAdmin	See the next table	See the next table	All Applications except ESS	Enables system administrator capability.

SysAdmin (continued)

Primary subordinate	Secondary subordinate	Application	Description
system.build		Administration	Grants access to old binary text format unload options. These are used to create a Service Manager file system from scratch and must always be available to someone.
programmer		RAD	Grants access to the application generator (ag).
programmer	**Debug**	Debugger	Enables a non-sysadmin user to do debugging for testing purposes. Normal users should not have this capability word unless there is debugging being done for that user.
AlwaysAdmin		Database Manager	Defaults administration mode to true when using the Database Manager.
ChMAdmin		Change Management	Grants access to Change Management administration.
ChMAdmin	**change request**	Change Management	Grants access to Change Management requests.
ChMAdmin	**change task**	Change Management	Grants access to Change Management tasks.
ChMAdmin	**expedite change**	Change Management	RFC management: this person may expedite changes.
GUIAdmin		User Interface	Grants access to GUI administration.
GUIAdmin	**chart.breakdown**	Charts	Enables the charting capability in the Service Manager client.
GUIAdmin	**public.favorites**	Favorites	Enables public favorites in the Windows / and Web client.
GUIAdmin	**user.favorites**	Favorites	Enables the user favorites in the Windows / and Web client.

menu.commands		Command line	Grants command line access to functions from menu.
ICMAdmin		Configuration Management	Grants access to Configuration Management administration.
inventory management		Configuration Management	Grants access to Configuration Management.
help		Help	Enables the user to add or update help records
IncidentAdmin		Incident Management	Grants access to Incident Management administration.
IncidentAdmin	**amend suspension**	Incident Management	Enables the capability to amend the Incident Management profile rights for suspend/unsuspend.
IncidentAdmin	**incident management**	Incident Management	Grants access to Incident Management.
KMAdmin		Knowledge Management	The KMAdmin capability word enables a user to perform all Knowledge Management and Knowledge Centered Support (KCS) tasks for all document categories including the administrative tasks associated with maintaining the Knowledge Management system.
knowledge engineer		IR Expert	Enables the capability to modify the adaptive learning records.
mobile.admin		Mobile	Grants access to the Portal for mobile user administration.
mobile.admin	**mobile.user**	Mobile	Grants access to the Portal for mobile users.
ProbAdmin		Problem Management	Grants access to Problem Management administration.
ProbAdmin	**problem management**	Problem Management	Grants access to Problem Management.

query.window		Queries	Grants access to query window function in query.window application.
QueryAdmin		Queries	Enables query administrator capability. Full access to query options/maintenance. Includes: query window, stored query usage and editing, etc. This is intended to be used like sysadmin, as an administrator, not for general usage. Please use caution assigning this as it gives access to create/modify stored queries, which can adversely affect system performance if not done correctly.
QueryAdmin	**partial.key**	Queries	Enables the user to execute a partial key query in Database Manager or Change Management.
QueryAdmin	**partial.key.msg.skip**	Queries	Enables the user to skip the partial key query message in Database Manager.
QueryAdmin	**query.stored**	Queries	Enable user to execute stored queries when F6 (query) key is pressed, from any application. The user will be able to perform any stored query assigned to him/her or a group he/she belongs to (allowable groups in operator record).
QueryAdmin	**query.stored.mod**	Queries	Enable user execute/modify access of stored queries when F6 (query) key is pressed from any application.
OCMAdmin		Request Fulfillment	Allows access to Request Fulfillment administration, such as the approval delegation application.

OCMAdmin	**OCML**	Request Fulfillment	Allows access to the legacy Request Management Line Item applications. Since Line Item is unsupported in Request Fulfillment in Server Manager Codeless mode, it can be considered obsolete if you don't use it anywhere else.
OCMAdmin	**OCMO**	Request Fulfillment	Allows access to Request Fulfillment order records.
OCMAdmin	**OCMQ**	Request Fulfillment	Allows access to Request Fulfillment request records in the approval delegation application.
svcCatAdmin		Service Catalog	Grants access to Service Catalog administration.
svcCatAdmin	**service catalog**	Service Catalog	Enables users to request items from catalog.
svcCatAdmin	**svcCatDeptRequester**	Service Catalog	Enables a department to request items from catalog.
svcCatAdmin	**svcCatEmployeeRequester**	Service Catalog	Enables an employee to request items from catalog.
svcCatAdmin	**svcCatManagerRequester**	Service Catalog	Enables a manager to request items from catalog.
svcCatAdmin	**svcCatTechRequester**	Service Catalog	Enables a technician to request items from catalog.
svcCatAdmin	**svcCatRequestOnBehalf**	Service Catalog	Enables an employee self-service (ESS) user to submit a service catalog request on behalf of another user.
SDAdmin		Service Desk	Grants access to Service Desk administration.
SDAdmin	**service desk**	Service Desk	Grants access to Service Desk.
data administrator		Service Desk, Incident Management	Enables save, add and update on supporting tables from within Service Management and Incident Management.
user		User	Grants basic access to an end user but doesn't grant the ability to add, delete, or reset.

6.4.8.2. Add a capability word

Applies to User roles: System Administrator

To add a capability word, follow these steps:

1. Click **System Administration** > **Ongoing Maintenance** > **Capability Words**.
2. Type the information for your new capability word.
3. Click **Add**.

Related topics

Application Setup
Capability word model
Capability words
Delete a capability word
Search for capability words

6.4.8.3. Delete a capability word

Applies to User roles: System Administrator

To delete a capability word, follow these steps:

1. Click **System Administration** > **Ongoing Maintenance** > **Capability Words**.
2. Type or select optional search criteria.
3. Click **Search**.
4. Select the capability word that you want to delete.
5. Click **Delete**.
6. Click **Yes** to confirm the deletion.

Related topics

Application Setup
Capability word model
Capability words
Add a capability word
Search for capability words

6.4.8.4. Search for capability words

Applies to User roles: System Administrator

To search for capability words, follow these steps:

1. Click **System Administration** > **Ongoing Maintenance** > **Capability Words**.
2. Type or select optional search criteria.
3. Click **Search**.

Related topics

Capability word model
Capability words
Add a capability word
Delete a capability word

6.4.9. Operator passwords

Passwords play a major part in controlling system security within Service Manager. To protect your system from unauthorized access, you can require passwords for every operator.

A System Administrator can change any operator password. Users can also change their own passwords, unless the System Administrator denies operators password modification privileges through extra tailoring.

A System Administrator can set password format restrictions and time limits from the System Information Record. The System Administrator can set the following password options:

- Format requirements
- Maximum and minimum lifetimes
- The default password for reset passwords
- Enable the forgot password function and configure the related settings

A System Administrator can also configure a series of security questions on the System Security Question page after the forgot password function is enabled.

Related topics

Adding users
Checklist: Adding a new user
Controlling user access and security
Creating operator records
Defining named users
Operator records
Operator templates
Change a user's password
Disable the password reset option
Enable password history
Enable the password reset option
Enable the forgot password option
Reset an operator's password
Set password format restrictions
Set password maximum lifetimes
Set password minimum lifetimes

6.4.9.1. Change a user's password

Applies to User roles: System Administrator

> **Note**
>
> Note A System Administrator can both change and reset a user password. Changing an operator password permanently changes the password. Resetting an operator password temporarily changes the password until the operator next logs in at which time Service Manager prompts the operator to change the password.

To change a user password, follow these steps:

1. Click **System Administration** > **Ongoing Maintenance** > **Operators**.
2. Type or select optional search criteria.
3. Click **Search**.
4. Select the operator with password to be changed.
5. Click the **Security** tab.
6. In the **Password** field, type the new password for the operator.
7. Click **Save**.

Related topics

Application Setup
Checklist: Adding a new user
Controlling user access and security
Creating operator records
Operator passwords
Operator records

Operator templates
Disable the password reset option
Enable the password reset option
Enable password history
Reset an operator's password
Set password format restrictions
Set password minimum lifetimes
Set password minimum lifetimes

6.4.9.2. Disable the password reset option

Applies to User roles: System Administrator

To disable the password reset option, follow these steps:

1. Click **System Administration** > **Base System Configuration** > **Miscellaneous** > **System Information Record**.
2. Click the **Passwords** tab.
3. Select the **Prevent Pwd Resets** option.
4. Click **Save**.

Related topics

Application Setup
Checklist: Adding a new user
Controlling user access and security
Creating operator records
Operator passwords
Operator records
Operator templates
Change a user's password
Enable the password reset option
Enable password history
Reset an operator's password
Set password format restrictions
Set password minimum lifetimes
Set password minimum lifetimes

6.4.9.3. Enable password history

Applies to User roles: System Administrator

To enable password history, follow these steps:

1. Click **System Administration** > **Base System Configuration** > **Miscellaneous** > **System Information Record**.
2. Click the **Password Standards** tab.
3. Select the **Keep Password History** option.
4. Click **Save**.

Related topics

Application Setup
Checklist: Adding a new user
Controlling user access and security
Creating operator records
Operator passwords
Operator records
Operator templates
Change a user's password
Disable the password reset option
Enable the password reset option
Reset an operator's password

Set password format restrictions
Set password minimum lifetimes
Set password minimum lifetimes

6.4.9.4. Enable the password reset option

Applies to User roles: System Administrator

The **Reset Operators Password** option provides temporary passwords. However, when you reset an operator password, Service Manager doesn't verify that the new password conforms to any of the password standards that you can define in the company record.

To enable the password reset option, follow these steps:

1. Click **System Administration** > **Base System Configuration** > **Miscellaneous** > **System Information Record**.
2. Click the **Password Standards** tab.
3. Select one of the following options.

Password reset option	Description
Prevent Pwd Resets	Select this option to remove **Reset Operators Password** from the **More Actions** menu. You can still manually change operator passwords from the operator record.
Reset to User Name	Select this option to reset the operator password to the log-in name whenever an administrator chooses **Reset Operators Password** from the **More Actions** menu.
Prompt for Value	Select this option to display a dialog box where an administrator can enter a new password whenever the administrator chooses **Reset Operators Password** from the **More Actions** menu.
Reset to Value	Select this option to reset an operator password to the value you type in the text field whenever the administrator chooses **Reset Operators Password** from the **More Actions** menu.

4. Click **Save**.

Related topics

Application Setup
Checklist: Adding a new user
Controlling user access and security
Creating operator records
Operator passwords
Operator records
Operator templates
More Actions menu
Change a user's password
Disable the password reset option
Enable password history
Reset an operator's password
Set password format restrictions
Set password minimum lifetimes
Set password minimum lifetimes

6.4.9.5. Reset an operator's password

Applies to User roles: System Administrator

To reset an operator password, follow these steps:

1. Click **System Administration** > **Ongoing Maintenance** > **Operators**.
2. Use search or advanced search to find one or more records.
3. Select the operator record from the record list.
4. Click **More** or the **More Actions** icon, and then select **Reset Operators Password**.
5. Click **Yes** to confirm the operation.

When you reset an operator password, Service Manager doesn't verify that the new password conforms to the password standards that you defined in the company record. Therefore, use the **Reset Operators Password** option to provide only temporary passwords.

Related topics

Checklist: Adding a new user
Controlling user access and security
Creating operator records
Operator passwords
Operator records
Operator templates
Change a user's password
Disable the password reset option
Enable the password reset option
Enable password history
Set password format restrictions
Set password minimum lifetimes
Set password maximum lifetimes

6.4.9.6. Set password format restrictions

Applies to User roles: System Administrator

To set password format restrictions, follow these steps:

1. Click **System Administration** > **Base System Configuration** > **Miscellaneous** > **System Information Record**.
2. Click the **Password Composition** tab.
3. Type or select the following options:
    - **Min Password Length**: type the minimum number of characters a password can contain.
    - **Max Password Length**: type the maximum number of characters a password can contain.
    - **Allow Alpha Characters**: select this option to allow users to create passwords with alphabetical characters.
    - **Allow Numeric Characters**: select this option to allow users to create passwords with numeric characters.
    - **Allow Symbolic Characters**: select this option to allow users to create passwords with symbolic characters.

      Note

      Note Service Manager prevents users from creating passwords that start with the following symbolic characters: pound (#), equal sign (=), tilde (~), greater than (>), and less than (<).

    - **Always Require a Password**: select this option to require all users to have a password defined in their operator record. This option is the only way to prevent users from having blank passwords.

      Caution

      Caution Users who currently have blank passwords defined in their operator record can still log in until they are required to change their passwords.

    - **Require Alpha Chars**: select this option to require users to create passwords that contain at least one alphabetical character.

        - **Min Required**: type the minimum number of alphabetical characters a password must contain. This value must be a positive integer if you select the **Require Alpha Chars** option.
    - **Require Mixed Case**: select this option to require users to create passwords that contain both upper case and lower case alphabetical characters.
    - **Require Non-Alpha Chars**: select this option to require users to create passwords that contain at least one numerical or symbolic character.
        - **Min Required**: type the minimum number of numerical characters a password must contain. This value must be a positive integer if you select the **Require Non-Alpha Chars** option.
    - **Prohibit Space Character**: select this option to prohibit users from creating passwords that contain the space character.
4. Click **Save**.

> **Caution**
>
> Caution The only way to prevent users from defining blank passwords is to select the **Always Require a Password** option.

Related topics

Checklist: Adding a new user
Controlling user access and security
Creating operator records
Operator passwords
Operator records
Operator templates
Change a user's password
Disable the password reset option
Enable the password reset option
Enable password history
Reset an operator's password
Set password maximum lifetimes
Set password minimum lifetimes

6.4.9.7. Set password maximum lifetimes

Applies to User roles: System Administrator

To set password maximum lifetimes, follow these steps:

1. Click **System Administration** > **Base System Configuration** > **Miscellaneous** > **System Information Record**.
2. Click the **Password Lifetime** tab.
3. Type or select the following options:
    - **No Max Lifetime**: select this option to disable password expiration.
    - **Use Time Period**: select this option to have passwords expire after a set time period.
        - **Max Pwd Lifetime**: type the number of days, hours, minutes, and seconds before user passwords expire.

Type the time period in the following format: *Dayshours*: *minutes*: *seconds*. For example, 4 03: 02: 01 expires user passwords after 4 days, 3 hours, 2 minutes, and 1 second.

1.
    - **Use Number of Logins**: select this option to have passwords expire after a set number of user logins.
        - **Max before change**: type the number of logins before user passwords expire.
    - **Notify by Email on Password Change**: select this option to notify users by email whenever another user, such as a System Administrator, changes their password.
2. Click **Save**.

Related topics

Checklist: Adding a new user
Controlling user access and security
Creating operator records
Operator passwords
Operator records
Operator templates
Change a user's password
Disable the password reset option
Enable the password reset option
Enable password history
Reset an operator's password
Set password format restrictions
Set password minimum lifetimes

6.4.9.8. Set password minimum lifetimes

Applies to User roles: System Administrator

To set password minimum lifetimes, follow these steps:

1. Click **System Administration** > **Base System Configuration** > **Miscellaneous** > **System Information Record**.
2. Click the **Password Lifetime** tab.
3. Type or select the following options:
    - **No Min Lifetime**: select this option to disable password expiration.
    - **Use Time Period**: select this option to have passwords expire after a set time period.
        - **Min Pwd Lifetime**: type the number of days, hours, minutes, and seconds before user passwords expire. Type the time period in the following format: *Dayshours*: *minutes*: *seconds*. For example, 4 03: 02: 01 expires user passwords after 4 days, 3 hours, 2 minutes, and 1 second.
    - **Use Number of Logins**: select this option to have passwords expire after a set number of user logins.
        - **Min before change**: type the number of logins before user passwords expire.
    - **Notify by Email on Password Change**: select this option to notify users by email whenever another user, such as a System Administrator, changes their password.
4. Click **Save**.

Related topics

Checklist: Adding a new user
Controlling user access and security
Creating operator records
Operator passwords
Operator records
Operator templates
Change a user's password
Disable the password reset option
Enable the password reset option
Enable password history
Reset an operator's password
Set password format restrictions

6.4.9.9. Configure system security questions

Applies to User roles: System Administrator

Security questions assist to verify a user's identity. After you enable the forgot password function, the system prompts to select and answer the security questions when a user is resetting the login password from the web client. A user must select the security questions and enter the answers before proceeding to create a new password. As a system administrator, you can refer to the following instructions to configure system security questions for users to select and answer when they set up their security questions.

From the System Navigator, click **System Administration** > **Base System Configuration** > **Miscellaneous** > **System Security Question**. You can review the out-of-the-box system security questions, add new questions, localize the questions, and delete any unused questions on this form.

Add a security question

To add a new security question, perform either of the following:

- Click **New**. Type a question in the **Title** field, and then click **Add** and **OK**. The system automatically populates the ID for this question.
- Click **Search** to display a list of existing security questions, and then click **Add**. Type a question in the **Title** field, and then click **Save** and **OK**.

Update a security question

To update a security question, click **Search** to display a list of existing security questions. Select a question, update the **Title** field as required. Click **Save** and **OK** to save your updates.

Localize a security question

To localize a security question, select a question as required and then click **Localize**. For each language that's active in your Service Manager system, you can localize the **Title** field in the corresponding text field. After you localize the question, click **Save** to return to the question list.

Deactivate a security question

To deactivate a security question, select a question as required and then select the **Inactive** checkbox. The subsequent users can no longer see this deactivated question when they set up the security questions, whereas the users who are previously using this deactivated question aren't affected.

Delete a security question

To delete a security question, select a question as required and then click **Delete**. Note that you can't delete any security questions that are in use.

6.4.9.10. Enable the forgot password function

Applies to User roles: System Administrator

The forgot password function enables a user to reset the password through the Service Manager web client on one's own. A system administrator must firstly enable the forgot password function, and then configure the related settings.

The forgot password function is only available for either of the following scenarios:

- The Service Manager server uses database authentication and the users' passwords exist in the database. In this scenario, the web client login page provides a default Forgot Password link after you enable the forgot password function. Users can click this link to go to an internal password reset page.
- The Service Manager server uses LDAP authentication. In this scenario, the web client login page doesn't provide a default Forgot Password link and you must configure Web parameter: externalForgotPasswordUrl. After you enable the forgot password function, users can click the Forgot Password link to go to an external password reset URL as you specified in the *externalForgotPasswordUrl* parameter.

Tip: We recommend you to enable SSL before enabling the forgot password function.

To enable the forgot password function, follow these steps:

1. Sign in to Service Manager as a system administrator by using either the Windows client or the web client.
2. Click **System Administration** > **Base System Configuration** > **Miscellaneous** > **System Information Record**.
3. Click **Passwords** tab > **Forgot Password** tab.
4. Select the **Enable Forgot Password** checkbox to enable the forgot password function.
5. Configure the following settings.

Forgot password option	Description
Token Expiration Time (mins)	Set the number of minutes that the password token will be valid. The default maximum value is 60 minutes.
Reset Password Lock Duration (hours)	Set the number of hours that the reset password function remains locked before automatically becoming unlocked. The default minimum value is 1 hour.
Number of Security Questions	Set the number of security questions that the users need to answer when they forget passwords. By default, the minimum value is 3 and the maximum value is 6. If you update this value, a user can still reset the password based on the security questions settings before. The system informs and prompts a user to update the security questions accordingly when this user logs on to the Service Manager web client.
Minimum Character of a Valid Answer	Set the minimum number of characters that an answer will be valid. If an answer consists of fewer characters, this answer is invalid to the security questions. The default minimum value is 3 and you can set the maximum value according to the database type for Answer: VARCHAR(1000).
Maximum Answer Retry Times	Set the maximum number of retry attempts when a user's answer is incorrect. If the number of a user's retry attempts exceeds this value, the reset password function is temporarily locked. By default, the minimum value is 1 and the maximum value is 10.

6. Click **Save** and **OK**.

Next, you may need to review the out-of-the-box security questions, and then add, update, or localize these security questions as required. Security questions assist to verify a user's identity. The system will prompt for the security questions when a user is resetting the login password from the web client. For more information, see Configure system security questions.

Related topics

Application Setup
Checklist: Adding a new user
Controlling user access and security
Creating operator records
Operator passwords
Operator records
Operator templates
More Actions menu
Change a user's password
Disable the password reset option
Enable password history
Reset an operator's password
Set password format restrictions

Set password maximum lifetimes
Set password minimum lifetimes

6.4.10. License tracking

Service Manager tracks floating and named licenses on the application level for reporting purposes.

The logon process determines which applications are authorized for each user, allocates a floating or named license for the duration of the session, and records usage information in the stathistory file. When the user logs out or ends the session, Service Manager deallocates the license. Because Service Manager tracks the highest daily license usage for each application, administrators can use this information to determine how many licenses are being used and when peak usage times occur.

The license tracking feature tracks the named and floating licenses on an application basis. Service Manager refreshes the license usage information whenever the user logs out, or the session terminates by inactivity or administrator action.

Service Manager calculates license usage for all types of client connections, including SOAP, Web Services, and REST. The calculation is based on the number of login user names and the number of client IP addresses. For example:

- Two sessions that use the same login user name and two different client IP addresses consume two licenses.
- Two sessions that use the same login user name and the same client IP address consume one license.
- Two sessions that use two different login user names consume two licenses.

Authorized applications

The following applications monitor licenses:

- Contract Management
- Change Management
- Incident Management
- Configuration Management
- Request Fulfillment
- Problem Management
- Service Desk
- Knowledge Management
- Service Level Management
- Smart Analytics

6.4.10.1. License types

There are two types of licenses: login licenses (named or floating, which only control initial logins), and application licenses (which control access to individual modules).

License type	Description
Login licenses	Control initial logins: • Floating: Users aren't on the Named user list for the application and they have update capabilities. • Named: Users are on the Named user list for the application and they have update capabilities.
Application licenses	Control access to individual modules. Logging in as any user with update capacity for an application consumes a license for that application. Update capacity includes permission to add, update, or delete any record associated with the application.

Login licenses are tightly controlled so that the product will not allow you to log in if you exceed your login licenses. Application licenses are lightly controlled, and you can exceed the licenses with only a warning message.

If the Named Application field in the operator record is empty and the operator logging in is a Named user, the user will consume one named login license and floating licenses for application license. If the Named Application field in the operator record is only filled with, for example, Incident Management, the same user will consume one named login license, one floating license for each other application/module, and one named license for the Incident Management application/module. In other words, by default the Application Licenses are floating licenses regardless of whether the operator is a named or floating user.

Login licensing has been separated from Module licensing to provide more flexibility. It is possible to change the default behavior to ensure the Named user consumes a Named license, instead of a Floating license for the Modules. To do so, follow these steps:

1. Open the Operator record.
2. Click the **Login Profiles** tab, add the Modules required for the Named user in the **Named Applications** field.

Related topics

License tracking
Named users
Named users for applications
Stathistory table
Self-service licenses
Determine the number of named users available
Make an individual operator a named user
Define named users for applications
Generate a user license report
Profile fields for Contract Management license tracking
Profile fields for Knowledge Management license tracking

6.4.10.1.1. Security right fields for Configuration Management license tracking

Users who have the capability to update records in Configuration Management will consume either a floating or a named license. Service Manager tracks these licenses for reporting purposes.

Users who have update capabilities for any of the following fields in the Configuration Management security rights will consume a Configuration Management license.

Area	Tab	Label of Settings	Value	Can do
Configuration Item Configuration Management Configuration	Rights	New	true	New
Configuration Item Configuration Management Configuration	Rights	Update	Always/When assigned/When assigned to workgroup	Update
Configuration Item Configuration Management Configuration	Rights	Delete/Close	Always/When assigned/When assigned to workgroup	Close/Inactivate/Mass Inactivate
Configuration Item Configuration Management Configuration	Rights	Expert	true	Can suspend/Can unsuspend/Can Mass Update/Can Complex Update
Configuration Item	Rights	Change CI Type	true	Can change CI Type

6.4.10.1.2. Profile fields for Knowledge Management license tracking

Users who have the capability to update records in Knowledge Management will consume either a Floating or Named license. Service Manager tracks these licenses for reporting purposes.

Users who have update capabilities for any of the following fields in the Knowledge Management security profile (kmprofile) will consume a Knowledge Management license.

KCS I

Label	Field Name in kmprofile
Contribute New Knowledge	contribute
Save a Draft Copy	save.draft
Edit working copy documents in Knowledge Management	edit.in.workflow

KCS II

Label	Field Name in kmprofile
Publish internal immediately	int.immediate
Retire / Archive a document	retire
Publish a working copy document internally	internal
Unretire a retired document	unretire
Modify a published document	modify
Delete a retired document	delete.doc
Revert a working copy to the published version	revert
Delete Feedback	delete.feedback

KCS III

Label	Field Name in kmprofile
Publish internal and external immediately	ext.immediate
Publish a working copy document internally and externally	external

Admin

Label	Field Name in kmprofile
Manage document types and views	doctypes
Manage Categories	manage.categories
Manage knowledgebases	knowledgebases
Integration Mapping	integration.mapping
Manage shared content	shared.content
Configure hit list	configure.hit.list
Manage environment and profiles	environment.profiles
Review/publish any draft	any.draft
Search in all categories	admin.search.all
Edit Adaptive Learning	edit.adaptive.learning

Related topics

License tracking
Named users
Named users for applications
Stathistory table
Self-service licenses
Determine the number of named users available
Make an individual operator a named user

Define named users for applications
Generate a user license report

6.4.10.1.3. Profile fields for Contract Management license tracking

Users who have the capability to update records in Contract Management will consume either a Floating or Named license. Service Manager tracks these licenses for reporting purposes.

Users who have update capabilities for any of the following fields in the Contract Management security profile (ctenv) will consume a Contract Management license.

General Privileges

Label	Field Name
Mass Add	mass.add
Mass Update	mass.update
Mass Delete	mass.delete

Contracts

Label	Field Name
Add Contract	add.contract
Update Contract	update.contract
Cancel Contract	cancel.contract
Renew Contract	renew

Payments

Label	Field Name
Schedule Payment	sched.payment
Gen Payment Sched	gen.payment.sched
Update Payment	update.payment
Submit Payment	submit.payment
Cancel Payment	cancel.payment
Delete Payment	delete.payment

Contract Templates

Label	Field Name
Create Template	create.template
Update Template	update.template
Delete Template	delete.template

Asset Allocation

Label	Field Name
Add Asset	add.items
Gen Asset Allocation	gen.item.allocation
Update Asset	update.items
Delete Asset	delete.items

Software Counters

Label	Field Name
Create Counter	add.counter
Update Counter	update.counter
Delete Counter	delete.counter
Compliance Check	compliance.check

Terms and Conditions

Label	Field Name
Add Term/Condition	add.term
Update Term/Condition	update.term
Delete Term/Condition	delete.term

Related topics

License tracking
Named users
Named users for applications
Stathistory table
Self-service licenses
Determine the number of named users available
Make an individual operator a named user
Define named users for applications
Generate a user license report

6.4.10.1.4. Security right fields for Service Desk license tracking

Users who have the capability to update records in Service Desk will consume either a floating or a named license. Service Manager tracks these licenses for reporting purposes..

Users who have update capabilities for any of the following fields in the Service Desk security rights will consume a Service Desk license.

Area	Tab	Label of Settings	Value
Service Desk	Rights	New	true
Service Desk	Rights	Update	Always/When assigned/When assigned to work group
Service Desk	Rights	Delete/Close	Always/When assigned/When assigned to work group

6.4.10.1.5. Security right fields for Incident Management license tracking

Users who have the capability to update records in Incident Management will consume either a floating or a named license. Service Manager tracks these licenses for reporting purposes.

Users who have update capabilities for any of the following fields in the Incident Management security rights will consume an Incident Management license.

Area	Tab	Label of Settings	Value	Can do
Incident Incident Tasks	Rights	New	true	New
Incident Incident Tasks	Rights	Update	Always/When assigned/When assigned to workgroup	Update
Incident Incident Tasks	Rights	Delete/Close	Always/When assigned/When assigned to workgroup	Close/Inactivate/Mass Inactivate
Incident Incident Tasks	Rights	Expert	true	Can suspend/ Can unsuspend

6.4.10.1.6. Security right fields for Problem Management license tracking

Users who have the capability to update records in Problem Management will consume either a floating or a named license. Service Manager tracks these licenses for reporting purposes.

Users who have update capabilities for any of the following fields in the Problem Management security rights will consume a Problem Management license.

Area	Tab	Label of Settings	Value
Problem Problem Tasks	Rights	New	true
Problem	Rights	Create Known Error	true
Problem Problem Tasks	Rights	Update	Always/When assigned/When assigned to workgroup
Problem Problem Tasks	Rights	Delete/Close	Always/When assigned/When assigned to workgroup

6.4.10.1.7. Security right fields for Change Management license tracking

Users who have the capability to update records in Change Management will consume either a floating or a named license. Service Manager tracks these licenses for reporting purposes.

Users who have update capabilities for any of the following fields in the Change Management security rights will consume a Change Management license.

Area	Tab	Label of Settings	Value	Can do
Change/Change Task	Rights	New	true	Copy and Open/New
Change/Change Task	Rights	Expert	true	Update when closed/Can Mass Approve
Change/Change Task	Rights	Admin	true	Change Category/Change Phase/Can override Approvals
Change/Change Task	Rights	Update Delete/Close	Always/When assigned/When assigned to workgroup	Update/delete/Close
Change/Change Task	Settings	Reopen/Can Approve	true	Reopen/Can Approve

6.4.10.1.8. Security right fields for Request Fulfillment license tracking

Users who have the capability to update records in Request Fulfillment will consume either a floating or a named license. Service Manager tracks these licenses for reporting purposes.

Users who have update capabilities for any of the following fields in the Request Fulfillment security rights will consume a Request Fulfillment license.

Area	Tab	Label of Settings	Value	Can do
Request/Request Tasks	Rights	New	true	New
Request/Request Tasks	Rights	Expert	true	• Update • Mass Approve
Request/Request Tasks	Rights	Admin	true	Override (Approve.Override)
Request/Request Tasks	Rights	Update Delete/Close	Always/When assigned/When assigned to workgroup	Close
Request	Settings	Can Approve	true	Can Approve
Request	Settings	Reopen	true	Reopen

Related topics

Security
Request Fulfillment security areas

6.4.10.1.9. Security right fields for Service Level Management license tracking

Users who have the capability to update records in Service Level Management will consume either a floating or a named license. Service Manager tracks these licenses for reporting purposes.

Users who have update capabilities for any of the following fields in the Service Level Management security rights will consume a Service Level Management license.

Area	Tab	Label of Settings	Value
Service Level Management	Rights	New	true
Service Level Management	Rights	Update	Always/When assigned/When assigned to workgroup
Service Level Management	Rights	Delete/Close	Always/When assigned/When assigned to workgroup
Service Level Management	Settings	Manage SLT Catalog Records	true

6.4.10.1.10. Fields for Self-service license tracking

Users who have following fields in the Service Desk security rights and users who have the capability to Order from Service Catalog via ESS portal (added with **service catalog** capability word) will consume a Service Catalog ESS license.

Area	Tab	Label of Settings	Value
Service Desk	Rights	New	true

6.4.10.1.11. Capability words for Smart Analytics license tracking

Users who have the capability to update records in Smart Analytics will consume either a floating or a named license. Service Manager tracks these licenses for reporting purposes.

Users who have the **SysAdmin** or **idol.assistant** capability word will consume a Smart Analytics license. If there is no Help Desk license, Smart Analytics can't work.

6.4.10.2. Named users

Service Manager uses the software license agreement, which provides for two types of licensed users: named users and floating users.

Named users are users who are always authorized to sign in to the Service Manager system when either floating user licenses or named user licenses are available. Floating users are limited to a maximum number of concurrent logins as defined by the floating user license. The Service Manager license key password file encodes the number of named users and floating users in the system as separate values. You can define as many named users as you want. If the number of named users signed in exceeds the limit of named user licenses, the surplus named users consume floating licenses to sign in to the Service Manager system. Floating users also consume floating licenses when they sign in. If all of your floating licenses have been used, Service Manager denies access to any additional unique floating users. Unique-named users will still be able to sign in if Named licenses are available. If all named licenses and floating licenses are used, no additional unique users (named or floating) will be able to connect.

Related topics

License tracking
Named users for applications
Stathistory table
Self-service licenses
Determine the number of named users available
Make an individual operator a named user
Define named users for applications
Generate a user license report
License types
Profile fields for Contract Management license tracking

6.4.10.2.1. Defining named users

You can use the following method to define named users on your Service Manager system:

- Define each named user in the user's operator record

Related topics

Application Setup
Adding users
Checklist: Adding a new user
Controlling user access and security
Creating operator records
Operator passwords
Operator records
Operator templates

6.4.10.2.2. Determine the number of named users available

Applies to User Roles: System Administrator

You must have access to the Service Manager server to perform this procedure.

To determine the number of named users available, follow these steps:

1. Open a Windows command prompt window.
2. Change directories to the RUN folder of your Service Manager installation.

3. Type the following command:

```
sm -reportlic
```

4. Press **Enter**.

Service Manager displays a report of your licensed features and your current usage. The Named Users field lists the number of named users that are active and inactive on your system.

Related topics

License tracking
Named users
Named users for applications
Stathistory table
Self-service licenses
Make an individual operator a named user
Define named users for applications
Generate a user license report
License types
Profile fields for Contract Management license tracking

6.4.10.2.3. Make an individual operator a named user

Applies to User roles: System Administrator

To make an individual operator a named user:

1. Click **System Administration** > **Ongoing Maintenance** > **Operators**.
2. Type or select optional search criteria.
3. Click **Search**.
4. Select the operator record from the record list.
5. Click the **Login Profiles** tab.
6. Select the **Named User** option.
7. Click **Save**.

Related topics

License tracking
Named users
Named users for applications
Stathistory table
Self-service licenses
Determine the number of named users available
Define named users for applications
Generate a user license report
License types
Profile fields for Contract Management license tracking

6.4.10.3. Named users for applications

Service Manager tracks floating and named licenses for each application based on the number of users with the ability to update records in the application. Administrators can define a user as a named User for the application in the operator record or in the nameduser table.

Triggers keep the operator records and named user records synchronized. When you update the named.modules field in the operator record, Service Manager also updates the respective named user record.If you update a named user record; Service Manager updates the corresponding operator record.

Note

Note The out of box SM Reports module includes license utilization reports.

Related topics

License tracking
Named users
Stathistory table
Self-service licenses
Determine the number of named users available
Make an individual operator a named user
Define named users for applications
Generate a user license report
License types
Profile fields for Contract Management license tracking

6.4.10.3.1. Define named users for applications

Applies to User Roles: System Administrator

You can grant access and update rights to named users.

To define named users for applications by using the Named Applications method, follow these steps:

1. Click **System Administration** > **Ongoing Maintenance** > **Operators**.
2. Type or select optional search criteria.
3. Click **Search**.
4. Select the operator record to update.
5. Click the **Login Profiles** tab.
6. In the **Named Applications** field, type or select the application for which you want to enable operator access.
7. Click **Save**.

Note

Note If you want to grant access and update rights for multiple named users in a specific application, you can do this using the Named User method in Database Manager.

To define named users for applications by using the Named User method, follow these steps:

1. Click **Tailoring** > **Database Manager**.
2. In the **Form** field, type "nameduser".
3. Type or select optional search criteria.
4. Click **Search**.
5. In the **Application** field, specify the application to be accessed by the operator.
6. In the **Named users** field, type or select the operators to be assigned update rights for the specified application.
7. Click **Save**.

Related topics

License tracking
Named users
Named users for applications
Stathistory table
Self-service licenses
Determine the number of named users available
Make an individual operator a named user
Generate a user license report

License types
Profile fields for Contract Management license tracking

6.4.10.4. Generate a user license report

Applies to User roles: System Administrator

You must have administrative access to the server operating system to perform this procedure.

To generate a user license report, follow these steps:

1. Change directories to the RUN folder of your Service Manager installation.
2. Type the following command:
   sm -reportlic:1
3. Press **Enter**.

Related topics

License tracking
Named users
Named users for applications
Stathistory table
Self-service licenses
Determine the number of named users available
Make an individual operator a named user
Define named users for applications
License types
Profile fields for Contract Management license tracking

6.4.10.5. Stathistory table

The stathistory table updates every time a user logs on to the system. A stathistory record tracks the named and floating licenses on an application basis.

The application creates a new stathistory record daily. If a stathistory record already exists, the record updates. If one doesn't exist, the application creates a new record. The stathistory record names are:

Application	License type
Application name	- named
Application name	- Floating

For example, to search for floating license Configuration Management records, type:

```
Configuration Management - Floating
```

Each stathistory record updates with the number of named or Floating licenses used at that moment. The number updates only when it is greater than the number already in the table. This enables the administrator to keep track of the highest number of licenses used at any time. Here is a scenario to further explain the logic:

Scenario	Stathistory record value	Remark
User A logs into SM	1	
User B logs into SM	2	
User A logs out of SM	2	
User C logs into SM	2	Only User B and C are logged in SM at this moment. The number is not increased to 3 as there are only two concurrent users logged into SM.
User D logs into SM	3	
User B logs out of SM	3	The number updates only when it is greater than the number already in the table, so the number is not decreased to 2.

To view the stathistory table, follow these steps:

1. Click **Tailoring** > **Database Manager**.
2. In the File field, type: **stathistory**.
3. Click **Search**.
4. Select any record to view it.

Related topics

License tracking
Named users
Named users for applications
Self-service licenses
Determine the number of named users available
Make an individual operator a named user
Define named users for applications
Generate a user license report
License types
Profile fields for Contract Management license tracking

6.4.10.6. Self-service licenses

The number of licenses that are active depends on the number of users signed in who can access an application to add or update records. Self-service users can sign in to an ess.do URL to perform certain activities (such as, searching the knowledgebase to find an answer or logging new interactions) without contacting the Service Desk.

Users who have the capability to Order from Service Catalog via ESS portal (added with **service catalog** capability word) will consume a Service Catalog ESS license. Service Manager tracks these licenses for reporting purposes. You will receive a warning message in the **sm.log** file if the Service Catalog ESS licenses are exceeded.

Note

Note License utilization reports are included out-of-box in the Service Manager SM Reports module.

Related topics

License tracking
Named users
Named users for applications
Stathistory table
Determine the number of named users available
Make an individual operator a named user
Define named users for applications
Generate a user license report
License types
Profile fields for Contract Management license tracking

6.4.11. Folder entitlement

Folder entitlement isolates company information in folders, ensuring that the right users have the right access to sensitive company data. For example, a company that manages more than one organization could create separate security folders for each organization. Users in organization A could be granted access to Folder A and users in organization B could be granted access to Folder B. The information in an organization is tied to a security folder when users open records, such as incident records. Therefore, users in organization A could not access incidents, changes, or requests made by organization B, and vice versa. When folder entitlement is enabled, a new field appears when users open records. This field allows users to select the security folder, allowing them to tie the information in the record to that security folder. Users outside the organization who have not been granted permission to access the data in the folder can't see the record. When users outside the organization search incidents, changes, or requests, the records aren't found.

In Service Manager, a System Administrator enables or disables folder entitlement. For more information on enabling and disabling folder entitlement, see the related topics.

The available out-of-box company security folders on the Folder Entitlement tab are DEFAULT and advantage. However, the System Administrator can add folders to meet your business needs.

For each security folder added, a System Administrator must grant permissions to a specific application security profile, thereby granting permissions to all operators associated with that profile. Permission settings are selected for operators based on roles that reflect their responsibilities. When defining roles, each role displays a top level view for all folders. You can then click on each folder and set rights on individual fields for a folder, or modify workgroups. If a user doesn't have specific rights defined for a folder, the default rights from the user profile are assigned.

The System Administrator can set the default folder for an individual operator. This setting allows the records opened by the operator to be automatically associated with the default security folder selected.

Related topics

Disable folder entitlement
Enable folder entitlement
Delete a folder
Specify use of the default folder from the operator record

6.4.11.1. Enable folder entitlement

Applies to User Roles: System Administrator

To enable folder entitlement, follow these steps:

1. Click **System Administration** > **Base System Configuration** > **Miscellaneous** > **System Information Record**.
2. Click the **General** tab.
3. Select the **Enable Folder Entitlement** checkbox.
4. Click **Save**.
5. Click **OK**.
6. Log out of Service Manager and log back in.
   Folder Entitlement is now enabled.

Related topics

Folder entitlement
Disable folder entitlement
Add a security folder
Delete a folder

6.4.11.2. Disable folder entitlement

Applies to User Roles: System Administrator

To disable folder entitlement, follow these steps:

1. Click **System Administration** > **Base System Configuration** > **Miscellaneous** > **System Information Record**.
2. Click the **General** tab.
3. Deselect the **Enable Folder Entitlement** checkbox.
4. Click **Save**.
5. Click **OK**.
6. Log out of Service Manager and log back in.
   Folder entitlement has been disabled. There are no longer any security folders assigned to operators.

Related topics

Folder entitlement
Enable folder entitlement
Add a security folder
Delete a folder

6.4.11.3. Add a folder

Applies to User roles: System Administrator

To add a security folder, follow these steps:

1. From the System Navigator, click **System Administration** > **Ongoing Maintenance** > **Security Folders**.
   The **Folder Definition** form opens.
2. Type the name of the new folder in the **Folder Name** field.
3. Type the **Description** of the security folder.
4. Click **Add**.
5. Click **OK**.

To use an existing security folder to add a new folder:

1. From the System Navigator, click **System Administration** > **Ongoing Maintenance** > **Security Folders**.
   The **Folder Definition** form opens.
2. Add optional search criteria, and then click **Search**.
3. Select a folder from the list that most closely matches the new folder you want to add.
4. Type the new name of the folder in the **Folder Name** field.
5. Add or change the **Description** for this new security folder.
6. Click **Add**.

   Caution

   Caution Make sure that you don't click Save because doing so will replace the existing folder with the new security folder you are attempting to add.

7. Click **OK**.

Related topics

Folder entitlement
Delete a folder

Enable folder entitlement
Disable folder entitlement

6.4.11.4. Delete a folder

Applies to User Roles: System Administrator

To delete a folder, follow these steps:

1. From the System Navigator, click **System Administration** > **Ongoing Maintenance** > **Security Folders**.
2. Type the folder name, or click **Search** to select the folder name from a record list.
3. Select the folder and click **Delete**.
4. Click **OK**.

Note

Note You can't delete a folder that's currently in use. An error message displays if you attempt to delete a folder that's in use.

Related topics

Folder entitlement
Add a security folder
Enable folder entitlement
Disable folder entitlement

6.4.11.5. Specify use of the default folder from the operator record

Applies to User Roles: System Administrator

The System Administrator can set the default folder for an individual operator. This setting allows the records opened by the operator to be automatically associated with the default folder set in the operator record.

To specify use of the default folder from the operator record, follow these steps:

1. From the System Navigator, click **System Administration** > **Ongoing Maintenance** > **Operators**.
   The Operator search form opens.
2. Type optional search criteria, and then click **Search**.
3. Select the operator record to be edited.
4. In the **General** tab, select the **Folder Entitlement** tab.
5. In the **Default Folder** field, select a security folder as the default folder for this operator.
6. Click **Save**.
7. Click **OK**.
8. Log out of Service Manager, and then log back in.
   The default folder selected will automatically be used the next time the operator opens a new record.

Related topics

Folder entitlement
Application profiles
Add a security folder
Delete a folder
Enable folder entitlement
Disable folder entitlement

6.4.11.6. Add folder permissions to a security role

User Role: System Administrator

To add folder permissions to a security role:

1. Click **System Administration** > **Security** > **Roles**.
2. Click **Search** to select a security role from the record list.

3. Click the security role to which you want to add folder permissions.
4. Select a security area from the table in the security role details pane.
5. In the Folder Rights section, click **Add new folder** to open the Add New Folder Rights wizard.
6. Select a folder from the drop-down list, and then click **Next**.
7. Specify the rights for the role in this folder, and then click **Finish**.

Related topics

Process Designer security model
Processes and Best Practices
Add security roles and settings

6.4.11.7. Modify the folder permissions of a security role

User Role: System Administrator

To modify the folder permissions of a security role:

1. Click **System Administration** > **Security** > **Roles**.
2. Click **Search** to select a security role from the record list.
3. Click the security role for which you want to modify the folder permissions.
4. Select a security area from the table in the security role details pane.
5. In the Folder Rights section, select the folder for which you want to modify the permissions.
6. Modify the rights as required, click **Save**, and the click **OK**.

Related topics

Process Designer security model
Processes and Best Practices
Add security roles and settings
Add folder permissions to a security role

6.5. Calendars

The Service Manager calendar is an optional feature that enables the System Administrator to define custom work schedules for each group in the organization. These custom calendars determine when alerts and notifications occur for particular groups.

By default, Service Manager uses a calendar with a 24-hour work day and a 7-day work week to determine when alerts and notifications occur. In the default 24-hour, 7-day calendar, Service Manager sends alerts and notifications without regard to operator work shifts. For example, if an operator schedules a notification to occur in 6 hours, then Service Manager sends the notification after 6 hours have passed regardless of whether any operator is present to receive it. A notification scheduled at 5:00 p.m. in the default calendar arrives at 11:00 p.m. the same day.

Using a custom calendar, however, the same 6-hour delay occurs only during the defined work schedule. For example, a System Administrator can define an 8-hour work day lasting from 8:00 a.m. to 5:00 p.m., which includes a 1-hour break from 12:00 p.m. to 1:00 p.m. If an operator then schedules a notification to occur 6 hours after 5:00 p.m., the notification actually occurs at 3:00 p.m. the next work day, or 6 working hours after the notification start time.

System Administrators can create custom calendar records to use throughout Service Manager. The caldutyhours table contains custom calendar records that a System Administrator can assign to Service Manager activities:

- Change Management work schedule
- Change Management group definitions
- Incident Management assignment group definitions
- Incident Management deadline alert group definitions
- Request Management target order times
- Request Management target completion times
- Service Level Agreement service hours
- Service Level Agreement availability schedules

- Service Level Agreement response time schedules
- Service Desk on call times
- Vendor schedules

Related topics

Holiday records
On-call schedules
On-call schedule exceptions
Work schedules
Time zones
Create a holiday group
Create an on-call schedule

6.5.1. Holiday records

Service Manager includes out-of-box records for common holidays in the United States. To add a holiday, you must create a new holiday record in the calholidays table and associate that holiday with a group in calholtable.

Service Manager uses calholtable to organize holidays into different groups. For example, one group of holidays might list all holidays observed in France. This group would include worldwide holidays like Christmas and New Year's Day and those unique to France. A second group might also list the worldwide holidays, but add those unique to North America.

Associated tables

Service Manager uses information in these tables:

- calholidays contains a record for each holiday observed by your company.
- calholtable contains a record with the name of each group of holidays.

Create a holiday group

Applies to User Roles: System Administrator

To create a holiday group:

1. Click **Tailoring** > **Database Manager**.
2. Type **calholtable** in the Form field.
3. Click **Search**.
4. Type the name of the new holiday group in the **Holiday Group Name** field.
5. Click **Add**.

Related topics

Calendars
On-call schedule exceptions
On-call schedules
Time zones
Work schedules
Add, delete, or update a primary data record
Create an on-call schedule

6.5.2. On-call schedules

On-call schedules help managers or administrators determine whether they have enough resources on call to cover all shifts for each day. The on-call schedule specifies who is to be notified, when, and the notification method. An administrator must complete the On Call Information form to identify individual operator or contact assignments before Service Manager can create an on-call schedule record.

At midnight, the problem background processor generates the on-call schedule for the next day. The background processor uses information in these tables to produce the on-call schedule for the next 24 hours:

- caldutyhours contains shift definition records.
- operator and contacts provide information about the operator.

- oncallsched defines on call assignments, notification methods, and time zone variations.
- tzfile defines time zones.

Time zones

The On Call Information form has a time zone field where you can note deviations from the server time zone for any operator. For example, consider this scenario:

- A European company has headquarters in London, where the corporate server is located.
- The company record specifies that Western European Time (WET) is the default time zone.
- Teresa Thompson is an employee who works remotely from California.
- The manager notes that Teresa Thompson's Local Time in the on-call schedule form is Pacific Standard Time (PST).
- When the manager sends Teresa Thompson to Chicago for two weeks of on-site customer support, the manager makes an entry in the Exception Time Zone field that temporarily changes Teresa Thompson's location to Central Standard Time (CST).
- When Teresa Thompson returns to California, the manager removes this exception entry from the on-call schedule form before Service Manager generates the next day's record.

Related topics

Calendars
Holiday records
On-call schedule exceptions
Work schedules
Time zones
Create a holiday group

6.5.2.1. On-call schedule exceptions

The On Call Information form enables you to define exceptions to the standard notifications and time intervals in the daily schedule. There is an Exceptions tab where you can note deviations from the server time zone and notification method for any operator. For example, consider this scenario and how the Exceptions tab captures schedule variations.

- A European company has headquarters in London, where the corporate server is located.
- The company record specifies that Western European Time (WET) is the default time zone.
- Teresa Thompson is an employee who works remotely from California.
- The manager notes that Teresa Thompson's Local Time in the On Call Information form is Pacific Standard Time (PST).
- When the manager sends Teresa Thompson to Chicago for two weeks of on-site customer support, the manager makes an entry in the Exception Time Zone field that temporarily changes Teresa Thompson's location to Central Standard Time (CST).
- When Teresa Thompson returns to California, the manager removes this exception entry from the on-call schedule form before Service Manager generates the next day's record.

Daily Schedule

Service Manager displays daily schedule information for regular on call dates and times using a relative date and time format that's not dependent on a specific time zone. The relative time format is *dd hh:mm:ss*. For example, the Servicedesk Agent always begins work at 8:00:00 and stops work at 17:00:00.

Exceptions

Service Manager displays on call exception dates and times using an absolute date and time format. The absolute date and time are for the time zone specified in the On Call Information form. If you omit a time zone, the default value is the time zone in the company record. The absolute time format is *mm/dd/yyyy hh:mm:ss*. For example, Teresa Thompson worked in the Central time zone only one day. Teresa Thompson began work on 03/14/2009 08:00:00, and stopped work on 03/14/2009 at 17:00:00.

Related topics

Calendars
Holiday records

On-call schedules
Work schedules

6.5.2.2. Create an on-call schedule

Applies to User Roles: System Administrator

To create an on-call schedule:

1. Click **Tailoring** > **Notifications** > **Daily On Call Records**.
2. Specify the name of the group in the **Group Name** field.
3. Fill in the following fields.

Daily Schedule tab: Use this tab to type the schedule for each contact in the group.

Field	Description
Contact	Type the contact or operator name of the person to receive the notification.
Start Time	Type the time that the person starts being on call. Type the time in the 24-hour time format. By default, Service Manager treats a blank start time as equivalent to 00:00:00 or midnight.
End Time	Type the time that the person ends being on call. Type the time in the 24-hour time format. By default, Service Manager treats a blank end time as equivalent to 23:59:59 or one second until midnight.
Days of the week	Type true for any day that the person is on call, and type false for days the person in not on call. By default, Service Manager treats a blank entry as false.
Notify Method	Specify the name of the message class used to notify this person.
Condition	Type `true` to enable the notification in all conditions. Type an expression that's true or false to define the condition for notification. Type `false` to disable the notification in all conditions. By default, Service Manager treats a blank entry as true.
Local Time Zone	Click the drop-down list to select the local time zone where the contact resides. This is optional if all contacts are in the time zone specified in the company record.

Exceptions tab: Use this tab to define alternative notification methods and conditions during exception dates such as holidays and vacations.

Field	Description
Contact	Type the contact or operator name of the person receiving the notification.
Start Date	Type the date and time on which the exception schedule begins. Type the time in the 24-hour time format. By default, Service Manager treats a blank start time as equivalent to 00:00:00 or midnight.
End Date	Type the date and time on which the exception schedule ends. Type the time in the 24-hour time format. By default, Service Manager treats a blank end time as equivalent to 23:59:59 or one second until midnight.
Notify Method	Specify the name of the alternate message class used to notify the operator during the exception schedule.
Condition	Type `true` to enable notification in all conditions. Type an expression that's true or false to define the condition when notification occurs. Type `false` to disable notification in all conditions. By default, Service Manager treats a blank entry as true.
Replace Daily	Type `true` to use notification conditions defined only on the **Exception** tab. Type `false` to use notification conditions defined on the Daily Schedule and Exception tab. By default, Service Manager treats a blank entry as false.
Exception Time Zone	Click the drop-down list to select an exception time zone if the contact is in a different location temporarily. For example, an operator might be deployed to a customer site for a week.

4. Click **Add**.

Related topics

Work schedules
Time zones
Create a holiday group

6.5.3. Work schedules

A work schedule defines the work hours for one or more operators. Service Manager can generate a complex 24x7 schedule that spans multiple time zones, includes all shift and break information, accommodates any regional shift to Daylight Savings time, and automatically accounts for local or national holidays. Service Manager uses the following information to create a work schedule:

- Shift and break information
- Holiday information (optional)

Service Manager uses work schedule information in various calculations. One of the more important calculations is when to trigger an alert that an incident should escalate to the next level.

The caldutyhours table

The caldutyhours table contains records that identify the time to start and stop work, and the time to start and stop a break. The caldutyhours table contains the following out-of-box records.

Shift name	Holiday table	Duration
Long	standard	50-hour week with no breaks
Short	standard	42.5-hour week with no breaks
Day shift	standard	40-hour week with one hour breaks
Day shift 2	standard	40-hour week with one hour breaks
Graveyard shift	none	40-hour week with no breaks
Managers	none	40-hour week with no breaks
Ops graveyard	none	40-hour week with no breaks
Second shift	none	50-hour week with no breaks
Standard	none	40-hour week with no breaks
Swing shift	none	50-hour week with no breaks

Work schedules can apply to a group, such as an assignment group, or to an individual named in the operator or contacts table. When you create schedule records, start and stop times must not overlap, and breaks must occur within the defined work shift.

Related topics

Calendars
Holiday records
On-call schedules
On-call schedule exceptions
Time zones
Create a holiday group
Create an on-call schedule

6.6. Clocks

Clocks are background processes that track the duration of specific conditions in Service Manager. You can add a clock to track almost any event or condition. By default, Service Manager creates clocks only for the following conditions:

- When an incident record changes status (for example, when an incident record changes from Open to Pending status, or from Open to Closed status)
- When an operator edits an incident record and the System Administrator has enabled the Track Operator Times option (for example, when an operator edits an incident record to add details or a solution)

To track additional Service Manager conditions, you can manually start or stop a clock using the following tools:

- Format control utility
- Command line
- Macro

Clock example

The following example illustrates how Service Manager uses clocks to track the duration of each phase of an incident record. In this example, the clocks **total.time** and **pending.time** track the elapsed time that an incident record is open and pending, respectively.

Date and time	Current incident record phase	Status of clock total.time	Status of clock pending.time
July 1 1: 00 p.m.	Open	clock starts Total 00: 00	Clock inactive
July 2 4: 00 p.m.	Pending	Clock stops Total 27: 00	Clock starts Total 00: 00
July 4 2: 00 p.m.	Open	Clock restarts Total 27: 00	Clock stops Total 46: 00
July 4 2: 30 p.m.	Closed	Clock stops Total 27: 30	Clock stopped Total 46: 00

Related topics

Scheduled processes
Add a clocks record
Add a clock to track incident record status changes
Start or stop a clock from a macro
Start or stop a clock from format control
View a clock record

6.6.1. Add a clocks record

Applies to User roles: System Administrator

To add a clocks record, follow these steps:

1. Click **System Administration** > **Base System Configuration** > **Monitoring** > **Clocks**.
2. Type or select the following information.

Field	Description
Type	Choose the **RAD** application that the clock tracks. You can choose from **Problem**, **Downtime**, **cm3r**, **SLA**, and **Incident**.
Char Key	Specify the name of the application record that starts and stops the clock.
Char Number	Specify the numeric ID of the record that starts and stops the clock.
Name	Specify the label for the clock.
Total	Display the total time that the clock has been running.
Closed Total	Display the total time that the clock used when it was last run.
Schedule	Specify the name of the work schedule for the clock to use.
Close Date	Displays the date and time when the clock stopped running.
Start array	Specifies the dates and times for the clock to start.
Stop array	Specifies the dates and times for the clock to stop.

3. Click **Add**.

Related topics

- Clocks
- Add a clock to track incident record status changes
- System quiesce: Login restrictions
- Start or stop a clock from a macro
- Start or stop a clock from format control
- View a clock record

6.6.2. Add a clock to track Incident record status changes

Applies to User roles: System Administrator

To add a clock to track incident record status changes, follow these steps:

1. Click **Tailoring** > **Database Manager**.
2. In the **Table** field, type or select the table name **pmstatus**.
3. Click **Search**.
4. Type or select optional search criteria.
5. Click **Search**.
6. Select the status to track with a clock.
7. In the **On Entering This Status** section, type the following information:
   • **Start These Clocks**: type the names of the clocks to start when an incident enters this status.
   • **Stop These Clocks**: type the names of the clocks to stop when an incident enters this status.
8. In the **On Exiting This Status** section, type the following information:
   • **Start These Clocks**: type the names of the clocks to start when an incident exits this status.
   • **Stop These Clocks**: type the names of the clocks to stop when an incident exits this status.

9. Click **Save**.

Related topics

Clocks
Add a clocks record
System quiesce: Login restrictions
Start or stop a clock from a macro
Start or stop a clock from format control
View a clock record

6.6.3. Start or stop a clock from a macro

Applies to User roles: System Administrator

To start or stop a clock from a macro, follow these steps:

1. Click **Tailoring** > **Tailoring Tools** > **Macros**.
2. Click **Add**.
3. Type or select the following information to describe your macro.

Field	Description
Macro Name	Type the name of your new macro
Applies When	Select the condition when your macro runs
Macro Type	○ To start a clock select **Start A Clock** ○ To stop a clock select **Stop A Clock**
Macro Condition	Type `true` , or a condition statement that evaluates to true or false, to determine when to run the macro

4. Click **Set Parameters**
5. Type or select the following parameters to describe the clock.

Field	Description
Name of Clock	This value describes the name of the clock where Service Manager stores duration information. This name must uniquely identify the clock. ○ Select **Fixed Key Field** to type a specific clock name. ○ Select **Evaluating Expressions** to define a dynamic name based on an expression. **Note** Note An evaluating expression must contain the `$L.name` variable.
Type of Clock	This value describes the type of clock to start. Service Manager automatically places a value in this parameter based on your **Macro Type** selection. ○ Select **Fixed Key Field** to type a specific clock type. ○ Select **Evaluating Expressions** to define a dynamic clock type based on an expression. **Note** Note An evaluating expression must contain the `$L.type` variable. The possible clock type values are: ○ problem ○ downtime ○ cm3r ○ SLA ○ incidents
Clock Key Field	This value identifies the record, event, or object to track. This value must be unique throughout Service Manager. For a record, you can type number in $L.new to enter the unique ID associated with the record. Service Manager automatically places a value in this parameter based on your **Macro Type** selection. ○ Select **Fixed Key Field** to type a specific clock key field. ○ Select **Evaluating Expressions** to define a dynamic key field based on an expression. **Note** Note An evaluating expression must contain the `$L.key` variable.

6. Click **Save**.

7. Click **OK**.

Related topics

Clocks
Add a clocks record
Add a clock to track incident record status changes
System quiesce: Login restrictions
Start or stop a clock from format control
View a clock record

6.6.4. Start or stop a clock from format control

Applies to User roles: System Administrator

To start or stop a clock from format control, follow these steps:

1. Click **Tailoring** > **Format Control**.
2. Type or select optional search criteria.
3. Click **Search**.
4. Select a form, and then click the **Subroutines** tab.
5. In the **Application Name** field, type one of the following options:
    - apm.start.clock to start a clock.
    - apm.stop.clock to stop a clock.

6. Type the following information into the **Names** and **Values** fields:

Names	Values	Description
Name	Type one of the following values: ○ problem ○ downtime ○ cm3r ○ SLA ○ incident	This value describes the type of clock to start. For Incident Management clocks, type problem or SLA.
prompt	Type the name of the clock.	This value describes the name of the clock where Service Manager stores duration information. This name must only be unique within the format control.
query	Type the Unique key for the clock to use.	This value identifies the record, event, or object for the clock to track. This value must be unique throughout Service Manager. For a record, you can type the value number in $file to enter the unique ID associated with the record.
string1	Type one of the following values: ○ stop ○ strobe	This value is only required if you are calling the **apm.stop.clock** application. The value stop stops the clock named in the format control. The value strobe forces the clock to recalculate the clock's current running time.
time1	Type the time the clock is to stop or start.	This value determines when the clock stops or starts. To stop the clock at the current date and time, type tod().

7. Type or select any additional format controls.
8. Click **Save**.

Related topics

Clocks
Add a clocks record
Add a clock to track incident record status changes
System quiesce: Login restrictions
Start or stop a clock from a macro
View a clock record

6.6.5. View a clock record

Applies to User roles: System Administrator

To view a clock record, follow these steps:

1. Click **System Administration** > **Base System Configuration** > **Monitoring** > **Clocks**.

2. Type or select optional search criteria.
3. Click **Search**.
4. Choose the clock record to view.

Related topics

Clocks
Add a clocks record
Add a clock to track incident record status changes
System quiesce: Login restrictions
Start or stop a clock from a macro
Start or stop a clock from format control

6.7. Self-service

Self-service enables any user to connect with a Service Manager application to request a service, provide information, or track previous requests. Self-service users can also be granted approval capabilities that enable the user to approve change requests. Typically, this capability is given to high-level managers with a need to approve special requests who may otherwise have no need to use Service Manager on a regular basis.

The self-service feature provides a way for administrators to grant access to infrequent users, field personnel, or any other user outside of the IT organization without consuming a user license. These users need no special training or software. If users can connect to the Service Manager server using a supported browser, they can open incidents or make service requests.

Note

Note Users that are logged in to the ESS client can only see records in which they are the service recipient, primary contact, or records for which they are an approver.

Related topics

Who uses self-service?
Self-service licenses
Self-service tailoring
Configuring the self-service working environment
Creating self-service users
Choosing a configuration item for self-service requests
Working with self-service requests

6.7.1. Working with self-service requests

Self-service has features that enable you to open, view, update, copy, resubmit, or close service requests. Users are able to create new requests, using a supported browser, that become service desk interaction records. Self-service users can review their pending requests and make changes to them as needed. Self-service users can also use the Service Catalog interface to order items and services from the service catalog.

6.7.1.1. Visible updates for self-service requests

In the Employee Self-Service User view, when a self-service user makes an update to an open request, the user can view the update text in the Interaction record's History section. When a Service Manager IT agent views the same record in the Power User view, the **Activities** tab consists of two tabs that relate to updates.

- Record new activities in the **Updates** field and view journal entries in the **Journal Updates** field when the administrator enables **Journal Updates** in the Environment record on the **Journal Updates** tab
- View the list of updates on the **Activity Logs** tab

When the IT agent makes an update to an open self-service request, Service Desk hides the update text from the self-service user. To show the update text to the self-service user, the IT agent must select the **Visible to Customer** checkbox on the **Activities** tab of the Interaction form.

Note

Note This option can be bypassed by setting the **Bypass Visibility Option** flag to **true** in the activityactions table. When this flag is set to **true**, automatic activities are invisible to customers, whether the **Customer Visible** option is selected or not. When the flag is set to **false**, all activities are visible to customers if the **Customer Visible** option is selected.

Related topics

Self-service
Who uses self-service?
Self-service licenses
Configuring the self-service working environment

6.7.2. Creating self-service users

There are different ways to create a self-service user:

- Manually create an operator and contact record for a single self-service user
  Service Manager has an out-of-box template record to assist you with this task.
- Create a self-service operator from a contact record with a single operation
- Create multiple self-service operators from a contact record list with a single operation

Related topics

Self-service
Who uses self-service?
Self-service licenses
Self-service tailoring
Configuring the self-service working environment
Self-service template record
What's a self-service power user?
Accessing Service Manager
Working with self-service requests
Add a user role record
Create a self-service user
Create a self-service user from an existing contact
Create multiple self-service users from a contact list
Grant self-service access

6.7.2.1. ESS and ESSM-Approval users

ESS users have a regular Service Desk profile that enables them to log on to Service Desk only through a self-service web client URL. A user with an ESSM-Approval profile (for example, a manager) can approve change requests specified by a dollar amount or company. Typically, this capability is given to high-level managers with a need to approve special requests who otherwise have no need to use Service Manager on a regular basis.

Related topics

Creating self-service users
Grant self-service access

6.7.2.2. Self-service template record

Service Manager has an out-of-box operator template record, Template_SelfService, that contains all required operator settings. The password field is null. Therefore, the first time a registered user logs in, no password (or a blank password) appears. Service Manager prompts the self-service user to create a password.

The self service out-of-box user role record lists thefollowing capability words:

- service desk
- svcCatEmployeeRequester
- service catalog
- change request

If you create self-service users manually, make sure that you include the applicable settings in the out-of-box records.

Related topics

Self-service
Who uses self-service?
Self-service licenses
Self-service tailoring
Configuring the self-service working environment
Creating self-service users
What's a self-service power user?
Accessing Service Manager
Working with self-service requests
Add a user role record
Create a self-service user
Create a self-service user from an existing contact
Create multiple self-service users from a contact list
Grant self-service access

6.7.2.3. Create a self-service user

Applies to User roles: System Administrator

A new self-service user requires an operator record and a contact record. We recommend that you also create a user role for all self-service users. Alternatively, you can use the out-of-box self-service user role record.

Create self-service user contact record and operator record in one operation

To create the contact record and the operator record in one operation, follow these steps:

1. Click **System Administration** > **Base System Configuration** > **Contacts**.
2. Specify information for the new contact that you want to create. The following fields are required for a self-service user:
    - **Contact Name**
    - **Full Name**
    - **Email**
      Service Manager uses the email address as the user's Service Manager ID when it creates the new operator record.
3. (Optional) Choose the **Primary Configuration Item** to be associated with this user.
4. Click **Add**.
5. When the new contact record is created, click **More** or the **More Actions** icon, and then click **Create Operator**.
6. Select **Self Service User** for the operator type, and then click **Next**.
7. Click **Save**.
8. Click **OK**.

Create self-service user manually

To create a self-service user manually, follow these steps:

> **Note**
>
> Note If you create the operator record manually, you can use the out-of-box template record (Template_SelfService) to create a self-service user. If you choose this method, you must also create a matching contact record.

1. Click **System Administration** > **Ongoing Maintenance** > **Operators**.
2. Type or select optional search criteria, and then click **Search**.
3. Select **Template_SelfService**.
4. Replace the **Login Name** and **Full Name**.
5. Verify the following:
    - **User Role** field contains **SELF SERVICE**.
    - **Service Profile** field contains **SELF SERVICE**.
6. Add optional information, such as **Language** and **Default Company** name.
7. Click the **Login/Contact Profiles** tab and choose a resource type from the **Resource Type** list.
8. Click the **Notification** tab and in the **Email Addr.** field, type the self-service user's email address.
9. Click the **Self Service** tab:
    - To limit the new user to only self-service requests, select the **Self Service Access Only** checkbox.
    - Type the name of the **Self Service Menu** to display when the user logs on using a self-service URL. The out-of-box menus are **ESSSM** and **ESSM-Approval**. ESSM-Approval provides the Approval Request link to the Approval view on the self-service starting page for those users with approval capability.
    - Click the list to choose the **Self Service Starting Page** to display when the user logs on using a self-service URL. For example, choose **Submit a Request**.
10. Add optional information to any tab.
11. Click **Add**.
12. Create a self-service user contact record:
    1. Click **System Administration** > **Base System Configuration** > **Contacts**.
    2. Type the new contact information.
    3. Click **Add**.

> **Note**
>
> Note
>
> - The Service Manager ID field in the contact record must match the operator Login Name in the operator record.
> - If you choose the User Quick Add Utility, Service Manager prompts you to create the contact record for the user if it doesn't exist.
> - You can apply the Template_SelfService template to a new operator record.

Related topics

User roles
Self-service
Creating self-service users
What's a self-service power user?
Add a user role record
Delete a user role record
Search for a user role record
Create a self-service user from an existing contact

Create multiple self-service users from a contact list
Grant self-service access

6.7.2.4. Create a self-service user from an existing contact

Applies to User roles: System Administrator

To create a self-service user from an existing contact, follow these steps:

1. Click **System Administration** > **Base System Configuration** > **Contacts**.
2. Use search or advanced search to find one or more records.
3. Select a contact record.
4. Click **More** or the **More Actions** icon above the form of the contact record, and then click **Create Operator**.
5. Select **Self Service User** for the operator type, and then click **Next**.
6. Click **Save**.
7. Click **OK**.

Note

Note The self-service user name is the user's email address.

Related topics

User roles
Self-service
Creating self-service users
What's a self-service power user?
Add a user role record
Delete a user role record
Search for a user role record
Create a self-service user
Create multiple self-service users from a contact list
Grant self-service access

6.7.2.5. Create multiple self-service users from a contact list

Applies to User roles: System Administrator

Caution

Caution This function will fail if anyone in the contact list already has an operator record. At that point it stops and doesn't create operator records for any remaining contacts in the list.

To create multiple self-service users from a contact list, follow these steps:

1. Click **System Administration** > **Base System Configuration** > **Contacts**.
2. Use search or advanced search to find one or more records.
3. Select the contacts for which you want to create self-service users.
4. Click **More** or the **More Actions** icon, and then select **Mass Create Operators**.
5. Select **Self Service User** for the operator type, and then click **Next**.
6. Click **OK**.

Related topics

User roles
Self-service
Creating self-service users
What's a self-service power user?
Add a user role record
Delete a user role record

Search for a user role record
Create a self-service user
Create a self-service user from an existing contact
Grant self-service access

6.7.2.6. What's a self-service power user?

A self-service power user has two ways to access Service Desk. The access method depends on the task to complete:

- They have a Service Desk profile that enables them to log on to Service Desk (or other Service Manager applications) using the Windows client connection dialog or any valid Web client URL to view, add, update, or delete records. For example, the user has a Service Desk profile that enables the user to take service requests and provide services to a user community.
- They have a self-service profile that enables them to initiate service requests through a self-service URL. For example, a self-service user can use this feature to request services.

The following table shows how a user can access Service Manager in self-service or power user mode.

URL	Description
http: //*server_name*: *port_number*/SM/ess.do	The standard Web client interface for self-service users. The viewrecordlist parameter is disabled.
http: //*server_name*: *port_number*/SM/index.do	The standard Web client interface for power users. Self-service mode using the parameters set in the web.xml file.
http: //*server_name*: *port_number*/SM/accessible_ess.do	The accessible Web client in self-service mode. The viewrecordlist parameter is disabled because it doesn't conform to accessibility requirements.

Related topics

Self-service
Who uses self-service?
Creating self-service users
Self-service template record
Add a user role record
Delete a user role record
Search for a user role record
Create a self-service user
Create a self-service user from an existing contact
Create multiple self-service users from a contact list
Grant self-service access

6.7.2.7. Grant self-service access

Applies to User Roles: System Administrator

To grant self-service access, follow these steps:

1. Click **System Administration** > **Ongoing Maintenance** > **Operators**.
2. Type or select optional search criteria.
3. Click **Search**.
4. Select an existing operator record.
5. On the **General** tab, click **Fill** to choose a Service Profile, or click **Find** to modify the existing profile.
   For example, you can add desired self-service privileges, such as Open, Update, or Close to the existing Service Profile.
6. Click the **Self Service** tab. Make sure that the operator record has these settings:
   - The **Self Service Access Only** checkbox must be cleared.

- Type the name of the **Self Service Menu** to display when the user logs on using a self-service URL. The out-of-box menus are **ESSSM** and **ESSM-Approval**.
  ESSM-Approval provides the Approval Request link to the Approval view on the self-service Starting page for those users with approval capability.
- Click the list to choose the **Self Service Starting Page** when the user logs on by using a self-service URL.
  For example, choose **Submit a Request**.

7. Click **Save**.

Note

Note The Login Name in the operator record must match the Service Manager ID in the contact record.

Related topics

User roles
Self-service
Creating self-service users
What's a self-service power user?
Add a user role record
Delete a user role record
Search for a user role record
Create a self-service user
Create a self-service user from an existing contact
Create multiple self-service users from a contact list

6.7.3. Configuring the self-service working environment

The self-service user experience requires special forms that the user sees, and security settings that enable user access. You can use the out-of-box records and forms created for self-service users, customize them, or create new ones. The following table lists the records that you must configure to define a user's self-service working environment.

Record	Content	Form (if applicable)	Description
User role record	A template record that combines a collection of application profiles and capability words into a single record.		Click **Tailoring** > **Database Manager**. Type `userrole` in the table field and click **Search**. Select **userrole** from the record list. Place your cursor in the **User role** field and click **Search** to display a record list of all out-of-box user role descriptions.
Service Desk profile record	Specifies which form to display when the self-service user chooses different tasks. You can change the user experience by creating a unique profile for all self-service users in a group, department, or company. Each profile identifies the forms that the user sees. For example, if you create separate company profiles, each form set would have unique corporate branding. You can view the Environment profile record by clicking **System Administration** > **Ongoing Maintenance** > **Profiles** > **Service Desk Profiles**. Click **Search** to generate a list of all existing profiles.	There should be a form listed for each task enabled for a self-service user. For example, the form to display to open, update, search, or display a list of requests.	Click **Tailoring** > **Database Manager**. Type `smenv` in the table field and click **Search**. Select **smenv** from the record list. Place your cursor in the Profile name field and click **Search** to display a record list of all out-of-box Service Desk environment profile records.
		ESS Initial Format identifies the form to open a request.	Click **Tailoring** > **Forms Designer**, type `ess.SD.open` in the **Form**field, and click **Search** to view the out-of-box form.
		ESS Edit Format identifies the form to update a request.	Click **Tailoring** > **Forms Designer**, type `ess.SD.update` in the **Form** field, and click **Search** to view the out-of-box form.
		ESS Search Formatidentifies the form to search for a request.	Click **Tailoring** > **Forms Designer**, type `ess.SD.search` in the **Form** field, and click **Search** to view the out-of-box form.
		ESS List Format identifies the form to display a list of requests.	Click **Tailoring** > **Forms Designer**, type `ess.SD.qbe` in the **Form**field, and click **Search** to view the out-of-box form.

Service Desk environment record	Specify on this record whether self-service users see a blank list screen when they issue a pre-defined search. This behavior is controlled at the system level.		Click **System Administration** > **Ongoing Maintenance** > **Profiles** > **Service Desk Profiles**. Place your cursor in the **Profile name** field and click **Search**. Select the profile name for which you would like to restrict from viewing records lists. Click the **Security** tab. Click the **Rights** tab. Select the **View** checkbox. Click **Save** and then click **OK**.
Operator record(Self-Service tab)	**Self Service Menu** field: Specifies the layout of the user's home page when they log on as a self-service user.		Describes the actions available to the user in the left navigation pane (ESSSM).
	Self Service Starting Page field: Specifies the first form that appears in the main window.		Identifies the default start page in the main browser window. For example, if you select **Display ESS Menu**, the starting page displays the same menu items that appear in the left navigation pane. If you select **Submit a Request**, the starting page is the form to open a new request.

6.7.3.1. Create a Service Desk self-service security role

Applies to User roles: System Administrator

The out-of-box self service security role is indented for self-service users. You can create a security role for self-service users based on this one.

To create a Service Desk self-service security role, follow these steps:

1. Navigate to **System Administration** > **Security** > **Roles**.
2. Locate the **self-service** security role by performing a search. This security role has only View rights to some modules.
3. Create a new security role from this role or update this role as needed.

Now, you can assign the new or updated security role to self-service users by using the Mass Update feature.

6.7.3.2. Who uses self-service?

Self-service access is useful for all individuals in an organization. It enables them to make requests through self-service Web pages. These requests include service disruptions, service requests, catalog item requests, requests for information (RFI), or complaints. Typically, these requests generate an interaction in Service Desk, or they are a request for specific item or service

in a service catalog. The following table shows typical organizational roles and how different employees might use the self-service feature.

User name	Tasks
Employee user	Uses self-service to make requests for self and on behalf of other employees and managers. Example, a user needs spreadsheet software upgraded to the newest version.
Administrative assistant	Uses self-service to make requests primarily for others. Example: a manager directs an assistant to request a color printer for the office.
Department Manager	Uses self-service to make requests for self and direct reports. Example: a manager requests new computer systems for each software developer in a group.
IT Technician	Makes service requests for self. Example: a technician needs to change an address in an employee record.
High level executive	Makes approvals for high dollar catalog orders. Example: a senior manager needs to approve a request at the business executive level.

Related topics

Self-service
Self-service licenses
Self-service tailoring
Configuring the self-service working environment
Creating self-service users
Accessing Service Manager
Working with self-service requests

6.7.4. Choosing a configuration item for self-service requests

Each self-service request can associate one Configuration Item (CI) with the request. For example, the request might be for service on your computer, or to request a new display device. To add a CI to a request, click **Search** to display a **Search Configuration Item** form. You can specify filtering criteria first, or just click Search again to view a list of available CIs.

Related topics

Self-service
Working with self-service requests

6.7.5. Self-service tailoring

A System Administrator can modify the look and feel of self-service forms to reflect their corporate branding by using the Forms Designer or by modifying the display options and menu selections. However, a self-service user is not able to customize the self-service environment.

Because self-service is part of the Web tier, you can also customize the Web tier presentation to include font changes, logos, graphics, or navigation pane alterations. Customizations can affect the top header, navigation pane, or main page area.

Related topics

Self-service
Who uses self-service?
Self-service licenses
Configuring the self-service working environment
Creating self-service users
Accessing Service Manager
Working with self-service requests
Forms Designer best practices
Customize the self-service interface

6.7.5.1. Customize the self-service interface

Applies to User Roles: System Administrator

You can customize the self-service interface by changing the menu and forms that Service Manager displays to the user.

Task 1: Change the menu record

The self-service menu record contains option numbers, descriptions, and the RAD application called by the option. If you change the contents of the menu record, especially the options, change the corresponding forms to match.

To change the menu record, follow these steps:

1. Click **Tailoring** > **Database Manager**.
2. In the **Form** field, type menu.
3. Click **Search**.
4. Double-click the **menu** form.
5. In the **Menu Name** field, type ESSSM.
6. Click **Search** to display the self-service menu record.
7. If you make changes, click **Save**.

Task 2: Change the forms

The self-service menu record contains option numbers that correspond to the Button ID on the form. If you changed the options in the menu record, the Button ID on the form must match the associated option number in the menu record.

To change the forms, follow these steps:

1. Click **Tailoring** > **Forms Designer**.
2. In the **Form** field, type menu.gui.ess.SD.
3. Click **Search**.
4. Click **Design**.
5. After you make the changes, click **OK**.

Related topics

Self-service
Self-service tailoring

6.8. Views and favorites administration

A view uses a pre-defined query to display a set of records. As a system administrator, you can use Views/Favorites to create a new view, edit an existing view, or delete a view. A view includes the following elements, which can be specified or modified when the view is created or edited.

- View definition:
  - View Fields - fields displayed in the view.
  - Group By - how the records are grouped.

    - Sort By - how the records are sorted.
- Audience specifies which users can use the view.
- Ownership specifies who is allowed to manage the view and who is allowed to manage the content displayed by the view.

System administrators access the Views/Favorites function from the System Administration menu:

System Administration > **Base System Configuration** > **Miscellaneous** > **Views/Favorites**

Create a view[edit]

Applies to User roles: System Administrator

You can create a view for one or more users to customize the default log-on view. For example, if a group of users regularly searches on the same query, you can provide them with a shared query view as their default whenever they log on to Service Manager.

This topic assumes that you are familiar with Service Manager table and field names.

To create a view, follow these steps:

1. Click **System Administration** > **Base System Configuration** > **Miscellaneous** > **Views/Favorites**.
2. Click **New**.
   The **New View** wizard opens.
3. Select the Area to create the view in and then click **Next**.
4. Specify the name and type for the view you wish to create, and whether it is a system view or personal view.
5. Select who the view will be available to, and then click **Next**:
    - Everyone
    - Selected Groups
    - Selected Roles
6. Select the fields you want in your view:
    1. Click **Fields**.
    2. To add fields, select items from the table menu and click **Add to List**.

       Note
       Note If you select a field that has a link to another table, select from the secondary menu to add fields you want in the view.
7. Select view properties for Group By, Sort By, Filter, and Autoformat as desired.
8. Click **Finish**.
   After the system adds the view, you can provide additional information for the view definition
9. In the **Audience** tab, select an audience for the view and then click **Save**.
10. You can also update or add additional properties for the view with the View definitions, Query definitions, and Ownership tabs.

Add a view for a new table[edit]

Applies to User roles: System Administrator

When you want to expand the list of views in a queue, you can use the **New View** wizard. However, if the table doesn't yet exist, you need to first configure the table as an object record. You can then add the view to the table.

Note

Note These procedures use the kmdocument table as an example for configuring and adding a Knowledge Documents view to a table.

To configure an object record for a new table, follow these steps:

1. Click **Tailoring** > **Document Engine** > **Objects**.
2. Type the file name (table name) in the **File name** field. For this example, type kmdocument.
3. In the **Manage Queues** tab, type an expression in the **Manage condition** field that evaluates to true. For example, type lioption("Knowledge Management").
4. Create a form to be used to display the view. Type your entry in the **Manage default view** field. For example, type Knowledge Documents.
5. Click **Fill** in the **Manage display format** field and choose **inbox.view**.
6. Click **Add**.
7. Log out of Service Manager, and then log back in. This enables the new object `kmdocument` table that you configured to be added to the view list. You can now add the new view to the table.

To add a view to a table, follow these steps:

1. Click **System Administration** > **Base System Configuration** > **Miscellaneous** > **Views/Favorites**.
2. Click **New**. The **New View** wizard opens.
3. Select an entry for Area (**Knowledge Documents**) and then click **Next**.
4. Type the name of the view in the **Name** field and then click **Next**.
5. Click **Fields**.
6. Select **Title** in the **Knowledge Documents** list entries and then click **Add to List**. The Title entry appears in the Destination Fields box.
7. Click **Finish**. You receive a message that states the view record was added. You also see the View Definition form to add additional properties to the view you created.
   The view appears under Favorites and Dashboards in the navigation pane.

Delete a view[edit]

Applies to User roles: System Administrator

To delete a system view, follow these steps:

1. Click **System Administration** > **Base System Configuration** > **Miscellaneous** > **Views/Favorites**.
2. Click **Search** to find a system view.
3. Select a system view from the records list.
4. Click **Delete**, and then click **Yes** to confirm.

Change fields in a view[edit]

Applies to User roles: System Administrator

You can add, delete, or change the order of fields in a system view. This topic assumes that you are familiar with Service Manager table and field names.

Note

Note You can only change fields for views with a Table view type.

To change fields in a system view, follow these steps:

1. Click **System Administration** > **Base System Configuration** > **Miscellaneous** > **Views/Favorites**.
2. Click **Search** to find the view you want to change.
3. Select the view from the records list.
4. On the View definitions tab, click **View Fields**.
5. To add fields, select items from the table menu and click **Add to List**.

> **Note**
> Note If you select a field that has a link to another table, select from the secondary menu to add fields you want in the view.

6. To delete fields, highlight the field you want to delete in the **Destination Fields** window and click **Remove Field**.
7. To move a field location up, highlight the field you want to move in the **Destination Fields** window and click **Move Field Up.**
8. To move a field location down, highlight the field you wan tot move in the **Destination Fields** window and click **Move Field Down**.
9. Click **Next**.
10. Click **Save**.
    Service Manager updates the View record.
11. Click **OK**.

Change roles for a view[edit]

Applies to User roles: System Administrator

To change roles for a view, follow these steps:

1. Click **System Administration** > **Base System Configuration** > **Miscellaneous** > **Views/Favorites**.
2. Click **Search** to find a view.
3. Select the view to updated from the records list.
4. In the **Audience** tab, make the applicable updated.
5. Click **Save**.
   Service Manager updates the View record.
6. Click **OK**.

Add a new autoformatting rule[edit]

Applies to User roles: System Administrator

Autoformatting is part of a view in that it is a view that has been customized by a set of rules that specify certain display conditions for records that satisfy the conditions. For example, you can create a view that displays all incident records older than one week and then in that view create an autoformatting rules that displays records as red if their urgency is critical.

System administrators access autoformatting by selecting **Customize Current View** from the **More Actions** menu for a view or queue.

To add a new autoformatting rule, follow these steps:

1. From the System Navigator, click **Service Desk** > **Interaction Queue**.
2. From the Queue menu, select **Approval** to open the Approval Queue.
3. Click **More** or the **More Actions** icon, and then **Customize Current View**.
   The **Configure View Properties** wizard opens.
4. Click the **Autoformat** button.
5. Click the **Add Rule** button to open a new autoformat rule.
    1. Type a name in the name field.
    2. Select a color from the list in the Color field.
    3. Select a field for your new rule in the Field list.
    4. Select an operator in the Operator field, such as `starts with`.
    5. In the next field, select a comparison value, such as *a* to establish the rule `starts with the letter a`.
6. Click **Next** to view a list of current rule definitions. The Configure Autoformat Rules form opens.
   You can use this form to perform the following operations:
    - Add, edit, or remove autoformatting rules
    - Move the order of rules up or down; the first in the order overrides the others

- Return to the main **Configure View Properties** wizard form

7. Click **Next** to return to Configure View Properties.
8. Click **Next** for the new rule to take effect.
   The records that match your rule criteria now display in the color selected for this rule.

Adjust the order of an autoformatting rule[edit]

Applies to User roles: System Administrator

To adjust the order of an autoformatting rule, follow these steps:

1. From the System Navigator, click **Service Desk** > **Interaction Queue**.
2. From the Queue menu, select **Approval** to display the Approval Queue.
3. From the next Queue menu, select **Incident** to view the Incident queue.
4. From the View menu, select a view, such as **Autoformat date view**.
   Note the existing autoformatting rules defined.
5. Click **More** or the **More Actions** icon, then **Customize Current View**.
   The **Configure View Properties** wizard opens.
6. Click **Autoformat**.
7. Select a rule to adjust the order of and click the **Move Up** or **Move Down** button to adjust the order of the rule.
   Note that the first rule in the order overrides all others in the record list.
8. Click **Finish** to return to the Incident queue.

Edit an autoformatting rule[edit]

Applies to User Roles: System Administrator

To edit an autoformatting rule, follow these steps:

1. From the System Navigator, click **Service Desk** > **Interaction Queue**.
2. From the **Queue** list, click **Approval** to open the Approval Queue.
3. From the next **Queue** menu, click **Incident** to view the Incident queue.
4. From the next **View** menu, select a view, such as **Autoformat date view**.
   Note the existing autoformatting rules defined.
5. Click **More** or the **More Actions** icon, then **Customize Current View**.
   The **Configure View Properties** wizard opens.
6. Click the **Autoformat** button.
7. Select a rule to edit and click the **Edit** button.
   The **Autoformat Rule** wizard opens.
8. In the name field, you can optionally change the name of the rule.
9. For the rule to be active, select the **Is Active** checkbox. For the rule to be inactive, clear the **Is Active** checkbox.
10. In the **Color** field, you can optionally modify the color in which the rule appears in the queue by using the color selections.
11. Optionally modify the field name by using the field selections.
12. In the **Operator** field, you can optionally modify the operator to use for this rule.
    Valid options for operators in autoformatting rules are:
    - On
    - On or After
    - On or Before
    - Is Between
13. You can optionally modify the date in the date field by selecting a date using the calendar widget or by manually typing in a new date.
14. Click **Next** to view your modified rule definition.
15. Click **Finish**.
16. From **Specify View Type**, select **Save as a system view** or **Save as a personal view**.

17. Click **Finish** to return to the Incident queue and view your changes.

Remove an autoformatting rule[edit]

Applies to User Roles: System Administrator

To remove an autoformatting rule, follow these steps:

1. From the System Navigator, click **Service Desk** > **Interaction Queue**.
2. From the Queue menu, select **Approval** to open the Approval Queue.
3. Click **More** or the More Actions icon, then **Customize Current View**.
   The Configure View Properties wizard opens.
4. Click the **Autoformat** button.
5. Select the rule to remove.
6. Click **Remove Rule**.
7. Click **Yes** to confirm your choice.

Customize current view[edit]

System administrators use the Customize Current View feature to modify the properties of an existing view. In this case, a view is a query that has been saved to generate a list of records. The out-of-box system provides a set views that may include My tasks or All my open request. The View field in an application lists the views available for the user.

System administrators can access Customize Current View from the **More Actions** menu on any of the display views, queues, or record lists for any of the applications. The system provides a wizard to configure the view properties.

Contents

- 1 Create a view
- 2 Add a view for a new table
- 3 Delete a view
- 4 Change fields in a view
- 5 Change roles for a view
- 6 Add a new autoformatting rule
- 7 Adjust the order of an autoformatting rule
- 8 Edit an autoformatting rule
- 9 Remove an autoformatting rule
- 10 Customize current view

6.9. ITIL Alignment: Terms

Service Manager is fully aligned with ITIL terms and definitions. The following features of Service Manager are good examples of alignment with ITIL:

- Detailed descriptions of the focus of each ITIL process and supporting module in Service Manager. This includes the scope of the function or process, the goals, the roles and thus profiles typically required to deliver the function and their responsibilities. This information is included in the help server.
- Process diagrams that define the out-of-box process flows to deliver the function are included in the help server.
- The screens, work flows, profiles and system logic is pre-defined in the Service Manager system to deliver the functions as described in the help server.

6.9.1. Access Control

Authorized users can create, modify, and close records through the different Service Manager modules. Service Manager security utilities govern access to functionality within each module and define access based on role. For example, one user may be able to create and modify, but not close, records.

Service Manager security utilities govern access to modules, functionality within modules, and underlying data.

Related topics

Controlling user access and security
Application profiles
Capability word model

6.9.2. Mandatory Fields

Service Manager provides two basic methods for setting field-level controls including making a field mandatory. The system provides two methods to indicate that a field is mandatory in the user interface.

If the field is only required to be mandatory on a screen-by-screen basis, the field can be set as mandatory using the format control validations. This control can be visually indicated by a property in Forms Designer.

If the field needs to be set as mandatory for all access points throughout the system, a data policy for the field can be set using the supplied configuration tools. In this case the visual indication is automatic.

On screen, Service Manager identifies mandatory fields with a red asterisk indicator.

Related topics

Fields and Keys tab
Access Data Policy
Add a Set Mandatory Fields rule

6.9.3. Reporting

Service Manager includes two options for generating management reports using key performance indicators: Service Manager Dashboard queries and charts, Reporting, and Database Reports. These tools allow you to generate predefined and ad hoc reports to display any information. The following describes these tools.

Service Manager Reports

The Service Manager Reports module provides reports and dashboards to enable faster analysis and improved time to resolution. These reports organize data into various chart formats, and the dashboards display one or more reports to provide global information about critical activities or metrics. These reports display relationships among categories of data. For example, one report might display the number of incidents per customer, while another displays the number of incidents by priority. Viewing these reports together as a dashboard enables you to make better business decisions, such as assigning resources to close incidents. Report Managers can share a report or dashboard.

The Service Manager Reports module aims to provide a lightweight reporting feature for active operational data, and the reports are therefore designed to retrieve, represent and visualize up to 100,000 active records out of millions. To define analytical reports against the entire data set, you need to use a third-party business intelligence tool.

Database Reports

Service Manager provides built-in report views in the database for operational reporting or regularly scheduled reporting requirements. You can either review the sample report views to determine which tables, fields, and filters are used for the development of the report or create your own management reports using these details. After your customization, you need to use a third-party reporting tool (such as Microsoft Power BI) to access the Service Manager database and then generate the reports as required.

6.9.4. Audit Trail

For all records in Service Manager, a date time stamp and a user stamp are updated in the record when it is created or saved.

Service Manager provides an audit trail capability that identifies each step taken in the resolution of a record. Each time the record is updated (whether manually or automatically), a separate historical activity is created for the record. An authorized user may review each historical activity to develop a comprehensive understanding of the history of the record and its resolution, including which operators took what actions at what times. These updates are displayed into an aggregated journal of all updates.

Service Manager also provides an auditing feature that records modifications to fields within the Service Manager database. Field modifications are detected by comparing the fields in the original version of a record to the updated version. When modifications are detected, an Audit Log entry is recorded for each changed field showing the name of the modified field, the old and new version of the data, the current date/time, and the current operator's user ID.

Related concepts

Audit processing
Invoke auditing from Format Control

6.9.5. Archiving

Closure of processed problems, incidents, changes, interactions, and other records is a standard part of the workflow for each module. Typically, the closure action not only supports saving the record but separates validations, rules, and possibly even separate forms to help ensure business success.

The most typical approach for archiving records is to leverage the chosen RDMS tools from Oracle or MS SQL to perform database administration for purging and archiving.

Service Manager provides an archive and purge tool that can be configured by table and criteria for archiving. This is a facility only the administrator would use. This capability allows the ability to archive either manually or on a scheduled basis.

One new capability in this area is the ability to take advantage of the company's archive tool suite. These tools allow data to be archived out of Service Manager into a system-accessible file system. The production or alternative instance of Service Manager can then be used to access these data records. See the link below for additional details.

Related topics

Revision control
Archive records

6.9.6. Notification and Escalation

Service Manager provides alerts and escalations in the following ways:

- Notifications
  Service Manager provides a broad and deep solution to deliver notifications to stakeholders whenever a record is opened, updated or closed. The notification engine allows administrators complete control over the following:
  • What notification vehicle and format to deliver
  • What conditions on which to send the notification
  • To whom the notification should be sent
- Alerts
  Modules include the ability to set criteria for the generation of alerts. For example, the administrator can set rules for three stages of alert and a deadline condition. When these targets are met, specified users receive alerts and notifications through the Service Manager tool, email, etc.

- Escalations
  Business rules can be used to trigger automated escalations of records under various conditions. Examples include: should the status remain unchanged for too long; the record reassigned too often between assignment groups; or for other rules as defined. In addition, to document the violation of these rules the administrator can configure Service Manager to act in certain ways should these rules be breached. For example, if a record remains in open state for too long, reassign it to the assigned user's manager.
- Service Level monitoring
  Alerts and escalations can be triggered through evaluation of Service levels as defined in the Service Level Management module. This approach is used to set Service Level Objectives and Service Level Agreements with associated actions should service levels approach breach conditions. In this way, escalation is accomplished proactively to avoid breach rather than react to breach.

6.10. Native release control for Service Manager

Release Control (RC) ties the change planning and analysis process with automated change execution and validation to help reduce the risk of service downtime and to increase the change success rate. It provides the means for accurate planning and approval decisions during the review process, as well as real-time visibility and cross-team communication to prevent service disruptions that occur due to unplanned changes, schedule delays, collisions, or failures during change execution. Teams can coordinate hand-offs and sharing of information between the Change Advisory Board (CAB) and the implementation team.

Now, Service Manager has a comprehensive collection of native built-in release control functionality and provides extended change management support without the need to integrate with the former product known as Release Control.

Risk assessment and calculation (new with version 9.62)

Service Manager provides out-of-box (OOB) risk assessment and calculation. After you enable the **Risk Assessment** function, click the **Calculate Risk** button or click **More** > **Calculate Risk** to calculate the risk involved of each Change based on the risk factors that you defined in Service Manager.

After the calculation is completed, the system opens the **Risk Result Report** and displays the latest version of the risk result. This report provides an overall summary of the risk analysis for the selected change request, including the total score, risk assessment result, all related risk factors and the corresponding answers, collision status, and individual score for each risk factor in this report. This information enables you to pinpoint the most significant factors contributing to the risk level of the selected change request. If there are any collisions, the system documents these details in the Change record, which is also visualized in the Change Calendar.

For more information, see Calculate risks.

CI visualization and impact analysis (provided since version 9.50)

The integration between Service Manager and UCMDB enables users to see the visualized CI relationships and impact results directly inside the Service Manager UI in the context of an affected service or configuration item (CI) in the Change, Configuration, Incident, Problem, and Interaction modules.

For more information, see Centralized CI Visualization and Impact Analysis.

Unplanned Change Management

The integration between Service Manager and UCMDB enables users to track unplanned Changes. Modifications to the infrastructure are detected and exposed to Service Manager. Service Manager receives the modified data, and in case differences to the planned values are recognized an event like a Change record or Incident record can be automatically created.

The purpose of Unplanned Change management is to provide IT organizations a formal process by which all modification that occur to the IT infrastructure are tracked and exposed to the organizations formal review process.

For more information, see Managing Unplanned Changes.

Time Period Management (provided since version 9.34)

You can use Time Period Management to manage time period definitions for your organization so that users have a better support in planning their changes in times with the least impact. If integrated with another module in Service Manager,

Time Period Management will enable you to directly view time periods from that module. For example, if integrated with Change Management, time period definitions will enable users to view time periods and changes in a Calendar widget and view time periods and changes in the context of a selected change.

For more information, see Time Period Management.

Service Manager Calendar (provided since version 9.34)

Service Manager Calendar displays time period and other records in a graphic and intuitive user interface, and enables users in different locations and departments of your organization to perform the following tasks and optimize their task planning accordingly:

- To view time conflicts of object records from either conflict icon tooltips and shadow bars.
- To view the business records such as Change and Incident. See Display records from an application module in the calendar for detailed instructions.
- To easily see how their activities will be affected in a specific time range.
- To easily see which records are scheduled for or associated with a specific time range.
- To directly view important details of time period records and associated records from tooltips.
- To directly open records from the calendar to view their details.

For more information, see Working with the calendar.

Alerting

You can configure Service Manager to send notifications whenever specific system events occur. For example, you can receive notifications when certain modifications are made to the change requests. By default, Service Manager supports notifications for every type of status change event in all applications.

For more information, see Notifications and Configuring Change Management notifications.

6.11. Supporting tags in Service Manager

Service Manager and later versions provides out-of-the-box tag fields to Knowledge Management, Interaction, and Incident and enables you to add one or more keywords or terms to a record. However, to use the tags function in other modules such as Request, you may refer to the following sample tasks.

If you are working with earlier applications versions from 9.50 to 9.70, Service Manager doesn't provides the tag feature for all modules by default. To use the tags function, you need to refer to the following sample tasks and perform customizations to the corresponding Service Manager modules such as Interaction, Incident, and Knowledge Management.

Important

Important: If you want to enable tag in more Service Manager modules or work with earlier applications versions, make sure to upgrade the Windows client to version or later so that you can manually update your forms to support tags.

Task 1: Update the database dictionary for the Request module

Follow these steps to update the database dictionary for the Request module:

1. Click **Tailoring** > **Database Dictionary**. Type `request` in the File Name field, and then click **Search**.
2. Click **New Field/Key**, and then add an array character tags field as follows:

Name	Type	Level	Index	SQL Name	SQL Type	SQL Table
tags	array	1	114			a2
tags	character	2	1	TAGS	VARCHAR(100)	a2

   You need to map both array and character fields to the alias table. Note that the Index values and the SQL Table values may vary depending on your actual implementation.
3. Click **OK** to save your changes.

4. Click **SM Alters**.

Task 2: Update the related formats for the Request module

You need to add the tag field to the related formats for the Request module. For example, to display the tag field when creating a new Request, you need to access Forms Designer, open the rm.request.logging.subform.detail format, and then add the tag control to this format. As another example, to search for a request by using its tags, you need to open the rm.advFind.search format and then add the tag control to this format.

Note: Service Manager provides the rm.request.logging.subform.detail and rm.advFind.search formats by default. The format's names may vary depending on your customization. A Request has several related formats and rm.request.logging.subform.detai is only one of these formats that can be used to create a Request.

(Optional) Task 3: Enable tag search in Knowledge Management search engine

You may need to search for records that are related to tags by using Smart Search. Follow these steps to enable the tag search in Knowledge Management search engine:

1. Click **Knowledge Management** > **Configuration** > **Knowledgebases**.
2. Type `Request_Library` in the Knowledgebase Name field, and then click **Search**.
   Note: The Request_Library record exists only if you have enabled IDOL.
3. Add the tags field to the Field Definitions list. See the following example:

Field Name	Alias	Type	Hitlist	Doc Body
tags	tags	String	false	true

4. Click **Save** and **Full Reindex**.

(Optional) Task 4: Enable the tag auto-suggestion list

When you are typing a tag, the system automatically opens the auto-suggestion list as long as your input matches one or more tags that are commonly used. Follow these steps to enable the tag auto-suggestion feature:

1. (Optional) Load the QCCR1E157823_SM964_SM950.unl unload file to your Service Manager server. This file is shipped in the Service Manager release package.
   Note: Apply this unload file only if you are working with earlier applications versions from 9.50 to .
2. Click **Tailoring** > **Document Engine** > **Objects** to open the Object Definition form.
3. Type `request` in the File name field, and then click **Search**.
4. Click **Configure Object Based Rule Sets and Actions** .
5. Click the **Object Based Rule Sets** tab, and then add the **common.taghints.sync** rule set to the **After successful add** and the **After successful update** tabs, respectively.
6. Click **Save** to save your changes.

7. Database Administration

This section contains information about converting, maintaining, tuning, and troubleshooting your Relational Database Management System (RDBMS). The topics are intended for experienced Service Manager system and database administrators responsible for installing and implementing databases or individuals hosting Service Manager data and assisting in database conversion.

Other information contained in this section:

- Information required to configure and use Service Manager databases
- Practical information to learn about various methods used to perform queries against information contained in the database
- Information and steps used to import, export, purge, and archive records
- Information and steps used to run the Service Manager out-of-box maintenance utilities
- Information about how to use IR Expert

- Database tuning and troubleshooting tips
- Best practices and examples using out-of-box data

Note
To migrate your database from Oracle to PostgreSQL, see https://softwaresupport.softwaregrp.com/doc/KM03319848.

Related topics

Database dictionary
Database performance tuning
Database record auditing
Exporting records
Importing records
IR Expert
Keys and indexes in Service Manager
Purging and archiving records
Queries
Record retrieval
Stored queries
Supported RDBMS
Time zones
Triggers
UTF-8 (Unicode) support
Work schedules
Backups

7.1. Supported RDBMS

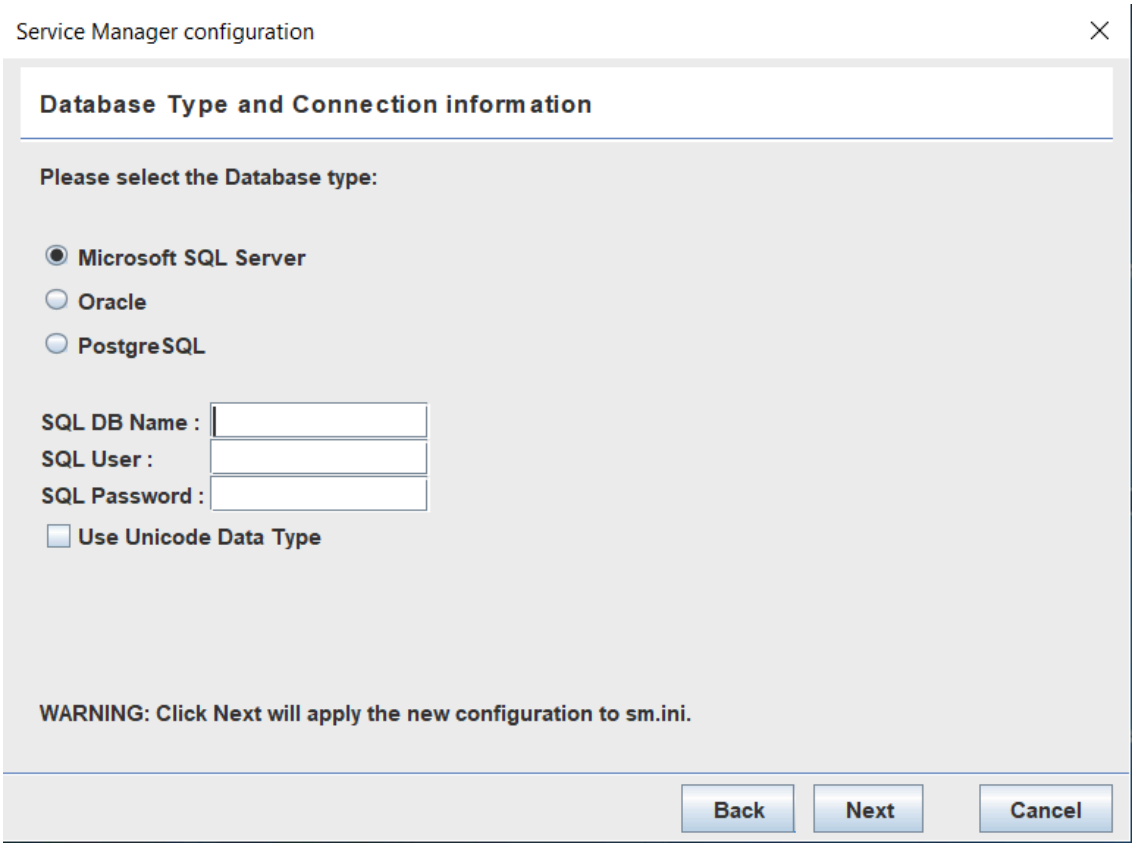
For a list of supported Relational Database Management Systems (RDBMS) for all versions of Service Manager, view the Support matrix.

Service Manager doesn't replicate vendor documentation or tasks used to maintain a Relational Database Management System (RDBMS). For the latest information, see your RDBMS Vendor documentation and the RDBMS Vendor support Web site.

7.2. Microsoft SQL Server database Unicode support

Prior to version 9.40, Service Manager used the VARCHAR, CHAR, and TEXT data types to store text strings in a Microsoft SQL Server database. Microsoft SQL Server relies on database collation to support a specific language. Although some languages share the same collation, most languages don't. As a result, Service Manager didn't support multiple languages when running on a SQL Server database.

We have enhanced Service Manager 9.40 to support the Unicode data type on SQL Server. You can store text strings in multiple languages by using the Unicode data type. When you run the server configuration utility, if you select the **Use Unicode Data Type** checkbox for SQL Server (see the following figure), Service Manager creates data with the Unicode data type when it loads the applications and demo data.



The following table provides the mapping between the old and new data types.

Non-Unicode	Unicode
VARCHAR	NVARCHAR
CHAR	NCHAR This occurs only when the CHAR field length is greater than 1.
TEXT	NVARCHAR(MAX)

The operation is not reversible, which means once you have enabled this option, you must use the Unicode data type when adding new data. When your SQL Server database uses the Unicode data type, the read-only flag **Use Unicode Data Type** in the **sqlserver** record of the **sqldbinfo** table is automatically turned on.

Note

Note The following four system tables will not use Unicode: **dbdict**, **licenseinfo**, **lock**, and **lockshared**.

The Unicode data type support is available only for new installations of the Service Manager 9.40 or later applications. If you are upgrading from a previous version of the applications, this option is unavailable.

For information about how to enable the Unicode support, see Install the Service Manager Server.

Related topics

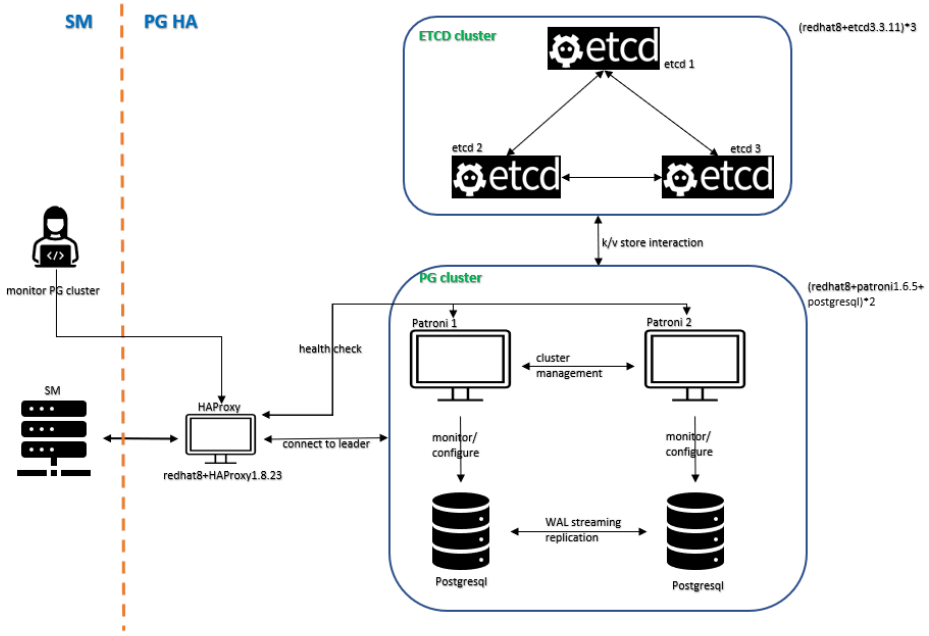
Database Administration
Prepare your RDBMS

7.3. High availability PostgreSQL with Patroni

Patroni is a tool for deploying PostgreSQL servers in high-availability mode.

Patroni architecture

The following diagram illustrates the installation of a highly available PostgreSQL cluster using Patroni and HAProxy. Alternatively, you can deploy a hardware load balancer to replace the HAProxy in the following diagram. In our lab environment, we have validated the configurations and deployments with an F5 load balancer.



Deployment

The following guide describes the installation of a highly available PostgreSQL cluster using Patroni and HAProxy/F5 load balancer on RHEL8 in a lab environment.
The parameters are set for 2000 concurrent users for Service Manager.

Prerequisites

1. Prepare 6 virtual machines(VMs) or physical hosts – In this guide, we use 6 VMs prepared with Red Hat Enterprise Linux release 8(or 5 VMs+load balancer):
    - 2 VMs for Patroni and Postgresql – Postgresql operation orchestrator: it supports bootstrapping, automatic failover, streaming replication.
    - 3 VMs for Etcd – Manages the cluster: an open-source distributed key-value store providing a hub for cluster coordination and state management. It handles leader elections during network partitions and it can tolerate machine failure.
    - 1 VM for HAProxy or a load balancer – Load Balancer: it offers load balancing and proxying for TCP and HTTP based applications.
2. Run the following commands for each VM allow the specific inbound port:
    ```
    sudo firewall-cmd --add-port={2379,2380,5432,5000,7000,8080}/tcp --permanent
    ```
    ```
    sudo firewall-cmd --reload
    ```
    ```
    sudo dnf -y update
    ```
    ```
    sudo reboot
    ```

3. In the following case, we use Red Hat Enterprise Linux release 8.2, Patroni 1.6.5, PG12, Etcd 3.3.11, and HAProxy1.8.23/F5 load balancer as example versions.
4. The parameters in the example are set for 2000 concurrent users for Service Manager.
   Note: Be sure to change the parameters according to your environment. These values are highlighted in red for your reference.

Install Etcd

1. Run the following command to install Etcd:
   `sudo dnf -y install http://mirror.centos.org/centos/7/extras/x86_64/Packages/etcd-3.3.11-2.el7.centos.x86_64.rpm`
2. Modify the `/etc/etcd/etcd.conf` file on all Etcd hosts based on your host addresses:

```
#[Member]

    #ETCD_CORS=""

    ETCD_DATA_DIR="/var/lib/etcd/default.etcd"

    #ETCD_WAL_DIR=""

    ETCD_LISTEN_PEER_URLS="http://<host address>:2380"

    ETCD_LISTEN_CLIENT_URLS="http://<host address>:2379,http://127.0.0.1:2379"

    #ETCD_MAX_SNAPSHOTS="5"

    #ETCD_MAX_WALS="5"

    ETCD_NAME="etcd1"

    #ETCD_SNAPSHOT_COUNT="100000"

    #ETCD_HEARTBEAT_INTERVAL="100"

    #ETCD_ELECTION_TIMEOUT="1000"

    #ETCD_QUOTA_BACKEND_BYTES="0"

    #ETCD_MAX_REQUEST_BYTES="1572864"

    #ETCD_GRPC_KEEPALIVE_MIN_TIME="5s"

    #ETCD_GRPC_KEEPALIVE_INTERVAL="2h0m0s"

    #ETCD_GRPC_KEEPALIVE_TIMEOUT="20s"

    #

    #[Clustering]

    ETCD_INITIAL_ADVERTISE_PEER_URLS="http://<host address>:2380"

    ETCD_ADVERTISE_CLIENT_URLS="http://<host address>:2379"

    #ETCD_DISCOVERY=""
```

```
#ETCD_DISCOVERY_FALLBACK="proxy"

#ETCD_DISCOVERY_PROXY=""

#ETCD_DISCOVERY_SRV=""

ETCD_INITIAL_CLUSTER="etcd1=http://<host address>:2380,etcd2=http://<host address>:2380,etcd3=http://<host address>:2380"

ETCD_INITIAL_CLUSTER_TOKEN="etcd-cluster"

ETCD_INITIAL_CLUSTER_STATE="new"
```

3. Run the following command to reload changed files:
   `sudo systemctl daemon-reload`
4. Repeat step 1 to step 3 for the other 2 VMs.
5. Start 3 VMs simultaneously:
   `sudo systemctl start etcd`
6. Check Etcd status:
   `sudo systemctl status etcd`
7. Show etcd member list:
   `etcdctl member list`
8. Show etcd cluster heath:
   `etcdctl cluster-health`

Note

Note
After you make any changes for the configurations of Etcd cluster, you need to run the following command to clear previous temporary configurations:
`sudo rm -rf /var/lib/etcd/default.etcd`

Install PostgreSQL

1. Run the following command to add the PostgreSQL Yum Repository:
   `sudo dnf -y install https://download.postgresql.org/pub/repos/yum/reporpms/EL-8-x86_64/pgdg-redhat-repo-latest.noarch.rpm`
2. Disable the built-in PostgreSQL module:
   `sudo dnf -qy module disable postgresql`
3. Install both client and server packages:
   `sudo dnf -y install postgresql12-server postgresql12 postgresql12-devel postgresql12-contrib`

Install Patroni

1. Run the following command to install Python, Python `pip` and other dependencies:
   `sudo dnf install python3-pip`

   ```
   sudo yum install python3-devel

   sudo dnf install gcc
   ```

2. Install Patroni by using Python `pip`.
   `sudo pip3 install patroni[etcd]`

   ```
   sudo dnf -y install https://github.com/cybertec-postgresql/patroni-packaging/releases/download/1.6.5-1/patroni-1.6.5-1.rhel7.x86_64.rpm
   ```

3. Patroni uses the YAML file to store its configuration. So, you need to run the following command to create a configuration file:
   `sudo cp -p /opt/app/patroni/etc/postgresql.yml.sample /opt/app/patroni/etc/postgresql.yml`

4. Next, edit the postgresql.yml file:
   `sudo vi /opt/app/patroni/etc/postgresql.yml`
5. Set the configuration parameters:

```
scope: postgres

    namespace: /pg_cluster/

    name: pg_node1
restapi:

     listen: <host address>:8008

    connect_address: <host address>:8008
etcd:

    hosts: <etcd1 host address>:2379, <etcd2 host address>:2379, <etcd3 host address>:2379
bootstrap:

    dcs:

      ttl: 30

      loop_wait: 10

      retry_timeout: 10

      maximum_lag_on_failover: 1048576

      synchronous_mode: false

      postgresql:

        use_pg_rewind: true

        use_slots: true

        parameters:

          shared_buffers: "16GB"  #Recommended value: 25% of physical memory

          work_mem: "4MB"

          maintenance_work_mem: "1GB"

          max_worker_processes: 16  #Recommended value: 200% of CPU number

          wal_buffers: "64MB"

          max_wal_size: "2GB"

          min_wal_size: "1GB"

          effective_cache_size: "32GB"   #Recommended value: 50% of physical memory
```

```
        fsync: on
        checkpoint_completion_target: 0.9
        log_rotation_size: "100MB"
    listen_addresses: "*"
    max_connections: 2000
    temp_buffers: "4MB"
initdb:
    - encoding: UTF8
    - data-checksums
pg_hba:
    - host replication replicator 127.0.0.1/32 md5
    - host replication replicator <Patroni host1 address> md5
    - host replication replicator <Patroni host2 address> md5
    - host all all 0.0.0.0/0 md5
users:
    admin:
      password: admin
      options:
        - createrole
        - createdb
postgresql:
    listen: <host address>:5432
    connect_address: <host address>:5432
    data_dir: /var/lib/pgsql/12/data
    bin_dir: /usr/pgsql-12/bin
    pgpass: /tmp/pgpass
    authentication:
      replication:
        username: replicator
        password: replicator
```

```
        superuser:
            username: postgres
            password: postgres
    rewind: # Has no effect on postgres 10 and lower
      username: pgrewind
      password: pgrewind
  tags:
        nofailover: false
        noloadbalance: false
        clonefrom: false
        nosync: true
```

6. Run the following command to start Patroni:

```
sudo systemctl daemon-reload
sudo systemctl enable patroni
sudo systemctl start patroni
sudo systemctl status patroni
```

7. Repeat step 1 to step 6 for the other Patroni host.

Install HAProxy

1. Run the following command to install HAProxy:

```
sudo yum -y install haproxy
```

2. Back up the existing HAProxy configuration file on the HAProxy instance:

```
sudo cp -p /etc/haproxy/haproxy.cfg /etc/haproxy/haproxy.cfg.bkp
```

3. Run the following command to update your HAProxy configuration file:

```
sudo vim /etc/haproxy/haproxy.cfg
```

4. Update the values highlighted in red according to your network setup:

```
global
        log 127.0.0.1 local2
  chroot /var/lib/haproxy
        pidfile /var/run/haproxy.pid
  maxconn 6000
        user haproxy
        group haproxy
        daemon
  stats socket /var/lib/haproxy/stats
defaults
        mode tcp
        log global
```

```
        retries 3
        timeout queue 1m
        timeout connect 10s
        timeout client 31m #Recommended value: Greater than session timeout value
        timeout server 31m #Recommended value: Greater than session timeout value
        timeout check 10s
        maxconn 3000
listen stats
        mode http
        bind *:7000
        stats enable
        stats uri /
listen postgres
        bind *:5000
        option httpchk
        http-check expect status 200
        default-server inter 3s fall 3 rise 2 on-marked-down shutdown-sessions
        server pg_node1 <PGnode1 IP address>:5432 maxconn 2000 check port 8008
        server pg_node2 <PGnode2 IP address>:5432 maxconn 2000 check port 8008
```

5. Create a `dir/file:/etc/systemd/system/haproxy.service.d/haproxy.conf` configuration file that contains the following information:

```
[Service]
        LimitNOFILE=6000
```

6. Run the following command to reload the system daemon:

```
sudo systemctl daemon-reload
```

7. Start the HAProxy service and enable it to start at system reboot:

```
sudo systemctl start haproxy
        sudo systemctl enable haproxy
```

8. Verify the status of the HAProxy service

```
sudo systemctl status haproxy
```

9. Check the maximum number of open files:

```
cat /proc/haproxy pid/limits|grep open
```

For example:

```
[sm972@haproxy ~]$ cat /proc/2893/limits|grep open
```

```
Max open files 20037 20037 files
```

Note: When the "Failed to bind to port" error occurs, run the following command to disable the SELinux boolean:

```
sudo setsebool -P haproxy_connect_any=1
```

Configure hardware load balancer

Alternatively, you can deploy a hardware load balancer to replace HAProxy.

In our lab environment, we have validated the configurations and deployments with an F5 load balancer.

See the following documents for F5 configurations and deployments:

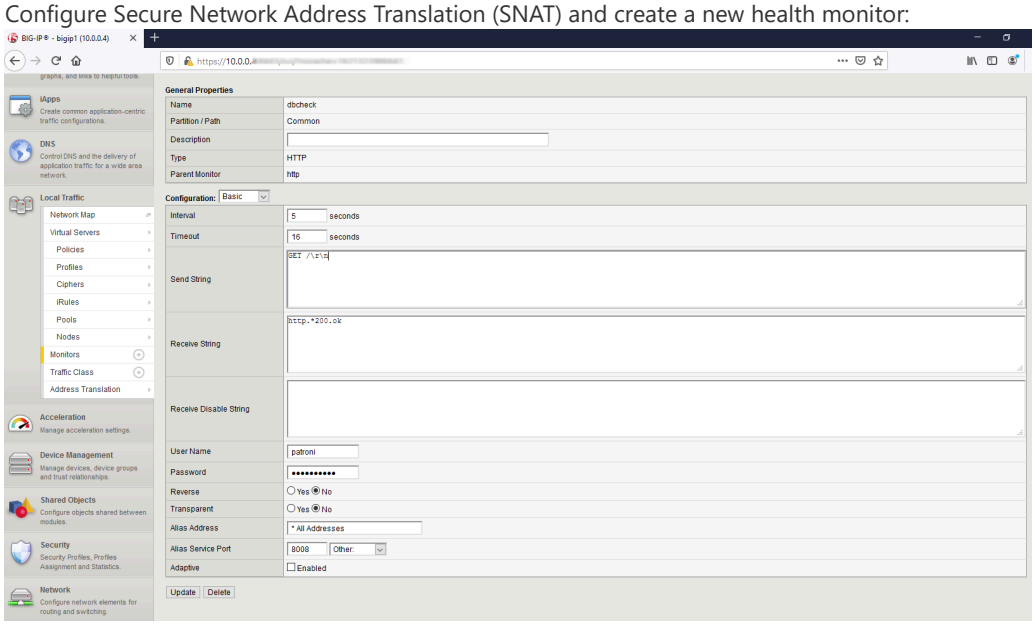
Configure an F5 hardware load balancer

Common F5 hardware load balancer deployments

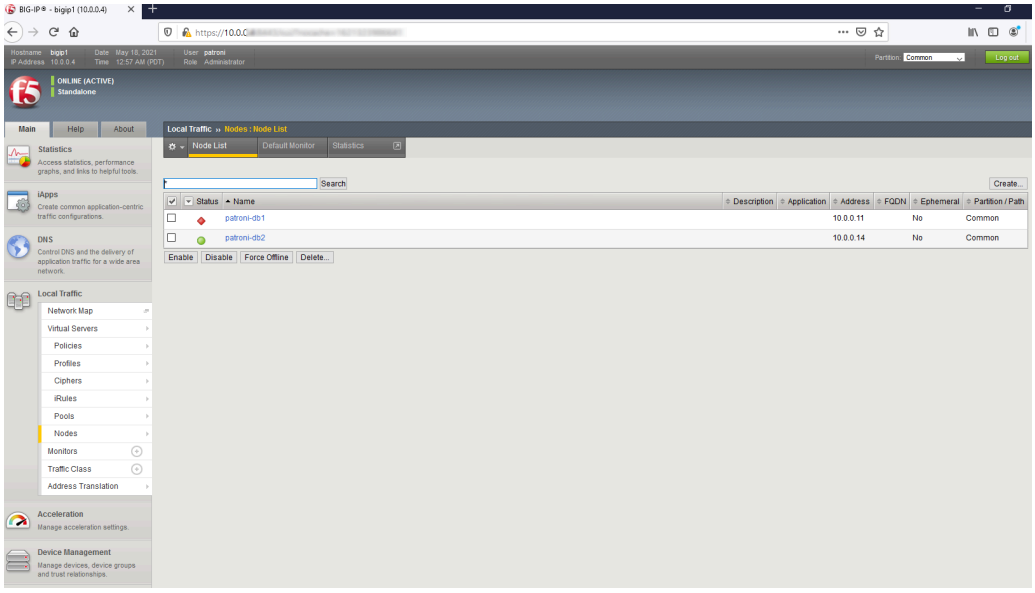
You may refer to the following screenshots as a typical example to connect to Postgres cluster:

Before you start, you need to assign an virtual address to the physical address of F5 network interface. The newly added virtial address will be used for the virtual server later on.

1. Configure Secure Network Address Translation (SNAT) and create a new health monitor:

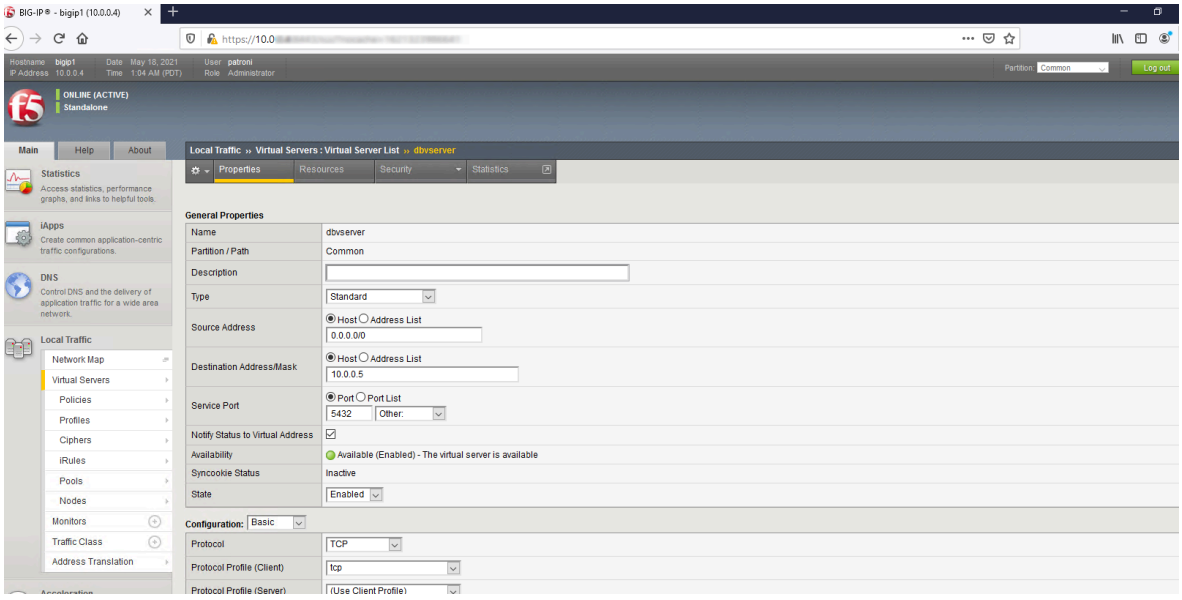


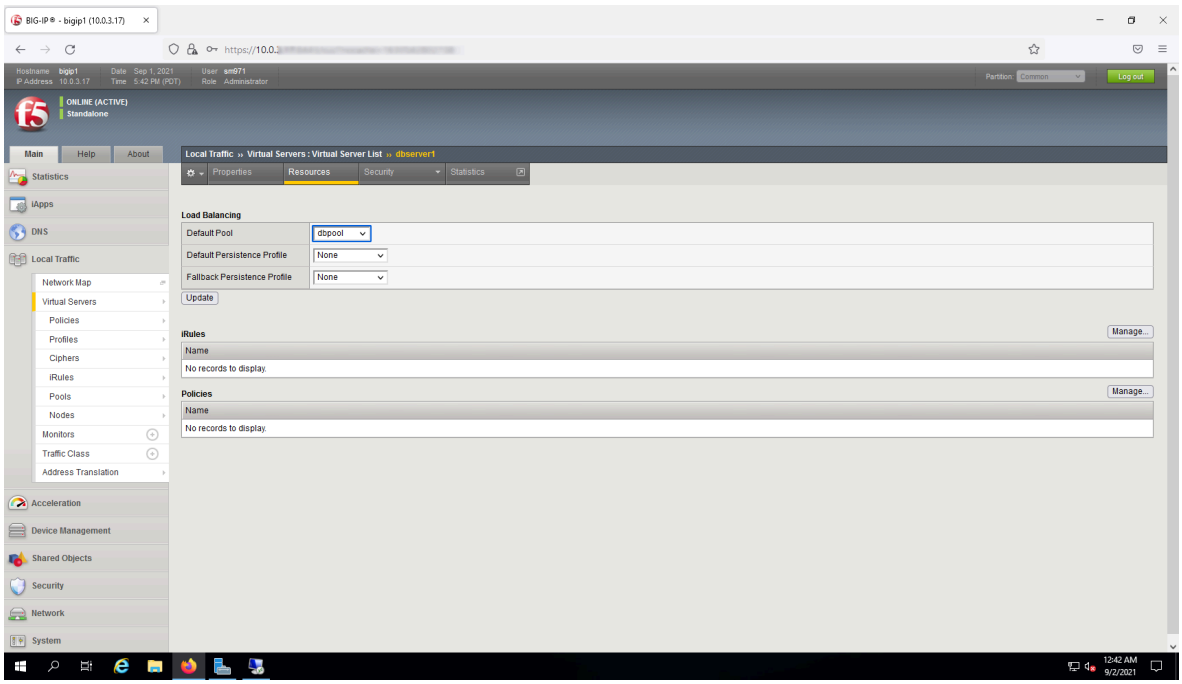
2. Create the node list based on the health monitor:

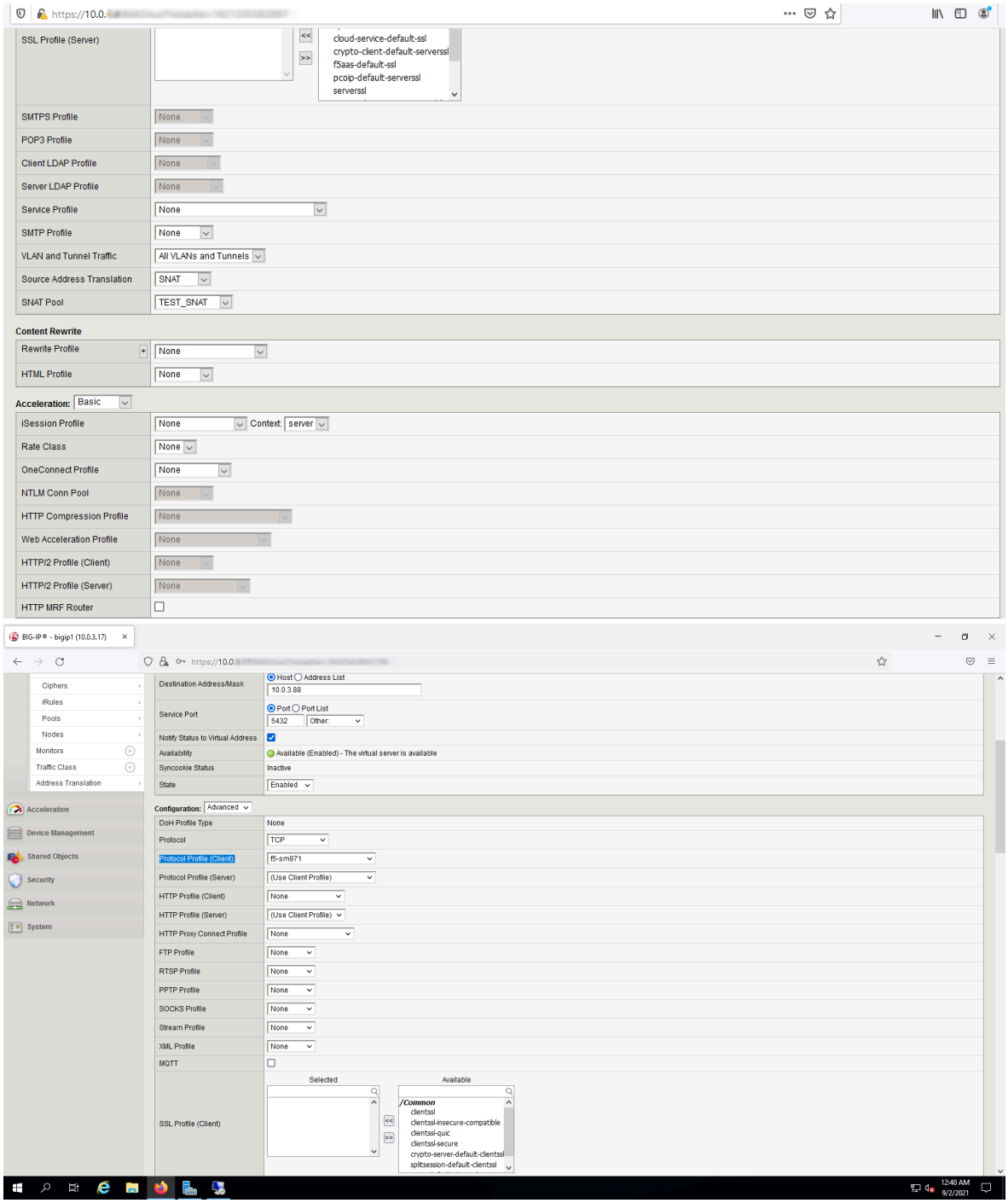


3. Configure a pool member

4. Create a TCP profile from **Local Traffic > Virtual Servers > Profiles > Protocol > TCP**
    - Set **Idle Timeout** to **Indefinite**
    - Set **Keep Alive Interval** to Indefinite
5. Create a new virtual server based on the pool member, the SNAT, and the TCP profile (Protocol Profile (Client) and Protocol Profile (Server)) created above.







7.4. Support of SQL Server AlwaysOn

Service Manager supports SQL Server AlwaysOn for SQL Server 2012 and later versions. As a high availability solution for SQL Server databases, AlwaysOn refers to two separate but related technologies: AlwaysOn Failover Clustering and AlwaysOn Availability Groups. For more information, refer to the Microsoft SQL Server documentation.

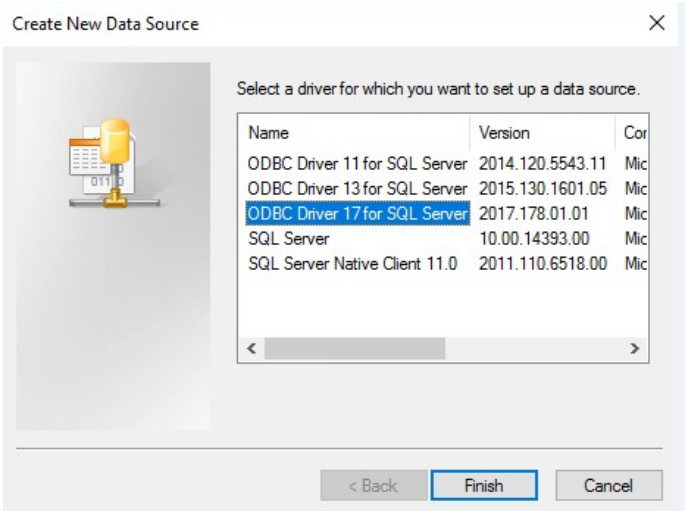
To support SQL Server AlwaysOn, make sure your Service Manager environment meets the following conditions.

- Microsoft ODBC Driver 17 for SQL Server (or a later version) is installed. You can download it from the Microsoft website.

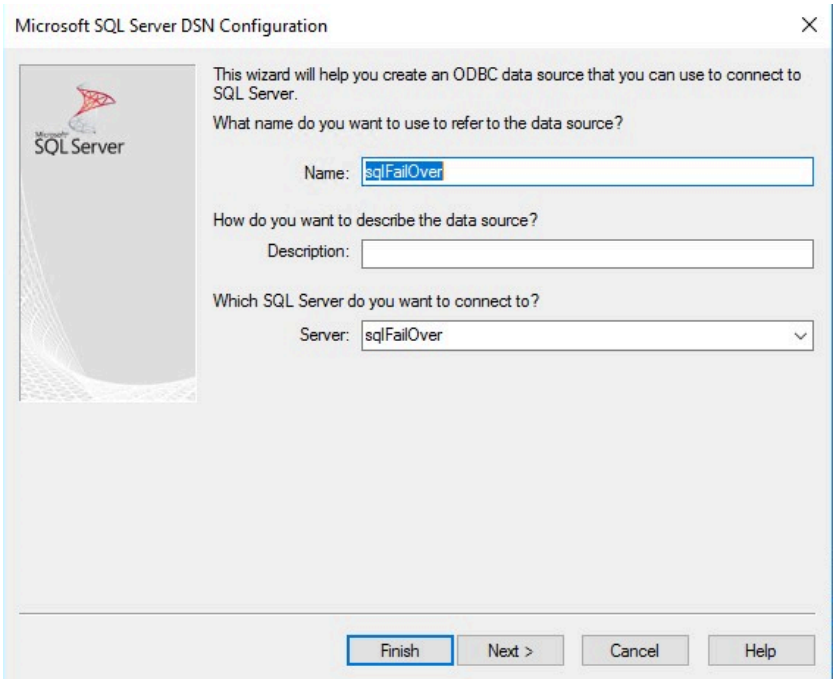
Caution

Caution Do not use the ODBC SQL Server Driver or SQL Server Native Client ODBC Driver.

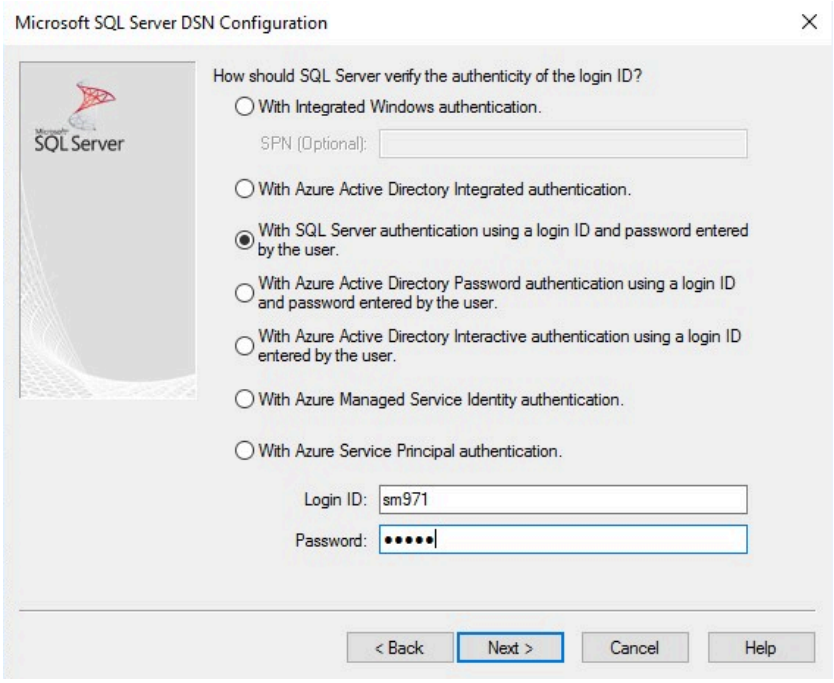
- A system DSN is added using Microsft ODBC Driver 17 for SQL Server (or later).
  Pay special attention to the following settings when creating the system DSN:
  - You must select ODBC Driver 17 for SQL Server (or later).

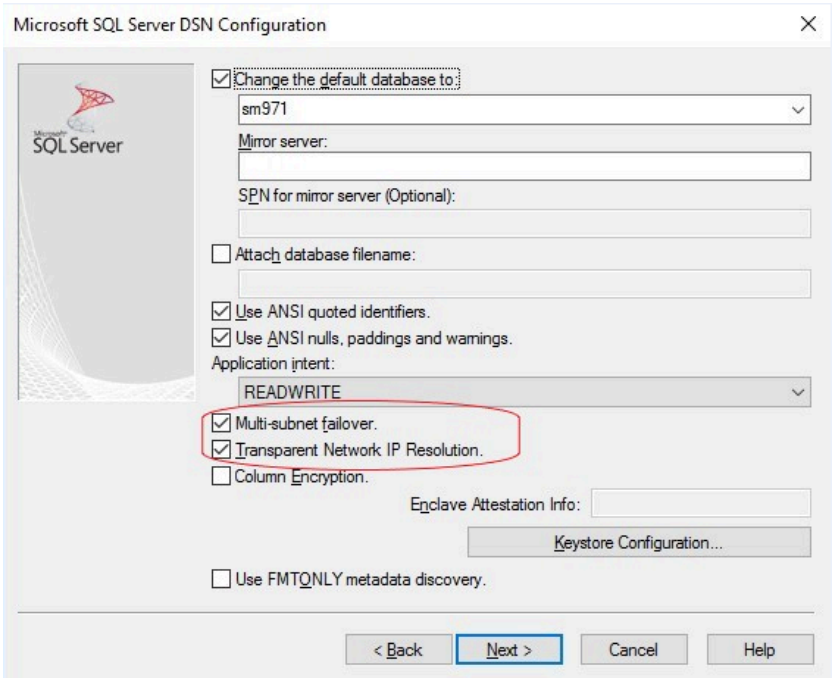


  - In the Server field, you must specify the SQL availability group client access point (for example, SqlFailOver), instead of the SQL Server instance. You need to create a client access point when enabling AlwaysOn for the SQL Server database cluster.

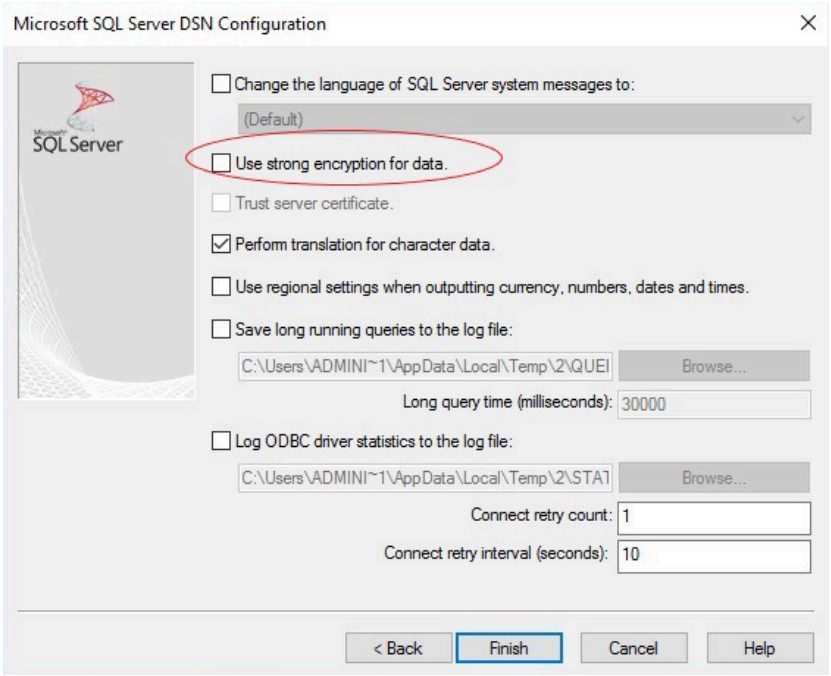


- You must select the **Multi-subnet failover** option.





- The **Use strong encryption for data** option is selected by default. If required, you may unselect this option based on business needs.



- The following lines are added in the Service Manager Server configuration file (sm.ini):

  ```
  sqlalwayson:1
  ```

  The sm.ini file should resemble the following example:

  ```
  [sqlserver]
  sqldb:xxxx
  *sqllogin:xxxxxx
  sqlalwayson:1
  ```

For detailed steps to set up SQL Server AlwaysOn for Service Manager, refer to the Enabling SQL Server AlwaysOn for Service Manager white paper.

Related topics

Parameter: sqlalwayson

7.5. Case-Sensitivity

Service Manager works with both case-sensitive and case-insensitive databases, enabling you to set things up the way that best meets your needs. Your users will be able to search for data case-sensitively or insensitively, as you choose. To achieve this, all servers and clients must be set to the same sensitivity.

Service Manager supports both case-sensitive and case-insensitive databases. You can configure the case-sensitivity of your database when you set it up. For more information, refer to Install.

- Prior to converting a system to be case-insensitive, you must first ensure that your RDBMS supports case-insensitivity.
- We recommend implementing case-insensitive searching on a test system prior to setting it up on your production system.

Related topics

JavaScript/RAD implications of a case-insensitive database environment
Load/unload implications of case-sensitivity
Oracle case-sensitivity
Prepare your RDBMS

7.5.1. Load/unload implications of case-insensitivity

When you load/unload data from a Service Manager system that connects to a case-sensitive database and load it into one that connects to a case-insensitive database, you can choose whether you want to Add and Update records or to Only add new records.

- If you select **Record Load Options - Add/update records**, the new records will be inserted into the table and the existing records will be updated.
    - **Example 1**: Suppose that you use a case-sensitive database to create an unload file that has two operator records - "operator01" and "operator02". You then load this unload file into a case-insensitive database which already has an operator record "OPERATOR01". In this case, "OPERATOR01" will be updated by "operator01" and "operator02" will be added.
      Note: If you try to load records with unique IDs that differ only in case from records that already exist in the case-insensitive database, the data in the unload will update the data in the current system.
    - **Example 2**: Suppose that you use a case-sensitive database to create an unload file that has two operator records - "operator01" and "OPERATOR01". You then load this unload file into a case-insensitive database which already has an operator record "OPERATOR01". In this case, "OPERATOR01" will be updated by "operator01" and subsequently by "OPERATOR01", or visa-versa depending on the order of those records in the unload file.
      Note: If you try to add/update two records with unique IDs that differ only in case to a case-insensitive database, the second record loaded will update the first record loaded.
- If you select **Record Load Options - Only add new records**, the new records will be inserted into the table.
    - **Example 1**: Suppose that you use a case-sensitive database to create an unload file that has two operator records - "operator01" and "operator02". You then load this unload file into a case-insensitive database which already has an operator record "OPERATOR01". In this case, "OPERATOR01" will remain unchanged and "operator02" will be added.
      Note: If you try to load records with unique IDs that differ only in case from records that already exist in the case-insensitive database, the data in the unload will add only the records which don't yet exist in the case-insensitive database.
    - **Example 2**: Suppose that you use a case-sensitive database to create an unload file that has two operator records - "operator01" and "OPERATOR01". You then load this unload file into a case-insensitive database which doesn't have either operator record "OPERATOR01" or "operator01". In this case, "operator01" will be added but "OPERATOR01" will not be.
      Note: If you try to add two records with unique IDs that differ only in case to a case-insensitive database, the second record will not be added.

Related topics

Case-Sensitivity
JavaScript/RAD implications of a case-insensitive database environment
Oracle case-sensitivity

7.5.2. JavaScript implications of case-insensitivity

In a case-insensitive environment, such as when Service Manager is set up to connect to Microsoft SQL Server with a case-insensitive code page or Oracle with case-insensitive settings, you should be aware that:

- Only Service Manager queries are evaluated case-insensitively.
- All Service Manager expressions are still evaluated case-sensitively.

This includes RAD expressions, JavaScript expressions as well as the expressions within DVD conditions in Service Manager forms.

For example, given a database that contains a contacts record where the actual last.name is stored in the database as "Smith," the following code will successfully select the record but fail the Java Script test:

```
var contactList = new SCFile( "contacts" );
var findContact = contactList.doSelect( "last.name=\"SMITH\"" );
var lastName = "SMITH";
if ( lastName == findContact.last_name )        // This expression is evaluated case-sensitively.
{
print("Hello!");                                // This code will never be executed.
}
```

In this example:

- The query in contactList.doSelect() will be evaluated case-insensitively, therefore findContact will get the record back.
- All other JavaScript expressions will evaluate case-sensitively. Therefore, statements within the "if" block will never be executed.

Related topics

Case-Sensitivity
Load/unload implications of case-sensitivity
Oracle case-sensitivity

7.5.3. Oracle case-sensitivity

You can set the Oracle database to the desired case-sensitivity when you create it. Refer to Prepare an Oracle database for instructions.

Refer to the Case-sensitivity section of the Service Manager Support Matrix for versions of Oracle for which Service Manager supports case-insensitive mode.

Requirements for using a case-insensitive Oracle database

If you want to use a case-insensitive Oracle database with Service Manager, you must meet the following requirements:

- NLS_SORT=BINARY_CI and NLS_COMP=LINGUISTIC must be set in the Oracle instance level.
  - If these values are set as given above, Service Manager will issue an ALTER SESSION statement to match the NLS_SORT and NLS_COMP session parameter with the NLS_SORT and NLS_COMP instance parameter.
  - If these values aren't set or are different from the given values, Service Manager will treat it as case-sensitive Oracle and issue an ALTER SESSION statement to modify those parameters to be NLS_SORT=BINARY and NLS_COMP=BINARY.

  This ensures that all Service Manager server processes within a horizontal or vertical scaling group have consistent Oracle case-sensitivity settings.
- All unique keys in Service Manager tables must have a matching case-insensitive unique index.

Limitations of case-insensitive Oracle

You will not be able to rename or change a column size of keyed fields from the Service Manager Dbdict Utility when using a case-insensitive Oracle instance. Instead, select generate DDL when modifying columns from Dbdict Utility. An Oracle database administrator can then drop affected function based index, apply the DDL, and recreate the function based index on the Oracle database.

Validate case-insensitive unique indexes

During the initial system load, Service Manager created a set of case-insensitive indexes for each table, based on the keys in that table. Service Manager logs these indexes first time reads a table when the `sqldebug:1` parameter is in the `sm.ini` file. You can review the settings created for each table by viewing the `sm.log file`.

The case-insensitive unique indexes should be created as Oracle function based indexes where the Column Expression is NLSSORT("<field name>",'nls_sort=*BINARY_CI*').

To verify whether the case-insensitive unique indexes are being with the correct column expression:

1. Set `sqldebug:1` in the `sm.ini` file, located in the `<SM_install_location>\Server\RUN` directory.

2. Start Service Manager.
3. Check the `sm.log` file, located in the <SM_install_location>\Server\logs directory.

The following entry in the log file indicates that the Oracle instance is set to case insensitive, and that you were able to connect to it successfully.

```
RTE I Oracle server settings for language, territory and character set: AMERICAN_AMERICA.AL32UTF8
(AL16UTF16)
RTE I OCI Client settings for language, territory and character set: AMERICAN_AMERICA.AL32UTF8 (UTF16)
....
RTE I Oracle instance setting for NLS_SORT is set to BINARY_CI
RTE I Oracle instance setting for NLS_COMP is set to LINGUISTIC
...
RTE I Oracle session is set up in CASE INSENSITIVE mode
```

The following information in the log file indicates that the Dbdict table has an index, DBDICTM1C989DE64, with a key called "NAME", which is case-insensitive.

```
RTE D Table Name: DBDICTM1
RTE D Schema Name                    Index Name                    Type Column Name
Column Expression
RTE D ---------------------------- ---------------------------- ---- ---------------------------- -
---------------------------------------
RTE D SMDB                           DBDICTM1C989DE64              U    SYS_NC00003$
NLSSORT("NAME",'nls_sort=''BINARY_CI''')
RTE D ---------------------------- ---------------------------- ---- ---------------------------- -
---------------------------------------
```

Related topics

Case-Sensitivity
JavaScript/RAD implications of a case-insensitive database environment
Load/unload implications of case-sensitivity

7.6. Backups

When deciding what files you need to back up, note that the following file types are essential to run Service Manager:

- sm.ini
- sm.cfg
- All RDBMS data files

Related topics

Database Administration

7.7. Purging out-of-the-box data

Before your system goes live, remove all testing data records and out-of-the-box demonstration data records. It's advisable that you use the predefined **PurgeOutofBoxData** unload script record to remove this data.

To purge the data, follow these steps:

1. Go to Database Manager, and open the **unload** table.
2. Select the **PurgeOutofBoxData** record.
3. Make sure to purge all the listed tables.
   • To avoid purging a table, remove the line from the unload script record.
   • Add or modify a query (for example, add name="failsave" to the **operator** line) as needed.
   • Add more tables (such as **knownerrortasks** and **rootcausetasks**) to the list of tables to purge if you have entered information during testing.

4. Click **Proceed**.

Related topics

Application Setup
Checklist: Adding a new user
Controlling user access and security
Creating operator records
Defining named users
Operator passwords
Operator records
Operator templates

7.8. Attachments

Service Manager stores each attachment as a set of separate records in the SYSATTACHMENTS table. The SYSATTACHMENTS table stores each attachment as a logical record consisting of:

- Unique identifier
- File name
- MIME type
- File size
- Application (the name of the table to which the attachment relates)
- Topic (the value of the primary key in the related table)

For example, Application would contain the name kmdocument for a Knowledge Management document attachment.

Each logical record consists of one or more segments where the server stores portions of the attachment. Administrators can set the attachment segment size with the `attachmentsegmentsize` system parameter.

The SYSATTACHMENTS table allows administrators to easily identify all the attachments stored in their system. In addition, it improves system performance since the server can read information about each attachment individually. The server no longer has to open all attachment records to generate a list of attachments. In addition, the server no longer has to open all other attachments for a user to make a change in any individual attachment.

Related topics

Parameter: attachmentsegmentsize

7.8.1. Support for allowing attachments with certain file extensions

Starting from Service Manager 9.50, a new attachment allowlist feature is supported for Service Manager clients (windows client, web client, and mobility client) and the clients that connect to RTE by web services. You can configure this attachment allowlist via **Tailoring** > **Allowlist** > **Attachment Allowlist** to allow only certain types of attachments to be submitted to Service Manager.

If the Service Manager clients retrieve an existing attachment whose extension is not in the allowlist or has no file extension, the attachment is appended with .UNSAFE when displayed in the Service Manager clients or the clients which connect to RTE by web services, so that the attachment is prevented from being automatically run on the user's machine.

Note

- Attachment allowlist defined via **Tailoring** > **allowlist** > **Attachment allowlist** takes effect only when the version of your Service Manager applications is no earlier than 9.50.
- If you set the value of denyattachwithoutext to 0, the approved attachments without file extensions aren't appended with .UNSAFE when displayed in the clients which connect to RTE by web services.

To enable the Attachment allowlist and define a list of allowed file extensions, follow these steps:

1. Sign in to Service Manager as a system administrator.
2. Click **Tailoring** > **allowlist** > **Attachment allowlist**.

3. Define the list based on your business needs. In the out-of-box system, the attachment allowlist includes the following file types, which are considered safe:

File Extension	Description
bmp	bitmap
jpg	jpg image
jpeg	jpeg image
png	portable network graphic
gif	graphical interchange format file
txt	plain text file
log	log file
unl	Service Manager unload file
csv	comma separated values file
eml	e-mail message saved

   By default, the **Enable Attachment allowlist** checkbox is selected. If you clear this checkbox, end-users can submit all types of attachments to Service Manager.

   Caution

   By selecting to change the allowed file types to be added to the system, you are disabling or bypassing security features, thereby exposing the system to increased security risks. By using this option, you understand and agree to assume all associated risks and hold the company harmless for the same.
   In case file types are changed from their default configuration, you may need to add relevant protection measures to protect against risks associated with these new file types, which is not provided by the company. By not implementing relevant protection measures you may be exposing the system to increased security risks. You understand and agree to assume all associated risks and hold the company harmless for the same. It remains at all times the Customer's sole responsibility to assess its own regulatory and business requirements. The company doesn't represent or warrant that its products comply with any specific legal or regulatory standards applicable to Customers in conducting Customer's business.

4. Click **Save** and **OK**.
5. Log out and then log on to the Service Manager client to make the customized attachment allowlist effective.

Additional protection

To provide additional protection, you can use Tika to analyze content by detecting and extracting metadata and text from attachments of various file types (such as PPT, XLS, and PDF). To learn about the formats from which Tika is able to extract metadata and textual content, visit https://tika.apache.org in your browser. Click your version of Tika, and then click the **Supported Formats** link. To determine the version of Tika in use by your SM Server, look up the JAR filename in your RUN/lib directory. For example, /RUN/lib/tika-core-<version>.jar.

For file types that are not listed in the Apache Tika **Supported Formats** documentation, you need to configure both custom-mimetypes.xml files under the web tier and the server installation directories if you want Tika to extract metadata and textual content from such file types:

- For Service Manager web tier, go to <web tier installation path>\WEB-INF\classes\custom-mimetypes.xml and configure as needed.
- For Service Manager server to to <server installation path>\RUN\custom-mimetypes.xml and configure as needed.

For more information, see https://tika.apache.org/<*Tika version*>/parser_guide#Add_your_MIME-Type. By default, file type UNL is already configured in custom-mimetypes.xml.

Related topics

Attachments
Startup parameter: denyattachwithoutext
Enter a parameter in the sm.ini file
Security parameters

7.8.2. View the attachment history

Every time that a user adds or deletes an attachment in Service Manager, the action is recorded in the activityactions table.

> **Note**
> Note If a file doesn't have a related object file or if the activity log table is undefined, the action is not logged.

To view the attachment history of a specific record, open that record and browse to the "Activity" section. If an attachment has been added to or deleted from the record, the following information is displayed in the "Activity" section.

Field Name	Values
Number	Record ID number
Type	• Attachment added • attachment deleted
Datestamp	Current date
Operator	Current operator
Description	• <*Attached file name*> added • <*Attachment file name*> deleted
Customer visible	FALSE

Related topics

Attachments

7.9. System Definition

The System Definition section of the System Navigator is an administrative tool that enables you to view and modify system tables and their components:

- Fields
- Keys
- Links
- Forms
- Records

You can use the System Definition section if you have the Admin plug-in installed and the SysAdmin capability word enabled in your user profile. If you have SysAdmin rights, but the Admin plug-in is not installed, you can view the System Definition section but you can't change it.

Administrators can do the following:

- View, add, change, or delete forms, links, records, triggers, and Data Policy records.
- Use the System Definition utility to add help or create an alias for any field.

- View, add, change, or delete database dictionary definitions.
- View, add, change, or delete SQL mappings to a back-end RDBMS.
- Open, reset, regenerate, and copy database dictionary definitions.
- Add, change, copy, and delete keys and fields.
- Count the number of records in logical tables

Each time you right-click a node in the System Definition section, the System Navigator displays a Context menu of available actions.

System Definition utility

The System Definition utility is a graphic view of existing table elements and properties and a quick way to define new tables. It enables you to accomplish the tasks traditionally handled by the database dictionary utility. The System Definition utility has two tabs: tables and fields. Service Manager displays SQL mapping information only when mapping information is available. Each time you double-click any table in the System Definition area of the System Navigator, the System Definition utility displays the Table tab first. You can access the other tabs by clicking the labeled tabs or links in the Editor.

If you change a table element, an asterisk marks the table and the element in the System Navigator until you click **Save**.

Note

Note

The System Definition utility is disable by default. You can enable this feature by adding the following line to the {workspace}\.metadata\.plugins\org.eclipse.core.runtime\.settings\com.hp.ov.sm.client.eclipse.base.prefs preference setting file:

```
showsystemdefinition=true
```

Access the System Definition utility

Applies to User Roles: System Administrator

To access the System Definition utility:

1. From the System Navigator, click **System Definition**.
2. Expand the name of any table in the record list.

Note

Note

You can type the first letter of a table name to jump to that alphabetic section. For example, type `o` to jump to the oncall table.

Related topics

Editing System Definition nodes
Access field help from the System Navigator

7.9.1. Table tab

The Table tab summarizes the properties of an Service Manager table, provides links to other table-related objects, and allows you to map a database dictionary record to an RDBMS.

The Table overview contains these sections:

- Table definition
- Table management
- SQL Properties
- Associated objects
    - Forms

- Links
- Triggers

Table definition

The Table definition section contains the following information:

Field	Contents	Notes
Caption	Optional. Caption information may appear as a label in the data policy rules for the table. Service Manager uses some Caption fields to specify an index value for a table of localized text.	An administrator can customize this field. For example, you can display the name of table.
Description	Optional. The Knowledgebase uses the first line of the table description.	An administrator can customize this field. For example, you can specify information about the table.
SQL base name	SQL mapping uses a unique base name for each table. The maximum length is 13 alphanumeric characters. When you create a new table, Service Manager creates a valid SQL base name and assigns it to the new table.	To avoid future upgrade conflicts, it is helpful to specify a base name that doesn't overlap Service Manager out–of–box table names.
Revision number	Specifies the cumulative number of changes to the table.	Read only.
Add, delete, or edit fields and keys	Click to advance to the Fields and Keys tab.	

If you make any changes to the table definition fields or other table information, click **Save**.

Table management

Use the Table management section to complete routine maintenance activities. When you select the action, the System Definition utility prompts you for the next response required to complete the task. You can:

- Delete the table
- Rename the table
- Copy the table definition (only)
- Copy the table definition and data
- Regenerate the table
- Reset the table data

Service Manager always asks for confirmation before you complete a task that makes irreversible changes. For example, you must confirm when you try to delete, regenerate, or reset a table.

Related topics

System Definition
Forms, triggers, and links
Fields and Keys tab
Add a field to a table
Edit a field in a table
Edit a record key

7.9.2. Editing System Definition nodes

When you double-click any node in System Definition:

- The System Definition utility displays fields, keys, links, and triggers.
- Forms Designer displays forms.
- Service Manager displays records collections in a record list or a single record in the associated display form.

Related topics

System Definition
Access field help from the System Navigator

7.9.3. Access field help from the System Navigator

Applies to User roles: System Administrator

To access field help from the System Navigator:

1. Click **System Definition > Tables > *tablename***.
2. Double-click a table name to view information about that table.
3. Click the **Fields and Keys** tab.
4. Select any field in the **Fields** section.
5. Select one of these links to create or edit the default help record.
   • **Create default help on this field**. This link appears when no field help exists for this field. Click this link to create field help.
   • **Edit default help on this field**. This link appears when field help already exists for this field. Click this link to edit existing field help.
6. Click **Search for specific help on this field in forms** to open and edit existing form specific help for this field.
7. If you make changes, click **Save**.
8. Click **OK**.

Related topics

Editing System Definition nodes
System Definition
Creating field help
Field help editor
Add or edit help records
Access the help table

7.9.4. Forms, triggers, and links

When you choose one of these components associated with the table, Service Manager opens the appropriate form or application to edit the component.

- If you double-click a form, Forms Designer displays the selected form.
- If you double-click a trigger, Service Manager displays the related trigger record.
- If you double-click a link, Service Manager displays the related link record.

Edit the form, trigger, or link. If you make any changes to forms, triggers, or links, click **Save**.

Related topics

System Definition
Table tab
Fields and Keys tab

7.9.4.1. Update a form

Applies to User roles: System Administrator

To update a form:

1. Click **Tailoring** > **Forms Designer** in the System Navigator, or type `fd` on the Service Manager command line and then press **Enter**.

2. Type a form name and click **Search**.
   Service Manager opens a list of forms. Click the form you want to update to open it.
3. Click **Design** to open design mode.
4. Add controls in any of the following ways:
   - Drag and drop the control from the toolbar to the drawing canvas to create a default–sized object.
   - Select a control in the toolbar and then click anywhere on the drawing canvas to create a default–sized object.
   - Drag and drop field(s) from a table in the System Definition onto the drawing canvas.
   - Cut, copy, or paste a control within the same form, from a different form, or from different Service Manager client.

   The properties for the control you selected appear in the properties view.
5. Specify the name of the form in the Value text box of the Name property.
6. Specify the database field to access in the Value text box of the Input property.
7. Specify the height, width, and other properties as desired.
8. To delete a control from the drawing canvas, select the control and press **Delete**.
9. Click **Web Preview** to see what the form will look like in the Web client.
10. Click **OK** to see what the form will look like in the Windows client.
11. Click **OK** to save your changes.
12. Open the form using Database Manager to test your changes:
    1. Click **Tailoring** > **Database Manager**.
    2. Click **Search**. Service Manager opens a list of forms.
    3. Click the form you just created to open it.

7.9.4.2. Triggers

A trigger monitors a file for certain types of activity, such as Adds, Updates, or Deletes, and launches a specified JavaScript or RAD application when the appropriate conditions occur. You can supply custom JavaScript or RAD code to launch a specific action when the trigger conditions apply.

Create a Trigger record

Applies to User Roles: System Administrator

To create a trigger record:

1. From the System Navigator, click **System Definition** > **Tables** > **triggers**.
2. From the Keys sections, click **New**.
3. From the General section, select the trigger type from the More Actions menu.
4. Click **Add**.
5. From the Fields section, modify the properties of the available fields to define the trigger. You can use the table below as a general guideline for the most commonly-used parameters.

Property	Description
trigger.name	Any unique, meaningful name.
table.name	Select the file to be monitored by this trigger.
trigger.type	Select the specific activity to react to: **1 - Before Add** - When adding a new record to this file, launch the application prior to committing the addition. **2 - After Add** - When adding a new record to this file, launch the application after committing the addition. **3 - Before Update** - When modifying a record in this file, launch the application prior to committing the change. **4 - After Update** - When modifying a record in this file, launch the application after committing the change. **5 - Before Delete** - When deleting a record in this file, launch the application prior to performing the deletion. **6 - After Delete** - When deleting a record in this file, launch the application after performing the deletion. **Note:** Only triggers of *type 4* (After Update), and *type 6* (After Delete) are supported by Cascade Updates.
application.name	The RAD application to launch. **Note:** The RAD application may not require parameters and will only have access to $L.old, the record prior to the change, and $L.new, the post-change record. **Note:** For most Cascade Updates, set to **cascade.update.wrapper**. The only exception is if the file you are monitoring is the device file or a related device attribute file, such as joincomputer. In those cases, set to **am.cascade.update.wrapper**.

Edit a trigger

Applies to User Roles: System Administrator

To edit a trigger:

1. Click **Tailoring** > **Database Manager**.
2. Type **triggers** in the Table field, and then click **Search**.
3. Click **Search** again.
   A list of records opens.
4. Select a record to edit.
5. Update the following information about the trigger:
   - **Trigger Name**
   - **Table Name**
   - **Trigger Type**
   - **Application**
   - **Script**
6. If you make changes, click **Save**.

Related topics

Cascade Updates

7.9.4.3. Modify an existing link

User Role: System Administrator

To modify an existing link.

1. Click **Tailoring** > **Tailoring Tools** > **Links**.
2. Type optional search criteria, and then click **Search**.
3. Double-click a link to view or change it. If necessary, press Ctrl+H to view help for each field.
   When the link record opens, the following options are available from the More Actions menu:

Option	Value
Insert Line	Opens a window to prompt for the number of lines to insert, then inserts them above the cursor position.
Delete Line	Opens a window to prompt for the number of lines to delete, then deletes them beginning with the line the cursor is on.
Select Line	Allows advanced link processing.
Check Field	When the cursor is positioned on a Source Field Name or Target Field Name, prompts for a file name and then checks the database dictionary of the file to determine whether a field of that name exists. If invalid, allows selection of a valid field.

4. If you make changes, click **Save**.

Related topics

Understanding links
Advanced link editing features
Specifying a link query
Copying fields by name during fill operations
Scalar/non-scalar field links
Keeping changes
Link dependencies within Service Desk
Specifying a link query
Add a new link file
Test a link
Delete a link
Access advanced link maintenance

7.10. Data persistence

Service Manager requires a relational database management system (RDBMS) for data persistence. See the support web site for a list of supported RDBMS vendors.

You can format and manage the data in your RDBMS from the database dictionary. Service Manager uses the database dictionary to keep a logical version of your RDBMS database dictionary and index.

Related topics

Database dictionary
Create a database dictionary record by importing RDBMS columns
Create an active database dictionary record
Create an inactive database dictionary record

7.10.1. Database dictionary

Service Manager maintains a logical view of your RDBMS tables and columns in the database dictionary. The database dictionary describes each table and column in your system and how they are mapped to logical entities within Service Manager. The Service Manager applications use the logical definitions in the database dictionary to query and manage the actual records in your RDBMS. You must create a database dictionary record for each table and column in your system.

Every time you add tables and columns for Service Manager to your RDBMS system, you must also create database dictionary records describing these new items before Service Manager can make use of them. The best practice we recommend is to create the database dictionary records first, and then have Service Manager create the back-end tables and columns for you automatically. Some environments, however, don't permit Service Manager to add or update records in the back-end RDBMS.

In restricted-access RDBMS environments, Service Manager can create database definition language (DDL) describing the changes proposed by your database dictionary records. The RDBMS administrator can then create the necessary tables and columns for Service Manager . After the RDBMS has the necessary tables and columns, the Service Manager administrator can then update the database dictionary records to map to the actual RDBMS objects. To simplify the mapping process, Service Manager administrators can also import columns from an RDBMS into a database dictionary record.

You can access database dictionary records either from the System Definition utility or from the database dictionary form. Both utilities require having the `SysAdmin` capability word or its equivalent. We recommend using the System Definition utility to create database dictionary records if Service Manager has table create-alter-drop-rights to the RDBMS. If your environment restricts access to the RDBMS, we recommend using the database dictionary form to create the initial null mappings and then update the database dictionary by importing the RDBMS tables and columns.

To provide Service Manager with table create-alter-drop-rights to your RDBMS, you must add the RDBMS connection parameters to the server configuration file (sm.ini). The RDBMS user name you provide Service Manager must at least have add and drop access rights. We recommend you grant the RDBMS user owner rights over the database so that Service Manager can manage the data as needed.

To have the server create DDL for database dictionary changes, you must add the `sqlddllogging` and `sqlddldirectory` parameters to the server configuration file (sm.ini). The server creates one DDL file for each logical table in the database dictionary. For example, the file prosummary.ddl contains DDL for changes made to the probsummary database dictionary record, which in turn, maps to the physical tables such as PROSUMMARYA1, PROSUMMARYM1, and PROSUMMARYM2 in your RDBMS. You can enable exporting to DDL independently of providing Service Manager with table create-alter-drop-rights to the RDBMS. You may want to export all RDBMS changes to DDL to keep a log of changes to your RDBMS.

Related topics

Data persistence
Automatic update of database dictionary records
Automatic update of RDBMS data when database dictionary records change
Database dictionary data types
Database dictionary records
Create a database dictionary record by importing RDBMS columns
Create an active database dictionary record
Create an inactive database dictionary record
Enable DDL logging of database dictionary changes
Import RDBMS columns into an existing database dictionary record

7.10.1.1. Database dictionary records

Database dictionary records define how RDBMS tables and columns map to logical entities in Service Manager. RDBMS tables map to database dictionaries. RDBMS columns map to fields. Database dictionary records also contain information about RDBMS indexes. You must create a database dictionary record for each table and column you add to Service Manager.

IMPORTANT: We recommend that you refrain from using the following table mapping options for the disposition of arrays:
--*Field in Alias Table
--*BLOB in Alias Table
Using these mapping options can result in potential performance-related issues and unexpected product behavior.

Database dictionary records contain the following information.

Database dictionary object	Brief description	Contents of object	Example database dictionary entry	Example RDBMS equivalent or mapping
Database dictionary name	A logical name for one or more related RDBMS tables	A list of the main and alias tables in the RDBMS that make up the logical table	probsummary – logical table containing incident records	In Microsoft SQL Server, the probsummary database dictionary maps to the following RDBMS tables. • PROBSUMMARYA1 – alias table 1 • PROBSUMMARYM1 – main table 1 • PROBSUMMARYM2 – main table 2
Field name	A logical name for an RDBMS column	A value that Service Manager uses to identify a field. Field names can only contain alphanumeric characters and periods.	category – logical column containing the type of interaction	In Microsoft SQL Server, the category field maps to the SQL name CATEGORY
Field type	An abstract data type describing the information in the RDBMS column	A value that defines the Service Manager data type. Service Manager uses the abstract data type to generate queries, perform calculations, and create SQL mappings.	The field category has the Service Manager data type character	In Microsoft SQL Server, the category field maps to the VARCHAR(50) SQL data type
Field level	A logical data type describing whether a given field is part of a structure or an array of structure	A numerical value that Service Manager uses to determine what level in the logical tree a given field occupies	The field category has the field level of 1	There is no RDBMS equivalent for field level

Field index	A parameter that describes what order a field occupies in the database dictionary record	A numerical value that Service Manager uses to determine the order of a field in the database dictionary record. Service Manager lists fields in the order they are added to the database dictionary. Fields that list the same index number are aliases.	The field category has the field index of 2 and is therefore the second field in the database dictionary record	In Microsoft SQL Server, the CATEGORY column is the second column in the PROBSUMMARYM1 table
Key type	A logical key type describing an RDBMS index	A data type value that Service Manager uses to determine what SQL index type to use when mapping RDBMS index data	The probsummary database dictionary uses the unique key number	In Microsoft SQL Server, the unique key number maps to the unique index NUMBER on the RDBMS column
Key fields	A list of fields that identify the logical key	A list of the fields that make up the key definition	The probsummary database dictionary has a unique key consisting of the number field	In Microsoft SQL Server, the unique index consists of the NUMBER column
SQL table alias	A logical name for a SQL main or alias table	An alphanumeric value that Service Manager uses to identify an RDBMS table	The probsummary database dictionary has three SQL table aliases a1, m1, and m2	In Microsoft SQL Server, the SQL table aliases map to the SQL tables PROBSUMMARYA1, PROBSUMMARYM1, and PROBSUMMARYM2
SQL table name	A listing of the actual RDBMS table names that map to the database dictionary	A RDBMS table name	When mapped to a Microsoft SQL Server, the probsummary database dictionary record lists the SQL table names PROBSUMMARYA1, PROBSUMMARYM1, and PROBSUMMARYM2	In Microsoft SQL Server, the SQL table aliases map to the SQL tables PROBSUMMARYA1, PROBSUMMARYM1, and PROBSUMMARYM2
SQL table type	A description of the RDBMS vendor	A code that identifies the RDBMS vendor	If you are mapped to a Microsoft SQL Server, the SQL type is sqlserver	There is no RDBMS equivalent for SQL type

Related topics

Database dictionary
Database dictionary data types

Create an active database dictionary record
Create an inactive database dictionary record
Enable DDL logging of database dictionary changes
Import RDBMS columns into an existing database dictionary record
Database dictionary data types

7.10.1.2. Making database dictionary changes

Before making database dictionary (dbdict) changes, consider the following behaviors and constraints.

A database dictionary (dbdict) update will fail if:

- A SQL data type is changed and it's not a change from CHAR to VARCHAR or from VARCHAR to CHAR (VARCHAR2 on Oracle).
- A SQL data type is changed and it's not a change from NCHAR to NVARCHAR or from NVARCHAR to NCHAR on Microsoft SQL Server or Oracle (NVARCHAR2).
- A SQL data type is changed and it is a change from NCHAR to CHAR or from NVARCHAR to VARCHAR on Microsoft SQL Server.
- A SQL data type CHAR or VARCHAR is shortened.
- ALTER TABLE statements were generated but failed to execute.

A full table copy is triggered for any of the following conditions:

- The RC flag changes for any field.
- The alias changes for any field (but not from or to a NULLTABLE alias).
- A new table is defined for a dbdict.
- The SQL dbtype changed significantly. (For example, from one sqlserver2k to sqlserver, if those two SQL dbtypes use different sqldb or sqllogin parameter values.)

An ALTER TABLE statement will be generated to accomplish any of the following changes:

- A SQL data type changed from CHAR to VARCHAR, or vice versa (VARCHAR2 on Oracle).
- A SQL data type changed from NCHAR to NVARCHAR, or vice versa on Microsoft SQL Server.
- A SQL data type changed from VARCHAR to NVARCHAR on Microsoft SQL Server.
- A SQL data type changed from NCHAR to NVARCHAR2, or vice versa on Oracle.
- A SQL data type was widened (for example, from CHAR(10) to CHAR(15)).
- A SQL column was renamed.
- A table was renamed.

A DROP TABLE statement will be generated if a table entry was completely removed from the SQL Tables tab.

If Service Manager determines it doesn't have rights to CREATE a TABLE, it will assume not having the rights of performing any of the above operations. In that case it will write all database definition language (DDL) into files in the <RUN directory>/ddl folder and accept changes to the dbdict without performing any of the above operations. It will, however, still verify that the SQL type changes are valid.

Related topics

Methods for updating database dictionary records
Methods for deleting database dictionary records
Adding a key
Modifying a key
Deleting a key

7.10.1.3. Database dictionary data types

Every field in a database dictionary record requires a data type value to determine the format of the data in the field. Database dictionary records use an abstract data type that the Service Manager server uses to determine how to map your system data to the SQL data type required by your RDBMS. For example, the database dictionary data type "number" maps to FLOAT in Microsoft SQL server and maps to NUMBER in Oracle. For a complete listing of how the database dictionary data types map to RDBMS data types, use the Database Manager utility to view the records in the `sqldbinfo` table.

Caution

Caution Do not change the data type mappings in the **sqldbinfo** table unless directed to do so by customer support. Changing these values can result in data loss or corruption.

Note

Note Starting with Service Manager 9.40, the **sqlserver** record in the **sqldbinfo** table contains a read-only flag named **Use Unicode Data Type** to indicate if Microsoft SQL Server uses the Unicode data type (nvarchar).

The Service Manager server doesn't automatically validate field data to ensure that it matches the data type defined in the database dictionary record. If you want to validate your data prior to submitting it to the RDBMS, you have several options:

- You can customize your forms so that the input field matches the data type. By default, Forms Designer uses the data type to determine the input object to use when you drag a field on to a form.
- You can create a data policy record for the table and then create validation rules for each field in the table you want to validate.
- You can create a format control record for the table and then create validation rules for each particular record state such as add, update, delete, or initial display.

Once you save data to Service Manager, the RDBMS controls the data type and any validation. If you change the data type on the RDBMS side, Service Manager can update the database dictionary record to match the RDBMS data type with a system restart of the server. We recommend you control data types from the database dictionary so you don't have to restart your system every time you make a change to a field.

Service Manager data types also allow you to take advantage of the following advanced features.

- You can move your system data from one RDBMS type to another
- You can map system data to multiple RDBMS types at the same time

You can use the SQL to SQL Conversion utility to enable one or both of these advanced features. The SQL to SQL Conversion utility allows you to move or map data on a table by table basis. It also allows you to customize how you want data in array fields to be mapped in the new RDBMS as well as offering several export options such as creating DDL or creating duplicate tables.

Related topics

Automatic update of database dictionary records
Database dictionary
Splitting SQL tables
Database dictionary records
SQL data type for localizable fields
Add a field to an existing database dictionary record
Create an active database dictionary record
Create an inactive database dictionary record
Update the data type of a database dictionary field
Database dictionary data types

7.10.1.3.1. Changing data types

A System Administrator can change the type of a scalar field. For example, from a character field to a number field.

If no data exists in the file, you can edit the field and change the field type quickly without any other modification required.

If data exists in the file, you must first change the type of the field in the database dictionary record, and then convert the existing data in the modified field to be consistent with the new field type. This must be done for all records in the file that contain data in the modified field.

Related topics

Modify field types: Single to concatenated
Modify field types: Character (scalar) to array

Change data types: Character to number

7.10.1.3.2. SQL data type for localizable fields

Consider the following if you know that a certain data field might contain localized data:

- **MS SQL Server:**

Microsoft's SQL Server doesn't offer a UTF-8 codepage. Regular data types such as VARCHAR, CHAR or TEXT use single byte code pages (such as MS Windows 1252) or double-byte code pages (such as Shift-JIS), but they can't hold characters out of multiple regions.
For example, Shift-JIS allows you to store English and Japanese characters, but not Cyrillic or Portuguese characters. On the other hand, MS WIN 1251 allows you to store Cyrillic characters but not Japanese.
If you have the requirement to store special characters from many different languages, you should use the SQL Server data types NVARCHAR, NCHAR, or NTEXT instead of VARCHAR, CHAR or TEXT. These data types use UTF-16 as a codepage and therefore are capable of storing special characters from many different languages. Be careful to use these data types only for fields that should be localized, and not for fields that will only use English characters, as they take up twice as much storage space. Also, MS SQL Server has a limitation of 8060 bytes per row in a single table.

- **Oracle:**

If you want to store localized data in Oracle, set up your database to use UTF-8 as the codepage. Service Manager internally uses UTF-8 as the code page and sends data to the RDBMS clients in UTF-8. By using UTF-8 as the database server's code page, code page translations aren't necessary, and VARCHAR, CHAR and CLOB data type can still hold all special characters from all languages.

Related topics

Database dictionary
Database dictionary data types
Create an active database dictionary record
Import RDBMS columns into an existing database dictionary record
Database dictionary data types

7.10.1.3.3. Change data types: character to number

Applies to User Roles: System Administrator

You can change data types by using System Definition, by using Mass update, or by using the Database Dictionary utility.

Change data types by using System Definition

To change data types by using System Definition:

1. In the System Navigator, click **System Definition** > **Tables** >*table_name*> **Fields** > *field_name*.
2. Double-click the field name.
3. In the General properties tab, select a new type from the **Data type** list.
4. Click **Save**.

Change data types by using Mass Update

In this example, records have a character in the zip field that's now defined as a number field.

If some of the records in the file contain data in the modified field that invalidates the new field type, the data is not converted and the system generates a "can't evaluate" message for each record failing the conversion.

Service Manager updates only those records that don't fail the conversion.

To change data types by using mass update, use the syntax: `zip in $file=nullsub{val(zip in $file,1),zip in $file}` .

Change data types by using the Database Dictionary utility

This example illustrates how to change the field called **zip** in the **vendor** file from a *character* type field to a *number* type field. Assume that data exists in the file.

Note: You must use the Windows client whenever you need to add or edit a field/key in a database dictionary table.

To change data types by using the Database Dictionary utility:

1. Click **Tailoring** > **Database Dictionary**.

2. Type **vendor** in File Name field, and then click **Search**.
3. Select the **zip** field name.
4. Click **Edit Field/Key**.
   The field editing window opens.
5. In the Type field, choose **number** in the Type list selection.
   This replaces *character* with *number* in the **Type** field.
6. Click **OK** to close the window and commit the change to the file.
7. Click **OK** to update your database dictionary record.
   The message *Record updated in the dbdict file* displays to confirm the update.
   Note: This may not be the end of the process. If records exist in the file, you need to perform a Mass Update to change the field type of the zip field from character to number. (The type of field, for example, character, number, or date/time is stored in the data record with each associated value).
   All records containing a value in the field zip need to be updated to reflect the new field type. For additional information on the Mass Update procedure, see the Database Manager Help.

Related topics

Methods for updating database dictionary records
Methods for deleting database dictionary records
Changing data types
Alias fields
Adding a key
Modifying a key
Deleting a key
Access the Database Dictionary utility
Modify field types: Character (scalar) to array
Modify field types: Single to concatenated
Open a database dictionary
Regen a database dictionary

7.10.1.3.4. Database dictionary field data types

Service Manager uses the following data types to describe field data.

Data type	Purpose	Typical RDBMS mapping
Number	Assumes field data consist of numeric data.	Service Manager maps number fields to RDBMS data types reserved for numerical data, such as FLOAT or NUMBER.
Character	Assumes field data consist of alphanumeric characters (numbers, letters, special characters).	Service Manager maps character fields to RDBMS data types reserved for alphanumeric data, such as VARCHAR or VARCHAR2.
Date/time	Assumes field data contain a Service Manager formatted date or time. Service Manager date/time formats include: • MM/DD/YY HH:MM:SS • MM/DD/YYYY HH:MM:SS • DD HH:MM:SS	Service Manager maps date/time fields to RDBMS data types reserved for date/time data, such as TIMESTAMP or DATE.
Logical	Assumes the field data contain one of the following logical values: true, false, NULL or unknown. Logical fields accept yes or no values for input, but the server converts them to true or false values when stored.	Service Manager maps logical fields to RDBMS data types reserved for single character data, such as CHAR.
Array	The field data are compound data elements that contain one or more elements of a single data type or a structure field.	Service Manager maps array fields to RDBMS data types reserved for character large object (CLOB) data, such as TEXT.
Structure	The field data are compound data elements that contain several subfields, each of which may have a different data type. Structures must be child elements of an array field.	Service Manager maps structure fields that are within an array field to RDBMS data types reserved for binary large object (BLOB) data, such as IMAGE.
Expression	The field data contain parsed Rapid Application Development (RAD) language expressions.	Service Manager maps expression fields to RDBMS data types reserved for binary large object (BLOB) data, such as IMAGE.
Label	The field data contain system data used and aren't intended for customer data.	Service Manager doesn't map label fields to any RDBMS data type.
Record	The field data contain system data and aren't intended for customer data.	Service Manager doesn't map record fields to any RDBMS data type.

Arrays

An array tracks multiple occurrences of a given field. The values in the field are all the same field type. For example, you can track all of the components for a printer in the computer file. To accomplish this, create an array field defined as a character

type, rather than creating multiple scalar fields. A scalar field is a simple data element comprised of a single occurrence of data (as opposed to an array).

Arrayed structures

An arrayed structure is comprised of more than one field with field type definitions that may or may not be the same. You can define an arrayed structure to your database dictionary record to ensure that, on any form (format) attached to a file, you have the ability to scroll more than one field at a time. When the cursor is in a scrolling field, associated information in one or more other fields will also scroll. For example, you can track contact names and phone numbers in a Configuration Item (CI) file. When scrolling the array of contact names, you can also designate the array of associated telephone numbers to scroll.

Caution: You can't use the SQL Query method against fields contained within an arrayed structure.

Structures

See Structures.

Related topics

Database dictionary
Database dictionary data types
Database dictionary records
Add a field to an existing database dictionary record
Update the data type of a database dictionary field
Service Manager Programming

7.10.1.3.5. Structures

Structures are used when you have a common group of fields that you want displayed on multiple formats. For example, Incident Management has the following three structures defined in the problem database dictionary record:

- Header
- Middle
- Action

The fields defined in the **header** and **action** structures are designed for use in every problem format, regardless of the category. Instead of using the same input fields for each category format, specific formats are created and then invoked as a subformat on each category dialog. The fields defined on a given subformat may be of different **type** definitions.

Note: On the primary format, the input field defined as a structure is associated with the subformat containing some or all of the fields defined within that structure. Fields defined on the given subformat must be defined within the associated structure, but not all fields defined within the associated structure need to be contained on the given subformat.

Create a file called **employee** with many formats attached to the file (many different views depending on the user profile). A common group of these fields is displayed on all associated formats. To avoid redundant typing of the same fields on all formats, define a structure in your database dictionary record and create a separate format to contain this common group of fields.

Fields within structures

It is important to define fields within structures when these fields exist on a format(s) associated with the input field structure name. To define fields within structures, your cursor (focus) must be on the structure name in the database dictionary record. If fields don't need to be defined within a given structure in a database dictionary record, the cursor should be positioned on the descriptor structure when adding the fields.

Organization of arrays of structure in the Database Dictionary

In each database dictionary, an array of structures appears in a group. The group is organized hierarchically, with the field of type **array** listed first. The field of type **structure** is listed next, and is indented once from the field of type array. The remaining fields are indented and listed below the field of type structure, and can be of types Character, Number, Location, or Date/Time.

Example: Arrayed structure

An example of an arrayed structure is an unavailability graph. This structure is defined in the standard Configuration Item (CI) files distributed with the Configuration Management application.

To create your own CI files and track availability, add the following availability fields to the CI files:

Database Dictionary field name	Field type
number.of.problems	number
explicit.unavailability	date/time
implicit.unavailability	date/time
perceived.unavailability	date/time
start.of.operating.hours	date/time
graph.of.unavailability	array
graph.of.unavailability	structure
date	date/time
graph	character

The field names must be spelled exactly as they are shown in the above table and defined with the specified field type.

In this example, there is a devicepc file for which you want each record within this file to contain statistical information on availability.

Notes:

- The graph fields must also be added to the CI formats.
- You must use the Windows client whenever you need to add or edit a field/key in a database dictionary table.

Add availability fields that pertain to the devicepc file

1. Sign in to the Service Manager system with a system administrator account.
2. Click **Tailoring** > **Database Dictionary**.
3. In the File Name field, type **devicepc**, and then click **Search**.
4. Place the cursor on the **descriptor** field, and then click **New Field/Key**.
   The field window opens.
5. In the Name field, type **graph.of.unavailability**.
6. Click the list in the Type field and choose **array**.
7. Click **Add**.
   The message *Enter data type of array's element* displays at the top of the screen and the window refreshes.
   The **Name** field still displays graph.of.unavailability, but the **Type** field is blank.
   To create an arrayed structure, define the type of array as a structure, such as a character, number, or date and time structure.
   Click the list in the Type field and choose **structure**.
8. Click **Add**.
   There are two entries in the database dictionary record for the field **graph.of.unavailability**. The first entry defines the field as an array. The second entry defines the data type for the array. Both entries are listed as **graph.of.unavailability**. The system default, which can be overridden, is to use the same field name for both entries.
   The two entries are added to the bottom of the list of fields in the devicepc database dictionary record.
   Note: The second entry is indented in the above example. This occurs only if the number of fields specified in the record is less than 50.

Associate the new field to the structure

1. Tab to the *second* occurrence of the **graph.of.unavailability** field, the structure type.
   For any field to be part of a structure, the cursor must be positioned on that structure field name before clicking New for the new field to be associated with it.

2. Click **New**.
   The Structure field contains the value **graph.of.unavailability**, which is the current structure to which you are adding fields.
3. In the Name field, type **date**.
4. Click the list in the Type field and choose **date/time**.
5. Click **Add**.
   The date field is displayed under the structure field **graph.of.unavailability**.
6. Select the *second* occurrence of the **graph.of.unavailability** field, the structure type.
   Note: For any fields to be part of a structure, the cursor must be positioned on that structure field name before clicking New.
7. Click **New**.
   Notice the Structure field contains the value **graph.of.unavailability**, which is the current structure to which you are adding fields.
8. In the Name field of the window, type **graph**.
9. Click the list in the Type field and choose **character**.
10. Click **Add**.
    The graph field is added as the last field under the structure field **graph.of.unavailability**, and the message *graph field added to the pcdevice file* is displayed.
    In this example, the graph field is behind the window at the bottom of the screen.

When you have completed adding fields to the database dictionary record

1. Click **OK**.
   This closes and automatically updates the file. The message *Record updated in the dbdict file* displays when complete.
2. Press **Back** to end the database dictionary session and return to the main menu.

You have now added all of the required fields comprising the **graph.of.unavailability** structure. You can now edit your database dictionary record again, update it, or cancel without updating.

Note: You must update your database dictionary record when adding a new field or the database dictionary record will remain the same. You must also add new fields to the appropriate formats in Forms Designer.

Create the unique and attribute subtables

Applies to User Roles: System Administrator

You must use the Windows client whenever you need to add or edit a new field/key in a database dictionary table.

To create the unique and attribute subtables:

1. Name the unique and attribute files.
    1. Search for the dbdict from which you want to create subtables. In this example, the **device** dbdict is used.
    2. Scroll down the **Type** column to locate an array of structures.
       Note: An array of structures is identified by an **Array** field type with a field of the same name.
    3. Type `Structure` directly below the Array field. The **Structure** field name is indented. In this example, the array of structures is **features**.
    4. Locate the **features** field of Type `Structure`.
    5. Insert your cursor in the **Type** column for the features structure field.
    6. Click **Edit Field/Key**.
    7. In the **field.window** dialog box, type a name for the **attribute** file. The standard naming convention for the attribute file is: *dbdict* + *array* + *attr*. In this example, the attribute file is named **devicefeaturesattr**.
       Note: No spaces or special characters are allowed in the subtable names.
    8. In the **field.window** dialog box, type a name for the **unique** file. The standard naming convention for the unique file is: *dbdict* + *array* + *uniq*. In this example, the unique file is named **devicefeaturesuniq**.
       Note: If the attribute file field is left blank, only the unique subtable will be created.
    9. Click **OK**.
2. Designate unique fields in the array of structure.
    1. Locate the field you want to designate as unique. In this example, the unique field is **feature.id**.

2. Insert your cursor in the **Type** column for the field you want to designate as unique.
3. Click **Edit Field/Key**.
4. Click the **Unique in Structured Array** checkbox.
5. Click **OK**.
6. Repeat steps a to e for each field you want to designate as unique.

3. From the database dictionary, click **OK**.

Service Manager updates the dbdict and creates the unique and attribute subtables from the array of structures.

Name the unique and attribute files

Applies to User Roles: System Administrator

To name the unique and attribute files:

1. Search for the dbdict from where you want to create subtables.
   In this example, the **device** dbdict is used.
2. Scroll down the **Type** column to locate an array of structures.
   Note: An array of structures is identified by an **Array** field type with a field of the same name.
3. Type `Structure` directly below the Array field. The **Structure** field name is indented.
   In this example, the array of structures is **features**.
4. Locate the **features** field of Type `Structure`.
5. Insert your cursor in the **Type** column for the features structure field.
6. Click **Edit**.
7. In the **field.window** dialog box, type a name for the **attribute** file.
   The standard naming convention for the attribute file is: *dbdict* + *array* + *attr*.
   In this example, the attribute file is named **devicefeaturesattr**.
   Note: No spaces or special characters are allowed in the subtable names.
8. In the **field.window** dialog box, type a name for the **unique** file.
   The standard naming convention for the unique file is: *dbdict* + *array* + *uniq*.
   In this example, the unique file is named **devicefeaturesuniq**.
   Note: If the attribute file field is left blank, only the unique subtable will be created.
9. Click **OK**.

View the unique and attribute subtables

Applies to User Roles: System Administrator

To view the unique and attribute subtables:

1. Click **Tailoring** > **Database Manager**.
2. Search for the dbdict that contains the array of structures from which the subtables were created. The unique and attribute files you named should appear in the record list.
3. Double-click on the file you want to view.

Related topics

Alias fields
Adding a key
Methods for updating database dictionary records
Modifying a key
Methods for deleting database dictionary records
Deleting a key
Add a structure to the database dictionary
Add fields to a structure
Create an active database dictionary record

7.10.1.3.6. Splitting SQL tables

The Service Manager server reads the row size limitation and column number limitation from file "sqldbinfo" and uses the information when making a decision about whether to split SQL tables or not. This applies to all supported database servers.

Note

Note By default, for SQL Server, all fields are put into M1 and will never be put into M2 or other “M” tables.

Caution

Caution Modifications to the "sqldbinfo" table require a restart of the Service Manager server.

Constraints when disabling the row size limitation on SQL Server

If you want to disable the row size limitation on SQL Server, you need to consider the following constraints:

- The length of individual columns must still fall within the limit of 8,060 bytes for varchar, nvarchar, varbinary, sql_variant, and CLR user-defined type columns. Only their combined lengths can exceed the 8,060-byte row limit of a table.
- The sum of other data type columns, including char and nchar data, must fall within the 8,060-byte row limit. Large object data is also exempt from the 8,060-byte row limit.
- After removing the row size limitation, if a file does have an M2 table, before you manually move all of the fields in M2 into M1, newly added fields will still be put into M2.
- If you want to disable table splitting, you need to manually set the row size limitation in the “sqldbinfo" table to a very large number, which is supported by SQLSERVER 2008 and later with ROW_OVERFLOW_DATA. The Row-Overflow feature is great for allowing occasionally rows longer than 8,060 bytes, but it's not well suited for the majority of rows being over-sized, which can lead to a drop in query performance.

Related topics

Automatic update of database dictionary records
Database dictionary
Database dictionary records
SQL data type for localizable fields
Add a field to an existing database dictionary record
Create an active database dictionary record
Create an inactive database dictionary record
Update the data type of a database dictionary field
Database dictionary data types

7.10.1.4. Alias fields

An alias field is a field that has a different name but the same index, level, and type as an existing field. Alias fields are used to streamline file maintenance; force fully keyed queries, such as those in the Incident Management and Service Desk applications; and facilitate virtual joining. Alias fields are only supported for scalar fields of the following data types: number, character, date/time, logical. The functionality is not designed for other data types like arrays and structures.

If regular purge and archive tasks are performed on files with different field names for the search criteria, such as `close.time` in the `problem` and `probsummary` files, `msg.time` in the msglog file, and `evtime` in the `eventin` file, you can build an alias in each file called **archive.time** and then perform all archive activity based on archive.time without regard to the file name.

Note

Note

For the table whose dbdict structure is similar to table cm3r, the performance of alias field query may be slow. For example, the alias field query of the field `header,number`.

In Service Desk and Incident Management, alias keys are used to ensure that a fully keyed query is performed each time an argument is entered from a record list.

Source fields are defined as an alias when they must be defined as both a normal link *and* a virtual join; *and* the target file or field is different. For example, in the **devicepc** database dictionary record and the **device.pc** format and link records, the **vj.logical.name** field is the alias of the **logical.name** field.

Related topics

Structures
Adding a key
Methods for updating database dictionary records
Modifying a key
Methods for deleting database dictionary records
Deleting a key
Add an alias field
Add a field to an existing database dictionary record
Create an active database dictionary record

7.10.1.5. Access the database dictionary utility

Applies to User Roles: System Administrator

Use one of the following methods to access the Database Dictionary utility:

- To access the Database Dictionary utility from the System Navigator, click **Tailoring** > **Database Dictionary**.
- To access the Database Dictionary utility from the Service Manager command line, type `dbdict` in the command line, and then press **Enter**.

Related topics

Copying a database dictionary
Rename a database dictionary
Regen a database dictionary

7.10.1.6. Enable DDL logging of database dictionary changes

Applies to User Roles: System Administrator

You must have administrative access to the server operating system to use this procedure.

To enable Database Definition Language (DDL) logging of database dictionary changes:

1. Sign in to the server running Service Manager.
2. Stop the Service Manager server.
3. Open the Service Manager initialization file (`sm.ini`) with a text editor.
4. Add the following parameters to enable DDL logging of database dictionary changes:

   ```
   sqlddllogging:1
   sqlddldirectory:<path>
   ```

   For `<path>` , specify the relative path to the directory where you want Service Manager to save DDL files. This path is relative to the server's `RUN` directory.
5. Save the Service Manager initialization file.
6. Restart the Service Manager server.

Related topics

Database dictionary
Database dictionary records
Create an active database dictionary record
Create an inactive database dictionary record

7.10.1.7. Open a database dictionary

Applies to User Roles: System Administrator

You can open a database dictionary (dbdict) from Forms Designer, by using System Definition, or by using the Database Dictionary utility. Choose one of the following methods to open a dbdict.

Open a dbdict by using System Definition

To open a dbdict by using System Definition:

1. From the System Navigator, click **System Definition** > **Tables** > ***<dbdict_name>***.
   Note: The files are listed alphabetically. You can type the first letter of a table name to jump to that alphabetic section. For example, type `o` to jump to the oncall table.
2. Click the plus (+) next to the file name to open the folder.
3. Click the plus (+) next to the *Fields* or *Keys* folder to open it.
4. Double-click a field or key to edit it.

Open a dbdict when using Forms Designer

To open a database dictionary when using Forms Designer:

1. Open an associated form using Forms Designer. For this example, access the *contacts* form.
2. Select **System Definition** from the More Actions menu. The dbdict for that form opens.

Open a dbdict by using the Database Dictionary utility

This example opens the contacts dbdict.

To open a database dictionary by using the Database Dictionary utility:

1. Click **Tailoring** > **Database Dictionary**.
2. Type the name of the dbdict file you want to edit into the **File Name** field.
   For this example, type: **contacts**.
3. Click **Search**.
   The dbdict definition record(s) with that name opens. If more than one file is displayed, select the correct file to view it.

To see more fields, use the scroll bar on the right of the field names list. To see more keys, use the scroll bar on the right of the keys list.

Related topics

Copying a database dictionary
Rename a database dictionary
Methods for updating database dictionary records
Methods for deleting database dictionary records

7.10.1.8. Copying a database dictionary

You can make copies of your database dictionary for backup purposes or to create another file with similar dictionary definitions.

When you copy a database dictionary, you have the option to copy the record only or the record and all individual data records contained in the file. If you copy the database dictionary and all associated records, data in the original file is inaccessible during the copy process. The amount of time it takes to copy the file and records will vary depending on how many records are in the original file.

Related topics

Rename a database dictionary
Copy a database dictionary and data records
Copy a database dictionary only
Access the Database Dictionary utility

7.10.1.8.1. Copy a database dictionary and data records

Applies to User Roles: System Administrator

You can copy a database dictionary by using System Definition or by using the Database Dictionary utility. Choose one of the following methods to copy a database dictionary and data records.

By using System Definition

To copy a database dictionary and data records, by using System Definition:

1. From the System Navigator, click **System Definition**.
2. Double-click a table.
   An overview of the table opens.
3. In the **Table management** section of the overview, click **Copy the definition and data**.
   A confirmation box opens.
4. In the **New table name** field, type the name of the table, and then click **OK**.
   A message displays, stating that the table definition and data were copied to *new name*.
5. Click **OK**.

By using the Database Dictionary utility

Example: Make a copy of all the records in the *assignment* dbdict by creating a new file with the same database dictionary structure and containing all records contained in the original file. To copy records from a file by copying the entire file, copy the associated data records as well.

To copy a database dictionary and data records, by using the Database Dictionary utility:

1. Open the assignment dbdict.
2. Open the More Actions menu and click **Copy/Rename**.
3. In the **New Name** field, enter **assignment1**.
   Note: File names must be one word. No blanks and no special delimiters are allowed.
4. Select the **Copy dbdict and data** option to copy the dbdict and all records from the **assignment** file to the **assignment1** file.
5. Click OK to copy the dbdict and data stored in the file.
   Note: The length of time it takes to copy a file with all records will vary depending on how many records exist in the file. Copy a file with all associated records during non-peak operation times.
   The database dictionary for the assignment1 file opens and Service Manager displays this message: **Dbdict successfully copied*
   Note: When an asterisk displays in front of a message, it means there are additional messages to display which may have relevant information regarding this operation.
   You can now make changes to the **enduserbackup** database dictionary.
6. Click OK to exit when you have completed modifications to this file.
   Note: You will need to update your database dictionary only if you have added, deleted, or modified fields and/or keys.

Related topics

Copy a database dictionary only

7.10.1.8.2. Copy a database dictionary only

Applies to User Roles: System Administrator

You can copy a database dictionary by using System Definition or by using the Database Dictionary utility. Choose one of the following methods to copy a database dictionary.

By using System Definition

To copy a database dictionary only, by using System Definition:

1. From the System Navigator, click **System Definition**.
2. Double-click a table.
   An overview of the table opens.
3. In the **Table management** section of the overview, click **Copy the definition**.
   A confirmation box opens.
4. In the **New table name** field, type the name of the table, and then click **OK**.
   A message displays, stating that the table definition was copied to *new name*.
5. Click **OK**.

By using the Database Dictionary utility

Example: Make a copy of the database dictionary in the assignment file. In this example, we will copy only the database dictionary, not the data records.

To copy a database dictionary only, by using the Database Dictionary utility:

1. Open the **assignment** file using database dictionary.
2. Open the More Actions menu and click **Copy/Rename**.
3. In the New Name field of the rename screen, type `assignment2`.
   Note: File names must be one word; no blanks and no special delimiters are allowed.
4. Select the **Copy dbdict only** radio button.
5. Click **OK** to begin file copy. The copied dbdict opens. Service Manager displays this message: *Dbdict successfully copied*.

Related topics

Copy a database dictionary and data records

7.10.1.9. Methods for adding database dictionary records

Your database dictionary records should match the tables and columns Service Manager manages in the RDBMS. We recommend that you use database dictionary records as your primary data definition source and allow Service Manager to make changes to RDBMS tables, columns, and indexes as needed. If you provide Service Manager with an RDBMS account with table create-alter-drop-rights, any time you make a change in a database dictionary record, the server automatically makes the necessary changes in the RDBMS.

If you don't provide Service Manager with table create-alter-drop-rights to the RDBMS, an RDBMS administrator must make all the necessary changes each time you add, remove, or update a database dictionary record. In such cases, we recommend you enable DDL logging to capture database dictionary changes. The Service Manager administrator can then provide the DDL to the RDBMS administrator who can approve or modify it as needed. After the RDBMS administrator has created the necessary tables, columns, and indexes, the Service Manager administrator can import the existing RDBMS columns.

The following table summarizes the methods you can use to add database dictionary records and the best practices we recommend.

Action	From the System Definition utility	From the Database Dictionary form	From RDBMS utilities
Add a database dictionary record	**Best practice**: Provide Service Manager table create-alter-drop-rights to the RDBMS, and create a new database dictionary record. Service Manager creates the required tables, columns, and indexes in the RDBMS.	• **Best practice**: Provide Service Manager table create-alter-drop-rights to the RDBMS, and create a new database dictionary record. Service Manager creates the required tables, columns, and indexes on the RDBMS. • Enable DDL logging and use the Database Dictionary form to create a new database dictionary record. Have the RDBMS administrator add tables, columns, and indexes based on the DDL, and then use the Database Dictionary form to import the existing RDBMS columns.	You can't create a Service Manager database dictionary record from RDBMS utilities.

Add a table	**Best practice**: Provide Service Manager table create-alter-drop-rights to the RDBMS, and then create a new database dictionary record. Service Manager creates the required table in the RDBMS.	• **Best practice**: Provide Service Manager table create-alter-drop-rights to the RDBMS, and create a new database dictionary record. Service Manager creates the required tables in the RDBMS. • Enable DDL logging and use the Database Dictionary form to create a new database dictionary record. Have the RDBMS administrator add a table based on the DDL, and then use the Database Dictionary form to import the existing RDBMS columns.	Have the RDBMS administrator add a table directly to the RDBMS, and then use the Database Dictionary form to import the existing RDBMS columns.

Add a field	**Best practice**: Provide Service Manager table create-alter-drop-rights to the RDBMS, and add a field to a database dictionary record. Service Manager creates the required column in the RDBMS.	• **Best practice**: Provide Service Manager table create-alter-drop-rights to the RDBMS, and use the Database Dictionary form to add a field to a database dictionary record. Service Manager creates the required column in the RDBMS. • Enable DDL logging and use the Database Dictionary form to add a field to a database dictionary record. Have the RDBMS administrator add a column based on the DDL, and then use the Database Dictionary form to import the existing RDBMS columns.	You can't create a Service Manager field from RDBMS utilities.

Add a column	**Best practice**: Provide Service Manager table create-alter-drop-rights to the RDBMS, and add a field to a database dictionary record. Service Manager creates the required column in the RDBMS.	• **Best practice**: Provide Service Manager table create-alter-drop-rights to the RDBMS, and use the Database Dictionary form to add a field to a database dictionary record. Service Manager creates the required column in the RDBMS. • Enable DDL logging and use the Database Dictionary form to add a field to a database dictionary record. Have the RDBMS administrator add a column based on the DDL, and then use the Database Dictionary form to import the existing RDBMS columns.	Have the RDBMS administrator add a column directly to the RDBMS, and then use the Database Dictionary form to import the existing RDBMS columns.
Add a key	**Best practice**: Provide Service Manager table create-alter-drop-rights to the RDBMS, and add a key to a database dictionary record. Service Manager creates the required index in the RDBMS.	• **Best practice**: Provide Service Manager table create-alter-drop-rights to the RDBMS, and use the Database Dictionary form to add a key to a database dictionary record. Service Manager creates the required index in the RDBMS. • Enable DDL logging and use the Database Dictionary form to add a key to a database dictionary record. Have the RDBMS administrator add an index based on the DDL.	You can't create a Service Manager key from RDBMS utilities.

Add an index	**Best practice**: Provide Service Manager table create-alter-drop-rights to the RDBMS, and add a key to a database dictionary record. Service Manager creates the required index in the RDBMS.	• **Best practice**: Provide Service Manager table create-alter-drop-rights to the RDBMS, and use the Database Dictionary form to add a key to a database dictionary record. Service Manager creates the required index in the RDBMS. • Enable DDL logging and use the Database Dictionary form to add a key to a database dictionary record. Have the RDBMS administrator add an index based on the DDL.	Have the RDBMS administrator add an index directly to the RDBMS, and then manage it from RDBMS utilities.

Related topics

Automatic update of database dictionary records
Automatic update of RDBMS data when database dictionary records change
Database dictionary
Database dictionary records
Create an active database dictionary record
Create an inactive database dictionary record
Create a database dictionary record by importing RDBMS columns
Import RDBMS columns into an existing database dictionary record

7.10.1.9.1. Field name requirements

You must follow these rules when creating field names:

- Valid characters include uppercase and lowercase letters of the alphabet, numeric characters 0 through 9, and the period character.
- Field names must begin with a letter of the alphabet.
- Spaces and all other special characters aren't permitted.
- Do not use the following Service Manager reserved words (uppercase or lowercase):
- - AND
  - BEGIN
  - DO
  - ELSE
  - END
  - F
  - FALSE
  - FOR
  - IF
  - IN

- ISIN
- LIKE
- MOD
- N
- NO
- NOT
- OR
- STEP
- T
- THEN
- TO
- TRUE
- U
- UNKNOWN
- WHILE
- Y
- YES

Related topics

Methods for adding database dictionary records
Methods for adding database dictionary records
Structures
Alias fields
Adding a key
Methods for updating database dictionary records
Modifying a key
Methods for deleting database dictionary records
Deleting a key
Search for a field in the database dictionary
Database dictionary data types

7.10.1.9.2. Create an inactive database dictionary record

Applies to User roles: System Administrator

This procedure assumes the Service Manager system doesn't have table create-alter-drop-rights to the RDBMS system. If your system does have table create-alter-drop-rights to the RDBMS system, then Service Manager automatically creates any necessary tables, columns, and indexes required by your database dictionary records.

We recommend you enable the option to create DDL to provide to your RDBMS administrator. You can then activate the database dictionary record after your RDBMS administrator has created the tables and columns specified in the DDL.

Note: You must use the Windows client whenever you need to add a new field/key to a database dictionary table.

To create an inactive database dictionary record:

1. Sign in to the Service Manager system with a system administrator account.
2. Click **Tailoring** > **Database Dictionary**.
3. Type the table name in the **File Name** field, and then click **New**.
4. In the Fields tab, click **New Field/Key** to add one field for each RDBMS column you want to add.
5. In the Keys tab, click **New Field/Key** to add a unique key to the database dictionary record.
6. Click **OK**.
   Service Manager creates the database dictionary record and maps each field to a null table alias.

The database dictionary record remains inactive until you activate it and remap the fields to actual RDBMS tables and columns.

Activate a database dictionary record

Applies to User roles: System Administrator

This procedure assumes the Service Manager system doesn't have table create-alter-drop-rights to the RDBMS system, and that you have not already created a database dictionary record for this change. If your system has table create-alter-drop-rights to the RDBMS system, then Service Manager automatically creates any necessary tables, columns, and indexes when you create a database dictionary record. If your Service Manager system doesn't have RDBMS table create-alter-drop-rights, we recommend you create the database dictionary record first and provide DDL to the RDBMS administrator to implement the necessary changes. The Service Manager administrator can then use this procedure to activate the database record when the RDBMS tables and columns become available.

To activate a database dictionary record:

1. Sign in to the Service Manager system with a System Administrator account.
2. Click **Tailoring** > **Database Dictionary**.
3. Type the existing database dictionary record name in the **Filename** field, and then click **Search**.
   The database dictionary record opens.
4. Click the More Actions and choose **Activate Nulltable Rows** to have the Service Manager server sign in to the RDBMS and check for new columns in the SQL tables that match the fields in database dictionary record.
   Service Manager automatically displays the new field mappings in the database dictionary record.
5. Click **OK** to save the database dictionary record.

Related topics

Database dictionary
Database dictionary records
Data persistence
Enable DDL logging of database dictionary changes

7.10.1.9.3. Create an active database dictionary record

Applies to User roles: System Administrator

This procedure assumes the Service Manager system has table create-alter-drop-rights to the RDBMS system. If Service Manager doesn't have table create-alter-drop-rights, your database dictionary record remains inactive and mapped to null tables until you activate it. If your system doesn't have table create-alter-drop-rights to the RDBMS, we recommend you enable the option to create DDL to provide to your RDBMS administrator. You can then activate the database dictionary record after your RDBMS administrator has created the tables and columns specified in the DDL.

For Windows clients:

To create an active database dictionary record:

1. Sign in to the Service Manager system with a system administrator account.
2. In the System Navigator, click **System Definition** > **Tables**.
3. Right-click **Tables**, and then click **New Table**.
4. Type the **Table name**.
5. Add one field for each RDBMS column you want to add.
6. Add a unique key to the database dictionary record.
7. Click **OK**.

For Web clients:

To create an active database dictionary record:

1. Sign in to the Service Manager system with a system administrator account.
2. In the System Navigator, click **Tailoring** > **Database Dictionary**.
3. Type the name of the new table in the **File Name** field, and then click the New icon.
4. Add one field for each RDBMS column you want to add.
5. Add a unique key to the database dictionary record.
6. Click **OK**.

Related topics

Database dictionary
Database dictionary records
Data persistence
Create a database dictionary record by importing RDBMS columns
Enable DDL logging of database dictionary changes

7.10.1.9.4. Add a structure to the database dictionary

Applies to User Roles: System Administrator

You can add a structure to the database dictionary by using System Definition or by using the Database Dictionary utility. Choose one of the following methods to add a structure.

Note: You must use the Windows client whenever you need to add a new field/key to a database dictionary table.

Add a structure by using System Definition

To add a structure by using System Definition:

1. In the System Navigator, click **System Definition** > **Tables** > ***filename*** > **Fields**.
2. Click **New Field**.
3. Type a name in the **Field Name** field.
4. Click **OK**. The new field name is now listed in its alphabetical place in the fields list.
5. Select the new field from the list, and in the **General properties** section select **Structure** in the **Data type** field, and then specify other general properties and SQL mappings properties for the field.
6. Select the structure field from the Fields list, and click **New Field**.
7. Type a name in the **Field name** field, and click **OK**. The field is added to the structure.
8. Specify general properties, aliases, and SQL mapping properties for the field.
9. Repeat steps 6 through 8 to add more fields to the structure.
10. Click **Save** to save the file.

Add a structure by using the Database Dictionary utility

This example illustrates how to add a structure that contains fields for basic employee information. The fields are name, telephone number, and address. These fields will appear on all formats or views of the employee file. Salary, hire.date, review.date, rehire, termination.date, position, and department will appear only on certain forms.

To add a structure by using the Database Dictionary utility:

1. From the database dictionary, open the **employee** file.
2. On the Fields tab, place the cursor on the **descriptor** structure.
3. Click **New Field/Key**.
4. In the Name field, type **basic.employee.info**.
5. In the Type field, choose **structure** from the Type list.
6. Click **Add**.
   You can now add fields within the structure you just created.
7. In the Name and Type fields, type a name and select a data type for the first field you want to add to the structure. For example, type **person.name** and select **character**.
8. Click **Add**.
   The structure with its first field is added to the Fields tab.
9. On the Fields tab, select the structure field you added, and click **New Field/Key** to add more fields to the structure field. For example, add two more fields named **telephone** (data type: **character**) and **address** (data type: **character**) .
10. Click **OK** to save your edits.
11. Click **SM Alters** when you are asked how you want to update the database dictionary.
    This returns the message: Record updated in the dbdict file.

Related topics

Forms Designer
Adding a key

Alias fields
Methods for deleting database dictionary records
Deleting a key
Methods for updating database dictionary records
Modifying a key
Structures
Access the Database Dictionary utility
Add fields to a structure
Open a database dictionary
Regen a database dictionary

7.10.1.9.5. Add an alias field

Applies to User Roles: System Administrator

You can add an alias field by using System Definition or by using the Database Dictionary utility. Choose one of the following methods to add an alias field.

Note: You must use the Windows client whenever you need to add or edit a field/key in a database dictionary table.

Add an alias field by using System Definition

1. In the System Navigator, click **System Definition** > **Tables**.
   Note: To expand any table in the record list, type the first letter of a table name to jump to that alphabetic section. For example, type `o` to jump to the oncall table.
2. Expand **Fields**, and then select a field.
   The Fields and keys definitions for the activity table opens.
3. In the Aliases section, click **New**.
4. In the **Alias name** field, type an alias name.
5. Click **OK**.
6. Click **Save** to save your changes.

Add an alias field by using the Database Dictionary utility

To add an alias field by using the Database Dictionary utility:

1. Click **Tailoring** > **Database Dictionary**.
2. In the File Name field type a file name, and then click **Search**.
   The record opens.
3. On the Fields tab, scroll through the fields and select one.
4. Click **Edit Field/Key**.
   The field window opens.
5. Click **Create Alias**.
6. In the **Name** field, type the name of the alias.
7. Click **OK**.
8. After the record has been updated, click **OK** again.

Related topics

Alias fields
Database dictionary data types
Structures
Adding a key
Methods for updating database dictionary records
Modifying a key
Methods for deleting database dictionary records
Deleting a key
Add a field to an existing database dictionary record
Open a database dictionary

Access the Database Dictionary utility
Regen a database dictionary

7.10.1.9.6. Add an array

Applies to User Roles: System Administrator

You can add an array by using System Definition or by using the Database Dictionary utility. Choose one of the following methods to add an array.

Note: You must use the Windows client whenever you need to add a new field/key to a database dictionary table.

Add an array by using System Definition

1. In the System Navigator, click **System Definition** > **Tables**.
   Note: To expand any table in the record list, type the first letter of a table name to jump to that alphabetic section. For example, type `o` to jump to the oncall table.
2. Double-click **Fields**.
   The Fields and keys definitions for the table opens.
3. Place the cursor on a non-structue field, and click **New array**.
4. Type a name in the **Field Name** field, and click **OK**.
   The array along with an element named after the array is now listed in its alphabetical place in the fields list.
5. Select a data type, and specify other properties for the array field.
6. Click **Save** to save your changes to the table.

Add an array by using the Database Dictionary utility

Example: Add an array to the **unitofmeasure** file.

1. Click **Tailoring** > **Database Dictionary**.
2. In the File Name field, type **unitofmeasure**, and then click **Search**.
3. Place the cursor in the **descriptor** structure.
4. Click **New Field/Key**.
   The field window opens.
5. In the Name field of the field window, type **language**.
   Note: When specifying field names, no blanks are allowed between words. Always use a period as a delimiter.
6. In the Type field, type **array**, or select array from the Type list.
7. Click **Add**.
   Note: Service Manager displays this message: *Enter data type of array's element*.
   A new window opens, requesting a data type for the array element you have created.
8. In the Type field, select **character**, and then click **Add** to add the new array element.
9. Use the scroll bar to locate the new array and element at the bottom of the fields list.
   There are two entries in the record for the new field: The first entry defines the field as an array. The second entry defines the data type of the array. Both entries are listed, using the field name for both entries. However, you can override this if you choose. The two entries are added to the bottom of the list of fields in the database dictionary record. The second entry is also indented from the first entry (this occurs only if the number of fields specified in the database dictionary record is less than 50).
   You can now reedit your database dictionary record, update it, or cancel without updating.
   Note: You must update your database dictionary record when adding a new field, or the database dictionary record will remain the same. You must also add a new field(s) to the applicable format(s) via Forms Designer. See the Forms Designer Help for additional information about coordinating record changes.
10. Once you are finished adding fields to the database dictionary, click **OK** to close and automatically update the file.

Related topics

Database dictionary data types
Methods for adding database dictionary records
Field name requirements
Structures
Alias fields

Adding a key
Methods for updating database dictionary records
Modifying a key
Methods for deleting database dictionary records
Deleting a key
Regen a database dictionary
Search for a field in the database dictionary
Open a database dictionary
Access the Database Dictionary utility
Create an active database dictionary record
Database dictionary data types

7.10.1.9.7. Add fields to a structure

Applies to User Roles: System Administrator

You can add fields to a structure with the Database Dictionary utility.

Note: You must use the Windows client whenever you need to add a new field/key to a database dictionary table.

To add a field to a structure:

Example: Add a field to the descriptor structure in the device table.

To add fields to a structure by using the Database Dictionary utility:

1. Open a file in the database dictionary, and then position the cursor on the structure **descriptor**.
2. Click **New Field/Key**. In the **Name** field, type `logical.example`. In the Type field, type `character`, or select a character from the Type list of valid data types.
3. Click **Add** to add the new field in this structure.
   The field appears in the bottom of the list of fields under the structure.
4. Double-click the new field to enter the remaining SQL information.
5. Click **OK** once more to save your edits and end, or click **Cancel** to quit without recording the edits.

You must also add any new fields to the appropriate formats using Forms Designer to make the field available to users. For additional information, see the Forms Designer Help.

When you add a new field to structures that are blobs such as an image type, you don't need the additional SQL information.

When creating categories in Incident Management, define any additional fields in the newly created categories within the middle structure in the `problem` database dictionary record.

Related topics

- Structures
- Database dictionary data types
- Alias fields
- Adding a key
- Methods for updating database dictionary records
- Modifying a key
- Methods for deleting database dictionary records
- Deleting a key
- Add a structure to the database dictionary
- Open a database dictionary
- Access the Database Dictionary utility
- Regenerate a database dictionary

7.10.1.9.8. Create a database dictionary record by importing RDBMS columns

Applies to User roles: System Administrator

This procedure assumes the Service Manager system doesn't have table create-alter-drop-rights to the RDBMS system, and that you have not already created a database dictionary record for this change. If your system has table create-alter-drop-rights to the RDBMS system, then Service Manager automatically creates any necessary tables, columns, and indexes when you create a database dictionary record. If your Service Manager system doesn't have RDBMS table create-alter-drop-rights, we recommend you create the database dictionary record first and provide DDL to the RDBMS administrator to implement the necessary changes. The Service Manager administrator can then activate the database record when the RDBMS tables and columns become available. This procedure provides an alternative means for you to keep your database dictionary records synchronized with an RDBMS.

In order for Service Manager to successfully import columns into a database dictionary record, the RDBMS tables must have the following features:

- The table names must be in uppercase
- The table must be owned by a Service Manager user

To create a database dictionary record by importing RDBMS columns:

1. Have your RDBMS administrator create the new tables and columns you want Service Manager to manage.
2. Click **Tailoring** > **Database Dictionary**.
3. Type the name you want to use for database dictionary record in the **File Name** field, and then click **New**.
   For example: `newtableone`.
   A new blank database dictionary record opens.
4. Click the **SQL tables** tab.
5. In the **Alias** field, type the short name you want fields in this database dictionary record to use to refer to this table.
   For example, `m1`.
   Note: Alias names are limited to three characters.
6. In the **Name** field, type the name of the RDBMS table to which you want this database dictionary record to map.
   For example: `NEWTABLEONEM1`.
7. In the **Type** field, type one of the following codes to identify the vendor of your RDBMS.
   - `oracle` - for supported versions of Oracle databases
   - `sqlserver` - for supported versions of Microsoft SQL Server databases
8. If this database dictionary maps to multiple RDBMS tables, repeat steps 5-7 for each RDBMS table.
9. Click **Import new columns from SQL** to have the Service Manager server log in to the RDBMS and load all the columns from the SQL tables defined for the database dictionary record.
   Service Manager automatically displays new columns as fields in the database dictionary record.
10. Click **OK** to save the database dictionary record.

Related topics

Database dictionary
Database dictionary records
Data persistence
Create an active database dictionary record
Create an inactive database dictionary record
Import RDBMS columns into an existing database dictionary record

7.10.1.9.9. Designate unique fields in the array of structures

Applies to User Roles: System Administrator

To designate unique fields in the array of structures:

1. Locate the field you want to designate as unique. In this example, the unique field is **feature.id**.
2. Insert your cursor in the **Type** column for the field you want to designate as unique.
3. Click **Edit**.
4. Click the **Unique in Structured Array** checkbox.
5. Click **OK**.
6. Repeat steps 1 to 5 for each field you want to designate as unique.

Related topics

Adding a key
Alias fields
Database dictionary data types
Methods for deleting database dictionary records
Deleting a key
Methods for updating database dictionary records
Modifying a key
Structures
Database dictionary
Access the Database Dictionary utility
Add an array
Open a database dictionary
Regen a database dictionary

7.10.1.10. Methods for deleting database dictionary records

Your database dictionary records should match the tables and columns Service Manager manages in the RDBMS. We recommend that you use database dictionary records as your primary data definition source and allow Service Manager to make changes to RDBMS tables, columns, and indexes as needed. If you provide Service Manager with an RDBMS account with table create-alter-drop-rights, any time you make a change in a database dictionary record, the server automatically makes the necessary changes in the RDBMS.

If you don't provide Service Manager with table create-alter-drop-rights to the RDBMS, an RDBMS administrator must make all the necessary changes each time you add, remove, or update a database dictionary record. In such cases, we recommend you enable DDL logging to capture database dictionary changes. The Service Manager administrator can then provide the DDL to the RDBMS administrator who can approve or modify it as needed. After the RDBMS administrator has created the necessary tables, columns, and indexes, the Service Manager administrator can import the existing RDBMS columns.

The following table summarizes the methods you can use to delete database dictionary records and the best practices we recommend.

Action	From the System Definition utility	From the Database Dictionary form	From RDBMS utilities
Delete a database dictionary record	**Best practice**: Provide Service Manager table create-alter-drop-rights to the RDBMS, and delete an existing database dictionary record. Service Manager deletes any tables, columns, and indexes in the RDBMS.	• **Best practice**: Provide Service Manager table create-alter-drop-rights to the RDBMS, and delete an existing database dictionary record. Service Manager deletes the tables, columns, and indexes in the RDBMS. • Enable DDL logging and use the Database Dictionary form to delete an existing database dictionary record. Have the RDBMS administrator delete tables, columns, and indexes based on the DDL.	You can't delete a Service Manager database dictionary record from RDBMS utilities.
Delete a table	**Best practice**: Provide Service Manager table create-alter-drop-rights to the RDBMS, and delete an existing database dictionary record. Service Manager deletes the table in the RDBMS.	• **Best practice**: Provide Service Manager table create-alter-drop-rights to the RDBMS, and delete an existing database dictionary record. Service Manager deletes the tables in the RDBMS. • Enable DDL logging and use the Database Dictionary form to delete an existing database dictionary record. Have the RDBMS administrator delete the tables based on the DDL.	Have the RDBMS administrator delete the tables directly in the RDBMS. Service Manager will automatically update any corresponding database dictionary records to map to NULLTABLE the next time the server restarts and there is a query against the changed table.

Delete a field	**Best practice**: Provide Service Manager table create-alter-drop-rights to the RDBMS, and delete an existing field in a database dictionary record. Service Manager deletes the column in the RDBMS.	• **Best practice**: Provide Service Manager table create-alter-drop-rights to the RDBMS, and use the Database Dictionary form to delete an existing field in a database dictionary record. Service Manager deletes the column in the RDBMS. • Enable DDL logging and use the Database Dictionary form to delete an existing field in a database dictionary record. Have the RDBMS administrator delete the column based on the DDL.	You can't delete a Service Manager field from RDBMS utilities.
Delete a column	**Best practice**: Provide Service Manager table create-alter-drop-rights to the RDBMS, and update an existing field in a database dictionary record. Service Manager updates the column in the RDBMS.	• **Best practice**: Provide Service Manager table create-alter-drop-rights to the RDBMS, and use the Database Dictionary form to update an existing field in a database dictionary record. Service Manager updates the column in the RDBMS. • Enable DDL logging and use the Database Dictionary form to update an existing field in a database dictionary record. Have the RDBMS administrator update the column based on the DDL.	Have the RDBMS administrator update the column directly to the RDBMS. Service Manager will automatically update any corresponding database dictionary records the next time the server restarts and there is a query against the changed table.

Update a key	**Best practice**: Provide Service Manager table create-alter-drop-rights to the RDBMS, and delete an existing key in a database dictionary record. Service Manager delete the index in the RDBMS.	• **Best practice**: Provide Service Manager table create-alter-drop-rights to the RDBMS, and use the Database Dictionary form to delete an existing key in a database dictionary record. Service Manager deletes the index in the RDBMS. • Enable DDL logging and use the Database Dictionary form to delete an existing key in a database dictionary record. Have the RDBMS administrator delete the index based on the DDL.	You can't delete a Service Manager key from RDBMS utilities.
Delete an index	**Best practice**: Provide Service Manager table create-alter-drop-rights to the RDBMS, and delete an existing key in a database dictionary record. Service Manager deletes the index in the RDBMS.	• **Best practice**: Provide Service Manager table create-alter-drop-rights to the RDBMS, and use the Database Dictionary form to delete an existing key in a database dictionary record. Service Manager deletes the index in the RDBMS. • Enable DDL logging and use the Database Dictionary form to delete an existing key in a database dictionary record. Have the RDBMS administrator delete the index based on the DDL.	Have the RDBMS administrator delete an index directly in the RDBMS.

Delete a database dictionary record

Applies to User Roles:

System Administrator

This procedure assumes the Service Manager system has table create-alter-drop-rights to the RDBMS system. If Service Manager doesn't have table create-alter-drop-rights, your database dictionary record changes remain inactive and mapped to null tables until you activate them. If your system doesn't have table create-alter-drop-rights to the RDBMS, we recommend you enable the option to create DDL to provide to your RDBMS administrator. You can then activate the database dictionary record changes after your RDBMS administrator has created the tables and columns specified in the DDL.

To delete a database dictionary record:

1. Sign in to the Service Manager system with a system administrator account.
2. From the System Navigator, click **System Definition** > **Tables**.
3. Double-click the table name you want to delete.
   Note: Each table name is actually a database dictionary record.
   Note: You can type the first letter of a table name to jump to that alphabetic section. For example, type `o` to jump to the oncall table.
   An overview of the table opens.
4. In the **Table management** section of the Table tab, click **Delete**.
5. Click **OK** to confirm the deletion.
6. Click **OK** to close the confirmation message.

Remove a field from an existing database dictionary record

Applies to User roles: System Administrator

This procedure maps any unwanted fields in your system to NULLTABLE. Service Manager doesn't actually remove data or delete the corresponding columns from the RDBMS.

Important: We recommend you leave any unwanted columns mapped to NULLTABLE rather than actually deleting them from the RDBMS to preserve any legacy data these columns may still contain and to provide you a means to remap the columns at a later date.

If you choose to remove the column from the RDBMS, Service Manager will recognize the change and update the database dictionary record to map the deleted column to NULLTABLE the next time you restart the server and query the RDBMS table that contained the deleted column.

To remove a field from an existing database dictionary record, add a table nulltable with alias n1, using the dbdict utility:

1. Sign in to Service Manager as a System Administrator.
2. Click **Tailoring** > **Database Dictionary**.
3. In the File Name field, enter the required file name, and then click **Search**.
4. When the correct file is displayed, select the **SQL Tables** tab.
5. Add the following line:

Alias	Name	Type
n1	NULLTABLE	sqlserver

6. Click **OK**. A message appears, stating that the record has been updated in the dbdict file.

7. Select the **Fields** tab and double-click the column to be moved to the Null Table. The field window opens.

8. In the SQL Table field, change the entry from **m1** to **n1**.

9. Click **OK** to return to the dbdict record. The 'SQL Table' column value for the changed field is now 'n1.'

10. Click **OK** to save the changes in the specified table. A message appears, stating that the record has been updated in the dbdict file.

When you select the table through the System Definition utility, you will see the change.

Related topics

Database dictionary
Making database dictionary changes

7.10.1.11. Methods for updating database dictionary records

Your database dictionary records should match the tables and columns Service Manager manages in the RDBMS. We recommend that you use database dictionary records as your primary data definition source and allow Service Manager to make changes to RDBMS tables, columns, and indexes as needed. If you provide Service Manager with an RDBMS account with table create-alter-drop-rights, any time you make a change in a database dictionary record, the server automatically makes the necessary changes in the RDBMS.

If you don't provide Service Manager with table create-alter-drop-rights to the RDBMS, an RDBMS administrator must make all the necessary changes each time you add, remove, or update a database dictionary record. In such cases, we recommend you enable DDL logging to capture database dictionary changes. The Service Manager administrator can then provide the DDL to the RDBMS administrator who can approve or modify it as needed. After the RDBMS administrator has created the necessary tables, columns, and indexes, the Service Manager administrator can import the existing RDBMS columns.

The following table summarizes the methods you can use to update database dictionary records and the best practices we recommend.

Action	From the System Definition utility	From the Database Dictionary form	From RDBMS utilities
Update a database dictionary record	**Best practice**: Provide Service Manager table create-alter-drop-rights to the RDBMS, and update an existing database dictionary record. Service Manager updates any tables, columns, and indexes in the RDBMS as well as remaps any existing columns.	• **Best practice**: Provide Service Manager table create-alter-drop-rights to the RDBMS, and update an existing database dictionary record. Service Manager updates the changed tables, columns, and indexes in the RDBMS. • Enable DDL logging and use the Database Dictionary form to update an existing database dictionary record. Have the RDBMS administrator update tables, columns, and indexes based on the DDL.	You can't update a Service Manager database dictionary record from RDBMS utilities.
Update a table	**Best practice**: Provide Service Manager table create-alter-drop-rights to the RDBMS, and update an existing database dictionary record. Service Manager updates the table in the RDBMS.	• **Best practice**: Provide Service Manager table create-alter-drop-rights to the RDBMS, and update an existing database dictionary record. Service Manager updates the tables in the RDBMS. • Enable DDL logging and use the Database Dictionary form to update an existing database dictionary record. Have the RDBMS administrator update the tables based on the DDL.	Have the RDBMS administrator update the tables directly in the RDBMS. Service Manager will automatically update any corresponding database dictionary records the next time the server restarts and there is a query against the changed table.

Update a field	**Best practice**: Provide Service Manager table create-alter-drop-rights to the RDBMS, and update an existing field in a database dictionary record. Service Manager updates the column in the RDBMS.	• **Best practice**: Provide Service Manager table create-alter-drop-rights to the RDBMS, and use the Database Dictionary form to update an existing field in a database dictionary record. Service Manager updates the column in the RDBMS. • Enable DDL logging and use the Database Dictionary form to update an existing field in a database dictionary record. Have the RDBMS administrator update the column based on the DDL.	You can't update a Service Manager field from RDBMS utilities.
Update a column	**Best practice**: Provide Service Manager table create-alter-drop-rights to the RDBMS, and update an existing field in a database dictionary record. Service Manager updates the column in the RDBMS.	• **Best practice**: Provide Service Manager table create-alter-drop-rights to the RDBMS, and use the Database Dictionary form to update an existing field in a database dictionary record. Service Manager updates the column in the RDBMS. • Enable DDL logging and use the Database Dictionary form to update an existing field in a database dictionary record. Have the RDBMS administrator update the column based on the DDL.	Have the RDBMS administrator update the column directly to the RDBMS. Service Manager will automatically update any corresponding database dictionary records the next time the server restarts and there is a query against the changed table.

Update a key	**Best practice**: Provide Service Manager table create-alter-drop-rights to the RDBMS, and update an existing key in a database dictionary record. Service Manager updates the index in the RDBMS.	• **Best practice**: Provide Service Manager table create-alter-drop-rights to the RDBMS, and use the Database Dictionary form to update an existing key in a database dictionary record. Service Manager updates the index in the RDBMS. • Enable DDL logging and use the Database Dictionary form to update an existing key in a database dictionary record. Have the RDBMS administrator update the index based on the DDL.	You can't update a Service Manager key from RDBMS utilities.
Update an index	**Best practice**: Provide Service Manager table create-alter-drop-rights to the RDBMS, and update an existing key in a database dictionary record. Service Manager updates the index in the RDBMS.	• **Best practice**: Provide Service Manager table create-alter-drop-rights to the RDBMS, and use the Database Dictionary form to update an existing key in a database dictionary record. Service Manager updates the index in the RDBMS. • Enable DDL logging and use the Database Dictionary form to update an existing key in a database dictionary record. Have the RDBMS administrator update the index based on the DDL.	Have the RDBMS administrator update an index directly to the RDBMS, and then manage it from RDBMS utilities.

Related topics

Automatic update of database dictionary records
Automatic update of RDBMS data when database dictionary records change

Database dictionary
Database dictionary records
Making database dictionary changes
Add a field to an existing database dictionary record
Import RDBMS columns into an existing database dictionary record
Update an existing database dictionary record

7.10.1.11.1. Automatic update of database dictionary records

Service Manager can detect changes you make to *existing* columns and data types in your back-end RDBMS and then automatically update the database dictionary to match these changes after a system restart. This mechanism allows the database dictionary to better match the actual columns within the RDBMS. To trigger the automatic update of database dictionary records, you must first restart Service Manager and then make a query to the changed table in the RDBMS. Since several queries are made as part of the log in process, simply logging in may be sufficient to trigger an automatic update. For tables not queried during login, you will need to directly query the changed table in some fashion. You can either search from a form that queries the table or create a query from the Database Manager.

For example, suppose an RDBMS administrator changes the length of the "NUMBER" column in the "PROBSUMMARYM1" table from VARCHAR2(90) to VARCHAR2(100). To pick up the change, the administrator restarts the Service Manager server, logs in, and opens the incident queue form to trigger a query against the probsummary table. This process causes Service Manager to compare the database dictionary record to the RDBMS column definition, recognize the change, and update the database dictionary record to match. Service Manager writes all such changes to the log file. For example:

```
5120( 5872) 02/05/2009 10:57:00 RTE I Change of SQL data type for field 'number' from 'VARCHAR2(90)'
to 'VARCHAR2(100)' detected has been saved to DBDICT for file 'probsummary'
```

Note: Service Manager can only change existing entries in database dictionary records, which are stored in the dbdict table. If your RDBMS administrator adds new columns intended for Service Manager to the RDBMS, the Service Manager administrator will have to manually add entries to the database dictionary to reflect those changes. If your RDBMS administrator removes columns managed by Service Manager, then Service Manager updates the database dictionary to map the file to a null table.

Note: In the above example, the "number" field is the unique key of the "probsummary" database dictionary, which maps to multiple tables in the out-of-box system. In an Oracle back-end RDBMS for example, the "probsummary" database dictionary maps to the "probsummarym1" and "probsummarya1" RDBMS tables. If you change the SQL data type of a unique key that's shared across multiple tables, be sure to make the same change in all associated tables. In our example above, the RDBMS administrator should also change the "number" column of the "probsummary **a** 1" table from VARCHAR2(90) to VARCHAR2(100). The Service Manager server displays a warning in the log file when it detects a difference between unique key definitions. For example:

```
4784( 4452) 02/05/2009 10:58:34 RTE W getSqldas: SQL data type for column 'NUMBER' in alias table a1
(VARCHAR2(90)) doesn't match SQL data type in main table (VARCHAR2(100))
```

Related topics

Automatic update of RDBMS data when database dictionary records change

7.10.1.11.2. Automatic update of RDBMS data when database dictionary records change

The Service Manager server can push changes you make in the database dictionary to your back-end RDBMS. If you provide the Service Manager server with table create-alter-drop-rights to the RDBMS, it automatically adds or updates any tables, columns, or indexes needed in the RDBMS. If you don't provide the Service Manager server table create-alter-drop-rights to the RDBMS, we recommend you enable exporting your database dictionary changes as DDL. This allows you to make the necessary changes to the database dictionary and then have your RDBMS administrator review and implement the DDL on the RDBMS.

To trigger the automatic update of RDBMS data, you add or update a database dictionary record and save it. When you save the database dictionary record, Service Manager consults the mapping format for your RDBMS to create the corresponding tables, columns, and indexes required by your addition or change. Service Manager then connects to the RDBMS with the

user account you provided in the configuration file and attempts to add or update the database. If the add or update is successful, you will see the updated mapping for the table, column, or index after you restart Service Manager. If the add or update fails, the Service Manager server changes the database dictionary mappings to map to null tables and writes an error message in the Service Manager log file.

For example, suppose the Service Manager administrator creates a new database dictionary called `testtable`. The testtable database dictionary consists of two fields: a number field called `test.id` and a character field called `test.name`. The administrator makes the test.id field the unique key. After saving the new database dictionary record, the administrator views the database dictionary record to see how they are mapped to tables and columns in the RDBMS. On a Microsoft SQL Server, the testtable database dictionary might map to the following tables and columns.

Sample database dictionary for the test.id number field

Database dictionary object	Value if RDBMS table create-alter-drop-rights granted	Value if RDBMS table create-alter-drop-rights denied
Database dictionary field name	test.id	test.id
Database dictionary data type	number	number
SQL column name	TEST_ID	TEST_ID
SQL data type	FLOAT	FLOAT
SQL alias	m1	N-m1
SQL table name	TESTTABLEM1	NULLTABLE

Sample database dictionary for the test.name character field

Database dictionary object	Value if RDBMS table create-alter-drop-rights granted	Value if RDBMS table create-alter-drop-rights denied
Database dictionary field name	test.name	test.name
Database dictionary data type	character	character
SQL column name	TEST_NAME	TEST_NAME
SQL data type	VARCHAR(60)	VARCHAR(60)
SQL alias	m1	N-m1
SQL table name	TESTTABLEM1	NULLTABLE

When Service Manager has table create-alter-drop-rights to the RDBMS, the database dictionary record contains mapping information to the actual tables and column names. When Service Manager doesn't have table create-alter-drop-rights to the RDBMS, the database dictionary record contains mapping information to the proposed table and column names. The RDBMS administrator can use Service Manager's generated DDL to create the proposed table and column names. Until the actual tables and columns exist, the database dictionary record points to a null table mapping. All null table aliases start with an "N-" prefix, and the SQL table name maps to the value NULLTABLE. After the RDBMS administrator has created the tables and columns for the database dictionary record, the Service Manager administrator can use the **Activate Nulltable Rows** option to update the SQL mappings to their new values. Service Manager will use the column names listed in the database dictionary record to match it to the proper table name in the RDBMS.

Related topics

Automatic update of database dictionary records
Database dictionary
Database dictionary records

Create an active database dictionary record
Create an inactive database dictionary record
Import RDBMS columns into an existing database dictionary record

7.10.1.11.3. Add a field to an existing database dictionary record

Applies to User roles: System Administrator

This procedure assumes the Service Manager system has the right to create, alter, and drop tables in the RDBMS system. If the system doesn't have such rights, your database dictionary record remains inactive and mapped to null tables until you activate it. If your system doesn't have such rights, it's advisable that you enable the option to create Database Definition Language (DDL) to offer rights to your RDBMS administrator. You can then activate the database dictionary record after your RDBMS administrator has created the tables and columns specified in the DDL.

Note: If you choose to add a new column directly from the RDBMS, Service Manager will not automatically recognize the new column. You must import the new column into the database dictionary record before Service Manager can manage it.

To add a field to an existing database dictionary record:

1. Sign in to the Service Manager system with a system administrator account.
2. Click **System Definition** > **Tables**.
3. Double-click the table name you want to edit.
   Note: Each table name is actually a database dictionary record.
4. Click the **Fields and Keys** tab.
5. In the Fields and keys definitions for the table, do one of the following:
   • Click **New Field** to create a scalar field
   • Click **New Array** to create an array or an array of structure
6. Type a name for the new field.
7. Click **OK**. The field name appears in the list of fields.
8. Select a value from the Data Type field.
   Note: Each field must have a data type.
9. Add any optional field properties, such as data policies or editing rules.
10. Click **Save**. Service Manager creates a new column in the RDBMS.

Related topics

Methods for deleting database dictionary records

7.10.1.11.4. Import RDBMS columns into an existing database dictionary record

Applies to User Roles: System Administrator

This procedure assumes the SM system doesn't have the right to create, alter, or drop tables in the RDBMS system. Meanwhile, you haven't already created a database dictionary record for this change. If your system has such rights in the RDBMS system, SM automatically creates necessary tables, columns, and indexes when you create a database dictionary record. If your SM system doesn't have such rights, it's advisable to:

1. Create the database dictionary record first
2. Offer Database Definition Language (DDL) to the RDBMS administrator to implement the necessary changes

The Service Manager administrator can then activate the database record when the RDBMS tables and columns become available. This procedure provides an alternative means for you to keep your database dictionary records synchronized with an RDBMS.

In order for Service Manager to successfully import columns into a database dictionary record, the RDBMS tables must have the following features:

- The table names must be in uppercase
- The table must have an owner who is a Service Manager user

To import RDBMS columns into an existing database dictionary record:

1. Sign in to the Service Manager system with a system administrator account.
2. Click **Tailoring** > **Database Dictionary**.

3. Type the existing database dictionary record name in the **Filename** field, and then click **Search**.
   The database dictionary record opens.
4. Click **Import new columns from SQL** to have the Service Manager server sign in to the RDBMS and check for new columns in the SQL tables defined for the database dictionary record.
   Service Manager automatically displays new columns as fields in the database dictionary record.

   Caution
   Caution Service Manager only supports importing new columns from main tables.

5. Click **OK** to save the database dictionary record.

Related topics

Automatic update of database dictionary records
Database dictionary records
Create an active database dictionary record
Create an inactive database dictionary record
Enable DDL logging of database dictionary changes

7.10.1.11.5. Modify field types character (scalar) to array

Applies to User roles: System Administrator

Example: Change a field with existing data, called address, from a character type scalar field to an array. For this example, assume that we created a new file called vendorview from the format vendor.

To modify field types, character (scalar) to array:

1. Select the **address** field name. Only one entry for **address** should appear, as it is currently a scalar field.
2. Click **Edit**.
3. In the **Name** field, type `old.address`.
4. Leave the **Type** field value as *character*.
5. Click **OK** to close the window and commit the change to the file.

The address field name is now old.address in the Field Name array.

1. Select the **descriptor** structure, and then click **New**.
2. In the **Name** field, type `address`.
3. In the **Type** field, type `array`.
4. Click **Add** to commit the new array to the file.

When adding an array field, you also need to specify the type of array (for example, character, number, date/time, etc.). Service Manager prompts you for this information now.
A new **field.window** opens with **address** displayed in the **Name** field and the **Type** field blank. The message *Enter data type of array's element* also appears at the bottom of the format window.

1. Select the **Type** field and type: `character`.
2. Click **Add**.

Notice two entries for address now exist in the database dictionary record since it has been defined as a character array.

1. Click **OK** to close the window and save the changes.
2. Click **OK** to end this session with the vendorview file.

The field type has been changed, but this is not the end of the process.

Because data exists in the file, you need to move the contents of the field now called **old.address** to the field called **address**. To accomplish this, perform a **Mass Update** in Database Manager on all records in the file that have data in the old.address field.

Once the data has been moved, we recommend that you also remove the **old.address** field from the database dictionary record.

The Mass Update instruction string to move the data from the field called **old.address** to the first element in the newly created array called **address** is:

```
1 in address in $file = old.address in $file
```

Related topics

Methods for adding database dictionary records
Methods for updating database dictionary records
Methods for deleting database dictionary records
Database dictionary data types
Structures
Alias fields
Adding a key
Modifying a key
Deleting a key
Access the Database Dictionary utility
Add fields to a structure
Modify field types: Single to concatenated
Open a database dictionary
Regen a database dictionary
Search for a field in the database dictionary

7.10.1.11.6. Modify field types: single to concatenated

Applies to User Roles: System Administrator

Example Modify the single key *wdManagerName* in a copy of the **assignment** database dictionary record to make it a concatenated key comprised of the fields *wdManagerName* and *company*.

To modify field types: single to concatenated:

1. Open the assignment dbdict.
2. Make a copy of the assignment dbdict named *assignment2*.
3. Position the cursor on the key type Nulls & Duplicates, above the key model.
4. Click **Edit**.
5. Add company in the second element of the field names array below *wdManagerName*.
6. Click **OK** to close the window.
7. You can now re-edit your database dictionary record, update it, or cancel without updating.
8. Click **OK**, to close and automatically update the file. This returns the message Record updated in the dbdict file.
9. When an IR key is added or updated, then Service Manager prompts you to regenerate the file. You can either click **OK** (regen file) to continue with the regen, click **Cancel** (F3) to abort the regen and restore the database dictionary record to the version prior to the latest changes, or schedule a time for the file regeneration, using Schedule (clock button).

Related topics

Database dictionary data types
Adding a key
Alias fields
Changing data types
Structures
Database dictionary
Access the Database Dictionary utility
Change data types: Character to number
Copy a database dictionary only
Modify field types: Character (scalar) to array

Open a database dictionary
Regen a database dictionary

7.10.1.11.7. Regenerate a database dictionary

Applies to User roles: System Administrator

You can regenerate an IR index in the database dictionary by using the Database Dictionary utility.

Or, you can regenerate all IR indexes by using the `sm -upgradeir` command.

Regenerate an IR index using Database Dictionary

You can regenerate IR for a table by using the Database Dictionary utility:

1. Sign in to the Service Manager Windows or web client.
2. Click **Tailoring** > **Database Dictionary**.
3. In the Table field, type the table name (for example, **incidents**), and click **Search**.
   The table details form opens.
4. From the table details form, click **More** or the More Actions icon, and select **Regen IR**.
5. Click **OK** to run the IR regeneration immediately.

Regenerate IR indexes using the `sm -upgradeir` command

1. Shut down your Service Manager system. For details, see Shut down Service Manager.
2. Run the `sm -upgradeir` command to regenerate IR indexes.
3. Restart your Service Manager system as required. For details, see Start Service Manager.

Related topics

Regenerate database keys
Regenerate IR keys

7.10.1.11.8. Search for a field in a database dictionary

Applies to User Roles: System Administrator

The ability to search the dictionary for a specific field name is particularly useful when dealing with a large database dictionary.

You can search for a field in the database dictionary by using System Definition or by using the Database Dictionary utility. Choose one of the following methods to search for fields.

Search for fields by using System Definition

To search for a field by using System Definition:

1. From the System Navigator, click **System Definition** > **Tables** > ***file_name*** > **Fields**
   Note: You can type the first letter of a table name to jump to that alphabetic section. For example, type `o` to jump to the oncall table. Double-click a table to open it. Double-click the table to open it.
2. Type the first letter(s) of the field you are looking for. The navigator will go to the first field starting with those letters.
   Note: If there is a delay between the letters you type, the Navigator will assume that you want a different search and jump out of the folder to find the matching letters.

Search for fields by using the Database Dictionary utility

Using the Database Dictionary utility when searching for fields enables you do a “contains” search.

To search for a field by using the Database Dictionary utility:

1. Click **Tailoring** > **Database Dictionary**, and then open the database dictionary file.
2. From within the database dictionary record you want to search, open the More Actions menu and click **Field Search**.
   This brings up the Field Search window prompt.
3. Type the field name to search, for example, **one**.
   - Click **Exact match** if you are sure that what you entered is a field in the dictionary record.
   - Click **Starts with** to find all fields starting with the string entered.

- Click **Contains** to search for fields containing the string entered anywhere in the field name. (Used in this example.)
- Click **End** to cancel the search.

4. If Service Manager finds a match, it responds by giving a message similar to the following: **Field "phone" found: type=character; level=3; index=2*.
   The message gives the field type, the level (how many structures down the field where it is located), and the index (location within its parent structure).
   For example, **one** appears in:
   - A character type field (phone)
   - In the third level structure (alternate.contact)
   - As the second field in the alternate.contact structure.
5. If multiple fields are found by the search, multiple messages are returned, one message for each field found. The message in the above example displays an asterisk (*), indicating it does have additional messages to display.
6. View all messages related to this operation.
   In this example, these messages reveal that **one** is present in at least two fields in this database dictionary.
   However, since the **Contains** button was used, the letters are found in their specific order, regardless of where they appear in the field name.
   For an exact match on the word **one**, you can use the Exact match or Starts with button. (Using these options against this example returns no positive hits using Exact match or Starts with.)

Related topics

Add a field to an existing database dictionary record
Modify field types: Single to concatenated
Modify field types: Character (scalar) to array

7.10.1.11.9. Update an existing database dictionary record

Applies to User Roles: System Administrator

This procedure assumes the Service Manager system has the right to create, alter, and drop tables in the RDBMS system. If Service Manager doesn't have such right, your database dictionary record changes remain inactive and mapped to null tables until you activate them. In this case, it's advisable that you enable the option to create Database Definition Language (DDL) for your RDBMS administrator. You can then activate the database dictionary record changes after your RDBMS administrator has created the tables and columns specified in the DDL.

To update an existing database dictionary record:

1. Sign in to the Service Manager system with a system administrator account.
2. Click **System Definition** > **Tables**.
3. Double-click the table name you want to edit.
   Note: Each table name is actually a database dictionary record.
4. From the **Table** tab, you can delete/rename/copy the database dictionary record and add triggers.
5. From the **Fields and Keys** tab, you can add, delete, or update fields and keys.
6. Make the changes you want to make, and then click **Save** to update the database dictionary record.

Related topics

- Automatic update of RDBMS data when database dictionary records change
- Database dictionary
- Database dictionary records
- Create a database dictionary record by importing RDBMS columns
- Create an active database dictionary record
- Create an inactive database dictionary record
- Update the data type of a database dictionary field

7.10.1.11.10. Update the data type of a database dictionary field

Applies to User roles: System Administrator

This procedure assumes the Service Manager system has the right to create, alter, and drop table to the RDBMS system. If Service Manager doesn't have such right, your database dictionary record changes remain inactive and mapped to null tables until you activate it. In this case, it's advisable you enable the option to create Database Definition Language (DDL) for your RDBMS administrator. You can then activate the database dictionary record changes after your RDBMS administrator has created the tables and columns specified in the DDL.

To update the data type of a database dictionary field:

1. Sign in to the Service Manager system with a system administrator account.
2. Click **System Definition** > **Tables**.
3. Double-click the table name you want to edit.
   Note: Each table name is actually a database dictionary record.
4. Click the **Fields and Keys** tab.
5. Click the field name you want to edit. The General properties pane opens.
6. Select a new value from the **Data type** field.
   A confirmation window opens.
   Note: You can't use the drop-down list to change a scalar field to an array or structure. Instead, you must add a new array or structure.
7. Click **OK** to verify the data type change.
8. Click **Save** to update the database dictionary record.

Related topics

Database dictionary
Create an active database dictionary record
Import RDBMS columns into an existing database dictionary record

7.10.1.11.11. Rename a database dictionary

Applies to User Roles: System Administrator

After you create a database dictionary, you can do one of the following:

- Change the name of the dictionary.
- Change the name of one or more of the files sent with the base system.

For example, you can rename the *communications* file to *interactions*. This changes the file name but retains all of the field and key definitions in the dictionary.

If records exist in the file, renaming the dictionary associates all of the records with the new dictionary name. This differs from copying a file because it takes only a few seconds to perform this function.

Database dictionary names must be one word. No blanks and no special delimiters are allowed.

You can rename a database dictionary by using System Definition or by using the Database Dictionary utility. Choose one of the following methods to rename a database dictionary.

By using System Definition

To rename a database dictionary using System Definition:

1. On the System Navigation, click **System Definition** > **Tables** > ***file_name***.
   Note: You can type the first letter of a table name to jump to that alphabetic section. For example, type `o` to jump to the oncall table. Double-click a table to open it.
2. Click **Rename**.
3. Type the desired file name in the **Rename To** text box.
4. Click **OK**. Service Manager displays the following message: *File renamed from <old name> to <new name>*.

By using the Database Dictionary utility

To rename a database dictionary using the Database Dictionary utility:

1. Open the old dbdict using the Database Dictionary utility.
   Click **Tailoring** > **Database Dictionary**.

2. Open the More Actions menu and click **Copy/Rename**.
3. Type the desired file name in the **New Name** text box.
4. Select the **Rename** option.
5. Click **OK**. The renamed database dictionary opens with the following message in the status bar: *Dbdict successfully renamed*.
6. Click **OK** to exit the record and save the file with the new name.

Related topics

Access the Database Dictionary utility

7.10.1.12. Resetting a database table

In some scenarios, you may need to reset a database table to remove all records. For example, when moving a table from a development environment to a production environment, you may want to start with an empty production table. You may also need to limit the size of files that continually increase in size, such as syslog or msglog.

> **Caution**
> Caution
> Use caution when choosing this option, as resetting removes all records from the table.

Resetting the database and all records

If the records in a dictionary are no longer needed but the file structure is to be retained for later use, perform a file reset.

The **Reset** option deletes all data records from a dictionary. In most instances, once a system is set up and is in operation, this option is not used. Most often it is utilized when clearing test data from a preimplemented system or test file.

> **Caution**
> Caution
> You should only use this option when you intend to permanently remove data file records. Once a reset is performed, the records can't be retrieved.

Reset a database table from Database Manager

Applies to User Roles: System Administrator

In this example we reset the syslog table. Resetting other tables works the same way.

To reset a database table from Database Manager:

1. Open the syslog form in Database Manager.
2. Click **More** or the More Actions icon.
3. Select **Reset**.

Service Manager displays a prompt, asking you to confirm the action and allowing you to schedule the Reset.

1. From the prompt, you can:
    - Click **OK** to reset the syslog table.
    - Click **Cancel** to leave the table intact and return to the blank syslog format.
    - Click **Schedule** to schedule a time to run the file reset operation, either once or repeatedly at a set interval.

Schedule a reset

Applies to User Roles: System Administrator

If you chose to schedule a reset by clicking the Schedule button on the reset prompt you are prompted with a schedule dialog, Service Manager displays a schedule form.

To schedule a reset:

1. Provide a date in a DD/MM/YYYY format in the Date/Time to run field.
2. Provide a time in an HH:MM:SS format the Date/Time to run field.

3. Select a Repeat Interval option to reset the file on a regular basis. This setting is optional. The interval period starts from the time and date set in steps 2 and 3.
    - Monthly – reset once a month.
    - Quarterly – reset every three months.
    - Semi-annually – reset every six months.
    - Annually – reset once a year.
    - Other – reset in the specified number of days, at the specified time.

    Use the dd hh:mm:ss format, where dd is the number of days from the initial date and time, and hh:mm:ss is the time of day. For example, 26 10:00:00 schedules the reset to run at 10 a.m., 26 days from the initial reset date and time set in the Date/Time to run field.
4. Click **Schedule** to confirm the reset action.

Related topics

Regenerate database keys

7.10.2. Keys and indexes in Service Manager

Service Manager keys are abstract entities that provide a logical view of the indexes in your RDBMS. When you create Service Manager logical keys, the server creates corresponding indexes on the back-end RDBMS. The benefits of using Service Manager logical keys are:

- They allow administrators a means to manage and move indexes from one environment to another (for example, from a test environment to a production environment)
- They allow administrators a means to move indexes from one RDBMS type to another (for example, from a SQL Server Express edition demonstration RDBMS to a supported RDBMS in a development environment)

In RDBMS terms, Service Manager keys provide constraints on column data as well as an index of the records in a table that improves query performance. From the Service Manager side, logical key definitions are one part of a database dictionary record. The full database dictionary record also includes field and column mappings. Typically, you will use database dictionary records to manage and move both logical keys and field mappings at the same time.

Currently, Service Manager can only push keys as indexes on an RDBMS. It can't read existing indexes (pull) from the RDBMS and create the corresponding logical keys. Service Manager can still use existing indexes on an RDBMS without the indexes being defined as logical keys. Any query issued against an indexed column still gets a performance benefit. However you can only take advantage of the features of logical keys if you create them from Service Manager.

Service Manager uses the following logic to create indexes:

Service Manager key type	Type of index created on the RDBMS
Primary	Primary
Unique	Index with unique constraint
No Nulls	Index
No Duplicates	Index
Nulls and Duplicates	Index
IR index	None Service Manager directly manages IR Index keys

You can define Service Manager keys from either the System Definition utility or the Table Definition utility.

Related topics

Adding a key
Methods for adding database dictionary records

Add a key as the first key
Insert a key between other keys
Key type attributes
Key type definitions

7.10.2.1. Key type definitions

Key type definitions allow you to specify the type of key and the names of the fields within the key. There may be more than one field in one key. A key with multiple fields is called a concatenated key.

Key Type	Description
Primary	No fields in the key can be null. The value of the complete key can't be null and must be unique in the index. A table can have at most one primary key.
Unique	At least one field in the key must not be null. The value of the complete key must be unique in the index.
No Nulls	At least one field in the key must not be null.
No Duplicates	The value of the complete key must be unique in the index or the values of all fields must be null. **Note:** Postgres 15 adds the clause `NULLS NOT DISTINCT`. See PostgreSQL 15 release notes If your SM server version is 9.80 and your PostgreSQL version is 15 or above, `NULLS NOT DISTINCT` is supported on No Duplicates key, which means you are not allowed to have multiple records with a null entry in this column/field. If your SM server version is 9.80p1 or above, `NULLS NOT DISTINCT` is only supported on No Duplicates key when the key contains multiple columns. For No Duplicates key with single column, there's no `NULLS NOT DISTINCT` constraints. If you have upgraded the server to 9.80 or 9.80p1 and later versions, and want to apply this new functionality, you need to recreate the existing indexes.
Nulls & Duplicates	All fields can be null. The complete key value can be in the index more than once.
IR key	The fields in the key are indexed by IR Expert. Only one IR key can be used per dbdict record or IR searches on that file don't work. You can concatenate several fields in an IR key.

Related topics

UTF-8 (Unicode) support

7.10.2.1.1. Key type attributes

The attributes of Unique keys differ by RDBMS type. For example, in an Oracle environment, the constraints are No Duplicates and Null. Due to the variety of constraints for each RDBMS, Service Manager only creates Unique key constraints on the RDBMS that allows Nulls.

You can choose to modify the Unique constraints to primary keys in your RDBMS. If you choose to do so, this will allow No Nulls and No Duplicates if the key is to contain those values.

Key type attributes:

Key Type	No Duplicates	Allows Duplicates
No Nulls	Unique Primary (If primary key mode is enabled)	No Nulls
Allows Nulls	No Duplicates	Nulls & Duplicates

Note: IR keys are indexed separately and aren't a part of the key types listed in this table.

Related topics

UTF-8 (Unicode) support
Service Manager Programming
Key type definitions

7.10.2.1.2. Tuning: Designing keys for queries

Fully keyed queries typically offer the best performance. You can design keys to ensure that every query is fully keyed, although it may not be practical in all instances. Running all queries fully keyed requires defining a large number of keys, which in turn can cause performance degradation when using the add, update, and delete operations. In general, the Service Manager System Administrator should discuss query performance with the Database Administrator and tune the system accordingly.

Points to consider when designing keys:

- Design keys for the most used queries.
- You can optionally force users to issue fully keyed queries by not allowing partially keyed and non-keyed queries.
- Specify the fields that are most commonly used at the beginning of the key in the query.
- Specify fields that have many possible values at the beginning of the key. A key with a Boolean field at the beginning is inefficient, unless the majority of your queries return only a small number of records. For example, flag=true in the probsummary table eliminates 90% of all records.
- Do not use the same field in multiple keys. If you update a record and then change the value of that field, update all of the indexes that contain that key.
- Do not define more than 25 keys for one file. The more keys defined for a file, the more time is required to add, update, and delete records in that file. Likewise, if you define too few keys, the operations run faster but the searches on the individual file run slower.
- There is no specific rule of how many keys you should define for a file. Logical consideration of these facts and your specific work environment are the best decision-making factors.

Related topics

Nonkeyed queries
Partially keyed queries
Stored queries
True queries
Key type definitions
Tuning: Fully keyed queries
Tuning: Improving query speed
Tuning: Key selection algorithms
Tuning: Number of fields in files

7.10.2.1.3. Tuning: Key selection algorithms

A key selection algorithm selects a key to perform a query based on the order of the fields in the query expression, and the order of the fields in the keys defined in the dbdict. The system then assigns a weight to each key based on the order in which the fields appear in the query for the key, and the order in which they appear in the query and key. For example, if the first field in a key matches the first field in the query, Service Manager assigns that key a higher weight than one that has that field as the second field in the key. The system makes the decision for all fields in the query and keys.

Example query and keys:

```
query:  a=1 and b=2 and c=3 and d=4
        key1: b,c,d
        key2: a,c,d
        key3: a,b
        key4: a
```

In this example, the key selection algorithm selects key3. The next highest weight is key2, followed by key4 and key1. The new key selection is based on the location database dictionary record with the keys {location}, {location,state}, {location,city}, {location.name,location}, and {location.code}.

Related topics

Key type definitions
Tuning: Designing keys for queries
Tuning: Improving query speed
Tuning: Number of fields in files

7.10.2.2. Working with primary keys

In versions of Service Manager earlier than 9.32, Service Manager did not support primary keys or `Not Null` constraints in the RDBMS tables. While the Service Manager logical representations did include support for unique keys with a `Not Null` constraint, this support was managed entirely within Service Manager and not in the RDBMS. Because of this lack of functionality, SQL queries generated by Service Manager added an `Is Null` condition in the `JOIN/WHERE` sub-clause as shown in the following example:

```
SELECT m1."RELATIONSHIP_NAME", m1."LOGICAL_NAME"
FROM CIRELATIONSHIPM1 m1 LEFT OUTER JOIN CIRELATIONSHIPA1 a1
ON
( ( (m1."LOGICAL_NAME" = a1."LOGICAL_NAME")
OR (m1."LOGICAL_NAME" IS NULL AND a1."LOGICAL_NAME" IS NULL) )
AND
( (m1."RELATIONSHIP_NAME" = a1."RELATIONSHIP_NAME")
OR (m1."RELATIONSHIP_NAME" IS NULL AND a1."RELATIONSHIP_NAME" IS NULL) ) )
WHERE …
```

This mechanism resulted in poor performance when you consider the speed advantages of leveraging the inherent integrity of the database.

In Service Manager 9.32 or later versions of Service Manager, primary keys and `Not Null` constraints on the unique keys are now supported.

Usage

From command line

The `sm -system_addconstraint` command line is used to add either `NOT NULL` or primary key constraint to the existing files in SM.

Note

Note For details on how to use the `system_addconstraint` command, see the Modify keys: Convert the first unique key to primary key and the Modify keys: Add Not Null constraints to unique keys topics.

There are three levels of operations you can perform by using this command:

1. Add the NOT NULL constraint only
   While Service Manager already has a `No Nulls` key type, this key type is in fact mapped to an normal index in RDBMS. The `NOT NULL` constraint added by `system_addconstraint` command adds a true constraint in the RDBMS (such as the `NOT NULL` constraint in and Oracle database.
   The `NOT NULL` constraint is invisible to end users; End users can't add, modify or delete it from anywhere in the Service Manager client.
2. Modify the first unique key to primary key
   By using this command line, the Service Manager server can convert the first unique key of the Service Manager files to true primary keys in RDBMS. However, if the data in the table is not qualified for the primary key constraint, the conversion from unique key to primary key will fail.
3. Modify the first unique key to primary key more aggressively
   If you can afford the risk of lost data in the RDBMS, you can use this level of operation. At this level, the `system_addconstraint` converts the first unique key to a primary key more aggressively. That is, a temporary table will be created, all the qualified data will be moved to this new table, and the old table is dropped, and the temporary table is renamed to the original one.
   This process is referred to as a full table copy; unqualified data is lost, and qualified data is left.

From SM client

From the Service Manager client, users can add, modify or remove the primary key type just like any other key type.

Considerations

The following behavior is subjected to the primary key feature:

- When there is data in a table, you can't modify the name of the primary key column.
- If there is a RECORD_KEY column and one unique key column in a table, the RECORD_KEY column is removed, and the first unique key is converted to a primary key when you run the `sm -system_addconstraint` command.
- Service Manager automatically creates a hidden RECORD_KEY field when there is no primary key or unique key in the Database Dictionary file. It is recommended that you explicitly define a primary key when creating a new dbdict record or convert the first unique key to a primary key if you think it necessary for better performance or other reasons.
- Once the database is converted to primary key mode by using the system_addconstraint parameter, the first unique key in the Database Dictionary is changed, and a hidden RECORD_KEY field may or may not have been created accordingly. This behavior could potentially impact any integration you may have implemented in Service Manager. For example, the legacy SCENTER integration and ODBC driver may rely on the primary key field directly. If such integration doesn't work as expected after you apply the primary keys, you should review and modify the integration accordingly.
- The type of a newly-created RECORD_KEY field is now VARCHAR2 or VARCHAR in Service Manager. However, any existing RECORD_KEY fields remain as INTEGER.

Related topics

Retrieve primary key values
Retrieve primary key mode
Modify keys: Add Not Null constraints to the first unique key in a table
Modify keys: Convert the first unique key to primary key

7.10.2.2.1. Retrieve primary key values

Caution

Caution: You can enable primary key mode only after you have upgraded to Service Manager 9.32 applications or later. Otherwise, the following procedures may have unexpected results.

You can use the following example to derive your own scripts to read the primary key values:

1. Start Service Manager and then log in as an administrator.
2. Navigate to dbdict.
3. Create a `testkey` table with a primary key defined.
4. Create a detailed form based on the `testkey` table.
5. Insert several data points into the `testkey` table.
6. Navigate to ScriptLibrary.
7. Create a new script by copying the code below:

```
var retValue;
var retCode;
var keyValues = new SCDatum();
var file = new SCFile("testkey");
file.doSelect( "true" );
file.getFirst();
retValue = system.functions.rtecall("getprimary", retCode, keyValues, file );
print( "file testkey is ", file );
print( "The return value of getprimary is ", retValue );
print( "The value of primary key is ", keyValues );
```

8. Execute the script. You should see output that resembles the following:

```
The value of primary key is  [C++ object SCDatum] - {'12/12/12 00:00:00', "54343"}
The return value of getprimary is  true
File testkey is  [C++ object SCFile] - testkey={['12/12/12 00:00:00', "fgdf", , "54343"]}
```

7.10.2.2.2. Retrieve primary key mode

You can run a script to verify whether Service Manager has the primary key feature enabled.

Application developers

Application developers can retrieve primary key mode information by using the `sysinfo.get("PKMode")` Rapid Application Development (RAD) function. To do this, use the following example:

1. Sign in to Service Manager as an administrator.
2. Navigate to **Script Library**.
3. Create a new script named `test`.
4. Copy the following code into the text box.

```
var retValue;
retValue = system.functions.sysinfo_get ("PKMode" );
print ( "Current value of PKMode is " + retValue );
```

5. Run the script.

If you have enabled the primary key feature, the following message appears in the Messages pane of the SM client:

`Current value of PKMode is true`

Client Developers

Client developers can verify whether they have enabled the primary key feature. To do this, use the following example:

1. Sign in to Service Manager as an administrator.
2. Navigate **System Navigator** > **System Definition** > **Tables**.
3. Double-click any table, and then click the **Fields and Keys** tab.

4. Click any key in the Keys list.
5. Click the **Type** drop-down list in the General section.

If you have enabled the primary key feature, you should see **Primary Key** as an option in the **Type** drop-down list.

7.10.2.3. Adding a key

A key is an identifying field in a file, used when queries are processed so all fields in a record don't need to be searched for set criteria. A field or fields in a key must be defined as a field or fields in the database dictionary record. A key may be added in a production system without a system restart.

When a database dictionary file is created through **Forms Designer**, Service Manager automatically uses the first field defined on the associated format to create a *Unique* key. Manually modify the database dictionary record to add the necessary keys to support on-line and reporting queries, as well as sort sequences. Many other default system files, set up in the base system, may need to be changed, based on your individual search and reporting requirements. This section details the manual addition and manipulation of database dictionary record/file keys.

Note: If records already exist in the file when you add a key, updating the database dictionary record will cause Service Manager to automatically perform an index regeneration.

If you add keys to a file that contains records, be careful when defining key types. For example, don't define a new key as **No Nulls** (no blank values) if records exist in the file that contain no data in the particular field(s) comprising the key.

Each record not containing a value in the **key** field would require an update, populating the field with data. The updates must be performed prior to adding the No Nulls key in order to satisfy the new key definition.

A CREATE INDEX statement is generated when a new key is defined in the dbdict.

Note: If Service Manager determines it doesn't have rights to CREATE a TABLE it writes all database definition language (DDL) into files in the `<RUN directory>/ddl` folder and accepts changes to the dbdict without performing the CREATE INDEX operation.

Related topics

Modifying a key
Deleting a key
Structures
Database dictionary data types
Alias fields
Methods for updating database dictionary records
Methods for deleting database dictionary records
Delete a key
Add a key as the first key
Insert a key between other keys
Add keys to bottom of the key list
Modify keys: Nulls and duplicates to unique

7.10.2.3.1. Add a key as the first key

Applies to User Roles: System Administrator

Example: Add the language.key field as a No Nulls key to the **unitofmeasure** database dictionary record.

In this example, you add the language.key field to the **unitofmeasure** record and assign it as the first key with a property of No Nulls. This field is a required field in the record.

To add a key as the first key:

1. Sign in to the Windows client.
2. From the System Navigator, click **System Definition** > **Tables** > **unitofmeasure** > **Keys**.
3. In the Keys section, click **New**.
4. In the General section, click the **Type** field. For this example, select **No nulls** in the Type list as the key type.
   The following key types are valid:
   • **Nulls and duplicates** - allows nulls and duplicates in all the fields in the key.
   • **No nulls** - allows duplicates, if selecting No nulls, the key must contain data in at least one of the fields in the key.

• **No duplicates** - allows null records but not duplicates. Although the data must be unique, but the field may be blank.
• **Unique** - must contain unique data and can't be blank.
• **IR key** - files that contain an IR key must also contain another key that's Unique, with No nulls, No duplicates or Nulls, and Duplicates. A Service Manager file can contain only one IR key.

5. Click **Add**.
6. From the **Choose a field to add to the key dialog**, select the **language.key** field, and then click **OK**.
7. In the Keys section, select the key you added (language.key in this example), and click **Up**.
   The key **language.key** now appears as the first key in the list of keys in the database dictionary record. All other keys are moved down the screen.
8. Once you are finished adding keys to the table, click **Save** and then click **OK** to confirm your changes.
   Service Manager automatically updates and regenerates the file.
9. To remove the key, select the key and then click **Delete**. Click **Save**.

Related topics

- Adding a key
- Alias fields
- Database dictionary data types
- Methods for deleting database dictionary records
- Deleting a key
- Methods for updating database dictionary records
- Modifying a key
- Structures
- Database dictionary
- Access the Database Dictionary utility
- Add keys to bottom of the key list
- Delete a key
- Insert a key between other keys
- Modify keys: Nulls and duplicates to unique
- Open a database dictionary
- Regenerate a database dictionary

7.10.2.3.2. Add keys to the bottom of a key list

Applies to User Roles: System Administrator

You can add keys to the bottom of the key list by using System Definition or the Database Dictionary utility. Choose one of the following methods to add keys.

Note: You must use the Windows client whenever you need to add a new field/key to a database dictionary table.

Add keys to the bottom of the key list by using System Definition

To add keys to the bottom of the key list by using the System Definition:

1. In the System Navigator, click **System Definition** > **Tables** > file_name > **Database Definition** > **Keys**.
   Note: To expand any table in the record list, type the first letter of a table name to jump to that alphabetic section. For example, type `o` to jump to the oncall table.
2. Right-click **New** > **Key**.
3. Select the key type, and then click **OK**.
   Your new key appears in the key list without a (+) sign to the left, because the key is still empty.
4. Drag the fields you want for that key from the Fields node and drop them on the key.
5. Right-click **Save** on the **Database Definition** node to save your changes.

Add keys to bottom of the key list by using the Database Dictionary utility

Example: Add a concatenated key including the fields **sysmodtime** and **sysmoduser** to the bottom of the key list.

To add keys to the bottom of the key list by using the Database Dictionary utility:

1. Open the file in the database dictionary, and then scroll down the keys until you reach the bottom of the list.
   The last key in the **unitofmeasure** database dictionary record is **language.key**.
   This is a single key, because it contains only one field. If the key is a concatenated key, regardless of how many fields are included in the key, only two fields are displayed in the key list.
   To view all fields contained in a concatenated key, you must use the scroll arrow buttons to the right of the array, or edit the key.
   The blank header represents the beginning of the next key. The number of lines displayed per key depends on the number of fields concatenated in that key.
   Position the cursor on the blank header field located below the key type, and then click **New Field/Key** to display the key definition window.
2. Type **Nulls & Duplicates** in the **Type** field, or select **Nulls & Duplicates** from the Type list.
3. Select the fields array, and then do the following:
   • Type **sysmodtime** in the first field of the array.
   • Type **sysmoduser** in the second field of the fields array.
   • Click **Add** to commit this new key to the database dictionary file. Service Manager displays the message: Key added to the **unitofmeasure** file . The concatenated key comprised of **sysmodtime** and **sysmoduser** now appears in the bottom of the list of keys in the **unitofmeasure** database dictionary record.
   • Place the cursor in the applicable position in the keys section of the database dictionary record, and then click **New** to add additional keys to the database dictionary record without closing the window.
4. You can reedit your database dictionary record, update it, or cancel without updating.
   Note: You must update your database dictionary record when adding a new key, or the database dictionary record will remain the same.
5. Press **OK** once you are finished adding keys to the database dictionary record, to close and automatically update the file.
   This returns the message Record updated in the dbdict file.
6. You can continue by doing one of the following:
   • Click **OK** (regen file) to continue with the regen.
   • Click **Cancel** (F3) to abort the regen and restore the database dictionary record to the version prior to the latest changes.
   • Schedule a time for the file regeneration, using **Schedule** (clock button).
7. To remove the key, select it and click **Edit**.
   The edit window appears.
8. Ignore the edit window and instead click **Delete** on the main record format.
9. The next viable key is displayed. If you don't modify this key, click **OK**.
   You are returned to the database dictionary record, minus the key.
10. Click **OK** to save the record.

Related topics

Adding a key
Alias fields
Deleting a key
Methods for deleting database dictionary records
Methods for updating database dictionary records
Modifying a key
Structures
Access the Database Dictionary utility
Add a key as the first key
Insert a key between other keys
Open a database dictionary
Regen a database dictionary

7.10.2.3.3. Insert a key between other keys

Applies to User Roles: System Administrator

Example: Insert the key company between the keys name and operators in a copy of the assignment file.

To insert a key between other key:

1. Open the assignment dbdict.
2. Make a copy of the assignment dbdict named assignment1.
3. Click on the key type of the second key, Nulls & Duplicates in this example, above the **operators** key.
4. Click **New**.
5. In the new key window, select a valid key in the **Type** field by either typing in the value or selecting it from the drop down menu under the arrow button. Use **No Nulls** for this example.
6. Enter **company** in the field names array.
7. Click **Add** to add this key to the assignment2 database dictionary record.

The key **company** now appears after the *first* key, in the second place in the list, and the **operators** key has moved down the list of keys in the assignment2 database dictionary record. All other keys are moved down the screen. You can scroll the keys array to view the other keys in the list.

1. You can now reedit your database dictionary record, update it, or cancel without updating.

You must update your database dictionary record when adding a new key, or the database dictionary record will remain the same.

1. Press **OK** once you are finished adding keys to the database dictionary record, to close and automatically update the file. This returns the message Record updated in the dbdict file.
2. You can either click **OK** (regen file) to continue with the regen, click **Cancel** (F3) to abort the regen and restore the database dictionary record to the version prior to the latest changes, or schedule a time for the file regeneration, using **Schedule** (clock button).
3. To remove the key, select it and click **Edit**.
4. The edit window appears, but ignore the window, and instead click **Delete** on the main record format.
5. The next viable key is displayed, if not modifying this key click **OK**.

You are returned to the database dictionary record minus the key.

1. Click **OK** to save the record.

Related topics

Adding a key
Modifying a key
Deleting a key
Structures
Alias fields
Methods for updating database dictionary records
Methods for deleting database dictionary records
Open a database dictionary
Copy a database dictionary and data records
Add a key as the first key
Add keys to bottom of the key list
Modify keys: Nulls and duplicates to unique
Delete a key
Access the Database Dictionary utility
Regen a database dictionary

7.10.2.4. Deleting a key

You can delete keys from a database dictionary record. Remember, there is a direct correlation between the number of keys defined in a database dictionary record and the length of time it takes to add a record to that file.

A DROP INDEX statement is generated if a key is removed from the dbdict.

Note: If Service Manager determines it doesn't have rights to CREATE a TABLE it writes all database definition language (DDL) into files in the `<RUN directory>/ddl` folder and accepts changes to the dbdict without performing the DROP INDEX operation.

Related topics

Adding a key
Modifying a key
Structures
Alias fields
Methods for updating database dictionary records
Methods for deleting database dictionary records
Add a key as the first key
Insert a key between other keys
Add keys to bottom of the key list
Modify keys: Nulls and duplicates to unique
Delete a key

7.10.2.4.1. Delete a key

Applies to User Roles: System Administrator

You can delete keys from System Definition and from the Database Dictionary utility. Choose one of the following methods to delete a key.

Delete a key by using System Definition

To delete a key by using System Definition:

1. From the System Navigator, click **System Definition** > **Tables** > ***file_name*** > **Fields**
   Note: You can type the first letter of a table name to jump to that alphabetic section. For example, type `o` to jump to the oncall table.
   An overview of the table opens.
2. Select the Fields and Keys tab.
3. In the Keys section, click on a key.
   Caution: When you select a key for deletion, you don't receive a confirmation message. The key is automatically deleted.

Delete a key by using the Database Dictionary utility

Example: Delete the Nulls & Duplicates key for **operators** in the **assignment1** database dictionary record.

To delete a key by using the Database Dictionary utility:

1. Click **Tailoring** > **Database Dictionary**.
2. Open the `assignment` file, and then make a copy of the assignment dbdict named assignment1.
3. Open the assignment1 dbdict record.
4. On the Keys tab, place the cursor in the Nulls & Duplicates section under which the **operators** key is located, and then click **Edit Field/Key**. The key window opens.
5. Click **Delete** from the tool tray.
6. Click **Yes** to confirm the delete.
   The screen redisplays with the key **operators** removed from the list of keys, and the keys below it moved up to take its former place.
7. Click **OK** to close and automatically update the file.
   You receive a message that states the record has been updated in the dbdict file.

Related topics

Deleting a key
Adding a key
Modifying a key
Structures

Alias fields
Adding a key
Methods for updating database dictionary records
Methods for deleting database dictionary records
Open a database dictionary
Copy a database dictionary and data records
Add a key as the first key
Insert a key between other keys
Add keys to bottom of the key list
Modify keys: Nulls and duplicates to unique
Access the Database Dictionary utility
Regen a database dictionary

7.10.2.5. Modifying a key

As new users are added to Service Manager and additional requirements are specified, you may find the need to modify keys in your database dictionary records in order to optimize performance.

Note: As with adding new keys, if records already exist in the file when you modify a key, Service Manager automatically performs a regeneration of file data when you update the database dictionary record.

A modification to a key may be a change in the field(s) comprising the key or a change in the key type definition.

Note: When you modify keys in a file that contains records, be careful when defining key types.

For example, don't change a key from a **Nulls & Duplicates** type to **No Nulls** if the field(s) comprising the key is **all null** (contains no value). In this case, each record that did not contain a value in the field(s) comprising the key would require an update, putting data in the field before the key could be changed to No Nulls.

A CREATE INDEX statement is generated when an existing key is changed. For example, if the key type is changed or fields are added to the key.

A DROP INDEX statement is generated when an existing key is changed. For example, if fields are removed from the key.

Note: If Service Manager determines it doesn't have rights to CREATE a TABLE, it assumes it also doesn't have the rights to perform any of the above operations either. In which case it writes all database definition language (DDL) into files in the `<RUN directory>/ddl` folder and accepts changes to the dbdict without performing the CREATE INDEX or DROP INDEX operations.

Related topics

Adding a key
Deleting a key
Structures
Alias fields
Methods for updating database dictionary records
Methods for deleting database dictionary records
SQL cache
Add a key as the first key
Insert a key between other keys
Add keys to bottom of the key list
Modify keys: Nulls and duplicates to unique
Delete a key
Create an active database dictionary record

7.10.2.5.1. Modify keys: Nulls and Duplicates to Unique

Applies to User Roles: System Administrator

You can modify keys by using System Definition or by using the Database Dictionary utility. Choose one of the following methods to modify keys.

Modify keys by using System Definition

To modify keys by using System Definition:

1. From the System Navigation, click **System Definition** > **Tables** > **Database Definition** > **Keys** > ***key name***
2. Right click **Edit**.
3. Make your changes.
4. Click **OK**.
5. Right click **Save** on the **Database Definition** node to save your changes.

Modify keys by using the Database Dictionary utility

Example: Change the key type for operators in a copy of the assignment file from Nulls & Duplicates to Unique. If data already exists in the file, ensure that a unique value also exists in the operator field for every record in the file. If this condition is not met, you will then have to update each record individually or perform a Mass Update.

To modify keys by using the Database Dictionary utility:

1. Open the assignment dbdict.
2. Make a copy of the assignment dbdict named assignment2.
3. Position the cursor on the key type Nulls & Duplicates above the key field **operators**.
4. Click **Edit**.
5. In the **Type** field, type: `unique`.
6. Click **OK** to close the window and commit the changes to the file.
7. You can now reedit your database dictionary record, update it, or cancel without updating.
8. Click **OK** to close and automatically update the file.

This returns the message *Record updated in the dbdict file*.

Related topics

Modifying a key
Adding a key
Deleting a key
Structures
Alias fields
Methods for updating database dictionary records
Methods for deleting database dictionary records
Open a database dictionary
Copy a database dictionary only
Add a key as the first key
Insert a key between other keys
Add keys to bottom of the key list
Delete a key
Open a database dictionary
Access the Database Dictionary utility
Regen a database dictionary

7.10.2.5.2. Modify keys: Convert the first unique key to a primary key

You can run the `sm` command together with `system_addconstraint` parameter to convert the first unique key of a table to a primary key. When you run this command, Service Manager first modifies the fields of the first unique keys to be `NOT NULL` (as if you ran `system_addconstraint:0` as described in Modify keys: Add Not Null constraints to the first unique key in a table) and then applies a primary constraint to those keys.

To convert the first unique key to a primary key by using the command line, use one of the following methods:

Note

Note: Make sure that Service Manager is offline before you execute this command.

Method 1

Run the following `system_addconstraint` command with a value of `1`. You can use this method with an Oracle database. You can't use this with a SQL Server database.

```
sm -system_addconstraint:<target>:1
```

In this example, *<target>* represents the name of the Service Manager table on which you wish to add the constraint, and `1` indicates that you wish to convert the first unique key of the *<target>* to a primary key.

Method 2

Caution

Caution If Service Manager fails to perform the conversion by using the command in Method 1, or if you have a Microsoft SQL Server database, use the following method. However, you should note the following dangers when using this method:

- Data loss will occur if you run this method on a unique index that contains null values.
- This method invokes a full-table-copy operation, which may take a long time.
- Any database exception error that occurs during the operation can cause unexpected results.

To use this method, follow these steps:

1. Make sure that you have no null values on any unique index.
2. Execute the following command: `sm -system_addconstraint:<target>:2`

Note

Note Regardless of which method you use, an attempt to revert from a primary key to a unique key may cause unpredictable results.

7.10.2.5.3. Modify keys: Add Not Null constraints to unique keys

To add a `Not Null` constraint on a key by using the command line, enter the following command from the operating system command prompt:

Note

Note

- Make sure that Service Manager is offline before you execute this command. Otherwise, the "Not Null" constraint may be lost after you modify the dbdict. In this case, you would need run the command again.
- Due to differences in how databases handle "Not Null" constraints, you should note the following behaviors:
  For Oracle database: a "Not Null" constraint is added to the unique index.
  For SQL Server database, a "Not Null" attribute is added to each field of the unique index.

```
sm -system_addconstraint:<target>:0
```

In this example, *<target>* represents the tables on which you wish to add the constraint, and 0 represents that you wish to add a `Not Null` constraint on the first unique key of the *<target>*.

After you run this script, you will see information that resembles the following in the `sm.log` file.

```
 11508( 12940) 12/11/2012 11:24:59  RTE I Adding NOT NULL constraint for table 'Alertlog'.
 11508( 12940) 12/11/2012 11:24:59  RTE I Adding NOT NULL constraint for mapped table 'ALERTLOGM1'.
 11508( 12940) 12/11/2012 11:24:59  RTE I Add NOT NULL constraint for field 'ALERT_ID' successfully!
 11508( 12940) 12/11/2012 11:24:59  RTE I Add NOT NULL constraint for mapped table successfully, updated
to DBDICT!
 11508( 12940) 12/11/2012 11:24:59  RTE I Adding NOT NULL constraint for table 'Approval'.
 11508( 12940) 12/11/2012 11:24:59  RTE I Adding NOT NULL constraint for mapped table 'APPROVALM1'.
 11508( 12940) 12/11/2012 11:24:59  RTE I Add NOT NULL constraint for field 'UNIQUE_KEY' successfully!
 11508( 12940) 12/11/2012 11:24:59  RTE I Add NOT NULL constraint for field 'FILE_NAME' successfully!
 11508( 12940) 12/11/2012 11:24:59  RTE I Add NOT NULL constraint for field 'NAME' successfully!
 11508( 12940) 12/11/2012 11:24:59  RTE E Error: SQL code=2293 message=ORA-02293: can't validate
(SM930GA.APPROVALM18C1CC5FB) - check constraint violated
 11508( 12940) 12/11/2012 11:24:59  RTE E API=OCIStmtExecute [in sqociExecOne], Statement=ALTER TABLE
APPROVALM1 ADD CONSTRAINT APPROVALM18C1CC5FB CHECK( "COMPONENT" IS NOT NULL )
 11508( 12940) 12/11/2012 11:24:59  RTE W Adding NOT NULL constraint for field 'COMPONENT' failed, please
change/remove the NULL values in records!
 11508( 12940) 12/11/2012 11:24:59  RTE I Adding NOT NULL constraint for mapped table 'APPROVALA1'.
 11508( 12940) 12/11/2012 11:24:59  RTE I Add NOT NULL constraint for field 'UNIQUE_KEY' successfully!
 11508( 12940) 12/11/2012 11:24:59  RTE I Add NOT NULL constraint for field 'FILE_NAME' successfully!
 11508( 12940) 12/11/2012 11:24:59  RTE I Add NOT NULL constraint for field 'NAME' successfully!
 11508( 12940) 12/11/2012 11:24:59  RTE E Error: SQL code=2293 message=ORA-02293: can't validate
(SM930GA.APPROVALA153D2D5D4) - check constraint violated
 11508( 12940) 12/11/2012 11:24:59  RTE E API=OCIStmtExecute [in sqociExecOne], Statement=ALTER TABLE
APPROVALA1 ADD CONSTRAINT APPROVALA153D2D5D4 CHECK( "COMPONENT" IS NOT NULL )
 11508( 12940) 12/11/2012 11:24:59  RTE W Adding NOT NULL constraint for field 'COMPONENT' failed, please
change/remove the NULL values in records!
```

Lines in blue mean Service Manager starts to add a `NOT NULL` constraint for a logical table in SM.

Lines in dark yellow mean Service Manager starts to add a `Not Null` constraint for a physical table in SM.

Lines in red mean Service Manager can't add a `Not Null` constraint to that field, because that field has `NULL` values in records. Therefore, you should remove these `NULL` values manually.

7.10.2.6. SQL cache

Service Manager uses caches for table descriptions and definitions. They can be flushed by using the Refresh SQL Cache utility (**Tailoring > SQL Utilities**). This utility can be executed at any time and may be used to force Service Manager to retrieve the table descriptions again from the relational database management system (RDBMS) in case the database administrator has changed a table while Service Manager was active. Service Manager doesn't cache information about keys because it doesn't know much about the keys in the RDBMS.

Related topics

Resetting a database table

7.10.2.7. Regenerating database keys

Database keys provide efficient, organized access to records in a table. The keys define a hierarchical tree of indexes associated with the actual data records. Regenerating database keys discards and then regenerates the existing index tree for a table. The regeneration process examines each data record in the table and then adds indexes to the new index tree for each record.

Note

Note When regenerating the keys to a file, ensure that other users aren't accessing the file. The regeneration process will interrupt all activity in progress on the file being regenerated.

Related topics

Resetting a database table

7.10.2.7.1. Regenerate database keys

In this example we regenerate the keys to the contacts table. Regenerating keys for other tables works the same way.

Applies to User roles: System Administrator

To regenerate database keys:

1. Click **Tailoring** > **Database Manager**.
2. Select **Administration mode**.
3. In the **Form** field, type `contacts`, and then click **Search**.
4. Select the form.
5. Open the More Actions menu and choose **Regen IR**.
   The Confirm Action window opens.
6. From the prompt, do one of the following:
   - Click **OK** to continue with the file/key regen.
   - Click **Cancel** to exit this screen without doing a regen.

Related topics

Regenerate IR keys
Regen a database dictionary

7.10.3. Data maps

Data maps are a means of identifying related records among separate tables in Service Manager. Service Manager applies data map queries to each record returned in a purge or archive selection query. For example, if during the purge and archive process you select 10 probsummary records, then Service Manager uses the probsummary data map to search for related records on each of the 10 records.

When you enable a data map, Service Manager returns a list of additional related records for each purge or archive record queried. Service Manager purges or archives these related records along with the original records you selected during the purge and archive process. For example, using a data map on 10 probsummary records may result in 30 actual records being purged or archived.

Note: Using data maps can greatly increase the number of records that you purge or archive. Since there is no limit on the number of data maps that you can define, there can potentially be a large number of mappings for any given table.

Tip: To prevent the accidental loss of substantial numbers of records, use the purge and archive function to save copies of your records to an external table before deleting them. This enables you to restore any records that you may have deleted accidentally.

By default, Service Manager has 155 pre-defined data mappings that you can use to purge or archive records and forms. These mappings contain the most commonly purged or exported records and tables. If you add custom tables and fields to your Service Manager system, you may want to create custom data mappings to purge or archive any related records based on these custom tables.

To create data mappings to related records, you must know the primary key fields and join fields used by your Service Manager tables. You must also know how to write Service Manager queries using escaped characters, special operators, and RAD variables.

Related topics

Alternatives to data maps
Types of data maps
Exporting records
Importing records
Purging and archiving records
Unload script utility
Add a data map record
Create data maps to related records
Delete a data map record

Update a data map record
View a data map record

7.10.3.1. Alternatives to data maps

Data maps are an optional administration feature that you can use as part of the health and maintenance of your system to identify related records to purge or archive. In addition to data mapping, the following table shows other methods to purge and archive records.

Feature	Description	Differences in data mapping
Unload script utility	Unload records from multiple tables using saved queries.	No restrictions on record queries (data map queries must identify related records). Save multiple queries as one script record (data maps can support only one type of query for each record). List forms that you want to protect from purging (data maps can only purge records).
Entity relationship diagram definition (Erddef)	Define joins by entity relationships.	Define multiple joins (Each data map record can only define one join). Not integrated with the purge/archive function. Cascading deletion of related records (data mapping can only delete records as part of a purge process).

Related topics

Data maps
Types of data maps
Exporting records
Importing records
Purging and archiving records
Unload script utility
Add a data map record
Create data maps to related records
Delete a data map record
Update a data map record
View a data map record

7.10.3.2. Types of data maps

You can define three types of data maps.

Type	Description	Required fields	Joins
0	A Type 0 data map defines the primary key of a file.	**Filename** — identifies the name of the table queried to find source records. This is typically a table to purge or archive. For example, **format**. **Field Name** — identifies the primary key of the table to query. For example, **name**. **Map Query** — identifies the query to select records. For example, **"name=\""+str($umapdata)+"\""**. **Note:** The map query often uses RAD variables to select information about a record rather than hard code a particular record value. Type 0 data map records are equivalent to the following SQL query: SELECT <Field Name> FROM <Filename> WHERE <Field Name>=<Value in Map Query>	None

1	A Type 1 data map defines a join between two tables using one field to query the external file.	**Filename** — identifies the name of the table queried to find source records. This is typically a table to purge or archive. Service Manager uses the records from this table as the selection criteria to identify related records in external tables. For example, **format**. **Field Name** — identifies the join field shared between the primary and external tables. Service Manager uses the records from this field as the selection criteria to identify related records in the external table. For example, **name**. **Query File** — identifies the name of the external table queried with the join field. Service Manager searches for related records in this table. For example, **formatctrl**. Type 1 data map records are equivalent to the following SQL query: SELECT [<Override Qry> | <Field Name>] FROM <Query File> WHERE <Field Name>=<Value in FileName>	Joins the Field Name from the *Filename* to the values in the *Query File*.

2	A Type 2 data map defines a join between two tables using a more restrictive query of the external table.	**Filename** — identifies the name of the table queried to find source records. This is typically a table to purge or archive. Service Manager uses the records from this table as the selection criteria to identify related records in external table. For example, **format**. **Field Name** — identifies the join field shared between the primary and external table. Service Manager uses the records from this field as the selection criteria to identify related records in the external table. For example, **name**. **Map Query** — identifies the additional selection criteria to use when querying the join field to select records from the external target table. For example, **"name=\""+name in $umapfile+".g\" or name=\""+name in $umapfile+".w\""**. **Note:** The map query field uses the Service Manager query format. **Query File** — identifies the name of the external table that Service Manager queries with the join field. Service Manager searches for related records in this table. For example, **formatctrl**. Type 2 data map records are equivalent to the following SQL query: SELECT [<Override Qry> | <Field Name>] FROM <Query File> WHERE <Field Name>=<Value in FileName> and <Value in Map Query>	Joins the *Field Name* from *Filename* to the values returned by the *Map Query* in the Query File.

Related topics

Data maps
Alternatives to data maps
Exporting records
Importing records
Purging and archiving records
Unload script utility
Add a data map record
Create data maps to related records
Delete a data map record
Update a data map record
View a data map record

7.10.3.3. Add a data map record

Applies to User Roles: System Administrator

To add a data map record:

1. Click **System Administration** > **Base System Configuration** > **Miscellaneous** > **DataMaps**.
2. Fill in the following fields.

Field	Description
Desc	Type a description of the data map record.
Filename	Type the name of the Service Manager table with the primary key field that you want to query. Service Manager purges or archives records from this table. This is a required field for all data map records.
Type	Service Manager automatically populates this field with the data map type 0, 1, or 2. Do not change the value of this field directly.
Field Name	Type the name of the primary key field for the table to create a Type 0 data map, or type the name of the join field to query for Type 1 and Type 2 data maps. This is a required field for all data map records.
Array?	Type `true` or **false** to specify whether the join field to query is an array. By default, this field is false. Choose **true** if the data map is to check every element in the array for related records in the external file. This field is only for Type 1 data maps.
Arrayed Structure?	Type `true` or **false** to specify whether the join field to query is an arrayed structure. By default, this field is false. Choose **true** if the data map is to check every element in the **AS Field No.** for related records in the external file. This field is only for Type 1 data maps.
AS Field No	Type the ID number of the join field to query in an Arrayed Structure. This ID number is defined in the dbdict.

Map Query	Type the query to identify source records in the primary table. Adding a query to this field makes the record a Type 0 or Type 2 data map, depending on the **Query File** value. You must write this query in the Service Manager query format, and you can use the following variables: $umapdata — This variable stores the value of the **Field Name** field of the current record. For example, the Field Name was **name**, the value could be an operator such as **System.Admin** or a form name such as **contacts** depending on the table you are querying. $umapfile — This variable reads the value of the **Filename** field in the purge or archive record. $umap.modtime — This variable reads the value of the current modification time in the purge or archive record. $umap.mod.date — This variable reads the value of the last modification date in the purge or archive record. If no value is listed, Service Manager displays the current date and time. $acntr — This variables reads the value of the element in an array or array of structures in the purge or archive record.
Query File	Type the name of the external file to query for related records. Adding an entry to this field makes the record a Type 1 or Type 2 data map, depending on the Map Query value.
Override Qry	Type the query used to replace the Field Name to identify related records in the external file. If this field is blank, Service Manager uses the Field Name as the join field to query for related records. Adding an entry to this field makes the record a Type 1 or Type 2 data map, depending on the Map Query value.
Map Cond	Type true, false, or a Boolean expression that evaluates to true or false to identify when Service Manager should use this data map record. By default, this field is true meaning that Service Manager should use the data map in all conditions.
Add to Map List	Type true, false, or a Boolean expression that evaluates to true or false to identify when Service Manager should add an entry to the data map list. By default, this field is true meaning that Service Manager should always add an entry to the data map list..

3. Click **Add**.

Related topics

Data maps
Alternatives to data maps
Types of data maps
Exporting records
Importing records
Purging and archiving records
Unload script utility
Create data maps to related records
Delete a data map record
Update a data map record
View a data map record

7.10.3.4. Create data maps to related records

Applies to User Roles: System Administrator

You can create data maps to related records across Service Manager tables to use with the purge and archive functions. The purpose of data maps is to identify related records of the records you select to purge or archive. All data maps use the same basic structure to search tables and fields for related records.

To create data maps to related records:

1. Create one Type 0 data map record for each table to purge or archive. The Type 0 data map record typically identifies the primary key field of the table.
   Type 0 data map records are equivalent to the following SQL query:
   `SELECT <fieldname> FROM <filename> WHERE <fieldname>=<Value in Map Query>`
2. Create one Type 1 data map record for each join field to query for related records. The Type 1 data map record uses only the join field to search for related records in the external file.
   - To specify additional selection criteria of records from the primary table, you can add a Map Query, which creates a Type 2 data map record.
   - To specify additional selection criteria for records in the external file, you can add an override query in the **Override Qry** field. Because this query overrides the field listed in the **Field Name** field, you must include the join field within this query.

   Type 1 data map records are equivalent to the following SQL query:
   `SELECT [<override_qry> | <fieldname>] FROM <queryfile> WHERE <fieldname>=<value in filename>`
3. Create one Type 2 data map record for each join field that has additional selection criteria for records from the primary table. Type 2 data map records use the **Map Query** field in addition to the **Field Name** field to identify records from the primary table.
   Type 2 data map records are equivalent to the following SQL query:
   `SELECT [<override_qry> | <fieldname>] FROM <queryfile> WHERE <fieldname>=<Value in filename> and <Value in Map Query>`

Related topics

Data maps
Alternatives to data maps
Types of data maps
Exporting records
Importing records
Purging and archiving records
Unload script utility
Add a data map record
Delete a data map record
Update a data map record
View a data map record

7.10.3.5. Delete a data map record

Applies to User Roles: System Administrator

To delete a data map record:

1. Click **System Administration** > **Base Configuration** > **Miscellaneous** > **Data Maps**.
2. Fill in optional search criteria, and then click **Search**.
3. Click the data map to delete.
4. Click **Delete**.
5. Click **Yes** to confirm the deletion.

Related topics

Data maps
Alternatives to data maps
Types of data maps
Exporting records
Importing records
Purging and archiving records
Unload script utility
Add a data map record
Create data maps to related records
Update a data map record
View a data map record

7.10.3.6. Update a data map record

Applies to User Roles: System Administrator

To update a data map record:

1. Click **System Administration** > **Base System Configuration** > **Miscellaneouus** > **Data Maps**.
2. Fill in optional search criteria, and then click **Search**.
3. Click the data map to update.
4. Fill in the following fields.

Field	Description
Desc	Type a description of the data map record.
Filename	Type the name of the Service Manager table that contains the primary key field to query. Service Manager purges or archives records from this table. This is a required field for all data map records.
Type	Service Manager automatically populates this field with the data map type 0, 1, or 2. Do not change the value of this field directly.
Field Name	Type the name of the primary key field for the table to create a Type 0 data map, or type the name of the join field to query for Type 1 and Type 2 data maps. This is a required field for all data map records.
Array?	Type `true` or `false` to identify if the join field to query is an array. By default, this field is false. Specify **true** to check every element in the array for related records in the external file. This field is only used for Type 1 data maps.
Arrayed Structure?	Type `true` or `false` to specify if the join field is an arrayed structure. By default, this field is false. Specify **true** if the data map is to check every element in the **AS Field No.** for related records in the external file. This field is used only for Type 1 data maps.
AS Field No	Type the ID number of the join field to query in an Arrayed Structure. This ID number is defined in the dbdict.

Map Query	Type the query to identify source records in the primary table. Adding a query to this field makes the record a Type 0 or Type 2 data map depending on whether you specify a **Query File** value. You must write this query in the Service Manager query format, and you can use the following variables: $umapdata — This variable stores the value of the **Field Name** field of the current record. For example, if the Field Name is **name**, the value could be an operator. $umapfile — This variable reads the value of the **Filename** field in the purge or archive record. $umap.modtime — This variable reads the value of the current modification time in the purge or archive record. $umap.mod.date — This variable reads the value of the last modification date in the purge or archive record. If no value occurs, Service Manager displays the current date and time. $acntr — This variables reads the value of the element in an array or array of structures in the purge or archive record.
Query File	Type the name of the external file to query for related records. Adding an entry to this field makes the record a Type 1 or Type 2 data map depending upon if you type a Map Query value.
Override Qry	Type the query to replace the **Field Name** to identify related records in the external file. If this field is blank, Service Manager uses the Field Name as the join field to query for related records. Adding an entry to this field makes the record a Type 1 or Type 2 data map depending upon if you type a Map Query value.
Map Cond	Type `true`, `false`, or a Boolean expression that's true or false to identify when Service Manager should use this data map record. By default, this field is true, which specifies that Service Manager should use the data map in all conditions.
Add to Map List	Type `true`, `false`, or a Boolean expression that's true or false to identify when Service Manager should add an entry to the data map list. By default, this field is true, which specifies that Service Manager should always add an entry to the data map list.

5. Click **Save**.

Related topics

Data maps
Alternatives to data maps
Types of data maps
Exporting records
Importing records
Purging and archiving records
Unload script utility
Add a data map record
Create data maps to related records
Delete a data map record
View a data map record

7.10.3.7. View a data map record

Applies to User Roles: System Administrator

To view a data map record:

1. Click **System Administration** > **Base System Configuration** > **Miscellaneous** > **Data Maps**.
2. Specify the data map search criteria.
3. Click **Search**.

Related topics

Data maps
Alternatives to data maps
Types of data maps
Exporting records
Importing records
Purging and archiving records
Unload script utility
Add a data map record
Create data maps to related records
Delete a data map record
Update a data map record

7.10.4. Exporting records

A system administrator can export Service Manager records to an external file using the following methods:

- Export an individual record as an unload file
  - As a binary unload
  - As a text unload
  - As a formatted text unload
- Export a record from a record list
  - As an unload file
  - As an Excel file
  - As a text file
- Export a list of records from multiple tables
  - As an unload file

Related topics

Data maps
Export descriptor records
Unload files
Importing records
Purging and archiving records

Unload script utility
Add an export descriptor record
Export a record list to a text file
Export a record list to an Excel file
Export a record list using an export description record
Export an individual record using an export descriptor record
Export selected records from a table using an export descriptor record
Schedule the exporting of records using an export descriptor record
Schedule the unloading of records
Unload a database dictionary only
Unload a record list
Unload an individual record
Unload records from multiple files
Unload selected records from a file
Update an export descriptor record
Unload error messages
Load error messages and notifications

7.10.4.1. Export descriptor records

You can create an export descriptor record to define the rules for exporting data from Service Manager tables to external files. Each export descriptor record describes the following information:

- The Service Manager source table to be exported
- The fields from the table to export
- Formatting information about each field
  - Field start position
  - Field width
  - Number of entries to extract from array fields
  - Date and time format
- The related Format Control record to use, if any, to validate and format data prior to export

Note: You can't export database dictionary field definitions using an export description record. To export database dictionary field definitions, use the native Service Manager unload format.

When you create an export descriptor record, Service Manager validates the table selected for exporting against the database dictionary. If the validation is unsuccessful, Service Manager displays the following message: `Validation failed`

You can click on the warning message to see the reasons why the data validation failed. Although Service Manager lets you save an invalid export description record, you can't use this record to export data. You need to correct any validation errors before exporting data. After Service Manager validates the table, it exports each record as a single line entry in the resulting external file.

Related topics

Data maps
Exporting records
Unload files
Importing records
Purging and archiving records
Unload script utility
Add an export descriptor record
Export a record list to a text file
Export a record list to an Excel file
Export a record list using an export description record
Export an individual record using an export descriptor record
Export selected records from a table using an export descriptor record
Schedule the exporting of records using an export descriptor record
Schedule the unloading of records

Unload a database dictionary only
Unload a record list
Unload an individual record
Unload records from multiple files
Unload selected records from a file
Update an export descriptor record
Unload error messages

7.10.4.2. Unload error messages

Service Manager displays an error message if it can't successfully unload an external file.

Unload error message	Description
Invalid unload append/create parameter.	Valid parameters are append or create for the unload append/create input field.
Select ONE export format: binary, text, or formatted text.	Valid entries are binary, text, and formatted text for the unload mode input field.
Invalid unload dbdict load parameter.	Valid entries are create, replace, and remove for the unload dbdict input field.
Unload mode must be export.	When you specify an export descriptor record, you must set the unload type to export for the unload mode input field.
Invalid unload record load parameter.	Valid entries are update and add only for the record load input field.
No unload filename.	Type an external file name for the unload file.
The unload format must be binary for Purge/Archives.	Select the binary unload format when purging or archiving records.
<*database name*> file unloaded to external file: <*Unix filename*>	Unload processing the named database to the Unix file complete.
External file: <*UNIX.filename*> is locked; try again later.	Another task has exclusive control over the named external file.
The config record (database load/unload) is missing.	The hardware configuration record database load/unload.unix or database load/unload.winnt required for database loads and unloads has been deleted or renamed. It must be replaced.
<*nn*> records unloaded so far from <*database name*> file to <*unload file*>	The unload utility generates this message every hundredth record it attempts to unload.
<*database name*> file can't be unloaded to external file: <*UNIX/NT.FILENAME*>.	Service Manager could not connect to the named external file.
unable to allocate.	This message occurs in two distinct situations: The unload utility attempted to connect to a non-existent external file causing the message to be generated. However, the unload utility will attempt to dynamically allocate the Unix/NT filename. If the dynamic allocation is successful, confirmation messages will follow. In this situation, it doesn't indicate an error condition. The unload utility attempted to connect to a non-existent external file causing the message to be generated. The subsequent attempt to dynamically allocate the Unix/NT filename failed causing the message to be generated a second time. Other error messages will follow indicating the error condition.

Application:file.unload has completed with errors; check other messages.	An error exit condition has occurred; check the message log for the error message indicating which exit.
Invalid unload options specified for unload, no records unloaded.	Since the unload options were invalid, no records were unloaded.

Related topics

Exporting records
Export descriptor records
Unload files
Importing records
Import descriptor records
Unload files
Export a record list to a text file
Export a record list to an Excel file
Export a record list using an export description record
Export an individual record using an export descriptor record
Export selected records from a table using an export descriptor record
Add an import descriptor record
Import a character-delimited text file
Import a fixed-width text file

7.10.4.3. Unload files

The native Service Manager export format is the unload file. An unload file stores the database dictionary of Service Manager tables in addition to records. The database dictionary embedded in an unload file determines where Service Manager loads records during import. The unload file also specifies whether Service Manager should update or ignore duplicate records when loading records. Service Manager can export unload files in the following file formats:

- Binary - The binary unload file format archives Service Manager tables. You can load binary unload files only from the Database Manager.
- Text - The text unload file format is human readable but only intended to be loaded from the Database Manager.
- Formatted text - The formatted text unload file format is intended to be compatible with external applications, although the actual format of the file is determined by the form you select as a template. You can create export forms to create formatted text unload files compatible with specific applications.

You can export records into an unload file from the following Service Manager utilities:

- Database Manager
- Unload Script Utility
- Directly from some record lists

After you load a .unl file to import data into Service Manager, you must stop and then restart the Service Manager server. This ensures that the table changes are initialized.

Related topics

Data maps
Exporting records
Importing records
Purging and archiving records
Using data maps to purge and archive records
Unload script utility
Archive records
Purge and archive records
Purge records
Schedule purge and archive processes

7.10.4.4. Using data maps to purge and archive records

You can use data maps to purge or archive related records in your database. A data map defines connections between various tables in your database. For example, when you create a record in the probsummary table, you typically create associated records in the problem and screlation tables. By default, Service Manager comes with a collection of data maps to common tables. For example, the following tables have default data maps defined in Service Manager:

Table	Table associated by a data map
probsummary	problem, screlation
cm3r	cm3t, screlation
ocmo	ocmalertlog, ocmapprlog, ocml, ocmlpage, ocmopage, ocmphaselog
ocmq	ocmalertlog, ocmapprlog, ocml, ocmlpage, ocmphaselog, ocmqpage

To purge or archive all records associated with a particular table quickly, enable **Use Data Map?** from the Purge and Archive function. You can add or update data maps to create new associations among records.

Related topics

Data maps
Exporting records
Importing records
Purging and archiving records
Unload files
Unload script utility
Archive records
Purge and archive records
Purge records
Schedule purge and archive processes

7.10.4.5. Archive records

Applies to User Roles: System Administrator

To archive records:

1. Click **System Administration** > **Ongoing Maintenance** > **System** > **Purge/Archive**.
2. Fill in the following fields:

Field	Description
Enter Format Name	Type the name of the form to archive.
Use Data Map?	Type true or false to enable the data map associated with this form. If enabled, Service Manager archives all associated records with the form you select.
Prompt Data Map?	Type true or false to enable a confirmation message to use a data map for each record archived. This option is only valid if the **Use Data Map?** option is also enabled.

3. Click **OK**.
4. Fill in optional search criteria.
5. Click **Query**.
6. Click **Execute Search**.

7. Click **Archive**.
8. Fill in the following fields:

Field	Description
External File Name	Type the name and path of the file used to save the records. By default, this file is in the RUN folder of the Service Manager server.
Member	This field is obsolete.
Unload Type	Select the operating system file format of the external file.
Append to File	Select this option to add records to the end of an external file of the same name. If an external file of the same name doesn't exist, Service Manager creates a new file. If this field is cleared, then Service Manager overwrites any existing file of the same name.
Use Data Map	Select this option to use a predefined data map when archiving the selected records. Using a data map may result in Service Manager archiving more records than what you originally selected by the query.
Prompt for Data Map	Select this option to enable a confirmation message to use a data map for each record archived. This option is only valid if the **Use Data Map** option is also enabled.
Suppress Messages	Select this option to hide all messages that the purge and archive function generates.
Export Mode — Binary	Select this option to save the external file in Service Manager's binary format. Only the Service Manager Database Manager can read this binary file format. You must select this option for archiving records.
Export Mode — Text	Select this option to save the external file in Service Manager's text format. This format is human readable but is not intended for export to other applications other than Service Manager's Database Manager. You can't select this option for archiving records.
Export Mode — Formatted Text	Select this option to save the external file in a user-defined text format. This format is intended for export to other applications. If you select this option, you must type a valid text format name in the **Form Name** field. You can't select this option for archiving records.
Form Name	Type the name of the form to use when exporting records as formatted text. This option is valid only when you select the **Formatted Text** option. If the text form doesn't already exist, Service Manager prompts you to create it.
Dbdict Load Options — Create file if not there, and update if present	Select this option if the data saved in this external file should replace existing data in the target Service Manager system.

Dbdict Load Options — Create file if not there, but don't update if present	Select this option to ignore data saved in the external file that already exists in the target Service Manager system.
Dbdict Load Options — Create file if not there, and update if present, including keys	Select this option if the data saved in this external file should replace existing data in the target Service Manager system including the database keys.
Dbdict Load Options — Create file if not there, and update if present, including SQL mapping	Select this option if the data saved in this external file should replace existing data in the target Service Manager system including the database SQL mapping.
Dbdict Load Options — Create file if not there, and update if present, including keys AND SQL mapping	Select this option if the data saved in this external file should replace existing data in the target Service Manager system including the database SQL mapping and the database keys.
Record Load Options — Add/update records	Select this option if the records saved in this external file are to replace any existing records in the target Service Manager system.
Record Load Options — Only add new records	Select this option to ignore records saved in the external file that already exist in the target Service Manager system.

9. Click **unload/export**.
   Service Manager saves the select records to an external file.

Related topics

Data maps
Exporting records
Importing records
Purging and archiving records
Unload files
Using data maps to purge and archive records
Unload script utility
Purge and archive records
Purge records
Schedule purge and archive processes
Load/unload implications of case-sensitivity

7.10.4.6. Purge and archive records

Applies to User Roles: System Administrator

To purge and archive records:

1. Click **System Administration** > **Ongoing Maintenance** > **System** > **Purge/Archive**.
2. Fill in the following fields.

Field	Description
Format Name	Type the name of the form to purge and archive.
Use Data Map?	Type true or false to enable the data map associated with this form. If enabled, Service Manager archives all associated records with the form you select.
Prompt Data Map?	Type true or false to enable a confirmation message to use a data map for each record purged and archived. This option is only valid if the **Use Data Map?** option is also enabled.

3. Click **OK**.
4. Fill in optional search criteria to select the records to purge and archive.
5. Click **Query**.
6. Click **Execute Search**.
7. Click **Purge/Arch**.
8. Fill in the following fields.

Field	Description
External File Name	Type the name and path of the file used to save the records. By default, this file is located in the RUN folder of the Service Manager server.
Member	This field is obsolete.
Unload Type	Select the operating system file format for the external file.
Append to File	Select this option to add records to the end of an external file of the same name. If a external file of the same name doesn't exist, Service Manager creates a new file. If this field is cleared, then Service Manager overwrites any existing file of the same name.
Use Data Map	Select this option to use a predefined data map when archiving the selected records. Using a data map may result in Service Manager archiving more records than what you originally selected by the query.
Prompt for Data Map	Select this option to enable a confirmation message to use a data map for each record archived. This option is only valid if the **Use Data Map** option is also enabled.
Suppress Messages	Select this option to hide all messages that the purge and archive function generates.
Export Mode — Binary	Select this option to save the external file in Service Manager's binary format. Only the Service Manager Database Manager can read this binary file format. You must select this option for archiving records.
Export Mode — Text	Select this option to save the external file in Service Manager's text format. This format is human readable but is not intended for export to other applications other than Service Manager's Database Manager. You can't select this option for archiving records.
Export Mode — Formatted Text	Select this option to save the external file in a user-defined text format. This format is intended for export to other applications. If you select this option, you must type a valid text format name in the **Form Name** field. You can't select this option for archiving records.
Form Name	Type the name of the text format to use when exporting records as formatted text. This option is only valid with the **Formatted Text** option is selected. If the text format description doesn't already exist, Service Manager prompts you to create it.
Dbdict Load Options — Create file if not there, and update if present	Select this option if the data saved in this external file should replace existing data in the target Service Manager system.

Dbdict Load Options — Create file if not there, but don't update if present	Select this option to ignore data saved in the external file that already exists in the target Service Manager system.
Dbdict Load Options — Create file if not there, and update if present, including keys	Select this option if the data saved in this external file should replace existing data in the target Service Manager system including the database keys.
Dbdict Load Options — Create file if not there, and update if present, including SQL mapping	Select this option if the data saved in this external file should replace existing data in the target Service Manager system including the database SQL mapping.
Dbdict Load Options — Create file if not there, and update if present, including keys AND SQL mapping	Select this option if the data saved in this external file should replace existing data in the target Service Manager system including the database SQL mapping and the database keys.
Record Load Options — Add/update records	Select this option if the records saved in this external file are to replace any existing records in the target Service Manager system.
Record Load Options — Only add new records	Select this option to ignore records saved in the external file that already exist in the target Service Manager system.

9. Click **unload/export**.
10. Click **Purge All**.

Related topics

Data maps
Exporting records
Importing records
Purging and archiving records
Unload files
Using data maps to purge and archive records
Unload script utility
Load/unload implications of case-sensitivity
Archive records
Purge records
Schedule purge and archive processes

7.10.4.7. Purge records

Applies to User roles: System Administrator

To purge records:

1. Click **System Administration** > **Ongoing Maintenance** > **System** > **Purge/Archive**.
2. Fill in the following fields.

Field	Description
Format Name	Type the name of the form to purge.
Use Data Map?	Type true or false to enable the data map associated with this form. If enabled, Service Manager purges all associated records with the form you select.
Prompt Data Map?	Type true or false to enable a confirmation message to use a data map for each record purged. This option is only valid if the **Use Data Map?** option is also enabled.

3. Click **OK**.
4. Fill in optional search criteria.
5. Click **Query**.
6. Click **Execute Search**.
7. Click **Purge**.
8. Click **Purge All**.

Related topics

Data maps
Exporting records
Importing records
Purging and archiving records
Unload files
Using data maps to purge and archive records
Unload script utility
Archive records
Purge and archive records
Schedule purge and archive processes

7.10.4.8. Schedule purge and archive processes

Applies to User Roles: System Administrator

To schedule purge and archive processes:

1. Click **System Administration** > **Ongoing Maintenance** > **System** > **Purge/Archive**.
2. Fill in the following fields.

Field	Description
Enter Format Name	Type the name of the form to purge and archive.
Use Data Map?	Type yes or no to enable the data map associated with this form. If enabled, Service Manager archives all associated records with the form you select.
Prompt Data Map?	Type yes or no to enable a confirmation message to use a data map for each record purged and archived. This option is only valid if the **Use Data Map?** option is also enabled.

3. Click **OK**.
4. Fill in optional search criteria.
5. Click **Query**.
6. Click **Execute Search**.

7. Click **Schedule**.
8. Fill in the following fields.

Field	Description
Date/Time to Run	Specify the date and time to run the archive, purge, or purge and archive process.
Repeat Interval — Enter Number of Days and Time	Specify the number of days, hours, minutes, and seconds to wait, if any, before repeating the archive, purge, or purge and archive process. Type the interval in the following format: <*days*><*hours*>:<*minutes*>:<*seconds*>. For example, 1 00:00:00 is a repeat interval of 1 day.
Repeat Interval — Monthly	Select this option to repeat the archive, purge, or purge and archive process every once every month from today's date.
Repeat Interval — Quarterly	Select this option to repeat the archive, purge, or purge and archive process every once every quarter from today's date.
Repeat Interval — Semi-Annually	Select this option to repeat the archive, purge, or purge and archive process every once every six months from today's date.
Repeat Interval — Annually	Select this option to repeat the archive, purge, or purge and archive process every once every year from today's date.
Archive Only — Ext File	Type the name and path of the external file to contain archive records.
Archive Only — Member	This field is obsolete.
Archive Only — Append to existing file or create new file (append/create)	Select **Append** to add records to the end of an external file of the same name. Select **Create** to overwrite an external file of the same name. With either option, if an external file of the same name doesn't exist, Service Manager creates a new file.
Use Data Map?	Type yes or no to enable the data map associated with this form. If enabled, Service Manager archives all associated records with the form you select.
Suppress Messages	Select this option to hide all messages that the purge and archive function generates.

Dbdict load options include	Select one of the following options: ○ Create — Select this option to create a new file if one is not present in the target Service Manager system and to preserve the existing file if there is a duplicate file in the external file. ○ Replace— Select this option to create a new file if one is not present in the target Service Manager system and to overwrite any existing files with the entry in the external file. ○ Remove — Select this option to create a new file if one is not present in the target Service Manager system and to replace any existing file with the entry in the external file.
Record load options include	Select one of the following options: ○ Add/Update — Select this option to create a new records if they're not present in the target Service Manager system and to overwrite existing records with entries from the external file. ○ Add Only — Select this option to create a new records if they're not present in the target Service Manager system and to preserve any existing records if there duplicates entries the external file. The records will be updated or inserted into the table regardless of case-sensitivity.

9. Click one of the following options:
   - **Archive**
   - **Purge**
   - **Purge/Arch**

Related topics

Data maps
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Unload script utility
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Purge and archive records
Purge records

7.10.4.9. Add an export descriptor record

Applies to User Roles: System Administrator

To add an export descriptor record:

1. Click **Tailoring** > **Database Manager**.
2. In the Form field, type **=export**, and click **Search**.
   A list of existing export descriptor records is displayed.

3. Select a record from the list, and update the following fields.

Field	Description
Export Descriptor Record	Type the unique name for this record.
Service Manager File Name	Type the name of the Service Manager database table exporting records.
Format Control Record Name	Type the name of the optional Format Control record to use with the export process. This Format Control record determines how Service Manager formats exported records.
Field Name	Type the database dictionary name of the fields exporting data. If the field is part of a structure, include the structure name before the field name. If the field is part of an arrayed structure, type only the field name.
Field Type	Service Manager fills this field automatically using the database dictionary definition of the field. You don't need to type any information in this field.
Start Pos	Type the number of space characters between the start of a line and the information Service Manager exports from the selected field. This information determines which column and line number Service Manager uses to export the field information. **Note**: Most external files use a standard of 80 characters per line. If you type a value greater than 80 then each record will use multiple lines.
Fld Lng	Type the number of characters Service Manager has to write information from the exported field into the external file.
Array Type	Service Manager fills in the value of this field automatically based on the database dictionary definition of the field. You don't need to manually type any information in this field.
Array Occ	Type the number of elements in an array that you want Service Manager to export. You can use the Array Occ value to determine the start position of the next field. For example, if you want to export five elements from an array and you allocated 80 character spaces for each element, this field requires a total of 400 character spaces. The next field should begin more than 400 characters spaces from the start of the file.

Date Format	Type the date and time format for date and time fields. You can use the following tokens to represent units of time: dd — two-digit day mm — two-digit month yy — two-digit year yyyy — four-digit year hh — hours ii — minutes ss — seconds You can use any character to separate these tokens. For example, mm/dd/yyyy hh:ii:ss exports dates and times with slashes between the month, day, and four-digit year and colons between the hours, minutes, and seconds. **Note**: If you don't include a time format, then Service Manager doesn't export any time information.

4. Click **Add**.

Related topics

Data maps
Export descriptor records
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Importing records
Purging and archiving records
Unload files
Unload script utility
Unload error messages
Export a record list to a text file
Export a record list to an Excel file
Export a record list using an export description record
Export an individual record using an export descriptor record
Export selected records from a table using an export descriptor record
Schedule the exporting of records using an export descriptor record
Schedule the unloading of records
Unload a database dictionary only
Unload a record list
Unload an individual record
Unload records from multiple files
Unload selected records from a file
Update an export descriptor record

7.10.4.10. Delete an export descriptor record

Applies to User Roles: System Administrator

To delete an export descriptor record:

1. Click **Tailoring** > **Database Manager**.
2. Type `=export` in the **Form** field.
3. Select **Administration Mode**. You must select this option for Service Manager to display the **Delete** button.
4. Click **Search**.
5. Click **Search** again to display a list of export descriptor records.
   Note: You may need to click **Document** > **Restore** to see the list of export descriptor records.

6. Select the export descriptor record to be deleted.
7. Click **Delete**.
8. Click **Yes** to confirm the deletion.

Related topics

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Export an individual record using an export descriptor record
Export selected records from a table using an export descriptor record
Schedule the exporting of records using an export descriptor record
Schedule the unloading of records
Unload a database dictionary only
Unload a record list
Unload an individual record
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7.10.4.11. Export a record list to a text file

Applies to User Roles: All users

To export a record list to a text file:

Open a list of records. For example, open a list of open incident records.

1. Click **Incident Management** > **Search Incidents**.
2. Select **Open** status, and then click **Search**.
   A list of incident records opens.
   Note: You can click **More** or the More Actions icon and then click **Modify Columns**, so that only those columns containing data that you intend to export are displayed. You must keep the column that contains the key field.
3. From the record list, click **More** or the More Actions icon, and then choose **Export to Text File**.
4. Fill in the following fields.

Field	Description
Filename	Type, or browse to, the path and name of the new text file.
Comma Separated Value (CSV)	Select this option to use the comma character as a field delimiter.
Semicolon Separated CSV	Select this option to use the semicolon character as a field delimiter.
Tab	Select this option to use the tab character as a field delimiter.
Asterisk	Select this option to use the asterisk character as a field delimiter.
Caret	Select this option to use the caret character as a field delimiter.
Other	Select this option to use a designated character as a field delimiter. Type the character delimiter.

5. Do one of the following to save the file:
    - *For Windows clients:*
      In the **Filename** field, specify a path and name for the text file, and then click **OK**.
    - *For Web clients:*
      Click **OK**. Save the file when a download job starts in your Web browser.

Related topics

Data maps
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Export a record list using an export description record
Export an individual record using an export descriptor record
Export selected records from a table using an export descriptor record
Schedule the exporting of records using an export descriptor record
Schedule the unloading of records
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7.10.4.12. Export a record list to an Excel file

Applies to User Roles: All users

Note: You must have the Microsoft Excel application program on your computer.

To export a record list to an Excel file:

Windows client:

1. Go to **Window** > **Preferences**, and make sure that the **Client side load/unload** option is selected.
2. Open a list of records. For example, open a list of open incident records.
   Note: You can use **Modify Columns**, so that only those columns containing data that you intend to export displays. You must keep the column that contains the key field.
3. From the record list, click the More Actions icon, and choose **Export to Excel**.
4. In the **Filename** field, specify a path and name for the file, and then click **OK**.
   Note: If you specify a file that doesn't exist, Service Manager will create it.
   Service Manager exports the record list to a CSV file and opens it in Excel.

Web client:

For a Web tier client, export the record list to a text file, and then open the text file in Excel. Do the following:

1. From the record list, click **More** and choose **Export to Text File**.
2. Fill in the following fields.

Field	Description
Filename	Type, or browse to, the path and name of the new text file.
Comma Separated Value (CSV)	Select this option to use the comma character as a field delimiter.
Semicolon Separated CSV	Select this option to use the semicolon character as a field delimiter.
Tab	Select this option to use the tab character as a field delimiter.
Asterisk	Select this option to use the asterisk character as a field delimiter.
Caret	Select this option to use the caret character as a field delimiter.
Other	Select this option to use a designated character as a field delimiter. Type the character delimiter.

3. Open the Excel application, and then click **Open**.
4. Select **All files** to see a list of all files in your directory.
5. Select the new text file of the records list that you just created.
6. Answer the Excel program prompts to import your text file. For example, select **Delimited**, so that characters such as commas or semicolons separate each field.
7. When you are done making your selections, click **Finish**.
8. Click **OK**.
   Your record list is imported into the Excel spreadsheet.

Related topics

Data maps
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Importing records

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Unload a record list
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7.10.4.13. Export a record list using an export description record

Applies to User Roles: System Administrator

To export a record list using an export description record:

1. Click **Tailoring** > **Database Manager**.
2. Type the form or table name that contains the records to be unloaded.
3. Select the **Administration Mode** option. You must select this option to ensure that Service Manager displays the unload options in the target form.
4. Click **Search**.
5. Select the form used to search for records.
6. Fill in optional search criteria, and then click **Search**.
7. A record list appears. Click **Mass Unload**.
8. Fill in the following fields.

Field	Description
External File Name	Type the name and path of the file used to save the record list. By default, this file is located in the RUN folder of the Service Manager server.
Member	This field is obsolete.
Unload Type	Select the operating system file type for the external file.
Append to File	Select this option to add records to the end of an external file of the same name. If an external file of the same name doesn't exist, Service Manager creates a new file. If this field is cleared, then Service Manager overwrites any existing file of the same name.
Use Data Map	Select this option to use a pre-defined data map when unloading the selected records. Using a data map may result in Service Manager unloading more records than you originally selected to unload.
Prompt for Data Map	Select this option to enable a confirmation message to use a data map for each record unloaded. This option is only valid if the **Use Data Map** option is also enabled.
Suppress Messages	Select this option to hide all messages that the unload process generates.
Export Mode — Formatted Text	Select this option to save the external file using an export descriptor record. This format is intended for export to other applications.
Form Name	Type the name of the export descriptor record. This option is valid only when the **Formatted Text** option is selected. If the export descriptor record doesn't exist, Service Manager prompts you to create it.

9. Click **Export Ext**.
10. If Service Manager displays a list of existing export descriptor records, select the export descriptor record to use.
11. Do one of the following:
    - To change the export descriptor record, click **Edit Record**to edit the fields.
    - To export the record, click **Export**.
      Service Manager saves the select records to an external file.

Related topics

Data maps
Exporting records
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Export an individual record using an export descriptor record
Export selected records from a table using an export descriptor record
Schedule the exporting of records using an export descriptor record
Schedule the unloading of records
Unload a database dictionary only
Unload a record list
Unload an individual record
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Unload selected records from a file
Update an export descriptor record
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7.10.4.14. Export an individual record using an export descriptor record

Applies to User Roles: System Administrator

To export an individual record using an export descriptor record:

1. Click **Tailoring** > **Database Manager**.
2. Type the form or table name that contains the record to be unloaded.
3. Select the **Administration Mode** option. You must select this option to ensure that Service Manager displays the unload options in the target form.
4. Click **Search**.
5. Select the form used to search for the record.
6. Fill in optional search criteria, and then click **Search**.
7. Select the record to unload.
8. Click **More** or the More Actions icon, and then select **Export/Unload**.
9. Fill in the following fields.

Field	Description
External File Name	Type the name and path of the destination file. By default, this file is located in the RUN folder of the Service Manager server.
Member	This field is obsolete.
Unload Type	Select the operating system file type for the external file.
Append to File	Select this option to add records to the end of an external file of the same name. If a external file of the same name doesn't exist, Service Manager creates a new file. If this field is cleared, then Service Manager overwrites any existing file of the same name.
Use Data Map	Select this option to use a pre-defined data map when unloading the selected records. Using a data map may result in Service Manager unloading more records than you originally selected to unload.
Prompt for Data Map	Select this option to enable a confirmation message to use a data map for each record unloaded. This option is only valid if the **Use Data Map** option is also enabled.
Suppress Messages	Select this option to hide all messages that the unload process generates.
Export Mode — Formatted Text	Select this option to save the external file using an export descriptor record. This format is intended for export to other applications.
Form Name	Type the name of the export descriptor record to use. This option is only valid with the **Formatted Text** option is selected. If the export descriptor record doesn't already exist, Service Manager prompts you to create it.

10. Click **Export Ext**.
11. If Service Manager displays a list of existing export descriptor records, select the export descriptor record to use.
12. Do one of the following:
    - To change the export descriptor record, click **Edit Record**.
    - To export the record, click **Export**.
      Service Manager saves the select records to an external file.

Related topics

Data maps
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Export an individual record using an export descriptor record
Schedule the exporting of records using an export descriptor record
Schedule the unloading of records
Unload a database dictionary only
Unload a record list
Unload an individual record
Unload records from multiple files
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7.10.4.15. Export selected records from a table using an export descriptor record

Applies to User Roles: System Administrator

To export selected records from a table using an export descriptor record:

1. Click **Tailoring** > **Database Manager**.
2. Type the form or table name that contains the records to unload.
3. Select the **Administration Mode** option.
   Note: You must select this option to ensure that Service Manager displays the unload options in the target form.
4. Click **Search**.
5. Select the form used to search for records. Do not type or select any search criteria from this form.
6. Click **More** or the More Actions icon, and then select **Export/Unload**.
7. Fill in the following fields.

Field	Description
Field Name	Type the name of the field containing the selection criteria.
Field Values	Type a unique value for each record unloaded from the **Field Name**.

8. Click **Unl Records**.
   Service Manager displays a record list of the records you selected.
9. From the record list, select the form, click **More** or the More Actions icon, and then select **Mass Unload**.
10. Fill in the following fields.

Field	Description
External File Name	Type the name and path of the destination file to contain the record list. By default, this file is located in the RUN folder of the Service Manager server.
Member	This field is obsolete.
Unload Type	Select the operating system file type to use for the external file.
Append to File	Select this option to add records to the end of an external file of the same name. If a external file of the same name doesn't exist, Service Manager creates a new file. If this field is cleared, then Service Manager overwrites any existing file of the same name.
Use Data Map	Select this option to use a pre-defined data map when unloading the selected records. Using a data map may result in Service Manager unloading more records than you originally selected to unload.
Prompt for Data Map	Select this option to enable a confirmation message to use a data map for each record unloaded. This option is only valid if the **Use Data Map** option is also enabled.
Suppress Messages	Select this option to hide all messages that the unload process generates.
Export Mode — Formatted Text	Select this option to save the external file using an export descriptor record. This format is intended for export to other applications.
Form Name	Type the name of the export descriptor record to use. This option is only valid with the **Formatted Text** option is selected. If the export descriptor record doesn't already exist, Service Manager prompts you to create it.

11. Click **Export Ext**.
12. If Service Manager displays a list of existing export descriptor records, select the export descriptor record to use.
13. Do one of the following:
    - To change the export descriptor record, click **Edit Record**.
    - To export the record, click **Export**.
      Service Manager saves the select records to an external file.

Related topics

Exporting records
Export descriptor records
Unload files
Export a record list to a text file
Export a record list to an Excel file
Export a record list using an export description record
Export an individual record using an export descriptor record
Unload error messages

7.10.4.16. Schedule the exporting of records using an export descriptor record

Applies to User Roles: System Administrator

To schedule the exporting of records using an export descriptor record:

1. Select the records to export.
2. From the file.prompt.dbu.g form, also known as the Service Manager unload/export facility, click **Export Ext**.
3. If Service Manager displays a list of existing export descriptor records, select the export descriptor record to use.
4. Do one of the following:
    - To change the export descriptor record, click **Edit Record**.
    - To schedule the export of records, click **Schedule**.
5. Fill in the following fields.

Field	Description
Date/Time to Run	Specify the date and time to unload the records.
Repeat Interval — Enter Number of Days and Time	Type the number of days, hours. minutes, and seconds to wait before repeating the unload process. Enter the interval with the following format: *<Days><Hours>:<Minutes>:<Seconds>* You can specify a repeat interval or select one of the pre-defined intervals. Any value you type in this field overrides the pre-defined intervals.
Repeat Interval — Monthly	Select this option to repeat the export process every once every month from today's date.
Repeat Interval — Quarterly	Select this option to repeat the export process every once every quarter from today's date.
Repeat Interval — Semi-Annually	Select this option to repeat the export process every once every six months from today's date.
Repeat Interval — Annually	Select this option to repeat the export process every once every year from today's date.

6. Click **Confirm**.

Related topics

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Export an individual record using an export descriptor record
Export selected records from a table using an export descriptor record
Schedule the unloading of records
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Unload a record list
Unload an individual record

Unload records from multiple files
Unload selected records from a file
Update an export descriptor record

7.10.4.17. Schedule the unloading of records

Applies to User Roles: System Administrator

To schedule the unloading of records:

1. Click **Tailoring** > **Database Manager**.
2. Type the form or file name that contains the records to unload, and then click **Search**.
3. Select a form, and then click **Search** again.
   A list of records opens.
4. Select a record.
5. Open the More Actions menu and choose **Export/Unload**.
6. Fill in optional search criteria.
7. From the file.prompt.dbu.g form, also known as the unload/export facility, click **Schedule**.
8. Fill in the following fields.

Field	Description
Date/Time to Run	Specify the date and time to unload the records.
Repeat Interval — Enter Number of Days and Time	Type the number of days, hours. minutes, and seconds to wait before repeating the unload process. Enter the interval with the following format: <*Days*><*Hours*>:<*Minutes*>:<*Seconds*> **Note**: You can either type in a repeat interval or select one of the pre-defined intervals. Any value you type in this field overrides the pre-defined intervals.
Repeat Interval — Monthly	Select this option to repeat the unload process every once every month from today's date.
Repeat Interval — Quarterly	Select this option to repeat the unload process every once every quarter from today's date.
Repeat Interval — Semi-Annually	Select this option to repeat the unload process every once every six months from today's date.
Repeat Interval — Annually	Select this option to repeat the unload process every once every year from today's date.

9. Click **Confirm**.

Related topics

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Export a record list to an Excel file
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Export an individual record using an export descriptor record
Export selected records from a table using an export descriptor record
Schedule the exporting of records using an export descriptor record
Unload a database dictionary only
Unload a record list
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Unload records from multiple files
Unload selected records from a file
Update an export descriptor record
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7.10.4.18. Unload a database dictionary only

Applies to User Roles: System Administrator

To unload a database dictionary only:

1. Click **Tailoring** > **Database Manager**.
2. Type the form or file name that contains the records to unload.
3. Select the **Administration Mode** option. You must select this option to ensure that Service Manager displays the unload options in the target form.
4. Click **Search**.
5. Select the form to use to search for records. Do not type or select any search criteria from this form.
6. Click **More** or the More Actions icon, and then select **Export/Unload**.
7. Click **Unl Dbdict**.
8. Fill in the following fields.

Field	Description
External File Name	Type the name and path of the file used to save the record list. By default, this file is located in the RUN folder of the Service Manager server.
Member	This field is obsolete
Unload Type	Select the operating system file type for the external file.
Append to File	Select this option to add records to the end of an external file of the same name. If a external file of the same name doesn't exist, Service Manager creates a new file. If this field is cleared, then Service Manager overwrites any existing file of the same name.
Use Data Map	Select this option to use a pre-defined data map when unloading the selected records. Using a data map may result in Service Manager unloading more records than you originally selected to unload.
Prompt for Data Map	Select this option to enable a confirmation message to use a data map for each record unloaded. This option is only valid if the **Use Data Map** option is also enabled.
Suppress Messages	Select this option to hide all messages that the unload process generates.
Export Mode — Binary	Select this option to save the external file in Service Manager's binary unload format. Only the Service Manager Database Manager can read this binary file format.
Export Mode — Text	Select this option to save the external file in Service Manager's text format. This format is human readable but is not intended for export to other applications other than Service Manager's Database Manager.
Export Mode — Formatted Text	Select this option to save the external file in a user-defined text format. This format is intended for export to other applications. If you select this option, you must type a valid Service Manager form name in the **Form Name** field.
Form Name	Type the name of the Service Manager form to use when exporting records as formatted text. This option is only valid with the **Formatted Text** option is selected. If the text format description doesn't already exist, Service Manager prompts you to create it.
Dbdict Load Options — Create file if not there, and update if present	Select this option if the data saved in this external file should replace existing data in the target Service Manager system.

Dbdict Load Options — Create file if not there, but don't update if present	Select this option to ignore data saved in the external file that already exists in the target Service Manager system.
Dbdict Load Options — Create file if not there, and update if present, including keys	Select this option if the data saved in this external file should replace existing data in the target Service Manager system including the database keys.
Dbdict Load Options — Create file if not there, and update if present, including SQL mapping	Select this option if the data saved in this external file should replace existing data in the target Service Manager system including the database SQL mapping.
Dbdict Load Options — Create file if not there, and update if present, including keys AND SQL mapping	Select this option if the data saved in this external file should replace existing data in the target Service Manager system including the database SQL mapping and the database keys.
Record Load Options — Add/update record	Select this option if the records saved in this external file are to replace any existing records in the target Service Manager system.
Record Load Options — Only add new records	Select this option to ignore records saved in the external file that already exist in the target Service Manager system.

9. Click **Unload/export**.
   Service Manager saves the select records to an external file.

Related topics

Data maps
Exporting records
Export descriptor records
Unload files
Importing records
Purging and archiving records
Unload script utility
Load/unload implications of case-sensitivity
Add an export descriptor record
Export a record list to a text file
Export a record list to an Excel file
Export a record list using an export description record
Export an individual record using an export descriptor record
Export selected records from a table using an export descriptor record
Schedule the exporting of records using an export descriptor record
Schedule the unloading of records
Unload a record list
Unload an individual record
Unload records from multiple files
Unload selected records from a file
Update an export descriptor record
Unload error messages

7.10.4.19. Unload a record list

Applies to User Roles: System Administrator

To unload a record list:

1. Click **Tailoring** > **Database Manager**.
2. Type the form or file name that contains the records to unload.
3. Select the **Administration Mode** option. You must select this option to ensure that Service Manager displays the unload options in the target form.
4. Click **Search**.
5. Click the form used to search for records.
6. Fill in optional search criteria.
7. Click **Search**.
8. Click **More** or the More Actions icon for record list, and then select **Mass Unload**.
9. Fill in the following fields.

Field	Description
External File Name	Type the name and path of the file used to save the record list. By default, this file is located in the RUN folder of the Service Manager server.
Member	This field is obsolete.
Unload Type	Select the operating system file format for the external file.
Append to File	Select this option to add records to the end of an external file of the same name. If an external file of the same name doesn't exist, Service Manager creates a new file. If this field is cleared, Service Manager overwrites any existing file of the same name.
Use Data Map	Select this option to use a pre-defined data map when unloading the selected records. Using a data map may result in Service Manager unloading more records than you originally selected to unload.
Prompt for Data Map	Select this option to enable a confirmation message to use a data map for each record unloaded. This option is only valid if the **Use Data Map** option is also enabled.
Suppress Messages	Select this option to hide all messages that the unload process generates.
Export Mode — Binary	Select this option to save the external file in Service Manager's binary unload format. Only the Service Manager Database Manager can read this binary file format.
Export Mode — Text	Select this option to save the external file in Service Manager's text format. This format is human readable but is not intended for export to other applications other than Service Manager's Database Manager.
Export Mode — Formatted Text	Select this option to save the external file in a user-defined text format. This format is intended for export to other applications. If you select this option, you must type a valid Service Manager form name in the **Form Name** field.
Form Name	Type the name of the Service Manager form to use when exporting records as formatted text. This option is only valid with the **Formatted Text** option is selected. If the text format description doesn't already exist, Service Manager prompts you to create it.
Dbdict Load Options — Create file if not there, and update if present	Select this option if the data saved in this external file should replace existing data in the target Service Manager system.

Dbdict Load Options — Create file if not there, but don't update if present	Select this option to ignore data saved in the external file that already exists in the target Service Manager system.
Dbdict Load Options — Create file if not there, and update if present, including keys	Select this option if the data saved in this external file should replace existing data in the target Service Manager system including the database keys.
Dbdict Load Options — Create file if not there, and update if present, including SQL mapping	Select this option if the data saved in this external file should replace existing data in the target Service Manager system including the database SQL mapping.
Dbdict Load Options — Create file if not there, and update if present, including keys AND SQL mapping	Select this option if the data saved in this external file should replace existing data in the target Service Manager system including the database SQL mapping and the database keys.
Record Load Options — Add/update records	Select this option if the records saved in this external file are to replace any existing records in the target Service Manager system.
Record Load Options — Only add new records	Select this option to ignore records saved in the external file that already exist in the target Service Manager system.

10. Click **Unload/export**.
    Service Manager saves the select records to an external file.

Related topics

Data maps
Exporting records
Export descriptor records
Unload files
Importing records
Purging and archiving records
Unload script utility
Load/unload implications of case-sensitivity
Add an export descriptor record
Export a record list to a text file
Export a record list to an Excel file
Export a record list using an export description record
Export an individual record using an export descriptor record
Export selected records from a table using an export descriptor record
Schedule the exporting of records using an export descriptor record
Schedule the unloading of records
Unload a database dictionary only
Unload an individual record
Unload records from multiple files
Unload selected records from a file
Update an export descriptor record
Unload error messages
Load error messages and notifications

7.10.4.20. Unload an individual record

Applies to User Roles: System Administrator

To unload an individual record:

1. Click **Tailoring** > **Database Manager**.
2. Type the form or file name that contains the record to unload.
3. Select the **Administration Mode** option. You must select this option to ensure that Service Manager displays the unload options in the target form.
4. Click **Search**.
5. Click the form used to search for the record.
6. Fill in optional search criteria.
7. Click **Search**.
8. Select the record to unload.
9. Click **More** or the More Actions icon for the record, and then select **Export/Unload**.
10. Fill in the following fields.

Field	Description
External File Name	Type the name and path of the file used to save the record. By default, this file is located in the RUN folder of the Service Manager server.
Member	This field is obsolete.
Unload Type	Select the operating system file type to use for the external file.
Append to File	Select this option to add records to the end of an external file of the same name. If a external file of the same name doesn't exist, Service Manager creates a new file. If this field is cleared, then Service Manager overwrites any existing file of the same name.
Use Data Map	Select this option to use a pre-defined data map when unloading the selected records. Using a data map may result in Service Manager unloading more records than you originally selected to unload.
Prompt for Data Map	Select this option to enable a confirmation message to use a data map for each record unloaded. This option is only valid if the **Use Data Map** option is also enabled.
Suppress Messages	Select this option to hide all messages that the unload process generates.
Export Mode — Binary	Select this option to save the external file in Service Manager's binary unload format. Only the Service Manager Database Manager can read this binary file format.
Export Mode — Text	Select this option to save the external file in Service Manager's text format. This format is human readable but is not intended for export to other applications other than Service Manager's Database Manager.
Export Mode — Formatted Text	Select this option to save the external file in a user-defined text format. This format is intended for export to other applications. If you select this option, you must type a valid Service Manager form name in the **Form Name** field.
Form Name	Type the name of the Service Manager form to use when exporting records as formatted text. This option is valid only with the **Formatted Text** option is selected. If the text format description doesn't already exist, Service Manager prompts you to create it.
Dbdict Load Options — Create file if not there, and update if present	Select this option if the data saved in this external file should replace existing data in the target Service Manager system.

Dbdict Load Options — Create file if not there, but don't update if present	Select this option to ignore data saved in the external file that already exists in the target Service Manager system.
Dbdict Load Options — Create file if not there, and update if present, including keys	Select this option if the data saved in this external file should replace existing data in the target Service Manager system including the database keys.
Dbdict Load Options — Create file if not there, and update if present, including SQL mapping	Select this option if the data saved in this external file should replace existing data in the target Service Manager system including the database SQL mapping.
Dbdict Load Options — Create file if not there, and update if present, including keys AND SQL mapping	Select this option if the data saved in this external file should replace existing data in the target Service Manager system including the database SQL mapping and the database keys.
Record Load Options — Add/update records	Select this option if the records saved in this external file are to replace any existing records in the target Service Manager system.
Record Load Options — Only add new records	Select this option to ignore records saved in the external file that already exist in the target Service Manager system.

11. Click **Unload/export**.

Related topics

Data maps
Exporting records
Export descriptor records
Unload files
Importing records
Purging and archiving records
Unload script utility
Load/unload implications of case-sensitivity
Add an export descriptor record
Export a record list to a text file
Export a record list to an Excel file
Export a record list using an export description record
Export an individual record using an export descriptor record
Export selected records from a table using an export descriptor record
Schedule the exporting of records using an export descriptor record
Schedule the unloading of records
Unload a database dictionary only
Unload a record list
Unload records from multiple files
Unload selected records from a file
Update an export descriptor record
Unload error messages

7.10.4.21. Unload records from multiple files

Applies to User Roles: System Administrator

To unload records from multiple files:

1. Click **Tailoring** > **Unload Script Utility**.

2. Fill in the following fields.

Field	Description
Unload Script	Type a name or label to identify your unload script. This name can include spaces.
Unload?	Select this option to create an external unload file from the records queried by this script.
Purge?	Select this option to delete the records queried by this script.
Show Unload Records	Select this option to display the Filename, Query, and Data Map fields. These fields enable you to specify the tables and queries to use when unloading records.
Filename	Specify the name of the table to unload records. This field is visible only if you select **Show Unload Records**.
Query	Type the SQL query to select records for unloading. You can use Service Manager operators and variables in this field. This field is visible only if you select **Show Unload Records**.
Datamap	Specify true or false to specify whether Service Manager should use the data map file to unload any associated records. You must have previously defined a data map for the listed file in order to use this feature. By default, Service Manager reads a blank entry in this field as false. This field is only visible if you have select **Show Unload Records**.
Show Protected Formats	Select this option to display the Protected Formats field. This field enables you to specify the forms to protect from purging.
Protected Formats	Type the names of the forms to protect when you purge records from this script.

3. Click **Add**.

Related topics

Data maps
Exporting records
Export descriptor records
Unload files
Importing records
Purging and archiving records
Unload script utility
Load/unload implications of case-sensitivity
Add an export descriptor record
Export a record list to a text file
Export a record list to an Excel file
Export a record list using an export description record
Export an individual record using an export descriptor record
Export selected records from a table using an export descriptor record

Schedule the exporting of records using an export descriptor record
Schedule the unloading of records
Unload a database dictionary only
Unload a record list
Unload an individual record
Unload selected records from a file
Update an export descriptor record
Unload error messages
Load error messages and notifications

7.10.4.22. Unload selected records from a file

Applies to User Roles: System Administrator

To unload selected records from a file:

1. Click **Tailoring** > **Database Manager**.
2. Type the form or file name that contains the records to unload.
3. Select the **Administration Mode** option. You must select this option to ensure that Service Manager displays the unload options in the target form.
4. Click **Search**.
5. Select the form used to search for records.
6. Click **More** or the More Actions icon, and then select **Export/Unload**.
7. Fill in the following fields.

Field	Description
Field Name	Type the name of the field used as the selection criteria. This field contains the values you list in the **Field Values** field.
Field Values	Type a unique value for each record to unload from the **Field Name**.

8. Click **Unl Records**.
9. Select the form, click **More** or the More Actions icon, and then select **Mass Unload**.
10. Fill in the following fields.

Field	Description
External File Name	Type the name and path of the file used to save the record list. By default, this file is located in the RUN folder of the Service Manager server.
Member	This field is obsolete.
Unload Type	Select the operating system file type to use for the external file.
Append to File	Select this option to add records to the end of an external file of the same name. If a external file of the same name doesn't exist, Service Manager creates a new file. If this field is cleared, then Service Manager overwrites any existing file of the same name.
Use Data Map	Select this option to use a pre-defined data map when unloading the selected records. Using a data map may result in Service Manager unloading more records than you originally selected to unload.
Prompt for Data Map	Select this option to enable a confirmation message to use a data map for each record unloaded. This option is only valid if the **Use Data Map** option is also enabled.
Suppress Messages	Select this option to hide all messages that the unload process generates.
Export Mode — Binary	Select this option to save the external file in Service Manager's binary unload format. Only the Service Manager Database Manager can read this binary file format.
Export Mode — Text	Select this option to save the external file in Service Manager's text format. This format is human readable but is not intended for export to other applications other than Service Manager's Database Manager.
Export Mode — Formatted Text	Select this option to save the external file in a user-defined text format. This format is intended for export to other applications. If you select this option, you must type a valid Service Manager form name in the **Form Name** field.
Form Name	Type the name of the form to use when exporting records as formatted text. This option is valid only when you select the **Formatted Text** option. If the text format description doesn't already exist, Service Manager prompts you to create it.
Dbdict Load Options — Create file if not there, and update if present	Select this option if the data saved in this external file should replace existing data in the target Service Manager system.

Dbdict Load Options — Create file if not there, but don't update if present	Select this option to ignore data saved in the external file that already exists in the target Service Manager system.
Dbdict Load Options — Create file if not there, and update if present, including keys	Select this option if the data saved in this external file should replace existing data in the target Service Manager system including the database keys.
Dbdict Load Options — Create file if not there, and update if present, including SQL mapping	Select this option if the data saved in this external file should replace existing data in the target Service Manager system including the database SQL mapping.
Dbdict Load Options — Create file if not there, and update if present, including keys AND SQL mapping	Select this option if the data saved in this external file should replace existing data in the target Service Manager system including the database SQL mapping and the database keys.
Record Load Options — Add/update records	Select this option if the records saved in this external file are to replace any existing records in the target Service Manager system.
Record Load Options — Only add new records	Select this option to ignore records saved in the external file that already exist in the target Service Manager system.

11. Click **Unload/export**.
    Service Manager saves the select records to an external file.

Related topics

Data maps
Exporting records
Export descriptor records
Unload files
Importing records
Purging and archiving records
Unload script utility
Load/unload implications of case-sensitivity
Add an export descriptor record
Export a record list to a text file
Export a record list to an Excel file
Export a record list using an export description record
Export an individual record using an export descriptor record
Export selected records from a table using an export descriptor record
Schedule the exporting of records using an export descriptor record
Schedule the unloading of records
Unload a database dictionary only
Unload a record list
Unload an individual record
Unload records from multiple files
Update an export descriptor record
Unload error messages
Load error messages and notifications

7.10.4.23. Update an export descriptor record

Applies to User Roles: System Administrator

To update an export descriptor record:

1. Select the records to export.
2. From the file.prompt.dbu.g form, also known as the unload/export utility, click **Export Ext**.
3. If Service Manager displays a list of existing export descriptor records, select the export descriptor record to use.
4. Click **Edit Record**.
5. Fill in the following fields.

Field	Description
Export Descriptor Record	Type the name used to identify this record uniquely.
File Name	Type the name of the Service Manager table used to export records.
Format Control Record Name	Type the name of the optional Format Control record to use with the export process. This Format Control record determines how Service Manager formats exported records.
Field Names	Type the database dictionary name of the Service Manager field used to export records. If the field is part of a structure, include the structure name before the field name. If the field is part of an arrayed structure, you only need to type the field name.
Field Type	Service Manager fills in the value of this field automatically based on the database dictionary definition of the field. You don't need to manually type any information in this field.
Start Pos	Type the number of space characters between the start of a line and the information Service Manager exports from the selected field. This information determines which column and line number used to export the field information. Most external files use a standard of 80 characters per line. If you type a value greater than 80 then each record uses multiple lines.
Fld Lng	Type the number of characters Service Manager has to write information from the exported field into the external file.
Array Type	Service Manager fills in the value of this field automatically based on the database dictionary definition of the field. You don't need to manually type any information in this field.
Array Occ	Type the number of elements from an array to be exported for this field. You can use the Array Occ value to determine the start position of the next field. For example, if you want to export five elements from an array and you allocated 80 character spaces per element, this field requires a total of 400 character spaces. The next field should start more than 400 characters spaces from the start of the file.

Date Format	Type the date and time format for date and time fields. You can use the following tokens to represent units of time: *dd* — two digit day *mm* — two digit month *yy* — two digit year *yyyy* — four digit year *hh* — hours *ii* — minutes *ss* — seconds You can use any character to separate these tokens. For example, mm/dd/yyyy hh:ii:ss exports dates and times with slashes between the month, day, and four-digit year and colons between the hours, minutes, and seconds. **Note**: If you don't include a time format, then Service Manager doesn't export any time information.

6. Click **Save**.

Related topics

Data maps
Exporting records
Export descriptor records
Unload files
Importing records
Purging and archiving records
Unload script utility
Add an export descriptor record
Export a record list to a text file
Export a record list to an Excel file
Export a record list using an export description record
Export an individual record using an export descriptor record
Export selected records from a table using an export descriptor record
Schedule the exporting of records using an export descriptor record
Schedule the unloading of records
Unload a database dictionary only
Unload a record list
Unload an individual record
Unload records from multiple files
Unload selected records from a file
Unload error messages
Load error messages and notifications

7.10.5. Importing records

A system administrator can import Service Manager records from an external unload file using the Database Manager. Service Manager can import records from all three formats of unload files (binary unload, text unload, and formatted text unload) as well as from two formats of text files (fixed-width and character-delimited text files). You can import records into Service Manager as a manual foreground process or as an automated scheduled background process.

How Service Manager imports records depends upon the format of the source file. When importing unload files, Service Manager uses the database dictionary stored in these files to determine where and how to import records into Service Manager. The database dictionary inside an unload file determines whether Service Manager updates or ignores duplicate records in the existing system. When importing text files, Service Manager requires an import descriptor record to tell it

where and how to import records. You can define an import descriptor record during the import process or prepare one in advance.

An import descriptor record defines:

- The external file source
- The internal destination table
- The format used to identify fields in the imported file
    - Fixed-width fields
    - Character-delimited fields
- The rules for handling duplicate records
    - Ignore duplicate records
    - Replace duplicate records
    - Update duplicate records
- The optional Format Control record to apply
- The names of destination fields for import records
- The date format to use
- The number of elements from an array to import

Related topics

Data maps
Exporting records
Import descriptor records
Unload files
Purging and archiving records
Unload script utility
Add an import descriptor record
Import a character-delimited text file
Import a fixed-width text file
Load an unload file
Schedule the importing of records using an import descriptor record
Schedule the loading of an unload file
View the contents of an unload file
Load error messages and notifications

7.10.5.1. Import descriptor records

You can create an import descriptor record to define the rules for importing data from external files into Service Manager. Each import descriptor record describes the following information:

- The external file source
- The internal destination table
- The format used to identify fields in the imported file
    - Fixed-width fields
    - Character-delimited fields
- The rules for handling duplicate records
    - Ignore duplicate records
    - Replace duplicate records
    - Update duplicate records
- The optional Format Control record to apply
- The names of destination fields for import records
- The date format to use
- The number of elements from an array to import

Note: You can't import the database dictionary definition of fields from an external file. To import database dictionary field definitions, you must use the native Service Manager unload format.

When you create an import descriptor record, Service Manager validates the records in the external file against the Service Manager database dictionary. If the validation is unsuccessful, Service Manager displays the message: `Validation failed`

You can click on the warning message to see the reasons why the data validation failed. Although Service Manager enables you to save an invalid import description record, you can't use this record to import data. You must correct any validation errors before importing data. After Service Manager validates the file, it imports one record for each entry in the external file.

Related topics

Data maps
Exporting records
Importing records
Unload files
Purging and archiving records
Unload script utility
Add an import descriptor record
Import a character-delimited text file
Import a fixed-width text file
Load an unload file
Schedule the importing of records using an import descriptor record
Schedule the loading of an unload file
View the contents of an unload file
Load error messages and notifications

7.10.5.2. Load error messages and notifications

Service Manager displays an error message if it can't successfully load an external file. The following list contains load error messages and notifications.

Load error message or notification	Description
<*WINDOWS.FILENAME*> *or* <*UNIX.FILENAME*> file loaded.	Service Manager has finished loading the named file.
File(s) loaded from external file: <*WINDOWS.FILENAME*> *or* <*UNIX.FILENAME*>	Service Manager has finished loading the named files.
<*Number*> records loaded into <*Database name*> from external file: <*WINDOWS.FILENAME*> *or* <*UNIX.FILENAME*>	Service Manager has finished loading <*Number*> records from the named file.
Loading <*Database name*> from external file: <*WINDOWS.FILENAME*> *or* <*UNIX.FILENAME*>.	Service Manager has started loading <*Database name*> from the named file.
<*Number*> records have been loaded so far into the <*Database name*> file.	The load utility generates this message for every hundredth record it attempts to load into the named Service Manager database.
The config record (database load/unload) is missing.	The hardware configuration record database load/unload.unix required for database loads and unloads has been deleted or renamed. It must be replaced. Refer to Unload/Load for more information.
The external file <*WINDOWS.FILENAME*> *or* <*UNIX.FILENAME*> is being written to and can't be read at this time.	The load utility could not access the named external file because another task has a lock on it.
Invalid end of file encountered in external file <*WINDOWS.FILENAME*> *or* <*UNIX.FILENAME*>.	The load utility encountered an incomplete file. Typically, this means that the unload process was interrupted before completion.
All null keys:file: <*Database name*>;rec no:<*Number*> in external file: <*WINDOWS.FILENAME*> *or* <*UNIX.FILENAME*>.	The load utility attempted to add record number <Number> to the named Service Manager database and encountered all null keys.
Invalid null key;file:<*database name*>;rec no:<*Number*> in external file <*WINDOWS.FILENAME*> *or* <*UNIX.FILENAME*>.	The load utility attempted to add record number <*Number*> to the named Service Manager database and encountered an invalid null key.
Invalid duplicate key:file:<*Database name*>;rec no: <*Number*> in external file <*WINDOWS.FILENAME*> *or* <*UNIX.FILENAME*>.	The load utility attempted to add record number <*Number*> to the named Service Manager database and encountered an invalid duplicate key.
External file: %S is not in Unload Format.	The named file is not in file format required by the load utility.
You have duplicated column headers: <column names>	When duplicate source columns are detected in the external source file, warns the user about the column names.
You have specified data mapping between the target fields and source columns. This will disable RAD expressions.	If the user has specified a correct mapping between target fields and source columns, when the user specifies a RAD expression, warns the user that RAD expressions will not work in this situation.
Column not found in external file, column: <columns names>	When a specified column is not found in the source file, warns the user about the missing column.

Line <line id> is empty. Skipped the line.	When the line being processed is empty, warns the user about the line number.
Error occurs when processing line <line id>.	When an error occurs during the processing of a line, warns the user about the line number.
Record already exists on line<line id>. Updated the record.	When a record already exists during the processing of a line, warns the user about the line number.
<processed rows count> records processed.	When every 100 rows have been processed, notifies the user about the current number of rows processed.

Related topics

Importing records

7.10.5.3. Add an import descriptor record

Applies to User Roles: System Administrator

Service Manager enables you to add import descriptor records for two types of text files: character-delimited, and fixed-width.

Add an import descriptor record for a character-delimited text file.

To add an import descriptor record for a character-delimited text file, follow these steps:

1. Click **Tailoring** > **Database Manager**.
2. Click **More** or the More Actions icon, and then select **Text Import Wizard**.
3. Fill in the following fields.

Field	Description
File to Import	Type or browse to the path and name of the external text file to import. **Note** Note Only Windows Dos text file format is supported.
Member	This field is obsolete.
Destination Table	Type the database dictionary name of the table receiving the imported records.
File Type	Select the source operating system of external file. **Note** Note The Winnt file type is recommended.

4. Click **Search**.
   Service Manager searches for the external file and displays one of the following messages:
   - If Service Manager finds the external file but there is not existing import descriptor record, Service Manager displays the form to create a new import descriptor.
   - If Service Manager finds the external file and also finds one or more import descriptor records, Service Manager displays a list of import descriptor records.

- If Service Manager can't find the external file, it displays an error message.

5. If Service Manager displays a list of import descriptor records, do one of the following:
   - Click an existing import descriptor record.
   - Click **New**.
6. From the **Creating a New Import Descriptor** form, select **Character-delimited** for **Input File Type**.
7. Click **Proceed**.
8. Fill in the following import data definitions.
   ;

Field	Description
Import Name	Type the name used to save the import descriptor record.
Import Table	This field lists the destination table you selected earlier in the wizard. This is a read-only field.
Import Mode	Select one of the following options: ○ **Add**: Select this option to only add new records. ○ **Add/Replace**: Select this option to both add new records and replace existing records. You may also select this option to update fields that you define in the **Advanced** tab. ○ **Replace only**: Select this option to only replace existing records. **Note** Note If you use field mapping for importing, Service Manager always uses the **Add/Replace** import mode regardless of the mode you specified.
Format Control	Type the name of the optional Format Control record used to format the incoming records. **Note** Note For the Format Control record, only the following sections are supported: **Calculations**, **JavaScript**, and **Validations**.

Delimiter	Service Manager only displays this field on character-delimited imports. Select one of the following options: ○ Comma: Select this option if the import file uses a comma (,) between fields. ○ Tab: Select this option the import file uses a tab character between fields. ○ Semicolon: Select this option if the import file uses a semicolon (;) between fields. ○ Other: Select this option to specify the character the import file uses between fields. **Note** Note **Delimiter** is set to **Comma** by default. However, when you add a descriptor for the import file, if your source file uses a comma as the delimiter, you must select other delimiters first and then reselect Comma. Alternatively, you can first click **Add** to add a descriptor, and then click **Back** to return to the descriptor. You need to do so to trigger the retrieval of the source columns from the source file using the specified delimiter. Otherwise, you will see a blank Source Column list.
Text Qualifier	Service Manager only displays this field on character-delimited imports. Specify the text qualifier the import file uses to include special characters such as text delimiters in the record data.
Array Separator	You can specify this field to split array data in your source file. Service Manager splits the array data based on the Array Type field. Select one of the following separators for your array data: ○ Semicolon: Select this separator if the source file uses a semicolon to split the array data. ○ Line Break: Select this separator if the source file uses a line break to split the array data. For example, when data is stored as 'va;vb', and **Semicolon** is selected in the **Array Separator** drop-down list, Service Manager parses the value to ['va','vb'].
Field Name	Specify the database dictionary field name receiving the imported record data. Type one field name for each source field in the import file.

Source Column	This is a list of column headers in the source file. Service Manager automatically retrieves the list according to the delimiter you specified. If you specify a wrong delimiter, the source column retrieval fails. You should map each source column to a field in the target table. ○ If the mapping table contains a field that is not mapped to a source column, all mappings are ignored. Instead, all specified fields are automatically mapped to the source columns in their display order in the source file. ○ One source column can be mapped to multiple target fields. For example, a Service Catalog item can have the same value for **Name** and **Display Name**. **Name:** PC Hardware **Display Name:** PC Hardware If you are importing Service Catalog items into Service Manager, you can map the **Name** column in your source file to the **name** and **displayName** fields in the **svcDisplay** table in Service Manager, as shown below. **Field Name:** name **Source Column:** Name **Field Name:** displayName **Source Column:** Name ○ You can't specify duplicate field names in the mapping table, that is, each field name can only occur once in the mapping table. Service Manager returns an error if it detects a duplicate field name. ○ Only the specified source columns are imported.
Field Type	This is a read-only field and can't be edited. If Service Manager validates the data from the import file, it displays a number describing the data type.

Date Format	Type the date and time format used to import date and time fields. You can use the following tokens to represent units of time: ○ dd: two digit day ○ mm: two digit month ○ yy: two digit year ○ yyyy: four digit year ○ hh: hours ○ ii: minutes ○ ss ;: seconds **Note** ○ When you specify a mapping between the source columns and target fields, **yyyy** is unsupported. ○ When you load date type records, the date format is converted to the same as the login user's. For example, the date format in the external file is "mm/dd/yy" with the value of "07/12/14", and defined in the mapping table. However, the login user uses a date format of "yy/mm/dd". In this case, the date format is converted to "14/07/12" and saved in Service Manager.
Array Type	This is a read-only field and can't be edited. If Service Manager validates any array data from the import file, it displays a number describing the array type.
Occurrences	Type the number of array elements to import for this field.
Query	Type a query used to select an existing record from the destination Service Manager database for updating. If this query returns a single record, Service Manager updates this record with the data from the import file. If the query returns more than one record, then Service Manager uses the **Import Mode** setting to determine what action to take. **Note** Note The import utility stores the values of imported fields in the `$L.db` variable. You can use this variable to construct queries. For example: `not null(name in $L.bd)`

Skip	Type a query that identifies records to skip when adding or updating from the import file. If this query returns one or more records, Service Manager ignores adds or updates from records matching the query. To use this field you must also provide an entry in the **Query** field. **Note** Note ;The import utility stores the values of imported fields in the `$L.db` variable. You can use this variable to construct a skip condition query. For example: `null(name in $L.bd)`
Expressions	Type any RAD expressions used to identify records for importing. **Note** Note If you have specified a correct mapping between the target fields and source columns, RAD expressions are disabled.
Javascript	Specify Javascript code to do advanced import. The JavaScript expressions run against each record in the source file. For example, when importing your Outlook contacts data into the **contacts** table in Service Manager, you can combine the First Name and Last Name, and then import the combined value into the **contact.name** field in the **contacts** table. **First Name:** ;John **Last Name:** ;Smith After the combination, a contact record **John, Smith** is inserted into the **contacts** table in Service Manager. To achieve this, you can define the following statement in the **JavaScript** field: `target['contact.name'] = source['First Name'] + ',' + source['Last Name'];` where the **target** object represents the Service Manager target field, and the **source** object represents one record in the source file.
Update Field Names	Type the list of field names from the destination table used to uniquely identify a record. If Service Manager finds existing records matching these fields, it updates the record with the data from the import file.

Case Conversion	Select one of the following options to convert record data to a specific case: ○ None: Select this option if you don't want to convert the case of imported data. ○ Upper case: Select this option to convert imported data into all upper case. ○ Lower case: Select this option to convert imported data into all lower case.
Skip Lines	Specify the optional number of lines to skip at the beginning of the file. **Note** Note When you specify a mapping between the source columns and target fields, you must set **Skip Lines** to **1** which means the column headers is specified as the first line.

9. Click **Add**.

Add an import descriptor record for a fixed-width text file.

To add a fixed-width import descriptor record, follow these steps:

1. Click **Tailoring** ;> **Database Manager**.
2. Click **More** or the More Actions icon, the form and select **Text Import Wizard**.
3. Fill in the following fields.
   ;

Field	Description
File to Import	Type or browse to the path and name of the external text file to import.
Member	This field is obsolete.
Destination Table	Type the database dictionary name of the table receiving the imported records.
File Type	Select the source operating system of external file.

4. Click **Search**.
   Service Manager searches for the external file and displays one of the following messages:
   - If Service Manager finds the external file but there is not existing import descriptor record, Service Manager displays the form to create a new import descriptor.
   - If Service Manager finds the external file and also finds one or more import descriptor records, Service Manager displays a list of import descriptor records.
   - If Service Manager can't find the external file, it displays an error message.
5. If Service Manager displays a list of import descriptor records, do one of the following:
   - Click an existing import descriptor record.
   - Click **New**.
6. From the **Creating a New Import Descriptor** form, select **Fixed-width** for **Input File Type**.

Field	Description
Fixed-width	Select this option if each field in the file occupies a fixed-width position. For example, the following are two records using fixed width positions. The number field starts at position 1 and ends at position 6. The open.time field starts at position 8 and ends at position 15.

Position	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Record 1	I	M	1	0	4	2		1	2	/	2	9	/	0	0
Record 2	I	M	1	0	5	8		1	2	/	2	9	/	0	0

7. Click **Proceed**.
8. Fill in the following import data definitions.
   ;

Field	Description
Import Name	Type the name of the new import descriptor record.
Import Table	This field lists the destination table you selected earlier in the wizard. This is a read-only field.
Import Mode	Select one of the following options: ○ Add: Select this option to only add new records. ○ Add/Replace: Select this option to both add new records and replace existing records. You may also select this option to update fields that you define in the **Advanced** tab. ○ Replace only: Select this option to only replace existing records.
Format Control	Type the name of the optional Format Control record used to format the incoming records.
Field Names	Specify the database dictionary field names receiving the imported record data. Type one field name for each source field in the import file.
Field type	This is a read-only field and can't be edited. If Service Manager validates the data from the import file, it displays a number describing the data type.
Start Position	Service Manager only displays this field on fixed-width imports. Type the column number where data for this field starts.
Field Length	Service Manager only displays this field on fixed-width imports. Type the number columns that the data from this field uses.
Array Type	This is a read-only field and can't be edited. If Service Manager validates any array data from the import file, it displays a number describing the array type.
Occurrences	Type the number of array elements to import for this field.

Date Format	Type the date and time format used to import date and time fields. You can use the following tokens to represent units of time: ○ dd: two digit day ○ mm: two digit month ○ yy: two digit year ○ yyyy: four digit year ○ hh: hours ○ ii: minutes ○ ss: seconds You can use any character to separate these tokens. For example, mm/dd/yyyy hh:ii:ss imports dates and times with slashes between the month, day, and four-digit year and colons between the hours, minutes, and seconds.
Query	Type a query to select one existing record from the destination Service Manager database for updating. If this query returns a single record, Service Manager updates this record with the data from the import file. If the query returns more than one record, then Service Manager uses the **Import Mode** setting to determine what action to take.
Skip	Type a query to identify records to skip when adding or updating from the import file. If this query returns one or more records, Service Manager ignores adds or updates from records matching the query.
Expressions	Type any RAD expressions used to identify records for importing.
Update Field Names	Type the list of field names from the destination table to uniquely identify a record. If Service Manager finds existing records matching these fields, it updates the record with the data from the import file.
Case Conversion	Select one of the following options to convert record data to a specific case: ○ None: Select this option if you don't want to convert the case of imported data. ○ Upper case: Select this option to convert imported data into all upper case. ○ Lower case: Select this option to convert imported data into all lower case.
Skip Lines	Specify the optional number of lines to skip at the beginning of the file.

9. Click **Add**.

Delete or update an import descriptor record.

To delete or update an import descriptor record, follow these steps:

1. Click **Tailoring** ;> **Database Manager**.
2. Type `=import` in the **Form** field.
3. Select **Administration Mode**. You must select this option for Service Manager to display the **Delete** button.
4. Click **Search**.
5. Click the form where both the **Format Name** and **File Name** equal **import**.
6. Click **Search** again to display a list of import descriptor records.

> **Note**
> Note You may need to click **Document** ;> **Restore** to see the list of import descriptor records.

7. Select the import descriptor record you want to update or delete.
8. To update the descriptor record, update the fields and then click **Save**.
9. To delete the descriptor record, click **Delete**, and then select **Yes**.

Related topics

Data maps
Exporting records
Importing records
Import descriptor records
Unload files
Purging and archiving records
Unload script utility
Load error messages and notifications
Import a character-delimited text file
Import a fixed-width text file
Load an unload file
Schedule the importing of records using an import descriptor record
Schedule the loading of an unload file
View the contents of an unload file

7.10.5.4. Import a character-delimited text file

Applies to User Roles: System Administrator

Before you import a character-delimited text file, you need to prepare your source file and add a character-delimited import descriptor record first.

To import a character-delimited text file, follow these steps:

1. Prepare the source file. The following table describes the rules that external source files must observe.

Rule	Note
Only plaint text files are supported.	Be sure to not load binary files, such as .JPG and .XLS files.
If the source data contains non-ASCII characters, the external text file must use UTF-8 or Unicode encoding.	
Column headers in the external file are mandatory and must be unique.	CSV files with duplicate column headers can't be imported. The wizard displays a warning on the duplicate columns and is not able to proceed. Make sure your source file contains column headers on the first line, and remove or rename duplicate columns as needed.
Column headers in source files should follow the JavaScript Identifier Naming conventions.	Column headers in source files are used as JavaScript identifiers in this feature.
If the number of values in a row doesn't match the number of the column headers, the row is skipped. For example, if your source file contains 6 column headers, a row with more or less than 6 values is skipped.	Make sure your source file doesn't contain invalid rows.
When a source file is huge (more than 15, 000 lines or 10 MB in size), it may take a long time to load the file.	You are recommended to split a huge file to smaller ones before loading.
Digit grouping is unsupported and only numbers in raw format can be imported.	For example, "1, 000.000, 3" can't be correctly recognized. You must change it to "1000.0003".
Unique keys aren't automatically generated during the import operation.	Make sure that the records to import match the data constraints that are defined in their dbdict.
Delimiters in external CSV files can be one of the following: ○ a comma ○ a tab ○ a semicolon ○ a user-defined character other than listed above	When a field value contains the delimiter character, enclose the value in double quotes. For example: **"abc,def"**
The following date formats are supported for external source files: ○ mm/dd/yy [hh:ii:ss] ○ yy/mm/dd [hh:ii:ss] ○ dd/mm/yy [hh:ii:ss] **Note** Note [hh:ii:ss] is optional.	Make sure all of your source data uses the correct date format. You will need to specify the date format later when adding an import descriptor for the source file. **Note** Note After importing, the imported data is converted to the login user's date format.

2. Resolve necessary validations or dependencies first.

> **Note**
> Note Triggers are enabled during importing. For this reason, you may need to resolve some validations or dependencies first. For example, before loading a contact record, an operator record should be loaded first. Otherwise, a message, for example, “No operator record to sync”, pops up.

3. Launch the Text Import Wizard, and add an import descriptor record for the text file. For details, see Add an import descriptor record.
4. Click **Test** to test the importing of records.
5. Review the test results on the **Test Import** tab.

> **Note**
> Note Only the first ten lines in the external file are tested.

6. Click **Proceed**.
7. Click **Run**.
   Service Manager imports the external file.

> **Note**
> - Scheduled import is unsupported for character-delimited files.
> - Some joinfiles need to be loaded completely to achieve desired data import results. For example, when loading service catalog items, you need to load the service catalog display records concurrently. Otherwise, the imported records aren't displayed correctly in the catalog.

Example: Import Outlook contacts into the contacts table

The following example describes how to import your Outlook contacts into the contacts table.

1. In Outlook 2010, export your contacts to a `.csv` file. To do this, follow these steps:
   1. Click **File** > **Open** > **Export** to open the Import and Export wizard form.
   2. Select **Export to a file**, and then click **Next**.
   3. Select **Comma Separated Values (Windows)**, and then click **Next**.
   4. Select **Contacts**, and then click **Next**.
   5. Click **Browse** to specify the location and file name for your .csv file, and then click **Next.**
   6. Click **Finish**.
2. In Service Manager, type db in the command line field, and then press **Enter**. Database Manager opens.
3. Click **More** or the More Actions menu, and then select **Text Import Wizard**. The Import wizard form opens.
4. In the **File to import** field, browse to your `.csv` file.
5. In the **Destination table** drop-down list, type or select contacts.
6. In the **File Type** drop-down list, select **winnt**.
7. Click **Search**.
8. If a descriptor for the contacts table exists, click **New** to create a descriptor for your import. Otherwise, the Creating a New Import Descriptor page automatically opens.

> **Note**
> Note The **Import Table** field is automatically populated with contacts.

9. In the **Input File Type** field, select **Character-delimited**, and then click **Proceed**.
10. In the **Import Name** field, specify a name for the descriptor (for example, Outlook Contacts Import), and then click **Add**.

Note

Note If you want to delete an existing descriptor, see Add an import descriptor record

11. In the **Import Mode** drop-down list, select **Add/Replace**.
12. In the **Delimiter** radio button, select **Other**, and then select **Comma**. This is because your Outlook contacts were exported as Comma Separated Values. If this is not the case, specify another delimiter that was used to export your Outlook data.

Note

Note The default Delimiter is Comma. However, you need to select another value and then select Comma again to activate the retrieval of the source columns in your .csv file.

13. In the **Format Control** drop-down list, select **fcContacts**.
14. On the **Fields** tab, complete the field mapping table. The following is an example.

Field	Source Column
title	Job Title
contact.name	First Name
email	E-mail Address
company	Company
manager	Manager's Name
location	Location
dept	Department
portable.phone	Mobile Phone

Note

Note Only the specified source columns are imported. Make sure that you follow these rules, otherwise, errors occur. You can't specify a duplicate field in the destination table. However, you can map a source column to multiple fields. A field can't have an empty source column mapped to it.

15. On the **Advanced** tab, add the following lines to the JavaScript box:

```
target['contact.name']=source['First Name']+'.'+source['Last Name'];
target['company']='HP1';
```

16. Click **Save**.
17. Click **Proceed**.
18. Click **Run**.
    The import starts. When it has finished, a message is displayed indicating how many records have been successfully inserted. You can go to the contacts table to check the import results.

Related topics

Data maps
Exporting records
Importing records
Import descriptor records

Unload files
Purging and archiving records
Unload script utility
Add an import descriptor record
Import a fixed-width text file
Load an unload file
Schedule the importing of records using an import descriptor record
Schedule the loading of an unload file
View the contents of an unload file
Load error messages and notifications

7.10.5.5. Import a fixed-width text file

Applies to User roles: System Administrator

Before you import a fixed-width text file, you need to add a fixed-width import descriptor record first.

To import a fixed-width text file, follow these steps:

1. Add an import descriptor for the fixed-width text file. For details, see Add an import descriptor record
2. Click **Test** to test the importing of records.
3. On the **Test Import** tab, review the test results.

   Note
   Note
   Only the first ten lines in the external file are tested.

4. Click **Proceed**.
5. Click **Run**.
   Service Manager imports the external file.

Related topics

Data maps
Exporting records
Importing records
Import descriptor records
Unload files
Purging and archiving records
Unload script utility
Add an import descriptor record
Import a character-delimited text file
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Schedule the loading of an unload file
View the contents of an unload file
Load error messages and notifications

7.10.5.6. Load an unload file

Applies to User roles: System Administrator

If your application version is 9.30 ap3, 9.31 or later, you are recommended to use Unload Manager to load an unload file, because Unload Manager can help you create a backup of your old data and reconcile conflicts during the installation of the unload; if your application version is other than any of these, Unload Manager is unavailable and you can use Database Manager instead.

Note

Note

You can use the Unload Manager on Windows client only.

To load an unload file using Unload Manager, follow these steps:

1. Go to **System Administration** > **Ongoing Maintenance** > **Unload Manager**.
2. Double-click **Apply Unload**. A wizard opens.
3. Select the unload file you want to apply, also specify a backup file, and then click **Next**. Details of the unload file appear.
4. Double-click a conflicting object in the table to open the merge tool:
    1. Merge the object, and then select the **Reconciled** checkbox.
    2. Click **Save** to go back to the wizard.
5. Click **Next** after all the conflicting objects are reconciled.
6. Click **Yes** on the confirmation window to apply the unload.
7. Click **Finish**.
   Now, the unload has been applied and at the same time your old data backed up.

To load an unload file using Database Manager, follow these steps:

1. Click **Tailoring** > **Database Manager**.
2. Click **More** or the More Actions menu, and select **Import/Load**.
3. Fill in the following fields.

Field	Description
File Name	Type the name and path of the file to load.
Import Descriptor	Type the name of the optional Import Descriptor record used to import records. Since unloads files don't require an Import Descriptor record, leave this field blank.
File Type	Select the source operating system of the unload file.
Messages Option — All Messages	Select this option to see all messages that Service Manager generates loading the file.
Messages Option — Totals Only	Select this option to see only the total number of files Service Manager loads.
Messages Option — None	Select this option to hide all messages that Service Manager generates loading the file.

   Note: You can view the contents of an unload file before importing it by clicking **List Contents**.
4. Click **Load FG**.

Related topics

Data maps
Exporting records
Import descriptor records
Importing records
Purging and archiving records
Unload files
Unload script utility
Add an import descriptor record

Import a character-delimited text file
Import a fixed-width text file
Schedule the importing of records using an import descriptor record
Schedule the loading of an unload file
View the contents of an unload file
Load error messages and notifications

7.10.5.7. Schedule the importing of records using an import descriptor record

Applies to User roles: System Administrator

Before you schedule the importing of records, you may want to add an import descriptor record first, refer to Add an import descriptor record.

To schedule the importing of records using an import descriptor record, follow these steps:

1. Open the import descriptor record you want to schedule, and then click **Proceed**.
2. In the **Expiration** field, type the date and time when the scheduled import runs.
3. To repeat at regular intervals, select one of the following options.

Field	Description
Repeat — Monthly	Select this option to repeat the import process every once every month from the Expiration date.
Repeat — Quarterly	Select this option to repeat the import process every once every quarter from the Expiration date.
Repeat — Semi-Annually	Select this option to repeat the import process every once every six months from the Expiration date.
Repeat — Annually	Select this option to repeat the import process every once every year from the Expiration date.

4. Click **Schedule**.

Related topics

Data maps
Exporting records
Importing records
Import descriptor records
Unload files
Purging and archiving records
Unload script utility
Load/unload implications of case-sensitivity
Add an import descriptor record
Import a character-delimited text file
Import a fixed-width text file
Load an unload file
Schedule the loading of an unload file
View the contents of an unload file
Unload error messages
Load error messages and notifications

7.10.5.8. Schedule the loading of an unload file

Applies to User roles: System Administrator

To schedule the loading of an unload file:

1. Click **Tailoring** > **Database Manager**.

2. Click **More** or the More Actions icon, and then select **Import/Load**.
3. Fill in the following fields.

Field	Description
File Name	Type the name and path of the file to load.
Member	This field is obsolete.
Import Descriptor	Type the name of the Import Descriptor record used to import records. Because unload files don't require an Import Descriptor record, leave this field blank.
File Type	Select the source operating system of unload file.
Messages Option — All Messages	Select this option to see all messages that Service Manager generates loading the file.
Messages Option — Totals Only	Select this option to see only the total number of files Service Manager loads.
Messages Option — None	Select this option to hide all messages that Service Manager generates loading the file.

4. Click **Load BG**.
5. Fill in the following fields.

Field	Description
Expiration Date	Specify the date and time the scheduled process runs. Service Manager loads the file at this date and time. By default, Service Manager uses the current date and time.
Repeat Interval	Select the time period to wait before repeating the load process. Type the interval in the following format: *<days><hours>:<minutes>:<seconds>*. For example, 1 00:00:00 is a repeat interval of 1 day.
External Filename	Type the name and path to the external file to load. By default, Service Manager looks for unload files in the server RUN folder.
Member	This field is obsolete.
Suppress Messages?	Type true or false to show or hide messages Service Manager generates loading the external file.

6. Click **Load BG**.

Related topics

Data maps
Exporting records
Importing records
Import descriptor records
Unload files
Purging and archiving records
Unload script utility

Add an import descriptor record
Import a character-delimited text file
Import a fixed-width text file
Load an unload file
Schedule the importing of records using an import descriptor record
View the contents of an unload file
Unload error messages
Load error messages and notifications
Load/unload implications of case-sensitivity

7.10.5.9. View the contents of an unload file

Applies to User roles: System Administrator

To view the contents of an unload file:

1. Click **Tailoring** > **Database Manager**.
2. Click **More** or the More Actions icon, and then select **Import/Load**.
3. Fill in the following fields.

Field	Description
File Name	Type the name and path of the file you want to load.
Member	This field is obsolete.
Import Descriptor	Type the name of an Import Descriptor record used to import records. Because unload files don't require an Import Descriptor record, you can leave this field blank.
File Type	Select the source operating system of the unload file.
Messages Option — All Messages	Select this option to see all messages that Service Manager generates while loading the file.
Messages Option — Totals Only	Select this option to see only the total number of files Service Manager loads.
Messages Option — None	Select this option to hide all messages that Service Manager generates loading the file.

4. From the More Actions menu, click **List Contents**.

Related topics

Data maps
Exporting records
Importing records
Import descriptor records
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Purging and archiving records
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Add an import descriptor record
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Import a fixed-width text file
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Unload error messages
Load error messages and notifications

7.10.6. Unload script utility

The unload script utility enables system administrators to create Service Manager unload files automatically. The unload script utility enhances the standard unload creation process in many ways. With unload scripts, you can:

- Save records from multiple tables into a single unload file.
- Specify a query for each source table that filters the records added to the unload file.
- Purge records during the unload process.
- Specify which formats to protect during a purge process.
- Add related records from the data map file to the unload file.

The unload script utility is available in Tailoring. By default, Service Manager includes a collection of unload scripts that you can use for common unload tasks. You can also use the default unload scripts as templates to create your own customized unload scripts.

To create and use unload scripts effectively, you must be familiar with the Service Manager Database Manager. You can create a query on any field in the file, but querying keyed fields improves system performance and response time. You can create record queries using Structured Query Language (SQL) syntax.

You can use the Purge/Archive utility to unload and purge records from one file; if you use datamaps, you can use Purge/Archive to unload and purge related records as well.

After you load a .unl file to import data into Service Manager, you must stop and then restart the Service Manager server. This ensures that the table changes are initialized.

Note: The Unload Script utility unloads tables using the binary unload file format.

Related topics

Tools to use for promoting customizations
Data maps
Environment configuration
Exporting records
Importing records
Purging and archiving records
Using data maps to purge and archive records
Add an unload script record
Protect forms from the unload script utility
Purge records with the unload script utility
Run an unload script
Validate an unload script record

7.10.6.1. Add an unload script record

Applies to User Roles: System Administrator

To add an unload script record:

1. Click **Tailoring** > **Unload Script Utility**.
2. Fill in the following fields.

Field	Description
Unload Script	Type a name or label to identify your unload script. This name can include spaces.
Unload?	Select this option to create an external unload file from the records queried by this script.
Purge?	Select this option to delete the records queried by this script.
Show Unload Records	Select this option to display the **Filename**, **Query**, and **Data Map** fields where you can specify the tables and queries for unloading or purging records.
Filename	Specify the name of the tables unloading or purging records. This field is visible only if you select **Show Unload Records**.
Query	Type the SQL query to identify the records to unload or purge. You can use Service Manager operators and variables in this field. This field is visible only if you select **Show Unload Records**.
Datamap	Specify true or false to specify whether Service Manager should use the data map file to unload or purge any associated records. You must have previously defined a data map for the listed file in order to use this feature. By default, Service Manager reads a blank entry in this field as false. This field is only visible if you select **Show Unload Records**.
Show Protected Formats	Select this option to display the Protected Formats field where you can specify the forms to protect from purging.
Protected Formats	Type the names of the forms to protect purging records.

3. Click **Add**.

To delete or update an unload script record:

1. Click **Tailoring** > **Unload Script Utility**.
2. Fill in optional search criteria, and then click **Search**.
3. Select the unload script to be deleted, and then click **Delete**.
4. Click **Yes** to confirm the deletion.

Related topics

Data maps
Exporting records
Importing records
Purging and archiving records
Unload script utility
Protect forms from the unload script utility
Purge records with the unload script utility
Run an unload script
Validate an unload script record

7.10.6.2. Protect forms from the unload script utility

Applies to User Roles: System Administrator

You can protect a form from being purged by the unload script utility by explicitly listing the form in the protected formats list.

1. Click **Tailoring** > **Unload Script Utility**.
2. Fill in optional search criteria, and then click **Search**.
3. Select the unload script to update.
4. Select the **Show Protected Formats** options.
5. In the **Protected Formats** field, type the names of the forms to protect when you purge records from this script.
6. Click **Save**.

Related topics

Data maps
Exporting records
Importing records
Purging and archiving records
Unload script utility
Add an unload script record
Purge records with the unload script utility
Run an unload script
Validate an unload script record

7.10.6.3. Purge records with the unload script utility

Applies to User Roles: System Administrator

To purge records with the unload script utility:

1. Click **Tailoring** > **Unload Script Utility**.
2. Fill in optional search criteria, and then click **Search**.
3. Select the unload script to run.
4. Select the **Purge?** option.
5. Select the **Show Protected Formats** options.
6. In the **Protected Formats** field, type the names of the forms to protect when you purge records from this script.
7. Click **Proceed**.
8. Click **Yes**.

Related topics

Data maps
Exporting records
Importing records
Purging and archiving records
Unload script utility
Add an unload script record
Protect forms from the unload script utility
Run an unload script
Validate an unload script record

7.10.6.4. Run an unload script

Applies to User Roles: System Administrator

To run an unload script:

1. Click **Tailoring** > **Unload Script Utility**.
2. Fill in optional search criteria.
3. Click Search.

4. Select the unload script to run.
5. Click **Proceed**.
6. Fill in the following fields.

Field	Description
External File Name	Type the name of the unload file to create. This name can include the path information but must use characters that are valid for the target operating system. By default, Service Manager saves the unload file in RUN folder of the server. If users have enabled client-side unloading, then the Service Manager saves the unload file on their Windows desktop.
Append to File?	Select this option to add records to an existing unload file.
When loading records into an existing database dictionary — Use existing database dictionary	Select this option to use the table and field definitions stored in the database dictionary of the target system.
When loading records into an existing database dictionary — Use database dictionary of loaded record	Select this option to use the table and field definitions stored in the unload file. This option will overwrite the existing database dictionary, if any, for the loaded file and records.
When loading records — Add new records and update existing records	Select this option to update existing records in the target system with records from the unload file. The new records will be inserted into the table and the existing records will be updated regardless of case-sensitivity.
When loading records — Add new records only	Select this option to ignore existing records and add only new records from the unload file. The new records will be inserted into the table regardless of case-sensitivity.

7. Click **Proceed**.
   Service Manager unloads the tables and records you selected in the unload script.

After you load a .unl file to import data into Service Manager, you must stop and then restart the Service Manager server. This ensures that the table changes are initialized.

Note: The unload script utility unloads tables using the binary unload file format.

Related topics

Data maps
Exporting records
Importing records
Purging and archiving records
Unload script utility
Load/unload implications of case-sensitivity
Add an unload script record
Protect forms from the unload script utility
Purge records with the unload script utility
Validate an unload script record
Unload error messages

7.10.6.5. Validate an unload script record

Applies to User Roles: System Administrator

To validate an unload script record:

1. Click **Tailoring** > **Unload Script Utility**.
2. Use search or advanced search to find one or more records.
3. Select the unload script to validate.
4. Click **More** or the More Actions icon for the record, and then select **Validate**.

Related topics

Data maps
Exporting records
Importing records
Purging and archiving records
Unload script utility
Add an unload script record
Protect forms from the unload script utility
Purge records with the unload script utility
Run an unload script
Unload error messages
Load error messages and notifications

7.10.7. Purging and archiving records

The purge and archive function lets you save Service Manager records to an external file and then delete those records from the database. You can use the purge and archive function to archive data as part of your backup plan or to free up space on your system. You can access the purge and archive function manually or as part of a scheduled process.

You can use the purge and archive function in one of four ways:

- Archive records only
- Purge records only
- Archive and then purge records
- Schedule an archive, a purge, or an archive and purge process

You can also use the purge and archive function in conjunction with data maps. When you define a data map, you declare certain records as having a mapping or connection to other tables and records. If you have created data maps, you can enable them during the purge and archive process to determine what additional records Service Manager should include in the process.

Related topics

Data maps
Exporting records
Importing records
Unload files
Using data maps to purge and archive records
Unload script utility
Archive records
Purge and archive records
Purge records
Schedule purge and archive processes

7.10.8. Database record auditing

Database record auditing allows you to check the Service Manager database for modifications to specified fields within a file. When you update records in a file, record auditing tracks the updates when Paging is unavailable, whereas Paging creates a unique record or page for every update.

Database Manager detects field modifications by comparing the input values in fields from the original version of a record to the values of the same fields in the updated version. When a modification is detected, Database Manager generates an Audit Log entry containing the following information:

- The name of the modified fields.
- The old and new version of the data.
- The date and time of the modification.
- The user ID of the operator who modified the record.

Related topics

Audit log file
Audit processing
Audit specifications file
Start database record auditing
Add an audit specifications record
Add lookup functionality to Format Control
Define an audit specifications entry
Invoke auditing
Open the audit specifications table
Set up event triggers
Test audit lookup functionality

7.10.8.1. Invoke auditing

Applies to User roles: System Administrator

You can invoke auditing from Format Control. If you are using the Change Management application, you can invoke auditing from Database Manager.

Select one of the following options to invoke auditing:

- Invoke auditing from Format Control
- Invoke auditing from the file in Database Manager (Change Management only)

7.10.8.2. Invoke auditing from Format Control

Applies to User roles: System Administrator

The topics below comprise an example of how to use Database record auditing. The steps in these topics must be followed in order. If you skip a step, Database record auditing may not work.

1. Open the audit specifications table.
2. Add an audit specifications record.
3. Define an audit specifications entry.
4. Invoke auditing.
5. Set up event triggers.
6. Add lookup functionality to Format Control.
7. Test audit lookup functionality.

To invoke auditing from Format Control, perform the following steps:

1. Create an Audit Specifications record for a particular file.
2. Open the Format Control record associated with the form and the file for which you created the audit specifications record in step 1.
3. Select the **Save Copy** option.
4. Click **Subroutines** to display the Subroutines form.
   The Subroutines panel opens.
5. Enter the desired format control. For this example, enter:

Field	Enter
Add	true
Upd	true
Before	true
Application	audit.compare
Error Message	Audit Processing could not complete.
Names	file second.file
Values	$file0 $file

Related topics

Database record auditing
Audit log file
Audit specifications file
Audit processing
Tools to use for promoting customizations
Environment configuration
The Subroutines form

7.10.8.3. Invoke auditing from the file in Database Manager (Change Management only)

Applies to User roles: System Administrator

The topics below comprise an example of how to use Database record auditing. Follow the steps in this order. If you skip a step, the example will not work.

1. Open the audit specifications table
2. Add an audit specifications record
3. Define an audit specifications entry
4. Invoke auditing
5. Set up event triggers
6. Add lookup functionality to Format Control
7. Test audit lookup functionality

This process is only valid for the Change Management application.

To invoke auditing from Database manager:

1. Open the **cm3rcatphase.main** form in Database Manager. A blank **cm3rcatphase.main** form opens.
2. Enter the name of any phase in the form. For this example, enter Analysis.
3. Click **Search**.
4. Set the **Audit Records** field value to true.
5. Click **Save**.

Related topics

Database record auditing
Audit log file
Audit specifications file
Audit processing
Invoke auditing from Format Control

Start database record auditing
Determine if a file has audit processing

7.10.8.4. Start database record auditing

Applies to User Roles: System Administrator

This list is an example of the sequence of events that must occur prior to initiating database record auditing. Each of these steps is required for auditing to function properly.

To start database record auditing:

1. Open the audit specifications table.
2. Add an audit specifications record.
3. Define an audit specifications entry.
4. Invoke auditing
5. Set up event triggers.
6. Add lookup functionality to Format Control.
7. Test audit lookup functionality

Related topics

Audit log file
Audit processing
Audit specifications file
Database record auditing
Add lookup functionality to Format Control
Add an audit specifications record
Define an audit specifications entry
Invoke auditing
Open the audit specifications table
Set up event triggers
Test audit lookup functionality
Key type definitions

7.10.8.5. Test Audit Lookup functionality

Applies to User roles: System Administrator

Some files may not be configured for Audit Lookup. In that case, you must add Audit Lookup functionality before Audit Lookup will work.

Service Manager invokes Audit Processing using the definitions provided in the Format Control, when a record is added or updated in a file for which auditing has been set up.

Note: Before using Audit Lookup, be sure that the Audit Specifications include all fields that need auditing.

To test Audit Lookup functionality:

1. Add a record to the contacts file using the values in the table below.

Field	Value
Contact Name	MILLER, JOHN
Employee ID	NEW00003
Last Name	Miller
First Name	John
Company	advantage
Email	john.miller@Advantage.com
Dept Name	Documentation

Note: Service Manager displays the following message: *"Contact Information record added."* The audit process is invoked. Service Manager doesn't open an audit record until a user changes the current field values.
Service Manager has recorded all fields specified in the Audit Specifications record. All Old values are NULL or contain no data because there is no previous version of the record; this is a new record.
Note: Audit Log entries are created in response to any changes made to the current values in this record. These audit records are displayed in the **audit.summ** form, when accessed using the Format Control option created earlier.
If you check Audit Lookup now, a message informing you that there are no audit records.

2. To complete the example, modify the record created in the previous steps by entering new values in the fields of the form that were specified as Unique A-D fields in the audit specifications file. For this example, change the Employee ID to NEW00004.
3. Click **Save** to commit the changes to the database.
   Service Manager displays the following message: *"Record updated in the contacts file. Audit Record successfully recorded and added."* when the record is saved and an Audit Log is created.
4. Open the More Actions menu and select **Audit Lookup**.
   A record list opens.
   Note: The updates you see may vary, depending on how you have your triggers set up for the file.
5. Select the desired record from the list.
   Note: Only those fields which were modified are recorded. Service Manager displays both the Old and the New versions of each modified field in the audit log record. If none of the fields defined in the Audit Specifications Table have been modified, then Service Manager doesn't generate an audit log entry for that database dictionary file.
6. For further information regarding the modifications to the **contacts** file, click **Show detail**.
   The same record is re-displayed in an expanded form audit.g), which shows more detailed information for the arrayed fields, as well as scrollable fields.
7. Click **End** to return to the summary form.
8. From the summary form, click the **Prev** and **Next** buttons to review any additional audit records in the log for this contacts record.
9. Click **End** to return to the Contacts record.

The topics below comprise an example of how to use Database record auditing. Follow the steps in this order. If you skip a step, the example will not work.

To use Audit Lookup:

1. Open the audit specifications table.
2. Add an audit specifications record.
3. Define an audit specifications entry.
4. Invoke auditing.
5. Set up event triggers.
6. Add lookup functionality to Format Control.
7. Test audit lookup functionality.

Related topics

Audit log file
Database record auditing
Audit specifications file
Audit processing

7.10.8.6. Audit processing

Maintenance of the Audit Specifications table used for audit processing follows normal Service Manager file maintenance procedures. For some tables, such as the contacts table, the out-of-box Format Control specifies that each time an update occurs, the audit.compare application is called. If you invoke auditing for a table for which Format Control was previously set up to invoke auditing, both trigger.invoke.auditor and Format Control will generate audit records, resulting in two audit records.

Note: If you are invoking Auditing from Format Control or a RAD call, ensure that you invoke Auditing *after* Validity Table processing but *before* the record add or update.

For additional information about regenerating database keys or resetting a database table, see the related topics.

Related topics

Audit log file
Audit specifications file
Database record auditing
Regenerating database keys
Resetting a database table
Add lookup functionality to Format Control
Determine if a file has audit processing
Invoke auditing
Invoke auditing from the file in Database Manager (Change Management only)
Invoke auditing from Format Control
Set up event triggers
Start database record auditing
Test audit lookup functionality

7.10.8.7. Determine if a file has audit processing

Applies to User Roles: System Administrator

Some files may not be configured for Audit Lookup. In that case, you must add Audit Lookup functionality before Audit Lookup will work.

To determine whether a file has audit processing:

1. Open the triggers file in Database Manager.
2. Click **Search**. A list of all files with triggers opens.
3. Search for the filename on the list.

Related topics

Audit log file
Database record auditing
Audit specifications file
Audit processing
Start database record auditing
Open a record using Database Manager
Invoke auditing from the file in Database Manager (Change Management only)
Invoke auditing
Add lookup functionality to Format Control
Set up event triggers
Test audit lookup functionality

7.10.8.8. Audit log file

The Audit Log File is the repository for data gathered during the audit. It displays the old and new values of input field data, who made the revisions, and when. The log is a record of all update transactions for the specified source file. The system adds one log record per source record update when you update one or more of the source record input fields defined in the Audit Specifications file.

Note: The following procedure displays a list of audit log records only if you have made modifications to contacts records after creating the audit specifications record. Otherwise, the log record is empty.

Open the audit log file

Applies to User Roles: System Administrator

To open the audit log file:

1. Use one of the following methods to open the log file.
   • Type `audlog` from the command line and press **Enter** or click **Execute Command**.
   • Open the `audit.summ` form from Database Manager.
   • Click **Tailoring** > **Audit** > **Audit Log**.
   Service Manager displays a blank audit log form.
2. Click **Search**. If log records exist, Service Manager displays them.

Related topics

Database record auditing
Audit specifications file
Audit processing
Invoke auditing
Define an audit specifications entry
Start database record auditing
Verify the field name
Verify the file name

7.10.8.9. Verify the field name

Applies to User Roles: System Administrator

The system validates all field names defined in the record (Unique A through D and all Field Names) against the selected database dictionary file. When it finds an invalid field name, Service Manager displays the following message: *"The Field 'field name' is invalid. Select one from the list."*, and a pop-up window opens showing a list of valid field names for the specified database dictionary.

To correct a field name:

1. Double-click on the correct Field name in the fieldname window, and then click **End**.

The system copies selected field name to the appropriate field.
- or -
Copy the correct field name, click **End**, and paste the field name into the field.

Service Manager displays the following message: *"Record added to the auditspecs file"*, which means that all fields have been validated and that the record has been saved.

1. Re-query the record before attempting to make any modifications.
2. Exit the form.
3. Re-open the **auditspec** form.
4. Query for the contacts specifications record.

Related topics

Audit log file
Database record auditing
Audit specifications file
Audit processing
Invoke auditing

Define an audit specifications entry
Verify the file name

7.10.8.10. Verify the file name

Applies to User Roles: System Administrator

When defining an audit specifications entry, if the File Name value is invalid, Service Manager displays the following message: `The filename "filename" - is invalid. Select one from the list.`

A record list opens and displays the valid database dictionary file names.

To correct or modify a file name, select an entry from the list, and then press **Enter**. The system copies the correct file name to the File Name field.

Related topics

Audit log file
Database record auditing
Audit specifications file
Audit processing
Define an audit specifications entry
Invoke auditing
Verify the field name

7.10.8.11. Add an audit specifications record

Applies to User roles: System Administrator

To add an audit specification record:

1. Click **Tailoring** > **Audit** > **Audit Specifications**.
2. Type a file name in the **Filename** field.
   In this example, type **contacts**.
3. Enter unique keys in the Unique A, B, C, or D field.
   The Unique key for the contacts file is contact.name. In this example, type **contact.name** in the Unique A field and leave Unique B through D as NULL.
4. Click **Add**.

When Service Manager records an Audit Log record, for example, the contact name Teresa Chan, the Filename field is recorded in the Audit Log as contacts and the Unique A field is recorded as TERESA CHAN.

The contacts Audit Log record then has a unique association with a contacts record. You can generate new Audit Logs based on the data found in the previous log record for a specific device.

Note

Note A *device* is also referred to as a *Configuration Item* (CI).

Related topics

Invoke auditing

7.10.8.12. Open the audit specifications table

Applies to User Roles: System Administrator

Use one of the following methods to open the audit specifications table:

- From the System Navigator, click **Tailoring** > **Audit** and then double-click **Audit Specifications**.
- From the Command line, type `audspec` from the command line and press **Enter** or click **Execute Command**.
- From the System Navigator, click **System Definition** > **Tables** > **auditspecs** > **forms** and then double-click **auditspecs**.

Related topics

Audit log file
Audit specifications file
Audit specifications file description
Audit processing
Database record auditing
Open a record using Database Manager
Start database record auditing

7.10.8.13. Define an audit specifications entry

Applies to User roles: System Administrator

The system validates Filename and Field Name values every time you update an existing record or add a new record. The Audit utility safeguard system prevents records with misspelled and incorrect file names or field names from being processed. Such errors potentially could cause faulty communication within the database.

Note: The name of the file and one field name of this example have been entered incorrectly to illustrate error-correction process built into the Audit utility.

To enter data in the Audit Specifications file:

1. Click **Tailoring** > **Audit** > **Audit Specifications**.
   A blank Audit Specifications Table form opens.
2. To select the file to create the specifications for.
   Enter information in the **Filename** field and **Unique A** through **Unique D** fields, as necessary, to parallel the Unique key in the source file with the Audit Log.
   Note: see the related topics and read the topic "Audit specifications file description" for more information on input fields.
   For this example, enter `contacts` in the **Filename** field, and then click **Search**.
   Caution: Making changes to the **contacts** file to cause it to invoke auditing will cause two audit records to be generated when a file is updated. To prevent this, use a backup, and restore the original file when finished with the example, or create a different file to practice on.
3. Define those fields you want Service Manager to audit and any aliases. If a field name is invalid, a list will open up allowing you to copy a valid name and use it to replace the invalid name.
   For this example, enter:

Field Name	Alias
contact.name	Contact Name
user.id	Employee ID
first.name	First Name
last.name	Last Name
dept.name	Department
email	Email Address
location	Location

4. Click **Add**. to retain this record and commit it to the database.

The topics below comprise an example of how to use Database record auditing. Follow the steps in this order. If you skip a step, the example will not work.

1. Open the audit specifications table.
2. Add an audit specifications record.
3. Define an audit specifications entry.

4. Invoke auditing.
5. Set up event triggers.
6. Add lookup functionality to Format Control.
7. Test audit lookup functionality.

Related topics

Audit log file
Audit specifications file
Audit processing

7.10.8.14. Audit specifications file

The Audit Specifications (audspecs) file provides instructions on how and when to perform an audit. Use it to define files and fields that you want the Audit application to monitor. There is one specification record for each Service Manager database dictionary (dbdict) record.

Related topics

Audit log file
Audit processing
Database record auditing
Start database record auditing
Invoke auditing
Open the audit specifications table
Audit specifications file description

7.10.8.15. Audit specifications file description

The Format Control record associated with the **auditspecs** form runs a routine that validates entries in the Filename and Field Name fields. Format Control runs this validation routine when you add or update Audit Specification records. It prevents users from entering invalid fields or file values and controls unpredictable run-time results.

Field	Definition
File Name	The name of a valid Service Manager database dictionary upon which Audit is to be performed. This is a required field on the table.
Unique A–D	Used to parallel data records in the Audit File to data records in the source file. Under most circumstances, the fields specified in Unique A through D are the same fields that are defined as Unique keys in the specified database dictionary file. These values need to be unique identifiers for the specific file. Unique A is the only required entry. Additional entries for Unique B through D are optional. Do not use Nulls & Duplicates keys to define Unique A through D, because if the fields in the key are NULL, then Service Manager creates Audit records that don't relate to any records in the Source File. You don't need to define the fields Unique A through D as keys in the Source File. However, you must ensure that any non-key fields are defined as unique identifiers, in order to avoid these errors. When defining Audit Specifications for the Incident and Change Management files, parallel the number field. It is unnecessary to parallel the last or page fields. See database dictionary for information on identifying fields in files.

Field Name	Defines the name of the field to be checked for modifications. The fields specified for Auditing can be of any data type except arrayed structures or fields within arrayed structures. You can specify any number of fields; however, Auditing overhead increases as the number of fields increase. Do not specify all fields within a file for Auditing. Rather, analyze the fields within the file to determine which are critical for the management of data records in the file. The critical fields in the **contacts** file are: • contact.name • first.name • last.name • dept.name • email • location The recommended maximum number of fields to audit is 20. Auditing more than 20 fields causes performance degradation. Under most circumstances, the recommended maximum will not impair the management of a file.
Alias	Defines the alias of a field name used in the audit log. When Service Manager records entries in the audit log (audit.summ.g form), the default is to record the actual field name. You can override the default by specifying an alias. For example, if the widgets file has a field named fd.ast.no, it may be more meaningful to define the alias field name fixed asset (configuration item) number. A one-to-one correlation exists between the Field Name and Alias input fields. Due to processing considerations, these are independent arrays. Therefore, you scroll one; you must also scroll the other to keep the definitions synchronized.

Related topics

Audit specifications file
Database record auditing
Audit log file
Audit processing
Add an audit specifications record
Define an audit specifications entry
Invoke auditing
Open the audit specifications table
Start database record auditing
Verify the field name

7.10.8.16. The Subroutines form

Consider each field in the Subroutines form in terms of the outcome you are seeking. The following fields are contained in the Subroutines form:

Field	Definition
Add	This field needs to be true, since you want audit data to be written whenever a new record is added.
Upd	(update) This field needs to be true to allow audit data to be written whenever there is a modification to an existing record.
Del	(delete) This field is empty, as that would write an audit record whenever a record is deleted.
Dis	(display) This field is empty.
Before	This setting on the subroutine call can be set to true or false. • If it is true, and the audit detects a key error, then an Audit record exists, but the update never occurs. • If it is false, then you receive the Record Updated message before the Audit Recorded message.
Application	This field states the application that compares the fields in the old and new data records, and if necessary, adds a record to the Audit Log file.
Error Message	This field provides the error message, which is displayed if the process can't be completed as planned.
Names	This field states which files are handled by the subroutine. Activate the Save Copy option on the Format Control Maintenance form to make this file variable available • Pass the Old (or original) version of the data record to the file parameter. • Pass the New (or updated) version of the data record to the second.file parameter.
Values	Values in this field correspond to each parameter. These values state to which variables Service Manager passes the data.

Related topics

Database record auditing
Audit specifications file
Audit log file
Audit processing
Add lookup functionality to Format Control
Invoke auditing

7.10.9. IR Expert

IR Expert is an intelligent, concept-based information retrieval engine that searches the Service Manager database for similar or related information based on a natural language query. For example, instead of relying on exactly matching

keywords to find similar incidents in the Incident Management database, Service Manager uses the IR Expert engine and the description of an incident to locate similar incidents. IR Expert assigns a probability of relevance to the query results, and ranks them by relevance to the original query.

Start from SM 9.62, The IR Expert solution leverages the native Full-Text Searching feature in DBMS, which enables rapid searching based on preprocessed documents in DBMS and reduces the configuration effort by SM administrators. IR Expert supports the Microsoft SQL Server, Oracle, and PostgreSQL databases. The main features of IR Expert include the following:

- Text parsing and tokenization of query
- Generating and managing IR key indexes
- Query with text search

Upgrade IR Expert

- If you are using the legacy IR search (version before 9.62), you may run the IR Upgrade utility to enable the new IR solution. See the **Upgrade the legacy IR to the new IR solution** section below for detailed steps.
- If you are using the new IR solution, after upgrading applications or after adding IR keys, you need to regenerate the IR indexes. See the **Regenerate the IR indexes** section below for detailed steps.

Upgrade the legacy IR to the new IR solution

To upgrade to the new IR solution, follow these steps:

1. Make sure that your Service Manager server is upgraded to version 9.62. Do not upgrade your Service Manager applications at this moment.
2. Map the IR key fields of the "help" file. In the out-of-box system, the "help" file is not mapped to DBMS correctly. There are three IR key fields defined in the file, but only the "topic" field is correctly mapped while the "brief" and "detail" fields are not. Check these fields in your system and map the fields if they are not correctly mapped. The following is a mapping example for Oracle.
    1. Click **Tailoring** > **Database Dictionary**, and search for the "help" file.
    2. Edit the following two fields.

Name	Type	Level	SQL Name	SQL Type	SQL Table
brief	character	2	BRIEF	CLOB	m1
detail	character	2	DETAIL	CLOB	m1

    3. Click **OK**. Wait for the mapping process to complete. The waiting time depends on the number of records in the "help" file.
3. Stop the Service Manager server.
4. If you use the Oracle database, you must use the DBA user "sys" to grant EXECUTE privileges on CTX PL/SQL packages to the Oracle database user account.

Note

Note: This step is required by Oracle. For more information, see "3. Grant EXECUTE privileges on the CTX PL/SQL package." at https://docs.oracle.com/en/database/oracle/oracle-database/18/ccapp/getting-started-with-oracle-text.html.

To do this, use the following sqlplus command:

```
$ sqlplus sys/<password_of_sys>@127.0.0.1:1521/orcl as sysdba
SQL> grant execute on CTXSYS.CTX_DDL to sm_db_user_name;
```

This command requires the password for the "sys" account. If you forget the password, reset it as follows:

```
$ sqlplus / as sysdba
SQL> alter user sys identified by "<new_password_for_sys>";
```

5. Go to the RUN directory.
6. Execute the following command to run the IR upgrade utility:
   `sm –upgradeir`
   The IR upgrade utility scans all tables, and regenerate IR indexes on DBMS for the tables with the IR keys defined.

> **Note**
> Note: For the Oracle database, because Oracle Full Text Index does not support table fields in the NCHAR, NVARCHAR2, and NCLOB types to define IR keys, the utility converts the IR key fields in these types to the supported CHAR, VARCHAR2, and CLOB types.
> In addition, if the original fields in these types also belong to some other keys besides the IR key, Oracle will remove the indexes that include such fields in DBMS. Before the conversion, you must back up the SQL statements for creating indexes on tables in DBMS. After the IR upgrade, you need to add the index back by executing the SQL statements from your backup.

If your Service Manager is integrated with version control (SVN or GIT), follow these steps to upgrade to the new IR solution:

1. Run the `sm –svc_export` command to export all the tables and data from source system as xml files.
2. Prepare an empty database, and then run the `sm –svc_import –svc_cleanbuild:1` command to import the source data to the target system.
   At the end of the importing process, IR indexes are generated on DBMS for the tables with IR keys defined.

Regenerate the IR indexes

you need to run the **sm -upgradeir** command to regenerate the IR indexes.

1. Stop the Service Manager server.
2. Go to the RUN directory.
3. Execute the following command to run the IR upgrade utility: **sm –upgradeir**

The IR upgrade utility scans all tables, and regenerate IR indexes on DBMS for the tables with the IR keys defined.

If you have uploaded the SM demonstration data when configuring the server, or if you do not use the IR Expert feature, there is no need to regenerate the IR indexes.

The new IR Expert supports the Microsoft SQL Server, Oracle, and PostgreSQL databases.

> **Note**
> Note:
> - If you use Microsoft SQL Server, Full-Text Search is an optional component of the Microsoft SQL Server Database Engine. If you did not select Full-Text Search when you installed SQL Server, run SQL Server Setup again to add it. The installation process does not require a shutdown of your SQL Server Database Engine.
> - If you use an Oracle database, IR requires the following additional privileges for $SM_DB_USER:
>
>   ```
>   GRANT EXECUTE ON CTXSYS.CTX_DDL TO $SM_DB_USER;
>   GRANT SELECT ON DBA_SCHEDULER_JOBS TO $SM_DB_USER;
>   GRANT CREATE JOB TO $SM_DB_USER;
>   ```
>
> - If you use a PostgreSQL database, when the PostgreSQL database owner is not the super user "postgres", the IR stop list might not be applied correctly and you might get unexpected search results. To resolve this issue, follow the workaround provided for Limitation 2 in IR solution comparison.

Related topics

Entity Relationship Diagram creation utility
IR Expert tasks
IR Query features
Standard record lists and IR Expert

7.10.9.1. IR Expert solution comparison

The new IR Expert solution is based on the Full-Text Searching feature in DBMS, instead of having its own IR engine for indexing and searching in Service Manager.

For a comparison of behavior changes between the new and the old IR Expert solutions, see the following table.

Performance / Function	Old IR (SM 9.61 and earlier)	New IR (in the current release)
Time to generate IR index	The system takes hours to regenerate 1M records and the performance goes lower with more records.	With the new solution, the time to regenerate 1M records is dramatically reduced to several minutes.
IR search	New, updated, or deleted records can only be searched out after regenerating the IR indexes manually in dbdict.	New, updated, or deleted records can be search out in Service Manager as soon as IR indexes are re-generated automatically in DBMS. For the Oracle database, you can configure the ir_pending_secs parameter to adjust the time for IR index synchronization.
IR language	Old IR uses the "ir_language" parameter together with the user-defined IR language files managed by Service Manager.	Microsoft SQL Server and PostgreSQL require the "ir_language" parameter for switching languages. Oracle doesn't require the "ir_language" parameter, and the new IR uses Oracle's multilingual feature (WORLD_LEXER) for indexing tables containing documents of different languages.
IR parameters	You need to configure many IR parameters for the old IR solution to work.	Many IR parameters are obsoleted, and you only need to configure the following parameters: ir_disable, ir_language, ir_pending_secs, and ir_stoplist.

Stop words	A user-defined list of stop words is managed by Service Manager.	By default, Service Manager uses the stop words defined in the default database stoplist. To see the default stoplist for different databases, see the following links: • Oracle: https://docs.oracle.com/cd/B28359_01/text.111/b28304/astopsup.htm#CCREF1400 • PostgresSQL: https://www.postgresql.org/docs/11/textsearch-dictionaries.html#TEXTSEARCH-STOPWORDS • MS SQL Server: https://docs.microsoft.com/en-us/sql/relational-databases/search/configure-and-manage-stopwords-and-stoplists-for-full-text-search?view=sql-server-2017 If you want to use your own database stoplist, first create your database stoplist according to your database documentation, and then specify your stoplist in the ir_stoplist parameter. After you create or update the stoplist, make sure that you run the `sm -upgradeir` command to regenerate IR indexes.
Query of IR search with the "*" wildcard Example: mail*	Old IR returns records containing words like mail(s)/mailbox(es).	New IR returns more results. That is, records containing words like e-mail(s)/email(s)/mail(s)/mailbox(es).
Query of IR search with different forms of a verb Example: run	Old IR returns records containing run/running.	New IR returns more results, including run/running/ran.
Generating and managing IR indexes	The ir.xxx (file name) file is in the scirexpert file in Service Manager.	The indexes starting with "SMIR_xxxx" (index name) are generated in DBMS.
Stem dictionary	A user-defined stem dictionary is managed by Service Manager.	New IR directly uses the stem dictionary of the DBMS. The new IR search always enables stemming if the search term is not followed by the "*" wildcard.
Suffix dictionary	A user-defined suffix dictionary is managed by Service Manager.	New IR directly uses the suffix dictionary of the DBMS.

HTML tags in IR search fields	Old IR strips off the HTML tags during the search.	New IR doesn't strip off the HTML tags during the search. The HTML tags are taken as meaningful characters during the search.
Relational operators	=: equal to #: starts with ~#: doesn't start with like: is similar to ~=: not equal to <>: not equal to ><: not equal to <: less than <= or =<: less than or equal to >: greater than >= or =>: greater than or equal to isin: is element in	=: equal to #: starts with ~#: doesn't start with like: is similar to ~=: not equal to <>: not equal to ><: not equal to

Limitation 1	PostgreSQL only supports the following languages for text search in Service Manager: English, French, German, Italian, Brazilian Portuguese, Russian, Spanish, and Turkish. PostgreSQL doesn't support following languages for text search: Arabic, Japanese, Simplified Chinese, and Korean. For more information on the PostgreSQL supported languages, see "List text search dictionaries" in: https://www.postgresql.org/docs/10/textsearch-psql.html

Limitation 2	When the PostgreSQL database owner is not the super user "postgres", the IR stop list might not be applied correctly and you might get unexpected search results. **Workaround**: 1. Modify all the owners of configurations in the "pg_catalog" namespace to specific DBUser($DBUser) previously owned by "postgres" . Run the following statement against your PostgreSQL, and change the $DBUser' to the actual DB User. `update pg_catalog.pg_ts_config set cfgowner=(select oid from pg_catalog.pg_roles where rolname='$DBUser') where oid in (select c.oid from pg_catalog.pg_ts_config c inner join pg_catalog.pg_roles r on c.cfgowner=r.oid inner join pg_catalog.pg_namespace n on c.cfgnamespace=n.oid where r.rolname='postgres' and n.nspname='pg_catalog');` 2. Modify all the owners of dictionaries in the "pg_catalog" namespace to specific DBUser($DBUser) previously owned by 'postgres' . Run the following statement against your PostgreSQL, and change the $DBUser' to the actual DB User. `update pg_catalog.pg_ts_dict set dictowner= (select oid from pg_catalog.pg_roles where rolname='$DBUser') where oid in (select d.oid from pg_catalog.pg_ts_dict d inner join pg_catalog.pg_roles r on d.dictowner=r.oid inner join pg_catalog.pg_namespace n on d.dictnamespace=n.oid where r.rolname='postgres' and n.nspname='pg_catalog');` **Notes:** • You must apply this workaround as the PostgreSQL super user. • It is recommended that you apply this solution in a completely new database. If you apply this solution in an existing database, check the database carefully and confirm this solution will not affect other existing design of the database. You need to Regen IR key in the existing database.

Limitation 3	When you use the "OR" operator to combine an IR query and a SQL query, while the SQL query contains unsearchable contents (such as Array fields, BLOB or CLOB objects, Image data-types, Text data-types, or Encrypted fields), then the search query won't return the proper results.
Limitation 4	See Limitations on cross-table join queries for details.

7.10.9.2. IR Expert tasks

IR Expert tasks involve:

- Access IR Expert
- Enable IR search for a file
- Regenerate IR keys
- Load data files with IR Expert keys

Related topics

IR Expert
Standard record lists and IR Expert
IR Query features

7.10.9.2.1. Access IR Expert

Applies to User roles: System Administrator, IT Agent

You can access the IR Expert application from many locations within Service Manager.

The following table contains some examples.

Application or Tool	To Access IR Expert:
Service Desk	Click **Service Desk** > **Interaction Queue** > **Search** > **Text Search**. 1. Click **Service Desk** > **Search Interaction Records**. 2. Click the **Text Search** tab. 3. Complete the Search Text text field. 4. Click **Search**.
Incident Management	Click **Incident Management** > **Incident Queue** > **Search** > **Text Search**. 1. Click **Incident Management** > **Search Incidents**. 2. Click the **Text Search** tab. 3. Complete the Search Text field. 4. Click **Search**.
Service Catalog	1. Click **Service Catalog** > **Administration** > **Order from Catalog** or **Non-cart Catalog Requests**. 2. Select a category. 3. Complete the **for** field. 4. Click **Search**.
Search Help Records	1. Type **help** in command line, and press **Enter**. 2. From the **Search Help Records** form, enter your search text. 3. Click **Search**.

Related topics

IR Expert
IR Expert tasks
More Actions menu
Enable IR search for a file

7.10.9.2.2. Enable IR search for a file

Applies to User Roles: System Administrator

To enable IR search for a file (table), you must create an IR key for the file, and then regenerate the IR index for the file.

Note

Note: If the search triggers the Cross Table Query, for example, Text Search + CI Type (e.g. Computer, a type has join file), there will be some limitations on cross-table join queries.

You can define an IR key using the Database Dictionary utility. As an example, the following procedure describes how to enable IR search for the `device` file.

1. Sign in to the Service Manager Windows client.
   Note: The web client doesn't allow you to add a new field/key for a Database Dictionary record.
2. Click **Tailoring** > **Database Dictionary**.
3. In the File Name field, type **device**.
4. Click **Keys**, place the cursor in the first blank key line, and click **Edit Field/Key**.
5. In the key window, do the following:

1. In the Type field, select **IR key**.
2. In the Fields section, type two fields: **comments**, and **title**. These are the names of the fields that you want to use as IR key fields.
   Note: For the rules that you must follow when selecting IR key fields, see IR keys and non-IR keys.
3. Click **OK**.
   Service Manager creates the IR key for the device file.

6. Click **OK** to save the file.
   Service Manager displays a message, stating that you are about to Regen the file and asking if you want to continue.
7. Click **OK** to run the IR Regen now.
8. Wait for the IR Regen to complete.
   A message displays, stating that the IR Regen has been completed.
9. (Optional) Verify that IR search is now available for the device file.
   1. Click **Configuration Management** > **Resources** > **Search CIs**.
   2. Do a true search, and locate two CIs whose Title or Description field is not empty. Make note of two words that display in any of these fields in the two CIs, one word for each CI.
   3. Click **Cancel**.
   4. Click **Text Search**.
   5. In the Search Text field, type the above two words, and select **Any of these words**.
   6. Click **Search**.
      The two CIs should appear in the returned search results.

Related topics

IR Expert
Access IR Expert

7.10.9.2.3. Regenerate IR keys

Applies to User roles: System Administrator

You need to run an IR Regen only in the following cases:

- If you are changing an IR key (adding or removing a field);
- If you are changing the IR language;
- If you have found an IR index corruption.

When needed, perform an IR Regen against a specific file or against all IR indexed files in the system.

You can perform an IR Regen for a table by using the Database Dictionary utility:

1. Sign in to the Service Manager Windows or web client.
2. Click **Tailoring** > **Database Dictionary**.
3. In the Table field, type the table name (for example, **incidents**), and click **Search**.
   The table details form opens.
4. From the table details form, click **More** or the More Actions icon, and select **Regen IR**.
5. Click **OK** to run the IR Regen immediately.

To perform an IR Regen for all tables using the `sm -upgradeir` command:

1. Shut down your Service Manager system. For details, see Shut down Service Manager.
2. Run the `sm -upgradeir` command to regenerate IR indexes.
3. Restart your Service Manager system as required. For details, see Start Service Manager.

Related topics

IR Expert
Regenerating database keys
Regenerate database keys
Regen a database dictionary

Access IR Expert
Enable IR search for a file

7.10.9.2.4. Configure IR Search Performance for the Oracle database

If you use the Oracle database for Service Manager, after you upgrade to the new IR solution and start Service Manager, two database procedures are created in the Oracle database to improve the performance of IR search for the Oracle database:

- SM_SYNC_IR_INDEX_SP
  This procedure is to schedule a database job to do index synchronization. You can configure the ir_pending_secs parameter to control the maximum pending period (in seconds) before an IR index gets synchronized after the base table is updated. For more information, see ir_pending_secs.
- SM_OPTIMIZE_IR_INDEX_SP
  This procedure is to schedule a database job to do index optimization. By default, the index optimization includes a full optimization that runs hourly from 1 a.m. to 5 a.m. and a fast optimization process that runs at other time hourly. If you want to change the time for index optimization, connect to the database and update the SM_OPTIMIZE_IR_INDEX_SP procedure as needed. For more information about index optimization in the Oracle database, see https://docs.oracle.com/en/database/oracle/oracle-database/18/ccref/CTX_DDL-package.html.

7.10.9.2.5. Load data files with IR Expert keys

Applies to User Roles: System Administrator

IR keys are created when loading data into Service Manager, such as when moving files from a test system to a production system. Any files containing one or more IR keys don't require an IR Regen of the new system.

When you use svc_import to load data from xml files, if the the file already exists in the new system and the file contains at least one record, you must add the IR key and perform the IR Regen before loading the records from your old system. Then Service Manager updates the IR indexes as it adds each record during the load.

If you are upgrading your system from a version earlier than SM 9.62, you must use the IR upgrade utility to perform the IR Regen. For detailed instructions, see the Upgrade IR Expert section.

Related topics

IR Expert
Database dictionary and IR Expert
IR keys and non-IR keys
How IR Expert evaluates documents for relevance
Access IR Expert

7.10.9.2.6. Manage IR stop words

A stop-word list is a list of terms that can be ignored when the search engine is searching or indexing. Typically, stop-word lists include short and common words or prepositions, such as "a," "the," or "with" in English. However, they may also include longer words, such as long number strings, or words that are too common to be useful as search targets, such as the term "internet."

Important

Important:

- This feature is for Oracle and MS SQL Server databases only. For PostgresSQL databases, you can use the ir_stoplist parameter to specify your stoplist.
- If you don't specify your own stoplist using this IR Stopwords tool, or if you don't migrate your customized stoplist from previous versions (before 9.64.p1 and 9.70), or if you don't apply the **IRStoplist_SM970_SM960.unl** unload file while your applications in version 9.6x, Service Manager will use the default database stoplist to get the stopwords for IR search.

You can refer to the following table and find the scenario that best fits your situation to migrate or use the IR Stopwords tool:

RTE version (before upgrade)	RTE version (after upgrade)	Applications version	Database type	Must apply platform unload?	Is migration tool applicable?
9.60, 9.61, 9.62	9.70	9.6x	Oracle / MSSQL	Yes	No
9.63, 9.64	9.70	9.6x	Oracle / MSSQL	Yes	Yes
New install	9.70	9.70	Oracle / MSSQL	No	No

- After upgrade, for the system in which IR Language is set as English (default), you can:
  - Use the default database stop words of Service Manager
  - Migrate your customized English stop words from previous versions
  - Manually add the stop words using the IR Stopwords tool
- After upgrade, for the system in which IR Language is set as non-English languages, you must apply the unload files (**IRStoplist_oracle_nonenglish.unl f**or Oracle and **IRStoplist_sqlserver_[language].unl** for MS SQL Server), then you can:
  - Use the default stop words defined in the unload files
  - Migrate your customized non-English stop words from previous versions
  - Manually add the stop words using the IR Stopwords tool

Manage IR stop words

To specify the stop words for IR search:

1. Sign in to Service Manager as the administrator.
2. Click **Tailoring** > **Tailoring tools** > **IR Stopwords.**
3. Add a new word or modify or delete an existing word.
4. Click **Save**.
5. Click the **More** or the **More Actions** icon, then click **IR Synchronize** to synchronize the stop word to your database. **Note:** After you update all stop words, make sure that you have regenerate IR indexes by running an IR Regen or running the `sm -upgradeir` command.

Migrate stop words

If you have already created your own stoplist for your database (which can be enabled by the ir_stoplist parameter) in old versions, you need to migrate the existing stoplist after you upgrade to 9.70.

Prerequisites

Before you start, you need to meet the following prerequisites:

- Apply the **IRStoplist_SM970_SM960.unl** unload file. For more information, see Install platform unloads.
- Optional: If you need to maintain your stoplist for non-English languages, you need to further apply the following unload files:
  - For Oracle: **IRStoplist_oracle_nonenglish.unl**
  - For MS SQL Server: **IRStoplist_sqlserver_[language].unl** for each language (For example, if your IR language is set as Simplified Chinese, apply the **IRStoplist_sqlserver_Simplified Chinese.unl** unload file)
- Install JDK 11 or later version, and then set JAVA_HOME to the location where the JDK was installed.
- Download the JDBC driver for Oracle and MS SQL Server according to the Java version.

Use the migration tool

1. Download and open the IR stop word migration tool, which contains the following files:
   - **irstopword-migrate-9.70.jar**: the function tool implemented by JAVA.

- **irstopword_migration.bat**: the startup script on Windows
- **irstopword_migration.sh**: the startup script on Linux
- **irstopword_migration.ini**: the configuration file to define the parameters

2. Modify the startup script according your environment (**irstopword_migration.bat** for Windows and **irstopword_migration.sh** for Linux):
   - Set the script variable *oracle_jdbc* to the directory path of Oracle JDBC driver.
     For example, the path of "ojdbc7.jar".
   - Set the script variable *mssql_jdbc* to the directory path of MS SQL Server JDBC driver.
     For example, the path of "mssql-jdbc-8.2.2.jre8.jar".
3. Modify the parameters in the **irstopword_migration.ini** configuration file as requried:
   1. **dstdburl**: Set the destination database by this parameter. the destination database must be the SM database which applied the IR Stop List Refactor.
      Example for Oracle: `dstdburl:jdbc:oracle:thin:@16.1xx.xx.xx:15xx:smcs`
      Example for SQL Server: `dstdburl:jdbc:sqlserver://16.1xx.xx.xxx:14xx;databaseName=sm96x`
   2. **srcdburl**: Set the source database by this parameter.
      Example for Oracle: srcdburl:jdbc:oracle:thin:@16.1xx.xx.xx:15xx:smcs
      Example for SQL Server: srcdburl:jdbc:sqlserver://16.1xx.xx.xxx:14xx;databaseName=sm96x
   3. **srcstoplist**: Set the source stoplist by this parameter.
      For example: `srcstoplist:SM_FULLTEXT_STOP_LIST`
   4. **ir_language**: Set the IR language for MS SQL Server, which is the same as the configuration in sm.ini. The collation of the destination database should support this language.
      For example: `ir_language:English`
4. Please note that the destination database and the source database should be of the same type. Here are the complete examples:
   - Example for Oracle:

     ```
     dstdburl:jdbc:oracle:thin:@16.1xx.xx.xx:15xx:smcs1
     srcdburl:jdbc:oracle:thin:@16.1xx.xx.xx:15xx:smcs2
     srcstoplist:customized_stoplist
     ```

   - Example for SQL Server:

     ```
     dstdburl:jdbc:sqlserver://16.1xx.xx.xxx:14xx;databaseName=sm96x1
     srcdburl:jdbc:sqlserver://16.1xx.xx.xxx:14xx;databaseName=sm96x2
     srcstoplist:customized_stoplist
     ir_language:English
     ```

5. Run the `sm -upgradeir` command to regenerate IR indexes.

7.10.9.3. Standard record lists and IR Expert

When you create a query, Service Manager checks the target file for IR keys. If you specify data that's part of an IR key, then Service Manager runs an IR query before any other query. By default, IR Expert returns a list of records for each search. The number of records selected is a function of the granularity of the IR parameters and the presence or absence of additional, non-IR query elements.

IR queries can be combined with other queries to help narrow the search for information. For example, in Incident Management you may want to find out about a particular application error code, but only for incidents in the sysops assignment group. If you type sysops in the assignment field, and the application fails with 00023.EXE, and you describe this in the incident description, IR Expert searches for incidents that are like the incident description first. For a match to occur, the description contains words related to the stems of each major component of the example message. Next, Service Manager eliminates any incidents found by the IR search if the assignment field doesn't begin with sysops.

You can also use an IR search in conjunction with the Find option by defining a link from a field normally used in an IR query in the source file to any field that's part of an IR key in the target file. For example, you can define a link from the

`Action.action` field of the `problem` file to the `description.structure.description` field of the `cm3r` file to detect similar change requests when opening or updating an Incident.

Related topics

IR Expert
IR Query features
IR Expert tasks

7.10.9.4. Database Dictionary and IR Expert

Installing IR Expert creates three changes in the Database Dictionary utility:

- The IR key types are available in the key list.
- The regenerate type IR Regen is available when keys in a file that contains an IR key has been modified.
- The Type field in the datadict.g record reflects the presence of an IR key in a file.

Note: An IR Regen doesn't necessarily rebuild any indexes other than IR indexes. If you modify any key other than an IR key, Service Manager may not recognize the change until you perform a full regen. The average IR Regen time is 1 minute for each 300,000 records.

Data Policy information (datadict file) is a critical factor in the IR Query application. It stores details on which files contain IR keys and which fields within those files are keyed for IR queries. You can access the datadict file by clicking **Tailoring** > **Data Policy**.

Related topics

IR Expert
IR keys and non-IR keys
How IR Expert evaluates documents for relevance
Access IR Expert
Load data files with IR Expert keys

7.10.9.5. IR query features

IR Expert requires a specific query to process against the target file that's a plain text query. It returns a record list containing related records sorted by relevance.

IR Expert offers these advantages when you design queries:

- You can reuse the same query against multiple files.
- You can tune, modify, and resubmit a query.
- You can copy information from a target record into the source field.
- You can use conditional logic to build other files (or IR-specific files) when you update conventional files.

Related topics

IR Expert
Standard record lists and IR Expert
IR Expert tasks

7.10.9.6. IR keys and non-IR keys

A Service Manager file can contain only one IR key, and must contain a non-IR key. All files that contain an IR key must also contain another key that's Unique, with No Nulls, No Duplicates, or Nulls & Duplicates. An attempt to build an index for a file that contains only an IR key results in a regen error. The system builds the external IR indexes, but removes the IR key from the Service Manager index.

Building IR keys

Applies to User Roles: System Administrator

Define IR keys using the standard Database Dictionary utility in Service Manager. IR keys are composed of one or more array or scalar text fields. IR keys that combine array and scalar fields should define an array field as the first element of the key.

Use only text fields in an IR key. A text field is an array or scalar field that contains arbitrary information that not used for traditional queries. For example, the device file contains a scalar field called description. This field contains descriptive text about a device, such as:

- Does not support the new drawing package.
- This modem needs ventilation on the top and sides.

The assignment file contains an array field called operators. The operators array contains the exact login name of Service Manager operators, and is not a good candidate for an IR key since IR searches are relevance searches.

Follow these rules when defining IR keys:

- If a query contains a field that's part of an IR key, Service Manager always performs an IR search. IR Expert selects and presents records based upon relevance.
- If a query contains a field that's part of an IR key, your sort criteria are honored. IR queries always sort by relevance.

You can combine IR queries with traditional queries to specifically limit the answer set.

Related topics

IR Expert
Load data files with IR Expert keys

7.10.9.7. How IR Expert evaluates documents for relevance

IR Expert queries return results based on relevance to the query. The relevance is controlled by the Full Text Search mechanism in DBMS.

See the Full Text Search documentation for your DBMS for details. For examples:

Oracle: https://docs.oracle.com/en/database/oracle/oracle-database/18/ccapp/indexing-with-oracle-text.html

MS SQL Server: https://docs.microsoft.com/en-us/sql/relational-databases/search/full-text-search?view=sql-server-2017

PostgreSQL: https://www.postgresql.org/docs/10/textsearch.html

Related topics

Access IR Expert

7.10.10. Record retrieval

Service Manager allows you to search for and retrieve database records based on particular user-provided search criteria (query). Searching is case-sensitive by default, unless your database has been set up for case-insensitive searching. For information on how to improve query speed, see the Help topic Tuning: Improving query speed.

There are three methods of querying records from a Service Manager database:

- Simple queries are defined using the record list method.
- More complex queries are defined using the query window method.
- Other queries can be performed using IR Expert information retrieval.

Related topics

Relational operators
Logical operators
Wildcard characters
Record list
Queries
Retrieving records by using the record list method
Retrieving records using the Query window method
Perform IR Expert queries
Service Manager Programming
Tuning: Improving query speed

7.10.10.1. Open a record using Database Manager

The Service Manager Database Manager utility runs in two modes, standard and administrative.

- In standard mode, behavior is determined by whatever security you have in place. Request Management, for example, uses Request security while the standard database uses Format Control. In standard mode, an administrator doesn't necessarily see all options. Options that are potentially troublesome for Service Manager, mass-updates, regenerating keys, and so on, are kept out of sight to prevent accidental use.
- Administration Mode is similar to root privileges on Linux systems. Administration Mode is powerful, in that you can make changes that affect Service Manager as a whole. In Administration Mode, (with the Administration Mode checkbox selected), a System Administrator will have rights to ALL options unless a different displayscreen is used based on the value of the Administration Mode checkbox.
  The value of the Administration Mode checkbox affects which displayscreen is used for certain files. The link file will use the link.view displayscreen when Admin Mode is disabled and db.view when Admin Mode is enabled. The globallists file will use the global.view displayscreen when Admin Mode is disabled and db.view when Admin Mode is enabled. When Admin Mode is enabled for the link and globallists files, the SysAdmin no longer has access to the Select Line and Rebuild Global List options respectively.

Note: Do day-to-day administration in standard mode.

Two capability words control whether a user administrator can use standard or administrative mode.

- Users with the *SysAdmin* capability word defined in their operator file will see the Administration Mode checkbox.
- Users with the *AlwaysAdmin* capability word defined in their operator file will be in Administration Mode by default.

To open a record using Database Manager:

1. Click **Tailoring** > **Database Manager**.
   Note: Alternatively, you can open Database Manager from the command line, by typing `db`, and then pressing **Enter** or clicking **Execute Command**.
2. Select a form or file by typing the name of a form in the **Form** field or the name of a file in the **Table** field.
   If one or both fields are left blank, the record list returns all records that match your criteria.
3. Click **Search**.
4. Enter your search criteria. For example, you can enter known information in key fields, such as names of operators, names of devices, or incident numbers.
   Note: Executing a search that includes a field that's not fully keyed slows response time.
5. Click **Search**.
   - If found, a record that matches your criteria opens.
   - If more than one match is found, a record list opens.
     For multiple returns, click a record from the record list to display it.
     To update the record, edit the record and then click **Save**.

Note: A *device* is also referred to as a Configuration Item (CI).

Related topics

Record retrieval

7.10.10.2. Determining System Navigator menu options

The function and contents of the items displayed in the More Actions menu for the System Navigator are determined by a combination of the following:

- Additional Options
- Display settings
- Format Control
- The RAD application currently running

Additional options are only available in select applications. You can use Format Control to create additional capabilities for users that allow extended user rights and additional access to data gathering, such as additional views and use of the Find and Fill functions.

Icons

More Actions menu: ▾
Find function:

Fill function:

For additional information about Additional Options and Format Control, see the related topics.

Related topics

Format Control
Multiple-record functions
Single-record functions

7.10.10.3. Record list

A record list is a specific form displaying pertinent field values from each record found in the search. If you have not created a customized record list form for this list, Service Manager will create one (<*filename*>.qbe) based on key fields defined in the database dictionary for the file.
Select a record from the list by positioning the cursor on the desired record and double-clicking or pressing Enter.

- If the search criteria uniquely define only one record, that record opens using the form used to define the search or the one defined by a form name in the record, depending on format control.
- If the search criteria don't find any records, Database Manager returns the message: *No records found to satisfy record list search argument(s)*.

Refresh a record list

Applies to User Roles: System Administrator

To refresh a record list:

1. Open the More Actions menu.
2. Select **Refresh List**.

Related topics

Relational operators
Logical operators
Wildcard characters
Retrieving records by using the record list method
Retrieving records using the Query window method
Using functions in a query
Stored queries
Format Control
Perform IR Expert queries

7.10.10.4. Multiple-record functions

Multiple-record functions allow you to perform add, update, delete, and print functions on multiple records within a database. The Mass functions (Mass Add, Mass Update, Mass Delete) become available whenever a query produces a record list, that is, the desired function acts upon the records in the record list.

The topics below comprise an example of how to use multiple-record functions. Follow the steps in this order. If you skip a step, the example will not work.

1. Add multiple records using Mass Add with a literal value
2. Add multiple records using Mass Add with a variable value
3. Update multiple records using a literal value
4. Update multiple records using a variable value
5. Avoid invalid duplicate or null key errors
6. Delete multiple records

Note
Note
The Mass functions may require use of the Administration Mode checkbox when starting Database Manager.

Related topics

Adding multiple records
Updating multiple records
Print multiple records

7.10.10.4.1. Adding multiple records

After opening a record list from a Service Manager file, the Mass Add function allows you to add a block of records to the database. The new records are exact duplicates of those in the original record list. You can also run format control triggers and macros by setting the **add** field to *true*. Mass Add processing statements use the $file parameter to access fields on records in the current record list. For example, *name* in the $file parameter or the number one (*1*) in the $file parameter.

You can choose to add exact duplicates only to databases where keys are defined as Nulls & Duplicates or there are No Nulls in the database dictionary. Adding duplicate records in any other manner causes Database Manager to issue an invalid duplicate key error message for each record. To avoid this error when running a Mass Add function against a file with Unique or No Duplicates keys, run processing statements during the Mass Add function that will manipulate the contents of the new records.

Add multiple records using a literal value

Applies to User Roles: System Administrator

Example: Duplicate all contact records for advantage. Retain the same information in each record, with the exception of changing the **Location** to North America. Since Contact Name (contact.name) is a Unique key and Employee ID (user.id) is a No Duplicates key in the contacts file, you must assign unique contact.name and user.id values to the new records. For additional information about key type definitions, see the related topics. In this example, append the value *-NA* to the contact.name and user.id values to maintain uniqueness.

To mass add records using a literal value:

1. Open the form in Database Manager. Ensure that you select the Administration Mode checkbox. In this example, open the following form: `contacts`. A blank contacts form opens.
2. Enter the desired value in the **Company** input field. For this example, select advantage from the Company list.
   Click **Fill**. to open the list.
3. Click **Search**.
   Database Manager performs the equal to search and displays the matching records, if any.
   If the query produces no matching records, Service Manager displays a message.
4. To display the Mass function buttons, use a query that returns more than one record.
   Select one of the listed records, and then select **Mass Add**. from the toolbar to start the process of duplicating the records shown in the list.
   Database Manager displays the Mass Add/Update Instruction screen.
5. Type the applicable statements in the input fields.
   • Statements in the first input field (**Instructions to be executed ONCE at the beginning of Mass Add/Update**) are executed once at the beginning of the add.
   • Statements in the subsequent input fields (**Instructions for action on EACH RECORD**) are executed against every record as it is added.
   For this example, leave the first input field blank and enter the following assignment statements in the **Instructions for action on EACH RECORD** input fields.

Statement	Purpose
contact.name in $file=contact.name in $file +"-NA"	Appends -NA to the end of all contact.name values.
location in $file="North America"	Changes the location value to North America.
user.id in $file=user.id in $file +"-NA"	Appends -NA to the end of all user.id values.

6. Click **Execute**,
   Database Manager performs the Mass Add in the foreground. Your session is devoted to the add task until it is completed. If you are updating a large number of records, this can take a while.

Upon completion, Service Manager returns terminal control to you and displays the following message: *<nnn>records added to the contacts file* where *nnn* is the number of records added.

7. Click **View Messages** to check the messages for errors.
   • A blue icon indicates a required action.
   • A yellow icon indicates an informational or warning message.
   • A red icon indicates an error message.

Close the Messages window to return to the contacts form.

- Click the More Actions icon and choose **Refresh**.
  In the new records, the Contact Name (contact.name) and Employee ID (user.id) values are the same as the original records with -NA appended. The Location (location) values are now North America.
  The original records listed before the Mass Add remain in the contacts file with no changes to the original data.
- Click **OK**. to return to the contacts form.
- On the contacts form, type **North America** in the Location field. Click **advantage** in the Company field. Type **Search**. Database Manager performs the equal to search and displays the matching record(s), if any, in the contacts.qbe form.

Add multiple records using a variable value

Applies to User roles: System Administrator

Example: Duplicate all "contacts" records with a company.name value of advantage. Include the same information in each record, except for changing the Company to NewAdvantage, creating sequential user.id values, and creating a Unique key for each record.

Since Contact Name (contact.name) is a Unique key and Employee ID (user.id) is a No Duplicates key in the contacts file, you must assign unique contact.name and user.id values to the new records. For additional information about key type definitions, see the related topics. For this example, suffix the user.id values in the new records with sequential numbers, starting with 1000 (1000, 1001, and so on.)

Note: Mass Add processing statements reference $file to access fields on records in the list. For example, name in $file or 1 in action in $file.

To add records using a variable value:

1. Open the form in Database Manager. Ensure that you select the Administration Mode checkbox. For this example, open the following form: `contacts`. A blank contacts form opens.
2. Enter the desired value in the **Location** field.
   For this example, type **North America**, and then click **Search**.
   Database Manager performs the equal to search and displays the Record List of matching record(s), if any. If the query produces no matching records, Service Manager displays a message.
3. To display the Mass function buttons, use a query that returns more than one record.
   Select **Mass Add** from the toolbar to start the process of duplicating the records shown in the Record List.
   Database Manager displays the Mass Add/Update Instruction screen.
4. Type the following assignment statement in the **Instructions to be executed ONCE at the beginning of Mass Add/Update** input field.

Statement	Purpose
$sequential.number=1000	Sets the start point for the variable at 1000 and will be executed only once for the entire function.

5. Enter the following statements on the **Instructions for action on EACH RECORD** input fields:

Statement	Purpose
user.id in $file="NewAdvantage"+str($sequential.number)	Sets the User Id to NewAdvantage with a sequential number as a suffix.
$sequential.number+=1	Causes one (1) to be added to all future numbers created from this variable.
contact.name in $file=first.name in $file + " " + last.name in $file	Makes the contact name the same as the first and last name.

6. Click **Execute**.
   Database Manager performs the Mass Add function in the foreground. Your session is devoted to the add task until it is completed. If you are updating a large number of records, this can take a while.
   Upon completion, Service Manager returns terminal control to you and displays the following message:*<nnn>records added to the contacts file* where *nnn* is the number of records added.
7. Click **View Messages** to check the messages for errors.
   • A blue icon indicates a required action.
   • A yellow icon indicates an informational or warning message.
   • A red icon indicates an error message.
8. Close the Messages window to return to the contacts form.

Related topics

Format Control
Service Manager macros
Multiple-record functions
Updating multiple records
Key type definitions
Service Manager Programming

7.10.10.4.2. Updating multiple records

The Mass Update functions let you modify the contents of each selected record in the list of records accessed from Database Manager. You enter one or more processing statements, which Service Manager executes during adds, updates, and deletes.

Mass Update processing statements use the `$file` parameter to access fields on records in the current record list. For example, `name in $file` or `1 in action in $file`.

Update multiple records with Mass Update

Applies to User Roles: System Administrator

Mass Update enables a user (ADMIN or SYSADMIN) to update one or several fields in a set of records. Mass Update consists of three related functions, template Mass Update, Simple Mass Update, and Complex Mass Update. In a template Mass Update, a template displays a list of fields that can be updated. The user enters a value for each field to be updated to complete the process. In a Simple Mass Update, Service Manager displays a record form in which the user can enter values for selected fields. These values are then applied to all of the records selected for the Simple Mass Update. In a Complex Mass Update, Service Manager displays a form in which the user enters RAD expressions containing variables and concatenated fields to populate another field in the selected records.

The Mass Update function can be used with the following applications:

- Change Management
- Configuration Management
- Incident Management
- Problem Management
- Service Desk

The Database Manager also uses Mass Update to update the following tables:

- contacts
- company
- department (dept)
- location
- operator

The Mass Update functions in Database Manager also let you modify the contents of each record in the list of records accessed from Database Manager, by entering one or more processing statements, which Service Manager executes during adds, updates, and deletes. When add is set to true, Service Manager also executes Format Control, triggers, and macros, if present for the form.

You can update multiple records with Mass Update by using the Mass Update template form, Simple Update, or Complex Update. Choose one of the following methods to update multiple records.

Options for Mass Update errors

The following are the available options for Mass Update if validations on format control fail:

Option	Action
Retry	Reruns the mass update function on the selected records. Service Manager once again validates each record in the mass update against its format control.
Skip	Skips the current record without making any changes.
Force	Updates the current record with whatever information that you entered, regardless of format control.

Update multiple records with Mass Update by using the Mass Update template form

To update multiple records with Mass Update by using the Mass Update template form:

1. Click **System Administration > Base System Configuration > Companies**.
2. Click **Search**.
3. Select the records in the record list you want to update.
4. Click **Mass Update**.
5. Type the value for the field in the text box or use the Find feature to display a list of potential values for the field. **Note:** The value(s) you enter in the form are propagated to all of the records you selected.
6. Click **Next**.
7. Continue updating fields and click **Execute** when you complete your updates.

Note: If you exit the Mass Update template form or Complex Update form without pressing **Execute**, all of your changes are cleared on the respective form.

Update multiple records with Mass Update by using Simple Update

To update multiple records with Mass Update by using Simple Update:

1. Click **System Administration > Base Configuration > Companies**.
2. Click **Search**.
3. Select the records in the list you want to update.
4. Click **Mass Update**.
5. Click **Simple Update**.
6. Type new values into the appropriate fields on the form.
7. Click **Execute**.

Update multiple records with Mass Update by using Complex Update

To update multiple records with Mass Update by using Complex Update:

1. Click **System Administration > Base Configuration > Companies**.
2. Click **Search**.
3. Select the records in the list you want to update.
4. Click **Mass Update**.
5. Click **Complex Update**.
6. Type the RAD code in the Complex Update form.
7. Click **Execute**.

Update multiple records using a literal value

Applies to User roles: System Administrator

Example: modifying the Service Contract field value to the company's VIP Service for the contacts records for all advantage employees in North America.

To update multiple records with a literal value:

1. Sign in to the Service Manager Windows client.
2. Open the contacts form in Database Manager. Ensure that you select the Administration Mode checkbox.
3. Enter the desired value in the **Company** input field. For this example, select **advantage** from the list.
4. Enter the desired value in the **Location** input field. For this example, type North America.
5. Click **Search**.
   Service Manager displays all matching records in a record list. If the query produces no matching records, Service Manager displays a message. To display the Mass function buttons, use a query that returns more than one record.
6. Select **Mass Update** from the tool bar to start the process of updating the records shown in the record list.
   Database Manager re-displays the initial form with new option buttons.
7. Select **Simple Update** from the toolbar.
8. Enter the desired changes. For this example, select **VIP Service** from the **Service Contract** drop-down list.
9. Click **Execute**.
   The system returns terminal control to you and displays the newly updated records in the record list, along with the message `*nnn* records updated in the contacts file`, where `*nnn*` is the number of records updated.

Update multiple records using a variable value

Applies to User Roles: System Administrator

Example: Change the email address for all Advantage contacts records for people located in North America to follow the pattern: firstname.lastname@advantage.com.

To update multiple records with a variable value:

1. Open the contacts form in Database Manager. Ensure that you select the Administration Mode checkbox.
2. Enter the desired value in the **Company** input field. For this example, select **advantage** from the Company list.
3. Enter the desired value in the **Location** input field. For this example, type `North America`.
4. Click **Search**.
   Database Manager performs the full search and displays the Record List of matching records, if any, in the **contacts** form.
5. Select **Mass Update** from the tool bar to start the process of updating the records listed.
   Database Manager re-displays the initial form with different option buttons.
6. Do not enter values in any field. Click **Complex Update**.
   Database Manager displays the Mass Add/Update Instruction screen.
7. Enter the following assignment statement in the **Instructions for action on EACH RECORD** input field, as shown:

Statement	Purpose
email in $file = first.name in $file + "." + last.name in $file + "@advantage.com"	Sets the email address to firstname.lastname@advantage.com.

The system executes this instruction for each record updated. Then it returns terminal control to you, and displays the message `nnn records updated in the contacts file`, where `nnn` is the number of records updated.

Template mass update and required fields

Template mass update processes one record at a time, updating the record with the values that you enter in the template form. If a record is missing required fields, then Service Manager asks for the missing field(s), one at a time.

For example, the change queue tasks out-of-box are missing required fields Initial Impact Assessment and Urgency. When you use template mass update to modify these records and you don't enter a value for Initial Impact Assessment and Urgency, Service Manager displays "The Initial Impact Assessment is required". This message pertains to the present record that's being updated. If you enter an Initial Impact Assessment in the template form, then Service Manager prompts you for the missing Urgency for that record.

If there are three out-of-box change queue task records to modify using template mass update, and the second and third records are missing two required fields (for example, the Initial Impact Assessment and Urgency field values) and you enter a new Downtime End in the template form, Service Manager responds by:

1. Updating the value of the Downtime End field you entered in the first record.
2. Prompting for the missing field(s) for each record. When the Initial Impact Assessment and Urgency field values are entered in the template form, the second record is updated with the Downtime End, Initial Impact Assessment, and Urgency.
3. Prompting for the missing fields for the next record. When the Initial Impact Assessment and Urgency are added to the template, the third record is updated with the Downtime End, Initial Impact Assessment, and Urgency.

In summary, the data you originally added, the Downtime End, is never lost. The third record might have a different Initial Impact Assessment and Urgency from the second record, and since you entered no Initial Impact Assessment and Urgency in the template at first, Service Manager asks you for the Initial Impact Assessment and Urgency for the second and for the third record, one at a time.

Avoid invalid or null key errors

Applies to User Roles: System Administrator

Records can't have duplicate Unique or No Duplicates keys. This example shows one such error and how to fix it.

To handle an invalid duplicate of NULL key error:

1. Open the form in Database Manager. For this example, open the following form: `contacts`. Be sure to select the Administration Mode checkbox.
   A blank contacts form opens.
2. Type **NewAdvantage** in the Employee ID field., and then click **Search**.
3. Select **Mass Update**. from the toolbar to begin updating the records shown in the Record List.
   Database Manager displays the initial form again.
4. On the displayed form, type **NewAdvantage** in the Contact Name field.
5. Click **Simple Update**.
   - The system accepts the first update, because the contact name is unique.
   - The system rejects the second update, since changing it would create a duplicate contact name.

   Database Manager opens the record for modification and displays an error message.
6. Click **View Messages** to check the messages for errors.
   - A blue icon indicates a required action.
   - A yellow icon indicates an informational or warning message.
   - A red icon indicates an error message.
7. At this point, there are several options concerning the outcome of this process.

- Enter a unique key and click **Retry** for each item on the list separately, to cause Service Manager to reexecute everything, including the Format Control.
- Click **Skip** to cause Service Manager to skip the record.
- Click **Force** to cause Service Manager to update the record with the information you entered, regardless of Format Control.

For this example, modify the **Contact Name** field value to make it unique for each item. (NewAdvantage2, NewAdvantage3, NewAdvantage4, and so on.)
If the update was successful, Service Manager displays the following message: `<n>records updated in the contacts file.`

8. Type **NewAdvantage** in the Employee ID (user.id) field, and then click **Search**.
   Database Manager displays the records you have just added in a record list.
   The Employee ID values consist of Advantage + 1000, 1001, and so on, and the Contact Name values the first and the last names. The original records listed before the Mass Add remain in the contacts file, with no changes to the original data.

7.10.10.4.3. Delete multiple records

Applies to User Roles: System Administrator

Example: Delete the contacts for advantage with a variant of NewAdvantage as the contact name.

To delete all records with a specified value:

1. Click **Tailoring** > **Database Manager**.
2. Select **Administration mode**.
3. For this example, open the `contacts` form.
4. Enter the desired value. For this example, type **NewAdvantage*** in the Contact Name field.
5. Click **Search**.
   Database Manager performs the search and displays the record list of matching records, if any, in the contacts.qbe form.
   If the query produces no matching records or only one matching record, then Service Manager doesn't display a record list.
   Note: To display the Mass function buttons, use a query that returns more than one record.
6. Select **Mass Delete** from the tool bar to remove the records shown in the record list from the database.
   Database Manager confirms the delete request with a prompt screen.
7. Do one of the following:
   - To cancel the delete, click **No**.
   - To confirm the delete, click **Yes**.
     Database Manager deletes all records in the list.
     Note: Service Manager performs the Mass Delete function in the foreground, which means your terminal is devoted to this function until complete. If you are deleting a large number of records, this can take a while.

Upon completion, Service Manager returns terminal control to you. A blank contacts form opens with the message: *"nnn records deleted from the contacts file"*, where *nnn* is the number of records deleted.

Related topics

Multiple-record functions
Open a record using Database Manager

7.10.10.4.4. View multiple records using an alternate form

Applies to User Roles: System Administrator

Windows client:

To view a list of records using a predefined alternate form:

1. From the Window Menu select **Preferences...**.
2. Select **Service Manager** > **Appearance**.
3. Verify that the option **Show detail pane with record list** is unchecked. If it isn't, uncheck it. Log out and log back in.
4. From a list of records click the **More Actions** menu.

5. Select **Alternate Forms**.
6. Select a form to use to display the list.

Note: This option will be available only when there are additional forms specified for the list of records on the formatcontrol and the condition(s) specified are met.

Web client:

To view a list of records using an predefined alternate form:

1. Locate the web.xml document belonging to your Service Manager installation. For example:

C:\Program Files\Apache Software Foundation\[tomcat directory]\webapps\[ServiceManager web directory]\WEB-INF\web.xml

1. Open it for edit.
2. Set the parameter "viewrecordlist" to false.
3. Restart the Apache Tomcat service (**Windows Control Panel** > **Administrative Tools** > **Services**)
4. From a list of records click the **More Actions** menu.
5. Select **Alternate Forms**.
6. Select a form to use to display the list.

Note: This option will be available only when there are additional forms specified for the list of records on the formatcontrol and the condition(s) specified are met.

On the Web client, you can also:

1. Logout.
2. Add the following parameter to the end of the URL:
   `?viewrecordlist=false`
3. Example:

```
http://localhost:8080/ [ServiceManager web directory]/index.do?viewrecordlist=false
```

Related topics

Multiple-record functions
Adding multiple records
Open a record using Database Manager

7.10.10.4.5. Mass close interaction records

Applies to User roles: System Administrator

The Mass Close function enables a user to select and then close a set of incident or interaction records simultaneously. For the out-of-box systems, the ADMIN and SYSADMIN profiles include the Mass Close capability. In Service Desk, you can use Mass Close for interaction lists and interaction queues. In Incident Management, you can close multiple incident records.

For the out-of-box systems, the ADMIN and SYSADMIN profiles include the Mass Close capability. You can use Mass Close for interaction lists and interaction queues. Choose one of the following methods to mass close interactions.

Mass close interaction records after searching for interactions[edit]

To mass close interaction records after searching for interactions:

1. Click **Service Desk** > **Search Interaction Records**.
2. Use Search to display a list of interaction records.
3. Select the records in the record list you want to close.
4. Click **Mass Close**.
5. Edit the text present for the resolution of the interaction or accept the text.
6. Click **Finish**.

Mass close interaction records from the Interaction Queue[edit]

To mass close interaction records from the Interaction Queue:

1. Click **Service Desk** > **Interaction Queue**.
2. Specify the Queue and the View.
3. Select the records in the record list you want to close.
4. Click Mass Close in the More Actions menu.
5. Specify the Resolution Code. Use the Fill icon to display a list of available codes.
6. Click **Next**.
7. Accept or edit the text in the Resolution text box.
8. Click **Finish**.

Related topics[edit]

Contents

- 1 Mass close interaction records after searching for interactions
- 2 Mass close interaction records from the Interaction Queue
- 3 Related topics

7.10.10.4.6. Determine the number of records in a record list

To determine the number of records in a record list:

1. Open the `contacts` form in Database Manager.
2. Select the contacts file from the record list.
   A blank contacts record opens.
3. Click **Search**.
   The contacts format opens with a record list at the top and the **Count** option in the toolbar and in the More Actions menu.
4. From the More Actions menu, choose **Count** to get the record count.
   Service Manager displays the record count on the **Count** button.
   Note: If you select **Count** from the tool bar, a prompt opens asking whether all records in the list should be counted. Confirm by clicking **Yes**.
   The system counts the records and displays the following message: *"There are<n>records in this list"*.

Related topics

Multiple-record functions
Open a record using Database Manager

7.10.10.4.7. Print multiple records

Applies to User roles: System Administrator

Example: Print all contact records.

Open the contacts form in Database Manager, don't enter any values and then click **Search**. Database Manager performs the true search and displays a complete list of all the contact records.

> **Note**
>
> Note If the query produces no matching records or only one matching record, then Service Manager doesn't display a record list.

Follow the instructions in Print options to print in the windows client or the web client.

Related topics

Multiple-record functions
Print options
Open a record using Database Manager

7.10.10.5. Single-record functions

Single-record functions include adding, updating, deleting, and printing individual records within a database.

Add a record

Applies to User roles: System Administrator

Example: Add a record for the new contact, Bob Hoskins, to the contacts file.

To add a contact:

1. Open a form in Database Manager. For this example, open the `contacts` form.
2. Select the contacts file from the record list, and then click **Search**.
3. Enter the required values in the Business Information section of the Business tab, and other values as needed. In this example, enter the following values.

Field	Record value
Contact Name	HOSKINS,BOB
Company	Advantage

4. Click **Add**.

Database Manager adds the record, retains the input from the screen displayed, and responds with the following message: `Contact Information record added.`

Clear an initial record

Applies to User roles: System Administrator

If you need to redo an incorrect record or need otherwise to clear all data from an initial record form, you can remove this data quickly and completely using the Clear command.

To clear all data from an initial screen:

1. Open the form in Database Manager. For this example, open the `contacts` form.
2. Select the record from the record list.
3. Open the More Actions menu.
4. Select **Clear**.

Delete a record

Applies to User roles: System Administrator

Example: Delete the contact record you added for Richard Butler.

To delete a record:

1. Open a form in Database Manager. In this example, open the **=contacts** form.
2. Search for the existing record by entering **BUTLER, RICHARD** in the Contact Name field and clicking **Search**.
3. Click **Delete**. Confirm the action at the prompt.
4. Click **Yes** to continue with the record deletion. If successful, the system displays the following message: `Contact Information record deleted`.

Duplicate an existing record

Applies to User roles: System Administrator

Example: Add a new record to the contacts table that's a near duplicate of an existing record. For this example, all information except the Primary Configuration item (CI) and Contact Name will be the same.

To duplicate an existing record:

1. Open the form in Database Manager. For this example, open the `contacts` form.
2. Select the contacts file from the record list.
   A blank contacts record opens.
3. Click **Search**.
4. Select CHAN, TERESA from the returned list.
5. Select the Business tab.
6. Type a new name in the **Contact Name** field.
   For this example, type **BUTLER, RICHARD**.
7. Enter a new **Primary Configuration Item** for this record.
8. Delete the **Employee ID**.
   Add your own at this point, or leave this field blank.
9. Click **Add**.

Database Manager adds the record, retains the input from the screen displayed, and responds with the following message: `Contact Information record added`.

Print a record

Applies to User roles: System Administrator

Example: Printing an existing contacts record for Richard Butler.

To print a record:

1. Open the contacts form in Database Manager.

2. Select the contacts file from the record list, and then press Enter. A blank contacts file opens.
3. Click **Search**.
4. Select Richard Butler from the returned list.

Search for newly added records

Applies to User Roles: System Administrator

To search for newly added records:

1. Click **Tailoring** > **Database Manager**.
2. Place the cursor in the **Table** field, and then click **Search**.
3. Click the **Last Updated** column header to sort the table of records returned by date and time.

Update an existing record

Applies to User roles: System Administrator

Example: Update the existing contacts record for Teresa Chan, whose primary Configuration Item (CI) has an identification number of *167*, and change the value to *adv-nam-desk-267*.

In this example, we use the **contacts** form to change the identification number of Teresa Chan's primary CI, from *adv-nam-desk-167* to *adv-nam-desk-267*.

Note: Audit Specification Records must be defined for Configuration Items prior to modifying.

To update an existing record:

1. Click **Tailoring** > **Database Manager**.
2. Type `=contacts` in the **Form** field, and then click **Search**.
3. On the Contact Information form, place the cursor in the Contact Name field, and then click **Search**.
4. Select **CHAN, TERESA** from the record list returned and select the **Business** tab.
5. In the **Primary Configuration Item** field, type `adv-nam-desk-267` to replace the current value.
6. Click **Save**.

View a single record using an alternate form

Applies to User Roles: System Administrator

To view a record using predefined alternate form:

1. From a detail record click the **More Actions** menu.
2. Select **Alternate Forms**.
3. Select a form to use to display the record.

Note: This option will be available only when there are additional forms specified for the record on the formatcontrol and the condition(s) specified is met.

Related topics

Determining System Navigator menu options
Multiple-record functions

7.10.11. Queries

Queries are logical expressions that evaluate to either true or false against the records queried. Click **Tailoring** > **Database Manager** to search for records matching your query. For records that evaluate to true, those records display in a table of matching records. You can click on each record for details. For records that evaluate to false, no records are returned and an error message is displayed.

Related topics

Logical operators
Record list
Relational operators
Retrieving records by using the record list method
Retrieving records using the Query window method

Excluding records from a query
Perform IR Expert queries

7.10.11.1. Simple query expressions

Service Manager offers several methods to perform queries. Simple queries using the Query Window method are most useful in the following situations:

- On forms where a record list query can't be performed.
- When performing range queries.
- When a field exists in the database dictionary record of the file, but is not displayed on the form.
- When the input field on the form is not long enough to contain the number of characters needed for the value you want to query.

Note: The Query Window Application form, named query.window, is used in Database Manager, Incident Management, and Configuration Management. In addition to the query string, you can optionally specify sort fields to further filter your results. You must have the query.window capability word enabled in your operator record to open the Query Window.

Related topics

Retrieving records using the Query window method
Retrieving records using the Query window method
Retrieving records using the Query window method
Complex query expressions
Logical operators
Relational operators
Wildcard characters
Create a query using a simple query expression
Perform IR Expert queries
Use like queries to search for a string of characters

7.10.11.2. Complex query expressions

Service Manager offers several methods to perform complex queries. These include:

- Logical operators, such as AND, OR, and NOT
- Relational Operators, such as greater than, less than, and equal to
- RAD functions, such as index() and tod()
- IR Expert searches

Related topics

Logical operators
Relational operators
Wildcard characters
IR Expert
Retrieving records using the Query window method
Simple query expressions
Create a query using not in statements
Perform IR Expert queries
Use like queries to search for a string of characters
Use the index function in a query
Use the lng function in a query
Use the tod function in a greater than or less than query
Service Manager Programming

7.10.11.2.1. Logical operators

Logical operators evaluate Boolean expressions and determine whether the entire expression is true or false based on the operator used.

You can use logical operators to do the following:

- Construct simple queries that return true, false, or unknown results using Boolean logic.
- Perform queries using combinations of logical operators that aren't available to use with the Record list query method, use the Retrieving records method from the query window.

Operator	Symbol	Usage	Operation
AND	&	*value* AND *value*	Returns TRUE if both component conditions are TRUE. Returns FALSE if either is FALSE. Otherwise returns UNKNOWN.
OR		*value* OR *value*	Returns TRUE if either component condition is TRUE. Returns FALSE if both are FALSE. Otherwise returns UNKNOWN.
NOT	~	not value	Returns TRUE if the following condition is FALSE. Returns FALSE if it is TRUE. If it is UNKNOWN, then it remains UNKNOWN.

Logical operators take precedence in the following order: (1) NOT, (2) AND, (3) OR.

Results of applying the NOT operation to an expression:

--

TRUE

FALSE

UNKNOWN

NOT

FALSE

TRUE

UNKNOWN

Results of combining two expressions by using AND:

AND

TRUE

FALSE

UNKNOWN

TRUE

TRUE

FALSE

UNKNOWN

FALSE

FALSE

FALSE

FALSE

UNKNOWN

UNKNOWN

FALSE

UNKNOWN

Results of combining two expressions by using OR:

OR

TRUE

FALSE

UNKNOWN

TRUE

TRUE

TRUE

TRUE

FALSE

TRUE

FALSE

UNKNOWN

UNKNOWN

TRUE

UNKNOWN

UNKNOWN

Related topics

Relational operators
Wildcard characters
Retrieving records by using the record list method
Retrieving records using the Query window method
Using ranges in queries
Create a query using an or statement or an and statement
Perform IR Expert queries
Use the not symbol with logical or relational operators

7.10.11.2.2. Relational operators

Relational operators compare values and return True, False, or Unknown. Whenever a comparison is executed against a null value, Service Manager returns Unknown. For example, if you are searching for all contact records matching the query `shift="day"` all contact records without a value in the shift field will return Unknown.

The most efficient relational operator in a query is the equal to (=) relational operator, for which exact results are returned. Although queries using this operator are valid for all field types, it is seldom used for queries to date or time fields. For date and time fields, the greater than (>) and less than (<) operators are more efficient because the values returned when using these operators include seconds.

Relational Operator	Definition
#	starts with
~#	doesn't start with
=	equal to
~=	not equal to
<>, ><	not equal to
<	less than
<= or =<	less than or equal to
>	greater than
>= or =>	greater than or equal to
isin	is element in
like	is similar to

Related topics

Equal to relational operator
Greater than relational operator
Less than relational operator
Like relational operator
Starts with relational operator
Logical operators
Retrieving records by using the record list method
Retrieving records using the Query window method
Use the equal to relational operator
Use the greater than relational operator
Use the less than relational operator
Use the like relational operator
Use the starts with relational operator
Perform IR Expert queries

7.10.11.2.3. Starts with relational operator

In a search screen, the starts with (#) relational operator is the default query type, unless you use a wildcard character. If the value you enter contains wildcard characters, the like relational operator is assumed. To use the starts with operator in conjunction with wildcard characters, enter the starts with (#) operator as the first character in your query.

In the query window, everything is interpreted literally as you type it in.

Related topics

Relational operators
Wildcard characters
Retrieving records by using the record list method
Retrieving records using the Query window method
Use the equal to relational operator
Use the greater than relational operator
Use the like relational operator
Use the less than relational operator
Use the starts with relational operator
Perform IR Expert queries

7.10.11.2.4. Equal to relational operator

The equal to (=) relational operator is the most efficient relational operator and should be used when possible. Queries that use the equal to relational operator are valid on all field types although it's not useful to query date and time fields because the results are literal.

Related topics

Relational operators
Wildcard characters
Retrieving records by using the record list method
Retrieving records using the Query window method
Use the equal to relational operator
Use the greater than relational operator
Use the less than relational operator
Use the like relational operator
Use the starts with relational operator
Perform IR Expert queries

7.10.11.2.5. Less than relational operator

The less than (<) operator can be used to query any field type. It is most frequently used to query date and time fields because the values returned include seconds, which makes using the equal to (=) operator for date and time fields impractical.

Related topics

Relational operators
Wildcard characters
Retrieving records by using the record list method
Retrieving records using the Query window method
Use the equal to relational operator
Use the greater than relational operator
Use the like relational operator
Use the less than relational operator
Use the starts with relational operator
Perform IR Expert queries

7.10.11.2.6. Greater than relational operator

The greater than (>) relational operator can be used to query any field type. It is most frequently used to query date and time fields because the values returned include seconds, which makes using the equal to (=) operator for date and time fields impractical.

Example Scenario:

Suppose you want to find records opened at 11:00 AM on June 1. Since it's not likely that the record was opened at *exactly* at 11:00 AM and zero seconds (11:00:00). You are more likely to get the results you want by using a range of times. For example, if you type `>06/01/2008 11:00` in the **open.time** field, all records opened after 11:00 AM, starting with 11:00:01, are returned.

Related topics

Relational operators
Wildcard characters
Retrieving records by using the record list method
Retrieving records using the Query window method
Use the equal to relational operator
Use the less than relational operator
Use the like relational operator
Use the greater than relational operator
Use the starts with relational operator
Perform IR Expert queries

7.10.11.2.7. Using ranges in queries

A range is defined by using a combination of less than (<) and greater than (>) relational operators between the *and* logical operator or the *or* logical operator.

To query of a range of data, use the Query Window.

Operator	Usage	Examples
AND and	>*value* and <*value*	`field_name > lesser_value and field_name < greater_value` For example, `open.time>'09/15/2008' and open.time<'09/27/2008'` The *and* logical operator is not case-sensitive. You can use (&) as a shortcut for the *and* logical operator.
OR or	<*value* or >*value*	`field_name < lesser_value or field_name > greater_value` For example, `open.time<'09/15/2008' or open.time>'09/27/2008'` The *or* logical operator is not case-sensitive. You can use (|) as a shortcut for the *or* logical operator.

Related topics

Logical operators
Relational operators
Retrieving records by using the record list method
Retrieving records using the Query window method
Open the Query Window
Perform IR Expert queries

7.10.11.2.8. Like relational operator

The like relational operator allows you to search for character strings similar to the values entered. The like operator is only valid on character fields.

Related topics

Relational operators
Wildcard characters
Retrieving records by using the record list method
Retrieving records using the Query window method
Use the equal to relational operator
Use the greater than relational operator
Use the less than relational operator
Use the like relational operator
Use the starts with relational operator
Perform IR Expert queries

7.10.11.2.9. Wildcard characters

By default, Service Manager uses the question mark (?) as the wildcard that represents a single character, the asterisk (*) to represent multiple characters, and the backslash (\) as the escape character. Therefore, for a literal search on the * or ? character, you must immediately precede that character with a backslash (\), such as `\?`.

Note: You can modify the wildcard characters and the escape character by using the **wildcardcharacters** parameter in the Service Manager server **sm.ini** file.

Example Scenario:

Suppose you want to find all records containing last names where the second character is**e**.

To search for any character followed by the letter e followed by the asterisk (*) character, you would type `?e\*` in the **Last Name** field to generate a*like* query. Although the question mark (?) is a wildcard character, it verifies that the last name ends with a literal *.

The query would retrieve and display records with last names that begin with:

```
Ae*
ae*
?e*
*e*
...
```

Related topics

Logical operators
Relational operators
Use like queries to search for a string of characters
Use the index function in a query

7.10.11.3. Create a query using not in statements

Applies to User Roles: System Administrator

Example: Retrieve all contacts records where the value of the manufacturer isn't `advantage` and the not symbol (~) is unavailable on the keyboard.

This example uses the `not` relational operator to perform an Advanced Search on records in the **contacts.qbe** form and return a list of all contacts that aren't employees of Advantage.

To create a query using not in statements:

1. Click **System Administration** > **Base System Configuration** > **Contacts**
2. On the Contact Information form, open the More Actions menu.
3. Select **Expert Search**. The Query window opens.
4. In the **Query** field, type `not (company="advantage")`, and then click **Search**.
5. Click any record in the list of returned records to view the details.

Related topics

- Complex query expressions
- Record retrieval
- Relational operators
- Retrieving records using the Query window method
- More Actions menu
- Create a query using an or statement or an and statement
- Create a query using isin in statements
- Use like queries to search for a string of characters

7.10.11.4. Create a query using an or statement or an and statement

Applies to User Roles: System Administrator

Example: Retrieve all `contacts` records where the Company is either `DEFAULT` or `advantage`, and the value in the **Contact Name** field begins with the letter `B`.

This example performs a complex query using a combination of logical operators to query the `contacts.qbe` form.

To create a query using an `or` statement or an `and` statement:

1. Click **Tailoring** > **Database Manager**.
2. Type `contacts` in the **Form** field, and then click **Search**.
3. Double-click the **contacts** form from the list of returned records.
4. On the **Contact Information** form, open the More Actions menu.
5. Select **Expert Search**.
6. In the **Query** field, type `(company="advantage" or company="DEFAULT") and contact.name#"B"`, and then click **Search**. This query may result in a partially keyed search. If so, you can optionally enter a time limit in your query before executing.
7. Click any record in the list of returned records to view the details.

Note: If you type the same expression without using parentheses, such as `company="advantage" or company="DEFAULT" and contact.name#"B"`, the `and` operator takes precedence over the `or` operator and returns a different set of records. This query is less restricted and returns records with a value of either `DEFAULT` or `advantage`.

Related topics

- Complex query expressions
- Partially keyed queries
- Record retrieval
- Relational operators
- Retrieving records using the Query window method
- Create a query using not in statements
- Use like queries to search for a string of characters

7.10.11.5. Use like queries to search for a string of characters

Applies to User roles: System Administrator

Example: Use a like query in combination with a field function to retrieve all records for vendors/suppliers that begin with the character string *DEFAULT*.

This example uses the `like` relational operator on the **vendor.qbe** form to query the **vendor** field for vendors/suppliers. These vendors/suppliers begin with *DEFAULT*, followed by any character or character string.

To use like queries to search for a string of characters:

1. Click **Tailoring** > **Database Manager**.
2. Type `=device` in the **Form** field, and then click **Search**.
3. On the Asset Information form, open the More Actions menu.
4. Select **Expert Search**.
5. In the **Query** field, type `vendor like "DEFAULT*"`, and then click **Search**.
6. Click any record in the list of returned records to view the details.

Related topics

- Relational operators
- Like relational operator
- Record retrieval
- Retrieving records using the Query window method
- More Actions menu
- Use the like relational operator
- Create a query using more than one field
- Create a query using array fields

7.10.11.6. Create a query using array fields

Applies to User Roles: System Administrator

Example: Retrieve all operator records where the operator has ability to execute tasks with the capability words of SysAdmin or problem management.

In queries using array fields, the position of an element within an array is not relevant. The logical operator *or* is used in array field queries and all records with any of the values entered in the array are retrieved.

In this example, we use the *or* logical operator to search for records in the **operator.g** form that contain the execute capability words of *SysAdmin* or *Problem Management*.

To create a query using array fields:

1. Click **Tailoring** > **Database Manager**.
2. Type `=operator` in the **Form** field, and then click **Search**.
3. Click **Search** again.
4. On the Operator Record form, click **Startup** tab.
5. Type `=SysAdmin` in the first **Execute Capabilities** array field.
6. Type `=problem management` in the second **Execute Capabilities** array field, and then click **Find**.

Click any record in the list of returned records to view the details.

Related topics

Record retrieval
Retrieving records by using the record list method
Using ranges in queries
Create a query using more than one field
Use like queries to search for a string of characters
Capability words

7.10.11.7. Create a query using more than one field

Applies to User Roles: System Administrator

Example: Retrieve all device records where the **logical.name** field in the Manage CIs form begins with the letter *H* and the **IP address** field starts with *196.76.209*.

In this example, we perform a search on the **Location** and **IP Address** fields in the **device** form and return a list of device records matching these criteria. Database Manager forms record list queries on scalar, or *non-arrayed* fields, with the *and* logical operator. Searches are case-sensitive, unless you first set up your database for case-insensitive searching.

To create a query using more than one field:

1. Click **Tailoring** > **Database Manager**.
2. Type **=device** in the **Form** field, and then click **Search**.
3. On the Search Configuration Records form, do the following:
   - Type **H** in the **Location** field.
   - Select the Advanced Filter tab, and then click **Add New Filter Criteria**.
     The Add Filter Criteria wizard opens.
   - In the **Field in Configuration Item** field, select the **IP Address** field.
   - In the **Comparison** field select **Starts With**.
   - In the **Value** field, type **196.76.209**.
   - Click **Next**.
     You are returned to the Search Configuration Records form.
4. Open the More Actions menu and choose **Expert Search**.
   You will see your query in the **Query** field.
5. Click **Search**.
   When a partially keyed query is executed, Service Manager searches the file for a specified time interval, or until the screen buffer is filled. When either threshold is met, it stops and displays the records retrieved up until that point.

A warning then displays, indicating an attempt to initiate a partially keyed query. If you receive a warning message of this type, you have limitations in your operator or user profile that prevent the execution of inefficient queries.

6. Add a time interval to your query, and then click **Search** to execute the query again.
7. Click any record in the list of returned records to view the details.

Note: A *device* is also referred to as a *Configuration Item* (CI).

Related topics

Record retrieval
Relational operators
Retrieving records by using the record list method
Using ranges in queries
Create a query using array fields
Use like queries to search for a string of characters
Use the not symbol with logical or relational operators

7.10.11.8. Perform IR Expert queries

Applies to User Roles: System Administrator

You can perform an *intelligent* query using an information retrieval engine called the *IR Expert utility*. With this utility, you can accurately define queries for a list of helpful topics filtered from your search criteria.

To perform an IR Expert query on the probsummary table:

1. Click **Tailoring** > **Database Manager**.
2. Type `=probsummary` in the **Form** field, and then click **Search**.
3. On the incident search form, click **Text Search**.
4. In the Search Text area, type your search word(s).
   Note: Be sure to insert a comma (,) or white space between your search words if you want to search for any of them. For example, type "**computer, reboot**" ' ***or*** ' "**computer reboot**".
5. Specify a search type:
   - All of these words - search for records that contain all of your search words.
     For example, if you entered "web browser hangs" in the Search Text field on the incident search form, selecting this option will only search for incident records whose Description or Title field contains "web browser hangs".
   - Any of these words - search for records that contain any of your search words.
     For example, if you entered "web browser hangs" in the Search Text field on the incident search form, selecting this option will search for incident records whose Description or Title field contains "web", or "browser" or "hangs".
     - Search only these words - search for your search words only;
     - Also search related words (slower) - search for your search words and also their related words.
6. Click **Search**.
   Service Manager displays the search results.

Related topics

IR Expert
Record retrieval
Retrieving records by using the record list method
Retrieving records using the Query window method
More Actions menu
Open a record using Database Manager

7.10.11.9. Use the equal to relational operator

Applies to User Roles: System Administrator

Example: Retrieve all device records where the value of the **Location** field is equal to *DEFAULT HQ*.

In this example, we execute an equal to (=) search from the **device** form and then displays a record list containing all exact matches.

To use the equal to relational operator:

1. Click **Tailoring** > **Database Manager**.
2. Type `=device` in the **Form** field.
3. Click **Search**.
4. Type `=DEFAULT HQ` in the **Location** field on the device form.
5. Click **Find**.
6. Click any record to view the details.

Note: A *device* is also referred to as an *Asset* or *Configuration Item* (CI).

Related topics

Equal to relational operator
Record retrieval
Relational operators

7.10.11.10. Use the greater than relational operator

Applies to User Roles: System Administrator

Example: Retrieve all device records where the value of the **logical.name** field is greater than *d*.

In this example, we execute a greater than (>) search from the **device** form and then displays a record list containing all matches with a value greater than the letter *d*.

To use the greater than relational operator:

1. Click **Tailoring** > **Database Manager**.
2. Type `=device` in the **Form** field.
3. Click **Search**.
4. Type `>d` in the **Asset Tag** field on the device form.
5. Press **Enter**.
6. Click any record to view the details.

Note: A *device* is also referred to as an *Asset* or *Configuration Item* (CI).

Related topics

Greater than relational operator
Record retrieval
Relational operators
Retrieving records by using the record list method

7.10.11.11. Use the less than relational operator

Applies to User Roles: System Administrator

Example: Retrieve all device records where the value of the **IP Address** field is less than *196*.

In this example, we execute a less than (<) search from the **device** form and then displays a record list containing all matches with a value less than the number *196*.

To use the less than relational operator:

1. Click **Tailoring** > **Database Manager**.
2. Type `=device` in the **Form** field.
3. Click **Search**.
4. Type `<196` in the **Serial number** field on the device form.
5. Press **Enter**.
6. Click any record to view the details.

Note: A *device* is also referred to as an *Asset* or *Configuration Item* (CI).

Related topics

Less than relational operator
Record retrieval
Relational operators
Retrieving records by using the record list method

7.10.11.12. Use the like relational operator

Applies to User Roles: System Administrator

Example: Retrieve all contact records that contain the letter *e* as the second letter of the last name.

In this example, we execute a like query where the **full.name** field is like *?e** and then displays a record list containing all matching records.

To use the like relational operator:

1. Click **System Administration** > **Base System Configuration** > **Contacts**.
2. Type `?e*` in the **Full Name** field.
3. Press **Enter**.
4. Click any record to view the details.

Note: This query contains two wildcard characters, which triggers Database Manager to automatically generate a like query. A like query doesn't automatically generate a starts with query. For example, to see all records that start with any character followed by *e** with any number of characters following, type `?e\**` in the **last.name** field. The returned records include those where the **last.name** is like *Ae**, *he*e*, *fe**, *Be*bcdefg*, and so on.

Related topics

Like relational operator
Record retrieval
Relational operators
Wildcard characters
Retrieving records by using the record list method

7.10.11.13. Use the not symbol with logical or relational operators

Applies to User Roles: System Administrator

Example: Retrieve all device records, excluding those that begin with the letter *C*, in the **Vendor Name** field.

Note: If your database has not first been set up to enable case-insensitive searching, this query returns records that begin with a lowercase letter *c*.

In this example, we use the not relational operator to exclude records in the **device.qbe** form.

To use the not symbol with logical or relational operators:

1. Click **Tailoring** > **Database Manager**.
2. Type `=device` in the **Form** field, and then click **Search**.
3. With the **device** form open, type `C` or `~C` in the **Vendor Name** field, and then click **Find**.
   Note: Using the tilde (~) character is platform-dependent.
4. Click any record in the list of returned records to view the details.

Note: A *device* is also referred to as an *Asset* or *Configuration Item* (CI).

Related topics

Logical operators
Relational operators
Retrieving records by using the record list method
Using ranges in queries

7.10.11.14. Use the starts with relational operator

Applies to User Roles: System Administrator

Example: Retrieve all capability words that begin with the letter *p*.

In this example, we execute a starts with query for all capability records where the **capability** field on the Capability Word form begins with the letter *p*.

To use the starts with relational operator:

1. Click **Tailoring** > **Database Manager**.
2. Type `capability` in the **Form** field, and then click **Search**.
3. Double-click the **capability** form in the records returned.
4. Type `p` in the **Capability** field on the Capability Word form, and then press **Enter**.
5. Click any record to view the details.

Note: Results returned are case-sensitive and return only matches for the case used in your query string, unless your database is first set up for case-insensitive searching.

Related topics

Record retrieval
Relational operators
Retrieving records by using the record list method
Starts with relational operator

7.10.11.15. Create a query using a simple query expression

Applies to User roles: System Administrator

In this example, a query is performed on a specific assignee for all contracts in Configuration Management. Once stored, operators or groups of operators can access the query and automatically generate a list of records without reentering the detailed search criteria.

Note: You must have the capability word **QueryAdmin** or **query.stored.mod** operator record to add or edit stored queries.

To create a query using a simple query expression:

1. Click **Configuration Management** > **Contracts** > **Contracts**.
2. Use the Fill function in the Assignee field to choose an assignee. For this example, Carlton Hulman was chosen.
3. Open the More Actions menu and choose **Expert Search**.
   The Query form opens. The new query is displayed in the **Query** field. For this example, the query for Carlton Hulman is displayed as follows: `assignee#"Carlton.Hulman"`
4. On the Query form, click **Keys**.
   Note: Do not modify the automatically-generated query string, `assignee#"Carlton.Hulman"`, as it appears in the **Query** field.
5. On The Key Window (keylist form), type a key number to select the number of positions for the **assignee.name** key in the query. For example, type the number `3` to select *3* positions.
6. Click **OK**.
   The Sort Fields field in the Query Window is now modified to agree with the key definition you selected.
7. On the Query form, click **Store** to create the **querystored** file.
8. Fill in the following fields:
   - In the **Access List** field, type in the Query Groups or Operator Names that you want to have access to this query. If you want all users to have access, leave the list blank.
   - In the **Name** field, type a unique name for your query.
   - In the **Description** field, type an explanatory description.
9. Click **Add**.
   The following message confirms your entry was successful: `Query added to querystored file`.
10. Continue to further refine your search criteria, or click **Search**. You can also click any record in the list of returned records to view the details.

Related topics

Complex query expressions
Record retrieval

Retrieving records using the Query window method
Simple query expressions

7.10.11.16. Create a query using isin in statements

Applies to User Roles: System Administrator

Example: Retrieve all contacts records where the Company name is either `DEFAULT` or `advantage` and the Contact Name begins with the letter `B`.

In this example, we perform a complex query on the **contacts.qbe** form using the `isin` relational operator. An Advanced Search is performed and records are returned where the value of the **Company** field is either `DEFAULT` or `advantage` and the value of the **Contact Name** field begins the letter `B`.

To create a query using `isin` in statements:

1. Click **System Administration** > **Base System Configuration** > **Contacts**.
2. On the Contact Information form, open the More Actions menu.
3. Select **Expert Search**.
4. In the **Query** field, type `company isin {"advantage", "DEFAULT"} and contact.name#"B"`, and then click **Search**.
5. Click any record in the list of returned records to view the details.

Related topics

Record retrieval
Relational operators
Retrieving records using the Query window method
Using functions in a query
More Actions menu
Create a query using an or statement or an and statement
Create a query using not in statements
Create a query using not in statements
Use like queries to search for a string of characters

7.10.11.17. Use the index function in a query

Applies to User Roles: System Administrator

Example: Retrieve all records where the character string *DEFAULT* occurs anywhere in the device name and the device name ends in the number string *002*.

This example uses a `like` query and wildcard characters with the **index** function to perform an Advanced Search on records on the **device.qbe** form.

To use the index function in a query:

1. Click **Tailoring** > **Database Manager**.
2. Type `=device` in the **Form** field, and then click **Search**.
3. On the Asset Information form, open the More Actions menu.
4. Select **Expert Search**.
5. In the **Query** field, type `logical.name like "*002" and index("DEFAULT", logical.name)>0`, and then click **Search**.
   Note: Wildcard characters are only valid in like queries. In other types of queries, the characters are literal interpretations.
6. Click any record in the list of returned records to view the details.

Note: A *device* is also referred to as an *Asset* or *Configuration Item* (CI).

Related topics

- Complex query expressions
- Record retrieval
- Relational operators
- Stored queries

- Using functions in a query
- More Actions menu
- Use like queries to search for a string of characters
- Use the lng function in a query
- Use the tod function in a greater than or less than query

7.10.11.18. Use the lng function in a query

Applies to User roles: System Administrator

Example: Retrieve operator records where the length of the operator name is longer than five (5) characters.

This example uses the `lng` function to specify the length of character strings in the **operator.qbe** form and then performs an Advanced Search for records matching that criteria.

To use the lng function in a query:

1. Click **Tailoring** > **Database Manager**.
2. Type `=operator` in the **Form** field, and then click **Search**.
3. On the Operator Record form, open the More Actions menu.
4. Select **Expert Search**.
5. In the **Query** field, type `lng(name)>5`, and then click **Search**.
   Note: This lng query is the same as typing `name like "??????*"`
6. Click any record in the list of returned records to view the details.

Related topics

- Record retrieval
- Relational operators
- Using functions in a query
- More Actions menu
- Use the index function in a query
- Use the tod function in a greater than or less than query

7.10.11.19. Use the tod function in a greater than or less than query

Applies to User roles: System Administrator

Example: Retrieve all operator records where the password was updated within the past *100* days, or where the password has not been updated in the past *100* days.

In this example, we use the greater than (>) or less than (<) operator, in conjunction with the **tod** function, to perform an Advanced Search on records in the **operator.qbe** form.

To use the tod function in a greater than or less than query:

1. Click **Tailoring** > **Database Manager**.
2. Type `=operator` in the **Form** field, and then click **Search**.
3. On the Operator Record form, open the More Actions menu.
4. Select **Expert Search**.
5. In the **Query** field, type `password.date>tod() - '100 00:00:00'`, and then click **Search**.
6. Click any record in the list of returned records to view the details.

Note: Typing `password.date<tod() - '100 00:00'` in Step 5 returns the opposite result, where the password has *not* been changed in 100 days or more.

Related topics

Record retrieval
Relational operators
Using functions in a query
More Actions menu
Create a query using not in statements

Create a query using an or statement or an and statement
Use like queries to search for a string of characters
Use the index function in a query
Use the lng function in a query

7.10.11.20. Retrieving records by using the record list method

The record list is a standard feature for accessing records in a database. You will use it most frequently at prompts where a form or file name is required. You enter the necessary information in the form and Database Manager processes the query by looking for similar records based on the values you entered. Search values from record lists can be made more efficient through the use of relational operators. Database Manager displays the records it finds in a record list.

Retrieve all records in a table

Applies to User Roles: System Administrator

Example: Retrieve all records in the Manage CI form.

In this example, we execute a *true* query, which returns all records contained in the **device.g** form.

To retrieve all records in a table:

1. Click **Tailoring** > **Database Manager**.
2. Type `=device` in the **Form** field, and then click **Search**.
3. On the **Manage CI** form, click **Search**.
4. Click any record in the list of returned records to view the details.

Retrieve records within a range

Applies to User Roles: System Administrator

Example: Retrieve all device records where the value of the **Part Number** field is greater than a character value of *500* and less than a character value of *1000*.

In this example we execute a greater than (>) and less than (<) search on the *device.g* form to search part numbers where the value is greater than the number *500* and less than the number *1000*.

Note: For a range query on numeric fields, use the Query window.

To retrieve records within a range:

1. Click **Tailoring** > **Database Manager**.
2. Type `=device` in the **Form** field, and then click **Search**.
3. On the Manage CI form, type `>500 AND <1000` in the **Part Number** field.
4. Click **Find**.
5. Click any record in the list of returned records to view the details

Note: A *device* is also referred to as a *Configuration Item* (CI).

Related topics

When should I use the Query Window instead of the record list query method?
Record retrieval

7.10.11.21. Retrieving records using the Query window method

The Query Window method allows you to retrieve records by typing a logical expression directly into the query field. When you press Enter, Database Manager searches for records that match your query and displays the list of matching records in a table. Additionally, you can define the querywindow capability word in your operator record to add the ability to specify sort fields in the Query Window application.

Related topics

Capability words
Logical operators
Record retrieval
Retrieving records by using the record list method

When should I use the Query Window instead of the record list query method?
Excluding records from a query
Perform IR Expert queries

7.10.11.21.1. When should I use the Query Window instead of the record list query method?

Use the Query Window instead of the record list query method in the following situations:

- When performing simple queries in situations where a record list query can't be used.
- For queries that include ranges, such as when the field to query exists in the database dictionary record of the file, but is not displayed on the form.
- When the input field on the form is not long enough to contain the desired value.

Related topics

Record retrieval
Retrieving records by using the record list method
Retrieving records using the Query window method
Perform IR Expert queries

7.10.11.21.2. Excluding records from a query

To exclude specific records from a query, use the not symbol with logical or relational operators.

Related topics

Logical operators
Relational operators
Perform IR Expert queries
Use the not symbol with logical or relational operators

7.10.11.21.3. Using functions in a query

Use the Query window to perform queries that involve functions, such as **index**, **tod**, or **lng**.

Related topics

Record retrieval
Retrieving records using the Query window method
Retrieving records by using the record list method
Perform IR Expert queries
Use the index function in a query
Use the lng function in a query
Use the tod function in a greater than or less than query
Service Manager Programming

7.10.11.21.4. Open the query window

Applies to User roles: System Administrator

To open the Query window from the operator record:

1. Click **Tailoring** > **Database Manager**.
2. Type `=operator` in the **Form** field, and then click **Search**.
3. On the Operator Record form, open the More Actions menu.
4. Select **Expert Search**.
5. The Query form is displayed. The display options on the form are dependent on the execution capability words assigned for the user, as follows:

1. - **Query Window access** – The Query window is displayed at the bottom of your screen.
   - **Stored query access** – If stored queries are defined, a list is displayed.
   - **Both** – The Query window with buttons defining query options is displayed.

Related topics

Complex query expressions
Record retrieval
Logical operators
Relational operators
Retrieving records using the Query window method
Stored queries
Using functions in a query
More Actions menu
Create a query using a simple query expression
Use like queries to search for a string of characters
Use the Query Window

7.10.11.21.5. Use the query window

Applies to User roles: System Administrator

Example: Retrieve all device records where the logical name begins with the character *A*.

This example uses the equal to (=) relational operator to perform an Advanced Search on records in the **device.qbe** form. The system returns all values in the **logical.name** heading that beginning with the letter *A*.

To use the Query Window:

1. Click **Tailoring** > **Database Manager**.
2. Type `=device` in the **Form** field, and then click **Search**.
3. On the Asset Information form, open the More Actions menu.
4. Select **Expert Search**.
5. In the **Query** field, type `logical.name#"A"`, and then click **Search**.
6. Click any record in the list of returned records to view the details.

Note: A *device* is also referred to as an *Asset* or *Configuration Item* (CI).

Related topics

- Complex query expressions
- Stored queries
- Record retrieval
- Relational operators
- Retrieving records using the Query window method
- Using functions in a query
- Create a query using a simple query expression
- Open the Query Window
- Use like queries to search for a string of characters

7.10.11.22. Stored queries

Stored queries enable users to retrieve and display current information efficiently by using predefined search parameters. Depending on permissions, users can run predefined queries, modify predefined queries before running them, or store a personal list of modified queries and run a query from their list.

Administrators can use stored queries to display lists of specific records or populate dynamic display objects such as charts and marquees. They can add new queries and fine tune the out-of-box queries to decrease response time or implement best practices in their environment.

Stored query option definitions

Option key	Function	Definition
F1	Append using the & character	Appends an additional query to the end of your original query using the word AND to separate the two queries.
F2	character	Appends an additional query to the end of your original query using the word OR to separate the two queries.
F3	Insert using the & character	Inserts an additional query at the beginning of your original query using the word AND to separate the two queries.
F4	character	Inserts an additional query at the beginning of your original query using the word OR to separate the two queries.

Add a stored query

Applies to User roles: System Administrator

You can add, edit, or run stored queries from the Query window, which is accessible from the form in Database Manager or from the Application search form. Data entered into the form is automatically entered into the Query window to form the query. Depending on your permissions, you can edit and run that query, or select a query from a list of stored queries.

In this example, a query is performed on a specific assignee for all contracts in Configuration Management. Once stored, operators or groups of operators can access the query and automatically generate a list of records without reentering the detailed search criteria.

Note

Note You must have the capability word **QueryAdmin** or **query.stored.mod** operator record to add or edit stored queries.

To add a stored query:

1. Click **Configuration Management** > **Contracts** > **Contracts**.
2. Use the Fill function in the Assignee field to choose an assignee. For this example, Carlton Hulman was chosen.
3. Click the More Actions icon and choose **Expert Search**.
   The Query form opens. The new query is displayed in the **Query** field. For this example, the query for Carlton Hulman is displayed as follows: `assignee#"Carlton.Hulman"`

   Note

   Note In Service Manager, you can use field name information in field help to identify the field names you want to use in your stored queries. To display field name information in field help, make sure one of the following conditions is fulfilled:

   - In the Windows client, the **Show context-sensitive help debug information preference** is enabled.
   - In the Web client, `viewcontexthelp=true` is appended to the URL before you log in.

4. On the Query form, click **Keys**.

Note: Do not modify the automatically-generated query string, `assignee#"Carlton.Hulman"`, as it appears in the **Query** field.

5. On The Key Window (keylist form), type a key number to select the number of positions for the **assignee.name** key in the query. For example, type the number `3` to select *3* positions.
6. Click **OK**.
   The Sort Fields field in the Query Window is now modified to agree with the key definition you selected.
7. On the Query form, click **Store** to create the **querystored** file.
8. Fill in the following fields:
   - In the **Access List** field, type in the Query Groups or Operator Names that you want to have access to this query. If you want all users to have access, leave the list blank.
   - In the **Name** field, type a unique name for your query.
   - In the **Description** field, type an explanatory description.
9. Click **Add**.
   The following message confirms your entry was successful: `Query added to querystored file`.
10. Continue to further refine your search criteria, or click **Search**. You can also click any record in the list of returned records to view the details.

Update a stored query

Applies to User roles: System Administrator

You can add, edit, or run stored queries from the Query window, which is accessible from the form in Database Manager or from the Application search form. Data entered into the from is automatically entered into the Query window to form the query. Depending on your permissions, you can edit and run that query or select a query from a list of stored queries.

In this example, we update and save a stored query for Incident Management (the probsummary file).

To update a stored query:

1. Open the query window.
   - To open the query window from a Service Manager application:
     1. Open the application Search form. For this example, click **Incident Management** > **Search Incidents**.
     2. Open the More Actions menu.
     3. Select **Expert Search**.
   - To open the query window from a Database Manager search form:
     1. Open the form in Database Manager. For this example, type `=probsummary` in the **Form** field, and then press Enter.
     2. Open the More Actions menu.
     3. Select **Advanced Search**.
        The Query window opens.
2. Click **Select** to display the list of stored queries for the associated file.
3. Double-click a query to select it. For this example, select **status.resolved**.
   The Stored Query record opens.
4. Click **Select** to edit the query as desired. In this example, change the query to `flag=true and problem.status<>"Resolved"`.

> **Note**
>
> Note In Service Manager, you can use field name information in field help to identify the field names you want to use in your stored queries. To display field name information in field help, make sure one of the following conditions is fulfilled:
>
> - In the Windows client, the **Show context-sensitive help debug information preference** is enabled.
> - In the Web client, `viewcontexthelp=true` is appended to the URL before you log in.

5. Specify any other controls you want.
6. Click **Store** to save your edits to the selected query.
   The Stored Query Maintenance form opens.
7. Give the query a unique **Name**. For this example, `not.resolved`.
8. Add a **Description**. For this example, type `Unresolved records`.
9. Click **Add.**
   Service Manager returns you to the Query window and displays the message: `Query added to querystored file`.

You can click **Execute Search** to run the saved query at this point, or click **Back** to exit.

Append stored queries

Applies to User roles: System Administrator

You can add, edit, or run stored queries from the Query window, which is accessible from the form in Database Manager or from the Application search form. Data entered into the from is automatically entered into the Query window to form the query. Depending on your permissions, you can edit and run that query or select a query from a list of stored queries.

Note: You must have the capability word **QueryAdmin** or **query.stored.mod** operator record to add or edit stored queries.

In this example, two stored queries for Incident Management (the probsummary file) are joined and saved.

To append stored queries:

1. To open the query window from a Service Manager application:
   1. Open the application Search form.
      For this example, click **Incident Management** > **Search Incidents**.
   2. Open the More Actions menu.
   3. Choose **Expert Search**.
      The query window opens.
2. To open the query window from a Database Manager search form:
   1. Open the form in Database Manager.
      For this example, type **=probsummary** in the **Form** field, and then press Enter.
   2. Open the More Actions menu.
   3. Choose **Expert Search**.
      The Query window opens.
3. Click **Select** to display the list of stored queries for the associated file, and then double-click a query to select it.
   For this example, select **deadline**. The Stored Query record opens.
4. Click **Select**.
   Service Manager returns you to the list of queries.
5. Select another query from the list.
   For this example, select **pri.1**. The Stored Query record opens with additional buttons on the toolbar.
6. Click one of these buttons to use this query in addition to the one you selected previously.
   - Click **Append using &** to add this query after the previous one using the *and* (&) operator.
   - Click **Append using |** to add this query after the previous one using the *or* (|) operator.
   - Click **Insert using &** to add this query before the previous one using the *and* (&) operator.
   - Click **Insert using |** to add this query before the previous one using the *or* (|) operator.

   For this example, click **Append using &**.
7. You can click **Execute Search** to run the query without saving it or click **Store** to save your modified query for future use.
   For this example, click **Store**. The Stored Query Maintenance form opens.

- The value in the **Name** field is that of the last appended or inserted stored query, pri.1.
- The value in the **Query** field is the new query you created. (flag=true and status#"DEADLINE" and problem.status~="Cleared" and flag=true and priority.code="1").
- For this example, enter the following information:
  - Give the query a unique **Name**. For this example, type **pri.1&deadline**.

    - Add a **Description**. For this example, type **Priority 1 records in deadline alert**.
    - Edit the **Query**, if necessary. For this example, delete the second `and flag=true`.
    - Specify any other controls you want.
- Click **Add.**
  Service Manager returns you to the Query window and displays the `Query added to querystored file` message.
- You can click **Execute Search** to run the saved query at this point, or click **Back** to exit.

Run a stored query

Applies to User roles: System Administrator

Depending on your permissions, you can add, edit, or run stored queries from the Query window, which is accessible from the form in Database Manager or from the Application search form. Data entered on the form is automatically entered into the Query window to form the query. Depending on your permissions, you can edit and run that query, or select a query from a list of stored queries.

Example: Selecting and running a stored query.

In this example, a stored query for Incident Management (the probsummary file) is selected and run.

1. Open the query window.
    - To open the query window from a Service Manager application:
        1. Open the application Search form. For this example, click **Incident Management** > **Search Incidents**.
        2. Open the More Actions menu.
        3. Select **Expert Search**.
    - To open the query window from a Database Manager search form:
        1. Open the form in Database Manager. For this example, type `=probsummary` in the **Form** field, and then press Enter.
        2. Open the More Actions menu.
        3. Select **Expert Search**.

   The Query window opens.
2. Click **Select** to display the list of stored queries for the associated file.
3. Double-click a query to select it. For this example, select **status.resolved**. The Stored Query record opens.
4. Click **Select**.
   Service Manager returns you to the Query window.
5. Click **Execute Search** to run the stored query.
   Records matching the query display in a record list.
    - If only one record matches your search criteria, that record opens.
    - If no records match, the server displays the following message: `No records found`.

Related topics

Using stored queries in display objects
Menu option searches
Grant access to stored queries
Create stored queries from the Query Maintenance form

7.10.11.23. Cross-table join query improvements

The performance when running a join across database tables has been vastly improved as compared to version of Service Manager prior to 9.32. Previously, a query running across two tables was performed in memory by Service Manager. For data sets that contain millions of records, this process could be extremely time consuming.

The mechanism by which Service Manager executes queries across tables was changed in Service Manager 9.32. With the new mechanism, the initial Service Manager query is parsed, translated into SQL, and then issued directly to the RDBMS. The RDBMS then handles the initial join operation and returns a subset of results to Service Manager. However, because the database now performs this work, a number of limitations are imposed. For more information on these limitations, see Limitations on cross-table join queries

Additionally, the new functionality is limited to the Service Manager functions on the Cross-table join functions.

7.10.11.23.1. Functions

Under the new mechanism, you can only use the following functions in your query:

- lng()
- null()
- nullsub()
- operator()
- substr()
- tod()
- toupper() and tolower()

Note

Note

It is highly recommended that you restrict queries to run only on indexed fields. Running these queries on non-indexed fields can result in poor performance.

lng()

The lng() function determines the length of a value specified by the parameter.

Example of the lng Function Translation

Service Manager function	Translated SQL equivalent:
`select ps.number, ps.title from probsummary ps where lng(ps.openen.by)=6`	`select t01.”NUMBER”, t01.”TITLE” from PROBSUMMARYM1 t01 where LENGTH(t01.”OPENED_BY”) = 6`

The parameter can only be a field name or an expression that can be evaluated beforehand; It can't be an expression that can't be evaluated before translation into SQL:

The following cases are valid:

```
lng ( ps.category )    //In this example, “category” is a field from file “probsummary”
lng ( $L.var )    //In this example, $L.var is a local variable defined beforehand.
```

The following cases are invalid:

```
lng ( ps.title + ps.brief )
lng ( ps.title + $L.var )
```

null()

The Null() function determines whether a value specified by the parameter is null or not.

Example of the NULL Function Translation

Service Manager function	Translated SQL equivalent:
`select ps.number, ps.title from probsummary ps where null(ps.assignment)`	`select t01.”NUMBER”, t01.”TITLE” from PROBSUMMARYM1 t01 where t01.”ASSIGNMENT” is NULL`

The parameter can only be a field name or an expression that can be evaluated beforehand; It can't be an expression that can't be evaluated before translation into SQL:

The following cases are valid:

```
null( ps.category )    //In this example, “category” is a field from “probsummary” file.
null( $L.var )    //In this example, $L.var is a local variable defined beforehand.
```

The following cases are invalid:

```
null ( ps.title + ps.brief )
null ( ps.title + $L.var )
```

nullsub()

The nullsub() function substitutes the variable with a default value if the variable specified by the first parameter is null:

Example of the nullsub Function Translation

Service Manager function	Translated SQL equivalent:
select ps.number, ps.title from probsummary ps where nullsub (ps.openen.by, “falcon”)=”falcon”	select t01.”NUMBER”, t01.”TITLE” from PROBSUMMARYM1 t01 where COALESCE (t01.”OPENED_BY”, ‘falcon’) = ‘falcon’

The first parameter can only be a field name or an expression that can be evaluated beforehand, it can't be an expression that can't be evaluated before translation into SQL. The other parameters can only be constant numbers or expressions that can be evaluated to numbers before translation into SQL:

The following cases are valid:

```
nullsub ( ps.category, “abc” )   //In this example, “category” is a field from file “probsummary”
nullsub ( $L.var, “abc” )  //In this example, $L.var is a local variable defined beforehand.
```

The following cases are invalid:

```
nullsub ( ps.title + ps.brief, “abc” )
nullsub ( ps.title + $L.var, “abc” )
```

operator()

When you use the operator() function in part of a query, the Service Manager server will replace the call to operator() with the appropriate value. Therefore, the following query is valid:

```
select ps.number, ps.title from probsummary ps where ps.opened.by = operator()
```

substr()

The substr() function extracts a substring from a value specified by the parameter.

Example of the lng Function Translation

Service Manager function	Translated SQL equivalent:
select ps.number, ps.title from probsummary ps where substr(ps.openen.by, 0, 3)=”fal”	select t01.”NUMBER”, t01.”TITLE” from PROBSUMMARYM1 t01 where SUBSTR(t01.”OPENED_BY”) = ‘fal’

The first parameter can only be a field name or an expression that can be evaluated beforehand, it can't be an expression that can't be evaluated before translation into SQL. The other parameters can only be constant numbers or expressions that can be evaluated to numbers before translation into SQL:

The following cases are valid:

```
substr ( ps.category, 0, 4 )   //In this example, “category” is a field from file “probsummary”
substr ( $L.var, 0, 4 )   //In this example,$L.var is a local variable defined beforehand.
```

The following cases are invalid:

```
substr ( ps.title + ps.brief, 0, 4 )
substr ( ps.title + $L.var, 0, 4 )
```

tod()

When you use the tod() function in part of a query, the Service Manager server will replace the call to tod() with the appropriate value. Therefore, the following query is valid:

```
select ps.number, ps.title from probsummary ps where ps.open.date<tod()
```

toupper() and tolower()

The totoupper() and tolower() functions convert the given parameter to upper or lower case as appropriate.

Example of the toupper() Function Translation

Service Manager function	Translated SQL equivalent:
`select ps.number, ps.title from probsummary ps where toupper(ps.openen.by)=”FALCON”`	`select t01.”NUMBER”, t01.”TITLE” from PROBSUMMARYM1 t01 where UPPER(t01.”OPENED_BY”) = ‘FALCON’`

The parameter can only be a field name or an expression that can be evaluated beforehand; It can't be an expression that can't be evaluated before translation into SQL:

The following cases are valid:

```
toupper ( ps.category )   //where “category” is a field from file “probsummary”
tolower ( $L.var )    //where $L.var is a local variable defined beforehand.
```

The following cases are invalid:

```
tolower ( ps.title + ps.brief )
toupper ( ps.title + $L.var )
```

7.10.11.23.2. Cross-table join query examples and use cases

The following examples detail various scenarios in which you will see the benefits of the cross-table join queries:

Note

Note The table alias are mandatory for cross-table query. For example, `select pro.* from probsummary pro` is correct, whereas `select probsummary.* from probsummary` is incorrect.

Reserved words

Reserved words have a special meaning in Service Manager that's defined in RAD. These special words can only be used for that purpose. Do not use them for any other purpose such as field names or table alias. The following is an alphabetical list of reserved words:

AND, BEGIN, DO, ELSE, END, FALSE, FOR, IF, IN, ISIN, NOT, NULL, OR, STEP, WHILE, THEN, TRUE, UNKNOWN, INNER, OUTER, JOIN, ON, SELECT, AS, FROM, WHERE, ASC, DESC, BETWEEN, GROUP, BY, HAVING, LIKE, TABLE, VIEW, WHERE, MIN, MAX, AVG, COUNT, SUM, ID.

In addition, all the RAD and JavaScript function names, and standard SQL reserved word in Oracle or SQL Server are similarly reserved.

Use case: Incident Analyst

As an Incident Analyst, I would like to show interactions that have an incident attached that's in open status.

Task 1: Configure the relationship between interaction and incident.

1. Sign in as a system administrator.
2. From the command line, enter `db` and then press the **Execute Command** button.
3. In **Form** enter `relatedObjectMap` and then click the **Search** button.
4. Type the following values in the corresponding fields and then click the **Search** button.

Field	Value
Source Table:	`incidents`
Source Field:	incident.id

5. Click **Delete** to delete the relation between `incident` and `incdepends`, which is not used in the Inbox.
6. Type the following values in the corresponding fields and then click the **Add** button.

Field	Value
Source Table:	incidents
Source Field:	`incident.id`
To Table:	screlation
To Field:	source

7. Type the following values in the corresponding fields and then click the **Add** button.

Field	Value
Source Table:	screlation
Source Field:	`depend`
To Table:	probsummary
To Field:	number

8. Sign out of Service Manager.

Task 2: Create the view

1. Sign in as a system administrator.
2. Switch to the Interaction queue.
3. Click **More** or the More Actions menu, and then click **Define Views**.
4. Click **New**.

5. In the name field, type the following, choose **System View**, and then click **Next**:

```
All interactions that have an open incident attached
```

6. Click **Next**.
7. Click **Filter**, click **Advanced Filter**, and then click **Add New Filter Criteria**.
8. In **Field in Interaction**, select Interaction ID, and then click **Use Multi Level Field Chooser**.
9. Select field **Depend** for `screlation`, and select **Status** for `Incident`.
10. Click **Next**.
11. Set **Comparison** to Equals, set **Value** to Open, and then click **Next**.
12. Click **OK**.
13. Click **Fields**.
14. Specify the following values in the corresponding fields and then click **Add to List**.

Field	Value
Interaction	Interaction ID
screlation	Depend
Incident	Status

15. Specify the following values in the corresponding fields and then click **Add to List**.

Field	Value
Interaction	Interaction ID
screlation	Depend

16. Click **Finish** and then click **Apply View**.

After you complete these tasks, you can sign in as an Incident Analyst, and then switch to the "All interactions that have an open incident attached" view.

Use case: Department manager

As a department manager, I would like to see all incidents that were closed by contacts from my department in the last 30 days.

Task 1: Create the view

1. Sign in as a system administrator.
2. Switch to the Incident queue.
3. Click **More** or the **More actions** menu, and then click **Define Views**.
4. Click **New**.
5. In the name field, type the following, choose **System View**, and then click **Next**:

```
All incidents closed by contacts from my department in last 30 days
```

6. Click **Next**.
7. Click **Filter**, click **Advanced Filter**, and then click **Add New Filter Criteria**.
8. In **Field in Incident**, select Closed By, and then click **Use Multi Level Field Chooser**.
9. Select field **Contact Name** for `Operator`, and then select **Dept Name** for `Contact`.
10. Click **Next**.
11. Set **Comparison** to Equals, set **Value** to North America - IT, and then click **Next**.

12. Click **OK**.
13. Click **Fields**
14. Specify the following values in the corresponding fields and then click **Add to List**.

Field	Value
Incident	Close Time
Incident	Closed By

15. Click **Finish** and then click **Apply View**.

Task 2: Define the restriction in the inbox definition

1. Sign in as a system administrator.
2. From the command line, enter db and then press the **Execute Command** button.
3. In **Form** enter `inbox` and then click the **Search** button.
4. Click **inbox.view**, click **Search**, and then click **All incidents closed by contacts from my department in last 30 days**.
5. Click **Query definition** and modify the query form as follows:
   From:

   ```
   probsummary;closed.by;operator;name;operator;contact.name;contacts;contact.name;contacts;dept.name=
   "North America - IT"
   ```

   To:

   ```
   probsummary;closed.by;operator;name;operator;contact.name;contacts;contact.name;contacts;dept.name=
   $lo.dept and close.time>tod() - '30 00:00:00'
   ```

6. Click **Save**.

After you complete these tasks, you can sign in as an Incident Manager, and then switch to the "All incidents closed by contacts from my department in last 30 days" view.

Use case: JavaScript Developer

The following use case describes how a developer counts the attachments related to Interactions related to an Incident.

1. Sign in as a system administrator.
2. Navigate to **Tailoring**> **Script Library**
3. Type `getAttachments` in the **Name** field.
4. Click **Add**.

5. Copy and paste the following code into the code window:

```
var start=new Date();
var file = new SCFile("probsummary");
var sql = "SELECT pro.*, scr.*, att.* FROM probsummary pro, screlation scr inner join
SYSATTACHMENTS att on (scr.source=att.topic ) where scr.depend=pro.number and
scr.depend=\"IM10002\" and att.segment=0";
var success=file.doSelect( sql ) ;
var findNextRecord=RC_SUCCESS;
var count=0;
//iterate to print the related attachment names
while ( success== RC_SUCCESS && findNextRecord == RC_SUCCESS && count<120)
{
print("find attachment No.: " + count + " " + file.filename + " in " + file.topic);
//print(system.functions.descriptor(file));
findNextRecord=file.getNext();
count++;
}
var end=new Date();
var timespent = end-start;
print ("select time is: " + timespent + "ms");
start=new Date();
//count how many attachments are related to this incidents
count=file.doCount(sql);
print ("There are " + count + " attachments");
end=new Date();
timespent = end-start;
print ("count time is: " + timespent + "ms");
```

6. Save and compile the JavaScript code.
7. Execute the compiled code.

Example: Query two tables

This query finds all open incidents for which your department is the service owner of the affected service:

```
var file = new SCFile("probsummary");
var sql = "select p.* from probsummary p LEFT OUTER JOIN device d ON (p.affected.item=d.logical.name)
where d.owner=$lo.dept and p.flag=true";
var success = file.doSelect(sql);
while (success == RC_SUCCESS)
{
print(file.affected_item)
success = file.getNext();
}
```

Example: Query more than two files

This query finds all open incidents that are assigned to my department:

```
var file = new SCFile("probsummary");
var sql = "SELECT ta01.*, ta03.dept.name AS dept.name FROM probsummary ta01 LEFT OUTER JOIN operator ta02
LEFT OUTER JOIN contacts ta03 ON  ( ta02.contact.name = ta03.contact.name )  ON  ( ta01.assignee.name =
ta02.name )  WHERE ( ta01.flag = true and ta03.dept.name=$lo.dept)";
var success = file.doSelect( sql ) ;
while (success == RC_SUCCESS)
{
success = file.getNext();
}
```

Example: Sort

This query finds all open incidents for which your department is the service owner of the affected service, and sorts by service name in ascending order and incident number in descending order:

Note

Note If possible, use the `order by` clause in the query instead of `setsort()`, because the `order by` clause sort fields before the `select` operation. `setsort()` requires another SQL operation to sort. `setOrderBy` is unsupported in cross table query.

```
var file = new SCFile("probsummary");
var sql = "select p.*, d.logical.name as lname from probsummary p LEFT OUTER JOIN device d ON
(p.affected.item=d.logical.name) where d.owner=$lo.dept and p.flag=true";
sql += " order by d.logical.name asc, p.number desc"
var success = file.doSelect(sql);
while (success == RC_SUCCESS)
{
findNextRecord = file.getNext();
}
```

Example: Count

This query counts all open incidents for which your department is the service owner of the affected service:

Note

Note For better performance, this example use `doCount` in the query instead of looping on the record list.

```
var file = new SCFile("probsummary");
var sql = "select p.* from probsummary p LEFT OUTER JOIN device d ON (p.affected.item=d.logical.name)
where d.owner=$lo.dept and p.flag=true";
var count = file.doCount(sql);
print(count);
```

Example: Handle two files that have the same field name

This query finds all open incidents for which your department is the service owner of the affected service, and then prints the category of the incidents and the category of the service.

Note

Note If two fields have same name, you must use an alias to distinguish the fields.

```
var file = new SCFile("probsummary");
var sql = "select p.*,d.category AS cate from probsummary p LEFT OUTER JOIN device d ON
(p.affected.item=d.logical.name) where d.owner=$lo.dept and p.flag=true";
var success = file.doSelect(sql);
print(file.category);
print(file.cate);
```

Example: Handle BLOB, CLOB, text, and images

This query shows all open Incidents, where current operator is a member of an assignment group (Support Groups) that's defined in the affected service.

Note

Note

- Because an RDBMS can't search on a BLOB field directly, if you want to use BLOB field as a query condition, you must map the BLOB field as multi-row array.
- You must map `support.groups` from `DEVICE2M1` to `DEVICE2A1` for this example to work. Otherwise, you will receive the following error:
  `Queries on Blob/Clob/Text/Image field is unsupported in cross table query. (display,process.option) Field "ta02.support.groups" in the query is not mapped or mapped to Blob/Clob/Text/Image. Queries on Blob/Clob/Text/Image field is unsupported in cross table query. (display,process.option)`

```
var file = new SCFile("probsummary");
var sql = "SELECT ta01.*, ta02.support.groups FROM probsummary ta01 LEFT OUTER JOIN device ta02 ON  (
ta01.affected.item = ta02.logical.name )  WHERE ( ta02.support.groups isin $lo.pm.assignments and
ta01.flag=true )";
var success = file.doSelect(sql);
while (success == RC_SUCCESS)
{
success = findNextRecord = file.getNext();
}
```

Example: Handle a one-to-many relationship

This query shows all Incidents that have an associated open interaction relation.

Note

Note All previous examples show a many-to-one relationship; that is, multiple rows in the first table are associated to the same row in another table. For example, multiple Incidents have the same owner. In this case, the query returns only one row for each row in the Incident table. In this example, the query returns more than one row for each row in the first table. Therefore, the results of this query may resemble the following:

Incident	Interaction
IM10003	SD10060
IM10003	SD10033
IM10003	SD10003
IM10002	SD10006
IM10002	SD10002

```
var file = new SCFile("probsummary");
var sql = "SELECT ta01.*, ta03.incident.id AS iid FROM probsummary ta01 LEFT OUTER JOIN screlation ta02
LEFT OUTER JOIN incidents ta03 ON  ( ta02.depend = ta03.incident.id )  ON  ( ta01.number = ta02.source )
WHERE ( ta03.open#\"Open\" )";
var success = file.doSelect(sql);
while (success == RC_SUCCESS)
{
print(file.number + " " + file.iid);
success = file.getNext();
}
```

7.10.11.23.3. Limitations on cross-table join queries

The new implementation of the Cross Table Join mechanism has the following limitations and behavior changes:

- You can only run a query on the eight functions that are described in Cross-table join functions
- The `DISTINCT` clause is unsupported.
- You can't run a join operation on the following:
  - Array fields
  - BLOB or CLOB objects (This is a limitation common to most RDBMSs)
  - Image data-types
  - Text data-types
  - Encrypted fields
  - LDAP files
  - Joined files
  - Merged files
  - Tables in different RDBMSs
- Queries on IR files now behave differently. Specifically, the following behaviors now occur:
  - Full text search no longer stems. The revised mechanism uses an SQL `LIKE` clause to retrieve results. Therefore, a search based on `LIKE "%printer%"` will no longer return stemmed variants such as "printing" or "printed." In the previous implementation, stemming was included because the query was run entirely in Service Manager and leveraged the stemming dictionary.
  - You can no longer perform a full text search in the Cross Table Query in SM. Full-Text Search can only be performed in the initial relational search.
  - Results are returned only from the exact fields that you specify. For example, if you searched for the term "printer" on the "Title" field by using the previous mechanism, you would receive a hit even if the term was not mentioned in the "Title" field, but only in the associated "Desc" field. With the new mechanism, you will only receive a hit if the term appears in the "Title" field.
- You can no longer `GROUP` or `SORT` by BLOB objects.

7.10.11.23.4. Error Messages

To support the new cross table join mechanism, the following error messages have been implemented:

```
Unable to locate field %s in any file
```

```
The %S function is unsupported in cross table queries.
```

```
Queries on Blob/Clob/Text/Image field %S is unsupported in cross table query.
```

```
Cannot locate field %s in join condition \"%.*s\"
```

```
The join condition \"%.*s\" can't be converted into an SQL expression. One or more fields might be a
Blob/Clob/Text/Image field.
```

```
The file alias \"%.*s\" is specified in the "order by" clause, but the definition can't be found.
```

```
The field \"%.*s\" in the "order by" clause can't be found.
```

```
Cannot locate field %s.
```

```
Cannot sort or group on array field %s.
```

```
Joining tables of derived file type like JOIN_FILE or MERGED_FILE is unsupported.
```

```
Joining tables from different RDBMS is unsupported.
```

```
Cannot locate field %s in SELECT list.
```

7.10.12. Database performance tuning

The topics in this section list suggestions for tuning your database to improve performance, such as query speed and memory usage. Click any of the topics in this section to learn more. For additional information, you can search the knowledgebase in Knowledge Management or visit Software Support for a variety of best practice documents and published white papers.

7.10.12.1. Tuning: Improving query speed

Most performance degradation occurs during queries. An inefficient query can adversely affect the performance of your system by allocating resources that aren't always needed. Queries can be tuned to perform more efficiently so that the engine is not required to search entire files to determine which records match the query. Other factors that can affect the performance of file system searches include key design, key selection algorithms, and number of records.

Tuning: searching operation

Issue: When you search for an incident, interaction, or change in a heavily-populated database, the operation takes more than 20 seconds.

Solution: In this case, change the first unique key to the primary key for the modules that have this issue.

Tuning: query execution time

Slow query execution time is usually related to hardware and network limitations, or report design. After eliminating hardware or network limitations as a source of the slow response time, you can apply the following suggested tips to your report designs to speed execution:

- Limit the number of records kept in a database table, such as avoid saving several years of incident records in a single database table. The more records that are kept in a file, the larger the index is. The larger the index is, the slower queries run.
- Limit sorting layers. If the raw SQL contains more than two *ORDER BY* statements, the processing time increases. Each *ORDER BY* statement significantly increases execution time.
- Use joins instead of subreports. Although subreports are universal and allow greater portability over different servers, they execute SQL queries against the database one time for every record. For example, a report with 100 records with a subreport in the Details section executes 101 SQL statements. A joined report of equivalent size executes only once.
- Use indexes where possible. Typically, the Unique key of any table is already indexed; however, in a join, placing a key on the field used to join the two tables dramatically increases the speed to execute. Add indexes only with the

guidance of your Service Manager administrator. Adding too many indexes to a Service Manager table can affect performance.

Related topics

Key type definitions
Tuning: Designing keys for queries
Tuning: Key selection algorithms
Tuning: Number of fields in files

7.10.12.2. Tuning: Number of fields in files

If you keep the number of fields in a database table low and limit the number of records, performance is improved in the following areas:

- Searching for fields in a dbdict.
- Compressing and writing records to disk.
- Decompressing and reading records from disk.
- Operating client and server communications.
- Exercising the Load and Unload options.

Related topics

Key type definitions
Tuning: Designing keys for queries
Tuning: Improving query speed
Tuning: Key selection algorithms

7.10.12.3. Tuning queries

This section offers information about tuning queries. The suggestions, though not required, are derived from best practices learned from previous applications of Service Manager in large working environments. For additional information and suggestions, you can search the knowledgebase in Knowledge Management or visit the Software Support web site for a variety of documents and published white papers.

Related topics

Database performance tuning
Relational operators
Retrieving records by using the record list method
Retrieving records using the Query window method
Using ranges in queries
Create a query using an or statement or an and statement
Perform IR Expert queries
Use the not symbol with logical or relational operators
Software Support

7.10.12.3.1. Tuning: Fully keyed queries

In fully keyed queries, all referenced fields are defined in the correct order using a single key to process the query. This is an efficient query type because the query engine determines precisely which records match the query by reading only part of the index tree and none of the data record.

Related topics

Nonkeyed queries
Partially keyed queries
Stored queries
True queries
Troubleshooting and tuning queries
Tuning: Improving query speed

7.10.12.3.2. True queries

A true query is a query that only uses the *true* expression. The query engine uses the first Unique key defined in the key list to process the query and then returns all records in the database. If no Unique key is defined, Service Manager uses the first key. Although a true query quickly and efficiently returns results, the results include all records in the table and don't specify only records you want to view.

Related topics

Nonkeyed queries
Partially keyed queries
Stored queries
Tuning: Fully keyed queries

7.10.12.3.3. Partially keyed queries

A partially keyed query is a query where some of the referenced fields are defined in the key that processes the query. Using the index, the query engine can only partially determine whether a record matches the query. For an efficient partially keyed query, the first field in the key must be included in the query expression. If it's not included, all index entries in the index must first be read to determine the results of the query.

Related topics

Nonkeyed queries
Stored queries
True queries
Create a query using more than one field
Tuning: Fully keyed queries
Tuning: Improving query speed

7.10.12.3.4. Nonkeyed queries

Non keyed queries are typically used for small files only and aren't efficient when used against a large file. The query engine must first read all index entries in an index and then read all records within those entries to determine which records match the query.

Related topics

Partially keyed queries
Stored queries
True queries
Tuning: Fully keyed queries
Tuning: Improving query speed

7.11. Entity relationship management

Service Manager defines relationships between files and fields, which are constraints that should be enforced for relevant data in your database. However, your database sometimes may contain invalid data that no longer obeys the constraints. For example, you have renamed or deleted an operator record in your database, while some existing records are still referencing the old operator name. You may want to find out these invalid records and clean them up as needed. Additionally, you may want to get a clear picture of all entity relationships in your database and identify the impact of potential data changes.

Service Manager provides three utilities that can help you manage your entity relationships, run data reference check reports for selected files, as well as generate entity relationship diagrams:

- Entity Relationship Diagram utility
- Missing Reference Report utility
- Relationship Manager

Related topics

Build relationships
Search for relationships between specific tables and fields

Add a relationship record
Access the relationship definition of a relationship record

7.11.1. Entity Relationship Diagram utility

The Entity Relationship Diagram utility presents entity relationships for selected files and fields in your database in interactive diagrams (called Entity Relationship Diagrams or ERDs) so that you can have a clear picture of your entity relationships and identify the impact of potential data changes. Additionally, the utility can export the diagrams to static PDF format for offline use.

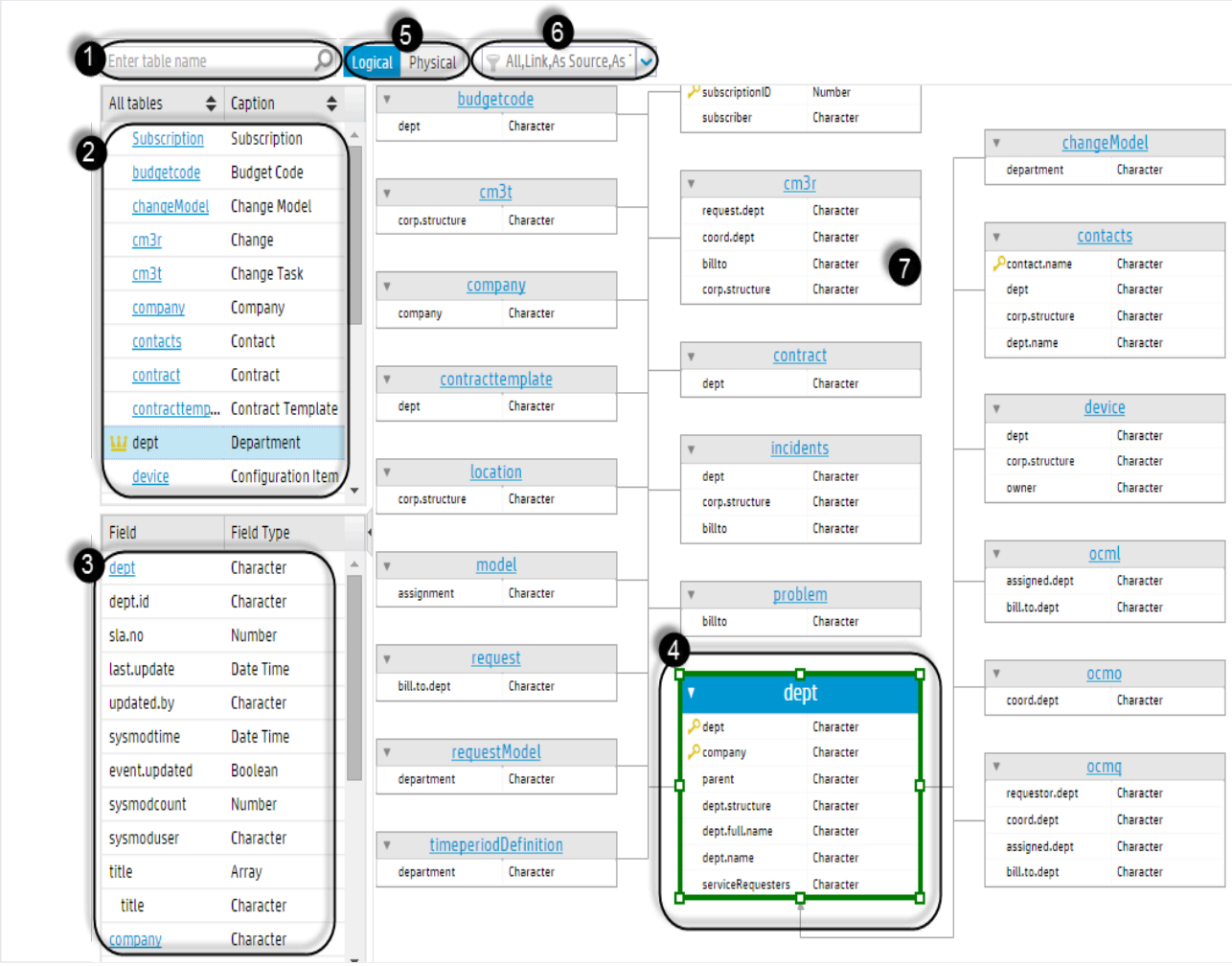
The utility processes only entity relationships that have a Definition Type of **link**, **erddef**, or **manual**.

Note
Note

- This functionality is available only from the web client.
- To use this utility, you must have the SysAdmin capability word.
- Screen readers aren't supported for entity relationship diagrams generated by this utility.
- The accuracy of relationships in the diagrams depends on the correctness of the relevant **link** and **erddef** records.

Entity Relationship Diagram interface

The following figure shows the Entity Relationship Diagram interface.



The following table explains different parts of the interface.

Number	Name	Description
1	Table name input box	Enter the name of a table for which you want to generate an ERD, and then click the magnifying glass icon. For details, see Generate an entity relationship diagram for a file.
2	Table list	This is a list of tables, including the current table and all its related tables. The current table is identified by a crown icon before its name (for example, **dept**).
3	Field or relationship list	Initially, this area lists all fields of the current table. However, if you select a field of the current table in area 4, this area switches to a list of relationship records of the selected field.
4	Current table and its fields	In the ERD, the current table is highlighted with a green border. Felds that are the unique key of a table have a key icon before their names.
5	View tabs	These two tables allow you to switch the diagram between two views: • **Logical**: This view displays the tables and fields using their names defined in the DBDICT records. • **Physical**: This view displays the tables and fields using their SQL Tables (for tables) and SQL Names (for fields) defined in the DBDICT records. By default, the Logical view is displayed.

6	Filter box	This box enables you to select filters to further filter information in the diagram. • **All**: Shows all of the relationships between the current table and other tables. By default, all diagrams are displayed using this filter. • **Link** > **As Source**: Shows only those relationships between the current table and other tables where the Definition Type is **link** and the Source Table is the current table. • **Link** > **As Target**: Shows only those relationships between the current table and other tables where the Definition Type is **link** and the Target Table is the current table. • **ERDDef**: Shows only those relationships between the current table and other tables where the Definition Type is **erddef**. • **Manual**: Shows only those relationships between the current table and other tables where the Definition Type is **manual**.
7	Related tables and fields	This area displays all related tables and fields of the current table.

Dynamic display

When you select another table or field in the diagram, the diagram is reloaded automatically. For details, see View relationships of a specific field in an entity relationship diagram.

Toolbar

The Entity Relationship Diagram interface also provides a toolbar, which allows you to interact with the diagram using the tools on it. For more information, see Interact with an entity relationship diagram.

Related topics

Generate an entity relationship diagram for a file
View relationships of a specific field in an entity relationship diagram
Interact with an entity relationship diagram

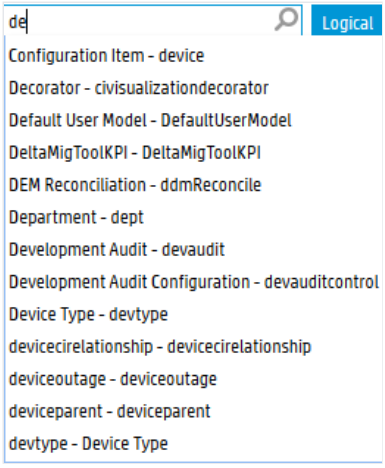
7.11.1.1. Generate an entity relationship diagram for a file

Applies to User roles: System Administrator

To generate an entity relationship diagram (ERD) for a file, follow these steps:

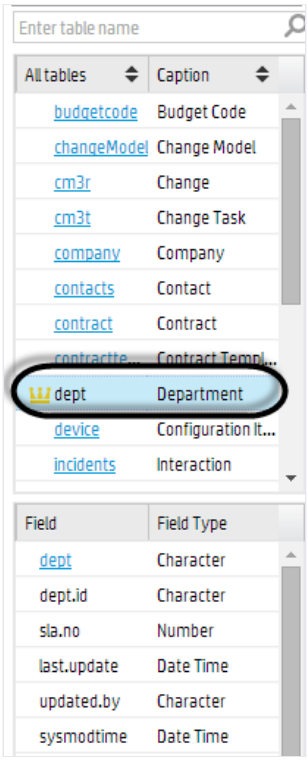
1. Navigate to **Tailoring** > **SQL Utilities** > **Entity Relationship Diagram** to open the Entity Relationship Diagram utility.
2. In the Table Name box at the top left corner, type the name of the file. For example, **dept**.

Auto Complete is available for table names. When you are typing, suggested items are automatically listed. The list items are in the following format and are sorted in alphabetical order by caption: *<caption>* - *<table name>*. Instead of typing the full table name, you can select the name from the list.



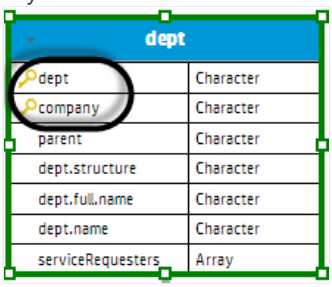
3. Press Enter, or click **Search** (the magnifying glass icon).
   The Entity Relationship Diagram utility creates a list of related tables and fields on the left pane, and generates a dynamic ERD that shows all relationships between the current file (**dept** in this example) and other files.
4. View the list of tables and fields on the left pane.
   In the upper area, a list of tables that have a relationship with the current file (**dept** in this example) is displayed along with their captions; in the lower area, a list of fields of the current file that have a relationship with other tables is displayed along with their field types.

   Tip
   Tip
   The current file is identified by a crown icon before its name. When you select a file in the list, a crown icon appears before the name of that file. You can use the up/down arrows to sort the table list in ascending or descending order by table name or by table caption.



5. View the ERD in the right pane.

In the ERD, the current table is highlighted with a green border, and fields that are the unique key of a table have a key icon before their names.



Next, you can work with the ERD as described in the following tasks:

Interact with an entity relationship diagram

View relationships of a specific field in an entity relationship diagram

Related topics

Entity Relationship Diagram utility
Interact with an entity relationship diagram
View relationships of a specific field in an entity relationship diagram

7.11.1.2. Interact with an entity relationship diagram

Applies to User roles: System Administrator

The Entity Relationship Diagram (ERD) utility generates dynamic diagrams, which allow numerous user interactions.

1. Generate an ERD. For details, see Generate an entity relationship diagram for a file.
2. You can interact with the table in the following ways.

Interaction	Description
Filter the diagram	Use the filters from the box.
View field specific relationships	For details, see View relationships of a specific field in an entity relationship diagram.
Drag and drop the diagram	You can drag and then drop the diagram to a desired place on the right pane.
Collapse all fields	By default, an ERD is displayed with all field names expanded under the table names. You can collapse the fields to get a clearer picture of the relationships between the tables. To collapse all fields in an ERD, click the **Collapse All** icon at the top right of the diagram pane. **Tip** Tip Once the fields are collapsed, you can click the **Expand All** icon to expand them.
Show a thumbnail	You can show a thumbnail of an ERD. To do so, click the **Show Thumbnail** icon at the top right of the diagram pane. A thumbnail appears at the upper right of the diagram pane. **Tip** Tip Once a thumbnail is displayed, you can click the **Hide Thumbnail** icon to hide it.
Zoom in/out	You can zoom an ERD in or out by dragging the zoom slider to the right or left or by clicking the plus (+) or minus (-) icon at the top right of the diagram pane.
Fit the diagram to the page	You can click the **Fit** icon at the upper right of the diagram pane to fit the diagram to the page.
Export the diagram to a PDF	You can export the diagram to a PDF file by clicking the **Export** icon at the top right of the diagram pane.

Related topics

Entity Relationship Diagram utility
View relationships of a specific field in an entity relationship diagram

7.11.1.3. View relationships of a specific field in an entity relationship diagram

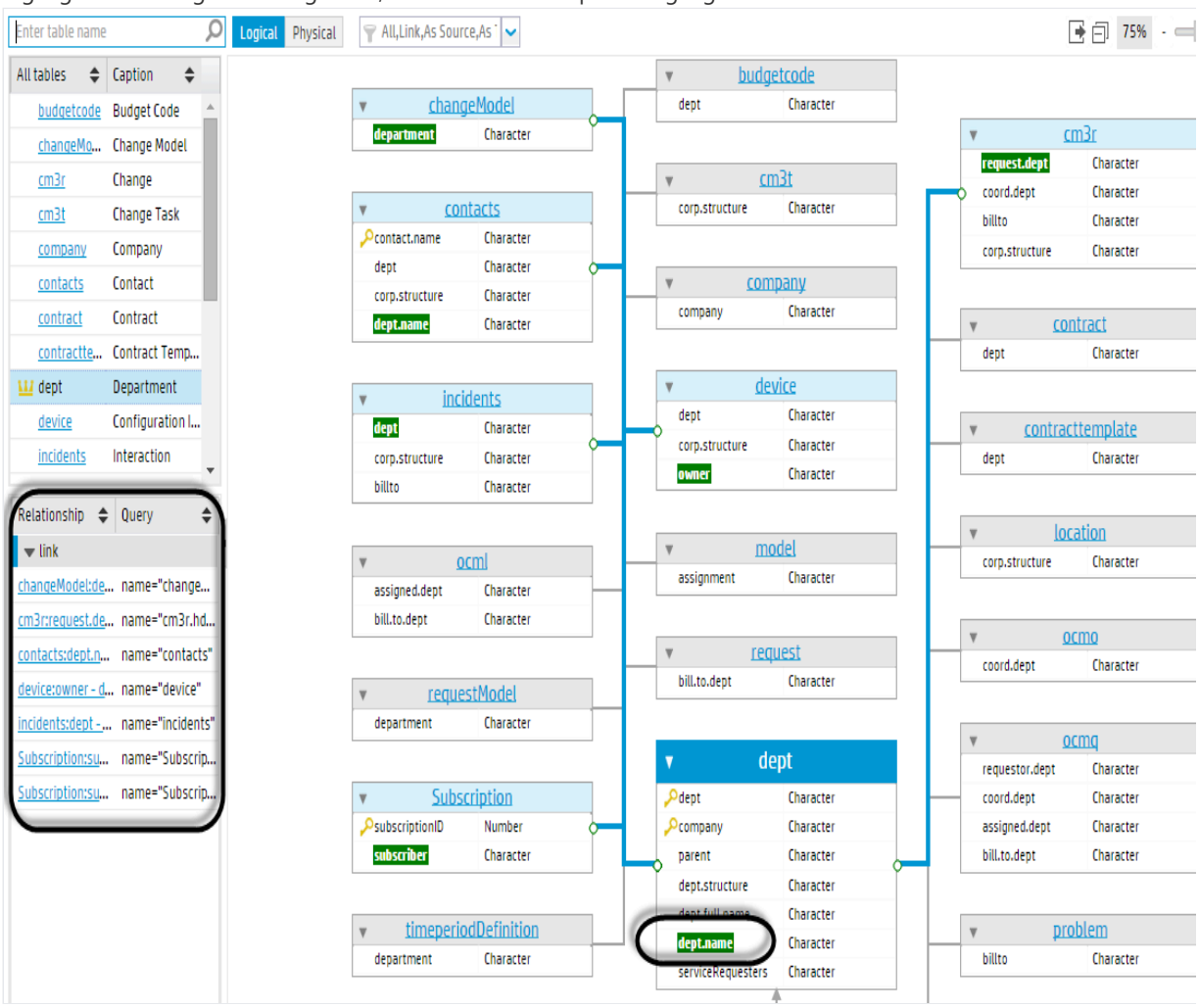
Applies to User roles: System Administrator

Before you make updates to a specific field in a table or to its values in the database, you may want to see the relationships of this specific field to understand the potential impact on other records.

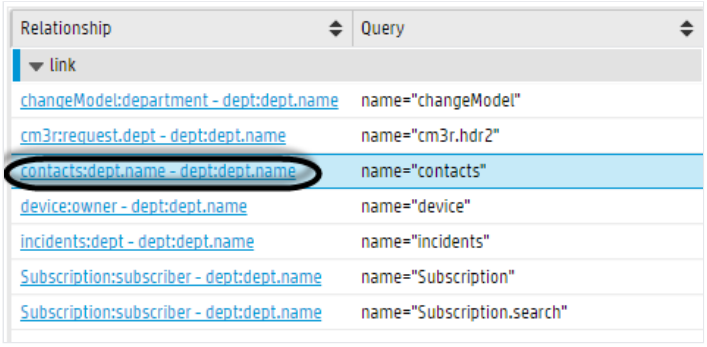
To view the relationships of a specific field, follow these steps:

1. Navigate to **Tailoring** > **SQL Utities** > **Entity Relationship Diagram**.
2. In the Table Name box, type the name of the table that contains the field.

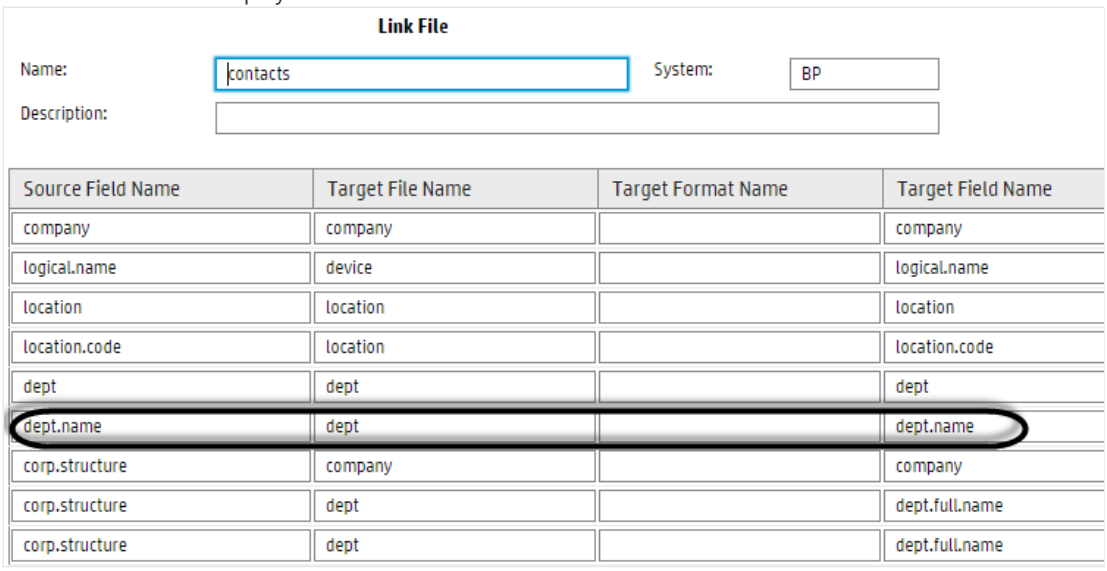
3. Click **Search**.
   Service Manager generates an ERD for the specified table.
4. In the diagram, select a field in the current table.
   The selected field and its related fields in other tables are highlighted in green, and the relationships between them are highlighted with blue connector lines. The following figure shows an example in which the relationships between the **dept.name** field in the **dept** table and related fields in other tables are highlighted. The related fields are highlighted with a green background, and the relationships are highlighted with blue connector lines.



   In the meantime, the lower part of the left pane is refreshed to display the relationship definitions for the current field, as well as their queries. The relationship definitions are displayed as links, which you can directly open. See the following figure for an example in which the relationship definitions are all **link** files.
   The display text of each relationship definition link uses the following format: *<source table name>:<source field alias>-<target table name>:<target field alias>*, where the aliases are the Source Field Name and Target Field Name that are defined in the relationship definition. The following is an example: **contacts:dept.name - dept:depart.name**.



If you click the example link, the **link** record is displayed, in which you can see the source and target table names and field names that are displayed in the link text above.



Link File

Name: contacts System: BP

Description:

Source Field Name	Target File Name	Target Format Name	Target Field Name
company	company		company
logical.name	device		logical.name
location	location		location
location.code	location		location.code
dept	dept		dept
dept.name	dept		dept.name
corp.structure	company		company
corp.structure	dept		dept.full.name
corp.structure	dept		dept.full.name

Related topics

Entity Relationship Diagram utility
Interact with an entity relationship diagram

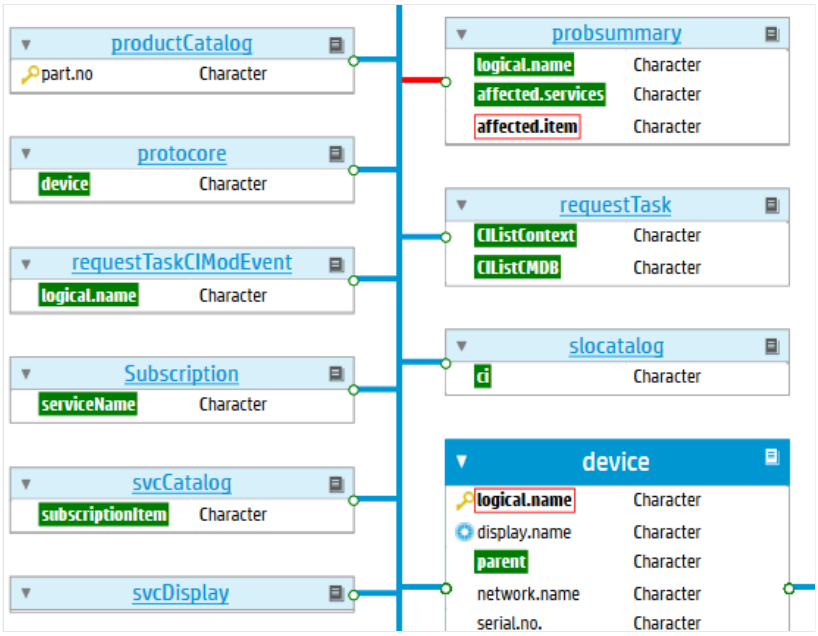
7.11.1.4. Identify misconfigured reference fields

Applies to User roles: System Administrator

A reference field retrieves data from another table instead of the table to which it belongs. An example reference field is the **affected.item** field in the **probsummary** table, which retrieves data from the **device** table. A reference field must have the correct referenced table configured in the datadict record. The Entity Relationship Diagram utility can help you identify reference fields whose referenced table is not configured correctly.

To identify misconfigured reference fields, follow these steps:

1. Navigate to **Tailoring** > **SQL Utities** > **Entity Relationship Diagram**.
2. In the Table Name box, enter the name of a referenced table. For example, enter device.
3. Click **Search**.
   Service Manager generates an ERD for the specified table.
4. Select the **Logical** view.
5. Select **Link** and then **As Target** as the filter.
6. Select the key of the referenced table. For example, select the **logical.name** field of the **device** table.
7. Check for red connector lines between the referenced table and other tables. If a red line connects the referenced table and another one, check that table for fields that are enclosed in a red box. Such fields are misconfigured reference fields.
   The following figure shows an example, which indicates the **affected.item** field in the **probsummary** table doesn't have a referenced table configured.



Related topics

Entity Relationship Diagram utility
Specify a referenced table for a field
Interact with an entity relationship diagram

7.11.2. Missing Reference Report utility

Service Manager uses the database to store all information. Relationships between tables are defined by certain tables (**dbdict**, **erddef**, **link**, and **triggers**) and applications other than by primary and foreign keys. There is a challenge to maintain database integrity when there is a change of a referenced value in the database.

Prior to version 9.40, although Service Manager can define relationships between Service Manager files through the **link**, **erddef**, and **relatedObjectMap** tables, there is no way to know if data stored in the files complies with the predefined relationships. As a result, orphan records and data redundancy may occur, which could lead to data corruption or data inconsistency. For example, renaming a Configuration Item or a department may result in orphan records, which still contain a reference to the old name (that is, a missing reference).

As of version 9.40, Service Manager provides a tool, Missing Reference Report, which can generate a report for selected files to help you find out and solve potential data integrity problems. This utility can help your organization mitigate system risk and save the total cost of ownership (TCO) of adopting Service Manager.

Missing Reference Report can report the following items:

- Records that don't follow their primary key constraint
- Records that reference values that don't exist in the target file.
- Records that reference nonexistent field values and contain referential fields that are mapped to a large object (LOB)

Note

Note This utility skips some relationships in the out-of-box system by default. For details, see List of skipped out-of-box relationships .

Limitations

This utility has the following limitations:

- This utility can detect only missing references to fields of either the **number** or the **character** data type (including number and character fields in arrays). Other data types, such as **logical** and **date/time**, aren't supported.
- This utility ignores **new** or **fixed** relationships in which the same source field has multiple parents (target fields) defined. The **location.full.name** field in the **contacts** table is an example, as shown in the following table.

Source Table	Source Field	Target Table	Target Field	Definition	Status
contacts	location.full.name	company	company	link	new
contacts	location.full.name	location	location.full.name	link	new
contacts	location.full.name	location	location.full.name	link	new

Using the utility

Note

Note To use this utility, you must have the SysAdmin capability word.

The following tasks describe how you can use this utility:

Start the refcheck.startup scheduler

Enable data reference checking for a file

Skip data reference checking for a relationship

Run a data reference check against selected files

View the status of existing data reference checks

Cancel a scheduled data reference checking job

View missing reference report results

Identify the root cause of a missing reference

Related topics

Relationship Manager

7.11.2.1. Start the refcheck.startup scheduler

Applies to User roles: System Administrator

The Missing Reference Report utility relies on the refcheck.startup scheduler to work. Before running this utility, start this scheduler.

To start this scheduler, follow these steps:

1. Navigate to **System Status**, and click **Start Scheduler**.
2. Double-click **refcheck.startup**.
   A message is displayed, indicating the scheduler is started.

Related topics

Missing Reference Report utility
Enable data reference checking for a file
Run a data reference check against selected files

7.11.2.2. Enable data reference checking for a file

Applies to User roles: System Administrator

The Missing Reference Report utility can detect records in your database that contain missing references. However, to run this utility on a file, you must first update the data policy record of this file to enable data reference checking for the file. There are several ways of accessing a data policy record. The following steps use the Database Dictionary utility.

Note

Note Data reference checking is enabled by default for the following files in the out-of-box system: **assignment**, **dept**, **device**, **company**, **operator**, **location**, and **contacts**.

To enable data reference checking for a file, follow these steps:

1. In the Service Manager command line, type dbdict and press Enter.
2. In the **File Name** field, enter the name of the file. For example, **probsummary**.
3. Click **Search**. A list of records is displayed.
4. Click **probsummary** in the list. The **probsummary** DBDICT record opens.
5. From the DBDICT record, click **More** or the More Actions icon, and select **Data Policy**.
   The data policy record opens.
6. Select the **Enable Data Reference Check** option.
7. Click **Save**.
   By default, enabling data reference checking for a file will apply to all relationships of the file, no matter whether the file is a source file or target file in the relationships. However, you can choose to skip data reference checking for a specific relationship of this file. For details, see Skip data reference checking for a relationship.
   Once you have enabled data reference checking for all desired files and specified the relationships that must be skipped for data reference checking, you are ready to run the Missing Reference Report utility against the files. See Run a data reference check against selected files.

Related topics

Relationship Manager
Missing Reference Report utility
Skip data reference checking for a relationship
Run a data reference check against selected files

7.11.2.3. Skip data reference checking for a relationship

Applies to User roles: System Administrator

The **Enable Data Reference Check** flag in a relationship record controls whether or not this relationship is skipped when you run a data reference check. The Missing Reference Report utility only scans relationships that have this flag turned on. This mechanism is useful when you want to skip data reference checking for a specific relationship. For example, you already know that a relationship is defined incorrectly, you don't care about the relationship, or you simply want to skip certain fields of a file for performance considerations.

Note

Note Some out-of-box relationships have data reference checking disabled by default. For more information, see List of skipped out-of-box relationships .

To skip data reference checking for a relationship, follow these steps:

1. Navigate to **Tailoring** > **Tailoring Tools** > **Relationship Manager**.
2. Perform a search to locate the relationship you want to configure.
3. If the relationship has the **Enable Data Reference Check** flag turned on, click **Skip Data Reference Check** on the task bar.
   The **Enable Data Reference Check** checkbox is cleared. Also, an **Enable Data Reference Check** button appears on the task bar; if you click this button, data reference checking is enabled again.
   Now, if you run a data reference check against the source file or target file of this relationship, this relationship is skipped.

Note

Note The **Enable Data Reference Check** setting is retained when running the Build Relationships functionality. For more information, see Build relationships .

Related topics

Relationship Manager
Missing Reference Report utility
Enable data reference checking for a file
Run a data reference check against selected files

7.11.2.3.1. List of skipped out-of-box relationships

Service Manager has a number of out-of-box relationships that don't really require data reference checking, such as relationships for operator names. For example, your Service Manager system may contain records that reference the names of some former employees who have already resigned from your organization. However, such missing references don't really require any remedial actions.

For this reason, Service Manager allows you to disable data reference checking for a specific relationship. For details, see Skip data reference checking for a relationship. Additionally, the Missing Reference Report utility doesn't scan some out-of-box relationships, to achieve optimized performance without compromising the functionality.

The following table lists the out-of-box relationships that are skipped when running the Missing Reference Report utility. These relationship records have their **Data Reference Check** flag turned off by default.

Source File	Source Field	Target File	Target Field	Relationship Type	Definition Type
assignment	company	company	company	Many to Many	link
budgetcode	manager	contacts	contact.name	Many to One	link
cm3t	header,requested.by	contacts	contact.name	Many to One	link
contacts	manager	contacts	contact.name	Many to One	link
incidents	contact.name	contacts	contact.name	One to One	erddef
incidents	alternate.contact	contacts	contact.name	Many to One	link
incidents	callback.contact	contacts	contact.name	Many to One	link
incidents	owner.name	operator	name	Many to One	link
incidents	opened.by	operator	name	Many to One	link
incidents	updated.by	operator	name	Many to One	link
incidents	closed.by	operator	name	Many to One	link
knownerror	closed.by	operator	name	Many to One	link
knownerror	updated.by	operator	name	Many to One	link
ocml	contact.name	contacts	contact.name	Many to One	link
ocml	employee.id	contacts	user.id	Many to Many	link
ocml	ship.to.code	location	location.code	Many to Many	link
ocml	ship.to.code	location	location	Many to One	link
ocml	coordinator	operator	name	Many to One	link
ocml	coordinator	operator	full.name	Many to Many	link
ocmo	bill.to.code	location	location.code	Many to Many	link
ocmo	ship.to.code	location	location.code	Many to Many	link
ocmo	orig.operator	operator	name	Many to One	link
ocmq	requested.for	contacts	contact.name	Many to One	link
ocmq	orig.operator	operator	name	Many to One	link
ocmq	work.manager	operator	name	Many to One	link
oncallsched	exception.schedule,exception.schedule,ex.contact	contacts	contact.name	Many to One	link
oncallsched	exception.schedule,exception.schedule,ex.contact	operator	name	Many to One	link

patcotask	rm.requestor	contacts	contact.name	Many to One	link
patcotask	cm.coordinator	operator	name	Many to One	link
patcotask	cm.work.manager	operator	name	Many to One	link
patcotask	rm.assigned.to	operator	name	Many to One	link
patcotask	cm.assign	operator	name	Many to One	link
patcotask	problem.owner	operator	name	Many to One	link
patcotask	rm.coordinator	operator	name	Many to One	link
problem	middle,assignee.name	operator	name	Many to One	link
probsummary	actor	operator	name	Many to One	link
probsummary	opened.by	operator	name	Many to One	link
probsummary	updated.by	operator	name	Many to One	link
rootcause	updated.by	contacts	full.name	Many to Many	link
rootcause	updated.by	contacts	contact.name	Many to One	link
timeperiodDefinition	created.by	operator	name	Many to One	link
timeperiodDefinition	updated.by	operator	name	Many to One	link

For more information about relationships, see Add a relationship record.

Related topics

Skip data reference checking for a relationship

7.11.2.4. Run a data reference check against selected files

Applies to User roles: System Administrator

Once you have enabled data reference check for desired files, you can run a data reference check against them to find out invalid records associated with them.

Note

Note The Missing Reference Report utility can detect only missing references to fields of either the **number** or **character** data type (including number or character fields in arrays). Other data types, such as **logical** and **date/time**, aren't supported.

To run a data reference check, follow these steps:

1. Navigate to **Tailoring** > **SQL Utilities** > **Missing Reference Report**.
2. Click **Fill** to select the files you want to check.
3. Optionally, select the **Check LOB fields** option to also check fields that are mapped to a large object (LOB).

Caution

Caution Enabling this option may impact your system performance if the selected files have a large number of LOB field values.

4. Click **Generate**.
   A message is displayed, indicating that the reference check job is scheduled. When the processing of a file is finished, a message is displayed, indicating the progress.
   You can continue to view the job status, as well as the status of other data reference check jobs. See View the status of existing data reference checks.

Related topics

Relationship Manager
Missing Reference Report utility
List of skipped out-of-box relationships
Skip data reference checking for a relationship
View the status of existing data reference checks

7.11.2.5. View the status of existing data reference checks

Applies to user roles: System Administrator

Once you have triggered a data reference check, you can view its status, along with the status of other existing data reference checks.

To view the status of a data reference check, follow these steps:

1. Navigate to **Tailoring** > **SQL Utilities** > **Missing Reference Report**.
2. Click **View Status** to view the status of existing data reference checks.
   - **Id**: This is the unique identification number of the report. If you click this field, the report opens.
   - **File Name**: This field are the names of the files against which the data reference check is run. If you selected multiple files, their names are separated by a forward slash, for example, **contacts/operator**.
   - **Status**: This column indicates the status of the check. For each check, the initial status is **Scheduled**. When the job starts to run, the status changes to **In Progress**. When the job is finished, the status becomes **Finished**.
   - **Invalid**: This column indicates the total number of invalid records that are detected for the files that are included in this report. Invalid records reference a nonexistent value.
   - **Lob Check**: This column indicates if the **Check LOB fields** option was selected when running the job (**true**: selected).
3. Click **Refresh** to refresh the status display.

Tip

Tip
If the status still remains **Scheduled** after a long period of time, go to **System Status** and make sure the **refcheck.startup** background scheduler is started.

If you no longer want a scheduled job to run, you can cancel the job; you can also delete the report of a finished job. See Cancel a scheduled data reference checking job.
You can also view the results of a finished job, and perform necessary data cleanup. See View missing reference report results.

Related topics

Relationship Manager
Missing Reference Report utility
Cancel a scheduled data reference checking job
View missing reference report results

7.11.2.6. Cancel a scheduled data reference checking job

Applies to User roles: System Administrator

Once you have triggered a data reference checking job by clicking the **Generate** button in the Missing Reference Report interface, the job is placed in the job queue with a status of **Scheduled**. You can cancel a scheduled job if you like.

> **Note**
>
> Note Once the job has started running (the status has changed to **In Progress**), you can't cancel it. If a job has finished, you can delete the job report.

To cancel a scheduled data reference check job, follow these steps:

1. Navigate to **Tailoring** > **SQL Utilities** > **Missing Reference Report**.
2. Click **View Status**.
3. Select a job whose status is **Scheduled**.
4. Click **Delete**.

   > **Note**
   >
   > Note If a job has a **Finished** status, clicking **Delete** will delete the report instead.

5. Select **Yes** in the confirmation dialog.
   The job is canceled and removed from the list.

Related topics

Relationship Manager
Missing Reference Report utility
Run a data reference check against selected files
View missing reference report results

7.11.2.7. View missing reference report results

Applies to User roles: System Administrator

Data changes may result in invalid records in your database. For example, a department name has changed in your database, which may cause previous incidents to disappear from an Incident Manager's "Incidents assigned to my group" view. In this case, you can run a data reference check on the **dept** table to find out the invalid records, and further view the detailed results so that you can clean up the invalid data accordingly.

> **Tip**
>
> Tip Instead of using the steps described here, you can directly view missing reference report results by clicking the **Id** field in the Missing Reference Report Status form. See View the status of existing data reference checks.

To view the results of a data reference check and fix the invalid data, follow these steps:

1. Navigate to **Tailoring** > **SQL Utilities** > **Missing Reference Report**.
2. In the **Report Set** box, click Fill to select the data reference check report.

   > **Note**
   >
   > Note The **Batch ID** field corresponds to the **Id** field that appears when you click **View Status**.

3. Optionally, complete one or more of the following fields to narrow down your results:
   - Table Name: the name of a file on which the data reference check was performed
   - Field: the name of a field in the file
   - Relationship Type: a type of relationships for the file
   - Definition Type: a type of relationship definitions for the file
4. Click **View Result**.

A list of relationships for the files is displayed. In these relationships, the files can be either a source file or a target file.

5. Select a relationship from the list. The reference check results of the relationship are displayed. The results consist of two sections: **Details** and **Record Analysis**.
6. View the information in the Details section. This read-only section displays the relationship details and statistic information about the source and target records.
   Field
   Description
   Notes
   Id
   The unique identifier number of the relationship record
   Source File
   The name of the source file of the relationship record
   rowspan='7'|For more information, see Add a relationship record
   Source Field
   TOBEREMOVED
   TOBEREMOVED
   TOBEREMOVED
   TOBEREMOVED
   TOBEREMOVED
   TOBEREMOVED
   The name of the source field of the relationship record
   Target File
   The name of the target file of the relationship record
   Target Field
   The name of the target field of the relationship record
   Relationship Type
   The type of the relationship record
   Definition Type
   The type of the relationship definition

   Note

   Note To view the relationship definition record, click the **Go to definition** arrow button next to this field.

   Definition Query
   The query of the relationship definition
   From Lob
   Indicates if the source field or target field is mapped to a large object (LOB)
   Report ID
   The unique identifier number of the date reference check report
   Report Time
   The time when the report was generated
   Null Records
   The number of null records based on the relationship
   Source Records
   The number of source records based on the relationship
   Target Records
   The number of target fields based on the relationship
   Missing Reference Records
   The number of records that contain a missing reference
   Source field in First unique key
   Indicates if the source field is, or is part of, the first unique key of the source file.
   rowspan='2'|The first unique key of a file may consist of multiple fields.
   Target field in First unique key

TOBEREMOVED
Indicates if the target field is, or is part of, the first unique key of the target file.

7. The Record Analysis section displays the numbers and percentages of valid and invalid records. To view or edit the invalid records, click **Show All Invalid Records**.
   A list of invalid records that use the default form of either the source file or the target file is displayed.

> **Note**
> Note If there is only one invalid record, the invalid record opens directly.

Next, you need to identify the root cause of each invalid record in the report and take necessary actions. For more information, see Identify the root cause of a missing reference.

Related topics

Relationship Manager
Missing Reference Report utility
View the status of existing data reference checks

7.11.2.8. Identify the root cause of a missing reference

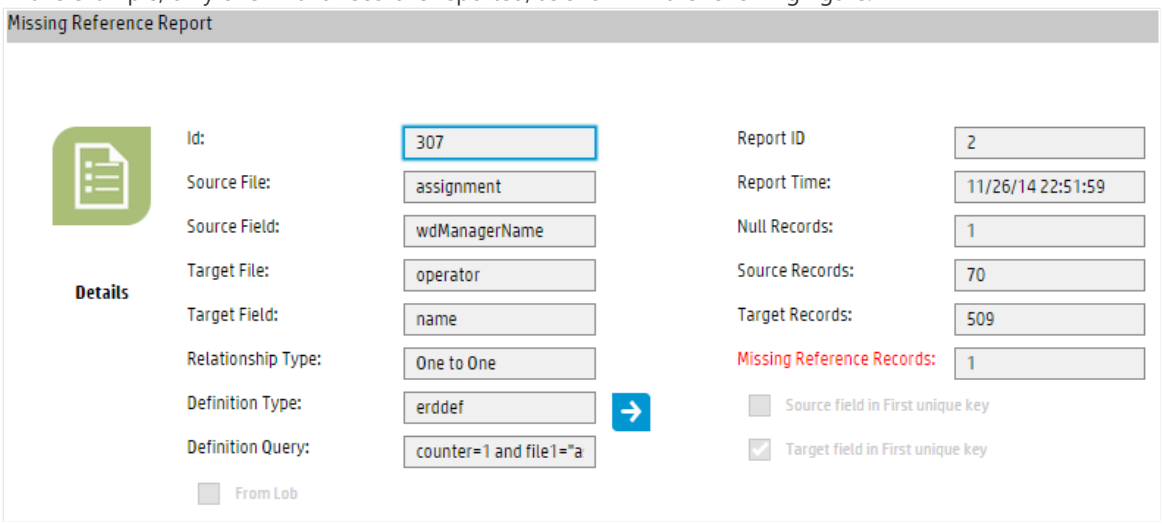
If a missing reference report has detected any invalid records, you need to identify their root causes. Remember that the data in the source file should also exist in the target file, otherwise the data is an "orphan" or there is a non-relational **link** or **erddef** definition behind that "orphan" data.

Example 1 : The missing reference is caused by a pure orphan value.

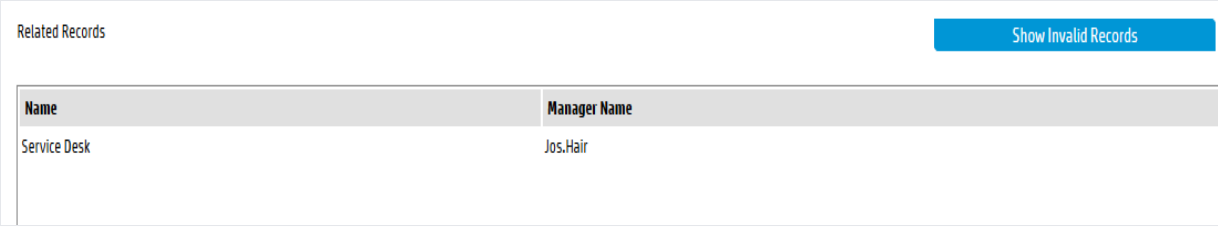
This example demonstrates the steps to handle a missing reference, using the **operator** file and out-of-box data in the system.

Task 1: Generate a missing reference report.

1. Make sure the background scheduler for data reference check is started. For details, see Start the refcheck.startup scheduler.
2. Generate a missing reference report for the **operator** file. For details, see Run a data reference check against selected files.
3. View the report results. For details, see View missing reference report results.
   In this example, only one invalid record is reported, as shown in the following figure.



   The Related Records section lists the invalid record (which is the Service Desk assignment group), as shown in the following figure.



By following the steps above, you can generate a missing reference report for any files. This is only the first step of handling a missing reference report, and your final goal is to resolve each “orphan” value or non-relational link or erddef definition.

Task 2: Understand the report results.

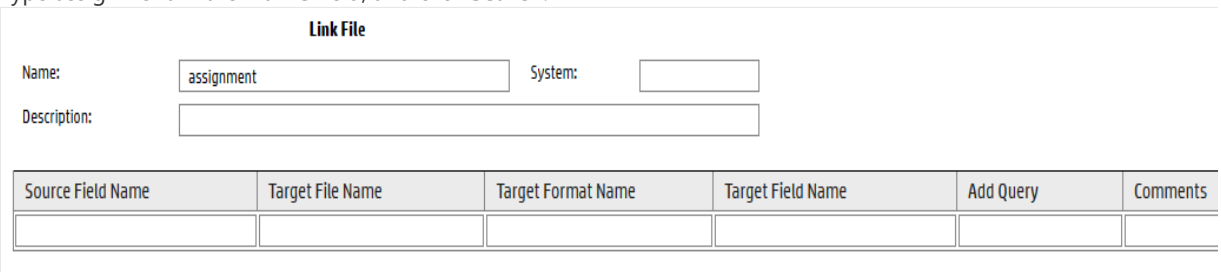
1. Verify that **Jos.Hair** doesn't exist in the **operator** file. To do this, follow these steps:
    1. Type `operator` in the command line, and press Enter.
    2. Type Jos.Hair in the **Login Name** field, and click **Search**.

    The search returns no results, which means that **Jos.Hair** doesn't exist in the **operator** file.
2. Open several existing operator records to understand what their Login Name values look like.

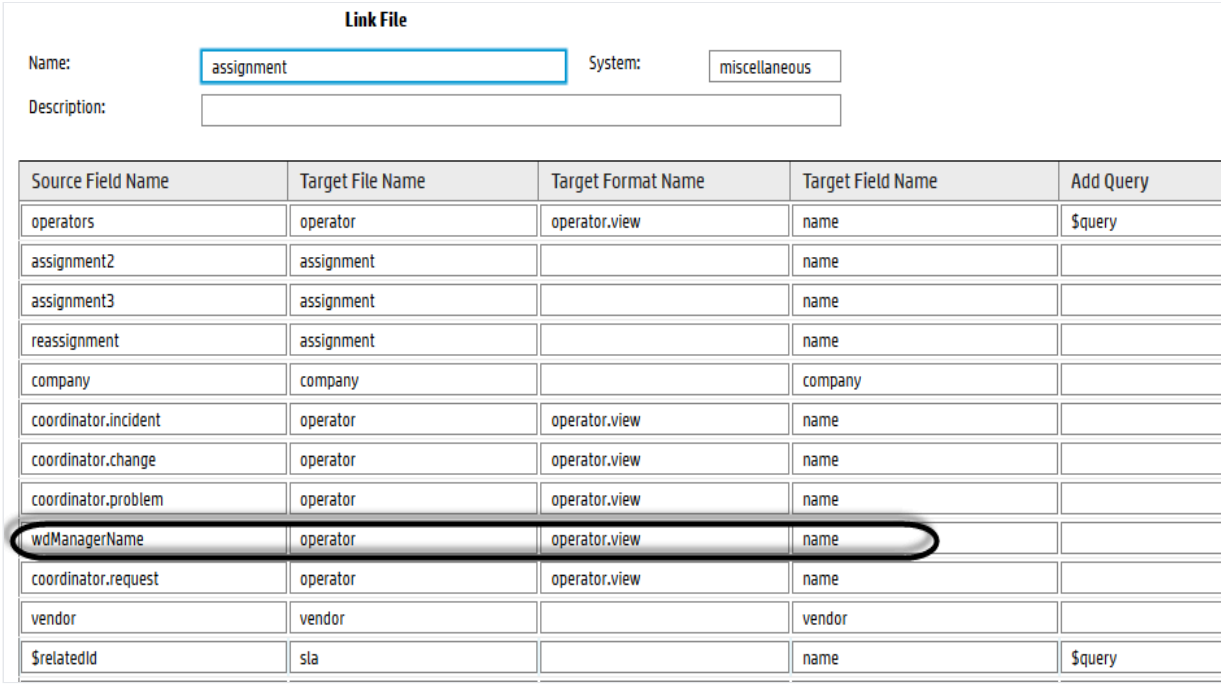
    Note

    Note This step can help you find out if all values in a field should follow certain rules.

3. In the command line, type link and press Enter. The Search Link Records form opens.
4. Type assignment in the **Name** field, and click **Search**.

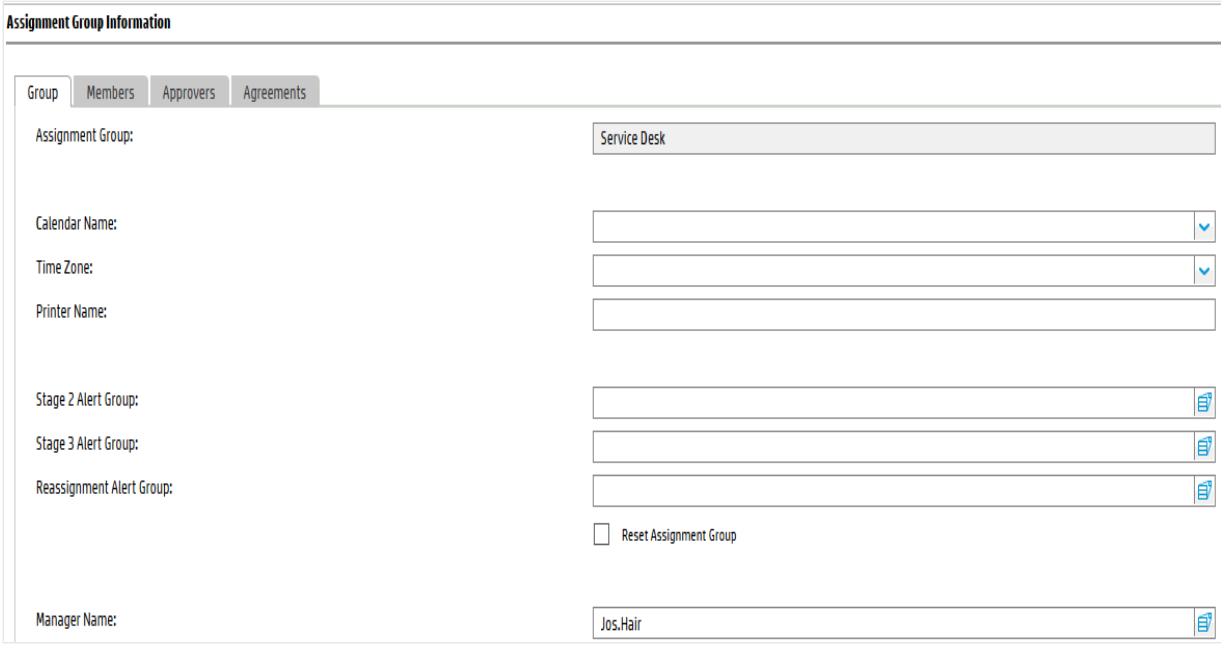


5. Locate the row in which the Source Field Name is **wdManagerName**.
   This is the relevant relationship definition. You should check to see if this link relationship complies with the business logic or is designed as desired. If not, the inappropriate link definition is probably the root cause of the “orphan” data; in this situation, you need to take actions on the **link** definition. If yes (which is true in most scenarios), the “orphan” data is then a pure “orphan” (that is, a real missing reference).



Source Field Name	Target File Name	Target Format Name	Target Field Name	Add Query
operators	operator	operator.view	name	$query
assignment2	assignment		name	
assignment3	assignment		name	
reassignment	assignment		name	
company	company		company	
coordinator.incident	operator	operator.view	name	
coordinator.change	operator	operator.view	name	
coordinator.problem	operator	operator.view	name	
wdManagerName	operator	operator.view	name	
coordinator.request	operator	operator.view	name	
vendor	vendor		vendor	
$relatedId	sla		name	$query

6. Find the record in the **assignment** table whose Manager Name is **Jos.Hair**.
   The Manager Name field is the **wdManagerName** field in the **assignment** table, which is mapped to the **name** field in the **operator** file.



7. According to the business logic, manager "Jos.Hair" should not have appeared in the assignment group record, because an assignment group manager should also be an operator in Service Manager. Furthermore, if you remove "Jos.Hair" from the Manager Name field and then click the Fill button, a list of operators is displayed. However, your previous search has already shown that "Jos.Hair" doesn't exist in the **operator** file. Therefore, it is true that "Jos.Hair" is an orphan value.
   In summary, the root cause of this orphan value is that the assignment group record defines a manager name that can't be found in the system.

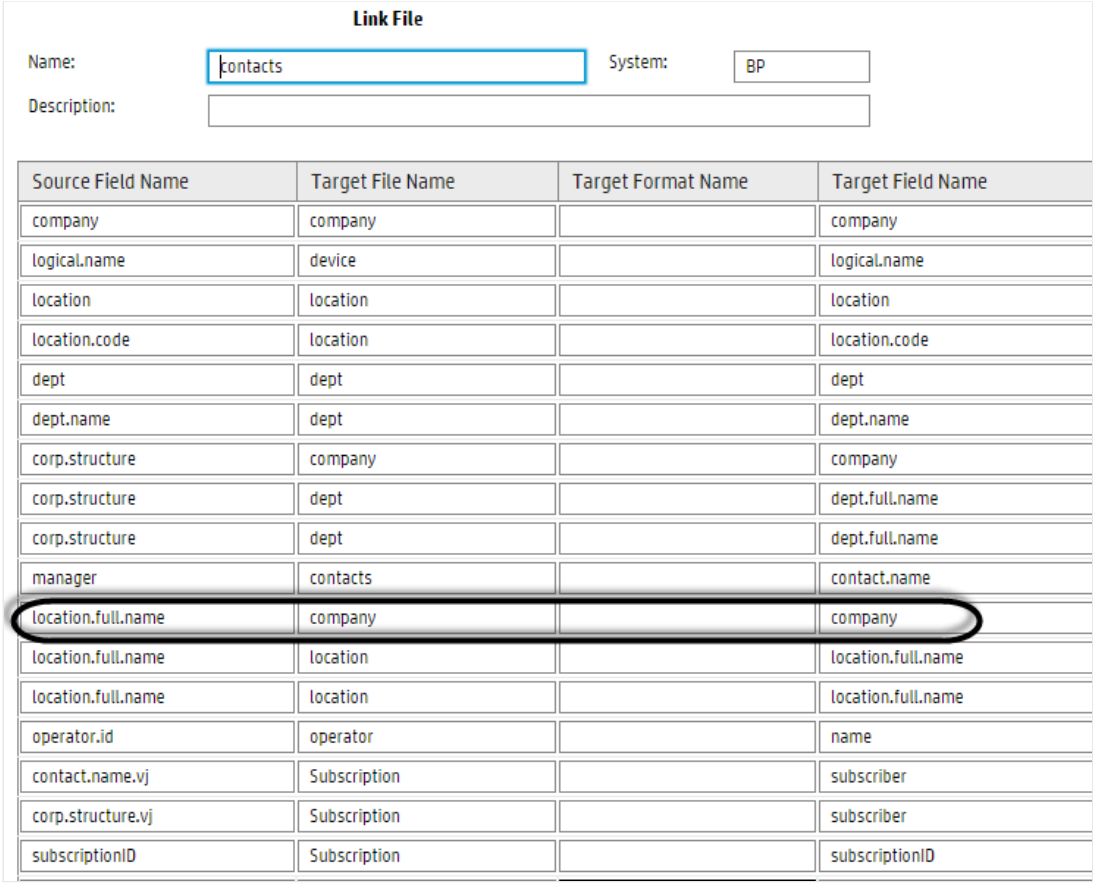
Example 2. The missing reference is caused by an inappropriate link definition

The following describes an example of an inappropriate link definition.

1. Navigate to **System Administration** > **Tailoring** > **Tailoring Tools** > **Relationship Manager**, locate the following relationship record by performing a search.

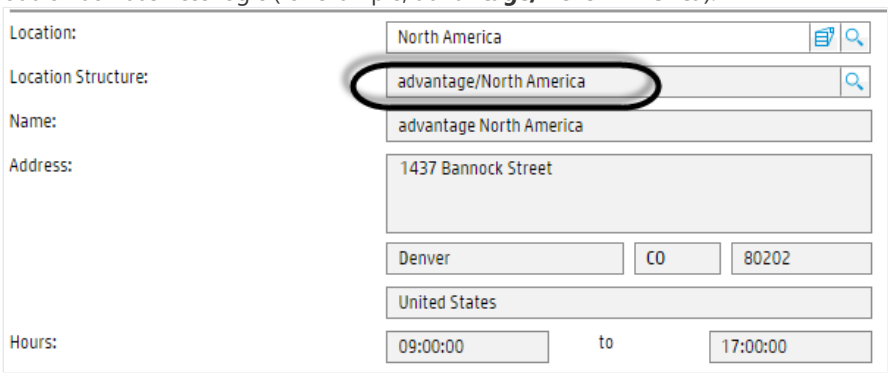
SOURCE_FILE	SOURCE_FIELD	TARGET_FILE	TARGET_FIELD	RELATION_TYPE	DEFINITION_TYPE
contacts	location.full.name	company	company	Many to Many	link

2. In the relationship record, click **Goto Definition** to open the link definition, and locate the row that has a **Source Field Name** of **location.full.name** and a **Target Field Name** of **company**.

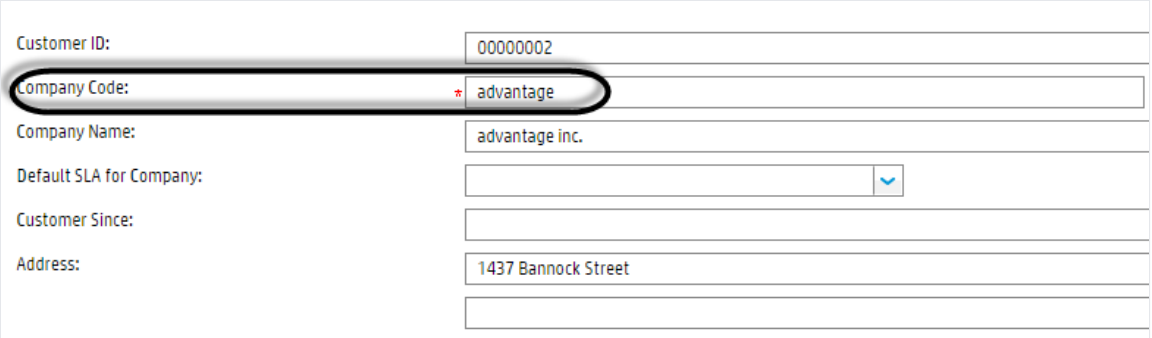


Source Field Name	Target File Name	Target Format Name	Target Field Name
company	company		company
logical.name	device		logical.name
location	location		location
location.code	location		location.code
dept	dept		dept
dept.name	dept		dept.name
corp.structure	company		company
corp.structure	dept		dept.full.name
corp.structure	dept		dept.full.name
manager	contacts		contact.name
location.full.name	company		company
location.full.name	location		location.full.name
location.full.name	location		location.full.name
operator.id	operator		name
contact.name.vj	Subscription		subscriber
corp.structure.vj	Subscription		subscriber
subscriptionID	Subscription		subscriptionID

3. Type contacts in the Service Manager command line, press Enter, and then open a contact record by performing a search.
   The **location.full.name** field in the **contacts** table has a label of **Location Structure** in contact records (the **Address** tab). As shown in the following figure, this field should be a combination of company and location according to the out-of-box business logic (for example, **advantage/North America**).



4. Open the **company** table in Database Manager, and locate a company record by performing a search.
   The **company** field in the **company** table has a label of **Company Code** in company records. The following figure shows an example value.



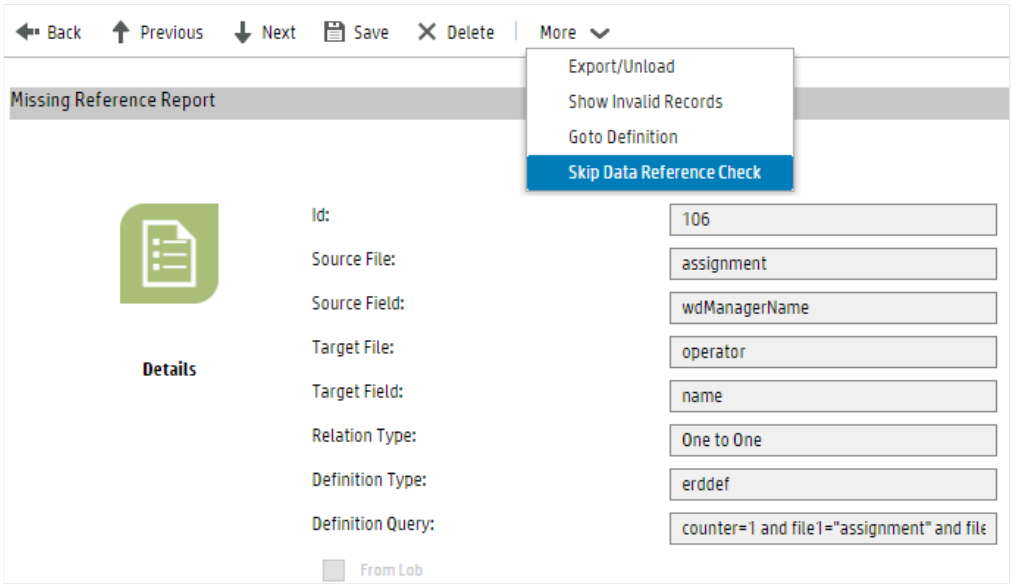
It is obvious that the **link** file defines a pair of source and target fields that have different value structures and therefore should not be mapped to each other. So this is an inappropriate **link** definition.

> **Note**
>
> Note Because of a limitation (see the Limitations section in Missing Reference Report utility), the Missing Reference Report utility doesn't report invalid records caused by this **link** definition when running a date reference check against the **contacts** table.

Steps to analyze a missing reference record

Normally, you can use the following process (which is similar to the example process described above) to analyze a missing reference record:

1. Search in the target file to make sure the orphan value can't be found in the target file, and examine the data in the target file to see what a correct value should look like.
2. Go to the **link** or **erddef** record to locate the relevant relationship definition.
3. Identify the problem:
   - Find the invalid "orphan" value in the source file.
   - Find the non-relational definition in the **link** or **erddef** file.
4. Analyze the problem based on the business logic.
5. Find out the root cause of the problem.
6. Perform the following tasks based on the root cause identified:
   - If the invalid record does contain a missing reference, correct the wrong information and then click **Save**. Alternatively, you can leave it as is if it is only historic data or you can click **Delete** to delete the record if you no longer need to keep it.
   - If the invalid record was caused by an inappropriate **link** or **erddef** definition, update the **link** or **erddef** definition.
   - Mark the corresponding relationship record so that it is skipped for future data reference checking. To do so, on the report details page of this relationship, click **More** or the More Actions menu and then select **Skip Data Reference Check** (see the following figure). If you open this relationship record from Relationship Manager, you will see the **Enable Data Reference Check** flag in the relationship is now turned off.



Related topics

Start the refcheck.startup scheduler
Run a data reference check against selected files
View missing reference report results

7.11.3. Relationship Manager

Relationship Manager is a tool that can automatically discover entity relationships between tables and fields in your Service Manager system by querying the **erddef**, **link**, and **relatedObjectMap** tables, and then save the results in the **relation** table. It also enables you to manually create relationships and manage them.

Also, Relationship Manager provides output that's consumed by the Missing Reference Report utility and Entity Relationship Diagram utility.

Note

Note To use Relationship Manager, you must have the SysAdmin capability word.

Using Relationship Manager, you can perform the following tasks:

- Automatically trigger a reload of all relationships in your system. See Build relationships .
- Find entity relationships for tables and fields with other tables. See Search for relationships between specific tables and fields.
- Manually add relationships. See Add a relationship record.
- Update invalid or fixed relationships. See Update an invalid or fixed relationship.
- Update the relationship definition of a relationship record. See Access the relationship definition of a relationship record.

Related topics

Missing Reference Report utility
Entity Relationship Diagram utility

7.11.3.1. Build relationships

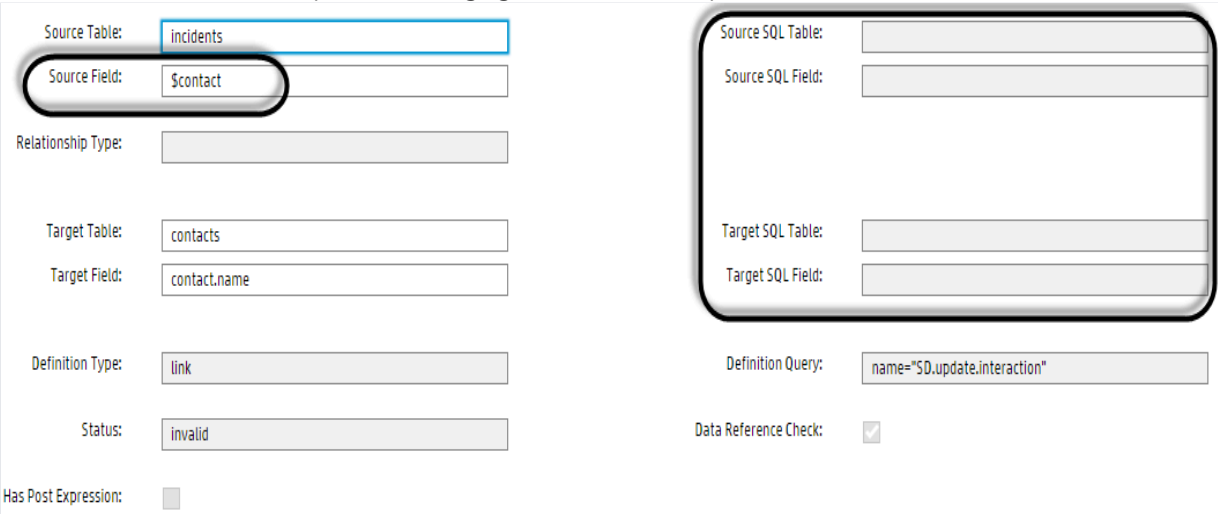
Applies to User roles: System Administrator

Relationship Manager can discover all relationships that are defined by the **erddef**, **link**, and **relatedObjectMap** tables. When upgrading to or installing the Service Manager applications, the **Build Relationships** functionality is automatically executed to load all relationships and save them in the **relation** table. When the **Build Relationships** functionality is executed, the following rules apply:

- Relationships that were manually created are retained, and all other previously built relationships are cleared and then rebuilt.
- Duplicate relationships from multiple sources are saved, and identified by the unique key, **id**.
- Relationship Manager generates a pair of relationship records for both relationship directions from one **erddef** definition. See the following table for an example.

Source Table	Source Field	Target Table	Target Field	Definition Type	Relationship
category	name	probsummary	category	erddef	One to Many
probsummary	category	category	name	erddef	Many to One

- Relationships without a SQL name for any of the **Source Table**, **Source Field**, **Target Table**, or **Target Field** fields are saved as an **invalid** relationship. The following figure shows an example, in which the source field uses a variable.



- Links that have a post expression are saved with the **Has Post Expression** field set to **true**.
- The **Enable Data Reference Check** setting (true or false) is retained when rebuilding each relationship. See Skip data reference checking for a relationship.
- While relationships are being built, messages are displayed, showing the building progress. If an error occurs, you are prompted to fix it and retry.

There may still be times when your system contains outdated relationship records. The **Build Relationships** option allows you to manually trigger a reload of all relationship records in your system. To manually build relationships, follow these steps:

1. Navigate to **Tailoring** > **Tailoring Tools** > **Relationship Manager** to open Relationship Manager.
2. Click the **Build Relationships** button.
   Service Manager starts to purge all relationships that were not manually created, and then reload all the relationships in your system. This process may take a while. When the building process is complete, a "done" message is displayed. If an error occurs, you are prompted to fix it and retry.

Once all relationships are discovered and saved in the **relation** table, you can search for all relationships or relationships between specific tables and fields. See Search for relationships between specific tables and fields.

Related topics

Relationship Manager
Search for relationships between specific tables and fields
Skip data reference checking for a relationship

7.11.3.2. Search for relationships between specific tables and fields

Applies to User roles: System Administrator

In Relationship Manager, you can search for relationships by one or more of the following fields: Table Name, Field Name, Definition Type, and Status.

To search for relationships between specific tables and fields, follow these steps:

1. Navigate to **Tailoring** > **Tailoring Tools** > **Relationship Manager** to open Relationship Manager.
2. In the **Table Name** field, type the name of either the source or the target table. For example, **operator**.
3. In the **Field Name** field, type the name of a field in the table you specified. For example, **name**.
4. In the **Definition Type** field, select a definition type: **link**, **erddef**, **relatedObjectMap**, or **manual**.
5. In the **Status** field, select a status: **new**, **invalid**, or **fixed**.
6. Click **Search**.
   A list of records is displayed, showing the relationships between the specified field and fields in other tables.

   Note

   Note You can also perform a search by leaving some or all of the search fields empty. We strongly recommend that you review all relationships to ensure they are correct, because some other features (such as the Entity Relationship Diagram utility and the Missing Reference Report utility) heavily rely on the correctness of your relationship records.

7. Double-click a record from the list to open the details page and view the details.
   A relationship record may contain the following fields.

Field	Description	Notes
Source Table	One of the following fields depending on the definition type of the relationship record (see Definition Type in this table): ○ **First Filename** in the **erddef** definition ○ **Name** in the **link** definition ○ **Source Table** in the **relatedObjectMap** definition	
Source Field	One of the following fields depending on the definition type of the relationship record: ○ **Field Names from First Filename** in the **erddef** definition ○ **Source Field Name** in the **link** definition ○ **Source Field** in the **relatedObjectMap** definition	
Relationship Type	The type of the relationship. It can have one of the following values: ○ One to Many ○ Many to One ○ One to One ○ Many to Many	System-generated
Target Table	One of the following fields depending on the definition type of the relationship record: ○ **Second Filename** in the **erddef** definition ○ **Target File Name** in the **link** definition ○ **To Table** in the **relatedObjectMap** definition	
Target Field	One of the following fields depending on the definition type of the relationship record: ○ **Field Names from Second Filename** in the **erddef** definition ○ **Target Field Name** in the **link** definition ○ **To Field** in the **relatedObjectMap** definition	

Definition Type	The type of the relationship definition, which indicates how the relationship is defined: ○ **link**: the relationship is defined in a **link** record. ○ **erddef**: the relationship is defined in an **erddef** record. ○ **relatedObjectMap**: the relationship is defined in a **relatedObjectMap** record. ○ **manual**: the relationship is manually defined.	System-generated
Status	The status of the relationship: ○ **new**: the relationship was automatically created and contains valid data. ○ **fixed**: the relationship was manually created or updated and contains valid data. ○ **invalid**: the relationship was automatically created and contains invalid data (any of the SQL table names or SQL field names can't be found).	System-generated
Has Post Expression	Indicates if the link record of the relationship has a post expression defined. Applicable only for relationships whose definition type is **link**: ○ true ○ false **Note** Note A post expression is an expression that's executed after the link function is triggered by clicking **Fill**.	System-generated
Source SQL Table	The SQL Table name of the source table	System-generated
Source SQL Field	The SQL Name of the source field	System-generated
Target SQL Table	The SQL Table name of the target table	System-generated
Target SQL Field	The SQL Name of the target field	System-generated

Definition Query	The query of the relationship definition **Note** Note This field is empty if the relationship is manually created.	System-generated
Enable Data Reference Check	This option indicates if this relationship is skipped when you run a data reference check against its source file or target file: ○ If this option is selected: not skipped ○ If this option is not selected: skipped	System-generated

Note

Note You can use one of the following methods if you have any invalid relationships:

- Update its relationship definition using the **Goto Definition** button. See Access the relationship definition of a relationship record.
- Add a new relationship for it by using the **Copy** button. See Add a relationship record.
- Update the relationship if the Definition Type is **link**. See Update an invalid or fixed relationship.

Related topics

Relationship Manager
Access the relationship definition of a relationship record
Add a relationship record
Update an invalid or fixed relationship

7.11.3.3. Access the relationship definition of a relation record

Applies to User roles: System Administrator

For relationships whose Definition Type is not **manual**, Relationship Manager provides a **Goto Definition** button, which enables you to view the relationship definition and update it if needed.

Note

Note Manually created relationships, that is, relationships whose Definition Type is **manual**, don't have a **Goto Definition** button.

To access the relationship definition of a relationship, follow these steps:

1. Navigate to **Tailoring** > **Tailoring Tools** > **Relationship Manager**.
2. Complete the **Table Name** and **Field Name** fields.

   Note

   Note This step is optional. You can leave one or both of the fields empty.

3. Click **Search**. A list of relationships is displayed.

4. Select the relationship record from the list, and click **Goto Definition** in the toolbar of the relationship detail pane.
5. View the relationship definition, and perform any of the following tasks if needed:
    - Update the relationship definition and then click **Add** to save it as a new relationship definition.
    - Update the relationship definition and click **Save** to save your changes to the existing relationship definition.
    - Click **Delete** to delete the relationship definition.

Related topics

Relationship Manager

7.11.3.4. Add a relationship record

Applies to User roles: System Administrator

Relationship Manager allows you to manage not only relationships that are defined by the **erddef**, **link**, and **relatedObjectMap** tables, but also relationships that you manually create. You can manually add a relationship record, and then update it or delete it as needed.

Note

Note Manually created relationship have a Definition Type of **manual**. You can update or delete this type of relationships .

To add a relationship record, follow these steps:

1. Navigate to **Tailoring** > **Tailoring Tools** > **Relationship Manager** to open Relationship Manager.
2. In the **Table Name** field, type the name of the table for which you want to add a relationship. For example, **operator**.
3. (Optional) Enter other search fields as needed.
4. Click **Search** to display a list of relationships.
5. Select a record from the list, and click **Copy**.
6. Change the values of the following fields as needed: **Source Table**, **Source Field**, **Target Table**, and **Target Fields**.
7. Click **Add**. Service Manager adds the relationship record.

Caution

Caution If your input contains wrong data, data validation fails and the system doesn't add the record.

   The **Relationship Type** and **Status** fields are automatically populated with **manual** and **fixed**, respectively.
8. The **Data Reference Check** option is automatically selected. If you want to skip this relationship for data reference checking, click **Skip Data Reference Check**, and then click **Save**. See Skip data reference checking for a relationship.
9. Verify the new record. To do so, click **OK** to return to Relationship Manager. Perform a search to locate the record you added. The record should have the following values:
   • **Relationship Type**: manual
   • **Status**: fixed
10. If needed, you can further update the record, and then click **Save**.
11. If you no longer want to keep the record, click **Delete** to delete it.

Related topics

Relationship Manager
Update an invalid or fixed relationship

7.11.3.5. Update an invalid or fixed relation

Applies to User roles: System Administrator

In Relationship Manager, you can update only relationship records that have either of the following:

- a Status of **invalid** and Definition Type of **link**
- a Status of **fixed**

Note

Note You can't update a relationship whose status is **new**.

To update a relationship record, follow these steps:

1. Navigate to **Tailoring** > **Tailoring Tools** > **Relationship Manager** to open Relationship Manager.
2. Optionally, enter or select a value in the **Table Name** and **Field Name** fields.
3. In the **Status** field, select **fixed** or **invalid**.
4. If you selected **invalid** in the **Status** field, select **link** in the **Definition Type** field.
5. Click **Search**. A list of relationships is displayed.
6. Select the record that you want to update, and update the relationship record as needed.
7. Click **Save** to save your changes.
   If the record had an **invalid** status, it now changes to a **fixed** status. If it had a **fixed** status, the status doesn't change.

   Caution

   Caution If your input contains incorrect data, data validation fails and the record is not updated.

   Note

   Note If you no longer want to keep the record, click **Delete** to delete the record.

Related topics

Relationship Manager
Add a relationship record
Access the relationship definition of a relationship record
Build relationships

8. System Security

Service Manager includes a number of security options you can configure after installation. The following table lists the security options you can enable or configure from a new development environment installation.

Tip

Tip

For overall instructions on how to implement your Service Manager environment in a secure manner, see the Service Manager Security Guide.

Feature	Description	Default state in new installations
Encryption of configuration file settings	A security option that protects values listed in the configuration file	Enabled
Encryption of operator passwords	A security option that protects the passwords listed in operator records	Enabled
Inactivity timer	A security option that automatically closes user sessions that have been idle for a specified period of time (except for those operators who are on the exception list)	Enabled
Lockout feature	A security option that automatically disables a user account if the user fails to provide the correct password after a specified number of attempts	Enabled
Mandanten file security	A security option that filters the data that operators can see when they query specific files	Disabled
Multicompany mode	A security option that filters the company information that service desk technicians see when creating service desk interactions and opening incidents	Disabled
Requirements for required SSL encryption	An implementation option that requires SSL encryptions for all connections	Disabled
Requirements for required SSL encryption and client authentication	An implementation option that requires SSL encryptions for all connections and validates the client's certificates	Disabled
Requirements for required SSL encryption and trusted clients	An implementation option that requires SSL encryptions for all connections and restricts connections to a list of trusted clients	Disabled
Script utilities	A security option that enables checksum calculation for Service Manager binaries and data security deletion.	Disabled
Secure Sockets Layer (SSL) encryption	An implementation option to encrypt communications between your Service Manager server and clients	Disabled

Trusted sign-on	An implementation option that enables Service Manager clients to automatically sign in using the same authentication information as users entered when they signed in to their client workstation's operating system.	Disabled
Common Access Card (CAC) sign-on	An implementation option that enables Service Manager clients to automatically sign in using authentication information from the user's personal certificate stored in the user's CAC card.	Disabled
SAML Single Sign-On	An implementation option that enables Service Manager clients to sign in to multiple web applications through single sign-on based on Security Assertion Markup Language (SAML) authentication.	Disabled
FIPS mode	A security option that requires data be encrypted using the FIPS-compliant AES encryption algorithm.	Disabled
Tokenization	A security option that enables tokenization in the web client to safeguard sensitive data.	Enabled
Enhanced query hash algorithm	A security option that enables the enhanced query hash algorithm.	Disabled

Related topics

Security tables
Generating Web tier URL queries
Upgrade Service Manager

8.1. Encryption of configuration file settings

You can encrypt values within the Service Manager configuration file (sm.ini) to protect passwords and authentication information. The encryption scheme is ideal for connection parameters such as RDBMS or LDAP user names and passwords. Using a command line utility you can convert any value to an AES256-256-CBC encrypted value. You can then copy the encrypted value into the configuration file and add an asterisk to the beginning of the parameter name. The asterisk is a required element that indicates to the server that the parameter value is encrypted.

For example, the unencrypted value:

```
sqllogin:rdbmsuser/mypassword
```

Becomes the following value when encrypted:

```
*sqllogin:D51CB23B379C873CBA055FB9A3798375AC93D48BB8AE2CC773D7317E4715EAE7
```

After you have encrypted a configuration file value there is no way to convert it back to clear text. The Service Manager server decrypts the value for administrative purposes, but it doesn't save or store the decrypted value.

Note

Note Only use the encryption scheme to encrypt server parameters in the Service Manager configuration file (sm.ini). Do not encrypt other values in other files as this may damage your system or cause data loss.

In addition, the values of the following security parameters in the sm.ini file are encrypted automatically during the server startup. You don't need to encrypt them manually.

- changeencrkey
- upgradeencralg
- encryptionkey
- sqllogin
- ldapbindpass
- smtppassword
- keystorePass
- truststorePass
- ssl_trustedClientsPwd
- idmsigningkey

Encrypt values in the Service Manager configuration file

Applies to User roles: System Administrator

You can use the **sm -encrypt** command to encrypt parameter values in the sm.ini file. You must have administrative access to the server operating system to use this procedure.

Note

Note The values of some security parameters are encrypted automatically during the server startup, including the sqllogin parameter that's used as the example in the following task.

To encrypt values in the Service Manager configuration file, follow these steps:

1. Open the Service Manager configuration file (sm.ini).
2. Copy the parameter name and parameter value you want to encrypt.
   For example, copy `sqllogin:RDBMS user/my password`.
3. Open the operating system's command prompt.
   For example, on Windows click **Start** > **Programs** > **Accessories** > **Command Prompt**.
4. Change directories to the RUN folder of your Service Manager installation.
   For example:

   ```
   cd C:\Program Files\OpenText\Service Manager\Server\RUN
   ```

5. Type the following command:

   ```
   sm -encrypt:<parameter name>:<parameter value>
   ```

   If the string containing the <parameter name>:<parameter value> includes spaces, you must enclose the entire parameter name and value in quotation marks. For example:

   ```
   sm -encrypt:"sqllogin:RDBMS user/my password"
   ```

Tip

Tip The sqllogin credentials are automatically encrypted and added to the sm.ini file if you configure the credentials by running the Server Configuration Utility instead of running this command.

6. Press Enter.
   Service Manager displays the new encrypted value.
   For example:

```
sqllogin:"RDBMS user/my password" is encrypted:
99E6136711576164187C6619C6DBA9DFFC89F7EDF186BC5827E3615BE8419CD4
```

7. Copy the new encrypted value.
8. Return to the Service Manager configuration file and add an asterisk to the beginning of the parameter name.
   For example:

```
*sqllogin:
```

9. Replace the clear-text value with the encrypted value.
   For example:

```
*sqllogin:99E6136711576164187C6619C6DBA9DFFC89F7EDF186BC5827E3615BE8419CD4
```

10. Save the configuration file.

Related topics

System Security

8.2. Encryption of operator passwords

The Service Manager server encrypts all operator passwords stored on the database using a SHA512 one-way encryption process that can't be decrypted. The server automatically encrypts existing passwords on your system the first time they are accessed. You can also do a mass update of the operator table to convert all passwords at once.

You need to update the SQL mapping for the password field to accept a larger character limit. The data policy settings are as follows.

Database object	Requirement
Table	operator
Field	password
Data type	VARCHAR
Size	136 characters

Automatic operator password encryption replaces the legacy data policy encryption option that was controlled by the `encryptionkey` parameter. To convert to the automatic operator password encryption scheme in legacy systems, you must first turn off the existing data policy encryption. The server updates the password encryption the next time it reads the operator record.

Service Manager clients use a two-way encryption process (PBE with MD5, DES in non-FIPS mode, and AES in FIPS mode) to secure operator passwords when communicating with the server. The server decrypts the password sent from the client and then one-way encrypts it to compare the results to the encrypted value stored in the database. The server never stores the operator password in an unencrypted form.

If your Service Manager implementation uses LDAP authentication, the server must still send an unencrypted operator password to the directory service because LDAP servers are unaware of Service Manager's encryption scheme. If you require encryption between Service Manager and the LDAP server, you can configure OpenSSL or another standard encryption scheme between the two servers.

Related topics

Lightweight Directory Access Protocol (LDAP)
System Security

8.3. Encryption of client keystore passwords

Service Manager supports encryption of keystore passwords in the Windows and web clients.

Windows client

The Windows client keystore password that you enter in **Window** > **Preferences** > **Service Manager** > **Security** is automatically encrypted and stored in the following file:

<Your user workspace dir>\ServiceManager\workspace\.metadata\.plugins\org.eclipse.core.runtime\.settings\com.hp.ov.sm.client.eclipse.base.prefs

Web client

You must enter your web client keystore password in *<Webtier>*/WEB-INF/webtier.properties file. If you have specified the **customize-folder** parameter in the web.xml file, enter your web client keystore password in *<Customize-Folder>*/config/webtier.properties (where *<Customize-Folder>* is the folder specified in the **customize-folder** parameter).

When the web application server is started, Service Manager changes this value to an encrypted string. The following steps describe how you set a new or update an existing web client keystore password.

To set a web client keystore password using the <Customize-Folder>, follow these steps:

1. Make sure the **customize-folder** parameter is configured in the web tier configuration file (web.xml).
2. Create a webtier.properties file in the <Customize-Folder>/config directory.

   Note
   Note Make sure the web application server has read/write access to this directory.

3. Open the file in a text editor, and add the following line to the file:
   `keystorePassword=` < **your keystore password in clear text format**>
4. Save and close the file.
5. Restart the web application server.

Related topics

Encryption of configuration file settings
Encryption of operator passwords

8.4. Randomly generated master keys

Service Manager (SM) uses randomly generated master keys to protect local sensitive data (for example, keystore passwords) of the SM Server, web tier, Windows Client, and Mobility Client.

Service Manager automatically generates a master key at the first startup of each of the following components and writes it to their corresponding configuration file.

The master keys serve as KEK (Key Encryption Key), which are used to encrypt all the security parameters.

Important

Important

The data contained in these files should be considered sensitive. Be sure to take appropriate measures to protect these files by ensuring only those users with appropriate access privileges are allowed to access and view the contents of these files.

	Master key parameter	Configuration file
Server	smmasterkey	For server upgraded from previous version: Server\RUN\sm.ini To intall, rotate, migrate, or change the storage location of `smmasterkey`, see Key management for smmasterkey.
Web tier	masterKey	*<customize-folder>*\config\webtier.properties or *<web tier>*\WEB-INF\webtier.properties Where: *<customize-folder>* represents the directory defined in the customize-folder parameter.
Windows Client	masterKey	%USERPROFILE%\ServiceManager\workspace\.metadata\.plugins\org.eclipse.core.runtime\.settings\com.hp.ov.sm.client.eclipse.base.prefs
Mobility Client	randomRawKey	<webapp>\WEB-INF\web.properties

Caution

Caution

Once a master key is generated, don't remove it or change its value, otherwise the Service Manager Server or other SM components (for example, the web tier) will not work.

You don't need to change the system-generated master keys; however, if you want the system to generate a new master key for Web tier, Windows Client, or Mobility Client, perform the following steps:
Note: For Service Manager Server, see Key management for smmasterkey.

1. Remove the master key parameter from the corresponding configuration file of the SM component.
2. Reset all encrypted parameter values in the configuration file to clear text.
3. Restart the Service Manager web application server, Windows Client, or Mobility application server.
   A new master key is automatically generated; the parameter values are encrypted again.

Related topics

Encryption of configuration file settings
Encryption of client keystore passwords

8.4.1. Key management for smmasterkey

The master key named `smmasterkey` serves as a KEK (Key Encryption Key), which is used to encrypt all the security parameters from the sm.ini file.

The `smmasterkey` key parameter is generated and stored in a secure position for new installations.

If you need to rotate, migrate, or change the storage location of `smmasterkey`, follow the instructions below.

Permission control

For security concerns, you may apply the permission control, thus only limited users can manage `smmasterkey`.

In Linux system, the permission control is automatically configured for `smmasterkey`.

In Windows system, you need to manually define and grant permission to a group:

1. Go to **Computer Management > Local Users and Groups**, create a user group for permission control.
2. Add users into this group. Make sure that the user who start the Service Manager server must be included in this group.
3. Go to the **`%programdata%\OpenText`** folder, right-click the **Service Manager** folder, then select **Properties**.
4. In the **Service Manager Properties** window, go to the **Security** tab, then click **Advanced**.
5. In the **Advanced Security Settigs for Service Manager** window, click **Disable inheritance**, then select **Convert inherited permissions into explicit permissions on this objection**.
6. Remove all the existing permission entries in the **Permissions** tab, then add the group created in step 1.
7. Sign in to the system as the user of the group with associated permission.

Change the storage location

The `smmasterkey` key parameter is generated and stored in Server\RUN\sm.ini when you upgrade Service Manager from previous versions.
The `sm -keyrotate:1` command can protect the cryptographic master key (KEK) in a way that is resistant to some of the most common remote file exfiltration attacks. You may use the `sm -keyrotate:1` command to store the master key in a secure position:

1. Stop the Service Manager server.
2. Run the `sm -keyrotate:1` command.
   Note that The `sm -keyrotate:1` command will also change the value of `smmasterkey`.
3. To ensure that the change has take effect, you can check the sm.ini file and verify that the `keystorage:1` parameter is included.
4. Start the Service Manager server.
5. Make sure the Service Manager server is started successfully.

Revert the location change

Important
Caution: It is not recommended that you move the `smmasterkey` back to the the sm.ini file.

If you want to revert this location change, follow these steps:

1. Stop the Service Manager server.
2. Run the `sm -keyrotate:0` command.
   Note that this command will also change the value of `smmasterkey`.
3. To ensure that the change has take effect, you can check the sm.ini file and verify that the `smmasterkey` parameters are included.
4. Start the Service Manager server.
5. Make sure the Service Manager server is started successfully.

Rotate the master key

For security concerns, you can rotate the master key on a regular basis:

1. Stop the Service Manager server.
   Note: In a horizontal scalability (HS) environment, there is no need to stop all nodes on each machine simultaneously. You can apply the key rotation on each SM node one by one.

2. Run the following command to update the value of `smmasterkey`:
   `sm -keyrotate:1`
   Note: After the key rotation, it is recommended that you use the `sm -keyexport` command to export `smmasterkey` and store the key in a secure location.
3. Start the Service Manager server.
4. Make sure the Service Manager server is started successfully.

Export the master key

You can use the `sm -keyexport` command to export the master key. For example, you may use this command to backup your `smmasterkey`.

It is recommended that you store the exported key in a secure location.

1. Run the following command to export the master key:
   `sm -keyexport`
   The key will be exported to the `<RTE installation folder>\RUN` directory and will be named as `smmasterkey.key`.
   - If you want to export the key as a specified file, you can run the following command:
     `sm -keyexport:"<export directory>\<file name>"`
     For example:
     `sm -keyexport:"CustomPath\mykey.key"`
     Make sure you have read-write permissions to the folder.

Import the master key

When you need to migrate the Service Manager system from one machine to another machine, you can first use the `sm -keyexport` command to export the master key, then use the `sm -keyimport` command to migrate the key to the target system.

1. Copy the sm.ini file and then paste the file to the target system.
2. Copy and paste the exported `smmasterkey` to the `<RTE installation folder>\RUN` directory or to a specified location.
3. Run the following command:
   `sm -keyimport`
   The key named `smmasterkey.key` will be imported from the `<RTE installation folder>\RUN` directory.
   - If you want to import a specified key file, you can run the following command:
     `sm -keyimport:"<import directory>\<file name>"`
     For example:
     `sm -keyimport:"CustomPath\mykey.key"`
     Make sure you have read-write permissions to the folder.

8.5. Inactivity timer

The inactivity timer functionality closes user sessions that have been idle for a specified period of time. This feature is enabled by default and the time-out value is 30 minutes. System administrators can configure required settings and start or stop the inactivity timer.

Note

Note You need the `SysAdmin` capability word to configure and start/stop the inactivity timer.

Configure inactivity timer settings

Administrators can configure required settings at a system level and if needed, also an operator level, as described in the following table.

Note

Note In a horizontal/vertical scaling environment, you need to configure the inactivity timer settings only once (that is, only on any one of your server nodes).

Configuration Level	Settings	Notes
Operator Level	• **Warning Time** • Allowed Inactive Time	• To configure the settings, click the **Security** tab in an operator record. • Operator level settings take precedence over system level settings; if an operator record has empty inactivity timer settings, the system level settings are used for the operator. • An allowed inactive time of zero (0) means inactivity session timeout will never happen to the operator. • A zero or empty warning time means no warning message will display to the operator.
System Level	• **Number of minutes to allow idle user** ("allowed inactive time") • **Warn user within <x> minutes of termination** ("warning time") • **DO NOT terminate these users** ("exception list")	Click **System Administration** > **Ongoing Maintenance** > **System** > **Start Inactive Timer** > **Start Inact**.

The following table is an explanation of the settings described above.

Setting	Description
Allowed inactive time	This is a time limit that users can remain inactive until the inactivity timer closes their user session.
Warning time	This is an amount of time between the appearance of a warning message and the termination of the session. **Caution** Caution The warning time should be no greater than the allowed inactive time.
Exception list	This is a list of operators that the inactivity timer ignores. This list should include all operators that should not be logged out regardless of the type of license (Named or Floating license).

When the inactivity timer settings take effect

Operator level settings and system level settings take effect the next time the user logs in.

Start the inactivity timer

> **Note**
>
> Note Starting with SM9.32, the **inactive.startup** background process has been deprecated. You no longer need to start this process in order for the inactivity timer functionality to work.

To enable the inactivity timer functionality for users, start the inactivity timer as follows:

1. Click **System Administration** > **Ongoing Maintenance** > **System** > **Start Inactivity Timer**.
   The inactive.setup.g form opens.
2. Type or select the inactivity timer record information. See *Configure inactivity timer settings*.
3. Click **Start Inact**.

If you want to disable the inactivity timer functionality for all users, you can stop it as follows.

1. Click **System Administration** > **Ongoing Maintenance** > **System** > **Start Inactivity Timer**.
   The inactive.setup.g form opens.
2. Click **Stop Inact**.
   The **Start Inact** button is displayed.

After modifying your system level inactivity timer settings, you need to restart the inactivity timer as follows.

1. Click **System Administration** > **Ongoing Maintenance** > **System** > **Start Inactivity Timer**.
   The inactive.setup.g form opens.
2. Modify the settings as needed.
3. Click **Save**. Service Manager displays the message: `The inactive process has been set up to look for users idle after <n> minutes.`

Respond to the inactivity warning message

Applies to User Roles: All users

You may receive an inactivity warning message depending on the inactivity timer settings your administrator has configured. When this happens, you can respond as needed to either time out or extend your session.

The following table describes your actions in the Windows/web client and the corresponding inactivity timer behavior.

Client	Warning Message	Action
Web	A warning message displays in a pop-up window.	Click **OK**: Your session extends and the inactivity timer restarts counting. Click **Cancel**, press **ESC**, or make no response: The inactivity timer continues the counting and terminates your session in the end.
Windows	A warning message displays in the toolbar.	Perform any operation: Your session extends and the inactivity timer restarts counting. Make no response: The inactivity timer continues the counting and terminates your session in the end.

Related topics

System Security

8.6. Lockout feature

The lockout feature automatically disables a user account if the user fails to provide the correct password after a specified number of attempts. Your user profile must include the system administration capability word to use this feature.

Enable the user lockout feature

Applies to User roles: System Administrator

The user lockout feature automatically disables a user account if the user fails to provide the correct password after a specified number of attempts.

> **Note**
>
> Note Enabling the user lockout feature is the only way to restrict access from the Windows client. The **Attempts Per Login Session** settings doesn't affect Windows clients because the Windows client creates a new session for each login attempt.

To enable the user lockout feature:

1. Click **System Administration** > **Base System Configuration** > **Miscellaneous** > **System Information Record**.
2. Click the **Logon Info** tab.
3. In the Use User Lockout section, select **Use User Lockout**.
4. In the **Attempts Until Lockout** field, type the number of login attempts the user has until Service Manager locks out the account.
5. Select one of the following options:
   - **Complete Lockout** — select this option to deny the user access to Service Manager until a system administrator resets the lockout.
   - **Lock Out Time Period** — select this option to deny the user access to Service Manager for a specified period of time.
6. Type the time period in the following format: *Dayshours*:*minutes*:*seconds*. For example, 4 03:02:01 locks out the user for 4 days, 3 hours, 2 minutes, and 1 second.
7. Click **Save**.

Lock out a user

Applies to User Roles: System Administrator

To lock out a user:

1. Click **System Administration** > **Ongoing Maintenance** > **Operators**.
2. Type or select optional search criteria.
3. Click **Search**.
4. Select the operator to lock out.
5. Click the **Security** tab.
6. Select the **Administrative Lockout** option
7. Click **Save**.

Reset a locked out user

Applies to User Roles: System Administrator

To reset a locked out user:

1. Click **System Administration** > **Ongoing Maintenance** > **Operators**.
2. Type or select optional search criteria.
3. Click **Search**.
4. Select the operator record.
5. Click the **Security** tab.
6. Click **More** or the More Actions icon, and then select **User Lockout Reset**.
7. Click **Save**.

Service Manager unlocks the user's operator record and clears the **Failed Login Count**, **Locked Until**, and **User has been Locked?** fields.

View a user's lockout history

Applies to User Roles: System Administrator

To view a user's lockout history

1. Click **System Administration** > **Ongoing Maintenance** > **Operators**.
2. Click **Search**.
3. Select the operator record to view.
4. Click the **Security** tab.

View system level user lockout history

If a system administrator has enabled the Use User Lockout feature in the System Information Record, a user may be locked out for several reasons. For example, the user has attempted to log in with an invalid password more than the number of times defined by the system administrator or the user is locked out by the system administrator.

A specific user's lockout status in displayed in the Security tab of the user's operator record. When a user is locked out, the **User has been locked out** option is selected and the Lockout Reason field displays the reason (for example, Max Failed Logins).

At the system level, the lockout history of all users is stored in the `operatorlockhistory` table in the database. To view system level user lockout history, use either of the following ways:

- Open Database Manager, enter operatorlockhistory in the Table field and then click the Search button.
- Navigate to **Miscellaneous** > **Dashboard** and then open the License Overview dashboard. If the "Operator Lockout History within 7 days" report is not displayed on the dashboard, click **Add Content** and then click **License** to add this report to the dashboard. This report uses the operatorlockhistory table as the data source.

Related topics

Application profiles
Environment record
User roles
Capability word model
Adding users
Checklist: Adding a new user
Controlling user access and security
Creating operator records
Defining named users
Operator passwords
Operator records
Operator templates
System quiesce: Login restrictions

8.7. System quiesce: Login restrictions

Quiesce mode sets login restrictions to prevent users from logging on to Service Manager processes. This gives System Administrators a way to stop users from logging on to Service Manager processes, and wait for existing users to gracefully log off before starting system maintenance or testing tailoring activities.

There are three levels of login restrictions. Quiesce level 1 restricts all users, except System Administrators, from logging on to Service Manager processes. Quiesce level 2 restricts all users from logging on to Service Manager processes. Quiesce level 0 (zero) sets Service Manager processes to allow user logins. In the vertical scaled or horizontal scaled environment, Service Manager load balancer doesn't forward any client connection requests to Service Manager processes that are in quiesce level 1 or 2. If System Administrators want to connect to a Service Manager process in quiesce level 1, they must connect directly to the Service Manager process without connecting through Service Manager load balancer.

Quiesce mode information is stored in the shared memory. When a Service Manager process sets a quiesce level in the shared memory, all other Service Manager processes that read from the same shared memory have the same quiesce level. So if one Service Manager process sets a quiesce level on a host, all Service Manager processes have the same level of quiesce on that host. In a horizontal scaled environment, the "-host:<host name or IP>" or "-group" option can be used with the "sm -quiesce:<quiesce level>" command. The "sm -quiesce:<1 or 2 or 0> -group" command sets the quiesce level to all Service Manager processes on all hosts within the horizontal scaled group.

In a vertical scaled environment when Service Manager processes are running and you issue the "sm -quiesce:1" or "sm -quiesce:2" command, all Service Manager processes on the local host are set to quiesce level 1 or quiesce level 2, respectively. The Service Manager load balancer stops forwarding any new client connection requests to the Service Manager processes, since they are quiesced. If any user tries to connect to Service Manager load balancer at this point, the user receives a message that states "max session exceed" from the load balancer, as there are no available Service Manager processes. Existing users on the system aren't affected. However, once existing users log off, they can't log back on until after the System Administrator changes the quiesce level back to 0 (zero). Once all users have logged off the system, the System Administrator can perform system maintenance. When system maintenance is complete, the System Administrator can issue the "sm -quiesce:0" command to set the quiesce level back to 0, so there are no login restrictions to all Service Manager processes. All Service Manager processes now accept user logins and Service Manager load balancer forwards client connection requests to these Service Manager processes.

In a horizontal scaled environment, assume there are two hosts in the horizontal scaled group, Host A and Host B. While the system is running, System Administrators can quiesce Host A for maintenance and keep Host B running by issuing the command "sm -quiesce:<1 or 2> -host:<Host A name or IP address>" on either Host A or Host B. Service Manager load balancer then forwards all client requests to Host B, as all Service Manager processes on Host A are quiesced. Existing users on Host A aren't affected until they log off. If users try to log back on, Service Manager load balancer redirects their connection requests to Host B. After maintenance is complete on Host A, the System Administrator issues the command "sm -quiesce:0 -host:<Host A name or IP address>" to bring Host A back to service. This way the System Administrator can maintain one of the hosts in a horizontal scaled group and avoid down time. The System Administrator can also quiesce all Service Manager processes in the group by issuing the "sm -quiesce:<1 or 2> -group" command. When the maintenance is complete, the System Administrator can then set all Service Manager processes in the group back to non-quiesce mode by issuing the "sm -quiesce:0 -group" command.

A System Administrator can restrict logins to Service Manager using the system.quiesce application. System maintenance tasks include the following:

- Upgrading from one version of Service Manager to another
- Tailoring forms, tables, or format controls

The system.quiesce application provides three levels of login restrictions:

Restriction level	Description
Level 0	Service Manager has no login restrictions and accepts all logins normally.

Level 1	Service Manager restricts login to operators who have the SysAdmin capability word. Service Manager denies login to all other operators and displays the message:
	`System quiesced, you can't login at this time.`
	When you issue `sm -quiesce:1 -group` on a primary or secondary host, all Service Manager processes (except Service Manager load balancer) are set to quiesce mode 1. Service Manager load balancer updates its available node list to 0. If users try to connect to Service Manager load balancer at this time, they receive the following message, since there are no available Service Manager processes.
	`Max session exceeded.`

Level 2	Service Manager denies login to all operators and displays the message: `System quiesced, login restricted at this time.`

Note: A quiesced system restricts new login attempts only. Currently logged on users can continue working until they log off.

Enable logging of user access

Applies to User Roles: System Administrator

If you enable system logging of user access, Service Manager records the time and user ID each time an operator logs on or logs off.

To enable logging of user access:

1. Click **System Administration** > **Base System Configuration** > **Miscellaneous** > **System Information Record**.
2. On the General tab, select the **Syslog Audit** option.
3. Click **Save**.

Enable login restrictions

Applies to User Roles: System Administrator

To enable login restrictions:

1. Click **System Administration** > **Ongoing Maintenance** > **System** > **Connection Restrictions**.
2. Click one of the following options:
   - **Set Level 0** — No login restrictions
   - **Set Level 1** — Only system administrators can log in
   - **Set Level 2** — No operators can login

   Service Manager displays the current access level.

Disable login restrictions

Applies to User Roles: System Administrator

To disable login restrictions:

1. Click **System Administration** > **Ongoing Maintenance** > **System** > **Connection Restrictions**.
2. Click **Set Level 0**.

Enable tracking of operator times

Applies to User roles: System Administrator

You can track how long each operator edits an Incident record by enabling a tracking option in the Incident Management environment.

To enable tracking of operator times:

1. Click **System Administration** > **Ongoing Maintenance** > **Environment Records** > **Incident Management Environment**.
2. Select the **Track Operator Times?** checkbox.
3. Click **Save**.

Enable tracking of operator times

Applies to User roles: System Administrator

You can track how long each operator edits an Incident record by enabling a tracking option in the Incident Management environment.

To enable tracking of operator times:

1. Click **System Administration** > **Ongoing Maintenance** > **Environment Records** > **Incident Management Environment**.
2. Select the **Track Operator Times?** checkbox.
3. Click **Save**.

Related topics

Application profiles
Environment record
User roles
Capability word model
Adding users
Checklist: Adding a new user
Controlling user access and security
Creating operator records
Defining named users
Operator passwords
Operator records
Operator templates
Lockout feature

8.8. Mandanten file security

Mandanten is an optional file security feature that filters the data that operators can see when they query specific files. Rather than having access to all the records in a file, operators who are members of a security group see only the records that meet the specific filtering criteria of their group. The system administrator defines the filtering conditions when creating a security group. The system administrator decides which operators belong to particular security groups and can assign operators to any number of security groups. Operators who are members of multiple security groups see only the records that match all their separate filtering conditions.

At login, Service Manager reads the operator record to determine the security groups of which the operator is a member and uses this information to determine the files to which the operator has limited access. When an operator queries a restricted file, Service Manager reads the security group records to determine the filtering conditions to apply to the query. Service Manager then returns only those records that match the filtering conditions in the security group records.

Unlike Format Control, which provides security at the application layer, Mandanten secures files at the database layer. Any file that a system administrator restricts from an operator with Mandanten always uses the filtering conditions regardless of the operator's user role and application profile. Only operators who aren't members of any security group can have unrestricted access to files protected by Mandanten.

Typically, a system administrator enables Mandanten file security in a multi-company environment where each company wants to ensure that only their users see the data relevant to their business. However, system administrators can also use Mandanten to conceal department records selectively within an organization. For example, a system administrator could create two filtering conditions for an operations and finance department that allow the operations personnel to see their own Incident records and devices but not those belonging to the finance department.

Note

Note

The users created in Service Manager by default don't belong to any security group and by default have access to all records of users from other groups. Hence, you shouldn't use users of this type. By selecting or creating users of this type, you are disabling or bypassing Mandanten security features, thereby exposing the system to increased security risks. By using this option, you understand and agree to assume all associated risks and hold the company harmless for the same.

Setting filtering conditions

To enable Mandanten file security, a system administrator must create records in two files:

- **scsecuritygroup** — The system administrator uses this file to define the security group name and the field values to be used as the filtering condition.
- **scmandant** — The system administrator uses this file to define the Service Manager file to be protected and the field to be read for the filtering condition. The field defined in this file is referred to as the Mandant field. You can only define one Mandant field for each file you want to protect, although the Mandant field can contain an array of values. The Mandant field you chose must be defined either in the file you want to protect or by a virtual join in another file.

System administrators can define additional filtering conditions on fields other than the Mandant field. These additional filtering conditions are referred to as restricting queries because they further restrict the data that an operator can access. To define a restricting query, a system administrator must create a record in the **scaccess** file.

Restrictions

- When you execute triggers, Mandanten security will be disabled.
- You can't enable Mandanten file protection on the following shared system files:
  - code
  - datadict
  - dbdict
  - environment
  - format
  - formatctrl
  - link
  - menu
  - operator
  - tzfile

Add a restricting query to a security group

Applies to User roles: System Administrator

Note: You must have the **SysAdmin** capability word to use this procedure.

You can add a restricting query to a security group to further limit the data that operators can see when they query a Mandanten protected file. To add a restricting query to a security group:

1. Click **Tailoring** > **Database Manager**.
   The Database Manager form (format.prompt.db.g) opens.
2. In the **Form** field, type **scaccess**.
3. Click **Search**.
   The Mandanten Restricting Query form (scaccess.g) opens.
4. In the **File Name** field, type or select the name of the Mandanten protected file you want to further restrict.
5. In the **Security Group ID** field, type the name of the security group that you want to further restrict.
6. In the **Restricting Query** field, type the SQL query you want to use to further restrict the data operators can access **Note**: Typically, this query is of the format *field*="*value*".
7. Click **Add**. Service Manager displays the message: `scaccess record added` .

Create a security group

Applies to User roles: System Administrator

Note: You must have the **SysAdmin** capability word to use this procedure.

You must create a security group to enable the Mandanten file security feature. To create a security group:

1. Click **Tailoring** > **Database Manager**. The Database Manager form (format.prompt.db.g) opens.

1. In the **Form** field, type **scsecurity**.
2. Click **Search**. The Mandanten Security Groups form (scsecuritygroup.g) opens.
3. In the **Security ID** field, type the name of your new security group.
4. In the **Include Value List** array, type the Mandanten field values that you want to use to grant access to the security group.
5. In the **Exclude Value List** array, type the Mandanten field values that you want to use to deny access to the security group.
6. Click **Add**.
   Service Manager displays the message:
   `scsecuritygroup record added.`

Define the Mandant field for a security group

Applies to User roles: System Administrator

Note: You must have the **SysAdmin** capability word to use this procedure.

You must define the Mandant field to enable the Mandanten file security feature.

To define the Mandant field for a security group:

1. Click **Tailoring** > **Database Manager**.
   The Database Manager form (format.prompt.db.g) opens.
2. In the **Form** field, type **scmandant.g**.
3. Click **Search**.
   The Mandanten Field Restriction form (scmandant.g) opens.
4. In the **File Name** field, type or select the name of the file you want to protect.
5. In the **Mandant Field Name** field, type the name of the field you want Service Manager to read when it checks whether an operator meets the filtering conditions.
   Note: This field must either be defined in the file you select or be defined in another file as a virtual join.
6. If the Mandant field is a virtual join field, type the following additional information:
   - In the **Linkage Field Name** field, type the name of the field that stores the virtual join reference.
   - In the **Source File Name** field, type the name of the external file that contains the Mandant field.
   - In the **Source Field Name** field, type the name of the external field that stores the Mandant field values you want Service Manager to read when it checks whether an operator meets the filtering conditions.

Note: If the source field contains an array of values, then Service Manager will search the array for valid filtering conditions. If any one of the filtering conditions is met, then the operator will have access to the file.

- In the **Exclude Field** field, type any values that you want Service Manager to exclude from queries against the Mandant field.

7. Click **Add**. Service Manager displays the message: `scmandant record added.`

Enable Mandanten security on a file

Applies to User roles: System Administrator

Note: You must have the **SysAdmin** capability word to use this procedure.

To enable Mandanten security on a file:

1. Create a security group to define the filtering conditions you want to use on the protected file.
2. Define the Mandant field that you want Service Manager to read when it checks whether an operator meets the filtering conditions.
3. Add any restricting queries you want to add to the security group.
   Restricting queries further restricts the data that operators in the security group can access.
4. Assign operators to security groups that match the data you want them to access.
   Note: If the system administrator doesn't assign an operator to a security group, then that operator can see an unfiltered view of all the files to which the operator's user role and application profile permit access.

Related topics

System Security

8.9. Multicompany mode

You can run Service Manager in multicompany mode to filter the company information that service desk technicians see when creating service desk interactions and opening incidents. Normally, support desk technicians can see information about all the companies that the service desk supports. However, when a system administrator enables multi-company mode, service desk technicians only see information relevant to the company they are currently supporting.

For additional data security, you can enable the Mandanten feature.

Enable multicompany mode

Applies to User Roles: System Administrator

You must have the **SysAdmin** capability word to use this procedure.

To enable multicompany mode:

1. Click **System Administration** > **Base System Configuration** > **Miscellaneous** > **System Information Record**.
2. On the General tab, select the **Run in Multi-Company Mode** option.
3. Click **Save**.
   Service Manager displays the message:
   `Information record updated.`

Related topics

Mandanten file security

8.10. Script utilities

Service Manager includes two new scripts:

- smfsdel.bat
  Provides a mechanism to securely delete files from the file system on which you have installed Service Manager.
- smfchecksum.bat
  Provides a mechanism to verify the Service Manager binaries.

These scripts were created to meet the Federal Service for Technical and Export Control (FSTEC) certification as per the government regulations of Russia. These two scripts are packaged together with Service Manager. Most customers will have no reason to run these scripts but you can run them freely as you like.

For more information about these two scripts and how to use them, see Scripts for FSTEC Certification on Service Manager.

Related topics

System Security

8.11. Security tables

Service Manager applications contain built-in security that enables an administrator to define the rights for individual users (operators). For example, some users may have the right to open a call report or incident, but not to close a call report or incident.

Related topics

Application profiles
Capability word model
License tracking
User roles

8.12. Secure Sockets Layer (SSL) encryption

Service Manager supports Secure Hypertext Transfer Protocol (HTTPS), which encrypts and decrypts message requests and responses. Service Manager uses Secure Sockets Layer (SSL) for encryption only and relies on the server to authenticate each operator's user name and password. Service Manager supports SSL for the following connections:

- SSL on the Service Manager server to encrypt all communications between clients and the server.
- SSL on Service Manager clients to verify the client's identity and limit server connections to these identified clients

Enabling SSL on the Service Manager server

The primary reason to enable SSL on the Service Manager server is to protect operator user names and passwords that Service Manager clients send with each request as part of an HTTP Basic Authorization header. You can enable SSL on the Service Manager server but not require each client to present an individual client certificate. When you enable SSL on the server only, clients connect to the server using anonymous SSL. This is referred to as one-way SSL later.

Enabling SSL on Service Manager clients

The primary reason to enable SSL on Service Manager clients is to restrict access to the server to only those clients known and identified by the server. Enabling client-side SSL requires creating or purchasing signed certificates for each Service Manager client. This is referred to as two-way SSL later. The Service Manager Web Tier can share a single signed certificate for all Web Client connections.

Client/server SSL handshake process

During the client/server handshake process, the client looks at the server certificate, determines which certificate authority signed the certificate, and compares the certificate signature to a list of trusted certificate authorities identified in the truststore.

The client also compares the IP address or host name of the server to the address encrypted in the server certificate. If they don't match, the handshake fails.

What is a keystore file?

A keystore, as the name implies, provides storage and container for keys. It stores private key entries, certificates with public keys, or just secret keys that you may use for various cryptographic purposes. It stores each by an alias for ease of lookup.

In Service Manager, the supported keystore format is PKCS12 in non-FIPS mode, and BCFKS in FIPS mode.

What's a cacerts file?

The `cacerts` file is a collection of trusted certificate authority (CA) certificates. Oracle includes a `cacerts` file with its SSL support in the Java™ Secure Socket Extension (JSSE) tool kit and JDK. It contains certificate references for well-known Certificate authorities, such as VeriSign™. Its format is the "keystore" format defined by Oracle. An administrator can edit the `cacerts` file with a command line tool (also provided by Oracle) called keytool. For more information about keytool, see the Oracle website.

Note

Note The default password for the cacerts file supplied by Oracle is `changeit`. You must use this password to view the contents or to import a new certificate. For security reasons, change the default password.

The essential requirement is that the certificate authority that signed the Service Manager server's certificate must be in the list of certificate authorities named in this file. To use a self-issued server certificate created with OpenSSL or a tool such as Microsoft Certificate Server™, you must import the certificate for this private certificate authority into the `cacerts` file that the client uses for SSL. If you don't import the certificate, SSL connections fail because the Java SSL implementation doesn't recognize the certificate authority.

Secure Sockets Layer (SSL) configuration options

Service Manager supports several different Secure Sockets Layer (SSL) configurations that you can use to protect communications between the server and clients. The following is a list of the most common SSL configurations in order of increasing security and setup required:

Configuration Type	SSL encryption?	Server credentials authenticated?	Client credentials authenticated?	Client can bypass login screen?	Additional certificates required
Required SSL encryption (One-way SSL)	Required Set from server	Yes	No	No	• Certificate authority certificate • Server certificate
Required SSL encryption and client authentication (Two-way SSL)	Required Set from server	Yes	Yes	No	• Certificate authority certificate • Server certificate • Web tier certificate • Windows client certificates • Mobility certificate • Chat certificates
Required SSL encryption and trusted clients (Two-way SSL)	Required Set from server	Yes	Yes	Yes	• Certificate authority certificate • Server certificate • Web tier certificate • Windows client certificates • Mobility certificate • Chat certificates

Use SSL connections to connect to a Web Service

- **Enable SSL encryption for published Web Services**
  If you want external Web Services clients to use an SSL connection with the Service Manager server, you must provide them with the CA certificate for the Service Manager server. If you purchased a server certificate, copy the CA certificate from the CA certificate keystore provided with your purchased certificate. If you generated your own server

certificate by using a self-signed private CA certificate, copy the CA certificate from your private CA certificate keystore instead.

- **Enable SSL encryption for external Web Services**
  When using SSL connections to an external Web service, the Service Manager server acts like a client and must be set up accordingly. The Web service provider must send the root certificate or the certificate authority's (CA) certificate to the Service Manager administrator.

See Use SSL connections to connect to a Web Service for details.

Related topics

Trusted sign-on
Common Access Card (CAC) sign-on

8.12.1. Work with the certificate generation toolkit

As a courtesy, we provide a certificate generation toolkit along with KM000018408 to help you generate certificates for a Service Manager deployment in an automated way. Since the scripts in this package are based on the keytool utility available in a standard Java Development Kit to create various certificates, you need to install OpenJDK 17+ or Oracle JDK 17+ first, and then set the `JAVA_HOME` environment variable properly.

To use these scripts on a headless Linux server, you need to install rng-tools first. For example, run the **sudo yum install rng-tools** command on Cent OS, or run the **sudo apt install rng-tools5** command on Ubuntu. The rng-tools are a set of utilities that helps to increase the quantity of entropy in the kernel to make the /dev/random file operate faster. Usually, the /dev/random file operates slowly because by default this file only collects entropy from device drivers and other sources that may respond slowly as well. Hence, the scripts may stop responding when generating certificates in the FIPS mode if you don't install rng-tools.

Configure toolkit settings

Configure global parameters

Before working with the toolkit, you must define the global parameters in the set-params.bat or set-params.sh file. You need to change the default passwords before generating the certificates.

Parameter	Default Value	Override	Description
FIPS_MODE	false	No	Define whether to generate certificates for the FIPS mode or not. **Note:** When the value for this parameter is `false`, the keystore type is pkcs12 (*.p12). When the value is `true`, the keystore type is bcfks (*.bcfks).
CA_KEY_OPTS	-keyalg RSA -keysize 2048	No	The key algorithm for the private root CA. **Note:** `RSA` is more efficient when verifying the digital signatures.
CERT_KEY_OPTS	-keyalg EC -groupname secp256r1	No	The key algorithm for server or client certificates. **Note:** `EC` is more efficient when performing SSL handshakes.
DEFAULT_CA_CERT_VALIDITY	3650	Yes	The validity (days) of the root CA certificate.
DEFAULT_CERT_VALIDITY	1095	Yes	The validity (days) of server or client certificates.
CACERT_PASSWD	changeit	No	The password of the `cacerts` truststore from the JRE. **Note:** Change this password only if you have changed the default password of the `cacerts` truststore of the JRE.
CAROOT_PASSWD	caroot	No	The password of the private root CA keystore or truststore.
SERVER_KEYSTORE_PASSWD	serverkeystore	Yes	The password of server certificate keystores.
CLIENT_KEYSTORE_PASSWD	clientkeystore	Yes	The password of client certiticate keystores.
TRUSTEDCLIENTS_KEYSTORE_PASSWD	trustedclients	No	The password of the trusted clients truststore.

Parameter	Default Value	Override	Description
CA_ALIAS	smrootca	Yes	The alias of the private root CA certificate
SIGN_BY_OPENSSL	false	No	Certificate signing tool setting. `false` - use the `keytool` command to sign certificate signing requests. `true` - use the `openssl` command to sign certificate signing requests.

Notes:

- The **Override** column indicates whether the parameter can be overridden by the certificate definition attribute or by the command line option.
- If you want to use the `EC` algorithm for the private root CA for development or test purpose and have signed a server certificate using the keytool for the external server (such as Apache HTTP server) that's visible to the end users, Chrome may report the `ERR_CERT_INVALID` error for the server certificate and can't establish the https connection. To avoid this issue, you can either create a self-signed certificate for the external server or use OpenSSL to sign the server certificate.

Define certificates

You can define certificates you want to generate in the certs.conf file.

Organization

Define your organization information in the `[organization]` section by specifying the following attributes. The `[organization]` section must be the first in the certs.conf file and you must define at least one of these attributes.

Attribute Name	Description
organizationUnit	Your organization unit (for example, department or section).
organizationName	Your organization name (for example, company).
localityName	Your locality name (for example, city).
stateName	Your state or province.
country	Your country code in two letters (for example, GB).

Private root CA

Define Service Manager private root CA in the `[root_ca]` section by specifying the following attributes. The `[root_ca]` section must be the second in the certs.conf file.

Attribute Name	Mandatory	Description
alias	No	The alias of the private root CA. If you don't specify this attribute, the system uses the value of the `CA_ALIAS` global parameter instead.
commonName	Yes	The common name of the private root CA subject.
validity	No	The validity of the private root CA in days. If you don't specify this attribute, the system uses the value of the `DEFAULT_CA_CERT_VALIDITY` global parameter instead.

Server or client certificates

Define the server or client certificates in the `[server_cert]` or `[client_cert]` section by specifying the following attributes.

Attribute Name	Apply To	Mandatory	Description
alias	`[server_cert]` `[client_cert]`	Yes	The alias of the certificate. The alias must be unique among all certificates, and can't contain spaces or other chars that are illegal in file names.
commonName	`[server_cert]` `[client_cert]`	Yes	The common name of the certificate subject. You can use a descriptive name as the common name, or use the FQDN as the common name if you don't want to add the FQDN to the `subjectAltName` attribute.
subjectAltName	`[server_cert]` `[client_cert]`	No	The subject alternative name extension of the certificate. You can specify the value as `type:value(,type:value)*`, where the type can be DNS or IP. For example, `DNS:server1.yourdomain.net, DNS:server2.yourdomain.net`.
validity	`[server_cert]` `[client_cert]`	No	The validity of the certificate in days. If you don't specify this attribute, the toolkit uses the value of the `DEFAULT_CERT_VALIDITY` global parameter instead.

Attribute Name	Apply To	Mandatory	Description
storepass	[server_cert] [client_cert]	No	The password for the keystore that will be created to store the certificate. If you don't specify this attribute, the toolkit uses the value of the SERVER_KEYSTORE_PASSWD global parameter for the server certificate, and uses the value of the CLIENT_KEYSTORE_PASSWD global parameter for the client certificate.
exportKey	[server_cert] [client_cert]	No	Whether to export the private key to a PEM-encoded file
selfsign	[server_cert]	No	Define whether to create a self-signed server certificate. The possible values are yes and no, and the default value is no.
clientAuth	[server_cert]	No	Define whether to add clientAuth to the extended key usage. The possible values are yes and no, and the default value is no. The extended key usage of the server certificate is serverAuth by default. Adding clientAuth to the extended key usage indicates that server certificate can also be used as a client certificate

Attribute Name	Apply To	Mandatory	Description
serverAuth	[client_cert]	No	Define whether to add serverAuth to the extended key usage. The possible values are yes and no, and the default value is no. The extended key usage of the client certificate is clientAuth by default. Adding serverAuth to the extended key usage indicates that the client certificate can also be used as a server certificate.
overwrite	[server_cert] [client_cert]	No	Define whether to overwrite the existing certificate. The possible values are yes and no. This attribute takes precedence over the --overwrite command line option in the gen-all-certs.bat or the gen-all-certs.sh file.

Generate certificates

Follow these steps to generate a series of certificates by using the certificate generation toolkit:

1. Modify global parameters in the set-params.bat or set-params.sh file. Set the FIPS_MODE parameter as necessary, and then update the password parameters.
2. Define certificates in the certs.conf file according to your actual deployment.
3. Run **gen-all-certs.bat** on Windows or run the **./gen-all-certs.sh** command on Linux to generate certificates.
4. Copy related truststore and keystore to target locations, and then update the settings. For more information, see Configure two-way SSL.

Scripts in the toolkit

The certificate generation toolkit consists of the following scripts.

gen-all-certs.bat / gen-all-certs.sh

```
Usage: gen-all-certs.bat [options]
Generates all certificates defined in the certs.conf file.
Example: gen-all-certs.bat --overwrite no

Options:
   --overwrite {yes|no}    whether to overwrite existing certificates. The overwrite
                           attribute defined in the certs.conf will override this option
                           for individual certificate. If it's not present neither in
                           the certificate definition, nor in the commandline options,
                           you will be asked for a choice for the existing certificate.
                           The private root CA certificate ignores this option, if you
                           want to regenerate it either, add the --clean option.
   --clean                 clean all certificates before generates new ones
   --printcert             print detailed info of the server or client certificates
   --help                  print this help message
```

Example 1: Generate all certificates for the first time

- Windows

  ```
  gen-all-certs.bat
  ```

- Linux

  ```
  ./gen-all-certs.sh
  ```

Example 2: Generate new certificates and ignore the existing ones

- Windows

  ```
  gen-all-certs.bat --overwrite no
  ```

- Linux

  ```
  ./gen-all-certs.sh --overwrite no
  ```

Example 3: Generate new certificates and overwrite the existing ones

- Windows

  ```
  gen-all-certs.bat --overwrite yes
  ```

- Linux

  ```
  ./gen-all-certs.sh --overwrite yes
  ```

Example 4: Re-generate the smwebtier certificate

First, add the `overwrite` attribute to the web tier certificate definition in the certs.conf file.

```
# Client certificate for the webtier.
[client_cert]
alias = smwebtier
commonName = Service Manager Webtier
subjectAltName = DNS:smwebtier.yourdomain.net
overwrite = yes
```

Next, run **gen-all-certs.bat(sh)** with the **--overwrite no** command line option to re-generate the smwebtier certificate only.

- Windows

  ```
  gen-all-certs.bat --overwrite no
  ```

- Linux

  ```
  ./gen-all-certs.sh --overwrite no
  ```

gen-root-ca.bat / gen-root-ca.sh

Use these two scripts to generate a self-signed private CA root. You need to run these scripts before generating a server or client certificate.

```
Usage: gen-root-ca.bat [options]
Generates a self-signed private root CA certificate. This private root CA certificate will be used
to sign certificates generated for the server or client applications.
Example: gen-root-ca.bat --dname "CN=Service Manager Private Root CA, OU=ITOM, O=OpenText, C=CA"

Options:
   --alias {alias}         the alias of the private root CA certificate
                           (defaults to smrootca, see set-params.bat)
   --dname {dname}         X.500 Distinguished Names to specify the subject and issuer
                           of the certificate. Following subparts are supported:
                             CN=commonName
                             OU=organizationUnit
                             O=organizationName
                             L=localityName
                             S=stateName
                             C=country
                           you must specify them in the order above. However, you can
                           use a subset, for example:
                             CN=Service Manager Private Root CA, OU=ITOM, O=OpenText, C=CA
                           if omitted, you will be prompted for subparts.
   --validity {days}       the validity in days of the private root CA certificate
                           (defaults to 3650, see set-params.bat)
   --clean                 clean all certificates before generates new ones
   --help                  print this help message
```

Example: Generate a private CA certificate

- Windows

  ```
  gen-root-ca.bat --alias smrootca --dname "CN=Service Manager Private Root CA, OU=ITOM, O=OpenText,
  C=CA"
  ```

- Linux

```
./gen-root-ca.sh --alias smrootca --dname "CN=Service Manager Private Root CA, OU=ITOM, O=OpenText,
C=CA"
```

gen-server-cert.bat / gen-server-cert.sh

```
Usage: gen-server-cert.bat [options]
Generates a certificate for the server application and sign it with the private root CA certificate.
Example: gen-server-cert.bat --alias smserver --dname "CN=Service Manager Server, OU=ITOM, O=OpenText,
C=CA"

Options:
   --alias {alias}         the alias of the server certificate (required)
   --dname {dname}         X.500 Distinguished Names to specify the subject of the
                           certificate. Following subparts are supported:
                             CN=commonName
                             OU=organizationUnit
                             O=organizationName
                             L=localityName
                             S=stateName
                             C=country
                           you must specify them in the order above. However, you can
                           use a subset, for example:
                             CN=Service Manager Server, OU=ITOM, O=OpenText, C=CA
                           if omitted, you will be prompted for subparts.
   --san {san}             the Subject Alternative Name extension of the certificate.
                           the value should be specified as type:value(,type:value)*,
                           where the type can be DNS or IP. For example:
                             DNS:server1.yourdomain.net,DNS:server2.yourdomain.net
                           if omitted, you will be prompted for one.
   --storepass {password}  the password used to protect the server keystore
                           (defaults to serverkeystore, see set-params.bat)
   --caalias {alias}       the alias of the private root CA certificate
                           (defaults to smrootca, see set-params.bat)
   --validity {days}       the validity in days of the private root CA certificate
                           (defaults to 1095, see set-params.bat)
   --clientauth            add clientAuth to the extended key usage as well
                           the extended key usage is serverAuth by default, adding
                           clientAuth to it means the certificate can also be used
                           as a client certificate
   --selfsign              generates a self-signed certificate only
   --exportkey             export the private key to a PEM-encoded file
   --overwrite {yes|no}    overwrite the existing certificate
                           if omitted, you will be asked for existing certificate
   --help                  print this help message
```

Example 1: Generate a server CA certificate that's used by two server nodes

- Windows

  ```
  gen-server-cert.bat --alias smserver --dname "CN=Service Manager Server, OU=ITOM, O=OpenText, C=CA"
  --san "DNS:smserver1.yourdomain.net,DNS:smserver2.yourdomain.net"
  ```

- Linux

  ```
  ./gen-server-cert.sh --alias smserver --dname "CN=Service Manager Server, OU=ITOM, O=OpenText,
  C=CA" --san "DNS:smserver1.yourdomain.net,DNS:smserver2.yourdomain.net"
  ```

Example 2: Generate a server CA certificate that's signed by a specific CA

- Windows

```
gen-server-cert.bat --alias smserver --dname "CN=Service Manager Server, OU=ITOM, O=OpenText, C=CA"
--san "DNS:smserver.yourdomain.net" --caalias myrootca
```

- Linux

```
./gen-server-cert.sh --alias smserver --dname "CN=Service Manager Server, OU=ITOM, O=OpenText,
C=CA" --san "DNS:smserver.yourdomain.net" --caalias myrootca
```

Example 3: Generate a self-signed server CA certificate for the Apache HTTP Server and export the private key

- Windows

```
gen-server-cert.bat --alias apache --dname "CN=Service Manager HTTP Server, OU=ITOM, O=OpenText,
C=CA" --san "DNS:apache.yourdomain.net" --selfsign --exportkey
```

- Linux

```
./gen-server-cert.sh --alias apache --dname "CN=Service Manager HTTP Server, OU=ITOM, O=OpenText,
C=CA" --san "DNS:apache.yourdomain.net" --selfsign --exportkey
```

gen-client-cert.bat / gen-client-cert.sh

```
Usage: gen-client-cert.bat [options]
Generates a certificate for the client application and sign it with the private root CA certificate.
The client certificate will also be imported into the trustedclients.keystore.
Example: gen-client-cert.bat --alias smwebtier --dname "CN=Service Manager Webtier, OU=ITOM, O=OpenText,
C=CA"

Options:
   --alias {alias}         the alias of the client certificate (required)
   --dname {dname}         X.500 Distinguished Names to specify the subject of the
                           certificate. Following subparts are supported:
                             CN=commonName
                             OU=organizationUnit
                             O=organizationName
                             L=localityName
                             S=stateName
                             C=country
                           you must specify them in the order above. However, you can
                           use a subset, for example:
                             CN=Service Manager Webtier, OU=ITOM, O=OpenText, C=CA
                           if omitted, you will be prompted for subparts
   --san {san}             the Subject Alternative Name extension of the certificate.
                           the value should be specified as type:value(,type:value)*,
                           where the type can be DNS or IP. For example:
                             DNS:smwebtier.yourdomain.net
   --storepass {password}  the password used to protect the client keystore
                           (defaults to clientkeystore, see set-params.bat)
   --caalias {alias}       the alias of the private root CA certificate
                           (defaults to smrootca, see set-params.bat)
   --validity {days}       the validity in days of the private root CA certificate
                           (defaults to 1095, see set-params.bat)
   --serverauth            add serverAuth to the extended key usage as well
                           the extended key usage is clientAuth by default, adding
                           serverAuth to it means the certificate can also be used
                           as a server certificate
   --exportkey             export the private key to a PEM-encoded file
   --overwrite {yes|no}    overwrite the existing certificate
                           if omitted, you will be asked for existing certificate
   --help                  print this help message
```

Example 1: Generate a client certificate for the web tier that's deployed on two worker nodes

- Windows

```
gen-client-cert.bat --alias smwebtier --dname "CN=Service Manager Webtier, OU=ITOM, O=OpenText,
C=CA" --san "DNS:smwebtier1.yourdomain.net,DNS:smserver2.yourdomain.net"
```

- Linux

```
./gen-client-cert.sh --alias smwebtier --dname "CN=Service Manager Webtier, OU=ITOM, O=OpenText,
C=CA" --san "DNS:smwebter1.yourdomain.net,DNS:smserver2.yourdomain.net"
```

import-cert.bat / import-cert.sh

```
Usage: import-cert.bat [options]
Import a CA certificate to a truststore. This script is a simple wrapper around the keytool -importcert
command.
Example: import-cert.bat --alias corprootca --file certs/corprootca.pem --keystore certs/corprootca.p12 -
-storepass caroot

Options:
   --alias {alias}          the alias of the CA certificate (required)
   --file {file}            the PEM-encoded X.509 certificate file (required)
   --keystore {keystore}    the keystore (required)
   --storepass {password}   the password used to protect the keystore (required)
   --storetype {storetype}  the type of the keystore
                            (jks, pkcs12 or bcfks, defaults to pkcs12)
   --help                   print this help message
```

Example: Import the corporate root CA certificate to a BCFKS truststore

- Windows

  ```
  import-cert.bat --alias corprootca --file certs/corprootca.pem --keystore certs/corprootca.bcfks --
  storetype bcfks --storepass caroot
  ```

- Linux

  ```
  ./import-cert.sh --alias corprootca --file certs/corprootca.pem --keystore certs/corprootca.bcfks -
  -storetype bcfks --storepass caroot
  ```

convert-keystore.bat / convert-keystore.sh

```
Usage: convert-keystore.bat [options]
Converts keystore type. Supported store types are jks, pkcs12 and bcfks. This script is a simple wrapper
around the keytool -importkeystore command.
Example: convert-keystore.bat --srckeystore certs/smwebtier.p12 --srcstorepass clientkeystore --
destkeystore certs/smwebtier.bcfks --deststoretype bcfks

Options:
   --srckeystore {keystore}        the source keystore (required)
   --srcstorepass {password}       the password used to protect the source keystore (required)
   --srcstoretype {storetype}      the type of the source keystore (defaults to pkcs12)
   --destkeystore {keystore}       the destination keystore (required)
   --deststorepass {password}      the password used to protect the source keystore
                                   if not specified, the srcstorepass is used
   --deststoretype {storetype}     the type of the source keystore (defaults to pkcs12)
   --trusted                       whether the srckeystore contains trusted certificates
   --help                          print this help message
```

Example 1: Convert the smwebtier keystore from JKS to PKCS12

- Windows

  ```
  convert-keystore.bat --srckeystore certs/smwebtier.keystore --srcstoretype jks --srcstorepass
  clientkeystore --destkeystore certs/smwebtier.p12
  ```

- Linux

```
./convert-keystore.sh --srckeystore certs/smwebtier.keystore --srcstoretype jks --srcstorepass
clientkeystore --destkeystore certs/smwebtier.p12
```

Example 2: Convert the smwebtier keystore from PKCS12 to BCFKS

- Windows

```
convert-keystore.bat --srckeystore certs/smwebtier.p12 --srcstorepass clientkeystore --destkeystore
certs/smwebtier.bcfks --deststoretype bcfks
```

- Linux

```
./convert-keystore.sh --srckeystore certs/smwebtier.p12 --srcstorepass clientkeystore --
destkeystore certs/smwebtier.bcfks --deststoretype bcfks
```

Example 3: Convert the smwebtier keystore from BCFKS to PKCS12

- Windows

```
convert-keystore.bat --srckeystore certs/smwebtier.bcfks --srcstorepass clientkeystore --
srcstoretype bcfks --destkeystore certs/smwebtier.p12
```

- Linux

```
./convert-keystore.sh --srckeystore certs/smwebtier.bcfks --srcstorepass clientkeystore --
srcstoretype bcfks --destkeystore certs/smwebtier.p12
```

Example 4: Merge the truststore corprootca.bcfks to the truststore smrootca.bcfks

- Windows

```
convert-keystore.bat --trusted --srckeystore certs/corprootca.bcfks --srcstorepass caroot --
srcstoretype bcfks --destkeystore certs/smrootca.bcfks --deststoretype bcfks --deststorepass caroot
```

- Linux

```
./convert-keystore.sh --trusted --srckeystore certs/corprootca.bcfks --srcstorepass caroot --
srcstoretype bcfks --destkeystore certs/smrootca.bcfks --deststoretype bcfks --deststorepass caroot
```

Export the certificate and the private key from a keystore

You can use the `openssl` utility to export the certificate and the private key from a keystore. For example:

- Windows

```
openssl\openssl.exe pkcs12 -in certs\myserver.p12 -passin pass:serverkeystore -nocerts -nodes -out
certs\myserver.key
openssl\openssl.exe pkcs12 -in certs\myserver.p12 -passin pass:serverkeystore -nokeys -out
certs\myserver.crt
```

- Linux

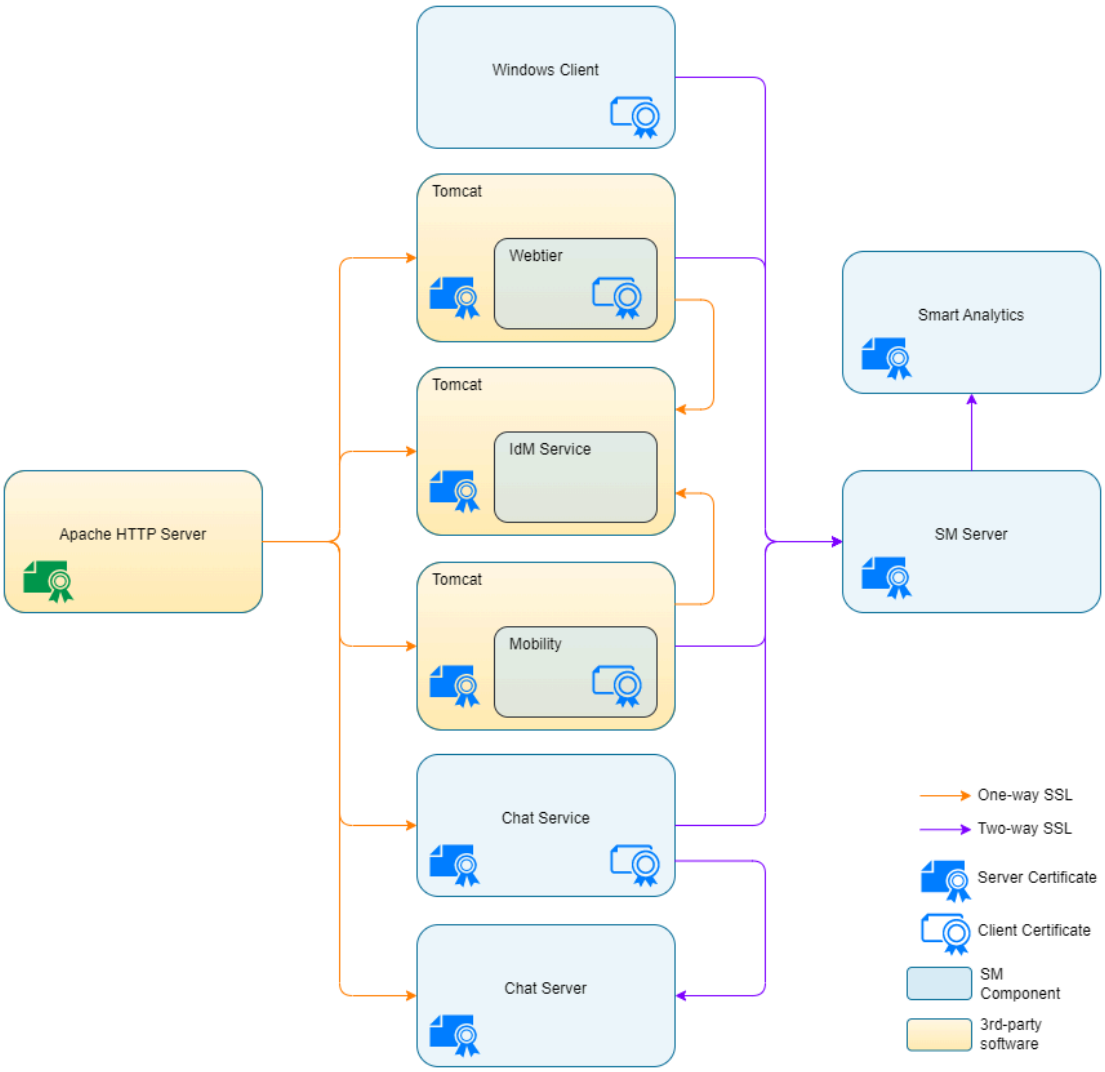
```
openssl pkcs12 -in certs/myserver.p12 -passin pass:serverkeystore -nocerts -nodes -out
certs/myserver.key
openssl pkcs12 -in certs/myserver.p12 -passin pass:serverkeystore -nokeys -out certs/myserver.crt
```

8.12.2. Configure two-way SSL

Two-way SSL is the prerequisite of Trusted sign-on (TSO) and Common Access Card (CAC) sign-on. The scope of two-way SSL is limited to the communication between the Service Manager server and the Service Manager clients (for example, web tier, mobility, Windows client, and chat service). Other communications are one-way SSL because the client certificate authentication is unnecessary.

Typical Service Manager deployment with SSL

The diagram below illustrates a typical SM deployment that uses the Apache HTTP Server as the reverse proxy and Tomcat as the web container. Certificates needed in the SSL configuration are illustrated in the diagram as well.



In this diagram:

- The Apache HTTP Server is visible to end users, and you must use a server certificate issued by the certificate authority of your company in the production environment. In the development and test environment, you can use a server certificate signed by the Service Manager private CA.
- Though the AJP protocol is more efficient than the HTTPS protocol in the integration between the Apache HTTP Server and Tomcat, the transferred data is not encrypted. You can use the AJP protocol only if the Apache HTTP Server and Tomcat are deployed in a trusted and secured environment. When using the AJP protocol, Tomcat doesn't need the server certificate.
- If you use the IBM HTTP Server (IHS) and the WebSphere Application Server (WAS) combination, the certificate generation toolkit can't generate server certificates used by IHS and WAS. Refer to the IBM documentation on how to

configure SSL in IHS and WAS.

- If you use a hardware load balancer as the reverse proxy, you should refer to the manufacturer's documentation on how to configure SSL in the hardware load balancer.

See the following table for a complete list of certificates. You may not deploy all Service Manager components in your environment, and you can create certificates that are necessary for your deployment only.

Alias	Common Name	Certificate type	Note
smserver	Service Manager Server	server	The extended key usage contains `clientAuth` as well as `serverAuth`.
smartanalytics	Service Manager Smart Analytics	server	The extended key usage contains `clientAuth` as well as `serverAuth`. The certificate and the private key will be exported to PEM-encoded files.
smchatserver	Service Manager Chat Server	server	
smchatservice	Service Manager Chat Service	server	
smchatserviceclient	Service Manager Chat Service Client	client	
idmservice	Service Manager IdM Service	server	
tomcat	Service Manager Webtier Tomcat	server	
smwebtier	Service Manager Webtier	client	
mobilitytomcat	Service Manager Mobility Tomcat	server	
smmobility	Service Manager Mobility	client	
apache	Service Manager HTTP Server	server	Use this certificate in the development and test environment only. The certificate and the private key will be exported to PEM-encoded files.
<domain user name>	<Full Name>	client	Certificate per user for the Windows client in TSO mode.

In this table:

- You can change the alias and the common name, and then use a related keystore when configuring SSL later.
- You must change the DNS name of each certificate according to your deployment.

- You can remove certificate definitions of components that you don't want to install.

In addition to the certificates listed above, the toolkit copies or generates the following truststore files in the `certs` folder.

Truststore	Comments
cacerts	The truststore copied from JRE, with the private CA certificate added.
smrootca.p12	The truststore contains the private CA certificate only.
smrootca_key.p12	The keystore contains the private key of the private CA.
smrootca.crt	The PEM-encoded private CA certificate. It can be imported to the system certificate store or the browser's certificate store.
smrootca.key	The PEM-encoded private key of the private CA.

Requirements for required SSL encryption and client authentication

This configuration is intended for customers who:

- Want to require SSL encryption for all connections
- Want to protect against complex SSL-related attacks
- Want to authenticate that the Service Manager server is a valid host
- Want to authenticate that the Service Manager clients are valid hosts

Certificates required

You must create or obtain the following certificates for SSL encryption.

- Certificate authority certificate *****
- Keystore containing the certificate authority certificate *****
- Service Manager server certificate
- Web tier certificate
- Windows client certificates

***** A typical SSL configuration uses a single certificate authority to issue certificates for all authenticated components. If, however, you use multiple certificate authorities to sign your certificates, then you need to obtain a certificate for each certificate authority.

Private keys required

You must create or obtain the following private keys for SSL encryption.

- Certificate authority's private key *****
- Service Manager server's private key

***** This key is only necessary if you are managing your own private certificate authority.

Parameters required in the server configuration file (sm.ini)

- keystoreFile - identify the keystore file containing the Service Manager server's certificate and private key
- keystorePass - identify the password to the keystore file containing the Service Manager server's certificate and private key
- ssl:1
- ssl_reqClientAuth:1
- sslConnector:1
- truststoreFile - identify the keystore file containing the certificate authority's certificate
- truststorePass - identify the password to the keystore file containing the certificate authority's certificate

Parameters required in the web tier configuration file (web.xml)

You can optionally set the customize-folder parameter in web.xml so that the <Customize-Folder>/config/webtier.properties file can be shared by multiple web tier instances on the same host.

- customize-folder - specify the absolute path to a folder on the web tier host in which the webtier.properties file is located

Parameter required in the <Customize-Folder>/config/webtier.properties or <WEBTIER>/WEB-INF/webtier.properties file

You must set the following web parameter:

- ssl:true
- cacerts - identify the truststore file containing the certificate authority that signed the server's certificate
- cacertsPassword - identify the password for the truststore
- keystore - identify the keystore containing the web tier's client certificate
- keystorePassword - identify the password for the web tier's keystore

For more information, see Encryption of client keystore passwords.

Windows client preferences required

You must set the following preferences from the **Window** > **Preferences** > **Service Manager** > **Security** menu.

- CA certificates file - identify the truststore file containing the certificate authority that signed the server's certificate
- CA certificates password - identify the password for the truststore
- Keystore file - identify the keystore containing the Windows client's certificate
- Keystore password - identify the password for the Windows client's keystore

Other requirements

You must do the following additional steps to ensure that Service Manager can use your private certificates.

- Add your private certificate authority's certificate to a keystore that your Web and Windows clients can access
- Ensure that the Service Manager client's host name matches the common name (CN) listed in the client's signed certificate
- Ensure that the Service Manager server's host name matches the common name (CN) listed in the server's signed certificate

Requirements for required SSL encryption and trusted clients

This configuration is intended for customers who:

- Want to require SSL encryption for all connections
- Want to protect against complex SSL-related attacks
- Want to authenticate that the Service Manager server is a valid host
- Want to authenticate that the Service Manager clients are valid hosts
- Want to restrict access to the server to only those clients whose certificates are in a list of trusted clients

Certificates required

You must create or obtain the following certificates for SSL encryption.

- Certificate authority certificate *****
- Certificate list containing the certificates of web and Windows clients that are allowed to connect to the server
- Keystore containing the certificate authority certificate *****
- Service Manager server certificate
- Web tier certificate
- Windows client certificates

***** A typical SSL configuration uses a single certificate authority to issue certificates for all authenticated components. If, however, you use multiple certificate authorities to sign your certificates, then you need to obtain a certificate for each certificate authority.

Private keys required

You must create or obtain the following private keys for SSL encryption.

- Certificate authority's private key *****
- Service Manager server's private key

***** This key is only necessary if you are managing your own private certificate authority.

Parameters required in the server configuration file (sm.ini)

- keystoreFile - identify the keystore file containing the Service Manager server's certificate and private key
- keystorePass - identify the password to the keystore file containing the Service Manager server's certificate and private key
- ssl:1
- ssl_reqClientAuth:2
- ssl_trustedClientsJKS - identify the keystore file containing the list of trusted client certificates allowed to connect to the server
- ssl_trustedClientsPwd - identify the password to the keystore file containing the list of trusted client certificates
- sslConnector:1
- truststoreFile - identify the keystore file containing the certificate authority's certificate
- truststorePass - identify the password to the keystore file containing the certificate authority's certificate

Parameters required in the web tier configuration file (web.xml)

Same as the corresponding section above.

Parameter required in the <Customize-Folder>/config/webtier.properties or <WEBTIER>/WEB-INF/webtier.properties file

Same as the corresponding section above.

Windows client preferences required

Same as the corresponding section above.

Other requirements

Same as the corresponding section above.

Configure two-way SSL

To configure two-way SSL, you need to generate the certificates first. Next, copy related truststore and keystore to target locations as described below, and then update the settings.

Notes:

- In the FIPS mode, change the keystore type to `BCFKS` and then change the file extension to `.bcfks` in the following sample settings.
- The following examples use default passwords for demo purposes. You must update these passwords to your own ones. Similarly, the FQDNs in the following examples are for demo purposes as well. You must update these FQDNs based on the actual deployment.
- Restart the related services after you update the settings.

Service Manager server

Copy the cacerts, smserver.p12, and trustedclients.p12 files to the <SM_SERVER_HOME>/RUN folder, and then edit <SM_SERVER_HOME>/RUN/sm.ini as follows.

```
sslConnector:1
ssl:1
ssl_reqClientAuth:2
sslEnabledProtocols:TLSv1.2,TLSv1.3
ciphers:TLS_AES_256_GCM_SHA384,TLS_AES_128_GCM_SHA256,TLS_DHE_RSA_WITH_AES_256_GCM_SHA384,TLS_ECDHE_RSA_W
ITH_AES_256_GCM_SHA384,TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256,TLS_ECDHE_ECDSA_WITH_AES_256_GCM_SHA384,TLS_
ECDHE_ECDSA_WITH_AES_128_GCM_SHA256

keystoreFile:smserver.p12
keystorePass:serverkeystore
truststoreFile:cacerts
truststorePass:changeit
ssl_trustedClientsJKS:trustedclients.p12
ssl_trustedClientsPwd:trustedclients
```

Note: If you don't want to restrict access to the server to only those clients whose certificates are in a list of trusted clients, set the `ssl_reqClientAuth` parameter to 1, and you don't need to configure the `ssl_trustedClientsJKS` and `ssl_trustedClientsPwd` parameters.

Chat server

Copy the smrootca.p12 and smchatserver.p12 files to the <CHAT_SERVER_HOME>/resources/security folder, and then edit <CHAT_SERVER_HOME>/conf/openfire.xml as follows.

```
<securityConfig>
  <ssl>
     <keyStoreType>PKCS12</keyStoreType>
     <keyStorePath>resources/security/smchatserver.p12</keyStorePath>
     <keyStorePassword>serverkeystore</keyStorePassword>
     <trustStorePath>resources/security/smrootca.p12</trustStorePath>
     <trustStorePassword>caroot</trustStorePassword>
     <clientTrustStorePath>resources/security/smrootca.p12</clientTrustStorePath>
     <clientTrustStorePassword>caroot</clientTrustStorePassword>
  </ssl>
  ...
</securityConfig>
```

Sign in to the Openfire Administration Console. Click **Server > Server Settings > Client Connections**. In the **Plain-text (with STARTTLS) connections** section, activate the **Advanced Configuration...** link. Set STARTTLS policy to **Required**, and then set Mutual Authentication to **Needed**.

Chat service

Copy the smrootca.p12, smchatservice.p12, and smchatsericeclient.p12 files to the <CHAT_SERVICE_HOME>/conf folder, and then copy the app.properties and config.yml files from the <CHAT_SERVICE_HOME>/conf/samples/https folder to the <CHAT_SERVICE_HOME>/conf folder. Edit the app.properties file as follows.

```
app.clientKeyStoreType=PKCS12
app.clientKeyStorePath=../conf/smchatserviceclient.p12
app.clientKeyStorePassword=clientkeystore

app.keyStoreType=PKCS12
app.keyStorePath=../conf/smchatservice.p12
app.keyStorePassword=serverkeystore

app.trustStoreType=PKCS12
app.trustStorePath=../conf/smrootca.p12
app.trustStorePassword=caroot
```

IdM admin

Copy the IdM client truststore to the `WEB-INF\classes\conf` folder. Edit the `WEB-INF\classes\conf\IdmConfig.xml` file, set the `trustAll` attribute to `false`, and then configure the client truststore. For example:

```
<client trustAll="false" ...>
  ...
  <trustStoreFile>WEB-INF/classes/conf/smrootca.p12</trustStoreFile>
  <trustStoreType>PKCS12</trustStoreType>
  <trustStorePass>your encrypted password for the client truststore</trustStorePass>
  ...
</client>
```

IdM service

Copy the idmservice.p12 file to the <TOMCAT_HOME>/conf folder, and then edit the <TOMCAT_HOME>/conf/server.xml file as follows.

```
...
<Connector port="8443" protocol="org.apache.coyote.http11.Http11NioProtocol"
           maxThreads="150" SSLEnabled="true"
           proxyName="apache.yourdomain.net" proxyPort="443">
    <SSLHostConfig>
        <Certificate certificateKeystoreFile="conf/idmservice.p12"
                     certificateKeystorePassword="serverkeystore"
                     certificateKeystoreType="PKCS12" />
    </SSLHostConfig>
</Connector>
...
```

In this example, `proxyName` is the FQDN of the Apache HTTP Server, and `proxyPort` is the SSL port of the Apache HTTP Server.

If you want to use the `AJP` protocol between the Apache HTTP Server and Tomcat, enable the AJP connector in the <TOMCAT_HOME>/conf/server.xml file as follows.

```
...
<Connector protocol="AJP/1.3"
           secret="MyT0mcAt"
           tomcatAuthentication="false"
           address="0.0.0.0"
           port="8009"
           redirectPort="8443" />
...
```

You must change `secret` to your own secret.

Web tier

Copy the smrootca.p12 and smwebtier.p12 files to <Customize-Folder>/config folder if the `customize-folder` parameter is set in the web.xml, and then append the following scripts to the <Customize-Folder>/config /webtier.properties file.

```
ssl=true
keystore=<Customize-Folder>/config/smwebtier.p12
keystorePassword=clientkeystore
cacerts=<Customize-Folder>/config/smrootca.p12
cacertsPassword=caroot
cookie.secure=true
```

Copy the smrootca.p12 and smwebtier.p12 files to <WEBTIER>/WEB-INF folder if the `customize-folder` parameter is not set in the web.xml, and then append the following scripts to the <WEBTIER>/WEB-INF/webtier.properties file.

```
ssl=true
keystore=/WEB-INF/smwebtier.p12
keystorePassword=clientkeystore
cacerts=/WEB-INF/smrootca.p12
cacertsPassword=caroot
cookie.secure=true
```

Mobility

Copy the smrootca.p12 and smmobility.p12 files to the <MOBILITY>/WEB-INF folder, and then edit the <MOBILITY>/WEB-INF/web.properties file as follows.

```
endpoint=https://smserver.yourdomain.net:13443/SM/ui
cacerts=/WEB-INF/smrootca.p12
cacertsPassword=caroot
keystore=/WEB-INF/smmobility.p12
keystorePassword=clientkeystore
```

Apache HTTP Server

The apache SSL config file differs depending on different platforms:

- **Windows**
  The apache SSL config file is `<APACHE24_HOME>/conf/extra/httpd-ssl.conf`, the `<APACHE24_HOME>` refers to the install directory of the Apache HTTP Server.
  You need to load necessary modules and include the `httpd-ssl.conf` file in the `<APACHE24_HOME>\conf\httpd.conf` file first, for example:

  ```
  ...
  LoadModule socache_shmcb_module modules/mod_socache_shmcb.so
  ...
  LoadModule ssl_module modules/mod_ssl.so
  ...
  # Secure (SSL/TLS) connections
  Include conf/extra/httpd-ssl.conf
  ...
  ```

- **Linux**
  The apache SSL config file is `<APACHE24_HOME>/conf.d/ssl.conf`, the `<APACHE24_HOME>` refers to the `/etc/httpd` directory.

Copy the smrootca.crt, apache.crt, and apache.key files to the <APACHE24_HOME>/conf/ folder, and then edit the apache SSL config file as follows.

```
<VirtualHost _default_:443>
  SSLEngine on
  ServerName apache.yourdomain.net:443
  SSLCertificateFile "${ServerRoot}/conf/ssl/apache.crt"
  SSLCertificateKeyFile "${ServerRoot}/conf/ssl/apache.key"
  SSLCACertificateFile "${ServerRoot}/conf/ssl/smrootca.crt"
  SSLProxyEngine on
  SSLProxyCACertificateFile "${ServerRoot}/conf/ssl/smrootca.crt"
  ...
</virtualhost>
```

Note: The `SSLCACertificateFile` directive is only needed if you set up the Common Access Card (CAC) sign-on in the web tier. It needs to be the root CA certificate of your organization in the production environment. In the test and development environment, you need to use `smrootca.crt` if the end-users' certificate is issued by the Service Manager private CA.

It's strongly recommended that you set secure TLS protocols and cipher suites in the apache SSL config file. For example:

```
SSLProtocol all -TLSv1.1 -TLSv1 -SSLv3
SSLProxyProtocol all -TLSv1.1 -TLSv1 -SSLv3
SSLCipherSuite ECDHE:!SHA1
SSLProxyCipherSuite ECDHE:!SHA1
```

Note:

- You can use the `openssl ciphers -v` command to view effective cipher suites of the given cipher-spec string.
  For example:
  Windows:

  `<APACHE24_HOME>\bin\openssl ciphers -v "ECDHE:!SHA1"`

  Linux:

  `openssl ciphers -v 'ECDHE:!SHA1'`

- You can also set explicit cipher suites instead of aliases.
  For example:

  ```
  SSLCipherSuite TLS_AES_256_GCM_SHA384:TLS_AES_128_GCM_SHA256:ECDHE-ECDSA-AES256-GCM-SHA384:ECDHE-
  RSA-AES256-GCM-SHA384
  ```

Depending on the protocol you want to use between the Apache HTTP Server and Tomcat, you need to add some additional settings to the apache SSL config file.

- **AJP**
  Use the `ajp://` prefix when defining the URL of the `ProxyPass` or `BalancerMember` directive. For example:

  ```
  # If the backend is a tomcat cluster
  <Proxy "balancer://webtierCluster">
    BalancerMember "ajp://smwebtier.yourdomain.net:8009" secret=MyT0mcAt route=1.1 timeout=300
    BalancerMember "ajp://smwebtier.yourdomain.net:8019" secret=MyT0mcAt route=1.2 timeout=300
    ProxySet lbmethod=bybusyness stickysession=JSESSIONID nofailover=On
  </Proxy>
  ProxyPass /webtier balancer://webtierCluster/webtier

  # If the backend is a single tomcat instance
  ProxyPass /mobility ajp://smmobility.yourdomain.net:8009/mobility secret=MyT0mcAt retry=0
  ```

- **HTTPS**

Use the `https://` prefix when defining the URL of the `ProxyPass` or `BalancerMember` directive. For example:

```
# If the backend is a tomcat cluster
<Proxy "balancer://webtierCluster>
  BalancerMember "https://smwebtier.yourdomain.net:8443" route=1.1 timeout=300
  BalancerMember "https://smwebtier.yourdomain.net:8453" route=1.2 timeout=300
  ProxySet lbmethod=bybusyness stickysession=JSESSIONID nofailover=On
</Proxy>
ProxyPass /webtier balancer://webtierCluster/webtier
<Location "/webtier">
  RequestHeader set X-Forwarded-Host expr=%{HTTP_HOST}
  RequestHeader set X-Forwarded-Proto expr=%{REQUEST_SCHEME}
</Location>

# If the backend is a single tomcat instance
ProxyPass /mobility https://smmobility.yourdomain.net:8443/mobility retry=0
```

Note: For the webtier, you need either set the X-Forwarded-* HTTP request headers as above, or set the webtierFullUrl parameter in the webtier.properties file. The webtierFullUrl parameter takes precedence over the X-Forwarded-* headers.

Note:

o The secret value must be the same as what defined for the AJP connector in the Tomcat configuration.

o If the backend is a single tomcat instance, use the simple ProxyPass directive.

o If the backend is a tomcat cluster, use the loading balancer configuration. Note that the route value must be the same as the jvmRoute attribute defined in the Tomcat configuration.

Tomcat

If you want to launch multiple Tomcat instances to form a cluster, you need to set the `jvmRoute` attribute in the `<Engine>` element in the <TOMCAT_HOME>/conf/server.xml file. The value must be unique among all Tomcat instances. For example:

```
<Engine name="Catalina" defaultHost="localhost" jvmRoute="1.1">
  ...
</Engine>
```

Depending on the protocol you want to use between the Apache HTTP Server and Tomcat, you can use either of the following settings.

- **AJP**
  Edit the <TOMCAT_HOME>/conf/server.xml file as follows.

  ```
  ...
  <Connector protocol="AJP/1.3"
             secret="MyT0mcAt"
             tomcatAuthentication="false"
             address="0.0.0.0"
             port="8009"
             redirectPort="8443" />
  ...
  ```

  You must update the value of `secret` to your actual secret value.
- **HTTPS**
  Copy the Tomcat certificate keystore file for the SM components (such as mobilitytomcat.p12 or tomcat.p12) to the <TOMCAT_HOME>/conf folder, and then edit the <TOMCAT_HOME>/conf/server.xml file as follows.

```
...
<Connector port="8443" protocol="org.apache.coyote.http11.Http11NioProtocol"
           maxThreads="150" SSLEnabled="true">
    <SSLHostConfig>
        <Certificate certificateKeystoreFile="conf/tomcat.p12"
                     certificateKeystorePassword="serverkeystore"
                     certificateKeystoreType="PKCS12" />
    </SSLHostConfig>
</Connector>
...
```

WebSphere Application Server Liberty

Check whether the ${wlp.user.dir}/servers/serverName/resources/security folder exists or not. If not, create this folder and then copy the WebSphere Application Server Liberty certificate keystore file for the SM components (such as mobilitywasliberty.p12 or wasliberty.p12) to it. Next, refer to the following example to update the ${wlp.user.dir}/servers/serverName/server.xml file.

```
<!-- Enable features -->
    <featureManager>
              <feature>servlet-6.0</feature>
              <feature>ssl-1.0</feature>
              <feature>pages-3.1</feature>
    </featureManager>

<httpSession idLength="128" useContextRootAsCookiePath="true"/>

<!-- To access this server from a remote client add a host attribute to the following element, e.g.
host="*" -->
<httpEndpoint id="defaultHttpEndpoint"
                  enabled="true"
                  host="*"
                  httpPort="9080"
                  httpsPort="9443" />

<!-- default SSL configuration is defaultSSLSettings -->
    <sslDefault sslRef="defaultSSLSettings"/>
    <ssl id="defaultSSLSettings"
         keyStoreRef="defaultKeyStore"/>
    <keyStore id="defaultKeyStore"
              location="wasliberty.p12"
              password="serverkeystore"
                     type=" PKCS12"/>
```

The example above only specifies the default SSL certificate, keystore, and configuration in Liberty. For more detailed instructions about how to enable SSL communication in Liberty, see Enabling SSL communication in Liberty.

JBoss Web Server

Refer to all configurations as described in the *Tomcat* section on this page.

Smart Analytics

Refer to Configure TSL/SSL for two-way authentication in the Smart Analytics administrator tasks documentation to set up a TLS/SSL connection between Service Manager and the Smart Analytics main server, the Connector Framework Server (CFS)/connectors, or the Media Server.

Windows Client

Copy the smrootca.p12 and <username>.p12 files to the <WINDOWS_CLIENT_HOME>\configuration folder, and then launch the Windows client. Click **Window > Preferences > Service Manager > Security**, and then edit the settings as

follows.

- **CA certificates file:** <WINDOWS_CLIENT_HOME>\configuration\smrootca.p12
- **CA certificates password:** caroot
- **Client keystore file:** <WINDOWS_CLIENT_HOME>\configuration\<username>.p12
- **Client keystore password:** clientkeystore

8.12.3. Configure one-way SSL

If you don't set up Trusted sign-on (TSO) or Common Access Card (CAC) sign-on, the two-way SSL is not mandatory. You can use one-way SSL to simplify the configuration.

Requirements for required SSL encryption

This configuration is intended for customers who:

- Want to require SSL encryption for all connections
- Want to protect against complex SSL-related attacks
- Want to authenticate that the Service Manager server is a valid host

Certificates required

You must create or obtain the following certificates for SSL encryption.

- Certificate authority certificate
- Keystore containing the certificate authority's certificate
- Service Manager server certificate

Private keys required

You must create or obtain the following private keys for SSL encryption.

- Certificate authority's private key (This key is only necessary if you are managing your own private certificate authority)
- Service Manager server private key

Parameters required in the server configuration file (sm.ini)

- keystoreFile – identify the keystore file containing the Service Manager server's certificate and private key
- keystorePass – identify the password to the keystore file containing the Service Manager server's certificate and private key
- ssl:1
- sslConnector:1
- truststoreFile – identify the keystore file containing the certificate authority's certificate
- truststorePass – identify the password to the keystore file containing the certificate authority's certificate

Parameters required in the web tier configuration file (web.xml)

You can optionally set the customize-folder parameter in web.xml so that the <Customize-Folder>/config/webtier.properties file can be shared by multiple web tier instances on the same host.

- customize-folder – specify the absolute path to a folder on the web tier host in which the webtier.properties file is located

Parameter required in the <Customize-Folder>/config/webtier.properties or <WEBTIER>/WEB-INF/webtier.properties file

You must set the following web parameter:

- ssl:true
- cacerts – identify the truststore file containing the certificate authority that signed the server's certificate
- cacertsPassword – identify the password for the truststore

Windows client preferences required

You must set the following preference from the **Window** > **Preferences** > **Service Manager** > **Security** menu.

- CA certificates file – identify the truststore file containing the certificate authority that signed the server's certificate
- CA certificates password – identify the password for the truststore

Other requirements

You must do the following additional steps to ensure that Service Manager can use your private certificates.

- Add your private certificate authority's certificate to a keystore that your web and Windows clients can access
- Ensure that the Service Manager server's host name matches the common name (CN) listed in the server's signed certificate

Configure one-way SSL

The overall one-way SSL configuration is similar to two-way SSL configuration. Refer to Configure two-way SSL but skip the following tasks:

- Generate client certificates.
- Configure the keystore related parameters in various client configuration files.
  - <SM_SERVER_HOME>/RUN/sm.ini
    - ssl_trustedClientsJKS
    - ssl_trustedClientsPwd
  - <CHAT_SERVICE_HOME>/conf/app.properties
    - app.clientKeyStoreType
    - app.clientKeyStorePath
    - app.clientKeyStorePassword
  - <WEBTIER>/WEB-INF/webtier.properties
    - keystore
    - keystorePassword
  - <MOBILITY>/WEB-INF/web.properties
    - keystore
    - keystorePassword
  - Windows Client security settings
    - Client keystore file
    - Client keystore password

Meanwhile, disable the client certificate authentication in the Service Manager server and the chat server.

- SM server
  Edit the <SM_SERVER_HOME>/RUN/sm.ini file as below.

  ```
  ssl_reqClientAuth:0
  ```

- Chat Server
  Sign in to the Openfire Administration Console. Click **Server > Server Settings > Client Connections**. In the **Plain-text (with STARTTLS) connections** section, activate the **Advanced Configuration...** link. Set STARTTLS policy to **Required**, and then set Mutual Authentication to **Disabled**

8.12.4. TLS 1.2 support and configuration

When the Service Manager web tier and the Mobility client act as TLS/SSL servers, they use the Oracle or IBM Java virtual machine (JVM) that's required by your third-party web applications server to securely manage and connect incoming client requests that use TLS or SSL protocols. Examples of client requests include requests that originate from web browsers that are running on end-user's workstations. Depending on the third-party web application server, these connections default to the highest supported version of the TLS/SSL protocol. We don't provide the software that hosts the Service Manager web tier or Mobility client products; and it is this software that controls the specific TLS/SSL protocols that are used. As such, we recommend that you consult the vendor of your browser and third-party web application server (such as IBM Websphere and Apache Tomcat) for information about how to configure TLS 1.2.

As of Service Manager 9.41, the web tier, Mobility client and Windows client use TLS 1.2 by default.

The Service Manager server uses Java to securely manage and connect incoming client requests that use TLS and SSL protocols. In this scenario, the Service Manager server is acting as a TLS/SSL server. As of Service Manager 9.41, the Service Manager server uses TLS 1.2 by default. To enforce the TLS 1.2 protocol, configure the following parameter in the sm.ini file:

```
sslEnabledProtocols:TLSv1.2
```

Note

Note TLS 1.0 and 1.1 are versions of the TLS protocol that are no longer considered secure and have been disabled by default. If you encounter issues, you can, at your own risk, re-enable the versions. See Re-enable the acceptance of a Java certificate algorithm for more information.

When the Service Manager server acts as a TLS/SSL client, it uses OpenSSL to connect to the Directory Services server via the secure LDAP protocol. Secure LDAP is also known as LDAP over SSL (LDAPS). Depending on the Directory Service server, these connections will default to the highest supported version of the TLS/SSL protocol. To force the Service Manager server to establish only TLS 1.2 connections with the LDAP server, set the following environment variable in the operating system:

```
LDAPTLS_PROTOCOL_MIN=3.3
```

Note

Note For information about how to set an environment variable, refer to your operating system's documentation or to your server administrator.

The Service Manager server may act as a TLS/SSL client to securely connect to SMTP servers or to consume third-party external Web Services over HTTPS, depending on the SMTP Server or the Web Services server. These connections will default to TLS 1.2. To enforce the TLS 1.2 protocol for these scenarios, configure the following parameter in the sm.ini file:

```
sslEnabledProtocols:TLSv1.2
```

8.12.5. TLS 1.3 support and configuration

The TLS/SSL protocol secures the connection between two communication endpoints, typically the client and the server. OpenText recommends using TLS 1.3 in Service Manager. For web clients, URLs that require a secure connection start with HTTPS instead of HTTP. You can enable TLS/SSL communications between:

- The browsers and the Service Manager web tier
- The browsers and the Service Manager Mobility client
- The Service Manager web tier and Service Manager server
- The Service Manager Mobility client and Service Manager server
- Service Manager server and optional Directory Services Server (LDAP server)
- Service Manager server and optional Smart Analytics Server (IDOL)
- Service Manager server and third-party Web Services integrations
- Service Manager Smart Email solution and a third-party email server

As of Service Manager 9.80, the Service Manager supports TLS 1.3. To enforce the TLS 1.3 protocol, follow these steps.

Upgrade binaries to Service Manager 9.80

Follow the documents on upgrading the Service Manager binaries:

- Upgrade the Service Manager server
- Upgrade the web tier
- Upgrade the Windows client
- Upgrade the Mobility client
- Upgrade Service Manager Collaboration
- Upgrade Smart Analytics

Prepare certifications

Make sure you have prepared the required certificates for your Service Manager deployment. See Work with the certificate generation toolkit for detailed information.

Set Service Manager server parameters

Make sure that the following parameters are configured with TLS 1.3 in the sm.ini file:

- sslEnabledProtocols
- ciphers
- pgsslEnableProtocols
- pgsslEnableCiphers

When the Service Manager server acts as a TLS/SSL client, it uses OpenSSL to connect to the Directory Services server via the secure LDAP protocol. Secure LDAP is also known as LDAP over SSL (LDAPS). Depending on the Directory Service server, these connections will default to the highest supported version of the TLS/SSL protocol. To force the Service Manager server to establish only TLS 1.3 connections with the LDAP server, set the following environment variable in the operating system:

```
LDAPTLS_PROTOCOL_MIN=3.4
```

Web application server requirements

You must have a supported web application server to deploy web tier. To enforce the TLS 1.3 protocol, the following configurations are required:

Application server	• Apache Tomcat 9.0.13 or above • IBM WebSphere Application Server Liberty • JBoss Web Server (JWS) 5.7
Web server	Apache HTTP Server 2.4.43 or above is required in order to operate a TLS 1.3 web server with OpenSSL 1.1.1 IIS (bundled with Windows Server 2022)

Meet web browser requirements

Make sure that you are using the latest version of the supported browsers:

- Edge
- Chrome
- Firefox

Integrations

Here is the compatibility matrix between TLS 1.3 and the integrated products.

Integrated Product	Support TLSv1.3?	Product Version
APM	Yes	APM 9.60
SAP	No	SAP 7.1
Teams	No	N/A
ALM	Yes	ALM17
OO	Yes	OO 2023.05
Octane	Yes	Octane 16.0.400
PPM	No	N/A
UCMDB Gateway	Yes	2023.05
Jira	Yes	2023
UCMDB	Yes	2023.05
OBM	Yes	2023.05

8.12.6. Re-enable the acceptance of a Java certificate algorithm

The Service Manager server no longer accepts Java certificates that are generated by using certain "weak" algorithms. If you used these algorithms to generate your Java certificates, you must now regenerate them by using a more complex algorithm, such as RSA.

We strongly recommend that you regenerate the Java certificates by using a more complex algorithm. If it's not practical to do so, you can disable this enhancement. However, disabling this enhancement lowers the security profile of Service Manager and you're not adhering to the known security best practices.

Follow these steps to disable this enhancement:

1. Open the extra.java.security file that's located in the RUN folder.
2. Comment out or delete the contents of the file.
3. Restart the Service Manager server.

Alternatively, you can re-enable the acceptance of specific algorithms. For more information about the security best practices for enabling the secure algorithms, refer to https://wiki.mozilla.org/Security/Server_Side_TLS. Note that disabling certain algorithms may cause connectivity issues between Service Manager and third-party web service clients. You may need to perform thorough connectivity testing after you update this file.

The following algorithms are no longer accepted:

- MD5
- DSA
- 3DES_EDE
- DES_CBC
- DHE, RC4
- SSLv3
- TLSv1
- ECDH_anon
- DH_anon
- NULL
- DH keySize < 768
- RSA keySize < 2048

8.13. Support of the HTTP Strict Transport Security protocol

HTTP Strict Transport Security (HSTS) is a web security policy mechanism, which helps protect web application users from some passive (eavesdropping) and active network attacks. As of Service Manager 9.72, the HSTS response header is automatically added on the web tier side when the web parameter `cookie.secure` is set to `true`. To enable HSTS for Service Manager Mobility Client, you only need to enable HSTS in the web server or the web application server. Hence, HSTS adds an HTTP header named `Strict-Transport-Security` when establishing an HTTPS session.

Important

- If you are using a web server (Apache or IIS) in front of your web application server (Tomcat), you only need to enable HSTS in the web server; otherwise, you need to enable HSTS in the web application server.
- HSTS works only for ports 80 and 443. Make sure you have configured the web server or web application server to use port 80 for HTTP and port 443 for HTTPS.
- To use HSTS, be sure to enable the `secureLogin` parameter in the Service Manager Mobility Client and configure SSL between the web application server and browser.

The following are instructions on how to enable HSTS in the supported web servers and web application servers.

How to enable HSTS in Tomcat

To enable HSTS in Tomcat, follow these steps:

1. Open the `<Tomcat>/conf/web.xml` file in a text editor.
2. Uncomment the `httpHeaderSecurity` filter definition and the `<filter-mapping>` section.
3. Add the `hstsMaxAgeSeconds` parameter and the `antiClickJackingOption` parameter as shown below.

```
<filter>
    <filter-name>httpHeaderSecurity</filter-name>
    <filter-class>org.apache.catalina.filters.HttpHeaderSecurityFilter</filter-class>
    <init-param>
      <param-name>hstsMaxAgeSeconds</param-name>
      <param-value>31536000</param-value>
    </init-param>
    <init-param>
      <param-name>antiClickJackingOption</param-name>
      <param-value>SAMEORIGIN</param-value>
    </init-param>
    <async-supported>true</async-supported>
</filter>
...
<filter-mapping>
    <filter-name>httpHeaderSecurity</filter-name>
    <url-pattern>/*</url-pattern>
    <dispatcher>REQUEST</dispatcher>
</filter-mapping>
```

4. Save the file.
5. Restart Tomcat.

How to enable HSTS in Apache

To enable HSTS in an Apache server, follow these steps:

1. Open the `<Apache>/conf/httpd.conf` file in a text editor.

2. Uncomment the header module:

```
LoadModule headers_module modules/mod_headers.so
```

3. Add a header setting in the `VirtualHost` section:

```
<VirtualHost _default_:80>
ServerName apache.yourdomain.net:80
Header always set Strict-Transport-Security "max-age=63072000; includeSubdomains; preload"
</VirtualHost>
```

4. Restart Apache.

How to enable HSTS in IIS

To enable HSTS in IIS, do the following:

1. Add a `Strict-Transport-Security` header to the `web.config` file under the IIS installation root directory:

```
<system.webServer>
<httpProtocol>
<customHeaders>
<add name="Strict-Transport-Security" value="max-age=31536000"/>
</customHeaders>
</httpProtocol>
</system.webServer>
```

2. Restart IIS.

8.14. Enable SSL for PostgreSQL

If your PostgreSQL database requires SSL, you can enable SSL connections between the Service Manager server and the PostgreSQL database server.

To do this, follow these steps:

Configure PostgreSQL connection in Service Manager server

1. Prepare the required certification.
   Note: Customized file names and locations of the client certificate, client key, and root certificate are used in this example. You can also follow the PostgreSQL SSL documentation to use the default file names and locations.
   1. Prepare the root certificate. For example, if you use self-signed root certification:

      ```
      openssl req -new -nodes -text -out pgroot.csr -keyout pgroot.key -subj
      "/C=CN/ST=Shanghai/L=Shanghai/O=BTO/CN=root.<your domain name>"
      openssl x509 -req -in pgroot.csr -text -days 3650 -extensions v3_ca -signkey pgroot.key -out
      pgroot.crt
      ```

      Note: In a production environment, you are recommended to use your own certificate that's signed by your corporate CA or a third-party CA.
   2. Prepare the PostgreSQL database server certificate and server private key. For example:

      ```
      openssl req -new -nodes -text -out pgserver.csr -keyout pgserver.key -subj
      "/C=CN/ST=Shanghai/L=Shanghai/O=BTO/CN=<db server FQDN>"
      openssl x509 -req -in pgserver.csr -text -days 365 -CA pgroot.crt -CAkey pgroot.key -
      CAcreateserial -out pgserver.crt
      ```

3. Prepare the client certificate and client private key. For example:

```
openssl req -new -nodes -text -out pgclient.csr -keyout pgclient.key -subj
"/C=CN/ST=Shanghai/L=Shanghai/O=BTO/CN=<Postgres DB User Name>"
openssl x509 -req -in pgclient.csr -text -days 365 -CA pgroot.crt -CAkey pgroot.key -
CAcreateserial -out pgclient.crt
```

For details, see the PostgreSQL documentation.

2. Configure PostgreSQL database files and restart postgreSQL.
    1. Configure pg_hba.conf by appending a line for hostssl. For example:

```
hostssl    all    all    <SM Server FQDN>    cert
```

Note: If your Service Manager server is configured as horizontal scaling, you need to add each host as an entry in the pg_hba.conf file.

    2. Add the root certificate, PostgreSQL server certificate, and PostgreSQL server private key (created in step 1.a and step 1.b) to `$PGDATA`. For example, `/var/lib/pgsql/16/data`.
    3. Configure postgresql.conf by updating the following parameters:
        - ssl=on
        - ssl_ciphers=<cipher name>
        - ssl_cert_file=<server certificate file name>
        - ssl_key_file=<server private key file name>
        - ssl_ca_file=< root certificate file name>
        - ssl_min_protocol_version = 'TLSv1.2'
        - ssl_max_protocol_version = 'TLSv1.3'
    4. Restart the PostgreSQL service.
    Note: If you need to configure SSL between Service Manager Collaboration and PostgreSQL, you can restart the PostgreSQL service later after the step 8.
3. Upload the following certificate files to the RUN folder of the Service Manager server.
    - Root certificate
    - Client certificate
    - Client private key

    Notes:
    - When Service Manager Server is on Linux, make sure that you set the file permission as 600 for certifications.
    - Optionally, you can set Certificate Revocation List (CRL) file.
4. Add the FQDN of the database server to the host file of the Service Manager server. Alternatively, you can add user mapping by following the PostgreSQL document.
5. Configure the PostgreSQL settings in sm.ini. For example:

```
sqldictionary:postgresql
[postgresql]
sqldb:host=<dbhost> port=5432 dbname=<dbname> sslmode=verify-full sslcert=pgclient.crt
sslkey=pgclient.key sslrootcert=pgroot.crt
```

    - sslmode: secure connections can be ensured by setting the parameter to `verify-full` or `verify-ca`
    - sslcert: specifies the client certificate
    - sslkey: specifies the client private key
    - sslrootcert: specifies the root certificate

    Tip: For more information, see the PostgreSQL documentation: https://www.postgresql.org/docs/current/libpq-ssl.html

6. Configure the pgsslEnableProtocols and pgsslEnableCiphers parameters in sm.ini.
    - **pgsslEnableProtocols**: specifies the supported SSL protocols for PostgreSQL SSL connections. TLSv1.2, TLSv1.3 is set by default. For details, see pgsslEnableProtocols.
    - **pgsslEnableCiphers**: specifies an array of cipher suites used for PostgreSQL SSL connections. For details, see pgsslEnableCiphers.
7. Restart the Service Manager server.

Configure PostgreSQL connection for Chat

1. If you use Service Manager Collaboration, configure the JDBC client for the Chat server.
    1. Run the following command to convert the private key created in step 1.c from the PEM format to the DER format, and then name it to `pgclient.pk8`.

        ```
        # openssl pkcs8 -topk8 -inform PEM -outform DER -in pgclient.key -out pgclient.pk8 -nocrypt
        ```

    2. Copy certificates to the Chat server.
       On the Chat server, copy the root certificate (pgroot.crt), JDBC client certificate (pgclient.crt) and key (pgclient.pk8) to the *<Chat Server Home>/bin* folder.
    3. On the PostgreSQL server, open the pg_hba.conf file with a text editor, and then add a new line for SSL. For example:

        ```
        hostssl    all    all    <Chat server FQDN>    cert
        ```

        Restart the PostgreSQL server after your update.
    4. Add the FQDN of the database server to the host file of the Chat server. Alternatively, you can add user mapping by following the PostgreSQL document.
    5. Construct the JDBC URL. When you log in to the PostgreSQL database by using the JDBC URL, you can use the following JDBC URL:

        ```
        jdbc:postgresql://<db server FQDN>:<PG DB Port>/<database name>?currentSchema=<Schema
        Name>&sslmode=verify-full&sslcert=<client certification>&sslkey=<client
        keystore.pk8>&sslrootcert=<root certificate>
        ```

        Note: If Chat Server is deployed on Linux, use absolute path for sslcert, sslkey,sslrootcert parameters in above JDBC URL.
        For example:

        ```
        jdbc:postgresql://dbserver.yourdomain.net:5432/smdb?currentSchema=smdb&sslmode=verify-
        full&sslcert=pgclient.crt&sslkey=pgclient.pk8&sslrootcert=pgroot.crt
        ```

Configure PostgreSQL connection in the IdM service

To configure PostgreSQL connection in the IdM service, follow these steps:

1. Open the <idm-service>/WEB-INF/spring/applicationContext.properties file with a text editor.
2. Update the following database connection parameters.

Parameter	Description
idm.persistence.flyway.type	Set it to **postgresql**.
hibernate.dialect	Set it to **org.hibernate.dialect.PostgreSQL9Dialect**.
idm.persistence.dialect	Set it to **org.hibernate.dialect.PostgreSQL9Dialect**.
idm.persistence.connection.driver.class	Set it to **org.postgresql.Driver**.
idm.persistence.connection.url	Specify a value that uses the following format: **jdbc:postgresql://<*PostgreSQL database server host*>:<*port*>/<*IdM database name*>?ssl=true&sslmode=verify-full&sslrootcert=C:\\apache-tomcat-9.0.80\\webapps\\idm-service\\WEB-INF\\spring\\pgroot.crt** Example value: **jdbc:postgresql://localhost:5432/IDM-SM?ssl=true&sslmode=verify-full&sslrootcert=C:\\apache-tomcat-9.0.80\\webapps\\idm-service\\WEB-INF\\spring\\pgroot.crt**
idm.persistence.connection.username idm.persistence.connection.password	Enter the login credentials that you created for the PostgreSQL database.

8.15. Trusted sign-on

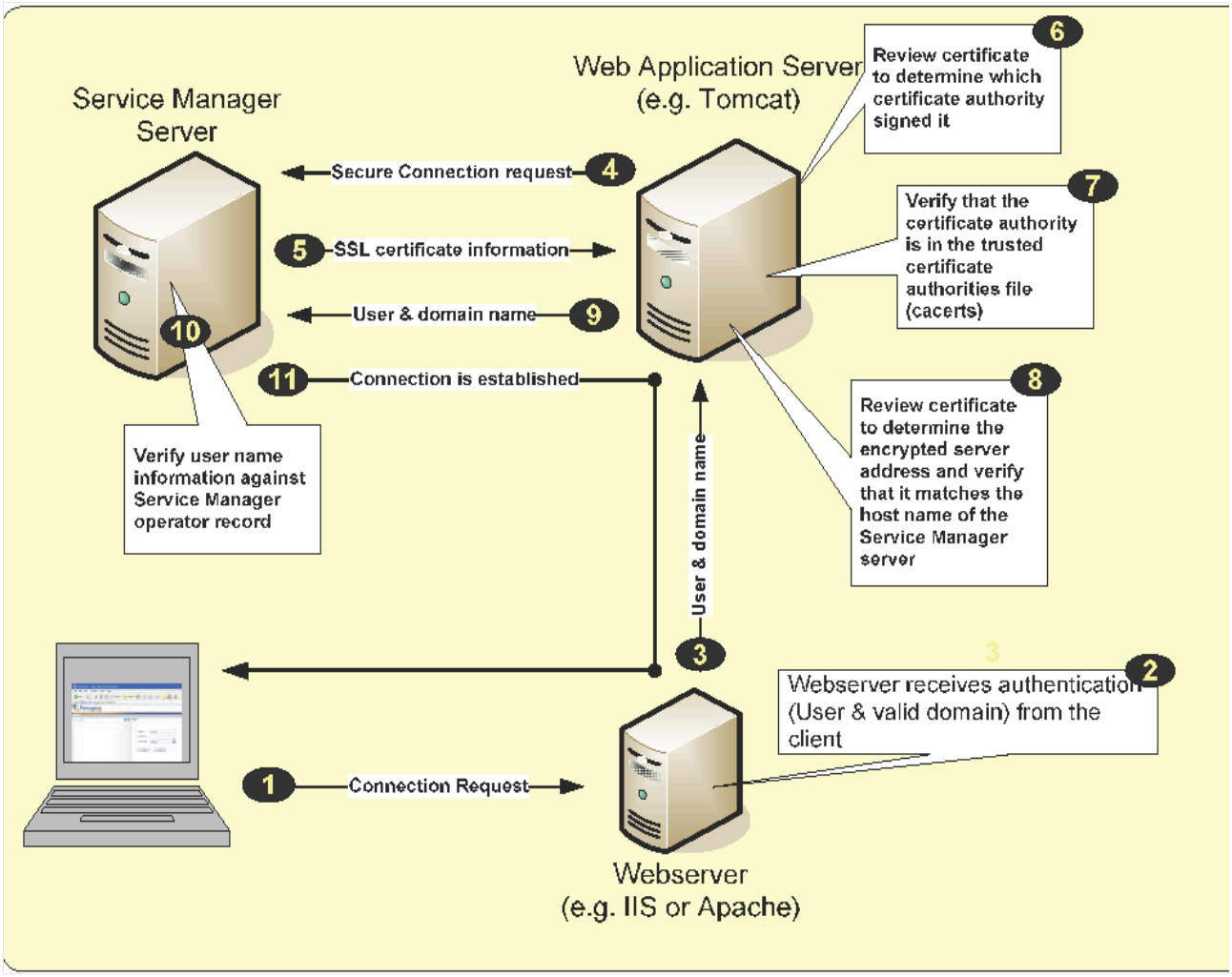
You can configure Service Manager clients to automatically log on using the same authentication information as users entered when they logged onto their client workstation's operating system. When you enable trusted sign-on, users bypass the Service Manager logon screen and directly enter the application.

In a trusted sign-on scenario the Service Manager server grants access to clients only if the following conditions are met.

- The user's logon credentials match an existing operator record in Service Manager or a valid LDAP source that Service Manager recognizes.
- A trusted authentication authority, such as the operating system, validates that the user's logon credentials are valid.
- The client (Service Manager Web Tier or Windows) must present a signed SSL certificate.

The following figure depicts the connection process between a Web server, a Web application server, and the Service Manager application server:

- The Web server receives the user information from the client via the browser, and passes the user name and domain name to the Web application server.
- The Web application server (such as Tomcat) acts as a client, and communicates with the Service Manager application server.
- The Service Manager application server also checks whether the user was authenticated by a valid domain. Local machine authentication is not accepted; if attempted the Service Manager server will reject such a request.



8.15.1. Requirements for trusted sign-on

This configuration is intended for customers who have an Service Manager server running in a secured environment and want users to be able to log on to the server without providing a user name and password.

Parameters required in the server configuration file (sm.ini)

```
trustedsignon:1
ssl:1
ssl_reqClientAuth:2
```

Parameters required in the web tier configuration file (web.xml)

`isCustomAuthenticationUsed` - Set the value to false to make Service Manager send the current user name in the HTTP header.

Other requirements

1. Configure the web tier to use a trusted authentication source.
    1. Stop the web application server running the web tier.
    2. Navigate to the folder `<web tier installation path>\WEB-INF\classes`, and open the file `application-context.xml` in a text editor.
    3. Search for the following string:
    `<sec:filter-chain pattern="/**" filters="securityContextPersistenceFilter,resourceFilter,anonymousAuthFilter"/>`

4. Add `preAuthenticationFilter` to use Microsoft Integrated Windows Authentication as your trusted authentication source:
   `<sec:filter-chain pattern="/**" filters="securityContextPersistenceFilter,resourceFilter,preAuthenticationFilter,anonymousAuthFilter"/>`
5. Configure the parameters in the bean definition of `preAuthenticationFilter` according to the username format used in Service Manager.

   ```
   keepDomain:
   true, the login name of the Service Manager operators contains the domain name;
   false (default), does not contain.

   userPrincipalNameFormat:
   true, User Principal Name (UPN) format - for example: UserName@DOMAIN is used;
   false (default), Down-Level Logon Name format - for example: DOMAIN\UserName is used.
   ```

   In the example above, `DOMAIN` is the "Domain name (pre-Windows 2000)" of your Active Directory.
6. Save the file and restart the web application server.

2. Create an operator record for each Windows user that you want to log on to Service Manager. Format of the username must match the configuration in *step 1* > *e*.
3. Install and configure an external authentication source, such as Microsoft Integrated Windows Authentication, to ensure that Service Manager can use your private certificates.
4. Refer to Example: Configuring the Web server for trusted sign-on to configure a web server for trusted sign-on.
5. Beginning with Service Manager 9.30, Trusted Sign-On (TSO) requires the use of `ssl_reqClientAuth:2`. You must create unique client SSL certificates for each Service Manager client that needs to access Service Manager with TSO. For example, if you have 20 Windows clients, you must create 20 unique client SSL certificates.

> **Tip**
> Tip If maintaining these unique client SSL certificates for Windows clients incurs unsustainable IT operation costs, you can consider the use of the `acceptsharedcert:1` parameter. See Parameter: acceptsharedcert.

Related topics

Example: Enabling trusted sign-on
Parameter: acceptsharedcert

8.15.2. Example: Enabling trusted sign-on

Trusted sign-on in Service Manager leverages Windows-integrated authentication that allows users on trusted clients who have logged in to a Windows domain to log on to Service Manager without providing a user name and password. The Windows client can directly retrieve the cached credentials from the Windows system to authenticate the operator.

For applications deployed on a Java EE application server, such as web tier and mobility, integration with a web server (such as Windows Internet Information Services (IIS) or Apache HTTP server) is required for the authentication.

> **Note**
> - Service Manager only supports trusted sign-on with SSL enabled and the `ssl_reqClientAuth` parameter set to "2".
> - To use trusted sign-on, you must first add your web server and Windows clients to a domain.

This example assumes that you are using Tomcat as the web application server and Apache or IIS as the web server.

To enable trusted sign-on, perform the following tasks:

Task 1: Create an operator record for each Windows user

You need to create an operator record for each Windows user so that they can log in to Service Manager.

There are two account formats for a Windows user in an Active Directory:

- sAMAccountName format.
  For example, DOMAIN\UserName. Here DOAMIN is the "Domain name (pre-Windows 2000)" of your Active Directory.
- UserPrincipalName format.
  For example, UserName@xxx.com.

Note that when you are using Windows integrated authentication, the account name passed to the web server is different based on the authentication type.

- For NTLM, the system always retrieves sAMAccountName as the account name even if you have logged on to the computer using the UserPrincipalName.
- For GSSAPI, the system returns the account name in the format of UserName@KerberosRealm. Usually the KerberosRealm is the same as the domain name.

The default configuration of trusted sign-on is to drop the domain part from the account name. Hence, when you create an operator in Service Manager, the login name doesn't contain domain name.

Suppose your Service Manager operators are from multiple domains. When you create an operator in Service Manager, the login name must contain domain name. You can use either the sAMAccountName format or the UserPrincipalName format. Note that in both formats the domain part is as follows.

- For NTLM, it is from sAMAccountName - the "Domain name (pre-Windows 2000)" of your Active Directory.
- For GSSAPI, it is the KerberosRealm.

Task 2: Enable required SSL encryption and trusted clients.

For detailed steps, see Configure two-way SSL.

Task 3: Configure the web tier to use trusted sign-on.

If the web server connects to Tomcat by using the AJP protocol, please follow below steps:

1. Stop the web application server running the web tier.
2. In the web tier's web.xml file, set `isCustomAuthenticationUsed` to false.
3. In the <Tomcat>/conf/server.xml file, insert **tomcatAuthentication="false"** in the following section as shown below.

```
<Connector protocol="AJP/1.3"
  secret="MyT0mcAt"
  address="0.0.0.0"
  port="8009"
  tomcatAuthentication="false"
  redirectPort="8443" />
```

4. Edit the web application server's application-context.xml file to enable pre-authentication.
   1. Open <web tier installation path>\WEB-INF\classes\application-context.xml in a text editor, and search for the following string:
      `<sec:filter-chain pattern="/**" filters="securityContextPersistenceFilter,resourceFilter,anonymousAuthFilter"/>`
   2. Insert `preAuthenticationFilter` in the string:
      `<sec:filter-chain pattern="/**" filters="securityContextPersistenceFilter,resourceFilter,preAuthenticationFilter,anonymousAuthFilter"/>`

Note: If you need to enable trusted sign-on for your web client users and also enable web tier lightweight single sign-on (LW-SSO) for integrations, add `lwSsoFilter` followed by `preAuthenticationFilter`, as shown in the following:

`<sec:filter-chain pattern="/**" filters="securityContextPersistenceFilter,resourceFilter,lwSsoFilter,preAuthenticationFilter,anonymousAuthFilter"/>`

For information about how to enable LW-SSO in the web tier, see Configure LW-SSO in the Service Manager Web tier.

3. Configure the parameters in the bean definition of `preAuthenticationFilter` according to the username format configured in Task 1.

```
keepDomain:
true, the login name of the Service Manager operators contains the domain name;
false (default), does not contain.

userPrincipalNameFormat:
true, UserPrincipalName format;
false (default), sAMAccountName format.
```

4. If the web server connects to Tomcat by using the HTTPS protocol and passes the user name by a specific request header (for example, `X-Remote-User`), you need to configure the web tier to get the remote user from the specific request header. Edit the <WEBTIER>/WEB-INF/classes/application-context.xml file to use the `HttpHeaderPreAuthenticationFilter` instead of the `PreAuthenticationFilter` for the `preAuthenticationFilter` bean. For example:

```
<bean id="preAuthenticationFilter"
class="com.hp.ov.cwc.security.spring.HttpHeaderPreAuthenticationFilter">
  <property name="authenticationManager">
    <ref bean="authenticationManager"/>
  </property>
  <property name="defaultRole">
    <value>ROLE_PRE</value>
  </property>
  <property name="usernameHeaderKey">
    <value>X-Remote-User</value>
  </property>
</bean>
```

5. Restart the web application server.

Task 4: Configure each Windows client to use trusted sign-on.

Do the following for each Windows client:

1. Make sure SSL encryption is enabled for the Windows client. See task 1.
2. Open a client connection, and on the Connection tab select **Use Trusted Sign-on**.
3. Make selections to the **Keep domain** and **User Principal Name format** checkboxes according to the username format set in Task 1.
4. Click **Apply**.

Task 5: Configure the Mobility client to use trusted sign-on.

If the web server connects to Tomcat by using the AJP protocol, please follow below steps:

1. Stop the web application server running the web tier.
2. In the web tier's web.xml file, set `isCustomAuthenticationUsed` to false.
3. In the <Tomcat>/conf/server.xml file, insert **tomcatAuthentication="false"** in the following section as shown below.

```
<Connector protocol="AJP/1.3"
  secret="MyT0mcAt"
  address="0.0.0.0"
  port="8009"
  tomcatAuthentication="false"
  redirectPort="8443" />
```

4. Edit the <MOBILITY>/WEB-INF/spring/security.xml file to enable `preAuthenticationFilter` for both the standard and ESS modes. Next, comment out `idmPreAuthFilter`, `lwSsoFilter`, `springSecurity2lwssoIntegrationFilter`, and `httpHeaderPreAuthenticationFilter`. Refer to the following examples.

```
<security:http pattern="/#{globalProps['standardModePath']}/**" use-expressions="true">
        <security:access-denied-handler ref="smAccessDeniedHandler"/>
        <security:custom-filter ref="logoutFilterStandard" position="LOGOUT_FILTER"/>
        <!--<security:custom-filter ref="idmPreAuthFilter" position="PRE_AUTH_FILTER"/> -->
        <security:custom-filter ref="preAuthenticationFilter" after="SECURITY_CONTEXT_FILTER"/>
        <!--<security:custom-filter ref="lwSsoFilter" before="BASIC_AUTH_FILTER"/>
        <security:custom-filter ref="springSecurity2lwssoIntegrationFilter" position="LAST"/> -->
        <!--<security:custom-filter ref="httpHeaderPreAuthenticationFilter"
after="SECURITY_CONTEXT_FILTER"/>-->
        <security:intercept-url pattern="/#{globalProps['standardModePath']}/**"
access="hasRole('STD_USER')"/>
        <security:intercept-url pattern="/#{globalProps['standardModePath']}/auth" method="POST"
access="hasRole('STD_USER')"/>
        <security:form-login login-page="/#{globalProps['standardModePath']}/login"
                             login-processing-url="/#{globalProps['standardModePath']}/auth"
                             username-parameter="user.id" password-parameter="old.password"
                             authentication-details-source-ref="detailSource"
                             authentication-failure-handler-ref="stdFailureHandler"
                             authentication-success-handler-ref="stdSuccessHandler"
                             default-target-url="/#{globalProps['standardModePath']}/"
                             always-use-default-target="true"/>
        <security:intercept-url pattern="/#{globalProps['standardModePath']}/logout"
method="POST"/>
</security:http>
```

```
<security:http pattern="/#{globalProps['essModePath']}/**" use-expressions="true">
    <security:access-denied-handler ref="smAccessDeniedHandler"/>
    <security:custom-filter ref="logoutFilterEss" position="LOGOUT_FILTER"/>
    <!--<security:custom-filter ref="idmPreAuthFilter" position="PRE_AUTH_FILTER"/> -->
    <security:custom-filter ref="preAuthenticationFilter" after="SECURITY_CONTEXT_FILTER"/>
    <!--<security:custom-filter ref="lwSsoFilter" before="BASIC_AUTH_FILTER"/>
    <security:custom-filter ref="springSecurity2lwssoIntegrationFilter" position="LAST"/> -->
    <security:intercept-url pattern="/#{globalProps['essModePath']}/**"
access="hasRole('ESS_USER')"/>
    <security:intercept-url pattern="/#{globalProps['essModePath']}/auth" method="POST"
access="hasRole('ESS_USER')"/>
    <security:form-login login-page="/#{globalProps['essModePath']}/login"
                             login-processing-url="/#{globalProps['essModePath']}/auth"
                             username-parameter="user.id" password-parameter="old.password"
                             authentication-details-source-ref="detailSource"
                             authentication-failure-handler-ref="essFailureHandler"
                             authentication-success-handler-ref="essSuccessHandler"/>
    <security:intercept-url pattern="/#{globalProps['essModePath']}/logout" method="POST"/>
</security:http>
```

According to the username format configured in Task 1, you may also need to configure the `keepDomain` and `userPrincipalNameFormat` parameters in the bean definition of `preAuthenticationFilter`.

```
keepDomain:
true, the login name of the Service Manager operators contains the domain name;
false (default), does not contain.

userPrincipalNameFormat:
true, UserPrincipalName format;
false (default), sAMAccountName format.
```

If the web server connects to Tomcat by using the HTTPS protocol and pass the user name by a specific request header (for example, `X-Remote-User`), you need to configure the mobility to get the remote user from the specific request header. Edit the <MOBILITY>/WEB-INF/spring/security.xml file to enable the security filter `httpHeaderPreAuthenticationFilter` instead of `preAuthenticationFilter`. For example:

```
<security:http pattern="/#{globalProps['standardModePath']}/**" use-expressions="true">
    ...
    <security:custom-filter ref="httpHeaderPreAuthenticationFilter"
after="SECURITY_CONTEXT_FILTER"/>
    ...
</security:http>

<security:http pattern="/#{globalProps['essModePath']}/**" use-expressions="true">
    ...
    <security:custom-filter ref="httpHeaderPreAuthenticationFilter"
after="SECURITY_CONTEXT_FILTER"/>
    ...
</security:http>

<bean id="httpHeaderPreAuthenticationFilter"
class="com.hp.sw.sm.client.web.auth.preauth.HttpHeaderPreAuthenticationFilter"
parent="preAuthenticationFilter">
    ...
    <property name="principalRequestHeader" value="X-Remote-User" />
    ...
</bean>
```

Task 6: Install and configure the web server (Apache or IIS) to use trusted sign-on.

To install and configure the web server, see Configure a web server to redirect requests to the web tier.

To configure the web server to use trusted sign-on, see Example: Configuring the web server for trusted sign-on.

Task 7: Configure web browsers to enable web client users to use trusted sign-on

You need to enable Integrated Windows Authentication (IWA) for browsers to use trusted sign-on.

For Edge or Chrome, follow these steps to enable IWA.

1. Open **Control Panel** > **Network and Internet** > **Internet Options**.
2. On the **Security** tab, add the FQDN of the web server (for example, apache.yourdomain.net) to the **Trusted sites** or the **Local intranet sites**.
3. Click the **Custom level** button of the **Trusted sites** or the **Local intranet sites**, scroll down to **User Authentication \ Logon**. Select the **Automatic logon with current user name and password** option.

For Firefox, follow these steps to enable IWA.

1. Open Firefox, navigate to the **about:config** URL, and then accept the warning page.
2. Type **trusted-uris** in the search field to filter options.

3. Add the FQDN of the web server (for example, apache.yourdomain.net) to each option of **network.automatic-ntlm-auth.trusted-uris** and **network.negotiate-auth.trusted-uris**, and then save your updates.

Task 8: Test your trusted sign-on setup

1. Start the Service Manager server, the web server (Apache or IIS), and the web application server (in this example, Tomcat).
2. Start a Windows client, and log in using trusted sign-on.
   Service Manager should automatically log you in with your NT account username.
3. Start the browser, and open the web tier login URL: https://apache.yourdomain.net/webtier/index.do
   Service Manager should automatically log you in without displaying the login screen.

Related topics

Trusted sign-on
Using LW-SSO with integrations
Requirements for trusted sign-on
Configure LW-SSO in the Service Manager Web tier

8.15.2.1. Example: Configuring the Web server for trusted sign-on

To enable trusted sign-on for web tier clients, you must install and configure a web server (for example, Microsoft Internet Information Services (IIS) or Apache), which receives the user information from the client through the browser, and passes the user name and domain name to the web application server.

> **Note**
>
> The following steps assume that the web server and web application server configurations are already established and that the only necessary changes to the configurations of these servers are those described below.

Apache configuration changes

> **Note**
>
> In this example, the mod_authn_ntlm module is used for Apache on Windows and the mod_auth_gssapi module is used for Apache on Linux.

Windows

1. Obtain the mod_authn_ntlm module from a binary distribution. For example, from Apache Haus.
2. Extract the downloaded *.zip file, and then add the `mod_authn_ntlm.so` file to the `<APACHE_HOME>/modules` directory.
3. Load the `mod_authn_ntlm` module in the `<APACHE_HOME>/conf/httpd.conf` file.

   ```
   # Load mod_authn_ntlm module
   LoadModule auth_ntlm_module modules/mod_authn_ntlm.so
   ```

4. Configure the authentication type for the web application (for example, web tier) by adding the following lines to the default virtual host.

```
<Location "/webtier">
  AuthName "Service Manager Web Client"
  AuthType SSPI
  NTLMAuth On
  NTLMAuthoritative On
  NTLMOfferBasic Off
  NTLMOmitDomain On
  <RequireAll>
    Require valid-user
    Require not user "NT-AUTHORITY\ANONYMOUS-LOGON"
  </RequireAll>
</Location>
```

Notes:
If the Apache HTTP Server connects to Tomcat by using the HTTPS protocol, you need to set the user name to a specific request header (for example, `X-Remote-User`). For example:

```
<Location "/webtier">
  ...
  RewriteEngine On
  RewriteCond %{LA-U:REMOTE_USER} (.+)
  RewriteRule .* - [E=RU:%1]
  RequestHeader set X-Remote-User %{RU}e
</Location>
```

Note:

- The name within the Location tag needs to be the path the user enters to open the Service Manager web client website. Usually, the name is taken from the *.war file of the web tier.
- On Windows, the default virtual host is `<VirtualHost _default_:80>` defined in the `<APACHE_HOME>/conf/extra/httpd-vhosts.conf` file or `<VirtualHost _default_:443>` defined in the `<APACHE_HOME>/conf/extra/httpd-ssl.conf` file, depending on whether you turn on SSL for the Apache HTTP Server.

Linux

1. On a Linux system, install the `mod_auth_gssapi` module and the `mod_session` module. For example:

```
$ sudo yum install mod_auth_gssapi
$ sudo yum install mod_session
```

2. Create a user account named apache in the Active Directory (AD) and use this account as the Service Principal for the Apache HTTP server (httpd).
Note: The user logon name is the hostname (without the domain part) where the Apache HTTP server is installed. On the **Account** tab, select **User password never expires** and **This account supports kerberos AES 256 bit encryption** options.
3. Open a command prompt in the Windows system of the AD and generate the keytab file for the Apache HTTP server with the following command.

```
> ktpass  -princ HTTP/apache.yourdomain.net@YOURDOMAIN.NET -mapuser apache -crypto AES256-SHA1 -
ptype KRB5_NT_PRINCIPAL -pass +rndPass -out c:/auth_apachehost_kerb.keytab
```

4. Configure the Apache HTTP server to use mod_auth_gssapi with the keytab generated.

Copy the keytab generated in *step 3* from the Windows system to `/etc/httpd/conf` in the Linux system where the Apache HTTP server is installed and set appropriate file permission. For example:

```
$ sudo chmod 755 /etc/httpd/conf/auth_apachehost_kerb.keytab

$ sudo klist -e -k /etc/httpd/conf/auth_apachehost_kerb.keytab
Keytab name: FILE:/etc/httpd/conf/auth_apachehost_kerb.keytab
KVNO Principal
---- --------------------------------------------------------------------------
   6 HTTP/apache.yourdomain.net@YOURDOMAIN.NET (aes256-cts-hmac-sha1-96)
```

5. Configure the authentication type for the SM web tier by adding the following lines to the default virtual host in `/etc/httpd/conf.d/ssl.conf`.

   ```
   <Location "/webtier">
       AuthType GSSAPI
       AuthName "Service Manager Web Client"
       GssapiCredStore keytab:/etc/httpd/conf/auth_apachehost_kerb.keytab
       GssapiUseSessions On
       Session On
       SessionCookieName gssapi_session path=/webtier;httponly;secure;
       Require valid-user
   </Location>
   ```

   Notes:
   If the Apache HTTP Server connects to Tomcat by using the HTTPS protocol, you need to set the user name to a specific request header (for example, `X-Remote-User`). For example:

   ```
   <Location "/webtier">
     ...
     RewriteEngine On
     RewriteCond %{LA-U:REMOTE_USER} (.+)
     RewriteRule .* - [E=RU:%1]
     RequestHeader set X-Remote-User %{RU}e
   </Location>
   ```

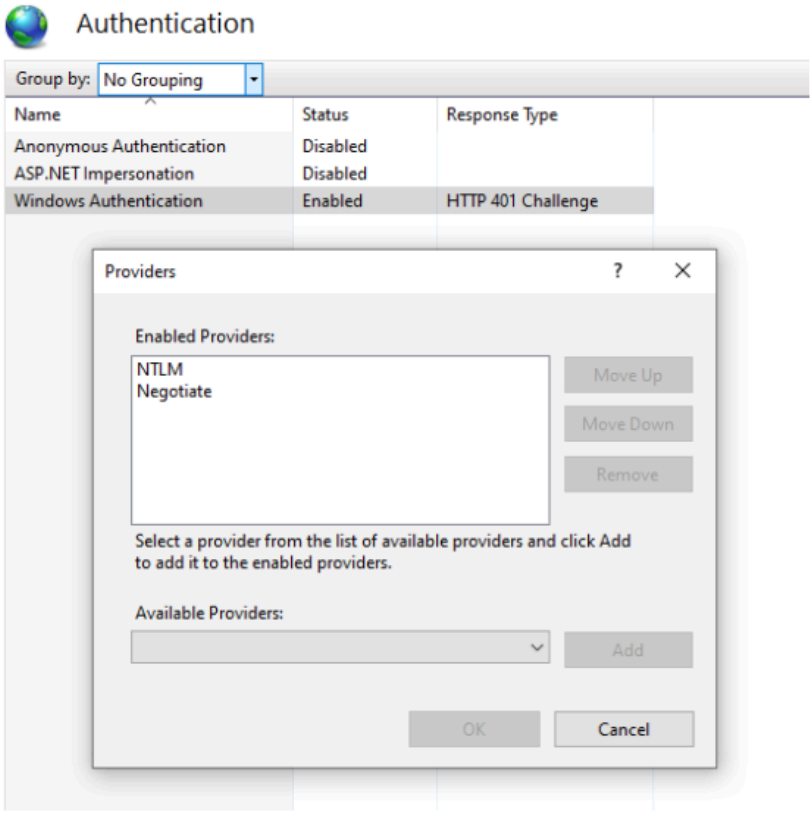
6. Check the configuration and restart the Apache HTTP server.

   ```
   $ sudo apachectl configtest
   Syntax OK
   $ sudo systemctl restart httpd
   ```

IIS configuration changes

> **Note**
>
> You need to use either the BonCode AJP 1.3 connector or the Apache JK connector for TSO.

1. Open the **Internet Information Services (IIS) Manager**.
2. Select **Default Web Site**, and then double-click **Authentication**. Disable **Anonymous Authentication**, and then enable **Windows Authentication**. Edit the providers of the Windows Authentication by moving **NTLM** to the top of the enabled providers.



Your IIS web server configuration changes are completed. You can continue to configure the web browser's security settings to enable trusted sign-on for web clients.

Related topics

Trusted sign-on
Example: Enabling trusted sign-on
Configure a web server to redirect requests to the web tier

8.15.2.2. Example: Configuring Single Sign-on with third party authentication on the web tier

Configuring the Web Client for third-party authentication

1. To activate single sign-on in Service Manager's web tier, navigate to the <path to Web application>/WEB-INF/classes/ directory.
2. Open the application-context.xml file with a text editor. Update `/**=httpSessionContextIntegrationFilter,anonymousProcessingFilter` to `/**=httpSessionContextIntegrationFilter,preAuthenticationFilter, anonymousProcessingFilter` as described in the comment in the application-context.xml file.
3. Save your changes and restart the web server on the Service Manager web tier.
4. When using IIS you need to configure an ISAPI connector for your Web application server, and you need to modify the virtual directory to use Integrated Windows Authentication.

See the Customer Support Web site for Knowledge Base articles about configuring Integrated Windows Authentication on common Web application servers or refer to Example: Configuring the Web server for trusted sign-on.

Defining a JavaBean® to handle authentication

All modifications for the single or trusted sign-on settings in the Service Manager Web Tier are done in the <path to web application>/WEB-INF/classes directory in the application-context.xml file.

For Windows-based authentication, out-of-box Service Manager ships with a JavaBean called preAuthenticationFilter, which is defined as follows:

```
<bean id="preAuthenticationFilter"
class="com.hp.ov.cwc.security.spring.PreAuthenticationFilter">
<property name="authenticationManager">
  <ref bean="authenticationManager"/>
</property>
<property name="defaultRole">
  <value>ROLE_PRE</value>
</property>
</bean>
```

To use another authentication source that passes the username in the HTTP header, you add or replace the preAuthenticationFilter block with code such as the following.

The following example shows how to add a new filter. To replace the filter, set the bean ID to preAuthenticationFilter (rather than SiteminderPreAuthenticationFilter as shown below).

```
<bean id="SingleSignOnpreAuthenticationFilter"
class="com.hp.ov.cwc.security.spring.SiteminderPreAuthenticationFilter">
<property name="authenticationManager">
  <ref bean="authenticationManager"/>
</property>
<property name="defaultRole">
  <value>ROLE_PRE</value>
</property>
<property name="keepDomain">
  <value>true</value>
</property>
<property name="siteminderUsernameHeaderKey">
  <value>USERNAME</value>
</property>
<property name="siteminderPasswordHeaderKey">
  <value>PASSWORD</value>
</property>
</bean>
```

In this definition, the siteminderUsernameHeaderKey property passes the user name value from the single sign-on tool to Service Manager. To successfully pass the user name value, set the <value> element to the name of the property in the HTTP header that stores the user name in the third-party authentication tool's configuration.

If the HTTP header is as follows, replace `<USERNAME>` with `REMOTE_USER`.

```
POST /sc61server/ws HTTP/1.1
Content-Type: text/xml; charset=utf-8
SOAPAction: "Retrieve"
Content-Length: 1474
Expect: 100-continue
[…]
REMOTE_USER: falcon
Cookie: $Version=1; SessionId=127.0.0.1:4819
```

To activate the sign-on filters, modify the following line in the application-context.xml file.

```
/**=httpSessionContextIntegrationFilter, SingleSignOnpreAuthenticationFilter,preAuthenticationFilter,
anonymousProcessingFilter
```

The SingleSignOnpreAuthenticationFilter has to match the bean ID from above. If you want multiple sign-on filters, add them in order to the filter chain. If you do this, the first to succeed will provide the authenticated username.

Note: The httpSessionContextIntegrationFilter always needs to be in the first position within the chain.

Integrating Custom Java Classes into the Java Bean

In Service Manager, it is possible to develop and use a custom Java class for third party authentication as well. You can either write a new custom Java class or modify the existing Java class to fit the new requirements for Service Manager. The requirements for the new Java class are described in the example of integrating an old Java class into the Service Manager system. The modified Java class then needs to be put into the web-inf/classes directory, or as a jar file into web-inf/lib.

In the <path to web application>/WEB-INF/classes directory in the application-context.xml file, your bean will then look as follows:

```
<bean id="preAuthenticationFilter"
class="com.hp.ov.cwc.security.spring.<custom java class>">
<property name="authenticationManager">
  <ref bean="authenticationManager"/>
</property>
<property name="defaultRole">
  <value>ROLE_PRE</value>
</property>
</bean>
```

The main modification you have to do to the old Java code is to remove all mention of the user object, which is not passed into the class in Service Manager any longer. Instead, we return the user name, if authentication succeeded, or return null if it failed. This username has to be returned after a successful run of the class.

Example: Creating a custom Java Class for Single-Sign-On using LDAP

This is an example of how to create a custom Java class for Single Sign-On using LDAP authentication. Actual implementation will vary from system to system, depending on the LDAP server, setup and configuration.

The following classes are required to compile this example:

Class jar file	Comments
jasypt-1.9.2.jar	Part of the jasypt package. Copy the jar to <web-inf>\lib
commons-lang-2.6.jar	Part of SM9 web client
commons-codec-1.4.jar	Part of SM9 web client
com.ibm.icu_4.4.2.v20110208.jar	From the full client plugins folder
cwc-9.x.jar	Part of SM9 web client
spring-core-3.2.15.RELEASE.jar	Part of SM9 web client
spring-beans-3.2.15.RELEASE.jar	Part of SM9 web client
spring-security-core-3.2.9.RELEASE.jar	Part of SM9 web client
spring-aop-3.2.15.RELEASE.jar	Part of SM9 web client
spring-context-3.2.15.RELEASE.jar	Part of SM9 web client
spring-expression-3.2.15.RELEASE.jar	Part of SM9 web client
spring-security-config-3.2.9.RELEASE.jar	Part of SM9 web client
spring-security-web-3.2.9.RELEASE.jar	Part of SM9 web client
spring-web-3.2.15.RELEASE.jar	Part of SM9 web client
servlet-api.jar	Copied from Tomcat, only required for compilation.

```
package com.hp.ov.cwc.security.spring;
// servlet-api.jar

import javax.servlet.http.HttpServletRequest;
import java.util.Hashtable;
import javax.naming.Context;
import javax.naming.NamingEnumeration;
import javax.naming.NamingException;
import javax.naming.directory.Attributes;
import javax.naming.directory.SearchControls;
import javax.naming.directory.SearchResult;
import javax.naming.ldap.InitialLdapContext;
import javax.naming.ldap.LdapContext;
// defined in jasypt-1.9.2.jar
// should copy to app-lib folder
import org.jasypt.util.text.*;

// require spring jars in classpath
public class LDAPPreAuthenticationFilter extends PreAuthenticationFilter {
  public LDAPPreAuthenticationFilter() {
    keepDomain = false;
    credentialProvider = null;
//implements upn lookup if set to true can bbe overridden in application-context.xml
    useUPN = true;
    ldapBindDn = "";
//for example CN=adm2hartley,OU=Admins,OU=Users and Groups,OU=GC
CTR,OU=CTR,OU=Organizations,DC=company,DC=com";
    ldapBindPassword = "";
    ldapBaseDn = ""; //OU=Admins,OU=Users and Groups,OU=GC
CTR,OU=CTR,OU=Organizations,DC=company,DC=com";
    ldapServerURL = ""; //ldap://ldap.company.com:389";
//to implement - the filter would be iserted in an AND clause
//ldapExcludeFilter = (!(dn=*Contractors*))
  }

  protected String getAuthenticatedUsername(HttpServletRequest request) {
    String username = null;
    if (credentialProvider == null || credentialProvider.getUserName(request) != null &&
credentialProvider.getUserName(request).equals("")) {
      username = request.getRemoteUser();
      System.out.println("Default remote user : " + username);
      if (username != null)
        if (username.length() == 0)
          username = null;
        else if (!keepDomain) {
          int i = username.indexOf('\\');
          username = username.substring(i + 1);
        }
    } else {
      username = credentialProvider.getUserName(request);
    }
    if (useUPN) {
      System.out.println("Using UPN. SamAcctName : " + username);
      String userUPNname = lookupUPN(username);
      System.out.println("Found this UPN : " + userUPNname);
      return userUPNname;
    } else {
      System.out.println("Not using UPN. Username : " + username);
      return username;
```

```
  }
}

public boolean useUPN() {
  return useUPN;
}

public void setuseUPN(boolean useUPN) {
  this.useUPN = useUPN;
}

public void setCredentialProvider(CredentialProvider credentialProvider) {
  this.credentialProvider = credentialProvider;
}

public String getldapBindDn() {
  return ldapBindDn;
}

public void setldapBindDn(String ldapBindDn) {
  this.ldapBindDn = ldapBindDn;
}

public String getldapBindPassword() {
  return ldapBindPassword;
}

public void setldapBindPassword(String ldapBindPassword) {
  this.ldapBindPassword = ldapBindPassword;
}

public String getldapBaseDn() {
  return ldapBaseDn;
}

public void setldapBaseDn(String ldapBaseDn) {
  this.ldapBaseDn = ldapBaseDn;
}

public String getldapServerURL() {
  return ldapServerURL;
}

public void setldapServerURL(String ldapServerURL) {
  this.ldapServerURL = ldapServerURL;
}

protected String lookupUPN(String samUserName)

String myUPNuser = "";

{
  try {
    Hashtable env = new Hashtable();
    String adminName = ldapBindDn;
    String adminPassword = ldapBindPassword;
    String ldapURL = ldapServerURL;
    env.put(Context.INITIAL_CONTEXT_FACTORY, "com.sun.jndi.ldap. LdapCtxFactory");
    //set security credentials, note using simple cleartext authentication
```

```
      env.put(Context.SECURITY_AUTHENTICATION, "simple");
      env.put(Context.SECURITY_PRINCIPAL, adminName);
      //decrypt the password if it starts with *** else it is clear text, use as is.
      if (adminPassword.indexOf("***") == 0) {
        adminPassword = adminPassword.substring(3);
        BasicTextEncryptor textEncryptor = new BasicTextEncryptor();
        // this password the same as that used to encrypt the adminpassword with FilterEncrypt
        textEncryptor.setPassword("PssWrd");
        adminPassword = textEncryptor.decrypt(adminPassword);
      }
      env.put(Context.SECURITY_CREDENTIALS, adminPassword);
      //connect to my domain controller
      env.put(Context.PROVIDER_URL, ldapURL);
      //Create the initial directory context
      LdapContext ctx = new InitialLdapContext(env, null);
      //Create the search controls
      SearchControls userSearchCtls = new SearchControls();
      //Specify the search scope
      userSearchCtls.setSearchScope(SearchControls.SUBTREE_SCOPE);
      //specify the LDAP search filter to find the user in question
      String userSearchFilter = "(&(objectClass=user)(sAMAccountName=" + samUserName + "))";
      //Specify the Base for the search
      String userSearchBase = ldapBaseDn;
      //Specify the attributes to return
      String userReturnedAtts[] = {"userPrincipalName"};
      userSearchCtls.setReturningAttributes(userReturnedAtts);
      //Search for objects using the filter
      NamingEnumeration userAnswer = ctx.search(userSearchBase, userSearchFilter, userSearchCtls);
      //Loop through the search results
      while (userAnswer.hasMoreElements()) {
        SearchResult sr = (SearchResult) userAnswer.next();
        Attributes attrs = sr.getAttributes();
        if (attrs != null) {
          myUPNuser = attrs.get("userPrincipalName").get().toString();
        }
      }
    } catch (Exception e) {
      System.err.println("Problem searching directory: " + e);
      System.out.println("Problem searching directory: " + e);
    }
    return myUPNuser;
  }

  // added for UPNLookup
  boolean useUPN;
  String ldapBindDn;
  String ldapBindPassword;
  String ldapBaseDn;
  String ldapServerURL;
}
```

This class contains a reference to FilterEncrypt, which is shown below:

```
import org.jasypt.util.text.*;
public class FilterEncrypt
{
  public static void main(String arg[])
  {
    try
    {
        String GivenPwd = arg[0];
        BasicTextEncryptor textEncryptor = new BasicTextEncryptor();
        textEncryptor.setPassword("PssWrd");
        String myEncryptedText = "***" + textEncryptor.encrypt(GivenPwd );
        System.out.println("here is the encrypted : " + myEncryptedText);
    }
  catch (Exception e)
  {
    e.printStackTrace();
  }
}
}
```

8.16. Common Access Card (CAC) sign-on

Starting with version 9.32, the Service Manager web client supports CAC sign-on. System Administrators can configure the Service Manager server and web client to automatically log on using CAC authentication. CAC sign-on enables users to log in to the web client directly with a smart card that stores a valid user certificate, and users only need to enter a card PIN, instead of a user name and password.

Note

Note The Service Manager Windows client doesn't support CAC sign-on.

Supported smart cards

During CAC sign-on, the Service Manager web tier gets access to the user authentication public certificate and its counterpart private key through the underlying client crypto architecture. In other words, Service Manager doesn't directly communicate with the card reader. Technically, Service Manager supports any smart cards that store an X.509 user authentication certificate and are designed to work with smart card middleware (such as ActivClient) that is installed on the user's computer. The term "Common Access Card (CAC)" in its origin refers to the Unites States Department of Defense (DoD) issued cards; however, in the Service Manager documentation the term CAC is used in a broader scope and refers to any of the supported smart cards. Two examples of such smart cards are the Common Access Card issued by DoD and Personal Identity Verification (PIV) cards issued by other US Government agencies.

CAC authentication

A CAC smart card enables two-factor authentication: the CARD being something you have, and the CAC PIN being something you know. The CAC stores, in a secure way, the card holder's public certificates and its corresponding private keys. One of those stored certificates with its private keys is used for user authentication. Service Manager uses SSL and CAC authentication, which utilizes the certificate in the CAC card to identify the CAC card holder details and leverages the built-in SSL user authentication abilities to ensure that the CAC owner possess the private key. The certificate key-pair proves that this person is who the person claims to be. For example, if the certificate claims the public key is owned by userX and it was signed by a Certificate Authority (CA), when the certificate is presented assuming Service Manager trusts the CA (which means the CA public key is put in Service Manager's trusted CA store), Service Manager would trust the information and then challenge the person presenting the certificate to prove the person owns the matching private key. If the person provides the proper PIN, that private key is unlocked and then Service Manager authenticates the client as the one detailed in the certificate.

Flexible mapping between distinguished name and login name

During CAC sign-on, a unique attribute in the user's certificate is retrieved as the user's Distinguished Name (DN) and sent to the SM server as the user's login name.

Different organizations may have different definitions for the same attribute in the user's certificate. For example, your organization may use employee ID for Subject.CN, while another may use user's full name for it. In addition, it is also possible that an organization needs to use an Subject Alternative Name (SAN) sub-attribute as DN. For this reason, the Service Manager web tier provides a configuration file (`<web tier installation path>\WEB-INF\classes\cacconf.properties`), which contains two parameters (certificateFieldExtractDN and certificateOIDMapping) to allow flexible mapping between DN and login name. For more information, see Example: enabling CAC sign-on.

Required configurations

CAC sign-on requires the following configurations:

- Both of the Service Manager server and web client have been configured to support CAC sign-on mode.
- Two-way SSL is configured between the web server (or the web application server if no web server) and browser.
- Two-way SSL is configured between the Service Manager server and all clients (the web and Windows clients, and web services clients such as Mobility, and other integrated products).

  Note

  Note Enabling CAC sign-on on the SM server side will affect all SM clients - all SM clients can only connect to the server through two-way SSL.

- JDK 1.6 or above is installed on the host machine running the web tier.

Compatibility with other sign-on modes

The following table describes whether or not users can use other sign-on modes when CAC sign-on is enabled in the SM server and web client.

Mode	Description
Basic Authentication	**Web client**: Users can no longer log in to the web client by entering a user name and password (that is, using basic authentication). It is impossible that some web client users use CAC sign-on while some others use basic authentication. All users must use CAC sign-on. **Windows client**: Users can log in to the Windows client only with basic authentication, but two-way SSL authentication is required between the server and Windows client.
Trusted Sign-On (TSO)	When both CAC sign-on and TSO are configured in the same web tier, CAC sign-on takes precedence. For this reason, don't enable CAC sign-on and TSO in the same web tier at the same time.
Lightweight Single Sign-On (LW-SSO)	The following discussion assumes a scenario where Product A (which doesn't support CAC access) is integrated with Service Manager and LW-SSO are enabled on both product sides. **Outgoing LW-SSO**: SM web client users can directly access product A through LW-SSO. In other words, outgoing LW-SSO from the web tier works well. **Incoming LW-SSO**: Incoming LW-SSO is ignored, and users can only access SM from product A with a CAC.

Conditions for CAC access

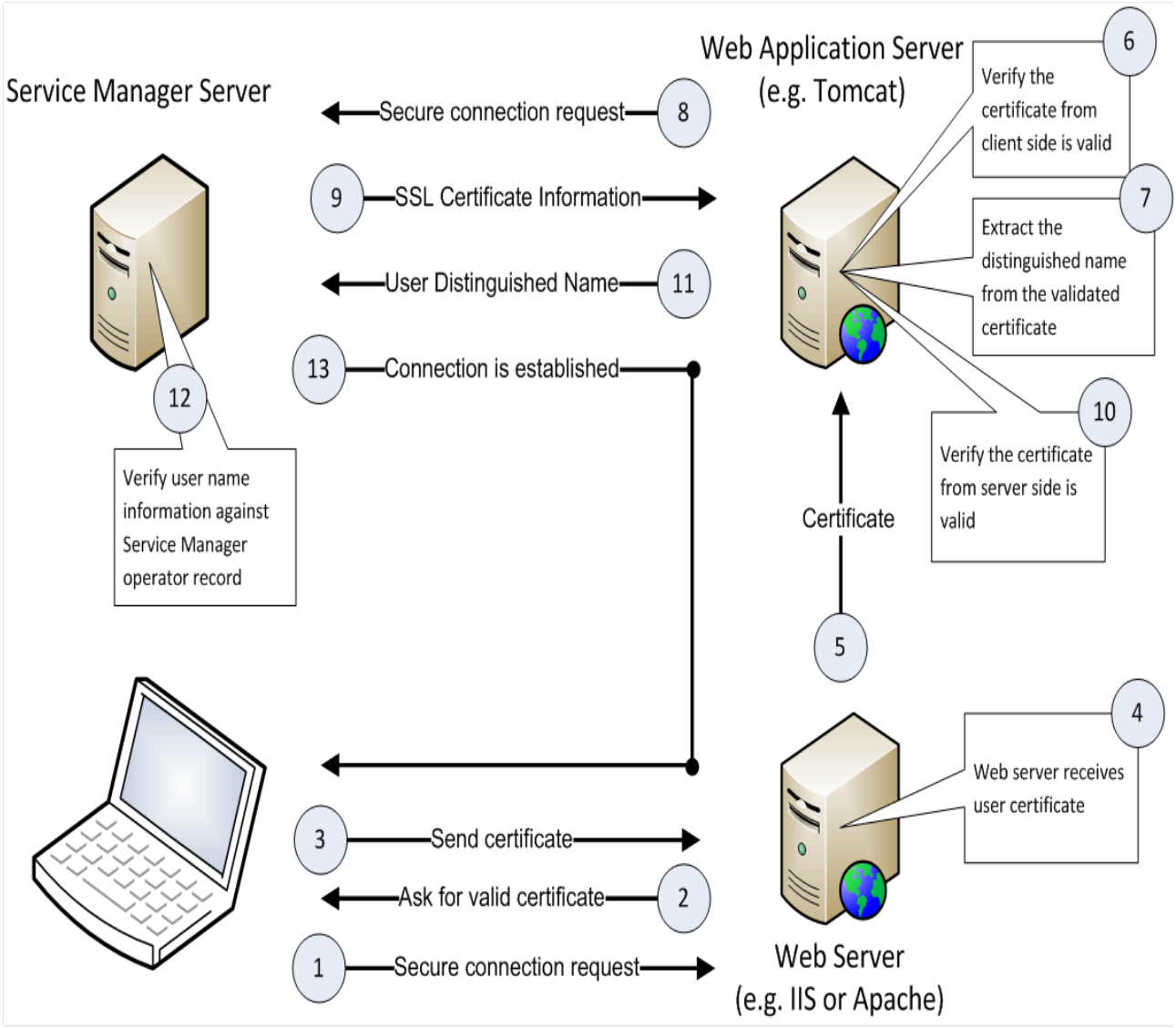
In a CAC sign-on scenario, the Service Manager server grants access to web clients only if the following conditions are met:

- The web browser from the client side sends the user certificate to the web server, and then the web server passes the certificate to the web application server.
- The user certificate passes the validation by the web tier.
- The user's logon credentials match an existing operator record in Service Manager or a valid LDAP source that Service Manager recognizes.
- The web client presents a signed SSL certificate.

> **Note**
> Note Each Windows client must also present a signed SSL certificate to the server.

CAC connection process

The following figure depicts the connection process between a web server, a web application server, and the Service Manager server in a CAC sign-on scenario.



The following describes how the process works:

1. The user inserts a CAC into a card reader.
2. The smart card middleware (such as ActivClient) automatically loads the user's personal certificate ("user certificate") from the CAC to the browser.
3. The browser prompts the user to select an installed certificate to log in.
4. The user selects the user certificate to log in.
5. The browser prompts the user to enter the PIN of the user's CAC.
6. The user enters the correct PIN.
7. The web server passes the user certificate to the web application server for validation based on a pre-configured validation strategy, which can include any combinations of the following checks except that at least one of the following options must be included: using a local CRL, using an online CRL, and using the OCSP.
   • Check if the issuer is from a trusted CA (Certificate Authority)
   • Check if the certificate is revoked using a local Certificate Revocation List (CRL)
   • Check if the certificate is revoked using an online CRL
   • Check if the certificate is revoked using the Online Certificate Status Protocol (OCSP)
   • Check if the certificate has expired
   • Check if the certificate type is Smart Card
8. The web application server extracts the Distinguished Name (DN) from the user certificate and sends the DN as the user's user name to the Service Manager server.

9. The server verifies the DN against the `name` field in the `operator` table (when LDAP is not configured) or against the LDAP attribute mapped to this field (if LDAP is configured). If the DN value matches an operator name, the server logs the user in; otherwise the server rejects the login request.

Related topics

Example: enabling CAC sign-on

8.16.1. Requirements for CAC sign-on

This configuration is intended for customers who have an Service Manager server running in a secured environment and want users to be able to log on to the server using a Common Access Card (CAC), without the need to enter a user name and password.

Parameters required in sm.ini

Service Manager 9.32 or later supports CAC sign-on with two-way SSL enabled and you must set the value of the `ssl_reqClientAuth` parameter to 2.

The server parameters required for CAC sign-on are listed as follows.

```
cacsignon:1

ssl:1
sslConnector:1
ssl_reqClientAuth:2
keystoreFile:<servercert.keystore>
keystorePass:<keystoreFile password>
truststoreFile:cacerts
truststorePass:<truststoreFile password>
ssl_trustedClientsJKS:<trustedclients.jks>
ssl_trustedClientsPwd:<ssl_trustedClientsJKS password>
```

Parameters required in web.xml

The following parameters are required in the web tier configuration file (`web.xml`):

- `isCustomAuthenticationUsed=false`
  Set the value to false to make Service Manager send the current user name in the HTTP header.
- `CACLogin=true`
  Set the value to true to enable the CAC logon mode in the web tier.

Other requirements

- Configure your web application server to enable CAC authentication. You do so by updating the web tier's `application-context.xml` file. For details, see Example: enabling CAC sign-on.
- When CAC logon is enabled in the server, you can set `ssl_reqClientAuth:2` in the `sm.ini` file. You must then create unique client SSL certificates for each Service Manager client wanting to access Service Manager with CAC. For example, if you have 20 Service Manager Windows clients, you must create 20 unique client SSL certificates. If you have 4 Service Manager Web Tier servers, you must create 4 unique client SSL certificates for them. In addition, you need to configure SSL in the web tier configuration file (`web.xml`) and also in the Windows client Preferences setting. For details, see Configure two-way SSL.

  Tip

  Tip If maintaining these unique client SSL certificates incurs unsustainable IT operation costs, you can consider the use of the `acceptsharedcert:1` parameter, which enables all clients to use a "shared certificate". For more information, see Parameter: acceptsharedcert

- CAC sign-on requires two-way SSL connections between the web server (or web application server if no web server) and the user's browser. You need to set up two-way SSL on the web server, or on the web application server (if you have no web server deployed).

- Each CAC user must have an operator record created in Service Manager.

Related topics

Parameter: acceptsharedcert

8.16.2. Example: enabling CAC sign-on

This example assumes that you are using Tomcat as your web application server and Apache as your web server.

Prerequisites

Before you proceed, make sure you have met the following prerequisites:

- Your organization has already a Common Access Card (CAC) security system in place; each Service Manager user owns a CAC card that stores the user's personal certificate ("user certificate") and you have downloaded your organization's CAC root CA certificate.
  Your CAC root CA certificate refers to the root CA certificate that signs the user certificates stored in the CACs. Later, you will need to:
  - copy it to your web tier host and specify its location in the `RootCertPath` parameter. See *Task 3: Configure the web tier to use CAC sign-on.*
  - let users import it to their web browser. See *Task 7: Configure web browsers to enable web client users to use CAC.*
- You have Keytool and OpenSSL installed on a machine on which you will generate SSL certificates.
- You have configured the web server and the web application server to use SSL. CAC sign-on requires two-way SSL authentication between the web server/web application server and the user's browser.For more information on configuring two-way SSL on your web server (Apache or IIS), see Configure two-way SSL.

Configuration Tasks

To enable CAC sign-on in your Service Manager system, complete the following tasks.

Task 1: Enable required SSL encryption and trusted clients.

Enable required SSL encryption between the SM server and Windows and web clients. For details, see Configure two-way SSL.

When the configuration is complete, verify that you can log in to the web client as usual and log in to Windows client with SSL encryption.

Task 2: Enable CAC sign-on in the SM server.

Set required parameters in the server configuration file (`sm.ini`). For details, see Requirements for CAC sign-on.

Task 3: Configure the web tier to use CAC sign-on.

1. Stop the web application server running the web tier.
2. Copy your CAC root certificate (s) to a directory on your web tier host. For example: <webtier>/WEB-INF/
3. Set the following parameter values in the web tier's `web.xml` file.

Parameter	Value
`isCustomAuthenticationUsed`	`false`
`CACLogin`	`true`

4. Configure your web application server to use CAC as the trusted authentication source.
   1. Navigate to the `<web tier installation direcotry>\WEB-INF\classes` folder, and open the `application-context.xml` file in a text editor.
   2. Search for the following string:
      `<sec:filter-chain pattern="/**" filters="securityContextPersistenceFilter,anonymousAuthFilter"/>`

3. Replace the search string with the following text to use CAC authentication as your trusted authentication source:
`<sec:filter-chain pattern="/**" filters="securityContextPersistenceFilter,cacFilter,`

`anonymousAuthFilter"/>`

Note

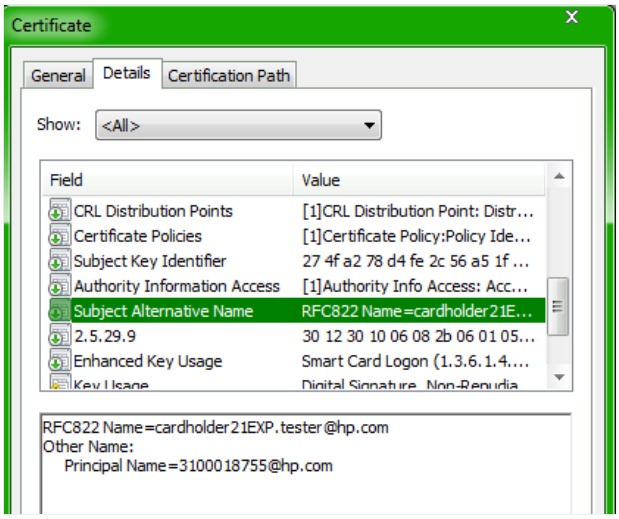
Note If you also want to enable lightweight single sign-on (LW-SSO) for integrations, add `cacFilter` followed by `lwSsoFilter`, as shown in the following:
`<sec:filter-chain pattern="/**" filters="securityContextPersistenceFilter,cacFilter,`

`lwSsoFilter,anonymousAuthFilter"/>`
Be aware that CAC sign-on can only work with outgoing LW-SSO (LW-SSO from SM to another product). For information about how to enable LW-SSO in the web tier, see Configure LW-SSO in the Service Manager Web tierauthenticationManager

```
<ref bean="authenticationManager"/>
    </property>
    <property name="defaultRole">
       <value>ROLE_PRE</value>
    </property>
    <property name="cacProps">
       <ref bean="cacConfiguration"/>
    </property>
</bean>
<bean id="cacConfiguration"
class="org.springframework.beans.factory.config.PropertiesFactoryBean">
    <property name="location"><value>classpath:cacconf.properties</value>
</property>
</bean>
```

4. Save the file.

5. Configure CAC authentication in the web tier's `cacconf.properties` file.
    1. Open `<web tier installation path>\WEB-INF\classes\cacconf.properties` in a text editor.
    2. Specify the following parameters.
        - **Paramter:** `certificateFieldExtractDN`
          Default: `Subject.CN`
          Description:
          This parameter specifies a unique attribute in the user certificate that the web application server extracts as a Distinguished Name (DN) and sends to the SM server as the user's user name.
          You can specify a Subject attribute in this parameter. By default, the Subject CN attribute (the certificate owner's Common Name) is extracted as the user's DN.
          You can also specify this parameter as one of the following Subject Alternative Name (SAN) sub-attributes: RFC822 Name, and Other Name (see the figure below).



- If using a sub-attribute of Other Name, specify the parameter in this format: `SAN_OTHERNAME.<Alias>`, where `<Alias>` is a name that you specify for the sub-attribute. The user certificate's SAN Other Name value that binds to the OID will be extracted as the DN. For example, if you use Other Name's Principal Name sub-attribute (also known as UPN, which should be mapped to OID "1.3.6.1.4.1.311.20.2.3"), you can specify the following value (where PrincipalName is an alias for Principal Name):
  `certificateFieldExtractDN=SAN_OTHERNAME.PrincipalName`

  Tip
  Tip As a best practice, avoid including white spaces in alias names. For example, for Principal Name, specify `PrincipalName` or `PN` as an alias.

  Note
  Note Next, you will specify a mapping between the alias and object ID (OID) of the specified sub-attribute of SAN Other Name, by using the *certificateOIDMapping* parameter.

- If using the RFC822 Name sub-attribute, specify the following value:
  `certificateFieldExtractDN=SAN.RFC822`

Examples:

```
certificateFieldExtractDN=Subject.E
certificateFieldExtractDN=SAN_OTHERNAME.PrincipalName
```

- **Paramter:** `certificateOIDMapping`
  Default: (empty)
  Description:
  This parameter specifies a mapping between the `<Alias>` you specified in `certificateFieldExtractDN` and an OID. Specify the parameter in this format: `<Alias>:<OID>`.
  For example, for SAN's Principal Name, specify this mapping:

  ```
  certificateOIDMapping=PrincipalName:1.3.6.1.4.1.311.20.2.3
  ```

- **Paramter:** `validationStrategy`
  Default: (empty)
  Description:
  This parameter specifies how the web application server verifies each user certificate.

You can specify a combination of the following values, each of which represents a checking method:

- `1`: Check the revocation status of the user certificate using a local Certificate Revocation List (CRL).
- `2`: Check the revocation status of the user certificate using an online CRL.
- `3`: Check the revocation status of the user certificate using the Online Certificate Status Protocol (OCSP).
- `4`: Check if the user certificate is of the Smart Card type.

Pay attention to the following:

- You can't leave this parameter empty.
- If you specify more than one value, separate them with a comma. In this case, all specified checking methods are used in the sequence they are listed in the parameter.
- You must specify at least one of the following values: 1, 2, and 3. If you don't do so, users will not be able to log in; instead, they will see a message in their browser that states "An error has occurred"; in the meantime, an error message will occur in the web application server's `sm.log` file.
- Smart Card checking (value 4) is mandatory for CACs, because CACs are smart cards.
- You can configure CRL or OCSP checking either in your web server or in this parameter, but not in both.

Example:

```
validationStrategy=3,4
```

- **Parameter:** `RootCertPath`
  Default: (empty)
  Description:
  This parameter specifies the file path (either absolute or relative) to the issuer's CA certificate.

  Note
  Note The issuer's CA certificate refers to the root CA certificate that signs the user certificates stored in the CACs (also referred to as the "CAC root CA certificate" in this document). If you need to specify multiple CAC root CA certificates, use a comma to separate them.

  Example:

  ```
  RootCertPath=/WEB-INF/HPCAB.cer,/WEB-INF/HPCA.cer
  ```

- **Parameter:** `CRLDownloadURL`
  Default: (empty)
  Description:
  Specifies the URL that identifies the location of your online certificate revocation list (CRL).
  You can leave this parameter empty, and then the CRL Distribution Point from the user certificate being validated is retrieved at runtime, which might cause a slightly slower performance; however, if FIPS mode is enabled in the web client (see FIPS mode), you must explicitly specify this parameter (can't leave it empty).
  If your root CA has multi-level intermediate CAs:
  If your root CA certificate specified in `RootCertPath` has multi-level intermediate CA certificates, you should either specify the multi-level CRLs of the root CA and intermediate CA certificates (comma-separated), or leave the parameter empty.

Examples:

```
CRLDownloadURL=http://onsitecrl.example.com/LatestCRL.crl
CRLDownloadURL=http://onsitecrl.example.com/LatestCRL.crl,http://onsitecrl.example.com/IntermediateCRL.crl
```

- **Parameter:** `OCSPResponderURL`
  Default: (empty)
  Description:
  Specifies the URL that identifies the location of the OCSP responder. By default, this parameter is not specified, and the OCSP responder location is retrieved from the user certificate being validated.
  If you specify multiple root CA certificates:
  If you specify multiple root CA certificates in `RootCertPath`, you must leave `OCSPResponderURL` empty. In this case, the OCSP Responder URL stored in the user certificate being validated is retrieved instead during runtime.
  Example:
  `OCSPResponderURL=http://ocsp.example.net:80`
- **Parameter:** `CRLStoreLocation`
  Default: (empty)
  Description:
  Specifies the location of your local CRL.

  > **Note**
  > Note This parameter also supports multi-level intermediate CRLs (comma-separated).

  Examples:

  ```
  CRLStoreLocation=/WEB-INF/classes/LatestCRL.crl
  CRLStoreLocation=/WEB-INF/classes/LatestCRL.crl,/WEB-INF/classes/IntermediateCRL.crl
  ```

3. Specify other parameters as needed.

   > **Note**
   > Note The validation strategy you specified determines which parameters are required, recommended, or optional (see the following table).

Method	How to Enable	Parameters
Local CRL checking	`validationStrategy=1`	▪ `RootCertPath` (Mandatory) ▪ `CRLStoreLocation` (Mandatory)
Online CRL checking	`validationStrategy=2`	▪ `RootCertPath` (Mandatory) ▪ `CRLDownloadURL` (Recommended): If it's not specified, the default URL stored in the user certificate is used. ▪ `CRLRefreshScheduler` (Optional): This parameter specifies an interval at which the online CRL stored in the browser cache is refreshed. The default is 1440 minutes.
OCSP checking	`validationStrategy=3`	▪ `RootCertPath` (Mandatory) ▪ `OCSPResponderURL` (Recommended): If it's not specified, the default URL stored in the user certificate is used. ▪ `OCSPServerCertPath` (Optional): This parameter specifies the file path to the OCSP server CA certificate. If it's not specified, the `RootCertPath` is used as the certificate for the OCSP server.
Smart Card checking	`validationStrategy=4`	No other parameters are required. To ensure greater security, Smart Card checking is required for CAC cards, which are smart cards.

4. Save the file.
   The following is an example configuration of the `cacconf.properties` file.

```
certificateFieldExtractDN=Subject.CN
certificateOIDMapping=PrincipalName:1.3.6.1.4.1.311.20.2.3
validationStrategy=2
RootCertPath=/WEB-INF/CA.cer
CRLRefreshScheduler=1440
CRLDownloadURL=http://onsitecrl.verisign.com/myCompany/LatestCRL.crl
CRLStoreLocation=
OCSPResponderURL=
OCSPServerCertPath=
```

6. If a proxy is needed to access your OCSP responder URL or online CRL URL, configure a proxy in your web application server (in this example, Tomcat). You can use either of the following approaches depending on your Tomcat installation.
   To configure a proxy using the Configure Tomcat shortcut:
   1. From your operating system's Start menu, click **All Programs** > **<Apache Tomcat>** > **Configure Tomcat**.
   2. On the **Java** tab, append your proxy configuration lines to the Java Options text box. For example:

      ```
      -Dhttp.proxyHost=<Proxy Server FQDN>
      -Dhttp.proxyPort=<Proxy Server Port, for example, 8080>
      -Dhttp.nonProxyHosts="<Service Manager Server Host FQDN>|<Tomcat Server Host FQDN>"
      ```

   To configure a proxy using the `catalina.bat` file:
   1. Open the `<Tomcat>/bin/catalina.bat` file in a text editor.
   2. Add the following lines:

      ```
      set JAVA_OPTS=%JAVA_OPTS%
      -Dhttp.proxyHost=<Proxy Server FQDN>
      -Dhttp.proxyPort=<Proxy Server Port, for example, 8080>
      -Dhttp.nonProxyHosts="<Service Manager Server Host FQDN>|<Tomcat Server Host FQDN>"
      ```

   3. Save the `catalina.bat` file.
7. Configure Tomcat to get the client certificate from the specific request header `SSL_CLIENT_CERT` that's set by the Apache HTTP Server. Add `SSLValve` to the `<Host>` element in the <TOMCAT_HOME>/conf/server.xml file as follows.

   ```
   <Host name="localhost" ...>
     <Valve className="org.apache.catalina.valves.SSLValve" />
     ...
   </Host>
   ```

Task 4: Configure each Windows client to use SSL.

When CAC sign-on is enabled for the server and web client, the Windows client must run in SSL mode.

Do the following for each Windows client:

1. Make sure that you have already configured SSL connection between the server and the Windows client. See *Task 1: Enable required SSL encryption and trusted clients.*
2. Open a client connection.
3. On the **Advanced** tab, select **Use SSL Encryption**, and click **Apply**.

Task 5: Connect your web application server to the web server.

First, connect Tomcat to Apache. Refer to Configure a web server to redirect requests to the web tier for detailed instructions.

Next, configure the web server and the web application server to use SSL. CAC sign-on requires two-way SSL authentication between the web server/web application server and the user's browser. For more information on configuring two-way SSL

on your web server (Apache or IIS), see Configure two-way SSL.

If the Apache HTTP Server connects to Tomcat by using the AJP protocol, you need to configure the Apache HTTP Server to require the client to present a valid certificate. For example, add the following additional settings to the <APACHE24_HOME>/conf/extra/httpd-ahssl.conf file.

```
<Location "/webtier">
  SSLVerifyClient require
</Location>
```

If the Apache HTTP Server connects to Tomcat by using the HTTPS protocol, you need to configure the Apache HTTP Server to require the client to present a valid certificate and configure the Apache HTTP Server to forward the client certificate to the backend Tomcat through the `SSL_CLIENT_CERT` request header. For example, add the following additional settings to the <APACHE24_HOME>/conf/extra/httpd-ahssl.conf file.

```
<Location "/webtier">
  SSLVerifyClient require
  SSLOptions +ExportCertData
  RequestHeader set SSL_CLIENT_CERT "%{SSL_CLIENT_CERT}s"
  ...
</Location>
```

Note: Make sure the `mod_headers` modules are loaded in the <APACHE24_HOME>/conf/httpd.conf file.

Task 6: Create operator records for CAC users.

The web server extracts information from the user certificate based on the `certificateFieldExtractDN` value defined in the `<web tier installation path>\WEB-INF\classes\cacconf.properties` file.

You must create an operator record for each CAC user. The configuration varies depending on whether Lightweight Directory Access Protocol (LDAP) is enabled on the SM server or not.

If LDAP is disabled for the SM server

For each CAC user certificate, create an operator record whose Login Name (`operator.name`) field has the same value as the distinguished name (DN) value in the certificate.

See the following table for an example, which indicates that you need to create an operator record whose Login Name is the email address stored in the user certificate.

In the cacconf.properties file	In the user certificate	In the operator record
certificateFieldExtractDN=Subject.E (DN=certificate owner's email address)	Subject.E=jack.smith@mycompany.com	Login Name=jack.smith@mycompany.com

If LDAP is enabled for the SM server

Configure LDAP mapping for the `operator` and `contacts` tables, and make sure that the `operator.name` and `contacts.operator.id` fields are mapped to the same LDAP attribute that corresponds to the certificate attribute defined in `certificateFieldExtractDN` .

The following table lists the fields in the `operator` and `contacts` tables that you must map to an LDAP attribute.

Note

Note The LDAP attribute names here are provided only as an example, which may differ from your actual mappings. In this example, both of the `name` field in the `operator` table and the `operator.id` field in the `contacts` table are mapped to the `sAMAccountName` attribute, which corresponds to the user certificate attribute defined in `certificateFieldExtractDN`.

operator file

Field Name	LDAP Attribute Name
contact.name	cn
full.name	cn
name	sAMAccountName

contacts file

Field Name	LDAP Attribute Name
contact.name	cn
first.name	givenName
full.name	cn
last.name	sn
operator.id	sAMAccountName

For details about LDAP configuration, see the Service Manager LDAP Best Practices Guide.

Task 7: Configure web browsers to enable web client users to use CAC sign-on.

Before a user can log on with a CAC, the CAC root CA certificate(s), the root CA certificate generated during your web server (or web application server) SSL configuration, as well as the user's personal certificate must be installed in the user's browser.

When users insert their CACs, the browser automatically loads their personal certificates. Users only need to manually install the root CA certificates to Windows Certificate Store.

Follow these steps:

1. Right-click your CAC root CA certificate (CAC_ca.crt), and then click the **Install Certificate** menu to launch the certificate import wizard.
2. Select the **Local Machine** option for the store location, then click the **Next** button.
3. Select the **Place all certificates in the following store** option, and then click the **Browse** button. Sselect the **Trusted Root Certification Authorities** from the prompted dialog.
4. Click the **Next** button, the click the **Finish** button to finish the wizard.

For Firefox, you need to perform the following additional steps to let Firefox use the Windows Trusted Root Certification Authorities.

1. Navigate to the **about:config** URL to open the settings editor page.
2. Set the **security.enterprise_roots.enabled** option to **true**.

Task 8: Test your CAC sign-on setup.

1. Start the Service Manager server, the web server and the web application server (in this example, Apache and Tomcat).
2. Verify that you can log in to the web client with your CAC.

1. Open the web tier login URL in your browser:

   ```
   http://<Your Apache Host FQDN>:<port>/<webtier>/index.do
   ```

   or if you have no Apache but only Tomcat deployed:

   ```
   http://<Your Tomcat Host FQDN>:<port>/<webtier>/index.do
   ```

   The system prompts you to select a certificate from a list of the certificates installed in your browser.
2. Insert your CAC. The browser automatically loads your personal certificate.

   Note

   Note At your first CAC login, a browser refresh might be needed to make your personal certificate appear in the list.

3. Select your personal certificate from the list.
4. Enter your CAC PIN when prompted.
   Service Manager automatically logs you in.

CAC sign-on is now successfully set up in your web tier.

3. Verify that you can log in to the Windows client through SSL encryption.
   1. Start the Windows client, and check that the **Use SSL Encryption** option is enabled.
   2. Enter a valid username and password to log in.
      Service Manager logs you in successfully. Two-way SSL is successfully set up in your Windows client.

Now, your CAC sign-on setup is complete.

Related topics

Common Access Card (CAC) sign-on
Requirements for CAC sign-on

8.17. Configure IdM authentication in Service Manager

To integrate with the IdM service, you must have an IdM service available.

- In the classic mode (a standalone Service Manager system without integrating with the SMA Service Portal), install a standalone IdM service first. For detailed instructions, see Install Identity Manager (IdM).
- In the mixed mode (SMA-SM, a Service Manager system that integrates with the SMA Service Portal), you don't need to install the standalone IdM service. Instead, leverage the IdM service deployed in the SMA suite and then integrate Service Manager with the IdM service.

Typically, you will deploy a reverse proxy (Apache HTTP Server or IIS) or a hardware load balancer on top of the IdM service in the production environment. This section uses the Apache HTTP Server as an example for the reverse proxy and assumes the reverse proxy URL is https://apache.yourdomain.net. The later tasks use this sample reverse proxy URL and you need to replace it with the actual URL in your environment.

Compatibility with legacy LW-SSO

The IdM authentication is compatible with the legacy LW-SSO solution. If the IdM authentication is enabled in Service Manager, and another legacy application has legacy LW-SSO enabled, users can access either product from the other product without the need to enter a user name and password.

Caution

Caution

- The IdM service generates a LWSSO_COOKIE_KEY cookie and can also authenticate a user using tokens in the cookie, which makes it compatible with the legacy LW-SSO solution. As a result of this, other web applications that have legacy LW-SSO enabled can still get integrated by using LW-SSO. Note that the web tier doesn't renew the LW-SSO cookie now when the IdM authentication is enabled. If you want to support LW-SSO compatibility with other products, you need to set the expiration period of the LW-SSO token to a long time in the global settings of the IdM admin console (for example, 480 minutes).
- The Service Manager Server can have both IdM authentication and LW-SSO enabled if needed. This allows other applications of the company to access the SM Server through LW-SSO when necessary.
- If both LW-SSO and the IdM authentication are enabled in the Service Manager server, you only need to enable the IdM authentication in the Service Manager web tier or Mobility client. Do not enable LW-SSO in these clients at the same time.

Note

Limitations

The LW-SSO compatibility has the following limitations.

- For backend integration, legacy applications using legacy LW-SSO must share an LDAP server with Service Manager. Otherwise, you need to synchronize users between legacy applications and Service Manager by using external tools. Legacy applications may use a dedicated integration user to log in to Service Manager, and you can either add the integration user to the LDAP directory service or add the user to Service Manager directly.
- For UI-based SSO, legacy applications using legacy LW-SSO and Service Manager (server, web tier, and Mobility) must be in the same domain as the IdM service. This is because the IdM service needs to read the LW-SSO cookie written by the applications. Legacy applications and the IdP must share an LDAP server, or you need to synchronize users between legacy applications and the IdP by using external tools. Depending on the authentication type you configured in the IdM service, the IdP can be the IdM service itself for the DB authentication or the LDAP authentication, or ADFS for SAML-based SSO, or Keycloak for OIDC-based SSO.

Related topics

Configure IdM authentication in the Service Manager server
Configure IdM authentication in the Service Manager web tier
Configure IdM authentication in the Mobility client
Configure IdM authentication in Collaboration
Configure IdM parameters

8.17.1. Configure IdM authentication in the Service Manager Server

To enable IdM authentication in the Service Manager Server, stop the Service Manager server and then add the IdM-related parameters to the `<SM_SERVER>/RUN/sm.ini` file as follows.

If you set **Signature Algorithm** as **HS256** in IdM administration console:

```
idmssoenabled:1
idmtenant:SM
idmsigningkey:<your IdM signing key>
```

If you set **Signature Algorithm** as **RS256** in IdM administration console:

```
idmssoenabled:1
idmtenant:SM
idmendpoint:https://apache.yourdomain.net/idm-service
idmhttpuser:idmTransportUser
idmhttppassword:<your password of idmTransportUser>
idmssltrustall:0
idmtruststore:smrootca.p12
idmtruststorepassword:<your password of smrootca.p12>
idmtruststoretype:pkcs12
```

Note

Note

- For more details about the IdM-related parameters, see IdM parameters for the Service Manager server.
- For more information about how to set the appropriate values for common IdM parameters, see How to get IdM-related settings.

Next, set up user synchronization by using either IdM or LDAP.

Caution

Caution:

- Don't set up user synchronization from both IdM and LDAP. Otherwise, performance issues may occur during user login.
- If you configured SAML-based Single Sign-On and selected **User-Principal-Name** (instead of **SAM-Account-Name** documented in step 10.b of Create a trusted relying party in ADFS) as **LDAP Attribute**, you need to do the following:
  - Synchronize users from LDAP.
  - Set the `ldapquerymapping:samAccountName=userPrincipalName` parameter in sm.ini. See Startup parameter: ldapquerymapping for details.
  - Make sure that the `user.principal.name` field of `operator` is mapped to `userPrincipalName` when you Define file and field-level mappings to an LDAP server.

Synchronize users from IdM

Add IdM parameters

You need to add additional IdM parameters to connect to the IdM service. For example:

```
idmsyncuser:1
idmendpoint:https://apache.yourdomain.net/idm-service
idmhttpuser:idmTransportUser
idmhttppassword:<your password of idmTransportUser>
idmssltrustall:0
idmtruststore:smrootca.p12
idmtruststorepassword:<your password of smrootca.p12>
idmtruststoretype:pkcs12
```

Now start the Service Manager server and log on to the Windows client to perform the next tasks.

Create operator template

You need to create an operator template named "Template_IdM", which will be applied to the newly synced users.

See Create an operator template for detailed instructions.

Note

Note:

- If you don't create the Template_IdM template, the newly synced users will be assigned with the self-service user role.
- The name of the operator template must be set as "Template_IdM".
- The "Template_IdM" template is not applicable for the existing users.
- See Create an operator record from a operator template for more information.

Configure attribute mapping

You need to define how to synchronize the IdM user attributes to Service Manager by configuring attribute mapping. Service Manager provides the out-of-box mappings for some basic IdM user attributes. You can customize the attribute mappings except the `contact.name` field.

1. From the System Navigator, click **System Administration** > **Ongoing Maintenance** > **System** > **IdM Mapping**.
2. The mappings for **Operator** table and **Contact** table are displayed.
3. You may click on an existing attribute mapping to update the value, or add new mappings as required.
4. Save the changes.

Important

Important

Here is the priority order of the IdM mapping, the Template_IdM template, and the existing user attributes:

IdM mapping > existing user attributes > Template_IdM template

This means the IdM attributes mapped to Service Manager can overwrite the values defined in the Template_IdM template and the existing user attributes. Hence, you need to be careful when configuring the attribute mappings.

File	Field name	IdM attribute name
operator	Contact Name	username
	Email	email
contacts	Contact Name	username
	Email	email
	First Name	firstname
	Last Name	familyName
	Contact Phone	mobilePhoneNumber
	Postcode	code
	Language	language

IdM user attributes are populated from the external IdP (for example, Keycloak) and all user attributes in the userinfo (such as defined by mappers in Keycloak) are synchronized to IdM. Some attributes are mapped by the IdM and get new names in IdM, whereas other attributes keep their original names. For example:

Attribute name in userinfo	Mapped by IdM	IdM attribute name	IdM display name
preferred_username	Yes	username	Username
given_name	Yes	firstName	First Name
family_name	Yes	lastName	Family Name
middle_name	Yes	middleName	Middle Name
phone_number	Yes	mobilePhoneNumber	Mobile Phone Number
claims_locales	Yes	language	Language
address	Yes	location	Location
email	Yes	email	User Email
custom_attr1	No	custom_attr1	

Synchronize users from LDAP

First, edit `sm.ini` to add necessary LDAP parameters. See LDAP parameters for more details.

Next, start the Service Manager server and log on to the Windows client to enable an integration to LDAP.

Related topics

SAML Single Sign-On
IdM parameters for the Service Manager server

8.17.2. Configure IdM authentication in the Service Manager web tier

To enable Identity Manager (IdM) authentication in the web tier, you need to edit the `<customize-folder>/config/webtier.properties` file if you have configured the `customize-folder` parameter, or edit the `<WEBTIER>/WEB-INF/webtier.properties` if you haven't configured the `customize-folder` parameter. Add IdM related settings and set the `isCustomAuthenticationUsed` parameter to `false`. For more information, see IdM parameters for the Service Manager web tier and mobility client.

See the following configuration as an example:

```
idm.sso.enabled=true
idm.tenant=SM
idm.token.signingKey=<your IdM signing key>
idm.cookieName=IDM_AUTH_TOKEN

idm.endpoint=https://apache.yourdomain.net/idm-service
idm.http.user=idmTransportUser
idm.http.password=<your password for the idmTransportUser user>

idm.truststore=<customize-folder>/config/smrootca.p12 or /WEB-INF/smrootca.p12
idm.truststore.password=<your password of smrootca.p12>
idm.truststore.type=pkcs12

idm.returnUri=https://apache.yourdomain.net/webtier

isCustomAuthenticationUsed=false
```

For more information about how to set the appropriate values for common IdM parameters, see How to get IdM-related settings.

Note

Note

- If there is no reverse proxy of load balancer exists on top of the web tier, you don't need to configure the `idm.returnUri` parameter.
- You MUST NOT enable LW-SSO at the same time. You need to revert the LW-SSO configuration if you have enabled LW-SSO before.

Related topics

SAML Single Sign-On

8.17.3. Configure IdM authentication in the Mobility Client

To enable Identity Manager (IdM) authentication in the Mobility client, you need to edit the `<MOBILITY>/WEB-INF/web.properties` file and add IdM related settings. For more information, see IdM parameters for the Service Manager web tier and mobility client.

See the following configuration as an example:

```
idm.sso.enabled=true
idm.tenant=SM
idm.token.signingKey=<your IdM signing key>

idm.endpoint=https://apache.yourdomain.net/idm-service
idm.http.user=idmTransportUser
idm.http.password=<your password for the idmTransportUser user>

idm.ssl.trustAll=false

idm.truststore=/WEB-INF/smrootca.p12
idm.truststore.password=<your password of smrootca.p12>
idm.truststore.type=pkcs12

idm.returnUri=https://apache.yourdomain.net/mobility

isCustomAuthenticationUsed=false
```

For more information about how to set the appropriate values for common IdM parameters, see How to get IdM-related settings.

Note

Note

- If there is no reverse proxy of load balancer exists on top of the web tier, you don't need to configure the `idm.returnUri` parameter.
- You MUST NOT enable LW-SSO at the same time. You need to revert the LW-SSO configuration if you have enabled LW-SSO before.

Related topics

SAML Single Sign-On

8.17.4. Configure IdM authentication in Collaboration

You only need to update the chat server and chat service file to enable Identity Manager (IdM) authentication in Service Manager Collaboration.

Enable IdM authentication in the chat server

To enable IdM authentication in the chat server, edit the `<CHAT_SERVER>/conf/openfire.xml` file. See the following configuration as an example:

```
<idmConfig>
  <ssoEnabled>true</ssoEnabled>
  <tokenSigningKey>your IdM signing key</tokenSigningKey>
  <cookieName>IDM_AUTH_TOKEN</cookieName>
  <tenant>SM</tenant>
  <endpoint>https://apache.yourdomain.net/idm-service</endpoint>
  <integrationUser>idmTransportUser</integrationUser>
  <integrationPassword>your password</integrationPassword>
  <trustAll>false</trustAll>
  <trustStorePath>../resources/security/smrootca.p12</trustStorePath>
  <trustStorePassword>caroot</trustStorePassword>
  <trustStoreType>pkcs12</trustStoreType>
</idmConfig>
```

For more information, see IdM Parameters for Service Manager Collaboration.

For more information about how to set the appropriate values for common IdM parameters, see How to get IdM-related settings.

Enable IdM authentication in the chat service

To enable IdM authentication in the chat service, edit the `<CHAT_SERVICE>/conf/app.properties` file. See the following configuration as an example:

```
idmConfig.ssoEnabled=true
idmConfig.tokenSigningKey=<your IdM signing key>
idmConfig.cookieName=IDM_AUTH_TOKEN
idmConfig.tenant=SM
idmConfig.endpoint=https://apache.yourdomain.net/idm-service
idmConfig.integrationUser=idmTransportUser
idmConfig.integrationPassword=<your password>
idmConfig.trustAll=false
idmConfig.trustStorePath=../conf/smrootca.p12
idmConfig.trustStorePassword=caroot
idmConfig.trustStoreType=pkcs12
```

For more information, see IdM Parameters for Service Manager Collaboration.

Related topics

SAML Single Sign-On
How to get IdM-related settings.

8.17.5. Configure IdM parameters

This topic describes the IdM-related parameters as well as where to get the IdM-related settings.

IdM parameters for the Service Manager server

For the Service Manager server, you can define the IdM parameters in the `<SM_SERVER>/RUN/sm.ini` file.

Idm parameters

Parameter	Default value	Description
idmssoenabled	0	Specifies if IdM SSO is enabled for the Service Manager Server.
idmsigningkey	N/A	Specifies the IdM signing key that is used to verify the IdM tokens. This value must match the value that is configured in the IdM service. **Note** This value is automatically encrypted after the Server is started. If you need to update the value later, remove the asterisk symbol (*) before the parameter name.
idmtenant	N/A	The tenant in which all Service Manager user accounts reside. This value must match the tenant you created for the Service Manager in the IdM service. **Note** If you don't specify this value, the server will not verify the tenant in the IdM auth token.
idmsyncuser	0	Specifies whether to sync user from IdM or not. To synchronize users from IdM, set this parameter to 1. Otherwise, keep the default value (0).
idmendpoint	N/A	This is the absolute URL of the IdM service.

Parameter	Default value	Description
idmhttpuser idmhttppassword	N/A	This is the integration user account used to access the IdM service. **Note** The value of the *idmhttppassword* parameter is automatically encrypted after the Service Manager server starts. If you need to update the value later, remove the asterisk symbol (*) before the parameter name.
idmssltrustall	0	Indicates whether to verify that the IdM server certificate is trusted. To verify the IdM server certificate, keep the default value (0).
idmtruststore idmtruststorepassword	N/A	This is the IdM client trust store and password. The IdM client uses this truststore to validate the IdM server certificate. **Note** You must specify these two parameters if the *idmssltrustall* parameter is set to 0. The value of the *idmtruststorepassword* parameter is automatically encrypted after the Service Manager server starts. If you need to update the value later, remove the asterisk symbol (*) before the parameter name.
idmtruststoretype	N/A	This is the IdM client trust store type. Values for this parameter include JKS, PKCS12, and BCFKS.

Parameter	Default value	Description
idmkeystore idmkeystorepassword	N/A	This is the IdM client keystore and password. You must set a value for this parameter if the IDM service validates the client certificate. **Note** The value of the *idmkeystorepassword* parameter is automatically encrypted after the Service Manager server starts. If you need to update the value later, remove the asterisk symbol (*) before the parameter name.
idmkeystoretype	N/A	This is the IdM client keystore type. Values for this parameter include JKS, PKCS12, and BCFKS.

IdM parameters for the Service Manager web tier and mobility client

For the Service Manager web tier, you can define the IdM parameters in the `<customize-folder>/config/webtier.properties` file if you have configured the `customize-folder` parameter, or in the `<WEBTIER>/WEB-INF/webtier.properties` if you haven't configured the `customize-folder` parameter.

For the Service Manager mobility client, you can define the IdM parameters in the `<MOBILITY>/WEB-INF/web.properties` file.

IdM parameters

Parameter	Default value	Description
idm.sso.enabled	false	Indicates if IdM SSO is enabled. To enable IdM SSO, set this parameter to **true**.
idm.authGroupName	N/A	Specifies the authentication group to authenticate users. If you don't specify this value, the system uses the default authentication group. It can be overwritten by the auth URL parameter, the precedence order is: • The *auth* URL parameter • The *idm.authGroupName* parameter • The default authentication group For example, you can sign in to the web tier by using the https://apache.yourdomain.net/webtier/index.do?auth=db URL. This URL uses the database authentication group even if the default authentication group is SAML or OAuth.
idm.tenant	N/A	The tenant in which all Service Manager user accounts reside. This parameter value must match the tenant configured on the IdM service side. You can find the tenant configuration in Install Identity Manager (IdM).
idm.token.signingKey	N/A	Specifies the IdM signing key that is used to verify the IdM tokens. This value must match the value that is configured in the IdM service.
idm.refreshBeforeExpiration	300000	Specifies when to refresh the token before it expires. The unit is millisecond.
idm.cookieName	IDM_AUTH_TOKEN	The cookie name of the IdM auth token. This parameter is only available for the web tier.

Parameter	Default value	Description
idm.endpoint	N/A	This is the absolute URL of the IdM service.
idm.http.user idm.http.password	N/A	This is the integration user account used to access the IdM service.
idm.ssl.trustAll	false	Indicates whether to verify that the IdM server certificate is trusted. To verify the IdM server certificate, keep the default value (false).
idm.truststore idm.truststore.password	N/A	This is the IdM client trust store and password. The IdM client uses this truststore to validate the IdM server certificate. **Note** You must specify these two parameters if the **idm.ssl.trustAll** parameter is set to **false**.
idm.truststore.type	N/A	This is the IdM client trust store type. Values for this parameter include JKS, PKCS12, and BCFKS. **Note** You can ignore this parameter if the IdM client trust store type is the same as the web client keystore type.
idm.keystore idm.keystore.password	N/A	This is the IdM client key store and password.

Parameter	Default value	Description
idm.keystore.type	N/A	This is the IdM client key store type. Values for this parameter include JKS, PKCS12, and BCFKS. **Note** You can ignore this parameter if the IdM client key store type is the same as the web client keystore type.
idm.returnUri	N/A	The absolute URL of the Web client to which the IdM service will return after successful authentication. It must not contain the servlet path (*.do). For example, https://apache.yourdomain.net/webtier. **Note** You must configure this parameter if there is a reverse proxy or a load balancer on top of the web tier or mobility.
idm.backAppOnLogout	true	Determines whether to return to the Web client goodbye page after sign out when SAML is enabled. **Note** This parameter is obsoleted from 9.80.p1.

IdM parameters for Service Manager Collaboration

For the chat server, you can define the IdM parameters in the `<idmConfig>` element in the `<CHAT_SERVER>/conf/openfire.xml` file.

IdM parameters

Parameter	Default value	Description
ssoEnabled	false	Indicates if IdM SSO is enabled. To enable IdM SSO, set this parameter to `true`.
tokenSigningKey	N/A	Specifies the IdM signing key that is used to verify the IdM tokens. This value must match the value that is configured in the IdM service.
cookieName	IDM_AUTH_TOKEN	The cookie name of the IdM auth token.
tenant	N/A	The tenant in which all Service Manager user accounts reside. **Note:** If not specified, the chat server will not verify the tenant in the IdM auth token.
endpoint	N/A	Specifies the integration endpoint.
integrationUser	N/A	Specifies the integration user.
integrationPassword	N/A	Specifies the integration password.
trustAll	false	Specifies whether to trust all certificates.
trustStorePath	N/A	Specifies the path to the keystore file.
trustStorePassword	N/A	Specifies the password of the trust store.
trustStoreType	pkcs12	Specifies the type of the trust store.

For the chat service, you can define the IdM parameters in the `<CHAT_SERVICE>/conf/app.properties` file.

IdM parameters

Parameter	Default value	Description
idmConfig.ssoEnabled	false	Indicates if IdM SSO is enabled. To enable IdM SSO, set this parameter to `true`.
idmConfig.tokenSigningKey	N/A	Specifies the IdM signing key that is used to verify the IdM tokens. This value must match the value that is configured in the IdM service.
idmConfig.cookieName	IDM_AUTH_TOKEN	The cookie name of the IdM auth token.
idmConfig.tenant	N/A	The tenant in which all Service Manager user accounts reside. **Note:** If not specified, the chat service will not verify the tenant in the IdM auth token.
idmConfig.endpoint	N/A	Specifies the integration endpoint.
idmConfig.integrationUser	N/A	Specifies the integration user.
idmConfig.integrationPassword	N/A	Specifies the integration password.
idmConfig.trustAll	false	Specifies whether to trust all certificates.
idmConfig.trustStorePath	../conf/smrootca.p12	Specifies the path to the keystore file.
idmConfig.trustStorePassword	N/A	Specifies the password of the trust store.
idmConfig.trustStoreType	PKCS12	Specifies the type of the trust store.

How to get IdM-related settings

Classic mode

The classic mode refers to a standalone Service Manager system that doesn't integrate with the SMA Service Portal.

Get the tenant

Use the name of the CONSUMER tenant you created when installing the IdM service.

Get the signing key

The signing key is the value you specified for the `idm.encryptedSigningKey` in the `system_resource_config__1.0.0.1__update_signingKey_and_initString.json` seeded file when installing the IdM service. You can also get the signing key from the IdM Admin Console. Go to **SYSTEM SETTINGS**, double-click the **Encrypted signing key** property, and then check the value in the edit dialog.

Get the cookie name of the auth token

Use the default cookie name `IDM_AUTH_TOKEN`.

Get the service endpoint

Use the URL exposed by the reverse proxy or the load balancer (for example, https://apache.yourdomain.net/idm-service). If there is no reverse proxy or load balancer on top of the IdM service, use the URL exposed by Tomcat (for example,

https://idmservice.yourdomain.net:8443/idm-service).

Get the password of idmTransportUser

The password is the value you entered in the **Password** field for the integration user when you created the SM tenant in the IdM service.

Get the client truststore

Import your corporate root CA that issues the server certificate for the reverse proxy or the load balancer to a truststore by using the certificate generation toolkit (for example, `corprootca.p12`). Use this truststore as the IdM client truststore.

```
import-cert.bat --alias corprootca --file <your corporate root CA cert file> --keystore
certs/corprootca.p12 --storepass <your password for corprootca.p12>
```

If you use the Service Manager private root CA to sign the server certificate for the reverse proxy in the dev environment, or there is no reverse proxy or load balancer on top of the IdM service, use the `smrootca.p12` as the IdM client truststore.

Mixed mode (SMA-SM)

The mixed mode refers to a Service Manager system that integrates with the SMA Service Portal.

Get the tenant

Use the ID of the Service Manager tenant you created when installing the SMA Service Portal and setting up the integration.

Get the signing key

If your SMA version is before 2023.05:
Get the signing key from Suite Administration. Go to **CONFIGURATIONS > Security**, and then check the value of the **Signing key** field under IDM configuration.

If your SMA version is 2023.05 or higher, follow the steps below to get the IDM signing key:

1. Log in to a control plane node or a worker node of SMA as the root or a sudo user.
2. Run the following command to get the name of the **bo-config** pod:

   ```
   BO_CONFIG_POD=$(echo `kubectl get pods -n <SMA namespace> | grep -m1 itom-bo-config- | head -1 | awk
   '{print $1}'`) && echo $BO_CONFIG_POD
   ```

3. Run the following command to get the IdM signing key:

   ```
   SIGNING_KEY=$(kubectl exec -it $BO_CONFIG_POD -n <SMA namespace> -c itom-bo-config -- bash -c
   "/bin/get_secret idm_token_signingkey_secret_key" | awk -F= '{print$2}') && echo $SIGNING_KEY
   ```

Get the cookie name of the auth token

Use the same cookie name defined by the SMA suite. For example, `SMAX_AUTH_TOKEN`.

Get the service endpoint

Use the URL exposed by the SMA suite. For example, https://smax.yourdomain.net/idm-service.

Get the password of idmTransportUser

Follow these steps to get the password of `idmTransportUser`:

1. Sign in to one control plane node of SMA as root or a sudo user.
2. Run the following command to identify the pod name for IdM.

   ```
   kubectl get pod -n <namespace> | grep ^idm
   ```

3. Run the following command to go into the IdM pod.

```
kubectl exec -it <pod-name> -n <namespace> -c idm -- /bin/sh
```

4. Run the following command to get the password of the transport user.

```
get_secret idm_transport_user_password_secret_key
```

5. Exit the pod.

Get the client truststore

Follow these steps to get the client truststore:

1. Refer to the *Get the suite CA certificate* topic in the Service Management Automation - SM (SMA-SM) Help Center and get the SMA suite CA certificate (`<SMA cert file>`).
2. Import the SMA suite CA certificate to a truststore by using the certificate generation toolkit (for example, `suiterootca.p12`), and then use this truststore as the IdM client truststore.

```
import-cert.bat --alias suiterootca --file <SMA cert file> --keystore certs/suiterootca.p12 --
storepass <your password for suiterootca.p12>
```

8.18. SAML-based Single Sign-On

Security Assertion Markup Language (SAML) is an XML-based, open-standard data format for exchanging authentication and authorization data between parties. In particular, between an Identity Provider (IdP) and a Service Provider (SP). The single most important requirement that SAML addresses is web browser single sign-on (SSO). SAML 2.0 is the industry standard for federated identity management based on Single Sign-On (SSO).

Tip

Tip SSO is a session or user authentication process that permits a user to enter the same name and password to access multiple web applications.

The SAML 2.0 specification defines an exhaustive list of profiles. By leveraging Identity Manager (IdM), Service Manager support two essential profiles: Web Browser SSO Profile, and Single Logout Profile. The SAML SSO solution uses IdM as a Service Provider and a third-party Identity Provider.

Benefits of using SAML SSO

By default, SAML SSO is disabled in Service Manager. When SAML SSO is enabled, if Service Manager shares the same IdP and LDAP Server with other applications (whether the other applications leverage IdM or not), the user needs to enter a user name and password only once to log in to all of these web applications. Additionally, this solution supports single logout for multiple web applications that leverage IdM.

Note

Note Service Manager web tier, mobility client, and Collaboration support SAML SSO.

Using this solution has the following benefits:

- Provides tighter security controls through consistent enforcement of security policies across all applications
- Reduces turnaround time for provisioning and deprovisioning of user accounts in applications
- Fosters identity data collection, access reviews, and security analytics
- Provides single sign-on experience for end users
- Enables new users to gain faster access to the resources needed to perform their jobs
- Eliminates or reduces duplicate user IDs

Note that enabling SAML SSO may slow down user logins. According to laboratory tests by the company, user logins may take approximately 15% more time.

Supported Identity Providers (IdPs)

Service Manager leverages the IdM service to support SAML-based SSO. The IdM service supports the following SAML 2.0-compliant IdPs:

- Microsoft Active Directory Federation Services (ADFS) 3.0 or higher

This section uses Microsoft ADFS as an example for the IdP.

Prerequisites

- You must have a working ADFS and have synchronized the system clock with a reliable time source.
- You must have a working IdM service and have already integrated Service Manager with the IdM service. Refer to Install the IdM service and Configure IdM authentication in Service Manager for detailed instructions.

Note

Note

Configurations in this section are for the classic mode. If you are working with the mixed mode (SMA-SM), refer to the *Set up SAML authentication* topic in the SMA-SM Help Center for detailed instructions.

Related topics

Overview of Service Manager SAML SSO
Create a trusted relying party in ADFS
Configure SAML authentication in IdM
Use a custom certificate for SAML authentication
Renew the IdM SAML SSO Certificate
Renew ADFS certificates

8.18.1. Overview of Service Manager SAML SSO

This section provides an overview of how the SAML SSO solution works for Service Manager (SM).

SP-initiated web browser SSO

The following diagram illustrates the web browser SSO process of Service Manager. In this process, the IdM service acts as an SP to the IdP.

Caution

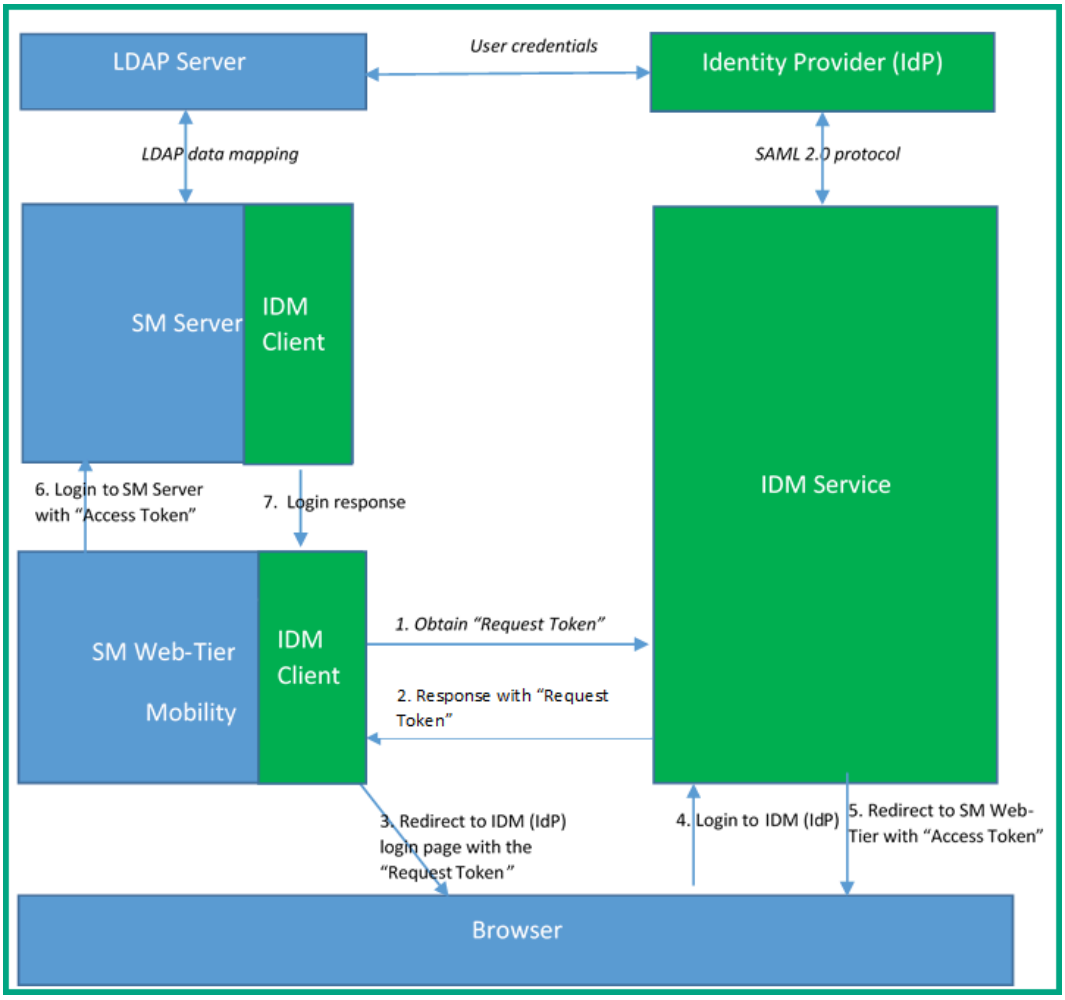
Caution

Use the standalone version of the IdM service released with Service Manager if you are not using SMA Service Portal. Otherwise, use the built-in IdM service in& SMA Service Portal.

The SM client (Web Tier or Mobility Client) uses a built-in IdM client as a filter to obtain an "IdM access token" from the IdM service and then passes the token to the SM Server. The SM Server then validates the IdM access token.

Note

Note LDAP authentication is replaced by IdM; however, the LDAP data mapping is retained in the SM Server.



The following table further explains each step in this process.

Step	Description
1	When the user launches the SM client (Web Tier or Mobility Client) web URL in the browser, the IdM Client embedded in the SM client sends a request to the IdM service for a request token.
2	The IdM service returns a response along with a request token to the built-in IdM client of the SM client (SM Web Tier or Mobility Client).
3	The IdM client redirects the request to the IdM service along with that token, and the IdM service then redirects it to the IdP login page with a SAML request.
4	The user enters a user name and password to log in to SM from the IdP login page. **Note** Note Users log in to Service Manager through the IdP page, and their IdP login name is determined by the IdP. That is, users need to enter their login name using the format required by the IdP. For example, when ADFS is used as the IdP, the login name must be either "userPrincipalName" (xxx@xxx.com) or "domain\samAccountName".
5	The IdP redirects the request to the IdM service with a SAML response, and then the IdM service redirects the request to the SM client (Web Tier or Mobility Client) along with an "Access Token". Additionally, the IdM service writes a cookie named LWSSO_COOKIE_KEY to the browser.
6	The SM client (Web Tier or Mobility Client) logs in to the SM Server with that "Access Token".
7	The user is logged in to SM.

Web browser single logout

The SAML SSO solution supports single logout for legacy HPE web applications that leverage Identity Manager (IdM). Once logged out of one application, the user is automatically logged out of the rest of the applications. This single logout behavior applies to the SM Web Tier and Mobility Client.

Non-browser login

The SM Windows client and web service clients aren't browser-based applications. Login to SM from these clients is referred to as "non-browser login", which doesn't support SAML. These clients log in to the SM Server through either a password or a token. The token is an LW-SSO token when using the legacy LW-SSO solution or an IdM token when using the IdM solution.

- For password login, the login logic is the same as before. That is, the login process doesn't involve IdM.
- For LW-SSO token login, the SM Server provides a new interface for third- party applications to log in to SM using an IdM access token. The SM Server adds a new http header named “X-Idm-Token”, which contains the string value of

the IdM access token.

Users must enter their login name that's determined by the LDAP field mapping on the SM side. For example, if the **name** field in the SM **operator** table is mapped to the **samAccountName** field in ADFS, users must enter their **samAccountName** value as the login name.

8.18.2. Create a trusted relying party in ADFS

ADFS helps you use single sign-on (SSO) to authenticate users to multiple, related web applications over the life of a single online session. Once ADFS is installed and configured to authenticate users from an LDAP directory, you are ready to add the IdM metadata to ADFS to add the IdM service as a trusted relying party.

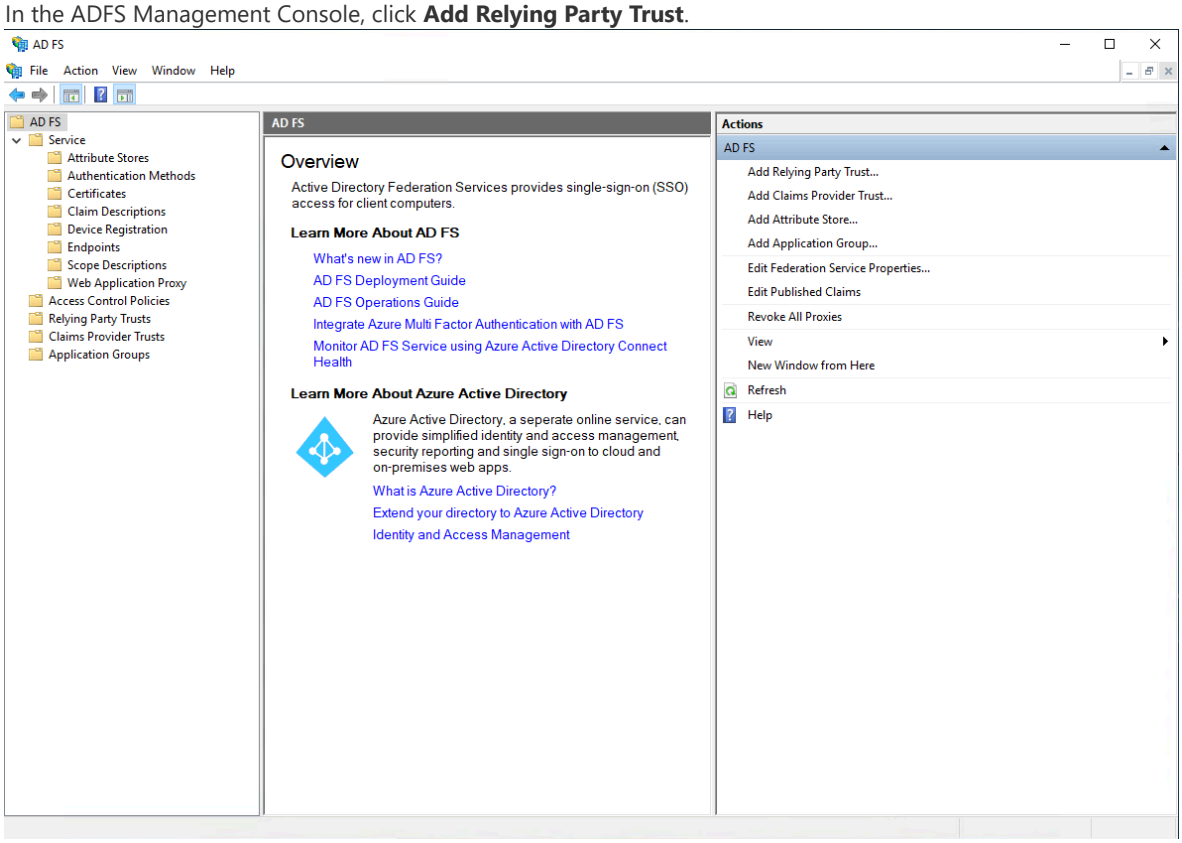
Note

Note

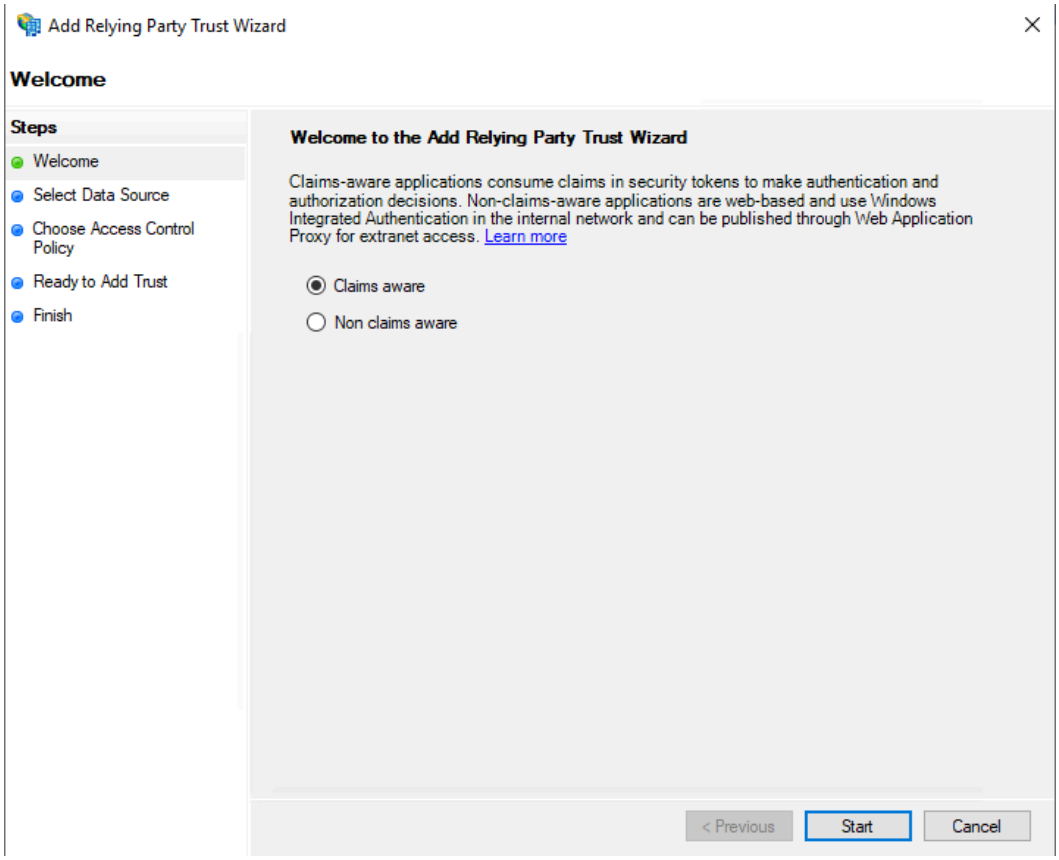
The IdM service will automatically generate a self-signed certificate with one-year validity at the first startup for encrypting the SAML token. If you want to use a custom certificate for SAML authentication, you can configure the custom certificate before downloading the IdM SAML metadata. See Use a custom certificate for SAML authentication for detailed instructions.

Follow these steps to add the IdM service as a trusted relying party to ADFS:

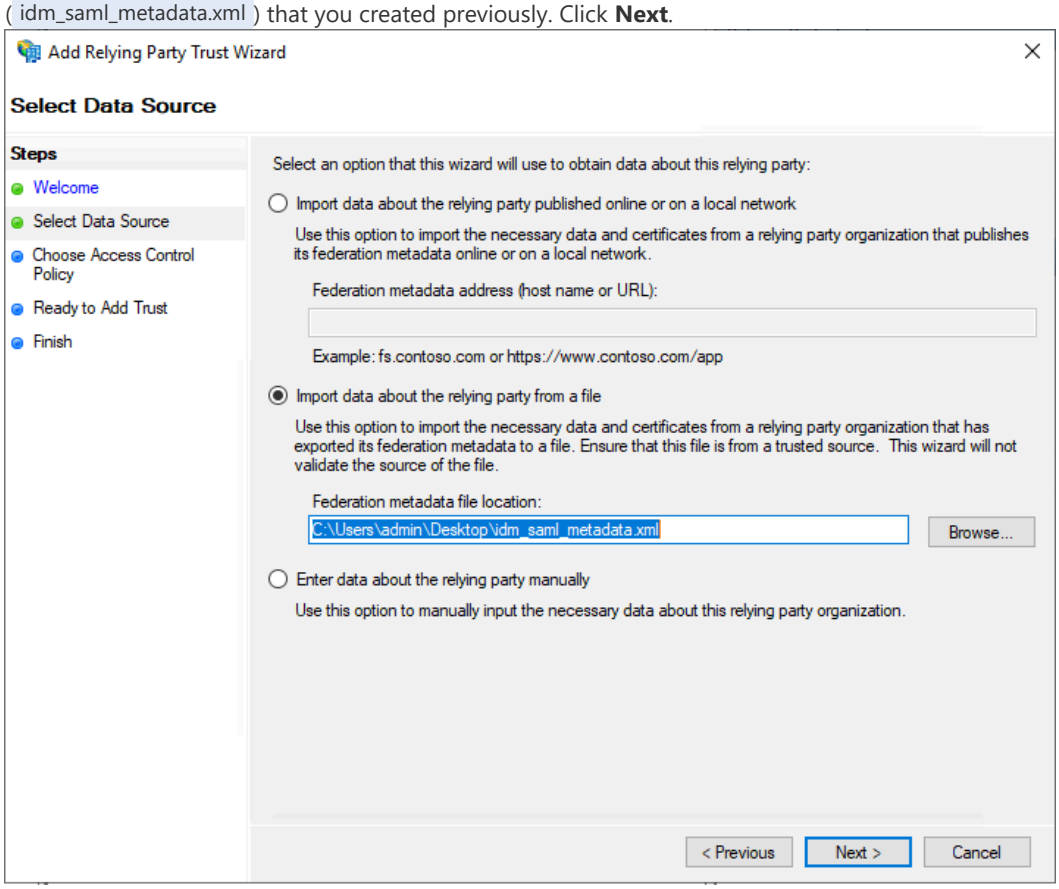
1. (Optional) Configure the SAML entity base URL.
   Skip this step if there is no reverse proxy or load balancer on top of the IdM service.
   1. Sign in to the IdM admin console (https://apache.yourdomain.net/idm-admin) as a system administrator.
   2. Click **GLOBAL SETTINGS**, then click the **SAML** > **Entity Base URL** property, and then set the value to https://apache.yourdomain.net/idm-service.
2. Download the IdM SAML metadata.
   Download the metadata from https://apache.yourdomain.net/idm-service/saml/metadata. Save this file as `idm_saml_metadata.xml`, and then copy this file to the ADFS server.
3. In the ADFS Management Console, click **Add Relying Party Trust**.



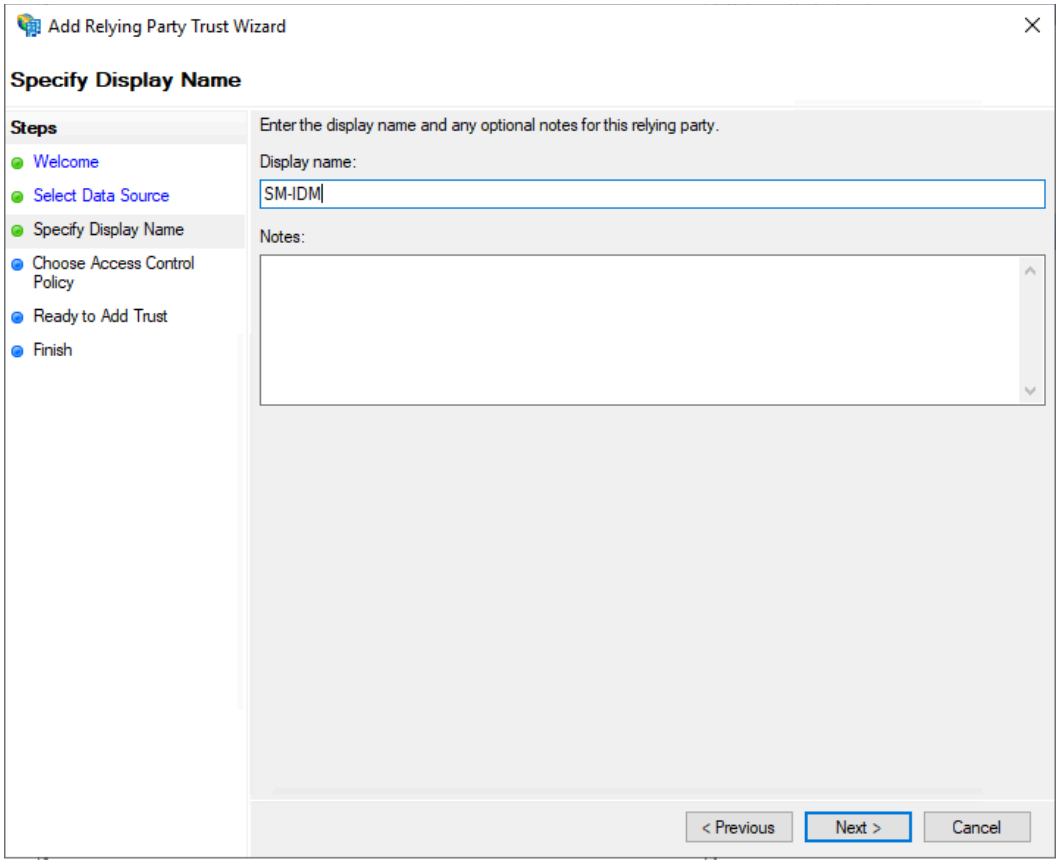
4. Click **Start**.



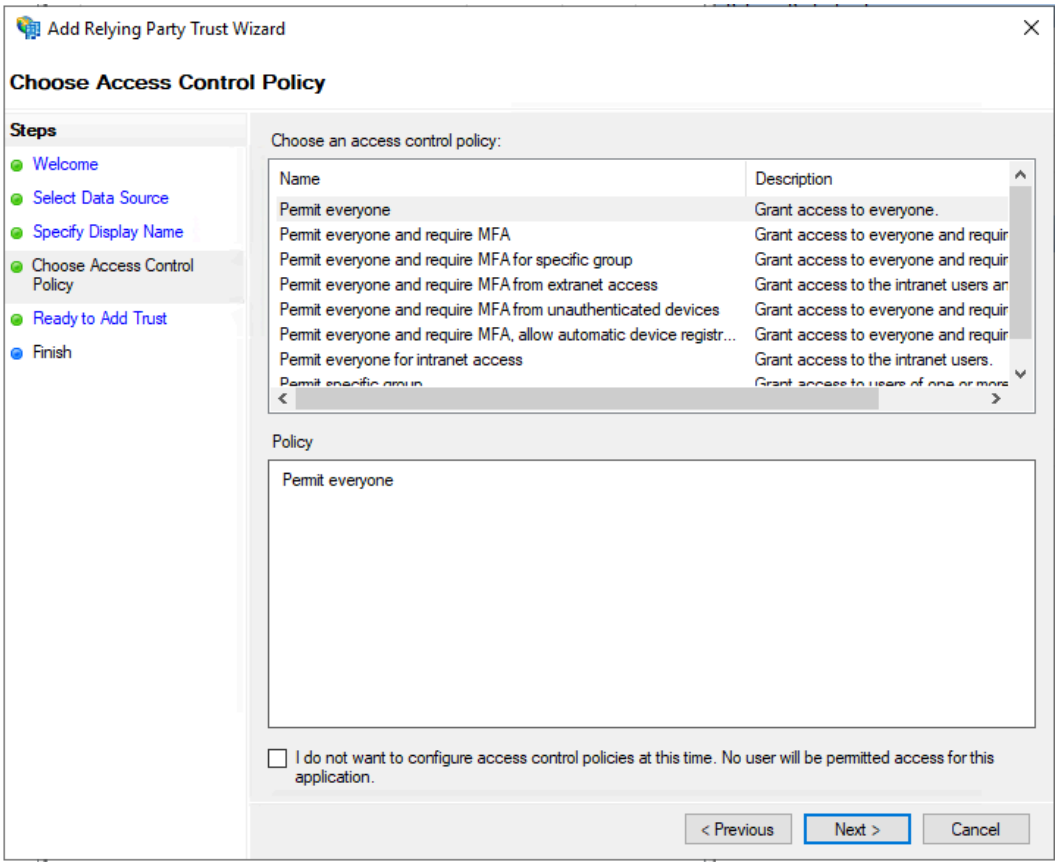
5. Select **Import data about the relying party from a file**, and then select the IdM metadata file (idm_saml_metadata.xml) that you created previously. Click **Next**.



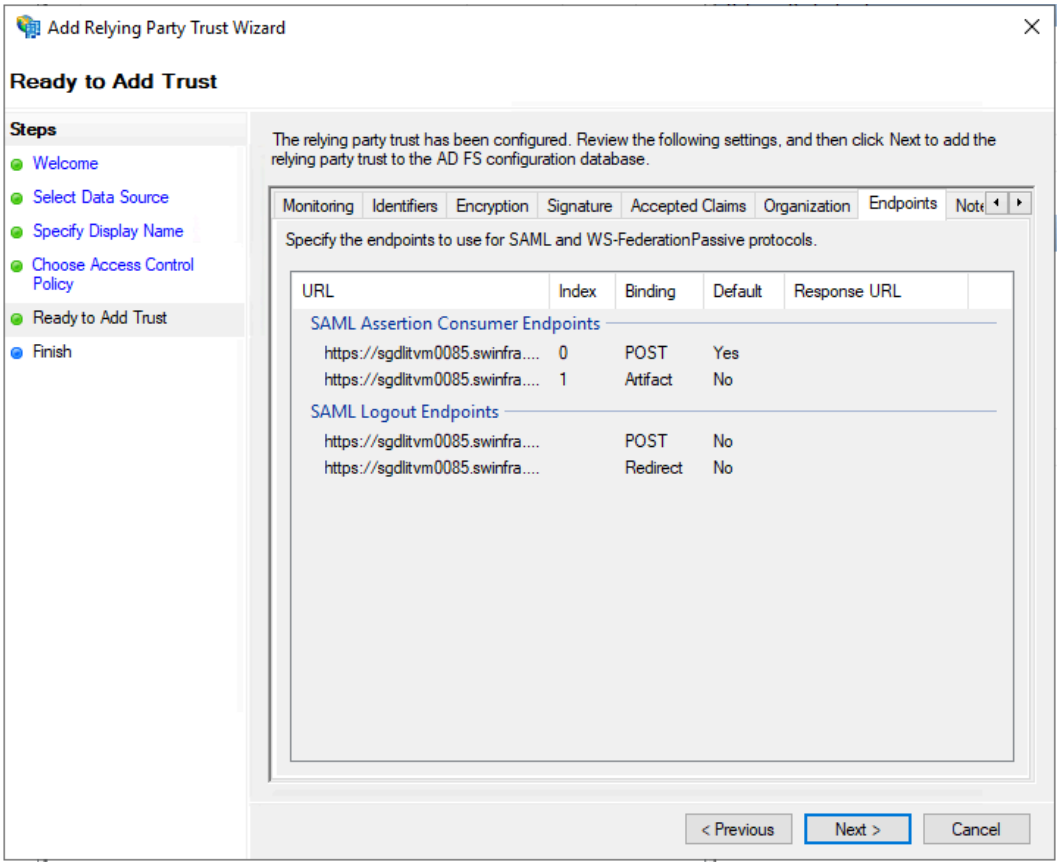
6. Specify a display name for the IdM service, and then add optional notes. Click **Next**.



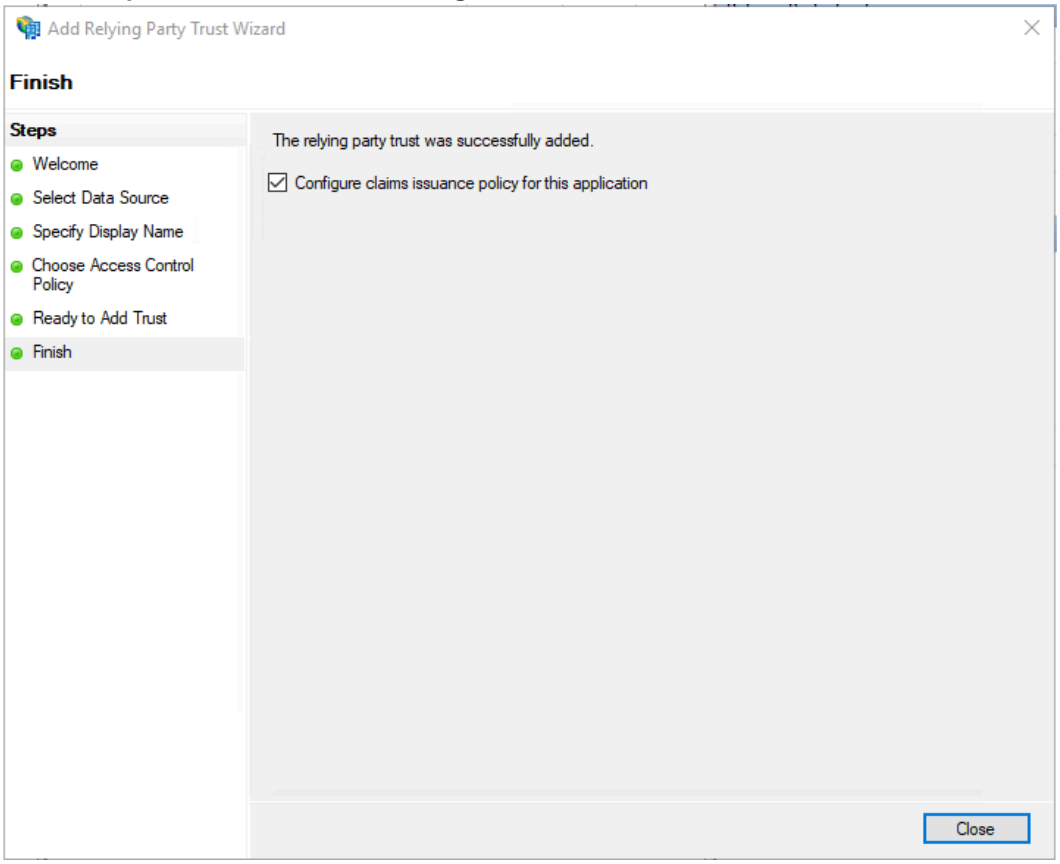
7. Click **Next**.



8. You are now in the Ready to Add Trust step. Check that the **Endpoints** tab contains multiple endpoint values. If not, verify that your metadata was generated with https protocol URLs.

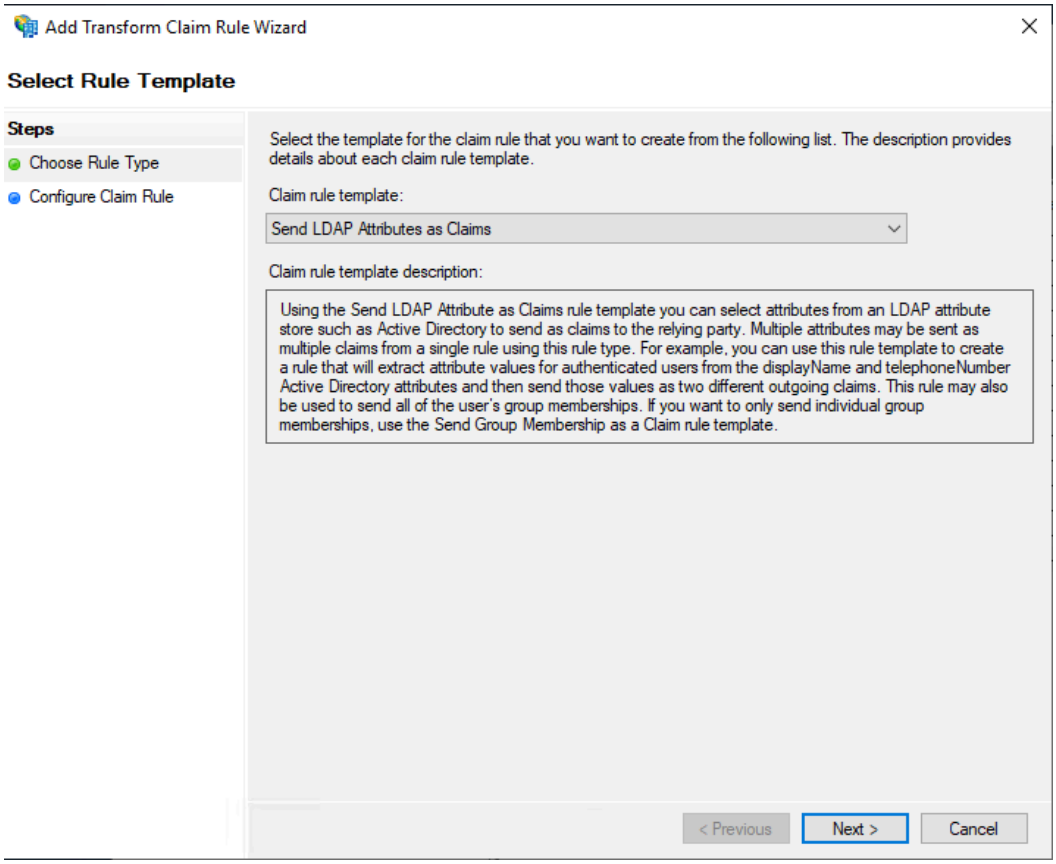


9. Leave the **Open the Edit Claim Rules dialog** checkbox selected, and click **Close** to close the wizard.

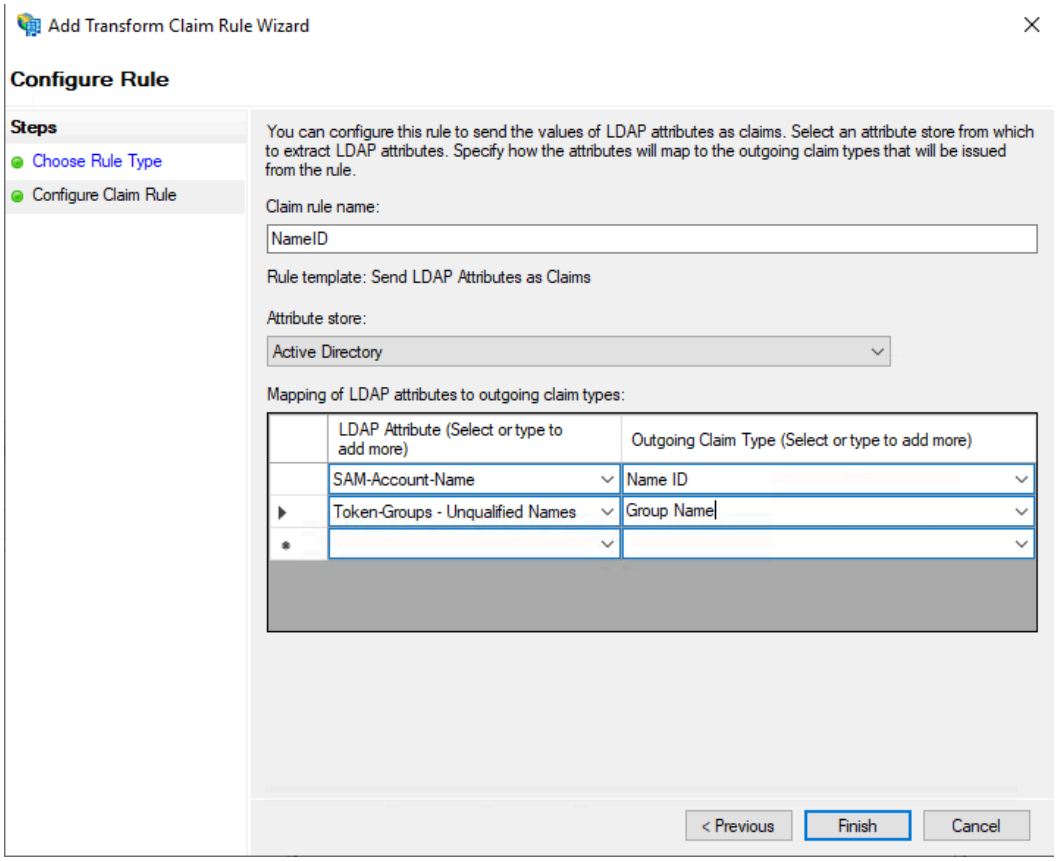


10. The Add Transform Claim Rule wizard opens. Perform the steps below to configure the NameID element as part of the Subject in the SAML Response message.

1. Select **Add Rule**, and then select **Send LDAP Attributes as Claims**. Click **Next**.



2. Specify the following fields to configure the claim rule:
   - Claim rule name: enter **NameID**
   - Attribute store: select **Active Directory**
   - LDAP Attribute: select **SAM-Account-Name**
   - Outgoing claim type: select **Name ID**
   - Token-Groups - Unqualified Names: select **Group Name**



Important

Important

The outgoing claim of **Name ID** is finally used as the username in the IdM access token. If IdM also integrates Service Manager Collaboration in your environment, it's mandatory that the login name of the authenticated operator is the same as the username in the IdM access token. Otherwise, Service Manager Collaboration will not work. In this case, if you choose to synchronize users from LDAP, you need to make sure that the LDAP attribute mapped to **Name ID** here is also mapped to the name field of the operator when you configure LDAP field mapping in Service Manager.

If you want the Service Manager server to synchronize user information from the IdM service instead of the LDAP, you should map additional LDAP attributes to the outgoing claim types as follows.

Outgoing Claim Type	IdM User Attribute Name
username	Username
firstName	First Name
familyName	Family Name
middleName	Middle Name
officePhoneNumber	Office Phone Number
homePhoneNumber	Home Phone Number
mobilePhoneNumber	Mobile Phone Number
zipCode	Zip Code
language	Language
location	Location
email	User Email

For example, you may need to add additional mappings as follows.

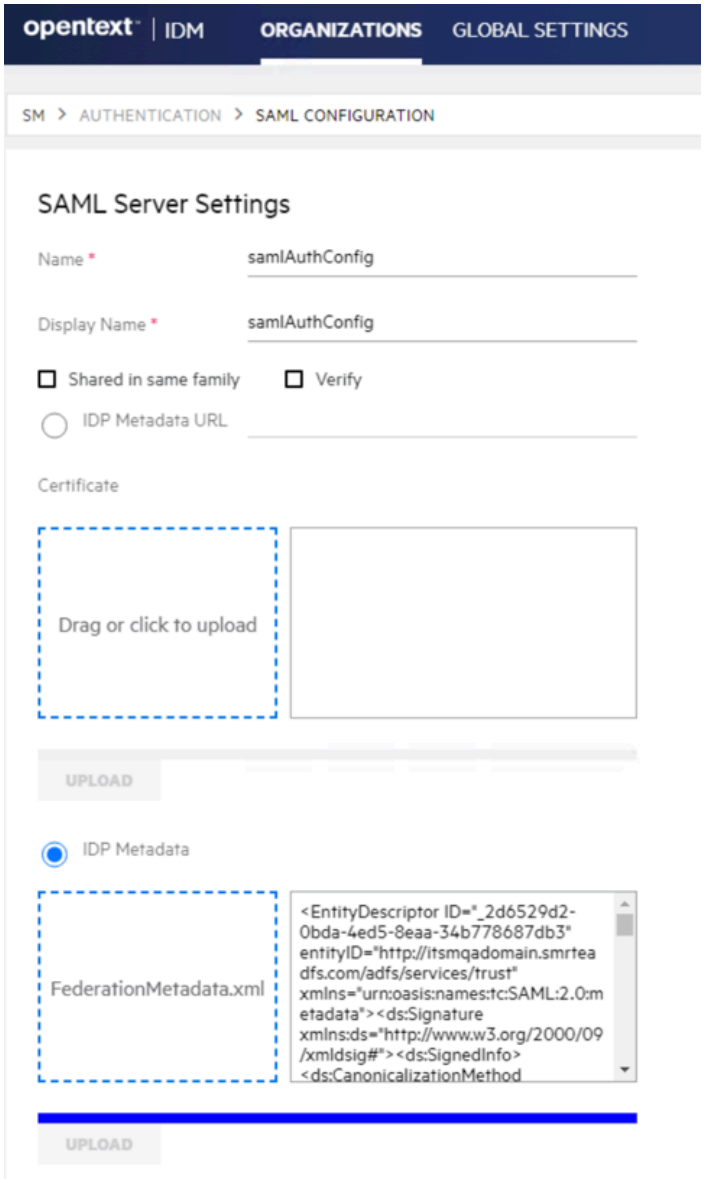
LDAP Attribute	Outgoing Claim Type
Given-Name	firstName
Surname	familyName
Telephone-Number	officePhoneNumber
User-Principal-Name	email

3. Close the wizard and the **Apply the claim rules** window.

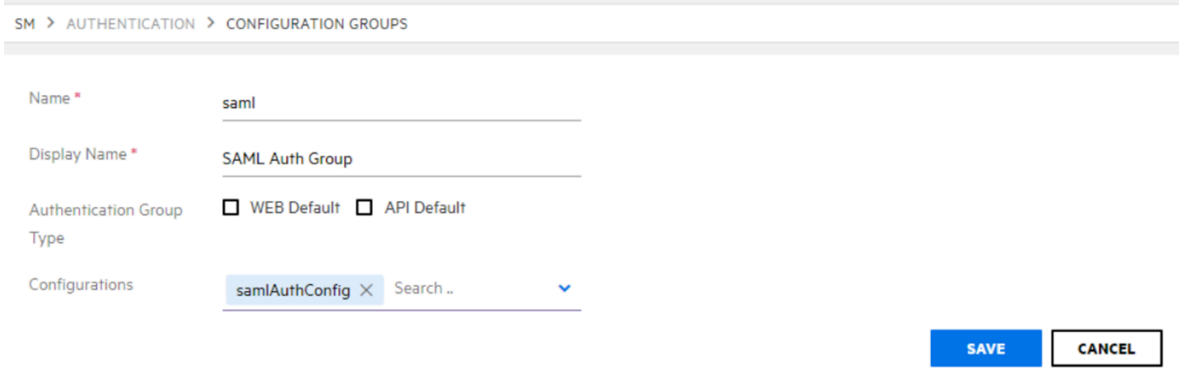
8.18.3. Configure SAML authentication in IdM

To configure SAML authentication in IdM, follow these steps:

1. Download the SAML metadata from the ADFS server (for example, https://adfs.yourdomain.net/FederationMetadata/2007-06/FederationMetadata.xml) and then save this file as `FederationMetadata.xml`.
2. Sign in to the IdM admin console (https://apache.yourdomain.net/idm-admin) as a system administrator.
3. Click the **SM** organization. Click **Authentication > CONFIGURATIONS > Add (+)**, and then select the **SAML** authentication type to create an SAML authentication configuration. For example:
   - **Display Name:** samlAuthConfig
   - **IDP Metadata:** upload the `FederationMetadata.xml` file or copy its content to the text area on the right.



4. After you create the SAML authentication configuration, click **Authentication > CONFIGURATION GROUPS > Add (+)** to create a configuration group. For example:
   - **Name:** saml
   - **Display Name:** SAML Auth Group
   - **Configurations:** samlAuthconfig



Verify the SAML SSO setup for Service Manager

Once you have completed all configurations required for SAML SSO, you are ready to verify if the setup is successful.

To verify the setup, follow these steps:

1. Make sure all components are started, including the SM Server, client (web tier or Mobility Client), the standalone IdM service, and the identity provider (IdP, that is, ADFS).
2. Launch the Service Manager web application URLs from a supported web browser (see the Service Manager Support Matrix).
    - SM web tier URL: http://<*web tier host*>:<*port*>/<*web tier name*>/index.do
    - Mobility Client URL: http://<*Mobility Client host*>:<*port*>/<*Mobility application name*>
3. Verify that the browser redirects you to the IdP login page.
4. Enter your username and password to sign in to one of the web applications.
5. Verify that you are automatically signed in to the rest of the web applications without the need to enter your credentials.

    Note
    Note
    ADFS doesn't allow you to select a login language. You are signed in to Service Manager with a language depending on which client you are using:
    - Web tier client: your browser language if the acceptBrowserLang parameter is set to true (default), or your operator's language if this parameter is set to false
    - Mobility Client: your browser language

6. Sign out of one of the Service Manager web applications.
7. Verify that you are automatically signed out of the rest of the web applications.

Tip
Tip
If any of the verification steps fails, see Troubleshoot security.

8.18.4. Use a custom certificate for SAML authentication

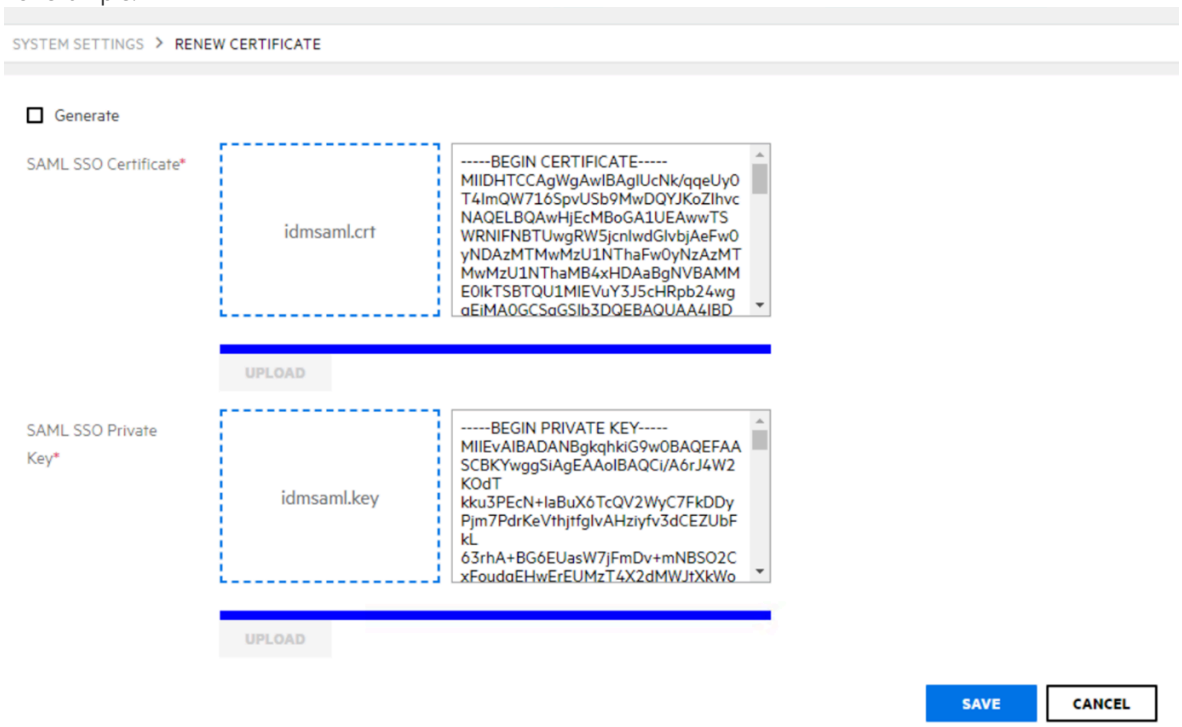
Follow these steps to configure a custom certificate for SAML authentication.

1. Create a self-signed certificate by using OpenSSL. For example, run the following command to create a self-signed certificate with 3 years validity.

    ```
    openssl req -x509 -newkey rsa:2048 -nodes -days 1095 -subj "/CN=IdM SAML Encryption" -keyout
    idmsaml.key -out idmsaml.crt
    ```

2. Sign in to the IdM admin console (`https://apache.yourdomain.net/idm-admin`) as a system administrator.

3. Click **GLOBAL SETTINGS**, and then click the **SAML / RENEW CERTIFICATE** button. Drag or click to upload the `idmsaml.crt` or `idmsaml.key` file, and then click the **UPLOAD** button, or you can copy its content to the text area on the right.
   For example:



4. Click **SAVE**.
5. Restart Tomcat.

8.18.5. Renew the IdM SAML SSO Certificate

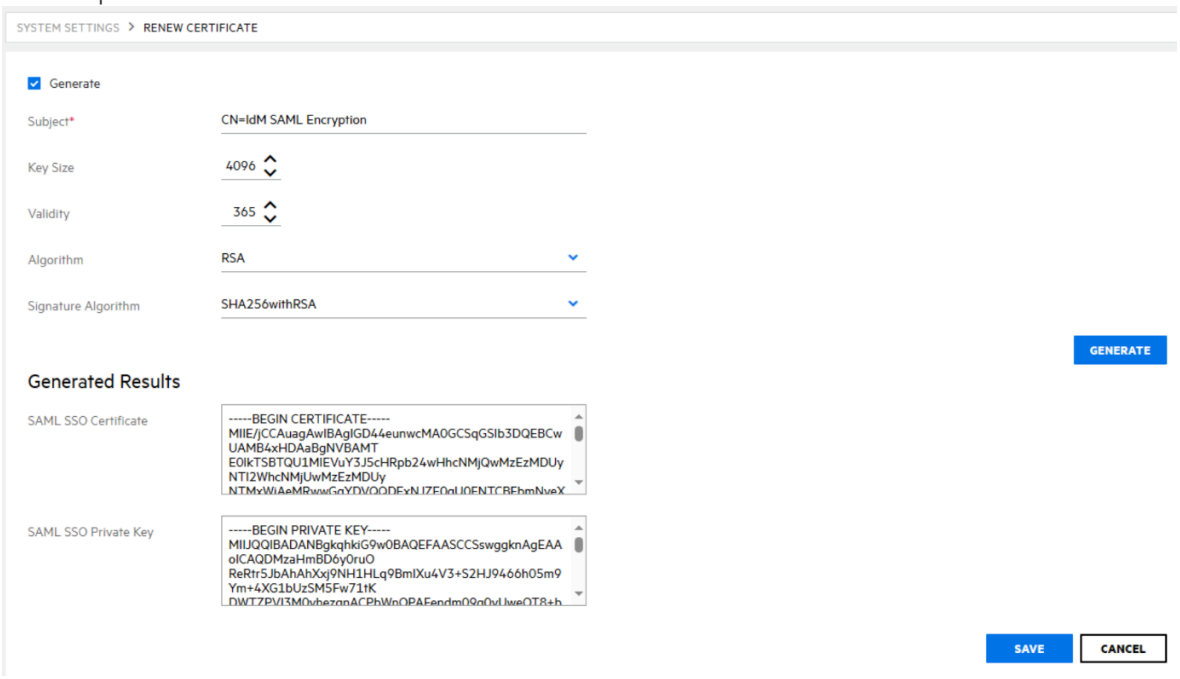
You need to renew the IdM SAML SSO certificate before it expires. Depending on the certificate type that the IdM service uses, you can renew the certificate in different ways.

Generate self-signed certificate automatically

Follow these steps:

1. Sign in to the IdM admin console (`https://apache.yourdomain.net/idm-admin`) as a system administrator.
2. Go to **GLOBAL SETTINGS**, clear the value for the **SAML / SSO Certificate** and **SSO PrivateKey** parameters.

3. Select the **Generate** checkbox and click the **GENERATE** button to generate a self-signed certificate, then click **SAVE**. For example:

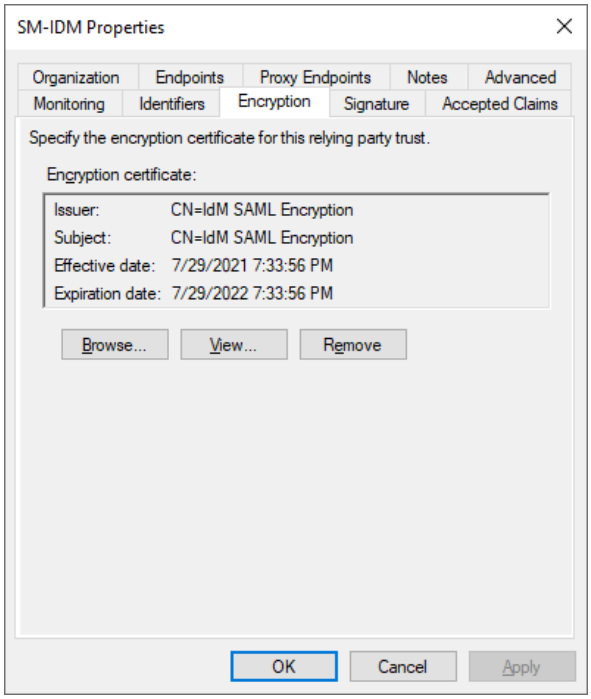


4. Restart Tomcat. The IdM service will automatically generate a self-signed certificate again.
5. Download new metadata from the IdM service, remove the old relying party trust in ADFS, and then create a new relying party trust. For detailed instructions, see Create a trusted relying party in ADFS.

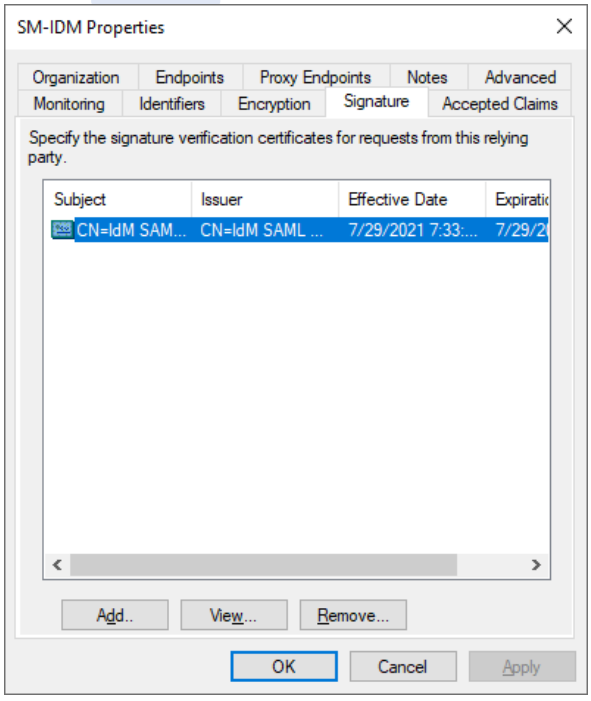
Custom certificate

Follow these steps:

1. Create a new self-signed certificate and upload it to the IdM service. For detailed instructions, see Use a custom certificate for SAML authentication.
2. Copy the `idmsaml.crt` to the ADFS server, then rename this file to `idmsaml.cer`.
3. Open the ADFS management console. Double-click the `SM-IDM` relaying party trust, and then switch to the **Encryption** tab page. Click the **Remove** button to remove the existing certificate, then click the **Browse** button to add the `idmsaml.cer`.



4. Switch to the **Signature** tab. Select the existing certificate, and then click the **Remove** button. Click the **Add** button to add the `idmsaml.cer`.



5. Click **OK**.

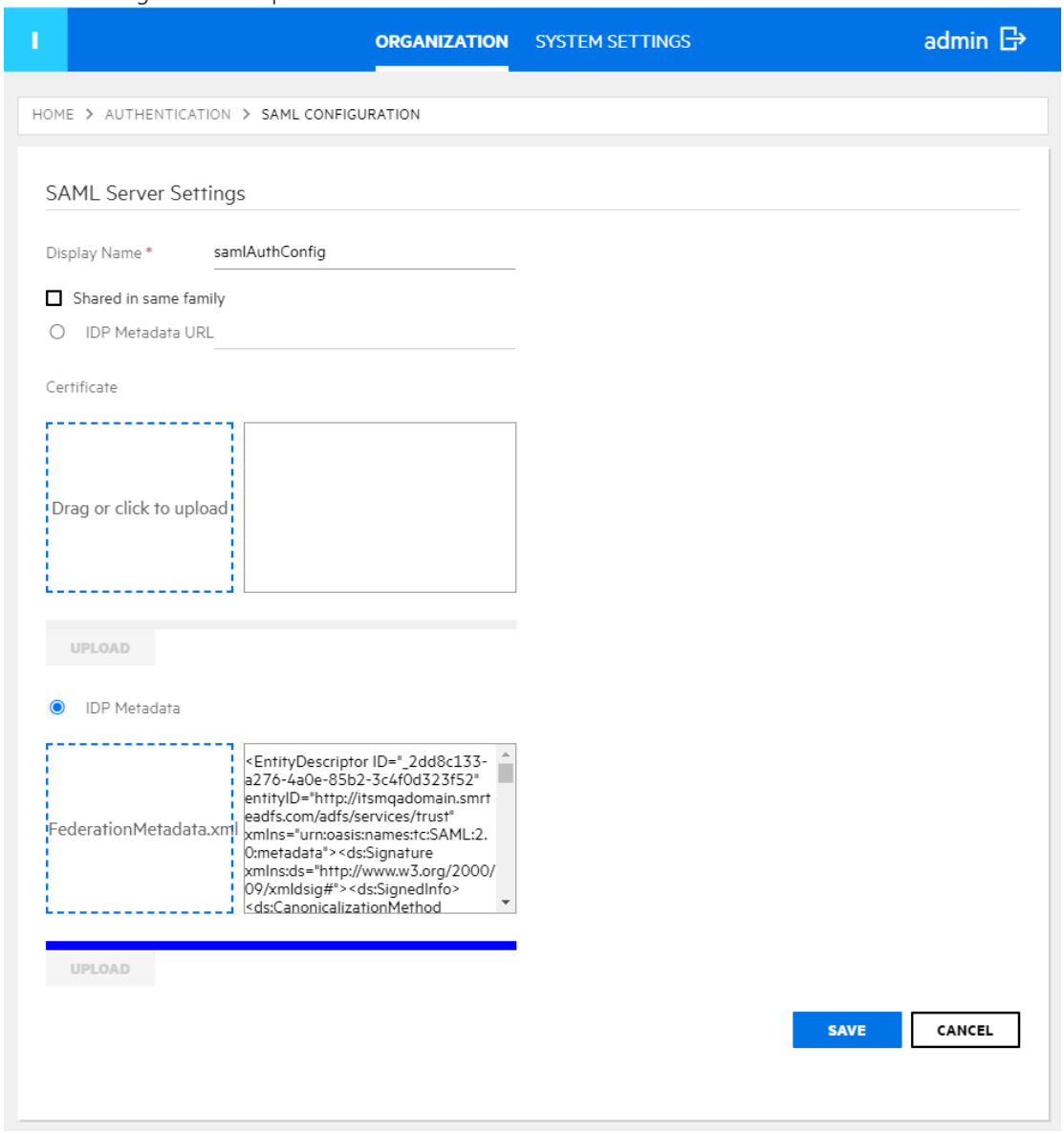
8.18.6. Renew ADFS certificates

The ADFS may automatically roll over its certificates, or you may manually renew certificates for the ADFS.

After you renew the certificates, you need to update the ADFS metadata for the SAML authentication by following steps below:

1. Download the SAML metadata from the ADFS server (for example, `https://adfs.yourdomain.net/FederationMetadata/2007-06/FederationMetadata.xml`). Save this file as `FederationMetadata.xml`.
2. Sign in to the IdM admin console (`https://apache.yourdomain.net/idm-admin`) as a system administrator.

3. Click the **SM** organization. Click **Authentication > CONFIGURATIONS > samlAuthConfig**, and then click the **Edit** button. Select the **IDP Metadata** radio button, upload the `FederationMetadata.xml` file or copy its content to the text area on the right. For example:



4. Click **SAVE**.

8.19. OIDC-based Single Sign-On

OpenID Connect (OIDC) is a simple identity layer on top of the OAuth protocol. It allows clients to verify the identity of the end-user based on the authentication performed by an authorization server, as well as to obtain basic profile information about the end-user in an interoperable and REST-like manner.

Users can access Service Manager after the OIDC configurations are completed. The user profile is synced to Service Manager after the user signs in for the first time. If the user profile is updated in the identity provider, the updated information will also be synced to Service Manager when the user signs in.

Step 1: Install Idm service and the IdM admin console

See Install Identity Manager (IdM) for detailed instructions.

Step 2: Install and configure IdP service

You can configure Service Manager to allow the users to sign in using Identity Provider (IdP) credentials.

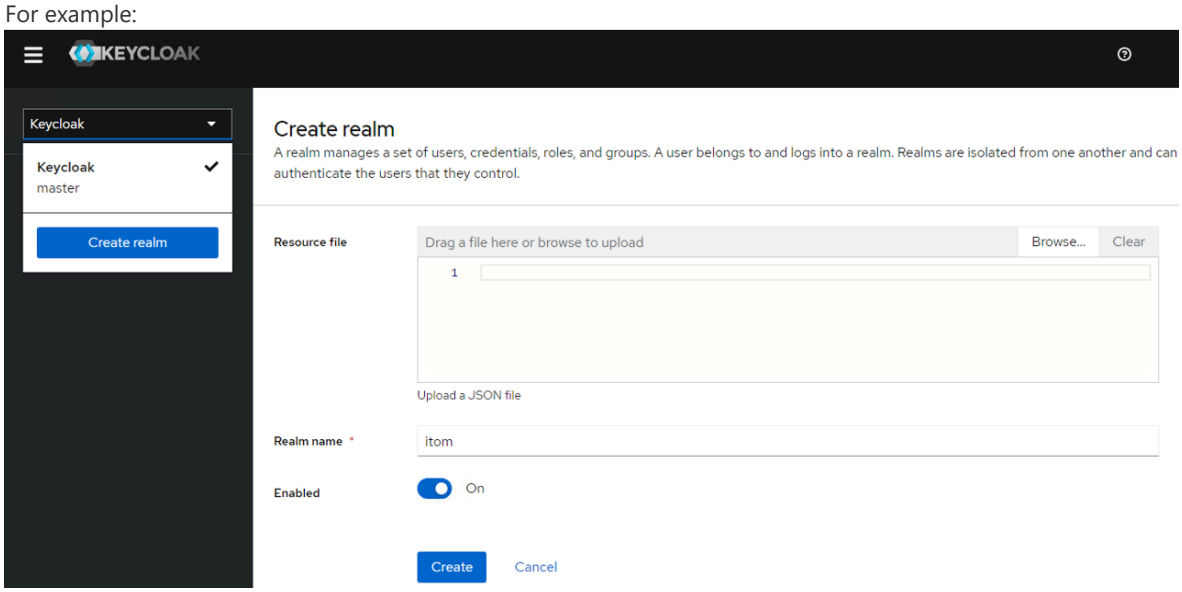
In the following settings, Keycloak performs as the IdP for user authentication as an example.

You may refer to the Keycloak documentation (https://www.keycloak.org/getting-started/getting-started-zip) for detailed information on Keycloak installation and setup.

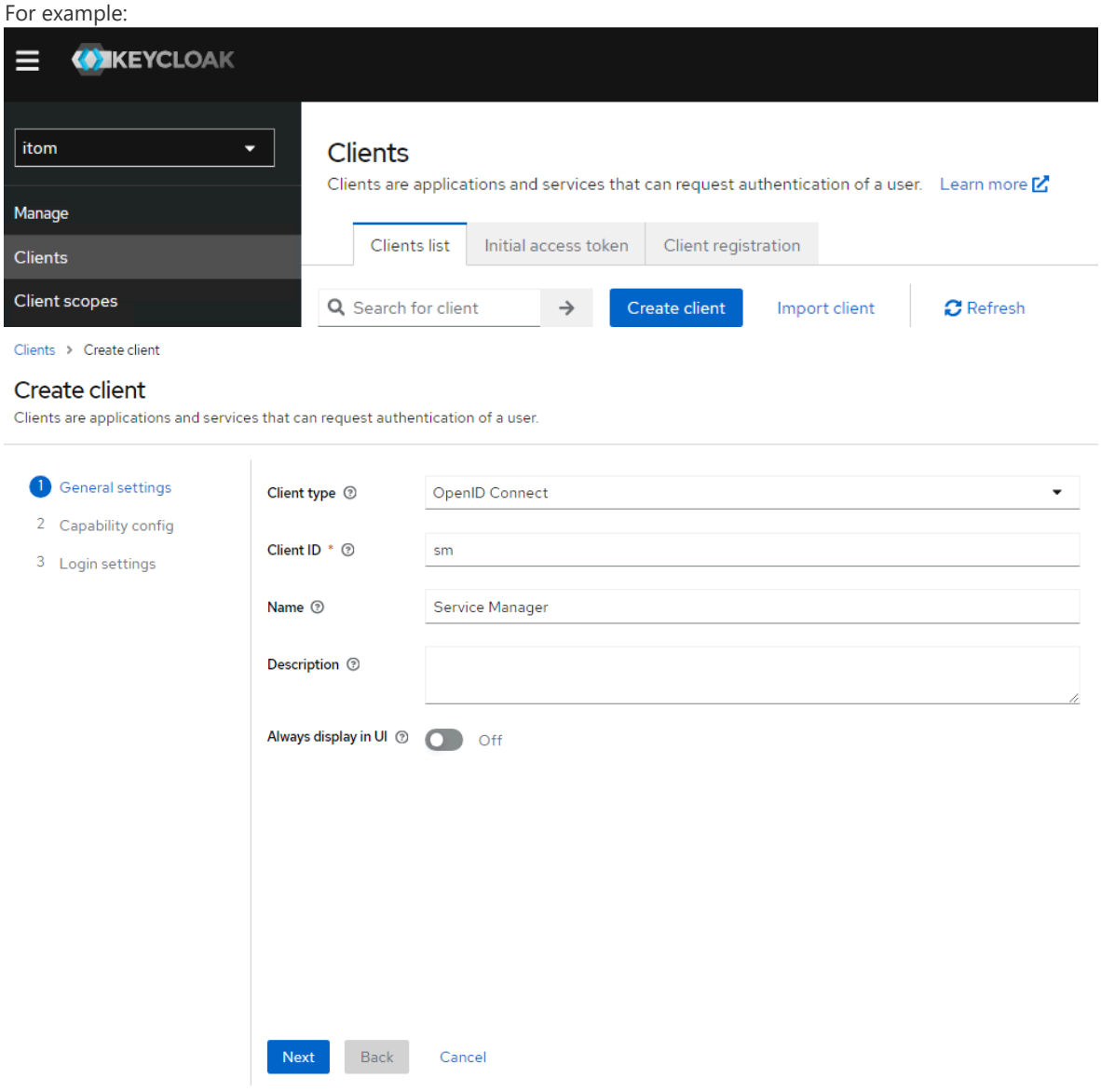
Create Client in IdP

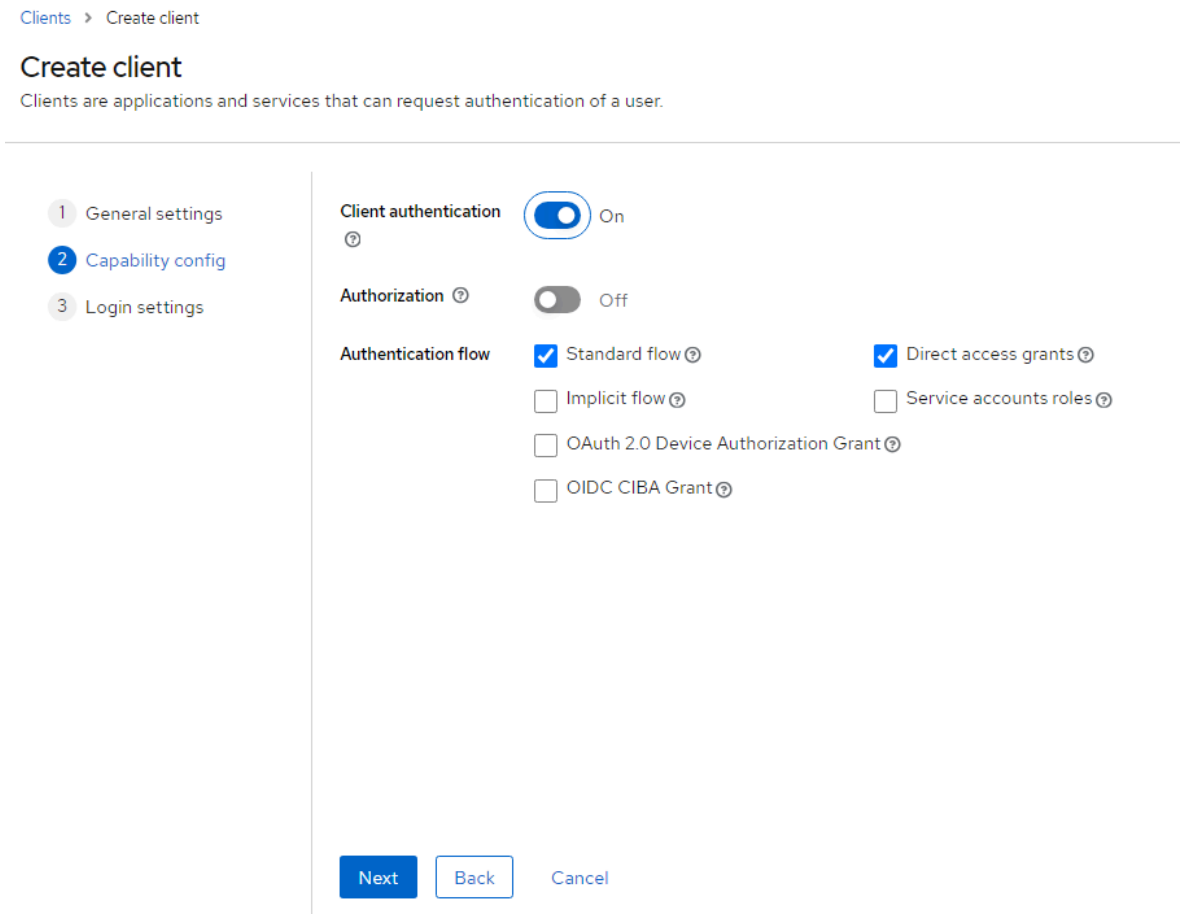
1. Open Keycloak admin console (http://<hostserver>:<port>) and sign in as `admin`.

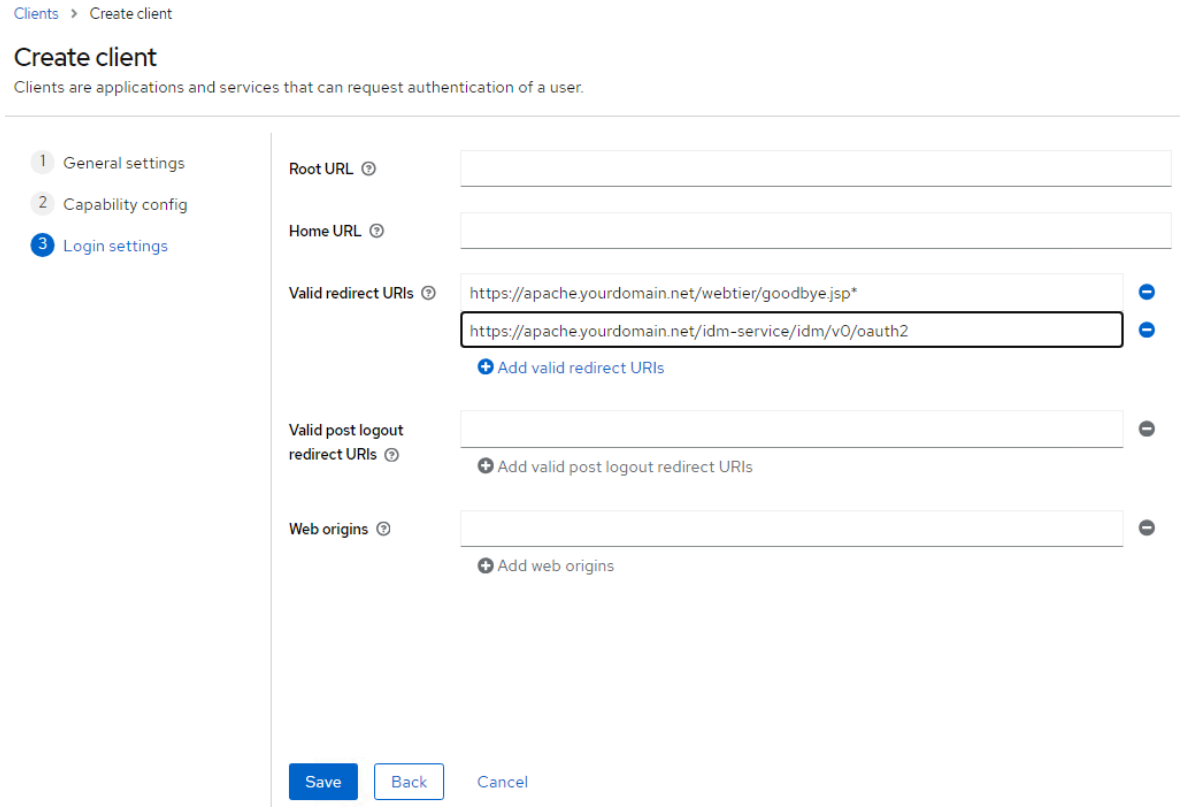
2. Click the **Create realm button** to create a realm.
   For example:



3. Select the realm in step 2, and then click the **Clients > Create client** button to create a client.
   For example:

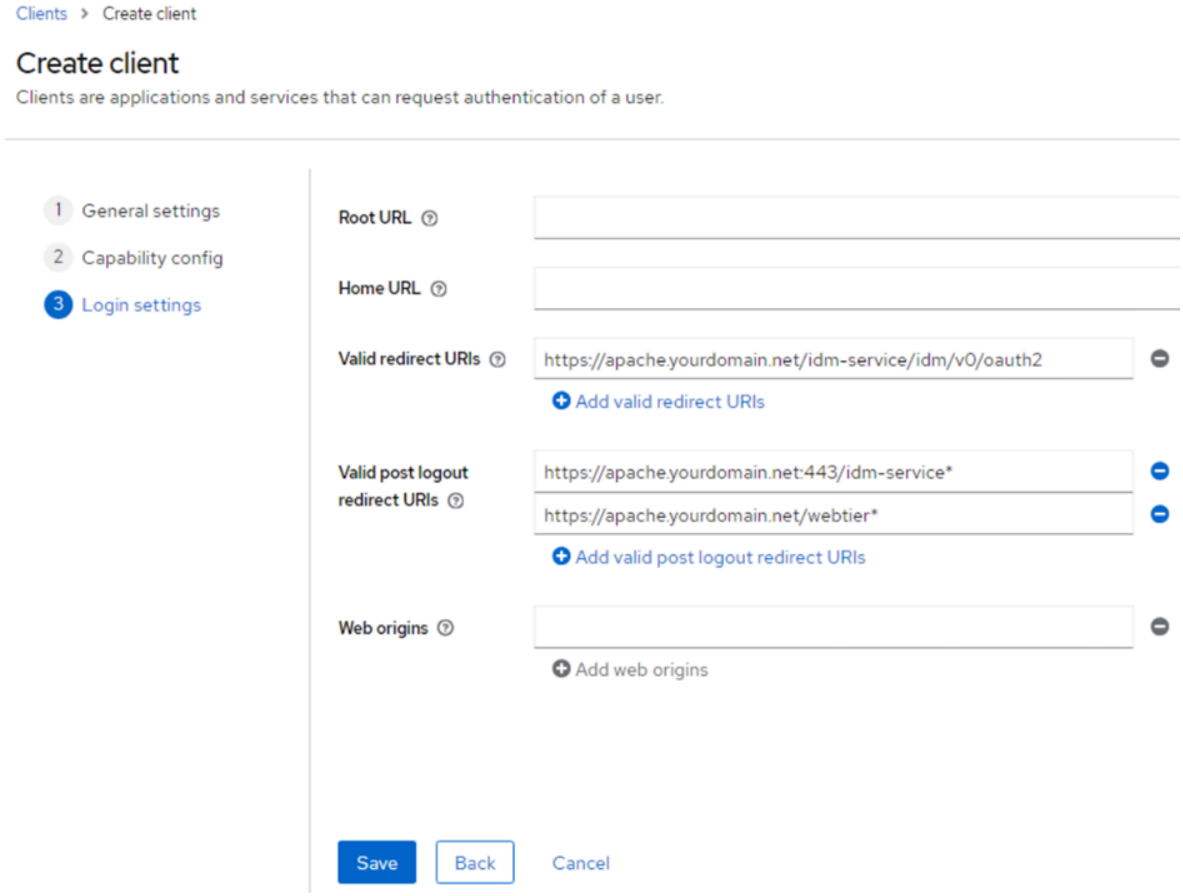




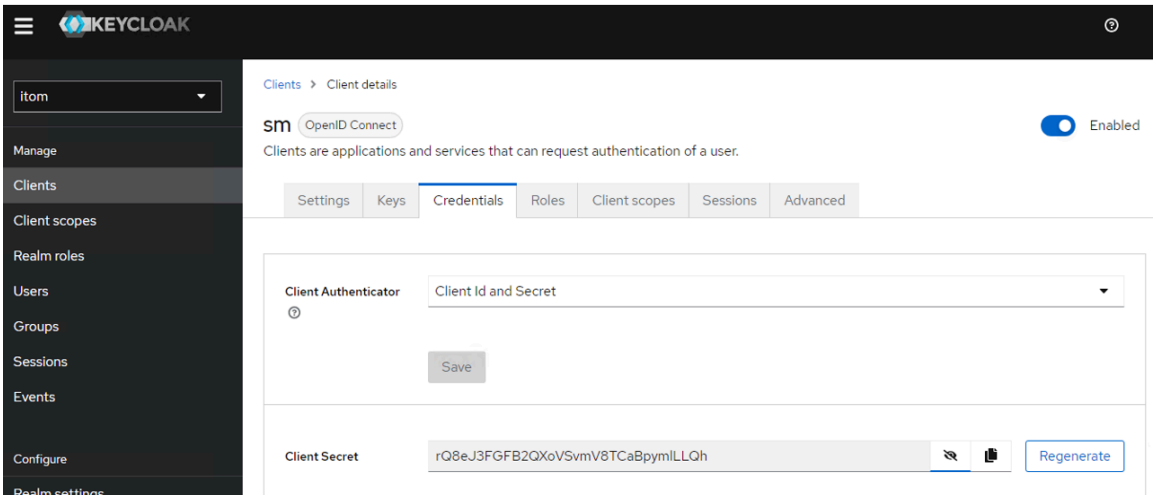


Note that if you have upgraded Service Manager to 9.80 p1 or above, which implements idm 24.2.1 and updated

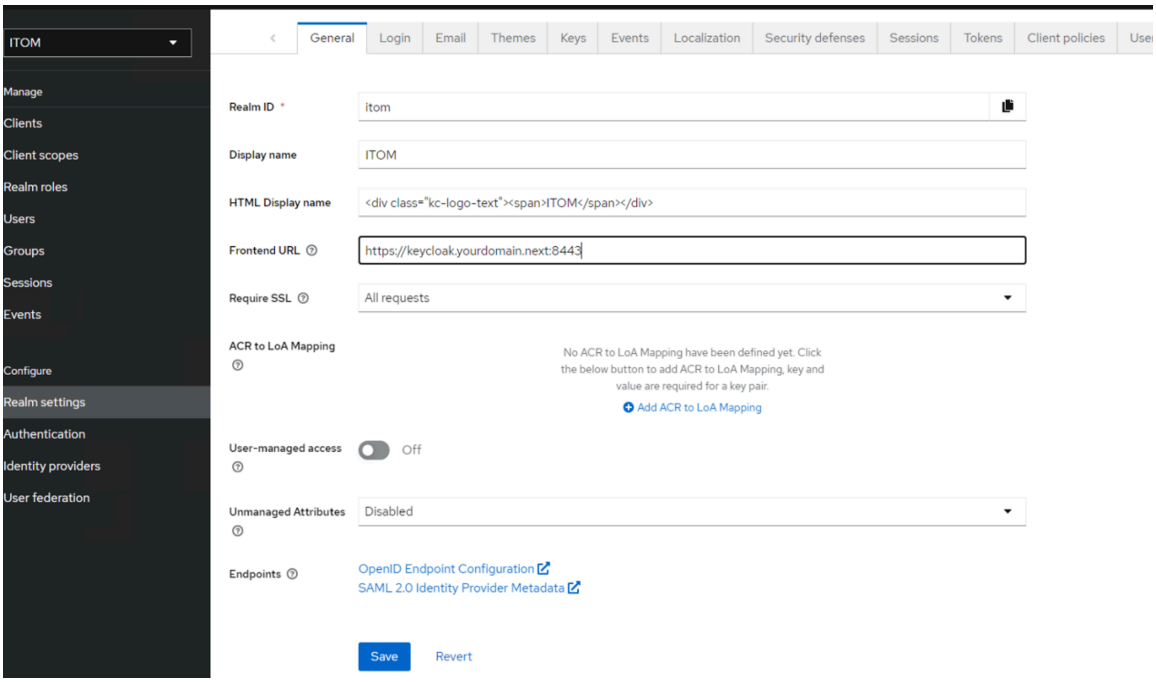
keycloak to 24.x version, then the URL configurations of above image need to be set as below:



4. After creation, edit the settings of the client sm as below:
   - **Consent Required:** ON
   - **Login Theme:** keycloak
     Note:
     - If your Keycloak version is 18 or above and you haven't applied the 9.80 p1 or later patch, you need to configure the following parameter in the **keycloak.conf** file, which is under the ***<keycloak installation>*/conf** path.
     `spi-login-protocol-openid-connect-legacy-logout-redirect-uri=true`
     Then, restart Keycloak for the changes to take effect.
     - It is assumed that you deployed the IdM service behind the Apache HTTP server. If there is no reverse proxy or load balancer exists in front of the IdM service, the URL exposed by the Tomcat should be used. For example: https://idmservice.yourdomain.net:8443/idm-service/idm/v0/oauth2.
5. Add mappers for custom user attributes if necessary.
6. Mark down the following settings which are required when you configure OAuth in the IdM:
   - The secret of the client `sm`. You can copy it from the **Credentials** tab.



- The endpoints of the realm `itom`. You can view endpoints by clicking the **OpenID Endpoint Configuration** link.



Step 3: Create an OAuth configuration in IdM

1. Sign in to the IdM admin console (https://apache.yourdomain.net/idm-admin) as a system administrator.
2. Click the **Organization** tab, and then click the organization you created for OAuth configuration.

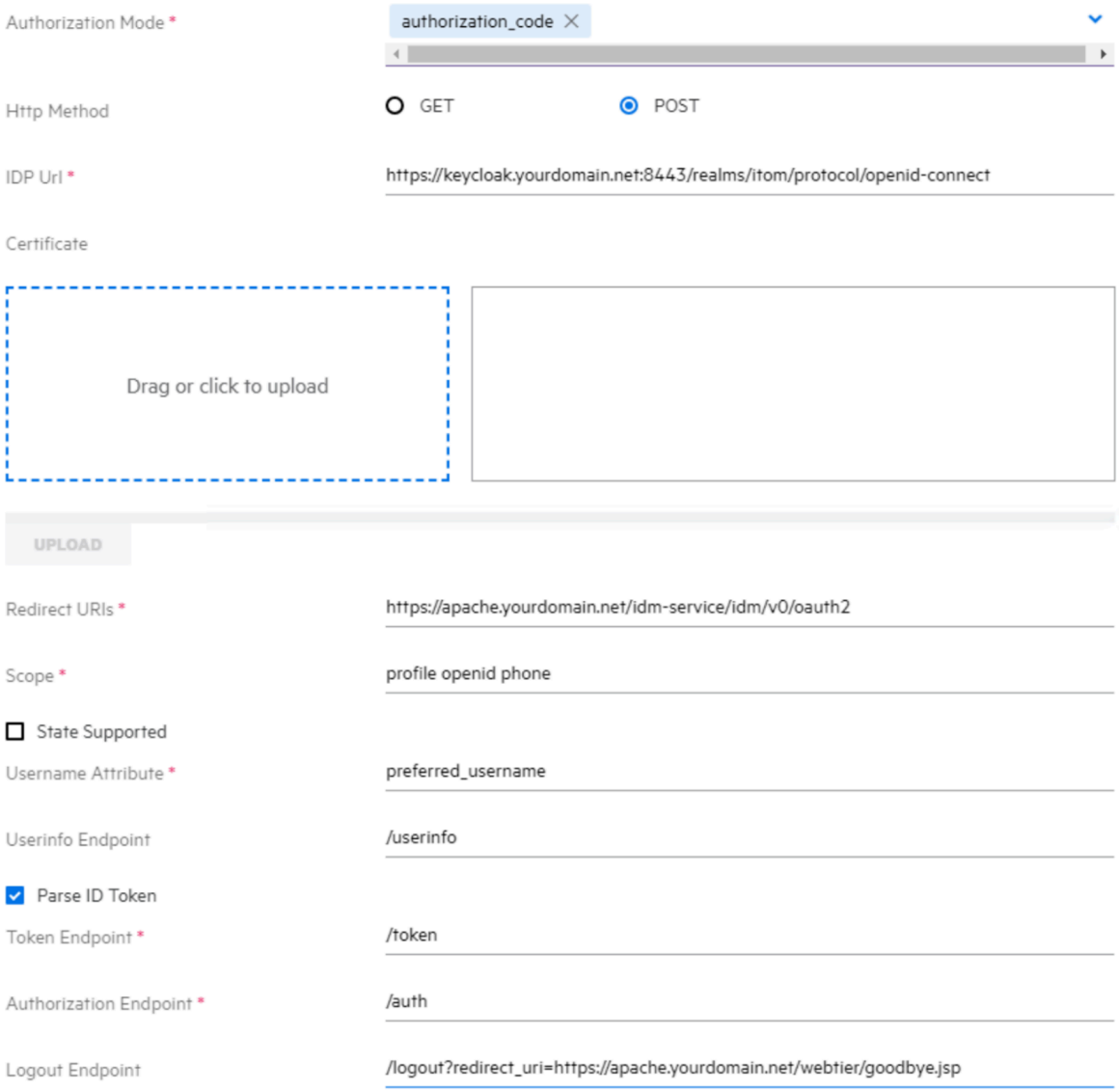
3. Click **Authentication > CONFIGURATIONS > Add (+)** , and select the authentication type as **OAUTH** to create an OAuth authentication confguration.

OAuth configuration

Field	Required	Description
Display Name	Yes	The display name of this configuration.
Shared in same family	No	Share the authentication settings within the same family. The supported values are "false" and "true''.
Client ID	Yes	The value of Client ID that you get from the OpenID identity provider.
Client Sercet	Yes	The value of Client Secret that you get from the OpenID identity provider.
HTTP Method	Yes	The HTTP method of getting a user's information from the endpoint. The supported values are "GET" and "POST". **Caution:** By selecting The GET option, you are disabling or bypassing security features, thereby exposing the system to increased security risks. By using this option, you understand and agree to assume all associated risks and hold the company harmless for the same.
IDP URL	Yes	The endpoint or URL path provided by the OpenID Identity Provider. The URL set for "Redirect URL" will be directed to the IDP URL.
Redirect URL	Yes	The value of redirect URL of the IDP URL for sign in.
Scope	Yes	The value of scope. For example, "profile openid phone". **Note:** Be sure to add the client scope `phone` if you want Keycloak to add phone number in the claim.
State Supported	No	Whether support the State Supported feature. The supported values are "false" and "true''.
Username Attribute	Yes	The attribute to define a username. For example, "preferred_username".
User Info Endpoint	Yes	An OAuth 2.0 Protected Resource that returns Claims about the authenticated end user. For example, /userinfo.

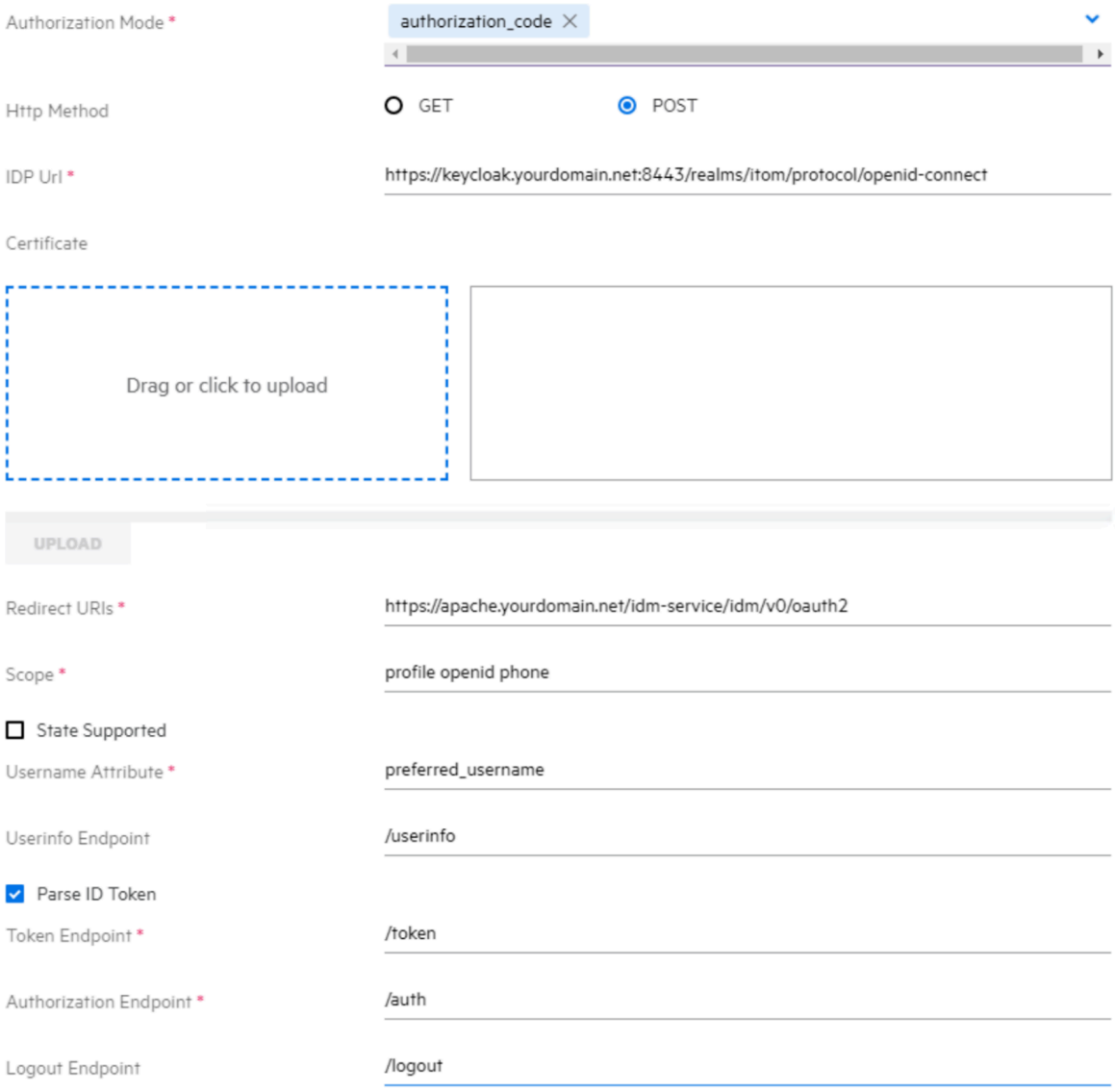
Token Endpoint	Yes	The token endpoint of the OpenID identity provider. The Token Endpoint is used to obtain a Token Response. For example, /token.
Authentication Endpoint	Yes	The Authorization Endpoint performs authentication of an end user. This is done by sending the user agent to the authorization server's endpoint for authentication and authorization, using request parameters defined by OAuth 2.0 and additional parameters and parameter values defined by OpenID Connect. For example, /auth.
Logout Endpoint	No	The token endpoint where you can end a session. For example: /logout? redirect_uri=https://apache.yourdomain.net/webtier/goodbye.jsp
Additional Parameter	No	The additional parameter for authentication.

- Here is an example:



Note that if you have upgraded Service Manager to 9.80 p1 or above, which implements idm 24.2.1 and updated

keycloak to 24.x version, then you don't need to add the "redirect_uri" parameter for **Logout Endpoint**:



Create a configuration group for OAuth

To create a configuration group for OAuth, follow these steps:

1. After you create an OAuth configuration, from the **CONFIGURATION GROUPS** section, click + to add an authentication group.
2. In the **Name** field, enter **oauth**.
3. In the **Display Name** field, enter a display name for the authentication group.
4. In the **Configurations** field, select the OAuth authentication configuration that you just created.

   Note
   You can add only one OAuth authentication configuration to the OAuth configuration group.

5. Click **SAVE**.

Step 4: Configure IdM authentication in Service Manger

See Configure IdM authentication in Service Manager for detailed instructions.

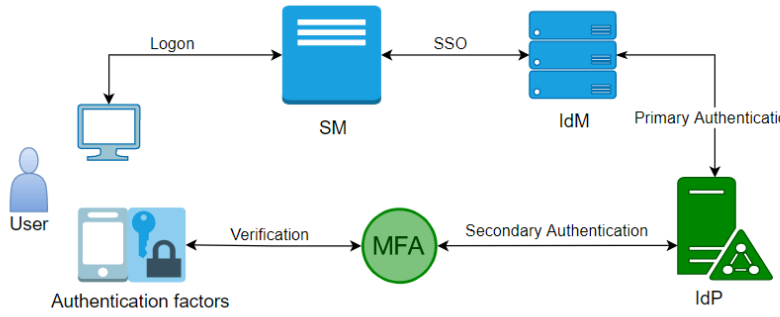
Step 5: Test your OAuth configuration

1. Access `https://apache.yourdomain.net/webtier` from your browser, and then you will be redirected to the IdP sign-in page.
2. After you successfully sign in using your IdP credentials, the web tier main page is displayed.

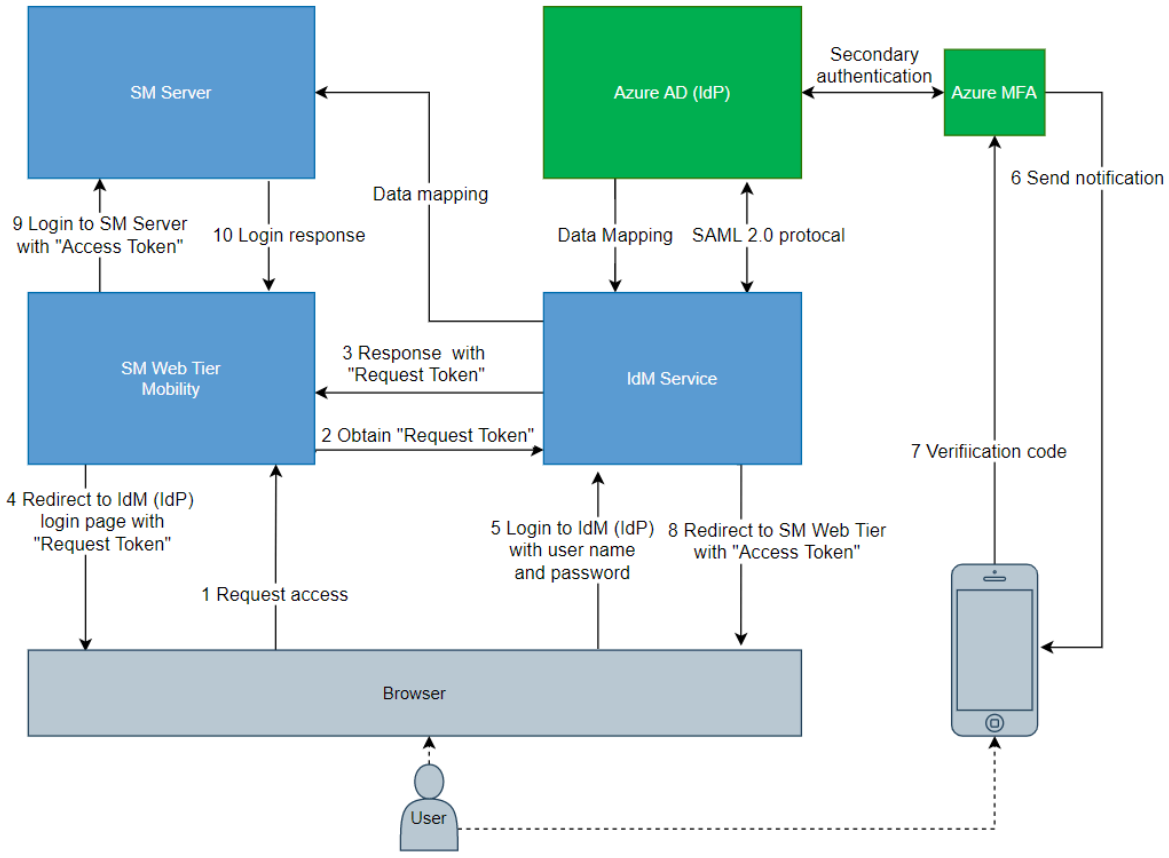
8.20. Two-Factor Authentication

Two-Factor Authentication (2FA) is an extra layer of protection that you can use to ensure the security of Service Manager accounts beyond just a username and password. It allows the Service Manager to verify the identity of the end-user based on the authentication performed by an authorization server.

Service Manager leverages IdM authentication, and IdM authentication in turn leverages a third-party identity provider to support 2FA authentication. The following diagram illustrates a simplified process of 2FA authentication.



The following diagram explicitly illustrates the 2FA authentication process of Service Manager with Azure AD as a sample IdP. In this example, all components are deployed on the Cloud.



The following table further explains each step in this process.

Step	Description
1	A user requests access to an SM client (web tier or Mobility client).
2	The SM client sends a request to the IdM service for a request token.
3	The IdM service returns a response along with a request token to the SM client.
4	The IdM client redirects the request to the IdM service along with that token, and then the IdM service redirects it to the Azure AD (IdP) login page with a SAML request.
5	The user enters a username and password to log in to SM from the Azure AD (IdP) login page. [1]
6	Azure MFA service sends a notification to the user's registered device (for example, a mobile phone).
7	The user approves the sign-in request (for example, the user enters the received verification code on the Azure login page).
8	The Azure AD (IdP) redirects the request to the IdM service with a SAML response, and then the IdM service redirects the request to the SM client along with an Access Token.
9	The SM client logs in to the SM Server with that Access Token.
10	The user logs on to SM.

[1] A user logs on to Service Manager through the Azure AD (IdP) login page, and the Azure AD (IdP) login name is determined by the Azure AD (IdP). That is, a user needs to enter the login name using the format required by the Azure AD (IdP).

Prerequisites

- Install SM server, SM client (web tier or Mobility client), and IdM server on Azure
  Refer to Install on Azure for detailed instructions.
- Configure Azure MFA
  Refer to the Azure documentation Enable Azure AD Multi-Factor Authentication for detailed instructions.
- Configure Azure as SAML IDP Server
  Refer to the IdM documentation Configure Azure as SAML IDP Server for detailed instructions.

Related topics

Install Service Manager server
Install Identity Manager (IdM)

8.21. FIPS mode

FIPS (Federal Information Processing Standards) are a set of standards that describe document processing, encryption algorithms and other information technology standards for use within U.S. non-military government agencies and by U.S. government contractors and vendors who work with the agencies.

FIPS 140-2, “Security Requirements for Cryptographic Modules,” was issued by the U.S. National Institute of Standards and Technology (NIST) in May, 2001. The standard specifies the security requirements for cryptographic modules utilized within a security system that protects sensitive or valuable data.

For FIPS 140-2 compliance, Service Manager (SM) supports the implementation of FIPS validated AES-256 data encryption, TLS connections and Lightweight Single Sign-On (LW-SSO), and FIPS compliant random number generation algorithms.

The following table describes two operation modes of the Service Manager server and clients.

Operation mode	Description	Notes
FIPS mode (FIPS 140-2 compliant mode)	Supports FIPS 140-2 compliant cryptographic functions.	
Non-FIPS mode (Non-FIPS 140-2 compliant mode)	Utilizes existing cryptography without the 3rd-party FIPS 140-2 validated cryptographic modules.	

In FIPS mode, Service Manager supports the same authentication methods as in non-FIPS mode. See the following table.

Authentication mechanism	Windows Client	Web Client	Mobility Client	Chat Server	Chat Service
Password-based mechanism (local and LDAP)	Yes	Yes	Yes	No	No
Trusted Sign-On (TSO)	Yes	Yes	Yes	No	No
LW-SSO	No	Yes	Yes	Yes	Yes
X.509 certificate authentication (CAC)	No	Yes	No	No	No
SAML-based SSO	No	Yes	Yes	Yes	Yes
OIDC-based SSO	No	Yes	Yes	Yes	Yes

Service Manager provides a set of parameters or options that determine whether the SM Server and other components are running in FIPS mode. See the following table.

Component	Parameter or option	Description
SM Server	fipsmode	Must be set to one of the following values to indicate whether the SM Server runs in FIPS mode: • 0 (default): non-FIPS mode • 2: FIPS mode
Windows Client	The **FIPS Mode** checkbox in the Preferences window	Determines whether the Windows Client runs in FIPS mode: • Selected: FIPS mode • Unselected (default): non-FIPS mode
Web Tier	The "fipsMode" parameter in the <Web Tier>/WEB-INF/webtier.properties file	Must be set to one of the following values to indicate whether the Web Tier runs in FIPS mode: • false (default): non-FIPS mode • true: FIPS mode
Mobility Client	The "fipsMode" parameter in the <Mobility Client>/WEB-INF/webtier.properties file	Must be set to one of the following values to indicate whether the Mobility Client runs in FIPS mode: • false (default): non-FIPS mode • true: FIPS mode
Chat Server	The "fipsmode" parameter in the <Openfire_home>/conf/openfire.xml file	Must be set to true or false to indicate whether the Chat Server runs in FIPS mode: • <fipsmode>false</fipsmode>: non-FIPS mode • <fipsmode>true</fipsmode>: FIPS mode
Chat Service	The securityConfig.fipsmode parameter in the <chat service>\conf\app.properties file	Must be set to true or false to indicate whether the Chat Service runs in FIPS mode: • false (default): non-FIPS mode • true: FIPS mode

FIPS validated AES-256 data encryption

Starting with version 9.32, Service Manager (SM) supports the implementation of FIPS validated AES-256 data encryption for encrypted fields in the Service Manager database.

FIPS validated TLS connections

Starting with version 9.41p3, Service Manager supports the implementation of FIPS validated TLS connections, managed by a FIPS validated cryptographic provider. The following table describes the cryptographic providers that Service Manager uses in FIPS mode.

Important

Important

- To enable FIPS mode in Service Manager, all truststore files and keystore files must use BCFKS format.
- SM Smart Analytics uses an OEM-licensed version of IDOL, which doesn't support FIPS mode. For this reason, it is impossible to implement FIPS validated TLS connections between the SM Server and the IDOL Server. However, when the SM Server is running in FIPS mode, the IDOL Server can still connect to SM through standard SSL and Smart Analytics can still work correctly.

TLS connections between	Cryptographic provider
SM Server and any of the following components: • Windows Client • Web Tier Client • Mobility Client • Chat Server • Chat Service • Web service integrations (when SM acts as a web services client or web services server)	Bouncy Castle FIPS-Java
SM Server	Bouncy Castle FIPS-Java
SM Server and LDAP/Directory Services Server	OpenSSL FIPS Object Module

FIPS validated LW-SSO

Service Manager supports FIPS validated LW-SSO. When FIPS mode is enabled on the Service Manager Server side, you have the option to enable an LW-SSO framework that's implemented by using a FIPS 140-2 validated security provider, such as the JsafeJCE provider or IBMJCEFIPS provider.

The LW-SSO configuration file (lwssofmconf.xml) in each of the following components has been updated to support LW-SSO in FIPS mode:

- SM Server
- Web Tier
- Mobility Client
- Chat Server
- Chat Service
- Identity Manager (IdM) service

FIPS compliant random number generation algorithms

When running in FIPS mode, Service Manager uses FIPS compliant random number generation algorithms, as described in the following table.

Component	Random number generation algorithm
SM Server	Uses RAND_bytes in OpenSSL.
Windows Client	Is hardcoded to use CTRAES256.
Web Tier, or Mobility Client	Uses CTRAES256 by default. You can configure the JRE's java.security file to use another FIPS compliant algorithm.
Chat Server or Chat Service	Uses CTRAES256 by default when deployed with Oracle JRE or ZULU JRE. You can configure the JRE's java.security file to use another FIPS compliant algorithm.

Related topics

Parameter: fipsmode

8.21.1. FIPS mode log messages

Service Manager writes FIPS mode related messages in the Server log and client log files.

Messages in the SM Server log

When FIPS mode is enabled on the Server side, the clients (Windows, Web Tier, Mobility, and Web Services) can connect to the Server through https (either FIPS validated TLS or normal TLS); however, the recommended best security practice is to run all of them in FIPS mode as well.

When FIPS mode is successfully enabled on the Server side, the following messages appear in the Server log (Server\logs\sm.log):

```
FIPS mode is set successfully.
FIPS provider was initialized successfully.
```

Additionally, when FIPS mode is enabled on the Server side and the Windows Client is running in non-FIPS mode, the following warning message appears in the Server log:

```
Your Client is running in non-FIPS mode, but the Service Manager Server is running in FIPS mode. For the
highest security, we recommend you enable FIPS mode for the Client.
```

Note

Note This message will also appear in the Windows Client user interface.

Messages in SM client logs

When FIPS mode is enabled successfully on the client sides, messages are written to the client logs. See the following table.

Client	Example messages	Log file
Web Tier	Examples: 2020-06-02 13:11:36,592 INFO [main] [-] [com.hp.ov.sm.client.common.util.JCEProviderHelper] JCE Provider has registered in java.security:BCFIPS 2020-06-02 13:11:36,592 INFO [main] [-] [com.hp.ov.sm.client.common.util.JCEProviderHelper] Fips mode enabled.	The sm.log file in the web application server installation directory. For example, <Tomcat>\bin\sm.log.
Mobility Client	Examples: [main] (02/06/2020 13:11:52,764) [INFO] client.util.JCEProviderHelper - JCE Provider has registered in java.security:BCFIPS [main] (02/06/2020 13:11:52,764) [INFO] client.util.JCEProviderHelper - Fips mode enabled.	The mobility.log file in the web application server installation directory. For example, <Tomcat>\bin\mobility.log.
Chat Server	Examples: 2020.06.04 15:24:47 com.hp.ov.sm.client.common.util.JCEProviderHelper - JCE provider BCFIPS has been initialized successfully. 2020.06.04 15:24:47 com.hp.ov.sm.client.common.util.JCEProviderHelper - Adding the new SunJSSEProviderName provider which is configured in FIPS mode. 2020.06.04 15:24:47 com.hp.ov.sm.client.common.util.JCEProviderHelper - FIPS JSSE provider has been initialized successfully. 2020.06.04 15:24:47 com.hp.ov.sm.client.common.util.JCEProviderHelper - Fips mode enabled. 2020.06.04 15:24:47 org.jivesoftware.util.AesEncryptor - AesEncryptor initialized for FIPS mode.	The info.log file in the chat server installation directory. For example, <Chat Server>\logs\info.log.

Chat Service	Examples: INFO [2020-06-04 15:56:28,419] com.hp.ov.sm.client.common.util.JCEProviderHelper: JCE provider BCFIPS has been initialized successfully. INFO [2020-06-04 15:56:28,419] com.hp.ov.sm.client.common.util.JCEProviderHelper: Adding the new SunJSSEProviderName provider which is configured in FIPS mode. INFO [2020-06-04 15:56:28,419] com.hp.ov.sm.client.common.util.JCEProviderHelper: FIPS JSSE provider has been initialized successfully. INFO [2020-06-04 15:56:28,419] com.hp.ov.sm.client.common.util.JCEProviderHelper: Fips mode enabled. INFO [2020-06-04 15:56:28,419] com.hp.ov.sm.chat.chatservice.util.AesEncryptor: AesEncryptor initialized for FIPS mode	Console

Related topics

FIPS mode

Parameter: fipsmode

8.21.2. Limitations of FIPS mode

The following limitation exist in the support of FIPS mode for Service Manager (SM).

Service Manager initialization might be slow in Linux environments

This issue may occur when fipsmode:2 is specified in the SM Server's sm.ini file. It is caused by a lack of entropy (randomness) available on the system used for random number generation and can commonly occur on systems that aren't receiving much direct user input, such as virtual machines.

As a workaround, you can direct the JVM to use /dev/urandom with the following directive:

- Add the following line to the Server's sm.ini file:

```
JVMOption0:-Djava.security.egd=file:/dev/./urandom
```

- Alternatively, you can add the following line to the jre/lib/security/java.security file:

```
securerandom.source=file:/dev/./urandom
```

For additional information, see the following important third-party information:

http://bugs.java.com/view_bug.do?bug_id=6521844

http://www.2uo.de/myths-about-urandom/

Related topics

FIPS mode

8.21.3. Server parameters for FIPS mode

The Service Manager Server uses the following parameters for enabling FIPS mode:

fipsmode

sslEnabledProtocols

ciphers

keystoreType

truststoreType

8.21.4. Configuring FIPS mode in Service Manager

For the best security, you can configure FIPS mode in Service Manager (SM) to implement FIPS 140-2 validated TLS connections in the entire system.

> **Note**
>
> Note Once FIPS mode is enabled on the SM Server side (that is, `fipsmode:2` is specified in the sm.ini file), clients running in non-FIPS mode are still allowed to connect to the SM Server through normal SSL. However, for the best security, this is not recommended.

To enable FIPS mode in Service Manager, perform the following steps in the listed order:

1. Configure Java and web application servers for FIPS mode
2. Generate FIPS validated certificates for the SM Server and other components
3. Configure FIPS mode in the Server
4. Configure FIPS mode in the Windows Client
5. Configure FIPS mode in the Web Tier
6. Configure FIPS mode in the Mobility Client
7. Configure FIPS mode in the Chat Server
8. Configure FIPS mode in the Chat Service
9. Configure FIPS mode in the IdM Service

Additionally, if you need to use LW-SSO, you can configure Service Manager to use FIPS validated LW-SSO. For detailed instructions, see Configuring LW-SSO in FIPS mode.

8.21.4.1. Configure Java and web application servers for FIPS mode

The Service Manager web tier, mobility, and IdM service are deployed on web application servers. You need to first configure Java security providers and web application servers for FIPS mode, and then configure FIPS mode for individual SM components.

> **Note**
>
> You don't need to manually configure Java security providers for the SM server, chat server, chat service, and Windows client. These components will automatically configure Java security providers when you enable FIPS mode for them.

Prerequisites

Before you proceed, make sure of the following:

1. You have installed Java 17 and web application servers.
2. You have already generated a server certificate file for web application servers. For details, see Generate FIPS validated certificates for the SM Server and other components.

Configure Java security provider

You need to configure the java security properties file to use the BCFIPS security provider as the default security provider.

1. Create a folder to store FIPS-related files (for example, `C:\FIPS`). This topic refers to this folder as `<FIPS_HOME>` in later steps.
2. Copy the `bc-fips-<version>.jar` file from the `<TOMCAT_HOME>/webapps/<appname>/WEB-INF/lib` folder to the `<FIPS_HOME>` folder. Note that the `<appname>` may be `webtier`, `mobility`, or `idmservice`.
3. Copy the `java.security` file from the `<JAVA_HOME>/conf/security` folder to the `<FIPS_HOME>` folder.
4. Edit the `<FIPS_HOME>/java.security` file.
    1. Add BCFIPS provider as the first security provider, and update provider ID for subsequent providers.
    2. Enable FIPS mode for SunJSSE by specifying a BCFIPS crypto provider for it.
    3. Set the default keystore type to `bcfks`.

    For example:

```
...
security.provider.1=BCFIPS C:DEFRND[CTRAES256];ENABLE{ALL};
security.provider.2=SUN
security.provider.3=SunRsaSign
security.provider.4=SunEC
security.provider.5=SunJSSE BCFIPS
security.provider.6=SunJCE
security.provider.7=SunJGSS
security.provider.8=SunSASL
security.provider.9=XMLDSig
security.provider.10=SunPCSC
security.provider.11=JdkLDAP
security.provider.12=JdkSASL
security.provider.13=SunMSCAPI
security.provider.14=SunPKCS11
...
keystore.type=bcfks
...
```

Configure Tomcat

You need to configure Tomcat to use the customized `java.security` file by setting appropriate JVM options.

If you start Tomcat by running the script file (`startup.bat` or `startup.sh`), you need to first create the `setenv.bat` or `setenv.sh` file in the `<TOMCAT_HOME>/bin` folder, and then add JVM options to the `setenv.bat` or `setenv.sh` file. For example:

Windows (`setenv.bat`):

```
set FIPS_HOME=<FIPS_HOME>
set FIPS_OPTS=--module-path=%FIPS_HOME% --add-
opens=java.base/sun.security.internal.spec=org.bouncycastle.fips.core
set FIPS_OPTS=%FIPS_OPTS% -Djava.security.properties==%FIPS_HOME%\java.security -
Dorg.bouncycastle.fips.approved_only=true
set JAVA_OPTS=%JAVA_OPTS% %FIPS_OPTS%
```

Linux (`setenv.sh`):

```
#!/bin/sh
FIPS_HOME=<FIPS_HOME>
FIPS_OPTS="--module-path=${FIPS_HOME} --add-
opens=java.base/sun.security.internal.spec=org.bouncycastle.fips.core"
FIPS_OPTS="${FIPS_OPTS} -Djava.security.properties==${FIPS_HOME}/java.security -
Dorg.bouncycastle.fips.approved_only=true"
JAVA_OPTS="${JAVA_OPTS} ${FIPS_OPTS}"
```

If you install Tomcat as a Windows service, start the tomcat9w.exe, switch to the **Java** tab, and then add the following JVM options to the **Java Options** list. For example:

```
--module-path=<FIPS_HOME>
--add-opens=java.base/sun.security.internal.spec=org.bouncycastle.fips.core
-Djava.security.properties==<FIPS_HOME>\java.security
-Dorg.bouncycastle.fips.approved_only=true
```

Next, copy the `cacerts` file and the Tomcat server certificate keystore file from `cert-utils` to the `<TOMCAT_HOME>/conf` folder. Edit the `<TOMCAT_HOME>/conf/server.xml` file to configure the truststore and the keystore for the https connector. For example:

```
...
<Connector port="8443" protocol="org.apache.coyote.http11.Http11NioProtocol"
           maxThreads="150" SSLEnabled="true">

    <SSLHostConfig truststoreFile="conf/cacerts"
                   truststorePassword="changeit"
                   truststoreType="bcfks">
        <Certificate certificateKeystoreFile="conf/tomcat.bcfks"
                     certificateKeystorePassword="serverkeystore"
                     certificateKeystoreType="bcfks" />
    </SSLHostConfig>
</Connector>
...
```

Note

You need to use the appropriate Tomcat server certificate keystore file for different SM components and then change the `certificateKeystoreFile` attribute accordingly. For example:

SM component	certificateKeystoreFile
web tier	conf/tomcat.bcfks
mobility	conf/mobilitytomcat.bcfks
idmservice	conf/idmservice.bcfks

Configure WebSphere Application Server Liberty

Important

The following sample configuration assumes that you are working with Open JDK or Oracle JDK. If you are working with IBM SDK, skip this section and refer to Setting up Liberty for FIPS compliance for more information.

You need to create jvm.options in the ${wlp.user.dir}/servers/serverName folder, and then update this file by adding configurations that are related to FIPS mode.

```
--module-path=<FIPS_HOME>
--add-opens=java.base/sun.security.internal.spec=org.bouncycastle.fips.core
-Djava.security.properties==<FIPS_HOME>\java.security
-Dorg.bouncycastle.fips.approved_only=true
```

Next, check whether the ${wlp.user.dir}/servers/serverName/resources/security folder exists or not. If not, create this folder and then copy the WebSphere Application Server Liberty certificate keystore file for the SM components (such as

mobilitywasliberty.bcfks or wasliberty.bcfks) from cert-utils to it. Refer to the following example to update the ${wlp.user.dir}/servers/serverName/server.xml file.

```
<!-- Enable features -->
    <featureManager>
           <feature>servlet-6.0</feature>
           <feature>ssl-1.0</feature>
           <feature>pages-3.1</feature>
    </featureManager>

<httpSession idLength="128" useContextRootAsCookiePath="true"/>

<!-- To access this server from a remote client add a host attribute to the following element, e.g.
host="*" -->
<httpEndpoint id="defaultHttpEndpoint"
                  enabled="true"
                  host="*"
             httpPort="9080"
             httpsPort="9443" />

<!-- default SSL configuration is defaultSSLSettings -->
    <sslDefault sslRef="defaultSSLSettings"/>
    <ssl id="defaultSSLSettings"
         keyStoreRef="defaultKeyStore"/>
    <keyStore id="defaultKeyStore"
              location="wasliberty.bcfks"
              password="serverkeystore"
              type="bcfks"/>
```

The example above only specifies the default SSL certificate, keystore, and configuration in Liberty. For more detailed instructions about how to enable SSL communication in Liberty, see Enabling SSL communication in Liberty.

Next step:

Generate FIPS validated certificates for the SM Server and other components

8.21.4.2. Generate FIPS validated certificates for the SM Server and other components

Before you can enable FIPS mode in the Service Manager (SM) system, you must obtain a CA certificates file and keystore files in BCFKS format (with the .bcfks file extension) for the SM Server and other components. These certificates are used for FIPS validated TLS connections between the SM Server and other components.

Note

Note In a production environment, you are recommended to use digital security certificates issued by a certificate authority provider, such as Verisign, Thawte, or your corporate certificate authority. The digital certificate contains a public key, the identity of the owner, and a matching private key. The certificate is required to encrypt data sent and received in a “trusted” environment. If you don't have a digital certificate from a provider listed above, you have the option of generating your own certificates to enable encrypted data transfer between the Service manager Server and the trusted clients, as described in the following.

- *Existing customers* need to either convert your existing certificates to BCFKS format or regenerate new BCFKS certificates.

Caution

Caution The SM Server truststore and trusted clients keystore don't contain a private key, and can work correctly in FIPS mode after conversion to BCFKS format. However, the SM server keystore and client keystores contain a private key. Before converting them to BCFKS format, make sure their private key was created using either "keytool -genkey -keyalg RSA" or "keytool -genkeypair -keyalg RSA"; otherwise they can't work correctly in FIPS mode after conversion to BCFKS format. Before you proceed, make sure the private key of the existing JKS file was created as noted above. If not, you will need to generate a new BCFKS certificate.

- *New customers* need to generate BCFKS certificates for the SM Server and clients (Windows, Web Tier, and Mobility).

As a courtesy, we provide a certificate generation toolkit to help you generate certificates for a Service Manager deployment in an automated way. For details, see Work with the certificate generation toolkit.

Next step:

Configure FIPS mode in the Server

8.21.4.3. Configure FIPS mode in the Server

By default, FIPS mode is disabled, and Service Manager uses the 64-bit DES data encryption algorithm and supports non-FIPS validated SSL/TLS connections. To run Service Manager in FIPS mode, you need to enable FIPS validated data encryption and TLS connections.

Prerequisites

Before you proceed, make sure you have already generated the following certificate files for the SM Server. For details, see Work with the certificate generation toolkit.

- The CA certificates file: \certs\cacerts
- The Server's trusted clients keystore file: \certs\trustedclients.bcfks
- The Server keystore file: \certs\smserver.bcfks

To enable FIPS mode in the SM Server, complete the following tasks.

Caution

To enable FIPS mode in a scaling environment, you must configure FIPS mode in each of the SM server nodes.

Steps to enable FIPS mode for data encryption

After enabling FIPS mode, you must upgrade your database's encryption algorithm from DES to AES by running the **sm -upgradeencralg** command.

1. Make a backup of your database.

   Caution

   FIPS mode requires the database to upgrade to the 256-bit AES encryption algorithm. Because the upgrade is irreversible, you need to make a backup of your database.

2. Check the length of each encrypted field in your database, and increase the length if needed.

   Tip

   Use the following formula to determine a safe length for an AES encrypted field (in bytes):

   ```
   Safe_Length=32 + 2 * source_string_length
   ```

   Where: source_string_length is the length of the source string (for example, if you enter an 8-character password, the source string length is 8 bytes).

3. Upgrade your applications to the AES data encryption algorithm.
   FIPS mode requires the use of the 256-bit AES data encryption algorithm for the database. By default, the Service Manager applications uses the 64-bit Data Encryption Standard (DES) encryption algorithm. You must upgrade the database from the DES encryption algorithm to AES by running the `sm -upgradeencralg` command.

   Caution

   This algorithm upgrade is required only when you want to enable FIPS mode.

   a. Make sure you can connect to the server successfully.
   b. Make sure all Service Manager server processes are stopped.
   c. Check that the server's configuration file (`sm.ini`) doesn't include the `fipsmode` parameter.
   d. From <Servcie Manager installation directory>/Server/Run, run the **sm -upgradeencralg** command.

   Note

   If you have many tables with encrypted fields, running this command may take quite a while; if you have already run this command on the database, running it again will have no effect and do no harm to the database.

   Tip

   You can specify a 32-character alphanumeric value for this parameter, as the new encryption key; if you don't specify any value, a default 32-character (256-bit) encryption key is used. You can also change your encryption key later using the `changeencrkey` parameter.

   e. Check the server log (sm.log) for any errors.

Steps to enable FIPS mode for TLS connections

Follow these steps:

1. Copy the certificate files listed in the previous step to the SM Server's RUN directory:
   • The CA certificates file: \certs\cacerts
   • The Server's trusted clients keystore file: trustedclients.bcfks
   • The Server keystore file: smserver.bcfks
2. Add the following parameters to the sm.ini file in the Server's RUN folder.

   Caution

   The example certificate file names and passwords used in the following lines are the default ones specified in the certificate generation batch files provided by the company. For details, see Generate FIPS validated certificates for the SM Server and other components. If you use different file names or passwords when generating your certificates, be sure to replace them with your own values.

   Tip

   The `ciphers` and `sslEnabledProtocols` parameters are specified here with their default values. Since Service Manager will use their default values when the two parameters aren't present in the sm.ini file, you can omit them safely. These parameters are provided just in case you don't want to use their default values.

```
fipsmode:2
ssl:1
sslConnector:1
ssl_reqClientAuth:2
keystoreFile:smserver.bcfks
keystorePass:serverkeystore
ssl_trustedClientsJKS:trustedclients.bcfks
ssl_trustedClientsPwd:trustedclients
truststoreFile:cacerts
truststorePass:changeit
ciphers:TLS_AES_256_GCM_SHA384,TLS_AES_128_GCM_SHA256,TLS_ECDHE_ECDSA_WITH_AES_256_GCM_SHA384,TLS_E
CDHE_ECDSA_WITH_AES_128_GCM_SHA256,TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384,TLS_ECDHE_RSA_WITH_AES_128
_GCM_SHA256
sslEnabledProtocols:TLSv1.2,TLSv1.3
JVMOption1:-Djdk.tls.ephemeralDHKeySize=2048
```

Note
When running under JDK 8, the JDK's SunJSSE TLS implementation is unable to negotiate a TLS handshake when using an ECDHE cipher suite.

3. Go to the <Service Manager server installation path>/RUN folder, and open `lwssofmconf.xml` in a text editor.
4. Locate the following section:

```
<lwssoValidation>
<domain>yourdomain.net</domain>
<crypto cipherType="symmetricBlockCipher"
engineName="AES" paddingModeName="CBC" keySize="256"
encodingMode="Base64Url"
initString="This is a shared secret passphrase"/>
<!-- When Service Manager is running in FIPS mode, the crypto configuration above needs to be
replaced with the configuration below. The initString value must be 32 characters long and contain
no spaces.  -->
<!--
<crypto cryptoSource="jce" cipherType="symmetricBlockCipher"
engineName="AES" paddingModeName="CBC" keySize="256"
pbeDigestAlgorithm="SHA1" encodingMode="Base64Url"
jcePbeAlgorithmName="AES" jcePbeMacAlgorithmName="AES"
macType="hmac"
macAlgorithmName="SHA1" directKeyEncoded="true" directKeyEncoding="Base64Url"
algorithmPaddingName="PKCS5Padding"
pbeCount="20" macKeySize="256" macPbeCount="20"
initString="rqa9mkdmfkvdvienfj4usldo9205mvid">
</crypto>
-->
</lwssoValidation>
```

Comment out the ordinary crypto configuration part, and then uncomment the FIPS mode crypto configuration part.
Note: Please make sure that the value of `initString` is complied with the encoding rule defined for `directKeyEncoding` in the section above.
When Service Manager starts, the system loads the LW-SSO related jar files and verifies the algorithms that are related to LW-SSO configuration. Based on the verification result, the system judges whether Service Manager is running in FIPS mode or not. Hence, to ensure Service Manager is running in FIPS mode, you must switch the crypto configuration in lwssofmconf.xml to the one as FIPS mode required.
Note: Do not change the domain setting in this file unless you are configuring LW-SSO.

```
<domain>yourdomain.net</domain>
```

5. Restart the SM Server.
   Next, you need to enable FIPS mode in the Windows Client to verify that your FIPS mode is successfully enabled on the Server side.

Next step:

Configure FIPS mode in the Windows Client

8.21.4.4. Configure FIPS mode in the Windows Client

Once you have enabled FIPS mode on the Service Manager (SM) Server side, you need to enable FIPS mode in the Windows Client and test if the Windows Client can connect to the SM Server successfully.

Prerequisites

Before you proceed, make sure of the following:

1. You have already generated a CA certificates file and a client keystore for the Windows Client host. For details, see Generate FIPS validated certificates for the SM Server and other components.
   - The CA certificates file: \certs\smrootca.bcfks
   - The Windows Client keystore file: \certs\<Windows Client alias defined in certs.conf>.bcfks (for example, james.jones.bcfks)
2. You have enabled FIPS mode on the SM Server side. For details, see Configure FIPS mode in the Server.

Steps to enable FIPS mode in the Windows Client

To configure FIPS mode in the Windows Client, follow these steps:

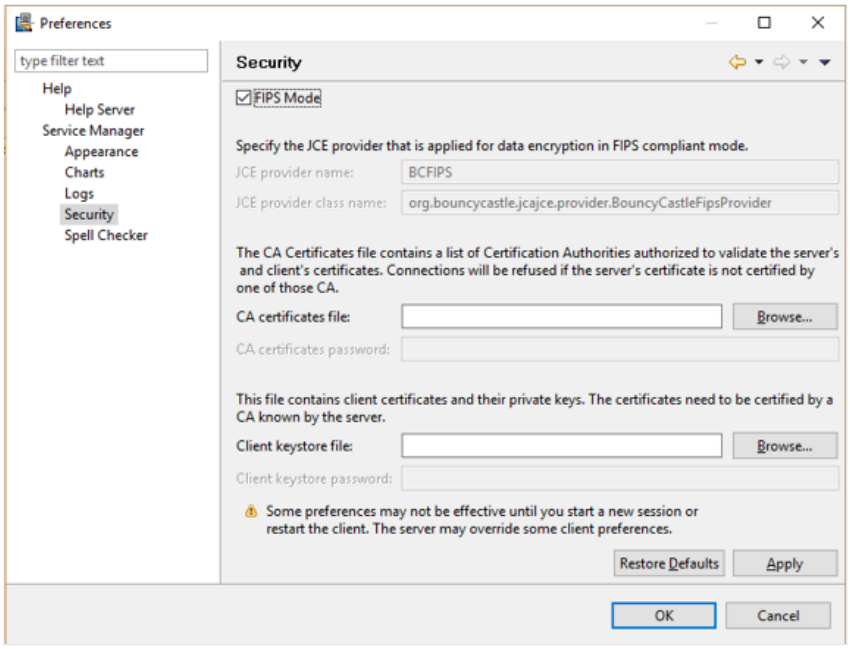
1. Copy the CA certificates file and client keystore file to a directory on the Windows Client host:
   - smrootca.bcfks
   - james.jones.bcfks
2. Configure the Windows Client preferences.
   1. Start the Windows client, and click **Window** > **Preferences** > **Service Manager** > **Security**.
   2. Select the **FIPS Mode** checkbox.
      A message appears, indicating that you need to restart the Windows Client to continue.
      Click **OK** to exit the message window.
      Because the Windows Client is hardcoded to use the RSA BSAFE JsafeJCE provider for FIPS mode, you no longer need to specify a cryptographic provider. For this reason, the following parameters are disabled (grayed out) when the **FIPS Mode** option is selected:
      - JCE provider name
      - JCE provider class name

      Tip

      Tip These parameters are used to specify a third-party Java Cryptography Extension (JCE) provider when the Windows Client is running in non-FIPS mode (that is, when the **FIPS Mode** checkbox is not selected).

      Note that the following parameters are also disabled at this point, because a restart of the Windows Client is needed after the **FIPS Mode** option is enabled. After a restart, these parameters will be available to configure in the Preferences window.
      - CA certificates password
      - Client keystore password

      The following screenshot illustrates the Preferences window when **FIPS Mode** is selected.



Click **OK** again to exit the Preferences window.

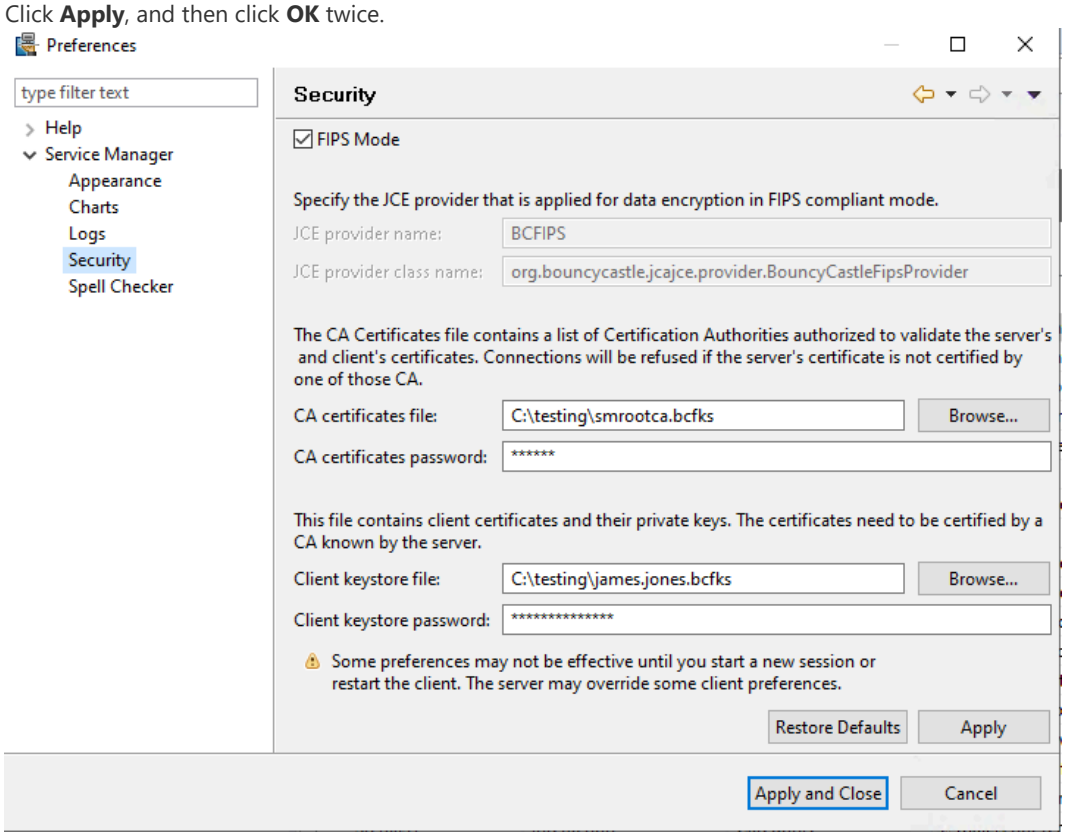
Close the Windows Client.

3. Restart the Windows Client.
4. Click **Window** > **Preferences** > **Service Manager** > **Security**.
5. Configure the parameters as described in the following.

   Click **Browse** to enter the CA certificates file, and then enter the password in the **CA certificates password** field.

   Click **Browse** to enter the client keystore file, and then enter the password in the **Client keystore password** field.

   Click **Apply**, and then click **OK** twice.



6. Click **File** to open your client connection, and do the following to verify the SM Server and Windows Client have FIPS mode enabled correctly:
   1. Select the **Use Login/Password** option on the Connection tab.

2. Check that the **Use SSL Encryption** option on the Advanced tab is selected and grayed out.

   Note
   Note When the **FIPS Mode** option is enabled, the Windows Client is forced to use SSL encryption.

3. Enter your user name and password.
4. In the **Server host name** field, enter the fully qualified domain name (FQDN) of the SM Server.
5. In the **Server port number** field, enter the HTTP port (for example, 13080) of the SM Server.
6. Click **Connect**.

If FIPS mode has been successfully enabled in the SM Server and Windows Client, you should be successfully logged in.

Next step:

Configure FIPS mode in the Web Tier

8.21.4.5. Configure FIPS mode in the Web Tier

Once you have enabled FIPS mode on the Service Manager (SM) Server side, you need to enable FIPS mode in the Web Tier.

Prerequisites

- You have configured Java security providers and web application servers for FIPS mode. For details, see Configure Java and web application servers for FIPS mode.
- You have already generated a CA certificates file and a client keystore for the Web Tier host. For details, see Work with the certificate generation toolkit.
- You have enabled FIPS mode on the SM Server side. For details, see Configure FIPS mode in the Server.

Steps to enable FIPS mode in the Web Tier

To configure FIPS mode in the web client, follow these steps:

1. Copy the smrootca.bcfks and smwebtier.bcfks files to <Customize-Folder>/config folder if the `customize-folder` parameter is set in the web.xml, and then append the following scripts to the <Customize-Folder>/config /webtier.properties file.

   ```
   fipsMode=true
   ssl=true
   keystore=<Customize-Folder>/config/smwebtier.bcfks
   keystorePassword=clientkeystore
   cacerts=<Customize-Folder>/config/smrootca.bcfks
   cacertsPassword=caroot
   cookie.secure=true
   ```

   Copy the smrootca.bcfks and smwebtier.bcfks files to <WEBTIER>/WEB-INF folder if the `customize-folder` parameter is not set in the web.xml, and then append the following scripts to the <WEBTIER>/WEB-INF/webtier.properties file.

   ```
   fipsMode=true
   ssl=true
   keystore=/WEB-INF/smwebtier.bcfks
   keystorePassword=clientkeystore
   cacerts=/WEB-INF/smrootca.bcfks
   cacertsPassword=caroot
   cookie.secure=true
   ```

2. Go to the <web tier root>/WEB-INF/classes folder, and open `lwssofmconf.xml` in a text editor.

3. Locate the following section:

```
<lwssoValidation>
<domain>yourdomain.net</domain>
<crypto cipherType="symmetricBlockCipher"
engineName="AES" paddingModeName="CBC" keySize="256"
encodingMode="Base64Url"
initString="This is a shared secret passphrase"/>
<!-- When Service Manager is running in FIPS mode, the crypto configuration above needs to be
replaced with the configuration below. The initString value must be 32 characters long and contain
no spaces.  -->
<!--
<crypto cryptoSource="jce" cipherType="symmetricBlockCipher"
engineName="AES" paddingModeName="CBC" keySize="256"
pbeDigestAlgorithm="SHA1" encodingMode="Base64Url"
jcePbeAlgorithmName="AES" jcePbeMacAlgorithmName="AES"
macType="hmac"
macAlgorithmName="SHA1" directKeyEncoded="true" directKeyEncoding="Base64Url"
algorithmPaddingName="PKCS5Padding"
pbeCount="20" macKeySize="256" macPbeCount="20"
initString="rqa9mkdmfkvdvienfj4usldo9205mvid">
</crypto>
-->
</lwssoValidation>
```

   Comment out the ordinary crypto configuration part, and then uncomment the FIPS mode crypto configuration part.
   Note: Please make sure that the value of `initString` is complied with the encoding rule defined for `directKeyEncoding` in the section above.
   When Service Manager starts, the system loads the LW-SSO related jar files and verifies the algorithms that are related to LW-SSO configuration. Based on the verification result, the system judges whether Service Manager is running in FIPS mode or not. Hence, to ensure Service Manager is running in FIPS mode, you must switch the crypto configuration in lwssofmconf.xml to the one as FIPS mode required.
   Note: Do not change the domain setting in this file unless you are configuring LW-SSO.

```
<domain>yourdomain.net</domain>
```

4. Restart your web application server for your configuration to take effect.
5. Launch your Web Tier client URL in a browser, and log in with your user account.
   If FIPS mode has been successfully enabled in the Web Tier, you should be successfully logged in.

Next step:

Configure FIPS mode in the Mobility Client

8.21.4.6. Configure FIPS mode in the Mobility Client

Once FIPS mode is enabled on the SM Server side, you need to enable FIPS-validated data encryption and TLS connections between the Service Manager Server and Mobility Client.

This topic provides both prerequisites and sample steps about how to enable FIPS mode in the Mobility Client if you have deployed the Mobility Client on Tomcat.

Prerequisites

Before you proceed, make sure of the following:

- You have configured Java security providers and web application servers for FIPS mode. For details, see Configure Java and web application servers for FIPS mode.
- You have already generated a CA certificates file and a client keystore for the Web Tier host. For details, see Work with the certificate generation toolkit.

- You have enabled FIPS mode on the SM Server side. For details, see Configure FIPS mode in the Server.

Steps to enable FIPS mode in the Mobility Client

To enable FIPS mode in the Mobility Client, follow these steps:

1. Copy the certificate files to the Mobility Client's WEB-INF folder (webapp-9.xx.xxxx.war\WEB-INF\):
   • The CA certificates file: smrootca.bcfks
   • The Mobility Client keystore file: smmobility.bcfks
2. Open the web.properties file in the Mobility Client's WEB-INF folder in a text editor.
3. Configure the following parameters in the file and then save the file:

```
endpoint=https://smserver.mycompany.net:13443/SM/ui
fipsMode=true
cacerts=/WEB-INF/smrootca.bcfks
keystore=/WEB-INF/smmobility.bcfks
keystorePassword=clientkeystore
cacertsPassword=caroot
```

4. Go to the Mobility Client's WEB-INF/classes folder, and open `lwssofmconf.xml` in a text editor.
5. Locate the following section:

```
<lwssoValidation>
<domain>yourdomain.net</domain>
<crypto cipherType="symmetricBlockCipher"
engineName="AES" paddingModeName="CBC" keySize="256"
encodingMode="Base64Url"
initString="This is a shared secret passphrase"/>
<!-- When Service Manager is running in FIPS mode, the crypto configuration above needs to be
replaced with the configuration below. The initString value must be 32 characters long and contain
no spaces.  -->
<!--
<crypto cryptoSource="jce" cipherType="symmetricBlockCipher"
engineName="AES" paddingModeName="CBC" keySize="256"
pbeDigestAlgorithm="SHA1" encodingMode="Base64Url"
jcePbeAlgorithmName="AES" jcePbeMacAlgorithmName="AES"
macType="hmac"
macAlgorithmName="SHA1" directKeyEncoded="true" directKeyEncoding="Base64Url"
algorithmPaddingName="PKCS5Padding"
pbeCount="20" macKeySize="256" macPbeCount="20"
initString="rqa9mkdmfkvdvienfj4usldo9205mvid">
</crypto>
-->
</lwssoValidation>
```

   Comment out the ordinary crypto configuration part, and then uncomment the FIPS mode crypto configuration part.
   Note: Please make sure that the value of `initString` is complied with the encoding rule defined for `directKeyEncoding` in the section above.
   When Service Manager starts, the system loads the LW-SSO related jar files and verifies the algorithms that are related to LW-SSO configuration. Based on the verification result, the system judges whether Service Manager is running in FIPS mode or not. Hence, to ensure Service Manager is running in FIPS mode, you must switch the crypto configuration in lwssofmconf.xml to the one as FIPS mode required.
   Note: Do not change the domain setting in this file unless you are configuring LW-SSO.

```
<domain>yourdomain.net</domain>
```

6. Restart the Mobility Client's web application server.

7. Make sure the SM Server is started.
8. Sign in to the Mobility Client.
   If FIPS mode has been enabled correctly in the Mobility Client, you should be signed in successfully.

Next step:

Configure FIPS mode in the Chat Server

8.21.4.7. Configure FIPS mode in the Chat Server

Note: The instructions here are supplementary to the steps that are provided in configure the chat server in HTTPS mode.

Before you proceed, make sure of the following:

1. You have already generated two CA certificate files and a keystore file for the Chat Server. For details, see Work with the certificate generation toolkit.
   • The CA certificates file for connections from other servers: \certs\smrootca.bcfks
   • The CA certificates file for connections from clients: \certs\smrootca.bcfks
   • The Server keystore file: \certs\smchatserver.bcfks
2. You have enabled FIPS mode on the SM Server side. For details, see Configure FIPS mode in the Server.
3. If you are configuring FIPS mode in an existing Chat Server, that is, the Chat Server ran in non-FIPS mode before, you must perform the following extra steps before you configure the chat server in HTTPS mode.
   a. Open the Openfire Administration Console in non-FIPS mode, and then click **System Properties**. Remove the following encrypted configuration items:
   `activemq.auth.password`
   `activemq.auth.userName`
   `admin.web.ssl.keypass`
   `admin.web.ssl.trustpass`
   `passwordKey`
   `xmpp.bosh.ssl.client.keypass`
   `xmpp.bosh.ssl.client.trustpass`
   `xmpp.component.keypass`
   `xmpp.component.trustpass`
   `xmpp.multiplex.keypass`
   `xmpp.multiplex.trustpass`
   `xmpp.socket.ssl.client.keypass`
   `xmpp.socket.ssl.client.trustpass`
   `xmpp.socket.ssl.keypass`
   `xmpp.socket.ssl.trustpass`
   b. Stop the Chat Server.
   c. Open <chat server>\conf\openfire.xml with a text editor. Update all the encrypted items with plain text values, and then update the value of `setup` to false: `<setup>false</setup>`.

Next, follow these steps to enable FIPS mode in the Chat Server:

Note: For a new installation, these are the prerequisite steps before you do further configurations through the Administration Console.

1. Copy the CA certificate files and keystore file (smrootca.bcfks and smchatserver.bcfks) to the <chat server>\resources\security folder.
2. Open the chat server configuration file (<chat server>\conf\openfire.xml) in a text editor.
3. Locate the `<securityConfig>` section, and then set the `fipsmode` parameter to true (default: false), as shown in the following.

```
<securityConfig>
        <ssl>
          <keyStoreType>BCFKS</keyStoreType>
          <keyStorePath>resources/security/smchatserver.bcfks</keyStorePath>
          <keyStorePassword>serverkeystore</keyStorePassword>
          <trustStorePath>resources/security/smrootca.bcfks</trustStorePath>
          <trustStorePassword>caroot</trustStorePassword>
          <clientTrustStorePath>resources/security/smrootca.bcfks</clientTrustStorePath>
          <clientTrustStorePassword>caroot</clientTrustStorePassword>
        </ssl>
      ...
        <fipsmode>true</fipsmode>
...
</securityConfig>
```

4. Save your changes and close this file.
5. Start the Chat Server.
6. Open the Openfire Administration Console and follow the setup process in configure the chat server in HTTPS mode.
   Note: If you are using Microsoft SQL server, be sure to select **Microsoft SQL server (legacy)** as the database driver.
7. Open the Openfire Administration Console, re-encrypt other encrypted items in System Properties if any.
8. In the Openfire Administration Console, delete all users except admin and publishadmin.

Next step:

Configure FIPS mode in the Chat Service

8.21.4.8. Configure FIPS mode in the Chat Service

Note: The instructions here are supplementary to the steps that are provided in configure the chat service in HTTPS mode.

Before you proceed, make sure of the following:

1. You have already generated a CA certificates file and a keystore file for the Chat Service. For details, see Work with the certificate generation toolkit.
   • The CA certificates file: \certs\smrootca.bcfks
   • The Chat Service server keystore file: \certs\smchatservice.bcfks
   • The Chat Service client keystore file: \certs\smchatserviceclient.bcfks
2. You have enabled FIPS mode on the SM Server side. For details, see Configure FIPS mode in the Server.
3. If you are configuring FIPS mode in an existing Chat Service, that is, the Chat Service ran in non-FIPS mode before, you must decrypt all the encrypted items first.
   a. Stop the Chat Service.
   b. Open <chat service>\conf\app.properties with a text editor, empty app.encryptionKey and then update all the encrypted items with plain text values.

Follow these steps to enable FIPS mode in the Chat Service:

1. Copy the CA certificate file and keystore file (smrootca.bcfks, smchatserviceclient.bcfks, and smchatservice.bcfks) to the <chat service>\conf folder.
2. Open the Chat Service configuration file (<chat service>\conf\app.properties) in a text editor.

3. Make configurations as shown below:

```
securityConfig.fipsmode=true
…
app.clientKeyStoreType=BCFKS
app.clientKeyStorePath=../conf/smchatserviceclient.bcfks
app.clientKeyStorePassword=clientkeystore

app.keyStoreType=BCFKS
app.keyStorePath=../conf/smchatservice.bcfks
app.keyStorePassword=serverkeystore

app.trustStoreType=BCFKS
app.trustStorePath=../conf/smrootca.bcfks
app.trustStorePassword=caroot
```

4. Save your changes and close this file.
5. Start the Chat Service for your configuration to take effect.

Next step:

Configure FIPS mode in the IdM Service

8.21.4.9. Configure FIPS mode in the IdM Service

To enable the FIPS mode in the IdM service, you need to make some additional configurations when you install the IdM service. Content in this topic is supplementary to Install Identity Manager (IdM).

Prerequisites

Before you proceed, make sure that you have configured Java security providers and Tomcat for FIPS mode. For details, see Configure Java and web application servers for FIPS mode.

Task 1: Configure the IdM Service

You need to configure the IdM service to use the BCFIPS security provider and configure the default truststore.

1. Edit the `WEB-INF/classes/idm-security.properties` file to enable FIPS mode and set the truststore. For example:

```
useExternalProvider=true
sslTruststore=<TOMCAT_HOME>/conf/cacerts
sslTruststorePassword=changeit
```

2. Edit the `WEB-INF/classes/security/key-configuration.properties` file to use the BCFIPS security provider. For example:

```
enckeytype.local.profile=strong256bit
enckeytype.local.jce=BCFIPS
enckeytype.local.keyfile=key-local-strong256.bin
enckeytype.local.passphrase.name=local strong256 passphrase
enckeytype.local.passphrase.provider=file
enckeytype.local.passphrase.file=key-local-strong256-passphrase.txt
enckeytype.local.passphrase.deleteFile=false

enckeytype.network.profile=strong256bit
enckeytype.network.jce=BCFIPS
enckeytype.network.keyfile=key-network-strong256.bin
enckeytype.network.passphrase.name=network strong256 passphrase
enckeytype.network.passphrase.provider=file
enckeytype.network.passphrase.file=key-network-strong256-passphrase.txt
enckeytype.network.passphrase.deleteFile=false
```

3. Add more LW-SSO settings to the `system_resource_config__1.0.0.1__update_signingKey_and_initString.json` seeded json file.
   For example:

```
[
  {
    "operation": "UPDATE",
    "type": "systemResourceConfig",
    "names":{
      "systemResourceConfigName":"idm.encryptedSigningKey"
    },
    "attributes": {
      "value": "<your IdM signing key, 32 characters>"
    }
  }, {
    "operation": "UPDATE",
    "type": "systemResourceConfig",
    "names":{
      "systemResourceConfigName":"lwssoConfig.domain"
    },
    "attributes": {
      "value": "microfocus.com"
    }
  }, {
    "operation": "UPDATE",
    "type": "systemResourceConfig",
    "names":{
      "systemResourceConfigName":"lwssoConfig.initString"
    },
    "attributes": {
      "value": "<your LWSSO init string, 32 characters, base64
-encoded>"
    }
  }, {
    "operation": "UPDATE",
    "type": "systemResourceConfig",
    "names":{
      "systemResourceConfigName":"lwssoConfig.crypto.cryptoSource"
    },
    "attributes": {
      "value": "jce"
    }
  }, {
    "operation": "UPDATE",
    "type": "systemResourceConfig",
    "names":{
      "systemResourceConfigName":"lwssoConfig.crypto.macAlgorithmName"
    },
    "attributes": {
      "value": "SHA256"
    }
  }, {
    "operation": "UPDATE",
    "type": "systemResourceConfig",
    "names":{
      "systemResourceConfigName":"lwssoConfig.crypto.directKeyEncoded"
    },
    "attributes": {
      "value": "TRUE"
```

```
    }
  }, {
    "operation": "UPDATE",
    "type": "systemResourceConfig",
    "names":{
      "systemResourceConfigName":"lwssoConfig.crypto.directKeyEncoding"
    },
    "attributes": {
      "value": "Base64"
    }
  }
]
```

Note: Please make sure that the `lwssoConfig.crypto.directKeyEncoded`, `lwssoConfig.crypto.directKeyEncoding`, and `lwssoConfig.initString` setings are consistent with the settings defined in **lwssofmconf.xml** for Server, Web Tier, and Mobility Client.

The `lwssoConfig.initString` must be base64-encoded and its original value must be 32 characters. Run the following commands to encode a value to base64:

- Windows (PowerShell)

  ```
  [convert]::ToBase64String([Text.Encoding]::UTF8.GetBytes('your LWSSO init string, 32 characters'))
  ```

- Linux

  ```
  echo -n 'your LWSSO init string, 32 characters' | base64
  ```

4. If the server certificate of the reverse proxy or the load balancer is issued by your corporate root CA, you need to import your corporate root CA certificate (PEM-encoded, such as `corprootca.crt`) to the truststore cacerts by using the certificate generation toolkit, then copy it to the `<TOMCAT_HOME>/conf` folder. For example:

   ```
   import-cert.bat --alias corprootca --file certs/corprootca.crt --keystore certs/cacerts --storetype bcfks --storepass changeit
   ```

Task 2: Configure the IdM admin console

Edit the `WEB-INF/classes/conf/IdmConfig.xml` file to use the bcfks truststore for the IdM client configuration. For example:

```
<client trustAll="false" ...>
  ...
  <trustStoreFile>WEB-INF/classes/conf/smrootca.bcfks</trustStoreFile>
  <trustStoreType>bcfks</trustStoreType>
  <trustStorePass>your encrypted password for the client truststore</trustStorePass>
  ...
</client>
```

If the server certificate of the reverse proxy or the load balancer is issued by your corporate root CA, you need to import your corporate root CA certificate (PEM-encoded, such as `corprootca.crt`) to a truststore by using the certificate generation toolkit (for example, `corprootca.bcfks`) and then use it as the IdM client truststore. For example:

```
import-cert.bat --alias corprootca --file certs/corprootca.crt --keystore certs/corprootca.bcfks -storetype bcfks --storepass <your password for the client truststore>
```

Task 3: Encrypt the passwords

Before encrypting the password, you need to set the JVM options to use the customized `java.security` file. For example:

```
set FIPS_HOME=<FIPS_HOME>
set FIPS_OPTS=--module-path=%FIPS_HOME% --add-
opens=java.base/sun.security.internal.spec=org.bouncycastle.fips.core
set FIPS_OPTS=%FIPS_OPTS% -Djava.security.properties==%FIPS_HOME%\java.security -
Dorg.bouncycastle.fips.approved_only=true
```

Run the following commands from the appropriate locations to encrypt the passwords by adding the JVM options.

- Encrypt the password for the IdM service

  ```
  java %FIPS_OPTS% -jar dataprotection-passwordtool-1.0.5-standalone.jar --encrypt --conf "
  <TOMCAT_HOME>\webapps\idm-service\WEB-INF\classes\security" local <your password>
  ```

- Encrypt the password for the IdM admin console

  ```
  java %FIPS_OPTS% -jar idm-client-core-1.10.37-41.jar crypto -e -a AES -m <your password> -f
  ..\classes\conf\IdmConfig.key
  ```

8.21.5. Configuring LW-SSO in FIPS mode

When running in FIPS mode, the Service Manager (SM) Server, Web Tier, Mobility Client, and Openfire Chat Server use a FIPS validated cryptographic provider for Lightweight Single Sign-On (LW-SSO).

When Service Manager is running in FIPS mode, the following rules apply for web services clients (integrations):

- It is recommended to enable FIPS mode for the web service clients as well.
- Web service clients that don't support FIPS mode can still integrate with SM through normal TLS connections.
- LW-SSO will not work if the web service client doesn't support FIPS mode.

For more information about the support of FIPS mode for the products that can integrate with SM, see the documentation of the specific products. This document covers only SM components.

The steps to configure LW-SSO in SM for FIPS mode are the same as for non-FIPS mode, except the cryptographic provider configuration part. The steps are described in the following tasks:

Configure LW-SSO in the Server for FIPS mode

Configure LW-SSO in the Web tier for FIPS mode

Configure LW-SSO in the Mobility Client for FIPS mode

Configure LW-SSO in the Chat Server for FIPS mode

Configure LW-SSO in the Chat Service for FIPS mode

8.21.5.1. Configure LW-SSO in the Server for FIPS mode

Applies to User roles: System Administrator

Enabling LW-SSO in the Service Manager (SM) Server enables web service integrations from legacy HPE products to bypass Service Manager authentication if the product user is already authenticated and a proper token is used; enabling LW-SSO in both the Service Manager Server and Web Tier enables users to bypass the login prompts when launching the Service Manager web client from legacy HPE applications.

Caution

Caution To use LW-SSO, your Service Manager Web Tier and Server must be deployed in the same domain; therefore you should use the same domain name for the Web Tier and Server. If you fail to do so, users who log in from another application to the web tier can log in but may be forcibly logged out after a while.

Prerequisite:

Before you proceed, make sure you have enabled FIPS mode on the Service Manager Server side. For detailed instructions, see Configure FIPS mode in the Server.

To configure LW-SSO in the Service Manager server for FIPS mode, follow these steps:

1. Go to the <Service Manager server installation path>/RUN folder, and open `lwssofmconf.xml` in a text editor.
2. Make sure that the `enableLWSSOFramework` attribute is set to `true` (default).
3. Locate the following section.

```
<lwssoValidation>
<domain>yourdomain.net</domain>
<crypto cipherType="symmetricBlockCipher"
engineName="AES" paddingModeName="CBC" keySize="256"
encodingMode="Base64Url"
initString="This is a shared secret passphrase"/>
<!-- When Service Manager is running in FIPS mode, the crypto configuration above needs to be
replaced with the configuration below. The initString value must be 32 characters long and contain
no spaces.  -->
<!--
<crypto cryptoSource="jce" cipherType="symmetricBlockCipher"
engineName="AES" paddingModeName="CBC" keySize="256"
pbeDigestAlgorithm="SHA1" encodingMode="Base64Url"
jcePbeAlgorithmName="AES" jcePbeMacAlgorithmName="AES"
macType="hmac"
macAlgorithmName="SHA1" directKeyEncoded="true" directKeyEncoding="Base64Url"
algorithmPaddingName="PKCS5Padding"
pbeCount="20" macKeySize="256" macPbeCount="20"
initString="rqa9mkdmfkvdvienfj4usldo9205mvid">
</crypto>
-->
</lwssoValidation>
```

4. Update this section by doing the following:
   - Comment out the ordinary crypoto configuration part.
   - Uncomment the FIPS mode crypto configuration part.
   - Configure the domain of the host.
   - Update the default `initString` value to your own value, if needed.

> **Caution**
>
> Caution The `initString` value is shared among all parties that participate in LW-SSO. In FIPs mode, this value must be 32 characters long and contain no spaces; otherwise LW-SSO will not work in FIPS mode.

The following is an example:

```
<lwssoValidation>
<domain>myhostdomain.com</domain>
<!--
<crypto cipherType="symmetricBlockCipher"
engineName="AES" paddingModeName="CBC" keySize="256"
encodingMode="Base64Url"
initString="This is a shared secret passphrase"/>
-->
<!-- When Service Manager is running in FIPS mode, the crypto configuration above needs to be
replaced with the configuration below.
The initString value must be 32 characters long and contain no spaces. -->
<crypto cryptoSource="jce" cipherType="symmetricBlockCipher"
engineName="AES" paddingModeName="CBC" keySize="256"
pbeDigestAlgorithm="SHA1" encodingMode="Base64Url"
jcePbeAlgorithmName="AES" jcePbeMacAlgorithmName="AES"
macType="hmac"
macAlgorithmName="SHA1" directKeyEncoded="true" directKeyEncoding="Base64Url"
algorithmPaddingName="PKCS5Padding"
pbeCount="20" macKeySize="256" macPbeCount="20"
initString="1qaz2wsxfkvdvienfj4usldo9205mvid">
</crypto>
</lwssoValidation>
```

Note

Note Here, `myhostdomain.com` refers to the domain of the SM Server host.

Note

Note

- You're not recommended to modify the `paddingModeName`, `keySize`, `encodingMode`, `engineName`, and `cipherType` attributes. If you need to modify them, be careful to ensure security and make sure that they are same with the LW-SSO settings of the other legacy HPE product that you want to integrate with Service Manager.
- Do not change the other configurations, such as the content in tag `<restURLs>`, and the attribute of tag `<service>`.

8.21.5.2. Configure LW-SSO on the web tier for FIPS mode

Applies to User roles: System Administrator

If Lightweight Single Sign-On (LW-SSO) is enabled on the Service Manager web tier, integrations from legacy HPE products will bypass Service Manager authentication when launching the Service Manager Web client, provided that the user of the legacy HPE product is already authenticated and a proper token is used.

Note

- To enable users to launch the Web client from another legacy HPE product using LW-SSO, you must also enable LW-SSO in the Service Manager server.
- Once you have enabled LW-SSO in the web tier, web client users should use the web tier server's fully-qualified domain name (FQDN) in the login URL: `http://<myWebtierHostName>.<myDomain>:<port>/webtier-x.xx/index.do`
- The following procedure is provided as an example, assuming that the Service Manager web tier is deployed on Tomcat.

Prerequisite

- Before you proceed, make sure you have enabled FIPS mode on the web tier. For detailed instructions, see Configure FIPS mode on the web tier.

Steps to configure LW-SSO on the Service Manager web tier for FIPS mode

To configure LW-SSO on the Service Manager web tier for FIPS mode:

1. Open the `<Tomcat>\webapps\<` Service Manager `web tier>\WEB-INF\web.xml` file in a text editor.
2. Modify the `web.xml` file as follows:
    1. Set the `<serverHost>` parameter to the fully-qualified domain name of the Service Manager server.

        > **Note**
        > This is required to enable LW-SSO from the web tier to the server.

    2. Set the `<serverPort>` parameter to the communications port of the Service Manager server.
    3. Change the value of context parameter **isCustomAuthenticationUsed** to false.

        ```
        <context-param>
        <param-name>isCustomAuthenticationUsed</param-name>
        <param-value>false</param-value>
        </context-param>
        ```

    4. Remove the comment tags (<!-- and -->) enclosing the following elements to enable LW-SSO authentication.

        ```
        <!--
        <filter>
        <filter-name>LWSSO</filter-name>
        <filter-class>com.hp.sw.bto.ast.security.lwsso.LWSSOFilter</filter-class>
        </filter>
        -->
        ......
        <!--
        <filter-mapping>
        <filter-name>LWSSO</filter-name>
        <url-pattern>/*</url-pattern>
        </filter-mapping>
        -->
        ```

    5. Save the `web.xml` file.
3. Open the `<Tomcat>\webapps\<Service Manager web tier>\WEB-INF\classes\lwssofmconf.xml` file in a text editor.
4. Update the lwssofmconf.xml file as described in the following steps.
    1. Set the value of `enableLWSSOFramework` to `true` (default is false).

        ```
        <enableLWSSO
        enableLWSSOFramework="true"
        enableCookieCreation="true"
        cookieCreationType="LWSSO"/>
        ```

    2. Locate the following section.

```
<lwssoValidation>
<domain>yourdomain.net</domain>
<crypto cipherType="symmetricBlockCipher"
engineName="AES" paddingModeName="CBC" keySize="256"
encodingMode="Base64Url"
initString="This is a shared secret passphrase"/>
<!-- When Service Manager is running in FIPS mode, the crypto configuration above needs to be
replaced with the configuration below. The initString value must be 32 characters long and
contain no spaces.  -->
<!--
<crypto cryptoSource="jce" cipherType="symmetricBlockCipher"
engineName="AES" paddingModeName="CBC" keySize="256"
pbeDigestAlgorithm="SHA1" encodingMode="Base64Url"
jcePbeAlgorithmName="AES" jcePbeMacAlgorithmName="AES"
macType="hmac"
macAlgorithmName="SHA1" directKeyEncoded="true" directKeyEncoding="Base64Url"
algorithmPaddingName="PKCS5Padding"
pbeCount="20" macKeySize="256" macPbeCount="20"
initString="rqa9mkdmfkvdvienfj4usldo9205mvid">
</crypto>
-->
</lwssoValidation>
```

3. Update this section by doing the following:
   - Comment out the ordinary crypoto configuration part.
   - Uncomment the FIPS mode crypto configuration part.
   - Configure the domain of the host.
   - Update the default `initString` value to your own value, if needed.

Caution

The `initString` value is shared among all parties that participate in LW-SSO. In FIPs mode, this value must be 32 characters long and contain no spaces; otherwise LW-SSO will not work in FIPS mode.

The following is an example:

```
<lwssoValidation>
<domain>myhostdomain.com</domain>
<!--
<crypto cipherType="symmetricBlockCipher"
engineName="AES" paddingModeName="CBC" keySize="256"
encodingMode="Base64Url"
initString="This is a shared secret passphrase"/>
-->
<!-- When Service Manager is running in FIPS mode, the crypto configuration above needs to be
replaced with the configuration below.
The initString value must be 32 characters long and contain no spaces. -->
<crypto cryptoSource="jce" cipherType="symmetricBlockCipher"
engineName="AES" paddingModeName="CBC" keySize="256"
pbeDigestAlgorithm="SHA1" encodingMode="Base64Url"
jcePbeAlgorithmName="AES" jcePbeMacAlgorithmName="AES"
macType="hmac"
macAlgorithmName="SHA1" directKeyEncoded="true" directKeyEncoding="Base64Url"
algorithmPaddingName="PKCS5Padding"
pbeCount="20" macKeySize="256" macPbeCount="20"
initString="1qaz2wsxfkvdvienfj4usldo9205mvid">
</crypto>
</lwssoValidation>
```

4. In the `<multiDomain>` element, set the trusted hosts connecting through LW-SSO. If the Service Manager web tier server and other application servers connecting through LW-SSO are in the same domain, you can ignore the `<multiDomain>` element; If the servers are in multiple domains, for each server, you must set the correct `DNSDomain` (domain name), `NetBiosName` (server name), `IP` (IP address), and `FQDN` (fully-qualified domain name) values. The following is an example.

```
<DNSDomain>yourdomain.net</DNSDomain>
<NetBiosName>myserver</NetBiosName>
<IP>1.23.456.789</IP>
<FQDN>yourserver.yourdomain.net</FQDN>
```

Note

As of version 9.30, Service Manager uses <multiDomain> instead of <protectedDomains>, which is used in earlier versions. The multi-domain functionality is relevant only for UI LW-SSO (not for web services LW-SSO). This functionality is based on the HTTP referrer. Therefore, LW-SSO supports links from one application to another and doesn't support typing a URL in a browser window, except when both applications are in the same domain.

5. Check the `secureHTTPCookie` value (default: true).

Note

- If you set `secureHTTPCookie` to `true` (default), you must also set `secureLogin` in the web tier configuration file (`web.xml`) to `true` (default); if you set `secureHTTPCookie` to `false`, you can set `secureLogin` to either `true` or `false`. In a production environment, you are recommended to set both parameters to `true`.
- If you don't want to use SSL, set both `secureHTTPCookie` and `secureLogin` to `false`.

6. Save the `lwssofmconf.xml` file.

5. Open the `<Tomcat>\webapps\<Service Manager web tier>\WEB-INF\classes\application-context.xml` in a text editor.

6. Modify the `application-context.xml` as follows:
    1. Add `lwSsoFilter` to the `filterChainProxy` bean:

        ```
        <sec:filter-chain pattern="/**"
        filters="securityContextPersistenceFilter,resourceFilter,lwSsoFilter,anonymousAuthFilter"/>
        ```

        Note
        If you need to enable web tier LW-SSO for integrations and also enable trusted sign-on for your web client users, add **lwSsoFilter** followed by **preAuthenticationFilter**, as shown in the following:

        ```
        <sec:filter-chain pattern="/**"
        filters="securityContextPersistenceFilter,resourceFilter,lwSsoFilter,preAuthentica
        tionFilter,anonymousAuthFilter"/>
        ```

        For information about how to enable trusted sign-on in Service Manager, see Example: Enabling trusted sign-on.
    2. Uncomment the `lwSsoFilter` bean by removing the comment tags:

        ```
        <bean id="lwSsoFilter" class="com.hp.ov.sm.client.webtier.lwsso.LwSsoPreAuthenticationFilter">
        <property name="authenticationManager">
        <ref bean="authenticationManager"/>
        </property>
        <property name="defaultRole">
        <value>ROLE_PRE</value>
        </property>
        </bean>
        ```

    3. Save the `application-context.xml` file.
7. Restart Tomcat so that the configuration takes effect.

8.21.5.3. Configure LW-SSO in the Mobility Client for FIPS mode

If Lightweight Single Sign-On (LW-SSO) is enabled on both the Service Manager (SM) server and the Mobility Client, Service Manager authentication will be bypassed if you have signed into another web application that also has LW-SSO enabled. The detailed bypass criteria includes both the protected domain/IP/DNS name and the initial LW-SSO string.

Prerequisites

Before you proceed, make sure of the following:

- You have already enabled FIPS mode in the SM Server. For detailed steps, see Configure FIPS mode in the Server.
- You have already configured LW-SSO for FIPS mode in both the SM Server and Web Tier. For detailed steps, see Configure LW-SSO in the Server for FIPS mode and Configure LW-SSO in the Web tier for FIPS mode.
- You have enabled FIPS mode in the Mobility Client. For detailed steps, see Configure FIPS mode in the Mobility Client.

As an example, the following procedure assumes that you are deploying the Mobility Client on Tomcat.

To enable LW-SSO on the Mobility Client for FIPS mode, follow these steps:

1. Extract the webapp-9.xx.xxxx.war file in an archive management program to the <Tomcat>/webapps directory.
2. Open the web.properties file located in the <Tomcat>/webapps/<Mobility>/WEB-INF directory in a text editor.
3. Update the web.properties file.
    1. Change the value of context parameter **isCustomAuthenticationUsed** to false.

        ```
        isCustomAuthenticationUsed=false
        ```

    2. Save the file.

4. Open the security.xml file located in the <Tomcat>/webapps/<Mobility>/WEB-INF/spring directory in a text editor.
5. Update the security.xml file.
    1. Locate the following scripts:

```
<security:custom-filter ref="idmPreAuthFilter" position="PRE_AUTH_FILTER"/>
<!-- <security:custom-filter ref="preAuthenticationFilter" after="SECURITY_CONTEXT_FILTER"/>
<security:custom-filter ref="lwSsoFilter" before="BASIC_AUTH_FILTER"/>
<security:custom-filter ref="springSecurity2lwssoIntegrationFilter" position="LAST"/> -->
```

    2. Comment out the first line, and then uncomment the third and fourth lines to enable LW-SSO, as shown in the following:

```
<!-- <security:custom-filter ref="idmPreAuthFilter" position="PRE_AUTH_FILTER"/>-->
<!-- <security:custom-filter ref="preAuthenticationFilter" after="SECURITY_CONTEXT_FILTER"/> -
->
<security:custom-filter ref="lwSsoFilter" before="BASIC_AUTH_FILTER"/>
<security:custom-filter ref="springSecurity2lwssoIntegrationFilter" position="LAST"/>
```

    3. Save the file.
6. Open the lwssofmconf.xml file located in the <Tomcat>/webapps/<Mobility>/WEB-INF/classes directory in a text editor.
7. Update the lwssofmconf.xml file.
    1. Change the enableLWSSOFramework parameter to true (default: false).

```
<enableLWSSO enableLWSSOFramework="true" enableCookieCreation="true"
cookieCreationType="LWSSO" enableSAML2Support="false"/>
```

    2. Locate the following section.

```
<lwssoValidation>
<domain>yourdomain.net</domain>
<crypto cipherType="symmetricBlockCipher"
engineName="AES" paddingModeName="CBC" keySize="256"
encodingMode="Base64Url"
initString="This is a shared secret passphrase"/>
<!-- When Service Manager is running in FIPS mode, the crypto configuration above needs to be
replaced with the configuration below. The initString value must be 32 characters long and
contain no spaces.  -->
<!--
<crypto cryptoSource="jce" cipherType="symmetricBlockCipher"
engineName="AES" paddingModeName="CBC" keySize="256"
pbeDigestAlgorithm="SHA1" encodingMode="Base64Url"
jcePbeAlgorithmName="AES" jcePbeMacAlgorithmName="AES"
macType="hmac"
macAlgorithmName="SHA1" directKeyEncoded="true" directKeyEncoding="Base64Url"
algorithmPaddingName="PKCS5Padding"
pbeCount="20" macKeySize="256" macPbeCount="20"
initString="rqa9mkdmfkvdvienfj4usldo9205mvid">
</crypto>
-->
</lwssoValidation>
```

    3. Update this section by doing the following:
        - Comment out the ordinary crypoto configuration part.
        - Uncomment the FIPS mode crypto configuration part.

- Configure the domain of the host.
- Update the default `initString` value to your own value, if needed.

> **Caution**
> Caution The `initString` value is shared among all parties that participate in LW-SSO. In FIPs mode, this value must be 32 characters long and contain no spaces; otherwise LW-SSO will not work in FIPS mode.

The following is an example:

```
<lwssoValidation>
<domain>myhostdomain.com</domain>
<!--
<crypto cipherType="symmetricBlockCipher"
engineName="AES" paddingModeName="CBC" keySize="256"
encodingMode="Base64Url"
initString="This is a shared secret passphrase"/>
-->
<!-- When Service Manager is running in FIPS mode, the crypto configuration above needs to be
replaced with the configuration below.
The initString value must be 32 characters long and contain no spaces. -->
<crypto cryptoSource="jce" cipherType="symmetricBlockCipher"
engineName="AES" paddingModeName="CBC" keySize="256"
pbeDigestAlgorithm="SHA1" encodingMode="Base64Url"
jcePbeAlgorithmName="AES" jcePbeMacAlgorithmName="AES"
macType="hmac"
macAlgorithmName="SHA1" directKeyEncoded="true" directKeyEncoding="Base64Url"
algorithmPaddingName="PKCS5Padding"
pbeCount="20" macKeySize="256" macPbeCount="20"
initString="1qaz2wsxfkvdvienfj4usldo9205mvid">
</crypto>
</lwssoValidation>
```

> **Note**
> Note Here, `myhostdomain.com` refers to the domain of the Mobility Client host. Beginning from the <creation> section to the end of the lwssofmconf.xml file, the variables are subject to change according to your actual deployment.

4. Save the file.

8. Restart the Mobility Client's web application server (in this example, Tomcat).
9. Make sure the SM Server is started.
10. Make sure the Web Tier's web application server is started.
11. Launch the Mobility Client URL in a browser.
    The URL resembles the following: https://myMobilityHostFQDN:8443/Mobility/std/
12. Sign in to the Mobility Client and verify that LW-SSO from the Mobility Client to the Web Tier works correctly.

8.21.5.4. Configure LW-SSO in the Chat Server for FIPS mode

Service Manager Collaboration uses LW-SSO to access and authenticate users. Service Manager Collaboration requires that you configure LW-SSO for the Service Manager web tier and the Openfire Chat Server. When FIPS mode is enabled on the SM Server, you need to configure LW-SSO in the Chat Server for FIPS mode.

Prerequisites

Before you proceed, make sure of the following:

1. You have already successfully enabled FIPS mode on the SM Server side. For details, see Configure FIPS mode in the Server.
2. You have already configured LW-SSO in the SM Server for FIPS mode. For details, see Configure LW-SSO in the Server for FIPS mode.
3. You have already configured LW-SSO in the SM Web Tier for FIPS mode. For details, see Configure LW-SSO in the Web tier for FIPS mode.

To configure LW-SSO in the Chat Server for FIPS mode, follow these steps:

1. Go to the <Openfire_home>\plugins\lwssoplugin folder, and open the lwssofmconf.xml file in a text editor.
2. Make configuration as follows:
    1. Comment out or delete the <crypto> element in line 15.
    2. Uncommont the <crypto> element in line 23, and then update the default `initString` value to the same as you used in the SM server and web tier.
    3. Configure the domain of the host. See the following example:

```
  <lwssoValidation>
    <domain>mycompany.net</domain>
        <crypto cryptoSource="jce" cipherType="symmetricBlockCipher"
        engineName="AES" paddingModeName="CBC" keySize="256"
        pbeDigestAlgorithm="SHA1" encodingMode="Base64Url"
        jcePbeAlgorithmName="AES" jcePbeMacAlgorithmName="AES"
        macType="hmac"
        macAlgorithmName="SHA1" directKeyEncoded="true" directKeyEncoding="Base64Url"
        algorithmPaddingName="PKCS5Padding"
        pbeCount="20" macKeySize="256" macPbeCount="20"
        initString="rqa9mkdmfkvdvienfj4usldo9205mvid"></crypto>
</lwssoValidation>
```

3. Save your changes and close this file.
4. Restart the Chat Server.
5. Verify that SM Collaboration can work correctly. For more information, see Install Service Manager Collaboration.

8.21.5.5. Configure LW-SSO in the Chat Service for FIPS mode

Service Manager Collaboration uses LW-SSO to access and authenticate users. Service Manager Collaboration requires that you configure LW-SSO for the Service Manager web tier and the Openfire Chat Service. When FIPS mode is enabled on the SM Server side, you need to configure LW-SSO in the Chat Service for FIPS mode.

Prerequisites

Before you proceed, make sure of the following:

1. You have already successfully enabled FIPS mode on the SM Server side. For details, see Configure FIPS mode in the Server.
2. You have already configured LW-SSO in the SM Server for FIPS mode. For details, see Configure LW-SSO in the Server for FIPS mode.
3. You have already configured LW-SSO in the SM Web Tier for FIPS mode. For details, see Configure LW-SSO in the Web tier for FIPS mode.
4. You have already configured LW-SSO in the Chat Server for FIPS mode. For details, see Configure LW-SSO in the Chat Server for FIPS mode.

To configure LW-SSO in the Chat Service for FIPS mode, follow these steps:

1. Go to the <Chat Service_home>\conf folder, and open the lwssofmconf.xml file in a text editor.
2. Make configuration as follows:
    1. Comment out or delete the <crypto> element in line 15.

2. Uncomment the <crypto> element in line 23, and then update the default `initString` value to the same as you used in the SM server and web tier.
3. Configure the domain of the host. See the following example:

```
<lwssoValidation>
        <domain>mycompany.net</domain>
        <crypto cryptoSource="jce" cipherType="symmetricBlockCipher"
        engineName="AES" paddingModeName="CBC" keySize="256"
        pbeDigestAlgorithm="SHA1" encodingMode="Base64Url"
        jcePbeAlgorithmName="AES" jcePbeMacAlgorithmName="AES"
        macType="hmac"
        macAlgorithmName="SHA1" directKeyEncoded="true" directKeyEncoding="Base64Url"
        algorithmPaddingName="PKCS5Padding"
        pbeCount="20" macKeySize="256" macPbeCount="20"
        initString="rqa9mkdmfkvdvienfj4usldo9205mvid"></crypto>
</lwssoValidation>
```

3. Save your changes and close this file.
4. Restart the Chat Service.
5. Verify that SM Collaboration can work correctly. For more information, see Install Service Manager Collaboration.

8.22. Tokenization

The Service Manager web tier has tokenization enabled by default. Enabling tokenization helps safeguard sensitive data. When tokenization is enabled, the web tier appends a token to URLs and checks requested URLs that contain any predefined actions or match any predefined patterns to see if they have the token appended; if the token is not found, the web tier returns an error.

Administrators can also disable this feature by turning off a parameter in the web tier configuration file (`web.xml`).

Related parameters

Three `web.xml` parameters are used for tokenization (see the following table). Out-of-the-box, the `simpleURLs` and `patterns` parameters aren't present in the `web.xml` file. If you don't add them, their default values take effect.

Note

Note To enable tokenization, enabling the `antiCSRFEnabled` parameter is enough. Adding the other parameters is unnecessary unless you need to change their values from the default ones.

Parameter Name	Parameter Value	Default Value	Description
antiCSRFEnabled	true or false	true	Set it to true to enable token validation or false to disable the feature.
simpleURLs	A list of comma-separated servlet actions that a URL can contain.	<![CDATA[service.do, detail.do, list.do, mashup.do, frames.do, printList.do, printDetail.do, uim.do, nav.menu, wf, ce, cm, /upload, /imageupload, /attachmentupload, /uniqueupload]]>	This parameter takes effect only when antiCSRFEnabled is set to true. If a requested URL contains any of the listed actions and also has no token appended, the SM web tier returns a 403 error. When antiCSRFEnabled is set to false, the web tier neither checks for a token in the requested URLs, nor appends a token to any URLs. **Note** Note We recommend that you use the default list. If needed, you can remove actions from the default list, but can't add new ones. **Note** Note Token validation is skipped for the following actions: index.do, ess.do, accessible.do, and accessible_ess.do.

patterns	A comma separated list of patterns	<![CDATA[attachments/, /download/, /colla/api/]]>	This parameter takes effect only when antiCSRFEnabled is set to true, and is ignored when the simpleURLs validation has passed. If a requested URL matches any of the listed patterns, the web tier performs token validation on it. **Note** Note We recommend that you use the default list. If you want, you can remove a pattern from the default list, but can't add new ones.

Enable or disable tokenization in the web client

By default, tokenization is enabled in the web client. You can customize the tokenization settings or disable tokenization.

To customize tokenization settings:

1. In the web.xml file, locate the following filter and check that antiCSRFEnabled is set to true:

```
<filter>
    <filter-name>antiCSRFFilter</filter-name>
    <filter-class>com.hp.ov.web.csrf.AntiCSRFFilter</filter-class>
    <init-param>
        <param-name>antiCSRFEnabled</param-name>
        <param-value>true</param-value>
    </init-param>
</filter>
```

2. Add the other two parameters if you want to assign them a custom value.

   Note
   Note We recommend that you use the default values for the two parameters; however, you are allowed to remove values from the default action/pattern list, but you can't add new ones.

   Tip
   Tip Ignore this step if you want to use the default values.

   You should add the parameters as shown in the following.

```
<filter>
    <filter-name>antiCSRFFilter</filter-name>
    <filter-class>com.hp.ov.web.csrf.AntiCSRFFilter</filter-class>
    <init-param>
        <param-name>antiCSRFEnabled</param-name>
        <param-value>true</param-value>
    </init-param>
    <init-param>
        <param-name>simpleURLs</param-name>
        <param-value><your custom URL action list></param-value>
    </init-param>
    <init-param>
        <param-name>patterns</param-name>
        <param-value><your custom pattern list></param-value>
    </init-param>
</filter>
```

3. Save the `web.xml` file.
4. If you want to disable tokenization, set `antiCSRFEnabled` to false.
5. Restart your web application server.

To disable tokenization:

1. Set `antiCSRFEnabled` to false.
2. Save the `web.xml` file.
3. Restart your web application server.

Related topics

Web parameter: antiCSRFEnabled

9. Server Performance Tuning

The Service Manager server includes a number of performance tuning options you can configure after installation. The following table lists the performance tuning options you can enable or configure from a new development environment installation.

Feature	Description	Default state in new installations
Shared memory	A configuration option that allows you to specify how much of the Service Manager host's system resources are available for processes and threads	Enabled

Related topics

Example: response time monitoring (RTM) output

Example: semaphore reports

Shared memory storage report

9.1. Defining batch sizes in the counters file

Administrators can allocate blocks of record identifiers for each servlet in horizontal and vertical scaling implementations. By allocating record IDs in advance, administrators can reduce traffic between the Service Manager server and the RDBMS, which improves overall system performance.

Note: Setting a batch size value replaces the fastcounters parameter from previous versions.

Administrators allocate record IDs in batches from the `counters` file. When you define a batch size, Service Manager retrieves that number of ID keys from the RDBMS and stores the keys in shared memory for later use. Each time someone creates a new record, the system assigns it an ID key from shared memory until there are no more IDs available. Once the system exhausts its supply of IDs, it requests the next batch of IDs from the RDBMS. Allocating IDs in advance improves performance, because retrieving the reserved ID keys from shared memory is more efficient than connecting to the RDBMS and requesting a new ID key with every new record.

Batch file size recommendation

We recommend you set the batch file size to the number of new records you expect users to create in an hour divided by the number of servers in the implementation. For example, if you expect to generate 200 incident records an hour and have only one server in your implementation, then set the **Batch Size** value to 200. If your system consists of multiple servers, such as in a horizontal scaling implementation, then divide the total records per hour by the number of servers. For example, a horizontal scaling implementation with four servers only needs a batch size of 50 for each server to equal 200 incidents per hour.

Note: In a horizontally scaled system, each server requests its own batch of ID keys. Since new records may be created from processes running on different servers in a horizontal scaling implementation, the ID numbers of new records may not be in sequential order. For example, process A on server 1 has reserved ID keys 1-100 and process B on server 2 has reserved ID keys 101-200. The first record a user creates on process A will have ID 1, but if the very next record created comes from process B, then the ID will be 101, not 2.

If you don't specify a batch size value, Service Manager uses the default batch size of 1 and fetches a new ID from the RDBMS for each new record you create.

Example data from a counters file

```
TABLE_NAME  COLUMN_NAME  CURRENT_VALUE  BATCH_SIZE
schedule    schedule.id  2587564        1000
```

Related topics

Defining record identifiers in batches using the numbers file

9.2. Defining record identifiers in batches using the numbers file

Administrators can define record identifiers and allocate those identifiers in batches from the `numbers` file for new interactions, incidents, and changes. By allocating record IDs in batches in advance, administrators can reduce traffic between the Service Manager server and the RDBMS, which improves overall system performance.

Note

Note

Currently the `counters` file is used by administrators to define the table and field where a new record number is going to be stored. However, the `numbers` file is still available for those who previously chose to define record numbers using the `numbers` file and are upgrading. This file has been enhanced to work more like the `counters` file. For more information on the `counters` file, see the related topics.

When using the `numbers` file, administrators define the following:

- Batch size.
- Starting number.
- Identifying prefix for each module. For example, IM for Incident Management records.
- Whether numbers are going to increment or decrement.
- The table and field where the new numbers are to be stored.

When you define a batch size, Service Manager retrieves that number of ID keys from the RDBMS and stores the keys in shared memory for later use. Each time someone creates a new record, the system assigns it an ID key from shared memory

until there are no more IDs available. Once the system exhausts its supply of IDs, it requests the next batch of IDs from the RDBMS.

We recommend you set the batch file size to the number of new records you expect users to create in an hour, divided by the number of servers in the implementation. For example, if you expect to generate 200 incident records an hour and have only one server in your implementation, then set the Batch Size value to 200. If your system consists of multiple servers, such as in a horizontal scaling implementation, then divide the total records per hour by the number of servers. For example, a horizontal scaling implementation with four servers only needs a batch size of 50 for each server to equal 200 incidents per hour.

In a horizontally scaled system, each server requests its own batch of ID keys. Since new records may be created from processes running on different servers in a horizontal scaling implementation, the ID numbers of new records may not be in sequential order. For example, process A on server 1 has reserved ID keys 1-100 and process B on server 2 has reserved ID keys 101-200. The first record a user creates on process A will have ID 1, but if the very next record created comes from process B, then the ID will be 101, not 2.

If you don't specify a batch size value, Service Manager uses the default batch size of 1 and fetches a new ID from the RDBMS for each new record you create.

Example data from a numbers file

```
TABLE_NAME  COLUMN_NAME  CURRENT_VALUE  BATCH_SIZE
schedule    schedule.id  2587564        1000
```

Related topics

Defining batch sizes in the counters file

9.3. Lock Management

The locking mechanism in versions of Service Manager (SM) prior to version 9.31 had a locking mechanism that used a multicasting to request and obtain a lock on a resource in Service Manager. This locking mechanism was implemented using a Peer Lock in the JGroups toolkit. However, there were several issues with this implementation, which are addressed with the new locking mechanism introduced in this document.

Issues in Service Manager Multicast Communication[edit]

Service Manager's previous multicast implementation suffers from the following limitations:

• All nodes must be contacted by the node that's requesting the resource. This leads to high overhead per request.

• All nodes must give approval to a request, regardless of whether a node is using a resource or not.

• A node that doesn't respond represents a single point of failure for the system, even if that node has no other connection to the resource or the requesting node.

• If one node doesn't respond, the request must be re-issued by the originating node, which increases overhead even further.

• Nodes that are slow to respond will eventually be removed from the node cluster.

• Scalability is very poor.

• The potential for netstorms is high (wherein every node is attempting to request permission from every other node, leading to n^2 requests).

New Locking Mechanism Overview[edit]

The new locking mechanism consists of a record entry for each locked resource in a database table. The new Lock table (for exclusive locks) and LockShared table (for shared locks) have been created to house these records. See the following tables for details on the structure and fields of the Lock and LockShared tables.

Note: The only difference between these two tables is the value in the TYPE field, and the primary key for the LockShared table is a combination of the LOCKID, pID, tID, and IP fields.

The Lock Table

Field	Type	Null	Key	Default	Extra
LOCKID	varchar(600)	N	PK		LockID is a hex value of ResourceName. This ensures that the ResourceName field can be either case-sensitive or case-insensitive.
RESOURCENAME	varchar(200)	N			Logic lock ID
TYPE	char(1)				Exclusive / Shared lock
PID	float				The Process ID that holds the lock.
TID	float				The Thread ID that holds the lock.
RADTHREADID	float				The Rad Thread ID that holds the lock.
SESSIONID	float				The session ID that holds the lock.
REASON	VARCHAR2(60)				Application private
USER	VARCHAR2(60)				SM login user (For example, "falcon")
HOSTNAME	VARCHAR2(60)				The host that holds the lock.
IP	VARCHAR2(60)				The IP address of the host that holds the lock.
DEVICENAME	VARCHAR2(60)				The device that holds the lock.
For the bg scheduler thread, the devicename is "SYSTEM"					

LOCKAT	datetime				When the lock is obtained.
STARTAT	datetime				When the lock was requested.
RETRYCOUNT	float				Number of times the lock was requested.
HEARTBEAT	float				Updated periodically by the lock holder.
SUSPECTED	float				Indicates another node suspects the lock holder has failed.
SYSRESTRICTED	CHAR(1)				Indicates no modification allowed for end-users
SYSMODCOUNT	float				
SYSMODUSER	VARCHAR(60)				
SYSMODTIME	datetime				

The LockShared Table

Field	Type	Null	Key	Default	Extra
LOCKID	varchar(600)	N	PK		LockID is a hex value of ResourceName. This ensure that the ResourceName field can be either case-sensitive or case-insensitive.
RESOURCENAME	varchar(200)	N			Logic lock ID
TYPE	char(1)	N	PK		Shared lock
PID	float		PK		The Process ID that holds the lock.
TID	float		PK		The Thread ID that holds the lock.
RADTHREADID	float				The Rad Thread ID that holds the lock.
SESSIONID	float				The session ID that holds the lock.
REASON	VARCHAR2(60)				Application private
USER	VARCHAR2(60)				SM login user (For example, "falcon")
HOSTNAME	VARCHAR2(60)				The host that holds the lock.
IP	VARCHAR2(60)				The IP address of the host that holds the lock.
DEVICENAME	VARCHAR2(60)				The device that holds the lock.
For the bg scheduler thread, the devicename is "SYSTEM"					

LOCKAT	datetime				When the lock is obtained.
STARTAT	datetime				When the lock was requested.
RETRYCOUNT	float				Number of times the lock was requested.
HEARTBEAT	float				Updated periodically by the lock holder.
SUSPECTED	float				Indicates another node suspects the lock holder has failed.
SYSRESTRICTED	CHAR(1)				Indicates no modification allowed for end-users
SYSMODCOUNT	float				
SYSMODUSER	VARCHAR(60)				
SYSMODTIME	datetime				

Lock Behavior[edit]

Locks may be either shared locks or exclusive locks. A shared lock allows multiple nodes to read the data from a resource. An exclusive lock may only be obtained if any shared locks on a resource have been released by the nodes that hold them.

Exclusive Lock[edit]

The process by which a node obtains an exclusive lock is as follows:

1. A node that requests an exclusive lock tries to insert a record into the Lock database table to see whether a resource is available.
2. If there is no record for this resource in the Lock table, the resource is available and the node holder obtains the lock and inserts a record into the Lock table.
3. If there is a record in the Lock table, the node will fail to obtain a lock.

To unlock an exclusive lock, the corresponding lock record is removed from the Lock table.

Shared Lock[edit]

The process by which a node obtains a shared lock is as follows:

1. A node that requests a shared lock tries to insert a record into the Lock database table to see whether a resource is available.
2. If there is no record for this resource in the Lock table, the node holder inserts a record into the Lock table and proceeds to step 4.
3. If there is a record in the Lock table, the TYPE field of that record is checked. If the TYPE field is “Exclusive” the lock requested is rejected. If the TYPE field is “Shared,” proceed to step 4.
4. The node tries to insert a record into the LockShared table. If the insertion is successful, the shared lock request is granted. If the insertion fails, the shared lock request is rejected.

To unlock a shared lock, the corresponding lock record is removed from the LockShared table. Then, the corresponding lock record is removed from the Lock table unless other records share the same resource ID.

Note: There is no mechanism to escalate a Shared lock to an Exclusive lock. To obtain an exclusive lock on a resource, all shared locks must be released.

Lock Retry, Timeout, and Heartbeat[edit]

Each process that requires a lock has a dedicated LockHandler thread to handle all lock related operations. When a process needs to execute a lock/unlock operation, the process places a request in queue. The LockHandler reads the queue and attempts to insert the appropriate records in to the Lock or LockShared tables. Additionally, the LockHandler will also return the response to the process that has requested the lock.

The LockHandler will attempt to retry a "wait" lock request every two seconds until the lock attempt succeeds. If a "no-wait" lock request is specified, the LockHandler will reattempt lock acquisition immediately.

- It is possible that a node may obtain a lock, and then fail to release the lock for several reasons. For example, the node could fail, or a problem with the network may prevent communication between the nodes. To prevent other nodes from waiting for an unresponsive lock owner, the following heartbeat mechanism has been implemented:
- When a process requests a lock and finds the lock is already held and the heartbeat value has not been updated, it sets the "Suspected" flag of the lock record in the database table to 1.
- The lock owner updates the heartbeat and checks the "Suspected" flag of the lock record in the database table every 60 seconds.
- If the "Suspected" flag is set to 1, the lock owner sets the "Suspected" flag back to 0 and then increases the heartbeat.
- If the lock owner has failed to update the "Suspected" flag and heartbeat, the process that's waiting will periodically recheck the "Suspected" flag. After 10 minutes (the default value specified in the deadnodelocktimeout parameter), the lock is forcibly removed by deleting the lock record from the database table.

New Parameter[edit]

This new locking mechanism implements the deadnodelocktimeout parameter. This parameter specifies the amount of time that must elapse before a process forcibly removes a lock from the Lock or LockShared table. By default, this parameter is set to 10, which indicates that 10 minutes must elapse before a record is forcibly removed. 10 minutes is also the minimum value for this parameter.

This parameter is specified in the sm.ini file. Changing the value of this parameter doesn't require a restart of the server.

Contents

- 1 Issues in Service Manager Multicast Communication
- 2 New Locking Mechanism Overview
- 3 Lock Behavior
  - 3.1 Exclusive Lock
  - 3.2 Shared Lock
  - 3.3 Lock Retry, Timeout, and Heartbeat
  - 3.4 New Parameter

9.4. Scheduled processes

You can start RAD applications, known as scheduled processes, as unattended background processes. Scheduled processes are also known as background agents, schedulers, processors, and background tasks. Scheduled processes run at set intervals and check for specific conditions before executing a RAD application. Each scheduled process has one or more schedule records that determine the conditions that must be met for the RAD application to run. You can start scheduled processes from the Scheduler form.

List of scheduled processes

By default, Service Manager provides the following scheduled processes:

Scheduler Name	Runs as Process Name or Names	Description
KMUpdate	KMUpdate	This process checks for update records and sends them to the index
SLA	sla	This process monitors service level agreements and processes alerts.
Sync	sync	This process identifies orphan processes that were shut down improperly and frees up any resources they are using before stopping them.
agent	agent	This process manages charts from stored queries.
alert.processor	alert	This process manages Incident Management log messages and alerts.
availability.startup	availability	This process manages asset availability and updates outage statistics.
change.startup	change	This process manages change events.
contract	contract	This process manages contract events.
event.startup	event	This process manages asynchronous incoming events from the Eventin Queue.
gie.startup	gie	This is a legacy process from earlier versions of Service Manager that manages incoming Prim Events.
inactive.startup	inactive	If using the SM applications earlier than version 9.32, this process automatically logs off users who are inactive for a period of time. If using the SM applications version 9.32 or later, this process is obsolete. A new inactivity timer mechanism is used, which no longer requires this process. Note that this new mechanism requires your SM server, applications, and web client to upgrade to version 9.32 or later. As long as your applications is earlier than version 9.32, the old mechanism is used.

linker.startup	linker	This process manages related calls to closed incidents.
lister.startup	lister	This process manages global variables in the *globallist* file.
marquee	marquee	This process manages static marquee messages.
ocm.startup	ocm	This process manages request events.
printer.startup	despooler	This process manages server-side print requests.
problem	problem	This process manages IM alerts and messages.
report.startup	report	This process manages Report Writer reports and importing and exporting of reports.
scauto.startup	scautod	This process manages incoming and outgoing ServiceCenter Automate connections.
schedule.monitor	schedule.monitor	This process monitors the invalid locks, clear these locks regularly, and send notification emails when encountering exceptions.

Related topics

Add an agent to the startup agent record
Configure the alert periodic schedule record
Restart stopped background processes automatically
Start a scheduled process
Stop a process

9.4.1. Access the schedule file

Applies to User Roles: System Administrator

You can access the schedule file to see a list of outstanding scheduled tasks your Service Manager system will perform. Scheduled tasks are created either from agent initialization records or from triggers.

To access the schedule file, type the following command into the Service Manager command line:

```
sch
```

The schedule.looksee.g form opens.

Related topics

Scheduled processes
Add an agent to the startup agent record
Configure the alert periodic schedule record
Restart stopped background processes automatically
Start a scheduled process
Stop a process

9.4.2. Add an agent to the startup agent record

Applies to User Roles: System Administrator

Adding an agent to the startup agent record allows Service Manager to automatically start a background process on startup.

To add an agent to the startup agent record:

1. Type the following command into the Service Manager command line:

   ```
   info
   ```

   The info.startup.g form opens.
2. In the Type field, type the following:

   ```
   startup
   ```

3. Click **Search**.
   The startup agent record opens.
4. Scroll to the first available open record in the Processor Information section.
5. Type or select the following agent information.

Field	Description
Name	Type the name you want to use for the agent. For example, `message1`. This name appears in the System Status list of background processes.
RAD Application	Type `scheduler`.
Class	Type the name for the type of objects this processor will handle. For example, `message`.
Wakeup Interval (secs.)	Type the number of seconds that you want the message processor to wait before checking for messages. For example, `60`.

6. Click **Add**.
   Service Manager adds the Information record.

Related topics

Start a scheduled process
Stop a process
Scheduled processes

9.4.3. Configure the alert periodic schedule record

Applies to User Roles: System Administrator

To configure the alert periodic schedule record:

1. Click **Tailoring** > **Database Manager**.
   The format.prompt.db.g form opens.
2. In the Form field, type schedule.
3. Click **Search**.
   Service Manager displays of a list of schedule forms.
4. Click **schedule**. The schedule.g form opens.
5. In the Name field, type Alert Periodic.

6. Click **Search**.
   Service Manager opens the details for the Alert Periodic schedule record.
7. In the Repeat Interval field, type the time interval you want Service Manager to wait between starting the alert periodic scheduled process.
   Type the time period in the following format: *Dayshours:minutes:seconds*. For example, 4 03:02:01 starts the alert periodic process every 4 days, 3 hours, 2 minutes, and 1 second.
8. In the Application field, verify that Service Manager displays us.alert.periodic for the application name.
9. Click **Save**
   Service Manager updates the schedule record updated.

Related topics

Scheduled processes
Add an agent to the startup agent record

9.4.4. Restart stopped background processes automatically

Applies to User Roles: System Administrator

You can use the following two parameters to restart stopped background processes automatically. Note that you must use these two parameters in conjunction.

- Startup parameter: enableAnubisMonitor: Start Anubis monitor to check the status of the background scheduler thread.
- Startup parameter: anubisPollInterval: Specify the time interval for the Anubis thread to check whether any registered scheduler is alive. The default value is 15 and the unit is second.

Note

Note

If the background scheduler is killed by either the system status or the operating system's Task Manager, this is considered as normal termination and Anubis will not take effect. Anubis will restart the background scheduler only when it terminates unexpectedly.

Related topics

Scheduled processes
Start a scheduled process

9.4.5. Start a scheduled process

Applies to User Roles: System Administrator

You can start a scheduled process from the system status scheduler. All scheduled processes run in the background at pre-defined intervals.

To start a scheduled process:

1. Click **Miscellaneous** > **System Status**. The system.status.list.g form opens.
2. Click **Start Scheduler**. Service Manager displays a list of RAD applications that you can start.
3. Double-click the RAD application you want to start.
   The system.status.lit.g form reopens, and Service Manager displays the message: System background scheduler: *Process* started at: *DateTime*.

Related topics

Scheduled processes
Restart stopped background processes automatically
Stop a process

9.4.6. Stop a process

Applies to User Roles: System Administrator

You can manually stop any running process from the system status form. If you want the server to automatically stop any process after a pre-defined period of inactivity, use the Start Inactivity Timer form.

To stop a process:

1. Click **System Status**. The system.status.list.g form opens.
2. To stop a process, type **k** in the Command field.
3. Click **Execute Commands**.
   Service Manager stops the selected process.
   Note: Service Manager stops the process without issuing a warning.

Related topics

Scheduled processes
Command line options for the system monitor
Inactivity timer
Restart stopped background processes automatically
Start a scheduled process

9.4.7. Unlock a deadlocked resource

Applies to User Roles: System Administrator

You can manually stop any running process from the System Status form. If you want the server to automatically stop any process after a pre-defined period of inactivity, use the Start Inactivity Timer form.

To unlock a deadlocked resource:

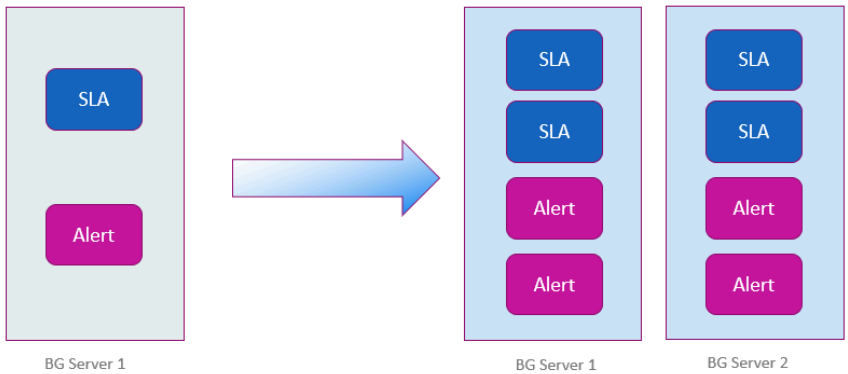
1. Click **System Status** > **Show Locks**.
2. Review the list of system locks, and locate the deadlocked resource.
3. To stop a process, type **k** in the Command field.
4. Click **Execute Commands**.
   Service Manager stops the selected process.
   Note: Service Manager stops the process without issuing a warning.

Related topics

Inactivity timer
Stop a process

9.4.8. Set up background processes with high availability

Service Manager background processes support high availability and multiple Service Manager threads for a scheduled class can work on multiple servers in parallel. With paralleled background processes, the Service Manager production system processes the tasks in the scheduled queue more efficiently. In the event of server restart or server failure, the other servers continue to work on the same class of tasks. For example, two SLA threads are running on background Server 1 and another two SLA threads are running on background Sever 2. All SLA threads are processing the SLA Task Queue in parallel. If Server 1 is restarting or runs to error, Server 2 continues processing the SLA threads.



Here are some guidelines for your reference when setting up background processes with high availability:

1. Add a set of backups of the original Service Manager servers that are running background process schedules.
2. At a minimum, start each background process separately on two Service Manager servers (for example, Server 1 and Server 2) by using any methods as follows.
   - Run the following command to start all background processes that are defined in the **Background Process Initialization Registry** with the name of **startup**. Go to **System Administration** > **Ongoing Maintenance** > **System** > **Startup Information** and then search for the **startup** schedule to view its contents.

     ```
     sm -bg system.start
     ```

   - Run the following command to start all background schedules that are defined in the **Background Processor Initialization Registry** with a specified name. Go to **System Administration** > **Ongoing Maintenance** > **System** > **Startup Information** to view all available schedules.

     ```
     sm -bg scheduler.start <name of the Background Processor Initialization Registry>
     ```

   - Run the following command to start a schedule of a specific class.

     ```
     sm -bg scheduler <class of the background process> 60 0 -log:../logs/<class of the background process>.log
     ```

3. On any two servers, use the **schedule.monitor** scheduler to monitor the background processes.
   When multiple background processes are running in parallel, a schedule record may be locked during processing. If any error occurs in the background processes, the record lock can't be cleared and hence becomes invalid. The schedule.monitor scheduler can monitor the invalid locks, clear these locks regularly, and send notification emails when encountering exceptions.
   To start the schedule.monitor background processes, follow these steps:
   1. (Optional) Customize the email notification conditions by updating the `ScheduleCustom` Javascript. By default, the system sends an email notification to a recipient after completing each of the following actions.
      - Clean up a deadlocked schedule
      - Clean up schedule records which status is running
      - Clean up a schedule that incorrectly stops

      In addition, you can perform customizations to the **EmailScheduleMonitor** email template.
   2. Click **Ongoing Maintenance** > **System** > **Schedule Monitor** to open the **Schedule Monitor Configuration** page.
   3. Type or select the following information.

Field	Description
Notify Email	Type an email address to which the system sends email notifications. **Note:** This function requires that you have an email infrastructure in place, including a properly configured SMTP server and configurations in some Service Manager files. See Email setup notes for more information.
Sleep Time	Type or select a repeat interval. For example, type `00:10:00` for a ten-minute repeat interval.
Scheduled Class	Type a name for the scheduled class. For example, `sch-monitor`.
Next Run	Next execution time of the schedule.

4. Click **Save** and **OK**.
5. Start the schedule.monitor scheduler by using any methods introduced in *step 2*.

9.4.9. Print background process information to log

Sometimes it's uneasy to troubleshoot background processes. Knowing the tasks that Service Manager processes is important for trace reading. In this case, in addition to debugging with the `rtm:3` parameter, you can use the `ScheduleInfo` JavaScript to print background process information to log when the scheduler processing a schedule record.

By default, the content of the `ScheduleInfo` JavaScript may resemble the following.

```
var log = getLog("ScheduleInfo");
function print(scheduleFile){
  log.info("The scheduler begin to run.
ScheduleName:"+scheduleFile["name"]+",ScheduleID:"+scheduleFile["schedule.id"]);
}
```

By using this JavaScript, Service Manager prints the following background process information to the log file.

```
$ cat sm.log|grep "The scheduler begin to run"
1612239(1653063) 06/12/2023 03:36:17.362   JS I ScheduleInfo The scheduler begin to run.
ScheduleName:Change Notify Sched,ScheduleID:6022090
```

To include more information, you can update the `ScheduleInfo` JavaScript by adding more field names from the `schedule` table in the database dictionary. For example:

```
var log = getLog("ScheduleInfo");
function print(scheduleFile) {
    var vString=scheduleFile["strings"];
    log.info("The scheduler begin to run. ScheduleName:" + scheduleFile["name"] + ",ScheduleID:" +
scheduleFile["schedule.id"]+ ", strings:  " + vString.getText() );
}
```

In this case, the printed background process information may resemble the following.

```
1612239(1653061) 06/12/2023 05:19:17.701   JS I ScheduleInfo The scheduler begin to run.
ScheduleName:message processor record,ScheduleID:6022113, strings:  {"1", "email", , "", , , "SOAP-Web",
, "C10066"}
```

9.5. Shared memory

Service Manager shared memory is a pool of memory available to all processes on the server. You can control how the server allocates and manages shared memory using system parameters in the Service Manager initialization file. Allocating too little shared memory can degrade system performance, and allocating too much wastes system resources.

The shared memory storage report generates a snapshot of shared memory that enables you to monitor usage. If you are trying to isolate a performance bottleneck, run a report at start-up and periodically throughout the monitoring period. As a general rule, if Unused falls below 25%, consider increasing the amount of shared memory (see Startup parameters: shared_memory). If you're not sure how to interpret the shared memory report, contact the customer support.

To run the shared memory report for a Linux or Windows system, issue the following command from a shell prompt (Linux) or a command prompt (Windows):

```
sm -reportshm
```

Related topics

Example: semaphore reports
Shared memory storage report
Debug a suspended or hung system
System performance parameters
Startup parameter: maxmsgsharedmemory
Special parameters: reportsem
Special parameters: reportshm
Special parameters: reportstatus
Startup parameter: shared_memory

9.5.1. Automatic selection of shared memory address

The Service Manager server can now automatically select a shared memory address based on your system's available memory. This change makes allocating shared memory easier on Windows systems because you no longer have to specify one particular address for your system's shared memory. The server automatically scans your system memory and selects an address range large enough to support your system's shared memory. If for some reason your system has insufficient free memory, the server will generate an error message telling you so.

The server will automatically select a shared memory address on Windows systems as long as you don't specify an address with the shared_memory_address parameter. On all Linux systems, the server uses the default values of the shared_memory_address parameter based on the operating system. If you previously specified a value for the shared_memory_address parameter, you can remove it from the sm.ini file to have your system automatically select a shared memory address.

Tip: We recommend you remove the shared_memory_address parameter on all Windows platforms so that the system automatically selects your shared memory address. This minimizes the chance that your system will fail to start due to a memory collision with a system resource assigned by Windows ASLR (Address space layout randomization).

Related topics

Shared memory
Enter a parameter in the sm.ini file
Startup parameter: shared_memory_address

9.5.2. Example: semaphore reports

The following illustrates the results of semaphore commands.

sm -reportsem

```
08/27/19 15:46:43   pid (22010) Service Manager diagnostic report follows:
Reporting pid = 22010


--- Mutexs ---


Name                 Last pid  Available  Use Count

Application cache       22010          1       1283

Shared memory           22010          1       7065

Licensing               22010          1          1

User chain              22010          1        230

Database Services       22010          1          1

Counter Services (counters)       22010          1          0

Counter Services (number)         22010          1          0

Diagnostics DB Hashtable          22010          1          0

Diagnostics APP Hashtable         22010          1          0

Diagnostics Process Hashtable     22010          1          0

Global cache            22010          1         50

9973 mutexes for Cache Manager



--- Semaphores ---


Name                 Last pid  Available  # Waiting

System                  21718          2          0

Application cache       21637          1          0

Shared memory           21637          1          0

Licensing               21637          1          0

User chain              21637          1          0

Database Services       21637          1          0

Counter Services (counters)       21637          1          0

Counter Services (number)         21637          1          0

Diagnostics DB Hashtable              0          0          0
```

```
Diagnostics APP Hashtable            0         0         0

Diagnostics Process Hashtable        0         0         0

Global cache               0          0          0
```

Related topics

Shared memory
Shared memory storage report
Special parameters: reportshm

9.5.3. Debug a suspended or hung system

Applies to User Roles: System Administrator

If Service Manager is unresponsive or appears hung, you can use the reportsem command to determine if any semaphores are locked.

To run the semaphore report type the following command from a shell prompt (Linux) or a command prompt (Windows):

```
sm -reportsem
```

Related topics

Example: semaphore reports

9.5.4. Shared memory storage report

In the shared memory storage report, the top “Shared memory” section summarizes shared memory allocation. The bottom “By type” section details areas of the system that use shared memory.

Shared memory

Shared Memory	Description
Current size	The current size of shared memory as specified in the sm.ini parameter file.
Occupied space	The amount of shared memory occupied by allocations less than 32K in size.
Big alloc space	The amount of shared memory occupied by allocations greater than 32K in size.
Unused	The amount of shared memory that has never been used.

By type

Allocation	Description
Not named	System overhead.
User blocks	Service Manager allocates a storage block for each logged on user.
Messages	Messages sent from one user to another.
Resource locks	Record locks.
Cache overhead	Overhead for the five cache categories (Application cache, DBDICT, Format, SQL descriptor, IR Expert).
Application cache	(Cache category.) RAD applications that have been read by users and that are in use.
DBDICT cache	(Cache category.) Database definitions.
SQL descriptor cache	(Cache category.) SQL definitions.
Join/ERD/Type cache	ERD and JOINDEFS definitions.
String Type Cache	Area in which formats, links and other definitions are stored.

Example: Shared Memory report

The following sections illustrates the results of the **sm -reportshm** command:

```
08/21/19 15:34:23   pid (20002) Service Manager diagnostic report follows:
------ Shared Memory ------

Shared Memory Release     9.63.2

Current Size             256000000

Segment Allocation         9346088
Large Block Allocation    11288576

Unused Space             235365336  (91%)
Free Space               235542624  (92%)


Shared Memory Type  Allocations    Frees     Allocated
------------------  ----------- ----------- -----------
Not named                   26           3      759744
User blocks                 24           3       43008
Messages                     0           0           0
Database Services          630           0       28848
Cache overhead              26           3       42656
Application cache         2158         703     1938832
DBDICT cache             11630        1146    15424064
SQL descriptor cache       530           0     1046432
Join/ERD/Type cache        402           0     1032320
String Type cache         1080         208      141472
Global cache                 0           0           0
Diagnostic Services          0           0           0
```

Type statistics

The following are definitions for the type statistics of a shared memory report.

Type	Definition
Allocations	The number of times that shared memory for this type has been allocated.
Frees	The number of times that shared memory for this type has been freed.
Allocated	The amount in bytes that's currently allocated for this type.

Related topics

Shared memory

9.5.5. Shared memory sizing considerations

Service Manager shared memory is divided into three main areas:

- User-related memory
- System table cache

The following topics explain how Service Manager uses shared memory as well as how each area must be included in the sizing calculation.

Creation and use of shared memory
Shared memory sizing
Horizontal scaling: special considerations
Shared memory related parameters

9.5.5.1. Creation and use of shared memory

Shared memory is a common area of storage that's used by all Service Manager server processes on a host machine. Using shared memory is an efficient means of sharing data between processes. For example, when a user or background process first executes a RAD application, Service Manager loads the application into shared memory. This allows other user processes to execute the same application without the overhead of storing a copy of their own into memory. Subsequent executions of the RAD application will result in Service Manager retrieving the application from shared memory thus decreasing the IO required by each process. This behavior of sharing certain data, such as applications, table definitions, and forms between all users significantly increases program performance because reading data from memory is more efficient than reading from disk memory. The total amount of memory used is additionally decreased because one copy of the application, dbdict or form is shared between all users.

The first process that initiates during the startup of Service Manager on the host machine creates the shared memory block in the size defined by the *shared_memory* parameter in the sm.ini file. The syntax of this parameter is:

```
shared_memory:<value>
```

Where <value> is the amount of memory in bytes Service Manager should allocate. This shared memory will be held in the machines virtual memory and thus should not be larger than the virtual memory the machine has available. Information about this allocation can be seen in the sm.log file by locating the line that reads “Creating Resources for system xxxxx with key 0xyyyyyyyyy” (where xxxxx represents the port number you are using and 0xyyyyyyyyy is a system key identifying the shared memory). Every process that connects afterwards attaches to the shared memory that was created by the first process. Information about this attachment can be seen in the sm.log file by locating the line that reads “Attaching to resources for system xxxxx with key 0xyyyyyyyyy.”. However, execution of some reports, like sm –reportsem, sm -reportshm and sm –reportlic, will not create shared memory because they are designed to only read data from existing shared memory.

Example: Shared Memory involvement during the login process[edit]

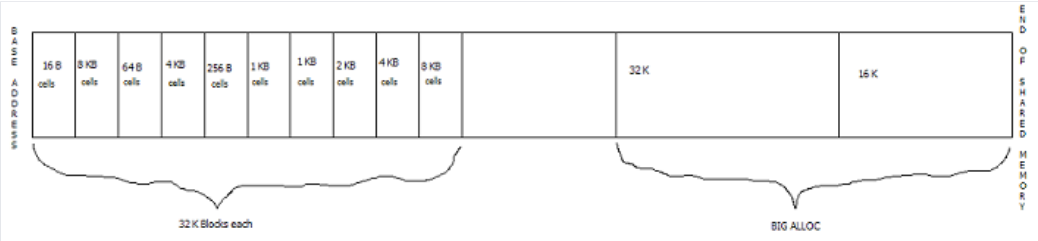
The login application is called. Service Manager will check shared memory, find that the login application is not currently stored there, retrieves the code record for the application from the database, and then loads it into shared memory. At this point, Service Manager will begin to execute the login application. Once it determines that the login.prompt form needs to be displayed, it will check shared memory for the form, find that it's not currently stored there, retrieve the login.prompt form from the database and load it into shared memory. Once the form is displayed, the user enters their userid and password, the application then attempts to validate this information against Service Manager's operator file. This method of checking shared memory, retrieving the item from the database, and storing it into shared memory continues throughout the life of the process. For this reason, the first user to log into Service Manager is responsible for loading most of the contents of shared memory, which allows each subsequent user to benefit.

Shared Memory Management[edit]

Because the memory in the allocated shared memory area is not released until the Service Manager server processes are stopped, the initial overhead of allocating doesn't depend on users logging in and out of Service Manager. It is assumed that while the server is operating, users will be logging in and out routinely and they will require continuous access to shared memory. Depending on the information, memory is released for reuse within the shared memory area at different times. For example, User Blocks are freed when the user logs out, and cached items are released when they are no longer needed. Cached information that has changed is flagged obsolete and a copy of the changed record is loaded into shared memory with the first user requesting the record after the change. The obsolete record is freed when the use count of the record is at 0, meaning all user processes using this record have released it from use.

For example, when a RAD application has changed, the old version is flagged obsolete if it is still in use and the new version is loaded into shared memory by the next user requesting the RAD application. This occurs because the system can't find a current copy of the record in shared memory at that time. The obsolete record is freed when the last user still using the old code record releases it.

Shared memory in Service Manager is divided into various size blocks of storage. The sizes available (in bytes) are 16, 32, 64, 128, 256, 512, 1024, 2048, 4096, 8192, and BIG_ALLOC. Any request for shared storage greater than 8192 bytes is satisfied from the BIG_ALLOC pool. Smaller sized blocks aren't allocated one at a time, but instead, come from a pool of similar sized requests. For example, when a request is made for 16 bytes of shared storage and there is no 16 byte block free, Service Manager will allocate a 32K byte block from the lower end of shared memory and divide that 32K area into 16 byte blocks. The first 16 byte block would be returned to the requestor and the remaining 16 byte blocks would remain on the 16 byte block free chain for later allocation. BIG ALLOC starts at the end of the allocated shared memory. When both ends (<8K and BIG ALLOC) meet, Service Manager is out of shared memory.



Contents

- 1 Example: Shared Memory involvement during the login process
- 2 Shared Memory Management

9.5.5.2. Shared memory sizing

Because the storage area of shared memory is not directly related to the number of users in the system, the per-user shared memory requirement can only be estimated. The user's total shared memory depends on the number of modules used, and the number of forms, dbdict records, links, format controls, and code records they access. The estimated amount of shared memory per user will be higher for a smaller user base, because the cached information in the system is usually relatively static and doesn't depend on the number of users in the system.

A sample calculation for shared memory is:

```
48 MB + 1MB per 10 users
```

Note

Note Shared memory in a horizontally scaled environment must be calculated per machine based on how many users this machine handles during peak times.

Shared Memory Adjustments

These suggestions offer a good baseline for most customer systems. However, it may still be necessary to periodically adjust the amount of shared memory. The amount of free storage should be monitored regularly using the Service Manager Shared Memory Report, **sm –reportshm**.

We recommend never letting the amount of free space fall below 25%. If continuous monitoring of the shared memory report indicates that free space is constantly above 75%, the **shared_memory** parameter in the sm.ini file can be set to a lower number to ensure memory is being used efficiently.

Note

Note Adjustments to the shared_memory parameter must not be made when the Service Manager server is running.

Cache Sizing

Cache is stored in slots. By default Service Manager uses 9973 cache slots. The cache report (sm -reportcache) shows the cache slots used and average and maximum depths.

```
C:\scs\sm7\server\RUN>sm -reportcache
---------------------- Cache Statistics      ----------------------

Slot use: 42%; Average Slot Depth: 1; Maximum Slot Depth: 13
```

The average slot depth should be below 3 and the maximum should be below 10. If the average slot depth is too deep, increase the cache_slots parameter to a larger prime number, such as 11001. Prime numbers are most efficient for cache_slots. This is due to the fact that we calculate a hash for each item in the cache resulting in numbers between 0 and 2^{32} -1 (32-bit integer). These hash numbers have to be distributed – if possible evenly – between the available cache slots. This is done with a modulus function against the cache_slots. Using a prime number has a higher probability of distributing the cache items evenly amongst the cache slots.

Unused items are removed from cache by the sync process. During the cache_clean_interval, the sync process enters a timestamp into all cache items that have a use count of 0. During the next cache_clean_interval, if it still shows the timestamp the item will be removed from cache. If the cache item was used after the first interval, the use count overwrites the timestamp indicating the item has been recently used. Thus cache items that were unused for cache_clean_interval (seconds) * 2 will be removed from the cache.

Shared Memory Reports

You can monitor the allocation and growth of shared memory by using the Service Manager Shared Memory Report, **sm –reportshm**. The following figure is an example of a shared memory report. The report contains a list of all the items

currently stored in shared memory in addition to the amount of storage each item is using.

```
------ Shared Memory ------

Shared Memory Release     9.63.2

Current Size             256000000

Segment Allocation         9346088
Large Block Allocation    11288576

Unused Space             235365336  (91%)
Free Space               235542624  (92%)


Shared Memory Type  Allocations    Frees     Allocated
------------------  ----------- ----------- -----------
Not named                    26           3      759744
User blocks                  24           3       43008
Messages                      0           0           0
Database Services           630           0       28848
Cache overhead               26           3       42656
Application cache          2158         703     1938832
DBDICT cache              11630        1146    15424064
SQL descriptor cache        530           0     1046432
Join/ERD/Type cache         402           0     1032320
String Type cache          1080         208      141472
Global cache                  0           0           0
Diagnostic Services           0           0           0
```

Current Size is the amount of shared memory allocated in bytes as defined by the shared_memory parameter in the sm.ini file.

Free Space is the amount of shared memory that's available for use by processes. Free Space is gained when previously used Shared Memory areas are made available again. Also, Free Space is the sum of "Unused Space" (memory never before allocated) and "Freed Space" (memory in the free list chain that was previously allocated) Free Space is shown as total bytes and as an overall percentage.

Unused Space represents available shared memory that has not yet been allocated. When Service Manager starts, the entire amount of memory defined by the shared_memory parameter is allocated, and blocks may then be allocated from within that shared memory as needed. Unused Space is shown in total bytes and as an overall percentage.

Using *sm -reportshm:1* to generate the Service Manager Shared Memory Report will produce a more detailed listing of shared memory allocation.

Messages

Every message that's sent from one user to another is put into the message block section of shared memory. The amount of shared memory allocated per message depends on the size of the message.

Database Services

Tuning statistics regarding executed queries and database access are stored in this area. Statistics indicating possible performance issues are written to the sc.alert.log.

Cache Overhead

Contains operators and other system information; It also contains overhead or miscellaneous items that need to be stored for multi-process access

Application Cache

This area in shared memory holds code records, pre-parsed and split up by panel.

DBDict Cache

This area contains cached Table Definitions (dbdict records) as well as other cached data records such as: format, link, format control, category, scmessage.

SQL Descriptor Cache

This area in shared memory is used by the RDBMS mapping engine and contains the mapping information between the dbdict and the RDBMS table definitions.

JOIN/ERD/TYPE Cache

This area contains records from joindef, erddef, scaccess, scmandant and sctypecheck

String Type Cache

This contains the fastcounters data structure, files and fields using counters, cascade delete definitions, cached charts, list of defined triggers, JavaScript triggers that for ease of Shared Memory storage are wrapped into a STRING.

JavaScript Members

This area contains information used by or required for JavaScript applications. It doesn't contain JavaScript code, but rather information on the current version of the Script member.

IR Expert and Shared Memory

Starting from version 9.62, IR no longer uses any shared memory.

9.5.5.3. Horizontal scaling: special considerations

Each horizontally scaled host machine has its own shared memory calculated based on the amount of users supposed to be running on that machine.

```
48 MB + 1 MB per 10 users
```

In the horizontally scaled system messages to users as well as Cache updates on changed cached records are communicated to all shared memory sections of all host machines in the group.

Locks aren't held in shared memory at all in horizontally scaled systems. Locks are handled differently depending on your version of Service Manager. For versions of Service Manager earlier than Service Manager 9.31, locks are communicated by using UDP Multicasting between the systems and the locking information is held in the JVM. For Service Manager 9.31 and later versions, locks are stored in the central RDBMS. For more information on how locks are managed in Service Manager 9.31, see the Lock Management topic in the Service Manager Help Server.

9.5.5.4. Shared memory related parameters

As soon as less than 25% of shared memory is unused, it is recommended that the amount of shared memory be increased by modifying the **shared_memory** parameter in the sm.ini file. Unused shared memory is the amount of memory that has never been allocated. The real percentage of available memory is always higher than what the unused percentage in the Service Manager Shared Memory Report depicts.

Because shared memory can become fragmented, leaving at least 25% of free space is recommended. This will ensure that fragmentation doesn't prevent new space from being allocated within the shared memory area.

The **shared_memory_address:<memory address>** parameter can be used on all operating systems to ensure that shared memory allocation is kept in an area that will not interfere with shared libraries, stacks, heaps or other memory related data structures. You can use this parameter to specify a new address that has enough contiguous memory for the Service Manager process.

If you encounter a shared memory allocation failure, the address that the Service Manager process is using doesn't have enough contiguous memory. On Windows, the sm.log file lists recommendations for possible **shared_memory_address** parameter values. If the debugvmmap parameter is active, the log also contains a map of virtual memory at the time of the failure.

For more details, see the following Service Manager online help topics:

- Startup parameter: shared_memory_address
- Failed to initialize or attach to shared memory environment (Windows only)

> **Note**
>
> Note The defaults should be working for all platforms. You should only use this parameter when instructed to do so by Customer Support.

Shared Memory Support Information

Shared memory problems can be some of the most difficult to resolve. However, if problems with shared memory arise, you can contact Customer Support with the following information ready to help expedite a satisfactory solution:

- Detailed description of the issue
- Copies of all Service Manager log files
- Copies of any core files that were generated
- Service Manager Shared Memory Report
- Service Manager Semaphore Report

10. Scheduled Maintenance

Scheduled Maintenance enables you to create a scheduled maintenance system to reduce unplanned outages and system failures. The objective is to service systems before they fail rather than after they fail.

Scheduled Maintenance helps you administer, configure, and customize scheduled maintenance. It installs without requiring an upgrade, and runs independent of Service Manager upgrades. It integrates into a pre-existing Service Manager installation without changing any tailoring and customization.

You should have an understanding of the Service Manager expression syntax and Database Manager to administer Scheduled Maintenance successfully.

A user can do the following tasks:

- Establish a formalized Scheduled Maintenance system to reduce unplanned outages and system failures by servicing systems before they fail
- Define and schedule recurring maintenance tasks, including Incident records, change requests, and Request Fulfillment requests, using closed loop integration with Service Manager
- Store all Scheduled Maintenance tasks in a single repository, ensuring that important maintenance occurs regularly
- Automatically notify staff of all maintenance items as they become due
- Enable users to create and update Scheduled Maintenance tasks even when they are unfamiliar with Service Manager
- Create audit information
- Create sophisticated or detailed maintenance tasks
- Conform to existing Service Manager tailoring

Related topics

Administrative access to Scheduled Maintenance
Scheduled Maintenance features

10.1. What's a scheduled task?

Scheduled Maintenance supports different types of scheduling. Select the one that best suits the task. Scheduled tasks run according to a defined schedule with no dependencies. For example, you can schedule a task to run every Tuesday at 4:00 a.m., or on the first Monday in January. Demand-based tasks run when the task relies on changes in the Configuration Management application. This can be defined by a trigger. For example, you can schedule a task to run whenever:

- The available drive space on the file server drops below 10 GB.

- The printer prints 8,000 pages.
- The available disk space on a server drops below 2 GB, require a log dump
- The number of devices attached to a particular router approaches the number of ports it has available.

Scheduled task delays

There may be a delay between the time you select a task to run and the time that it runs. Scheduled Maintenance checks for tasks to execute every minute. If it checked at 11:59:30 and 12:00:30, then the incident record, change request, or Request Fulfillment request would be opened at 12:00:30, not at 12:00:00.

There can be a few seconds delay as Scheduled Maintenance generates and saves the incident record, change request, or Request Fulfillment request. You can change the frequency of the Scheduled Maintenance Hook to balance the load.

If your server is disabled at the time a task is scheduled to run, the task executes at the next scheduled time after the server resumes operation.

Related topics

Scheduled Maintenance
Running a Scheduled Maintenance task
Cost Estimate tool

10.2. Running a Scheduled Maintenance task

When a Scheduled Maintenance task runs, Scheduled Maintenance generates an incident record, change request, or Request Fulfillment request as though a technician at your service desk created it. Scheduled Maintenance tracks the execution of the incident record, change request, or Request Fulfillment request to store a complete service history for each task.

Verifying a task

After you save a task, you can verify the task (or tasks) that Scheduled Maintenance created using one of these techniques.

- Force a task to execute immediately
- Check the execution history
- Check the execution details of a task

Defining the effect and details of a Scheduled Maintenance task

Scheduled Maintenance enables you to create several types of tasks. When you create a task, Scheduled Maintenance respects all normal default values, sends messages, adheres to Format Control rules, executes macros, and runs the usual validations. If an automatically generated Incident record, change request, or Request Fulfillment request doesn't match the validation rules, Scheduled Maintenance tries to ignore the validation rules and save the record anyway.

Scheduled Maintenance also enables you to open a collection of Incident records, change requests, or Request Fulfillment requests for every Configuration Item (CI) in Configuration Management that meets certain predefined criteria, or matches a specific user-defined query. Using one of these methods, you can create a maintenance task for any arbitrary subset of Configuration Items.

Incident records for scheduled maintenance tasks

Incident records can specify any of the following.

- Assignment group
- Category
- Expression
- Format control
- Incident description
- Incident title
- Maintained device
- Owner

- Priority code
- Status

Change requests for scheduled maintenance tasks

Change requests can specify any of the following.

- Expressions
- Format control
- Maintained device
- Assignee
- Category
- Coordinator
- Description
- Priority
- Work manager

Request Fulfillment requests for scheduled maintenance tasks

Request Fulfillment requests can specify any of the following.

- Title
- Description
- Category
- Subcategory
- Request Model
- Requester
- Assignee
- Priority
- Coordinator
- Expressions
- Format control

The Scheduled Maintenance Advanced Query option has a built in anti-spam feature. By default, the system generates 50 records, regardless of how many records the advanced query returns. This threshold is user definable under the Administrative Options section of the Scheduled Maintenance Menu.

Regular scheduling

Use this option to select a simple repeat interval or to make a task run at a certain time of day. For example, starting at 11/30/01 00:00:00 execute this task every hour on the hour.

To run a task run at the same time every day, set the start time and then a set a repeat interval of 24 hours. Specify a 24-hour interval with 1 00:00:00, not 24:00:00.

Repeat criteria

Demand based maintenance supports three different types of repeat criteria in an effort to model real-world maintenance activity as effectively as possible:

- Incremental repetition
- Quiescent repetition
- Manual repetition

Repeat criteria limitations

Many fields within Configuration Management that contain numeric values are character fields. This enables you to store unit values with the numeric data. A typical value can be 1,024 MB or 2,000 lbs. Scheduled maintenance can perform

comparative operations on these fields by removing non-numeric characters and treating the result as a number. Thus 1,024 MB becomes 1024 when Service Manager compares it to a threshold value.

Scheduled Maintenance doesn't consider the source of a change when it monitors the requested field on a device. Scheduled Maintenance reacts to any change from an interactive user, a background task, or a remote agent.

It is possible to have multiple demand maintenance tasks linked to the same device. If a single update can trigger more than one demand task, then each task runs independently. You can potentially create two or more maintenance tasks as a result of a single change to a device.

Scheduled Maintenance reacts to changes in the device record, not changes in the maintenance task. If you create or modify a maintenance task that depends on the current state of a monitored record, the task doesn't run until there is a change to the monitored device.

Task execution results

There are four potential results when you view the execution history of a task.

Error code	Meaning
Success	Executed Successfully. Scheduled Maintenance generated an incident record, change request, or Request Fulfillment request. No errors occurred. This is the result you want to see.
Reject	The incident record or change request was rejected by Format Control. Scheduled Maintenance populates fields in a record as a user would. If a field is required when a user creates an Incident record, change request, or Request Fulfillment request, it is also required when Scheduled Maintenance creates the same incident record, change request, or Request Fulfillment request. If the required information is unavailable by default, you can use expressions to populate it.
Error	Encountered an Error while executing task expressions. Although you completed the Expressions field on the Details tab of the task, one or more of the expressions did not execute properly. Identify and fix the error.
Error	Encountered an unknown error. An unhandled error occurred inside the Service Manager code, not Scheduled Maintenance Code. This is a very rare event.

Related topics

Scheduled Maintenance
What's a scheduled task?
Cost Estimate tool

10.3. Cost Estimate tool

The Cost Estimate tool enables you to select a subset of your maintenance tasks, and then calculate the estimated cost to run those tasks within a specified time frame, such as from 01/01/04 until 03/31/04.

Using the tool is a two-step process. First, you select a set of task records from within scheduled maintenance. Second, invoke the tool and provide it the criteria it requires.

Cost Estimate tool limitations

The tool estimates the number of times a task runs by using a mathematical function. These functions are fast and highly accurate. However, rounding errors or can cause the estimate to be one more or one less than the actual number of runs.

The tool makes no provision for tasks that expire halfway through the listed time frame. It assumes that all listed tasks continue throughout the time series.

If the tool uses a different currency conversion, it uses the current exchange rate or the most recent exchange rate it has on file. Actual exchange rates fluctuate, and may result in tasks later in the time frame performing under an actual exchange rate that's different from the rate used in the estimation process.

The cost estimates are only as good as the parts and labor information provided. Inaccurate or incomplete part and labor descriptions for maintenance tasks result in inaccurate or incomplete cost estimates.

Related topics

Scheduled Maintenance
Running a Scheduled Maintenance task
What's a scheduled task?

10.4. Access Scheduled Maintenance

Applies to User roles: System Administrator

To access Scheduled Maintenance, click **System Administration** > **Base System Configuration** > **Scheduled Maintenance** and then Select an activity.

- **Administrative Options** to select options for the template, or to generate Scheduled Maintenance tasks from Incident records, change requests, or Request Fulfillment requests.
- **Automation Rules** to select options for automatically running the predefined rules.
- **Execution Details** to display the results of a Scheduled Maintenance task's execution. The result may generate an Incident record, Change Management request, or Request Fulfillment request that you can open and view.
- **Execution History** to display a schedule when Scheduled Maintenance task executes.
- **Scheduled Maintenance Tasks** to create and edit Scheduled Maintenance tasks.

Related topics

- Scheduled Maintenance
- Check the execution details of a task
- Check the execution history of a task
- Create a change request from a Scheduled Maintenance task
- Create a Request Fulfillment request from a Scheduled Maintenance task
- Create a schedule for a task
- Create a Scheduled Maintenance task
- Create a Scheduled Maintenance task for an asset?
- Create a Scheduled Maintenance task from an open record
- Create an Incident record from a Scheduled Maintenance task
- Define the effect and details of a Scheduled Maintenance task
- Force a task to run immediately
- Schedule intervals
- Set the demand criteria for a task
- Specify incremental repetition
- Specify manual repetition
- Specify quiescent repetition
- Use expressions and Format Control only in a Scheduled Maintenance task
- Use the Cost Estimate tool

10.5. Check the execution details of a task

Applies to User roles: System Administrator

To check the execution details of a task:

1. Click **System Administration** > **Base System Configuration** > **Scheduled Maintenance** > **Execution Details**.
2. Type or select the search criteria. You can search by one of the following.

Field name	Description
ID	A unique identifier that Scheduled Maintenance gives to every task it creates.
Class	One of three values: incident, change, or request.
Number	The incident number, change number, or Request Fulfillment request number of the created task.
Opened at	The time the task opened.
Closed at	The time the task closed.
Task ID	The Service Manager ID of an Incident record, change request or Request Fulfillment request.
Logical Name	The Service Manager asset against which scheduled maintenance runs. For example, all maintenance history for the email server.

3. Click **Find** to locate the record.
4. If you don't have search criteria, insert the cursor in the field and click **Search** to locate the record, or use the drop-down calendar to specify a date.

If you have the task open in Scheduled Maintenance, you can click the **History** tab > **Execution Details**.

Related topics

Scheduled Maintenance
Access Scheduled Maintenance
Check the execution history of a task
Create a change request from a Scheduled Maintenance task
Create a Request Fulfillment request from a Scheduled Maintenance task
Create a schedule for a task
Create a Scheduled Maintenance task
Create a Scheduled Maintenance task for an asset?
Create a Scheduled Maintenance task from an open record
Create an Incident record from a Scheduled Maintenance task
Define the effect and details of a Scheduled Maintenance task
Force a task to run immediately
Schedule intervals
Set the demand criteria for a task
Specify incremental repetition
Specify manual repetition
Specify quiescent repetition
Use expressions and Format Control only in a Scheduled Maintenance task
Use the Cost Estimate tool

10.6. Check the execution history of a task

Applies to User roles: System Administrator

To check the execution history of a task:

1. Click **System Administration** > **Base System Configuration** > **Scheduled Maintenance** > **Execution History**.
2. Type the search criteria. You can search by Task ID, Event Date, Result, and Creation Count.
   Specify a value for one or more parameters.
3. Click **Search** to locate the record.
4. If you don't have search criteria, insert the cursor in the field and click **Search** to locate the record, or use the drop-down calendar to specify a date.

If you have the task open in Scheduled Maintenance, select the History tab and then click **Execution History**.

Related topics

- Scheduled Maintenance
- Access Scheduled Maintenance
- Check the execution details of a task
- Create a change request from a Scheduled Maintenance task
- Create a Request Fulfillment request from a Scheduled Maintenance task
- Create a schedule for a task
- Create a Scheduled Maintenance task
- Create a Scheduled Maintenance task for an asset?
- Create a Scheduled Maintenance task from an open record
- Create an Incident record from a Scheduled Maintenance task
- Define the effect and details of a Scheduled Maintenance task
- Force a task to run immediately
- Schedule intervals
- Set the demand criteria for a task
- Specify incremental repetition
- Specify manual repetition
- Specify quiescent repetition
- Use expressions and Format Control only in a Scheduled Maintenance task
- Use the Cost Estimate tool

10.7. Create a change request from a Scheduled Maintenance task

Applies to User roles: System Administrator

To create a change request from a Scheduled Maintenance task:

1. Click **System Administration** > **Base System Configuration** > **Scheduled Maintenance** > **Scheduled Maintenance Tasks**.
   Service Manager displays a blank Scheduled Maintenance form.
2. Specify the required information on the **Description** and **Schedule** tabs
3. Click the **Effect** tab.
4. Specify the type of task you would like to create by selecting a task effect.
   - If you select the fourth, fifth, or sixth option (one ending with **for each Device of Class X at location Y** you will also need to:
     - Select the asset type in the **Open for this Device Type** field.
     - Select the location in the **At Location** field.
     - You can specify the selection of assets to be maintained. Click **Use Advanced Query**. Service Manager replaces the location and asset based selection boxes with All Devices Matching This Query.

- Type the query in the **All Devices Matching This Query** field. Scheduled Maintenance requires that you specify a query in **All Devices Matching This Query** if you select **Use Advanced Query**. It doesn't validate this value. It is recommended that you test all queries using Asset Management to ensure that the query returns the correct records in a production environment.

Note: Unless you change **Max Generation Count**, Service Manager generates a maximum of 50 records.

5. Click the **Details** tab and complete the following fields:
   - **Open Change of Category**
   - **Coordinator**
   - **Work Manager**
   - **Set Priority to**
   - **Assign to**
   - **With a Company of**
6. If you are creating one change request, select the asset using the appropriate drop-down list. If you are creating a group of change requests, **Work on This Device** is unavailable. Instead, Scheduled Maintenance automatically sets a value as it works its way though your Configuration Items (CIs) to create a change request for each relevant asset.
7. Type optional Expressions for the record.
8. Choose a Format Control record for the record.
9. Click the **Parts & Labor** tab. When Service Desk generates the change request, it automatically copies any parts and labor estimates that you specify here. Cost estimating tools also use this information.

Related topics

Scheduled Maintenance
Access Scheduled Maintenance
Check the execution details of a task
Check the execution history of a task
Create a Request Fulfillment request from a Scheduled Maintenance task
Create a schedule for a task
Create a Scheduled Maintenance task
Create a Scheduled Maintenance task for an asset?
Create a Scheduled Maintenance task from an open record
Create an Incident record from a Scheduled Maintenance task
Define the effect and details of a Scheduled Maintenance task
Force a task to run immediately
Schedule intervals
Set the demand criteria for a task
Specify incremental repetition
Specify manual repetition
Specify quiescent repetition
Use expressions and Format Control only in a Scheduled Maintenance task
Use the Cost Estimate tool

10.8. Create a Request record from a Scheduled Maintenance task

Applies to User roles: System Administrator

To create a Request Fulfillment request from a Scheduled Maintenance task:

1. Click **System Administration** > **Base System Configuration** > **Scheduled Maintenance** > **Scheduled Maintenance Tasks**.
   Service Manager displays a blank Scheduled Maintenance form.
2. Specify the required information on the **Description** and **Schedule** tabs
3. Click the **Effect** tab.
4. Specify the type of task you would like to create by selecting one of the following choices.
   - Create One RM Request

- Create One RM Request for each Device of Class X at location Y

5. If you selected Create One RM Request for each Device of Class X at location Y, do the following
    1. Select the asset type in the **Open for this Device Type** field.
    2. Select the location in the **At Location** field.
    3. You can specify the selection of assets to be maintained. Click **Use Advanced Query**. Service Manager replaces the location and asset based selection boxes with All Devices Matching This Query.
    4. Type the query in the **All Devices Matching This Query** field. Scheduled Maintenance requires that you specify a query in **All Devices Matching This Query** if you select **Use Advanced Query**. It doesn't validate this value. It is recommended that you test all queries using Asset Management to ensure that the query returns the correct records in a production environment.

    Note: Unless you change **Max Generation Count**, Service Manager generates a maximum of 50 records.
6. Click the Details tab to display a list of fields for the Request Fulfillment request.
7. Complete the following fields.
    - **Open RM Request of Category**
    - **Open RM Request of Subcategory**
    - **Open RM Request of Request Model**
    - **Requestor Name**
    - **Assign to**
    - **Set Priority to**
    - **Coordinator**
    - **With a Company of**
8. Type optional Expressions for the request.
9. Choose a Format Control record for the request.

Related topics

Scheduled Maintenance
Access Scheduled Maintenance
Check the execution details of a task
Check the execution history of a task
Create a change request from a Scheduled Maintenance task
Create a schedule for a task
Create a Scheduled Maintenance task
Create a Scheduled Maintenance task for an asset?
Create a Scheduled Maintenance task from an open record
Create an Incident record from a Scheduled Maintenance task
Define the effect and details of a Scheduled Maintenance task
Force a task to run immediately
Schedule intervals
Set the demand criteria for a task
Specify incremental repetition
Specify manual repetition
Specify quiescent repetition
Use expressions and Format Control only in a Scheduled Maintenance task
Use the Cost Estimate tool

10.9. Create a schedule for a task

Applies to User roles: System Administrator

To create a schedule for a task:

1. Create a new Scheduled Maintenance task or locate an existing task.
2. Click the **Schedule** tab.

3. From **Execute Task at:**, select the date and time, then choose the correct time zone. If you omit the time zone, the default value is your current time zone.
4. Choose a repeat interval in the **Repeat Task** frame.
    - **Regularly** schedules a simple repeat interval
    - **Weekly** schedules a weekly interval
    - **Monthly** by date
    - **Monthly** by day
    - **Annually**
    - **Annually**
    - **An Arbitrary Schedule** (user defined)

    The available settings change depending on the interval you select.
5. Select one of the following duration options in the **Range of Recurrence** frame.
    - **No end date** for a task that should recur forever.
    - **End After *nn* Occurrences** where *nn* is a numeric value.
    - **End By *date*** where you select an end date from the calendar

Related topics

Scheduled Maintenance
Access Scheduled Maintenance
Check the execution details of a task
Check the execution history of a task
Create a change request from a Scheduled Maintenance task
Create a Request Fulfillment request from a Scheduled Maintenance task
Create a Scheduled Maintenance task
Create a Scheduled Maintenance task for an asset?
Create a Scheduled Maintenance task from an open record
Create an Incident record from a Scheduled Maintenance task
Define the effect and details of a Scheduled Maintenance task
Force a task to run immediately
Schedule intervals
Set the demand criteria for a task
Specify incremental repetition
Specify manual repetition
Specify quiescent repetition
Use expressions and Format Control only in a Scheduled Maintenance task
Use the Cost Estimate tool

10.10. Create a Scheduled Maintenance task

Applies to User roles: System Administrator

To create a Scheduled Maintenance task:

1. Click **System Administration** > **Base System Configuration** > **Scheduled Maintenance** > **Scheduled Maintenance Tasks**. Service Manager displays a blank Scheduled Maintenance Tasks form.
2. Type a name for the task in the **Name** field. The name you select can offer some insight into what the task does and you may use any naming convention, as long as each task name is unique.
3. Select **Active?** to activate the task. Every task must be active or inactive. Inactive tasks are in the system, but don't create any work orders or take any other action.
4. Select **Demand Based?** when the task relies on changes in the Configuration Management application. For example, you can schedule a task to run whenever the available drive space on the file server drops below 10 Gigabytes (GB), or every time the printer prints 8,000 pages. If the task is not demand based, don't check this box.
5. Click the **Description** tab. Type a description for the task in the text box. A description is optional, but it can be useful to document the task.

6. Click the **Schedule** tab to create a schedule for the task.
7. Click the **Effect** tab to create one or more Incident records, change requests, or Request Fulfillment requests.
8. Click the **Details** tab to add expressions to the task.
9. Click the **History** tab to view the history of the task.
10. Click **Add**.

If this is a scheduled task, Scheduled Maintenance calculates the next time this task should run and displays it in the Next Scheduled at field. The task runs according to this schedule; you may want to run the task immediately to verify that it works properly. If this is a demand based task, it runs as soon as an update causes it to meet the trigger criteria.

You can also create a Scheduled Maintenance task from:

- A record in Incident Management, Change Management, or Request Management.
- A CI in Configuration Management.

Related topics

Scheduled Maintenance
Access Scheduled Maintenance
Check the execution details of a task
Check the execution history of a task
Create a change request from a Scheduled Maintenance task
Create a Request Fulfillment request from a Scheduled Maintenance task
Create a schedule for a task
Create a Scheduled Maintenance task for an asset?
Create a Scheduled Maintenance task from an open record
Create an Incident record from a Scheduled Maintenance task
Define the effect and details of a Scheduled Maintenance task
Force a task to run immediately
Schedule intervals
Set the demand criteria for a task
Specify incremental repetition
Specify manual repetition
Specify quiescent repetition
Use expressions and Format Control only in a Scheduled Maintenance task
Use the Cost Estimate tool

10.11. Create a Scheduled Maintenance task for an asset

Applies to User roles: System Administrator, Configuration Manager, Configuration Administrator

To create a Scheduled Maintenance task for an asset:

1. Click **Configuration Management** > **Resources** > **Search CIs**. Service Manager displays a blank Configuration Item record.
2. To generate a record list of CIs, click **Search**.
3. Click a CI to select it.
4. Open the More Actions menu.
5. Click **Scheduled Maintenance** > **Generate Recurring** > **Incidents**, **Changes**, or **Requests**.

Service Manager displays a new Scheduled Maintenance task and imports data from the CI record and the template record.

Related topics

Scheduled Maintenance
More Actions menu
Access Scheduled Maintenance
Check the execution details of a task
Check the execution history of a task
Create a change request from a Scheduled Maintenance task

Create a Request Fulfillment request from a Scheduled Maintenance task
Create a schedule for a task
Create a Scheduled Maintenance task
Create a Scheduled Maintenance task from an open record
Create an Incident record from a Scheduled Maintenance task
Define the effect and details of a Scheduled Maintenance task
Force a task to run immediately
Schedule intervals
Set the demand criteria for a task
Specify incremental repetition
Specify manual repetition
Specify quiescent repetition
Use expressions and Format Control only in a Scheduled Maintenance task
Use the Cost Estimate tool

10.12. Create a Scheduled Maintenance task from an open record

Applies to User roles: System Administrator

To create a Scheduled Maintenance task from an open record:

1. Open a new record, or locate an existing record, in Incident Management, Change Management, or Request Management.
2. Open the More Actions menu and choose **Generate Maintenance**.

Service Manager displays a new Scheduled Maintenance record with some data automatically filled in from the original record and from the template record.

Related topics

Scheduled Maintenance
Access Scheduled Maintenance
Check the execution details of a task
Check the execution history of a task
Create a change request from a Scheduled Maintenance task
Create a Request Fulfillment request from a Scheduled Maintenance task
Create a schedule for a task
Create a Scheduled Maintenance task
Create a Scheduled Maintenance task for an asset?
Create an Incident record from a Scheduled Maintenance task
Define the effect and details of a Scheduled Maintenance task
Force a task to run immediately
Schedule intervals
Set the demand criteria for a task
Specify incremental repetition
Specify manual repetition
Specify quiescent repetition
Use expressions and Format Control only in a Scheduled Maintenance task
Use the Cost Estimate tool

10.13. Create an Incident record from a Scheduled Maintenance task

Applies to User roles: System Administrator

To create an Incident record from a Scheduled Maintenance task:

1. Click **System Administration** > **Base System Configuration** > **Scheduled Maintenance** > **Scheduled Maintenance Tasks**.
   Service Manager displays a blank Scheduled Maintenance form.
2. Specify the required information on the **Description** and **Schedule** tabs

3. Click the **Effect** tab and select one of the following.
    - **Create One Incident Ticket**
    - **Create one Incident Ticket for each Device of Class X at location Y.**
4. If you selected Create One Incident Ticket for each Device of Class X at location Y, do the following:
    1. Select the asset type in the **Open for this Device Type** field.
    2. Select the location in the **At Location** field.
    3. You can specify the selection of assets to be maintained. Click **Use Advanced Query**. Service Manager replaces the location and asset based selection boxes with All Devices Matching This Query.
    4. Type the query in the **All Devices Matching This Query** field. Scheduled Maintenance requires that you specify a query in **All Devices Matching This Query** if you select **Use Advanced Query**. It doesn't validate this value. It is recommended that you test all queries using Asset Management to ensure that the query returns the correct records in a production environment.

    Note: Unless you change **Max Generation Count**, Service Manager generates a maximum of 50 records.
5. Click the Details tab to display a list of fields for the Incident records.
6. Complete the following fields.
    1. **Open Incident Tickets of Category**
    2. **Assign to**
    3. **Set Status to**
    4. **With Ticket Owner of**
    5. **With Priority Code of**
    6. **With a Company of**
7. If you are creating one Incident record, select the asset using the appropriate drop-down list. If you are creating a group of Incident records, **Work on This Device** is unavailable. Instead, Scheduled Maintenance automatically sets a value as it works its way though your Configuration Items (CIs) to create records for each relevant asset.
8. Type optional Expressions for the record.
9. Choose a Format Control record for the record.
10. Click the **Parts & Labor** tab. When Service Desk generates the Incident record, it automatically copies any parts and labor estimates that you specify here. Cost estimating tools also use this information.

Related topics

Scheduled Maintenance
Access Scheduled Maintenance
Check the execution details of a task
Check the execution history of a task
Create a change request from a Scheduled Maintenance task
Create a Request Fulfillment request from a Scheduled Maintenance task
Create a schedule for a task
Create a Scheduled Maintenance task
Create a Scheduled Maintenance task for an asset?
Create a Scheduled Maintenance task from an open record
Define the effect and details of a Scheduled Maintenance task
Force a task to run immediately
Schedule intervals
Set the demand criteria for a task
Specify incremental repetition
Specify manual repetition
Specify quiescent repetition
Use expressions and Format Control only in a Scheduled Maintenance task
Use the Cost Estimate tool

10.14. Define the effect and details of a Scheduled Maintenance task

Applies to User roles: System Administrator

To define the effect and details of a Scheduled Maintenance task:

1. Click **System Administration** > **Base System Configuration** > **Scheduled Maintenance** > **Scheduled Maintenance Tasks**.
2. Specify the required information on the **Description** and **Schedule** tabs
3. Click the **Effect** tab.
4. Specify the type of task you would like to create by selecting one of the following choices.
    - **Create One Incident Ticket**
    - **Create One Change Request**
    - Create One RM Request
    - **Create one Incident Ticket for each Device of Class X at location Y**
    - **Create one Change Request for each Device of Class X at location Y**
    - Create One RM Request for each Device of Class X at location Y
    - **Create Nothing: Only Execute Expressions and Format Control**

    You can limit the creation of multiple tasks to only those assets at a particular location by filling in the **Location** field. If you don't specify a location, the task creates an Incident record, change request, or Request Fulfillment request for every Configuration Item (CI) of the appropriate type in your Configuration Management database, regardless of its location.
5. Specify a title. This is the title on the Incident record, change request, or Request Fulfillment request.
6. Type descriptive text. This is the description on the Incident record, change request, or Request Fulfillment request that Scheduled Maintenance creates.
7. Click the **Details** tab.
8. Specify the task details depending on whether you are creating an Incident record, change request, or Request Fulfillment request.

Related topics

Scheduled Maintenance
Access Scheduled Maintenance
Check the execution details of a task
Check the execution history of a task
Create a change request from a Scheduled Maintenance task
Create a Request Fulfillment request from a Scheduled Maintenance task
Create a schedule for a task
Create a Scheduled Maintenance task
Create a Scheduled Maintenance task for an asset?
Create a Scheduled Maintenance task from an open record
Create an Incident record from a Scheduled Maintenance task
Force a task to run immediately
Schedule intervals
Set the demand criteria for a task
Specify incremental repetition
Specify manual repetition
Specify quiescent repetition
Use expressions and Format Control only in a Scheduled Maintenance task
Use the Cost Estimate tool

10.15. Force a task to run immediately

Applies to User roles: System Administrator

To force a task to run immediately:

1. Click **System Administration** > **Base System Configuration** > **Scheduled Maintenance** > **Scheduled Maintenance Tasks**.

2. Click **Search** to run a query of Scheduled Maintenance tasks. Scheduled Maintenance displays a record list of Scheduled Maintenance tasks.
3. Double-click the selected task.
4. Locate **Next Scheduled At:**. This field contains a calculated time when Scheduled Maintenance will run the task.
5. Change the date to an earlier date, such as 01/01/1990.
6. If necessary, select the **Active** checkbox. (Only active tasks run.)
7. Click **Save**.
8. Click **OK**.

By default, Scheduled Maintenance checks for tasks to run every 60 seconds. You might wait up to one minute before your task executes. After enough time elapses for the task to run, repeat the steps to locate the Scheduled Maintenance task. If the task ran, the **Next Scheduled At** field changes.

After a task runs, Scheduled Maintenance automatically reschedules it for the next appropriate time.

Demand based tasks

1. Click **Configuration Management** > **Resources** > **Search CIs**.
2. Click **Search**.
3. Select a record from the record list.
4. Update the device record to meet the trigger criteria for the maintenance task. The maintenance task runs as soon as you update the record.

Related topics

Scheduled Maintenance
Access Scheduled Maintenance
Check the execution details of a task
Check the execution history of a task
Create a change request from a Scheduled Maintenance task
Create a Request Fulfillment request from a Scheduled Maintenance task
Create a schedule for a task
Create a Scheduled Maintenance task
Create a Scheduled Maintenance task for an asset?
Create a Scheduled Maintenance task from an open record
Create an Incident record from a Scheduled Maintenance task
Define the effect and details of a Scheduled Maintenance task
Schedule intervals
Set the demand criteria for a task
Specify incremental repetition
Specify manual repetition
Specify quiescent repetition
Use expressions and Format Control only in a Scheduled Maintenance task
Use the Cost Estimate tool

10.16. Schedule intervals

Applies to User roles: System Administrator

To schedule intervals:

1. Click **System Administration** > **Base System Configuration** > **Scheduled Maintenance** > **Scheduled Maintenance Tasks**.
2. Click **Search** to run a query of Scheduled Maintenance tasks. Scheduled Maintenance displays a record list of Scheduled Maintenance tasks.
3. Double-click the selected task.
4. Select the interval option you want to use:
    - Regular repeat interval

1. Click **Regularly**.
2. Choose the date and time for the first occurrence in the **Every** field.

- Weekly repeat interval
  1. Click **Weekly**.
  2. Select the day of the week from the **Every** drop-down list.
- Monthly repeat interval
  - Date in the month
    1. Click **Monthly**.
    2. Select a date from the **Every** drop-down list.
  - Week and day of the month
    1. Click **Monthly**.
    2. Select a week of the month from the **Every** drop-down list.
    3. Select a day of the week from the second drop-down list.
- Annual repeat interval
  - Date in the year
    1. Click **Annually**.
    2. Select the sequential day of the year from the **Every** drop-down list.
  - Month and day of the year
    1. Click **Annually**.
    2. Select a month from the **Every** drop-down list.
    3. Select a date from the second drop-down list.
- Arbitrary schedule
  1. Click **On an Arbitrary Schedule**.
  2. Click a row in the Recurrence Schedule: table. Choose a date and time. You can repeat this step for each scheduled occurrence.

Related topics

Scheduled Maintenance
Access Scheduled Maintenance
Check the execution details of a task
Check the execution history of a task
Create a change request from a Scheduled Maintenance task
Create a Request Fulfillment request from a Scheduled Maintenance task
Create a schedule for a task
Create a Scheduled Maintenance task
Create a Scheduled Maintenance task for an asset?
Create a Scheduled Maintenance task from an open record
Create an Incident record from a Scheduled Maintenance task
Define the effect and details of a Scheduled Maintenance task
Force a task to run immediately
Set the demand criteria for a task
Specify incremental repetition
Specify manual repetition
Specify quiescent repetition
Use expressions and Format Control only in a Scheduled Maintenance task
Use the Cost Estimate tool

10.17. Set the demand criteria for a task

Applies to User roles: System Administrator

Demand-based tasks run when the rask relies on a change in the Configuration Management application. You define the criteria, or trigger, that causes the task to be executed.

To set the demand criteria for a task:

1. Click **System Administration** > **Base System Configuration** > **Scheduled Maintenance** > **Scheduled Maintenance Tasks**.
2. Type a name for the task in the **Name** field. Every task needs a unique name. This name can offer some insight into what the task does, but you may use any naming convention, as long as each name is unique.
3. Select **Active?** to activate the task. Every task must be active or inactive. Inactive tasks are still in the system, but create no work orders or take any other action.
4. Select **Demand Based?**.
5. Type a description of the task.
6. Click the **Demand Trigger** tab.
7. Choose the device you want to monitor. Click **Fill** to display a list of available devices. Double-click the device name. Ensure that the name you specify is an exact match to the device name in the database.
8. Define the criteria which triggers this task. Type the name of the field to monitor in the **Trigger When this Field:** text box. This name must match a field name for the device you chose in the previous step.
9. Specify a threshold symbol, such as greater than (>), less than (<), or equal (=).
10. Specify a numeric trigger value that triggers this maintenance task when the value is true in the **This Value:** text box.

The combination of a field name, a sign, and a threshold value can be viewed as a simple equation:
Execute task if *field name* is *symbolthreshold*.

Related topics

Scheduled Maintenance
Access Scheduled Maintenance
Check the execution details of a task
Check the execution history of a task
Create a change request from a Scheduled Maintenance task
Create a Request Fulfillment request from a Scheduled Maintenance task
Create a schedule for a task
Create a Scheduled Maintenance task
Create a Scheduled Maintenance task for an asset?
Create a Scheduled Maintenance task from an open record
Create an Incident record from a Scheduled Maintenance task
Define the effect and details of a Scheduled Maintenance task
Force a task to run immediately
Schedule intervals
Specify incremental repetition
Specify manual repetition
Specify quiescent repetition
Use expressions and Format Control only in a Scheduled Maintenance task
Use the Cost Estimate tool

10.18. Specify incremental repetition

Applies to User roles: System Administrator

Some demand-based maintenance activities depend on a steadily increasing (or decreasing) value. For example, you want to change the oil in your car every 3,000 miles. Therefore, you expect a maintenance task to trigger when your car reaches 3,000 miles, then again at 6,000 miles, and again at 9,000 miles.

To specify incremental repetition:

1. Click **System Administration** > **Base System Configuration** > **Scheduled Maintenance** > **Scheduled Maintenance Tasks**.
2. Follow the procedure for setting the demand criteria for a task, as described in the related topics.

3. In the **Determine when to run next by:** field, select **Adding to the previous trigger value** or **Subtracting from the previous trigger value** from the drop-down list.
4. Specify a value in the Repeat Increment field.

The system automatically recalculates a new threshold value by adding or subtracting your repeat increment from the previous threshold value when your task runs.

Related topics

Scheduled Maintenance
Access Scheduled Maintenance
Check the execution details of a task
Check the execution history of a task
Create a change request from a Scheduled Maintenance task
Create a Request Fulfillment request from a Scheduled Maintenance task
Create a schedule for a task
Create a Scheduled Maintenance task
Create a Scheduled Maintenance task for an asset?
Create a Scheduled Maintenance task from an open record
Create an Incident record from a Scheduled Maintenance task
Define the effect and details of a Scheduled Maintenance task
Force a task to run immediately
Schedule intervals
Set the demand criteria for a task
Specify manual repetition
Specify quiescent repetition
Use expressions and Format Control only in a Scheduled Maintenance task
Use the Cost Estimate tool

10.19. Specify manual repetition

Applies to User roles: System Administrator

Use manual repetition when your demand based task doesn't fit into quiescent or incremental repetition categories.

To specify manual repetition:

1. Click **System Administration** > **Base System Configuration** > **Scheduled Maintenance** > **Scheduled Maintenance Tasks**.
2. Follow the procedure for setting the demand criteria for a task, as described in the related topics.
3. From the **Determine when to run next by:** field, select **Do Not Run Again**.

This task runs once when conditions satisfy the trigger criteria. The task remains inactive until you activate it manually . You can use this model for one-time maintenance events or tasks that require human intervention to recalculate the trigger conditions.

Related topics

Scheduled Maintenance
Access Scheduled Maintenance
Check the execution details of a task
Check the execution history of a task
Create a change request from a Scheduled Maintenance task
Create a Request Fulfillment request from a Scheduled Maintenance task
Create a schedule for a task
Create a Scheduled Maintenance task
Create a Scheduled Maintenance task for an asset?
Create a Scheduled Maintenance task from an open record
Create an Incident record from a Scheduled Maintenance task
Define the effect and details of a Scheduled Maintenance task

Force a task to run immediately
Schedule intervals
Set the demand criteria for a task
Specify incremental repetition
Specify quiescent repetition
Use expressions and Format Control only in a Scheduled Maintenance task
Use the Cost Estimate tool

10.20. Specify quiescent repetition

Applies to User roles: System Administrator

Choosing quiescent repetition creates a pause interval. For example, a maintenance task that runs whenever the available disk space on a file server drops below 10 GB can be set to pause for a week to enable IT to put a new drive on the server. Within this one week window, if the available space on the server continues to fall below the threshold value, the task doesn't run. At the end of the pause interval, if the available space is still beneath the threshold, the task runs again to create another change request.

Quiescent repetition creates a pause that enables you to resolve the issue that triggered the maintenance without creating multiple maintenance tasks on the issue.

To specify quiescent repetition:

1. Click **System Administration** > **Base System Configuration** > **Scheduled Maintenance** > **Scheduled Maintenance Tasks**.
2. Follow the procedure for setting the demand criteria for a task, as described in the related topics.
3. From the **Determine when to run next by:** field, select **Stop monitoring device for a specified time**.
4. Choose a beginning date and time in the **Pause For:** field.
5. Choose an ending date and time in the **Currently Paused Until:** field.

Related topics

Scheduled Maintenance
Access Scheduled Maintenance
Check the execution details of a task
Check the execution history of a task
Create a change request from a Scheduled Maintenance task
Create a Request Fulfillment request from a Scheduled Maintenance task
Create a schedule for a task
Create a Scheduled Maintenance task
Create a Scheduled Maintenance task for an asset?
Create a Scheduled Maintenance task from an open record
Create an Incident record from a Scheduled Maintenance task
Define the effect and details of a Scheduled Maintenance task
Force a task to run immediately
Schedule intervals
Set the demand criteria for a task
Specify incremental repetition
Specify manual repetition
Use expressions and Format Control only in a Scheduled Maintenance task
Use the Cost Estimate tool

10.21. Use expressions and Format Control only in a Scheduled Maintenance task

Applies to User roles: System Administrator

To use expressions and format control in a Scheduled Maintenance task:

1. Click **System Administration** > **Base System Configuration** > **Scheduled Maintenance** > **Scheduled Maintenance Tasks**. Service Manager displays a blank Scheduled Maintenance form.

2. Specify the required information on the **Description** and **Schedule** tabs
3. Click the **Effect** tab.
4. Click **Create Nothing: Only execute expressions and Format Control**.
5. Click the **Details** tab.
6. Complete **With a Company of**.
7. Type optional Expressions for the record.
8. Choose a Format Control record for the record.

Related topics

Scheduled Maintenance
Access Scheduled Maintenance
Check the execution details of a task
Check the execution history of a task
Create a change request from a Scheduled Maintenance task
Create a Request Fulfillment request from a Scheduled Maintenance task
Create a schedule for a task
Create a Scheduled Maintenance task
Create a Scheduled Maintenance task for an asset?
Create a Scheduled Maintenance task from an open record
Create an Incident record from a Scheduled Maintenance task
Define the effect and details of a Scheduled Maintenance task
Force a task to run immediately
Schedule intervals
Set the demand criteria for a task
Specify incremental repetition
Specify manual repetition
Specify quiescent repetition
Use the Cost Estimate tool

10.22. Use the Cost Estimate tool

Applies to User roles: System Administrator

To use the Cost Estimate tool:

1. Click **System Administration** > **Base System Configuration** > **Scheduled Maintenance** > **Scheduled Maintenance Tasks**.
   Service Manager displays a blank Scheduled Maintenance form.
2. Click **Search** to run a query of Scheduled Maintenance tasks. Scheduled Maintenance displays a record list of Scheduled Maintenance tasks.
3. From the record list More Actions menu, choose **Estimate Costs**. Scheduled Maintenance displays a form where you can specify the cost estimate parameters.
4. Specify a start and an end date for your estimation period using the drop-down calendars.
5. The default currency value is US Dollar. Click the row to choose a different value from the drop-down list.
6. Click **Proceed**.

Scheduled Maintenance calculates the number of times that each task will run within the time frame you specified and the cost to run each task once. The results of these calculations display in a spreadsheet.

Related topics

Scheduled Maintenance
Access Scheduled Maintenance
Check the execution details of a task
Check the execution history of a task
Create a change request from a Scheduled Maintenance task

Create a Request Fulfillment request from a Scheduled Maintenance task
Create a schedule for a task
Create a Scheduled Maintenance task
Create a Scheduled Maintenance task for an asset?
Create a Scheduled Maintenance task from an open record
Create an Incident record from a Scheduled Maintenance task
Define the effect and details of a Scheduled Maintenance task
Force a task to run immediately
Schedule intervals
Set the demand criteria for a task
Specify incremental repetition
Specify manual repetition
Specify quiescent repetition
Use expressions and Format Control only in a Scheduled Maintenance task

10.23. Administrative access to Scheduled Maintenance

As an application administrator, you can create Scheduled Maintenance templates and manage task limits. You can access Scheduled Maintenance administration by clicking **System Administration** > **Base System Configuration** > **Scheduled Maintenance** > **Administrative Options**.

Related topics

Scheduled Maintenance
Scheduled Maintenance features

10.24. Scheduled Maintenance features

Scheduled Maintenance enables you to create and store as many maintenance tasks as you need for your organization. You can:

- Create and schedule recurring maintenance tasks, including Incident records, change requests and Request Fulfillment requests, using closed loop integration with Service Manager.
- Keep all scheduled maintenance tasks in a central repository to ensure that important maintenance occurs on time. Stored maintenance tasks automatically generate Incident records, change requests, or Request Fulfillment requests as they become due.
- Automatically use Service Manager to notify users about maintenance tasks as they are due.
- Create and update Scheduled Maintenance tasks even if you are unfamiliar with Service Manager.
- Make audit information available as necessary. The maintenance history and the auditing information for each task is available for review.
- Create sophisticated and detailed maintenance tasks.
- Maintain existing Service Manager customization and tailoring.
- Track scheduled maintenance tasks.

Related topics

Scheduled Maintenance
Administrative access to Scheduled Maintenance

10.25. Automated task generation

There are two ways that Scheduled Maintenance automatically generates tasks.

- Create the scheduled maintenance task from an existing incident record, change request, or Request Fulfillment request.
- Create the scheduled maintenance task from a default incident, change, or request template identified in Administrative Options for a Configuration Item (CI) in Configuration Management.

Generating tasks from an existing record[edit]

Generating scheduled maintenance tasks can be tedious, especially if you must generate a large number of tasks to do approximately the same thing. For example, you might want to generate 10 scheduled maintenance tasks to back up 10 servers. When tasks are similar, you can use a standard template. For example, you can create a default Change Management template for server backups.

Scheduled Maintenance enables you to access Incident Management, Change Management, or Request Management, find a record, and generate a scheduled maintenance task. Therefore, instead of creating 10 scheduled maintenance tasks manually that are variations on the default template, you can access the template from Change Management and select Generate Maintenance 10 times.

You must create a separate recurrence schedule for each task, but Scheduled Maintenance automatically copies many of the details from the change request. The following table lists the fields that migrate to the maintenance task.

Incidents	Changes	Requests
category*	description	description
brief.description	category	category
assignment	logical.name**	requestor.name
logical.name**	coordinator	assigned.to
ticket.owner	work.manager	coordinator
priority.code	assigned.to	priority
problem.status	priority	action

* Override if the device has a default Incident Management category.
** Override with the device name of the selected item in the Configuration Management application when you choose this option.

Contents

- 1 Generating tasks from an existing record

10.26. Scheduled Maintenance commands in Configuration Management

Service Manager integrates Scheduled Maintenance with Configuration Management. You can find these commands on the More Actions menu that you access from an asset record.

Command	Description
Maintenance Tasks	Show any maintenance tasks that are bound to a specified device. Scheduled Maintenance doesn't display any task that references a range of devices, such as routine maintenance for every server in a geographic area.
Maintenance History	List how many incidents, changes, or requests generated when a maintenance task runs. From this list, you can view the individual incident records, change requests and Request Fulfillment requests in detail. Maintenance tasks that create one incident, change, or request are a single entry. Maintenance tasks that create a collection of incidents, changes, or requests for every Configuration Item (CI) matching certain parameters can generate different information each time the task runs. For example, on May 1, there might be two servers in a geographic area, but by June 1, there might be a third server installed. Therefore, a task that generates a change for every server in a geographic area would generate two changes in May and three changes in June.
Generate Recurring >Incidents	Generate a Scheduled Maintenance task for the current device, which creates an Incident record. This Scheduled Maintenance task uses a template Incident record and other information from Configuration Management.
Generate Recurring >Changes	Generate a Scheduled Maintenance task for the current device, which creates a change request. This Scheduled Maintenance task uses a template change request and other information from Configuration Management.
Generate Recurring >Requests	Generate a Scheduled Maintenance task for the current device, which creates a Request Fulfillment request. This Scheduled Maintenance task uses a template Request Fulfillment request and other information from Configuration Management.

Record limitations[edit]

The Scheduled Maintenance Advanced Query has a built in anti-spam feature. By default, Scheduled Maintenance can generate a maximum of 50 records, regardless of how many records the advanced query returns. This is to prevent accidentally creating a Scheduled Maintenance task that would open a record for every Configuration Item (CI) listed in your Configuration Management database. However, if you want to generate a larger number of records, you can increase that number.

Access Scheduled Maintenance from Configuration Management[edit]

Applies to User roles: System Administrator, Configuration Manager, Configuration Administrator

To access Scheduled Maintenance from Configuration Management:

1. Click **Configuration Management** > **Resources** > **Search CIs** (or **Configuration Management** > **Search CIs** if you don't have the Resources folder in your System Navigator).
2. Click **Search** to generate a record list.
3. Click a record to display it in the Manage CI Types information form
4. Open the More Actions menu.
5. Click **Scheduled Maintenance**.
6. Click one of the following:
    - **Maintenance Tasks**
    - **Maintenance History**
    - **Generate Recurring** > **Incidents**, **Changes**, or **Requests**

Access Scheduled Maintenance from Configuration Management[edit]

Applies to User roles: System Administrator, Configuration Manager, Configuration Administrator

To access Scheduled Maintenance from Configuration Management:

1. Click **Configuration Management** > **Resources** > **Search CIs** (or **Configuration Management** > **Search CIs** if you don't have the Resources folder in your System Navigator).
2. Click **Search** to generate a record list.
3. Click a record to display it in the Manage CI Types information form
4. Open the More Actions menu.
5. Click **Scheduled Maintenance**.
6. Click one of the following:
    - **Maintenance Tasks**
    - **Maintenance History**
    - **Generate Recurring** > **Incidents**, **Changes**, or **Requests**

Change the maximum number of records to be generated by Scheduled Maintenance[edit]

1. Click **System Administration** > **Base System Configuration** > **Scheduled Maintenance** > **Administrative Options**.
2. The Scheduled Maintenance Administrative Options form appears. Change the **Max Generation Count** field to a new value.
3. Click **OK**.
4. Click **System Administration** > **Base System Configuration** > **Scheduled Maintenance**.
5. Click **Search**.
6. Click the selected task to view it in the Scheduled Maintenance Tasks form.
7. If there are any Scheduled Maintenance tasks running, stop and restart them using the System Scheduler. This ensures that these tasks use the new Max Generation Count value.

Contents

- 1 Record limitations
- 2 Access Scheduled Maintenance from Configuration Management
- 3 Access Scheduled Maintenance from Configuration Management
- 4 Change the maximum number of records to be generated by Scheduled Maintenance

10.27. Scheduled Maintenance exception models

Scheduled Maintenance includes a system of exception models that enables an administrator to schedule tasks around weekends or holidays. You can choose one of these models:

Model	Restrictions
No Exceptions model	Scheduled Maintenance tasks run whenever they are scheduled, regardless of weekends or holidays. After the task runs, it immediately reschedules for the next occurrence. For example, the task runs every hour on the half hour at 8:30 AM, 9:30, 10:30, ... 5:30 PM and so on.
Only Calendars model	Scheduled Maintenance tasks run only if the assigned time occurs within a specific calendar. For example, a task scheduled to run only during the day shift (9:00AM - 5:00PM) can run at 2:00 PM, but not at 7:00 PM.
Except Calendars model	Scheduled Maintenance tasks run according to schedule unless the designated time falls within a specific calendar. For example, a task scheduled to run at any time, except during the day shift, can't run between 9:00AM and 5:00PM, but can run at all other times.

Exceptions do not stop a task from recalculating normally, but if its scheduled time falls within an exception window, the task doesn't run. Scheduled Maintenance displays a message in the task history log to indicate that the task triggered, but did not run because it occurred during an exception window.

Notes:

- Exceptions apply only to calendar-based tasks. They do not apply to demand-based maintenance tasks.
- Exceptions apply to any type of scheduled task. Tasks with weekly, monthly, or yearly schedules can use exception calendars. As a practical matter, they're probably most useful for tasks with simple recurrence schedules, but the system will support their use with any schedule you set up.
- All tasks default to the No Exceptions model.
- Tasks which expire and attempt to run during an exception window do not "push" to the next available acceptable execution time. They simply do nothing and reschedule.

Examples: Scheduled Maintenance exception models[edit]

Assume that you have a Day Shift calendar that specifies 9AM – 5PM, Monday – Friday. Saturday and Sunday aren't included in the calendar. There is a scheduled task that runs every hour beginning at 8:30 AM.

Model	Result
No Exceptions model	The task runs every hour on the half-hour 8:30 AM, 9:30, 10:30, ... 5:30 PM and so on.
Only Calendars model	The task runs only during the Day Shift. The task runs every hour on the half-hour 8:30 AM, 9:30, 10:30, ... 4:30 PM. The task triggers again at 5:30 PM, but doesn't run because it can't run outside of the defined calendar hours.
Except Calendars model	The task run at any time except during the Day Shift. The task triggers at 8:30 AM, 9:30, 10:30, ... 4:30 PM, but it doesn't run until the Day Shift ends. When the task triggers at 5:30 PM, it runs normally.

Contents

- 1 Examples: Scheduled Maintenance exception models

10.28. Using a Scheduled Maintenance template

Scheduled Maintenance follows a series of steps when it applies a default template to a new maintenance task.

- A user selects Generate Recurring Incidents, Changes, or Requests from the More Actions menu in Incident Management, Change Management, or Request Management.
- Scheduled Maintenance uses the default incident template specified in the Scheduled Maintenance Administration form.
- Scheduled Maintenance creates a Scheduled Maintenance task. When the task runs, it creates a new Incident record that looks like the template record.
- A user can subsequently modify the record.

The new record doesn't copy the template exactly. The default template record may reference another piece of hardware. Scheduled Maintenance overrides the template record category with the default Incident Management category for the Configuration Item (CI) as it appears in the Configuration Management record. It also replaces the device ID with the correct device ID.

Specify Scheduled Maintenance template records[edit]

Applies to User roles: System Administrator

To specify Scheduled Maintenance template records:

1. Click **System Administration** > **Base System Configuration** > **Scheduled Maintenance** > **Administrative Options**.
2. The Scheduled Maintenance Administrative Options form appears with a list of default templates. A template is an existing record that can be reused with different data. Replace any default template name with a different record name.
3. Click **Save**.
4. Click **OK**.

Contents

- 1 Specify Scheduled Maintenance template records

10.29. Adding Scheduled Maintenance data using expressions

You can use expression syntax to auto-fill fields in the Incident record, change request, or Request Fulfillment request generated by a Scheduled Maintenance task.

Expressions run to populate data fields after Scheduled Maintenance supply simple information in the Incident record, change request, or Request Fulfillment request. You can use expressions to override the category, assignee, or another field.

Add data using expression syntax[edit]

Applies to User roles: System Administrator

To add data using expression syntax:

1. Click **System Administration** > **Base System Configuration** > **Scheduled Maintenance** > **Scheduled Maintenance Tasks**..
   Service Manager displays a blank Scheduled Maintenance form.
2. Click **Search**.
3. Click the selected task to view it in the Scheduled Maintenance Tasks form.
4. Click the **Details** tab.
5. In the text box labeled **Use These Expressions to Fill the Task.....**, type expressions that set selected fields to new values. The Incident record, change request, or Request Fulfillment request are referred to as $L.file in these expressions. For example, to set the subcategory and priority code on incident records, add these two lines of expression code:

   ```
   subcategory in $L.file="application"
   priority.code in $L.file="4"
   ```

   where application is the name of the subcategory and 4 is the priority code.
6. Click **Save**.

Contents

- 1 Add data using expression syntax

10.30. Scheduled Maintenance overhead

Scheduled Maintenance relies on the problem background scheduler to call into the Scheduled Maintenance code at regularly scheduled intervals. If the problem scheduler is not running, no Scheduled Maintenance tasks run. When a Scheduled Maintenance task runs, it increases the load on the problem scheduler by a marginal amount.

Scheduled Maintenance doesn't put a large load on your system unless you are using the Scheduled Maintenance system to generate extraordinarily large numbers of incident records, change requests, or Request Fulfillment requests. If Scheduled Maintenance creates twenty tasks a day, the impact is approximately identical to one user opening twenty tasks a day.

If a Scheduled Maintenance task creates 20,000 Incident records at 2:00 AM, the task would slow the Service Manager system down somewhat. Scheduled Maintenance creates all 20,000 records consecutively, as though one user were opening 20,000 incident records consecutively. The increased load may be noticeable, but it will not be crippling.

Load balancing[edit]

You should not notice any change in your system load when you use Scheduled Maintenance. However, if you have an unusually large number of tasks opening every day, such as 1000 each day, or over 100 in a 10-minute period, consider creating a private scheduler that runs only Scheduled Maintenance tasks. When you create and run this scheduler, change the class on the Scheduled Maintenance Hook record to match your new scheduler. You can also change the repeat interval on the inhook schedule record.

Access the Scheduled Maintenance Hook record[edit]

Applies to User Roles: System Administrator

To access the Scheduled Maintenance Hook record:

1. Click **Tailoring** > **Database Manager**.
2. Type schedule.looksee in the **Form** field.
3. Type schedule in the **Table** field.
4. Click the large **Search** icon.
5. Click **Search**.
6. Click the Scheduled Maintenance Hook record to view it in the Schedule File form.

Note: The minimum effective repeat interval is one minute. This interval causes the server to check for tasks every minute. With adjustment, you can reduce this to about 10 seconds. You can't reduce the interval to less than 10 seconds because of the internal structure of the Service Manager scheduling system.

Contents

- 1 Load balancing
- 2 Access the Scheduled Maintenance Hook record

10.31. Running an extra Format Control record

Scheduled Maintenance includes a hook to enable you to run a particular Format Control record before it transfers control to the Service Manager application code. The Format Control record associated with an Incident record, change request, or Request Fulfillment request always runs. However, you can add an additional Format Control record. If the Format Control field on the Details tab of the Scheduled Maintenance Tasks form is blank, only the normal Format Control runs. If you specify a Format Control record on the Details tab, all normal Format Control runs after the named Format Control runs.

Adding a Format Control record enables you to manage Format Control, calculations, validations, or calls to subroutines before the normal Format Control record passes into the regular Service Manager incident, change or request processing.

Note

Note Task expressions can accomplish many of the same calculation tasks.

Specify an extra Format Control record[edit]

Applies to User roles: System Administrator

To specify an extra Format Control record:

1. Click **System Administration** > **Base System Configuration** > **Scheduled Maintenance** > **Scheduled Maintenance Tasks**.
2. Click **Search**.
3. Click the selected task to view it in the Scheduled Maintenance Tasks form.
4. Click the **Details** tab.
5. Click **Fill** on the text box labeled **Call out to this format control record**.
6. Choose a format control from the Related Records list.
7. Click **Save**.

The Add processes on this extra Format Control record will execute after the expressions are parsed, but before the regular incident record, change request, or Request Fulfillment request code.

Contents

- 1 Specify an extra Format Control record

10.31.1. Specify an extra Format Control record

Applies to User roles: System Administrator

To specify an extra Format Control record:

1. Click **System Administration** > **Base System Configuration** > **Scheduled Maintenance** > **Scheduled Maintenance Tasks**.
2. Click **Search**.
3. Click the selected task to view it in the Scheduled Maintenance Tasks form.
4. Click the **Details** tab.
5. Click **Fill** on the text box labeled **Call out to this format control record**.
6. Choose a format control from the Related Records list.
7. Click **Save**.

The Add processes on this extra Format Control record will execute after the expressions are parsed, but before the regular incident record, change request, or Request Fulfillment request code.

Related topics

Running an extra Format Control record

11. Status and Notifications

The Service Manager server includes a number of status and notifications options you can configure after installation. The following table lists the status and notification options you can enable or configure from a new development environment installation.

Feature	Description	Default state in new installations
Alerts	The server can automatically generate messages when a system event meets predefined criteria.	Enabled
Broadcast messages	Administrators can send messages to all logged on users.	Disabled
Message processors	A scheduled process dedicated to managing Service Manager messages.	Disabled
Notifications	The server can automatically generate messages when status change events occur.	Enabled
System bulletins	A utility that allows you to post system information and status messages to your Service Desk operators.	Disabled

Related topics

Alerts
Broadcast messages
Log switching
Message processors
Notifications
System bulletins
System status form

11.1. Alerts

An alert is a system event that occurs when the event meets predefined criteria. They signal a checkpoint, warning, or reminder to keep an activity on schedule. Service Manager calculates alert times by subtracting (or adding) a known interval from a specified time.

Alert calculations depend on the work schedule, time zones, or other defined input. A schedule might cover seven days a week and 24 hours a day. When Service Manager processes alerts, it excludes scheduled holidays and break times in its calculations. Service Manager can make the necessary adjustments to deliver alerts at the correct time, regardless of time zone differences. For example:

- The work shift starts at 8:00 a.m. and ends at 5:00 p.m.
- The lunch break starts at 12:00 p.m. and ends at 1:00 p.m.

If you schedule an alert to occur four and a half hours after the start of the shift, then the alert occurs at 1:30 p.m. because the system excludes the lunch break interval when calculating the alert. If a holiday occurs, Service Manager postpones the alert until the next regularly scheduled day. For example:

- Independence Day holiday starts at 07/04/2004–2005 00:00:00 and ends at 07/05/2004–2005 00:00:00.
- If you schedule an alert at 5 p.m. July 3 to occur six hours later, the alert occurs at 3:00 p.m. on July 5. The alert ignores all nonscheduled hours outside of work shift, the July 4 holiday, and scheduled break times.

You can also set an alert as a TO-DO alert, which enables this alert to appear on your home page.

Associated tables

Service Manager uses information from various tables to process alerts. The tables vary, depending on whether Service Manager processes the alert from the AlertDef or category table:

- AlertDef can use assignment, contacts, device, location, and myGroups
- category uses assignment (for Incident Management alerts)

Create an alert definition record

Applies to User roles: System Administrator, Change Manager, Problem Manager

To create an alert definition record, follow these steps:

1. Change Management: Click **Change Management** > **Configuration** > **Alerts**.
   Incident Management: Click **Incident Management** > **Configuration** > **Alerts Definitions**.
   Problem Management: Click **Problem Management** > **Administration** > **Alert Definitions**.
   Request Fulfillment: Click **Request Fulfillment** > **Configuration** > **Alert Definitions**.
   The **Alert Definition** form opens.
2. Type the alert name in the **Alert Name** field.
3. Type the alert description in the **Description** field.
4. Complete the remaining fields in the record form. If necessary, press **Ctrl+H** in the Windows client or **F1** in the Web client for each field.
   - The **Scheduling** tab describes the calculations that trigger when the alert occurs.
   - The **Update Info** tab describes Format Control considerations and notification messages to be sent.
   - The **Work Schedule** tab specifies the work schedule that the alert belongs to.
     - Holiday Group specifies the holiday group used when calculating an interval. The holiday group defined here overrides the value of holiday group in the work schedule.
   - The **Time Zone** tab enables you to choose one of the following:
     - Define a specific time zone by choosing from the drop-down list
     - Define a RAD expression to set the time zone alert
     - Obtain the time zone from a table lookup
   - The **To-Do Alert** tab enables you to set this alert as a To-Do alert.
5. Click **Add**.

Create a To-Do alert definition record

Applies to User roles: System Administrator, Change Manager, Problem Manager, Incident Manager, Request Manager

To create a To-Do alert definition record, follow these steps:

1. Change Management: Click **Change Management** > **Configuration** > **Alerts**.
   Incident Management: Click **Incident Management** > **Configuration** > **Alerts Definitions**.
   Problem Management: Click **Problem Management** > **Administration** > **Alert Definitions**.
   Request Fulfillment: Click **Request Fulfillment** > **Configuration** > **Alert Definitions**.
   The **Alert Definition** form opens.
2. Type the alert name in the **Alert Name** field.
3. Type the alert description in the **Description** field.
4. Complete the remaining fields in the record form. If necessary, press **Ctrl+H** in the Windows client or **F1** in the Web client for each field.
   - The **Scheduling** tab describes the calculations that trigger when the alert occurs.
   - The **Update Info** tab describes Format Control considerations and notification messages to be sent.
   - The **Work Schedule** tab specifies the work schedule that the alert belongs to.
     - Holiday Group specifies the holiday group used when calculating an interval. The holiday group defined here overrides the value of holiday group in the work schedule.
   - The **Time Zone** tab enables you to choose one of the following:
     - Define a specific time zone by choosing from the drop-down list
     - Define a RAD expression to set the time zone alert
     - Obtain the time zone from a table lookup
   - The **To-Do Alert** tab enables you to choose one of the following:
     - Define this alert as a To-Do alert
     - Assign an alert group

       Note

       Note We provide four out-of-box alert groups and you can add new alert groups in the **To-Do Alert Type** global list. These groups are used to show the types of alerts when you look at your To-Do Alerts window.

       Note

       Note For the SLT Alert group, the response target is supported in the out-of-box system, whereas the service target is unsupported.

     - Specify the alert message template for this alert group
       The message content is displayed in the **To-Do Alerts** window. You can use fixed or dynamic expressions as the message template. For example, the `1 in action in $L.file` expression means that the information of the ticket description is displayed.
5. Click **Add**.

Enable a To-Do alert for an operator

Applies to User roles: System Administrator, Change Manager, Problem Manager, Incident Manager, Request Manager

To enable To-Do alert for an operator, follow these steps:

1. Click **System Administartion** > **Ongoing maintenance** > **Operators**.
2. Open an operator record.
3. From the **To-Do Alert** tab, select the checkbox to enable To-Do alert for this operator.
4. Complete the remaining fields in the record form. If necessary, press **Ctrl+H** to view help for each field.
   - The **Display records from previous days** field specifies the days for the records to be displayed since the alert occurs.

- The **Maximum number of records in the list** field specifies the maximum number of the records to be displayed on the To-Do alert window.
- The **Display records assigned to** field specifies the assigned group for the records displayed on the To-Do alert page. There are two available fields: Me and My Group.
- The **Display records reported by the selected companies** field specifies the company for the records to be reported. You can select and add the available company names from the company list on the left panel.

  Note

  - If no company is specified on the right panel, Service Manager applies all the companies available by default.
  - This field is only available when the multi-company mode has been enabled. To enable the multi-company mode, click **System Administration** > **Base System Configuration** > **Miscellaneous** > **System Information Record**.

- The **Display records from the selected folders** field specifies the folder for the records to be reported. You can select and add the available folder names from the folder list on the left panel.

  Note

  - If no folder is specfied on the right panel, Service Manager applies all the folders available by default.
  - This field is only available when the folder entitlement has been enabled. To enable the folder entitlement, click **System Administration** > **Base System Configuration** > **Miscellaneous** > **System Information Record**.

5. Click **Save**.

Note

Note Out-of-box, To-Do alert is enabled for Incident, Incident Task, Request, Request Task, and Change Approval.

View the To-Do alerts

If the administrator has set To-Do alerts for you, the **To-Do Alerts** floating window appears on the top immediately after you log in. To view all the To-Do alerts in a Service Manager session, click the alerts icon to open the **To-Do Alerts** window.

Note

- The language of To-Do alert list refers to the operator's preferred language (the Language settings in the operator record). However, the login language is also referred when the operator logs in to the system. If the login language and operator's language setting aren't the same value, there might be two different languages displayed in the To-Do alert list.
- The To-Do alerts feature is disabled for the users who log in to Service Manager ESS portal.

Field	Description
Alert category	Displays the alert group for the current To-Do alert. The **To-Do Alerts** window displays four alert groups. • Approval: Lists the tickets that are pending on the approval of the current operator for more than 1 day. • Reminder: Lists the high priority (1 or 2) **In Progress** tickets which aren't updated for 6 hours. • Follow Up: Lists the tickets which are pending for customer actions for more than 1 day. • SLT: Lists the tickets whose SLT will breach within 30 minutes.
Record ID	Click the record ID to open this record in the main window. You can check the record details and perform required actions without closing the **To-Do Alerts** window. **Note** Note When you click the record ID link in a non-Approval alert, the alert that you selected and all other non-Approval alerts linked to the same record are automatically acknowledged and removed from the alert window.
Acknowledge	Click this button to acknowledge this alert, which removes this alert from the alert window. If there are more alerts than the maximum number set by the administrator, a new alert in the system will be added to the bottom of the list. **Note** Note Once the alert is acknowledged, the Ack flag is marked as true in the corresponding record of TodoAlertList table.
Display	Shows the number of alerts displayed in the list.
Total	Shows the number of total alerts in the system.
Close	Click this button to close the **To-Do Alerts** floating window.

Related topics

Clocks
Notifications
Time zones

11.2. Broadcast messages

As a system administrator, you can broadcast messages to the users currently logged on to the server from the **System Status** form. This form allows you to broadcast messages to all users or to pick which users you want to send a message to. Broadcast messages appear in the message bar and message view of the user's client.

Broadcast a message to a particular user

Applies to User Roles: System Administrator

To broadcast a message to a particular user, follow these steps:

1. Click **System Status**. The system status form opens.
2. Type the following command in the Command field for the user process to which you want to send a message:

   ```
   s
   ```

3. Click **Execute Commands**.
   The operator status display form (operator.status.g) opens.
4. Click **Send Msg**.
5. Type the message you want to broadcast.
6. Click **Send Msg**.
   Service Manager sends your message to the selected user.

Broadcast a message to all users

Applies to User Roles: System Administrator

You can broadcast a message to all users from the system status form.

To broadcast a message to all users, follow these steps:

1. Click **System Status**. The system status form opens.
2. Click **Broadcast**. The system.status.broadcast.g form opens.
3. Type the text of your message in the field provided.
4. Click **Send Msg**.
   Service Manager broadcasts your message to all currently logged on users.

Related topics

System bulletins

11.3. Dynamic debugging of user sessions or schedulers

The dynamic debugging feature enables administrators to enable or disable the debugging or tracing information for a user/scheduler session, without the need to restart the server. When enabled, relevant debugging information of that session/scheduler will be written to the server log (sm.log). Once you have finished troubleshooting, you can then disable some debugging information for that user session or scheduler, using dynamic debugging; however some debugging information can't be disabled using dynamic debugging, and in this case the user needs to reconnect to the server or an administrator needs to manually stop and then restart the scheduler, to disable such debugging information.

Caution

Caution If multiple sessions exist for one user account or scheduler (for example, one user logs in simultaneously from different clients or multiple instances of one scheduler have been started), the dynamic debugging settings specified for one of these sessions will take effect for all of the sessions of this user account or scheduler.

Dynamic debugging supported parameters

The following table lists the debugging parameters that can be enabled and disabled or can only be enabled in dynamic debugging.

Parameters can be enabled and disabled in dynamic debugging	Parameters can be enabled but can't be disabled in dynamic debugging
dbmonitorfiles: [file1],[file2]...	dbtriggertrace
debugca:n	dbstats
debughttp	debugattachments
debugdiagnostics	debugjni
debugscauto	debugjavascript
debugvmmap	debugadhocsql
debuglk	debugfileio
debugstartup	debugshutdown
debugrs	debugprocesses
ir_trace	debugdbtypes
rtm	debugdbquery
sqldebug	

Note

Note Once you have completed debugging of a user session or scheduler using any of those parameters that can't be disabled in dynamic debugging, you are recommended to re-connect the session or restart the scheduler to disable the parameters.

Set debugging parameters using dynamic debugging

Applies to User roles: System Administrator

Note

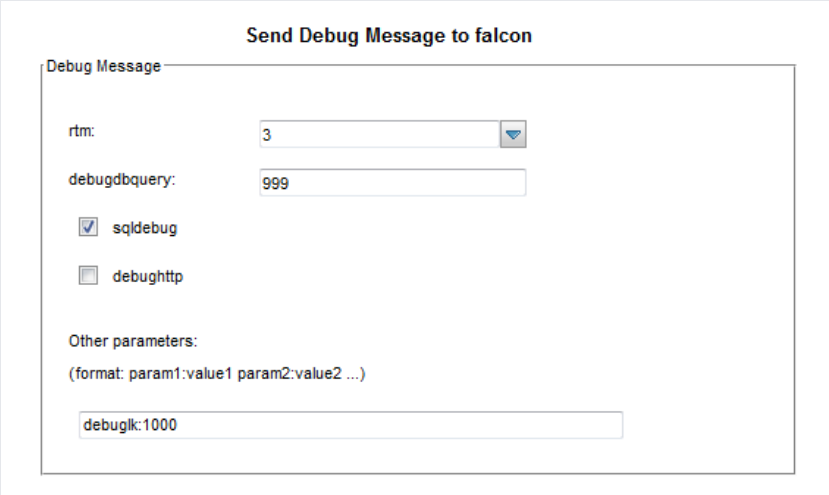
Note Some debugging parameters can be enabled and disabled using the dynamic debugging feature, while some can be enabled but can't be disabled using this feature.

To set debugging parameters for a user session or scheduler, follow these steps:

1. Click **System Status**. The system status form opens.
2. Type the following command in the Command field for the user session or scheduler:

   ```
   s
   ```

3. Click **Execute Commands**.
   The **Operator Status Display** form (operator.status.g) opens.
4. Click **Send Debug Msg**. The **Send Debug Message** form opens.
5. Set debugging parameters as needed. The following figure shows an example.



6. Click **Send Debug Message**. A message is displayed: `Message sent to all users specified`.
7. Click **End** to exit.
8. Check the server log file for relevant debugging information of the session or scheduler.
9. Once you have completed troubleshooting, you are recommended to disable the debugging parameters for the user session or scheduler. To do this, repeat the steps above and disable the parameters.

> **Note**
> Note Some parameters can't be disabled using dynamic debugging. To disable such debugging information, disconnect the user session or manually restart the scheduler.

Alternatively, you can send the dynamic debugging message using Service Manager JavaScript. To do this, follow these steps:

1. Click **Tailoring** > **Script Library**.
2. Add a new debugging JavaScript using the following format:

```
var debugmsg="<parameter> log:<directory and log file name>";
var rc=system.functions.rtecall("setdebug",vars.$L_rc, debugmsg );
```

You can specify one or more space-delimited parameters in `debugmsg`. In addition, you can also customize the output directory and the log file name for the debugging messages. If `log:<directory and log file name>` is omitted, the system outputs the debugging messages to the default server log file (sm.log).
In the following example, the system writes debug messages about RDBMS connections of the current session and prints this information to the mysm.log file:

```
var debugmsg="sqldebug:1 log:../logs/mysm.log";
var rc=system.functions.rtecall("setdebug",vars.$L_rc, debugmsg );
print(rc)
```

In the following example, the system writes debug messages about RDBMS connections and Response Time Monitor (RTM) performance statistics of the current session and prints this information to the sm.log file.

```
var debugmsg="sqldebug:1 rtm:3";
var rc=system.functions.rtecall("setdebug",vars.$L_rc, debugmsg );
print(rc)
```

3. Click **Execute**, and then click **OK** to exit.

Note

Note Your changes affect the current session only. The debugging process starts immediately after you executed the JavaScript.

4. Check the log file for relevant debugging information of the session or scheduler.

11.4. Server log file

The Service Manager server log file provides detailed information and error messages about the server and clients, which can assist support engineers and System Administrators in troubleshooting.

By default, the server log file (sm.log) is located in <Service Manager home>\Server\logs.

You can change the log file path by setting the startup parameter `log`, and you can also enable log switching for the server.

Related topics

Log switching
Stack traces
Trace JavaScript execution
View the Web clients connected to the server

11.4.1. Log switching

A system administrator can enable log switching to have Service Manager write log information to a new file when the current log file reaches a pre-defined size limit

Log switching is particularly useful for systems that run continuously and don't have downtime for system administrators to back up or delete logs. If you don't enable log switching, Service Manager will add all log messages to the one log file defined in the initialization file.

Configure log switching by the maximum log size

Applies to User Roles: System Administrator

To configure log switching by the maximum log size, follow these steps:

1. Stop the Service Manager server.
2. Open the sm.ini file in a text editor.
   This file is in your Service Manager RUN directory.
3. Type the following system parameter:

System parameter	Description
maxlogsize: *n*	For *n*, type the maximum log size in bytes you want Service Manager to use as the log size threshold. The default value is 5MB. Service Manager switches the log when the log size reaches the maximum size defined by this parameter.

Caution

Caution You must have configured the `numberoflogfiles` parameter to enable log switching.

4. Save your changes.
5. Restart Service Manager.
   Service Manager uses your new log file settings.

Enable log switching

Applies to User roles: System Administrator

To enable log switching, follow these steps:

1. Stop the Service Manager server.
2. Open the sm.ini file in a text editor.
   This file is in your Service Manager RUN directory.
3. Type the following system parameter.

System parameter	Description
numberoflogfiles: *n*	For *n*, type the number of log files you want Service Manager to maintain. The minimum value is 0 (disables log switching) and the maximum value is 100. The default value is 0. If the specified number exceeds the maximum value, Service Manager uses a value of 7. Service Manager switches the log when the log size reaches the maximum size defined by the `maxlogsize` parameter. When this happens, the current log file is archived to the log file ending with 1, and the existing log files are renamed to the next higher number, so that the archive log file n+1 always contains older data than archive file n.

4. Save your changes.
5. Restart Service Manager.
   Service Manager uses your new log file settings.

Disable log switching

Applies to User roles: System Administrator

To disable log switching, follow these steps:

1. Stop the Service Manager server.
2. Open the sm.ini file in a text editor.
   This file is in your Service Manager RUN directory.
3. Locate the *numberoflogfiles* parameter, and set its value to "0".
4. Save your changes.
5. Restart Service Manager.
   Service Manager uses your new log file settings.

Related topics

Server log file

11.4.2. Stack traces

Service Manager automatically copies any stack trace produced by a general protection fault (GPF) to the Service Manager log file (by default, sm.log). You can use this information to troubleshoot your server and to pass information to Customer Support.

Related topics

Log switching
Startup parameter: log

11.4.3. View the Web clients connected to the server

When users log on to Service Manager through Web clients, Service Manager automatically detects the web browsers and web application servers accessing the server, and logs web browser name and version, web application server (or web server) name and version, as well as Web Services client agent information (if available) to the server log file (by default,

sm.log). Support personnel can view information about the connected web clients from the server log without the need to consult users.

See the following for some examples (where `xxx.xx.xx.xx` represents IP addresses).

```
RTE I SOAP client information scguiwweb 9.71.1006 (P1) at xxx.xx.xx.xx Browser Firefox 106.0 AppServer
Apache Tomcat 9.0.65 ESSToolbarButton Disabled(Client Enabled, Server Disabled)
RTE I SOAP client information scguiwweb 9.71.1006 (P1) at xxx.xx.xx.xx Browser Chrome 106.0.0.0 AppServer
Apache Tomcat 9.0.65 ESSToolbarButton Disabled(Client Enabled, Server Disabled)
RTE I SOAP client information scguiwweb 9.71.1006 (P1) at xxx.xx.xx.xx Browser Edge 106.0.1370.47 AppServer
Apache Tomcat 9.0.65 ESSToolbarButton Disabled(Client Enabled, Server Disabled)
```

Related topics

Server log file
Log switching
Startup parameter: log

11.4.4. View diagnostic counter information

As a System Administrator or support engineer, you can view diagnostic counter information in the console or server log (default:sm.log) to assist in investigating system stability or performance issues.

Service Manager provides the following diagnostic counter information:

- Web services transaction count / size (max size of a web service transaction / average size)
- Session login/logout count
- Database client transmission volume / size (network traffic)
- Heartbeat interval - request response count (only servlet container process information is printed)
- event-in / event-out count
- email-in / email-out count

> **Note**
>
> Note All the above diagnostic counter information is retrieved for the time since system startup.

Retrieval of diagnostic counter information

Service Manager retrieves diagnostic counter information as follows:

- When you run the **sm -reportdiagnostics** command, Service Manager prints diagnostic counter information to the console and also writes the information to the server log.

  > **Note**
  >
  > Note By default, the Diagnostic Service is disabled. Before running the **sm -reportdiagnostics** command, administrators must specify `debugdiagnostics:1` to enable the Diagnostic Service.

- When the servlet node or SCAuto listener or emailout process terminates, Service Manager writes diagnostic counter information to the server log.

Sample diagnostic counter information

See the following scripts for sample information.

```
Web Service Diagnostics Summary(Bytes):
Pid:4010 Command:/home/fpeSM/sm930/RUN/smserver  Name:ThreadControllerId-16080
Count:1 Request Total(MIME/MTOM Attachment):721(0/0) Request Avg(MIME/MTOM Attachment):721(0/0) Response
Total(MIME/MTOM Attachment):238(0/0) Response Avg(MIME/MTOM Attachment):238(0/0)
Pid:4095 Command:/home/fpeSM/sm930/RUN/smserver system.start  Name:ThreadControllerId-background
Count:0 Request Total(MIME/MTOM Attachment):0(0/0) Request Avg(MIME/MTOM Attachment):0(0/0) Response
Total(MIME/MTOM Attachment):0(0/0) Response Avg(MIME/MTOM Attachment):0(0/0)
Pid:14273 Command:/home/fpeSM/sm930/RUN/smserver -reportdiagnostics  Name:
Count:0 Request Total(MIME/MTOM Attachment):0(0/0) Request Avg(MIME/MTOM Attachment):0(0/0) Response
Total(MIME/MTOM Attachment):0(0/0) Response Avg(MIME/MTOM Attachment):0(0/0)
Sessions Diagnostics Detail:
Pid:4010 Command:/home/fpeSM/sm930/RUN/smserver  Name:ThreadControllerId-16080
Session Login: 22, Logout: 18
Pid:4095 Command:/home/fpeSM/sm930/RUN/smserver system.start  Name:ThreadControllerId-background
Session Login: 0, Logout: 0
Pid:14273 Command:/home/fpeSM/sm930/RUN/smserver -reportdiagnostics  Name:
Session Login: 0, Logout: 0
Database Statistics:
Pid:4010 Command:/home/fpeSM/sm930/RUN/smserver  Name:ThreadControllerId-16080
FileName               Selects  Inserts  Updates  Deletes  Counts   Sorts    Finds    Fetchs   Inits
Cache Inits  Terms    Cache Terms  Cache Finds
dbdict                     2        0        0        0        12       0        58       1084     53
2            44       48           6
triggers                   111      0        0        0        0        0        0        334      44
0            38       38           0
datadict                   19       0        0        0        0        0        566      0        88
783          74       865          4732
scldapconfig               44       0        0        0        0        0        0        0        88
0            76       88           0
code                       0        0        0        0        0        0        57       0        54
0            46       48           0
format                     3        0        0        0        0        0        344      61       75
118          64       185          1146
link                       0        0        0        0        0        0        110      0        96
167          78       253          207
ScriptLibrary              0        0        0        0        0        0        14       0        48
0            41       42           520
info                       28       0        0        0        0        0        3        15       67
119          56       181          116
tzfile                     0        0        0        0        0        0        2        0        67
88           56       151          132
joindefs                   2        0        0        0        2        0        1        68       4
1            0        3            0
erddef                     4        0        0        0        0        0        0        121      5
0            3        5            0
sctypecheck                1        0        0        0        0        0        0        7        2
0            0        2            0
scmandant                  1        0        0        0        0        0        0        0        1
0            0        1            0
scaccess                   1        0        0        0        0        0        0        134      2
0            0        2            0
licenseinfo                1        0        56       0        1        0        79       0        47
0            36       42           0
userinfo                   1        22       19       18       0        0        59       0        104
0            95       40           0
SYSPUBLISH                 0        0        0        0        0        0        0        0        44
0            38       38           0
scversion                  2        0        0        0        2        0        0        3        4
0            0        3            0
```

```
scmessage                0       0       0       0       0       0       3451    0       35
0           30      30          29997
operator                 0       0       22      0       0       0       100     0       88
107         72      183         0
scsecuritygroup          0       0       0       0       0       0       22      0       22
0           18      22          0
cmcontrol                22      0       0       0       0       0       0       0       44
0           36      40          0
currency                 0       0       0       0       0       0       1       0       23
0           19      23          22
environment              0       0       0       0       0       0       11      0       264
45          216     261         308
smenv                    0       0       0       0       0       0       2       0       24
0           20      20          22
tableAccess              0       0       0       0       0       0       0       0       352
22          288     310         0
pmenv                    0       0       0       0       0       0       2       0       24
0           20      20          22
category                 0       0       0       0       0       0       0       0       22
0           18      18          0
globallists              22      0       0       0       0       0       72      2860    66
50          54      116         0
locallist                0       0       0       0       0       0       44      0       66
0           54      58          0
TodoMap                  22      0       0       0       0       0       0       242     44
0           36      44          0
Object                   19      0       0       0       0       0       34      437     80
114         70      192         256
dataaccess               0       0       0       0       0       0       108     0       22
224         18      246         492
cm3profile               44      0       0       0       0       0       0       0       44
22          36      58          0
stathistory              171     0       16      0       0       0       0       0       41
215         35      256         0
language                 34      0       0       0       0       0       0       80      68
0           60      64          0
kmprofile                22      0       0       0       0       0       0       0       66
0           54      58          0
sapconfig                22      0       0       0       0       0       0       0       44
0           36      40          0
ApprovalDelegation       88      0       0       0       0       0       0       0       88
0           72      72          0
slacontrol               22      0       0       0       0       0       0       0       44
0           36      40          0
slaprofile               0       0       0       0       0       0       22      0       44
0           36      40          0
rcenv                    0       0       0       0       0       0       2       0       24
0           20      20          22
ocmprofile               66      0       0       0       0       0       0       0       66
22          54      76          0
ctenv                    0       0       0       0       0       0       2       0       24
0           20      20          22
icmenv                   0       0       0       0       0       0       2       0       24
0           20      20          22
syslog                   0       22      0       0       0       0       0       0       44
1           36      45          0
counters                 6       0       7       0       0       0       7       3       9
0           6       7           0
mail                     22      0       0       0       22      0       0       0       29
```

```
5           25      34          0
formatctrl              47      0       0       0       0       0       12      0       46
79          38      125         98
contacts                0       0       0       0       0       0       24      0       50
2           42      48          0
dept                    22      0       0       0       0       0       0       0       22
0           18      22          0
menu                    3       0       0       0       0       0       15      0       37
111         32      146         139
inbox                   466     0       0       0       399     0       0       205     57
49          51      104         0
inboxgrid               0       0       0       0       0       0       6       0       21
0           18      19          38
docenv                  0       0       0       0       0       0       0       0       38
5           34      41          0
ViewDefault             0       0       0       0       0       0       19      0       19
0           17      17          0
Todo                    19      0       0       0       0       0       0       26      38
38          34      74          0
displaymaster           0       0       0       0       0       0       2       0       21
0           18      19          19
displayscreen           3       0       0       0       0       0       0       0       25
9           22      32          0
displayoption           3       0       0       0       0       0       0       0       21
13          19      34          0
displaycache            0       0       0       0       0       0       6       0       25
8           20      31          29
displayevent            2       0       0       0       0       0       0       2       22
8           18      28          0
systext                 23      0       0       0       0       0       0       69      28
9           24      37          0
distgroup               0       0       0       0       0       0       0       0       3
0           3       3           0
eventregister           4       0       0       0       6       0       0       189     6
1           4       5           0
eventout                3       2       0       1       0       0       0       2       9
10          4       19          0
number                  0       0       2       0       0       0       2       0       4
1           0       3           0
eventmap                173     0       0       0       376     0       0       5675    175
1           4       5           0
typecheck               0       0       0       0       0       0       0       0       2
2           2       4           0
config                  0       0       0       0       0       0       1       0       2
0           2       2           0
uimpages                2       6       0       0       0       0       6       13      7
0           7       7           0
schedule                0       7       0       0       0       0       0       0       3
0           3       3           0
uimuserpreferences      3       1       2       0       0       0       5       6       5
6           5       11          0
uimcompdefinitions      3       0       0       0       0       0       0       16      4
1           4       5           0
uimpagecategories       1       0       0       0       0       0       0       0       1
0           1       1           0
uimpagecatspages        8       0       0       0       0       0       0       0       1
7           1       8           0
uimuserpagecontent      6       0       0       0       0       0       0       0       2
4           2       6           0
```

```
extaccess                6        0        0        0        0        0        0        0        4
4            4        8           0
States                   0        0        0        0        0        0        13       0        11
13           7        24          21
help                     3        0        0        0        0        0        382      525      4
383          4        387         0
probsummary              2        0        0        0        0        0        0        0        14
13           14       15          0
sqlsystemtables          0        0        0        0        0        0        2        0        2
0            2        2           0
status                   0        0        0        0        0        0        0        0        2
0            2        2           0
incidents                1        0        0        0        0        0        0        0        5
5            5        6           0
SearchConfig             1        0        0        0        0        0        1        28       2
1            0        3           0
AdvFilter                0        0        0        0        0        0        0        0        1
0            0        1           0
eventin                  0        1        0        0        0        0        1        0        3
1            0        2           0
Pid:4095 Command:/home/fpeSM/sm930/RUN/smserver system.start  Name:ThreadControllerId-background
FileName                 Selects  Inserts  Updates  Deletes  Counts   Sorts    Finds    Fetchs   Inits
Cache Inits  Terms    Cache Terms  Cache Finds
dbdict                   0        0        0        0        0        0        53       0        19
0            2        2           0
triggers                 55       0        0        0        0        0        0        187      19
0            2        2           0
datadict                 0        0        0        0        0        0        61       0        1150
222          950      1186        1734
scldapconfig             19       0        0        0        0        0        0        0        38
0            6        38          0
code                     0        0        0        0        0        0        71       0        28
0            4        11          0
format                   0        0        0        0        0        0        27       0        30
0            5        13          135
link                     0        0        0        0        0        0        1        0        35
0            3        4           15
ScriptLibrary            0        0        0        0        0        0        6        0        22
0            3        5           192
info                     57       0        1        0        0        0        1        0        60
76           8        100         20
tzfile                   0        0        0        0        0        0        0        0        30
96           5        118         82
SYSPUBLISH               0        0        0        0        0        0        0        0        19
0            2        2           0
scmessage                0        0        0        0        0        0        112      0        16
0            2        2           4941
schedule                 5889     33       2649     40       0        0        0        0        33
16           5        21          0
counters                 5        0        26       0        0        0        26       3        7
0            2        4           0
anubissystemstate        2        0        2        0        0        0        0        0        2
1            2        3           0
syslog                   0        17       0        0        0        0        0        0        2
16           2        18          0
marquee                  31       0        0        0        0        0        0        0        2
29           1        31          0
shutdown                 1        0        0        0        0        0        0        0        2
0            2        2           0
```

```
cmcontrol                15      0       0       0       0       0       0       0       30
0            3       16          0
currency                 0       0       0       0       0       0       1       0       16
0            2       16          15
environment              0       0       0       0       0       0       11      0       145
32           13      45          175
smenv                    14      0       0       0       0       0       1       0       29
0            3       15          14
tableAccess              0       0       0       0       0       0       0       0       197
10           16      32          0
operator                 0       0       0       0       0       0       229     0       54
165          6       189         0
pmenv                    0       0       0       0       0       0       1       0       16
0            1       2           15
category                 298     0       0       0       0       0       0       1788    18
296          3       300         0
globallists              1505    0       1192    0       0       0       15      41882   51
2080         7       2129        0
locallist                0       0       0       0       0       0       30      0       45
0            6       19          0
TodoMap                  15      0       0       0       0       0       0       165     30
0            4       30          0
Object                   0       0       0       0       0       0       8       0       18
111          2       129         160
dataaccess               156     0       0       0       0       0       110     0       15
145          2       160         164
cm3profile               30      0       0       0       0       0       30      0       30
15           2       17          0
stathistory              0       10      0       0       0       0       0       0       16
40           2       56          0
sapconfig                15      0       0       0       0       0       0       0       30
0            3       16          0
ApprovalDelegation       60      0       0       0       0       0       0       0       60
0            8       8           0
slacontrol               15      0       0       0       0       0       0       0       30
0            3       16          0
slaprofile               0       0       0       0       0       0       15      0       30
0            3       16          0
formatctrl               0       0       0       0       0       0       16      0       16
0            1       2           30
contacts                 0       0       0       0       0       0       15      0       15
0            1       1           0
dept                     15      0       0       0       0       0       0       705     30
0            4       30          0
rcenv                    0       0       0       0       0       0       2       0       11
0            1       3           9
ocmprofile               0       0       0       0       0       0       54      0       27
9            3       12          0
ctenv                    9       0       0       0       0       0       2       0       18
0            3       10          9
icmenv                   9       0       0       0       0       0       2       0       18
0            3       10          9
patcotask                298     0       0       0       0       0       0       0       1
297          0       298         0
msgclass                 40      0       0       0       0       0       0       40      2
39           0       41          0
msgtype                  0       0       0       0       0       0       40      0       2
39           0       41          0
msglog                   0       25      0       0       0       0       0       0       2
```

```
24          0       26          0
erddef                  1       0       0       0       0       0       0       1
0           0       1           0
applicationfields       298     0       0       0       0       0       11026   2
297         1       299         0
eventin                 298     0       1       0       0       0       1       1       2
298         0       300         0
contract                60      0       0       0       0       0       0       0       1
59          0       60          0
expline                 30      0       0       0       0       0       0       0       1
29          0       30          0
agent                   269     0       0       0       0       0       0       0       1
268         0       269         0
outage                  10      0       0       0       0       0       0       0       1
9           0       10          0
sloavail                10      0       0       0       0       0       0       0       1
9           0       10          0
slo                     10      0       0       0       0       0       0       790     2
9           0       11          0
slaresponse             840     0       790     0       0       0       0       790     3
18          0       20          0
clocks                  0       0       0       0       0       0       0       0       1
9           0       10          0
sloresponse             790     0       0       0       0       0       0       0       2
1578        0       1580        0
sla                     10      0       0       0       0       0       0       50      2
9           0       11          0
slamonthly              50      0       0       0       0       0       0       0       1
9           0       10          0
slamonthlyag            0       0       50      0       0       0       50      0       2
9           0       11          0
ocmq                    0       0       0       0       0       0       0       0       1
0           0       1           0
ocml                    1       0       0       0       0       0       0       0       1
0           0       1           0
eventregister           1       0       0       0       0       0       0       1       2
0           0       2           0
mail                    0       0       0       0       0       0       0       0       3
0           0       2           0
eventmap                1       0       0       0       0       0       0       5       2
0           0       2           0
SYSATTACHMENTS          1       0       0       0       0       0       0       0       1
0           0       1           0
Pid:14273 Command:/home/fpeSM/sm930/RUN/smserver -reportdiagnostics  Name:
FileName              Selects  Inserts  Updates  Deletes  Counts   Sorts    Finds    Fetchs   Inits
Cache Inits  Terms    Cache Terms  Cache Finds
dbdict                  0       0       0       0       0       0       0       0       1
0           0       0           0
triggers                0       0       0       0       0       0       0       0       1
0           0       0           0
datadict                0       0       0       0       0       0       0       0       1
0           0       0           7
scldapconfig            1       0       0       0       0       0       0       0       2
0           0       2           0
code                    0       0       0       0       0       0       0       0       1
0           0       0           0
format                  0       0       0       0       0       0       0       0       1
0           0       0           0
link                    0       0       0       0       0       0       0       0       1
```

```
0           0       0           0
ScriptLibrary           0       0       0       0       0       0       0       0       1
0           0       0           0
info                    0       0       0       0       0       0       0       0       1
1           0       2           2
tzfile                  0       0       0       0       0       0       0       0       1
1           0       2           2
SYSPUBLISH              0       0       0       0       0       0       0       0       1
0           0       0           0
Apps Statistics:
Pid:4010 Command:/home/fpeSM/sm930/RUN/smserver  Name:ThreadControllerId-16080
Module                  Name                    Count
Pid:4095 Command:/home/fpeSM/sm930/RUN/smserver system.start  Name:ThreadControllerId-background
Module                  Name                    Count
Event Services          emailin                 1
Event Services          eventin                 1
Pid:14273 Command:/home/fpeSM/sm930/RUN/smserver -reportdiagnostics  Name:
Module                  Name                    Count
Heartbeat Diagnostics Detail:
Pid:4010 Command:/home/fpeSM/sm930/RUN/smserver  Name:ThreadControllerId-16080
Heart Beat Count: 1028     Heart Beat Interval: 15
```

Related topics

Server log file
Log switching
Startup parameter: log
Special parameters: reportdiagnostics
Parameter: debugdiagnostics

11.4.5. Trace JavaScript execution

If you are a Service Manager JavaScript developer and want to trace JavaScript execution invoked from within RAD, specify `rtm:3` in the sm.ini configuration file. This enables Service Manager to log tracing information in the Service Manager server log file (default: <Service Manager home>\Server\logs\sm.log).

The following is an example of the log information:

```
RTE D SCRIPTTRACE: localizeTable.getLocalizedValues entered, line 210
RTE D Parameter type:STRING value:categories
RTE D Parameter type:ARRAY value:{"complaint", "incident", "problem",  "request for change", "request for
information"}
RTE D SCRIPTTRACE: localizeTable.getLocalizedValues exited, line 219  elapsed: 110 ms
```

This example indicates that the getLocalizedValues function in script `localizeTable` has been invoked, starting from line 210; the execution completes at line 219, which might be the last line of this function; The execution consumes 110 ms in total. In addition, the parameters passed to this function are also dumped into the log file: the first one is STRING type, with a value of "categories"; the other one is an ARRAY, with its parameters enclosed in braces by their order in the array.

Related topics

Server log file
Enter a parameter in the sm.ini file

11.4.6. View the system log file

Applies to User Roles: System Administrator

The System log (Syslog) file contains information about system events, such as failed logon attempts.

To view or edit system events in the Syslog file, follow these steps:

1. Click **Tailoring** > **Database Manager**. Or, type db in the command line, and then press **Enter** or click **Execute Command**.
2. Type syslog in the **Form** field, and then press **Enter**.
3. Type any search criteria in the appropriate fields, and then click **Search**.
   A list of events in the Syslog file is displayed.
4. Select a system event in the list to display detailed information about the event. To edit the event, make the desired changes to the detailed information, and then click **Save**.

The Syslog file records the following types of event:

- Successful logon
  The following event is an example of a successful logon.

Event Type	Start Time	Stop Time	User Name	Terminal Name	Current Status	IP Address	PID
user	7/15/2013 1:27:56		falcon	SOAP-Windows 7	logged on	16.158.154.213	4

- Failed logon
  The following event is an example of a failed logon.

Event Type	Start Time	Stop Time	User Name	Terminal Name	Current Status	IP Address	PID
user	7/10/2013 20:12		ddd	SOAP-Windows 7	login failed	16.158.154.213	4

Note

Note Failed logon events are recorded both when a user enters an incorrect password and when a user enters an incorrect user name.

- Logout
  The following event is an example of a logout.

Event Type	Start Time	Stop Time	User Name	Terminal Name	Current Status	IP Address	PID
user	7/7/2013 20:36	7/7/2013 22:33	falcon	SOAP-Windows 7	logged off	16.158.154.213	4

- Scheduler starts
  The following event is an example of a scheduler starting.

Event Type	Start Time	Stop Time	User Name	Terminal Name	Current Status	IP Address	PID
scheduler	7/14/2013 0:25		availability	availability	started		3

Note

Note The IP address column is populated only if the event is triggered from a machine that's not the Service Manager server.

- Scheduler terminates
  The following event is an example of a scheduler terminating.

Event Type	Start Time	Stop Time	User Name	Terminal Name	Current Status	IP Address	PID
scheduler	7/14/2013 0:25	7/14/2013 0:25	KMUpdate	SYSTEM	stopped		18

> **Note**
> Note The IP address column is populated only if the event is triggered from a machine that's not the Service Manager server.

11.4.7. How to interpret http.log

The http.log may contain encoded or compressed messages from the Windows or Web Client communication. See below to turn off SOAP compression and FastInfoset encoding. Web Services communications aren't encoded nor compressed.

The http.log contains information on all Processes/Threads that connect to the traced servlets. If more than one Process/Thread is traced, information in the log may overlap from different sessions, resulting in multiple POST or GET messages together rather than the POST – GET pair you would expect.

To turn off FastInfoset and compression for the clients, follow these steps:

1. On the shortcut starting the Windows client, add –DFastInfoset=false to the target.
2. The command line argument for the Service Manager Windows client is:
   -vmargs -option1 -option2
3. As an example, if -DFastInfoset=false is the only command line option ensure your command line is:
   -vmargs -DFastInfoset=false
4. Turn off Compress SOAP Messages on the File - Connect – Connections screen's Advanced tab.
5. For the Web client, turn off FastInfoset by setting the JAVA Option -DFastInfoset=false. As an example, to turn off FastInfoset on Tomcat 5.5 or later, follow these steps:
   1. Go to the Apache Tomcat x.x folder on the Start Menu.
   2. Start the Configure Tomcat application.
   3. Select the Java tab in the configuration dialog.
6. Turn off the Compress SOAP Messages by editing the ...\WEB-INF\web.xml file.

```
<!-- Compress network communication between the application server
and the HP Service Manager server -->
<init-param>
<param-name>compress_soap</param-name>
<param-value>false</param-value>
</init-param>
```

11.5. Client log file

Each Service Manager client has a log file, which stores client session information for troubleshooting.

For each client, the log file is defined as follows:

- Windows client
  Go to **Window** > **Preferences** > **Service Manager** > **Logs**. The **Log file** field specifies the log file path.
- Web client
  By default, the log file is located in the Web application server's home directory: <Web application server home directory>/sm.log.

For example: C:\Program Files\Apache Software Foundation\<Tomcat version>\sm.log.

View SOAP fault information in the client log

When a SOAP fault or an exception occurs on the Service Manager server, the SOAP fault information is written to the client log file. This information can assist support engineers and System Administrators in troubleshooting.

Service Manager logs SOAP fault information in the Windows or Web client log file (sm.log) as follows:

- When a SOAP FAULT occurs, Service Manager logs the SOAP request, SOAP response, and SOAP fault information.
- When a server session ends unexpectedly and an exception is thrown, Service Manager logs the SOAP request, SOAP response (if any), and the exception information.

Example:

If a user is performing operations (such as generating charts) on the client side and then a System Administrator kills the user's process from System Status, SOAP fault information like the following is logged in the client log file (sm.log):

```
ERROR Timer-0 com.hp.ov.sm.client.webtier.SCLogging - Jan 10, 2022 17:58:32 GMT+08:00 [ERROR] SOAP
Failure - Session no longer valid
ERROR Timer-0 com.hp.ov.sm.client.webtier.SCLogging - Jan 10, 2022 17:58:32 GMT+08:00 [ERROR] request:
<getMessages />
ERROR Timer-0 com.hp.ov.sm.client.webtier.SCLogging - Jan 10, 2022 17:58:32 GMT+08:00 [ERROR] response:
<Fault>
<faultcode>SOAP-ENV:Server</faultcode>
<faultstring>Session no longer valid</faultstring>
<faultactor>Server</faultactor>
</Fault>
ERROR Timer-0 com.hp.ov.sm.client.webtier.SCLogging - Jan 10, 2022 17:58:32 GMT+08:00 [ERROR] The soap
fault is : SOAP-ENV:Server
Session no longer valid
```

Related topics

Server log file

11.6. Message processors

Message processors are scheduled processes dedicated to managing Service Manager messages. By default, the `problem` background process manages all Service Manager messages. If your system generates enough messages, it may create a backlog in your message queue. You can remove the message backlog by adding additional message processors.

Each message processor must have an agent initialization record definition that defines the name of the processor and the RAD application to call. The message processor name must match the name listed in the System Wide Company Record for the message processor. All message processors must call the `scheduler` RAD application.

To determine if the message queue is affecting your system performance, you can review the schedule file. If you see schedule records with the name **message processor record** with an expiration time greater than a minute old, then you should consider adding additional message processors.

Define which system processes manage message traffic

Applies to User Roles: System Administrator

To define which system processes manage message traffic, follow these steps:

1. Click **System Administration** > **Base System Configuration** > **Miscellaneous** > **System Information Record**.
2. Click the **Message Processors** tab.
3. In the **Processor Name** field, type the names of the background or scheduler processes that are to manage message traffic. The processor name must match the class name specified for the message class processor.
4. Click **Save**.
5. Start the message processors.

Service Manager rotates messages among all the processes you define. If you don't define any message processors, then Service Manager uses the problem processor to manage all message traffic.

Add additional message processors

Applies to User Roles: System Administrator

By default, Service Manager uses the `problem` background process to manage all message traffic. If your message queue grows too large, you can add additional message processors to handle the message load.

To add additional message processors, follow these steps:

1. Create a message processor agent record.
2. Add the message processors to the System Wide Company record.
3. Start the message processor processes.

> **Tip**
> Tip
> You can also add your message processors to the `startup` agent record to have Service Manager automatically start them.

Create a message processor agent record

Applies to User Roles: System Administrator

You can create message processor agent records to divide the message load between multiple processes.

To create a message processor agent record, follow these steps:

1. Click **System Administration** > **Base System Configuration** > **Miscellaneous** > **Agent Registry**. The **Background Processor Initialization Registry** form opens.
2. Click **Search**.
3. Type or select the information record information.
4. Click **Add**.
   Service Manager displays the message: Information record added.
5. Repeat steps 2-3 for each message processor you wish to add.

View the messages

Service Manager notifies you when you have activity messages to review. The messages can appear at the top of the detail window.

Information type	Meaning
Information	Information about the most recent action
Warning	A warning about the most recent action
Error	The most recent action caused an error to occur

> **Note**
> Note
> By default, these three types of messages all appear on the top of the detail window. You can change the default settings by modifying the related parameters.

To view all the messages in a Service Manager session, click the message icon to open the **Messages** window.

Related topics

Scheduled processes
Access the schedule file

11.7. Notifications

You can configure Service Manager to send notifications whenever specific system events occur. By default, Service Manager supports notifications for every type of status change event in all applications. You can also define custom notifications that Service Manager sends when specific events occur.

You can configure the following properties for notifications:

- Conditions under which Service Manager sends the notification
- Content of the notification
  - Record ID of the triggering event
  - Current phase of the triggering event
  - Operator assigned to the triggering event
  - E-mail formatting
- Delivery method of the notification
  - Active Note
  - E-mail
- Recipient list
  - Contact name
  - Group
  - Operator
  - Operator assigned to
  - Operator coordinator
  - Operator requested by

Related topics

Alerts
Message processors
Add a message record
Add a notification delivery method
Create a custom notification
Create a distribution group
Create an on-call schedule
Send a notification using Format Control
Default message classes
Default notifications

11.7.1. Default message classes

By default, Service Manager provides the following message classes.

- agent
- alert
- am
- approval
- cau
- cib
- cm
- cm3
- contract

- copyds
- ct
- da
- dbdict
- dcu
- dde
- de
- debug
- dewiz
- diffupg
- displayscreen
- doceng
- ds
- dt
- erd
- erp
- error
- es
- fc
- fin
- fm
- ga
- help
- icm
- ins
- ir
- itil
- kpak
- macro
- mail
- misc
- news
- ns
- object
- ocm
- patcort
- pm
- print
- ps
- queue
- rad
- rc
- rca
- rsm
- rw
- sap
- sc
- sc.email

- sc24x7
- scballon
- scbase
- scbutton
- sccaption
- scev
- scgui
- scir
- scjs
- scjsdesc
- scmenu
- scmenugroup
- scpolicy
- scrad
- scscd
- scsql
- sctitle
- scutility
- scxmlapi
- sla
- sm
- SmartEmail
- sql
- swcompliance
- trigger
- upg
- us
- validity
- wizard
- wm
- work

Related topics

Notifications
Add a message record
Add a notification delivery method
Create a custom notification
Create a distribution group
Create an on-call schedule
Send a notification using Format Control
Default notifications

11.7.2. Default notifications

By default, Service Manager includes the following notifications.

Notification record	Notification event
Activity Added	Adding an activity update with the "Visible to Customer" option selected
ChM Change Close	Closing a change phase in Change Management
ChM Change Open	Opening a change phase in Change Management
ChM Change Update	Updating a change phase in Change Management or opening a phase that has been deferred
ChM Task Close	Closing a task phase in Change Management
ChM Task Open	Opening a task phase in Change Management
ChM Task Update	Updating a task phase in Change Management or opening a deferred phase
ChM Retract All	Reversing approval for all levels of a change phase in Change Management
ChM Retract One	Reversing approval for one level of a change phase in Change Management
Clone Relation	Copying associations of a cloned incident or service desk interaction record
IM Action Alert	Updating the alert status of an incident in Incident Management
IM Alert Reassign	Exceeding the reassignment limit for an incident in Incident Management
IM Close	Closing an incident in Incident Management
IM Edit Close Cascade	Closing an incident and its related service desk interaction records in Incident Management
IM Edit Close Linked	Closing an incident but not its related service desk interaction records in Incident Management
IM Open	Opening an incident in Incident Management
IM Reopen	Reopening an incident in Incident Management
IM Resolved	Resolving an incident in Incident Management
IM Save Relation	Opening a related incident for a service desk interaction and linking the record to an existing incident
IM Updated	Updating an incident in Incident Management
RM Approval	Approving a request or order in Request Management
RM Denial	Denying request or order in Request Management
RM Final Approval	sent when the request is completely approved in Request Management

RM Final Denial	Denying a request in Request Management
RM Line Item Change Category	Manually changing the category of a line item in Request Management
RM Line Item Close	Manually closing a line item in Request Management
RM Line Item Drop Avail	Marking a line item as unavailable in Request Management
RM Line Item Mark Avail	Marking a line item as available to order in Request Management
RM Line Item Open	Opening a new line item in Request Management
RM Line Item Reopen	Reopening a line item in Request Management
RM Line Item Update	Updating a line item in Request Management
RM Order Change Category	Manually changing the category of the order in Request Management
RM Order Close	Closing an order in Request Management
RM Order Open	Opening an order in Request Management
RM Order Update	Updating an order in Request Management
RM Request Change Category	Manually changing the request category in Request Management
RM Request Close	Closing an request in Request Management
RM Request Deferred	Deferring a request in Request Management
RM Request Open	Opening a request in Request Management
RM Request Open Next Phase	Moving a quote to the next phase in Request Management
RM Request Phase Change	Manually changing the phase of the request in Request Management
RM Request Reopen	Reopening a request in Request Management
RM Request Update	Updating a request in Request Management
Request Late Notice	Expired due date in Request Management
Request Not Ordered	Too much elapsed time between the date that the first line item is ordered and moving to the ordering phase
SM Add	Opening a service desk interaction record in an active state in Service Desk
SM Close	Closing a service desk interaction record in Service Desk
SM Save	Opening a service desk interaction record in an inactive state in Service Desk
SM Update	Updating a service desk interaction record in Service Desk
Save Association	Linking two associated records

Save Relation	Opening a related record

Related topics

Notifications
Add a message record
Add a notification delivery method
Create a custom notification
Create a distribution group
Create an on-call schedule
Send a notification using Format Control
Default message classes

11.7.3. Add a message record

Applies to User Roles: System Administrator

To add a message record, follow these steps:

1. Click **Tailoring** > **Notifications** > **Messages**. The **Search Message Records** form opens.
2. Type or select the following information.

Field	Description
Language Code	You can leave this field blank to have Service Manager use the language code defined in the company record. To display the message in another language, type the ISO code of that language.
Class	Type a name or label to describe the Service Manager applications to which this notification applies. For example, you can make a message relating to Change Management part of the cm3 class.
Message Number	Type a unique ID to identify this message. For more information about how to set this ID, see RAD function: scmsg.
Severity	This field is reserved for future use and can be ignored.
Text	Type the text of the message to display. You can use the arguments %S and %S[*n*] to include information from the triggering event.
Comments	This is an optional field you can use to type a description of the message.

3. Click **Add**.

Related topics

Notifications
Add a notification delivery method
Create a custom notification
Create a distribution group
Create an on-call schedule
Send a notification using Format Control
Default message classes
Default notifications

11.7.4. Add a notification definition record

Applies to User roles: System Administrator

The Notification Engine is primarily responsible for sending messages that are generated by Service Manager events, such as opening or closing a quote or order. System Administrators can edit these messages, add new messages, change the conditions under which the messages will be sent out, and select who will receive the messages.

To add a notification definition record, click **Tailoring** > **Notifications** > **Notifications** to open the **Notification Definition** form. Type or select the following information and then Click **Add**.

Field	Description
Name	Type the name of the notification record.
Condition	Type true to enable the notification in all conditions. Type an expression that's true or false to specify a condition when Service Manager should send the notification. Type false to disable the notification in all conditions. By default, Service Manager treats a blank entry as true.
Message tab	Use this tab to define the message type, delivery method, and conditions under which Service Manager should send the notification.
Msg Class	Type the message class for the notification. This value must match a message class definition in the message record.
Msg No	Type the message number for the notification. This value must match a message number definition in the message record.
Arguments	Type any RAD expressions to display information from the record or event that triggered the notification. Service Manager displays this information in the text of the notification. You can enter an array of arguments by enclosing all arguments in brackets and separating each argument with a comma. For example: {argument1,argument2,argument3} **Note** Note Arguments typically take the following form: `<field name> in $L.file` where `<field name>` is the name of a field in the record that triggered the notification.
Condition	Type true to enable the notification in all conditions. Type an expression that can be true or false to define a condition when Service Manager should send the notification. Type false to disable the notification in all conditions. By default, Service Manager treats a blank entry as true.
Format	Type the format control record used to display the triggering record as part of the notification.

Notify Method	Type or select the delivery method to be used to send the notification. This value must match a delivery method defined in the msgtype table. **Note** Note • The notification applications listed below must be defined in RAD code and exist in Service Manager to work. • The message type must be enabled (true) by selecting "Active?" in the msgtype table. See the following list for valid message types. • **connect:** Send a message to an external device. • **email:** Send an email. • **log:** Create a log entry for a message. • **mail:** Send internal mail. • **msg:** Send an internal message to a user's current Service Manager session (for example, "Incident IM1011 has been assigned to your monitor list since..."). **Note** Note If the user is not currently connected to a Service Manager session, the message is not sent. • **print:** Print message. • **teams:** Send a Teams message.
Recipients	Type one of the following to identify who should receive the notification: • A contact name • An operator name • Array of approval group names • A RAD expression that gathers a contact or operator name from some field in the triggering record
Group File	Type the name of the group receiving the notification. You can specify the On Call group to send the notification to for those operators who meet the On Call criteria.
Group Area	Type the name of the group area receiving the notification. You can use this field only with ocmgroups and cm3groups group tables. Valid choices include: • ocmgroups: **All**, **Line Items**, **Orders**, and **Quotes** • cm3groups: **All**, **Changes**, and **Tasks**

Subgroup	Type the name of the subgroup area receiving the notification. You can use this field with assignment, ocmgroups, and cm3groups tables. Valid choices include **All**, **Approvers**, and **Members**. **Note** Note If you have set a valid group email of members/approvers in the Assignment Groups definition and then select **Members/Approvers** here, the system only sends emails to this group email address when sending notifications. Otherwise, the system separately sends the emails to each member in the assignment group. However, this group email function doesn't support all the Smart Emails with action buttons (such as the **Approve**, **Deny**, or **Close** buttons, and so on). The system separately sends the Smart Emails with action buttons to each member in the assignment group even though you have set a valid group email address in the **Approver Group Email** field.
Email/Mail Subject Line tab	Use this tab to type any custom information that should appear in e-mail notifications.
Msg Class	Type the message class for the subject of the notification. This value must match a message class definition in the message record.
Msg No	Type the message number to use as the subject of the notification. This value must match a message number definition in the message record.

Arguments	Type any RAD expressions used to display information from the record or event that triggered the notification. Service Manager displays this information in the subject of the notification. You can enter an array of arguments by enclosing all arguments in brackets and separating each argument with a comma. For example:
	`{argument1,argument2,argument3}`
	Note
	Note Arguments typically take the following form:
	`<field name> in $L.file`
	where `<field name>` is the name of a field in the record that triggered the notification.

Email Cc/Bcc tab	Use this tab to define the email Cc and Bcc recipients. Configurations on this tab apply only if you select **email** as the Notify Method. **Note:** If a Smart Email consists of tokens, the Cc and Bcc functions don't work.
Recipient(s) (Cc) Recipient(s) (Bcc)	Type one of the following to identify the email Cc and Bcc recipients. • A contact name • An operator name • Array of approval group names • A RAD expression that gathers a contact or operator name from some field in the triggering record
Group File (Cc) Group File (Bcc)	Type the name of the Cc and Bcc groups receiving the notification.

Subgroup (Cc) Subgroup (Bcc)	Type the name of the Cc and Bcc subgroups area receiving the notification. Valid choices include **All**, **Approvers**, and **Members**. **Note** Note If you have set a valid group email of members/approvers in the Assignment Groups definition and then select **Members/Approvers** here, the system only sends emails to this group email address when sending notifications. Otherwise, the system separately sends the emails to each member of the assignment group. However, this group email function doesn't support all the Smart Emails with action buttons (such as the **Approve**, **Deny**, or **Close** buttons, and so on). The system separately sends the Smart Emails with action buttons to each member in the assignment group even though you have set a valid group email address in the **Approver Group Email** field.
Email Attachment tab	Use this tab to define the conditions and query filters under which Service Manager should add attachments from the original record to the notification. When the condition is evaluated as true, the system filters the attachments in a record and then sends the attachments that match the query filter along with the notification email.
Condition	Type `true` to add attachments to the notification in all conditions. Type `false` to disable the attachments in all conditions. Alternatively, type an expression to define a condition when Service Manager should add the attachments. For example, `severity in $L.file=2 and not same(severity in $L.file, severity in $L.file.save)`.
Query Filter	Type `true` to add all types and sizes of attachments to the notification. Type `false` to disable attachments of all types and sizes. Alternatively, type query criteria for the allowed attachment types and sizes. For example, `size<10000000 and not (filename like "*.eml") and (filename like "*.pdf" or filename#"High Severity Notification")`.

Related topics

Notifications
Add a message record
Add a notification delivery method
Create a custom notification

Create a distribution group
Create an on-call schedule
Send a notification using Format Control
Default message classes
Default notifications

11.7.5. Add a notification delivery method

Applies to User Roles: System Administrator

To add a notification delivery method, follow these steps:

1. Click **Tailoring** > **Notifications** > **Message Types**. The **Message Type File** form opens.
2. Type or select the following information.

Field	Description
Type	Type a name or label to describe the notification method. For example, to send an e-mail message, you can create a type called e-mail.
Description	This is an optional field you can use to type a description of the notification method.
Application	Type the name of the RAD application that sends the notification.
Active?	Select this option to enable the notification method for use by notification records.

3. Click **Add**.

Related topics

Notifications
Add a message record
Create a custom notification
Create a distribution group
Create an on-call schedule
Send a notification using Format Control
Default message classes
Default notifications

11.7.6. Create a custom notification

Applies to User Roles: System Administrator

To create a custom notification, follow these steps:

1. Create the custom message for the notification.
2. Define the delivery methods available for notifications.
3. Define the on call schedules for the groups in your organization.
4. Create a notification definition record.
5. Configure your Service Manager applications to use your notification.

Related topics

Notifications
Add a message record
Add a notification delivery method
Create a distribution group

Create an on-call schedule
Send a notification using Format Control
Default message classes
Default notifications

11.7.7. Create a distribution group

Applies to User roles: System Administrator

You can create distribution groups that the Report Exerciser and Event Services applications can use to send reports to pre-defined group of users, or to automatically open change and incident records. Service Manager saves changes that you make to distribution groups in the distgroup table.

To create a distribution group, follow these steps:

1. Click **System Administration** > **Base System Configuration** > **Miscellaneous** > **Distribution**.
   The **Distribution Group File** form opens.
2. In the Group Name field, type the name of the group.
3. Select the type of distribution group.

Type	Description
Change	Creates a change request and assigns it to the distribution group
Email	Sends an e-mail message to the distribution group
Print	Sends a print job to the printer on behalf of the distribution group
Problem	Creates an incident and assigns it to the distribution group

4. In the Operators/Contacts field, type the operator name or names of contacts who belong to the distribution group.
5. Click **Add**.
   Service Manager adds the distribution group record.

Related topics

Notifications
Add a message record
Add a notification delivery method
Create a custom notification
Create an on-call schedule
Send a notification using Format Control
Default message classes
Default notifications

11.7.8. Send a notification using Format Control

Applies to User Roles: System Administrator

To send a notification using Format Control, follow these steps:

1. Click **Tailoring** > **Format Control**.
2. Search for and open the format control record for which you want to enable notification.
3. Select an existing notification definition record or create a custom notification and note the name of the record. This notification is the target of the Format Control call.
4. Click the **Subroutines** tab, or click **Subroutines** from **More** or the **More Actions** menu to create a Format Control subroutine with the following properties:

Name	Value
Add	The expression that's evaluated to determine whether to execute the specified RAD application every time that the corresponding format is used to add a record. Example: `true` (if you want a notification to be sent on any Add operation)
Upd	The expression that's evaluated to determine whether to execute the specified RAD application every time that the corresponding format is used to update a record. Example: `true` (if you want a notification to be sent on any Update operation)
Del	The expression that's evaluated to determine whether to execute the specified RAD application every time that the corresponding format is used to update a record. Example: `true` (if you want a notification to be sent on any Delete operation)
Dis	The expression that's evaluated to determine whether to execute the specified RAD application every time that the corresponding format is displayed. Example: `true` (if you want a notification to be sent every time the form is displayed) **Note** **Note:** This option is not typically used to create a notification.
Initial	The expression that's evaluated to determine whether to execute the specified RAD application every time that the corresponding format initially opens. Example: `true` (if you want a notification to be sent the first time the form displays) **Note** Note This option is not typically used to create a notification.
Before	Type false (to send a notification *after* an operation is made)
Application	Type us.notify

Msg ID	The message you want to display if the system can't run the application or one of the Name/Value inputs is incorrect. **Note** Note This option is not typically used to create a notification.
Error Message	The error message you want to display if the system can't run the application or one of the Name/Value inputs is incorrect. **Note** Note This option is not typically used to create a notification.
Names and **Values**	The names and values of the parameters to be passed into the RAD application.For the us.notify application, we can add the following parameters: **name**: The name of the notification definition record to be the target of the Format Control call. For example: `CM Deny` **record**: The record variable the notification should use. For example: `$file` **names** (optional): The array of values to display in the notification.

5. To save your changes, click **Save**.
6. Click **OK**.

Related topics

Notifications
Add a message record
Add a notification delivery method
Create a custom notification
Create a distribution group
Create an on-call schedule
Default message classes
Default notifications

11.7.9. Notifications in Process Designer

Process Designer includes the following notifications which are sent in response to certain events.

Interaction Notifications

Incident Notifications

Problem Notifications

Incident Task and Problem Task Notifications

Request Notifications

Request Task Notifications

11.7.9.1. Interaction Notifications

The following Interaction Notifications are present in Process Designer.

New Interaction[edit]

Status	Open
Rule Set	Notification is triggered when an interaction is created
Recipients	Contact, Service Recipient
Subject	Interaction %S (Interaction number) has been created.
Format	Service Manager Interaction %s (hyper link of interaction number) has been created. Details Description: Status: Sent from Service Manager. Do not reply to this email.

Cancellation[edit]

Status	Canceled
Rule Set	Notification is triggered when the status is changed to "Canceled"
Recipients	Contact, Service Recipient, Assignee (for complaint or compliment)
Subject	Interaction %S (Interaction number) has been canceled.
Format	Service Manager Interaction %s (hyper link of interaction number) has been canceled. Details Description: Status: Sent from Service Manager. Do not reply to this email.

Closure[edit]

Status	Closed
Rule Set	Notification is triggered when interaction status is changed to "Closed"
Recipients	Contact, Service Recipient, Assignee (for complaint or compliment)
Subject	Interaction %S (Interaction number) has been closed.
Format	Service Manager Interaction %s (hyper link of interaction number) has been closed. Details Description: Status: Sent from Service Manager. Do not reply to this email.

Pending Approval[edit]

Status	Pending
Rule Set	Notification is triggered when the approval status of a service catalog request is “pending”
Recipients	Approver
Subject	Interaction %S (Interaction number) is pending on your approval.
Format	Service Manager Interaction %s (hyper link of interaction number) is pending on your approval. Details Description: Status: Sent from Service Manager. Do not reply to this email.

Approved or Denied[edit]

Status	Approved or denied
Rule Set	Notification is triggered when the approval status of a service catalog request is changed from "pending" to "approved" or "denied"
Recipients	Contact, Service Recipient
Subject	Interaction %S (Interaction number) has been %S (Approval status).
Format	Service Manager Interaction %s (hyper link of interaction number) has been {$approval.status}. Details Description: Status: Sent from Service Manager. Do not reply to this email.

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- 1 New Interaction
- 2 Cancellation
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- 4 Pending Approval
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11.7.9.2. Incident Notifications

The following Incident Notifications are present in Process Designer.

Assignment Group

Status	All
Rule Set	Notification is triggered when only Assignment group is populated or updated and Assignee field is Blank
Notification name(notification definition table)	Incident Assignment
Recipients	Assignment Group members
Subject	Incident %S (Incident number) has been Assigned to you
Format	NOTE: SYSTEM GENERATED EMAIL, PLEASE DO NOT REPLY. FOR MORE INFORMATION REFER TO THE INCIDENT RECORD IN THE SYSTEM. Dear Sir/Madam, Incident %S (Incident number) is assigned to %S (assignment group name) Group. Identify an assignee to work on the Incident. Incident details are as follows: Title: Status: Category: Subcategory: Priority: Description: Click here to go to the Incident Record (provide the link to Incident record) Thanks and regards, Service Manager

Assignee

Status	All
Rule Set	Notification is triggered when Assignee field populated or updated
Notification name(notification definition table)	Incident Assignment
Recipients	Incident Assignee
Subject	Incident %S (Incident number) has been Assigned to you
Format	NOTE: SYSTEM GENERATED EMAIL, PLEASE DO NOT REPLY. FOR MORE INFORMATION REFER TO THE INCIDENT RECORD IN THE SYSTEM. Dear Sir/Madam, Incident %S (Incident number) is Assigned to you Incident details are as follows: Title: Status: Category: Subcategory: Priority: Description: Click here to go to the Incident Record (provide the link to Incident record) Thanks and regards, Service Manager

Major Incident Arrival

Status	All
Rule Set	Notification is triggered when Major Incident checkbox is checked and saved
Notification name(notification definition table)	Major Incident Created
Recipients	Incident Manager
Subject	Major Incident %S (Incident number) has been created
Format	NOTE: SYSTEM GENERATED EMAIL, PLEASE DO NOT REPLY. FOR MORE INFORMATION REFER TO THE INCIDENT RECORD IN THE SYSTEM. Dear Sir/Madam, Major Incident %S (Incident number) has been created Incident details are as follows: Title: Status: Category: Subcategory: Priority: Description: Click here to go to the Incident Record (provide the link to Incident record) Thanks and regards, Service Manager

Escalation

Status	All
Rule Set	Notification is triggered when Escalation checkbox is checked and saved
Notification name(notification definition table)	Escalated Incident
Recipients	Incident Manager, Incident Assignee
Subject	Incident %S (Incident number) has been escalated
Format	NOTE: SYSTEM GENERATED EMAIL, PLEASE DO NOT REPLY. FOR MORE INFORMATION REFER TO THE INCIDENT RECORD IN THE SYSTEM. Dear Sir/Madam, Incident %S (Incident number) has been escalated Incident details are as follows: Title: Status: Category: Subcategory: Priority: Description: Click here to go to the Incident Record (provide the link to Incident record) Thanks and regards, Service Manager

Pending with Customer

Status	Pending Customer
Rule Set	Notification is triggered when the status of the record is changed to "Pending Customer"
Notification name(notification definition table)	Incident pending customer
Recipients	Requestor
Subject	Incident %S (Incident Number) pending for Requestor Input
Format	NOTE: SYSTEM GENERATED EMAIL, PLEASE DO NOT REPLY. FOR MORE INFORMATION REFER TO THE INCIDENT RECORD IN THE SYSTEM. Dear Sir/Madam, The Incident %S (Incident number) is pending for your inputs. Please find the reason below and provide necessary input to enable resolution Title: Priority: Description: Click here to go to the Incident Record (provide the link to Incident record) Thanks and regards, Service Manager

Suspension

Status	Suspended
Rule Set	Notification is triggered when the status of the record is changed to “Suspended”
Notification name(notification definition table)	Incident suspended
Recipients	Requestor, Incident Coordinator, Incident Assignee
Subject	Incident %S (Incident Number) has been Suspended
Format	NOTE: SYSTEM GENERATED EMAIL, PLEASE DO NOT REPLY. FOR MORE INFORMATION REFER TO THE INCIDENT RECORD IN THE SYSTEM. Dear Sir/Madam, The Incident %S (Incident number) is suspended. Please find the reason and duration of suspension. The Incident may be unsuspended manually before the suspension duration ends. Title: Description: Click here to go to the Incident Record (provide the link to Incident record) Thanks and regards, Service Manager

Un-suspension

Status	Any status other than Suspended
Rule Set	Notification is triggered when the status of the record is changed from “Suspended” to previous status
Notification name(notification definition table)	Incident Unsuspended
Recipients	Requestor, Incident Coordinator, Incident Assignee
Subject	Incident %S (Incident Number) has been Resumed
Format	NOTE: SYSTEM GENERATED EMAIL, PLEASE DO NOT REPLY. FOR MORE INFORMATION REFER TO THE INCIDENT RECORD IN THE SYSTEM. Dear Sir/Madam, The Incident %S (Incident number) is now unsuspended. The status has been moved to %S (status). Title: Description: Status: Click here to go to the Incident Record (provide the link to Incident record) Thanks and regards, Service Manager

Recovery

Status	Status other than Resolved
Rule Set	Notification is triggered when Phase is moved to Recovery
Notification name(notification definition table)	Incident Recovery
Recipients	Incident Coordinator
Subject	Incident %S (Incident number) has moved to Recovery phase
Format	NOTE: SYSTEM GENERATED EMAIL, PLEASE DO NOT REPLY. FOR MORE INFORMATION REFER TO THE INCIDENT RECORD IN THE SYSTEM. Dear Sir/Madam, The Incident %S (Incident number) has been moved to Recovery Phase. Incident details are as follows: Title: Status: Description: Solution: Thanks and regards, Service Manager

Resolution

Status	Resolved
Rule Set	Notification is triggered when Status is Resolved
Notification name(notification definition table)	Incident Resolution
Recipients	To: Requestor cc: Incident Coordinator
Subject	Incident %S (Incident number) has been Resolved
Format	NOTE: SYSTEM GENERATED EMAIL, PLEASE DO NOT REPLY. FOR MORE INFORMATION REFER TO THE INCIDENT RECORD IN THE SYSTEM. Dear Sir/Madam, Incident %S (Incident number) is Resolved. Please verify the resolution and update the Incident record. Incident details are as follows: Title: Status: Description: Solution: Click here to go to the Incident Record (provide the link to Incident record) Thanks and regards, Service Manager

Closure

Status	Closed
Rule Set	Notification is triggered when Status is closed
Notification name(notification definition table)	Incident Closure
Recipients	Incident Assignee, Requestor
Subject	Incident %S (Incident number) has been Closed
Format	NOTE: SYSTEM GENERATED EMAIL, PLEASE DO NOT REPLY. FOR MORE INFORMATION REFER TO THE INCIDENT RECORD IN THE SYSTEM. Dear Sir/Madam, Incident %S (Incident number) is Closed Incident details are as follows: Title: Status: Category: Subcategory: Priority: Description: Closure Code: Click here to go to the Incident Record (provide the link to Incident record) Thanks and regards, Service Manager

11.7.9.3. Problem Notifications

The following Problem Notifications are present in Process Designer.

New Problem

Status	Categorize
Rule Set	Notification is triggered when a new problem record is created and saved
Notification name(notification definition table)	Problem record created
Recipients	Problem Manager, Problem Coordinator
Subject	Problem record %S (Problem number) has been created
Format	NOTE: SYSTEM GENERATED EMAIL, PLEASE DO NOT REPLY. FOR MORE INFORMATION REFER TO THE PROBLEM RECORD IN THE SYSTEM. Dear Sir/Madam, A Problem record %S (Problem number) has been created Problem details are as follows: Title: Category: Description: Priority: Click here to go to the Problem record (provide the link to Problem record) Thanks and regards, Service Manager

Known Error Created

Status	All
Rule Set	Notification is triggered when Known Error checkbox is checked
Notification name(notification definition table)	KE created
Recipients	To: Problem Manager, Problem Coordinator, Assignee, Requestor (Opened By)
Subject	Known Error %S (Problem number) has been created
Format	NOTE: SYSTEM GENERATED EMAIL, PLEASE DO NOT REPLY. FOR MORE INFORMATION REFER TO THE PROBLEM RECORD IN THE SYSTEM. Dear Sir/Madam, A Known Error record %S (Problem number) has been created Known Error details are as follows: Title: Category: Description: Root Cause: Workaround: Click here to go to the Known Error record (provide the link to Problem record) Thanks and regards, Service Manager

Assignment Group

Status	All
Rule Set	Notification is triggered when only Assignment Group is populated or updated and Assignee field is blank
Notification name(notification definition table)	Problem Assignment
Recipients	Assignment Group members
Subject	Problem %S (Problem number) has been assigned to you
Format	NOTE: SYSTEM GENERATED EMAIL, PLEASE DO NOT REPLY. FOR MORE INFORMATION REFER TO THE PROBLEM RECORD IN THE SYSTEM. Dear Sir/Madam, Problem %S (Problem number) is assigned to %S (Assignment Group name) Group. Identify an assignee to work on the Problem. Problem details are as follows: Title: Phase: Category: Subcategory: Priority: Description: Click here to go to the Problem record (provide the link to Problem record) Thanks and regards, Service Manager

Assignee

Status	All
Rule Set	Notification is triggered when Assignee field populated or updated
Notification name(notification definition table)	Problem Assignment
Recipients	Problem Assignee
Subject	Problem %S (Problem number) has been assigned to you
Format	NOTE: SYSTEM GENERATED EMAIL, PLEASE DO NOT REPLY. FOR MORE INFORMATION REFER TO THE PROBLEM RECORD IN THE SYSTEM. Dear Sir/Madam, Problem %S (Problem number) is assigned to you Problem details are as follows: Title: Phase: Category: Subcategory: Priority: Description: Click here to go to the Problem record (provide the link to Problem record) Thanks and regards, Service Manager

Workaround Found

Status	All
Rule Set	Notification is triggered when Workaround field is populated
Notification name(notification definition table)	Workaround Identified
Recipients	To: Problem Coordinator, Requestor (Opened By)
Subject	Workaround has been identified for Problem %S (Problem number)
Format	NOTE: SYSTEM GENERATED EMAIL, PLEASE DO NOT REPLY. FOR MORE INFORMATION REFER TO THE PROBLEM RECORD IN THE SYSTEM. Dear Sir/Madam, Workaround has been identified for the Problem %S (Problem number). Problem and Workaround details are as follows: Title: Category: Description: Root Cause: Workaround: Click here to go to the Problem record (provide the link to Problem record) Thanks and regards, Service Manager

Root Cause Found

Status	All
Rule Set	Notification is triggered when Root Cause field is populated
Notification name(notification definition table)	Root Cause Identified
Recipients	To: Problem Coordinator, Requestor (Opened By)
Subject	Root Cause has been identified for Problem %S (Problem number)
Format	NOTE: SYSTEM GENERATED EMAIL, PLEASE DO NOT REPLY. FOR MORE INFORMATION REFER TO THE PROBLEM RECORD IN THE SYSTEM. Dear Sir/Madam, Root Cause has been identified for the Problem %S (Problem number). Problem and Root Cause details are as follows: Title: Category: Description: Root Cause: Click here to go to the Problem record (provide the link to Problem record) Thanks and regards, Service Manager

Resolution

Status	Resolved
Rule Set	Notification is triggered when the status is moved to Resolved
Notification name(notification definition table)	Problem Resolution
Recipients	To: Problem Coordinator, Requestor (Opened By)
Subject	Problem %S (Problem number) is resolved
Format	NOTE: SYSTEM GENERATED EMAIL, PLEASE DO NOT REPLY. FOR MORE INFORMATION REFER TO THE PROBLEM RECORD IN THE SYSTEM. Dear Sir/Madam, The Problem %S (Problem number) has been resolved. Kindly verify the same to confirm. Problem details are as follows: Title: Category: Description: Root Cause: Solution: Click here to go to the Problem record (provide the link to Problem record) Thanks and regards, Service Manager

Abandonment

Status	Abandoned
Rule Set	Notification is triggered when the status is moved to Abandoned
Notification name(notification definition table)	Problem Abandonment
Recipients	Problem Coordinator, Requestor (Opened By)
Subject	Problem %S (Problem number) is abandoned
Format	NOTE: SYSTEM GENERATED EMAIL, PLEASE DO NOT REPLY. FOR MORE INFORMATION REFER TO THE PROBLEM RECORD IN THE SYSTEM. Dear Sir/Madam, The Problem %S (Problem number) has been abandoned. Problem details are as follows: Title: Category: Phase: Description: Click here to go to the Problem record (provide the link to Problem record) Thanks and regards, Service Manager

Deferred

Status	Deferred
Rule Set	Notification is triggered when the status is moved to Deferred
Notification name(notification definition table)	Problem Deferral
Recipients	Problem Coordinator, Requestor (Opened By)
Subject	Problem %S (Problem number) is deferred
Format	NOTE: SYSTEM GENERATED EMAIL, PLEASE DO NOT REPLY. FOR MORE INFORMATION REFER TO THE PROBLEM RECORD IN THE SYSTEM. Dear Sir/Madam, The Problem %S (Problem number) has been deferred. Problem details are as follows: Title: Category: Phase: Description: Click here to go to the Problem record (provide the link to Problem record) Thanks and regards, Service Manager

Closure

Status	Closed
Rule Set	Notification is triggered when the status is moved to Closed
Notification name(notification definition table)	Problem Closure
Recipients	Problem Coordinator, Requestor
Subject	Problem %S (Problem number) is closed
Format	NOTE: SYSTEM GENERATED EMAIL, PLEASE DO NOT REPLY. FOR MORE INFORMATION REFER TO THE PROBLEM RECORD IN THE SYSTEM. Dear Sir/Madam, The Problem %S (Problem number) has been Closed. Problem details are as follows: Title: Category: Description: Solution: Closure Code: Click here to go to the Problem record (provide the link to Problem record) Thanks and regards, Service Manager

11.7.9.4. Incident Task and Problem Task Notifications

The following Incident Task and Problem Task Notifications are present in Process Designer.

Assignment Group

Status	All
Rule Set	Notification is triggered when only Assignment Group is populated or updated and Assignee field is blank
Notification name(notfication definition table)	Task Assignment
Recipients	Assignment Group members
Subject	Task %S (Task ID) has been assigned to you
Format	NOTE: SYSTEM GENERATED EMAIL, PLEASE DO NOT REPLY. FOR MORE INFORMATION REFER TO THE PROBLEM RECORD IN THE SYSTEM. Dear Sir/Madam, Task %S (Task ID) is assigned to %S (Assignment Group name) Group. Identify an assignee to work on the Problem. Problem details are as follows: Title: Description: Priority: Click here to go to the Task record (provide the link to Task record) Thanks and regards, Service Manager

Assignee

Status	Active
Rule Set	Notification is triggered when Assignee field populated or updated
Notification name(notification definition table)	Task Assignment
Recipients	Task Assignee
Subject	Task %S (Task ID) has been assigned to you
Format	NOTE: SYSTEM GENERATED EMAIL, PLEASE DO NOT REPLY. FOR MORE INFORMATION REFER TO THE PROBLEM RECORD IN THE SYSTEM. Dear Sir/Madam, Task %S (Task ID) is assigned to you Task details are as follows: Title: Description: Priority: Click here to go to the Task record (provide the link to Task record) Thanks and regards, Service Manager

Closure

Status	Closed
Rule Set	Notification is triggered when the status is moved to Closed
Notification name (notification definition table)	Task Closure
Recipients	Task Assignee
Subject	Task %S (Task ID) is closed
Format	NOTE: SYSTEM GENERATED EMAIL, PLEASE DO NOT REPLY. FOR MORE INFORMATION REFER TO THE PROBLEM RECORD IN THE SYSTEM. Dear Sir/Madam, Task %S (Task ID) is Closed. Task details are as follows: Title: Description: Priority: Task Outcome: Closure Code: Click here to go to the Task record (provide the link to Task record) Thanks and regards, Service Manager

Cancellation

Status	Cancelled
Rule Set	Notification is triggered when the status is moved to Cancelled
Notification name (notification definition table)	Task Cancellation
Recipients	Task Assignee
Subject	Task %S (Task ID) is cancelled
Format	NOTE: SYSTEM GENERATED EMAIL, PLEASE DO NOT REPLY. FOR MORE INFORMATION REFER TO THE PROBLEM RECORD IN THE SYSTEM. Dear Sir/Madam, Task %S (Task ID) has been cancelled. Task details are as follows: Title: Description: Priority: Closure Code: Click here to go to the Task record (provide the link to Task record) Thanks and regards, Service Manager

11.7.9.5. Request Notifications

The following out-of-box Request Notifications are provided in the Request Fulfillment module.

Off-catalog Request Creation[edit]

Status	All
Rule Set	Notification is triggered when off-catalog request is registered
Notification name(notification definition table)	Request Open
Recipients	requestor
Subject	Request %s (hyper link of request number) has been registered.
Format	Service Manager Request %s (hyper link of request number) has been registered. Details Title (or Description): Category: Subcategory: Request Model: Status: Priority: Sent from Service Manager. Do not reply to this email.

Assignment Group[edit]

Status	All
Rule Set	Notification is triggered when assignment group is populated but assignee is empty
Notification name(notification definition table)	Request Assignment
Recipients	members of the assignment group
Subject	Request %s (hyper link of request number) has been assigned to you or your group.
Format	Service Manager Request %s (hyper link of request number) has been assigned to you or your group. Details Title (or Description): Category: Subcategory: Request Model: Status: Priority: Sent from Service Manager. Do not reply to this email.

Assignee[edit]

Status	All
Rule Set	Notification is triggered when assignee is populated
Notification name(notification definition table)	Request Assignment
Recipients	assignee
Subject	Request %s (hyper link of request number) has been assigned to you or your group.
Format	Service Manager Request %s (hyper link of request number) has been assigned to you or your group. Details Title (or Description): Category: Subcategory: Request Model: Status: Priority: Sent from Service Manager. Do not reply to this email.

Request Pending Approval[edit]

Status	All
Rule Set	Notification is triggered when the request approval status is “pending”
Notification name(notification definition table)	Request Pending Approval
Recipients	members of current.pending.groups
Subject	Request %s (hyper link of request number) is pending on your approval.
Format	Service Manager Request %s (hyper link of request number) is pending on your approval. Details Title (or Description): Category: Subcategory: Request Model: Status: Priority: Sent from Service Manager. Do not reply to this email.

Request is Approved or Denied[edit]

Status	All
Rule Set	Notification is triggered when request approval status is changed from “pending” to “approved” or request approval status is “denied”
Notification name(notification definition table)	Request is Approved or Denied
Recipients	requestor, request coordinator, assignee
Subject	Request %s (hyper link of request number) is {$approval.status}.
Format	Service Manager Request %s (hyper link of request number) is {$approval.status}. Details Title (or Description): Category: Subcategory: Request Model: Status: Priority: Sent from Service Manager. Do not reply to this email.

Request Escalation[edit]

Status	All
Rule Set	Notification is triggered when the “Escalated” flag is selected and the “Escalation Manager” is specified.
Notification name(notification definition table)	Request Escalation
Recipients	Request Manager, Assignee or members of Assignment Group if Assignee is null
Subject	Request %s (hyper link of request number) has been escalated.
Format	Service Manager Request %s (hyper link of request number) has been escalated. Details Title (or Description): Category: Subcategory: Request Model: Status: Priority: Sent from Service Manager. Do not reply to this email.

Request Suspension[edit]

Status	Suspended
Rule Set	Notification is triggered when Request status is “Suspended”
Notification name(notification definition table)	Request Suspension
Recipients	requestor, request coordinator, assignee
Subject	Request %s (hyper link of request number) is suspended. The request may be unsuspended manually before the suspension duration ends.
Format	Service Manager Request %s (hyper link of request number) is suspended. The request may be unsuspended manually before the suspension duration ends. Details Title (or Description): Category: Subcategory: Request Model: Status: Priority: Sent from Service Manager. Do not reply to this email.

Request Unsuspension[edit]

Status	All except “Suspended”
Rule Set	Notification is triggered when Request status is changed from “Suspended” to previous status
Notification name(notification definition table)	Request Unsuspension
Recipients	requestor, request coordinator, assignee
Subject	Request %s (hyper link of request number) is unsuspended. Now the status moves to {$status}.
Format	Service Manager Request %s (hyper link of request number) is unsuspended. Now the status moves to {$status}. Details Title (or Description): Category: Subcategory: Request Model: Status: Priority: Sent from Service Manager. Do not reply to this email.

Request Pending Customer[edit]

Status	Pending Customer
Rule Set	Notification is triggered when Request status is “Pending Customer”
Notification name(notification definition table)	Request Pending Customer
Recipients	requestor, request coordinator
Subject	Request %s (hyper link of request number) is pending customer’s input. Please contact customer to get required information.
Format	Service Manager Request %s (hyper link of request number) is pending customer’s input. Please contact customer to get required information. Details Title (or Description): Category: Subcategory: Request Model: Status: Priority: Sent from Service Manager. Do not reply to this email.

Request Resolution[edit]

Status	Resolved
Rule Set	Notification is triggered when Request status is “Resolved”
Notification name(notification definition table)	Request Resolution
Recipients	requestor, request coordinator
Subject	Request %s (hyper link of request number) has been resolved.
Format	Service Manager Request %s (hyper link of request number) has been resolved. Details Title (or Description): Category: Subcategory: Request Model: Status: Priority: Sent from Service Manager. Do not reply to this email.

Request Withdrawal[edit]

Status	Closed
Rule Set	Notification is triggered when Request status is “Closed” and closure code is “Withdrawal requested by customer”
Notification name(notification definition table)	Request Withdrawal
Recipients	requestor, request coordinator, assignee
Subject	Request %s (hyper link of request number) has been withdrawn by customer.
Format	Service Manager Details Title (or Description): Category: Subcategory: Request Model: Status: Priority: Sent from Service Manager. Do not reply to this email.

Request Closure[edit]

Status	Closed
Rule Set	Notification is triggered when Request status is “Closed” and closure code is not “Withdrawal requested by customer”
Notification name(notification definition table)	Request Closure
Recipients	requestor, request coordinator, assignee
Subject	Request %s (hyper link of request number) has been closed. The closure code is {$closure.code}.
Format	Service Manager Request %s (hyper link of request number) has been closed. The closure code is {$closure.code}. Details Title (or Description): Category: Subcategory: Request Model: Status: Priority: Closure Comments: Sent from Service Manager. Do not reply to this email.

Request Reopen[edit]

Status	Fulfilled
Rule Set	Notification is triggered when the request is re-opened.
Notification name(notification definition table)	Request Re-open
Recipients	Requestor, Request Coordinator, members of Assignment Group if Assignee is null, otherwise Assignee
Subject	Request %s (hyper link of request number) has been re-opened.
Format	Service Manager Request %s (hyper link of request number) has been re-opened. Details Title (or Description): Category: Subcategory: Request Model: Status: Priority: Sent from Service Manager. Do not reply to this email.

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- 10 Request Resolution
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- 13 Request Reopen

11.7.9.6. Request Task Notifications

The following out-of-box Request Task Notifications are provided in the Request Fulfillment module.

Assignment Group[edit]

Status	All
Rule Set	Notification is triggered when assignment group is populated but assignee is empty
Notification name(notification definition table)	Request Task Assignment
Recipients	members of the assignment group
Subject	Request task %s (hyper link of request task number) has been assigned to you or your group.
Format	Service Manager Request task %s (hyper link of request task number) has been assigned to you or your group. Details Title (or Description): Category: Status: Priority: Sent from Service Manager. Do not reply to this email.

Assignee[edit]

Status	All
Rule Set	Notification is triggered when assignee is populated
Notification name(notification definition table)	Request Assignment
Recipients	assignee
Subject	Request task %s (hyper link of request task number) has been assigned to you or your group.
Format	Service Manager Request task %s (hyper link of request task number) has been assigned to you or your group. Details Title (or Description): Category: Status: Priority: Sent from Service Manager. Do not reply to this email.

Cancellation[edit]

Status	Cancelled
Rule Set	Notification is triggered when request task status is “Cancelled”
Notification name (notification definition table)	Request Task Cancellation
Recipients	assignee
Subject	Request task %s (hyper link of request task number) has been cancelled.
Format	Service Manager Request task %s (hyper link of request task number) has been cancelled. Details Title (or Description): Category: Status: Priority: Sent from Service Manager. Do not reply to this email.

Closure[edit]

Status	Closed
Rule Set	Notification is triggered when request task status is “Closed”
Notification name (notification definition table)	Request Task Closure
Recipients	assignee
Subject	Request task %s (hyper link of request task number) has been closed. The closure code is {$closure.code}.
Format	Service Manager Request task %s (hyper link of request task number) has been closed. The closure code is {$closure.code}. Details Title (or Description): Category: Status: Priority: Closure Comments: Task output data <Task output data, e.g. CI information will be provided when ordering task is closed> Sent from Service Manager. Do not reply to this email.

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11.7.10. Sample notifications

Service Manager comes with sample notifications for the following modules:

- User Interaction Management
- Incident Management
- Problem Management
- Change Management

Notification Title	Notification Subject	Notification BodyText
Interaction Registration notification to ESS User	Interaction,(number) has been registered	Dear Sir/Madam, Interaction,(number); has been registered for you with the following details: Title:(Title) Service: (Service) Urgency: (Urgency) If you would like to view the actual status of this Interaction, please click this URL: (URL) If you have any queries you can always contact the Service Desk. Kind regards, ICT Service Desk tel: #8888
Interaction Registration notification to User	Interaction (Interaction #) has been registered	Dear Sir/Madam, Interaction (Interaction#) has been registered for you with the following details: Title: (Title) Service: (Service) Category: (Category) Area: (Area) Sub-area: (Sub-area) Priority: (Priority) Status: (Status) Target Date: (SLA target date) If you would like to view the actual status of this Interaction, please click this URL: (URL) If you have any queries you can always contact the Service Desk. Kind regards, ICT Service Desk tel: #8888

Web interaction processed by Service Desk	Interaction (Interaction #) has been processed by the Service Desk	Dear Sir/Madam, Interaction (Interaction#) has been processed by the Service Desk. Title: (Title) Service: (Service) Category: (Category) Area: (Area) Sub-area: (Sub-area) Priority: (Priority) Status: (Status) Target Date: (SLA target date) If you would like to view the actual status of this Interaction, please click this URL: (URL) If you have any queries you can always contact the Service Desk. Kind regards, ICT Service Desk tel: #8888
Interaction updated by User notification to Service Desk	Interaction (Interaction #) has been updated by User	Dear (Assignee), (Category) (Incident#) has been updated by (name of person who did the update (if possible)). (Category) details: Title: (Title) Service: (Service) Area: (Area) Sub-area: (Sub-area) Priority: (Priority) Status: (Status) Open Date: (Open Date) Target Date: (SLA target date) Assignment Group: (Assignment Group) (URL) Description: (Description) Last update: (text of the last update made (if possible))

Interaction Closure notification to User	Interaction (number) has been closed	Dear Sir/Madam, Interaction, (number) has been Closed for you. Title: (Title) Service: (Service) Category: (Category) Area: (Area) Sub-area: (Sub-area) Priority: (Priority) Open Date: (Open Date) SLA Target Date: (SLA target date) Close Date: (Close Date) If you disagree with the solution below, you can resubmit the Interaction by clicking this URL: (URL) Description: (Description) Solution: (Solution) If you have any queries you can always contact the Service Desk. Kind regards, ICT Service Desk tel: #8888

Notification Title	Notification Subject	Notification BodyText
Complaint Assigned notification to Service Desk Manager	(Category) (Incident #) has been assigned to you	Dear (Assignee), (Category) (Incident#) has been assigned to you. (Category) details: Title: (Title) Service: (Service) Area: (Area) Sub-area: (Sub-area) Priority: (Priority) Status: (Status) Open Date: (Open Date) Target Date: (SLA target date) Assignment Group: (Assignment Group) (URL) Description: (Description)
Incident SLA breached notification to Incident Manager	(Category) (Incident #) has breached its SLA	Dear (Assignee), (Category) (Incident#) has breached its SLA. (Category) details: Title: (Title) Service: (Service) Area: (Area) Sub-area: (Sub-area) Priority: (Priority) Status: (Status) Open Date: (Open Date) Target Date: (SLA target date) Assignment Group: (Assignment Group) Assignee: (Assignee) (URL) Description: (Description)

Incident Assigned notification to Assignee	(Category) (Incident #) has been assigned to you	Dear (Assignee), (Category) (Incident#) has been assigned to you. (Category) details: Title: (Title) Service: (Service) Area: (Are)) Sub-area: (Sub-area) Priority: (Priority) Status: (Status) Open Date: (Open Date) Target Date: (SLA target date) Assignment Group: (Assignment Group) (URL) Description: (Description)
Incident updated by NOT(Assignee) notification to Assignee	(Category) (Incident #) has been updated by IT specialist	Dear (Assignee), (Category) (Incident#) has been updated by (name of person who did the update (if possible)). (Category) details: Title: (Title) Service: (Service) Area: (Area) Sub-area: (Sub-area) Priority: (Priority) Status: (Status) Open Date: (Open Date) Target Date: (SLA target date) Assignment Group: (Assignment Group) (URL) Description: (Description) Last update: (text of the last update made (if possible))

Notification Title	Notification Subject	Notification BodyText
Problem Assigned notification to Assignee	Problem (Problem #) has been assigned to you	Dear (Assignee), Problem (Problem#) has been assigned to you. Problem details: Title: (Title) Service: (Service) Area: (Area) Sub-area: (Sub-area) Priority: (Priority) Status: (Status) Open Date: (Open Date) Target Date: (SLA target date) Assignment Group: (Assignment Group) (URL) Description: (Description)
Problem Rejected notification to Problem Coordinator	Problem (Problem #) has been Rejected by (Problem Analyst)	Dear (Problem Coordinator), Problem (Problem#) has been rejected by (name of person who did the update (if possible)). Problem details: Title: (Title) Service: (Service) Area: (Area) Sub-area: (Sub-area) Priority: (Priority) Status: (Status) Open Date: (Open Date) Target Date: (SLA target date) Assignment Group: (Assignment Group) (URL) Description: (Description)

Known Error Closed notification to Problem Coordinator	Known Error (Known Error#) has been Closed by (Operator)	Dear (Problem Coordinator), Known Error (Known Error ID#) has been closed by (Operator). Known Error details: Title: (Title) Service:(Service) Area: (Area) Sub-area: (Sub-area) Priority: (Priority) Assignment Group: (Assignment Group) (URL)
Problem Registered notification to Problem Coordinator	Problem (Problem #) has been registered	Dear (Problem Coordinator), Problem (Problem#) has been registered. Problem details: Title: (Title) Service: (Service) Area: (Area) Sub-area: (Sub-area) Priority: (Priority) Status: (Status) Open Date: (Open Date) Target Date: (SLA target date) Assignment Group: (Assignment Group) Problem Coordinator: (Problem Coordinator) (URL) Description: (Description)

Problem Assigned to group notification to Problem Coordinator	Problem (Problem #) has been assigned to your group	Dear (Problem Coordinator of Assignment Group), Problem (Problem#) has been assigned to your group. Problem details: Title: (Title) Service: (Service) Area: (Area) Sub-area: (Sub-area) Priority: (Priority) Status: (Status) Open Date: (Open Date) Target Date: (SLA target date) Assignment Group: (Assignment Group) (URL) Description: (Description)
Problem for Review & Closure notification to Problem Manager	Problem (Problem #) is ready for Review & Closure	Dear (Problem Manager), Problem (Problem#) has now entered the Problem closure and review phase. Problem details: Title: (Title) Service: (Service) Area: (Area) Sub-area: (Sub-area) Priority: (Priority) Status: (Status) Open Date: (Open Date) Target Date: (SLA target date) Assignment Group: (Assignment Group) (URL) Description: (Description)

Problem Workaround published notification to Service Desk Manager	Workaround published for Problem (Problem#)	Dear (Service Desk Manager), Known Error (Known Error#) now has a published workaround. Known Error details: Title: (Title) Service: (Service) Area: (Area) Sub-area: (Sub-area) Priority: (Priority) Status: (Status) Open Date: (Open Date) Assignment Group: (Assignment Group) (URL) Description: (Description) Workaround: (Workaround)
Problem Task Closed notification to Problem Coordinator	Problem Task (Problem Task#) has been Closed	Dear (Problem Coordinator of Assignment Group), Problem Task (Problem Task#) has been closed. Problem Task details: Title: (Title) Assignee: (Assignee) Problem details: Title: (Title) Service: (Service) Area: (Area) Sub-area: (Sub-area) Priority: (Priority) Status: (Status) Open Date: (Open Date) Target Date: (SLA target date) Assignment Group: (Assignment Group) (URL) Description: (Description)

Known Error proposed solution notification to Problem Manager	Known Error (Known Error#) has a proposed solution	Dear (Problem Manager), Known Error (Known Error#) now has a solution proposed Known Error details: Title: (Title) Service: (Service) Area: (Area) Sub-area: (Sub-area) Priority: (Priority) Status: (Status) Open Date: (Open Date) Assignment Group: (Assignment Group) (URL) Description: (Description) Solution: (Solution)
Known Error Assigned notification to Assignee	Known Error (Known Error#) has been assigned to you	Dear (Assignee), Known Error (Known Error#) has been assigned to you. Known Error details: Title: (Title) Service: (Service) Area: (Area) Sub-area: (Sub-area) Priority: (Priority) Status: (Status) Open Date: (Open Date) Assignment Group: (Assignment Group) (URL) Description: (Description)

Solution identification target exceeded notification to Problem Manager	Known Error (Known Error#) solution identification target has been exceeded	Dear (Problem Manager), Known Error (Known Error#) solution identification target has been exceeded. Known Error details: Title: (Title) Service: (Service) Area: (Area) Sub-area: (Sub-area) Priority: (Priority) Status: (Status) Open Date: (Open Date) Assignment Group: (Assignment Group) (URL) Description: (Description)

Notification Title	Notification Subject	Notification BodyText
Task rejected notification to Change Coordinator	Task (Task#) of Change (Change#) has been rejected by (Assignee)	Dear (Change Coordinator of parent Change), Task (Task#) of Change (Change#) has been rejected by (Assignee). Task details: Category: (category) Brief Description: (brief description) Rejection reason: (rejection reason) Change details: Category: (category) Brief Description: (brief description)
Change information incomplete or incorrectly assigned notification to Change Initiator	Change (Change#) is incorrectly assigned or has incomplete information	Dear (Change Initiator), Change (Change#) has been returned to you because of incomplete information or incorrect assignment. Please update the Change. Change details: Category: (category) Brief Description: (brief description)
Task assigned notification to Assignee	Task (Task#) of Change (Change#) has been assigned to you	Dear (Assignee), Task (Task#) of Change (Change#) has been assigned to you. Task details: Category: (category) Planned Start: (Planned Start) Planned End: (Planned End) Brief Description: (brief description) Change details: Category: (category) Change Coordinator: (Change Coordinator) Brief Description: (brief description)

Related topics

Notifications

11.7.11. Example: Creating an email notification for change updates

Applies to User roles: System Administrator

The following example describes how to set up a notification that meets the following requirements:

- The notification sends an email message when a change record is updated.
- The email message displays "Change [Num] Updated by [Operator]" where [Num] represents the number of the updated change and [Operator] represents the name of the operator who updated the change.

- The email message is sent to the Change Manager.

Task 1: Create a custom message record

To create a custom message record, follow these steps:

1. Click **Tailoring** > **Notifications** > **Messages**.
2. Specify the properties as listed in the following table.

Name	Value
Language Code	en
Class	ndp
Message Number	9001
Severity	3
Text	Change %S Updated by %S
Comments	Sent when a change is updated.

3. Click **Add**.

Task 2: Create a custom notification definition record

To create a custom notification record, follow these steps:

1. Click **Tailoring** > **Notifications** > **Notifications**.
2. Type NDP Change Update in the **Name** field, and then click **Add**. An ID is automatically created.
3. Type true in the **Condition** field.
4. On the **Message** tab, add a message entry in the list with properties listed in the following table:

Name	Value
Msg Class	ndp
Msg No.	9001
Arguments	{number in $L.file,operator in $L.file}
Condition	true
Notify Method	email
Recipient(s)	$G.chmmanagerEmail

Note

Note Set **Msg No** to an unassigned number. Use a custom message class for the **Msg Class** field.

5. (Optional) On the **Email Attachment** tab, define the conditions and query filters under which Service Manager should add attachments from the original change record to the notification.

> **Tip**
> Tip
> We recommend to keep the **scheduled.action.processor** background process running when working with attachments in notifications. Hence, the **Run Scheduled Action** rules work at run time and Service Manager performs scheduled automatic cleaning of the redundant attachments that accumulate in the current record.

6. Click **Save**.

Task 3: Enable the notification in the object record

To enable the notification in the object record, follow these steps:

1. Click **Tailoring** > **Document Engine** > **Objects**.
2. Type =cm3r in the **File name** field, and click **Search**.
3. Click the **Notifications** tab, verify that the **Update** field is empty, and then type NDP Change Update in the **Update** field.
4. Click **Save**.

Task 4: Test the notification

To test the notification, follow these steps:

1. Temporarily turn off the email agent that you are using, such as JavaMail, or switch it to debug mode. This is to make sure that newly created email messages remain in the eventout table.
2. Click **Change Management** > **Search Changes**.
3. Use the Search feature to locate a change record.
   Example: C10001.
4. Modify the **Description** field.
5. In the **Activities** section, select **Update** for the **New Update Time** field, and write a description in the **New Update** field.
6. Click **Save** to save the changes.
7. Click **Tailoring** > **Event Services** > **Output Events**.
8. Type email in the **Event Code** field, and click **Search**.
9. Verify that an event record is created that has an External Information String that resembles the following.
   `change.manager@advantage.com^falcon^change.manager@advantage.com^Change C10001 Updated by falcon^Change C10001 Updated by falcon`

Related topics

Notifications
Add a message record
Add a notification delivery method
Create a custom notification
Create a distribution group
Create an on-call schedule
Send a notification using Format Control
Default message classes
Default notifications

11.7.12. Configure email notifications for CI changes in reference modules

Applies to User roles: System Administrator

You can configure email notifications that are sent when the affected service or CIs in a reference record (for example, a Change record) gets changed. The `device` table stores CI names in the **display.name** field. You need to update your email notification definitions and HTML notification templates to use CI display names in CI related notifications.

To configure email notifications for CI changes, follow these steps:

1. Go to **Tailoring** > **Notifications** > **Notifications**.
2. Open a notification definition record, for example, **ChM Change Approval**.
3. On the **Message** and **Email/Mail Subject Line** tabs, wrap the **logical.name** filed with the get.display.value function to show CI display names in messages and email/mail subject lines. The following is an example:
   `{get.display.value($L.file,"logical.name",-1,true), $lo.ufname}`
   For details about the `get.display.value` function, see get.display.value.
4. Continue to follow *step 2* and *step 3* for all CI related notifications.
5. Configure HTML notification templates.
   a. Go to **Tailoring** > **Notifications** > **HTML Templates**.
   b. Open an HTML template, for example, **ChM Change Approval Pending**.
   c. On the **Expressions** tab, enter an expression that calls the `get.display.value` function to obtain the CI display name and assigns the value to a variable. For example:
   `var logical_name_disp_new= system.functions.get_display_value($RECORD, "logical.name",-1,true);`

11.7.13. Language for notification emails

When the system sends out a notification email to one recipient only, the system uses the language specified in the contact record of the recipient.

> **Note**
>
> Note The system first look for the specified in the contact record of the recipient. If no language is specified in the contact record, the system looks for the language specified in the operator record associated with the recipient.

When the system sends out a notification email to more than one recipient, if all the recipients have the same language defined in their contact or operator records, the system uses this language; however, if more than one languages are defined in the contact or operator records of the recipients, the system uses the system language (that is, the language code specified in System Information Record) for notification emails.

Related topics

Alerts
Message processors
Add a message record
Add a notification delivery method
Create a custom notification
Create a distribution group
Create an on-call schedule
Send a notification using Format Control
Default message classes
Default notifications

11.8. System bulletins

You can post system information and status messages to your Service Desk operators using system bulletins. System bulletins are a good way to make service desk operators aware of global service conditions such as a network outage or a pending change request. Service Desk operators can access system bulletins from their main menu using the Bulletin display option.

Create a system bulletin[edit]

Applies to User roles: System Administrator

To create a system bulletin, follow these steps:

1. Click **System Administration** > **Base System Configuration** > **Miscellaneous** > **System Bulletin**.
   The **System Bulletin** form opens.

2. In the **Date** field, type or select the date you want listed on the system bulletin.
3. Select **Activate Hot Tics?** if you want the system bulletin to display a list of the day's hot records.
4. Select **Default?** if you want to display the system bulletin to all users.

> **Note**
> Note Users can only see one bulletin at a time. The bulletin that has a date set to the current date with the time of 00:00:00 is what users see. If none exists, then the first selected Default record is displayed. If the date on the bulletin is set to the current date with a specified time, then it's not used at all. The Default option is supported for each day, as only one bulletin can be displayed at a time.

5. In the text field provided, type the text of the bulletin you want to create.
6. Click **Add** to create the bulletin.
   Service Manager creates the Daily Bulletin record.

Hide a system bulletin[edit]

Applies to User roles: System Administrator

You can hide any system bulletin that you don't want users to see from the bulletin.g form. Hidden system bulletins are saved for later use.

To hide a system bulletin, follow these steps:

1. Click **System Administration** > **Base System Configuration** > **Miscellaneous** > **System Bulletin**.
   The **System Bulletin** form opens.
2. Click **Search**.
   Service Manager displays a list of system bulletins.
3. Select the bulletin you want to hide.
4. Clear the **Default?** option.

> **Note**
> Note Users can only see one bulletin at a time. The bulletin that has a date set to the current date with the time of 00:00:00 is what users see. If none exists, then the first selected Default record is displayed. If the date on the bulletin is set to the current date with a specified time, then it's not used at all. The Default option is supported for each day, as only one bulletin can be displayed at a time.

5. Click **Save**.
   Service Manager displays the message: Daily Bulletin record updated.

Update a system bulletin[edit]

Applies to User roles: System Administrator

You can update system bulletins from the bulletin.g form, however users can only see one system bulletin at a time from the system.bulletin.g form. You can hide system bulletins to save them for later use.

To update a system bulletin, follow these steps:

1. Click **System Administration** > **Base System Configuration** > **Miscellaneous** > **System Bulletin**.
   The **System Bulletin** form opens.
2. Click **Search**.
   Service Manager displays a list of system bulletins.
3. Select the bulletin you want to update.
4. Type the new system bulletin text.
5. Click **Save**.
   Service Manager updates the Daily Bulletin record.

Contents

- 1 Create a system bulletin
- 2 Hide a system bulletin
- 3 Update a system bulletin

11.9. System status form

The system status form allows administrators to view and manage user sessions and processes.

Actions possible

From the system status form a user with system administrator capabilities can:

- View a list of users logged in to the server
- View a list of processes running on the server
- Schedule a process to run at a later date and time
- Broadcast a message to all connected users
- Display a list of users who are locked out of the system
- Filter the system events displayed in the status list
    - View all tasks: users and processes
    - View active processes only
    - View inactive processes only
    - View users only
    - View system processes only
- View the System Monitor for information about the system resources consumed by and connection properties used by each user and process running on the server
- View the list of system status command line options
- View the system status summary form

Note

Note There can be only one kmupdate process running at any time. Starting more than one kmupdate process causes unpredictable behavior on the search engine server.

System status list

The system status list contains the following columns.

Column	Description
Command	The text field where you can enter the commands **s** for status or **k** for kill process.
User Name	The login and process names currently running on the server.
PID	The unique numeric process identification Service Manager assigns to each user and process.
Device ID	The type of client or process running on the server.
Login Time	The date and time the user session or process started.
Idle Time	The length of time in *days hours*:*minutes*:*seconds* since the user session or process was active.

Save system monitor information

Applies to User roles: System Administrator

You can create a schedule process that exports information from the system monitor on a regular basis.

To save system monitor information, follow these steps:

1. Click **System Status**.
   The system status form opens.
2. Click **System Monitor**.
   The **System Monitor - Main User Info** form opens.
3. Click **Capture**.
   The **Schedule Monitor** form opens.
4. In the **Begin Time** field, type or select the date you want to start capturing system monitor information.
5. In the **Repeat Interval** field, type or select the next date you want to repeat capturing system monitor information.
6. In **Schedule Class** field, type the name of the schedule class under which you want the background process to run.
7. Click **Schedule** to start the background process.

View all processes running on the server

Applies to User roles: System Administrator

To view all processes running on the server from another form, follow the step below:

- Click **System Status**. The system status form opens showing a list of all processes.

To view all processes running on the server from the system status form, follow these steps:

1. Click **Display Options**.
2. Click **All Tasks**.
   Service Manager displays a list of all processes running on the server.

View process CPU usage

Applies to User roles: System Administrator

To view process CPU usage, follow these steps:

1. Click **System Status**.
   The system status form opens.
2. Click **System Monitor**.
   The **System Monitor - Main User Info** form opens.
3. Click **CPU Info**.
   Service Manager displays CPU usage including:
   - Process name
   - Last active time
   - CPU time in seconds
   - Priority
   - Application running
   - Memory used in bytes
   - Statements evaluated

View process input and output usage

Applies to User roles: System Administrator

To view process input and output usage, follow these steps:

1. Click **System Status**.
   The system status form opens.
2. Click **System Monitor**.
   The **System Monitor - Main User Info** form opens.
3. Click **IO Info**.
   Service Manager displays input and output usage including:
   - Process name

- Statements evaluated
- Physical reads
- Physical writes
- Non-keyed queries
- Screen input and output

View saved system monitor information

Applies to User roles: System Administrator

You can create a schedule process that exports information from the system monitor on a regular basis.

To view saved system monitor information, follow these steps:

1. Click **System Status**.
   The system status form opens.
2. Click **System Monitor**.
   The **System Monitor - Main User Info** form opens.
3. Click **View Capture**.
   A list of captured data opens.

View shared memory statistics

Applies to User roles: System Administrator

To view shared memory statistics, follow these steps:

1. Click **System Status**.
   The system status form opens.
2. Click **System Monitor**.
   The **System Monitor - Main User Info** form opens.
3. Click **Shared Memory Info**.
   Service Manager displays shared memory statistics of the server:
   - Current Size
   - Segment Allocation
   - Large Block Allocation
   - Unused Space
   - Free Space Percent

View the active processes running on the server

Applies to User roles: System Administrator

To view the active processes running on the server, follow these steps:

1. Click **System Status**.
   The system status form opens.
2. Click **Display options**.
3. Click **Active**.
   Service Manager displays a list of active processes running on the server.

View the inactive processes running on the server

Applies to User roles: System Administrator

To view the inactive processes running on the server, follow these steps:

1. Click **System Status**.
   The system status form opens.
2. Click **Display options**.
3. Click **Inactive**.
   Service Manager displays a list of inactive processes running on the server.

View the status of a process

Applies to User roles: System Administrator

To view the status of a process, follow these steps:

1. Click **System Status**.
   The system status form opens.
2. Type the letter s in the Command field of the process whose status you want to see.
3. Click **Execute Commands**.
   The operator status display form (operator.status.g) opens.

View the system monitor

Applies to User roles: System Administrator

To view the system monitor, follow these steps:

1. Click **System Status**.
   The system status form opens.
2. Click **System Monitor**.
   The system monitor opens displaying the following information:
   - Process name
   - Process identification
   - Device name – displays the type of client connection or process
   - Idle Time
   - Application running
   - Network address of the server
   - Percent usage of physical memory
   - Percent usage of CPU
   - Thread identification
   - Session identification
   - License type

View the system processes running on the server

Applies to User roles: System Administrator

To view the system processes running on the server, follow these steps:

1. Click **System Status**.
   The system status form opens.
2. Click **Display options**.
3. Click **System**.
   Service Manager displays a list of system processes running on the server.

View the total server usage

Applies to User roles: System Administrator

To view the total server usage, follow these steps:

1. Click **System Status**.
   The system status form opens.
2. Click **System Monitor**.
   The **System Monitor - Main User Info** form opens.
3. Click **Grand Totals**.
   Service Manager displays total server usage of:
   - CPU time
   - Memory

- Statements evaluated
- Non-keyed query reads
- Physical Reads
- Physical writes
- Records retrieved
- C Code Strcpys (C Code string copies)

View the users logged into the server

Applies to User roles: System Administrator

To view the users logged into the server, follow these steps:

1. Click **System Status**.
   The system status form opens.
2. Click **Display options**.
3. Click **User**.
   Service Manager displays a list of users logged into the server.

Related topics

Status and Notifications

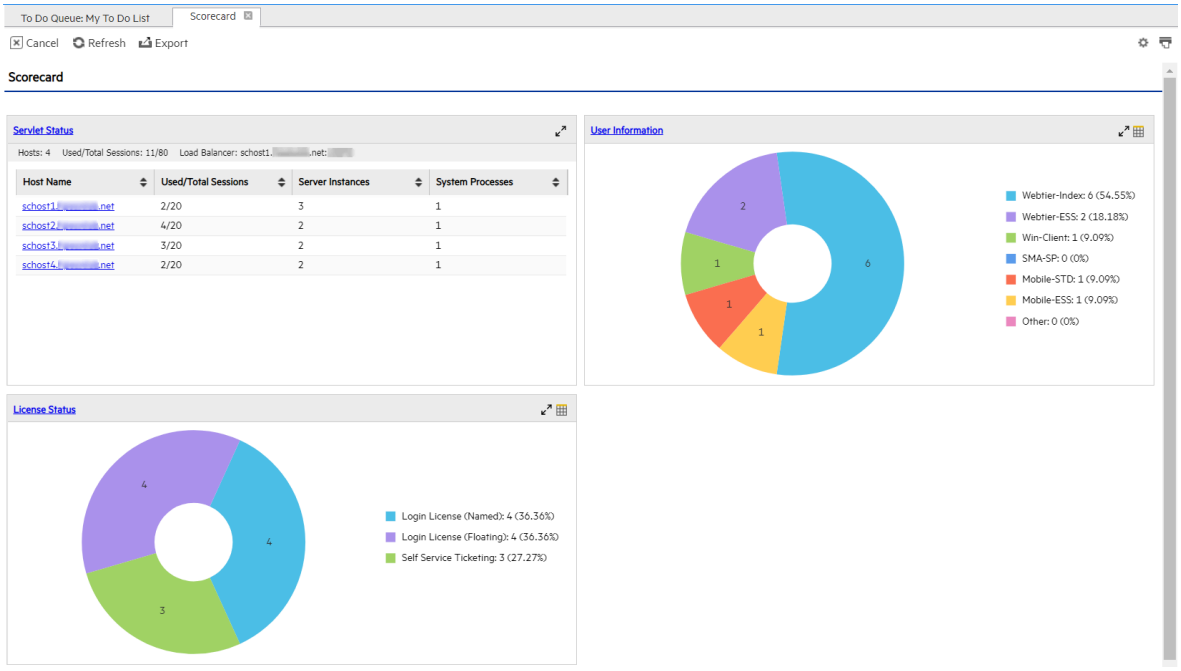
11.10. Scorecard

Scorecard can provide administrators an overview of the in-time and visualized Service Manager Server health information.

From the scorecard interface, a user with system administrator capabilities can view the following information:

Landing page

The landing page is available for Web Client only.



Possible actions:

- Click **Servlet Status** to view the detailed system scaling structure and the health information
  - Click a specific host name to view the detailed information of **Server Instances** and **System Processes** for the selected host
- Click **License Status** to view the detailed information of license usage and total available licenses for each license type

    - Click a specific license type to view the user login history with the selected license type
- Click **User Information** to view the detailed information of online user distributions and user login history
    - Click a specific client type to view the user login history with the selected client type
- Switch between the view data table and the bar charts by clicking the and icon (for Web Client only)
- Click the **Cancel** icon to exit
- Click the **Refresh** icon to synchronize the latest data and information
- Click the **Export** icon to export the servlet information as a formatted report
- Click the **Settings** icon to configure the Scorecard settings
- Click the icon to print the current page
- Click the icon or the icon to enlarge or reduce the size of the area

Scorecard settings

- For Web Client, you can click the **Settings** icon in the landing page to access the Scorecard settings.
- For Windows Client, you can click **Settings** in the Servlet status page to access the Scorecard settings.

You can configure the following settings for Scorecard:

Enable scorecard auto-refresh for audience
When enabled, the system automatically refreshes the data at certain minutes of the interval.
By default, this setting is disabled.

Auto-refresh interval (minutes)
Set the refresh interval for the system to automatically refresh the data.
The default value is set as 10 (minutes).

Max number of records displayed in History of Online Users
Set the maximum number of records that can be displayed in History of Online Users.
The default value is set as 12.

Auto-save interval for User Information snapshot (minutes)
Set the save interval for the system to automatically save the User Information snapshot.
The default value is set as 30 (minutes).

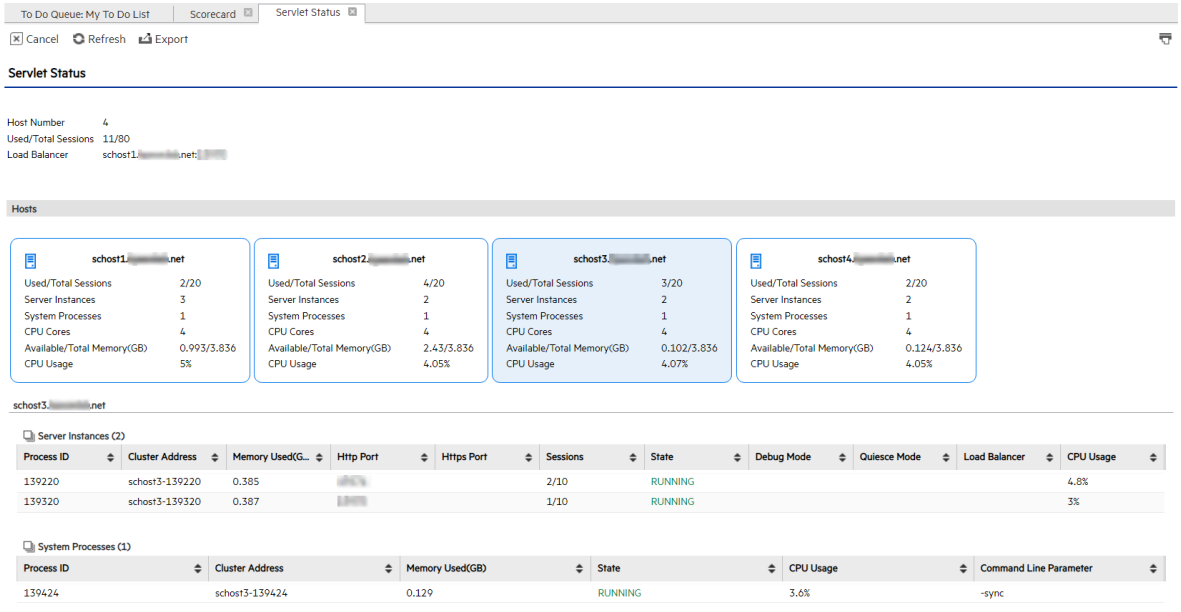
Auto-save interval for Servlet/License Status snapshot (minutes)
Set the save interval for the system to automatically save the Servlet/License Status snapshot.
The default value is set as 60 (minutes).

Alarm threshold for floating license usage (%)
If the floating licenses consumed exceed the set threshold value, the Floating license State value will be highlighted in red.
The default value is set as 80 (%).

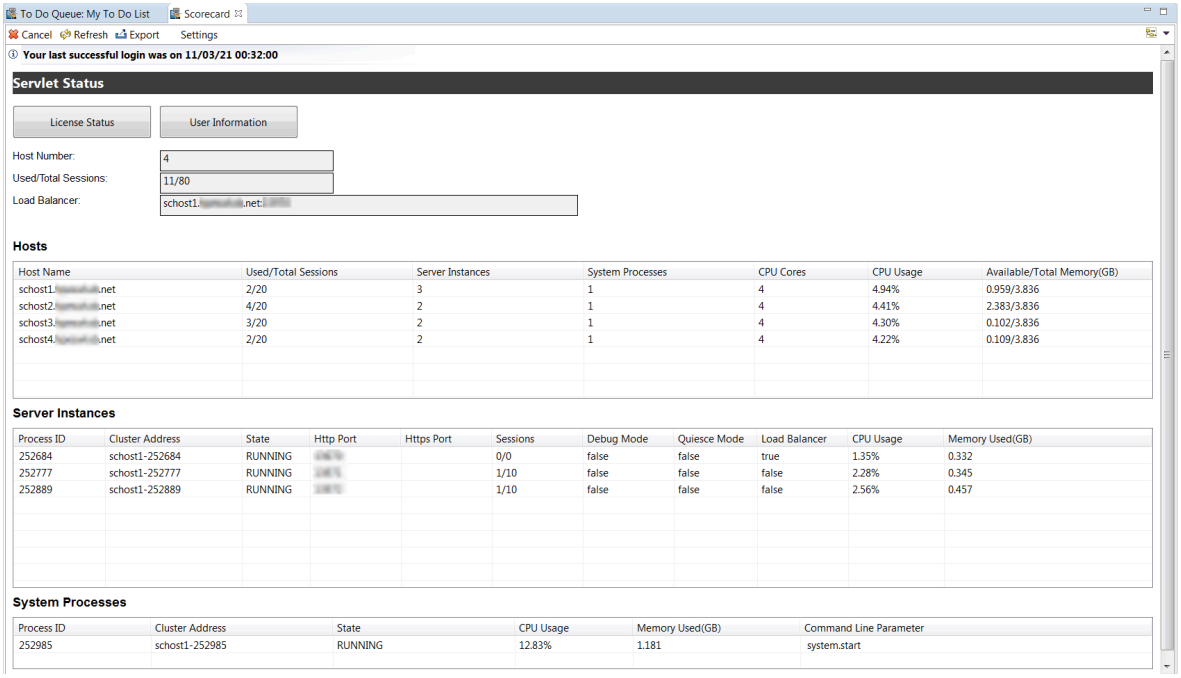
Alarm threshold for named license usage (%)
If the named licenses consumed exceed the set threshold value, the Named License State value will be highlighted in red.
The default value is set as 95 (%).

Alarm threshold for session usage (%):
If the sessions consumed exceed the set threshold value, the Used/Total Sessions value will be highlighted in red.
The default value is set as 85 (%).

Servlet status



Web Client



Windows Client

You can have an overview of the system scaling structure and the health information for each host in the **Servlet Status** tab:

- **Host number**
- **Used/Total Sessions**
- **Load Balancer**
- **Hosts**
  If you have multiple hosts, there will be multiple host cards displayed under the **Hosts** section. You can select and switch between different hosts. The **Server Instances** and **System Processes** information for the selected host are displayed.
  - In the Web Client view, you can single-click to switch the host card. The selected host card is displayed with white background color and the other host cards are displayed with grey background color.

- In the Windows Client view, you can double-click the host card. The **Server Instances** and **System Processes** information will be refreshed upon your selection.

Tip

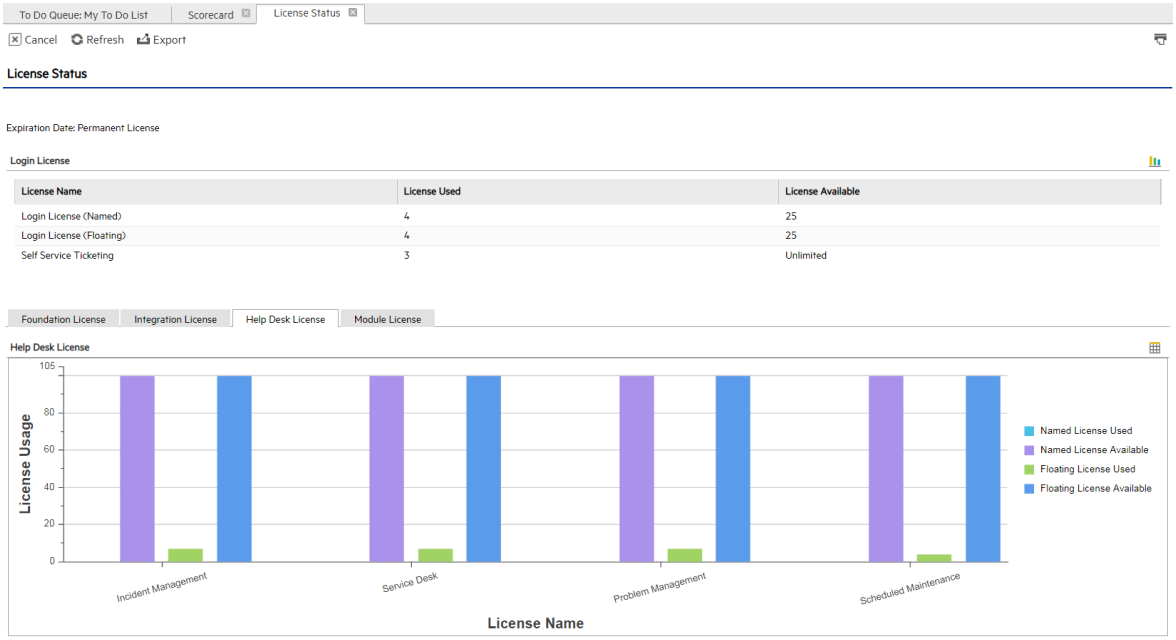
Here are some detailed definitions for your reference:

- **CPU Cores**: The number of CPU Cores for the current host
- **CPU Usage** (on host card): The percentage of CPU usage which indicates how much CPU capacity is in use by the current host
- **Available/Total Memory(GB)**: Available memory refers to how much memory is not already being used by the host; Total memory refers to the total amount of memory that can be used by the host
- **CPU Usage** (on instances/processes): The percentage of CPU usage which indicates how much CPU capacity is in use by the current instance or process
- **Memory Used**: The memory usage of the current instance or process
  Note: The external load balancers aren't monitored or included in SM processes.

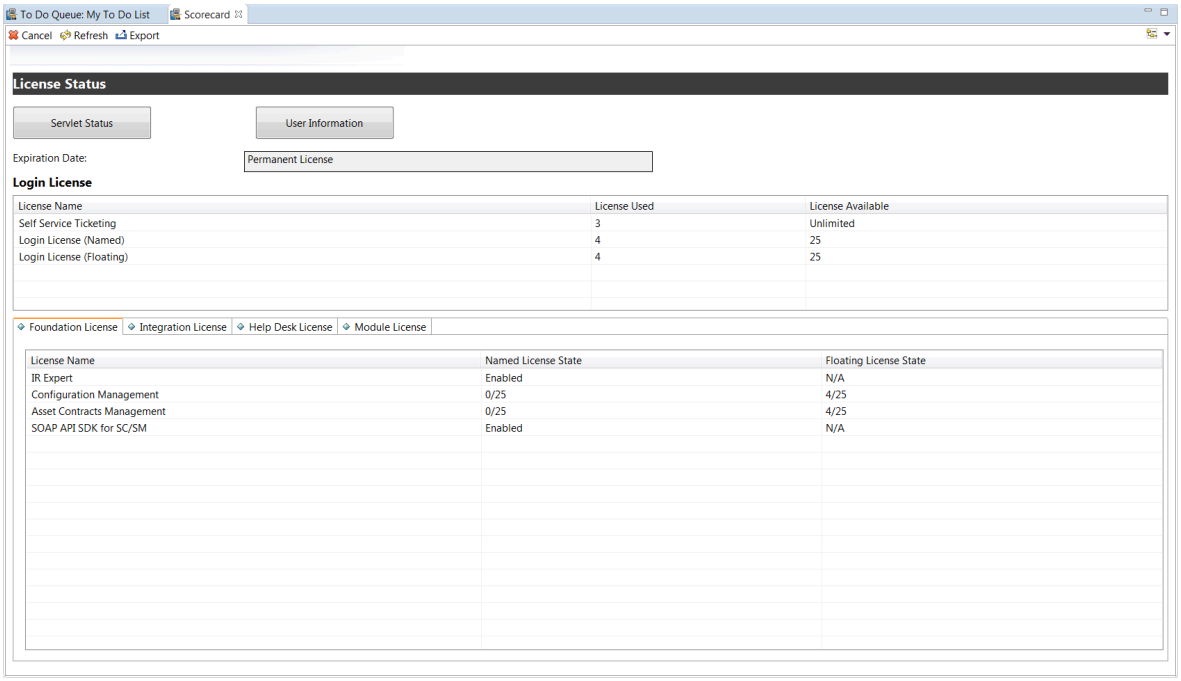
Possible actions:

- You can find out the number of available sessions for each instance
- Click the **Cancel** icon to exit
- Click the **Refresh** icon to synchronize the latest data and information
- Click the **Export** icon to export the servlet information as a formatted report
- Click the icon to print the current page

License status



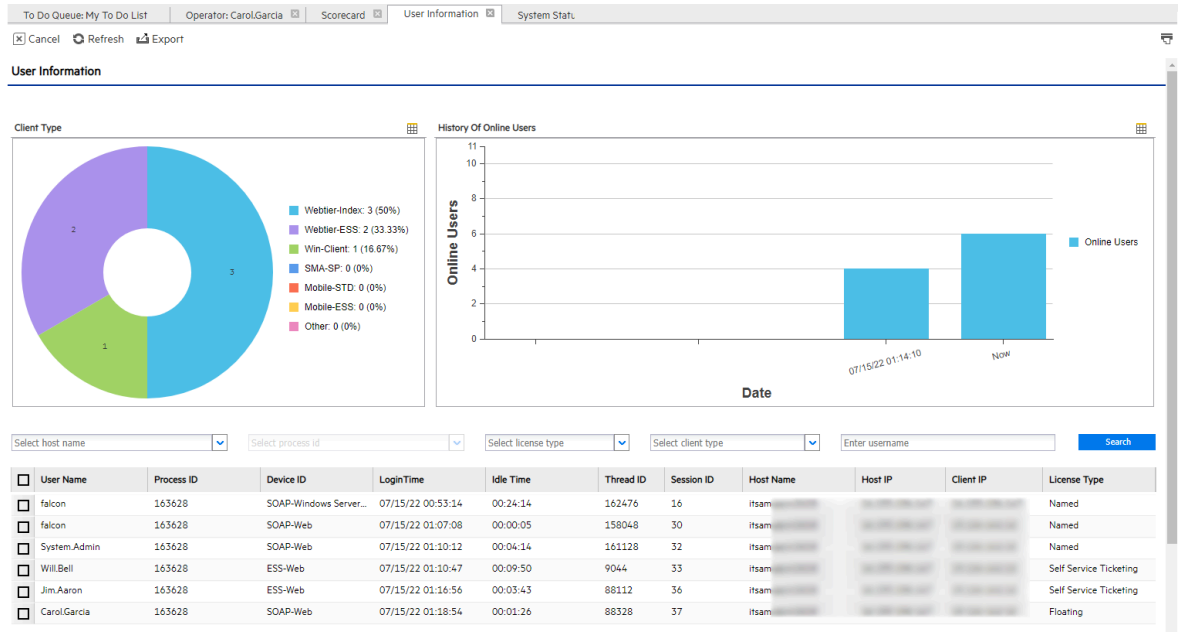
Web Client



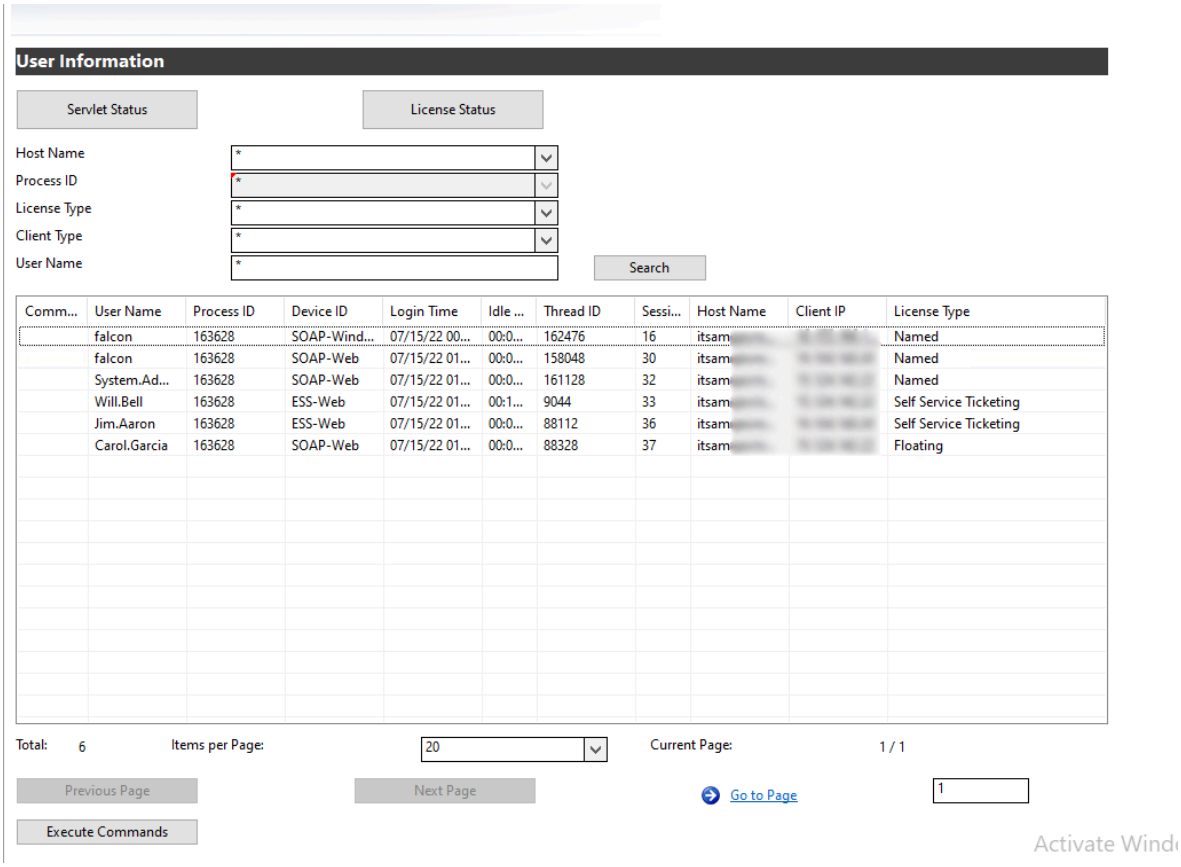
Windows Client

- You can have an overview of applied license information under the **License Status** tab.
  - Current license usage for each license type
  - Total number of available license for each license type
- **Possible actions:**
  - Click the **Cancel** icon to exit
  - Click the **Refresh** icon to synchronize the latest data and information
  - Click the **Export** icon to export the license information as a formatted report
  - Click the icon to print the current page
  - Switch between the view data table and the bar charts by clicking the and icon (for Web Client only)

User information



Web Client



Windows Client

- You can have an overview of online user distributions and user login history under the **User Information** tab.
  - Client type: The number of users that logged in to the system based on each client type
  - History of online users: The total number of users logged in at particular point of time
    - When you click a specific record, the data of Client type will display the online user distributions at the selected point of time
  - User login history: When any user logs in, the information will be recorded in the login history for tracking purposes
- **Possible actions:**
  - Use fuzzy search to filter the user login history by selecting:
    - Host name
    - Process ID
    - License type
    - Client type
    - Username
  - Select one or multiple login records and kill the processes
  - Click the **Cancel** icon to exit
  - Click the **Refresh** icon to synchronize the latest data and information
  - Click the **Export** icon to export the license information as a formatted report
  - Click the icon to print the current page
  - Switch between the view data table and the bar charts by clicking the and icon (for Web Client only)

11.11. Command line options for the system monitor

Users with the system administrators capability word can use the system monitor form to stop or view additional information about any running process on the server. The system monitor command line has the following options:

Command line option	Description
k	Use this command to stop a process.
s	Use this command to view status information about the process or user. You can also use this command to broadcast a message to a particular user.

Related topics

Stop a process
Broadcast messages

11.12. System alerts

System alerts are system events that have passed a pre-defined threshold and that Service Manager writes to the server log (default: sm.log). System alerts contain information about the health and performance of the Service Manager installation. System administrators can define the conditions that trigger system alerts from the Service Manager initialization file.

The server log lists all the system events that have passed a pre-defined threshold. System administrators can review the server log to determine the health and performance of their Service Manager installation. By default, the server log is located in the <Service Manager>\Server\logs folder. System administrators can change what information Service Manager writes to the server log as well as the location of the log from the Service Manager initialization file.

System alert message format

Service Manager writes system alerts to the server log using the following format:

[*Process ID*] [*Date and time*] [*Alert category*]-[*Alert type*]-[*Alert item*], [*Alert text*]

Alert format	Description
[Process ID]	This is the unique numerical identification of the process that generated the system event.
[Date and time]	This is the date and time that Service Manager recorded the system alert.
[Alert category]	This is one of the four basic alert categories. For more information about alert categories see the list of system alerts.
[Alert type]	This is a numerical value that describes the exact conditions that caused Service Manager to record the system alert. For more information about alert types see the list of system alerts.
[Alert item]	This is the application, resource, or user that generated the system event.
[Alert text]	This is a text description of the user and actions that generated the system event.

Sample system alert messages

The following are sample system alert messages:

```
1712 11/03/2003 07:59:00 Performance-1-company, Full File Scan for
query involving fields {show.company} ; user(falcon),
application(display), panel(show.rio)
1712 11/03/2003 07:59:06 Performance-2-cm3t, Partial File Scan for query
involving fields {header,parent.change, header,open} ; user(falcon),
application(display), panel(show.rio)
```

Set a system alert parameter

Applies to User roles: System Administrator

To set a system alert parameter, follow these steps:

1. Stop the Service Manager server.
2. Open the sm.ini file in a text editor.
   This file is in your Service Manager RUN directory.
3. Type the system alert parameters you want to add.
4. Save your changes.
5. Restart Service Manager.
   Service Manager uses your new system alert parameters.

List of system alerts

Service Manager uses the following alert categories.

- Limits
- Mapping
- Performance
- Stalled

Each alert category has a list of alert types that describe what system event conditions triggered the system alert.

Limits

Alert type	Description
Limits-1	The database pool is nearly full. The alert item lists the pool number. This alert type is obsolete in Service Manager.
Limits-2	A user's virtual memory usage exceeds the *alertvirtuallimit* parameter. The alert item lists the name of the user.
Limits-3	A user's CPU usage exceeds the *alertcpulimit* parameter. The alert item lists the name of the user.
Limits-4	The system shared memory is critically close to full.

Mapping

Alert type	Description
Mapping-1	A database field has been truncated and mapped in a compressed format. The alert item lists the truncated file and field names.
Mapping-2	There is a duplicate mapping for a single SQL field. The alert item lists the file name with the duplicate mapping. The alert text lists the field names with the duplicate mapping.
Mapping-3	A query could not be translated into SQL. The alert item lists the file name. The alert text lists the query and the function that could not be translated into SQL.
Mapping-4	There is a field that can't be used in an SQL query because of its data type. The alert item lists the file and field names.

Performance

Alert type	Description
Performance-1	There has been a non-keyed query request that exceeds the *alertquerylimit* parameter. The alert item lists the name of the file that was the target of the query. You can avoid this alert by creating a key to satisfy the query.
Performance-2	There has been a partially keyed query request. The alert item lists the name of the file that was the target of the query. You can avoid this alert by creating a key to satisfy the query or set a reasonable value of environment parameter *alertquerylimit*.
Performance-3	The system has been waiting for a query to return a result but the query has exceeded the *alertwaitlimit* parameter. The alert item lists the name of the lock.
Performance-4	The system has been waiting for a lock to release a resource but the lock has exceeded the *alertholdlimit* parameter. The alert item lists the name of the lock.
Performance-5	There has been a query request that exceeds the *alertquerylimit* parameter. The alert item lists the name of the file that was the target of the query. You can avoid this alert by creating a key to satisfy the query.
Performance-6	There has been a query request that exceeds the *alerthitratio* parameter. The alert item lists the name of the file that was the target of the query. You can avoid this alert by creating a key to satisfy the query.
Performance-7	When building a global list, the system displays an information message if the global list contains more than 500 items.

Stalled

Alert type	Description
Stalled-1	Obsolete alert. This functionality has been removed.
Stalled-4	Obsolete alert. This functionality has been removed.

System alert filters

System administrators can create system alert filters to hide alert messages about system events that they deem aren't detrimental to the system. System administrators can define system alert filters from the Service Manager initialization file. To filter out a system event, system administrators can add the `alertfilters` parameter to the particular event in the initialization file.

Related topics

Server log file

11.13. System events

Service Manager continually monitors its performance and uses the server log (default: sm.log) to track events. These events include information about the following activities and databases:

- SQL database
- Locked resources
- Virtual or shared storage

Service Manager writes information about system events to the server log. Each system event has its own condition that triggers Service Manager to write a message to the server log. System administrators can configure these system event conditions from the Service Manager initialization file. In addition, system administrators can also define how often Service Manager checks for system events by configuring the alert periodic schedule record.

Related topics

System alerts
Configure the alert periodic schedule record

11.14. Service Manager email solutions

The following table describes how the Smart Email solution handles Service Manager emails.

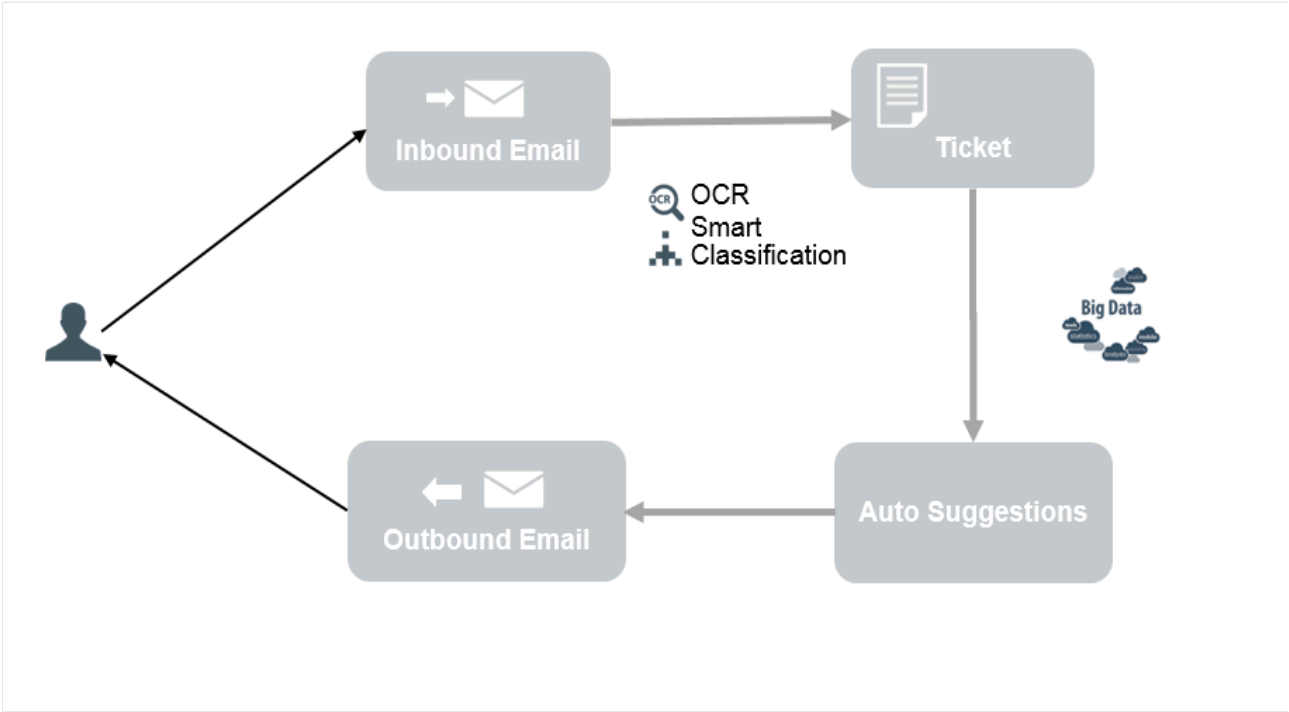
Smart Email	Internal/External Product	Running environment	Connection to SM server	Routing direction	Operating system	Fetch data from	Supported email format
Inbound email	Internal	In Service Manager Server	Connects directly to the SM server	Receives emails to other mail servers	Windows and Linux	External mail servers through Service Manager Integration Suite (SMIS)	Plain text, HTML email
Outbound email (`JavaMail`)	Internal	In Service Manager Server	Connects directly to the SM server	Sends SM emails to other email systems	Windows and Linux	`eventout` table	Plain text, HTML email

11.14.1. Smart Email Overview

The Smart Email solution provides the ability for users to interact with Service Manager through email messages. The supported features are as follows:

- Automatic record creation based on an inbound email
- Automatic reply with links to solutions suggested by virtual agent Sage
- Email-based approval for service catalog items
- Email-based commenting
- Integrating with Smart Ticket
- Supporting Hot Topic Analytics for interactions or incidents created from email

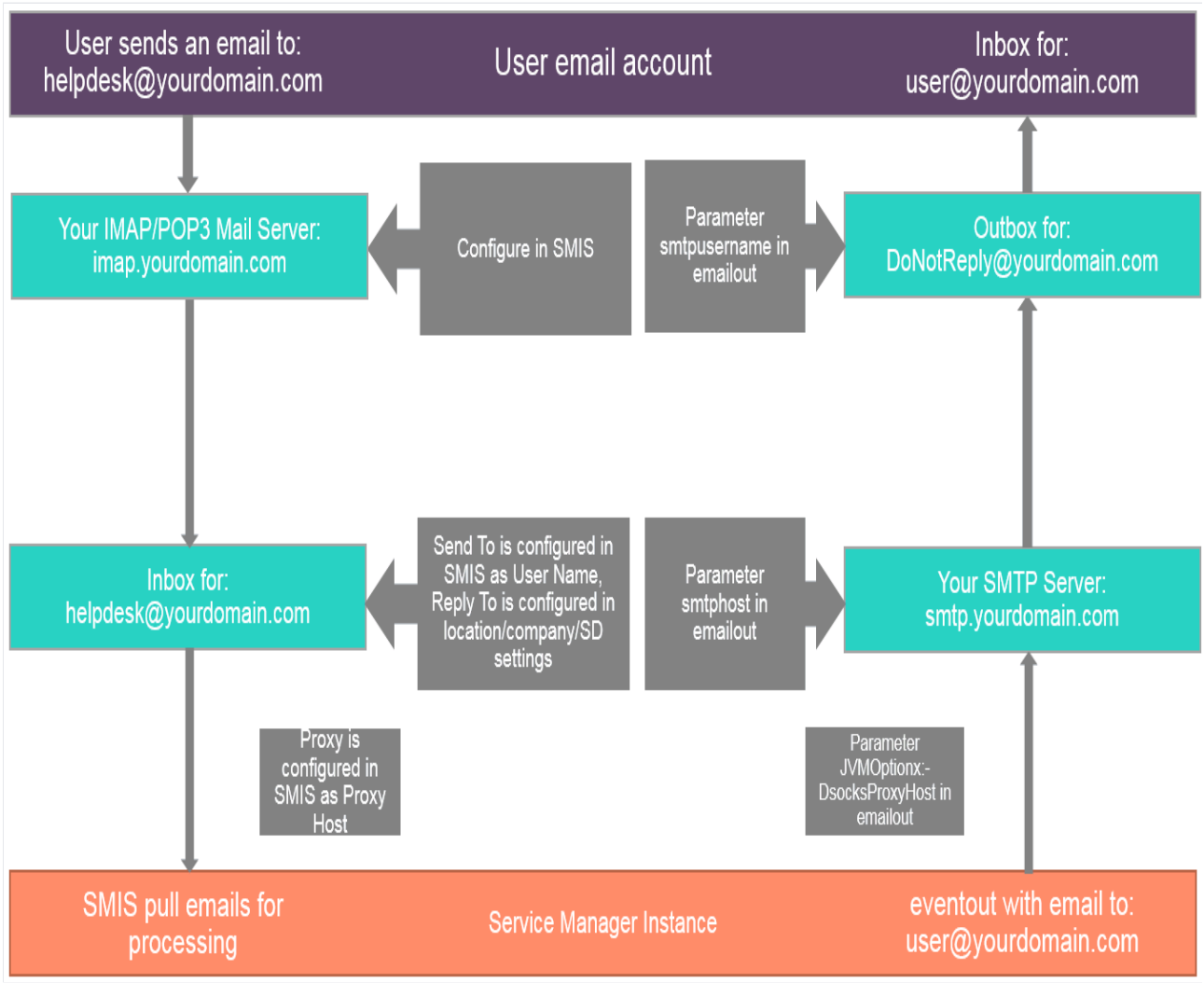
The following diagram provides an overview of the Smart Email solution.



As shown in the diagram, users can interact with Service Manager by using emails:

- When a user sends an email to Service Manager, the email is regarded as an inbound email to the system. Based on the inbound email, Service Manager can create or update a record. If Smart Analytics is enabled, the system automatically categorizes and assigns the record, and then the virtual agent replies to the user with the suggested solutions. For more information, see Automatic record creation.
- When Service Manager sends an email to a user, the email is regarded as an outbound email from the system. In the outbound email, suggested solutions and action buttons are embedded for the user to perform any appropriate task by replying the email. For more information, see Automatic record update.

The following diagram is an example about how the smart email solution is configured and how it works:



Related topics

Automatic record creation
Automatic record update
Smart Email best practices
Set up inbound email
Configure Smart Email settings

11.14.1.1. Set up inbound email

Applies to User roles: System Administrator

To set up inbound email for Smart Email, you must add and enable an instance of the Smart Email integration in Integration Manager.

To add and enable a Smart Email integration instance, follow these steps:

1. Sign in to Service Manager as a system administrator.
2. Click **Tailoring** > **Integration Manager**. Integration Instance Manager is displayed.
3. Click **Add**. The Integration Template Selection wizard opens.
4. Select **SmartEmail** from the Integration Template list, and then select the **Import Mapping** checkbox.
5. Click **Next**. The Integration Instance Information page is displayed.
6. Complete the following settings in the Integration Instance Information page as necessary.

Setting	Value	Description
Name (required)	User-defined. Default: `SmartEmail`.	Name of the integration instance.
Version (required)	1.0	Version number of the integration.
Interval Time (s) (required)	User-defined. Default: `300`.	Polling interval (in seconds) for synchronization.
Max Retry Times (required)	User-defined. Default: `5`.	Maximum allowed number of retries if the background scheduler fails to run.
SM Server	Enter a name that identifies your Service Manager server host. For example, `sm_host1`.	Display name for your Service Manager server host.
Endpoint Server	Enter a name that identifies your mail server host. For example, `mail_server1`.	Display name for your mail server host.
Log Level	Select a level from the drop-down list. Default: `INFO`.	Level of diagnostic information that the Service Manager server logs to the log file directory. The possible log levels are: DEBUG, INFO (default), WARNING, ERROR, and OFF.
Log File Directory	Specify a directory. For example: `c:\smartemail_logs`.	This is a directory that exists on the Service Manager server host where log files of the Service Manager and mail server integration are stored. The log includes information for receiving and processing emails. If this field is empty, the system uses the Service Manager sever log (sm.log) for Smart Email. If you set the log parameter for the Smart Email processor separately, the log file specified here only contains the logs for receiving emails while the log file set for the processor contains the logs for processing emails.
Run at system startup	True or false.	Automatically enable or disable the integration instance when the Service Manager server is started.
Description	User-defined.	Description of the integration instance.

7. Click **Next**. The Integration Instance Parameters page is displayed.
8. Complete the following settings in the **Connectivity** tab as necessary.

Basic Configuration

Setting	Value	Description

Mail Protocol	Select a protocol from the drop-down list.	The mail protocol for receiving emails. The supported protocols are IMAP, POP3, EWS, and Microsoft Graph. To use a secure protocol, select IMAPS or POP3S, or use HTTPS for EWS. **Note** ○ If you use a secure protocol, Service Manager only connects to the application or site that Service Manager trusts. That is, you must have a CA-signed certificate within the Java keystore that Service Manager uses. If you use a self-signed certificate, you need to import the certificate into the Java keystore (<SM installation>\Server\RUN\jre\lib\security\cacerts). Otherwise, Service Manager fails to connect to the mail server. ○ Using IMAP or POP3 together with STARTTLS is unsupported. ○ After you enable the FIPS mode, you can use the POP3 protocol only.

		Note Microsoft has announced the deprecation of Basic authentication in Exchange Online. Based on this fact, we recommend customers who are connecting to Exchange Online mail server for Smart Email select the Microsoft Graph or EWS with OAuth2 authentication as the mail protocol.
Shared Inbox	True or false. Default: `false`	This option is only available when you select the EWS or IMAPS mail protocol. If you enable this option, you can configure a shared mailbox in the **Shared Mailbox** field as the mailbox for inbound emails.
OAuth Authentication	True or false. Default: `false`	This option enables OAuth2 authentication. It is only available when you select the EWS or IMAPS mail protocol. If you enable this option, the system displays the **OAuth Authentication** tab and you can configure the related settings. **Note** The system displays the **OAuth Authentication** tab as well if you select the Microsoft Graph mail protocol.

IMAP/POP3 Mail Server	Specify an IP address or domian name. For example: `mail.example.com`.	The IP address or the fully qualified domain name (FQDN) of the IMAP or POP3 server. **Note** ○ If you select the EWS mail protocol, this field will be **EWS URL**. Enter the Exchange Web Service URL. For example, https://outlook.office365.com/EWS/Exchange.asmx ○ If you select the Microsoft Graph mail protocol, Service Manager hides this field since it requires no input.
IMAP/POP3 Port	Specify a port number. For example: `143`	The port number of the email server. The defaults are: ○ IMAP (non-secured): 143 ○ IMAP (secured): 993 ○ POP3 (non-secured): 110 ○ POP3 (secured): 995 ○ EWS: port not required

User Name	Type an email account. For example: `abc@exmaple.com`	The email account that's used to connect to the mail server. When a Service Manager user sends an email to this email account, the system processes the email based on the settings in the Smart Email integration instance. When you are configuring multiple Smart Email integration instances on one Service Manager server, we recommend using a unique mailbox for each instance. **Tip** If you select EWS in the Mail Protocol drop-down list, you can also type a domain account as the user name. For example, `example.com\abc`.
Password	Type a password.	The password of the email account that's used to connect to the mail server. This field is unavailable when you check **OAUTH Authentication**. **Note** According to Google's announcement, as of end of September 2024, user can no longer use a password for access gmail. For customers who are using Gmail as mailbox, currently the workaround is to use "app password" instead of user password for this field.

Shared Mailbox	Type an email address.	This field is only available when you select the **Shared Inbox** checkbox for the "EWS" or "IMAPS" mail protocol. Configure your shared mailbox in this field as the mailbox for receiving inbound emails. The email account in **User Name** must be an authorized user to access this shared mailbox.

Advanced Configuration

Setting	Value	Description
Proxy Host	Specify an IP address or domain name.	If you need to use a proxy server to connect to the mail server from Service Manager server, type the FQDN or IP address of the proxy server in this field. Only the SOCKS proxy (Socket 4 and Socket 5)is supported. The HTTP proxy is unsupported because JavaMail doesn't support direct access through an HTTP proxy web server. The SOCKS proxy configured in SMIS takes effect on the inbound email only without any impact on the outbound email.
Proxy Port	Specify a port number.	The port number of the proxy server.
Proxy Username	Type an account name.	The user name that's used to connect to the proxy server.
Proxy Password	Type a password.	The password that's used to connect to the proxy server
Authorized Domains	Specify one or multiple domain names (separated by semicolons). For example: `example.com` .	The domains that are authorized to send emails to the email account specified for the integration. The Smart Email processor rejects the emails from unauthorized domains and triggers the corresponding Smart Email error handling case. You must specify at least one authorized domain. Otherwise, no incoming email can be successfully processed by the Smart Email processor.

Inbox Folder	Specify the inbox folder. Default: INBOX	The folder where e-mail messages are received in the e-mail account. **Note** Only the root folder in the mailbox is supported. Using a subfolder is unsupported. **Note** If you leave it empty, Smart Email use INBOX as the default inbox folder.
Error Email Folder	Specify a folder name. For example: `Error`.	The folder in the email account to which the messages with errors are moved. **Note** Only the root folder is supported. Using a subfolder is unsupported. **Note** If you use the POP3 or POP3S protocol, this option doesn't take effect due to protocol limitations and the system doesn't display this option.
Delete Email when successfully processed	True or false. Default: `true`	Whether or not to delete the emails from the mail server once they are successfully processed.

Processed Email Folder	Specify a folder name. For example: `Processed`.	The folder in the email account to which the successfully processed emails are moved. If you select **Delete Email when successfully processed**, this option doesn't take effect and the system doesn't display this option. **Note** Only the root folder in the mailbox is supported. That is, a sibling folder to the Inbox folder. Using a subfolder is unsupported. **Note** If you use the POP3 or POP3S protocol, this option doesn't take effect due to protocol limitation and the system doesn't display this option.

Tip
To verify the connection, after you set up the email integration, send an email to the configured mailbox. If the connection fails, as configured in Smart Email error handling, the administrator will receive a "Failed to connect to the mail server" email notification and the system will create an incident.

Note
You can also configure the `smartemailTimeout` parameter in sm.ini to define the timeout value that the Smart Email process waits for a connection to the mail server to be established. The default value is 45 seconds. See Parameter: smartemailTimeout.

9. Click the **OAuth Authentication** tab, and then configure the following settings as necessary.
For IMAPS mail protocol, you may access Gmail via OAUTH protocol. To enable gmail and get Client ID/Client Secret from google cloud console, see https://console.cloud.google.com/welcome.

Setting	Value	Description
Authorization Grant Type	Authorization Code	The OAUTH Authenticate grant type
Client ID	The Client ID	The Client ID you received from google cloud console
Client Secret	The Client Secret string	The Client Secret you received from google cloud console

Clicking **Get Access Token** will jump to google console to get the access token.

For EWS mail protocol and Microsoft Graph, fill in the **OAuth Authentication** settings as below:

Setting	Value	Description
Authorization Grant Type	○ Client credentials if you select the EWS mail protocol together with OAuth Authentication ○ Authorization Code or Resource Owner Password Credentials if you select the Microsoft Graph mail protocol	The OAUTH Authenticate grant type
Application (Client) ID	The Application (Client) ID	The Application (Client) ID you received when registering the application
Directory (Tenant) ID	The Directory (Tenant) ID	The Directory (Tenant) ID you received when registering the application
Client Secret	The Client Secret string	A secret string that the application uses to provides its identity when requesting a token.
Permission Type	○ Application Permissions if you select the EWS mail protocol together with OAuth Authentication ○ Delegated Permissions if you select the Microsoft Graph mail protocol	The permission type of authentication.

For **Microsoft Graph** with **Authorization Grant Type** as **Authorization code**, the **Get Access Token** option is available. Clicking **Get Access Token** will jump to Azure portal to get the access token.

For more information for EWS protocol and Microsoft Graph, see the **Register your application** section on the Authenticate an EWS application by using OAuth page.
To use Delegated Permissions for Smart Email, you need to perform these additional steps in the Azure Active Directory admin center and Azure portal.
In the Azure Active Directory admin center:

1. Click **App Registrations** > <*Name of your application*> > **API permissions** and then click **Add a permission**.
2. Click **Microsoft Graph** and then click **Delegated permissions**.
3. Expand **Mail** and select **Mail.ReadWrite**. If you are using a Shared Inbox, select **Mail.ReadWrite.Shared** as well.
4. Grant admin consent to the delegated permissions as described in the previous step.
5. Add `https://<SM Webtier Host>/<Context Path>/oauth/callback` as the **Web Redirect URIs**. Where `<SM Webtier Host>/<Context Path>` should match the Web Server URL defined in System Information Record. For example: `https://myserver.mydomain.com:myport/webtier/oauth/callback`

Make sure that you have added Web Server URL for Service Manager clients. See Configure the Service Manager Web tier server URL.

In Azure portal, disable security defaults on Azure and disable multifactor authentication (MFA). For detailed instructions, see Security defaults in Azure AD.

10. Click the **Options** tab, and then configure the following Smart Email options as necessary.

Basic Options

Setting	Value	Description
New Email Action	Select an action from the drop-down list.	The action of Smart Email upon the receipt of a new email message: ○ Create Interaction: Each initial message automatically results in the creation of a new Interaction. ○ Create Incident: Each initial message automatically results in the creation of a new Incident. ○ Create Change: Each initial message automatically results in the creation of a new Change. **Note** This setting takes effect in the current Smart Email instance only. When you are configuring multiple Smart Email integration instances on one Service Manager server, we recommend using a unique mailbox for each instance. Hence, when an email is sent to the mailbox, the system can create an interaction, an incident, or a change as defined in the New Email Action in the current Smart Email instance.

Allow to update records by incoming emails	True or false. Default: `false`	Once enabled, the system automatically updates a record upon the receipt of a new email message of a particular format. Depending on your selection in the New Email Action field, the types of records that can be updated are Incident, Interaction, and Change. Suppose you have selected Create Interaction in the New Email Action field, the system automatically updates an Interaction record upon the receipt of a new email message. **Note** This setting takes effect in the current Smart Email instance only. Only the fields that are configured in *step 13* can be updated by incoming emails.
Enable email size restrictions	True or false. Default: `false`	Once enabled, you can define the email size (in KB) so that only the emails under the specified size are processed. If the user's email is larger than this value, the system triggers the corresponding Smart Email error case and notifies the sender with an email. For more information, see Smart Email error handling. **Note** If the size specified in this field is bigger than any other attachment size configured in Service Manager, the system ignores the value specified in this field.
Enable email archiving	True or false. Default: `true`	Once enabled, the system stores the inbound emails in a database table (emailarchivingstore).

Use default operator to create or update records	True or false. Default: `false`	If an end user (contact) is not associated with an operator, the system uses the default operator specified here to create or update records. **Note** If you enable this option, when Smart Analytics is used for smart email, similar recommendation or classification might be generated for different end users because the same default operator is used.

Smart Analytics Options

Setting	Value	Description
Enable Smart Ticketing	True or false. Default: `false`	Once enabled, the system automatically classifies the record created from an email by using Smart Analytics. The OCR (image to text) feature of Smart Analytics is also enabled. **Note** This option is only visible if Smart Analytics is enabled.

11. Click the **Error Handling** tab, and then configure the error handling settings in each sub-tab as necessary.
    1. On the **Configuration** sub-tab, specify the configurations for each error case.

Setting	Value	Description
Create Incident	Select Yes or No.	Select **Yes** to create an incident record for this error case. You can configure the field values of the incident in **Incident Options**.
Email to Admin	Select Yes or No.	Select **Yes** to send an email to the Admin Email specified in **Email Options**.
Email to Sender	Select Yes or No.	Select **Yes** to send an email to the sender of the initial email.
Email to Others	Select Yes or No.	Select **Yes** to send an email to the operators or email addresses specified in **Email Options**.
Error Message	SC message ID or plain text.	The error message in the mail. You can specify an SC message or type plain text. **Note** If you want to use the SC message, define it as scmsg: *<scmessage_id>*. The scmsg class should be SmartEmail.

2. On the **Incident Options** sub-tab, define the default field values in the incident record that the system creates when an error occurs during email processing.
3. On the **Email Options** sub-tab, specify the admin email address and other email addresses or operators to receive email notifications configured on the **Configuration** sub-tab.

Setting	Value	Description
Admin Email	Email address or operator name	Specify an email address or operator name for the administrator. When the **Email to Admin** option is set **Yes**, the system sends out an email to the email address specified here.
Other Emails	Email address or operator name	Specify one or multiple email addresses or operator names. When the **Email to Others** option is set to **Yes**, the system sends out an email to the email addresses specified here.

12. Click **Next**. The Integration Instance Fields page is displayed.
13. Configure the SM fields and the endpoint fields as necessary, and then click **Next**. The Integration Instance Mapping page is displayed.

14. Configure the field mapping between the message from the mail server and the SM record (interaction or incident) to be created based on the inbound email.

    Note

    The out-of-box field mapping in the SMIS template is based on interaction fields. You can use this mapping when you choose to create a new interaction for an inbound email. However, if you selected **Create Incident** in the **New Email Action** option, you need to manually re-configure the field mapping for incident fields.

    Note

    If format control "login.DEFAULT" is not executed properly in your environment, some variables can't be initialized. You need to reinitialize these variables in SMIS ("Post Script" in field mapping area) based on the current operator.

15. Click **Finish**. Service Manager creates the instance.
16. Enable the integration instance. A processor for the integration instance (<Smart Email SMIS instance name +instance ID>) that receives inbound emails now runs in the background.

    Tip

    When the status of the SMIS instance for inbound email goes to "Error", the system administrator needs to navigate to **Miscellaneous** > **System Status**, locate the "<Smart Email SMIS instance name +instance ID>" process and kill it. After that, go back to Integration Manager and edit the Smart Email SMIS instance as necessary.

17. Start the Smart Email processor that processes the received emails:

    Note

    You can configure the number of SmartEmail processors in Smart Email settings. If the configured number is more than one, you can start different debug nodes to run different processors.

    Note

    If the server running the debug node has multiple IP addresses, make sure that you set a public IP address for the groupbindaddress parameter in sm.ini. Otherwise, the Smart Email solution can't provide solutions in the emails because the server can't connect to the Smart Analytics Web Service API.

    1. Add a debug node in sm.cfg. For example:

       ```
       sm -httpPort:13081 -httpsPort:13082 -log:../logs/sm.13081.log -debugnode
       ```

    2. Restart Service Manager and log on to Service Manager with the debug node:
       Window Client: In the **Server port number** field, enter the debug node port (for example, 13081).
       Web Tier client: Append **serverPort=<*debug node port*>** to the web tier URL. For example:
       `http://myhost:8080/webtier-9.70/index.do?serverPort=13081`
    3. Navigate to **Miscellaneous** > **System Status**, click **Start Schedulers**, and then start SmartEmail_Processor<X>.

Related topics

Integration Manager
Integration Instance Information fields
Enable or disable an integration instance
Monitor integration instance status

11.14.1.2. Configure Smart Email settings

Applies to User roles: System Administrator

System administrators can configure Smart Email global settings such as outbound email archiving, one-time token, testing email, the number of acknowledge documents and service offerings proposed by Smart Analytics, and so on.

To configure the Smart Email settings, follow these steps:

1. Click **System Administration** > **Ongoing Maintenance** > **Smart Email** > **Settings**. The Smart Email Settings form is displayed.
2. Specify the settings on the form, as described in the following table.

Setting	Description	Default Value
Activate HTML template audit	Enable or disable the audit application to monitor field value changes in HTML templates. When this setting is set to true, the audit application monitors the following fields in the ltemplates" table: module, language, name, mail.title, content, active, expression, htmlcode, and cap.exec. **Tip** Tip You can view the audit log from **Tailoring** > **Audit** > **Audit Log**.	True
Activate outbound email archiving	Specify whether you want to archive outbound emails. If you select this option, the system automatically stores all the outbound emails in the emailarchivingstore table. To view the archived email, you can also go to **System Administration** > **Ongoing Maintenance** > **Smart Email** > **View Archived Emails**.	False
Save the email attachments to records	Specify whether you want to save the attachments of a smart email to the corresponding record. If you select this option, the system automatically stores the attachments of a smart email to an interaction, an incident, or a change record that is created based on the initial email. In the meantime, these attachments will not be removed from the attached smart email in a record. **Note:** If the email attachment's size exceeds the **Max Attachment Size** or the total attachment size of a record exceeds the **Total Attachment Size**, Service Manager no longer stores the email attachments to the record regardless of this option's setting.	False

Use one-time token	Specify whether you want to use a one-time token in smart emails. If you select this option, only the first reply that uses this token is valid. The system ignores any subsequent emails that use the same token.	False
Clean up temporary data store for emails received before this date	The system removes the received emails earlier than the specified date from the temporary data store in Service Manager. The selected date must be earlier than current day and in the *YYYY-MM-DD hh:mm:ss* format. **Note** Note If you use the POP3 protocol, make sure that the received emails older than the specified date have been manually deleted from the mailbox before you clean up the email store in Service Manager. Otherwise, the system might handle these emails once again.	None

Number of SmartEmail processors	Specify the number of processors to process inbound emails. For example, you may want to specify more than one processor when your system is in a horizontal or vertical scaling implementation. Possible values: no less than 1 **Note** Note Before you modify the number for this setting, make sure that you have terminated all the running "SmartEmail_Processor " schedulers from **Miscellaneous** > **System Status**. After you modify the value and click the **Save** button, you need to log in to Service Manager with a debug node and start the Smart Email processors. For details, see Set up inbound email.	1

Number of schedules per SmartEmail processor	Specify the number of background schedules for one SmartEmail processor. Possible values: 1 - 10 **Note** Note You need to terminate all the running "SmartEmail_Processor " schedulers before you modify the value here and restart the processors after you make the change. **Caution** Caution If you are using the POP3 or POP3S protocol, you must set this value to 1 to avoid the concurrency in Smart Email.	1
Recipient email address for testing	Specify an email address with which the system replaces all the recipient email addresses, so that the system only send emails to this email address for testing purpose.	None
Reply with this number of knowledge documents proposed by Smart Analytics	Specify the number of knowledge documents that you want to include in a smart email. This option only takes effect when you enable Smart Analytics in your system. The knowledge documents are automatically suggested by the virtual agent Sage. Besides the internal KM library, the virtual agent Sage can also propose documents from external libraries such as a SharePoint repository, a file system, or a web site if the corresponding connectors are set up in Smart Analytics. Possible values: 0 - 5	5

Reply with this number of service offerings proposed by Smart Analytics	Specify the number of service offerings that you want to include in a smart email. The service offerings are automatically suggested by the by virtual agent Sage. This option only takes effect when you enable Smart Analytics in your system. Possible values: 0 - 5	5
Retention period for archived emails	Specify how long the system archives the emails. This setting takes effort for both inbound and outbound emails.	7 00:00:00
Token expiration period	Specify the expiration period of the smart email token. If the token is expired, the system ignores the email that uses this token.	1 00:00:00

3. Click **Save**.

11.14.1.3. Automatic record creation

After a Service Manager user sends an email to the email account configured in the Smart Email integration instance, Service Manager processes the email and automatically creates an interaction, an incident, or a change record based on the initial email.

The system administrator can configure the **New Email Action** setting in the Smart Email integration instance from SMIS to determine if an interaction, an incident, or a change is created automatically. For the system to successfully create the record, the sender of the email must have the "New" security right for the corresponding module.

Service Manager creates the record in the system by completing the following:

- Service Manager automatically populates the fields of the new record according to the field mapping defined in the email integration instance. For example, the system populates the **Title** field and the **Description** field by using the content in the mail subject and the mail body respectively.
- Service Manager populates the **Source** field of the record as "5 - E-mail". If you enable Smart Analytics, the **Source** field is used as an advanced filter option in Hot Topic Analytics.
- If Smart Analytics is enabled and the "Enable Smart Ticketing" option is selected in SMIS, the smart classification and OCR take effect in the new record.
- The inbound email is saved as a .eml file under the **Attachments** tab from the record. If the email contains attachments, the attachments are also saved in the .eml file. However, the system ignores the inbound email if the total size of the email (including attachments) exceeds that size limit configured in the Smart Email integration instance.
- The system adds an activity log for each received email. The activity type is "Email Received" and the activity description is copied from the subject of the inbound email.

After the record is created, Service Manager generates an email to notify the email sender that a new record is created based on the information in the initial email. Within the email reply, the "Add a Comment" and "Mark as Solved" action buttons are available for further interaction with the Service Manager user.

If Smart Analytics is enabled, the administrator can configure the Smart Email setting to include links to knowledge documents and service offerings suggested by the virtual agent Sage, which is empowered by Smart Analytics, in the email reply. Depending on the configuration of Smart Analytics, the proposed knowledge documents can be KM articles, SharePoint pages, documents on a file system, or web pages.

In addition, if the corresponding assignment group is out of work schedule, the out-of-office message will be displayed in the email.

Note

Note

- For the Smart Analytics features (including smart classification and recommendation of knowledge documents and service offerings) to take effect, the end user (that is, email sender) must have the RESTful API capability word. In the out-of-box system, the end users normally don't have this capability word assigned in their operator records.
- Smart Analytics and the related features (such as smart classification) are unnecessary for change creation.

Related topics

Smart Email actions in HTML templates
Set up inbound email
Configure Smart Email settings

11.14.1.4. Automatic record update

In the Smart Email solution, users can use the action buttons in an email to reply to the system. Then, Service Manager automatically updates the record based on the selected action and the email message.

Action in Smart Email	Description
• Add a Comment • Mark as Solved	If an IT agent performs a "visible to customer" activity update, Service Manager automatically sends out an email that contains the **Add a Comment** and **Mark as Solved** buttons in addition to the update message. When the user receives the email, the user can click **Add a Comment**, add comments in the email body, and then reply the email. Then, Service Manager automatically updates the corresponding record with the comments in the email. **Note** Note If the email content is empty without any comments, Service Manager doesn't update the record. Or, the user can click **Mark as Solved** and reply the email to inform the IT agent that the issue has been resolved. In this case, Service Manager automatically changes the status of record to "Resolved" and populates the solution field with the email content. **Note** Note If the email content is empty, Service Manager populates the **Solution** field with a default value “Mark as solved by end user”.
• Accept • Reject	If an IT agent provides a solution in the record, Service Manager automatically sends out an email that contains the **Accept** and **Reject** buttons in addition to the solution message. The user can click the **Accept** or **Reject** button to reply the email. Then, the system automatically updates the record as follows: • If the user accepts the solution, Service Manager changes the status of the record to "Closed". • If the user rejects the solution, Service Manager changes the status of the record to "Work In Progress".

• **Approve** • **Deny**	If a service catalog request, a change, or a request is pending approval, Service Manager automatically sends out an email that contains the **Approve** and **Deny** buttons to the approver. The approver (or approval delegate) can click the **Approve** or **Deny** button to reply the email. Then, the system automatically updates the record as follows: • If the approver approves the approval request, Service Manager changes the approval status of the record to "approved" and copies the email content to the approval log. If the email content is empty, a fixed string "Approved by email – [email subject line]" is added to the approval log. • If the approver denies the approval request, Service Manager changes the status of the record to "denied" and copies the email content to the comments of approval log. If email content is empty, a fixed string "Denied by email – [email subject line]" is added to the approval log.
• **Close**	Service Manager automatically sends out an email that contains the **Close** button to the assignee when either of the following conditions are met: • When the **Close** button is already available in a Change task ticket, an operator updates the value in the **Assignee** field. • When the **Close** button is unavailable in a Change task ticket, an operator updates the phase of a Change task ticket so that the **Close** button becomes available. If the assignee closes the Change task, Service Manager changes the status of the record to **Closed** and copies the email content together with the "Closed from email" string to the Closure Comments field. If the email content is empty, only the fixed string "Closed from email" is added to the Closure Comments field. The Closure Code is "1-Successful".

In addition, when the system automatically updates the record, it also completes the following tasks:

- In the activity log, the system adds an update with a specific activity type such as "Accepted from Email" or "Rejected from Email". The activity description is copied from the subject of the inbound email.

  Note

  Note If the **Add a Comment** button is clicked and the email content is not empty, the activity description will be the email content.

- The entire email is saved as a .eml file under the **Attachments** tab from the record.
- If the record is associated with other records, the system also updates the status of the related records accordingly.

Reply email address

After the user clicks an action button, the reply email address is automatically populated. Likely the reply email address is different from the email account that the administrator set in the Smart Email SMIS instance, because the reply email address is calculated for each email receiver (end user) as follows:

1. The system looks for the location of the contact that's associated with the email receiver and uses the **Service Desk Email** field of location as the reply email address.

   > **Note**
   > Note
   > Locations are hierarchical. The loop starts from the contact's current location and checks whether there's an email address or not. If not, the loop goes to the parent level and performs a check again. This cycle continues till the loop reaches the top level of the hierarchy and will stop the loop as soon a location has the **Service Desk Email** field populated.

2. If the **Service Desk Email** field is always empty in locations, the system looks for the company of the contact and uses the **Service Desk Email** field in company record of the contact as the reply email address.
3. If **Service Desk Email** is also empty in company, the system uses the "Service Desk Email" documented in Service Desk Environment.

Related topics

Smart Email actions in HTML templates
Set up inbound email
Configure Smart Email settings

11.14.1.5. Automatic record fields update

After a Service Manager user sends an email to the email account configured in the Smart Email integration instance, Service Manager processes the email and automatically updates the record based on the email message.

The system administrator can configure the **New Email Action** setting in the Smart Email integration instance from SMIS to determine whether an interaction, an incident, or a change is updated automatically based on an incoming email. Meanwhile, a system administrator must set the **Allow to update records by incoming emails** option to true. For more information about the inbound email configuration, see Set up inbound email.

Permission settings

For the system to successfully update the record, the sender of the email must have the "Update" permission. If the system administrator has enabled folder entitlement in System Information Definition, the sender of the email must have the folder update permission. In addition, after folder entitlement is enabled, the System Administrator security role must have the "New", "View", and "Update" security rights for the folders in the interaction, incident, and change modules. Hence, the SmartEmail_Processor background process is authorized to update the tickets in the corresponding folders.

Inbound email format

For the system to successfully update the record, you must define the email format as follows:

Subject

The email subject must be `[Update <ticket ID>] <Description>`. The system processes the content in the square brackets only and the <Description> section is optional.

Content

The email content must be `[Field name]=<value>`. `[Field name]` is the name of the field that's defined in the database dictionary and only the character, boolean, number, datetime, simple array types (such as array-character, array-boolean, array-number, and array-datetime) are supported.

You can clear a field by setting the `<value>` to `NULL`. For example, `[Field name]=NULL`. If you set the value to an empty string, the field is not updated.

If a field name doesn't exist in the database dictionary, or the system can't resolve the value, or the value doesn't match the type, the system ignores this line in the email.

- If the field is array type, don't wrap the lines but use `\n` instead to separate multiple array elements. For example, `[Field name]=<value1>\n<value2>\n<value3>`.
- If the field is character type and there are double quotation marks at the beginning and end of the value, the system automatically filters out the double quotation marks.
- If the field is character type in the database dictionary but has the number type in the SMIS Integration Instance Fields setting, the system processes this field as the number type.
  For example, the severity (Urgency) field in a Change record is defined as the character type in the database dictionary, whereas the Smart Email Instance Fields setting defines this field as the number type. In this case, the system processes the severity (Urgency) field as the number type when updating this field. You must use the actual value instead of the display value. For example, use `[severity]=3` instead of `[severity]= 3 - Average`.
- If the field is boolean type, `<value>` can be `true`, `t`, `false`, or `f`. Your input is case-insensitive.
- If the field is datetime type, the system processes the value according to the time format and time zone setting in the corresponding operator profile of the email sender in Service Manager. Meanwhile, if the value is an offset duration, the format must be `+'NN hh:mm:ss'` or `-'NN hh:mm:ss'`. For example, if you set the value to `+'1 00:00:00'`, the system will set the time value by adding one day forward when the mail is processed.

After the value of a field is updated by an email, Service Manager may automatically set the value of the field based on certain mechanisms. Consequently, when you open the updated ticket, the value you see is the one that's automatically set instead of the value as you defined in the email.

After the record is updated, Service Manager generates an email to notify the email sender that the corresponding record is updated based on the information in the initial email. Within the email reply, the Service Manager user is able to review the list of updated fields and values.

Inbound email example

See the following example for an inbound email. The user requests to update the Change record C10066.

Subject: [Update C10066] Request to upgrade SAP Kernel

Content:

```
[brief.description]= SAP Kernel Upgrade - Hasn't upgraded yet
[description]= Hello,\n I request the change to upgrade SAP Kernel on adv-Server-106 and adv-Server-108
servers from patch level 23 to patch level 69 during the CPU replacement maintenance window on Oct 25th
 from 6 AM to 6 PM one week ago. But the current status is that still on Patch 23 which hasn't been
upgraded. And our project is blocked. Please provide the reason and the next step action. \n Best Regards
[initial.impact]= 1
[severity]=2
[requestedDate]= 20/11/23 11:58:34
[assigned.to]= Change.Manager
```

Service Manager email notification example

See the following email notification example after Service Manager updates the C10066 record.

Subject: Your email has updated the ticket successfully.

Content:

```
You have updated the ticket via email successfully.

The originating email information:
Subject         [Update C10066]
Module          Change
ID                  C10066
Action          Update

You have requested to update the following fields:
Field                       Value
brief.description               SAP Kernel Upgrade - Hasn't upgraded yet
description                     Hello,\n I request the change to upgrade SAP Kernel on adv-Server-106 and
adv-Server-108
                        servers from patch level 23 to patch level 69 during the CPU replacement
maintenance window
                        on Oct 25th from 6 AM to 6 PM one week ago. But the current status is that still
on Patch 23
                        which hasn't been upgraded. And our project is blocked. Please provide the reason
and the next
                        step action. \n Best Regards
initial.impact              1
severity                        2
assigned.to                     Change.Manager
```

Note that the system skips the requestedDate field because the format doesn't match the user's time format. Hence, the system neither updates this field nor lists this field in the email notification.

11.14.1.6. Email archiving

In the Smart Email solution, inbound and outbound emails can be archived for a specified period of time. If email archiving is enabled, the system stores the email in the emailarchivingstore table after an email is received or sent out.

The system administrator can enable or disable email archiving for inbound or outbound email separately:

- To enable email archiving for the inbound emails, select the **Enable email archiving** checkbox in the Smart Email integration instance.
- To enable email archiving for the outbound emails, select the **Activate outbound email archiving** checkbox in the Smart Email settings (**System Administration** > **Ongoing Maintenance** > **Smart Email** > **Settings**).

The system administrator can also specify how long the system stores the archived emails by configuring **Retention period for archived emails** in the Smart Email settings (**System Administration** > **Ongoing Maintenance** > **Smart Email** > **Settings**). This configuration takes effect for both the inbound and outbound emails.

View archived emails[edit]

If email archiving is enabled, users can click **More** > **View Archived Emails** from a record to view the archived emails related with this specific record. In addition, you can also view all the archived emails from **System Administration** > **Ongoing Maintenance** > **Smart Email** > **View Archived Emails**.

Contents

- 1 View archived emails

11.14.1.7. Smart Email error handling

During the Smart Email integration instance setup, the system administrator can configure the error handling settings to define the error handling behaviors:

- Whether or not the system sends out an notification to the sender, or the predefined administrators or operators. The system uses the "SmartEmailErrorHandling" HTML template for the error handling notification.
- Whether or not the system creates an incident for the email processing error.

To modify the error handling configuration, see the step for error handling in Set up inbound email.

The following table describes the default error cases in the out-of-box system.

Error Case	Cause
Failed to process the email as it comes from an unauthorized email domain.	The user's email address doesn't belong to an authorized email domain.
Failed to create or update the record due to validation error.	The creation or update of a record fails due to validation errors. The system requires more information to recreate the record. **Note** Note If you find a smart email task log (in the SMISTaskLog table) is in the Error status but the Internal Record ID is not empty, it is likely that Smart Analytics is enabled and returns an obsolete category or affected service. In this case, the system still creates the record by using the default category but the status of the task is set to Error. You can delete the obsolete category or service by using the following workaround: 1. Search for this obsolete category or affected service in the cate2idol2 table and log its ID. 2. Run the `saa` command. 3. Type http://<Smart Analytics Server host>:<Smart Analytics Server port>/action=CategoryDelete&Category=<id> in the field before the **Run** button 4. Click **Run**.
Failed to create or update the record due to insufficient user right.	The creation or update of a record fails because a user doesn't have the correct permission to create or update a request. By default, the user should have the "New" or "Update" right in the "Service Desk", "Incident", or "Change" area.

Failed to update the record because the record is no longer in a phase or status that allows the update.	This error occurs when a user attempts to execute an action from an email when the record is no longer in the correct phase. For example: a user adds a comment after the record has been closed. **Note** Note If this error handling doesn't work, you may be using a customized or localized phase name in your workflow. In this case, you need to change the following phase names in the SmartEmailSMHelper script library to the actual phase names in your workflow: `if (scfile['current.phase'] == 'Closure' || scfile['current.phase'] == 'Abandonment') { promise.rejectPromise(ReturnCode.ACTION_ERROR); return promise; }`
The token is expired.	SM fails to process an email action because the token contained in the email has expired.
The token has already been used.	SM fails to process an email action because the token contained in the email has already been used (the user already performed the action).
The token is invalid.	SM fails to process an email action because the token is invalid or may have been altered.
The email address is not associated with any registered user in the system.	A user with the sender's email address is not recognized as a registered SM user.
The email address doesn't match the original address in the token.	The sender of the email and the user in the email's token don't match. This might occur if a user forwarded the email containing a token to someone else.
Failed to process the email due to email integration error.	There might be problems with the email integration.
A task to process is expired in the SMIS task queue.	A task to process is expired in the SMIS task queue.
Failed to connect to the mail server.	The connection to the mail server can't be established.
The total email size exceeds the limit.	The total email size (including email body and all attachments) exceeds the limit configured in the SMIS instance.

11.14.1.8. Spam filtering in mail server

As most of the email servers provide the anti-spam feature, the SM Smart Email solution doesn't include an extra spam filter for inbound emails. Instead, system administrators are encouraged to configure spam filters in their own email servers.

The following is an example for how to set up the anti-spam protection in Office 365:

Connection filtering checks the reputation of the sender before allowing a message to get through. You can create an allow list, or safe sender list, to make sure you receive every message sent to you from a specific IP address or IP address range. You can also create a list of IP addresses from which to block messages, called a block list.

Spam filtering checks for message characteristics consistent with spam. You can change what actions to take on messages identified as spam, and choose whether to filter messages written in specific languages, or sent from specific countries or regions. You can also turn on advanced spam filtering options if you want to pursue an aggressive approach to spam filtering. Additionally, you can configure end-user spam notifications to inform users when messages intended for them were sent to the quarantine instead. (Sending messages to the quarantine is one of the configurable actions.) From these notifications, end users can release false positives and report them to Microsoft for analysis.

For more details, refer to the Microsoft Office help.

If you are using an email server other than Office 365, refer to the official guide of your email server to set up the spam filters.

11.14.1.9. Smart Email best practices

This topic describes some best practices when you implement the Smart Email solution.

Resolution email

You can either enable resolution email for interaction only or for fulfillment record only. Smart Email may not work properly if the resolution email is enabled for both the interaction and fulfillment records.

When multiple interactions are related to one fulfillment record, we highly recommend that you disable the resolution email to end user in fulfillment record and enable the resolution email in interaction only. By doing this, multiple end users recorded in the interactions can be informed by email instead of just the end user in the fulfillment record.

For the ordering from catalog use case (that is, one interaction are associated with multiple fulfillment records), it is recommended that the system sends out the email from the fulfillment records only. This is because in such case, the fulfillment records provide more details than the interaction record.

Auto-closure of resolved records

When auto-closure is configured in a workflow, the interaction or fulfillment record is automatically closed when the condition is met. In this case, it is possible that an end user is out-of-office and is not able to reply a resolution email in time before the record is automatically closed. Therefore, we recommend that you add an auto-closure notification email along with commenting and a RESUBMIT action button in the email to remind the end user to open a new interaction if the user is unsatisfied with the provided solution.

Performance consideration

For better performance on processing inbound emails, the administrator can adjust the following two settings in **System Administration** > **Ongoing Maintenance** > **Smart Email** > **Settings** to specify the number of SmartEmail processors and background schedules.

- Number of SmartEmail processors
- Number of schedules per SmartEmail processor

For benchmark information and recommendations, see Sizing.

Manage inbound email volume

Service Manager processes only emails that have already been received and made available by the configured mail server. Service Manager does not control or enforce the rate at which emails are delivered to the mail server.

To manage email volume and prevent excessive email traffic, configure the mail server to implement controls such as:

- Maximum emails received from an individual user within a specified time period.
- Maximum overall inbound email volume.
- Concurrent connection or request limits.
- Mail-flow throttling and queuing.

- Controls for excessive or abnormal email traffic.
- Sender-, mailbox-, or domain-level restrictions.

When a configured limit is exceeded, the mail server handles the email according to its configured policy, such as throttling, queuing, temporarily rejecting, or rejecting the message. Emails that are not successfully received by the mail server are not available for processing by Smart Email.

11.14.1.10. Smart Email actions in HTML templates

If a record is initially created from a user email (that is, the **Source** field is "5 - E-mail"), Service Manager uses the HTML templates that contain Smart Email actions.

The following Smart Email actions can be embedded in the HTML templates for different modules.

Action button	Modules of the HTML templates that can embed the action buttons
• **Add a Comment** • **Mark as Solved**	Interaction, Incident, Request, Problem
• **Approve** • **Deny**	Interaction, Change, Request **Note** Note Even if a Change is not created from a user email, Service Manager uses the HTML template that contain the **Approve** actions for a Change when this Change is pending approval. Meanwhile, you must enable the Smart Email instance in SMIS. Otherwise, Service Manager will not send out approval or denial emails for a Change that is pending approval.
• **Accept** • **Reject**	Interaction, Incident, Request, Problem
• **Close**	Change task

In the out-of-box system, the following Smart Email actions aren't enabled for the Incident, Request, and Problem modules:

- "Add a Comment" and "Mark as Solved" actions when an incident, request, or problem is updated.
- "Accept" and "Reject" actions when the status of an incident, request, or problem is set as Resolved.

For the system to send out an email with these actions, you need to tailor your system and enable the feature manually. For instructions, see Enable Smart Email actions for incident, problem, and request.

Note

Note

- If you use the HTML templates with action buttons, be sure to enable SmartEmail in SMIS. Otherwise, after users click an action button and reply the email, Service Manager can't process the email so that the action will not take effect.
- When an end user clicks an action button from a mail client (such as Outlook, Gmail, and Office365), normally a new email window appears with the predefined information including email recipient, subject, and authentication token. However, some mail clients might fail to open the new email window or miss the token information in the mail body because these mail clients handle the mailto tag in their own ways. In this case, you need to use other mail clients such as Outlook.
- If an end user uses Gmail, the images such as the company logo in the SM HTML templates may not be displayed in the Gmail mail client.

Out-of-box HTML templates with action buttons

The following tables list the HTML templates that contain action buttons in the out-of-box system.

- **Add a Comment** and **Mark as Solved** buttons

Embedded in HTML template	Defined in ruleset	Defined in notification
incident.open.for.smart.email	incident.notification.open	
sd.registered.for.smart.email	sd.notification.registered sd.svc.notification.registered	
sd.steamlined.open.for.smart.email	sd.streamlined.notification.open	
sd.require.comment		Activity Added

- **Approve** and **Deny** buttons

Embedded in HTML template	Defined in ruleset	Defined in notification
sd.pending.approval		Approval Added
cart.item.pending.approval		Approval Added
cm3r.pending.notification.for.smart.email		Approval Added Approval Updated
request.pending.notification.for.smart.email		Approval Added Approval Updated

- **Accept** and **Reject** buttons

Embedded in HTML template	Defined in ruleset	Defined in notification
sd.fulfilled.with.action	sd.notification.fullfilled.resolved	

- **Close** button

Embedded in HTML template	Defined in ruleset	Defined in notification
ChM Task Pending Close	chm.task.notification.pending.close chm.task.notification.pending.close.risk	ChM Task Pending Close

Tip

Tip To access HTML templates, from the system navigator, click **Tailoring** > **Notifications** > **HTML Templates**, and then search for the templates.

11.14.1.10.1. Enable Smart Email actions for incident, problem, and request

Applies to User roles: Administrator

In the out-of-box system, the following Smart Email actions aren't enabled for the Incident, Request, and Problem modules:

- "Add a Comment" and "Mark as Solved" actions when an incident, request, or problem is updated.
- "Accept" and "Reject" actions when an incident, request, or problem is set as Resolved.

However, you can tailor your system to manually enable these Smart Email actions for incident, problem, and request.

To enable the "Add a Comment" and "Mark as Solved" actions, follow these steps:

1. Add an HTML template for incident, problem, or request with the action buttons embedded. You can refer to the sd.require.comment template for interaction as an example.
2. In the notification table, add a notification. You can refer to the "Activity Added" notification for interaction. You at least need to modify the **Arguments** and **Activity Added** field.
3. In the activityactions table, add the notification that you created in the previous step to the "Update Activities" record for a module:
    1. Open the activityactions table, search for the "Update Activities" record for the corresponding module (the probsummary, rootcause, or request table).
    2. Locate the **Name** field as $apm.activity(for probsummary) or $pm.activity(for problem or request) and the **Condition** field as "Condition: not ($G.bg) ..."
    3. Add the new notification to the **Notification** field and save your changes.

To enable the "Accept" and "Reject" actions, follow these steps:

1. Add a new rule set for incident, problem, or request.
2. Add a rule for the rule set and set the rule type as Send HTML Email.
3. Edit the condition for the rule. For example:
   `Source in CurrentRecord = "5" AND Status in CurrentRecord = "Resolved"`
4. Select the following HTML template for the rule:
    - incident: IM.Resolution.with.action
    - problem: PM.Resolution.with.action
    - request: RM Fulfilled with Action
5. Select the **Recipients** field as Current Record and Select **Field** and **Filed Type** to define the people who will receive emails with the Accept/Reject buttons.
6. Save the rule and the rule set.
7. Add the new rule set to the incident, problem, or request workflow under the **Rule Sets** tabs, and then save the workflow. For example, you can add the new incident rule set in the transition between the Recovery and Review phase of the Incident workflow.

Note

Note If you can't save the workflow, you may need to copy the workflow (**Tailoring** >**Process Designer** >**Copy Existing Workflow**), add the new rule set, and then save. Also you need to change the workflow for the incident, problem, or request category to the new copied workflow.

11.14.1.10.2. Add a custom action in email

Applies to User roles: Administrator

In the out-of-box system, predefined Smart Email actions can be embedded in the HTML templates for different modules. In addition, you can tailor the system to add a new custom action for Smart Email.

To add a custom action (for example, an action called ABC) in the HTML template, follow these steps:

1. Open the HTML template.
2. In the **Expressions** tab of the HTML template, add the following line. This is the action display name.
   `var ABC_action_label = localscmsg(null, "SmartEmail", "ABC_action_label");`
3. In the **HTML Source** tab of the HTML template, add the following line:
   `email_action(ABC,{ABC_action_label},{$RECORD.number},{$RECORD.brief_description},color:#ffffff;font-weight:semibold;text-decoration:none;)`

   > **Note**
   > Note Among the code, the `color:#ffffff;font-weight:semibold;text-decoration:none;` part is optional as it is for the display style of the custom action.

4. In the MailService._genReplyBody() JavaScript, add the following line to the "default_action_body" array:
   `"ABC": _localscmsg(lang, "SmartEmail", "ABC_default_body")`
5. Add the corresponding scmessage record used in *step 2* and *step 4*.
6. Modify the MailUtil.calcReplyTo() JavaScript if the logic of the reply email address needs to be changed. See Reply email address for the default "reply to" logic.
7. Implement the logic of the new custom action as needed. You need to add the action to the SmartEmailSMAdapter function sendRecord.

11.14.1.11. View Smart Email reports

Applies to User roles: IT manager

The Reporting tool in Service Manager provides four out-of-box reports for Smart Email. You can view these reports through a dashboard named Smart Email Overview (Global).

To view the Smart Email reports, follow these steps:

1. Click **System Administration** > **Ongoing Maintenance** > **Smart Email** > **Smart Email Overview**. The Smart Email Overview dashboard is displayed.
2. View the reports on the dashboard. The following table describes the out-of-box reports for Smart Email.

Report	Description	Chart type
Email interactions resolved using email vs. email interactions resolved using SM client	This report compares the number of email interactions that are resolved through email and by using SM client during within six months.	Vertical bar
Average resolution time for email interactions vs. average resolution time for other interactions	This report compares the resolution time for email interactions and the resolution time for other interactions within six months.	Line
Customer satisfaction of email interactions within 6 months	This report indicates customer satisfaction of email interactions within six months based on the analysis of the surveys from email records. It shows the average score of each survey questions for the surveys.	Line
First-contact resolution rate of email interactions	This report shows the rate of email interactions that are closed at first contact.	Number

In the dashboard, a "Hot Topic Analytics for Email Interactions" section is also displayed. You can view the hot topics for email interactions if Smart Analytics is enabled.

11.14.2. Outbound Email

Applies to User roles: System Administrator

The outbound email (also known as JavaMail) for the Smart Email solution provides a monitor to handle Service Manager e-mail events. This monitor connects Service Manager into standard e-mail facilities and allows Service Manager operators and applications to send HTML emails. Any mail system that supports Simple Mail Transfer Protocol (SMTP) or has an SMTP gateway or bridge can receive e-mail from Service Manager if the outbound email is enabled.

> **Note**
>
> If you have already implemented a different outbound email solution such as SCAuto Mail or Connect-IT, you can also continue to use it as the outbound email solution for Smart Email.

To set up outbound email for the Smart Email solution, you need to configure the emailout parameter in the sm.ini file or SMIS integration. For detailed information on configuring the emailout parameter, see Startup parameter: emailout or Configure Emailout parameters in SMIS.

> **Note**
>
> Only messages enclosed within HTML tags can be sent out as HTML emails, otherwise messages are sent as plain text.

Related topics

Event Services

11.14.2.1. Event Services and Service Manager e-mail

The Service Manager Mail Utility checks the operator file for valid operator names before allowing mail to be sent. The Event Services version of this application expands the checking for valid users to those defined in the Service Manager contacts file.

The purpose of checking is to obtain the e-mail address from the e-mail field for the operator or contacts file. If the name for the addressee doesn't select a record from either file, Service Manager assumes that there is no such addressee and doesn't send mail. You can override this default by creating a login.event Format Control record and, in the Calculations section, setting the add condition to true and the calculation expression to the following: `$email.noaddr.ok=true`.

This causes Service Manager to assume that whatever name is passed to the e-mail event as the addressee is the complete e-mail address and attempts to send mail using that address.

Related topics

Event Services

11.14.2.2. Support of cc and bcc fields for outbound emails

The Service Manager Server now supports the use of the **cc**, **bcc**, and **reply.to** fields when sending email out through the eventout queue.

In email eventout, the **evsysopt** field is used to determine whether you want to use these fields:

- If this field is set to 0 or empty, the format for the evfields field is user.to^user.from^user.array^subject^text.
- If this field is set to 1, the format for the evfields field is user.to^user.from^user.cc^user.bcc^reply.to^subjext^text.

Sample scripts

The following are two sample scripts for your reference.

Sample 1

```
var eventout = new SCFile("eventout");
eventout.evtype = "email";
eventout.evtime = system.functions.tod();
eventout.evexpire = system.functions.tod();
eventout.evsysseq = String(Number(new Date()));
eventout.evsepchar = "^";
eventout.evfields =
"tim@microfocus.com^smith@microfocus.com^John@microfocus.com^marry@microfocus.com^smith@microfocus.com
;John@microfocus.com ^hello^hello all"; eventout.evsysopt = "1"; eventout.doSave();
```

Sample 2

```
var GroupTypes = ['assignment', 'cm3groups', 'ocmgroups', 'oncall'];
var GroupInformations = {
'assignment': {'file': 'assignment', 'nameField': 'name', 'memberField': 'operators'},
'cm3groups':  {'file': 'cm3groups',  'nameField': 'name', 'memberField': 'members'},
'ocmgroups':  {'file': 'ocmgroups',  'nameField': 'name', 'memberField': 'members'},
'oncall':     {'file': 'oncall',     'nameField': 'name', 'memberField': 'schedule', 'subMemberField':
'contact'}
};
function testSendEmail()
{
var from = 'falcon';
var toUsers = ['falcon'];
var toGroups = {'assignment': ['Application', 'ASSET MANAGEMENT'], 'cm3groups': 'Application'}; //2 group
types
var ccUsers = ['Adam.Cailat','Admin.General']; //2 operators
var ccGroups = {'ocmgroups': ['ASSET MANAGEMENT'], 'oncall': 'HP Admin'}; //2 group types. 1 group for
each type
var bccUsers = ['Adrian.Baxt'];
var bccGroups = {'assignment': ['Application', 'ASSET MANAGEMENT']}; //2 assignment groups
var replyTo = 'falcon';
var title = 'this is a test email';
var content = 'this is the email content';
sendEmail(from, toUsers, toGroups, ccUsers, ccGroups, bccUsers, bccGroups, replyTo, title, content);
}
testSendEmail();
/**
* Send email by creating a record in the eventout file
*
* toUsers, ccUsers, bccUsers, replyTo can be an JS array of operator/contact/email names or one
operator/contact/email name.
* toGroups, ccGroups, bccGroups should be an JS object. The key of the object is the group type
('assignment' or 'cm3groups' or 'ocmgroups' or 'oncall').
* The value is an array of groups or one group of that group type.
*/
function sendEmail(from, toUsers, toGroups, ccUsers, ccGroups, bccUsers, bccGroups, replyTo, title,
content)
{
if(from == null)
{
from = "";
}
if( title == null)
{
title = "";
}
if(content == null)
{
content = "";
}
var to = generateEmailAdress(toUsers, toGroups);
var cc = generateEmailAdress(ccUsers, ccGroups);
var bcc = generateEmailAdress(bccUsers, bccGroups);
replyTo = generateEmailAdress(replyTo);
var sep = '^';
var eventout = new SCFile("eventout");
eventout.evtype = "email";
eventout.evtime = system.functions.tod();
eventout.evexpire = system.functions.tod();
eventout.evsysseq = String(Number(new Date()));
```

```
eventout.evsepchar = sep;
eventout.evfields = to + sep + from + sep + cc + sep + bcc + sep + replyTo + sep + title + sep + content;
eventout.evsysopt = "1";
eventout.doSave();
}
function generateEmailAdress(users, groups)
{
if(groups == null)
{
groups = {};
}
var emails = {};
var members = {};
checkUsers(emails, members, users);
var length = GroupTypes.length;
for(var i = 0; i < length; i++)
{
var groupType = GroupTypes[i];
var groupNames = groups[groupType];
if(groupNames != null)
{
getGroupMembers(members, groupNames, groupType);
}
}
queryUserEmail(emails, members);
var res = "";
for( var email in emails )
{
if(emails[email] === true)
{
res = res + email + ';';
}
}
return res;
}
function queryUserEmail(emails, usersObj)
{
var users = objToArray(usersObj);
var length = users.length;
if(length === 0)
{
return;
}
var query;
if( length === 1 )
{
query = 'name="' + users[0] + '"';
}
else
{
query = 'name isin ' + system.functions.str(users);
}
var operator = new SCFile('operator', SCFILE_READONLY);
operator.setFields(['name','email']);
res = operator.doSelect(query);
for(; res === RC_SUCCESS; res = operator.getNext())
{
var email = operator.email;
if(email != null)
```

```
{
emails[email] = true;
delete usersObj[operator.name];
}
}
users = objToArray(usersObj);
length = users.length;
if( length === 1 )
{
query = 'contact.name="' + users[0] + '"';
}
else
{
query = 'contact.name isin ' + system.functions.str(users);
}
var contacts = new SCFile('contacts', SCFILE_READONLY);
contacts.setFields(['email']);
var res = contacts.doSelect(query);
for(; res === RC_SUCCESS; res = contacts.getNext())
{
var email = contacts.email;
if(email != null)
{
emails[email] = true;
}
}
}
function getGroupMembers(members, groupNames, groupType)
{
if(groupNames == null)
return members;
var groupInfo = GroupInformations[groupType];
if(groupInfo == null)
return members;
return _getGroupMembers(members, groupNames, groupInfo.file, groupInfo.nameField, groupInfo.memberField,
groupInfo.subMemberField);
}
function _getGroupMembers(members, groupNames, fileName, nameField, memberField, subMemberField)
{
var query;
if (system.functions.type(groupNames) == 8)
{
query = nameField + ' isin ' + system.functions.str(groupNames);
}
else
{
query = nameField + ' = "' + groupNames + '"';
}
var file = new SCFile(fileName, SCFILE_READONLY);
file.setFields([memberField]);
var res = file.doSelect(query);
for(; res === RC_SUCCESS; res = file.getNext())
{
var temp = file[memberField];
if(temp == null)
{
continue;
}
if(subMemberField != null)
```

```
{
var length = temp.length();
for(var i = 0; i < length; i++ )
{
var userName = temp[i][subMemberField];
if(userName != null && userName != '')
{
members[userName] = true;
}
}
continue;
}
temp = temp.toArray();
var length = temp.length;
for(var i = 0; i < length; i++ )
{
var userName = temp[i];
if(userName != null && userName != '')
{
members[userName] = true;
}
}
}
return members;
}
function isEmailAdress(name)
{
return name.indexOf('@') != -1;
}
function checkUsers(emails, members, users)
{
if(users == null)
return;
if(users instanceof Array)
{
var length = users.length;
for( var i = 0; i < length; i++ )
{
var value = users[i];
if(value != null)
{
if(isEmailAdress(value))
            {
              emails[value] = true;
            }
            else
            {
              members[value] = true;
            }
}
}
}
else
{
if(isEmailAdress(users))
{
emails[users] = true;
}
else
```

```
{
members[users] = true;
}
}
}
function objToArray(obj)
{
var res = [];
for( var name in obj)
{
if(obj[name] === true)
{
res.push(name);
}
}
return res;
}
```

Support of the CC and BCC fields in HTML Email notifications

Out-of-box, the **CC** and **BCC** fields are available when sending an HTML Email notification from a record. For details, see Sending HTML email messages manually.

11.14.2.3. Append an S/MIME digital signature to outbound emails

Secure/Multipurpose Internet Mail Extensions (S/MIME) is a method of exchanging emails securely. With the S/MIME technology, a digital signature can be appended to email messages to ensure non-repudiation and data integrity.

Starting with SM 9.33, you can enable a mechanism in Service Manager to allow for the signing of outbound email messages using the S/MIME technology. Once you enable this feature, the recipients can verify the signature on their mail system (for example, Microsoft Outlook), to make sure that the email messages are truly originated from Service Manager without being intercepted in transit.

Enable S/MIME signatures for non-FIPS environment

To enable an S/MIME digital signature for outbound emails for a non-FIPS environment, you need to obtain or generate a PKCS12 certificate and then specify three server parameters in the `sm.ini` file or the `emailout` parameter: `SMIMEKeystore`, `SMIMEKeystorePass`, and `SMIMEKeyAlias`.

The following is an example of enabling S/MIME digital signatures for outbound emails:

1. Generate the keystore in PKCS12 format. For example, with the OpenSSL toolkit, you can use the following openssl command:
   `keytool -genkey -keystore smemailkey.p12 -storepass smemailkeystorepass -alias smemailkeyalias -storetype pkcs12 -keyalg rsa`
2. Place the generated keystore file (smemailkey.p12) in the <SM Server>/RUN directory.
3. Configure these parameters in sm.ini:
   `SMIMEKeystore:smemailkey.p12`
   `SMIMEKeystorePass:smemailkeystorepass`
   `SMIMEKeyAlias:smemailkeyalias`

   Tip

   Tip An alternative way to set these parameters is to add the parameters to the `emailout` process in `sm.cfg`, as shown in the following example:
   `sm -emailout -mailFrom:xx -smtphost:xx -smtpport:xx -SMIMEKeystore:smemailkey.p12 -SMIMEKeystorePass:smemailkeystorepass -SMIMEKeyAlias:smemailkeyalias`
4. Restart the Service Manager server.

Enable S/MIME signatures for FIPS environment

To enable an S/MIME digital signature for outbound emails for a FIPS environment, you need to obtain or generate a BCFKS certificate and then specify three server parameters in the `sm.ini` file or the `emailout` parameter: `SMIMEKeystore`, `SMIMEKeystorePass`, and `SMIMEKeyAlias`.

Note

Note: If you have generated the keystore in PKCS12 format before, you can run the following command to covert the keystore to BCFKS format:

```
jre/bin/keytool -importkeystore -srckeystore smemailkey.p12 -destkeystore smemailkey.smemailkey.BCFKS -srcstoretype PKCS12 -deststoretype BCFKS -deststorepass smemailkeystorepass -srcstorepass smemailkeystorepass -providerclass org.bouncycastle.jcajce.provider.BouncyCastleFipsProvider -providerpath lib/bc-fips-1.0.2.4.jar
```

The following is an example of enabling S/MIME digital signatures for outbound emails:

1. Generate the keystore in BCFKS format. For example, you can use the following keytool command:
   ```
   keytool -genkey -keystore smemailkey.BCFKS -storepass smemailkeystorepass -alias smemailkeyalias -storetype BCFKS -providerclass org.bouncycastle.jcajce.provider.BouncyCastleFipsProvider -providerpath bc-fips-1.0.2.4.jar -keyalg rsa
   ```
   Note: The **bc-fips-1.0.2.4.jar** file can be located in the `RUN/lib` folder.
2. Place the generated keystore file (smemailkey.BCFKS) in the <SM Server>/RUN directory.
3. Configure these parameters in sm.ini:
   ```
   SMIMEKeystore:  smemailkey.BCFKS
   SMIMEKeystorePass:smemailkeystorepass
   SMIMEKeyAlias:smemailkeyalias
   ```

   Tip

   Tip An alternative way to set these parameters is to add the parameters to the `emailout` process in `sm.cfg`, as shown in the following example:

   ```
   sm -emailout -mailFrom:xx -smtphost:xx -smtpport:xx -SMIMEKeystore:smemailkey.BCFKS -SMIMEKeystorePass:smemailkeystorepass -SMIMEKeyAlias:smemailkeyalias
   ```
4. Restart the Service Manager server.

Related topics

Startup parameter: emailout

11.14.2.4. Configure Emailout parameters in SMIS

The Emailout instance is available in SMIS integration from Service Manager 9.81.

The existing email out settings in sm.ini configured through Startup parameter: emailout are still available.

If you choose to use SMIS to configure emailout settings, you need set the emailoutConfigFromSmisOnly:1 server parameter. After this parameter is enabled, the parameters defined in the emailout process through sm.cfg and sm.ini won't work. Instead, those parameter values such as smtphost will only be get from SMIS configuration.

SMTP OAuth 2.0 authentication supports both Google Gmail and Microsoft Azure accounts. If you want to use OAuth 2.0 authentication for SMTP, configure the EmailOut settings in an SMIS integration instance. Before configuring OAuth 2.0 authentication, ensure that the Web Server URL is defined in the System Information Record. See Configure the Service Manager Web tier server URL.

To configure the parameters, follow the below steps:

1. Sign in to Service Manager with a System Administrator account.
2. Select **Tailoring** > **Integration Manager**. Integration Manager opens.
3. Click **Add** to open the wizard.
4. Select **EmailOut**, click **Next**.
5. Fill in the information and click **Next**.
6. Configure the emailout parameters as required. See Startup parameter: emailout.

7. If you need to configure the related settings for OAuth2 authentication, select **OAuth authentication**.
8. Configure the OAuth provider as Google Gmail or Microsoft Azure.
    1. For **Google Gmail**:
        1. Configure the redirect URI in Google cloud console: `https://<SM Webtier Host>/<Context Path>/oauth/callback`. Where `<SM Webtier Host>/<Context Path>` should match the Web Server URL defined in System Information Record. For example: `https://myserver.mydomain.com:myport/webtier/oauth/callback`
        2. Obtain the **Client ID** and **Client Secret** from Google cloud console.
        3. Select **Get Access Token** to open the Google authentication page and authorize access.
        4. After authentication is completed successfully, the **Token Available** message appears.
    2. For **Microsoft Azure**:
        1. Register an application in Microsoft Azure.
        2. In the Azure application registration, configure Authentication settings and add the following redirect URI: `https://<SM Webtier Host>/<Context Path>/oauth/callback`. Where `<SM Webtier Host>/<Context Path>` should match the Web Server URL defined in System Information Record.
        3. Obtain the **Tenant ID**, **Client ID** and **Client Secret** from the Azure application registration.
        4. Select Microsoft Azure as the OAuth provider, and provide the required OAuth configuration values.
        5. Select **Get Access Token.**
        6. Sign in with the Microsoft account that will be used for Emailout and grant the requested permissions.
        7. After authentication is completed successfully, the **Token Available** message appears.
9. Click **Finish** to save the configurations.

Note

Microsoft Azure OAuth 2.0 authentication enables modern authentication for EmailOut and supports environments that require multi-factor authentication (MFA). Basic username/password authentication may not be supported in all Microsoft 365 environments.

Once SMIS configurations are changed, you need to restart the whole SM Server Cluster to apply the updates. Restarting SMIS instance only won't take effect. Restarting one host will apply the new changes on this host only, while other hosts won't get the updated values.

11.14.3. HTML Email

The Service Manager HTML Email Solution enables Service Manager to send HTML email messages automatically or manually on demand. This email solution is based on HTML templates that can be customized and localized.

This document describes how to install this solution, how it works, how to customize HTML email notifications, and how to send on-demand notifications.

Audience

HTML Email is intended for the following audiences:

- System administrators who manage notifications
- System administrators who manage Service Manager email setup
- Engineers who send HTML emails to notify users

Prerequisites

This solution requires the following prerequisites:

- Service Manager 9.70 (server runtime environment and applications)
- SMTP for email notifications
- Email configurations

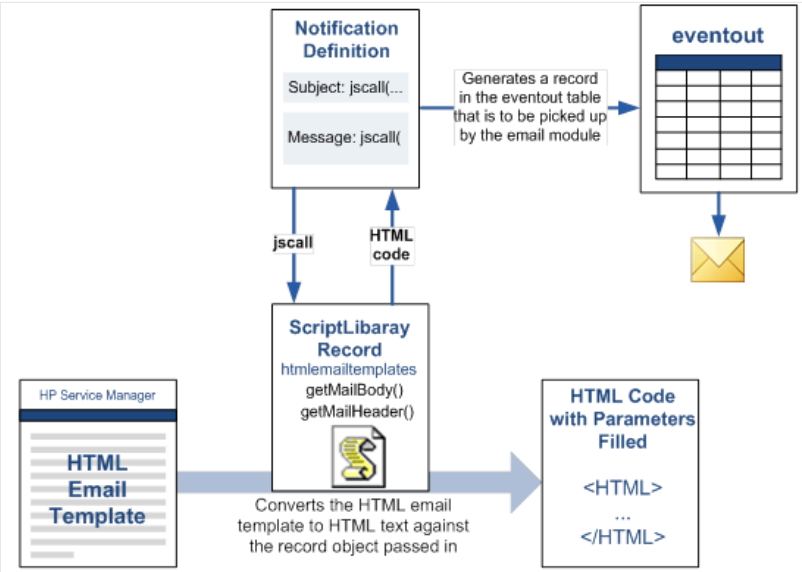
How the solution works

The Service Manager HTML Email Solution works as follows.

Components of the solution

The HTML Email Solution uses the following components:

- Notification definitions (in the notifications table)
- HTML templates (in the htmltemplates table)
- JavaScript functions (in the htmlemailtemplate record in the ScriptLibrary)
- The eventout table (available in the existing email notification mechanism)



HTML templates

HTML templates are records saved in the htmltemplates table. Each record defines the common HTML elements that are included in each email message and parameters that are to be replaced with specific values. Each htmltemplates record includes general information, HTML source with parameters, and expressions that are used to evaluate the parameters.

Each HTML template can have multiple language versions. These language versions are defined in separate htmltemplates records with the same name but different language codes, containing localized content.

For information about how the system chooses the language in a notification email, see Language for notification emails

Note

Note To manage or create HTML templates, users must have either the SysAdmin or the HTMLTemplatesAdmin capability word.

Note

Note The auditing feature is enabled by default for the htmltemplates table. When modifications are detected, an Audit Log entry is recorded for each changed field.

Notification definitions

HTML email notifications are also defined in notification records. However, instead of referencing a plain-text message or a predefined format, each HTML email notification record always references a message record with an msg class of HTMLTemplate and an id of 1. This message record is only a wrapper. The actual message body and subject are generated by a jscall expression in the **Arguments** field that returns HTML text by evaluating the corresponding HTML template.

The Arguments expression for the message body calls the getMailBody function to obtain the body HTML code, and the Arguments expression for the message subject calls the getMailHeader function to obtain the subject HTML code. The template name and the record that triggers the notification are passed in as arguments.

Below is an example for notification **ChM Change Approval**:

The **Message** tab

Field	Value
Msg Class	HTMLTemplate
Msg No.	1
Arguments	jscall("htmlemailtemplates.getMailBody", "ChM Change Approval Approved", $L.file, $L.file.save, requested.by in $L.file)
Condition	category in $L.file~="Subscription"
Notify Method	email
Recipient(s)	requested.by in $L.file

The **Email/Mail Subject Line** tab

Field	Value
Msg Class	HTMLTemplate
Msg No.	1
Arguments	jscall("htmlemailtemplates.getMailHeader", "ChM Change Approval Approved", $L.file, $L.file.save, requested.by in $L.file)

ScriptLibrary record

Both the getMailBody and getMailHeader functions are defined in an htmlemailtemplates ScriptLibrary record.

getMailBody

The getMailBody function returns the HTML body for the email message that's sent for a specific record. This function takes the following arguments:

Argument	Required	Description
templateName	Yes	Specifies the name of the HTML template.
$RECORD	Yes	Specifies the record that triggers the email notification. This record contains data that's used to construct the email message. Example: $L.file
$RECORDOLD	Yes	Specifies the pre-update version of the record that triggers the email notification. Example: $L.file.save
recipient	No	This argument is reserved. The email message will always be sent to the recipient defined in the notification record.
lang	No	Specifies the language code of a specific language version of the HTML template. This argument is typically used if you always send this notification in a certain language. If this argument is not specified, Service Manager takes the recipient's default language when determining which language version of the HTML template to use. For each recipient, Service Manager detects: • The language setting in the recipient's contact record. If this doesn't exist, then • The language setting in the recipient's operator record. If this doesn't exist, then • The system default language. If the recipient is a distribution group and all its members have the same default language, the common default language is used. If the members of that distribution have different default languages, the system default language is used. Example: de

Note

Note The **Arguments** field in the notification record takes only RAD expressions. Therefore, the expression must use a jscall function to call the getMailBody function in the ScriptLibrary. A typical expression resembles the following:

```
jscall("HTMLTemplates.getMailBody", "SM Add", $L.file, $L.file.save)
```

getMailHeader

The getMailHeader function returns the subject for the email message that's sent for a specific record. This function takes the following arguments:

Argument	Required	Description
templateName	Yes	Specifies the name of the HTML template.
$RECORD	Yes	Specifies the record that triggers the email notification. This record contains data that's used to construct the email message. Example: $L.file
$RECORDOLD	Yes	Specifies the pre-update version of the record that triggers the email notification. This record contains data that's used to construct the email message. Example: $L.file.save
recipient	No	This argument is reserved. The email message will always be sent to the recipient defined in the notification record.

lang	No	Specifies the language code of a specific language version of the HTML template. This argument is typically used if you always send this notification in a certain language. If this argument is not specified, Service Manager takes the recipient's default language when determining which language version of the HTML template to use. For each recipient, Service Manager detects: • The language setting in the recipient's contact record. If this doesn't exist, then • The language setting in the recipient's operator record. If this doesn't exist, then • The system default language. If the recipient is a distribution group and all its members have the same default language, the common default language is used. If the members of that distribution have different default languages, the system default language is used. Example: de

The **Arguments** field in the notification record takes only RAD expressions. Therefore, the expression must use a jscall function to call the getMailHeader function in the ScriptLibrary. A typical expression resembles the following:

```
jscall("HTMLTemplates.getMailHeader", "SM Add", $L.file, $L.file.save)
```

Eventout table

A record is created in the eventout table for each email notification. The eventout record remains until it is picked up by the JavaMail agent.

11.14.3.1. Customizing HTML email notifications

This chapter describes the most typical scenarios of customizing HTML email notifications. Instead of using the out-of-box set of email notifications and templates, your organization is likely to have its own email templates and include its own fields in email messages. For example, a company may include its company logo and follow its own color schemes in email messages.

In most scenarios, you only need to make some minor changes to an out-of-box HTML template, such as changing the formatting of the message body and including some additional fields in the email message. Therefore, you can use the out-of-box HTML template as a basis and create your own HTML templates by copying from that out-of-box template.

Create your own HTML templates

To create a custom HTML template, follow these steps:

1. Click **Tailoring** > **Notifications** > **HTML Templates**.
2. Search for and select an existing HTML template as the base of the new template.
   Example: SM Add
3. On the **General** tab, specify a name for the new template in the **Name** field.
4. Change the **Mail Title** field if necessary.

5. In the **Language** field, select a language for the template.
6. Describe the template in the **Comment** text box.
7. The **Module** and **Test Record** fields combined determine which record is passed in for preview. Select a module in the **Module** field and specify the id of the record in the **Test Record** field.

   > **Note**
   > Note The module name is designed to be a general name, it may include multiple tables or refer to nonexistent table. Therefore, not all previews are supported currently.

8. In the text box on the **Expressions** tab, edit the JavaScript code that evaluates the variables that you include in the email message.

   > **Note**
   > Note To post certain field values of the record that's passed in, you can add these fields directly in the HTML Source, such as $RECORD.incident_id. However, if you want to add variables computed based on record fields, you must evaluate these variables using JavaScript code on the **Expressions** tab and then post the variables in the HTML Source.
   > Only some simple data types are supported, including String, Date/Time, Logical, and Number.

9. The **HTML Source** tab displays an HTML editor where you can design the message body of your HTML template. You can click **Source** to toggle between the normal view and the source code view. To reference a variable or expression in the message, enclose your variable or expression in braces, such as `{$RECORD.title}`.
10. (Optional) Click the **Attachments** tab, and then click **Add Files...** to add one or more attachments to the template. If this HTML template has localized versions, the attachments are available to all localized versions as well. When a user receives a notification email that's using this HTML template, the system adds these attachments to this email.
11. To generate a preview using the **Module** and **Test Record** fields that you specified in step 7, click the **Preview** tab.
12. Click **Add** to create the new template.

Localize a template

To create a language version of an HTML template, follow these steps:

1. Click **Tailoring** > **Notifications** > **HTML Templates**.
2. Use search or advanced search to find the template that you want to localize.
3. On the **General** tab, select a language in the **Language** field for the localized template.
4. Keep the name in the **Name** field as is, because HTML template records with the same name are considered language versions of one template.
5. In the **Mail Title** field, replace the original text with the appropriate translation and make sure all the variables are in the correct positions.
6. On the **HTML Source** tab, replace the original text with the appropriate translation.
7. Click the **Preview** tab to verify that your localized version works as expected.
8. Click **Add** to create the localized template.

Link an HTML template to a notification

To link an HTML template to a notification, follow these steps:

1. Click **Tailoring** > **Notifications** > **Notifications**.
2. Search for the notification record to which you want to link the HTML template.
3. On the **Message** tab, locate the message with an **Msg Class** of **HTMLTemplate** and an id **Msg No.** of **1**.
4. In the **Arguments** field, replace the second argument of the jscall function with the name of the template that you want to use for this notification.
5. On the **Email/Mail Subject Line** tab, locate the message with an **Msg Class** of **HTMLTemplate** and an id **Msg No.** of **1**.

6. In the **Arguments** field, replace the second argument of the jscall function with the name of the template that you want to use for this notification.
7. Click **Save** to save the changes.

Add commonly used elements in templates

This section describes how to add certain elements that you are likely to include in your email message.

Images

You can use either of the following methods to post images in an message.

- Store the images in a location that your email recipients can access and add image tags referencing the corresponding URLs.
- Add one or more images as attachments to the HTML template, and then post these images in the message body. Follow these steps:
    1. Click the **Attachments** tab, and then click **Add Files...** to add one or more images to the template. If this HTML template has localized versions, the attachments are available to all localized versions as well.
    2. Save the HTML template.
    3. Click the **HTML Source** tab, and then place the cursor in the message body where you want to insert the image.
    4. Click the **Insert an Image** button. The system opens the **Select Content Attributes** wizard.
    5. Type a brief description in the **Link Text** field, and then specify the width and height of the image to be displayed.
    6. Click **Next**. The system lists all image attachments that are associated with the HTML template. Click an attachment and the system automatically add this image to the message body.
    Note that this operation affects the current HTML template only. If this HTML template has localized versions and you hope to add the same image, you need to repeat *step c* to *step f* to post the image in the message body of the localized templates one-by-one.

Note

Note If you post an image in the message body, the system no longer displays this image in the attachment list of a notification email. For example, suppose an HTML template has a *.txt file and a *.png file as attachments and you have posted *.png in the message body. Later when you receive a notification email that's using the HTML template, the email attachment list consists of the *.txt file only.

Web URL

It is a common practice to include a Web URL in the email message pointing to the corresponding record so that the recipient can follow the link to view the record in a Service Manager web client. You can reference the `web_url` variable that's defined in the expressions of the out-of-box templates.

You can keep one of the following statements and comment out the other one, depending on whether the email message is sent to an ESS user, a Service Portal user, or a regular Web-tier user.

For ESS users:

```
var web_url = lib.urlCreator.getESSURLFromQuery(file_name,record_query,record_title);
```

For Service Portal users:

```
var web_url = lib.urlCreator.getSMSPURL(file_name, id);
```

For regular Web-tier users:

```
var web_url = lib.urlCreator.getURLFromQuery(file_name,record_query,record_title);
```

11.14.3.2. Disabling individual notification definitions

HTML and non-HTML notifications can coexist. For example, you can:

- Enable all HTML notification definitions and disable their corresponding non-HTML ones
- Enable certain HTML and non-HTML notification definitions

By default, all HTML notification definitions are enabled. You can select to disable individual notification definitions, either non-HTML or HTML.

How to disable an individual notification definition[edit]

To disable an individual notification definition, follow these steps:

1. Click **Tailoring** > **Notifications** > **Notifications**.
2. In the **ID** field, type the ID of the notification definition you want to disable. For example, ChM Change Approval_1. Each non-HTML notification definition has a corresponding HTML version. The two definitions have similar names differentiated by an HTML prefix. For example, ChM Change Approval_1 (non-HTML) and HTML ChM Change Approval_1 (HTML).
3. Click **Search**. The notification definition record opens.
4. In the **Condition** field, make the following modifications as appropriate:
    - If the **Condition** field is empty, type false.
    - If the **Condition** field is not empty, insert false and to the left of the existing expression text. For example: false and category in $L.file={"Release Management", "Subscription", "CI Group", "KM Document"}
5. Click **Save** to save the record.

Contents

- 1 How to disable an individual notification definition

11.14.3.3. Sending HTML email messages manually

In addition to setting up notifications, engineers can manually send HTML email messages for a specific record by using the Notify option when viewing a record.

Send an HTML email message using the Notify option

To send an HTML email message using the Notify option, follow these steps:

1. Open a record.
   Example: Click **Service Desk** > **Search Interaction** to search for an interaction record.
2. From the **More** menu option, click **Notify**.
3. To send this message to individuals, specify the names of the recipients in the **To** field. To send this message to a distribution group, specify the name of the distribution group in the **Group** field.
4. Specify the names of the recipients in the **CC** and **BCC** fields if necessary.
5. Leave the **Type** field as **Email**.
6. If multiple language versions of the associated HTML template are available, select a language in the **Language** field to indicate which language version of the template you want to use to send the email message.
7. The **Subject** field displays the default subject text that's generated based on the HTML template associated with your current record type. You can modify the subject as appropriate.
8. The message body area (the **Content** tab) displays the default message body that's generated based on the HTML template associated with your current record type. You can edit the message body as appropriate. You can click **Source** to toggle between the normal view and the source code view.
   Note: If you click the **Insert a link toward an attached file** or the **Insert an image** button and then select a file, the system sends this file as an email attachment.
9. From the **Attachment** tab, click **Add Files** to add attachments to the email message.
10. Click **Send** to send the message.

HTML templates used by the Notify option

The Notify option selects an appropriate HTML template to generate the default subject and message body, depending on the type of the record from which you launch the Notify option.

The following table lists the HTML templates that the Notify option uses to generate email messages:

Module	Record type	Template
Service Desk	Interaction records (incidents)	SD.incident.notify
Incident Management	Incident records (probsummary)	IM.update.incident
Change Management	Change records (cm3r)	The templates named after the Default and Close views of the Change Phase record (on the **Scripts/Views** tab of each Change Phase record). **Note** Note If the out-of-box Default and Close views have the same name, only one HTML template is used. For example, for change records in the Change Review phase, the **CM.change.review** HTML template is used. If you have modified the out-of-box view settings of a Change Phase record, you need to create one or two (depending on whether the Default and Close views have the same name) HTML templates named after your customized names of the Default and Close views.

Change Management	Change task records (cm3t)	The templates named after the Default and Close views of the Task Phase record (on the **Scripts/Views** tab of each Task Phase record). **Note** Note If the out-of-box Default and Close views have the same name, only one HTML template is used. For example, for change task records in the Software phase, the **CM.change.review** HTML template is used. If you have modified the out-of-box view settings of a Task Phase record, you need to create one or two (depending on whether the Default and Close views have the same name) HTML templates named after your customized names of the Default and Close views.

11.14.3.4. Email setup notes

The HTML Email solution requires that you have an email infrastructure in place, including a properly configured SMTP server and configurations in some Service Manager files.

Configurations in the sm.ini file

Open the sm.ini file using a text editor and verify that the following parameters are correctly configured.

Parameter	Description
querysecurity:1	Enforces the Service Manager server to require a security hash with Web tier URL queries.
queryhashcode:hostname:port	Enables all Service Manager processes to consistently generate a unique hash code, which can be accepted and properly decoded by the Web tier. The parameter value must be a combination of the Service Manager hostname and communications port. If you provide the fully qualified domain name (FQDN) of a server host in web.xml, you need to provide the same value in this parameter. **Example:** `queryhashcode:smserverhost:13080`
smtphost:smtpserverhost	Specifies the name of the SMTP server host for client requests. The value for the parameter can be the IP address, machine name, or DNS name of the SMTP server uses for sending notifications. **Example:** `smtphost:smtpserverhost`
smtpport:port	Specifies the communications port the SMTP server uses. **Example:** `smtpport:25`
mailFrom:EmailAddress	Specifies the descriptive name or other identifier of the sender of an e-mail. This parameter should be set in the format of email address. **Example:** `mailFrom:accountname@microfocus.com`

For more information about the JavaMail connection parameters, see System configuration parameters.

Configuration in the web.xml file

Open the web.xml file that's located in the $TOMCAT_HOME\webapps\webtier-9.70\WEB-INF folder using a text editor, and add the following text at the end of the file if it doesn't exist.

```
<init-param><br />
<param-name>querySecurity</param-name><br />
<param-value>true</param-value><br />
</init-param>
```

When this parameter is enabled, the Service Manager Web tier embeds a security key in all queries generated by the Web client. The Service Manager server verifies the security key and, if valid, authorizes the query.

12. Delta migration tool

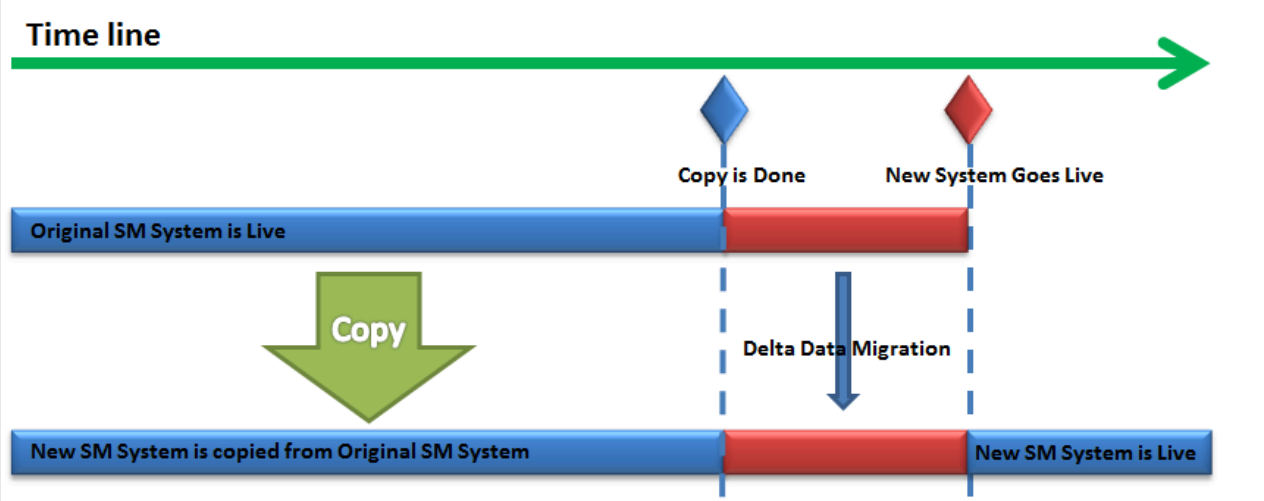
Service Manager provides a delta migration tool to transfer the “delta data” from one Service Manager system (original system) to another (new system).

Typical scenario

There are times where you need to migrate data from one Service Manager server to another. Take the following scenario:

- The original system is running live
- You make a copy of it
- You make changes to the workflow in the copy
- You replace the original live system with the copy that has the new workflow

Of course from the time you made a copy until the time the new system goes live, end users are making changes. The following figure depicts this scenario.



Tip

Tip You can also use this tool to keep the data in a "test" system in sync with a "production" system or a "development" system in sync with a "test" system.

Data migration methods

The delta migration tool enables you to keep the original system and the copy in sync in either of the ways:

- Allowing manual export or import of changed records
- Scheduling an automated export or import

Caution

Caution For a table without the "sysmodtime" field, the tool can't migrate "delta data" from the original system.

Prerequisites

There are some prerequisites for using this tool. For details, see Installing the delta migration tool .

Supported applications versions

You can find the latest **DeltaMigrationTool.unl** in the latest Service Manager Upgrade package root folder.

Related topics

Delta migration methods
Installing the delta migration tool
Troubleshooting delta migration

12.1. Installing the delta migration tool

To install the latest delta migration tool, you need to deploy the **DeltaMigrationTool.unl** package. This package is available in the upgrade folder, along with the other Service Manager application upgrade utility files. To install the delta migration tool, follow these steps:

1. Sign in to the Service Manager system as an administrator.
2. Open **Database Manager**, right-click, and then select **Import/Load**.
3. Select **DeltaMigrationTool.unl**, and then click **Load FG** to deploy it.

Related topics

Delta migration tool
Delta migration methods
Troubleshooting delta migration

12.2. Delta migration methods

There are two methods to complete the delta migration:

Method 1: Manually export and import a single UNL file

You need to use the delta migration tool to generate an unload script, and then use this unload script to export and import data manually. This method is simple and you can use it to migrate delta data between two systems with similar versions (for example, from Service Manager 9.60 to Service Manager 9.61).

Note

Note If the table structures of the new and original Service Manager systems are different, use method 2 to migrate the delta data.

Method 2: Automatically export and import multiple UNL files

You need to create background schedules to export the tables that contain the delta data into the UNL files in a specified original folder. Then, move the entire folder to the new Service Manager system and create background schedules to import them.

Migration by using this method uses the background schedule tasks to export and import the delta data, so it has minimal performance impact on the running production system. You can apply this method to migrate delta data between two systems with different versions (for example, from Service Manager 9.41 to Service Manager 9.61). Most of Service Manager upgrade projects should use method 2.

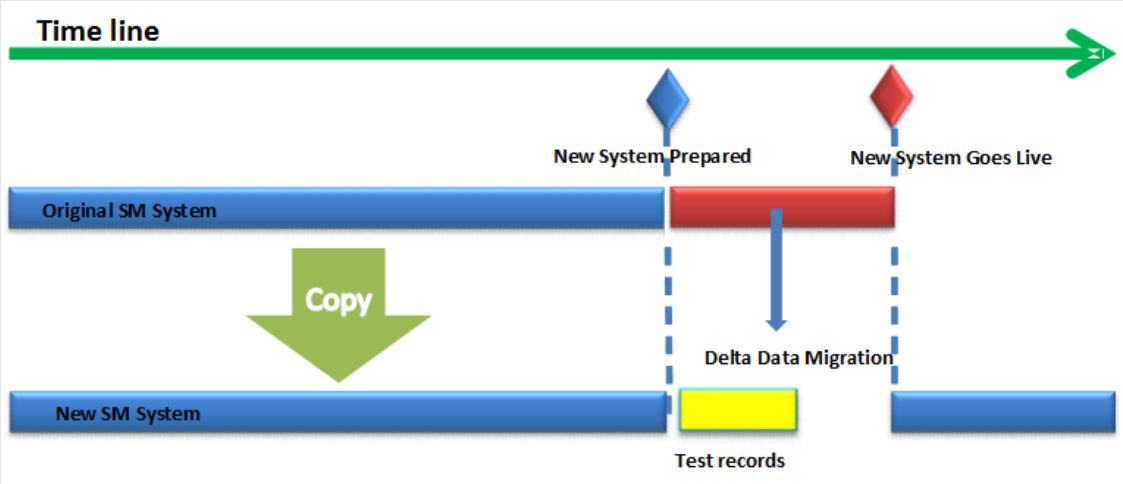
Prerequisites for using the delta migration tool

Before you use the delta migration tool to migrate the delta data, the following prerequisites must be met.

Note

Note Only the first requirement applies to both method 1 and method 2, while the other four requirements apply only to method 2.

- Make sure that the new Service Manager system is clean and doesn't contain any test records.
  Before the delta migration, the new Service Manager system should include only the records or modifications from the original system. You must remove any test data that was created or modified on the new system during the delta data period.
  If there are some test records added to the new system during the delta data period, as shown in the following figure, then the delta data from the original Service Manager system can't be migrated to the new system because of a data conflict. Even if you decide to use the delta data from the original Service Manager system to overwrite the test records, there is a risk of a data integrity violation.

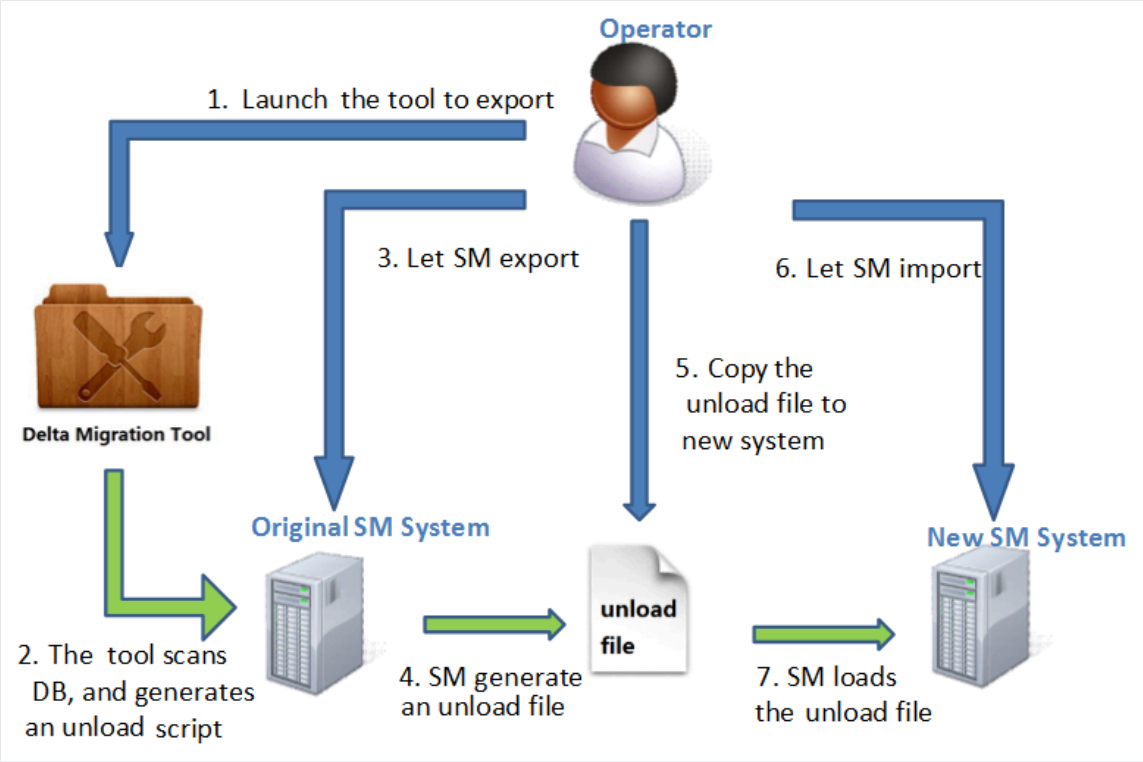


For example, a new incident IM12345 (called IM_A) is created in the original system during delta data period, and another test record IM12345 (called IM_B) is created in the new system, with several activity logs. If you then use IM_A in the original system to overwrite IM_B in the new system, you will find that IM_A in the new system contains some incorrect activity logs, which belong to IM_B.

- Make sure that the date format of the original Service Manager system doesn't have the y2k problem. To do this, follow these steps:
  1. Open **System Navigator** > **Base System Configuration** > **Miscellaneous** > **System Information Record**.
  2. Go to **Data Info**.
     If the date format contains only two digits for the year value, such as "mm/dd/yy", change it to four digits (for example, "mm/dd/yyyy").
  3. Save the modification and restart the Service Manager service.
- Create a new administrator's account in the original Service Manager system. The date format and time zone settings of this account should be the same as those in **System Information Record**.
  Because background tasks will use the default time zone and date format in the **System Information Record**, creating a new admin account with the same time zone setting and date format of **System Information Record** will avoid missing data.
- Make sure that the five background schedulers (bg_load_unload 1, ..., bg_load_unload 5) have been started in both of the Service Manager systems. If the background schedulers haven been started, they should be visible in the **System Status** record. If they're not there, click **Start Scheduler** and select **dmt.unload.load**.
- Make sure that an empty folder is created on the original Service Manager system for exporting the delta data.

Method 1: Manually export/import a single UNL file

The following flow chart illustrates how the operator migrates the delta data by using method 1.



To export the delta data by using method 1, follow these steps:

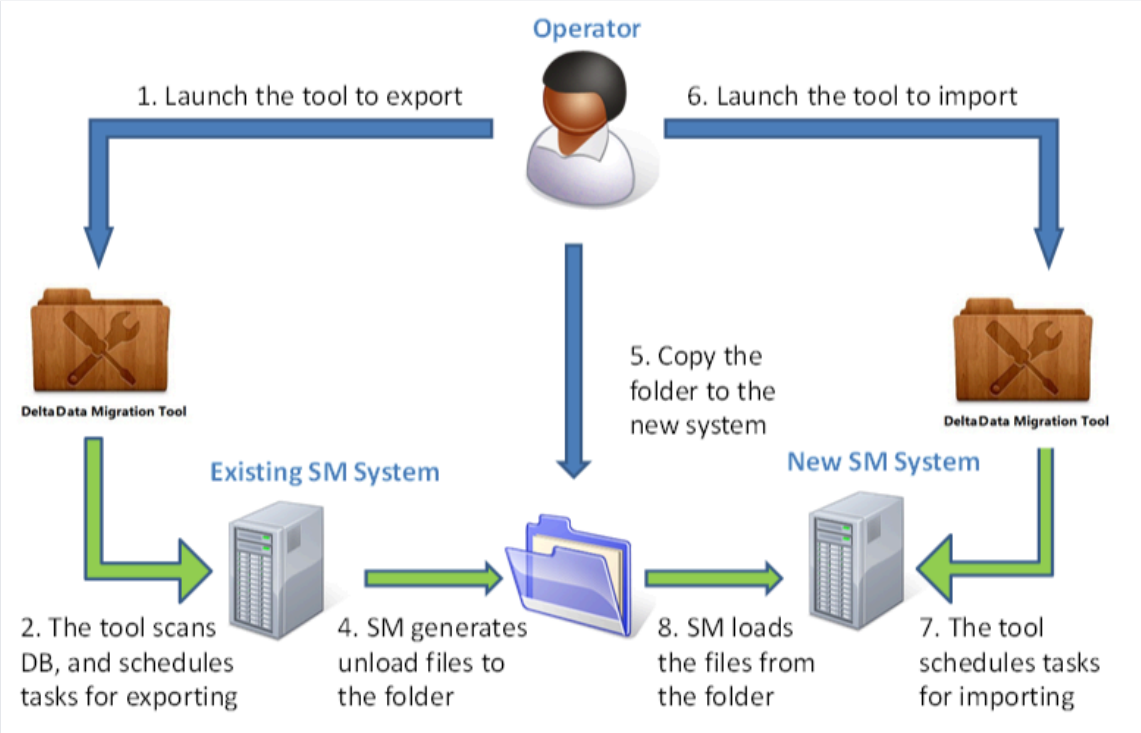
1. Sign in to the original Service Manager system.
2. Click **System Administraton** > **Ongoing Maintenance** > **Upgrade Utility** > **Delta Migration Export**. A welcome page appears showing the instructions for migration.
3. Click **Next** on the welcome page and select **Method 1: Manually export/import a single UNL file**.
4. Click **Next** to check the steps of using method 1.
5. Click **Next** to specify the time range of the delta data. The delta migration tool will generate an unload script to extract all the delta data within this range.
6. Click **Next** to generate the unload script.
7. Open the generated unload script, and then run it to generate an UNL file.
8. Copy the UNL file to the new system.

To import the delta data by using method 1, follow these steps:

1. Sign in to the new Service Manager system as an administrator.
2. Click **System Administraton** > **Ongoing Maintenance** > **Upgrade Utility** > **Delta Migration Import**.
3. Select **Method 1: Manually import a single UNL file**, and then select the unl file to import.
4. Click **Next** to import the unload data.
5. Sign out and then sign in to the new Service Manager system again.

Method 2: Automatically export and import multiple UNL files

The following flow chart illustrates how the operator migrates the delta data by using method 2.



To export the delta data by using method 2, follow these steps:

1. Sign in to the original Service Manager system.
2. Click **System Administraton** > **Ongoing Maintenance** > **Upgrade Utility** > **Delta Migration Export**. A welcome page appears showing the instructions for migration.
3. Click **Next** on the welcome page, and then select **Method 2: Automatically export/import multiple UNLs**.
4. Click **Next** to check the steps of using method 2.
5. Click **Next** to specify a folder for the exported delta data, and then specify the time range of the delta data.
6. Click **Next** to verify the provided migration suggestions after the tool checks the tables. You can ignore some tables even if they contain some delta data.
7. Click **Next** to verify the final confirmation for the migration. The message displays the number of tables that will be migrated and the time that the exporting will take.
8. Click **Next** to start background schedule tasks to export the delta data.
9. Click **Finish** to exit the tool when the overall progress reaches 100%.

> **Note**
> Note To check the export status, go to the selected folder and open **Overall Export Status Report.html**.

10. Copy the whole folder to the new Service Manager system.

To import the delta data by using method 2, follow these steps:

1. Sign in to the new Service Manager system as an administrator.
2. Click **System Administration** > **Ongoing Maintenance** > **Upgrade Utility** > **Delta Migration Import**. A welcome page appears showing the instructions for migration.
3. Select **Method 2: Automatically import multiple UNL files**, and then click **Next** to specify the folder that contains the exported delta data.
4. Click **Next** to select the tables to migrate. If the table structures are different, we recommend that you click **Click here to check the table structure mismatch...** link, to acknowledge the detailed difference, the risk, and then decide if you migrate or not.

Note

Note If the table structure mismatch may introduce risks, and you decide not to migrate, then try to migrate the data in these tables by other methods to avoid data loss. For more information about how to do this, see Cannot handle skipped tables.

5. Click **Next** to check the final confirmation for the migration. The message displays the number of tables that will be imported.
6. Click **Next** to start background schedule tasks to import the delta data.
7. Click **Finish** to exit the tool when the overall progress reaches 100%.

Note

Note To check the importing status, go to the selected folder and open **Overall Import Status Report.html**.

Note

Note If tables are ignored during the export operation because they have no last modification time field, then check if a full-table migration is needed.

Related topics

Delta migration tool
Installing the delta migration tool
Troubleshooting delta migration

12.3. Delta migration FAQ

Where can I check the status of the background export/import tasks?

During the import and export operations, there is a real-time report Overall Export Status Report.html or Overall Import Status Report.html located under the specified folder. Open the file and check the overall status at the top of it. If your browser warns you about the script or ActiveX controller on the webpage, allow it. When the overall status changes to 100%, all the background export/import tasks have finished.

What files are included as the output materials?

After a successful export operation, the materials in the user specified folder should include the following files.

File name	Description
readme.txt	Records of the basic information of this folder This file includes the time range of the exported UNL files.
Overall Export Status Report.html	A real-time report for the export operation
Overall Import Status Report.html	A real-time report for the import operation **Note** Note This file only appears during the import operation.
<SM table name>_unload_result.txt	A record of how many UNLs are to be exported for a specified table
<SM table name>_unload_structure.txt	A description of the table structure of the original Service Manager system This information is used for comparison with the table structure in the new Service Manager system, and to help to decide if the delta data in table should be migrated here.
<SM table name>_unload_structure_new.txt	Describe the table structure of the new Service Manager system **Note** Note These files only appear during the import operation.
<SM table name>_<start time>_<end time>.UNL	Unloaded UNL files The format of the start and the end time is "yyyymmdd_hhMMss"
table_exported.txt	A list of tables for which the delta data is exported The **timestamp** field in each table is also recorded in this file

How can I decide if the data in a specified table should be unloaded/loaded or not?

If the table has a **sysmodtime** field, and if the structure of the table is the same in the original system and in the new system, then the delta data can be migrated without problem. Simply migrate the data according to the suggestions made by the tool.

If a table has no **sysmodtime** field, the tool will suggest that you don't migrate the data in it. If the information in the table is not time-sensitive or is not changed very often, you can migrate all data in it to the new system by using a Service Manager unload script or by using the database functions.

If a table structure has changed, but the tool reports no risk to the import operation, then you can migrate the table. However, if the tool reports risks, don't import the table. For more information, see Cannot handle skipped tables.

13. Calendar Administration

As of version 9.34, Service Manager additionally provides a calendar (referred to as Service Manager Calendar) that's based on a calendar widget.

Service Manager Calendar can display time period records and associated business records in a graphic and intuitive user interface. It enables users in different locations and departments of your organization to perform the following tasks and optimize their task planning accordingly:

- To easily see how their activities will be affected in a specific time range. For example, they can see if a performance degradation or service outage will happen during the first week of May.
- To easily see which business records (changes, incidents, interactions, and so on) are scheduled for or associated with a specific time range. For example, they can see if a change is planned to start or end during the first week of May.
- To directly view important details of time period records and associated records from tooltips.
- To directly open records from the calendar to view their details.
- To view time conflicts of object records from conflict icon tooltips and shadow bars.

Service Manager Calendar also allows high-level tailoring through a set of configurations. For time period records and associated records, you can set color preferences, maximum number of records to display, field mappings, filters, and so on.

Related topics

Full calendar and embedded calendar
Calendar security area and role

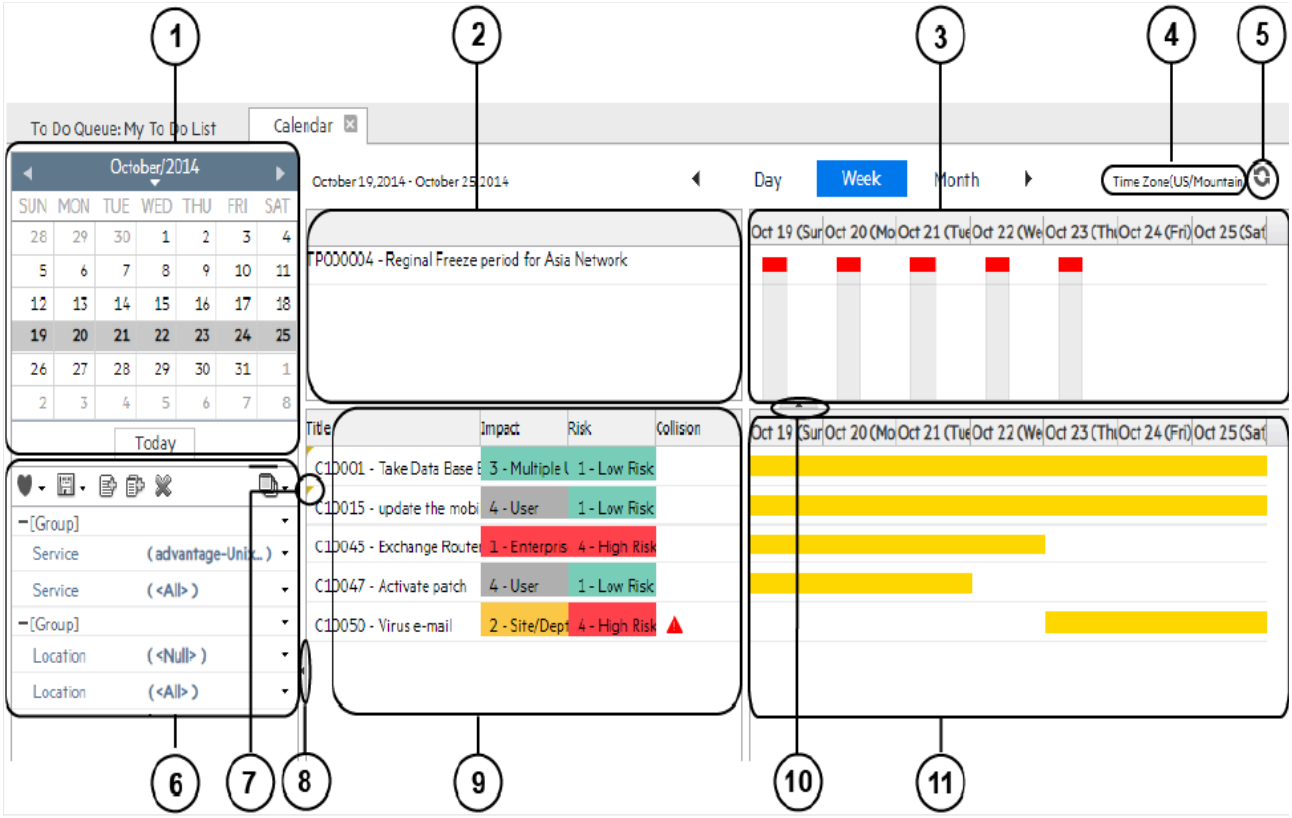
13.1. Full calendar and embedded calendar

Service Manager Calendar has two views: Full Calendar and Embedded Calendar.

Full calendar

This is a complete view of the calendar, accessible from **Miscellaneous** > **Calendar**, or from a button in the embedded calendar.

The following figure illustrates the full calendar.



The following table describes the major areas of the calendar.

Callout	Description
1	The date picker panel. You can use the arrow buttons on this panel to quickly jump to a specific year/month/date.
2	The Time Period record list panel, which is the upper part of the record list area. Note that the record list area is also referred to as the "Title area" of the calendar. This panel lists the time period records being displayed in the current calendar view, based on the settings in the **Title Configuration** section in **System Administration** > **Ongoing Maintenance** > **Calendar Administration** > **Calendar Mappings**.
3	The Time Period graphic view panel, which is the upper part of the right-side area of the calendar ("calendar entry area"). This panel displays time period occurrences as a single-color bar, and the length of each bar represents an occurrence duration. **Note** Note This panel displays only active time period occurrences.
4	The time zone area, which displays the time zone that's defined in the current user's operator record, or the one defined in the System Information Record if the former doesn't exist. **Note** Note If neither exists, this area displays `Time Zone (undefined)` .
5	The Refresh button, used to fresh the calendar display.
6	The filter panel, used to manage filtered views of the calendar. This filtering feature helps users locate data of their interest. On the filter panel, users can add individual filter fields and field groups.

7	Conflict icons. Each of these icons indicates that the current record has conflicts with one or more time periods of the Freeze Period or Maintenance Window type, or has one or more custom messages configured, or both. These icons have tooltips that show specific conflict information and custom messages. A record that has conflicts is one that exceeds, or misses, or overlaps with a Freeze Period or Maintenance Window time period. When it has conflicts only with maintenance windows, a yellow icon is displayed; as long as it has conflicts with freeze periods, a red icon is displayed; if it has no conflicts but has only custom messages configured, a blue icon is displayed. Conflicts and overlaps are also indicated by gray background bars ("shadow bars") behind the records bars. The more time periods a record has conflicts with, the darker the shadow bar becomes.
8	This button collapses the entire left-side panel (which consists of the date picker panel and the filter panel).
9	The object record list panel, which is the lower part of the record list panel ("Title area"). This panel lists object records that are associated with the current time range. For example, this panel may list a Change record whose Scheduled Implementation Start or Scheduled Implementation End time falls into the currently displayed time range. The Calendar Settings and Calendar Mappings configurations determine which records are displayed in this area.
10	This button collapses the entire time period panel.
11	The graphic view panel of object records, which is the lower part of the "calendar entry area". This panel displays relevant object records as a single-color bar to provide an intuitive view of the start and end times of these records. Administrators can use different colors to identify different records.

Embedded calendar

The embedded calendar is a simplified view of the calendar, accessible only from certain objects. For example, from the calendar section in a Change or a Change Task record. The embedded calendar displays information about time periods and object records in a way that's more relevant to the current record. This is implemented through Embedded Calendar Configuration, which allows you to configure a filter query to display related records in the embedded calendar.

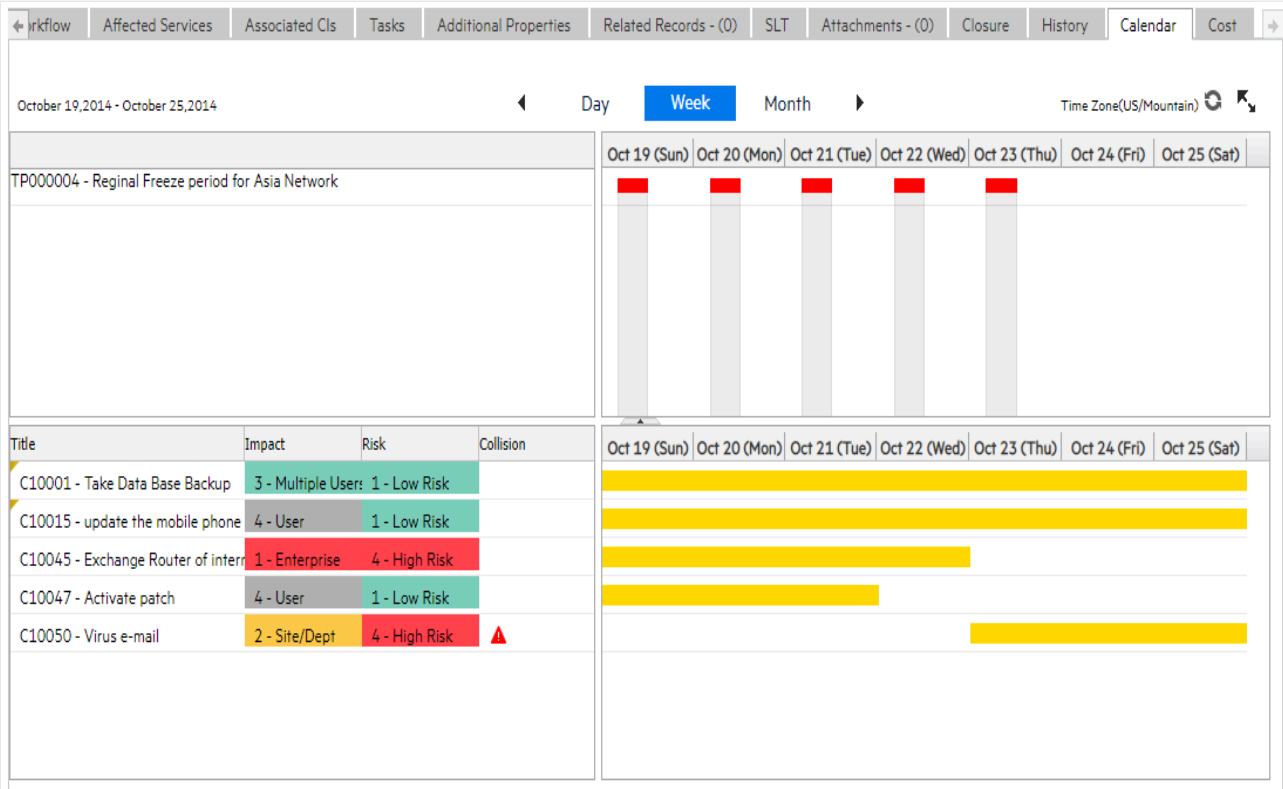
Note

Note The embedded calendar doesn't contain the date picker panel and filter panel that exist in the full calendar; however, if you want to view the same data in the full calendar, click the **Launch full calendar** button ( ) at the top right corner. The full calendar inherits the data filter of the embedded calendar.

Caution

Caution Calendar consumes a certain amount of system resources (memory and CPU). Therefore, if you plan to embed the calendar in forms that are heavily used in your organization's daily operation (such as incident or interaction forms), and if your server already runs at high capacity, we recommend that you analyze and understand the impact before deployment.

The following figure shows the embedded calendar that's displayed in a Change record.



Related topics

Calendar Administration
Calendar security area and role
Working with the calendar
Calendar administration tasks

13.2. Calendar security area and role

The default Calendar Administration security area and Calendar Administrator security role are defined for Service Manager Calendar.

Security area

The default Service Manager Calendar uses the following security area.

Area	Description
Calendar Administration	This area contains the default security rights that will be applied to all of the roles created for the Calendar area.

To access the Calendar Administration security area, follow these steps:

1. Navigate to **System Administration** > **Security** > **Areas**.
2. In the **Name** field, select **Calendar Administration**.

3. Click **Search**.

Security rights and settings

The following table describes the security rights and settings used in Service Manager Calendar.

Setting/Right	Description
View	Can view Calendar Filter Mapping, and Calendar Mapping records as well as their related records (such as filter field, filter mapping, embedded configuration records)
New	Can add Calendar Filter Mapping, and Calendar Mapping records as well as their related records (such as filter field, filter mapping, and embedded configuration records)
Delete/Close	Can delete Calendar Filter Mapping, and Calendar Mapping records as well as their related records (such as filter field, filter mapping, and embedded configuration records)
Expert	Unused
Admin	Unused
Modify Template	Unused
Add Public Favorite	Can create public favorites
Delete Public Favorite	Can delete public favorites
Update Public Favorite	Can update public favorites

Security role

Out-of-the-box, the following security role is available for Service Manager Calendar.

Role	Description	Default Rights/Setting in the Calendar Administration Area
Calendar Administrator	Full rights	• View • New • Update (Always) • Delete/Close (Always) • Expert • Admin • Modify Template • Add Public Favorite • Delete Public Favorite • Update Public Favorite

To access the Calendar Administrator security role, follow these steps:

1. Navigate to **System Administration** > **Security** > **Roles**.
2. In the **Name** field, select **Calendar Administrator**.
3. Click **Search**.

Related topics

Calendar Administration
Full calendar and embedded calendar

13.3. Calendar administration tasks

Before users can use the calendar, administrators must set up the following configurations.

- Enable or disable the calendar
  Enable or disable the Service Manager calendar.
- Configure calendar settings
  Configure the settings of the calendar, such as the objects (for example, Change and Incident) that can be mapped with time periods, and the maximum number of records returned in each calendar object query or calendar time period query.
- Display records from an application module in the calendar
  Define a mapping record for the Time Period object, and for each object mapped with it. Each mapping record defines a field mapping between the object and Time Period, and determines how the calendar displays records of the object.
- Display an embedded calendar on an incident form, an interaction form, or a problem form
  Display an embedded calendar on an incident form, an interaction form, or a problem form.
- Configure search filters for the full calendar
  Configure a single filter mapping record for the full calendar and the embedded calendar.
  For the full calendar, specify some fields of the specified objects as filter fields so that users can create filtered views of the full calendar. For the embedded calendar, define filters for the specified objects so that the embedded calendar automatically displays different filtered views in different records.
- Localize a filter field
  Create localized versions of the field labels if the environment needs to support multiple languages.
- Test your calendar
  Test the calendar to see if it works properly.

13.3.1. Enable or disable the calendar

The **Enable Calendar and Time Period Management** checkbox enables or disables Service Manager Calendar. By default, this checkbox is selected, which means Service Manager Calendar is enabled. To disable Service Manager Calendar, follow these steps:

1. Click **System Administration** > **Base System Configuration** > **Miscellaneous** > **System Information Record**.
2. Click the **General** tab.
3. Clear the **Enable Calendar and Time Period Management** checkbox.
4. Click **Save**.

Related topics

Calendar Administration
Full calendar and embedded calendar
Calendar security area and role
Configure calendar settings
Display an embedded calendar on an incident form, an interaction form, or a problem form
Display records from an application module in the calendar
Configure search filters for the full calendar
Localize a filter field
Test your calendar

13.3.2. Configure calendar settings

Administrators configure the calendar settings to define the overall look and feel of Service Manager Calendar.

To configure the calendar settings, follow these steps:

1. Click **System Administration** > **Ongoing Maintenance** > **Calendar Administration** > **Settings**.
2. Edit the fields as needed.
   The following table describes each field.

Field	Description	Default	Notes
Module	The name of the module to which this record belongs	Calendar	This setting is system generated and is not editable.
Area	The security area to which this record belongs	Calendar Administration	This setting is system generated and is not editable.
First Day of Week	The first day of the week	Sunday	This setting takes effect only on the calendar, not on the date-picker.
Max Records Returned in Calendar Object Query	Determines the maximum number of object records that a calendar query can return. This setting limits the number of object records (for example, Change and Incident records) that can be displayed in the calendar.	50	Must be a positive integer.
Max Records Returned in Time Period Query	Determines the maximum number of time period records that a calendar query can return. This setting limits the number of time period records that can be displayed in the calendar.	25	Must be a positive integer.
Time Range in Day View	Determines how the calendar daily view divides a day. By default, one day is divided into four sessions: • 00:00-06:00 • 06:00-12:00 • 12:00-18:00 • 18:00-24:00	Every 6 hours	Select a value from the list.
Available objects for Calendar Mapping	The table names of the objects that can be mapped with the Time Period object. The names are comma-separated.	cm3r,cm3t, incidents,probsummary,root cause	You can specify any tables (objects). This setting takes effect for both the full calendar and the embedded calendar. **Example Values**: cm3r, probsummary, rootcause, incidents

3. Save the record.

Now, you have completed the global settings for the calendar. Next, you need to configure calendar mapping. See Display records from an application module in the calendar.

Related topics

Enable or disable the calendar
Display an embedded calendar on an incident/interaction/problem form
Display records from an application module in the calendar
Configure search filters for the full calendar
Localize a filter field
Test your calendar

13.3.3. Display records from an application module in the calendar

The purpose of calendar mapping is to ensure that certain objects of Service Manager (such as Incident, Changes, and Time Periods) are properly displayed in the calendar. This feature also links the Time Period object with other objects in the system. Before proceeding with this task, you, as a system administrator, should have already specified the objects that you want to map with the Time Period object. See Configure calendar settings.

Now, you need to create a mapping record for each of these objects, including the Time Period object.

Caution

Caution Create only one mapping record for each object; otherwise, a duplicate record error occurs.

The Calendar Mapping form is accessible from **System Administration** > **Ongoing Maintenance** > **Calendar Administration** > **Calendar Mappings**. By default, there are three mapping records for Time Period, Change, and Change Task, respectively.

The following table describes the sections of the calendar mapping form.

Section	Description
Header	Displays general information of the mapping record, such as the ID, associated object, active or inactive state, and description.
Restricting Query Configuration	This is a default query that allows only relevant records of the selected object to be displayed in the calendar. The query must use this format: `field="value"`, where `field` is the name of a field in the object table, and `value` can be a static value, a RAD expression (for example, `affected.item in $L.file`), or a JavaScript call (for example, `jscall("JS method");`); or you can enter `true` for no restrictions. **Example value**: `status="closed" and category="Application"`
Title Configuration	Specifies which fields of the object to display in the Title area (record list area) of the calendar. All available fields of the selected object are displayed in the Available Fields box, and the fields that are used in the Title area are displayed in the Selected Fields box. **Note** Note This is a mandatory setting.
Tooltip Configuration	Specifies which fields are displayed in the tooltip when you hover your mouse over a record (or a record bar) of the selected object in the calendar. All available fields of the selected object are displayed in the Available Fields box, and the fields that are used in the tooltip are displayed in the Selected Fields box.
Color Preferences	Specifies color settings for the selected object. To cover different requirements, you can enter a condition for each color. The conditions are evaluated in top-to-bottom order. As soon as one condition evaluates to true, the evaluation of the color preferences is stopped. If you specify no color preferences, a default color is used. In the Default Color field, you can configure a default color by using HTML color codes (for example, `#000000`).
Default Sort Preferences	Specifies the default sort preferences for the Title area and calendar entry area in the calendar. If you specify no default sort preferences, records will be sorted by record ID in ascending order.

Time Period Field Mapping	This section is available only for objects other than Time Period. It creates a link between Time Period and the object. The mapping is used for conflict detection and overlapping functionality in the calendar.
	• Field for Start Date and Field for End Date (Mandatory): Specify which fields of the object are mapped with the Start Date and End Date of the Time Period object. The calendar displays records of the object as a record bar whose length is calculated based on the Start Date and End Date values.
	• Time Period Mapping Query: This query defines which time period objects are used for detecting the conflict for the object. When a conflict is detected, the calendar displays a conflict icon for a record. It also displays a gray background bar ("overlap shadow bar") behind the record bar when the record either overlaps with, exceeds, or misses a freeze window or maintenance window (see the *Conflict Icon Custom Messages* row for more information). You can use `$L.file` as a reference to the current record.
	The following is a query example:
	`current.phase="published" and (affected.services=affected.item in $L.file or affected.services="all") and (location=location.full.name in $L.file or location="all")`
	• Query for Time Period Conflict Calculation: You can enter a query against the selected object to restrict the records for conflict calculation with Time Period.
	The following is aquery example :
	`category in $L.file ~={"CI Group","Default","KM Document","Unplanned Change","Subscription"} and open in $L.file=true`

Conflict Icon Custom Messages	A conflict icon refers to a small red or yellow triangle icon that's displayed at the top left corner of an object record cell (see Full calendar and embedded calendar). The red triangle indicates the current record overlaps with one or more time periods of the Freeze Period type, while the yellow triangle indicates the record either exceeds or misses one or more time periods of the Maintenance Window type. When both types of conflicts occur, only the red triangle is displayed. The icon tooltip displays specific conflict information of the record. You can configure additional custom messages to be displayed as part of the tooltip of a conflict icon. For example, you can specify the following configuration so that the "This is an emergency change" message is displayed in the conflict icon tooltip for each Emergency Change record. • Condition: `emergency in $L.file=true` • Message: `scmsg(1,"timeperiodCalendarMapping")` **Note** Note If a record has no conflicts with any time periods but has one or more custom messages configured, a blue triangle is displayed instead of the red or yellow one.

Example: Display Time Period, Change, and Incident in the calendar

This example configuration assumes that you have specified the `cm3r` (Change) and `probsummary` (Incident) objects for calendar mapping in the calendar global settings. Now you need to configure a mapping record for each of the objects: Time Period, Change, and Incident.

By default, there are two mapping records for Time Period and Change, respectively. You may want to edit the existing mapping records to better suit your needs, and add one new mapping record for Incident.

To display Time Period, Change, and Incident in the calendar, follow these steps:

1. Click **System Administration** > **Ongoing Maintenance** > **Calendar Administration** > **Calendar Mappings**.
2. Click **Search**.
   A list of existing mapping records is displayed. Out-of-the-box, only the mapping records for Time Period, Change and Change task are displayed.
3. Edit the CALMAP01 (for Time Period), the CALMAP02 (for Change), and the CALMAP03 (for Change Task) records as needed.
   1. In the **Header** section, edit the optional Description field and make sure the **Active** option is selected.
      Only active objects can be displayed in the calendar. Canceling this option will disallow records of the object to appear in the calendar.
   2. In the **Restricting Query Configuration** section, enter a query. See the following table for examples.

Object	Restricting Query Configuration	Note
Time Period	`true`	This query filters the records to be retrieved from the `timeperiodDefinition` table.
Change	`true`	This query filters the records to be retrieved from the `cm3r` table.
Change Task	`true`	This query filters the records to be retrieved from the `cm3t` table.

3. In the **Title Configuration** section, enter a field separator or leave the field empty to use the default (a hyphen), and move the fields that you want to display in the Title area to the Selected Fields box. See the following table for examples.

Object	Selected Fields
Time Period	`ID` `Name`
Change	`Change ID` `Title`
Change Task	`Incident ID` `Title`

4. In the **Tooltip Configuration** section, move the fields that you want to display in the object's tooltip (hover-over text) to the Selected Fields box, and use the up/down arrow button to change the sort order of the fields. See the following table for examples.

Object	Selected Fields

Time Period	`ID`
	`Type`
	`Category`
	`Scope`
	`Name`
	`Owner Group`
	`Services`

Change	`Change ID`
	`Title`
	`Description`
	`Service`

Change Task	`Status`
	`Priority`
	`Affected Services`

5. In the **Color Preferences** section, click **Add** to add color preferences. See the following table for examples, where you specify three colors for three types of time periods, and one color for all change records.

Object	Color	Name	Condition
Time Period	`#FF0000`	`Freeze Period (Red)`	`(type in CurrentRecord = "nogo")`
Time Period	`#018424`	`Maintenance Window (Green)`	`(type in CurrentRecord = "go")`
Time Period	`#0197D6`	`Neutral Period Period (Blue)`	`(type in CurrentRecord = "neutral")`
Change	`#FFD800`	`Hardware (Yellow)`	`( assign.dept in CurrentRecord = "Hardware")`
Change Task	`#FFB11B`	`Hardware (Yellow)`	(assign.dept in CurrentRecord = "Hardware")

6. In the **Default Sort Preferences** section, select the fields by which you want to sort records of the object, and select **Ascending** or **Descending** order. See the following table for example.

Object	Sort Field	Sort Type
Time Period	ID	Ascending
Change	Change ID	Ascending
Change Task	Task ID	Ascending

7. In the **Time Period Field Mapping** section, select the fields to map with the Start Date and End Date in the calendar, and specify a mapping query as necessary.

   Note
   Note The Time Period Field Mapping section is unavailable for Time Period objects.

   See the following table for an example.

Object	Field for Start Date	Field for End Date	Time Period Mapping Query	Query for Time Period Conflict Calculation
Change	Planned Start	Planned End	current.phase="published" and (affected.services=affected.item in $L.file or affected.services="all") and (location=location.full.name in $L.file or location="all")	category in $L.file ~={"CI Group","Default","KM Document","Unplanned Change","Subscription"} and open in $L.file=true

8. Configure conflict icon custom messages for the Change mapping record. See the following table for examples.

Condition	Message
emergency in $L.file=true	scmsg(1,"timeperiodCalendarMapping")
foreign.id in $L.file~=NULL	scmsg(2,"timeperiodCalendarMapping") + foreign.id in $L.file

9. Save the mapping records.

4. follow these steps to add a mapping record for Incident.
   1. Click **System Administration** > **Ongoing Maintenance** > **Calendar Administration** > **Calendar Mappings**.
   2. In the **Object** field, select **Incident**.
   3. In the **Description** field, enter a description of the mapping (optional).
   4. Click **New**, and then click **Save**.
      The record is added and automatically assigned with an ID.
   5. Select the **Active** checkbox.

> **Note**
>
> Note By default, each mapping record is created as Inactive. Only active mappings take effect. Once a mapping is created, you can update it but can no longer delete it. If a mapping is no longer needed, disable it by clearing the **Active** checkbox.

6. In the **Restricting Query Configuration** section, enter a query. For example, use the default value: `true`.
7. In the **Title Configuration** section, enter a field separator or leave the field empty to use the default (a hyphen) , and move the fields that you want to display in the Title area to the Selected Fields box. See the following table for an example.

Object	Selected Fields
Incident	`Incident ID` `Title`

8. In the **Tooltip Configuration** section, move the fields that you want to display in the object's tooltip (hover-over text) to the Selected Fields box, and use the up/down arrow to change the sort order of the fields. See the following table for an example.

Object	Selected Fields

Incident	`Status`
	`Priority`
	`Affected Services`

9. In the **Color Preferences** section, configure a default color for the Incident object, and if needed add more color preferences.
   To configure a default color, in the Default Color field, enter an HTML color code (for example, `FF8000`).
   To add more color preferences, click **Add**, select a color from the color palette, enter a name for the selected color, if needed click **Edit** to enter a condition (RAD Expression) using the Condition Editor, and then click **OK**. See the following table for an example, where you specify one color for incident records that are in Closed status, which implies that incident records in any other status will display in the default color.

Object	Color	Name	Condition
Incident	`#01DF01`	`Closed Incidents (Green)`	`( Expression: problem.status in $L.file = "Closed")`

10. In the **Default Sort Preferences** section, select the fields by which you want to sort records of the object, and select the **Ascending** or **Descending** order. See the following table for an example.

Object	Sort field	Sort type
Incident	`Incident ID`	`Ascending`

11. In the **Time Period Field Mapping** section, select the fields to map with the Start Date and End Date of the Time Period object, and if needed specify a mapping query for conflict detection. See the following table for an example.

Object	Field for Start Date	Field for End Date	Time Period Mapping Query
Incident	Outage Start	Outage End	true

12. In the **Conflict Icon Custom Messages** section, configure messages and their conditions for conflict icon tooltip display.
13. Save the mapping record.

5. Similarly, add a mapping record for other objects (Problem, Interaction, and so on) as necessary.

Related topics

Calendar Administration
Full calendar and embedded calendar
Calendar security area and role
Enable or disable the calendar
Configure calendar settings
Display an embedded calendar on an incident/interaction/problem form
Configure search filters for the full calendar
Localize a filter field
Test your calendar

13.3.4. Display an embedded calendar on an incident/interaction/problem form

Caution: Calendar consumes a certain amount of system resources (memory and CPU). Therefore, if you plan to embed the calendar in forms that are heavily used in your organization's daily operation (such as incident or interaction forms), and if your server already runs at high capacity, we recommend that you analyze and understand the impact before deployment.

The default embedded calendar is accessible from the calendar section in a Change or a Change Task record only. To display an embedded calendar on an incident form, an interaction form, or a problem form, follow these steps:

1. Click **System Administration** > **Ongoing Maintenance** > **Calendar Administration** > **Calendar Filter Mappings** to open the **Calendar Filter Mapping** form.
2. Go to the **Embedded Calendar Configuration** section, and add predefined filters for the embedded calendar. You may want to add one or more filters for each object. The default embedded calendar is available only for the Change object and the Change task object.
    1. Click **Add Configuration**.
    2. Complete the fields and select the options as described in the following table.

Field/Option	Description
Active	Indicates if the configuration is active. Only active configurations take effect. By default, this option is selected when a new configuration is added. **Caution** Caution If one or more configuration records of the same module is active, the embedded calendar is visible in the current record. Otherwise, it will be invisible in the current record.
Object	Select an object to filter against. **Example**: Change
Default View	Select one value from the list: **Day**, **Week**, or **Month**. This field determines the default display view of the embedded calendar. If you leave it empty, the embedded calendar is displayed in the **Week** view.
Field for Initial Date	Select a date/time field from the list. Initially, the embedded calendar displays data based on the day or week of this field's value in the current record. For example, if you select **Scheduled Implementation Start** for the Change module, when users view the embedded calendar from a Change record whose Scheduled Implementation Start date is 22 May 2013, the embedded calendar will initially display data for the day or week of 22 May 2013 (the Default View value determines whether a daily or weekly view is initially displayed). This issue also occurs when users view the embedded calendar from another record, such as an Incident record. **Note** Note If you leave the Field for Initial Date field empty or if the specified field of the current record has an empty value, the embedded calendar uses the user's current date.

Filter Query	Type a query that filters Time Period records and records of the selected objects in the Calendar Entries field. Pay attention to the following: ▪ The query field should be one of the filter fields defined in the Filter Fields Configuration. ▪ Use "=" as the operator between the field and the value, and don't miss the space before and after the "=" operator. ▪ You can use `$L.file` in the value as a reference to the current record. ▪ The query supports "or" inside a field group and "and" between groups and single fields. Use lower case for the " `and` " and " `or` " operators. ▪ If you are using an RAD expression, you must insert a space or a semicolon (;) at the end of the expression. **Tip** Tip Click the **Analyze** button to see the logical structure of the query. This helps you make sure the query is exactly what you want. The analysis result is displayed in the Analysis Result box. **Example**: `assign.group = assignment in $L.file and (service = affected.item in $L.file or service = "all") and (location = location.full.name in $L.file or location = "all")`
Condition	The condition under which the filter query is executed. **Example**: `Always` (default)
Calendar Entries	Select the objects that you want to display in the embedded calendar. Only objects that have an active Calendar Mapping record defined (in **System Administration** > **Ongoing Maintenance** > **Calendar Administration** > **Calendar Filter Mappings**) are displayed and available for your selection.

3. Click **Save**.

Tip
Tip Immediately after you add a filter field and click **Save**, an **Add** button appears in the field form, allowing you to quickly add another new field.

4. Add more filters as needed.
The records you added appear in the list.

Tip

Tip To delete an existing filter record from the list, click the record to open it, and then click **Delete**.

3. Save the calendar filter mapping.

Related topics

Enable or disable the calendar
Configure calendar settings
Display records from an application module in the calendar
Configure search filters for the full calendar
Localize a filter field
Test your calendar

13.3.5. Configure search filters for the full calendar

The Calendar Filter Mapping feature enables users to manage filtered views of Service Manager Calendar, so that Service Manager Calendar displays only information of their interest.

Before users can do so, you, as an administrator, should configure filter mapping through the Calendar Filter Mapping form. This form is accessible from **System Administration** > **Ongoing Maintenance** > **Calendar Administration** > **Calendar Filter Mappings**. The following table describes the sections of this form.

Section	Description
Header	This section includes a description and data modification information of the filter mapping record.
Filter Fields Configuration	This section specifies the fields for the filter form in the calendar, and maps the fields in the filter form to a real field in the database. A list of specified fields appear on the **Add Filter Field** window, which is accessible from the filter area of the calendar. The following figure is an example of the **Add Filter Field** window. Add Filter Field Field ID | Label | Panel | Field Type | Data Source service | Service | Common | Advanced Selector | Queryselect logicalname,sm.device,displayname from device where type='bizservice' order by logicalname asc environment | Environment | Time Period | Box Selector | Globallist:environment department | Department | Time Period | Advanced Selector | Queryselect dept from dept owning.group | Owning Group | Time Period | Advanced Selector | Queryselect name from assignment where operators=operator() category | Category | Time Period | Box Selector | Queryselect name from timeperiodCategory where active=true order by name asc type | Type | Time Period | Box Selector | Globallist:timeperiodCategoryTypes scope | Scope | Time Period | Box Selector | Globallist:timeperiodScope current.phase | Current Phase | Change | Advanced Selector | Globallist:cm3rcatphase approval.status | Approval Status | Change | Box Selector | Globallist:Change Approval Status assign.group | Assignment Group | Change | Advanced Selector | Globallist:Assignment Groups priority | Priority | Change | Box Selector | Globallist:Calculated Priority impact | Impact | Change | Box Selector | Globallist:Initial Impact urgency | Urgency | Change | Box Selector | Globallist:Urgency affected.CI | Affected CI | Change | Advanced Selector | Queryselect logicalname,sm.device,displayname from device where type='bizservice' change.category | Category | Change | Box Selector | Globallist:cm3rcategory location | Location | Common | Advanced Selector | Globallist:locations (full) Each panel represents an object. Fields that appear on the Common panel are those configured for all available objects.
Embedded Calendar Configuration	This section specifies the filters for displaying the correct records in the embedded calendar when the calendar is displayed initially. The predefined filters are executed as soon as you launch the embedded calendar. With this filter setting, you can configure a query to display other related records in the calendar.

Example: Configure search filters for the full calendar

Before proceeding with this configuration task, make sure that you have already completed the following tasks:

- Configure calendar settings
- Display records from an application module in the calendar

This example assumes that you have already configured a calendar mapping record for Time Period and Change. In this example, you will define filter fields and field mappings for each object so as to filter which related records are displayed on the full calendar.

1. Click **System Administration** > **Ongoing Maintenance** > **Calendar Administration** > **Calendar Filter Mappings** to open the **Calendar Filter Mapping** form.
2. In the **Header** section, type a description in the **Description** field (optional).
3. In the **Filter Fields Configuration** section, add filter fields and field mappings.
    1. Click **Add Filter Fields** to open the **Calendar Filter Field** form. By default, the **Active** option is selected.

        Note

        Note Only active filter fields are used for filtering.

    2. Complete the fields on this form.

Field	Description
Field ID	Type a unique identification name for the field. This internal ID is used for filter mapping in the next step. **Example**: `Change ID`.
Label	Type a display name for the field ID. This field is mandatory. **Example**: `Change ID`.
Panel	Select a panel from the list. This is the name of a tab of the **Add Filter Fields** window, which is displayed when you click the **Add Filter Fields** button. The list includes the following panels: ▪ Time Period ▪ Common ▪ Objects specified in the **Available Objects for Calendar Mapping** field (**Calendar Administration** > **Calendar Settings**). **Example**: `Change`.
Sequence	This is the order in which the field is displayed in the **Add Filter Fields** window. **Example**: `1`.
Field Type	Select a field type to define the layout of the values of the field on the filter form. The following types are supported: ▪ Text: Allows you to enter a text string as a value for the field. ▪ Box Selector: Allows you to select a value for the field using a list of checkboxes. **Note** Note We recommend that you use this type for a field that has no more than 10 values. Displaying a large number of values in a Box Selector may cause usability issues. ▪ Advanced Selector: Allows you to select one or more values for the field from a list of available values. ▪ Date Time: Allows you to specify a date/time or date/time range value for the field. **Example**: `Advanced Selector`.

Data Source Type	This field is available only when the Field Type is **Box Selector** or **Advanced Selector**. The following types are supported: ■ Global List ■ RAD Expression ■ Query
Data Source Value	This field is available only when the Data Source Type is **Global List** or **RAD Expression**. ■ Global List: Specifies the global list to store the value list and display list. ■ RAD Expression: Specifies a RAD expression that returns a JSON object that contains the value list and display list. The format of the returned JSON object is as follows: {"values":["value1","value2"], "labels": ["display1", "display2"]} Below is an example RAD expression: `jscall("CalendarFieldValueJSCall.getMyAssignmentGroupOperators")`
Query Configuration	These settings are available only when the Data Source Type is Query. It specifies a query against the data in a field of a table, which retrieves a value list and display list for the field. ■ File: the name of the table to be queried. For example, `device`. ■ List Field: the name of the field that stores the value list. For example, `logical.name`. ■ Display Field: the name of the field that stores the display list. If it is specified, the calendar displays the values of this field for users to select from. For example, `id`. If it's not specified, the calendar uses the List Field instead. ■ Limiting SQL: a condition of the query. For example, `type="bizservice"`. ■ Sort By: the name of the sort-by field of the query results. If it's not specified, the List Field is used.
Multiple Select	This option is available only when the Data Source Type is **Box Selector**. Select this option if the filter field supports the selection of multiple values.

Extended Values Extended Labels	An additional user-defined list of values and their display labels. The labels are displayed at the top of the list of values to select for the filter field. For example, you can define the following extended value and label for the Time Period module so that each user can choose to display only time period records valid for the user's own department. ▪ Extended Value: `$lo.dept` ▪ Extended Label: `My department`

3. Click **Save**. An ID is automatically assigned to the filter field.
4. Click **Add Field Mapping**to open the **Calendar Field Mapping** form.
5. Select a module and a mapping field, and click **Save** to add the calendar filter field mapping. By default, the **Active** option selected.
6. If needed, continue to follow *step d* and *step e* to map this field to multiple modules.
7. Click **Save & Exit**.

4. If needed, continue to follow *step 1* to *step 3* to add multiple fields to the filter form.

> **Tip**
> Tip If you want to specify a filter field for multiple modules (for example, a filter field for `affected.item` in Change and `affected.services` in Time Period), you can specify the field for the Common panel, and then add mappings for each relevant module.

5. Save the calendar filter mapping.

Now, you have completed your calendar configurations. Next, you can test your calendar. See Test your calendar.

Related topics

Enable or disable the calendar
Configure calendar settings
Display an embedded calendar on an incident/interaction/problem form
Display records from an application module in the calendar
Localize a filter field
Test your calendar

13.3.6. Localize a filter field

Once you have added a filter field in the Calendar Filter Field form, you may need to create localized versions of the field labels if your environment needs to support multiple languages.

To create a localized version of a filter field, follow these steps:

1. Identify the message ID of the filter field.
    1. Click **System Administration** > **Ongoing Maintenance** > **Calendar Administration** > **Calendar Filter Mappings**.
    2. Click the field ID in the **Filter Fields Configuration** section to open the field detail form.
    3. Click **Localize**.
    4. Make a note of the message ID, and then click **Cancel**.
2. Make sure the target language you need is activated.
    1. In Database Manager, type language in the **Table** field, and then double-click **language** to open the **Language Identification** form.
    2. In the **Language** field, type the language name. For example, Japanese.
    3. Select **Active for login** option.
    4. Click **Save** and **OK**.

3. In Database Manager, type scmessage in the **Table** field, and click **Search**.
4. Click **message.edit**.
5. Add a record for the language.
    1. Type the required field values. See the following table.

Field	Value
Language Code	The language code of the target language. For example, jp.
ID	The original En message ID you recorded in step 1.
Class	Always type local:calendarFilterEditForm
Text	Type the field label in the target language.

    2. Click **Add**.
6. Go back to the filter field detail form, and click **Localize**.
   The localized version is now displayed in the list.
7. Restart the Service Manager web application server or the Windows client for the changes to take effect.

Related topics

Calendar Administration
Full calendar and embedded calendar
Calendar security area and role
Enable or disable the calendar
Configure calendar settings
Display an embedded calendar on an incident form, an interaction form, or a problem form
Display records from an application module in the calendar
Configure search filters for the full calendar
Test your calendar

13.3.7. Test your calendar

Once you have completed your Service Manager Calendar setup, test your calendar to see if it works properly.

Prepare data

Prepare data so that you can see records of your configured objects in the calendar and test them.

1. Create time period records for testing.

> **Tip**
> Tip You may want to create time periods with a daily recurrence type so that you can easily verify your test results.

2. Create new records or update existing records for objects that you have configured for calendar mapping.

Test the full calendar

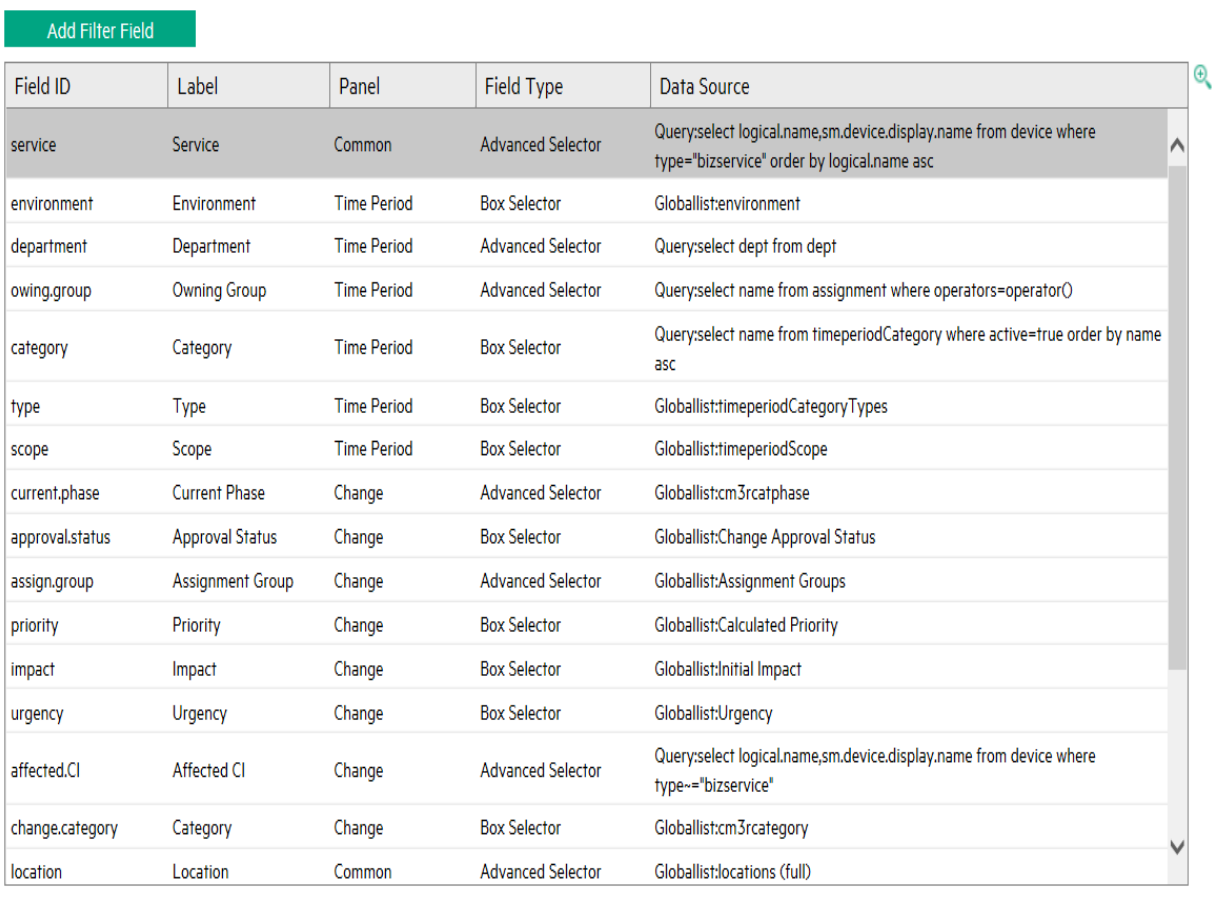
To test the full calendar, follow these steps:

1. Click **Miscellaneous** > **Calendar**.
2. Switch the calendar to a desired day, week, or month view, where you expect to see some Time Period, Incident, and Change records you previously created.
3. Go to the left-side filter area, click the **Choose Objects** button, and check that all objects you configured are displayed and selected in the list.

Tip

Tip If an object is missing from the list, clear your browser cache and try again.

4. Select all or some objects, and check that the calendar automatically refreshes to display the records as expected, based on their Restricting Query Configuration, Title Configuration, Tooltip Configuration, and Color Preferences settings.
5. Click the **Add Filter Fields** button, and in the pop-up window, check that the filter fields and panels are displayed as expected.



Field ID	Label	Panel	Field Type	Data Source
service	Service	Common	Advanced Selector	Query:select logical.name,sm.device.display.name from device where type="bizservice" order by logical.name asc
environment	Environment	Time Period	Box Selector	Globallist:environment
department	Department	Time Period	Advanced Selector	Query:select dept from dept
owing.group	Owning Group	Time Period	Advanced Selector	Query:select name from assignment where operators=operator()
category	Category	Time Period	Box Selector	Query:select name from timeperiodCategory where active=true order by name asc
type	Type	Time Period	Box Selector	Globallist:timeperiodCategoryTypes
scope	Scope	Time Period	Box Selector	Globallist:timeperiodScope
current.phase	Current Phase	Change	Advanced Selector	Globallist:cm3rcatphase
approval.status	Approval Status	Change	Box Selector	Globallist:Change Approval Status
assign.group	Assignment Group	Change	Advanced Selector	Globallist:Assignment Groups
priority	Priority	Change	Box Selector	Globallist:Calculated Priority
impact	Impact	Change	Box Selector	Globallist:Initial Impact
urgency	Urgency	Change	Box Selector	Globallist:Urgency
affected.CI	Affected CI	Change	Advanced Selector	Query:select logical.name,sm.device.display.name from device where type~="bizservice"
change.category	Category	Change	Box Selector	Globallist:cm3rcategory
location	Location	Common	Advanced Selector	Globallist:locations (full)

Test the embedded calendar

Note

Note Out-of-the-box, the embedded calendar is available only for Change and Change Task.

To test the embedded calendar, follow these steps:

1. Open a record.
2. Expand the **Calendar** section.
3. Check that the embedded calendar displays the correct information (based on your Embedded Calendar Configuration for the corresponding module).

Related topics

Calendar Administration
Full calendar and embedded calendar
Calendar security area and role
Enable or disable the calendar
Configure calendar settings

Display an embedded calendar on an incident/interaction/problem form
Display records from an application module in the calendar
Configure search filters for the full calendar
Localize a filter field

13.4. Working with the calendar

Once your system administrator has set up the calendar, you can perform the following tasks. For an overview of the calendar and its components, see Calendar Administration.

- Launch the calendar
- Understand the calendar data
- Switch to a desired time slot
- Select a year, month, or date from the date picker (full calendar only)
- Collapse and expand panels (full calendar only)
- View tooltip information
- View time conflicts and custom messages
- Access a record from the calendar
- Drill down to the Day view
- Select the objects (full calendar only)
- Create a filter (full calendar only)
- Clear the current filter
- Manage favorites (full calendar only)

Note

Note Some tasks are allowed only in the full calendar or the embedded calendar, while some are allowed in both. Unless otherwise noted in the task heading, a task is allowed in both.

Tip

Tip Before you work with the calendar, click the **Refresh** button ( ) at the upper right corner of the calendar, to get the up-to-date data.

13.4.1. Launch the calendar

Full calendar

To launch the full calendar, click **Miscellaneous** > **Calendar**.

Embedded calendar

To launch the embedded calendar, follow these steps:

1. Open a change record or a change task record.
2. Expand the **Calendar** section.

Tip

Tip To view the same calendar data in the full calendar, click the **Launch full calendar** button ( ) on the upper right corner of the embedded calendar. The full calendar displays the same data from the embedded calendar (because it inherits the data filter from the embedded calendar).

Related topics

Calendar Administration
Full calendar and embedded calendar

13.4.2. Understand the calendar data

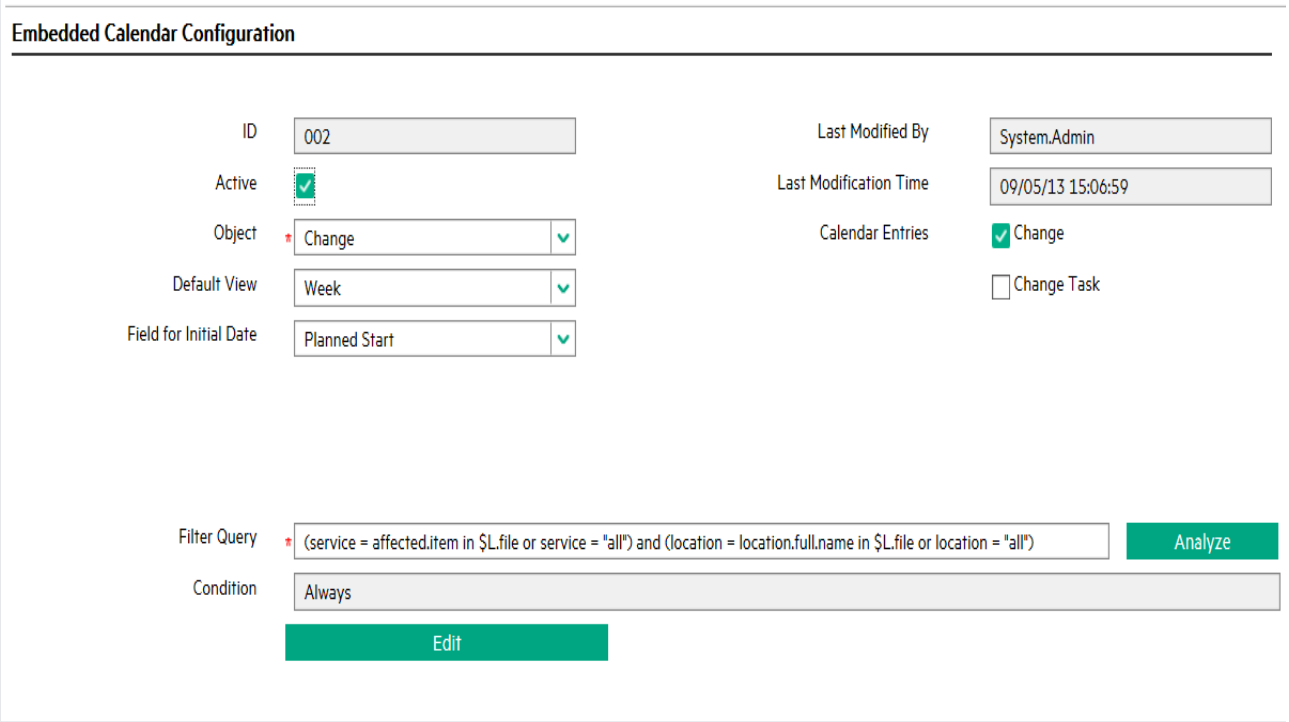
Full calendar

When initially opened, the full calendar displays information about published time periods and object records (such as Change records) for the current week.

Embedded calendar

Initially, the embedded calendar displays data according to the **Field for Initial Date** value in the current record. For example, if your administrator has configured the **Scheduled Implementation Start** field as the **Field for Initial Date** for the Change object (see Configure search filters for the full calendar), when you view the embedded calendar from a Change record whose Scheduled Implementation Start date is 22 May 2013, the embedded calendar will initially display data for the Day, Week, or Month of 22 May 2013 (the Default View value determines whether a Day, Week or Month view is initially displayed).

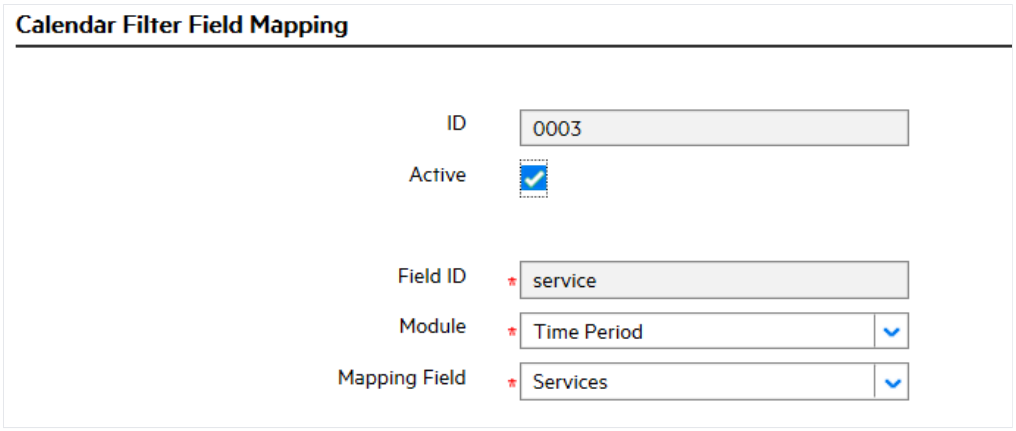
What record data to display in the embedded calendar is determined by the Embedded Calendar Configuration (**Calendar Administration** > **Calendar Filter Mappings**). See the following figure for an example. The example filter query indicates that the embedded calendar will display only records whose mapping fields for `service` and `location` either have the same value as the current Change record's `affected.item` field and `location.full.name` field respectively or have the " `all` " value.



The embedded calendar displays data only for Time Period and for objects that are selected as **Calendar Entries** (see the figure above). The way an object's records display depends on the whether an active field mapping of each Filter Query field has been defined for the object.

- If there is an active mapping for each Filter Query field defined for the object, records of this object are displayed in the current record only when they match the Filter Query;
- If there is not an active mapping record, the current record will not display the embedded calendar widget.

To see if the query fields are mapped to an object, navigate to **Calendar Administration** > **Calendar Filter Mapping**, and then open the filter query field in the Filter Fields Configuration section. The following figure shows an active field mapping of query field `service` for the Time Period object.



Related topics

Calendar Administration
Full calendar and embedded calendar
Calendar security area and role

13.4.3. Switch to a desired time slot

You can display the calendar in one of these time range views: **Day**, **Week**, and **Month**.

Tip

Tip In the full calendar, in addition to the steps below, you can directly select a year, month, and date in the Date Picker panel. See Select a year, month, or date from the date picker (full calendar only).

To switch the calendar to a desired time slot, follow these steps:

1. Click **Day**, **Week**, or **Month** to select a time range view as needed.
   The full calendar is displayed in the Week view by default, while the embedded calendar is initially displayed in the Default View defined in the Embedded Calendar Configuration. The full calendar displays a view based on today's date, while the embedded calendar displays data based on the Field for Initial Date value of the current record. See Embedded Calendar Configuration.
2. Click the left or right arrow button next to the **Day** and **Month** buttons respectively to switch to the desired time slot.

Related topics

Calendar Administration
Full calendar and embedded calendar
Calendar security area and role
Working with the calendar
Launch the calendar
Understand the calendar data
Select a year, month, or date from the date picker (full calendar only)
Collapse and expand panels (full calendar only)
View tooltip information
View time conflicts and custom messages
Access a record from the calendar
Select the objects (full calendar only)
Drill down to the Day view
Create a filter (full calendar only)
Clear the current filter
Manage favorites (full calendar only)

13.4.4. Select a year, month, or date from the date picker (full calendar only)

The calendar provides a Date Picker panel at the upper left corner, which allows you to quickly select a year, month, and date.

To select a year, month, or date from the date picker, follow these steps:

1. On the Date Picker panel, click the down arrow to the right of the currently displayed month and year (for example,

).
   The Date Picker panel switches to a month and year selection window, where the current month and year are selected by default.
2. Select a month and year, and then click **OK**.
3. Select a date.
   The calendar automatically switches to the selected year, month, or date.

Related topics

Launch the calendar
Understand the calendar data
Switch to a desired time slot
Collapse and expand panels (full calendar only)
View tooltip information
View time conflicts and custom messages
Access a record from the calendar
Select the objects (full calendar only)
Drill down to the Day view
Create a filter (full calendar only)
Clear the current filter
Manage favorites (full calendar only)

13.4.5. Collapse and expand panels (full calendar only)

By default, the calendar is displayed with all panels expanded. You can collapse the following panels to provide more space to display other parts of the calendar:

- The time period panel, which consists of the time period list area and graphic view area
  To collapse it, click the button.
- The left-side panel, which consists of the date picker panel and the filter panel
  To collapse it, click the button.

Once a panel is collapsed, an expand button displays instead to allow you to expand it again.

For the locations of the two collapse buttons, see Full calendar and embedded calendar.

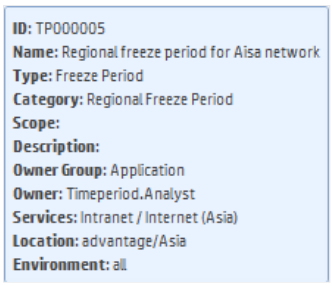
Related topics

Launch the calendar
Understand the calendar data
Switch to a desired time slot
Select a year, month, or date from the date picker (full calendar only)
View tooltip information
View time conflicts and custom messages
Access a record from the calendar
Select the objects (full calendar only)
Drill down to the Day view
Create a filter (full calendar only)
Clear the current filter
Manage favorites (full calendar only)

13.4.6. View tooltip information

To view the tooltip information of a record (time period, change, and so on), hover your mouse over the record in the record list area or over the record bar in the graphic view area.

The following screen shot illustrates the tooltip information of a time period record.



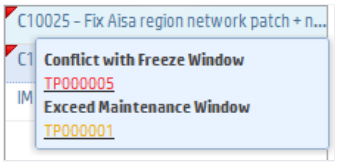
Related topics

Launch the calendar
Understand the calendar data
Switch to a desired time slot
Select a year, month, or date from the date picker (full calendar only)
Collapse and expand panels (full calendar only)
View time conflicts and custom messages
Access a record from the calendar
Select the objects (full calendar only)
Drill down to the Day view
Create a filter (full calendar only)
Clear the current filter
Manage favorites (full calendar only)

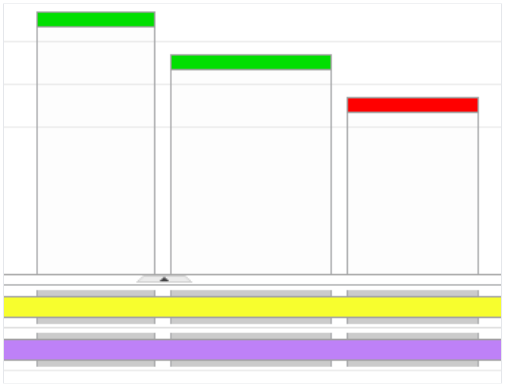
13.4.7. View time conflicts and custom messages

When the calendar detects that an object record has time conflicts with a time period record of the Freeze Period or Maintenance Window type, a red or yellow conflict icon is displayed for the record; if an object record has no conflicts but has one or more custom messages configured, a blue icon is displayed instead.

To view time conflict information for a record, hover your mouse over the conflict icon of the record. Custom messages, if any, are also displayed at the bottom of the tooltip.



You can also view the conflicts indicated by gray shadow bars, directly from the record bar area.



For more information, see Full calendar and embedded calendar and Display records from an application module in the calendar.

Related topics

Launch the calendar
Understand the calendar data
Switch to a desired time slot
Select a year, month, or date from the date picker (full calendar only)
Collapse and expand panels (full calendar only)
View tooltip information
Access a record from the calendar
Select the objects (full calendar only)
Drill down to the Day view
Create a filter (full calendar only)
Clear the current filter
Manage favorites (full calendar only)

13.4.8. Access a record from the calendar

You can access a record (for example, a time period or change record) from the calendar, to view its details or update it.

1. In the calendar, click the record from the record list area or from the graphic view area.
2. View or update the record as needed.

Related topics

Working with the calendar
Launch the calendar
Understand the calendar data
Switch to a desired time slot
Select a year, month, or date from the date picker (full calendar only)
Collapse and expand panels (full calendar only)
View tooltip information
View time conflicts and custom messages
Select the objects (full calendar only)
Drill down to the Day view
Create a filter (full calendar only)
Clear the current filter
Manage favorites (full calendar only)

13.4.9. Drill down to the Day view

You can drill down to the Day view of a specific day from a Week or Month view.

- In the Week view, click the name of the day on the header.
- In the Month view, click the number of the day.

Related topics

Launch the calendar
Understand the calendar data
Switch to a desired time slot
Select a year, month, or date from the date picker (full calendar only)
Collapse and expand panels (full calendar only)
View tooltip information
View time conflicts and custom messages
Access a record from the calendar
Select the objects (full calendar only)
Create a filter (full calendar only)
Clear the current filter
Manage favorites (full calendar only)

13.4.10. Select the objects (full calendar only)

By default, the calendar displays data for all available objects. You can select to display only objects of your interest.

1. On the upper right corner of the filter panel, click the **Select Objects** button ( ).
2. Select the objects you want.
   The calendar is refreshed and displays only the data of the selected objects.

Related topics

Launch the calendar
Understand the calendar data
Switch to a desired time slot
Select a year, month, or date from the date picker (full calendar only)
Collapse and expand panels (full calendar only)
View tooltip information
View time conflicts and custom messages
Access a record from the calendar
Drill down to the Day view
Create a filter (full calendar only)
Clear the current filter
Manage favorites (full calendar only)

13.4.11. Create a filter (full calendar only)

To view specific data only, you can create a filter by adding individual filter fields, or filter field groups, or both.

> **Note**
>
> Note When filtering data, the calendar performs a logical AND operation on all individual filter fields and field groups, while performs a logical OR operation on all fields within each group.

To create a filter, follow these steps:

1. Add individual filter fields as needed.
   1. Click **Miscellaneous** > **Calendar** to launch the calendar.
   2. Click the **Add Filter Fields** button.
   3. Click each tab, and select the fields you want.
   4. Click **OK**. The selected fields are displayed in the filter panel.
2. Add field groups as needed.
   1. Click the **Add Field Group** button, select the fields that you want to add to a group, and then click **OK**. The fields are added to the filter panel as a group.
      Alternatively, if you want to add an existing individual filter field to a group, click the down arrow button of the field, and select **In Group**. The field is added to the filter panel as a group.

      > **Note**
      >
      > Note Each group is identified by **- [Group]**, under which the group fields are nested. Clicking the minus icon will collapse the group.

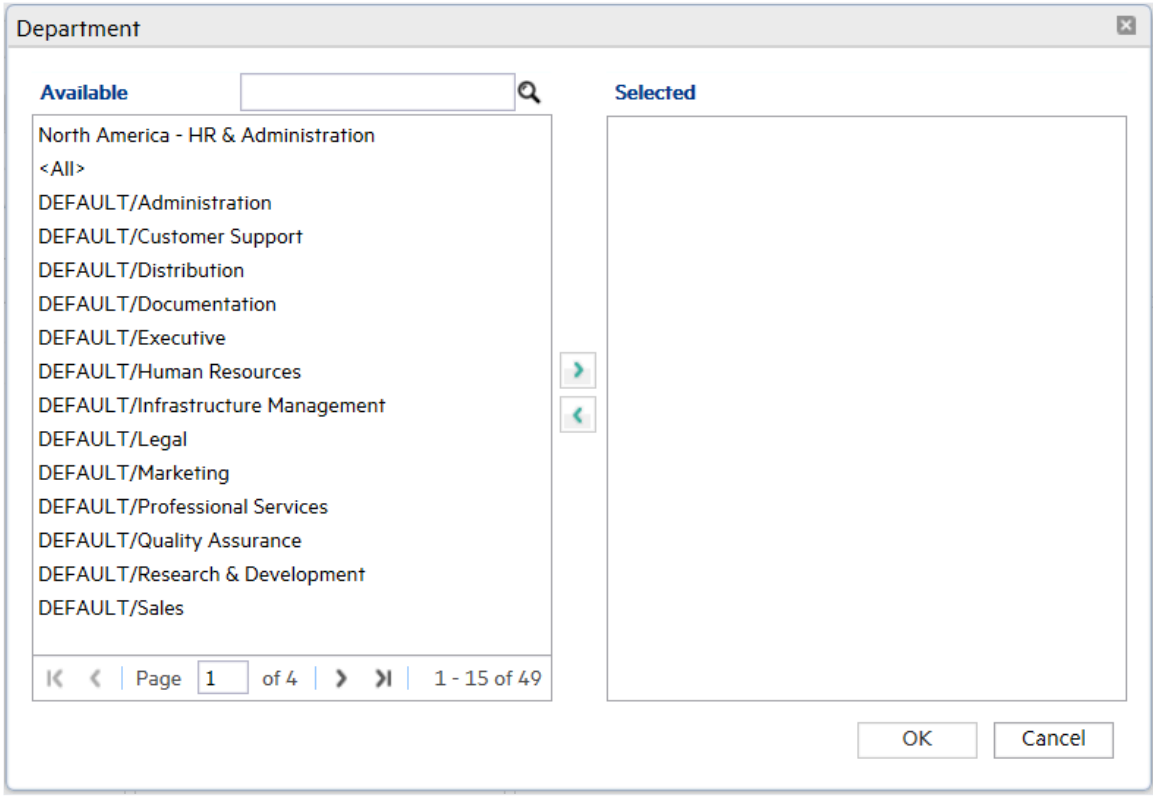
   2. If you still want to add more fields to the group, click the down arrow button of the group, select **Add Fields**, and then select more fields.

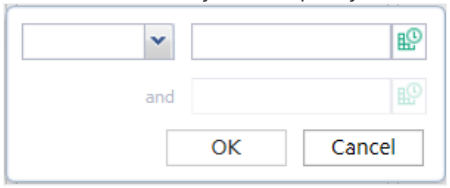
      > **Note**
      >
      > Note To delete a field from the group, click the down arrow button of the field, and then click **Delete**.

> **Note**
> Note To delete a group, click the down arrow button of the group, and select **Delete Group**.

3. Repeat the steps above to add more fields or groups as needed.
4. Specify one or more values as needed for each individual filter field and group field.
    1. Click a field.
    A value selection window opens, which can be a text input box, a Box Selector, an Advanced Selector, or a Date Picker, depending on the Field Type your administrator configured when adding the filter field.
    Text input box:

    Box Selector:

    Advanced Selector:
    

    Date Picker (where you can specify a date range):
    

    2. Enter a text value (for text type only), or select one or multiple values for the field.
    3. Repeat the steps above for the rest of the fields.

The filter is created, and the calendar automatically refreshes based on the filter.

You can further save the current filtered view as a private or public favorite for later reuse. For details, see Manage favorites (full calendar only).

Related topics

Launch the calendar
Understand the calendar data
Switch to a desired time slot
Select a year, month, or date from the date picker (full calendar only)
Collapse and expand panels (full calendar only)
View tooltip information
View time conflicts and custom messages
Access a record from the calendar
Select the objects (full calendar only)
Drill down to the Day view
Clear the current filter
Manage favorites (full calendar only)

13.4.12. Clear the current filter

To clear the current filter, click the **Clear Filter** button and then click **Yes** to confirm the deletion. All filter fields and groups are removed from the filter panel.

Related topics

Launch the calendar
Understand the calendar data
Switch to a desired time slot
Select a year, month, or date from the date picker (full calendar only)
Collapse and expand panels (full calendar only)
View tooltip information
View time conflicts and custom messages
Access a record from the calendar
Select the objects (full calendar only)
Drill down to the Day view
Create a filter (full calendar only)
Manage favorites (full calendar only)

13.4.13. Manage favorites (full calendar only)

Once you have created a filter, you can save the filtered data view as a favorite for reuse. The Manage Favorites feature enables you to add, update, delete, and open favorites.

Note

Note To add, delete, or update a public favorite, you need the following security rights respectively, which are defined in the Calendar Administration security area (see Calendar security area and role):

- Add Public Favorite
- Delete Public Favorite
- Update Public Favorite

To add a favorite, follow these steps:

1. Open the calendar, and create a filter. For details, see Create a filter (full calendar only).
2. Click **Manage Favorites** > **Add**.
3. Enter a name and description for the favorite, and select **Public** to make it public or leave **Public** unselected to make it private.

> **Note**
> Note A public favorite is accessible to all calendar users, while a private one is accessible only to the user who created it.

4. Click **Save**.

To update a favorite, follow these steps:

1. Click **Select Favorite**, and then select a favorite from the list.
   Alternatively, click **Manage Favorites** > **Organize**, click the **Refresh** button, select a favorite, and then click the **Open** button.
   The calendar switches to the data view of the favorite.
2. On the filter panel, update the filter as needed. For example, add or delete filter fields/groups or update filter field values. For details, see Create a filter (full calendar only).
3. Click **Manager Favorites** > **Update**.
4. Optionally, update the name, description, and privacy setting.
5. Click **Save**. The favorite is updated.

To delete a favorite, follow these steps:

1. Click **Manage Favorites** > **Organize**.
2. Click **Refresh** to refresh the favorite list.
   A refresh is needed in case other users create or update public filters while you are working with the calendar.
3. Select a favorite, and click the **Delete** button.
4. Click **Yes** to confirm the deletion. The favorite is deleted.

Related topics

Launch the calendar
Understand the calendar data
Switch to a desired time slot
Select a year, month, or date from the date picker (full calendar only)
Collapse and expand panels (full calendar only)
View tooltip information
View time conflicts and custom messages
Access a record from the calendar
Select the objects (full calendar only)
Drill down to the Day view
Create a filter (full calendar only)
Clear the current filter

14. Time Period Management

You can use Time Period Management to manage time period definitions for your organization so that end users can see how their activities will be affected during different time periods. A time period definition is a set of settings that defines a type of time windows based on a set of recurrence rules, including its category, current phase (based on a pre-configured workflow), approval status, scope, assignment group, owner, affected services, affected departments, and affected locations. You can define Local, Regional, or Global time periods that apply to a specific set of or all locations (countries or regions) in your organization; you can further define which departments, business services, environment type (for example, production or test) the time periods apply to. In addition, you can define the scope of each time period to indicate whether it will cause performance degradation or service outage to end users or it will have no impact on end user's activities.

If integrated with another module in Service Manager, Time Period Management will enable you to directly view time periods from that module. For example, if integrated with Change Management, time period definitions will enable users to view time periods and changes in a Calendar widget and view time periods and changes in the context of a selected change. This is helpful for change planning.

Note

Note If you disable the Service Manager Calendar, Time Period Management is disabled simultaneously.

Related topics

Time Period Management concepts
Time Period Management process
Time period workflow
Time period occurrences
Time period recurrence rules
Time period security areas and roles
Working with time periods
Time period configuration tasks

14.1. Time Period Management concepts

The following table lists the concepts used in Time Period Management.

Concept	Description
Time Period Definition	A set of settings that defines a type of time period, including its category, current phase (based on a pre-configured workflow), approval status, scope, owning group, owner, affected services, affected departments and locations, and so on.
Time Period Category	Time period categories are templates for creating different kinds of time period definitions. Each time period category is associated with a workflow that defines a life cycle for time periods. When creating a new time period, users are prompted to select a category from a list. The list of categories are grouped by type so that users can easily find the one they want. Category grouping is defined in Time Period Settings.
Time Period Rule	A rule is the central element in a time period to create time period occurrences. It reflects the recurrence of future start and end times of time period occurrences, and stores all the recurrence patterns.
Time Period Occurrence	An occurrence is the real time (start and end) when a time period window happens. Each occurrence contains a start and end date/time. A time period definition can have many occurrences.
Time Period Workflow	A time period workflow defines a life cycle for time periods. Each workflow is associated with a time period category. Out-of-the-box, the Timeperiod workflow includes four phases: Planning, Approval, Published, and Retired.
Time Period Approval	Administrators define approvals required for time periods using the Approval Definition utility. When the time period creator submits it for approval, the time period enters the Approval phase and goes through the pre-configured approvals. The Time Period Management module uses assignment groups and operators as approvers.
Time Period Notification	For time period approvals, standard approval notifications are defined in the timeperiodDefinition object record (**Tailoring** > **Document Engine** > **Objects**). Notifications can be automatically triggered and sent to relevant approvers or owners based on configured conditions.
Time Period Security Area and Role	The Time Period Management module uses security areas and roles to control user access to the module. By default, there are two areas (Timeperiod and Timeperiod Configuration), and four roles (Timeperiod Administrator, Timeperiod Analyst, Timeperiod Approver, and Timeperiod Manager).

Related topics

Time Period Management
Time Period Management process

Time period workflow
Time period occurrences
Time period recurrence rules
Time period security areas and roles

14.2. Time Period Management process

The following is an sample process for Time Period Management:

1. A System Administrator configures time period global settings.
2. A Time Period Administrator configures time period categories, workflows, and approvals.
3. A Time Period Analyst creates the following time period definitions:
    1. Creates the rules.
    2. Sets the ownership.
    3. Sets the affected services, locations, and departments.
    4. Checks the time windows.
    5. Submits the time periods for approval.
4. An approver (or an approval delegate) approves the time period definitions. Once approved, the time periods are automatically published.
5. A Time Period Administrator/Manager updates time period categories and types.
6. An owner updates, maintains, or retire the time period definitions assigned to the owner.
7. A Time Period Administrator/Manager unretires time period definitions if needed.

Related topics

Time Period Management
Time Period Management concepts
Time period workflow
Time period occurrences
Time period recurrence rules
Time period security areas and roles

14.3. Time period workflow

Each time period has a workflow, which is defined in the time period category. Out-of-the-box, a workflow (named **Timeperiod**) is available, which includes four phases. This workflow drives the creation, approval, publication, and retirement of time period definitions including their occurrences.

Note

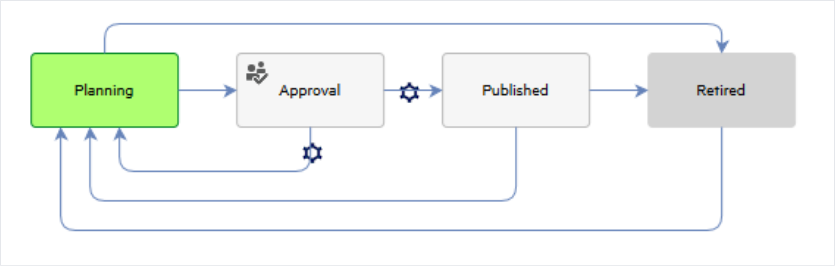
Note You can create your own workflows if needed.

To view the name of a time period's workflow, open the time period's category record (**Time Period Management** > **Configuration** > **Time Period Categories**), and then locate the Workflow field.

To access a time period workflow, navigate to **Time Period Management** > **Configuration** > **Workflows**.

Phases

The following table describes each phase in the **Timeperiod** workflow.



Phase	Description	Notes
Planning	Planning a time period • Create the rule • Specify the ownership • Specify the impacted services, locations, or departments • Check the time windows	
Approval	Approve the time period based on the configured approval rules.	An automatic transition is configured from the Approval phase to each of these phases: Planning, and Published. The transition triggers are as follows: • From Approval to Planning: (Expression: approval.status in $L.file="denied") • From Approval to Published: (Expression: approval.status in $L.file="approved")
Published	The time period, the rules and occurrences are displayed in read-only. Some occurrences can be manually canceled. In this phase the time period is approved and officially published.	
Retired	The time period is retired and the record is inactive (closed).	

Actions and activities in each phase

The following table summarizes the actions and activities available or allowed in each phase of the workflow: Planning (P1), Approval (P2), Published (P3), and Retired (P4). A cross mark indicates the item is available or allowed.

Button

Item	Description	P1	P2	P3	P4
OK	Saves the record and exits.	x	x	x	x
Cancel	Exits the record without saving.	x	x	x	x
Save	Saves the record.	x	x	x	
Submit for Approval	Moves the record to the Approval phase.	x			
Retire	Moves the record to the Retired phase and deletes all occurrences.	x		x	
Change Time Zone	Changes the time zone to display date/time values from the user's time zone to a different one.	x	x	x	x
Apply Template	Applies a time period template.	x			
Submit for Re-planning	Moves the record back to the Planning phase.			x	

Menu Option (More >)

Item	Description	P1	P2	P3	P4
Set Reminder	Calls the out-of-the-box reminder feature.	x	x	x	
Copy	Copies the record.	x	x	x	x
Change Type / Category	Changes the category and type of the current record. Works only in the first phase.	x			
Create Template from Record	Creates a new time period template from the record.	x	x	x	
Notify	Sends a notification.	x	x	x	
Approval > Approve/Deny/Retract	Approve: Approves the record. Once approved, the record's approval status changes to approved, and the record automatically moves to the Published phase. Deny: Denies the record. Once denied, the record's approval status changes to denied, and the record automatically returns to the Planning phase. Retract: Retracts a denial of the record. Once retracted, the record moves back to pending approval status.		x		

Unretire	Moves the record to the first phase and activates it again.				x

User Action

Item	Description	P1	P2	P3	P4
View rules	Allows users to view the existing rule(s) of the record.		x	x	
Create / Modify rules	Allows users to create new rules and modify the existing rule(s).	x			
Filter activities	Allows users to filter the activities of the record.	x	x	x	
Filter occurrences	Allows users to filter the occurrences of the record.	x	x	x	
Deactivate a single occurrence	Allows users to deactivate one single occurrence, when the date is no longer required.			x	

System Background Activity

Item	Description	P1	P2	P3	P4
Write activities	As soon as activities are created.	x	x	x	x
Update occurrences	Updates occurrences according to the defined rules.	x		x	x
Set create/update fields	Set the created by / updated by and related time stamps. These values are saved in the Rules. The form is fully editable.	x			
Set update fields	Sets the updated by and related time stamps.		x	x	
Move to previous/next phase	If the record is denied, the system automatically moves the record back to the Planning phase. If the record is approved, the system automatically moves the record to the next phase.		x		

Related topics

Time Period Management
Time Period Management concepts
Time Period Management process
Time period occurrences
Time period recurrence rules
Time period security areas and roles

14.4. Time period recurrence rules

A time period rule reflects the recurrence of future start and end times of the time period occurrences. The time period rule stores all the recurrence patterns. The occurrences are created based on the rule.

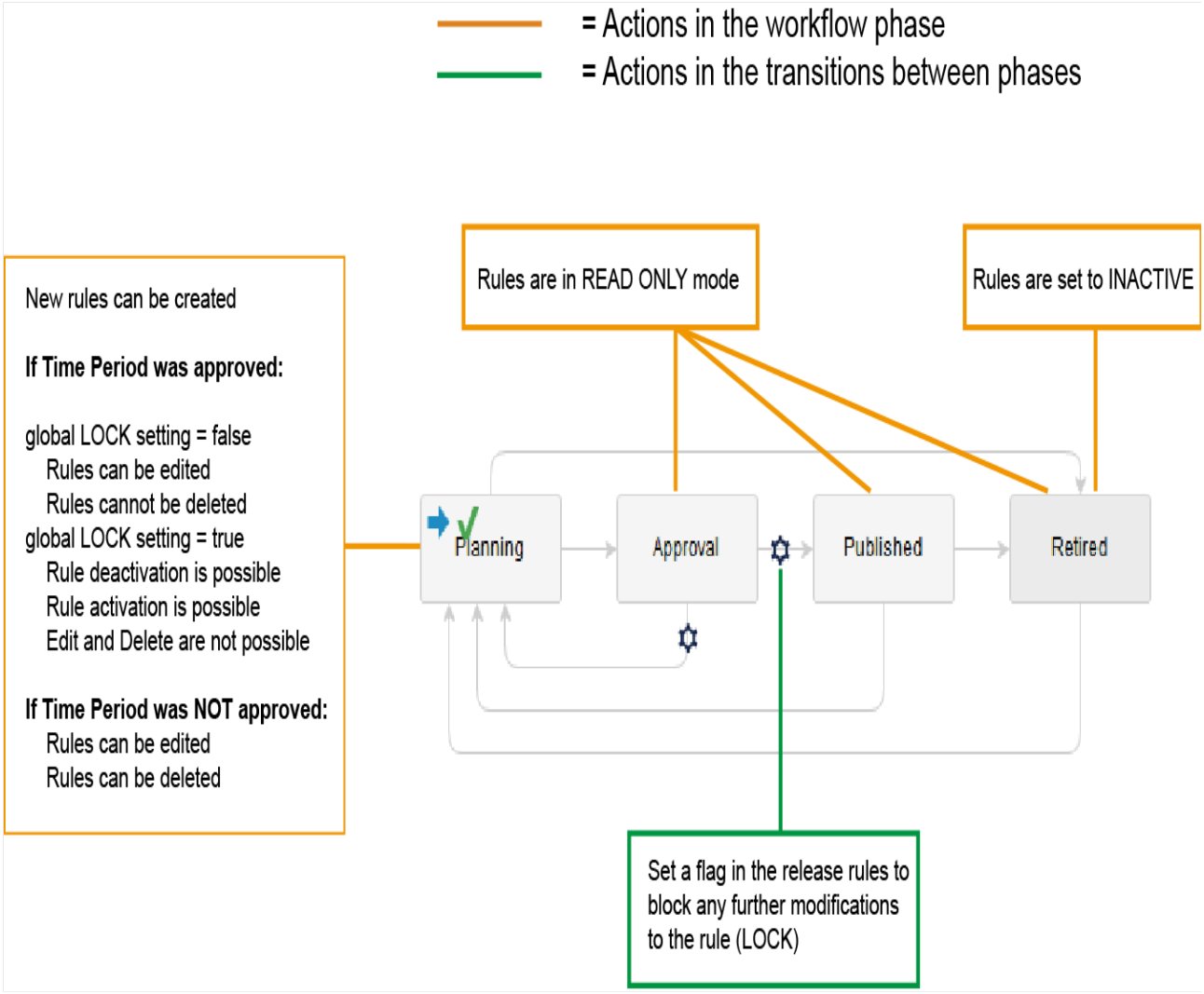
There are five rule types available: Arbitrary, Daily, Weekly, Monthly, and Yearly. All types, except for Arbitrary, can be configured with several recurrence rules. Arbitrary allows the configuration of specific start and end times, in case some occurrences don't follow a specific rule.

Principles of Rules

The following principles apply to time period rules.

Time Period Phase	Principles
Planning	You can create, change, or remove a rule when the time period was never approved before, and the following happens: • As soon as you create a rule, the occurrences are created in Inactive state. • As soon as you change a rule, the occurrences are deleted and new ones recreated in Inactive state. • As soon as you remove a rule, the occurrences are removed. Once the time period was approved, its rules are *locked* and can't be changed anymore if it's not in the first phase. • When the global LOCK setting is set to true, a locked rule allows only activation/deactivation in the Planning phase, and all its future occurrences are deleted. • When the global LOCK setting is set to false, a locked rule allows deletion / modification of the rule only in the Planning phase.
Approval	Rules are displayed in read-only.
Published	Rules are displayed in read-only.
Retired	When a Time Period is retired, the associated rules are set to Inactive. In case a Time Period is unretired, the associated rules remain Inactive and need to be reactivated manually if the rules are still valid.

Once a rule is configured, various actions are started to prevent users from unauthorized editing, as shown in the following figure.



Global lock setting

The time period global settings contains this option: **Lock Rules once approved**. This global lock setting allows administrators to decide whether or not users can edit, delete, activate, or deactivate rules after approval. By default, this global lock is disabled.

The following table describes the behaviors when this option is enabled (set to true) and disabled (set to false).

Global Lock Setting	Behavior
false	Update Rules Updating rules is only possible in the first phase of the workflow. To update a rule, access an existing time period in the “planning” phase, click the Rule Id in the **Rules** section, and then update the rule. When your edits are complete, click **Save**. The following functionality is implemented in case a rule was changed: • If the time period was NOT approved before, all occurrences are recalculated. • If the time period was approved before, all future occurrences are recalculated. Delete Rules Deleting rules is only possible in the first phase of the workflow. Delete is only possible when the time period was not approved before. To delete a rule, access an existing time period in the “planning” phase, click the Rule Id in the **Rules** section, and then click **Delete**. When a rule is deleted, all occurrences for this rule are deleted. Add Rules Adding rules is only possible in the first phase of the workflow. To add a rule, open a time period and click **New** in the **Rules** section.

true	Update Rules Updating rules is only possible in the first phase of the workflow if the time period was NOT approved. To update a rule, follow the steps described above. Once the time period was approved and the time period is again in the first phase, the rules are displayed in read-only, but a **Deactivate Rule** or **Activate Rule** button is available on the top of the screen to allow you to deactivate or activate the rule. Delete Rules Deleting rules is only possible in the first phase of the workflow if the time period was NOT approved. Once the time period was approved and the time period is again in the first phase, you can't delete its rules anymore. To delete a rule, access an existing time period in the "planning" phase, click the Rule Id in the **Rules** section, and then click **Delete**. When a rule is deleted, all occurrences for this rule are deleted. Add Rules Adding rules is possible only in the first phase of the workflow. To add a rule, open a time period and click **New** in the Rules section.

Rule security

Rules aren't secured by a separate area or role, and therefore the security of a rule is closely tied to the security of its time period definition. Basically, if you can update a time period, you can also update its rules.

Note

Note The security of the rules is implemented in function `getEditRights(record)` in the `timeperiodRule` ScriptLibrary record.

Related topics

Time Period Management
Time Period Management concepts
Time Period Management process
Time period workflow
Time period occurrences
Time period security areas and roles

14.5. Time period occurrences

An occurrence is the real time when a time period window happens. Each occurrence contains a start and end time. When you add an active rule to a time period, occurrences are automatically created based on the rule.

Occurrence details

In the Occurrences section of a time period, you can view the details of each occurrence as described in the following table.

Label	Description
Rule Id	The ID of the rule associated with the occurrence.
Start Date	The start time of the occurrence.
End Date	The end time of the occurrence
Time Zone	The time zone of the occurrence Start Date and End Date. See Change the time zone of a time period for more information.
Status	The status of the occurrence: **Active**, or **Inactive**. You can change the status only when the time period is in the Published phase.
Activate after approval	Indicates whether or not the occurrence will be activated after approval: **Yes**, or **No**. In the first phase or in case the time period is pending approval, an occurrence can be marked to stay inactive after approval. This allows users or approvers to disable specific occurrences and approve the rest of the occurrences for a time period.

Number of occurrences for initial display

The global setting **Number of occurrences for initial display** defines the maximum number of occurrences that can be displayed in the Occurrences section. When the number of occurrences exceeds this limit, the following note displays at the top of the Occurrences section:

Not all occurrences are displayed. You can use the filter functions to locate specific occurrences.

When this happens, you can use the following fields to filter occurrences: Rule Id, Start Date, End Date, Status, and Activate. Use the little magnifying glass to search for records, and use the little button with a red cross to clear the filter.

Note

Note The actual number of occurrences that are displayed is also limited by another global setting: **Maximum number of months for calculation of future occurrences**. The recommended value is 12 (months).

For information about these two settings, see Configure time period global settings.

Occurrence state

Time period occurrences store the calculated start and end times of time period definitions based on their configured rules. Occurrences are accessible as long as the time period definitions are active.

Each occurrence record contains a checkbox to indicate whether it is active or inactive, as described in the following table.

Condition	Occurrence State
When an occurrence is created	A new occurrence is created as **inactive** (false).
When a time period is created (in the planning phase)	Occurrences are **inactive**.
When a time period definition moves to the published phase	• All future occurrences are set to **active**, while occurrences in the past remain **inactive**.
When a time period definition moves back to the planning phase	• Future occurrences become **inactive**, while occurrences in the past remain **active**.
When a time period definition moves to the retired phase	• Past occurrences remain **active**, while future occurrences are removed.
When a time period definition is directly retired (was never approved)	All occurrences are removed.

Occurrence security

The following security rules apply to time period occurrences:

Time Period Phase	Security Rule
Planning	• Users can't activate or deactivate an occurrence, but can check whether or not an occurrence should be kept active after approval.
Published	• All users with update rights to the time period can deactivate occurrences, but can't reactivate them. Only users with Approval and Expert rights to the time period can reactivate occurrences.

Occurrence persistence

To ensure that customers can easily link changes and time periods together, the time stamps on a time period are made persistent. This allows the implementation of validations and additional reporting to ensure changes are linked with the right time period.

Related topics

Time Period Management
Time Period Management concepts
Time Period Management process
Time period workflow
Time period recurrence rules
Time period security areas and roles

14.6. Time period security areas and roles

The following security areas and roles are used to ensure proper security in the Time Period Management module.

Security areas

Out-of-the-box, the Time Period Management module has the following security areas. To view their default rights and default settings, navigate to **System Administration** > **Security** > **Areas**, and open these records by performing a search.

Area	Description
Timeperiod	For security related actions in time period definitions, rules and occurrences.
Timeperiod Configuration	For security related actions in time period categories, approvals and workflows.

Note

Note For the Timeperiod Configuration security area, don't select **When assigned** or **When assigned to workgroup** from the **Update** or **Delete/Close** drop-down list since the objects related to this security area have no assignee or assignment group fields.

Security rights and settings

The following table describes the security rights and settings used in the Time Period Management module.

Setting / Right	Description
View	Can view time period records
New	Can create new time period records
Update	Can update time period records
Delete / Close	Can retire new time period records
Expert	Can activate/deactivate occurrences
Admin	Is admin
Modify Template	Can create templates
Can apply template	Can apply templates
Can approve	Can approve time period records
Can delegate approvals	Can delegate approvals for time period records
Can override approvals	Can override approvals
Can copy record	Can copy time period records
Can send notifications	Can send notifications from time period records
Can set reminder	Can set a reminder for time period records
Can change category / type	Can change the category/type of a time period record
Can create global inbox	Can create global time period inboxes
Initial Inbox	Initial inbox for the time period module
Manage format	Sc manage format
Can create personal Inbox	Can create personal inboxes for time period records
Lock on Display	When displaying a record the user, Service Manager locks the record regardless of whether the user is modifying the record or not
New thread List -> Edit	When the user accesses a record from a record list, Service Manager keeps the record list open and displays the record in a new tab window
New thread Search -> List	When the user searches in the Time Period area, Service Manager keeps the search form open and displays the search results in a new tab window
New thread View -> Edit	When the user accesses a record from a Time Period view, Service Manager keeps the view form open and displays the record in a new tab window
New thread View -> Search	When the user runs a search from a Time Period view, Service Manager keeps the view form open and displays the search form in a new tab window

Un-retire	Can unretire (reopen) time period records

Security roles

Out-of-the-box, the following security roles are available for the Time Period Management module.

To access these rules, click **System Administration** > **Security** > **Roles**, and then open the roles by performing a search.

Role	Description	Default Rights/Setting in the Timeperiod Area	Default Rights/Setting in the Timeperiod Configuration Area
Timeperiod Administrator	Full rights.	• View • New • Update (Always) • Delete/Close (Always) • Expert • Admin • Can apply Template • Can approve • Can delegate approvals • Can override approvals • Can copy record • Can send Notifications • Can set reminder • Can change category / type • Can create global Inbox • Can create personal Inbox • Unretire • Allowed Categories: all categories	• View • New • Update (Always) • Delete/Close (Always) • Expert • Admin

Timeperiod Analyst	Partial rights; can update only when assigned.	• View • New • Update (When assigned to workgroup) • Update (When assigned to workgroup) • Can apply Template • Can copy record • Can send Notifications • Can set reminder • Can create global Inbox • Can create personal Inbox • Allowed Categories: all categories	No rights
Timeperiod Approver	Partial rights; can approve and delegate approvals.	• View • Update (Never) • Delete/Close (Never) • Can approve • Can delegate approvals • Can send Notifications • Can set reminder • Allowed Categories: all categories	No rights

Timeperiod Manager	Expert rights; can always update.	• View • New • Update (Always) • Delete/Close (Always) • Expert • Can apply Template • Can approve • Can delegate approvals • Can copy record • Can send Notifications • Can set reminder • Can change category / type • Unretire • Allowed Categories: all categories	• View • New • Update (Always) • Delete/Close (Always) • Expert

User roles

Out-of-the-box, four users (operators) were created for testing purposes based on the security roles described above.

Operator	Description
Timeperiod.Admin	Can perform all Time Period Management tasks.
Timeperiod.Analyst	Partial rights; can perform Time Period Analysis tasks.
Timeperiod.Approver	Partial rights; can approve time periods.
Timeperiod.Manager	Expert rights; can perform all Time Period Management tasks except for certain administration tasks.

Related topics

Time Period Management
Time Period Management concepts
Time Period Management process
Time period workflow
Time period occurrences
Time period recurrence rules

14.7. Working with time periods

Users perform the following tasks when working with time periods that are in different phases.

- Create a time period
- Submit a time period for approval
- Deactivate or activate a time period occurrence
- Deactivate a time period rule
- Delegate a time period approval
- Approve a time period

- Change the time zone of a time period
- Retire a time period
- Submit a time period for replanning

14.7.1. Create a time period

A time period must contain one or more rules in order to create occurrences. A time period rule stores all the recurrence patterns, and reflects the recurrence of the future start and end times of the time period occurrences. Rules are the central element in the Time Period module to create time period occurrences.

Tip

Tip You can create a time period from scratch or copy from an existing one.

To create a time period from scratch, follow these steps:

1. Click **Time Period Management** > **Create New Time Period**.
   A list of time period categories is displayed. The categories are grouped by Type.

Time period type	Description
Availability Hours	A period of time in which a service should be available.
Maintenance Window	A period of time in which Change requests may be implemented.
Neutral	A period of time indicating an external event, such as a holiday, which has no direct bearing on request implementation.
Freeze Period	A period of time in which Change requests may not be implemented.

2. Select a category.
   The **New Time Period** form opens, with certain field values automatically populated.
3. Complete the fields in the **Details** and **Applies to** sections as necessary. The following tables describe the two sections, and the Activities section.

Details

Attribute	Description	Notes
Id	A system-generated unique ID for the time period definition.	The ID is generated based on the number file.
Current Phase	The phase that the time period definition is currently in, depending on the pre-configured workflow.	The workflow is defined in the time period category.
Approval Status	The approval status of the time period definition.	Same approval states as in Change Management.
Type	The type of the selected category.	Read-only
Category	The selected category.	Read-only
Next Review Date	Enter the next review date of this time period definition.	Must be a future date.
Scope	The acceptable scope of the time period. **Note** Note This field is available only for a time period of the Maintenance Window type.	A Global List with three values: ○ No Impact ○ Degradation ○ Outage
Owning Group	The assignment group that owns this time period.	Linked to the assignment table. Values must be a valid assignment group.
Owner	The owner (person) of this time period.	Must be a member of the assignment group (Owning Group).
Name	A display name of the time period definition.	Any name is acceptable.
Description	A description of the time period definition.	Any textual description.

Activities

Attribute	Description
New Update Type	The type of the new update. The out-of-the-box types are **Update**, **Communication with Stakeholders**, and **Communication with Service Owners**.
New Update	The description of the new update for the activity log.
Activity Logs	Lists all activities performed on the record.
Activity Type filter	Allows a filtered view of the activity list per Activity Type.
User filter	Allows a filtered view of the activity list per User.

Applies to

Attribute	Description	Notes
Environments	An attribute to classify the time period definition even further.	
Services	Services for which this time period definition is created.	Must be a Business Service configuration item from the `device` file.
Locations	The locations this time period applies to.	Must be a valid record from the `location` file.
Departments	The departments this time period applies to.	Must be a valid record from the `department` file.

Note
Note By default, the time period is valid for all Locations, Departments, Services, and Locations (these fields have a default value of `<All>`). When you specify a specific item, the `<All>` value is cleared and replaced with your specified value.

4. Click **Save**.

Note
Note Before you can add a rule, you need to save the time period.

The time period is added.

5. In the **Rules** section, continue to add rules to the time period.
    1. Click **New**. The **New Time Period** form opens.
    2. In the **Rule Name** field, type a descriptive name. For example, **My Rule 1**.

Note
Note The rule is Active by default. If you want to add an inactive rule, deselect the **Active** option. Inactive rules will not create occurrences for the time period.

    3. In the **Time period** field, select a time zone. By default, the time zone defined in your operator record or the system time zone defined in the System Information Record is selected. If you are creating a time period

definition for users located in another time zone, you may want to select their time zone. Occurrences will be generated based on the specified time zone.

4. Select a Recurrence Type. For example, **Monthly**.
5. Specify when the rule will start to take effect and when it will expire. For example, you can specify February 1, 2013 as the effective date, and the rule will expire after 10 occurrences.

   Note

   Note The maximum allowed value in the **Expires after** field is defined in the Time Period Settings. See Configure time period global settings. If you specify a value greater than the maximum, an error occurs.

6. Select a recurrence pattern. For example, you can specify the rule recurs every 1 month, on every fourth of the month or on every first Monday of the month.

   Note

   Note The value of the recurrence pattern has the following limitations:

   - Daily: 1 to 1000
   - Weekly: 1 to 100
   - Monthly: 1 to 25
   - Yearly: 1 to 3

7. Specify the start and end times of the rule. For example, you can specify the rule starts at 00:00 (hh:mm) and ends on the same day at 08:00 (hh:mm).
8. If you want to set an arbitrary schedule for the rule, select the **On an arbitrary schedule** checkbox, and then specify the start day and end date for each individual occurrence.
9. Click **Save**.
   The rule is added.
10. Click **Save & Exit**.
    You are returned to the time period record. The new rule now displays in the **Rules** section, and its occurrences during the specified effective period are listed in the **Occurrences** section. When there are a large number of occurrences, you can use the filters (Rule ID, Start Time, End Time, Status, and Activate) at the top of this section to view specific rules.
    The maximum allowed value in the **Expire after** field and the number of months to calculate future occurrences are defined in the time period global settings. See Configure time period global settings.

    Note

    Note The occurrences are displayed in your time zone or the system time zone if yours is unavailable, instead of the time zone specified in the rule. You can change the occurrence display time zone only for this session by clicking the **Change Time Zone** button (if you log off and then log back in, the time zone change no longer takes effect). For details, see Change the time zone of a time period.

6. If needed, continue to add more rules as described above.
7. Click **Save**.
   The time period is created.
   The **Workflow** section shows the workflow linked with the time period category you selected previously, and the current phase of the time period is Planning.
   The **Approvals** section is now empty. Next, you can further update the time period or submit it for approval.

To create a time period from an existing one, follow these steps:

1. Click **Time Period Management** > **Search Time Periods**.
2. Open the existing time period by performing a search.

3. Click **More** or the **More Actions** menu, and select **Copy**.
   The new time period is created, populated with data from the original time period, except for the **Impact and Current Phase** field values. The new record always has an empty Scope and a Current Phase of **planning**.
4. Click **Save**.

> **Note**
> Note The **Copy options for Rules and Occurrences** setting specified in the Time Period Settings determines whether or not and how rules and occurrences are copied. When you click **Save**, rules and occurrences are copied regardless of their state (active or inactive).

5. Update the time period as needed, and click **Save**.

Update a time period

You can't update an existing time period unless it is in the Planning phase.

To update a time period, follow these steps:

1. Navigate to **Time Period Management** > **Search Time Periods**.
2. Click **Search**, or enter several fields on the search form and then click **Search**.

> **Note**
> Note An entry in the ID field must include the prefix TP. For example, TP000003.

3. Locate the time period record you want to update, and update it as needed.
4. Save the record.

Related topics

Submit a time period for approval
Deactivate or activate a time period occurrence
Deactivate a time period rule
Delegate a time period approval
Approve a time period
Change the time zone of a time period
Retire a time period
Submit a time period for replanning
Time period configuration tasks

14.7.2. Submit a time period for approval

Once you have finalized a time period definition, you can submit it for approval. By default each time period will go through the time period approvals defined in the Timeperiod approval definition, which is a global definition that takes effect on all time periods. Your administrator can create their own approval definitions and associate them to different time period categories.

To submit a time period for approval, follow these steps:

1. Navigate to **Time Period Management** > **Search Time Periods**, and open the time period by performing a search.
2. Click **Submit for Approval**.
   The time period moves to the **approval** phase, and its Approval Status changes to **pending**.
   The **Approvals** section now lists all currently required approvals. Once an approval is completed, the approval activity will be displayed in the Approval Log.

Related topics

Create a time period
Deactivate or activate a time period occurrence
Deactivate a time period rule
Delegate a time period approval

Approve a time period
Change the time zone of a time period
Retire a time period
Submit a time period for replanning
Time period configuration tasks

14.7.3. Activate or deactivate a time period occurrence

Once you have added rules to a time period, the occurrences are automatically calculated and displayed in the record. By default, all occurrences are automatically activated after the time period is approved; however you can manually deactivate an individual occurrence and activate it later if needed.

To cancel an occurrence in a time period, follow these steps:

1. Navigate to **Time Period Management** > **Search Time Periods**, to open the time period record.
2. In the **Occurrences** section, locate an occurrence and do the following according to the current phase of the time period:

Phase	
Planning or Approval	In the **Activate after approval** column, click **Yes** or **No** to change to the opposite value. **Yes** means the occurrence is activated after approval while **No** means the occurrence stays inactive after approval.
Published	In the **Status** column, click **Active** to deactivate or click **Inactive** to activate, and then enter an explanation about the status change if required. The value now changes from Active to Inactive or from Inactive to Active. **Note** Note After approval the **Activate after approval** column is no longer displayed. Instead the Status column is displayed as a link. Clicking the **Active** or **Inactive** link deactivates or activates the occurrence. **Caution** Caution As a normal user (with only update rights), you can always deactivate occurrences, but you can't reactivate occurrences once the record is published; to reactivate an occurrence after the approval phase, you must have Expert or Approval rights to a time period record. Normal users are disallowed to play with the status of the occurrences.

Related topics

Create a time period
Submit a time period for approval

Deactivate a time period rule
Delegate a time period approval
Approve a time period
Change the time zone of a time period
Retire a time period
Submit a time period for replanning
Time period configuration tasks

14.7.4. Deactivate a time period rule

By default, time period rules are active when created. If a time period rule no longer works for you, you can permanently remove it from the time period, or you can temporarily deactivate it. You can do so only when the time period is in the first phase of the workflow.

To deactivate a time period rule, follow these steps:

1. Navigate to **Time Period Management** > **Search Time Periods**, and then open the time period record that contains the rule.
2. In the **Rules** section, click the rule and then deselect the **Active** option.

   Note

   Note : If the time period was approved and back to the first phase, the existing rules are read-only, and you should click the **Deactivate Rule** button instead.

3. Click **Save** to save the rule, and then click **Save & Exit**.
   In the **Occurrences** section of the time period, all future occurrences based on this inactive rule are now removed.

Related topics

Create a time period
Submit a time period for approval
Deactivate or activate a time period occurrence
Delegate a time period approval
Approve a time period
Change the time zone of a time period
Retire a time period
Submit a time period for replanning
Time period configuration tasks

14.7.5. Delegate a time period approval

Approval Delegation is possible for time periods in the same way as for other modules in Service Manager. Before you can delegate your approvals to other users, your system administrator should have given you the rights to delegate approvals. For information about how to grant approval delegation rights, see Time period security areas and roles.

To delegate approvals, follow these steps:

1. Navigate to **Miscellaneous** > **Approval Delegation**.
   The **List Delegates** wizard opens.
2. Click **Add New Delegation.**
   The **Delegate Selected Approvals** option is selected by default.
3. Click **Next**.
4. Select the **Timeperiod** module, and click **Next**.
5. Select your group (for example, **Service Manager**) or operator (for example, **Timeperiod.Approver**), move it into the right pane, and then click **Next**.
6. Complete the following fields, and click **Next**.

Field	Description
Delegate to	The user to whom the approvals will be delegated. Click **Fill** to select a user from the list. **Note** Note If a desired user is not in the list, the user doesn't have the rights to access or approve time periods. Contact your system administrator for assistance. If your Service Manager system has a Process Designer Content Pack applied, the desired user must be a member of the ChM Membership group.
Valid From	The delegation start time.
Valid To	The delegation end time.

7. Click **Finish**.

Related topics

Create a time period
Submit a time period for approval
Deactivate or activate a time period occurrence
Deactivate a time period rule
Approve a time period
Change the time zone of a time period
Retire a time period
Submit a time period for replanning
Time period configuration tasks

14.7.6. Approve a time period

If you are the approver or a member of the approval group, or a delegate of a time period approval, you receive a notification when a time period is pending your approval. You can then complete your approval. Once all approvals of the time period are complete, the time period automatically advances to the published phase, and the approval status changes from pending to approved. If a time period is denied during an approval, it needs to be resubmitted for approval.

Note

Note You can Deny a pending approval or Retract a completed approval.

If you are the approver or a member of the approval group, follow these steps to approve or reject a time period:

1. Navigate to **Time Period Management** > **Search Time Periods**, to open the time period record.
2. In the **Approvals** section, click the **Timeperiod** approval in the **Currently Approvals** subsection.
3. Put the cursor in your approval, and click **More** > **Approval** > **Approve/Deny** to approve or reject the record.
   If you're not the approver but have approval override rights, you can also approve by clicking **More** > **Override** > **Approve Current**.
   • To approve, click **Approve**. If you want to redo your approval, click **Retract**.
   • To reject, click **Deny**.
4. Enter your approval comments, and click **OK**.
   If denied, the time period then returns to the planning phase, and its approval status becomes denied.

If there are still pending approvals, the time period remains in the approval phase with an approval status of pending; otherwise, it advances to the published phase with an approval status of approved.

If you are a delegate, follow these steps to approve or reject a time period:

1. In the To Do Queue, select **Approval Delegation** from the **Queue** list, and **Active approval delegations assigned to me** from the **View** list.
2. Click the approval for the time period.
3. Click **Approve** or **Deny**.

Related topics

Create a time period
Submit a time period for approval
Deactivate or activate a time period occurrence
Deactivate a time period rule
Delegate a time period approval
Change the time zone of a time period
Retire a time period
Submit a time period for replanning
Time period configuration tasks

14.7.7. Change the time zone of a time period

When you are viewing or updating a time period definition, its occurrences are displayed in your time zone or the system time zone if yours is undefined. In some cases, for example, if a time period is created for users in another time zone, you may want to change the current display time zone to that one.

Note

Note The time zone change takes effect only for your current session (it will no longer take effect if you log off and then log back in).

To change the time zone of a time period, follow these steps:

1. Navigate to **Time Period Management** > **Search Time Periods**, and then open the time period record you want to update.
2. Click **Change Time Zone**.
3. In the **New session's Time Zone** field, select another time zone.
4. Click **Finish**.
   The occurrences are now displayed in the new time zone.

Related topics

Create a time period
Submit a time period for approval
Deactivate or activate a time period occurrence
Deactivate a time period rule
Delegate a time period approval
Approve a time period
Retire a time period
Submit a time period for replanning
Time period configuration tasks

14.7.8. Retire a time period

Once a time period is no longer needed, you can retire it. When retired, the time period record is inactive and closed.

Note
Note
You can retire a time period only when it is in the planning or published phase.

To retire a time period, follow these steps:

1. Navigate to **Time Period Management** > **Search Time Periods**, and then open the time period record you want to retire.
2. Make sure the record is in the planning or published phase.
3. Click **Retire**.
   The time period advances to the retired phase, and becomes read-only. All occurrences in the Occurrences section have been removed.

To unretire a time period, follow these steps:

Note
Note
To unretire a retired time period, you need one of the following security roles: system administrator, Timeperiod Administrator, or Timeperiod Manager. These roles have the rights to unretire time periods.

1. Open the time period.
2. Click **More** or the **More Actions** menu, and then select **Unretire**.
   The time period returns to the planning phase.

   Note
   Note
   Although the rules are active at this time, the occurrences aren't automatically added back. You need to manually save the rules to get them back.
3. In the **Rules** section, open each rule, and click **Save** and then **Save & Exit**.
   The occurrences now display in the **Occurrences** section.

Related topics

Create a time period
Submit a time period for approval
Deactivate or activate a time period occurrence
Deactivate a time period rule
Delegate a time period approval
Approve a time period
Change the time zone of a time period
Submit a time period for replanning
Time period configuration tasks

14.7.9. Submit a time period for replanning

Once a time period is published, its rules and occurrences become read-only. You can only manually cancel an individual rule. If you need to update the rules or occurrences, you can submit it for replanning. Doing so returns the time period back to the planning phase.

To submit a time period for re-planning, follow these steps:

1. Navigate to **Time Period Management** > **Search Time Periods**, to open the time period record.
2. Click **Submit for Replanning**.

Note

Note

- Existing rules are read-only if the **Lock Rules once approved** global setting is enabled, and therefore you can no longer update them, but you can still deactivate/activate them using the **Deactivate Rule** or **Activate Rule** button that appears on the top of the rule detail form; if **Lock Rules once approved** is disabled (set to false), you can still update existing rules.
- Deactivating a rule will trigger the deletion of all future occurrences based on the rule; activating a rule triggers the creation of the occurrences as of today.
- The system will display a warning message if any existing changes conflict with the submitted time period.

Related topics

Create a time period
Submit a time period for approval
Deactivate or activate a time period occurrence
Deactivate a time period rule
Delegate a time period approval
Approve a time period
Change the time zone of a time period
Retire a time period
Time period configuration tasks

14.8. Time period configuration tasks

Administrators can perform the following configuration tasks for the Time Period module.

- Configure time period global settings
  Configure some global settings for the Time Period Management module. This configuration includes updating or deleting existing settings and adding new settings.
- Configure a time period workflow
  Define the workflow in the time period category.
- Configure time period categories
  Create new time periodcategories, and update or remove existing ones to manage the data structures of different kinds of time period definitions.
- Configure time period approvals
  Define approvals for the Time Period module using approval definitions. The Timeperiod approval definition is linked to the Approval phase.
- Configure time period approval notifications
  Define and trigger the time period approval notifications. When a time period is added or updated, notifications are generated.
- Calculate time period conflicts
  Calculate time period conflicts between the current changes record or change task record and the existing time period.
- Trigger an approval for time period conflicts
  Define an additional approval for a module and then associate the approval definition to the corresponding phase of a record.
- Configure time period conflict notifications for Change Management
  Configure time period conflict notifications for Change Management. When a change with the Conflict with Freeze Window conflict type is approved, a notification is sent to the owner of the conflicted time period.

Related topics

Time Period Management
Time Period Management concepts

Time Period Management process
Time period workflow
Time period occurrences
Time period recurrence rules
Time period security areas and roles
Working with time periods

14.8.1. Configure time period global settings

Administrators can configure some global settings for the Time Period Management module. This configuration includes updating or deleting existing settings and adding new settings.

To configure time period settings, follow these steps:

1. Click **Time Period Management** > **Administration** > **Settings**.
   The Time Period Settings form opens, in which the Module and Area fields are auto-populated with these values: **Timeperiod**, and **Timeperiod Configuration**.
2. Specify the settings on the form, as described in the following table.

Attribute	Description	Validation
Time period category group-by fields (comma separated)	The fields by which the time period categories are grouped when users creates a new time period definition.	Maximum: 4 fields
Time period category sort-by fields (comma separated)	The fields by which the time period categories are sorted when users creates a new time period definition.	Maximum: 4 fields
Link for copy	The link record used to copy the fields from the current record to the target. The default link is timeperiodDefinition.copy.	None
Copy options for Rules and Occurrences	These are the options when users copy time period definitions: ○ Do not copy Rules/Occurrences ○ Copy Rules only ○ Copy Rules/Occurrences	None
Lock Rules once approved	If this option is set to true, users can't edit or delete their rules anymore when time periods are approved and then return to the first phase of the workflow. Users can only deactivate or activate the rules. If this option is set to false, users can always edit or delete their rules as long as time periods are in the first phase. **Note** Note A time period rule reflects the recurrence of future start and end times of the time period occurrences. A time period rule stores all the recurrence patterns. For more information, see Time period recurrence rules.	None

Maximum number of occurrences before a rule expires	The maximum allowed value for the **Expires after** field in which the user enters the number of recurrences within a time period rule. For more information, see Create a time period.	No greater than 100
Maximum number of months for calculating future occurrences	The number of months to calculate future occurrences.	Maximum: 12 months The default value (12) is recommended for best performance.
Number of occurrences for initial display	The maximum number of occurrences that are displayed when users view a time period definition.	Maximum: 100
Force activity entry for occurrence status changes	If this option is selected, users are required to give an explanation when changing the status of an occurrence. The explanation is logged in the activity log.	None
Disable the Pop-up Window to warn time period conflicts	If this option is set to true, the time period conflicts warning is displayed in system message. If this option is set to false, a window is opened to warn time period conflicts.	None
Send an email notification to the time period owner when the conflicting change is approved	If this option is selected, an email will be sent to the time period owner of a freeze window when the conflicting change is approved.	None

3. Click **Save**.

To add a new time period setting, follow these steps:

1. Click **Time Period Management** > **Administration** > **Settings**.
   The Time Period Settings form opens.
2. Click **More** or the More Actions menu, and select **Add New Setting.**
   The New Setting Information wizard opens.
3. Compete the following fields, and click **Next**.

Field Name	Description	Example Value
Setting Id	Specify an ID for the setting. **Note** Note Once saved, the setting ID becomes non-editable.	categoryGroupBy
Display Label	Enter a display name of the setting.	Time period category sort-by fields
Description	Enter a description of the setting.	
Setting Type	Select a type from the list: ○ Number: a numeric field. ○ String: a String field. ○ Date/Time: a Date/Time field. ○ Boolean: a Boolean field. A checkbox will appear before the display label. ○ Manual List: a list of options. ○ Global List: a global list. ○ Record ○ Condition	String
Mandatory	Indicates if this setting is mandatory.	Not selected

4. If the setting requires validation, enter a validation script; otherwise leave it empty.
5. Click **Finish**.
6. Open **Time Period Settings** from the navigator. The new setting now appears on the form.

To edit or remove an existing time period setting, follow these steps:

1. Click **Time Period Management** > **Administration** > **Settings**.
2. Click **More** or the More Actions menu, and select **Edit Settings.**
   A wizard opens.
3. Click the **Setting Id** of the setting you want to edit or remove.
4. Update the field values as needed, and click **Finish**.
5. If you want to remove the setting, click **Remove This Setting**.

Related topics

Time Period Management
Time Period Management concepts
Time Period Management process
Time period workflow
Time period occurrences
Time period recurrence rules
Time period security areas and roles
Configure a time period workflow
Configure time period categories
Configure time period approvals
Configure time period approval notifications

Calculate time period conflicts
Trigger an approval for time period conflicts
Configure time period conflict notifications for Change Management

14.8.2. Configure a time period workflow

Each time period has a workflow, which is defined in the time period category.

You can find one OOB workflow (named **Timeperiod**) from **Time Period Management** > **Configuration** > **Workflows**.

> **Note**
>
> Note The workflow phases contain special rule sets to ensure that occurrences and rules are managed correctly.

To configure your own time period workflow, follow these steps:

1. Navigate to **Tailoring** > **Process Designer** > **Workflows**.
2. Click **New**.
3. Type a name and description for the workflow, and in the **Table name** field select **timeperiodDefinition**.
4. Click **OK**.
5. Define the phases and transitions for the workflow.
6. Save the workflow.

Related topics

Time Period Management
Time Period Management concepts
Time Period Management process
Time period workflow
Time period occurrences
Time period recurrence rules
Time period security areas and roles
Configure time period global settings
Configure time period categories
Configure time period approvals
Configure time period approval notifications
Calculate time period conflicts
Trigger an approval for time period conflicts
Configure time period conflict notifications for Change Management

14.8.3. Configure time period categories

You use time period categories to manage the data structures of different kinds of time period definitions. You can create new categories, and update or remove existing ones.

Category types

All categories fall into a set of types defined in globallist `timeperiodCategoryTypes` based on the `codes` table. The out-of-the-box types are described in the following table. You can configure the types to suit your needs.

Type	Description
Maintenance Window	Maintenance period.
Neutral	Information period.
Freeze Period	Blackout period.

Category workflows

Each category has an associated workflow, which is created for the `timeperiodDefinition` table. Before creating a time period, make sure you have already created a workflow for it. For details, see Configure a time period workflow.

Category configuration

To create a new time period category, follow these steps:

1. Click **Time Period Management** > **Configuration** > **Time Period Categories**.
   The Time Period Categories search form opens.
2. Click **New**.
   An ID is automatically assigned to the new category.
3. Complete the following fields.

Attribute	Description	Notes
Id	An automatically generated unique ID.	
Type	Select one from the pre-defined types. For example: ○ Maintenance Window ○ Neutral ○ Freeze Period	○ This is a mandatory field. ○ Cannot be changed as long as there are active time periods related to the category.
Name	A display name for the time period category.	○ This is a mandatory field. ○ Cannot be changed as long as there are active time periods related to the category.
Workflow	The workflow that's associated with this category. Time periods of this category will use this workflow.	○ A workflow created for the `timeperiodDefinition` table.
Active	Active flag. If the flag is set to false, the category will not be selectable when users create a new time period.	
Color	The color in which this time period category is displayed in the calendar.	○ Global List with predefined values as HTML color code
Usage Condition	A condition that allows the user an additional filtering on the categories. When a time period is created and this condition evaluates to false, the category is not selectable. To add a usage condition, Click **Edit** and then add a RAD expression in the Condition Editor.	○ Any RAD condition is acceptable. ○ Whether a user can use a category or not is also dependent on the Allowed Categories setting in the user's Security Role. For more information, see Time period security areas and roles.
Description	A description of the category.	

4. Click **Save**.
   The new category is created.

To update an existing category, follow these steps:

> **Note**
>
> Note You can't rename a category if there are active time period definitions linked with it.

1. Navigate to **Time Period Management** > **Configuration** > **Time Period Categories**.
   The Time Period Categories search form opens.
2. Complete some fields to narrow down your search results or leave the fields empty, and then click **Search**.
3. Select the category, and update its field values.
4. Click **Save**.

To delete an existing category, follow these steps:

> **Note**
>
> Note You can delete a category only when there are no active time period definitions linked with it.

1. Navigate to **Time Period Management** > **Configuration** > **Time Period Categories**, and open the category.
2. Click **Delete**.

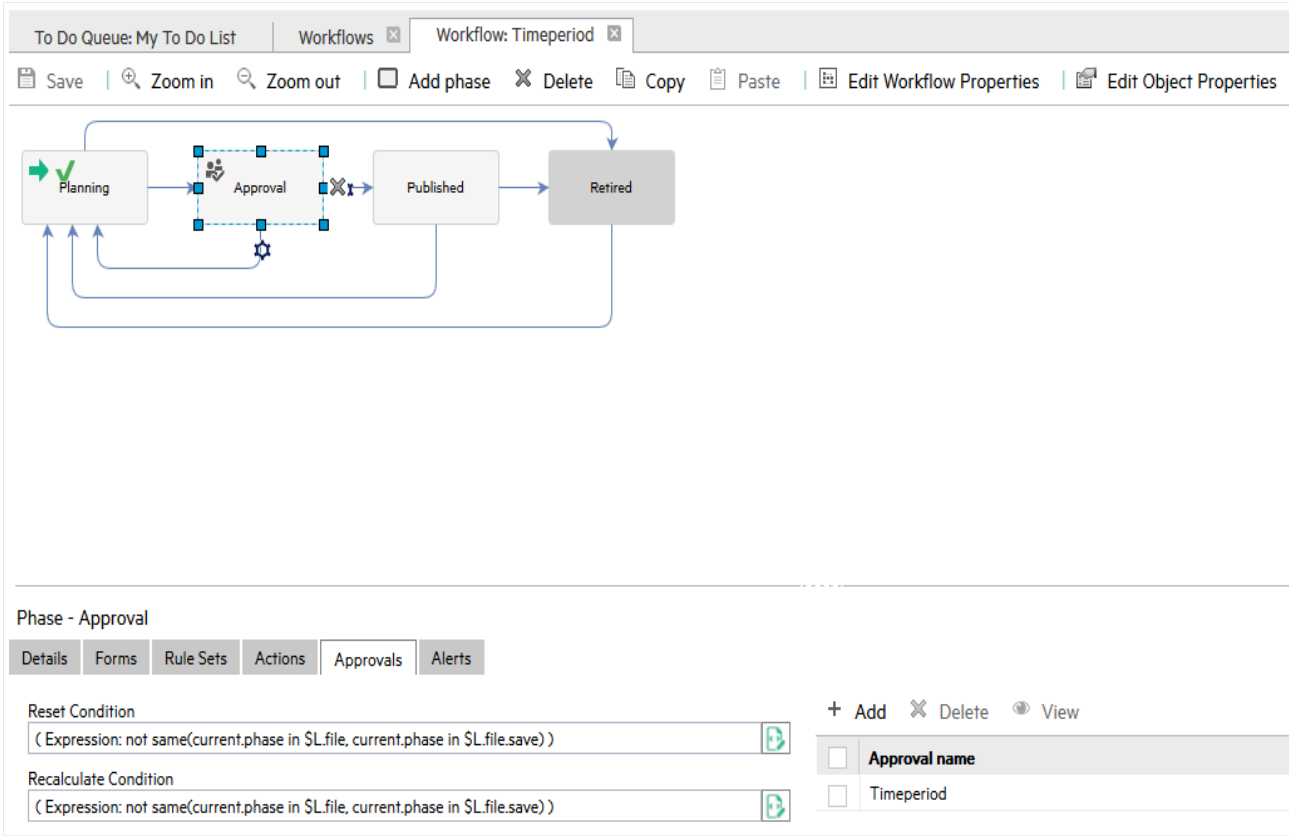
Related topics

Time Period Management
Time Period Management concepts
Time Period Management process
Time period workflow
Time period occurrences
Time period recurrence rules
Time period security areas and roles
Configure time period global settings
Configure a time period workflow
Configure time period approvals
Configure time period approval notifications
Calculate time period conflicts
Trigger an approval for time period conflicts
Configure time period conflict notifications for Change Management

14.8.4. Configure time period approvals

You define approvals for the Time Period module using approval definitions. The out-of-box approval definition is **Timeperiod**, which contains sample configurations only for testing purposes. The Timeperiod approval definition is linked to the Approval phase.

You can update the Timeperiod approval definition according to your needs; you can also create your own approval definitions and link them with their corresponding phase in the workflow.



To configure time period approvals, follow these steps:

1. Navigate to **Time Period Management** > **Configuration** > **Approval Definitions**.
   The timeperiod **Approval Definition** form is displayed. You can update this definition according to your needs. The following example shows how you create a new approval definition and link it to the Approval phase in the workflow.
2. Change the name of the Timeperiod definition, for example, change it to Timeperiod1.
3. Click **Add**.
4. In the **Approval Condition** field, enter a condition expression. For example, a true condition means the approval definition takes effect on all time periods.
5. In the **Approval Type** field, select an approval type.

   Note

   Note For a list of available approval types and their descriptions, see What's an approval type?.

6. In the **Approval Description** field, enter a description of the approval definition.
7. Enter one or more approval groups/operators, as well as their sequence and condition.

Column	Description	Example
Group/Oper	Enter the name of an assignment group or operator. **Note** Note A member of the assignment group will be able to perform the approval. Change Approval and Request Approval groups can't perform time period approvals.	Service Manager or Change.Coordinator
Sequence	The sequence of the approval.	1
Condition	The condition for the approval.	true

8. Save the approval definition.
9. Navigate to **Tailoring** > **Process Designer**, and associate the approval definition to the Approval phase of a time period workflow. For example, associate it to the out-of-box Timeperiod workflow.

Related topics

Time Period Management
Time Period Management concepts
Time Period Management process
Time period workflow
Time period occurrences
Time period recurrence rules
Time period security areas and roles
Configure time period global settings
Configure a time period workflow
Configure time period categories
Configure time period approval notifications
Calculate time period conflicts
Trigger an approval for time period conflicts
Configure time period conflict notifications for Change Management

14.8.5. Configure time period approval notifications

Notifications are generated when a Time Period is added or updated. The time period approval notifications are defined in the `timeperiodDefinition` Object record, and are triggered through the standard approval workflow in Service Manager. You can customize the notifications as needed.

To configure time period approval notifications, follow these steps:

1. Navigate to **Tailoring** > **Document Engine** > **Objects**.
2. In the File name field, type timeperiodDefinition.
3. Click **Search**.
   The object record opens.
4. Click the **Approvals** tab. Refer to the following table for the default notifications configuration.

Notification Name	Notification Record	Description
Approval notification	Timeperiod Approval	Executed when a time period approval happened. ○ An approval message is displayed in the client. ○ Emails are sent to the owner, owning group, delegates and next pending approval group.
Denial notification	Timeperiod Denial	Executed when a time period denial happened. ○ An approval message is displayed in the client. ○ Emails are sent to the owner, and owning group.
Retraction notification	Timeperiod Retract	Executed when a time period approval / denial was retracted ○ An approval message is displayed in the client. ○ Emails are sent to the owner, and owning group.
Final approval notification	Timeperiod Final Approval	Executed when a time period was fully approved. ○ An approval message is displayed in the client. ○ Emails are sent to the owner, and owning group.
Final denial notification	Timeperiod Final Denial	Executed when a time period was fully denied. ○ An approval message is displayed in the client. ○ Emails are sent to the owner, and owning group.

5. Navigate to **Tailoring** > **Notifications** > **Notifications**, and click **Search**.
6. Update the notifications listed above as necessary.

Related topics

Time Period Management
Time Period Management concepts
Time Period Management process
Time period workflow
Time period occurrences
Time period recurrence rules
Time period security areas and roles
Configure time period global settings

Configure a time period workflow
Configure time period categories
Configure time period approvals
Calculate time period conflicts
Trigger an approval for time period conflicts
Configure time period conflict notifications for Change Management

14.8.6. Calculate time period conflicts

Out-of-the-box, Service Manager only calculates time period conflicts for changes or change tasks of certain categories, including Emergency, Normal and Standard. If the start date and the end date are specified or updated without null, the system will calculate the conflicts between the current record and existing time periods when you trigger the save operation. Time period conflict calculation has the following five results available.

- 0: The time period conflict calculation is completed without conflicts.
- 1: The current record conflicts with a Freeze window.
- 2: The current record misses a Maintenance window.
- 3: The current record exceeds a Maintenance window.
- null: The time period conflict calculation is not performed.

You can type d tpconflict in $file in the RAD Debugger and then press **Enter** to view the time period conflict result on a change form or a change task form.

Calculate time period conflicts automatically

The Change Management module is used as an example in the sample task. To calculate the conflicts between a change and a time period, follow these steps:

1. Refer to Enable or disable the calendar to enable the calendar.
2. Refer to Display records from an application module in the calendar to configure the calendar mapping for a change.
3. Refer to Create a time period to create a new time period, and then refer to Approve a time period to approve the newly created time period.
4. Create a change with a category that can be used for time period conflict calculation. Fill in the Scheduled Implementation Start date and Scheduled Implementation End date if the two fields are null. The two date fields are configured in calendar mappings.
5. Fill in all mandatory fields, and then click **Save**. If the change conflicts with an existing time period, the system automatically selects the **Time Period Conflict** checkbox in the change record and displays the following warning message.
   The Change you scheduled conflicts with the existing freeze period or maintenance period.

   Note
   Note If you don't select the **Disable the Pop-up Window to warn time period conflicts**, the time period conflict warning message is displayed in a pop-up window. See Configure time period global settings for more information.
6. Locate and click the calendar to locate the change record, which is displayed with a red or yellow conflict icon. Hover your mouse over the conflict icon of the change record to view the detailed time conflict information.

Calculate time period conflicts manually

Sometimes when you are adding, retiring or updating time periods, or invoking other process, the time period conflicts may be out of synchronization because these updates don't trigger the save operation of a change.

To get the latest conflict status on a change form, click **More** or the **More Actions** icon and then select **Calculate Time Period Conflicts**. After the manual recalculation, the tpconflict flag in $file in the RAD Debugger is updated accordingly.

Note

Note

The **Calculate Time Period Conflicts** option is available only when all the following requirements are met:

- The Service Manager Calendar is enabled.
- The current change is configured for time period conflict calculation. Refer to Display records from an application module in the calendar for more information.
- The current change is in not closed.

Note

Note

Manual time period conflict calculation is not applicable to Problem Management, Incident Management and Interaction.

Related topics

Time Period Management
Time Period Management concepts
Time Period Management process
Time period workflow
Time period occurrences
Time period recurrence rules
Time period security areas and roles
Configure time period global settings
Configure a time period workflow
Configure time period categories
Configure time period approvals
Configure time period approval notifications
Trigger an approval for time period conflicts
Configure time period conflict notifications for Change Management

14.8.7. Trigger an approval for time period conflicts

You can define an additional approval for a module which has time period conflicts and also has the approval functionality. The default approval definition is Approval Depending on Timeperiod Conflict, which is linked to the Approval phase. You can either update the approval definition as necessary or create your own approval definitions and link them to their corresponding phases.

The Change Management module is used as an example in the sample task. Follow these steps to trigger an additional approval for a change record.

1. Click **Change Management** > **Configuration** > **Change Workflows**.
   The workflow list opens.
2. Select a workflow to view its detail. For example, Normal.
3. Click the **Approvals** tab.
4. Add the **Approval Depending on Time period Conflict** approval definition to the corresponding approval phase.
5. Click **Save**.

Note

Note In the **Approval Depending on Time period Conflict** approval definition, the **tpconflict in $L.file>0** condition indicates that an approval is triggered when there is a conflict between a change and an existing time period. You can customize this condition as necessary.

Related topics

Time Period Management
Time Period Management concepts
Time Period Management process
Time period workflow
Time period occurrences
Time period recurrence rules
Time period security areas and roles
Configure time period global settings
Configure a time period workflow
Configure time period categories
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14.8.8. Configure time period conflict notifications for Change Management

When a change with the Conflict with Freeze Window conflict type is approved, a notification is sent to the owner of the conflicted time period (type="Freeze Period") to inform that there is a change scheduled in the freeze period.

The steps to configure time period conflict notifications may vary, depending on the type of change category for which you want to configure.

For legacy change categories such as Hardware and Networks, follow these steps to configure time period conflict notifications:

1. Type db in the Service Manager command line, and then press **Enter**.
2. Type Object, and then press **Enter** to open the Search Object Records form.
3. Type cm3r in the **File name** field, and then click **Search** to open the object record.
4. On the **Approvals** tab, click **Find Related Information** in the **Approval process** field to open the change.approved Process Definition form.
5. On the **Final Javascript** tab, find the JS function that's called by Final Javascript.
6. Update this JS function.
7. Click **Save**.

Note

Note

Alternatively, you can directly update lib.timeperiodConflict.processTpConflictNotification to configure time period conflict notifications.

For new change categories such as Standard, Normal, and Emergency, follow these steps to configure time period conflict notifications:

1. Click **Change Management** > **Configuration** > **Change Workflows**.
2. Select a workflow. For example, Hardware.
3. Select a transition which leads to the approval process, and then add the **chm.notificiton.freeze.timeperiod** rule set to the transition.
4. Click **Save**.

Note

Note

If you don't select **Send an Email notification to the time period owner when the conflicting change is approved**, the system will not send email notification to the time period owner of the conflicted time period (type="Freeze Period"). See Configure time period global settings for more information.

Related topics

Time Period Management
Time Period Management concepts
Time Period Management process
Time period workflow
Time period occurrences
Time period recurrence rules
Time period security areas and roles
Configure time period global settings
Configure a time period workflow
Configure time period categories
Configure time period approvals
Configure time period approval notifications
Calculate time period conflicts
Trigger an approval for time period conflicts

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